TEACHING & REVISION GUIDE · UK PRIMARY CARE · 20 OCTOBER 2026

Primary Care Dermatology for Pharmacist Prescribers

Diagnose, differentiate & manage — the Derm Teach library in print

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Pharmacist independent prescriber · personal study notes

23 chapters · 154 condition pages · exam review appendix · real clinical photographs © DermNet or named holders · AI teaching diagrams are labelled

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Personal study resource, not clinical guidance. Doses, licences and pathways change: always check the current BNF / BNF for Children, NICE CKS, SmPC, MHRA alerts and your local formulary before prescribing.

How to use this edition

This PDF is generated directly from the Derm Teach website, so every condition page has the same 13-part textbook structure as the site: overview, epidemiology, aetiology, step-by-step pathophysiology, clinical features, diagnosis, differentials (with real photographs), management with drug cards and a treatment ladder, red flags and referral, complications, prescriber pearls, a self-check quiz, references and the full photo gallery.

- **Real photographs** carry their holder and source under each image. **AI diagrams** are labelled "AI diagram" and are teaching illustrations only.
- Coloured boxes: **red** = red flags / urgent referral, **green** = prescriber pearls, **blue** = key facts.
- Page references in the contents and cross-links between conditions are clickable.
- Appendix A reproduces the module 2, 4 and 5 exam reviews; appendix B is a one-page red-flag summary for quick revision.

Core prescribing principles for the pharmacist IP

These cross-cutting rules apply to every chapter (0–Z, 1B–2F). Learn them once and the chapters become variations on a theme.

Principle	What it means in practice
Formulate for the patient	Choose the vehicle as carefully as the drug. Ointments suit very dry, thick skin. Creams and gels suit weeping or hairy areas and patients who dislike greasiness. Use lotions, gels, foams or shampoos on the scalp. A product the patient will not use is the wrong product.
The steroid ladder	Match potency to **site** and **severity**. Face, flexures and genitals need lower potency for shorter courses. Thick-skinned palms, soles and chronic plaques tolerate higher potency. Always state the potency, site, frequency and duration.
Infection vs inflammation	Weeping crust on eczema is often inflammation. Bilateral red legs are rarely cellulitis. Antibiotics are not automatic.
Antimicrobial stewardship	No antibiotic monotherapy in acne. Use short, defined courses for skin and soft-tissue infection, avoid repeated topical fusidic acid, and set a review date for every course.
Skin of colour	Erythema may look violaceous or brown, or be hard to see. Judge severity by scale, induration, swelling, warmth, pigment change and symptoms, not redness alone. Post-inflammatory pigment change is often the patient's main concern.
Psychosocial impact	Acne, psoriasis and eczema carry real burdens on mood, sleep and self-esteem. Ask about them, record them, and treat or refer.
Neonates & infants	Ask first: "Is this baby well?" Thin, absorptive skin: mild potency, small quantities, short courses. Check licences and doses in the **BNF for Children**. Complicated haemangiomas, suspected EB and deep aplasia cutis are **recognise-and-refer** conditions.
Older people	Use a STOPP/START lens: avoid sedating, anticholinergic antihistamines; check renal function, falls risk and polypharmacy; suspected skin cancer goes via NICE NG12.
Aesthetics	Counsel, safety-net and refer. Face-to-face examination is required before prescribing non-surgical cosmetic medicines (GPhC). Any visual change or dusky/blanched skin after filler is an emergency.
Skin lesions & surgery	Use the NICE 7-point checklist and NG12 routes for suspected skin cancer; never partially remove a pigmented lesion; send every specimen for histology and track the result. Know the BNF lidocaine maximum (200 mg plain, 500 mg with adrenaline) and that most anticoagulants continue for minor skin surgery (BSDS).
Check before you sign	Current BNF / BNFC, SmPC, MHRA alerts and local formulary / microguide: licence, age limits, pregnancy, interactions.

Topical corticosteroid potency: worked examples used in this guide

Potency	Example(s) in this guide	Typical primary-care use
Mild	Hydrocortisone 1%	Mild eczema, face/flexures, short adjunct with antifungal for inflamed tinea (max 7 days)
Moderate	Clobetasone butyrate 0.05%; betamethasone valerate 0.025%	Moderate eczema; short courses on face/flexures for severe flares; flexural psoriasis
Potent	Betamethasone valerate 0.1%	Severe eczema on trunk/limbs; first-line (with vitamin D analogue) for trunk/limb and scalp psoriasis

Potency	Example(s) in this guide	Typical primary-care use
Very potent	Specialist territory	Generally specialist-led. Never in children without specialist advice. Psoriasis: 4 weeks per site.

Teach the **fingertip unit (FTU)** for every topical steroid prescription, so the patient knows how much to apply and the quantity you prescribe matches the area treated (check FTU tables in the BNF).

Application counselling that applies to every chapter

- Emollients go on liberally, in the direction of hair growth, ideally after bathing while the skin is still damp.
- Leave a gap between emollient and active topical, so the active is not diluted or spread to the wrong site. Follow local advice on the interval.
- **Fire risk:** paraffin-based (and other) emollients soak into clothing and bedding, which then catch fire easily. Warn patients about naked flames and smoking.
- Agree a clear stop or step-down plan. Under-treatment and abrupt stops cause rebound as often as over-use causes harm.

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CHAPTER 1

Foundations: describing skin

Start here: the language of dermatology (primary and secondary lesions, configuration, distribution), skin layers, Fitzpatrick skin types and why dermoscopy helps.

1. Describing skin lesions: the language of dermatology
2. Skin layers, Fitzpatrick skin types & dermoscopy

Foundations: describing skin Module 1 · Week 1

Describing skin lesions: the language of dermatology

The core vocabulary for describing a rash: primary lesions, secondary changes, configuration and distribution. A precise description drives the differential and makes referrals and photo requests useful.

ALSO CALLED primary lesions · secondary lesions · macule · papule · plaque · vesicle · +7

RELATED PAGES Skin layers, Fitzpatrick skin types & dermoscopy · Teledermatology & lesion photography · Annular (ring-shaped) rashes: differential · Approach to blistering diseases

KEY FACTS

- Describe: **site, number, distribution, configuration, size, shape, colour, surface, palpation** (SCALP-style).
- Primary lesions: macule/patch (flat), papule/plaque (raised, solid), nodule (deep), vesicle/bulla (fluid), pustule (pus), wheal (transient oedema).
- Secondary changes: scale, crust, erosion, ulcer, excoriation, lichenification, atrophy, scar.
- Configurations: annular, linear (Koebner, Blaschko), grouped (herpetiform), target, reticulate, serpiginous.
- Size thresholds: macule/papule under 1 cm (some use 0.5 cm); vesicle under 0.5 cm vs bulla.

Overview

This page is a **reference for dermatological terms**: how to describe lesions precisely so that a description alone points to a diagnosis.

Primary lesions (macule cyst)

A **primary lesion** is the initial morphological change produced directly by the disease process, before scratching, infection, treatment or time alter it. Correctly naming it is the first step to a differential diagnosis.

TERM	DEFINITION (PARAPHRASED; DERMNET TERMINOLOGY)	PALPABLE?	SIZE CUT-OFF
Macule	Flat, circumscribed change in colour/texture only	No	<1 cm
Patch	Flat area of colour/texture change (a large macule)	No	>1 cm
Papule	Solid, raised, palpable lesion	Yes	<1 cm
Plaque	Raised, palpable, flat-topped area, wider than it is tall (papules may coalesce into plaques)	Yes	>1 cm
Nodule	Solid, palpable lump, often deeper (dermis/subcutis) — may be felt more than seen	Yes	>1 cm (some: >0.5 cm)

TERM	DEFINITION (PARAPHRASED; DERMNET TERMINOLOGY)	PALPABLE?	SIZE CUT-OFF
Vesicle	Small fluid-filled (clear) blister	Yes	<1 cm
Bulla	Large fluid-filled blister	Yes	>1 cm
Pustule	Visible collection of pus (neutrophils ± debris) — can be infective or sterile	Yes	Usually <1 cm
Wheal (weal)	Transient, itchy, oedematous, pale-centred/pink papule or plaque from dermal oedema; individual lesions last <24 h	Yes	Any
Cyst	Closed sac lined by epithelium containing fluid or semi-solid material (e.g. keratin)	Yes (fluctuant/ firm)	Any

Related terms worth knowing: **purpura/petechiae** (non-blanching bleeding into skin: petechiae <2–3 mm, purpura larger, ecchymosis = bruise), **telangiectasia** (visible dilated superficial vessels), **burrow** (scabies), **comedone** (blocked follicle: open/blackhead, closed/whitehead), **abscess** (deep localised pus collection), **erythema** (blanching redness), **tumour** (large nodule — avoid alarming patients with the word).

Secondary lesions (scale scar)

Secondary lesions develop from primary lesions over time or because of external factors — scratching, rubbing, infection, healing or treatment. They tell you about **chronicity, itch, and depth of tissue damage**.

TERM	DEFINITION (PARAPHRASED)	DEPTH / KEY POINT
Scale	Visible flakes of excess or abnormally shed stratum corneum (hyperkeratosis/parakeratosis)	Epidermal surface
Crust	Dried serum, blood or pus on the skin surface ("scab")	Surface — often after vesicles/pustules/ erosions break
Erosion	Partial loss of epidermis (dermis not breached)	Heals **without** scarring
Ulcer	Full-thickness loss of epidermis plus part of dermis ± deeper	Heals **with** scarring
Fissure	Linear crack or split through epidermis, often into dermis	Painful; thick/dry skin over joints, heels, angles of mouth
Lichenification	Thickened, leathery skin with exaggerated normal skin markings from chronic rubbing/scratching	Epidermal hyperplasia (acanthosis) + hyperkeratosis
Atrophy	Thinning of epidermis (fine wrinkling, translucency, visible vessels) and/or dermis/subcutis (depression)	Steroid atrophy, striae, lichen sclerosus, ageing
Excoriation	Linear or punctate erosions/ulcers caused by scratching or picking	Marker of itch (or skin picking)
Scar	Fibrous tissue replacing normal dermis after injury that breached the dermis	Atrophic, hypertrophic (within wound) or keloid (beyond wound)

Other secondary/descriptive terms: **maceration** (soggy white skin in occluded folds/webs), **xerosis** (dryness), **post-inflammatory hyper-/hypopigmentation**, **eschar** (black adherent necrotic crust).

Distribution & configuration

- **Distribution** = where lesions are on the body and how widespread.
- **Configuration (arrangement/shape)** = how individual lesions are shaped or grouped relative to one another.

Lichenification in chronic atopic eczema

Case: an 8-year-old boy with longstanding atopic dermatitis, worst in the **flexures**, troubled by itch and poor sleep. Image shows thickened skin, accentuated skin creases and hyperpigmentation.

Q	CORRECT ANSWER	WHY (SHORT EXPLANATION)
Q1 — term for thickened skin, accentuated creases, hyperpigmentation	**Lichenification**	Secondary change from chronic rubbing/scratching; epidermal thickening (acanthosis/hyperkeratosis) exaggerates normal skin lines. Distractors: atrophy = thinning; excoriation = scratch erosions; PIH = colour change only (flat); xerosis = dryness without thickening
Q2 — where are spongiotic changes?	**Epidermis**	Spongiosis = intercellular oedema between keratinocytes (stretched desmosomes give a "sponge" look) — hallmark of eczema/dermatitis. Not basement membrane, dermis, stratum corneum alone, or subcutis
Q3 — primary lesions of eczema herpeticum	**Vesicles**	HSV-1 spreading through eczematous skin produces clustered monomorphic vesicles/pustules that rupture into punched-out erosions crusts. Crusts, erosions, ulcers are **secondary**; bullae are not typical; fissures belong to chronic dry eczema

Aetiology: causes and risk factors

Primary lesions (macule cyst)

> **MACULE/PATCH**
> change only in pigment or vascularity without thickening — more/less melanin (freckle, lentigo, café-au-lait; vitiligo) or vascular change (erythema, port-wine stain).

> **PAPULE/PLAQUE**
> epidermal thickening (acanthosis, hyperkeratosis) and/or dermal inflammatory infiltrate (psoriasis, lichen planus, eczema) or cellular proliferation.

> **NODULE**
> deeper dermal/subcutaneous infiltrate, proliferation or deposit (tumour, granuloma, prurigo).

> **VESICLE/BULLA**
> fluid accumulates within or beneath the epidermis when cell–cell (desmosome) or cell–basement-membrane (hemidesmosome) adhesion fails, or when keratinocytes are damaged (viral ballooning, spongiosis). - **Pustule:** neutrophils collect in the epidermis or follicle — bacteria (folliculitis, impetigo), or sterile inflammation (pustular psoriasis, acne, PPP).

> **WHEAL**
> mast-cell degranulation (histamine) transient vasodilation and leaky superficial dermal capillaries dermal oedema without epidermal change (hence no scale, resolves without mark).

> **CYST**
> epithelial-lined cavity, e.g. epidermoid cyst from infundibular epithelium producing keratin.

Secondary lesions (scale scar)

> **SCALE**
> accelerated epidermal turnover (psoriasis — keratinocytes reach the surface in days and retain nuclei = parakeratosis) or abnormal desquamation (ichthyosis, eczema, tinea).

> **CRUST**

serous exudate from damaged epidermis dries on the surface; golden/honey colour suggests S. aureus/ streptococcal impetigo.

EROSION/ULCER
loss of a blister roof, trauma, ischaemia (arterial), venous hypertension, neuropathy (diabetic foot), infection or malignancy. Depth relative to the basement membrane decides scarring.

FISSURE
dry, inelastic thickened stratum corneum splits under movement/tension.

LICHENIFICATION
repeated mechanical trauma (rubbing/scratching) keratinocyte proliferation, thickened epidermis and dermal fibrosis — the visible footprint of the **itch–scratch cycle**.

ATROPHY
reduced keratinocyte proliferation and collagen synthesis (potent topical/systemic corticosteroids, ageing, photo-damage) or inflammatory destruction (lichen sclerosus, discoid lupus).

EXCORIATION
scratching removes epidermis; may be the only visible sign in itch without rash (e.g. cholestasis, iron deficiency, scabies).

SCAR
fibroblasts lay down collagen to repair dermis; excess collagen hypertrophic/keloid; insufficient atrophic (acne, varicella).

Distribution & configuration

Pattern reflects mechanism: nerve (dermatomal = reactivated VZV in a sensory ganglion), external contact (sharp lines, geometric shapes = contact/irritant/artefact), light (photo-exposed sites with sparing under chin/behind ears), trauma (Koebner), blood/lymph vessels, embryological lines (Blaschko), friction/moisture (flexures), centrifugal fungal spread (annular).

Lichenification in chronic atopic eczema

Barrier dysfunction (e.g. filaggrin loss-of-function, reduced ceramides) + type-2 inflammation (IL-4/IL-13/IL-31 itch signalling) itch scratch further barrier damage and inflammation keratinocyte hyperproliferation **lichenification**. Pigment change follows inflammation (PIH), more marked in darker phototypes.

Myths to address

- **"RASH" IS A DESCRIPTION**
  precise terms change the differential diagnosis.

Lesions in cross-section

Each primary lesion reflects **what sits at which depth** of the skin.

1 **Flat vs raised**

Macules and patches are colour change only; papules, plaques and nodules are solid and raised, deeper as they grow.

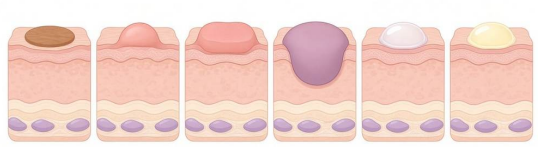

Fig. 1 · Primary lesions in profile: macule, papule, plaque, nodule, vesicle and pustule.

AI-generated illustration

2 **Fluid**

Vesicles and bullae hold clear fluid; pustules hold pus.

Clinical features

Primary lesions (macule cyst)

PRIMARY LESION	EXAMPLE DIAGNOSES (2–3)	TEACHING CLUE
Macule	Freckles (ephelides); lentigo; flat drug-eruption/viral exanthem macules	Close eyes and run a finger over it — no change
Patch	Vitiligo; café-au-lait patch; solar lentigo; pityriasis versicolor	6 café-au-lait >0.5 cm (child)/ >1.5 cm (adult) consider NF1
Papule	Lichen planus (flat-topped, polygonal, purple); molluscum contagiosum (umbilicated); acne papules; viral warts	Look at the top: flat, dome, umbilicated, scaly, warty
Plaque	Chronic plaque psoriasis; discoid (nummular) eczema; tinea corporis; mycosis fungoides (rare, persistent)	Psoriasis = well-demarcated + silvery scale + extensors
Nodule	Nodular basal cell carcinoma; nodular prurigo; dermatofibroma; nodular melanoma/SCC; erythema nodosum	Firm, persistent, growing, bleeding nodule think malignancy
Vesicle	Herpes simplex/eczema herpeticum; varicella/zoster; pompholyx; acute contact dermatitis	Grouped vesicles on red base = herpes until proven otherwise
Bulla	Bullous pemphigoid (tense); bullous impetigo; burns/friction blister; pemphigus (flaccid, rapidly erode); bullous drug reaction	Tense vs flaccid tells you the split level
Pustule	Bacterial folliculitis; acne vulgaris; palmoplantar pustulosis / pustular psoriasis (sterile); rosacea	A pustule is not automatically infection
Wheal	Urticaria (spontaneous/inducible); angio-oedema overlap; insect bites (papular urticaria); dermographism	Ask: do individual lesions last <24 h and leave no mark?
Cyst	Epidermoid cyst (central punctum); pilar (trichilemmal) cyst on scalp; acne cysts; milia (tiny keratin cysts)	Punctum + cheesy keratin = epidermoid

Secondary lesions (scale scar)

SECONDARY LESION	EXAMPLE DIAGNOSES (2–3)
Scale	Psoriasis (thick silvery); pityriasis rosea (collarette); tinea (scaly active edge); seborrhoeic dermatitis (greasy yellow)
Crust	Impetigo (honey-coloured); eczema herpeticum (haemorrhagic crusts after vesicles); infected eczema
Erosion	Bullous impetigo; pemphigus vulgaris (flaccid blisters rupture); erosive oral lichen planus; burst vesicles of herpes
Ulcer	Venous leg ulcer (gaiter area); diabetic neuropathic foot ulcer (pressure points); ulcerated BCC/SCC; pyoderma gangrenosum
Fissure	Hand/foot eczema; juvenile plantar dermatosis; angular cheilitis; cracked heels; eczema craquelé
Lichenification	Chronic atopic eczema (flexures — quiz case); lichen simplex chronicus (nape, shins, ankles, genitals); chronic contact dermatitis
Atrophy	Topical steroid atrophy and striae; lichen sclerosus; striae distensae; discoid lupus (with scarring)
Excoriation	Atopic eczema; scabies; nodular prurigo; skin-picking disorder / acne excoriée
Scar	Acne scarring (atrophic); hypertrophic surgical/burn scar; keloid (earlobe, chest, shoulders)

Distribution & configuration

Distribution terms

TERM	MEANING	CLASSIC EXAMPLES
Localised / generalised / widespread	Extent	—
Symmetrical	Both sides alike — often endogenous	Psoriasis, atopic eczema, vitiligo
Asymmetrical / unilateral	Often exogenous or infective	Tinea, contact dermatitis, zoster
Flexural	Skin folds (antecubital/popliteal fossae, axillae, groin, inframammary)	Atopic eczema (children/adults), flexural (inverse) psoriasis, intertrigo, candida
Extensor	Outer surfaces of elbows/knees	Plaque psoriasis; infantile atopic eczema; dermatitis herpetiformis
Dermatomal	Band in one sensory nerve territory, not crossing midline	Herpes zoster
Acral	Hands, feet, ears, nose	Erythema multiforme, chilblains, hand-foot-mouth
Photo-distributed	Face, V of neck, dorsal hands, forearms; sparing shaded areas	Polymorphic light eruption, drug photosensitivity, lupus
Seborrhoeic	Scalp, eyebrows, nasolabial folds, ears, central chest	Seborrhoeic dermatitis
Follicular	Centred on hair follicles	Folliculitis, keratosis pilaris
Intertriginous	Opposed skin surfaces	Candidal intertrigo, erythrasma

Configuration terms

TERM	MEANING	EXAMPLES
Annular	Ring with clearer centre	Tinea corporis (scaly edge), granuloma annulare (smooth edge), urticaria, erythema migrans
Arcuate / polycyclic	Part-rings / merged rings	Urticaria, subacute cutaneous lupus
Target (iris)	Concentric zones (three in typical targets)	Erythema multiforme
Discoid / nummular	Coin-shaped, uniformly involved	Discoid eczema
Linear	In a line	Koebner (psoriasis, lichen planus, vitiligo, warts), contact with plants (phytophotodermatitis), excoriations, Blaschko-linear naevi
Grouped / herpetiform	Clustered	Herpes simplex, dermatitis herpetiformis
Reticulate	Net-like	Livedo reticularis, Wickham striae (oral LP)
Serpiginous	Snake-like, wavy	Cutaneous larva migrans
Satellite lesions	Outlying lesions near a main patch	Candida
Confluent / coalescing	Lesions merging	Viral exanthems, psoriasis
Guttate	Drop-like	Guttate psoriasis (post-streptococcal)

Lichenification in chronic atopic eczema

- Age pattern: infants — face/cheeks, scalp, extensors (napkin often spared); children — **flexures** (antecubital/popliteal fossae, wrists, ankles, neck); adults — flexures, hands, face/neck, generalised.
- Chronic phase: lichenification, excoriations, xerosis, fissures, PIH/hypopigmentation; acute phase: erythema, papulovesicles, weeping, crust.
- Sleep disturbance = marker of severity and impact (POEM, CDLQI).

Fig. 2 · PLAQUE — raised, flat-topped, >1 cm (broader than tall). Well-demarcated salmon-pink plaques with silvery scale = chronic plaque psoriasis.

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Types to recognise

Macule

Freckles (ephelides); lentigo; flat drug-eruption/viral exanthem macules Site: Close eyes and run a finger over it — no change.

Patch

Vitiligo; café-au-lait patch; solar lentigo; pityriasis versicolor Site: 6 café-au-lait >0.5 cm (child)/ >1.5 cm (adult) consider NF1.

Papule

Lichen planus (flat-topped, polygonal, purple); molluscum contagiosum (umbilicated); acne papules; viral warts Site: Look at the top: flat, dome, umbilicated, scaly, warty.

Plaque

Chronic plaque psoriasis; discoid (nummular) eczema; tinea corporis; mycosis fungoides (rare, persistent) Site: Psoriasis = well-demarcated + silvery scale + extensors.

Nodule

Nodular basal cell carcinoma; nodular prurigo; dermatofibroma; nodular melanoma/SCC; erythema nodosum Site: Firm, persistent, growing, bleeding nodule — think malignancy.

Vesicle

Herpes simplex/eczema herpeticum; varicella/zoster; pompholyx; acute contact dermatitis Site: Grouped vesicles on red base = herpes until proven otherwise.

Bulla

Bullous pemphigoid (tense); bullous impetigo; burns/ friction blister; pemphigus (flaccid, rapidly erode); bullous drug reaction Site: Tense vs flaccid tells you the split level.

Pustule

Bacterial folliculitis; acne vulgaris; palmoplantar pustulosis / pustular psoriasis (sterile); rosacea Site: A pustule is not automatically infection.

Wheal

Urticaria (spontaneous/ inducible); angio-oedema overlap; insect bites (papular urticaria); dermographism Site: Ask: do individual lesions last <24 h and leave no mark?.

Cyst

Epidermoid cyst (central punctum); pilar (trichilemmal) cyst on scalp; acne cysts; milia (tiny keratin

Diagnosis and investigations

Primary lesions (macule cyst)

- Inspect in good light with magnification; palpate (wear gloves where appropriate).
- Decide **flat vs raised vs fluid-filled vs pus-filled vs deep**; then measure (ruler/photo with scale) to choose the correct size-based term.
- Use **diascopy** (press a glass slide): blanching = erythema/vascular dilation; non-blanching = purpura.
- Document primary lesion first, then the secondary change (e.g. "erythematous plaques with silvery scale").

Secondary lesions (scale scar)

- Describe in order: **primary lesion + secondary change** (e.g. "lichenified, excoriated plaques in both antecubital fossae").
- Gently lift scale/crust — bleeding points under psoriatic scale (Auspitz sign; not routinely elicited).
- Assess ulcer: site, size, edge (sloping = venous; punched-out = arterial/neuropathic; rolled = BCC; undermined violaceous = pyoderma), base, exudate, surrounding skin, pulses/ABPI (per local wound pathway).
- Atrophy: shiny, wrinkled "cigarette-paper" surface, visible vessels; ask about steroid use, potency, duration.

Distribution & configuration

Undress adequately and inspect the whole skin (plus scalp, nails, mouth) — distribution clues are missed in partial examinations. Note sparing as well as involvement (e.g. napkin-area sparing in infant atopic eczema; under-chin sparing in photosensitivity).

Lichenification in chronic atopic eczema

Clinical diagnosis (NICE CG57 / UK Working Party criteria: itchy skin + 3 of flexural involvement history, asthma/hay fever history, generally dry skin, onset <2 years, visible flexural eczema). Assess severity (clear/mild/moderate/severe — NICE) noting lichenification and excoriation as features of moderate–severe disease; check for infection (bacterial and herpetic).

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
Macule vs patch	Size (<1 cm vs >1 cm) — both flat
Papule vs nodule	Size and depth; nodules are deeper/firmer
Plaque vs patch	Plaque is palpable (thickened); patch is flat
Vesicle vs bulla	Size (<1 cm vs >1 cm)
Pustule vs vesicle	Contents: yellow/white pus vs clear fluid (vesicles may become pustular with time)
Wheal vs plaque	Wheals are transient (<24 h), shifting, no surface change
Cyst vs nodule/lipoma	Punctum, fluctuance, epithelial lining; lipoma is soft, rubbery, no punctum
Erosion vs ulcer	Depth — erosion is epidermal only (no scar); ulcer reaches dermis (scar)

CONDITION	DISTINGUISHING FEATURES
Scale vs crust	Scale = keratin flakes (white/silver/yellow); crust = dried exudate (honey, brown, haemorrhagic)
Lichenification vs plaque	Lichenification = secondary thickening with exaggerated skin lines from rubbing; plaque is the primary raised lesion
Lichenification vs PIH	Both often coexist in skin of colour; PIH is colour only (flat), lichenification is palpable thickening
Excoriation vs fissure	Excoriation = linear scratch marks (patient-made); fissure = crack from dryness/tension
Hypertrophic scar vs keloid	Keloid extends beyond wound and rarely regresses; hypertrophic stays within and may improve
Atrophy vs atrophic scar	Atrophy can be reversible (steroid epidermal thinning early); scars are permanent dermal repair

Distribution & configuration

- Annular + scale at edge + asymmetry → **tinea** (scrape before steroids); annular smooth, no scale → **granuloma annulare**; annular evanescent → **urticaria**.
- Flexural red, well-demarcated, little scale → **inverse psoriasis**; flexural with satellite pustules → **candida**; flexural lichenified → **atopic eczema**.
- Dermatomal pain before rash → **zoster** (early antivirals).

Lichenification in chronic atopic eczema

Lichen simplex chronicus (localised, adult, no atopy needed); psoriasis (well-demarcated, silvery scale, extensors); scabies (burrows, family itch, web spaces); contact dermatitis; tinea (asymmetric, scaly edge).

Management

Primary lesions (macule → cyst)

- The primary lesion steers the **differential** and therefore whether you can treat (e.g. plaque psoriasis, acne papules, urticarial wheals) or must **refer** (new growing nodule, unexplained bullae, rapidly evolving pigmented patch).
- Bullae, widespread pustules with systemic upset, or eczema herpeticum vesicles → same-day medical review (not OTC management).
- Suspicious lesions: follow NICE NG12 suspected-cancer recognition and local 2WW/urgent pathways (melanoma, SCC); BCC usually routine unless high-risk site/behaviour.

Secondary lesions (scale → scar)

> **SCALE**
> keratolytics (e.g. salicylic acid preparations, urea) and emollients; choose ointments for thick scale.

> **CRUST (IMPETIGO)**
> hydrogen peroxide 1% cream for localised non-bullous impetigo per NICE NG153; topical/oral antibiotics per extent, local guidance. (Pharmacy First in England covers impetigo in eligible patients — check current service specification.)

EROSIONS/ULCERS

wound care, infection review, **refer** unexplained erosions/ulcers (pemphigus, malignancy, non-healing >2 weeks/ suspicious edge). Leg ulcers local leg-ulcer/tissue-viability pathway; diabetic foot ulcer urgent foot protection/MDT service.

FISSURES

greasy emollients, occlusion overnight, treat underlying eczema; angular cheilitis — consider candida/staph cover, dentures, iron/B12/folate.

LICHENIFICATION

needs **potent** topical corticosteroid for adequate duration under emollient regimen (ointment, occlusion may help), break itch–scratch cycle; refer if refractory.

ATROPHY

prevention — right potency, right site, right duration, fingertip units; avoid potent steroids on face/flexures/ genitals unless specialist directed.

EXCORIATION

treat the itch cause (eczema, scabies, systemic causes); keep nails short; consider infection.

SCAR/KELOID

counsel early treatment of inflammatory acne; keloid refer (intralesional steroid/specialist).

Distribution & configuration

Pattern changes management: e.g. **zoster** (antiviral within 72 h, ophthalmic branch same-day eye review), **tinea** (antifungal; avoid steroid-alone "tinea incognito"), **photo-distributed** (review photosensitising drugs), **contact pattern** (identify/avoid trigger; patch test referral).

Lichenification in chronic atopic eczema

Management for pharmacist/IP (UK — NICE CG57 / CKS)

Emollients liberally and often (250–500 g/week for a child, as soap substitute; leave-on).

TOPICAL CORTICOSTEROID BY SEVERITY AND SITE

lichenified flexural areas in a child usually need a **moderate** (or short-course potent, per CKS/specialist guidance on limbs, not face) steroid once or twice daily until clear, then step down; consider weekend "proactive" maintenance for frequent flares. Mild potency for face/neck (moderate short courses only if severe).

Topical calcineurin inhibitors (tacrolimus/pimecrolimus) as second line for face/flexures in children 2 years when steroids unsuitable — prescriber/specialist initiation per local policy.

ITCH AND SLEEP

sedating antihistamine short-term only (6 months; not routine), cool environment, cotton clothing, short nails, bandaging/wet wraps under specialist advice.

INFECTION

bacterial — see NICE NG190 (secondary bacterial infection of eczema); **eczema herpeticum** **same-day referral**, start aciclovir per guidance (ophthalmology if periocular).

Refer if uncontrolled despite optimal therapy, diagnostic doubt, recurrent infections, significant sleep/ psychosocial impact, or suspected contact allergy.

Referral

SAME DAY	**Non-blanching** spots or purpura in an unwell patient: think meningococcal sepsis or vasculitis and escalate the same day.
ROUTINE	Blisters or erosions with **mouth, eye or genital** involvement: SJS/TEN or immunobullous disease.
SAME DAY	Punched-out erosions on eczema: **eczema herpeticum**, same-day assessment.
ROUTINE	A **new or changing pigmented lesion**: refer on the suspected-cancer pathway (NICE NG12).

Red flags and when to refer

- **Non-blanching** spots or purpura in an unwell patient: think meningococcal sepsis or vasculitis and escalate the same day.
- Blisters or erosions with **mouth, eye or genital** involvement: SJS/TEN or immunobullous disease.
- Punched-out erosions on eczema: **eczema herpeticum**, same-day assessment.
- A **new or changing pigmented lesion**: refer on the suspected-cancer pathway (NICE NG12).

Complications and prognosis

Complications

- Miscommunication in referrals and teledermatology.

Prognosis

- Not applicable.

Prescriber pearls (UK pharmacist IP)

- Write the description **before** you write the diagnosis — it protects against anchoring.
- Use plain language with patients ("small raised spots", "flat brown mark") but precise terms in notes and referrals.
- In skin of colour, erythema may look violaceous, grey, dark brown or be subtle — rely more on texture, warmth and palpation.

Secondary changes tell you **chronicity**: lichenification and atrophy take weeks–months to develop and to reverse — set expectations.

Crust or weeping on eczema = think infection (bacterial **or** herpetic). Punched-out erosions/ monomorphic vesicles = eczema herpeticum urgent same-day review.

Explain atrophy risk honestly but avoid "steroid phobia": under-treatment drives lichenification.

Ask "where did it start and how did it spread?" Geometric or "outside-in" shapes suggest something external. Bilateral symmetrical = more likely endogenous.

Show the family the **difference between lichenification and PIH**: thick skin should soften with adequate steroid + emollient over weeks; pigment change takes months and is not a sign of steroid damage.

Lichenified skin is less permeable — ointments and adequate potency/duration are needed; under-treatment due to steroid fear prolongs the itch–scratch cycle.

Fingertip-unit teaching; written eczema action plan; "steroids first, then emollient after ~15–30 min" (order matters less than adequate quantities).

Red flag script: "If the eczema suddenly becomes painful with clusters of small blisters or punched-out sores, or your child has a fever, seek same-day help."

Primary lesions (macule cyst)

Secondary lesions (scale scar)

Distribution & configuration

Lichenification in chronic atopic eczema

Self-check quiz

5 questions · optional

Q1 **When should you think of Describing skin lesions: the language of dermatology?**

✓ You need to describe a rash in notes, a referral, an advice-and-guidance request or a photo request.

Q2 What drives it?

✓ The level of the skin involved (epidermis, dermis, subcutis) decides what you see and feel.

Q3 What is the backbone of care?

✓ Describe lesion type colour surface configuration distribution, then build the differential from that description.

Q4 What must you never miss?

✓ Non-blanching purpura, blisters or erosions with mucosal involvement, and any changing pigmented lesion.

Q5 Name a red flag that needs escalation.

✓ **Non-blanching** spots or purpura in an unwell patient: think meningococcal sepsis or vasculitis and escalate the same day.

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Photo gallery

Primary lesions 16

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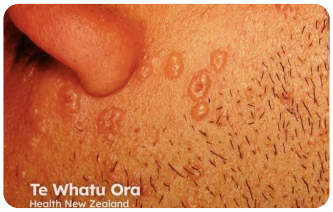

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Secondary lesions 14

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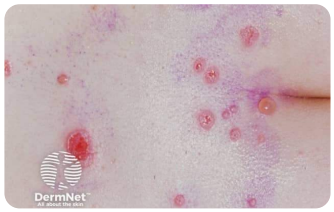

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Configuration and distribution

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Lichenification in atopic eczema

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Red flag: eczema herpeticum

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Foundations: describing skin Module 1 · Week 1

Skin layers, Fitzpatrick skin types & dermoscopy

How the layers of the skin explain each lesion type, why Fitzpatrick skin type matters for sunburn risk, pigment change and treatment, and what dermoscopy adds to naked-eye examination.

ALSO CALLED skin structure · epidermis · dermis · histology · Fitzpatrick · phototype · +3

RELATED PAGES Describing skin lesions: the language of dermatology · Hypopigmentation & skin of colour · Melanoma: recognition & referral · Polymorphic light eruption

KEY FACTS

- Skin layers: **epidermis** (keratinocytes, melanocytes, Langerhans cells, Merkel cells), **dermis** (collagen, vessels, nerves, adnexa) and **subcutis** (fat).
- Epidermis renews about every **28 days**: basal spinous granular stratum corneum.
- **Fitzpatrick types I–VI** describe burning vs tanning: I always burns, never tans; VI never burns.
- Fitzpatrick type predicts sunburn and skin-cancer risk, and risk of pigment change, keloids and procedure complications.
- The depth of a process explains colour: superficial pigment looks brown, deep pigment blue-grey (Tyndall).

Overview

This page covers **skin structure and skin typing**: the basis for interpreting lesions, dermoscopy and treatment risk.

Skin layers linked to lesion type

Skin = **epidermis** (stratified squamous keratinising epithelium) **basement membrane zone (dermo-epidermal junction)** **dermis** (papillary and reticular; collagen, elastin, vessels, nerves, appendages) **subcutis** (fat). DermNet: The structure of normal skin.

LAYER	KEY CELLS/STRUCTURES	FUNCTION
Stratum corneum	Flattened anucleate corneocytes in a lipid "bricks and mortar" matrix	Barrier to water loss, irritants, microbes
Stratum granulosum	Keratohyalin granules (incl. filaggrin precursor), lamellar bodies secreting lipids	Builds barrier lipids; filaggrin natural moisturising factor
Stratum spinosum	Keratinocytes joined by **desmosomes** ("prickles"); Langerhans cells	Mechanical strength; immune surveillance
Stratum basale	Dividing basal keratinocytes, **melanocytes** (~1 per 10 basal cells), Merkel cells; anchored by **hemidesmosomes**	Renewal (~4 weeks transit to surface); pigment

LAYER	KEY CELLS/STRUCTURES	FUNCTION
Basement membrane zone	Lamina lucida/densa, anchoring fibrils (type VII collagen), BP180/BP230 antigens	Glues epidermis to dermis
Dermis	Fibroblasts, collagen/elastin, blood vessels, lymphatics, nerves, mast cells, hair follicles, sebaceous/eccrine/apocrine glands	Strength, elasticity, thermoregulation, sensation, immune response
Subcutis	Adipocytes, septa, larger vessels	Insulation, energy store, cushioning

(Palms/soles have an extra **stratum lucidum**.)

Fitzpatrick skin types I–VI

The **Fitzpatrick skin phototype** classifies skin by its response to sun (burning vs tanning) after an initial exposure, originally (1975) to dose PUVA therapy. It is self-reported/clinically assessed and is the classification most widely used in practice and publications.

TYPE	TYPICAL DESCRIPTION	BURNS / TANS
I	Very fair/pale, often red hair, freckles, blue eyes	Always burns, never tans
II	Fair	Usually burns, tans minimally/with difficulty
III	Medium/light-olive	Sometimes mild burn, tans gradually/uniformly
IV	Olive/moderate brown	Rarely burns, tans easily
V	Brown/dark brown	Very rarely burns, tans very easily
VI	Deeply pigmented dark brown–black	Never burns

Dermoscopy: why it adds value

Dermoscopy (dermatoscopy, epiluminescence microscopy) uses a hand-held ×10 magnifier with a light source, either **polarised** (no contact; cross-polarised filters cancel surface glare) or **non-polarised** (contact with a liquid/gel interface). It reveals pigment and vascular structures in the epidermis, dermo-epidermal junction and upper dermis that are invisible to the naked eye.

Aetiology: causes and risk factors

Skin layers linked to lesion type

LESION	LEVEL / MECHANISM
Macule/patch	Change in melanin (basal melanocytes, melanosome transfer) or dermal blood flow — no thickening
Scale	Disordered stratum corneum: hyper-/parakeratosis (fast turnover in psoriasis)
Papule/plaque	Epidermal acanthosis ± dermal infiltrate (psoriasis: epidermal hyperproliferation + neutrophils; LP: band-like lymphocytes at the dermo-epidermal junction)
Nodule	Deep dermal/subcutaneous infiltrate, tumour or granuloma
Vesicle (spongiotic)	Eczema: inflammatory fluid separates keratinocytes **intercellular oedema = spongiosis** within the **epidermis** microvesicles (most visible on palms/soles as pompholyx)

LESION	LEVEL / MECHANISM
Vesicle (viral)	HSV/VZV infect keratinocytes → ballooning degeneration, multinucleate giant cells → intraepidermal vesicles
Flaccid bulla/erosions — intraepidermal split	Pemphigus: autoantibodies to desmogleins (desmosomes) → acantholysis → thin roof that breaks easily. Bullous impetigo/staph scalded skin: exfoliative toxin cleaves desmoglein-1 high in the epidermis (subcorneal)
Tense bulla — subepidermal split	Bullous pemphigoid: autoantibodies to BP180/BP230 (hemidesmosomes) → whole epidermis lifts off thick roof, blisters stay intact. Friction/burns: mechanical/thermal separation
Pustule	Neutrophils migrate into epidermis (spongiform pustule of Kogoj in psoriasis) or follicle (folliculitis/acne)
Wheal	Mast-cell histamine → upper-dermal oedema, epidermis normal
Lichenification	Chronic rubbing → acanthosis, hyperkeratosis, papillary dermal fibrosis
Erosion vs ulcer	Erosion = epidermis lost above basement membrane (heals by re-epithelialisation, no scar); ulcer = through basement membrane into dermis (heals by scar)
Atrophy	Thin epidermis ± loss of dermal collagen (steroids inhibit fibroblasts/keratinocytes)
Cyst	Epithelial-lined cavity in dermis

Why does a blister develop? a blister forms when the "glue" between cells fails or cells are damaged, and tissue fluid fills the gap. The level of failure predicts the clinical picture: high (intraepidermal) splits give fragile flaccid blisters and erosions; low (subepidermal) splits give tense, often haemorrhagic, durable bullae.

Functions to link: barrier (physical, chemical — acid mantle, antimicrobial peptides, lipids; microbiological — commensal flora competes with pathogens; immunological — Langerhans cells), thermoregulation (eccrine sweat, dermal blood flow; hair traps heat), sensation, vitamin D synthesis, UV protection (melanin caps over nuclei), social/communication. Commensals rarely cause bacteraemia because the intact stratum corneum, acid pH, sebum fatty acids and innate immunity prevent invasion — breaches (cannulas, ulcers, eczema) raise risk.

Ageing: intrinsic (thinner dermis, fewer fibroblasts, slower turnover, flattened dermo-epidermal junction → fragility) vs extrinsic/photo-ageing (UV + smoking → solar elastosis, wrinkles, lentigines). Twin studies show smoking and sun exposure produce visibly older skin in the exposed twin.

Fitzpatrick skin types I–VI

- **Melanocyte number is broadly similar across all phototypes** (type IV does not have more melanocytes than type I). Differences arise from melanosome **size, number, packaging and distribution**, the **eumelanin (brown-black) : pheomelanin (red-yellow)** ratio, and how efficiently melanin is transferred to keratinocytes.
- Larger, singly dispersed eumelanin-rich melanosomes in darker skin absorb/scatter more UV → greater natural photoprotection (approx. SPF-equivalent differences), but also a more reactive melanocyte response to inflammation → **post-inflammatory hyperpigmentation (PIH)**.
- UVB converts 7-dehydrocholesterol in the epidermis to pre-vitamin D3 → liver (25-OH D) → kidney (1,25-(OH)2 D). Melanin competes for UVB, so darker skin needs **longer exposure** to make the same amount of vitamin D.

Dermoscopy: why it adds value

Surface reflection from the stratum corneum normally hides deeper structure. Immersion fluid or cross-polarisation removes that glare, so light penetrates and returns from deeper layers — each colour/pattern corresponds to pigment depth (black/brown = epidermis/junction; grey/blue = dermis) and vessel morphology.

Myths to address

- **"DARK SKIN DOESN'T NEED SUN PROTECTION"**
  risk is lower but not zero; acral melanoma and pigmentation still matter.

Pathophysiology and pathology

Each layer has **its own cells and diseases**, so knowing depth explains what you see.

1. **Epidermis**
   Keratinocytes mature upwards and shed; melanocytes pass pigment to them.
2. **Dermo-epidermal junction**
   Basement membrane anchors epidermis to dermis (target in pemphigoid).
3. **Dermis**
   Collagen and elastin give strength; vessels, nerves, follicles and glands sit here.
4. **Subcutis**
   Fat lobules separated by septa (panniculitis sites).

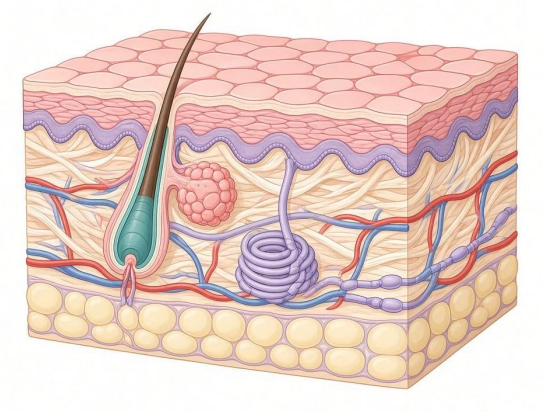

Fig. 1 · Normal skin: epidermis, wavy junction, dermis with follicle, glands, vessels and nerve, and subcutaneous fat.
AI-generated illustration

Clinical features

Skin layers linked to lesion type

- Thick-roofed tense bullae on flexural limbs/trunk in older adults BP.
- Mucosal erosions + flaccid blisters pemphigus (refer urgently).
- Itchy palmar/digital vesicles pompholyx (spongiotic).

Fitzpatrick skin types I–VI

DOMAIN	LOWER PHOTOTYPES (I–II)	HIGHER PHOTOTYPES (IV–VI)
Skin cancer (melanoma, BCC, SCC)	Highest risk — strict sun protection, skin checks	Lower UV risk but cancers often diagnosed later; acral lentiginous melanoma (palms/soles/nails) not UV-related
Photo-ageing	Earlier wrinkles, lentigines	Relative protection from photo-ageing; ageing may present as dyschromia, DPN, melasma
Erythema recognition	Redness obvious	Erythema may look violaceous/grey/brown — underestimates eczema severity (EASI/SCORAD) and cellulitis
PIH / hypopigmentation	Less common, shorter	Common, long-lasting (months–years); major patient concern after acne, eczema, LP, procedures
Keloid	Less common	More common (V–VI)
Phototherapy / lasers / peels	Lower starting UV doses (burn easily)	Higher risk of dyspigmentation with lasers/peels — specialist parameters
Vitamin D	Synthesise more efficiently	**Higher risk of deficiency at UK latitude**

Is phototype protective against ageing at higher latitude? — Partly: higher phototypes have more photoprotection and typically show fewer wrinkles and lentigines, but intrinsic ageing still occurs, and lifestyle (smoking, sunbeds) matters more than latitude alone. At UK latitude, darker phototypes gain photo-ageing protection but pay a vitamin D cost.

Vitamin D at UK latitude:

- UK lies ~50–60°N: from roughly **October to March** UVB is too weak for meaningful cutaneous vitamin D synthesis in anyone.
- People with darker skin (African, African-Caribbean, **South Asian** heritage), those who cover skin for cultural/religious reasons, the housebound/care-home residents, and those who use high-factor sunscreen constantly are at highest risk; UK South Asian populations consistently show high prevalence of deficiency (e.g. Darling; Correia et al. 2014).
- UK advice (SACN 2016 / UK government / NICE PH56 & CKS vitamin D deficiency): everyone aged 1 year should consider **10 micrograms (400 units) vitamin D daily** in autumn and winter; at-risk groups including people with dark skin should consider it **all year round**. Babies/young children have specific advice (Healthy Start). Check current guidance for treatment doses in confirmed deficiency (CKS/local formulary).

Dermoscopy: why it adds value

No — evidence supports real diagnostic value **in trained hands**:

- Meta-analyses and a Cochrane review (Dinnes et al. 2018) show dermoscopy improves **melanoma detection accuracy** versus naked-eye examination alone, with better sensitivity without loss of specificity when used by trained clinicians.
- Reduces unnecessary excisions of benign lesions (better benign:malignant excision ratio) and helps triage referrals.
- Helps distinguish seborrhoeic keratosis (milia-like cysts, comedo-like openings, "brain-like" fissures), solar lentigo (homogeneous light-brown, moth-eaten border, fingerprint pattern) from lentigo maligna (asymmetric

follicular pigmentation, rhomboidal structures, grey dots), BCC (arborising vessels, blue-grey ovoid nests), dermatofibroma (central white patch, peripheral network), molluscum (central white polylobular structures with crown vessels), scabies ("delta-wing jet" mite), and nail/scalp conditions (trichoscopy).

- Allows digital monitoring of atypical naevi and teledermatology image quality (UK referral pathways increasingly request dermoscopic images).

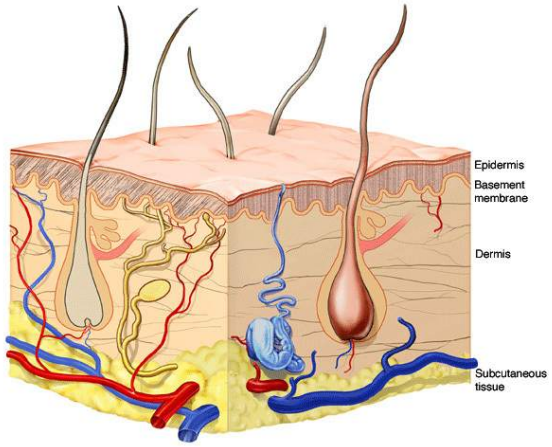

Fig. 2 · Layers: epidermis basement membrane dermis subcutis. Use to explain level of split (vesicle/bulla), wheal (dermal oedema), nodule (deep dermis).

Open access image · **Source**

Diagnosis and investigations

Skin layers linked to lesion type

Histology (punch biopsy) and **direct immunofluorescence** (perilesional skin) are specialist tests for blistering disorders; Nikolsky sign (lateral pressure shears off epidermis) suggests intraepidermal split (pemphigus, SSSS, TEN). Pharmacists/IPs recognise and refer; they do not need to perform these.

Fitzpatrick skin types I–VI

Ask: "If you went out in strong sun for 30–45 minutes at the start of summer without protection, would you burn, tan, or both?" Skin colour alone is not the same as phototype — record both self-reported burning/tanning and observed skin colour. Recognise limitations: Fitzpatrick was designed for white skin types and lumps diverse darker skin into V–VI; alternative descriptive scales exist (e.g. Monk scale, individual typology angle) but Fitzpatrick remains the clinical standard.

Dermoscopy: why it adds value

Dermoscopy is an adjunct to history + naked-eye examination, not a replacement. Structured algorithms exist (e.g. ABCD rule, 7-point checklist, 3-point checklist, "chaos and clues") — beyond the scope of this primer.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Pemphigus vulgaris; bullous impetigo; SSSS; TEN	Bullous pemphigoid; bullous drug reaction; friction blister; burns; insect-bite bullae; porphyria cutanea tarda

Skin layers linked to lesion type

FLACCID / ERODES EASILY	TENSE / PERSISTS
Pemphigus vulgaris; bullous impetigo; SSSS; TEN	Bullous pemphigoid; bullous drug reaction; friction blister; burns; insect-bite bullae; porphyria cutanea tarda

Fitzpatrick skin types I–VI

- PIH vs melasma vs lichenification-associated pigmentation vs drug-induced pigmentation.

Dermoscopy: why it adds value

Accuracy depends on **training**; untrained use may be no better (or worse) than naked eye. Featureless or amelanotic lesions remain challenging — biopsy if in doubt.

Management

Skin layers linked to lesion type

Histology level guides urgency: widespread blistering, mucosal involvement, skin pain/fever, or Nikolsky-positive skin **emergency/same-day referral** (possible SJS/TEN, pemphigus, SSSS). Localised friction blisters leave roof intact, protect, monitor for infection.

Fitzpatrick skin types I–VI

> **Sun protection counselling** by phototype: SPF 30 with high UVA rating (4–5 stars), shade, clothing; avoid sunbeds — especially I–II.

> **PIH PREVENTION**
> treat inflammation early and effectively (acne, eczema); avoid picking; daily broad-spectrum sunscreen (tinted iron-oxide sunscreens help visible-light-driven pigmentation); caution with irritant topicals (retinoids/benzoyl peroxide — start low, slow).

> **VITAMIN D**
> recommend OTC 10 microgram daily supplementation per UK advice for at-risk groups; refer/test per CKS if symptoms of deficiency (bone pain, proximal weakness) or high-risk.

> Severity assessment: in darker skin, use palpation, thickness, lichenification, and patient-reported symptoms (POEM) rather than redness alone.

Dermoscopy: why it adds value

Pharmacist/IPs without accredited dermoscopy training should **not** use dermoscopy to reassure about a suspicious pigmented lesion; refer per NICE NG12 (weighted 7-point checklist; any lesion of concern). With training, dermoscopic photos support teledermatology referrals.

Referral

ROUTINE — Changing or atypical pigmented lesion, including **acral and nail** sites in darker skin types: NICE NG12 referral.

ROUTINE — Severity under-estimated in skin of colour: look for thickening, scale, pigment change and symptoms rather than redness alone.

Red flags and when to refer

- Changing or atypical pigmented lesion, including **acral and nail** sites in darker skin types: NICE NG12 referral.
- Severity under-estimated in skin of colour: look for thickening, scale, pigment change and symptoms rather than redness alone.

Complications and prognosis

Complications

- Not applicable.

Prognosis

- Not applicable.

Prescriber pearls (UK pharmacist IP)

- Use histology to explain treatment logic: e.g. emollients restore stratum corneum lipids; potent steroids in lichenification target the thickened epidermis; why steroids cause atrophy (dermal collagen suppression).
- Patients with skin of colour often prioritise pigment change over the active rash — address it explicitly and set timelines (PIH fades over months with sun protection).
- Keloid-prone (V–VI) patients: counsel before piercings/elective procedures; treat inflammatory acne promptly.
- Don't assume darker skin "can't burn" or doesn't need skin checks — acral melanoma awareness.
- "When in doubt, cut it out (or refer)" — a "benign-looking" dermoscopic pattern in a changing lesion still warrants referral. Document dermoscopy findings and whether images were stored.

Skin layers linked to lesion type

Fitzpatrick skin types I–VI

Dermoscopy: why it adds value

Self-check quiz

5 questions · optional

Q1 When should you think of Skin layers, Fitzpatrick skin types & dermoscopy?

✓ Linking a lesion to its level in the skin; judging sun risk, PIH risk or erythema in skin of colour; deciding whether dermoscopy helps.

Q2 What drives it?

✓ Epidermal processes give scale and colour change; dermal processes give induration, nodules and scarring; melanin biology sets the phototype.

Q3 What is the backbone of care?

✓ Record the Fitzpatrick type, assess severity by more than redness, and counsel photoprotection to match.

Q4 What must you never miss?

✓ Under-calling inflammation in darker skin; melanoma in any skin type (acral and nail sites in darker skin).

Q5 Name a red flag that needs escalation.

✓ Changing or atypical pigmented lesion, including **acral and nail** sites in darker skin types: NICE NG12 referral.

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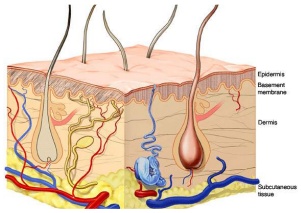
dermnetnz.org

Photo gallery

Skin structure

1



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Fitzpatrick phototypes

4

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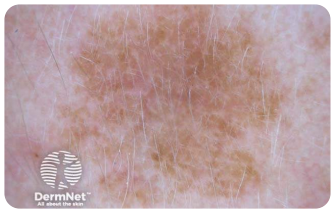
Dermoscopy examples

2



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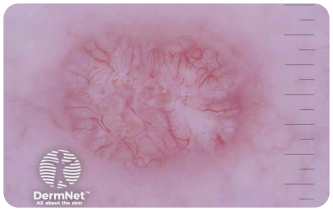

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CHAPTER 2

Inflammatory skin

Eczema, psoriasis, contact dermatitis and urticaria, plus the inflammatory spot diagnoses: erythema multiforme, annular rashes, lichen planus and discoid lupus.

1. Eczema / dermatitis
2. Psoriasis
3. Urticaria & angioedema
4. Contact dermatitis
5. Erythema multiforme
6. Annular (ring-shaped) rashes: differential
7. Lichen planus
8. Discoid lupus erythematosus

Inflammatory skin

Eczema / dermatitis

Itchy, barrier-driven inflammation. Treat with generous emollients and a topical steroid matched to site and severity, and never miss eczema herpeticum.

ALSO CALLED atopic eczema · atopic dermatitis · contact dermatitis · irritant dermatitis · allergic contact dermatitis · hand eczema · +2

RELATED PAGES Emollients & soap substitutes · Topical corticosteroids: potency, quantities & safety · Hand eczema (chronic, occupational & pompholyx) · Contact dermatitis · Itch & xerosis in older adults

KEY FACTS

- **Atopic eczema** is the commonest form: a chronic, itchy, relapsing inflammatory skin disease with a **barrier defect** and **type 2 (Th2) inflammation**.
- Affects up to **20% of children** and 2–10% of adults; about 80% start before age 6.
- Infants: cheeks and extensor surfaces; children and adults: **flexures**; chronic disease shows **lichenification**.
- UK care follows **NICE CG57 / CKS**: generous emollients, topical steroids matched to severity and site, and treat flares early.
- Never miss **eczema herpeticum** (punched-out erosions, unwell): same-day assessment.

Overview

Atopic eczema (atopic dermatitis) is a chronic, relapsing, itchy inflammatory skin disease built on a faulty skin barrier. It is the commonest inflammatory skin disease. Most cases start in early childhood, usually before age 5, and a sizeable minority carry on into adult life or come back later. It clusters with asthma, hay fever and food allergy.

Contact dermatitis is eczema caused by something touching the skin. **Irritant** contact dermatitis is the more common type. **Allergic** contact dermatitis is a delayed (type IV) hypersensitivity reaction. Contact dermatitis is a major cause of occupational skin disease in cleaners, healthcare staff, caterers and hairdressers.

NOTE

Other eczema patterns you will meet

Discoid (nummular), asteatotic (eczema craquelé), pompholyx (dyshidrotic), seborrhoeic dermatitis and venous (gravitational) eczema.

Epidemiology

- **Prevalence**
  - Lifetime prevalence over **15%** in high-income countries; about 230 million people worldwide.
  - Affects up to **20% of children** in the UK; 5–15% of young adults still have it.
- **Age**
  - Usually starts in **infancy**: about 80% before age 6.
  - Typically worst at 2–4 years; may recur in adult life with irritant or occupational exposure.
- **Atopy**
  - Clusters with **asthma, hay fever and food allergy** (the atopic march).
- **Ethnicity**
  - All ethnic groups; prevalence and pattern vary between populations (e.g. more extensor and follicular eczema in skin of colour).

Aetiology: causes and risk factors

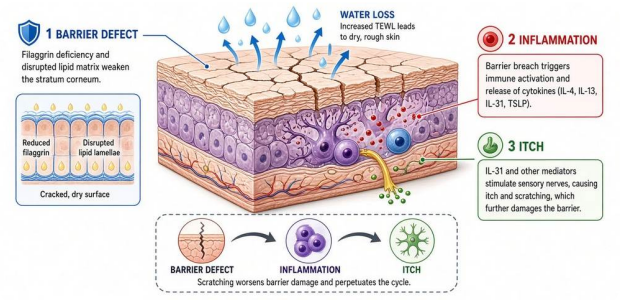

Fig. 1 · The barrier–inflammation–itch cycle in atopic eczema. A defective stratum corneum lets water out and irritants and allergens in, which drives immune activation, itch and scratching.

AI-generated illustration

Atopic eczema: a gene–environment problem

- **Barrier defect.** Filaggrin-related dryness and a disordered lipid layer increase water loss and let irritants and allergens in.
- **Immune dysregulation.** Th2-skewed inflammation drives the itch.
- **TRIGGERS**
  soaps and detergents, S. aureus, allergens, heat, wool, climate and stress.
- **Scratching** perpetuates the cycle, causing lichenification and secondary infection.

Contact dermatitis

Irritant: direct chemical, toxic or frictional damage from wet work, soaps and solvents. It depends on dose and exposure, so anyone can get it, but atopic skin gets it more easily.

Allergic: needs prior **sensitisation**. On re-exposure a delayed eczematous reaction appears, often a day or more later, and it can spread beyond the contact site. Common culprits are nickel, fragrance, rubber accelerants, preservatives and topical medicines themselves.

Risk factors and associations

> **GENETIC**
> loss-of-function **filaggrin (FLG)** mutations weaken the barrier; strong family history of atopy.

> **IRRITANTS**
> soaps, detergents, bubble bath, wool, heat, sweating, hard water.

> **ALLERGENS**
> house dust mite, pets, pollen; food allergy is a trigger in a minority of infants with moderate to severe eczema.

> **INFECTION**
> Staphylococcus aureus colonisation and flares; herpes simplex.

> Stress, hormonal changes, dry winter weather.

Myths to address

> **"IT IS CAUSED BY FOOD ALLERGY"**
> most eczema is not food-driven; avoid restrictive diets without specialist advice.

> **"STEROID CREAMS THIN THE SKIN, SO AVOID THEM"**
> used correctly (right potency, site and duration) they are safe; steroid phobia undertreats flares.

> **"ECZEMA IS CONTAGIOUS"**
> it is not, although infected eczema can spread bacteria or herpes.

> **"Emollients are optional once the skin looks better"**: they are the foundation of treatment and should continue.

Pathophysiology and pathology

Atopic eczema comes from a **leaky skin barrier** and an **overactive type 2 immune response** that feed each other through the itch–scratch cycle.

1. **A faulty barrier**
   Reduced **filaggrin** and ceramides weaken the stratum corneum, so water escapes (**raised transepidermal water loss**) and the skin becomes dry, rough and easily irritated.

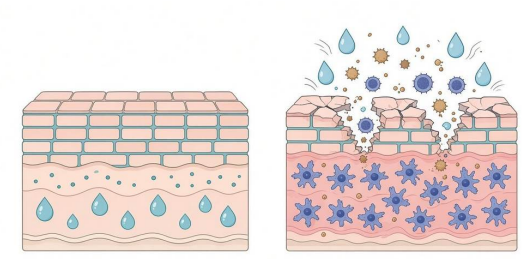

Fig. 2 · Healthy barrier (left) vs eczema: gaps between corneocytes let water out and allergens, irritants and bacteria in.
AI-generated illustration

2. **Allergens and microbes get in**
   Irritants, allergens and S. aureus penetrate the damaged barrier; keratinocytes release alarmins (**TSLP, IL-33, IL-25**).

3 **Type 2 inflammation**

Dendritic cells drive **Th2** cells and ILC2s to release **IL-4, IL-13** and **IL-31**. IL-4/IL-13 further suppress filaggrin and antimicrobial peptides; IgE rises. In chronic eczema Th1, Th17 and Th22 also contribute.

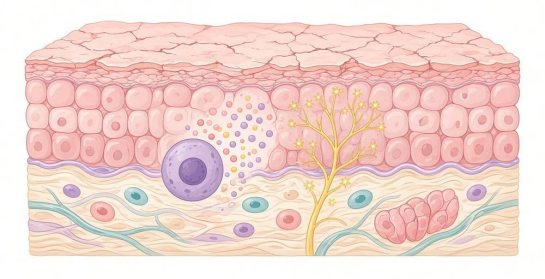

Fig. 3 · Type 2 inflammation: an immune cell releases cytokines into the epidermis while itch nerve fibres branch upwards.

AI-generated illustration

4 **Itch and scratching**

IL-31 and other mediators activate itch nerves. Scratching damages the barrier further and releases more alarmins: the **itch–scratch cycle**.

5 **Chronic changes**

Repeated rubbing thickens the epidermis (**lichenification**), with exaggerated skin markings, excoriations and pigment change.

HISTOPATHOLOGY

ACUTE

spongiosis (intercellular oedema in the epidermis), sometimes vesicles; dermal lymphocytes and eosinophils.

SUBACUTE

spongiosis with acanthosis and parakeratosis.

CHRONIC

marked acanthosis, hyperkeratosis and dermal fibrosis (lichen simplex pattern).

Biopsy is rarely needed; consider it when cutaneous T-cell lymphoma or another diagnosis is suspected.

Clinical features

PATTERN	TYPICAL FEATURES	COMMON SITES
Acute atopic flare	Red, weeping, vesicles or crust; intense itch	Varies with age (below)
Chronic atopic	Dry, scaly, lichenified, excoriated; pigment change	Flexures; hands in adults
Infant	Red, rough cheeks, scalp and trunk; nappy area often spared	Face and extensors
Child	Flexural eczema; eyelids and neck	Elbow and knee creases
Adult	Flexural or diffuse; hand and nipple eczema common	Hands, face, flexures
Irritant contact	Dry, fissured or red, sore skin at exposure sites	Hands especially
Allergic contact	Eczema matching the contact pattern; may spread	Hands, face, eyelids, under jewellery

PRESCRIBER TIP

Skin of colour

Erythema can look violaceous or grey-brown and is easily underestimated. Papular and follicular patterns are more common, and post-inflammatory hypo- or hyperpigmentation is often the most visible sign. Judge severity on thickening, scale and symptoms.

⚠ CAUTION

Clues to infected eczema

Weeping, honey-coloured crusts, pustules, rapid worsening, fever or malaise suggest secondary bacterial infection. Punched-out erosions with systemic illness suggest **eczema herpeticum**, which is an emergency.

Fig. 4 · Acute flexural eczema: ill-defined red, oozing/excoriated patch in the antecubital fossa — the classic adult/older-child site.

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Distribution

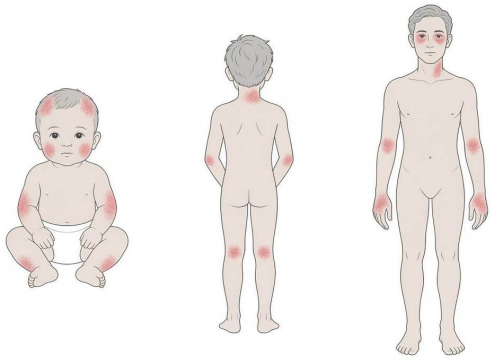

Fig. 5 · Pattern by age: infant cheeks, scalp and outer limbs (nappy area spared); child elbow and knee creases, neck; adult hands, flexures, neck and eyelids.

AI-generated illustration

Patterns to recognise

Infant

Red, rough cheeks, scalp and trunk; nappy area often spared. Site: Face, extensors.

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Child

Flexural eczema; eyelids and neck. Site: Elbow and knee creases.

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Adult / chronic

Lichenified, excoriated; hand and nipple eczema common. Site: Hands, face, flexures.

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Hand

Dry, red, scaly dorsal hands and fingers. Site: Hands.

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Irritant contact

Dry, fissured, sore skin at exposure sites. Site: Hands especially.

Allergic contact

Matches the contact pattern; may spread. Site: Hands, face, eyelids, jewellery.

Special groups

- **Skin of colour**
  - Redness looks **violaceous, grey or dark brown**, so severity is often underestimated.
  - Follicular (papular) and extensor patterns are more common; **post-inflammatory hyper- and hypopigmentation** is marked.
- **Infants**
  - Face, scalp and extensors; nappy area usually spared.
- **Older adults**
  - Asteatotic (eczema craquelé) and discoid patterns; exclude scabies, drug eruptions and lymphoma.

Diagnosis and investigations

Diagnosis is **clinical** in most cases: itch, plus typical morphology and distribution, plus personal or family atopy.

> **HISTORY**
> onset and course, triggers, occupation and hobbies, products used on the skin, response to past treatment, sleep and quality of life.

> **EXAMINATION**
> look at the whole skin for excoriation, lichenification and infection; check nails and eyes.

> **INVESTIGATE WHEN**
> the diagnosis is uncertain or treatment fails. Consider **patch testing** for suspected contact allergy, a swab if infection is suspected or recurrent, and rarely a biopsy.

> **SEVERITY**
> EASI and SCORAD are specialist and audit tools. In practice, grade as mild, moderate or severe using area, itch, sleep loss, exudate and thickening.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Psoriasis	Well-demarcated silvery plaques on extensors and scalp; less itchy; nail pits.
	Seborrhoeic dermatitis	Greasy yellow scale: scalp, nasolabial folds, eyebrows, chest.
	Tinea corporis	Asymmetrical annular plaque, active scaly edge; scrapings positive.
	Scabies	Night itch, burrows in web spaces, wrists, genitals; itchy contacts.
	Contact dermatitis	Matches exposure in time and place; often asymmetrical or occupational.
	Cutaneous T-cell lymphoma	Rare. Persistent atypical plaques in an older adult: refer.

Management

Self-care and OTC

- Apply **emollients** liberally and often. Use an ointment for very dry skin and a cream or gel if greasiness puts the patient off. Use a soap substitute, and do not use aqueous cream as a leave-on product.
- Avoid triggers such as fragranced soaps, bubble baths and wool next to the skin. Keep nails short.
- OTC **hydrocortisone 1%** for short courses on suitable sites, within product licence age and site limits.
- Non-sedating antihistamines do little for eczema itch. A short course of a sedating one is sometimes used when sleep is badly disturbed.

Stepped prescribing (NICE/CKS atopic eczema)

SEVERITY	EMOLLIENT	TOPICAL CORTICOSTEROID (EXAMPLE CLASS)
Mild	Always	Mild, e.g. hydrocortisone 1%
Moderate	Always, more liberally in a flare	Moderate, e.g. clobetasone butyrate 0.05%, betamethasone valerate 0.025%
Severe	Always	Potent, e.g. betamethasone valerate 0.1%. On the face and flexures, use moderate potency at most, for a short course

> Continue the steroid for about **48 hours after the flare settles**. Then return to emollient maintenance, with or without intermittent (e.g. weekend) steroid on hotspots. Do not use this approach on the face, genitals or axillae without specialist advice.

> **FACE AND FLEXURES**
> mild to moderate potency for short courses.

> **INFANTS UNDER 12 MONTHS**
> avoid potent steroids without specialist advice. Never use very potent steroids in children without specialist input.

> **Topical calcineurin inhibitors** (tacrolimus, pimecrolimus) are second line and often started by a specialist or GPwER.

Infected eczema

- If the patient is not systemically unwell, do **not** add an antibiotic routinely. Treating the eczema is often enough.
- If an antibiotic is needed, **flucloxacillin** is first line. Use clarithromycin for penicillin allergy, or erythromycin if pregnant and allergic. For small localised areas, consider topical fusidic acid.
- Replace emollient and steroid pots that may be contaminated.

Contact dermatitis

- Identify and **avoid** the irritant or allergen. Improvement can take weeks.
- Emollients and soap substitutes. Wear gloves for wet work, with cotton liners under rubber if needed.
- A short steroid course matched to site and severity.
- Refer for **patch testing** if the problem is chronic or recurrent, occupational, on the face or hands, or not responding to treatment.

Medicines at a glance

Mild steroid

Mild

DOSE	e.g. hydrocortisone 1% · Until 48 h after flare settles

Moderate steroid

Moderate

DOSE	e.g. clobetasone butyrate 0.05%, betamethasone valerate 0.025% · Until 48 h after flare settles
CAUTIONS	More emollient during a flare.

Potent steroid

Severe

DOSE	e.g. betamethasone valerate 0.1% · Short course
CAUTIONS	Face and flexures: moderate at most.

By site or situation

INFECTED ECZEMA

Not unwell: no routine antibiotic. If needed: **flucloxacillin**; clarithromycin if penicillin-allergic (erythromycin in pregnancy). Small area: consider topical fusidic acid. Replace contaminated pots.

CONTACT DERMATITIS

Identify and avoid the cause; gloves (cotton liners) for wet work; short steroid course. Patch test if chronic, occupational, face or hands.

ITCH

Non-sedating antihistamines do little. Short sedating course only if sleep badly disturbed.

Treatment ladder

Every patient

Emollient liberally and often; soap substitute. No aqueous cream as leave-on. Avoid fragranced soaps, bubble baths, wool. Keep nails short.

2 **Mild**

Mild steroid

3 **Moderate**

Moderate steroid

4 **Severe**

Potent steroid

5 **Second line / specialist**

Topical calcineurin inhibitors (tacrolimus, pimecrolimus): often specialist or GPwER started. Phototherapy or systemic treatment: refer.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Maintenance**: Back to emollient; ± weekend steroid on hotspots (not face, genitals, axillae without advice).
- **Infants < 12 months**: No potent steroid without specialist advice; never very potent in children.

Referral

SAME DAY	**Eczema herpeticum**: clustered punched-out erosions or vesicles, often with fever. Hospital + aciclovir.
SAME DAY	Severe or spreading infection, or a systemically unwell patient.
URGENT	Severe disease not controlled after about a week of stepped treatment.

ROUTINE	Diagnostic doubt, recurrent deep infections, major psychosocial impact.
ROUTINE	Possible phototherapy or systemics; occupational disease needing workplace advice.

Red flags and when to refer

- **SUSPECTED ECZEMA HERPETICUM**
  clustered punched-out erosions or vesicles, often with fever. Needs **same-day hospital assessment** and aciclovir.
- Severe or spreading infection, or a systemically unwell patient.
- Severe disease not controlled after about a week of appropriate stepped treatment.
- Diagnostic uncertainty, recurrent deep infections, or a major psychosocial impact.
- Possible need for phototherapy or systemic treatment, or occupational disease needing workplace advice.

Complications and prognosis

Complications

- **Bacterial infection** (S. aureus, impetiginisation): weeping, crusting, pustules.
- **Eczema herpeticum** (HSV): clustered punched-out erosions, fever; an emergency.
- Molluscum contagiosum and eczema coxsackium spread more widely.
- Sleep loss, poor school or work performance, anxiety and depression.
- **Contact dermatitis** (irritant or allergic, including to treatments) and occupational hand eczema.
- Erythroderma (rare); growth effects only with very severe disease or excessive steroid use.
- Steroid side effects (atrophy, striae, perioral dermatitis) with misuse.

Prognosis

- Many children improve with age: in a large meta-analysis about **20% still had eczema 8 years later**, and under 5% at 20 years.
- The skin stays dry and sensitive for life; adult relapse is common with irritant exposure.
- Earlier onset (before age 2) carries a lower risk of persistence.

Prescriber pearls (UK pharmacist IP)

Prescribe enough emollient. Adults often need several hundred grams a month, and under-prescribing is the commonest reason treatment fails. Apply in the direction of hair growth, ideally after bathing while the skin is still damp.

Fire risk: dressings, clothing and bedding soaked in paraffin-based (and other) emollients can catch fire easily. Warn patients to keep away from naked flames.

- Teach **fingertip units** (FTUs) so steroid doses are understood.
- Tackle **steroid phobia** with clear advice on potency, site and duration. Under-treatment and rebound are far more common than harm.
- **Inflamed is not the same as infected.** Weeping crust does not automatically mean an antibiotic.
- **Pregnancy:** most emollients and mild to moderate steroids are acceptable. Check the BNF for each product.
- Counselling: eczema is not caused by poor hygiene, and the barrier tendency is lifelong even when the skin is clear.
- Hands and feet: see hand eczema for occupational dermatitis, pompholyx and patch-test referral.
- Potency ladder, fingertip units and steroid side-effects: topical corticosteroids; choosing and quantifying moisturisers: emollients.

Self-check quiz

4 questions · optional

Q1 **How long should a topical steroid continue after a flare?**

✓ About 48 hours after the flare settles, then back to emollient maintenance.

Q2 **First-line oral antibiotic for infected eczema, if one is needed?**

✓ Flucloxacillin (clarithromycin if penicillin-allergic; erythromycin if pregnant and allergic).

Q3 What suggests eczema herpeticum, and what do you do?

✓ Clustered punched-out erosions or vesicles, often with fever. Same-day hospital assessment and aciclovir.

Q4 Commonest reason eczema treatment fails?

✓ Under-prescribing emollient: adults often need several hundred grams a month.

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Photo gallery

Site by age: infant, child, adult 4

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Chronic eczema: lichenification

2

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Hand eczema

2

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Skin of colour

2

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Infected eczema

2

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Red flag: eczema herpeticum

1

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AI teaching illustrations

2

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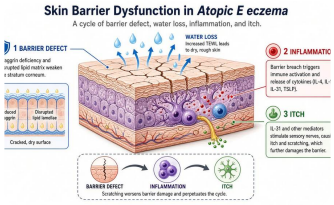

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Inflammatory skin Adult core · deepened with Modules 3 & 4

Psoriasis

Sharply defined, scaly plaques caused by IL-23/IL-17 inflammation. Manage with topical steroid and vitamin D, steroid breaks, and screening for joints and comorbidities.

ALSO CALLED plaque psoriasis · guttate psoriasis · scalp psoriasis · nail psoriasis · flexural psoriasis · inverse psoriasis · +1

RELATED PAGES Vitamin D analogues, coal tar, dithranol & phototherapy · Biologics, JAK inhibitors & conventional systemics · Topical corticosteroids: potency, quantities & safety · Emollients & soap substitutes · Palmoplantar pustulosis & palmoplantar psoriasis

KEY FACTS

- Chronic, relapsing, **immune-mediated** inflammatory disease of skin, nails and joints.
- Affects about **2–3%** of the UK population; onset peaks at **15–25** and **50–60** years.
- Driven by the **IL-23 Th17 IL-17** axis: epidermis turns over in **3–5 days** instead of about 28.
- Classic sign: **well-demarcated salmon-pink plaques with silvery scale** on elbows, knees, scalp and lower back.
- UK care follows **NICE CG153**: emollient + potent steroid + vitamin D analogue, with planned steroid breaks; screen yearly for **psoriatic arthritis** (PEST).

Overview

Psoriasis is a chronic, non-contagious, **immune-mediated inflammatory disease** that relapses and remits. It mainly causes sharply defined, red, scaly plaques, but it is a **multisystem** disease affecting the nails and joints and linked with cardiometabolic disease and depression.

- **Chronic plaque psoriasis** is the most common type (about 80–90%).
- Other types: guttate, flexural (inverse), scalp, palmoplantar, nail; rare emergencies are **erythrodermic** and **generalised pustular** psoriasis.
- Up to **30%** of people with psoriasis develop **psoriatic arthritis**.
- Diagnosis is **clinical**; management is long term and stepped, from topicals to phototherapy, systemic drugs and biologics.

Epidemiology

- **Prevalence**
  - About **1–3%** of the UK population (2–4% in many Western populations); up to about 4% worldwide.
  - Less common in tropical regions and in people of African or East Asian descent.

- **Age**
  - Can start at any age, including childhood (prevalence 0.5–2% in children).
  - Two onset peaks: **15–25 years** (early onset, often familial, HLA-C*06:02) and **50–60 years**.
- **Sex**
  - Males and females are affected about equally.
- **Family history**
  - About **one third** of patients have an affected relative.
- **Severity**
  - Roughly 60% of patients have mild disease, 30% moderate and 10% severe.

Aetiology: causes and risk factors

Psoriasis is **multifactorial**: a genetic predisposition plus environmental triggers that switch on an immune response in the skin.

Genetic factors

- **HLA-C*06:02** (HLA-Cw6) is strongly linked with early-onset and guttate psoriasis (not with nail or late-onset disease).
- Other susceptibility genes affect skin barrier (e.g. LCE3B/LCE3C deletion) and the IL-23/IL-17 pathway.

Triggers and aggravating factors

- **Infection**
  - **Streptococcal sore throat** (guttate psoriasis); **HIV** can cause severe or atypical disease.
- **SKIN INJURY**
  the **Koebner phenomenon**
  - New plaques at sites of cuts, scratches, surgery, burns, sunburn or friction.
- **Drugs**
  - **Beta-blockers**, **lithium**, **antimalarials** (hydroxychloroquine, chloroquine), interferons, sometimes NSAIDs, ACE inhibitors and terbinafine.
  - **Abrupt withdrawal of systemic (or potent topical) corticosteroids**: rebound and pustular flares.
- **Lifestyle**
  - **Smoking**, excess **alcohol**, **obesity**, stress.
- Cold, dry weather; sun usually helps, but about 10% of patients are worse with sun.

PRESCRIBER TIP

Drug-triggered flare after an MI

With post-MI polypharmacy the beta-blocker is the prime suspect, but **do not stop cardiology-critical drugs** without discussing with the GP or cardiologist.

Fig. 1 · Drug-triggered psoriasis: beta-blockers, lithium and antimalarials head the list.
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Associated conditions

- **Psoriatic arthritis** and spondyloarthropathy.
- **Cardiovascular disease** and **metabolic syndrome** (obesity, hypertension, dyslipidaemia, type 2 diabetes); psoriasis is an independent cardiovascular risk factor, especially when severe.
- Inflammatory bowel disease (Crohn's, ulcerative colitis), uveitis, coeliac disease, non-alcoholic fatty liver disease.
- **Depression and anxiety**; higher alcohol use.
- NICE notes a higher **VTE** risk in severe psoriasis.

Pathophysiology and pathology

Psoriasis is a self-amplifying loop between immune cells and keratinocytes.

1 **A trigger activates the innate immune system**
Injury, infection or drugs stress keratinocytes. Plasmacytoid dendritic cells, keratinocytes and macrophages release interferon-α, TNF-α and other signals that **activate myeloid dendritic cells** in the dermis.

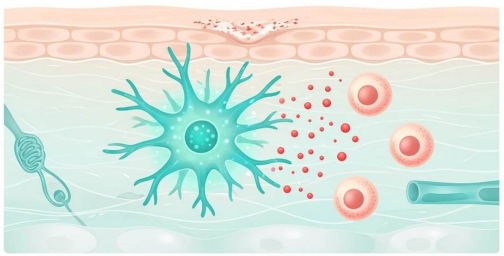

Fig. 2 · Step 1: a trigger (here a graze) activates a dendritic cell, which signals to nearby T cells.
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2 **Dendritic cells release IL-23 and IL-12**

Activated dendritic cells produce **IL-23**, which drives the survival and expansion of **Th17** (and Th22) cells, and IL-12, which drives Th1 cells. The **IL-23/Th17 pathway** is the dominant one.

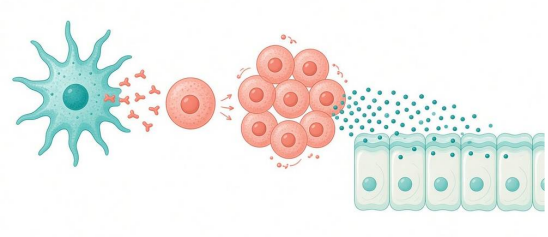

Fig. 3 · Step 2: dendritic-cell IL-23 expands Th17 cells, which release IL-17 and IL-22 towards the epidermis.

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3 **IL-17 and IL-22 drive keratinocytes**

Th17 cells (and γδ T cells, ILC3s, CD8 cells, neutrophils and mast cells) release **IL-17A, IL-17F and IL-22**. Keratinocytes **proliferate** and release chemokines (e.g. CCL20) and antimicrobial peptides that recruit more immune cells: a **feed-forward loop**. TNF-α amplifies every step.

4 **The epidermis thickens and capillaries dilate**

Keratinocytes cycle in **3–5 days instead of about 28**. The epidermis thickens (**acanthosis**) with long, regular rete ridges; papillary capillaries become dilated and tortuous, which explains the **Auspitz sign** (pinpoint bleeding when scale is removed).

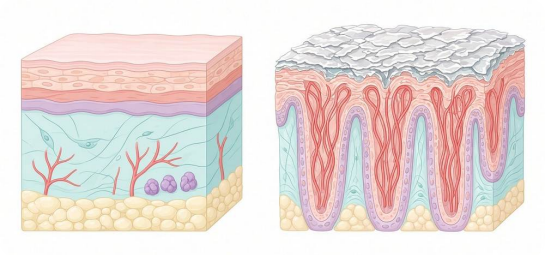

Fig. 4 · Step 4: normal skin (left) vs psoriatic plaque (right): thick epidermis, elongated rete ridges, dilated looping capillaries and silvery scale.

AI-generated illustration

5 **Immature cells make silvery scale**

Cells reach the surface still holding their nuclei (**parakeratosis**) and fail to release the lipids that bind corneocytes, so the stratum corneum flakes off as **silvery scale**. Neutrophils collect in small pockets (**Munro microabscesses**); IL-36 overactivity tips the balance towards **pustular** psoriasis.

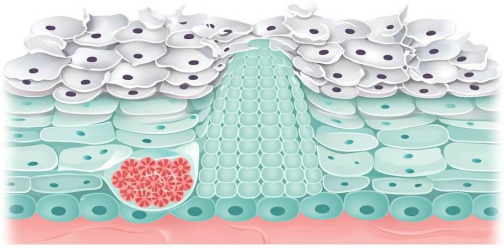

Fig. 5 · Step 5: fast-rising cells keep their nuclei (parakeratosis) and peel off as scale; neutrophils cluster in small pockets.

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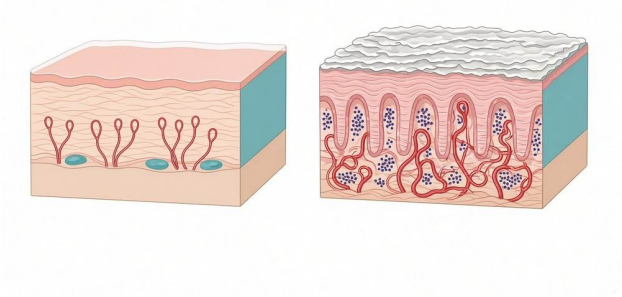

Fig. 6 · Psoriasis plaque in section: thickened epidermis, dilated capillaries and immune-cell infiltrate beneath silvery scale.

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How treatments map onto the mechanism

TOPICAL STEROIDS
broad anti-inflammatory action on the whole loop.

Vitamin D analogues (calcipotriol): slow keratinocyte proliferation and normalise differentiation.

COAL TAR, DITHRANOL, SALICYLIC ACID
antiproliferative and keratolytic.

PHOTOTHERAPY (NB-UVB)
induces T-cell apoptosis in the skin.

BIOLOGICS
TNF-α inhibitors, **IL-17** inhibitors, **IL-23** (p19) and IL-12/23 (p40) inhibitors block the central cytokines; TYK2 and PDE4 inhibitors act on intracellular signalling.

HISTOPATHOLOGY

- **Regular acanthosis** with elongated, club-shaped rete ridges.
- **Parakeratosis** (retained nuclei in the stratum corneum) with loss of the granular layer.
- **Munro microabscesses** (neutrophils in the stratum corneum) and **spongiform pustules of Kogoj**.
- Thinned suprapapillary plates and **dilated, tortuous papillary capillaries** (Auspitz sign).
- Perivascular lymphocytic infiltrate. Biopsy is only needed if the diagnosis is uncertain.

Clinical features

Symptoms

- **Itch** is usually mild to moderate but can be severe; scratching causes lichenification.
- Painful **fissures**, especially on palms, soles and in flexures.
- Scaling and shedding; **joint pain, stiffness or swelling** suggests psoriatic arthritis.
- Major **psychosocial burden**: embarrassment, low mood, impact on work and relationships.

Lesions and morphology

Well-demarcated plaque

Sharply defined, raised, **salmon-pink** plaque, usually symmetrical. The sharp edge

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Silvery scale

Thick, white, micaceous scale; removing it causes pinpoint bleeding (**Auspitz**

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Koebner phenomenon

New lesions appear in lines along scratches or scars (also seen in lichen planus

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Nail changes

Pitting, onycholysis, **oil-drop** discolouration, subungual hyperkeratosis, crumbling.

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Post-inflammatory change

Plaques may leave pale or dark marks that fade over months, especially in skin of

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Distribution

- **EXTENSOR SURFACES**
  elbows and knees.
- **Scalp**, often extending just beyond the hairline and behind the ears.
- **Lower back**, natal cleft and umbilicus.
- **Nails** (in up to half of patients, more with arthritis).
- Flexures, genitals, palms and soles in specific types.

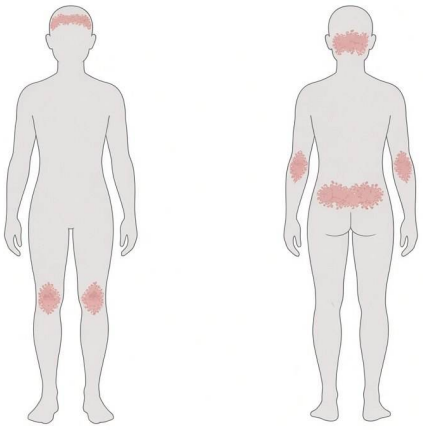

Fig. 12 · Typical sites: scalp and hairline, elbows, knees, lower back, natal cleft, umbilicus and nails; flexures in inverse psoriasis.

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Fig. 13 · Symmetrical scaly plaques on the knees.

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Fig. 14 · Multiple plaques on the trunk: estimate body surface area.

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Variants and subtypes

Chronic plaque

Persistent plaques > 3 cm on extensors, scalp and lower back; from mild to very

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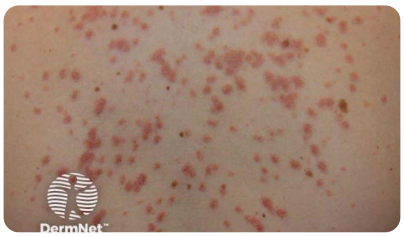
Guttate

Many small **raindrop** plaques on trunk and limbs, often **2–3 weeks after a strep throat**; may clear in months or



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Scalp

Thick adherent scale; often the first or only site; extends beyond the hairline.

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Flexural (inverse)

Smooth, shiny, well-defined red patches with **little or no scale** in axillae, groin, natal

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Palmoplantar

Well-defined scaly plaques or diffuse hyperkeratosis with painful **fissures**; disproportionately disabling.

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Nail psoriasis

Pitting, onycholysis and oil-drop sign; a marker of higher **psoriatic arthritis** risk.

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Sebopsoriasis

Overlap with seborrhoeic dermatitis on the scalp, face, ears and chest.

Unstable / pustular

Rapid extension, new plaques or pustules, often after steroid withdrawal or

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Erythrodermic

Confluent redness and scaling over **> 90%** of the skin; temperature dysregulation, fluid loss, heart failure. **Same-day** assessment.

Generalised pustular (von Zumbusch)

Sheets of sterile pustules on red skin with fever and malaise. **Same-day** assessment.

Special groups

- **Skin of colour**
  - Plaques are often thicker with more prominent grey or silvery scale and look **violaceous, grey or dark brown** rather than pink; erythema is harder to judge.
  - Post-inflammatory hyper- or hypopigmentation is marked and can last longer than the plaques.
  - Involvement is often more extensive; palmoplantar, pustular and erythrodermic forms may be more frequent.
- **Children**
  - Guttate and scalp/flexural/napkin disease are common; NICE recommends **specialist referral at presentation**.
- **Pregnancy**
  - Psoriasis often improves but may flare after delivery; acitretin and methotrexate are teratogenic (specialist planning).
- **HIV**
  - Can be extensive and recalcitrant; consider HIV testing in sudden severe psoriasis.

Fig. 22 · Skin of colour: violaceous plaques with grey scale.
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Fig. 23 · Flexural psoriasis: shiny, sharply defined, no satellites.
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Fig. 24 · Psoriatic nail pitting from matrix involvement.
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Diagnosis and investigations

Diagnosis is **clinical**, based on morphology, distribution, nails and joints.

History

- Onset, course, previous treatments and response; **triggers** (sore throat, drugs, stress, steroid withdrawal).
- **JOINT SYMPTOMS**
  pain, swelling, morning stiffness, heel pain, dactylitis.

- Family history; smoking, alcohol, weight; mood; impact on work and relationships.

Examination

- Morphology and **body surface area (BSA)**: the patient's palm with fingers is about 1% BSA.
- **HIGH-IMPACT SITES**
  face, scalp, hands, feet, nails, genitals.
- Nails, scalp, natal cleft and umbilicus for supporting clues; joints for swelling or dactylitis.

Severity and impact scoring

TOOL	WHAT IT MEASURES
BSA	Extent: the patient's palm 1%. > 10% suggests moderate to severe disease.
PASI	Psoriasis Area and Severity Index (0–72): redness, thickness, scale × area; used for systemic and biologic decisions.
PGA	Static physician's or patient's global assessment (clear to severe); NICE suggests patient and clinician assessment.
DLQI	Dermatology Life Quality Index (0–30); CDLQI for children.
PEST	Psoriasis Epidemiology Screening Tool: annual screen for psoriatic arthritis; score 3 rheumatology.

Investigations

- **Usually none.**
- **Skin biopsy** if the diagnosis is uncertain (e.g. to exclude mycosis fungoides or lupus).
- **Throat swab** (or ASO titre) if guttate psoriasis follows a sore throat.
- **Skin scrapings and nail clippings** for mycology when tinea or onychomycosis is possible.
- **Cardiovascular risk** assessment (NICE: at presentation, and at least every 5 years if severe): BP, BMI, lipids, HbA1c.
- Baseline bloods before systemic therapy are arranged by the specialist team.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Atopic eczema	Ill-defined, very itchy, **flexural** in adults; lichenification; atopic history.
	Seborrhoeic dermatitis	Greasy yellow scale on the central face, eyebrows, scalp and chest; less sharply defined; can overlap (sebopsoriasis).
	Tinea corporis	Asymmetrical, with an **advancing scaly edge** and central clearing; scrapings positive.
	Discoid eczema	Coin-shaped, very itchy, oozing or crusted plaques, often on the legs.

	CONDITION	DISTINGUISHING FEATURES
	Lichen planus	Violaceous, flat-topped polygonal papules with **Wickham striae**; wrists, mouth.
	Pityriasis rosea	Herald patch then oval patches with a **collarette** in a Christmas-tree pattern; clears in 6–12 weeks.
	Candidal intertrigo	Macerated, beefy red folds with **satellite papules and pustules** (flexural psoriasis has none).
	Hyperkeratotic hand eczema	Hardest to separate from palmoplantar psoriasis; patch testing or biopsy may be needed; they can coexist.
	Palmoplantar pustulosis	Sterile yellow and brown pustules on palms and soles; strongly linked to smoking.
	Mycosis fungoides	Persistent, atypical, often itchy patches in sun-protected sites that do not respond: biopsy.
	Onychomycosis (nails)	Asymmetric, yellow, crumbly nails, often toenails; mycology positive. Psoriasis has pits.

Management

Principles

- Follow **NICE CG153** and the CKS summary. Psoriasis is **chronic**: treatment is intermittent and most patients relapse after stopping.
- Agree goals; consider severity, sites, impact, patient preference and comorbidities.
- Explain **how much** to apply and **when** (fingertip units), planned **steroid breaks**, and how long until response (about 4 weeks for topicals).
- Review **4 weeks** after starting a new topical (2 weeks in children).
- Screen yearly for **psoriatic arthritis** and assess cardiovascular risk, mood and lifestyle.

Severity

- **MILD**
  limited plaques managed with topicals.
- **Moderate to severe** (e.g. BSA > 10%, PASI 10, DLQI > 10, high-impact sites, failure of topicals): consider referral for phototherapy or systemic therapy.

Conservative and self-care

- **Emollients** regularly and generously: reduce scale and itch and help active topicals work.
- Stop smoking; moderate alcohol; support weight loss (it can reduce severity).
- Avoid skin trauma and picking; manage stress.

- Seek prompt review for unexplained joint pain or swelling.
- Prescribing detail for emollients: emollients.

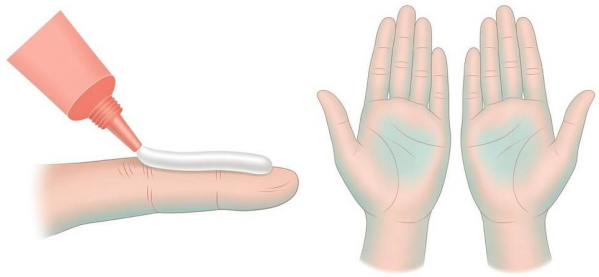

Fig. 25 · One fingertip unit (about 0.5 g) treats an area the size of two adult palms.

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Medical treatment: topical (adults)

Potent topical corticosteroid + vitamin D analogue

First line, trunk and limbs

DOSE	Potent steroid **once daily** plus vitamin D analogue **once daily**, applied at different times of day, for **up to 4 weeks**.
USE	First-line for chronic plaque psoriasis on trunk and limbs.
CAUTIONS	Potent steroid no longer than **8 weeks** at any one site; leave a **4-week break** between courses.
COUNSELLING	Apply to plaques only; separate the two products in time (e.g. morning and evening).

Vitamin D analogue (e.g. calcipotriol)

Second step

DOSE	**Twice daily** for 8–12 weeks if not controlled after 4 weeks of the combination above.
CAUTIONS	Irritation, especially on the face and flexures; avoid excessive amounts (hypercalcaemia).
COUNSELLING	Wash hands after use; can be used between steroid courses.

Calcipotriol / betamethasone dipropionate

Fixed combination (gel, foam, ointment)

DOSE **Once daily** for up to 4 weeks.

USE Third step (or alternative to a potent steroid twice daily or coal tar) when the vitamin D analogue fails; helps adherence.

CAUTIONS Contains a potent steroid: same duration limits and breaks. Brands vary by formulary.

COUNSELLING Foam and gel suit the scalp and hairy areas; shake foam well.

Very potent topical corticosteroid

Specialist advice

CAUTIONS Usually on specialist advice; no longer than **4 weeks** at a site; avoid face and flexures.

COUNSELLING Explain atrophy and rebound risk.

Coal tar

Antiproliferative, anti-scaling

USE Alternative step on trunk, limbs and scalp (shampoo).

CAUTIONS Messy, odour, staining; coal tar shampoo alone is not enough for severe scalp disease.

COUNSELLING Use in the evening; may stain clothes and bedding.

Dithranol (short-contact)

Antiproliferative

USE Hospital or specialist settings for large plaques.

CAUTIONS Burns and stains skin and fabrics; avoid face and flexures.

Scalp

- Start with a **potent steroid** in a suitable vehicle (lotion, gel or shampoo) for up to 4 weeks.
- Next: **calcipotriol/betamethasone** combination or a vitamin D analogue.
- Use a **scale softener** (e.g. salicylic acid or oil-based preparations) first for thick scale.

Face, flexures and genitals

- **Mild or moderate** potency steroid once or twice daily for **a maximum of 2 weeks**; flexures are at high risk of atrophy (about 1–2 weeks a month).
- **Do not** use potent or very potent steroids here.
- If response is poor or continuous steroid is needed: **topical calcineurin inhibitor** (tacrolimus or pimecrolimus) twice daily for up to 4 weeks, started by a clinician with psoriasis expertise (off-label).
- Vitamin D analogues may help but can irritate; tar, dithranol and salicylic acid are usually too irritant in folds.

Fig. 26 · Flexural psoriasis: mild or moderate steroid short course only.

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Fig. 27 · Steroid absorption varies by site: thin skin (face, eyelids, flexures, genitals) absorbs far more.

AI-generated illustration

Palms, soles and nails

- Emollients and keratolytics (e.g. **salicylic acid**, urea); **potent or very potent** steroid ± vitamin D analogue or combination, with occlusion overnight.
- Nail disease responds poorly to topicals; refer if severe or with joint symptoms.

Medical treatment: systemic and phototherapy (specialist)

Phototherapy (narrowband UVB; PUVA)

Hospital-based

SPECIALIST ONLY

USE Plaque or guttate psoriasis not controlled with topicals; hand–foot PUVA for palmoplantar disease.

CAUTIONS Cumulative skin cancer risk, especially PUVA.

Methotrexate

Conventional systemic (often first choice)

SPECIALIST ONLY

USE Moderate to severe psoriasis and psoriatic arthritis.

CAUTIONS **Once weekly** dosing; folic acid; teratogenic; interactions (e.g. trimethoprim, NSAIDs).

MONITORING FBC, LFTs, renal function; liver fibrosis assessment (specialist).

Ciclosporin

Conventional systemic

SPECIALIST ONLY

USE Rapid control, or when planning conception.

CAUTIONS Nephrotoxicity, hypertension, many interactions.

MONITORING BP and renal function.

Acitretin

Oral retinoid

SPECIALIST ONLY

USE Especially pustular and palmoplantar psoriasis.

CAUTIONS **Teratogenic for 3 years after stopping**: pregnancy prevention required; lipids and LFTs.

Apremilast, dimethyl fumarate, deucravacitinib

Small molecules (PDE4, fumarate, TYK2)

SPECIALIST ONLY

USE	Under NICE technology appraisals.

Biologics

TNF-α, IL-17, IL-23 and IL-12/23 inhibitors

SPECIALIST ONLY

USE	Severe psoriasis that has failed or cannot use conventional systemic therapy and phototherapy (NICE TAs).
CAUTIONS	Infection risk; screen for TB and hepatitis; **no live vaccines**.
COUNSELLING	Patients on biologics: see biologics and systemics.

⚠ CAUTION

Avoid oral steroids

Systemic corticosteroids are best avoided in psoriasis: withdrawal can trigger rebound, erythrodermic or pustular flares.

Treatment ladder

1 **Everyone**

Emollients, lifestyle support, explanation and arthritis screening.

2 **Step 1 (4 weeks)**

Potent steroid OD + vitamin D analogue OD (different times of day) to trunk and limbs; potent steroid for scalp; mild/moderate steroid for face and flexures.

3

Step 2 (8–12 weeks)

Vitamin D analogue BD if not controlled.

4

Step 3

Potent steroid BD (up to 4 weeks) or **coal tar**, or **calcipotriol/betamethasone OD** (up to 4 weeks). Plan 4-week steroid breaks.

5

Specialist

Phototherapy, methotrexate, ciclosporin, acitretin, then small molecules and biologics.

Psychological impact

- Psoriasis is linked with **depression, anxiety and suicidal ideation**; quality-of-life impact can equal other major chronic diseases.
- Ask about mood, alcohol and work at reviews; use the DLQI.
- Signpost support (e.g. the Psoriasis Association) and refer for psychological support when needed.

Referral (NICE CG153 / CKS)

SAME DAY	**Erythrodermic** or **generalised pustular** psoriasis.
URGENT	Rapidly worsening or unstable psoriasis.
ROUTINE	Suspected **psoriatic arthritis**: rheumatology (urgent if active synovitis).
ROUTINE	Diagnostic uncertainty; extensive disease or failure of topicals; major impact on quality of life; severe nail disease; all **children**.
ROUTINE	Eligible for phototherapy or systemic therapy.

Red flags and when to refer

SAME DAY	Widespread redness over **90%** of the skin (erythroderma), especially with shivering, feeling unwell or fluid loss.
SAME DAY	Sheets of **pustules** with fever (generalised pustular psoriasis), often after steroid withdrawal.
URGENT	Hot, swollen joints or dactylitis (active psoriatic arthritis).
ROUTINE	Atypical plaques not responding to treatment (exclude mycosis fungoides or lupus).

Complications and prognosis

Complications

Psoriatic arthritis (10–30%): joint damage if untreated.

CARDIOMETABOLIC DISEASE
ischaemic heart disease, stroke, hypertension, diabetes, chronic kidney disease (severe disease).

ERYTHRODERMA
fluid and electrolyte loss, hypothermia, infection, high-output heart failure.

TREATMENT-RELATED
steroid atrophy, striae and rebound; phototherapy skin cancer risk; systemic drug toxicity; infection on biologics.

Secondary infection in flexures (Candida, bacteria); lichenification from scratching.

Psychological: depression, alcohol misuse, social isolation.

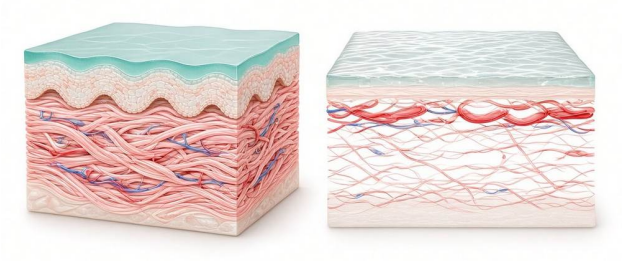

Fig. 28 · Steroid atrophy: thinning of epidermis and dermis with visible vessels after prolonged potent steroid use.
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Prognosis

- A **lifelong** relapsing and remitting condition; about 17–55% of patients have remissions of varying length.
- Guttate psoriasis often clears within months but may later evolve into plaque disease.
- Severe psoriasis is associated with reduced life expectancy (about 3–4 years), largely through cardiovascular disease.

Prescriber pearls (UK pharmacist IP)

- Explain that psoriasis is **chronic** and treatment is intermittent; most patients relapse after stopping.
- Demonstrate application; cosmetic acceptability drives adherence, so **gels and foams** often beat ointments for daytime use.
- Spell out **steroid breaks** (4 weeks between potent courses) to prevent atrophy and rebound.
- Calcipotriol/betamethasone products vary by formulary: check local brands.
- **Avoid oral steroids**: withdrawal can trigger rebound or pustular flares.
- Check the drug history for **beta-blockers, lithium and antimalarials**, but do not stop cardiology-critical drugs alone.
- Counsel on vaccination and infection risk if the patient later moves to immunosuppression.
- NICE notes a higher **VTE** risk in severe psoriasis.
- Related: vitamin D analogues, tar and phototherapy, biologics and systemics, topical corticosteroids.

Self-check quiz

8 questions · optional

Q1 Which cytokine axis is central to psoriasis?

✓ IL-23 Th17 IL-17 (with TNF-α amplifying).

Q2 How fast do keratinocytes turn over in a plaque?

✓ About 3–5 days instead of about 28.

Q3 What is the Auspitz sign?

✓ Pinpoint bleeding when scale is removed, from dilated papillary capillaries.

Q4 First-line NICE topical regimen for trunk and limbs?

✓ Potent steroid once daily plus vitamin D analogue once daily, at different times, for up to 4 weeks.

Q5 Maximum potent steroid duration at one site?

✓ 8 weeks (very potent 4 weeks), with a 4-week break between courses.

Q6 Which steroid potency for face and flexures?

✓ Mild or moderate only, for up to 2 weeks.

Q7 Which tool screens for psoriatic arthritis, and how often?

✓ PEST questionnaire, every year.

Q8 Two psoriasis emergencies?

✓ Erythrodermic and generalised pustular psoriasis: same-day assessment.

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Photo gallery

Chronic plaque psoriasis: morphology and extensor sites 4

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Scalp 2

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Nails 3

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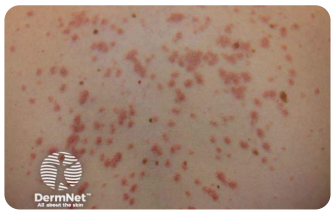
Guttate 2



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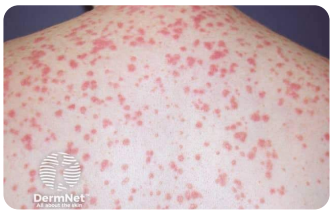

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Flexural (inverse): no scale 2

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Skin of colour 1

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Differential pairing: flexural psoriasis vs candida vs tinea 3

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AI teaching illustrations 2

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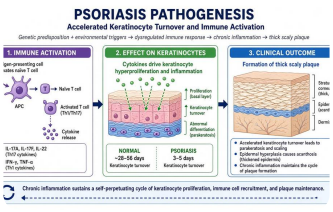

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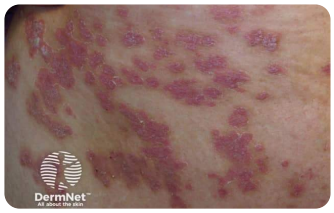
Drug-induced psoriasis 1



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Guttate psoriasis (Module 4 Week 1) 3

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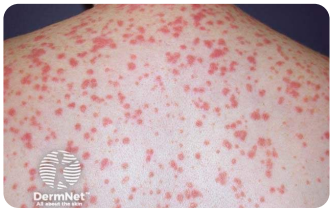

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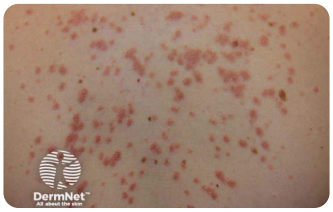

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Flexural (inverse) psoriasis (Module 3 Week 5) 4

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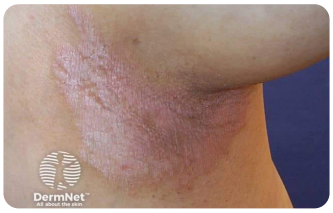

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Palmoplantar psoriasis (Module 3 Week 6) 3

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Nail psoriasis (Module 3 Week 2) 2

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Inflammatory skin Module 4 · Week 5

Urticaria & angioedema

Wheals that itch and fade within 24 hours, with or without angioedema. Regular non-sedating antihistamines (up-dosed if needed); never miss anaphylaxis or ACE-inhibitor angioedema.

ALSO CALLED urticaria | hives | wheals | CSU | chronic spontaneous urticaria | inducible urticaria | +5

RELATED PAGES ACE-inhibitor (bradykinin) angioedema | Urticarial vasculitis & HUVS | Papular urticaria & insect bites

KEY FACTS

- Urticaria itchy, transient **weals** (each lasting < 24 h) from mast-cell mediator release; **angioedema** is deeper swelling lasting up to 72 h.
- About **1 in 5** people have acute urticaria in their lifetime; chronic spontaneous urticaria affects **0.5–2%**.
- Acute (< 6 weeks) is often triggered by infection, drugs or food; chronic (6 weeks) is usually spontaneous or inducible.
- UK first line: a **non-sedating second-generation antihistamine**, increased up to 4 times the licensed dose (off-label) under guidance.
- Angioedema **without weals** (ACE inhibitor, hereditary) is bradykinin-mediated and does not respond to antihistamines; airway swelling is an emergency.

Overview

Urticaria is a transient, itchy eruption of wheals. **Acute** lasts under 6 weeks; **chronic spontaneous urticaria (CSU)** lasts 6 weeks or more, most days. **Inducible (physical)** urticaria is triggered by pressure, heat, cold, sweat or stroking (dermographism).

Angioedema is deeper subcutaneous/submucosal swelling. It may accompany histaminergic urticaria or occur alone (ACE-inhibitor bradykinin angioedema; rare hereditary angioedema).

Epidemiology

- **Acute urticaria**
  - About 20% lifetime prevalence; all ages and both sexes.
  - In children usually triggered by **viral infection**; in adults often idiopathic.
- **Chronic urticaria**
  - Chronic spontaneous urticaria affects **0.5–2%**, about two thirds women, peak 20–40 years.
  - Chronic inducible (physical) urticarias such as dermographism are more common.
- **Duration**
  - Chronic spontaneous urticaria lasts on average 2–5 years; about half clear within a year.

Aetiology: causes and risk factors

- Mast-cell degranulation (IgE, autoimmune, idiopathic, or direct — NSAIDs, opioids, contrast).
- CSU is usually not an IgE food allergy — restrictive diets and patch tests do not help.
- ACE inhibitors cause bradykinin angioedema that can start months to years after initiation.
- Inducible triggers: dermographism, delayed pressure, cholinergic, cold, solar, aquagenic.

Risk factors and associations

> **ACUTE TRIGGERS**
> viral or bacterial infection, **drugs** (NSAIDs, aspirin, antibiotics, opioids), IgE-mediated food allergy (milk, egg, peanut, shellfish), insect stings.

> **CHRONIC SPONTANEOUS**
> autoimmune (IgG autoantibodies to IgE or its receptor FcεRI, or IgE to self-antigens) in about half; associated with autoimmune thyroid disease.

> **INDUCIBLE**
> dermographism, cold, cholinergic (heat, sweat), pressure, solar, aquagenic, vibratory.

> **AGGRAVATORS**
> NSAIDs, alcohol, heat, stress, tight clothing.

Myths to address

> **"CHRONIC HIVES MUST BE AN ALLERGY"**
> chronic spontaneous urticaria is rarely allergic; extensive allergy testing is not helpful.

> **"ALL SWELLING RESPONDS TO ANTIHISTAMINES"**
> bradykinin angioedema (ACE inhibitor, hereditary angioedema) does not.

Pathophysiology and pathology

Weals come from **mast cells** in the dermis releasing histamine and other mediators; angioedema is the same process deeper down (or bradykinin-driven).

1. **Mast cells are activated**
   Triggers (IgE cross-linking by allergen, autoantibodies to FcεRI, complement, drugs, physical stimuli) activate dermal **mast cells** and basophils.

2. **Mediators are released**
   Mast cells degranulate, releasing **histamine**, prostaglandin D2, leukotrienes, platelet-activating factor and cytokines.

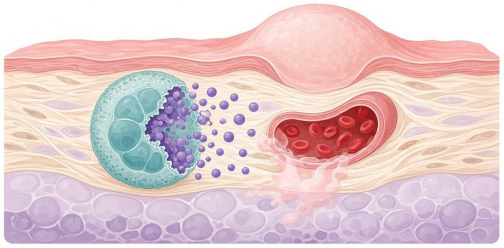

Fig. 1 · A dermal mast cell releases histamine-containing granules; the nearby vessel leaks fluid, raising a weal.
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3 **Vessels leak, nerves itch**

Histamine dilates and opens small vessels (**oedema and redness**: the weal and flare) and stimulates sensory nerves (**itch**).

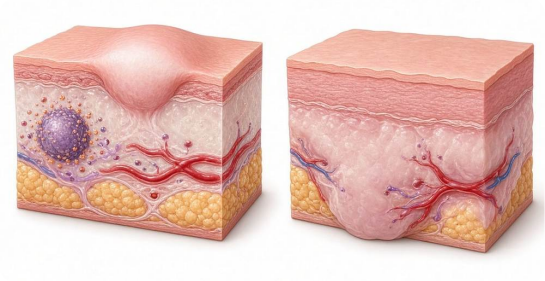

Fig. 2 · Weal (superficial dermal oedema) vs angioedema (oedema of deep dermis and subcutis).

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4 **Deeper swelling: angioedema**

The same leak in the **deep dermis, subcutis or mucosa** gives angioedema; **bradykinin** causes non-itchy angioedema without weals (ACE inhibitors, C1-inhibitor deficiency).

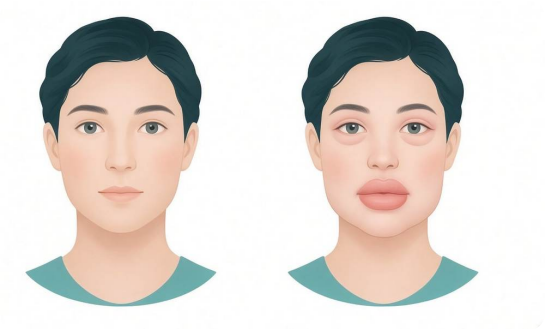

Fig. 3 · Angioedema of the lips and eyelids.

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5 **Weals resolve**

Mediators are cleared within hours, so each weal fades within 24 hours without marks; lesions lasting longer or bruising suggest **urticarial vasculitis**.

HISTOPATHOLOGY

- Dermal oedema with dilated vessels and a sparse perivascular infiltrate (neutrophils, eosinophils, lymphocytes); epidermis normal.
- Biopsy only if weals last over 24 hours, burn or bruise (to exclude urticarial vasculitis).

Clinical features

- Wheals: itchy, pink or skin-coloured, migrate, **individually fade within 24 hours**, leave normal skin.
- In skin of colour, feel for oedema when erythema is subtle.
- Angioedema: lips, eyelids, hands, genitals — may burn more than itch.
- Vasculitis clue: lesions >24 h, burning/pain, purpura or residual bruise.

Fig. 4 · Acute urticaria — itchy, raised, blanching wheals with surrounding flare; each wheal fades within 24 h without a mark.

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Types to recognise

Acute urticaria

Raised blanching wheals with flare. Site: Anywhere.

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With angioedema

Wheals + periorbital swelling. Site: Eyelids.

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Dermographism

Linear wheals where skin is stroked. Site: Trunk.

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Cold urticaria

Wheals on cold-exposed skin; warn about cold water. Site: Exposed skin.

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Cholinergic

Small 2–5 mm wheals with flare on heat or exercise. Site: Trunk.

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Urticarial vasculitis

> 24 h, burning, leaves bruising. Site: Trunk, limbs.

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Special groups

- **Skin of colour**
  - Weals may look skin-coloured or slightly darker rather than red; rely on raised, itchy, transient swellings.
- **Pregnancy and breastfeeding**
  - Loratadine or cetirizine are usually preferred (check BNF).

Diagnosis and investigations

- Clinical history ± phone photos. Acute urticaria: **no routine tests**.
- CSU: keep tests minimal — FBC and CRP/ESR reasonable; thyroid tests only if indicated. **Patch testing has no role** in urticaria.
- Activity tools: UAS7 / UCT for severity and omalizumab eligibility discussions.
- Suspected inducible disease: challenge testing in clinic/specialist settings.

History

- Timing of each wheal; phone photos.
- Drugs: **ACE inhibitor**, NSAIDs, opioids.

Acute

- **No routine tests.**

Chronic

- Keep minimal: FBC, CRP / ESR; thyroid only if indicated.
- **Patch testing has no role.**

Score it

- **UAS7 / UCT** for severity and omalizumab discussions.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Urticarial vasculitis	> 24 h, pain, purpura or bruise.
	Erythema multiforme	Fixed targets, acral, lasting days.
	Papular urticaria / bites	Grouped papules lasting days.
	Contact eczema	Scaly, lasting days; patch testing relevant.

	CONDITION	DISTINGUISHING FEATURES
	Mastocytosis	Fixed brown macules; **Darier sign**.
	ACE-inhibitor angioedema	Swelling **without wheals** on an ACEI.

Management

General

- Remove triggers; avoid NSAIDs/aspirin/opioids if they aggravate; prefer paracetamol.
- Cool measures; avoid overheating and tight clothes.

Antihistamines (CKS / guideline themes)

- First-line: **cetirizine, loratadine or fexofenadine** at licensed daily dose — take **every day** in CSU, not only as needed.
- If uncontrolled after 2–4 weeks: increase up to **four times** the licensed dose (off-label; counsel sedation risk at higher doses).
- Switch agent if one fails at high dose. Short sedating antihistamine at night only when sleep is wrecked — not routine with high-dose non-sedating drugs.
- Severe acute flare: short oral prednisolone (e.g. up to 7 days per CKS) **plus** antihistamine — avoid repeated courses in CSU.
- Montelukast sometimes added on specialist advice (NSAID-aggravated/pressure). H2 blockers are no longer routine.

Special situations

- ACE-inhibitor angioedema ± without wheals: **stop the ACEI**; do not rechallenge; antihistamines/steroids help poorly — emergency care if airway threatened; switch to ARB only with caution and specialist advice.
- Uncontrolled CSU dermatology/allergy for **omalizumab** (NICE TA) then other immunosuppressants.
- Pregnancy/breastfeeding/renal/hepatic dosing: check BNF carefully (cetirizine often favoured in pregnancy counselling themes — verify current advice).

⚠ **RED FLAG**

Anaphylaxis

Airway, breathing or circulation compromise with urticaria/angioedema intramuscular adrenaline and emergency services (RCUK). Do not treat as “just hives”.

Medicines at a glance

Cetirizine, loratadine or fexofenadine

First line

DOSE	Licensed daily dose · Every day in CSU, not as needed
CAUTIONS	Step down slowly once controlled.

Treatment ladder

1 **General**

Remove triggers; avoid NSAIDs, aspirin, opioids if they aggravate; paracetamol preferred. Cool measures; avoid overheating and tight clothes.

2 **First line**

Cetirizine, loratadine or fexofenadine

3 **Uncontrolled after 2–4 weeks**

Increase up to **4× licensed dose** (off-label); counsel sedation. Switch agent if one fails at high dose. Sedating antihistamine at night only if sleep is wrecked.

4 **Severe acute flare**

Short oral prednisolone (up to 7 days per CKS) **plus** antihistamine. Avoid repeated courses in CSU.

5 **Uncontrolled CSU**

Dermatology / allergy for **omalizumab** (NICE TA), then other immunosuppressants. Montelukast sometimes on specialist advice; H2 blockers no longer routine.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **ACE-inhibitor angioedema**: Stop the ACEI, don't rechallenge; antihistamines and steroids help poorly. ARB only with caution and advice.
- **Pregnancy / renal / hepatic**: Check BNF carefully.

Referral

999 / A&E	**Anaphylaxis** features.
999 / A&E	**Airway-threatening angioedema** (ACEI or other).
URGENT	Lesions > 24 h with purpura: vasculitis work-up.
URGENT	Fever, arthralgia, weight loss with urticarial rash: systemic disease / Schnitzler.
ROUTINE	CSU uncontrolled on up-dosed antihistamine.

Red flags and when to refer

- Anaphylaxis features.
- Airway-threatening angioedema (ACEI or other).
- Lesions lasting >24 h with purpura vasculitis work-up.
- Fever, arthralgia, weight loss with urticarial rash systemic disease / Schnitzler consideration.
- CSU uncontrolled on up-dosed antihistamine refer.

Complications and prognosis

Complications

- **Anaphylaxis** (with respiratory or circulatory compromise).
- Airway compromise from laryngeal or tongue angioedema.
- Sleep loss, poor quality of life, anxiety and depression in chronic urticaria.

Prognosis

- Acute urticaria usually settles within days to weeks.
- Chronic spontaneous urticaria: about 50% remit within a year; some persist for more than 5 years.

Prescriber pearls (UK pharmacist IP)

- "It isn't an allergy to something you keep eating" — most CSU is internal.
- Take the antihistamine every day, even on good days; step down slowly once controlled.
- Patch testing investigates eczema, not wheals.
- Always check BNF for residual renal/hepatic dosing and pregnancy.
- Swelling without wheals on an ACE inhibitor: see ACE-inhibitor angioedema.
- Wheals lasting more than 24 hours that bruise or burn: see urticarial vasculitis.

Self-check quiz

5 questions · optional

Q1 How long does an individual urticarial wheal last?

✓ Less than 24 hours, leaving normal skin. Longer with bruising suggests urticarial vasculitis.

Q2 First-line treatment for CSU and how to take it?

✓ A second-generation H1 antihistamine (cetirizine, loratadine or fexofenadine) every day, not as needed.

Q3 Next step if uncontrolled after 2–4 weeks?

✓ Up-dose to up to four times the licensed dose (off-label), then refer for omalizumab if still uncontrolled.

Q4 Lip swelling without wheals on ramipril?

✓ ACE-inhibitor (bradykinin) angioedema: stop the ACEI; emergency care if the airway is threatened.

Q5 Is patch testing useful in urticaria?

✓ No: it investigates delayed contact eczema, not wheals.

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Photo gallery

Acute urticaria and CSU

7

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Inducible urticaria & dermographism 8

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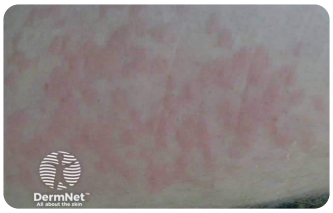

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Cutaneous mastocytosis 5

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Angioedema 5

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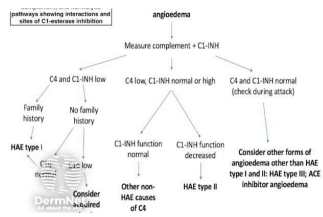

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Red flag: urticarial vasculitis

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Inflammatory skin Module 4 · Week 3

Contact dermatitis

Irritant vs allergic contact eczema — avoidance, emollients, stepped topical steroids, and when to refer for patch testing. Deepens the contact themes on the eczema page.

ALSO CALLED allergic contact dermatitis · irritant contact dermatitis · ACD · ICD · patch testing · occupational dermatitis · +2

RELATED PAGES Hand eczema (chronic, occupational & pompholyx) · Eczema / dermatitis · Emollients & soap substitutes · Topical corticosteroids: potency, quantities & safety

KEY FACTS

- Eczema caused by something touching the skin: **irritant (ICD, about 80%)** or **allergic (ACD, type IV delayed hypersensitivity)**.
- Hands and face are the commonest sites; occupational exposure is frequent (hairdressers, healthcare, cleaners, builders).
- ACD: about **25 allergens** cause half of cases; nickel is the leading cause worldwide.
- Identify and avoid the cause (**patch testing** for suspected ACD), emollients and soap substitutes, topical steroid.
- Persistent occupational dermatitis may need specialist and occupational health input.

Overview

Contact dermatitis is eczema caused by something touching the skin. **Irritant contact dermatitis (ICD)** is commoner and dose-dependent. **Allergic contact dermatitis (ACD)** needs prior sensitisation and is confirmed by **patch testing**.

Atopic eczema pathways and general steroid/emollient skills sit on eczema / dermatitis. This page deepens ICD vs ACD, occupational disease, cosmetics, tattoos and patch-testing referral.

Epidemiology

- **Common**
  - Contact dermatitis accounts for most **occupational skin disease**; irritant contact dermatitis is about four times commoner than allergic.
- **Who**
  - Any age; more common in adults with wet work; atopic people are at higher risk of irritant dermatitis.
  - Nickel allergy is commonest in women (jewellery, piercings) but falling in Europe since the EU Nickel Directive.

Aetiology: causes and risk factors

	ICD	ACD
Mechanism	Direct toxic/physical barrier injury	Type IV T-cell hypersensitivity
Sensitisation	Not required	Required
Onset	Minutes–hours or cumulative	Typically 24–72 h after re-exposure
Dose	Concentration/duration matter	Tiny amounts can elicit once sensitised

- Common irritants: water, detergents, solvents, cutting oils, friction, heat.
- Common allergens: nickel, fragrance mixes, rubber accelerators, preservatives (MI/MCI), PPD hair dye, topical antibiotics/steroids, epoxy resins.
- Mixed ICD+ACD is frequent on hands. About four-fifths of occupational hand dermatitis starts as irritant.

Risk factors and associations

> **IRRITANTS**
> water and wet work, soaps and detergents, solvents, friction, cutting oils, alkalis and acids.

> **COMMON ALLERGENS**
> nickel, cobalt, chromate (cement, leather), fragrances, preservatives (**methylisothiazolinone**, formaldehyde releasers), **PPD** hair dye, rubber accelerators, topical medicines (neomycin, corticosteroids), plants (Primula, Compositae).

> A damaged barrier (e.g. atopic eczema, leg ulcers, perianal disease) increases sensitisation to topical medicines.

Myths to address

- **"I've used it for years, so I can't be allergic"**: sensitisation can develop after years of exposure.
- **"HYPOALLERGENIC PRODUCTS ARE SAFE"**
  they can still contain relevant allergens.

Pathophysiology and pathology

Irritant dermatitis is direct damage; **allergic** dermatitis is a learned T-cell response.

1 **Irritant damage**
Irritants strip lipids and damage keratinocytes directly, causing barrier loss and non-specific inflammation in anyone exposed enough; no prior sensitisation needed.

Fig. 1 · Irritant dermatitis (direct damage) vs allergic dermatitis (sensitised T cells).
AI-generated illustration

2 **Sensitisation phase (allergic)**

Small reactive chemicals (**haptens**, under 500 Da) bind skin proteins; Langerhans cells carry them to lymph nodes and prime specific **T cells** (10–15 days).

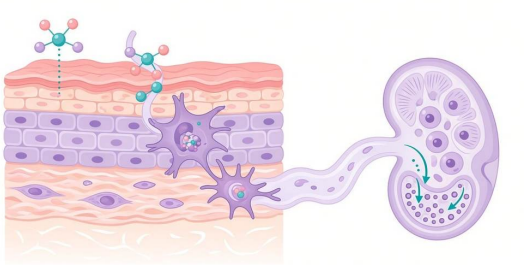

Fig. 2 · Sensitisation: a small chemical (hapten) binds skin protein; a Langerhans cell carries it to the lymph node, where specific T cells multiply.

AI-generated illustration

3 **Elicitation phase**

On re-exposure, memory T cells release cytokines at the contact site within **24–72 hours**, causing eczema (type IV hypersensitivity).

4 **Spread and chronicity**

Eczema can spread beyond the contact area (autosensitisation); continued exposure leads to chronic, lichenified, fissured dermatitis.

HISTOPATHOLOGY

- Spongiotic dermatitis (intercellular oedema, vesicles) with lymphocytes; eosinophils favour allergic dermatitis.
- Irritant: keratinocyte necrosis and neutrophils in acute damage. Histology cannot identify the cause: **patch testing** does.

Clinical features

- Both look eczematous: erythema, vesicles, scale, fissures, lichenification.
- ICD: burning/stinging often > itch; sharp borders early; interdigital “sentinel” wet-work dermatitis.
- ACD: itch dominates; geometric patterns (glove, plaster, jewellery); may spread beyond contact.
- Facial/cosmetic ACD: eyelids, hairline; airborne fragrance can affect exposed face.
- Tattoo ink allergy: localised eczema or lichenoid reaction in a colour distribution.
- Skin of colour: erythema less obvious; violaceous pigment and lichenification more prominent.

Distribution

- **Hands** (wet work, gloves, occupational).
- **Face, eyelids and neck** (cosmetics, airborne allergens, nail varnish).
- **Earlobes, wrists and umbilicus** (nickel).
- **Scalp margin and face** (hair dye PPD).
- **Feet** (shoes: chromate, rubber).

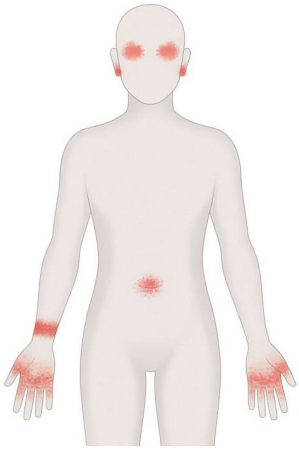

Fig. 3 · Classic allergic contact sites: eyelids (nail varnish, cosmetics), earlobes (nickel), wrist (watch strap), below the navel (jeans stud), hands (gloves, products).
AI-generated illustration

Diagnosis and investigations

- History: occupation, hobbies, cosmetics, topical medicines, holiday improvement.
- ICD: clinical exposure story; no confirmatory lab test.
- Suspected ACD that is chronic, recurrent, occupational, facial or treatment-resistant **refer for patch testing** (British baseline ± relevant series + patient's own products).
- Photopatch testing when sunscreen filters or photoallergens are suspected.
- Prick testing / specific IgE is for immediate contact urticaria (e.g. latex), not delayed ACD.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Atopic eczema	Flexural history; not limited to a contact site.
	Contact urticaria	Wheals within **minutes** (e.g. latex).
	Hand psoriasis	Sharp plaques, nail changes.
	Tinea	Annular scaly edge; scrape for mycology.

	CONDITION	DISTINGUISHING FEATURES
	Photocontact	Sun-exposed sites; shadow areas spared.

Management

1. Identify and **avoid** the irritant/allergen — improvement can take weeks.
2. Soap substitute + liberal emollient; cotton liners under gloves for wet work.
3. Short topical corticosteroid matched to site and severity (face/flexures: mild–moderate).
4. Refer for patch testing when indicated (CKS/BAD/PCDS).
5. Occupational cases: fit-note/workplace adjustments; RIDDOR awareness is employer-led — IPs recognise and signpost.

⚠ **CAUTION**

Emollient cautions

Do not use aqueous cream as a leave-on. Counsel paraffin fire risk and slippery bathroom floors.

ⓘ **NOTE**

Pigmented contact dermatitis

Grey-brown facial pigment from cosmetics/fragrances in darker skin — see contact & lichenoid pigmentation.

Treatment ladder

1 **Every patient**

Identify and **avoid** the irritant or allergen. Soap substitute + liberal emollient.

2 **Calm the flare**

Short topical corticosteroid matched to site and severity. Face and flexures: mild to moderate potency.

3 Find the allergen

Refer for **patch testing** when indicated (CKS / BAD / PCDS).

4 Occupational

Fit note and workplace adjustments. RIDDOR reporting is employer-led: recognise and signpost.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Aqueous cream**: Not as a leave-on emollient.
- **Paraffin emollients**: Fire risk on fabrics; slippery floors.
- **Facial pigment**: Grey-brown pigment from cosmetics in darker skin: see contact & lichenoid pigmentation.

Referral

URGENT	Severe **facial or eyelid swelling**, or systemic contact dermatitis.
ROUTINE	**Occupational** disease threatening livelihood: early patch testing and workplace advice.
ROUTINE	Suspected **airborne** allergy continuing despite stopping perfume.
ROUTINE	Not healing on treatment: think allergy to the **medicament** itself.

Red flags and when to refer

- Occupational disease threatening livelihood — early patch testing and workplace advice.
- Severe facial/eyelid swelling or systemic contact dermatitis.
- Suspected airborne allergy continuing despite “stopping perfume”.
- Non-healing dermatitis on medicaments — think contact allergy to the treatment itself.

Complications and prognosis

Complications

- Chronic hand dermatitis with loss of work; secondary bacterial infection.
- Severe facial and scalp swelling after PPD hair dye.
- Spread (autosensitisation) and erythroderma (rare).

Prognosis

- ICD improves when exposure stops, but barrier recovery takes weeks to months.
- ACD is lifelong: avoidance prevents recurrence; occupational ACD can persist even after avoidance.

Prescriber pearls (UK pharmacist IP)

- Many “allergies” on the hands are **irritant** — still treat the barrier aggressively.
- Atopics get ICD more easily and can still develop ACD.
- Hair-dye (PPD) and nail-varnish eyelid patterns are classic teaching cases.
- Always check BNF/SmPC for topical steroid site limits.

Self-check quiz

5 questions · optional

Q1 Irritant or allergic: which needs prior sensitisation?

✓ Allergic contact dermatitis (type IV). Irritant needs no sensitisation and is dose-dependent.

Q2 When does allergic contact dermatitis appear after re-exposure?

✓ Typically 24–72 hours.

Q3 Which test confirms allergic contact dermatitis?

✓ Patch testing (prick tests / IgE are for immediate contact urticaria).

Q4 Who should be referred for patch testing?

✓ Chronic, recurrent, occupational, facial or treatment-resistant suspected allergic contact dermatitis.

Q5 A rash worsens under a cream you prescribed. Think?

✓ Contact allergy to the medicament itself (e.g. topical antibiotic or steroid).

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Photo gallery

ACD vs ICD 4

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Facial and cosmetic contact 3

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Airborne / fragrance

3

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Tattoo ink allergy

3

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Occupational hand dermatitis

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Patch testing

4

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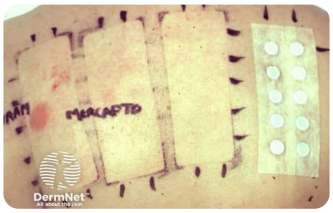

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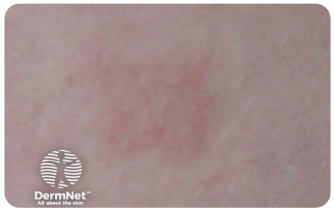

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Device / drug contact

1

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Inflammatory skin Module 4 · Week 1 · Module 3 · Week 6

Erythema multiforme

An acute immune reaction, usually to herpes simplex, giving fixed target lesions on the hands, feet and limbs, sometimes with mouth erosions. It is not the same as SJS/TEN, which is drug-driven and life-threatening.

ALSO CALLED EM | target lesions | EM major | EM minor | herpes-associated EM | Mycoplasma | +1

RELATED PAGES Herpes simplex labialis (cold sores) | SJS/TEN and drug-induced blistering | Urticaria & angioedema | Fixed drug eruption | Annular (ring-shaped) rashes: differential

KEY FACTS

- **Erythema multiforme (EM)** is an acute immune reaction with **target lesions** (three zones), mostly triggered by **herpes simplex**.
- Acral (hands, feet, elbows, knees), symmetrical, appears over 24–72 hours; EM major also involves mucosa.
- Not part of the SJS/TEN spectrum: SJS has atypical flat targets, widespread blistering and is usually drug-induced.
- Supportive care; treat HSV; recurrent HSV-associated EM may need antiviral prophylaxis (specialist).
- Urgent same-day care if mucosal involvement impairs eating, eyes are affected or skin is blistering.

Overview

Erythema multiforme (EM) is an acute, immune-mediated mucocutaneous reaction characterised by **typical target (iris / bull's-eye) lesions**. Most cases are infection-triggered, especially **HSV**. **EM minor** = skin only; **EM major** = skin plus 1 mucosal surface (oral, ocular, genital) — still usually infection-related, and is **not** the same disease as drug-induced SJS/TEN (which has different morphology, drug trigger, and severity).

Epidemiology

- **Who**
  - Young adults 20–40; recurs in some with recurrent HSV.

Aetiology: causes and risk factors

- Predominantly a **type IV (cell-mediated / delayed)** hypersensitivity reaction.
- **HSV-1/2** is the leading association (often 1–3 weeks after herpes labialis/genitalis; recurrent EM with recurrent HSV).

- Other infections: Mycoplasma pneumoniae (esp. children/young adults with respiratory prodrome), less often other viruses/bacteria.
- Drugs can rarely trigger EM-like eruptions, but severe drug blistering disease should be managed as SJS/TEN pathway until proven otherwise.
- Circulating immune response to microbial antigen cytotoxic T-cell attack on keratinocytes epidermal necrosis in the target centre.

Risk factors and associations

- HSV (most), Mycoplasma pneumoniae (especially children), other infections; drugs less often.

Myths to address

- **"EM IS MILD SJS"**
  they are different diseases with different causes and outlooks.

Pathophysiology and pathology

EM is a **T-cell reaction to viral DNA** deposited in the skin.

1. **HSV DNA in skin**
   Fragments of HSV DNA are carried to distant skin by blood cells.
2. **T-cell response**
   HSV-specific T cells release interferon-γ, damaging keratinocytes.
3. **Target zones**
   Dusky or blistered centre (necrosis), pale oedematous ring, red outer ring.

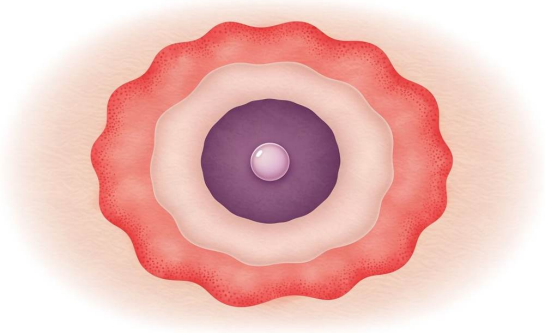

Fig. 1 · The three zones of a typical target lesion.
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4 **Acral distribution**

Lesions favour the hands, feet and extensor limbs.

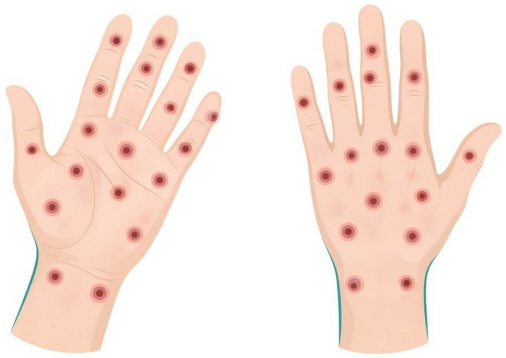

Fig. 2 · Acral target lesions on the hands.

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HISTOPATHOLOGY

- Interface dermatitis with necrotic keratinocytes; HSV PCR from a cold sore may help.

Clinical features

FEATURE	DETAIL
Typical target	Three zones: dusky/purpuric or blistered centre pale oedematous ring outer erythematous ring
Atypical targets / papules	May coexist; may coalesce
Distribution	Symmetrical; acral preference — hands/feet dorsa, extensor limbs, then trunk; face possible
Mucosa	Oral erosions/ulcers common in EM major; lips crusted; eyes/genitals less often
Prodrome	Mild fever/malaise optional; history of recent cold sore is a strong clue
Course	Lesions appear over days; resolve over 1–4 weeks; may recur with HSV

Skin of colour: Central duskiness and erythema may be harder to appreciate — rely on zone architecture and palpation.

Erythema multiforme on the palms and soles

Full EM chapter lives in the inflammatory rash pages notes (pathophysiology, EM major vs SJS/TEN, mucosal care, recurrent HSV prophylaxis). Use this section as a **cross-link box** in the palmar/plantar week.

Why EM belongs in a palmar/plantar week

EM has an **acral predilection** — it typically begins on the **dorsal hands and feet**, palms, soles and extensor forearms, then spreads centripetally to limbs and (less) trunk. Palm/sole **typical target lesions** are often the clinching sign.

Reading a palmar target

> **TYPICAL TARGET**
>
> 3 cm, round, **three zones** — dark-red/dusky (sometimes blistered or crusted) centre paler oedematous ring outer red ring with a sharp edge.

> **RAISED ATYPICAL TARGET**
> two zones, palpable — still EM.

> **Flat atypical targets / purpuric macules, widespread on trunk** think **SJS/TEN** (drug-related), especially with mucosal erosions or skin pain.

> Palms: thick stratum corneum can make targets look flatter and more "bruise-like"; the iris pattern is often easiest to see here.

Worked case

1. Diagnosis erythema multiforme **minor** (typical acral targets, polymorphous papules, **no mucosal involvement**).
2. Trigger most likely the **preceding viral infection** ("flu" 2 weeks earlier) — EM usually follows infection by 1–3 weeks; **HSV** is the commonest overall trigger (ask about cold sores, even subclinical) and Mycoplasma pneumoniae in young adults with respiratory symptoms. Drugs (sulfonamides, antiepileptics, NSAIDs, antibiotics) account for a minority of true EM; **paracetamol** and **chlorphenamine** are rare/doubtful culprits — note them, but the timing and morphology favour infection.
3. Treatment self-limiting over ~2–4 weeks. Supportive: emollient, moderate–potent topical steroid for symptomatic lesions, oral non-sedating antihistamine for itch/burning, analgesia. Treat identified infection (e.g. aciclovir only if active HSV; macrolide if Mycoplasma confirmed). Stop non-essential new drugs. **Recurrent** HSV-associated EM dermatology may use continuous oral aciclovir/valaciclovir prophylaxis. Systemic steroids only under specialist advice in severe mucosal EM.
4. Safety-net / urgent referral mucosal (oral, eye, genital) erosions, blistering/skin detachment, skin pain, fever/systemic upset, widespread flat atypical targets same-day assessment for EM major vs SJS/TEN; eye involvement ophthalmology.

Palm/sole target-like or macular differentials

CONDITION	CLUES
Secondary syphilis	Copper-red scaly macules on palms/soles, no true targets, mucous patches, lymphadenopathy, sexual history serology
Hand, foot and mouth disease	Grey oval **vesicles** on palms/soles, oral ulcers, child contacts
Urticaria (urticaria multiforme in children)	Weals with dusky centres that **move within 24 h**, no epidermal change
SJS/TEN	Drug 4–28 days before, flat atypical targets, mucosa, pain, detachment
Fixed drug eruption	Recurrent dusky plaque at same site with re-exposure
Acral erythema / hand–foot syndrome	Chemotherapy (capecitabine etc.) — painful symmetrical palmar erythema
Kawasaki disease (child)	Fever 5 days, palmoplantar erythema/oedema later periungual peeling
Vasculitis	Palpable purpura, lower legs

Photos

- Mucosal/eye EM: .
- DermNet — Erythema multiforme: dermnetnz.org

Fig. 3 · Classic target (iris) lesions on limbs — dusky centre, oedematous ring, outer erythema; post-HSV context.

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Types to recognise

Classic targets

Three-zone iris lesions. Site: Limbs.

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Palm

Targets with dusky centres. Site: Palms.

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Mucosal (EM major)

Crusted lips, oral erosions. Site: Lips, mouth.

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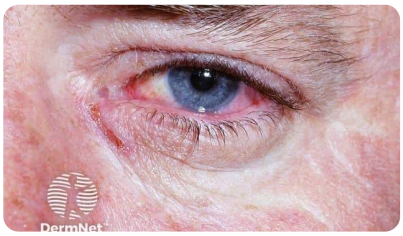
Eye involvement

Urgent ophthalmology. Site: Eyes.



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Special groups

- **Children**
  - Mycoplasma-induced rash and mucositis (MIRM) causes severe mucositis with few skin lesions.

Diagnosis and investigations

- **Clinical** in classic post-HSV target pattern.
- History: recent HSV, Mycoplasma/respiratory symptoms, drugs, mucosal symptoms, eye symptoms.
- Exam: full skin + oral + eyes + genitals.
- Investigate if atypical / severe: Mycoplasma testing if respiratory; HSV swabs of cold sores; consider biopsy if uncertain.
- **Do not** label as SJS solely because mucosa is involved — target morphology + infection trigger favour EM.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	SJS / TEN	Flat atypical targets, widespread purpuric macules, sheet-like detachment; **drugs**; very unwell.
	Urticaria / urticaria multiforme	Migratory wheals < 24 h; no fixed dusky centre.
	Fixed drug eruption	Round dusky plaque recurring at the same site.
	Secondary syphilis	Risk history; serology.
	Vasculitis / EM-like lupus	Atypical course; ANA, systemic features.

EM vs SJS vs TEN

Also consider: Mycoplasma pneumoniae-induced rash and mucositis (RIME), Kawasaki disease, toxic shock syndrome, acute GVHD, linear IgA bullous dermatosis (vancomycin), acute cutaneous lupus (TEN-like), paraneoplastic pemphigus.

Management

Self-care / supportive

- Emollients; bland oral care (soft diet, antiseptic/analgesic mouth rinses per local advice) for oral EM.
- Analgesia; rest; treat concurrent HSV cold sore if active (topical/oral antivirals per licence/BNF).
- Document drug history; stop any newly started suspect drug pending specialist advice if morphology atypical.

Prescribing / referral pathway

- Mild cutaneous EM: often supportive only in primary care.
- Recurrent HSV-associated EM: dermatology may consider continuous or intermittent **oral antiviral prophylaxis** (e.g. aciclovir — specialist/experienced initiation).
- **Urgent same-day / emergency** if: extensive mucosal disease impairing intake, ocular involvement, suspected SJS/TEN, systemic toxicity, inability to distinguish from severe drug reaction.
- Ophthalmology review for eye involvement.
- IP: do **not** start systemic steroids for presumed EM without specialist advice (may be used in selected severe cases under derm).

Treatment ladder

1 **Supportive**

Emollients; soft diet and antiseptic / analgesic mouth rinses for oral EM; analgesia; rest. Document drug history; stop a newly started suspect drug pending advice if morphology atypical.

2 **Prescriber**

Potent topical steroid to itchy / inflamed skin lesions. Treat an active cold sore (topical / oral antiviral per BNF).

3 **Recurrent HSV-EM**

Continuous or intermittent oral antiviral prophylaxis (e.g. aciclovir): specialist / experienced initiation.

4 **Specialist**

Systemic steroids only in selected severe cases under dermatology.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Systemic steroids**: Don't start for presumed EM without specialist advice.
- **Pregnancy**: Check BNF for antivirals / steroids.

Referral

999 / A&E	Mucosal erosions with **skin pain, blistering or skin loss**: possible **SJS / TEN**; stop suspect drugs, admit same day.
SAME DAY	Eye involvement: same-day ophthalmology.
SAME DAY	Unable to eat or drink: admission for fluids and analgesia.

Red flags and when to refer

- Mucosal erosions (mouth, eyes, genitals) with skin pain, blistering or skin loss: possible **SJS/TEN**, so stop suspect drugs and admit the same day.
- Eye involvement: same-day ophthalmology.
- Unable to eat or drink: admission for fluids and analgesia.

Complications and prognosis

Complications

- Eye scarring in EM major; dehydration from painful mouth.

Prognosis

- Resolves in 2–4 weeks without scarring; recurrences common with HSV.

Prescriber pearls (UK pharmacist IP)

- Teach patients with recurrent cold sores that EM can follow HSV — early antiviral for prodromal HSV may reduce triggers (individualise).
- Emphasise difference between EM major and SJS — reduces panic but does not delay referral when severe.
- Warn that scars are uncommon but pigment change can linger.
- Pregnancy: check BNF for any antivirals/steroids contemplated.

Self-check quiz

4 questions · optional

Q1 Commonest trigger?

✓ Herpes simplex virus.

Q2 Three zones of a typical target?

✓ Dusky / blistered centre, pale oedematous ring, red outer ring.

Q3 EM major vs minor?

✓ Major involves 1 mucosal surface.

Q4 Recurrent EM with each cold sore: prevention?

✓ Suppressive oral antiviral (specialist).

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Photo gallery

Erythema multiforme 4

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Acral targets (palms and hands) 4

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EM vs SJS/ TEN 3

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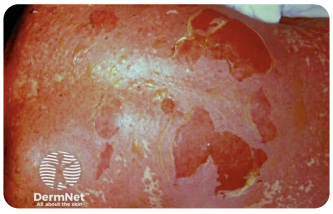

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Inflammatory skin Module 4 · Week 1

Annular (ring-shaped) rashes: differential

Not every ring is ringworm. Tinea, granuloma annulare, discoid eczema, pityriasis rosea, erythema annulare centrifugum and subacute lupus all form rings. Scale, its position and a skin scraping sort most of them.

ALSO CALLED ringworm or not | annular scaly patch | granuloma annulare | erythema annulare centrifugum | herald patch | nummular eczema | +1

RELATED PAGES Tinea: corporis, cruris & pedis | Pityriasis rosea | Lyme disease & erythema migrans | Urticaria & angioedema | Erythema multiforme

KEY FACTS

- Ring-shaped (**annular**) rashes have many causes: **tinea** (scaly active edge), granuloma annulare (no scale), erythema migrans, urticaria, pityriasis rosea herald patch, erythema multiforme, discoid eczema, subacute lupus, psoriasis, sarcoidosis.
- Key questions: **scale?** (where), duration of each lesion, itch, symptoms, sun exposure, tick bites, drugs.
- Scale at the advancing edge with central clearing: scrape for fungus before steroid.
- Trailing scale inside the edge: pityriasis rosea or erythema annulare centrifugum.
- Annular lesions lasting <24 h: urticaria; >24 h with bruising: urticarial vasculitis.

Overview

(Module 4 Case 3 teaching frame: solitary annular scaly erythema on chest later widespread eruption diagnosed as **pityriasis rosea**. Use the initial lesion to practise annular DDx.)

An **annular (ring-shaped) scaly erythematous patch/plaque** is a common primary-care puzzle. Leading differentials: **tinea corporis**, **annular / plaque psoriasis**, **granuloma annulare** (usually non-scaly), **herald patch of pityriasis rosea**, **erythema annulare centrifugum (EAC)**, discoid eczema, and less often subacute cutaneous lupus or erythema migrans.

Aetiology: causes and risk factors

CONDITION	MECHANISM (PLAIN LANGUAGE)
Tinea corporis	Dermatophyte infection of stratum corneum
Annular psoriasis	Same immune axis as plaque psoriasis; ring morphology
Granuloma annulare	Granulomatous dermal inflammation; often idiopathic; DM association sometimes discussed
Pityriasis rosea	Likely viral (HHV-6/7 hypothesis); self-limiting reactive eruption

CONDITION	MECHANISM (PLAIN LANGUAGE)
EAC	Reactive figurate erythema; trailing scale; hunt for triggers (infection, drugs, rarely malignancy)

Myths to address

- **"EVERY RING IS RINGWORM"**
  granuloma annulare and other rings are commonly mistaken for tinea.

Pathophysiology and pathology

Rings form when a process **spreads outwards while the centre heals**.

1. **Centrifugal spread**
   Fungi, bacteria or immune activity move outwards; the centre resolves.

2. **Edge vs centre**
   Active inflammation and organisms sit at the edge.

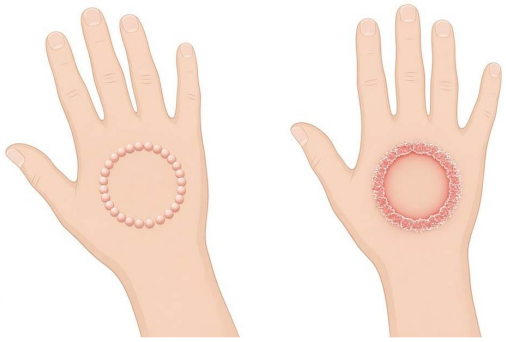

Fig. 1 · Granuloma annulare (smooth, no scale) versus tinea (scaly edge).
AI-generated illustration

3. **Scale position**
   Leading-edge scale (tinea) vs trailing collarette (pityriasis rosea).

Fig. 2 · Collarette scale versus tinea edge.

AI-generated illustration

Clinical features

CONDITION	SCALE / EDGE	ITCH	CLUES
Tinea corporis	Active raised scaly edge; central clearing	Variable	Asymmetry; pet/soil exposure; feet/nails reservoir
Annular psoriasis	Silvery scale; may be symmetric	Mild–mod	Other plaques, nails, scalp, family Hx
Granuloma annulare	**Usually smooth, non-scaly** rings	Mild/ none	Dorsum hands/feet, ankles; kids/young adults

CONDITION	SCALE / EDGE	ITCH	CLUES
PR herald patch	Oval plaque with collarette scale	Mild	Days–weeks later Christmas-tree secondary rash
EAC	Trailing (inner) scale; slow centrifugal spread	Mild	Trunk/thighs; chronic/recurrent

Case 3 arc: Herald patch misread as isolated annular lesion antifungal or mild TCS may "fail" as secondary PR eruption appears.

Fig. 3 · Granuloma annulare — flesh-coloured/ violaceous rings; usually NON-scaly (key vs tinea/psoriasis).

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Types to recognise

Tinea corporis

Active scaly edge. Site: Trunk, limbs.

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Granuloma annulare

Smooth flesh / violaceous rings. Site: Hands, feet.

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PR herald patch

Oval, collarette scale. Site: Trunk.

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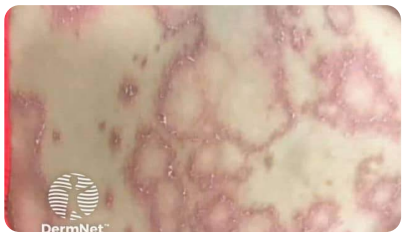
EAC

Trailing scale inside the edge. Site: Trunk, thighs.



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Diagnosis and investigations

- History: pets, steroids already used, herald then cascade, joint/nail clues, Lyme geography/tick bite.
- **Mycology scrape** before committing to prolonged TCS if tinea plausible (single most useful primary-care test).
- Wood's lamp limited value for most corporis species.
- Review in 1 week if solitary lesion — watch for PR secondary rash or tinea expansion.
- Biopsy if persistent atypical annular disease (EAC, SCLE, GA confirmation).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Tinea vs PR herald	Mycology; PR is followed by widespread ovals.
	Tinea vs GA	Scale vs no scale; KOH.
	Annular psoriasis	Symmetry, silvery scale, nails / scalp; KOH –.
	Erythema migrans	Expanding after tick bite.
	Nummular eczema	Coin plaques, very itchy, not a true ring.
	SCLE	Photosensitive annular plaques; ANA / Ro.

Management

SUSPECTED TINEA

topical terbinafine or imidazole (duration per site/BNF); oral terbinafine if extensive/failed topical — check interactions/LFTs per guidance; treat foot reservoir.

SUSPECTED PR

reassure self-limiting (often 6–12 weeks); emollient ± mild–moderate TCS for itch; BAD/PCDS PIL; atypical (palms/soles, very symptomatic) consider syphilis serology / derm.

GA

many localised cases observe; TCS/tacrolimus sometimes tried; extensive derm.

PSORIASIS

NICE topical pathway.

EAC

treat underlying trigger if found; topical steroids symptomatic; derm if persistent.

Treatment ladder

1 **Psoriasis / EAC**

Psoriasis: NICE topical pathway. EAC: treat trigger; topical steroid; dermatology if persistent.

2 **Tinea**

Topical terbinafine or imidazole (duration per BNF); oral terbinafine if extensive / failed (interactions, LFTs); treat the foot reservoir.

3 **PR**

Reassure (6–12 weeks); emollient ± mild–moderate steroid; atypical syphilis serology / dermatology.

4 **GA**

Localised: observe; topical steroid / tacrolimus sometimes; extensive dermatology.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Potent steroid on an undiagnosed ring**: Risk of **tinea incognito**.

Referral

SAME DAY — Expanding ring after a tick bite: **erythema migrans**, treat clinically (NG95).

URGENT — Photosensitive annular plaques with systemic features or a new drug (PPI, terbinafine, thiazide): **SCLE**.

ROUTINE — Persistent fixed non-healing annular plaques in an older adult: biopsy (cutaneous lymphoma).

Red flags and when to refer

- Expanding ring after a tick bite: **erythema migrans**, treat on clinical grounds (NICE NG95).
- Photosensitive annular plaques with systemic features or a new drug (PPI, terbinafine, thiazide): think **subacute cutaneous lupus**.
- Persistent, fixed, non-healing annular plaques in an older adult: biopsy to exclude cutaneous lymphoma.

Complications and prognosis

Complications

- Misdiagnosis (tinea incognito after steroids).

Prognosis

- Depends on cause.

Prescriber pearls (UK pharmacist IP)

Never use potent TCS long-term on an undiagnosed annular plaque — risk of **tinea incognito**.

Explain PR herald secondary rash so patients do not think treatment "caused" spread.

Recheck feet and nails whenever you diagnose body tinea.

Self-check quiz

4 questions · optional

Q1 Best primary-care test for a scaly ring?

✓ Mycology scrape (KOH / culture).

Q2 Smooth beaded ring on the back of the hand?

✓ Granuloma annulare.

Q3 Trailing scale inside a slowly advancing edge?

✓ Erythema annulare centrifugum.

Q4 Risk of potent steroid on undiagnosed tinea?

✓ Tinea incognito.

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Photo gallery

Tinea corporis 2

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Granuloma annulare

2

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Pityriasis rosea

2

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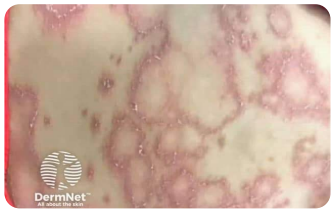
Erythema annulare centrifugum

1



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Inflammatory skin Module 4 · Week 1

Lichen planus

Itchy, flat-topped, purple, polygonal papules on the wrists, shins and lower back, often with white lacy streaks in the mouth. Mostly self-limiting on the skin; oral and genital erosive disease needs specialist care.

ALSO CALLED LP · oral lichen planus · Wickham striae · erosive lichen planus · lichenoid drug eruption · purple papules

RELATED PAGES Psoriasis · Scarring alopecia overview · Nail signs: psoriatic nails, Beau's lines & clubbing · Approach to a suspected drug eruption · Contact & lichenoid pigmentation

KEY FACTS

- **Lichen planus** is a T-cell–mediated inflammatory disease of skin, mucosa, hair and nails.
- The **6 Ps**: purple, polygonal, planar, pruritic papules and plaques, with white **Wickham striae**; flexor wrists, lower back, ankles.
- Check the **mouth**, genitals, nails and scalp; oral LP is often lace-like and white.
- Ask about drugs (lichenoid drug eruption) and consider **hepatitis C** testing according to risk.
- Potent topical steroids for skin; refer mucosal, nail, scalp (scarring) or extensive disease.

Overview

Lichen planus (LP) is an idiopathic inflammatory disorder of skin and/or mucosa presenting with intensely itchy **flat-topped, violaceous, polygonal papules**. Classic cutaneous LP favours **flexor wrists**, lumbar back, ankles. Mucosal, nail, scalp (lichen planopilaris), and genital variants exist. Association with **hepatitis C** is described in some populations — screen selectively by risk/prevalence.

Epidemiology

- **Who**
  - Adults 30–60; cutaneous LP affects about 1% of people, oral LP 1–2%.
  - Rare in children.
- **Course**
  - Skin LP usually clears within 1–2 years; oral and genital LP are often chronic.

Aetiology: causes and risk factors

- T-cell mediated attack on basal keratinocytes (lichenoid interface dermatitis).

- Triggers: idiopathic; drugs (**lichenoid drug eruption** — e.g. some antihypertensives, antimalarials, gold, TNF inhibitors — take drug history); dental amalgam sometimes linked to oral LP; HCV association variable by region.
- Koebnerisation along scratch lines possible.

Risk factors and associations

- Autoimmune attack on basal keratinocytes, possibly triggered by altered self-antigens (viral, drug).
- Drugs: beta-blockers, ACE inhibitors, thiazides, antimalarials, gold, checkpoint inhibitors.

Myths to address

> **"LP IS CONTAGIOUS"**
> it is not.

> **"ORAL LP IS JUST A MOUTH ULCER"**
> chronic erosive oral LP carries a small risk of SCC and needs follow-up.

Pathophysiology and pathology

Cytotoxic T cells attack the **basal layer** of the epidermis (an **interface dermatitis**).

1. **Antigen on keratinocytes**
   An altered antigen on basal keratinocytes is presented to T cells.

2. **T-cell attack**
   CD8+ T cells form a dense **band** along the dermo-epidermal junction and kill basal cells (apoptotic **Civatte bodies**).

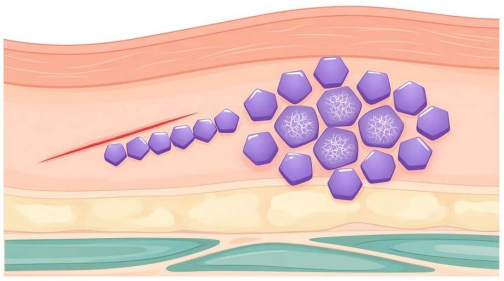

Fig. 1 · Flat-topped violaceous polygonal papules with Wickham striae and a Koebner line.
AI-generated illustration

3. **Thickened granular layer**
   The epidermis responds with a thick granular layer and **saw-tooth** rete ridges: seen clinically as Wickham striae.

4. **Pigment drop-out**
   Melanin falls into the dermis (pigment incontinence), causing long-lasting **post-inflammatory hyperpigmentation**, especially in darker skin.

5 **Mucosal disease**

In the mouth the same process causes white reticular lace and painful erosions.

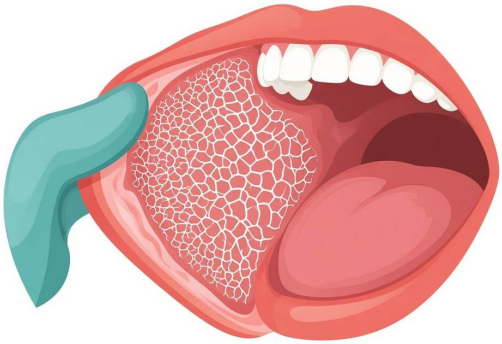

Fig. 2 · Reticular oral lichen planus on the buccal mucosa.

AI-generated illustration

HISTOPATHOLOGY

- Band-like lymphocytic infiltrate at the dermo-epidermal junction, hypergranulosis, saw-tooth acanthosis, Civatte bodies; direct immunofluorescence shows fibrinogen along the basement membrane.

Clinical features

FEATURE	DETAIL
Morphology	Violaceous, planar, polygonal, pruritic papules ± plaques (“6 Ps”); Wickham striae (white lace) on surface
Sites	Flexor wrists, lumbar, ankles, shins; genitals; oral (reticular white network, erosions)
Nails	Thinning, ridges, pterygium, shedding
Scalp	Scarring alopecia if lichen planopilaris
Post-inflammatory	Marked hyperpigmentation, especially skin of colour

Fig. 3 · LP on flexor wrists — intensely itchy; Case 4 classic distribution (wrists, lumbar, ankles).

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Types to recognise

Polygonal papules

6 Ps. Site: Wrists, shins.

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Flexor wrists

Intensely itchy. Site: Wrists.

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Oral LP

Wickham white reticulation. Site: Buccal mucosa.

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Koebner

Papules along a scratch. Site: Any.

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Special groups

- **Skin of colour**
  - Papules look grey-brown or purple-black; hyperpigmentation can be the main complaint.
- **Variants**
  - Hypertrophic (shins), annular, linear, bullous, actinic, lichen planopilaris (scarring alopecia), nail LP (pterygium).

Diagnosis and investigations

- Often **clinical**.
- Full exam: skin, mouth, nails, scalp, genitals.
- **Punch biopsy** if uncertain (band-like lymphocytic infiltrate, saw-tooth acanthosis, civatte bodies).
- Consider HCV testing based on risk/local guidance; drug review for lichenoid drug eruption.

Clinical

- Often clinical.

Full exam

- Skin, **mouth**, nails, scalp, genitals.

Biopsy

- If uncertain: band-like lymphocytic infiltrate, saw-tooth acanthosis, Civatte bodies.

Context

- HCV testing by risk / local guidance; **drug review**.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Lichenoid drug eruption	Drug timeline; sometimes photo-distributed.
	Discoid eczema	Coin plaques, less violaceous.
	Psoriasis	Silvery scale, extensors.
	Flat warts / lichen nitidus	Smaller, less itch.
	Secondary syphilis	Serology if atypical.
	Graft-vs-host disease	Transplant context.

Management

- Potent / very potent **TCS** ointments to itchy papules (short courses; caution thin skin); emollients.
- Antihistamines for itch (limited disease-modifying effect).
- Oral / genital / nail / extensive / scarring scalp disease **dermatology** (may need systemic steroids, phototherapy, or other immunosuppressants — specialist).
- Oral LP: topical steroid mouth preparations per dental/oral medicine pathways; check malignancy vigilance for persistent ulcers.
- Stop culprit drug if lichenoid drug eruption suspected (with prescriber of that drug).

Treatment ladder

1

Skin care

Emollients; avoid vigorous scrubbing of the scale.

2

First line

Potent / very potent topical steroid ointment to itchy papules, short courses (caution thin skin). Antihistamine for itch (sedating at night).

3

Drug-induced

Stop the culprit drug with its prescriber if lichenoid drug eruption suspected.

4

Specialist

Oral, genital, nail, extensive or scalp disease dermatology / oral medicine: steroid mouth preparations, systemic steroids, phototherapy, immunosuppressants.

Referral

URGENT	Non-healing oral ulcer, thickening white patch or lump in long-standing oral LP: **refer urgently** (oral SCC).
ROUTINE	Genital erosions, scarring or dyspareunia: refer (vulval / penile LP).
ROUTINE	Scalp redness with scarring hair loss: early referral (lichen planopilaris).

Red flags and when to refer

- Non-healing oral ulcer, white patch that thickens or a lump in long-standing oral LP: **refer urgently** (oral SCC).
- Genital erosions, scarring or dyspareunia: refer (vulval or penile LP).
- Scalp redness with scarring hair loss: lichen planopilaris, early referral.

Complications and prognosis

Complications

- Post-inflammatory hyperpigmentation.
- Scarring alopecia and permanent nail loss.
- Vulval/penile scarring; small risk of SCC in chronic erosive oral or genital LP.

Prognosis

- Cutaneous LP: most resolves in 1–2 years; hypertrophic LP persists longer. Mucosal LP is often lifelong.

Prescriber pearls (UK pharmacist IP)

- Warn about post-inflammatory pigment lasting months.
- Advise against vigorous scrubbing of Wickham-striae scale.
- Sexual health / genital itch: examine and refer early if erosive.
- Document HCV risk discussion where relevant.

Self-check quiz

4 questions · optional

Q1 Fine white lines on LP papules?

✓ Wickham striae.

Q2 Classic sites?

✓ Flexor wrists, lumbar back, ankles / shins, mouth.

Q3 Red flag in long-standing oral LP?

✓ Non-healing ulcer, thickening white patch or lump: urgent referral (SCC).

Q4 Nail sign of destructive LP?

✓ Pterygium.

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Photo gallery

Lichen planus 3

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Koebner phenomenon 2

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Inflammatory skin Module 4 · Week 1

Discoid lupus erythematosus

Chronic cutaneous lupus causing red, scaly plaques on the face, ears and scalp that scar, depigment and cause permanent hair loss. Strict photoprotection, potent steroids and early referral prevent disfigurement.

ALSO CALLED DLE | chronic cutaneous lupus | scarring plaques | carpet-tack scale | hydroxychloroquine | scarring alopecia lupus

RELATED PAGES Scarring alopecia overview | Dermatomyositis | Polymorphic light eruption | Sarcoidosis (incl. Löfgren syndrome) | Psoriasis

KEY FACTS

- **Discoid lupus erythematosus (DLE)** is chronic cutaneous lupus: scaly red plaques that heal with **scarring, atrophy and pigment change**.
- Face, ears (conchal bowl), scalp (scarring alopecia); photosensitive; follicular plugging ('carpet-tack' scale).
- About 5–10% progress to systemic lupus; check FBC, ANA, ENA, urine.
- Strict photoprotection, **stop smoking**, potent topical steroid; **hydroxychloroquine** via dermatology.
- Long-standing scars can develop SCC.

Overview

Discoid lupus erythematosus is the most common form of **chronic cutaneous lupus**. Scarring inflammatory plaques favour head and neck; scalp disease causes **scarring alopecia**. Most patients have skin-limited disease, but a minority evolve toward systemic lupus erythematosus (SLE). Module 4 Formative Quiz-1 answer was DLE (vs alopecia areata, LPP, tinea capitis, trichotillomania).

Epidemiology

- **Who**
  - Women 20–40; more common and more scarring in people of African descent.

Aetiology: causes and risk factors

- Autoimmune interface dermatitis with follicular involvement.
- UV exposure flares lesions.
- Risk factors emphasised in teaching: **darker skin phototypes** have higher rates of DLE and more disfiguring dyspigmentation/scarring; smoking associated with DLE activity/treatment resistance; female predominance overall (quiz noted male sex is not the higher-risk factor).

Myths to address

- **"IT'S JUST ECZEMA ON THE FACE"**
  scarring, follicular plugs and ear involvement point to DLE.

Pathophysiology and pathology

DLE is an **interface attack** on basal keratinocytes and hair follicles, triggered by UV.

1. **UV injury**
   UV causes keratinocyte apoptosis and exposure of nuclear antigens.
2. **Interferon response**
   Plasmacytoid dendritic cells produce type I interferon, recruiting cytotoxic T cells.
3. **Interface damage**
   T cells attack the basal layer and follicular epithelium.
4. **Scarring**
   Follicles are destroyed, leaving permanent scarring alopecia and atrophy.

HISTOPATHOLOGY

- Interface dermatitis, follicular plugging, thickened basement membrane, perivascular and periadnexal lymphocytes, dermal mucin; DIF: lupus band.

Clinical features

FEATURE	DETAIL
Plaques	Well-defined erythematous discs; adherent scale; follicular plugging; “carpet tack” scale when removed
Scarring	Central atrophy, permanent hair loss on scalp, pigment change
Sites	Face, ears, scalp, neck; occasionally below neck
Symptoms	May burn/itch; cosmetic distress high (quiz vignette)

Fig. 1 · Facial DLE — erythematous plaques with adherent scale; scarring risk; Formative Quiz-1 answer.

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Special groups

- **Skin of colour**
  - Marked hyper- and depigmentation; high scarring risk.

Diagnosis and investigations

- Clinical + **skin biopsy** (most important confirmatory test in quiz cluster).
- Bloods: ANA, ENA, complement — useful to screen for systemic disease but **may be negative** in purely cutaneous DLE.
- Mycology if tinea capitis plausible; trichoscopy helps vs AA/LPP.

Key points

- Clinical + **skin biopsy** (most important confirmatory test in quiz cluster).
- Bloods: ANA, ENA, complement — useful to screen for systemic disease but **may be negative** in purely cutaneous DLE.
- Mycology if tinea capitis plausible; trichoscopy helps vs AA/LPP.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Alopecia areata	Non-scarring; exclamation hairs; no discoid scale plaques
	Lichen planopilaris	Scarring; perifollicular scale; less classic facial discoid plaques
	Tinea capitis	Broken hairs, lymphadenopathy, mycology+

	CONDITION	DISTINGUISHING FEATURES
	Trichotillomania	Bizarre geometric; hairs of different lengths
	Psoriasis / seb derm scalp	Non-scarring usually; different morphology

Management

- Strict **photoprotection** (SPF, hats); smoking cessation.
- Potent/very potent TCS or intralesional steroid (often derm); topical calcineurin inhibitors on face sometimes.
- Antimalarials (hydroxychloroquine) — **specialist/shared-care** with monitoring.
- Early derm referral for scalp/face to minimise irreversible scarring.
- Screen for SLE symptoms (arthritis, serositis, renal, photosensitivity beyond skin, cytopenias).

Referral

ROUTINE	Joint pain, mouth ulcers, fatigue, chest pain or Raynaud's: screen for **SLE** (ANA, FBC, urine dipstick) and refer.
URGENT	Scalp DLE with hair loss: **urgent** dermatology (scarring alopecia).
ROUTINE	Hydroxychloroquine: baseline and annual eye screening after 5 years (RCOphth).

Red flags and when to refer

- Joint pain, mouth ulcers, fatigue, chest pain or Raynaud's: screen for **SLE** (ANA, FBC, urine dipstick) and refer.
- Scalp DLE with hair loss: **urgent** dermatology (scarring alopecia).
- Hydroxychloroquine: baseline and annual eye screening after 5 years (RCOphth).

Complications and prognosis

Complications

- Scarring alopecia, disfigurement, SCC in scars, progression to SLE.

Prognosis

- Chronic; early treatment limits scarring.

Prescriber pearls (UK pharmacist IP)

- Emphasise scarring risk — early treatment matters.
- Camouflage / psychological support for visible disease.
- Check BNF interactions if hydroxychloroquine co-prescribed later (retinopathy monitoring is specialty).

Self-check quiz

5 questions · optional

Q1 When should you think of Discoid lupus erythematosus?

✓ Red, scaly, coin-shaped plaques on the face, ears or scalp that leave scarring, pigment change or bald patches.

Q2 What drives it?

✓ Autoimmune interface dermatitis triggered by UV; smoking makes it worse and blunts hydroxychloroquine.

Q3 What is the backbone of care?

✓ Broad-spectrum sunscreen and sun avoidance, stop smoking, potent topical or intralesional steroid; specialist hydroxychloroquine.

Q4 What must you never miss?

✓ Scarring alopecia, systemic lupus (screen symptoms and bloods), and SCC in long-standing scars.

Q5 Name a red flag that needs escalation.

✓ Joint pain, mouth ulcers, fatigue, chest pain or Raynaud's: screen for **SLE** (ANA, FBC, urine dipstick) and refer.

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Photo gallery

Discoid lupus 3

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CHAPTER 3

Blistering diseases

How to approach blisters, then pemphigus, bullous pemphigoid, dermatitis herpetiformis and drug-induced blistering including SJS/TEN, with same-day referral routes.

1. Approach to blistering diseases
2. Pemphigus vulgaris & the pemphigus family
3. Bullous pemphigoid (incl. MMP & pemphigoid gestationis)
4. Dermatitis herpetiformis & coeliac disease
5. SJS/TEN and drug-induced blistering

Blistering diseases Module 4 · Week 6

Approach to blistering diseases

Sort a blistering rash by where the skin splits (subcorneal, intraepidermal, subepidermal), whether it is immunobullous, and how fast to refer. Biopsy for immunofluorescence is done by specialists, but the referral speed is decided in primary care.

ALSO CALLED blisters · bullae · split level · immunobullous · Nikolsky · direct immunofluorescence · +2

RELATED PAGES Bullous pemphigoid (incl. MMP & pemphigoid gestationis) · Pemphigus vulgaris & the pemphigus family · Dermatitis herpetiformis & coeliac disease · SJS/TEN and drug-induced blistering · Impetigo

KEY FACTS

- Sort blisters by **level of split**: intraepidermal (flaccid, erosions: pemphigus, impetigo, SSSS, SJS/TEN) vs subepidermal (tense: pemphigoid, DH, PCT, EBA).
- Ask: age, acute or chronic, mucosa, itch vs pain, drugs, sites, systemic illness.
- Emergencies: **SJS/TEN, SSSS, extensive pemphigus, eczema herpeticum, neonatal blistering**.
- Investigations: swab (bacteria, HSV PCR), skin biopsy of a fresh blister edge plus perilesional skin for **direct immunofluorescence**, serology.
- Common causes in primary care: impetigo, insect bites, friction, contact dermatitis, pompholyx, herpes.

Overview

A **vesicle** is a fluid-filled lesion under 5 mm (some texts say 10 mm) and a **bulla** is a larger one. An **erosion** is loss of epidermis only, often all that remains of a fragile, superficial blister. The depth of the split explains what you see. The higher the split, the more flaccid the blister and the more erosions there are. The deeper the split, the more tense the blister, and haemorrhage, scarring and milia can follow.

Aetiology: causes and risk factors

LEVEL	MECHANISM	EXAMPLES
Subcorneal / granular	Toxin or antibody against **desmoglein 1** (Dsg1, superficial epidermis)	Bullous impetigo and SSSS (exfoliative toxin cleaves Dsg1), pemphigus foliaceus, subcorneal pustular dermatosis, AGEP (pustules), miliaria crystallina
Intraepidermal, suprabasal	**Acantholysis**, i.e. loss of keratinocyte adhesion (anti-Dsg3 ± Dsg1)	Pemphigus vulgaris; Hailey-Hailey, Darier and Grover (genetic or acquired, **DIF-negative**)
Intraepidermal, spongiotic / ballooning	Intercellular oedema; viral cytopathic effect	Pompholyx, acute ACD, bullous insect bites; HSV/VZV

LEVEL	MECHANISM	EXAMPLES
Intraepidermal, basal	Keratin 5/14 defect	**EB simplex**
Subepidermal, lamina lucida	Antibodies to BP180 (collagen XVII)/ BP230; laminin-332; laminin-332 gene defect	BP, pemphigoid gestationis, MMP, linear IgA disease; **junctional EB**
Subepidermal, sublamina densa ("dermolytic")	Antibody to or defect in **collagen VII**	EBA, bullous SLE; **dystrophic EB**
Subepidermal, dermal papillae	IgA against **epidermal (TG3)** transglutaminase with neutrophil microabscesses	Dermatitis herpetiformis
Full-thickness epidermal necrosis	Cytotoxic T-cell and Fas-FasL/ granulysin keratinocyte apoptosis	SJS/TEN, bullous FDE
Other subepidermal	Porphyrin phototoxicity; oedema; ischaemia	PCT, pseudoporphyria; bullae from oedema, diabetes (bullosis diabeticorum), coma or barbiturates; friction or burns

Myths to address

- **"BLISTERS ARE ALWAYS INFECTION"**
  autoimmune and drug causes need biopsy and urgent referral.

Pathophysiology and pathology

Where the skin **splits** decides how a blister looks.

1. **Subcorneal / intraepidermal**
   Fragile roof breaks easily: erosions and crusts (bullous impetigo, pemphigus foliaceus).
2. **Suprabasal**
   Desmosome loss in pemphigus vulgaris: flaccid blisters, mucosal erosions.
3. **Subepidermal**
   Basement membrane damage (pemphigoid, DH, PCT, EBA): tense blisters.

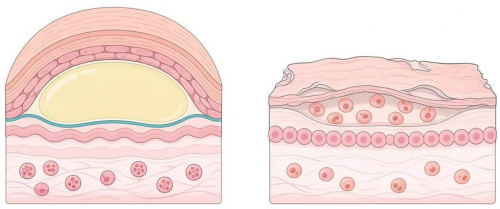

Fig. 1 · Intraepidermal (flaccid) versus subepidermal (tense) blister.
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4 **Full-thickness necrosis**

SJS/TEN: the whole epidermis dies and sloughs.

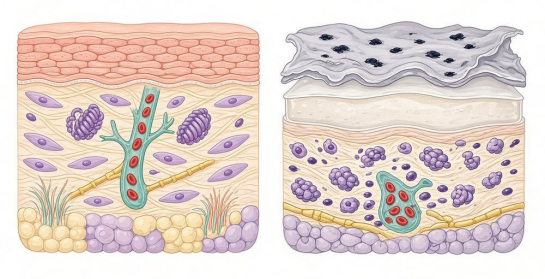

Fig. 2 · TEN: full-thickness epidermal necrosis.

AI-generated illustration

Clinical features

- **Flaccid blisters or erosions only, with oral ulcers** suggest pemphigus (PV), or SJS/TEN if acute and the patient is unwell.
- **Tense blisters, sometimes haemorrhagic, on an urticarial or eczematous base in an elderly person with itch** suggest BP.
- **Excoriations and tiny grouped vesicles on extensors and buttocks, intensely itchy** suggest DH.
- **Fragile blisters, scars and milia on the backs of the hands with hypertrichosis** suggest PCT (or pseudoporphyria, or EBA).
- **A "string of pearls" annular arrangement** suggests linear IgA disease (often drug-induced, e.g. vancomycin).
- **Flexural erosions with fissured maceration** suggest Hailey-Hailey. **Sudden, painful, targetoid or dusky skin with mucosal involvement** suggests SJS/TEN.

The **Nikolsky sign** (shearing the epidermis off with lateral pressure) is positive in PV, SSSS and TEN, and usually negative in BP. The **Asboe-Hansen sign** (pressure spreads the blister edge sideways) points to PV.

Fig. 3 · BP: large tense, clear, intact bulla. The subepidermal split gives a full-thickness roof, so it survives. Older adult; itch dominant.

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Diagnosis and investigations

(specialist workup that the IP should know)

- **TWO BIOPSIES**
  Lesional, from the edge of a **fresh** (<48 h) blister, for H&E (split level, acantholysis, eosinophils/neutrophils), sent in formalin; **Perilesional**, from normal-looking skin about 1 cm from a blister (or uninvolved skin for DH), for **direct immunofluorescence (DIF)**. Send it in **Michel's medium** or fresh on saline gauze, **never formalin**. Biopsy of an eroded or ulcerated site gives a false negative
- **DIF patterns** (use photos A06 and C05): intercellular **"chicken-wire/fishnet" IgG ± C3** = pemphigus; **linear IgG/C3 at the BMZ** = BP/MMP/PG; linear **IgA** = LAD; **granular IgA** in the dermal papillae = DH; perivascular IgG/C3 and "doughnut" deposits = PCT
- **Salt-split skin** separates BP (epidermal roof) from EBA (dermal floor).
- **SEROLOGY**
  ELISA for anti-Dsg1/Dsg3 (titres roughly follow activity) and anti-BP180/BP230; indirect IMF; for DH, **tTG-IgA + total IgA** (± EMA); for paraneoplastic pemphigus, anti-envoplakin/periplakin
- **Baseline for systemic therapy** (the IP will be asked about these): FBC, U&E, LFT, glucose/HbA1c, BP, weight, bone profile + vitamin D, **TPMT** (azathioprine), **G6PD** (dapsone), hepatitis B/C, HIV, VZV serology, TB screen (IGRA/chest X-ray) before rituximab or high-dose steroids, a pregnancy test, and immunoglobulins before rituximab.

Differential diagnoses

"Not everything that blisters is immunobullous":

- bullous impetigo and SSSS
- HSV/VZV (eczema herpeticum)
- bullous insect bites (**Module 4 exam Q15**)
- acute contact dermatitis and phytophotodermatitis
- pompholyx
- friction, burns and oedema blisters
- bullous drug reactions
- EM
- bullous tinea
- bullous LP
- genodermatoses (EB, Hailey-Hailey)

Management

- **Do not start oral steroids "to see" before biopsy unless the specialist advises it.** They reduce the diagnostic yield, and in DH/coeliac a gluten-free diet would do the same to serology.
- Supportive care while waiting for an urgent appointment: bland emollients and a soap substitute; puncture large tense blisters at the base with a sterile needle but leave the roof on; non-adherent dressings (silicone/

paraffin gauze) and no adhesive tape on fragile skin; a potent or superpotent TCS **only** for suspected BP once dermatology agrees; analgesia, plus benzydamine/chlorhexidine mouthwash or an antiseptic mouthwash for oral erosions

- **MEDICATION REVIEW IS THE IP'S CORE SKILL**
  gliptins, loop diuretics, spironolactone, PD-1 inhibitors, thiol drugs (penicillamine, captopril), vancomycin, NSAIDs, and recent new drugs (SJS/TEN window 4–28 days).

Referral

ROUTINE **IP do's:** stop suspected drugs; document; take photos; give fluid, analgesia and dressing advice; check for steroid or immunosuppressant use; Yellow Card.

ROUTINE **IP don'ts:** don't start oral prednisolone for undiagnosed blistering without a specialist discussion; don't put adhesive dressings on fragile skin; don't de-roof tense blisters; don't advise a gluten-free diet before coeliac tests.

Red flags and when to refer

> **IP DO'S**
> stop suspected drugs; document; take photos; give fluid, analgesia and dressing advice; check for steroid or immunosuppressant use; Yellow Card.

> **IP DON'TS**
> don't start oral prednisolone for undiagnosed blistering without a specialist discussion; don't put adhesive dressings on fragile skin; don't de-roof tense blisters; don't advise a gluten-free diet before coeliac tests.

Blistering red flags and referral routes

PRESENTATION	WHY IT MATTERS	ROUTE / TIMEFRAME
Widespread painful skin, dusky/target lesions, **detachment**, **2 mucosal sites**, fever; new drug in the last 4–28 days	**SJS/TEN** (mortality up to ~30% with TEN)	**999 / ED now.** Stop culprit drugs. Ask ED to involve on-call dermatology and ophthalmology; burns or ITU transfer
Widespread erosions **± oral ulcers**, unable to eat or drink, dehydrated or hypotensive (Case 1)	Severe PV, fluid/ protein loss, sepsis	**Same-day acute medical admission** + on-call dermatology
Fever, unwell, **facial oedema**, widespread rash, eosinophilia, abnormal LFT/U&E, 2–8 weeks after a new drug	DRESS	Same-day acute medicine / dermatology
Pustular sheets on erythema with fever 1–2 days after an antibiotic	AGEP (also consider GPP)	Same-day dermatology advice; admit if unwell
Child, febrile, tender red skin with superficial peeling, mucosa spared	**SSSS**	**Same-day paediatrics** (IV flucloxacillin)
Eczema with sudden clustered punched-out erosions, fever, unwell	**Eczema herpeticum** (Module 4 exam Q29)	Same-day: start oral aciclovir if well; if unwell, eye involvement or infant admit (see the eczema notes)

PRESENTATION	WHY IT MATTERS	ROUTE / TIMEFRAME
Eye pain, red eye, photophobia, reduced vision, or symblepharon with blistering	Ocular SJS/MMP/HSV	**Same-day ophthalmology** (eye casualty)
Hoarseness, dysphagia or stridor with mucosal blistering	Laryngeal MMP/PV	**999 / ED**
Pregnant woman with periumbilical urticated plaques and blisters	Pemphigoid gestationis	**Urgent (same/next day) dermatology + obstetrics**
Elderly person with new tense blisters, not systemically unwell	BP	**Urgent dermatology** (within ~1–2 weeks; teledermatology photos speed triage). Medication review now
Persistent oral erosions (>3 weeks) ± skin erosions	PV/MMP/LP/**SCC**	Urgent dermatology or oral medicine. **Suspected-cancer (2WW) pathway** if a single non-healing ulcer
Grouped itchy excoriations on extensors ± gut symptoms	DH	Routine dermatology + gastroenterology; coeliac serology **on gluten** first
Neonate or infant with blisters	HSV, SSSS, EB, mastocytosis	**Same-day paediatrics** (neonatal HSV is an emergency)
Immunosuppressed patient with blisters	Disseminated HSV/ VZV, sepsis	Same-day assessment; low threshold to admit

Complications and prognosis

Complications

- Infection, fluid loss, scarring, mucosal strictures.

Prognosis

- Depends on cause.

Prescriber pearls (UK pharmacist IP)

Ask about **mouth, eyes and genitals** at every blistering consultation. Mucosal involvement changes the urgency and the referral route.

This helps triage by teledermatology.

Infection risk: crusting, pain, fever or a sudden spread of erosions may mean secondary S. aureus or HSV. Swab and treat.

Common misconceptions corrected

POST	STATEMENT	CORRECTION
Worked case	Split levels: "dermolytic: EB"	EB occurs at **three** levels. **EB simplex** is intraepidermal (basal keratinocyte, keratin 5/14). **Junctional EB** is in the lamina lucida (laminin-332, collagen XVII). Only **dystrophic EB** (collagen VII) is dermolytic, below the lamina densa. EBA (acquired, collagen VII) is also dermolytic. See the infant skin pages notes
Worked case	Pemphigoid drug triggers: etanercept, furosemide	Furosemide is a fair example, and etanercept is a case-report association. **Add DPP-4 inhibitors (gliptins), the best-established and most relevant to primary care**, plus spironolactone and PD-1 inhibitors
Worked case	Subcorneal: impetigo, PF	Correct. SSSS also belongs here (exfoliative toxin cleaving Dsg1, the same target as PF).
Short notes	Pemphigus = vulgaris, foliaceus, erythematosus, vegetans, paraneoplastic	Good. Add **IgA pemphigus** and **drug-induced** pemphigus. Erythematosus is a localised form of PF, and vegetans is a variant of PV
Formative quiz	QA answered "intraepidermal acantholysis" or "granular IgA"	That is PV or DH. BP is **subepidermal with linear BMZ IgG/C3**
Overview	"DPP-4 inhibitors are a common cause of drug-induced bullous **pemphigus**; lisinopril can prevent it"	Gliptins cause **pemphigoid**. ACE inhibitors (including lisinopril) are a weak cause of pemphigus, not prevention. There is no evidence that lisinopril prevents gliptin-associated BP
Overview	MIG works by blocking CD95 (Fas) in TEN	That was the in-vitro rationale. Clinical evidence is mixed, and the UK 2016 guideline does not routinely recommend MIG. Ciclosporin is often preferred

Where the course differs from UK guidance

TOPIC	COURSE TEACHING	UK GUIDANCE / PRACTICE	PRACTICAL IMPACT FOR THE IP
PV first line (Case 1)	Prednisolone + azathioprine/MMF; rituximab/MIG only if these fail	**BAD 2017 + Ritux 3:** rituximab + prednisolone **first line** for new moderate–severe PV. NHS England commissions rituximab	Expect rituximab plans. Support hep B screening, vaccines and infection counselling
Case 1 acute care	Systemic steroids, monitor BP	Clinically hypovolaemic with extensive erosions, so **same-day admission**	Treat as an emergency, not an outpatient
PV antibodies	Dsg3 explains the oral ulceration	Mucocutaneous PV has **Dsg3 + Dsg1**	Teaching nuance
BP drug trigger	DPP-4 inhibitors linked to "pemphigus"; lisinopril preventive	Gliptins cause **pemphigoid** (SmPCs list BP); no preventive role for ACE inhibitors	Gliptin review in every elderly patient with BP
BP treatment	(Quiz: doxycycline)	Matches BAD/BLISTER. **Whole-body clobetasol** is also first-line for extensive disease. High-dose prednisolone (1 mg/kg) is **not** recommended	Quantify the clobetasol supply; doxycycline counselling

TOPIC	COURSE TEACHING	UK GUIDANCE / PRACTICE	PRACTICAL IMPACT FOR THE IP
SJS/TEN IVIG	IVIG blocks CD95, so it is used	UK 2016: supportive care in a burns or ITU setting is the cornerstone; no immunomodulator is routinely recommended (ciclosporin is used at many centres)	Awareness only
DH diagnosis	(No model answer given)	NICE NG20: serology **on gluten**; DIF on **perilesional** skin; refer to gastroenterology	Don't let the "IBS dietitian" start a GFD first
PCT HCQ	Low-dose HCQ (as in the photodermatoses notes)	**Twice weekly** 100–200 mg; daily dosing causes hepatotoxic flares	Check the dose on the script
Burow's solution	Aluminium acetate compresses	Not routinely stocked in UK community pharmacy (specials or extemporaneous). Saline or potassium permanganate 1:10,000 soaks are the usual UK alternatives	Use permanganate with care (staining, toxic if ingested, NPSA alert: dilute properly)

Self-check quiz

5 questions · optional

Q1 When should you think of Approach to blistering diseases?

✓ Blisters or erosions without an obvious burn, bite, friction or infection.

Q2 What drives it?

✓ Autoantibodies against desmosomes (intraepidermal, flaccid) or hemidesmosomes (subepidermal, tense); drugs; infection; genetics (EB).

Q3 What is the backbone of care?

✓ Identify the pattern, refer at the right speed, look for a drug cause, and prescribe and monitor within the specialist plan.

Q4 What must you never miss?

✓ SJS/TEN, pemphigus with mouth erosions, eye involvement, pemphigoid in pregnancy, eczema herpeticum and SSSS in children.

Q5 Name a red flag that needs escalation.

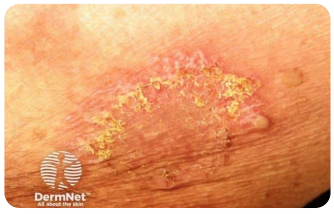
✓ **IP do's:** stop suspected drugs; document; take photos; give fluid, analgesia and dressing advice; check for steroid or immunosuppressant use; Yellow Card.

References

1. DermNet: Blistering skin conditions — https://dermnetnz.org/topics/blistering-skin-conditions
   dermnetnz.org
2. DermNet: Immunofluorescence — https://dermnetnz.org/topics/direct-immunofluorescence
   dermnetnz.org
3. PCDS Bullous pemphigoid and Blistering pages — https://www.pcds.org.uk/clinical-guidance
   pcds.org.uk

Photo gallery

Mimics: Hailey-Hailey disease and SSSS 2



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Blistering diseases Module 4 · Week 6 · Module 4 · Week 1

Pemphigus vulgaris & the pemphigus family

A rare, serious autoimmune disease with painful mouth erosions and fragile flaccid blisters that quickly become raw areas. It needs urgent or same-day dermatology; UK first-line treatment is rituximab with prednisolone.

ALSO CALLED PV | pemphigus vulgaris | pemphigus foliaceus | paraneoplastic pemphigus | drug-induced pemphigus | desmoglein | +2

RELATED PAGES Bullous pemphigoid (incl. MMP & pemphigoid gestationis) | Approach to blistering diseases | Dermatitis herpetiformis & coeliac disease | SJS/TEN and drug-induced blistering

KEY FACTS

- **Pemphigus vulgaris** is a rare autoimmune blistering disease with **flaccid blisters and erosions**, almost always involving the **mouth**.
- Autoantibodies to **desmogleins 3 (± 1)** separate keratinocytes (acantholysis): **Nikolsky sign** positive.
- Oral erosions often precede skin lesions by months; pemphigus foliaceus (Dsg1) spares the mucosa.
- Extensive or unwell: **same-day admission**; mucosal only: urgent dermatology.
- Specialist first line: **rituximab plus prednisolone**; untreated mortality is high.

Overview

Pemphigus vulgaris is a rare, potentially fatal autoimmune intraepidermal blistering disease of skin and mucosa, caused by IgG against desmogleins. In the UK it typically affects people aged 40–60, and it is commoner in people of Ashkenazi Jewish, Indian subcontinent and Mediterranean descent. Before steroids the mortality was very high. Modern mortality is mostly from treatment (sepsis and steroid complications), around the ~5–10% the course quotes.

Pemphigus vulgaris: primary-care spot diagnosis

Pemphigus vulgaris (PV) is a rare, potentially life-threatening **autoimmune intraepidermal blistering** disease. Autoantibodies against desmogleins disrupt keratinocyte adhesion **flaccid blisters** that rupture to painful erosions and crusts. Mucosal disease is common; oral erosions often precede skin. Module 4 Case 5: months of itchy crusted trunk sores + **hoarseness** (laryngeal mucosal involvement) after failed antibiotic courses discussion diagnosis PV.

Epidemiology

- **Who**
  - Adults 40–60; more common in Ashkenazi Jewish and Mediterranean, South Asian populations.

Aetiology: causes and risk factors

- **DESMOGLEIN COMPENSATION THEORY**
  Dsg3 dominates in mucosa and the lower epidermis, and Dsg1 in the upper epidermis: Anti-Dsg3 alone gives **mucosal-dominant PV**; Anti-Dsg3 + anti-Dsg1 gives **mucocutaneous PV**; Anti-Dsg1 alone gives **pemphigus foliaceus** (superficial skin erosions and no mucosal disease, because Dsg3 compensates in mucosa)
- Case 1 has **severe oral disease and widespread skin erosions**, so it is mucocutaneous, with **both** Dsg3 and Dsg1. "Dsg3" alone is half the story.
- **Drug-induced pemphigus** is mainly due to **thiol drugs** (penicillamine, captopril) and less often non-thiol drugs (ACE inhibitors such as ramipril/enalapril, rifampicin, NSAIDs). Thiol-induced disease is usually foliaceus-like and often remits when the drug is stopped. In Case 1, ramipril is worth noting and discussing with the specialist, but it is a **weak** association.
- **Paraneoplastic pemphigus** is linked to lymphoproliferative disease (NHL, CLL, Castleman), with intractable stomatitis, polymorphous skin lesions and bronchiolitis obliterans. It needs a malignancy workup.

Pemphigus vulgaris: primary-care spot diagnosis

- IgG autoantibodies to **desmoglein 3** (± desmoglein 1) acantholysis.
- Flaccid blisters; **Nikolsky sign** may be positive (shear extension of blister/erosion).
- Without treatment historically high mortality (infection, fluid/protein loss); modern immunosuppression transformed prognosis.
- Triggers/associations: idiopathic; drugs rarely (e.g. penicillamine, some ACEIs — uncommon); paraneoplastic pemphigus is a separate severe entity.

Myths to address

- **"RECURRENT MOUTH ULCERS ARE JUST APHTHOUS"**
  persistent oral erosions in an adult need biopsy.

Pathophysiology and pathology

Antibodies **cut the glue (desmosomes)** between epidermal cells, so the epidermis falls apart.

1. **Autoantibodies**
   IgG binds desmoglein 3 (mucosal) and desmoglein 1 (skin).
2. **Acantholysis**
   Keratinocytes lose adhesion and separate just above the basal layer.

3 **Flaccid blister**

The thin roof ruptures easily, leaving painful erosions.

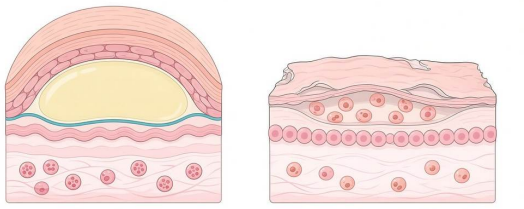

Fig. 1 · Pemphigus splits within the epidermis (flaccid blister); pemphigoid splits beneath it (tense blister).

AI-generated illustration

4 **Desmoglein compensation**

Which desmogleins are targeted explains mucosal versus skin involvement.

HISTOPATHOLOGY

- Suprabasal acantholysis with 'tombstone' basal cells; DIF: intercellular IgG and C3 ('fishnet'); serum Dsg1/Dsg3 ELISA.

Clinical features

- **ORAL EROSIONS FIRST IN ~50–70%**
  painful, persistent buccal/palatal/gingival erosions, often present for months before the skin is involved and treated as "aphthous" or "candida".
- **Flaccid blisters** rupture quickly, so the patient usually presents with **erosions and crusts** (Case 1: "no blisters seen"). Sites are scalp, face, chest, back, axillae and groin. The erosions are painful, not itchy.
- Positive Nikolsky and Asboe-Hansen signs.
- Severe disease brings fluid and protein loss, poor oral intake, **hypovolaemia** and sepsis (Case 1: clinically dry, BP 125/60 on two antihypertensives).
- **VARIANTS**
  vegetans (flexural vegetating plaques); **foliaceus** (scaly crusted "cornflake" erosions on the seborrhoeic trunk, face and scalp; photos A07/A08); **erythematosus/Senear-Usher** (malar, LE-like); **IgA pemphigus**; **paraneoplastic**

Pemphigus vulgaris: primary-care spot diagnosis

FEATURE	DETAIL
Skin	Flaccid blisters erosions/crusts on trunk, flexures, scalp, face; may look “infected”
Mucosa	Oral almost always; larynx **hoarseness**; genitals, conjunctiva possible
Pain	Often painful more than itchy (Case described itch — still consider PV in crusted widespread disease)
Systemic	Weight loss, poor oral intake, secondary infection

Fig. 2 · PV: flaccid blisters that rupture almost at once, leaving raw erosions. Case 1 says 'no blisters', which is typical, because the roof is too thin to survive.

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Special groups

- **Drug-induced and paraneoplastic pemphigus**
  - Penicillamine, ACE inhibitors; paraneoplastic pemphigus with lymphoma causes severe stomatitis.

Diagnosis and investigations

Case 1 answers, with the UK detail added:

- **HISTOLOGY**
  suprabasal acantholysis. The basal cells stay attached to the BMZ like a **row of tombstones**.
- **DIF** of perilesional skin: **intercellular IgG ± C3** in a "chicken-wire" pattern. DIF is the most sensitive test.
- **SEROLOGY**
  indirect IMF on monkey oesophagus, and **ELISA anti-Dsg3/Dsg1**, which are used for monitoring.
- **Tzanck smear** (acantholytic cells) is historical and not used in UK practice.
- Baseline before therapy: especially **TPMT** before azathioprine, hepatitis B before rituximab, and bone protection.

Pemphigus vulgaris: primary-care spot diagnosis

- **Urgent dermatology.**
- Gold standard: lesional biopsy for histopathology **plus** perilesional **direct immunofluorescence (DIF)**; serum ELISA for desmoglein antibodies.
- Do not rely on antibiotics for crusted erosions without considering immunobullous disease.
- ENT input if hoarseness / airway concern.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Bullous pemphigoid	Tense blisters, older adults, less oral; different DIF
	Severe eczema / psoriasis / impetigo	Lack true flaccid acantholytic pattern; cultures; biopsy if unsure
	Mucous membrane pemphigoid	Mucosa scarring focus; different immuno
	SJS/TEN / erythema multiforme major	Acute drug/infection course; different tempo
	Hailey–Hailey / other acantholytic	Flexural chronic family Hx

> **ORAL**
> aphthae; erosive LP; MMP; EM/SJS; HSV; candidiasis; Behçet's (**Module 5 exam Q22**); SCC (single non-healing ulcer)

> **SKIN**
> BP; PF; SJS/TEN; impetigo/SSSS; Hailey-Hailey (flexural, family history, DIF-negative); Darier and Grover (DIF-negative acantholysis)

Management

1. Recognise and refer. Erosions + mucosal disease + unwell, dehydrated or extensive skin **same-day admission** (acute medicine, with on-call dermatology). Case 1's hypovolaemia puts him here; Mucosal-only, stable disease **urgent dermatology** (days, not weeks) with oral medicine input
2. **Specialist treatment (BAD 2017, paraphrased; for awareness, not IP initiation): First line for new moderate–severe PV: rituximab** (2 × 1 g IV, 2 weeks apart) **plus oral prednisolone** (typically 0.5–1 mg/kg/day, tapered quicker than steroid-alone regimens). This follows the Ritux 3 trial, and **NHS England commissions rituximab for PV**; Steroid-sparing adjuvants: **azathioprine** (after TPMT) or **mycophenolate mofetil**. Use them when rituximab isn't possible or for maintenance; Refractory disease: **IVIG**, immunoadsorption, cyclophosphamide; Mild disease: superpotent topical or intralesional steroids; for oral disease, steroid mouthwash/spray (betamethasone soluble tablets as a mouthwash, or beclometasone inhaler sprayed onto the mucosa)
3. IP supporting roles **Steroid safety:** a **steroid emergency card**, **bone protection** (e.g. alendronic acid + calcium/vitamin D when long-term prednisolone 7.5 mg for 3 months is expected, per NOGG/BNF), a PPI if GI risk factors, glucose and BP monitoring (Case 1 BP: **consider pausing or reducing antihypertensives while volume-depleted**), and PJP prophylaxis per the specialist (co-trimoxazole with high-dose steroid + immunosuppressant); **Vaccines:** inactivated influenza/COVID/pneumococcal **before rituximab** where possible; **no live vaccines** while on immunosuppression; **Shared-care monitoring** for azathioprine/MMF (FBC/LFT per

the local protocol); **Oral care:** soft diet, avoid spicy or acidic food, chlorhexidine or benzydamine mouthwash, treat candida (miconazole oral gel; **check interactions, especially warfarin**, or use nystatin), and refer to the dentist

Pemphigus vulgaris: primary-care spot diagnosis

- **Recognition + urgent referral** — not primary-care disease-modifying prescribing.
- Supportive: emollients, wound care, analgesia, nutrition, treat secondary infection if present.
- Specialty care: systemic corticosteroids historically foundational; **rituximab** (anti-CD20) now major disease-modifying option in specialist protocols; steroid-sparing agents as directed by derm.
- Vaccination/infection risk counselling once on immunosuppression (specialist-led).
- Same-day acute if extensive denudation, inability to drink, or airway symptoms.

Referral

SAME DAY	Extensive erosions, unwell, dehydrated or unable to eat: **same-day admission**.
ROUTINE	Oral erosions persisting more than 2–3 weeks without a cause: refer (oral medicine or dermatology).
ROUTINE	Severe stomatitis with a known tumour: paraneoplastic pemphigus.

Red flags and when to refer

- Extensive erosions, unwell, dehydrated or unable to eat: **same-day admission**.
- Oral erosions persisting more than 2–3 weeks without a cause: refer (oral medicine or dermatology).
- Severe stomatitis with a known tumour: paraneoplastic pemphigus.

Complications and prognosis

Complications

- Dehydration, malnutrition, sepsis, steroid side effects; death without treatment.

Prognosis

- Controlled in most with modern treatment; relapses occur.

Prescriber pearls (UK pharmacist IP)

Months of "mouth ulcers" plus new skin erosions means PV until proven otherwise. Don't keep repeating antifungals or mouthwashes.

Drug-induced pemphigus: list every thiol or ACE inhibitor. Swapping ramipril for a non-ACE alternative (e.g. amlodipine alone, or an ARB) is reasonable to discuss with the specialist, but don't expect it to cure idiopathic PV.

- **Rituximab patients:** infection risk lasts 6–12 months, with hepatitis B reactivation and poor vaccine response. Coordinate any surgery or dental work.
- Patient support: **PEMFriends** (UK pemphigus and pemphigoid patient charity).
- Crusted "sores" failing antibiotics + mucosal clues (oral/hoarse) = red-flag immunobullous.
- Avoid attributing everything to infection or poorly controlled eczema without exam of mouth.
- IP: escalate early; do not trial prolonged systemic steroids alone outside shared-care.

Pemphigus vulgaris: primary-care spot diagnosis

Self-check quiz

5 questions · optional

Q1 When should you think of Pemphigus vulgaris & the pemphigus family?

✓ Persistent painful mouth ulcers with or without fragile blisters and raw erosions on the scalp, face, chest or back.

Q2 What drives it?

✓ IgG against desmoglein 3 (mucosal) ± desmoglein 1 (skin) causes acantholysis and an intraepidermal split.

Q3 What is the backbone of care?

✓ Recognise and refer urgently; specialists start rituximab + prednisolone (BAD 2017); the IP supports steroid safety, infection prevention and monitoring.

Q4 What must you never miss?

✓ Dehydration and sepsis from extensive erosions; paraneoplastic pemphigus; drug-induced pemphigus (thiol drugs: penicillamine, captopril).

Q5 Name a red flag that needs escalation.

✓ Extensive erosions, unwell, dehydrated or unable to eat: **same-day admission**.

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Photo gallery

Pemphigus vulgaris 7

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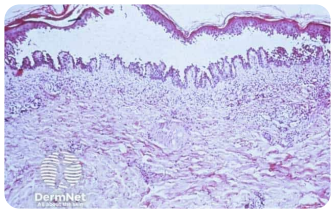
Histology and immunofluorescence

2



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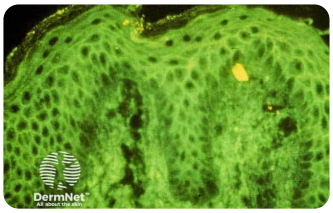

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Pemphigus foliaceus and drug-induced pemphigus

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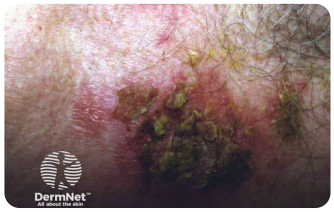

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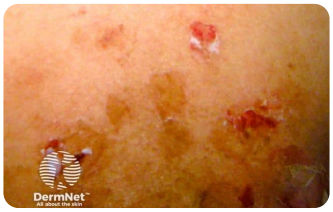

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Blistering diseases Module 4 · Week 6

Bullous pemphigoid (incl. MMP & pemphigoid gestationis)

The commonest autoimmune blistering disease in the UK: tense blisters on an itchy, urticarial base in older people, often after months of unexplained itch. Look for a gliptin, refer, and prefer doxycycline or clobetasol over high-dose prednisolone.

ALSO CALLED BP · pemphigoid · tense blisters elderly · gliptin pemphigoid · DPP-4 inhibitor · doxycycline BLISTER · +3

RELATED PAGES Approach to blistering diseases · Pemphigus vulgaris & the pemphigus family · Dermatitis herpetiformis & coeliac disease · SJS/TEN and drug-induced blistering · Itch & xerosis in older adults

KEY FACTS

- **Bullous pemphigoid (BP)** is the commonest autoimmune blistering disease: tense blisters in older people.
- Often starts with weeks to months of **itch** and urticarial plaques before blisters (prodrome).
- Autoantibodies against **BP180 and BP230** at the basement membrane; confirm with biopsy and immunofluorescence.
- Localised: clobetasol; mild–moderate: **doxycycline 200 mg once daily** (BLISTER trial: safer than prednisolone).
- Check drugs: **DPP-4 inhibitors (gliptins)**, diuretics, checkpoint inhibitors; associated with neurological disease (stroke, dementia, Parkinson's).

Overview

Bullous pemphigoid is the **commonest autoimmune blistering disease in the UK**. UK incidence is roughly 40 per million per year and rising, mostly in people **over 70–80**. It is strongly associated with **neurological disease** (dementia, Parkinson's, stroke, MS). One-year mortality is high (UK cohorts ~20–40%), driven by age, comorbidity and **treatment toxicity**. This is why UK and European guidelines favour superpotent topical steroid or doxycycline over high-dose oral prednisolone.

Epidemiology

- **Who**
  - Mostly over 70; incidence rising with age and drugs; UK about 4,000 new cases a year.

Aetiology: causes and risk factors

IgG (± IgE) autoantibodies to **BP180 (collagen XVII; NC16A domain)** and **BP230** in the hemidesmosome trigger complement activation, then eosinophil/neutrophil recruitment, proteases, and a **subepidermal** split in the lamina lucida.

DRUG TRIGGERS (THE IP'S JOB)

DPP-4 inhibitors (gliptins, especially vildagliptin, also sitagliptin/linagliptin/alogliptin/saxagliptin) are the best-established trigger, often months to years after starting. They often cause a **non-inflammatory** BP with less erythema; **Loop diuretics** (furosemide, bumetanide); spironolactone; **PD-1/PD-L1 checkpoint inhibitors**; penicillins and other antibiotics; TNF inhibitors (etanercept is an anecdotal association); psoralen-UVA and radiotherapy

QUIZ RELEVANCE

the 85-year-old with **poorly controlled T2DM** is the classic patient on a gliptin. **Check the repeat list.**

Risk factors and associations

- Autoimmunity to hemidesmosome proteins; drug triggers (gliptins, loop diuretics, spironolactone, PD-1 inhibitors).

Myths to address

- **"BLISTERS IN OLDER PEOPLE ARE JUST FRICTION"**
  itchy urticarial plaques with tense blisters need biopsy.

Pathophysiology and pathology

Antibodies attack the **anchors holding the epidermis to the dermis**, so it separates as a tense blister.

1. **Autoantibodies**
   IgG autoantibodies bind BP180 (collagen XVII) and BP230 in hemidesmosomes.
2. **Complement and eosinophils**
   Complement activation recruits eosinophils and neutrophils that release proteases.
3. **Subepidermal split**
   The whole epidermis lifts off: a **tense, roofed blister** (unlike flaccid pemphigus).

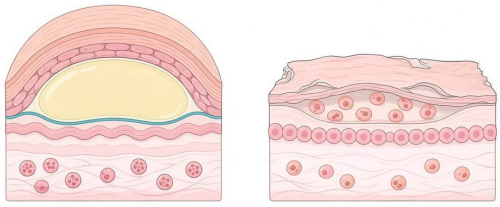

Fig. 1 · BP splits below the epidermis (tense blister); pemphigus splits within it (flaccid).
AI-generated illustration

4 **Distribution**

Flexures, limbs and trunk; mucosa involved in a minority.

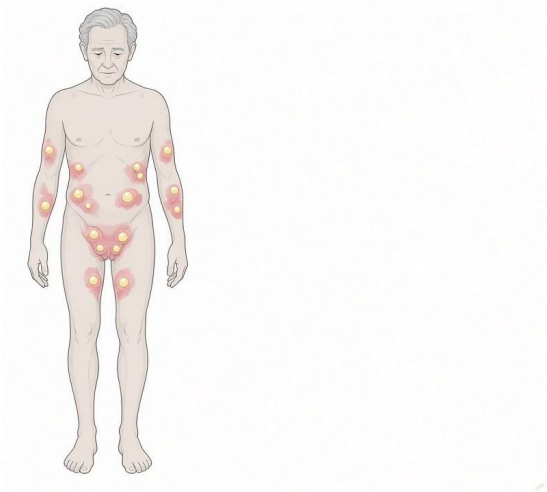

Fig. 2 · Typical BP distribution.

AI-generated illustration

HISTOPATHOLOGY

- Subepidermal blister with eosinophils; direct IF: linear IgG and C3 along the basement membrane; serum BP180/BP230 ELISA.

Clinical features

PRODROME (NON-BULLOUS PHASE)

weeks to months of **itch** with eczematous or **urticarial plaques** or excoriations alone. This is often misdiagnosed as "asteatotic eczema" or "scabies" in a care home.

BULLOUS PHASE

tense, fluid-filled (clear or **haemorrhagic**) bullae up to several cm, on an urticarial base or normal skin. Sites are the **flexural limbs, inner thighs, groin, lower abdomen and trunk**. The bullae heal without scarring, sometimes leaving milia.

Mucosal involvement in only ~10–20%, and it is mild. Prominent mucosal disease should make you think of **MMP**.

Nikolsky sign is usually negative. Eosinophilia is common.

Fig. 3 · BP: large tense, clear, intact bulla. The subepidermal split gives a full-thickness roof, so it survives. Older adult; itch dominant.

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Phases and variants

Non-bullous phase

Itchy urticated plaques for weeks to months. Site: Trunk, limbs.

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Tense bullae

Clear and haemorrhagic blisters on an urticated base. Site: Flexural limbs.

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Mucous membrane pemphigoid

Oral, ocular, genital erosions that **scar**. Site: Mouth, eyes.

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Pemphigoid gestationis

Starts **around the umbilicus**, 2nd–3rd trimester or postpartum. Site: Abdomen.

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Special groups

- **Frail older people**
  - Whole-body clobetasol is effective but impractical without carers; doxycycline is often preferred.

Diagnosis and investigations

> **QUIZ QA, THE CORRECT ANSWER IS 4**
> a **subepidermal** blister with eosinophils, and **linear IgG and/or C3 along the basement membrane zone** on perilesional DIF. Students who picked "intraepidermal/acantholysis" (that is PV) or "granular IgA" (that is DH) mixed up the patterns.

> **SEROLOGY**
> ELISA anti-BP180 (which roughly follows activity) and anti-BP230; indirect IMF on salt-split skin (epidermal side).

> Before treatment: glucose/HbA1c, BP, bone health, renal function (doxycycline is fine in CKD; check other drugs), and a baseline weight.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Pemphigus vulgaris	Flaccid blisters that rupture at once; mucosa prominent; Nikolsky positive.
	Scabies	Pre-bullous mimic in care homes: burrows, other residents itching.
	Asteatotic / discoid eczema	Pre-bullous mimic: dry, crazy-paving shins.
	Urticaria	Wheals fade within 24 h; no blisters.
	Bullous impetigo	Flaccid blister, collarette erosions; often children.
	Linear IgA disease	'String of beads' annular blisters; can be drug-induced.

Also: drug eruption, prurigo, MMP, EBA (scarring, milia), bullous insect bites, bullous cellulitis, oedema blisters, bullous tinea.

> **PRE-BULLOUS**
> asteatotic or nummular eczema; scabies (especially in care homes); urticaria; drug eruption; prurigo

> **BULLOUS**

> PV; MMP; EBA (scarring/milia, salt-split floor); linear IgA; bullous insect bites; bullous cellulitis; oedema blisters; bullous tinea; bullous impetigo

Management

Diagnosis is confirmed and treatment led by dermatology. The **IP prescribes and monitors under specialist advice**, often in care homes.

SEVERITY	UK SPECIALIST APPROACH (BAD 2012, BLISTER TRIAL, EADV 2022, PARAPHRASED)
Localised / mild	**Superpotent TCS** (clobetasol propionate 0.05%) to the lesions, **or doxycycline**
Mild–moderate generalised (**Quiz QC = 3, doxycycline**)	**Doxycycline 200 mg once daily** + a superpotent TCS to the lesions for the first weeks. In BLISTER, doxycycline was **non-inferior with respect to short-term blister control** (acceptable margin) and **significantly safer** than prednisolone 0.5 mg/kg (fewer severe or life-threatening adverse events, including deaths). It is therefore the best choice for an 85-year-old with hypertension, osteoporosis and poorly controlled T2DM.
Extensive	**Whole-body clobetasol** cream (~20–40 g/day split over the body, sparing the face; reduce over 4–12 months), which a French RCT showed was safer than prednisolone 1 mg/kg; **or** prednisolone 0.5 mg/kg/day if topical application isn't feasible
Refractory	Specialist adjuvants: azathioprine, MMF, methotrexate, dapsone; **dupilumab** or omalizumab (off-label or emerging; dupilumab has US approval for BP since 2025, but check UK commissioning); rituximab or IVIG

IP-specific actions:

> **MEDICATION REVIEW**
> stop or switch a **gliptin** after discussing with the patient and team (switch to another class for T2DM, and check eGFR, hypoglycaemia risk and frailty). Review loop diuretics and spironolactone.

> **DOXYCYCLINE COUNSELLING**
> take with a full glass of water, sitting or standing, not just before bed (oesophagitis). Keep 2–3 h away from calcium, iron, antacids and zinc (relevant with the **calcium/vitamin D** given for osteoporosis). It causes photosensitivity. Warfarin INR may rise. Nicotinamide 500 mg–2 g/day is sometimes added by specialists.

> **WHOLE-BODY CLOBETASOL IN A CARE HOME**
> quantify it (~30 g/day needs **large 500 g packs** and nurse application). Count fingertip units, warn about skin thinning and purpura, and taper as the specialist plans.

> If oral prednisolone is used: steroid card, bone protection (already osteoporotic, so check the bisphosphonate), glucose monitoring and insulin/oral agent titration (**steroid-induced hyperglycaemia**), and BP.

> **SKIN CARE**
> puncture tense blisters and leave the roof on; non-adherent dressings; emollients; watch for infection.

Medicines at a glance

Clobetasol propionate 0.05%

Localised / mild

DOSE To lesions · As specialist plans

CAUTIONS Or doxycycline.

Doxycycline 200 mg

Mild–moderate generalised

DOSE Once daily · With superpotent TCS for the first weeks

CAUTIONS BLISTER: non-inferior for short-term control and significantly safer than prednisolone.

Whole-body clobetasol

Extensive

DOSE ~20–40 g/day split over the body (spare face) · Taper over 4–12 months

CAUTIONS Safer than prednisolone 1 mg/kg (French RCT). 500 g packs, nurse application.

Prednisolone

Extensive

DOSE 0.5 mg/kg/day · If topical not feasible

CAUTIONS Steroid card, bone protection, glucose and BP monitoring.

Treatment ladder

1

Every patient

Stop or switch a gliptin with the patient and team; review loop diuretics. Puncture tense blisters, leave the roof on; non-adherent dressings; emollients.

2

Localised / mild

Clobetasol propionate 0.05%

3 Mild–moderate generalised

Doxycycline 200 mg

4 Extensive

Whole-body clobetasol; Prednisolone

5 Refractory · specialist

Azathioprine, MMF, methotrexate, dapsone. Dupilumab or omalizumab (check UK commissioning); rituximab or MIG.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Doxycycline counselling**: Full glass of water, upright, not at bedtime. 2–3 h from calcium, iron, antacids, zinc. Photosensitivity; INR may rise.
- **Clobetasol in care homes**: Quantify (~30 g/day), count FTUs, warn about thinning and purpura.

Referral

999 / A&E	Hoarseness or stridor: airway involvement.
SAME DAY	Extensive blistering or secondary infection in a frail patient.
SAME DAY	Eye redness, grittiness or scarring with oral erosions (**MMP**): same-week ophthalmology, same day if vision affected.
URGENT	Blistering in **pregnancy** starting round the umbilicus: dermatology + obstetrics.

Red flags and when to refer

- Eye redness, grittiness or scarring with oral erosions: **mucous membrane pemphigoid**, same-week ophthalmology (same day if vision affected).
- Hoarseness or stridor: airway involvement, emergency.
- Blistering rash in **pregnancy** starting round the umbilicus: urgent dermatology and obstetrics.
- Extensive blistering or secondary infection in a frail patient: same-day assessment.

Complications and prognosis

Complications

- Infection, sepsis, side effects of steroids; mortality about 20–40% in the first year in frail patients.

Prognosis

- Usually remits over 1–5 years with treatment; relapses common.

Prescriber pearls (UK pharmacist IP)

"Itchy elderly patient with eczema that won't settle" may be BP before the blisters appear. Refer for biopsy with DIF when the itch is out of proportion and standard treatment fails.

In care homes, **exclude scabies** (burrows, other residents itching) in parallel, since the two can coexist.

Variants to recognise: Mucous membrane pemphigoid presents mainly with oral, gingival ("desquamative gingivitis"), ocular, genital or laryngeal erosions that **scar**. **Ocular MMP** (symblepharon, cicatrising conjunctivitis) can cause blindness. **Same-week ophthalmology** is needed, or same-day if there is visual loss. Airway involvement (hoarseness, stridor) means emergency referral; **Pemphigoid gestationis** appears in the 2nd–3rd trimester or postpartum as urticarial plaques starting **around the umbilicus**, then vesicles and bullae. It needs **urgent** dermatology and obstetric input. It can recur in later pregnancies or with the COCP, and the baby can have transient blisters or be small for dates. Polymorphic eruption of pregnancy is the main DDx: it **spares the umbilicus** and starts in the striae

Patient support: PEMFriends; BAD leaflet Bullous pemphigoid.

Self-check quiz

5 questions · optional

Q1 Why are BP blisters tense?

✓ The split is subepidermal, so the whole epidermis forms a thick roof.

Q2 Which drug class is the best-established trigger?

✓ DPP-4 inhibitors (gliptins), especially vildagliptin.

Q3 What does perilesional DIF show?

✓ Linear IgG and/or C3 along the basement membrane zone.

Q4 Treatment for mild–moderate generalised BP in a frail 85-year-old?

✓ Doxycycline 200 mg once daily plus superpotent topical steroid at first (BLISTER: safer than prednisolone).

Q5 Gritty red eyes with oral erosions suggest what?

✓ Mucous membrane pemphigoid: same-week ophthalmology, same day if vision is affected.

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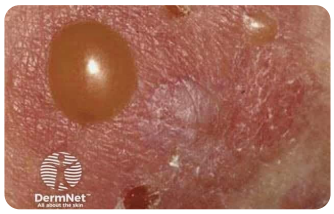
Photo gallery

Bullous pemphigoid 6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Mucous membrane pemphigoid 2

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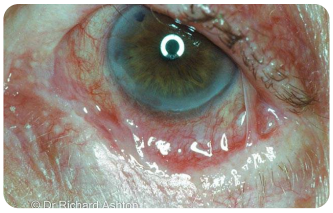
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Pemphigoid gestationis 1

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Blistering diseases Module 4 · Week 6

Dermatitis herpetiformis & coeliac disease

An intensely itchy, symmetrical eruption of tiny blisters on the elbows, knees, buttocks and scalp that is the skin form of coeliac disease. Test for coeliac disease while still eating gluten; a strict gluten-free diet is the long-term treatment.

ALSO CALLED DH · coeliac skin · gluten rash · itchy elbows buttocks · tTG · dapsone · +2

RELATED PAGES Bullous pemphigoid (incl. MMP & pemphigoid gestationis) · Pemphigus vulgaris & the pemphigus family · Scabies · Approach to blistering diseases · Eczema / dermatitis

KEY FACTS

- **Dermatitis herpetiformis (DH)** is the skin manifestation of **coeliac disease**: intensely itchy grouped vesicles.
- Extensor elbows, knees, buttocks and scalp; usually seen as excoriations rather than intact blisters.
- Confirm with **biopsy and direct IF** (granular IgA in dermal papillae) and coeliac serology **before** a gluten-free diet.
- **Lifelong gluten-free diet** treats skin and gut; **dapsone** controls itch within days (specialist; check G6PD).
- Refer to gastroenterology (NICE NG20); screen for associated autoimmunity.

Overview

Dermatitis herpetiformis is the **skin form of coeliac disease**. It is an intensely itchy, chronic, grouped papulovesicular eruption on the extensor surfaces, caused by gluten sensitivity. Almost all patients have gluten-sensitive enteropathy on biopsy, even though **most have few or no gut symptoms** (or symptoms labelled "IBS", as in Cases 2 and 4). It mostly affects people of Northern European descent aged 30–40 and is slightly commoner in men. It is rare in children, and in Japanese and African-heritage populations.

The **skin manifestation of coeliac disease (gluten-sensitive enteropathy)**: an intensely itchy, symmetrical papulovesicular eruption on the extensor surfaces.

- It affects ~10–20% of coeliac patients; adults aged 30–40 are typical; M F (unlike coeliac overall).
- **Most have coeliac enteropathy on biopsy**, though GI symptoms are often absent or mild. **This case has bloating, malaise and weight loss.**

Epidemiology

- **Who**
  - Adults 30–40; men slightly more; about 1 in 10,000 in the UK; most have enteropathy, often silent.

Aetiology: causes and risk factors

Genetic background is **HLA-DQ2/DQ8**. Gliadin is deamidated by tissue transglutaminase (TG2), which drives mucosal IgA anti-TG2 and then, by epitope spreading, IgA against **epidermal transglutaminase (TG3)**. IgA–TG3 complexes deposit in the **dermal papillae**, neutrophils gather into papillary microabscesses, and a subepidermal vesicle forms. Iodide (e.g. iodinated contrast or seaweed) and NSAIDs can make it flare.

Myths to address

- **"NO GUT SYMPTOMS, SO NOT COELIAC"**
  most people with DH have silent enteropathy.

Pathophysiology and pathology

Gut-made **IgA antibodies to transglutaminase** deposit in the skin.

1. **Gluten in the gut**
   Gluten triggers an immune response in genetically susceptible people (HLA-DQ2/DQ8).

2. **IgA to epidermal TG3**
   IgA antibodies against tissue and epidermal transglutaminase (TG3) circulate.

3. **Papillary deposits**
   IgA–TG3 complexes deposit at the tips of dermal papillae and attract neutrophils.

Fig. 1 · Dermatitis herpetiformis: granular IgA deposits (yellow) and neutrophils at the dermal papillary tips create tiny blisters beneath the epidermis.
AI-generated illustration

4. **Clearance**
   On a gluten-free diet, deposits fade over months to years.

How it is caused

- In genetically predisposed people (**HLA-DQ2/DQ8**), gluten triggers IgA autoantibodies to tissue transglutaminase (**TG2**, gut) and **epidermal transglutaminase (TG3)**.
- **IgA–TG3 complexes deposit in the dermal papillae** → neutrophil recruitment → subepidermal vesicles.
- Iodine and NSAIDs can flare it.
- Associations: autoimmune thyroid disease, T1DM, pernicious anaemia, vitiligo, alopecia areata, Sjögren's; **small-bowel (enteropathy-associated T-cell) lymphoma** (risk reduced by a gluten-free diet).

HISTOPATHOLOGY

- Neutrophilic microabscesses in dermal papillae with subepidermal clefts; DIF of perilesional skin: granular IgA in papillae.

Clinical features

- **SYMMETRICAL EXTENSOR DISTRIBUTION**
  elbows, forearms, knees, buttocks/sacrum, shoulders, scalp.
- **Intense itch or burning**, often before lesions appear. Tiny vesicles are scratched off quickly, so patients usually present with **excoriations, crusted erosions and post-inflammatory change only** (Case 4: "excoriations without blisters"; photo C04).
- Associated features: **gut:** bloating, loose stool, "IBS" (Cases 2 and 4); **iron-deficiency anaemia or pallor** (Case 2); aphthae, dental enamel defects, fatigue; personal or family history of autoimmunity (thyroid disease, T1DM, pernicious anaemia, vitiligo)
- Case 2's atopic history is a **red herring** that steers people toward "eczema". Case 4's **failure of Betnovate** (TCS barely touches DH) and the **normal routine bloods** are also typical.

Clinical features & sites (visible abnormalities)

- Grouped ("herpetiform") **small vesicles and papules**, usually seen as **excoriations, erosions and crusts** because the itch is intense — patients scratch the vesicles off. Post-inflammatory pigment change.
- Symmetrical on the **elbows/extensor forearms, knees, buttocks/sacrum**, shoulders, and **scalp/posterior hairline**; facial and palmar purpura (rare).
- Mucosa is usually spared.
- **SYSTEMIC/GI**
  bloating, diarrhoea, weight loss, malaise; iron/folate deficiency; dental enamel defects; osteoporosis.

Fig. 2 · DH classical distribution: elbows, sacrum and buttocks, symmetrical (Case 2 and Case 4). Intense itch or burning, so vesicles are often scratched away before you see them.

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Special groups

- **Dapsone safety**
  - Check G6PD; monitor FBC and LFT for haemolysis, methaemoglobinaemia and agranulocytosis.

Diagnosis and investigations

Case 2 and Case 4 answers:

1. **Coeliac serology while the patient is eating gluten** (NICE NG20): **IgA tissue transglutaminase (tTG-IgA) + total IgA** first line. Use IgA **EMA** if tTG is weakly positive, and IgG-based tests (IgG tTG/DGP/EMA) if IgA-deficient. Advise eating gluten in **more than one meal every day for at least 6 weeks** before testing. In DH, serology is positive in ~75–90%, so a negative result doesn't exclude it.
2. **Skin biopsy for DIF from perilesional, uninvolved skin** (about 1 cm from a lesion, ideally on the buttock or forearm; not lesional skin, where inflammation destroys the IgA). It shows **granular IgA in the dermal papillae**, which is the **diagnostic gold standard**. Lesional H&E (if an intact vesicle exists) shows a subepidermal blister with papillary neutrophil microabscesses.
3. Gastroenterology NICE NG20 says to refer anyone with DH for specialist assessment. Whether a duodenal biopsy is needed in proven DH is decided by gastroenterology and dermatology together (many UK centres skip it when skin DIF is positive).
4. Baseline bloods FBC, ferritin, folate, B12, calcium/vitamin D/ALP, LFT, TFT, glucose; **G6PD before dapsone**. Consider a DXA scan (NG20 criteria).

> **SKIN BIOPSY IS THE GOLD STANDARD**
> **perilesional, normal-looking skin** (~1 cm from a lesion) for **DIF** — **granular IgA in the dermal papillae**. Send it in **Michel's transport medium or saline, not formalin**; ± **lesional biopsy in formalin** for H&E: subepidermal blister with **neutrophil microabscesses in the dermal papillae**; Usually done by dermatology

> **Coeliac serology** (NICE NG20): **tTG-IgA + total IgA**. If IgA-deficient, use IgG-based tests (tTG-IgG, DGP-IgG or EMA-IgG); EMA-IgA if tTG is equivocal; epidermal TG3 antibodies (specialist); **The patient must be eating gluten — in more than one meal every day for at least 6 weeks** before testing. Ask before any gluten-free trial (course teaching: "prep to the test")

> **OTHER INVESTIGATIONS**
> FBC, **ferritin, folate, B12**, LFT, U&E, calcium, **vitamin D**, TFT (autoimmune thyroid), HbA1c; **DEXA** if risk factors or at diagnosis per BSG; **G6PD** before dapsone

> A **duodenal biopsy is usually not required** when DH is confirmed on skin biopsy (BSG 2014), though GI referral may still be needed for significant GI symptoms, red flags or uncertainty.

Differential diagnoses

- **Scabies** (burrows, web spaces, genitals, contacts; Case 4 specifically had **no burrows**)
- **atopic or nummular eczema** (Case 2's atopic history)
- **neurotic excoriations or prurigo**
- **papular urticaria / insect bites**
- **linear IgA disease** (linear IgA on DIF, no coeliac link, often drug-induced)

- **BP** (older patient, tense bullae)
- **folliculitis**
- **Grover's** (trunk, older men)
- **EM**
- **scabies + DH both possible**

Scabies (burrows, finger webs, genitals, contacts itchy), **atopic/nummular eczema**, **papular urticaria/insect bites**, **linear IgA disease** (linear IgA on DIF; drug-induced — vancomycin), **bullous pemphigoid** (older, tense bullae, linear IgG/C3), neurotic excoriations/prurigo, folliculitis, **eruptive xanthoma** (see Eruptive xanthomas — new diabetes — same sites, the course DDx), psoriasis (elbows/knees), herpes zoster/simplex (unilateral, grouped).

Management

Case 4:

> **IMMEDIATE MANAGEMENT**
>
> **do not start a gluten-free diet yet**; keep eating gluten until serology and biopsy are done. **Refer to dermatology** for biopsy with DIF, and to **gastroenterology** (NICE NG20). Meanwhile, symptomatic relief with emollients, a sedating antihistamine at night (limited benefit), and a potent TCS for short-term itch relief only (expect little response).

> **Best skin drug: dapsone** (specialist-initiated, then shared care): Typical start is 25–50 mg daily, titrated to the lowest dose that controls itch (often 50–100 mg). Itch responds within **48–72 h**, which also supports the diagnosis; **Pre-checks:** G6PD (risk of severe haemolysis if deficient), FBC, LFT, U&E; **Monitoring:** FBC **weekly for 4 weeks, then every 2 weeks to 12 weeks, then 3-monthly**, plus LFT and reticulocytes per the local shared-care protocol. Every patient has **dose-related haemolysis** and **methaemoglobinaemia** (blue-grey lips, breathlessness, headache). Rarer but serious: **agranulocytosis** (usually weeks 3–12, so counsel about sore throat, fever and mouth ulcers urgent FBC), **dapsone hypersensitivity syndrome / DRESS** (weeks 2–6; HLA-B*13:01), hepatitis, and peripheral motor neuropathy; Interactions: trimethoprim raises levels, rifampicin lowers them, and other oxidant drugs add to haemolysis; **Alternatives: sulfapyridine** (unlicensed, import or specials) or **sulfamethoxypyridazine**; **sulfasalazine** is sometimes used. Superpotent TCS as an adjunct for lesions

> **LIFESTYLE**
>
> **lifelong strict gluten-free diet (GFD)**, which is the definitive treatment. It lets dapsone be reduced or stopped, usually after **1–2 years** (the skin lags behind the gut), and lowers the risk of lymphoma. Wheat, barley, rye and cross-contaminated oats are excluded; uncontaminated oats are allowed for most. Refer to a **dietitian**, and offer **Coeliac UK** membership. Gluten-free food prescribing varies across the UK (e.g. England restricted it to bread and mixes in 2018, and ICBs vary).

> **COMPLICATIONS TO SCREEN FOR OR COUNSEL ABOUT**
>
> **enteropathy-associated T-cell lymphoma and other NHL** (risk falls with strict GFD); small-bowel adenocarcinoma (rare); **osteoporosis/osteopenia** (calcium, vitamin D, DXA per NICE); **iron, folate and B12 deficiency**; **autoimmune thyroid disease, T1DM**, pernicious anaemia, vitiligo, alopecia areata; **hyposplenism** (NG20: consider pneumococcal and other vaccines); reduced fertility and adverse pregnancy outcomes if

> untreated; **dapsone toxicity**; psychosocial burden of the diet; first-degree relatives have ~10% risk (offer serology if symptomatic; NG20)

1. **Lifelong strict gluten-free diet (GFD)** — first-line and essential. It treats the enteropathy, eventually controls the rash (**itch improves over months; full skin remission can take 6 months–2 years**), lets dapsone be withdrawn, and **lowers lymphoma risk**. Refer to a **dietitian**; Coeliac UK membership. Gluten-free food prescriptions vary by ICB in England (many have withdrawn them) and are free in Wales/Scotland.
2. **Dapsone** for rapid symptom control (itch settles within 48–72 h), specialist-initiated: **Check G6PD first.** Start 25–50 mg daily and titrate (usually 100–200 mg); **Haemolysis and methaemoglobinaemia** (dose-related, near universal) — breathlessness, cyanosis, fatigue; **Agranulocytosis** (mainly in the first 3 months — sore throat, fever, mouth ulcers urgent FBC); **Dapsone hypersensitivity syndrome** (DRESS-like, weeks 3–6; HLA-B*13:01 in Asian patients), hepatitis, peripheral neuropathy; **Monitoring (BNF/local shared care):** FBC (incl. reticulocytes/ methaemoglobin as indicated) and LFT, e.g. weekly for the first month, then monthly for 6 months, then 6-monthly; Alternatives if intolerant: sulfapyridine/sulfamethoxypyridazine (limited availability), potent topical steroids, antihistamines (minor benefit)
3. Coeliac follow-up (NICE NG20/BSG) annual review (weight, symptoms, diet adherence, **tTG** to monitor adherence, FBC, ferritin, folate, B12, LFT, TFT, bone health); **pneumococcal vaccination** (hyposplenism — BSG); calcium/vitamin D; **offer serological testing to first-degree relatives**.
4. **Avoid** iodine-containing preparations, ideally, NSAIDs if they trigger flares.

IP scope

Coeliac serology (while the patient is still eating gluten), baseline bloods, dietary counselling and referral, symptomatic topical steroid/antihistamine, shared-care dapsone monitoring (where locally agreed), and family testing.

Referral

DON'T — Do not start a gluten-free diet before serology and biopsy: it gives false-negative results.

ROUTINE — On dapsone: check G6PD first; watch for haemolysis, methaemoglobinaemia (blue lips, breathlessness), agranulocytosis and DRESS.

ROUTINE — Weight loss, anaemia or diarrhoea: gastroenterology (NICE NG20).

Red flags and when to refer

- Do not start a gluten-free diet before serology and biopsy: it gives false-negative results.
- On dapsone: check G6PD first; watch for haemolysis, methaemoglobinaemia (blue lips, breathlessness), agranulocytosis and DRESS.
- Weight loss, anaemia or diarrhoea: gastroenterology (NICE NG20).

Red flags / refer

Suspected DH (dermatology for biopsy with DIF); weight loss, anaemia, dysphagia or persistent GI symptoms despite GFD (gastroenterology — exclude refractory coeliac disease/lymphoma); symptoms of dapsone toxicity (urgent bloods); new lymphadenopathy, night sweats or abdominal pain (lymphoma).

Complications and prognosis

Complications

- Small-bowel lymphoma risk (reduced by gluten-free diet); osteoporosis, anaemia; associated thyroid disease and type 1 diabetes.

Prognosis

- Lifelong; remission on strict gluten-free diet.

Prescriber pearls (UK pharmacist IP)

"Itchy elbows and buttocks plus 'IBS' or anaemia" means DH. Test before the patient tries a gluten-free diet on their own (Case 4's dietitian referral could start this by accident; tell the dietitian).

Check excipients: wheat starch in tablet excipients is usually at negligible gluten levels (per BNF/Coeliac UK), but counsel patients who ask.

Dapsone: **record the G6PD result before the first dose**, give written agranulocytosis and methaemoglobinaemia warnings, and do not issue repeats without up-to-date bloods.

An annual coeliac review covers weight, symptoms, diet adherence, dietitian referral if needed, and bloods.

"**Itchy elbows, knees and buttocks with excoriations and no intact blisters = think DH — and don't let them go gluten-free before testing.**"

Dapsone brings quick relief, but **the diet is the treatment**.

Total IgA, always — IgA deficiency (~2–3% of coeliac patients) gives false-negative tTG-IgA.

Self-check quiz

5 questions · optional

Q1 When should you think of Dermatitis herpetiformis & coeliac disease?

✓ Intense itch with grouped tiny vesicles or just excoriations on the elbows, knees, buttocks and scalp, not responding to steroids.

Q2 What drives it?

✓ Gluten sensitivity: IgA against epidermal transglutaminase deposits in the dermal papillae (granular IgA on DIF).

Q3 What is the backbone of care?

✓ Coeliac serology (tTG-IgA with total IgA) before any gluten-free trial; refer for biopsy with DIF; lifelong gluten-free diet; specialist dapsone for itch control.

Q4 What must you never miss?

✓ Anaemia, osteoporosis and small-bowel lymphoma risk; G6PD check and haemolysis/ methaemoglobinaemia on dapsone.

Q5 Name a red flag that needs escalation.

✓ Do not start a gluten-free diet before serology and biopsy: it gives false-negative results.

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Photo gallery

Dermatitis herpetiformis 5

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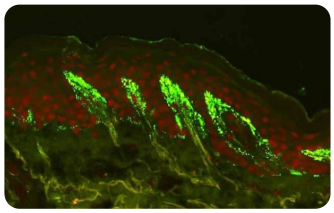

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Dermatitis herpetiformis 1

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Blistering diseases Module 4 · Week 6

SJS/TEN and drug-induced blistering

Stevens–Johnson syndrome and toxic epidermal necrolysis are life-threatening drug reactions with skin detachment and mucosal erosions. This page also orients to other drug-induced blistering (fixed drug eruption, AGEP, pseudoporphyria, drug-induced pemphigoid). Stop the culprit early, refer the same day, record the allergy and Yellow Card.

ALSO CALLED SJS · TEN · Stevens-Johnson · toxic epidermal necrolysis · fixed drug eruption · AGEP · +6

RELATED PAGES Fixed drug eruption · AGEP (acute generalised exanthematous pustulosis) · DRESS (drug reaction with eosinophilia and systemic symptoms) · Erythema multiforme · Approach to a suspected drug eruption

KEY FACTS

- **SJS/TEN** are severe drug reactions with widespread keratinocyte death: **SJS <10%**, overlap 10–30%, **TEN >30%** body surface detachment.
- Prodrome of fever and sore eyes/throat, then painful dusky macules, atypical flat targets, blisters, **Nikolsky sign** and **2 mucosal sites**.
- Culprits started **4–28 days** earlier: allopurinol, aromatic anticonvulsants, lamotrigine, sulfonamides (co-trimoxazole), nevirapine, oxicam NSAIDs.
- Stop all non-essential drugs; **999 / same-day** transfer to a burns or dermatology intensive unit; ophthalmology within 24 h.
- SCORTEN predicts mortality (about 10% in SJS, 30%+ in TEN); record the allergy and Yellow Card.

Overview

This is a group of reactions in which a drug causes vesicles, bullae, pustules or epidermal necrosis. They range from trivial (localised FDE) to life-threatening (TEN). For the IP, **drug causation** is the main reason to know them.

SJS/TEN in depth

SJS and TEN are one disease spectrum. They are rare, life-threatening **mucocutaneous** reactions, almost always drug-induced, with widespread **keratinocyte death and epidermal detachment**. They are classified by the **maximum % BSA detached** (Bastuji-Garin):

CATEGORY	DETACHED / DETACHABLE BSA
SJS	**<10%**
SJS/TEN overlap	10–30%
TEN	**>30%**

Incidence is ~1–6 per million per year (higher in HIV). Mortality is ~**5–10% for SJS** and **~30% (25–50%) for TEN**; it is higher in older people and those with comorbidity.

Erythema multiforme (EM) is a different disease. It is usually triggered by **HSV** (or Mycoplasma), and drugs cause it only rarely. It gives **typical raised 3-zone target lesions on the acral skin**, with or without mild mucosal involvement ("EM major"), and has a good prognosis (see the inflammatory rash pages notes).

Related pages: Fixed drug eruption, AGEP, DRESS, bullous pemphigoid.

Epidemiology

- **Who**
  - Rare (1–6 per million per year); higher risk with HIV, lupus, cancer and HLA-B*15:02 (carbamazepine in South-East Asian people).

Aetiology: causes and risk factors

COURSE CATEGORY	TYPICAL CULPRITS (UK TEACHING LIST)	MECHANISM / TIMING
Spongiotic (eczematous) bullous	ACD to topicals (neomycin, fragrance), systemic contact dermatitis	Type IV, 2–7 days
AGEP	**Aminopenicillins, macrolides**, quinolones, hydroxychloroquine, diltiazem, terbinafine	T-cell/IL-8/neutrophils. **Hours to 2 days** (short latency). Hundreds of non-follicular pustules on erythema, flexural first, with fever and neutrophilia; resolves within ~2 weeks of stopping
FDE, bullous or generalised	NSAIDs, **paracetamol**, co-trimoxazole, tetracyclines, **sildenafil**, laxatives (phenolphthalein historically)	Resident CD8 T cells. Recurs at the **same site** (lips, genitals, hands) within ~30 min to 8 h of re-exposure; heals with violaceous-brown pigmentation
EM / SJS / TEN	**Allopurinol**, **carbamazepine, lamotrigine, phenytoin, phenobarbital**, **sulfonamides (co-trimoxazole)**, sulfasalazine, NSAIDs (oxicams), nevirapine	Cytotoxic T cells, granulysin, Fas-FasL. Latency **4–28 days** (lamotrigine risk is higher with fast titration or valproate). HLA-B*15:02 (carbamazepine; South-East Asian ancestry) and HLA-B*58:01 (allopurinol). EM is usually HSV-related, not drug-related (see the inflammatory rash pages notes)
Drug-induced pemphigus	**Thiols: penicillamine, captopril**; ACE inhibitors, rifampicin, piroxicam	Often foliaceus-like and remits on stopping (more with thiols)
Drug-induced **pemphigoid**	**DPP-4 inhibitors (gliptins)**, loop diuretics, spironolactone, PD-1/PD-L1 inhibitors, penicillins, etanercept (case reports)	Months to years of latency for gliptins
Drug-induced **linear IgA**	**Vancomycin** (classic), penicillins, cephalosporins, captopril, NSAIDs, amiodarone, lithium	Within days; annular "string of pearls" and haemorrhagic/TEN-like forms possible
Pseudo-PCT	**Naproxen**, other NSAIDs, tetracyclines, furosemide, nalidixic acid, isotretinoin; sunbeds; dialysis	Phototoxic, porphyrins normal

COURSE CATEGORY	TYPICAL CULPRITS (UK TEACHING LIST)	MECHANISM / TIMING
(Also) photo-bullous phototoxic	Doxycycline, thiazides, amiodarone, psoralens	See the photodermatoses notes

SJS/TEN in depth

Myths to address

- **"BLISTERS ON LIPS AND EYES ARE JUST COLD SORES"**
  mucositis with a new drug and skin pain is SJS until proven otherwise.

Pathophysiology and pathology

Drug-specific **cytotoxic T cells** trigger mass **keratinocyte apoptosis**, so the whole epidermis peels away.

1. **Drug–HLA presentation**
   The drug binds specific HLA molecules and activates CD8+ T cells and NK cells.
2. **Cytotoxic mediators**
   Granulysin, perforin/granzyme and Fas–FasL kill keratinocytes.
3. **Full-thickness necrosis**
   The entire epidermis dies and separates from the dermis.

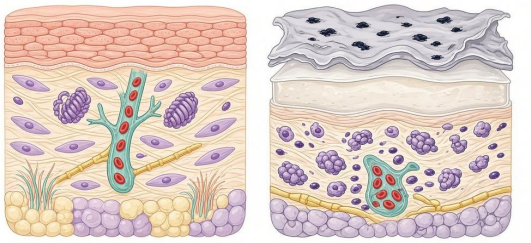

Fig. 1 · Normal skin (left) versus toxic epidermal necrolysis (right): full-thickness epidermal necrosis detaching as a sheet with sparse dermal inflammation.
AI-generated illustration

4. **Mucosal involvement**
   Eyes, mouth, genitals and airway are affected the same way, causing scarring.

How it is caused

- A **type IVc** reaction: drug-specific **cytotoxic CD8+ T cells and NK cells** release **granulysin** (the key mediator), perforin/granzyme B and soluble Fas ligand widespread **keratinocyte apoptosis/necrosis** full-thickness epidermal necrosis with subepidermal separation.
- **HLA associations** (pharmacogenomics), with UK screening implications:

DRUG	ALLELE	POPULATION	UK PRACTICE
Carbamazepine (also oxcarbazepine, phenytoin, fosphenytoin)	**HLA-B*15:02**	**Han Chinese, Thai, Malaysian, other SE Asian**	MHRA/BNF/SmPC: **test before starting carbamazepine** in people of these ancestries; do not start if positive unless no alternative
Carbamazepine	**HLA-A*31:01**	Europeans, Japanese	Associated with DRESS, MPE and SJS/TEN; testing considered (SmPC) — not routine in the UK
Allopurinol	**HLA-B*58:01**	Han Chinese, Korean, Thai; also African ancestry	Start low (100 mg or less in CKD) and titrate; testing where available in high-risk groups (BSR gout guidance)
Abacavir	HLA-B*57:01	All	Mandatory pre-treatment test (hypersensitivity syndrome)

- **High-risk ("notorious") drugs** (EuroSCAR/RegiSCAR, BAD 2016): **allopurinol** (the commonest in Europe), **carbamazepine**, **lamotrigine** (especially fast titration or co-prescribed with **valproate**), **phenytoin**, **phenobarbital**, **sulfonamides** (co-trimoxazole, sulfasalazine), **oxicam NSAIDs** (meloxicam, piroxicam), **nevirapine**. Lower risk: other antibiotics (aminopenicillins, cephalosporins, quinolones), paracetamol (controversial), PPIs. ~5–15% of cases are idiopathic or Mycoplasma-related (especially in children).
- **LATENCY**
  typically **4–28 days** after the first dose (BAD 2016; ALDEN "suggestive" window **5–28 days**). It is shorter on re-exposure (hours to 2 days). Drugs used for many months are rarely culprits, and anticonvulsants mostly cause it within the first 8 weeks.

HISTOPATHOLOGY

- Full-thickness epidermal necrosis with sparse dermal inflammation; frozen section can distinguish from SSSS (split high in epidermis).

Clinical features

SJS/TEN classification by detached BSA: **SJS <10%**, **SJS/TEN overlap 10–30%**, **TEN >30%** (**Module 4 exam Q27**, 40% = TEN).

- Prodrome: fever, sore throat and eye symptoms 1–3 days before the skin.
- Then **painful** dusky macules or atypical flat targets, which form bullae and sheet-like detachment with a positive Nikolsky sign.
- **2 mucosal sites** are involved (oral haemorrhagic crusting of the lips, conjunctivitis, genital; photo E02).

SSSS is the toxin-mediated mimic in young children: superficial peeling, mucosa **spared**, and the child is irritable and tender.

SJS/TEN in depth

> **PRODROME (1–3 DAYS)**
> fever, malaise, sore throat, conjunctivitis/photophobia, painful swallowing — often treated as an infection, which leads to new drugs (NSAIDs, antibiotics) that confuse the timeline.

SKIN

painful dusky-red macules and **atypical flat targets** (two zones, purpuric centres) on the trunk and face, spreading and coalescing. Flaccid blisters and **sheet-like detachment** follow ("wet cigarette paper"); **Nikolsky positive**.

MUCOSAE (2 SITES IN >90%)

haemorrhagic crusted lips, oral erosions, conjunctivitis/pseudomembranes, genital and urethral erosions; also respiratory/GI epithelium.

SYSTEMIC

fever, tachycardia, fluid and protein loss, hypothermia, sepsis (skin barrier lost), acute kidney injury, hepatitis (raised ALT), anaemia, **lymphopenia** (common), neutropenia (poor prognosis), and respiratory involvement (bronchial sloughing).

Sequelae (in up to 50–90% of survivors): **ocular** dry eye, trichiasis, **symblepharon** and corneal scarring/ blindness; pigmentary change; **nail loss/dystrophy**; scarring; genital adhesions; oesophageal/urethral strictures; psychological trauma.

Fig. 2 · SJS: dusky atypical targets and blisters with 2 or more mucosal sites, under 10% BSA. Prodrome of fever and sore eyes. High-risk drugs are carbamazepine, lamotrigine, allopurinol, sulfonamides, nevirapine and oxicam NSAIDs.

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Special groups

- **Children**
  - Mycoplasma causes mucositis-predominant disease (MIRM); also consider SSSS (painful red skin, superficial peeling, mucosa spared).

Diagnosis and investigations

Diagnosis is mostly clinical, with urgent specialist confirmation and frozen-section or standard biopsy. **SCORTEN** (calculated in the first 24 h) predicts mortality. Its 7 points are:

1. age 40

2. malignancy
3. HR 120
4. BSA 10%
5. urea >10 mmol/L
6. bicarbonate <20
7. glucose >14

For drug causation use **ALDEN** (for SJS/TEN). Patch testing is useful later for FDE and AGEP. For DRESS (the systemic DDx), RegiSCAR.

SJS/TEN in depth

- **CLINICAL DIAGNOSIS + URGENT SKIN BIOPSY**
  H&E shows full-thickness epidermal necrosis with sparse dermal infiltrate. Take a second sample for **direct immunofluorescence (negative)** to exclude autoimmune bullous disease.
- **Severity at admission — SCORTEN** (calculate within 24 h, repeat on day 3), 1 point each:

SCORTEN ITEM	POINT IF
Age	**40 years**
Malignancy	Present (active)
Heart rate	**120/min**
Epidermal detachment at day 1	**>10% BSA**
Serum urea	**>10 mmol/L**
Serum glucose	**>14 mmol/L**
Serum bicarbonate	**<20 mmol/L**

SCORTEN	0–1	2	3	4	5
Predicted mortality	**3.2%**	**12.1%**	**35.3%**	**58.3%**	**>90%**

(Bastuji-Garin et al., 2000. It is endorsed by BAD 2016 and tends to over-predict mortality in modern care; ABCD-10 is an alternative.)

- **Drug causality — ALDEN** (Algorithm of Drug causality for Epidermal Necrolysis; Sassolas et al., 2010). Score **every** drug taken in the 8 weeks before the **index day** (the first day of symptoms, including the prodrome):

CRITERION	SCORING (SUMMARY)
1. **Delay** from first intake to index day	**5–28 days +3**; 29–56 days +2; 1–4 days +1; >56 days –1; **started on or after the index day –3** (previous reaction to the same drug: 1–4 days +3, 5–56 days +1)
2. **Drug present** in the body on the index day	Definite 0; doubtful –1; **excluded –3** (stopped >5 half-lives before the index day, with normal clearance)
3. **Prechallenge/rechallenge**	SJS/TEN with same drug +4; similar drug or other reaction with the same drug +2; other reaction with a similar drug +1; no known exposure 0; **previous exposure without reaction –2**

CRITERION	SCORING (SUMMARY)
4. **Dechallenge**	Stopped (or unknown) 0; **continued without harm –2**
5. **Type of drug (notoriety)**	**High-risk ("strongly associated") +3**; associated +2; suspected +1; unknown 0; not suspected –1
6. **Other cause**	If any drug has an intermediate score >3, subtract 1 from each of the other drugs
Interpretation	**<0 very unlikely; 0–1 unlikely; 2–3 possible; 4–5 probable; 6 very probable**

- **BASELINE TESTS**
  FBC, U&E, LFT, glucose, bicarbonate/venous gas, CRP, coagulation, Mg/phosphate, albumin; skin, mucosal and blood cultures; Mycoplasma serology/PCR (in children/young adults); CXR (respiratory); HIV test.

Differential diagnoses

- SSSS (child, superficial, no mucosa)
- EM major (HSV, acral targets, <10%)
- generalised bullous FDE
- AGEP (pustules, not detachment)
- DRESS (facial oedema, eosinophilia, organ involvement, latency 2–8 weeks)
- GVHD
- paraneoplastic pemphigus
- linear IgA (TEN-like)
- phototoxic reactions
- burns

SJS/TEN in depth

	EM MAJOR	SJS	TEN	SSSS	AGEP	GBFDE	PEMPHIGUS / BP
Trigger	HSV, Mycoplasma	Drugs	Drugs	S. aureus toxin	Drugs (<48 h)	Drug re-exposure	Autoimmune
Lesion	Typical raised targets, acral	Atypical flat targets, trunk	Confluent erythema, sheet detachment	Tender erythema, superficial peel, flexures; children/renal failure	Pustules on erythema	Few large dusky plaques at recurring sites	Erosions (PV) / tense bullae (BP)
Mucosa	± mild	**Yes, severe**	**Yes, severe**	**No**	Rare	Mild	PV yes
Histology	Interface	Full-thickness necrosis	Full-thickness necrosis	**Subcorneal** split	Subcorneal pustules	Similar to SJS	DIF positive

Also consider: Mycoplasma pneumoniae-induced rash and mucositis (RIME), Kawasaki disease, toxic shock syndrome, acute GVHD, linear IgA bullous dermatosis (vancomycin), acute cutaneous lupus (TEN-like), paraneoplastic pemphigus.

Management

> **SUSPECTED SJS/TEN OR EXTENSIVE OR MUCOSAL BLISTERING**
>
> **STOP all non-essential drugs immediately**, especially the likely culprits started 4–28 days ago; **999 / same-day ED** for transfer to a **burns or dermatology intensive unit** (UK SJS/TEN guideline, Creamer 2016). Don't wait for a GP appointment; **Ophthalmology within 24 h**, and urology/gynaecology/oral medicine input; Specialist adjuncts: ciclosporin, MIG (the evidence for blocking Fas-FasL is mixed and UK guidance does not routinely recommend it), steroids, and anti-TNF (etanercept, being studied)

> **AGEP**
>
> stop the drug, give supportive care and a potent TCS. Refer the same day if febrile or extensive.

> **FDE**
>
> stop and avoid the culprit; a potent TCS for 1–2 weeks; document the specific drug.

> **DRUG-INDUCED BP OR PEMPHIGUS**
>
> stop the culprit and follow the specialist pathway. Recovery may take months.

> **ALLERGY DOCUMENTATION**
>
> Record the exact drug and reaction in the GP record (SNOMED) and on the patient's allergy list; **Report to MHRA Yellow Card** (all serious reactions; anything involving a black-triangle drug); For SJS/TEN: lifelong avoidance of the culprit and **structurally related drugs**, and **advice for first-degree relatives** about HLA risk. A MedicAlert bracelet is reasonable

> **PREVENTION (PHARMACIST ROLE)**
>
> **HLA-B*15:02** screening before carbamazepine in people of Han Chinese or Thai ancestry (BNF/MHRA); Start allopurinol low (100 mg) and titrate in CKD; Follow the **lamotrigine** titration schedule (halve doses with valproate), and counsel all new starters to **stop and seek help with any rash, fever or mouth ulcers in the first 8 weeks**

SJS/TEN in depth

Primary-care/community actions (you will see the prodrome or early rash):

1. Suspect it a new drug in the past 8 weeks + skin pain + dusky/target lesions or blisters + **mucositis** + fever.
2. **Stop all suspect drugs immediately.** Delay in withdrawal raises mortality; early withdrawal of drugs with a short half-life improves survival (Garcia-Doval 2000).
3. **Same-day emergency admission** (999/ED with a call to on-call dermatology). The BAD advises transfer to a **specialist centre (burns unit/ITU or dermatology with specialist MDT)** when BSA >10% or the patient is unwell. Do **not** start oral steroids or antibiotics in primary care "to cover".
4. **Document** the culprit(s) prominently, **report via Yellow Card**, and tell the patient and family that the drug and **structurally related drugs** must be avoided for life (aromatic anticonvulsants carbamazepine, phenytoin, phenobarbital, oxcarbazepine or lamotrigine cross-react; avoid all sulfonamide antibiotics after co-trimoxazole SJS). Advise a **medical alert** bracelet/card.
5. **First-degree relatives** may share the HLA risk — mention this for carbamazepine/allopurinol.

Hospital care (BAD 2016 UK guideline — for teaching):

> **SUPPORTIVE CARE IS THE CORNERSTONE**

warmed environment (25–28 °C); pressure-relieving/air-fluidised mattress; barrier nursing; fluids guided by urine output (avoid over-resuscitation); nutrition (NG feed); analgesia (incl. for dressing changes); VTE prophylaxis; stress-ulcer prophylaxis; repeated swabs, with **no prophylactic antibiotics**

SKIN

leave detached epidermis in place as a biological dressing; decompress blisters; whole-body greasy emollient (e.g. 50% white soft paraffin/50% liquid paraffin); non-adherent silicone dressings.

EYES

ophthalmology review within 24 h of admission, then daily. Lubricants, topical steroid/antibiotic drops, lysis of adhesions (amniotic membrane in severe disease).

MOUTH

white soft paraffin to the lips, benzydamine/mucoprotectant mouthwashes, antiseptic chlorhexidine mouthwash, and a corticosteroid mouthwash (e.g. betamethasone soluble tablet in water) — doses per BAD/BNF.

UROGENITAL

review; catheter; potent topical steroid to non-eroded mucosa; dressings to prevent adhesions.

IMMUNOMODULATION

no treatment is proven (BAD 2016, Cochrane 2022). Options used in UK centres, decided by the specialist team: **ciclosporin** (~3 mg/kg/day for 7–10 days), which has the most observational support; MIg; short pulsed corticosteroids early; **etanercept** (emerging RCT data)

IP scope

Recognition, **stopping drugs, emergency referral, documentation and Yellow Card** are core IP responsibilities. All treatment is in secondary care. After discharge: lifelong avoidance list, eye follow-up, psychological support, and checks on all new prescriptions and OTC sales (e.g. no ibuprofen if an NSAID was implicated).

Referral

999 / A&E	**Skin pain, fever, dusky targets, mucosal erosions or skin detachment**: SJS/TEN, stop all non-essential drugs and call 999 / admit (burns unit or ICU).
SAME DAY	Widespread sterile pustules with fever after a new drug (AGEP): same-day assessment.
ROUTINE	Rash with fever, facial swelling, lymph nodes and abnormal liver tests: **DRESS**.

Red flags and when to refer

- **Skin pain, fever, dusky targets, mucosal erosions or skin detachment**: SJS/TEN, stop all non-essential drugs and call 999 / admit (burns unit or ICU).
- Widespread sterile pustules with fever after a new drug (AGEP): same-day assessment.
- Rash with fever, facial swelling, lymph nodes and abnormal liver tests: **DRESS**.

Red flags / refer

Any suspected SJS/TEN **same-day admission**. Red flags: skin pain, 2 mucosal sites, dusky/atypical targets, Nikolsky sign, fever, conjunctivitis with rash, BSA detachment, dehydration, inability to eat/drink.

Complications and prognosis

Complications

- Sepsis, fluid loss, respiratory failure; eye scarring and blindness, strictures, nail and hair loss, pigmentation; death.

Prognosis

- Mortality depends on SCORTEN; survivors often have long-term eye and skin problems.

Prescriber pearls (UK pharmacist IP)

Ask **"What new medicine did you start in the last 8 weeks?"** at every acute rash with fever, skin pain or mucosal involvement.

Skin pain out of proportion + mucosal erosions + dusky skin = SJS/TEN until proven otherwise.

Paracetamol can cause FDE, so don't assume over-the-counter means innocent. Always ask about OTC and herbal products.

"**Pain out of proportion + mouth/eye involvement + a new drug = SJS until proven otherwise.**"

The prodrome is treated as an infection, so the drugs given for it (NSAIDs, antibiotics) appear on the chart just before the rash — **ALDEN's "started after index day –3"** stops you blaming them.

Prevention at prescribing: test for HLA-B*15:02 before carbamazepine in people of SE Asian ancestry. Start allopurinol low (100 mg, lower in CKD) and titrate. **Titrate lamotrigine exactly as in the SmPC** (halve the dose with valproate) and warn about rash in the first 8 weeks. Avoid co-trimoxazole where alternatives exist.

Patient leaflet: "If you develop a rash with blisters, sore mouth or sore eyes, stop the medicine and seek urgent help."

SJS/TEN in depth

Self-check quiz

5 questions · optional

Q1 When should you think of Drug-induced blistering (SJS/TEN, FDE, AGEP)?

✓ Blistering, dusky or pustular rash 4–28 days after starting a drug (days for AGEP; hours for repeat fixed drug eruption).

Q2 What drives it?

✓ T-cell-mediated keratinocyte death (SJS/TEN, FDE), neutrophilic reactions (AGEP) or drug-triggered autoantibodies (linear IgA, pemphigoid, pemphigus).

Q3 What is the backbone of care?

✓ Identify and stop the culprit, admit SJS/TEN, document the allergy clearly, Yellow Card, and screen high-risk drugs (HLA-B*1502 before carbamazepine in at-risk groups).

Q4 What must you never miss?

✓ SJS/TEN: skin pain, fever, mucosal erosions and Nikolsky-positive skin detachment.

Q5 Name a red flag that needs escalation.

✓ **Skin pain, fever, dusky targets, mucosal erosions or skin detachment**: SJS/TEN, stop all non-essential drugs and call 999 / admit (burns unit or ICU).

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Photo gallery

SJS / TEN 2

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Fixed drug eruption, AGEP and linear IgA 3

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SJS/TEN and sequelae 6

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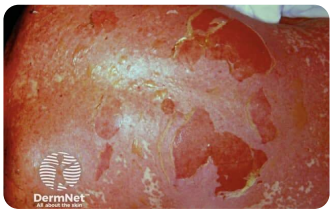

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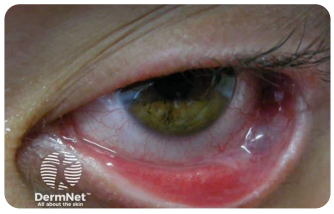

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CHAPTER 4

Hands, feet & skin folds

Hidradenitis suppurativa and intertrigo in the folds; hand eczema, pompholyx, keratolysis exfoliativa and palmoplantar pustulosis on the palms and soles.

1. Hidradenitis suppurativa
2. Intertrigo & the flexural differential
3. Hand eczema (chronic, occupational & pompholyx)
4. Keratolysis exfoliativa
5. Palmoplantar pustulosis & palmoplantar psoriasis

Hands, feet & skin folds Module 3 · Week 5

Hidradenitis suppurativa

A chronic inflammatory disease of hair follicles in the armpits, groin and under the breasts, with recurrent painful nodules, abscesses, tunnels and scarring. Treat early and refer before scarring sets in.

ALSO CALLED HS · acne inversa · Hurley · boils armpit · groin abscess · sinus tracts · +2

RELATED PAGES Boils & carbuncles · Folliculitis · Acne vulgaris · Intertrigo & the flexural differential · Biologics, JAK inhibitors & conventional systemics

KEY FACTS

- **Hidradenitis suppurativa (HS)** is a chronic inflammatory disease of the **hair follicle** in the flexures, not an infection of sweat glands.
- Recurrent painful nodules, abscesses, sinus tracts and **rope-like scars** in the axillae, groin, buttocks and under the breasts; **double-ended comedones**.
- Diagnosis is clinical: typical lesions, typical sites, recurrence (at least twice in 6 months).
- Stop smoking, lose weight, analgesia, topical clindamycin 1%, then a **12-week tetracycline course**; screen for metabolic syndrome and depression.
- Refer early for Hurley II–III or failed primary care treatment; biologics (adalimumab, secukinumab) are specialist.

Overview

Chronic autoinflammatory disease of **apocrine-bearing** / intertriginous skin (acne inversa). Painful deep nodules and abscesses that may form **draining sinus tracts** and irreversible scarring. Not caused by poor hygiene; not contagious. Marked quality-of-life and comorbidity burden.

Epidemiology

- **Who**
  - About 1% of people; women about 3× more; starts after puberty, average delay to diagnosis around 7–10 years.
  - Strong links with **smoking** and **obesity**; a third have a family history.

Aetiology: causes and risk factors

Follicular **occlusion** **rupture** innate immune cascade tunnels and fibrosis. Drivers: genetics (~⅓ family history), obesity/friction, **smoking**, hormones, microbial dysbiosis. Female predominance (~3:1) in Western cohorts; onset typically after puberty (2nd–3rd decade).

Follicular occlusion triad (classic teaching):

1. **Hidradenitis suppurativa**
2. **Acne conglobata**
3. **Dissecting cellulitis of the scalp** (perifolliculitis capitis abscedens et suffodiens)

Tetrad adds **pilonidal disease**. Syndromic overlap possible (e.g. PASH) — specialist territory.

Risk factors and associations

- Follicular occlusion with genetic susceptibility (γ-secretase mutations in familial HS).
- Smoking (nicotine promotes follicular plugging), obesity, friction, hormones.

Myths to address

> **"IT'S CAUSED BY POOR HYGIENE"**
> it is not.

> **"REPEATED INCISION AND DRAINAGE CURES IT"**
> lancing gives brief relief but scars and recurs; treat the disease.

Pathophysiology and pathology

HS starts with a **blocked hair follicle** that ruptures and drives chronic inflammation.

1. **Follicular occlusion**
   Hyperkeratosis plugs the hair follicle infundibulum in apocrine-gland–bearing skin.
2. **Dilation and rupture**
   The follicle swells and ruptures, spilling keratin and bacteria into the dermis.
3. **Inflammatory cascade**
   Innate immunity (IL-1β, **TNF-α**, **IL-17**) causes painful nodules and abscesses: the targets of biologic drugs.
4. **Tunnels and scars**
   Repeated inflammation forms epithelialised **sinus tracts**, bridged and **tombstone** comedones, and rope-like scars.

Fig. 1 · Double-ended (tombstone) comedones and sinus tracts in HS.
AI-generated illustration

5 **Staging**

Hurley I: abscesses without tracts; II: recurrent abscesses with tracts and scars; III: diffuse interconnected tracts.

Fig. 2 · Hurley stages I–III.

AI-generated illustration

HISTOPATHOLOGY

- Not needed for diagnosis. Follicular occlusion, perifolliculitis, abscesses and sinus tracts lined by epithelium.

Clinical features

- Open/closed comedones (often **paired/"tombstone"**), painful firm nodules, abscesses, **draining sinuses**, rope-like scars.
- Sites: **axillae**, groin, inframammary/submammary, buttocks, perianal/perineal, neck.
- Typical case: recurrent axillary lumps since puberty coalescent inflamed nodules, sinus tracts, pus, fibrosis ± smaller indurated nodules.

Hurley staging (clinical workhorse)

STAGE	MEANING
I	Single or multiple abscesses **without** sinus tracts or scarring
II	Recurrent abscesses **with** sinus tracts and scarring (separated by normal skin)
III	Diffuse involvement with **multiple interconnected** sinuses/abscesses across the region

Once scarring/sinuses exist, Hurley stage does not "step down" with medical therapy alone. Dynamic scores (IHS4, HiSCR) used more in secondary care/trials.

Typical staging: **at least Hurley II** (sinuses + scarring after failed long antibiotic courses).

Fig. 3 · Axillary HS — inflamed nodules coalescing into sinus tracts with purulent discharge.

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Types to recognise

Early nodule

Acute painful nodule before tunnels. Site: Axilla.

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Tombstone comedones

Paired comedones: follicular occlusion. Site: Axilla.

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Hurley II–III

Scars and tunnels. Site: Axilla.

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Milder HS

Scars and post-inflammatory redness. Site: Axilla.

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Special groups

- **Pregnancy and contraception**
  - Tetracyclines are avoided in pregnancy; specialist advice on alternatives (e.g. clindamycin, some biologics).
- **Skin of colour**
  - More common and often more severe in people of African descent; scarring and keloids.

Diagnosis and investigations

Clinical — modified Dessau criteria (typical lesions + typical sites + chronicity/recurrence 2 flares in 6 months). Swabs often negative or colonised (supportive if sterile abscess picture). Biopsy rarely needed (exclude SCC in chronic ulcerated anogenital disease). Ultrasound/MRI for extent when planning surgery.

Clinical (Dessau)

- Typical lesions + typical sites + **2 flares in 6 months**.

Swabs

- Often negative or colonised: supports a sterile picture.

Imaging / biopsy

- Ultrasound / MRI before surgery; biopsy chronic anogenital ulcers (SCC).

Document

- Hurley stage, sites, smoking, BMI, prior antibiotics.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Boils / carbuncles	Discrete pointing abscesses; no tunnels or rope scars.
	Epidermoid / Bartholin cyst	Single cyst; no multi-site comedonal disease.
	Perianal Crohn's	GI symptoms, fistulae: IBD work-up.
	Lymphadenitis / deep infection	Acute, unilateral, systemically unwell.

Management

Themes from DermNet and the BAD HS guideline (Ingram et al.):

Primary care / IP first steps

> **EDUCATE**
> not infectious, not "dirt"; validate pain and QoL.

> **LIFESTYLE**

smoking cessation; weight management; loose clothing; gentle antiseptic washes (e.g. chlorhexidine) as adjunct; wound dressings that don't tear skin.

PAIN

paracetamol/NSAID first-line; escalate thoughtfully.

TOPICAL (MILD DISEASE)

clindamycin 1% BD to active lesions (short adjunct).

Oral antibiotics (immunomodulatory courses, not one-off "boil" scripts): tetracycline-class (e.g. doxycycline/lymecycline) often first medical step for mild–moderate disease — typically reviewed ~12 weeks; follow local antimicrobial stewardship.

SCREEN COMORBIDITIES

metabolic syndrome, diabetes, PCOS, CVD risk, depression/anxiety.

Refer dermatology early for Hurley **II–III**, failed primary-care antibiotics, multi-site disease, or diagnostic doubt — waiting lists are long; do not delay.

Specialist / secondary care (know when to escalate)

Combination **clindamycin + rifampicin** (typically up to ~12 weeks) for more severe/refractory disease — specialist initiation/monitoring (interactions, resistance, contraception counselling).

Other medical: short steroid rescue for flares; acitretin (males / non-fertile females); dapsone; metformin/anti-androgens in selected women — specialist.

BIOLOGICS (SPECIALIST ONLY)

UK-licensed options include **adalimumab** (usually first biologic), **secukinumab**, and (more recently) **bimekizumab** — for inadequate control after conventional therapy. Pre-biologic infection screen (CXR, BBV, etc.). Pregnancy plans constrain many options.

SURGERY

incision & drainage = temporary relief only (almost always recurs); **deroofing**, local excision, or wider excision for chronic tunnels — combine with medical/lifestyle care.

Treatment ladder

1 **Educate and support**

Not infectious, not 'dirt'; validate pain and QoL. **Stop smoking**, weight management, loose clothing, chlorhexidine washes, non-adherent dressings.

2 **Mild**

Clindamycin 1% BD to active lesions (short adjunct).

3 Mild–moderate

Tetracycline-class course (doxycycline / lymecycline), review ~12 weeks; stewardship.

4 Specialist

Clindamycin + rifampicin (~12 weeks), steroid rescue, acitretin, dapsone, metformin / anti-androgens. **Biologics** (adalimumab, secukinumab per NICE TAs); surgery for tunnels.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Comorbidities**: Screen metabolic syndrome, diabetes, PCOS, CVD risk, depression.
- **Pregnancy**: Intent changes antibiotic, retinoid and biologic choices: ask early.

Referral

SAME DAY	Fluctuant abscess with **systemic upset**: drainage.
URGENT	**Hurley II–III**, or failed 12-week tetracycline: dermatology for systemic or biologic therapy.
URGENT	Chronic non-healing ulcer in long-standing perianal / buttock HS: exclude **SCC**.
ROUTINE	Low mood or suicidal thoughts: high psychological burden, support and refer.

Red flags and when to refer

- Fluctuant abscess with **systemic upset**: same-day drainage.
- **Hurley II–III** disease (tunnels, scarring) or failure of a 12-week tetracycline course: refer to dermatology for systemic or biologic therapy.
- Low mood or suicidal thoughts: HS has a high psychological burden.
- Chronic non-healing ulcer within long-standing perianal or buttock HS: exclude **SCC**.

Complications and prognosis

Complications

- Chronic pain, odour, depression and suicide risk; impaired work and relationships.
- Strictures, lymphoedema, anaemia, fistulae.
- SCC in long-standing buttock or perineal disease (rare); associated Crohn's disease, spondyloarthritis, PCOS.

Prognosis

- Chronic, relapsing; early treatment and lifestyle change limit scarring. Surgery (deroofing, excision) for persistent tracts.

Prescriber pearls (UK pharmacist IP)

- Stop calling every axillary lump a “boil” after the third recurrence in the same fold — think HS.
- Document **Hurley stage**, sites, smoking, BMI, prior antibiotics.
- Pregnancy/breastfeeding intent changes antibiotic, retinoid, and biologic choices — ask early.
- Abscess drainage alone is not disease-modifying.

Self-check quiz

4 questions · optional

Q1 Hurley stage with tunnels and scarring separated by normal skin?

✓ Hurley II.

Q2 First topical for mild HS?

✓ Clindamycin 1% twice daily to active lesions.

Q3 Classic clue that separates HS from boils?

✓ Paired (tombstone) comedones, tunnels and rope-like scars in flexures.

Q4 Biggest modifiable risk factors?

✓ Smoking and obesity.

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Photo gallery

Hidradenitis suppurativa 5

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Hands, feet & skin folds Module 3 · Week 5

Intertrigo & the flexural differential

Inflammation of opposed skin folds from friction, heat and moisture. It is a description, not a diagnosis: sort out candida, tinea, erythrasma, flexural psoriasis and seborrhoeic dermatitis before reaching for combination creams.

ALSO CALLED intertrigo | skin fold rash | inframammary rash | groin rash | maceration | erythrasma | +2

RELATED PAGES Candidal intertrigo | Tinea: corporis, cruris & pedis | Psoriasis | Napkin (nappy) dermatitis | Perineal dermatoses

KEY FACTS

- **Intertrigo** is inflammation of skin folds from friction, heat and moisture; it is a pattern, not one disease.
- Breasts, groin, axillae, abdominal folds, between toes; obesity, diabetes, incontinence and hot weather predispose.
- Look for secondary Candida (satellite papules), dermatophyte (active edge), **erythrasma** (coral-red Wood's light) or streptococcus.
- Keep folds dry, reduce friction, barrier paste; hydrocortisone 1% for 1–2 weeks; topical azole if candida likely.
- Persistent flexural rashes: think inverse psoriasis, seborrhoeic dermatitis, contact allergy, Hailey–Hailey or extramammary Paget's.

Overview

Superficial **inflammation of opposed skin surfaces** (flexures/folds) driven by heat, moisture, friction, and poor evaporation. May be purely irritant or complicated by yeast/bacteria; often overlaps inflammatory dermatoses (psoriasis, seborrhoeic dermatitis, contact allergy).

Epidemiology

- **Who**
  - Any age; common in obesity, diabetes, immobility, incontinence and infants (napkin and neck folds).

Aetiology: causes and risk factors

Occluded flexures trap sweat → maceration → barrier failure → inflammation ± overgrowth of resident flora (Candida, Corynebacteria, dermatophytes, Staph/Strep). Risk: obesity, diabetes, hyperhidrosis, immobility, incontinence, tight synthetics, hot/humid environments. Infant napkin dermatitis is a related fold problem.

Risk factors and associations

- Skin-on-skin friction, trapped sweat, maceration, and alkaline irritants (urine, faeces).

Myths to address

- **"TALC KEEPS FOLDS DRY"**
  it can clump and irritate; dry gently and use a barrier.

Pathophysiology and pathology

Folds trap heat and moisture, so the barrier **macerates** and rubs raw.

1. **Occlusion**
   Skin-to-skin contact traps sweat and raises temperature.

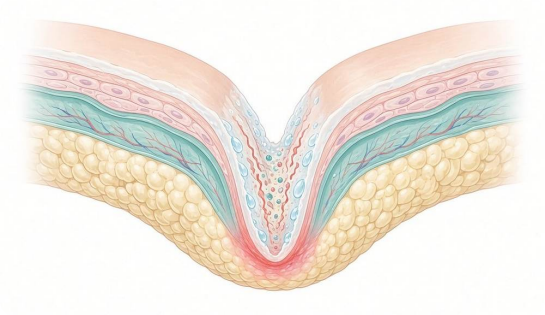

Fig. 1 · Inside a skin fold: trapped moisture and friction break down the barrier.
AI-generated illustration

2. **Maceration and friction**
   Wet stratum corneum softens and erodes with movement.

3. **Secondary infection**
   Damaged skin is colonised by Candida, dermatophytes, Corynebacterium (erythrasma) or streptococci.

4. **Typical sites**
   Submammary, axillary, inguinal, abdominal and interdigital folds.

Fig. 2 · Common intertrigo sites.
AI-generated illustration

Clinical features

- Erythema, itch/burn, moist/macerated skin, fissures in crease depth, peeling.
- Sites: inframammary, axillae, groin, abdominal apron, natal cleft, neck, finger/toe webs.
- **WORKED CASE**
  50-year-old; 3-month itchy red **symmetrical inframammary** rash; smooth surface, well-defined borders, mild peripheral scale; **KOH negative**; on HRT; no other sites or new topicals diagnosis **intertrigo** (inflammatory/irritant-predominant).

Secondary infection clues: satellite pustules (candida), foul odour (Gram-negatives), honey crusts (impetigo), coral-red Wood lamp (erythrasma).

Fig. 3 · Inframammary fold erythema (intertrigo morphology) — KOH-negative case: keep dry; screen secondary infection.

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Fold rashes to tell apart

Candida

Beefy red with satellite papules / pustules. Site: Folds.

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Tinea cruris

Raised active border, central clearing; often asymmetrical. Site: Groin.

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Erythrasma

Red-brown patch, little scale or itch. Site: Axilla, groin.

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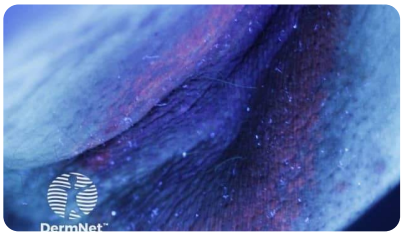
Erythrasma, Wood's lamp

Coral-red fluorescence. Site: Wood's lamp.



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Seborrhoeic

Ill-defined salmon-pink; check scalp and face. Site: Axilla.

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Hailey–Hailey

Painful erosions, fissures; familial: refer. Site: Axilla, groin.

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Special groups

- **Skin of colour**
  - Erythema may look brown or purple; post-inflammatory pigmentation is common.

Diagnosis and investigations

Clinical. KOH/mycology if scale, satellites, or treatment failure; bacterial swab if pustules/crust/odour; consider patch testing if allergen suspected; biopsy if atypical or refractory (granular parakeratosis, Hailey–Hailey, etc.).

Clinical

- Most are clinical.

KOH / mycology

- If scale, satellites or treatment failure.

Swab / Wood's lamp

- Swab if pustules, crust, odour; Wood's lamp for erythrasma.

Biopsy / patch test

- Atypical, one fold that won't clear, or suspected allergy.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Candida intertrigo	Satellite pustules; KOH positive.
	Tinea cruris	Annular, raised active edge.

	CONDITION	DISTINGUISHING FEATURES
	Inverse psoriasis	Shiny, well-demarcated; relapses as treatment stops.
	Seborrhoeic dermatitis	Salmon-pink; scalp, face, ears.
	Hidradenitis	Nodules, sinuses, rope scars.
	Extramammary Paget	One fold that fails to clear in an older adult: biopsy.

A KOH-negative picture still overlaps early inverse psoriasis and seborrhoeic dermatitis — reassess if not settling with dry-care + mild steroid.

Flexural differential table

Key takeaway: when treating flexural rashes, **keep dry**, address incontinence/sweating/obesity, and **screen secondary bacterial/fungal infection**.

Management

Cornerstone — keep dry and reduce friction:

- Shower after sweating; **dry folds thoroughly** (cool hairdryer useful under breasts/axillae).
- Loose natural-fibre clothing; address incontinence and obesity where relevant.
- Barrier pastes (e.g. zinc oxide) for weepy irritant folds; avoid irritating talc as primary strategy.
- Mild topical corticosteroid short course (e.g. **hydrocortisone 1%** 1–2 weeks) for inflammatory intertrigo once infection unlikely/treated.
- If candida likely or empiric after failed dry-care: topical azole (± mild steroid combo products used cautiously in primary care).
- Bacterial secondary infection: topical or oral antibiotic per local guidance.
- Avoid potent/very potent steroids in folds (atrophy, striae, rebound).

Further management: follow-up response; if persists repeat mycology/culture; reconsider inverse psoriasis, contact allergy, seborrhoeic dermatitis; review HRT/medications only if clinically relevant contributors.

Treatment ladder

1

Keep dry, reduce friction

Shower after sweating; **dry folds thoroughly** (cool hairdryer). Loose natural fibres; address incontinence and weight.

2 **Mild steroid**

Hydrocortisone 1%, 1–2 weeks, once infection unlikely or treated.

3 **Treat the organism**

Candida likely: topical azole (± mild steroid combo, used cautiously).
Bacterial: topical or oral antibiotic per local guidance.

4 **Not settling**

Repeat mycology / culture; rethink inverse psoriasis, contact allergy, seborrhoeic dermatitis.

⚠ CAUTION

Prescribing limits and pitfalls

- **Potent steroids in folds**: Atrophy, striae and rebound: avoid.
- **Combo creams**: Pragmatic, but reassess if no infection signs and no improvement.

Referral

SAME DAY	Spreading redness, pain and fever: **cellulitis**.
URGENT	Single fold that won't clear, especially older adult: biopsy for **extramammary Paget**.
ROUTINE	Relapses as soon as treatment stops: think **inverse psoriasis**.

Red flags and when to refer

- Rash that fails to clear in one fold, especially in an older adult: biopsy for **extramammary Paget disease**.
- Spreading redness, pain and fever: cellulitis.
- Repeated 'intertrigo' that relapses as soon as treatment stops: think **inverse psoriasis** (see the psoriasis page).

Complications and prognosis

Complications

- Secondary infection, fissuring, odour, pain; cellulitis in diabetes.

Prognosis

- Recurs unless predisposing factors are managed.

Prescriber pearls (UK pharmacist IP)

- KOH negative "never fungal" — sampling error and mixed pictures exist; re-test if course atypical.
- Combination clotrimazole–HC / similar: pragmatic in primary care but reassess if no clear infection signs and no improvement.
- Counsel that **prevention** (dryness) beats endless cream cycles.

Self-check quiz

4 questions · optional

Q1 Coral-red fluorescence under Wood's lamp?

✓ Erythrasma (Corynebacterium).

Q2 First step for KOH-negative irritant intertrigo?

✓ Dry-care and friction reduction, plus a short course of hydrocortisone 1%.

Q3 Why avoid potent steroids in folds?

✓ Natural occlusion increases absorption: atrophy, striae, rebound.

Q4 One fold that never clears in an older woman?

✓ Biopsy to exclude extramammary Paget disease.

References

1. DermNet: Intertrigo — https://dermnetnz.org/topics/intertrigo
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2. DermNet: Erythrasma — https://dermnetnz.org/topics/erythrasma
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emedicine.medscape.com

Photo gallery

Simple intertrigo 3

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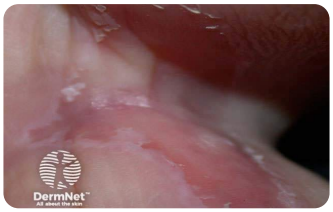

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Flexural differentials 6

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Hands, feet & skin folds Module 3 · Week 6

Hand eczema (chronic, occupational & pompholyx)

Chronic hand eczema is usually irritant (wet work, detergents, friction) with or without allergy (chromate, rubber, preservatives). Protect the barrier, use potent steroids correctly, and patch test anything persisting beyond 3 months.

ALSO CALLED hand dermatitis · chronic hand eczema · occupational dermatitis · irritant hand dermatitis · wet work · pompholyx · +4

RELATED PAGES Contact dermatitis · Eczema / dermatitis · Palmoplantar pustulosis & palmoplantar psoriasis · Keratolysis exfoliativa

KEY FACTS

- Common inflammatory dermatitis of the hands: **irritant**, **allergic**, **atopic** and **vesicular (pompholyx)** patterns, often mixed.
- Chronic hand eczema affects about **1 in 10** adults; frequent cause of time off work.
- High-risk jobs: healthcare, cleaning, catering, hairdressing, construction, metalwork (wet work and chemicals).
- Management: avoid irritants, **gloves**, frequent emollients, soap substitute, **potent topical steroid** short courses; patch test if allergy possible.
- Refer if severe, not responding, occupational or diagnostic doubt (tinea, psoriasis).

Overview

Approach to the palm and sole rash

Ask five questions — they separate most of these conditions:

QUESTION	POINTS TOWARDS
Itch?	Itchy: eczema (ICD/ACD/atopic), pompholyx, tinea (mild). Non-itchy: keratolysis exfoliativa, palmoplantar psoriasis (often sore/fissured rather than itchy), EM (may burn)
Inflammation?	None (just peeling collarettes): KE. Red ± scale: eczema, psoriasis, tinea. Pustules: PPP. Deep vesicles: pompholyx. Targets: EM
Symmetry / one hand?	One palm (+ both feet) **tinea manuum** ("two feet, one hand"). Dominant hand worse irritant/frictional. Symmetrical endogenous eczema, psoriasis, PPP, pompholyx, KE
Border?	Sharp edge, silvery/thick scale psoriasis. Ill-defined, fissured, vesicles at edges eczema. Raised advancing scaly edge tinea
Clues elsewhere?	Nails (pits/onycholysis = psoriasis; dystrophy/subungual debris = tinea); toe webs; scalp/elbows; smoking (PPP, psoriasis); job/hobbies & gloves (contact dermatitis); recent infection/drugs (EM); mucosa

Investigations a pharmacist IP can arrange/advise: skin scrapings for mycology (always before oral antifungal; any unilateral or treatment-resistant palm/sole scale); swab of crusted/weeping areas; refer for **patch testing** via dermatology when chronic hand eczema persists or a contact allergen is suspected.

Hand eczema (hand dermatitis) is inflammation of the skin of the hands/wrists; it is **chronic** when it lasts **> 3 months** or relapses **2 times a year** despite adequate treatment/adherence (ESCD 2022 definition). It is the commonest occupational skin disease in the UK and Europe. It is usually **multifactorial**: irritant ± allergic contact ± atopic tendency, and the morphology often doesn't tell you the cause.

ESCD aetiological categories: **irritant contact**, **allergic contact**, **atopic hand eczema**, **protein contact dermatitis/ urticaria** (e.g. food handlers, latex) — and clinical subtypes: **hyperkeratotic (palmar)**, **recurrent vesicular (pompholyx)**, nummular, pulpitis, and chronic fissured.

Epidemiology

- **Prevalence**
  - Hand dermatitis accounts for 20–35% of all dermatitis; chronic hand eczema affects 10–15% of adults.
  - More common in **young adult women** and people with **atopic eczema**.
- **Occupational**
  - Wet work (over 2 hours a day or frequent handwashing) is the main risk factor; leading cause of occupational skin disease.

Aetiology: causes and risk factors

Irritant contact dermatitis (ICD) — non-immunological, no prior sensitisation needed:

- Repeated exposure (wet work, detergents, solvents, alkaline cement, friction, abrasion, occlusive gloves) strips stratum-corneum lipids and damages keratinocytes **transepidermal water loss**, barrier failure.
- Damaged keratinocytes release IL-1α, TNF-α and other cytokines innate inflammation; cumulative sub-threshold insults add up ("cumulative ICD").
- Repetition drives **hyperkeratosis, acanthosis and lichenification**; in darker skin, **post-inflammatory hyperpigmentation** is prominent.
- Risk modifiers: atopy/filaggrin loss-of-function, low humidity/winter, hot water, dominant-hand exposure.

Allergic contact dermatitis (ACD) — **type IV (delayed, T-cell mediated)**: sensitisation phase (hapten + skin protein Langerhans cells lymph node memory T cells) then elicitation 24–72 h after re-exposure. A damaged irritant-barrier makes sensitisation more likely — so ICD and ACD frequently coexist.

Construction-worker exposures:

- **Wet cement** — alkaline (pH ~12–13) and abrasive ICD; contains **hexavalent chromium (chromate)** and cobalt ACD. EU/UK REACH restricts soluble Cr(VI) in cement to 2 ppm (ferrous-sulphate reduction), which reduced but did not abolish cement ACD.
- **Rubber/leather gloves** — thiurams, carbamates, mercaptobenzothiazole (rubber accelerators); chromate in leather.
- **Epoxy resins, acrylates, isothiazolinone preservatives, solvents/degreasers, hand cleansers with abrasives.**

Risk factors and associations

IRRITANTS

water, soaps, detergents, alcohol rubs, solvents, cutting oils, dusts, cold and friction.

ALLERGENS

rubber accelerators (gloves), preservatives, fragrances, chromate, nickel, hair dyes, acrylates (nails), plants.

ENDOGENOUS

atopy, filaggrin mutations, pompholyx (often triggered by heat, sweating, stress, nickel).

Myths to address

"ALCOHOL GEL IS WORSE THAN SOAP"

alcohol rubs are usually less irritant than repeated soap and water.

"GLOVES ALL DAY PROTECT THE SKIN"

long occlusion causes sweat irritation; use cotton liners.

Pathophysiology and pathology

Hand eczema is driven by **barrier damage** with or without an **allergic** (type IV) response.

1 **Barrier damage**

Repeated wet work and detergents strip lipids from the stratum corneum, increasing water loss and irritation.

Fig. 1 · Patterns of hand eczema: irritant (backs and finger webs), allergic (glove or contact pattern), atopic and pompholyx (vesicles on palms and sides of fingers).

AI-generated illustration

2 **Inflammation**

Damaged keratinocytes release cytokines; in allergic cases, sensitised T cells respond to specific allergens.

3 **Vesicles**

In pompholyx, intense spongiosis in thick palmar skin forms deep itchy **vesicles** that do not easily rupture.

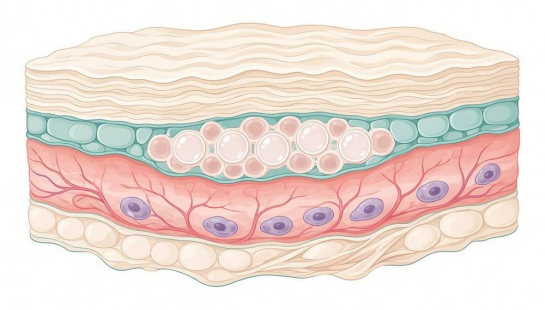

Fig. 2 · Pompholyx: deep, tapioca-like vesicles held within thick palmar epidermis.

AI-generated illustration

4 **Chronic changes**

Ongoing exposure causes hyperkeratosis, painful **fissures** and lichenification.

HISTOPATHOLOGY

- Spongiotic dermatitis; pompholyx shows intraepidermal vesicles in acral skin; chronic disease shows acanthosis and hyperkeratosis.
- Scrapings exclude tinea; patch testing identifies allergens.

Clinical features

STAGE	FEATURES
Acute	Erythema, oedema, papules, **vesicles** (finger sides/palms) that can coalesce into bullae; oozing, crusting; itch, burning, pain
Subacute	Scaling, crusting, fewer vesicles
Chronic	Dryness, scaling, **hyperkeratosis**, **lichenification / palmar hyperlinearity**, painful **fissures** (fingertips, knuckles, finger webs), pigment change (hyper/hypo), nail-fold inflammation and nail ridging/dystrophy

Pattern clues to cause (only clues — patch testing decides):

ICD

finger webs, dorsal hands, fingertips ("pulpitis"), ring site ("ring dermatitis"), dominant hand worse; glazed, fissured.

ACD

sites of contact with the allergen (e.g. dorsal hands/wrists with gloves, fingertips with acrylates, palms/fingers with tool handles), may spread beyond contact area, vesicular; flares 1–2 days after exposure, improves on holidays.

ATOPIC HAND ECZEMA

dorsal hands/wrists, history of flexural eczema/asthma/hay fever, onset in childhood/youth.

HYPERKERATOTIC PALMAR ECZEMA

thick, fissured central palms, middle-aged men, little itch — closely mimics psoriasis.

Worked case: itchy erythematous palms + scale + lichenification + PIH + dominant-hand predominance in a 20-yr cement worker chronic occupational hand eczema; ICD most likely, ACD to chromate/rubber must be tested.

Fig. 3 · Chronic palmar hand eczema — scale, fissuring and erythema on both palms; contrast with non-inflamed KE.

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Patterns to recognise

Chronic irritant

Dry, fissured, glazed; webs and ring site. Site: Dorsal hands, webs.

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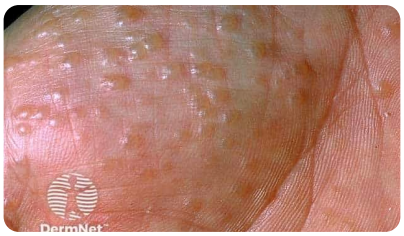
Pompholyx

Recurrent crops of deep, very itchy vesicles; peels in 2–3 weeks. Site: Finger sides, palms, soles.



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Infected

Crusting, weeping, pain: swab and treat. Site: Anywhere.

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Special groups

- **Healthcare and catering workers**
  - Involve occupational health early; alcohol rubs plus emollients reduce irritation.

Diagnosis and investigations

1. History onset vs job change; occupational tasks (wet work definition: hands wet or in occlusive gloves > 2 h/day, or > 20 hand-washes/day); products handled (ask for **safety data sheets**); gloves (type, duration, lining); hobbies (DIY, gardening, hair dye); improvement on holidays/weekends; atopy; previous treatment and response.

2. Examination both hands (dorsal/palmar/webs/nails), wrists, feet (tinea, pompholyx, psoriasis), scalp/elbows/nails (psoriasis), flexures (atopy).
3. **Skin scrapings** for mycology if any scale is asymmetric or treatment-resistant (tinea manuum is a real possibility in a manual worker).
4. **Patch testing** (dermatology): indicated for **all chronic hand eczema (> 3 months)** or suspected ACD; baseline series (incl. **potassium dichromate**, cobalt, nickel, rubber mixes, fragrance mixes, isothiazolinones) plus extended/occupational series and patient's own products. Allergens applied on the back for 48 h, read at ~D2 and D4 (± D7); a positive must be judged for **relevance** to the exposure history.
5. **Prick test / specific IgE** only if immediate (type I) symptoms — latex or food proteins (protein contact dermatitis/urticaria). Prick testing is for type I reactions; patch testing for type IV.
6. Diagnosis of ICD is largely **by exclusion** (negative/irrelevant patch tests + irritant history).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Palmoplantar psoriasis	Sharp margins, thick silvery scale, nails.
	Tinea manuum	'Two feet, one hand'; white scale in creases.
	Keratolysis exfoliativa	No itch, no redness; air-filled peeling collarettes.
	Palmoplantar pustulosis	Sterile pustules brown dots; smokers.
	Scabies	Burrows in webs and wrists; night itch; contacts.

Pompholyx (recurrent vesicular hand eczema)

> **WHAT**
>
> recurrent, symmetrical, **intensely itchy** eruption of deep-seated 1–2 mm **fluid-filled** vesicles ("tapioca"/"sago") on the sides of fingers, palms and soles; resolves with peeling in 2–3 weeks. Accounts for ~1 in 5 hand-dermatitis patch-test referrals. Not caused by blocked sweat glands despite the name.

> **ASSOCIATIONS/TRIGGERS**
>
> atopy (mixed evidence); **contact allergy** (nickel, chromium, cobalt — oral nickel/cobalt can flare); irritants; **heat, hyperhidrosis and season**; **tinea pedis** (id/dermatophytide reaction — treat the feet); stress; smoking; MIG; UVA.

> **DIAGNOSE**
>
> clinical; **scrapings** if unilateral or feet involved; patch test if recurrent.

> **DDX**

> KE (air not fluid, no itch), PPP (pustules not vesicles), tinea (vesicular/bullous, unilateral), bullous pemphigoid (older, tense bullae elsewhere), scabies, HFMD.

> **MANAGE (IP)**
> potassium permanganate 1:10,000 soaks or wet compresses for acute weeping; **potent/very potent TCS** short course (ointment; cream if very wet); emollients and soap substitute; drain large blisters with sterile needle leaving roof; treat infection; treat tinea pedis; manage hyperhidrosis. Refer if frequent/severe: patch testing, PUVA, alitretinoin, delgocitinib, systemic immunosuppressants.

Management

Management for pharmacist/IP

Step 1 — Skin protection & avoidance (all patients):

- Identify and **reduce/replace** irritants and allergens (hierarchy of control: eliminate substitute engineering PPE). For cement: dry handling where possible, low-chromate cement, prompt washing of splashes.
- **GLOVES**
  appropriate material for the chemical (e.g. nitrile; avoid rubber if accelerator-allergic; leather work gloves can contain chromate); keep inside dry; **cotton liners**; limit occlusion time; don't wear gloves on wet or contaminated skin.
- **HAND WASHING**
  lukewarm water, **soap substitute/emollient wash**, rinse and pat dry thoroughly (including webs); in healthcare/food settings alcohol hand rub is often better tolerated than repeated soap and water.
- **Emollients** liberally and frequently (after every wash, at breaks, and greasy ointment under cotton gloves overnight); fragrance-free; preservative-allergic patients need tolerated products. "Barrier creams" are no substitute for gloves/emollients.
- Remove rings for wet work.

Step 2 — Topical anti-inflammatory (IP prescribing):

- **Potent topical corticosteroid** (e.g. mometasone furoate 0.1% or betamethasone valerate 0.1% ointment) once daily for 2–4 weeks for a flare; **very potent** (clobetasol propionate 0.05%) often needed for thick palmar skin — short courses, then taper to twice-weekly **proactive maintenance** on usual sites. Ointments preferred on hands. A history of "topical dexamethasone" with partial temporary relief = undertreatment + continuing exposure.
- **Topical calcineurin inhibitors** (tacrolimus 0.1% ointment, off-label for hands) as steroid-sparing maintenance, especially dorsal skin.
- Treat **secondary infection** (swab; oral flucloxacillin or per local guidance if impetiginised). Potassium permanganate 1:10,000 soaks for weeping/vesicular phases (stains; never ingest).
- Oral antihistamines don't treat the eczema; a sedating one at night only if sleep is disturbed.

Step 3 — Refer to dermatology (routine; sooner if severe or work at risk):

- Not controlled after ~4–6 weeks of optimised potent/very potent TCS + protection; or chronic > 3 months (for **patch testing**); or diagnosis uncertain.
- Specialist options: **delgocitinib 2% cream** (topical pan-JAK inhibitor; **NICE TA1107, 5 Nov 2025** — moderate–severe CHE in adults when TCS ineffective/unsuitable; started by secondary care); **alitretinoin** oral 30 mg (10

mg if not tolerated) for 12–24 weeks (**NICE TA177** — severe CHE unresponsive to ~4 weeks potent TCS; dermatologist-initiated; teratogenic — Pregnancy Prevention Programme; lipids/headache); hand **PUVA / narrowband UVB**; systemic ciclosporin, methotrexate, azathioprine (off-label); dupilumab helps atopic hand eczema within its atopic dermatitis licence. (DELTA FORCE, Lancet 2025: delgocitinib cream outperformed oral alitretinoin with fewer adverse events.) Short oral prednisolone only for severe acute flares under specialist advice — not for chronic disease.

Occupational/UK specifics:

- Encourage **occupational health** referral and workplace skin surveillance (HSE guidance; COSHH duty on employers to assess and control skin exposures).
- **RIDDOR**
  occupational dermatitis is a reportable disease **by the employer** when a doctor diagnoses it in writing and the work involves significant exposure to a known irritant/sensitiser — advise patient to inform employer/OH; the clinician's role is accurate written diagnosis.
- Industrial Injuries Disablement Benefit can apply to prescribed occupational dermatitis (PD D5).
- Fit note if needed; avoid premature advice to quit the job — modification often succeeds, but prognosis is worse with persistent exposure/ACD.

Medicines at a glance

Potent TCS (mometasone 0.1% or betamethasone valerate 0.1% ointment)

Flare

DOSE	Once daily · 2–4 weeks
CAUTIONS	Then twice-weekly proactive maintenance on usual sites.

Very potent TCS (clobetasol 0.05%)

Flare

DOSE	Short courses · Thick palmar skin
CAUTIONS	Taper; review every 4 weeks; dorsal skin thins easily.

Treatment ladder

1 **Protect · every patient**

Reduce or replace irritants and allergens. Right gloves, **cotton liners**, short occlusion; remove rings for wet work.

2 **Flare**

Potent TCS (mometasone 0.1% or betamethasone valerate 0.1% ointment); Very potent TCS (clobetasol 0.05%)

3 Add-ons

Tacrolimus 0.1% ointment (off-label) as steroid-sparing maintenance.
Potassium permanganate 1:10,000 soaks for weeping or vesicular phases.

4 Refer · dermatology

Not controlled after ~4–6 weeks optimised TCS, chronic > 3 months, or uncertain. Specialist: **delgocitinib 2% cream** (NICE TA1107), **alitretinoin** (NICE TA177; Pregnancy Prevention Programme), phototherapy, systemics.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Antihistamines**: Do not treat eczema; sedating one at night only for sleep.
- **Barrier creams**: No substitute for gloves and emollients.
- **Oral prednisolone**: Severe acute flares only, under specialist advice.

Referral

SAME DAY	Painful punched-out erosions or sudden worsening: **eczema herpeticum**.
URGENT	Spreading infection, fever or lymphangitis.
ROUTINE	**Work-related**: occupational health, document exposures, patch-test referral.
DON'T	One-sided scaly hand + tinea pedis: **scrape before steroids**.

Red flags and when to refer

- Painful punched-out erosions or sudden worsening: **eczema herpeticum**, same-day.
- Spreading infection, fever or lymphangitis.
- Work-related dermatitis: advise occupational health, document exposures, patch-test referral.
- One-sided scaly hand with tinea pedis ('two feet, one hand'): scrape for **tinea manuum** before steroids.

Complications and prognosis

Complications

- Painful fissures, secondary infection (S. aureus), eczema herpeticum.
- Loss of work and income; long-term disability.
- Sensitisation to treatments (gloves, preservatives).

Prognosis

- Often chronic and relapsing; only about half are clear years later, especially with atopy, allergy or continued exposure.
- Early diagnosis, glove use and job adaptation improve outlook.

Prescriber pearls (UK pharmacist IP)

- Most chronic hand eczema relapses if exposure continues — success = protection + adequate steroid + maintenance, not steroid alone.
- Fingertip-unit guide: one FTU covers both sides of one hand.
- Thin palmar skin is uncommon — palms tolerate very potent TCS better than other sites, but dorsal hands thin easily; review every 4 weeks.
- Dominant-hand predominance, ring-site eczema, and finger-web involvement irritant. Dorsal hand + wrist cut-off at glove cuff think glove allergy.
- Always look at the **feet** in a hand rash.
- Healthcare hand hygiene: prefer alcohol rub when hands not visibly soiled, dry hands fully before gloving, emollient at breaks and end of shift, report skin problems early (NHS staff hand-dermatitis pocket guides support this).
- Concise answers to the six key questions, drawn from UK guidance.
- One-liner: acute = wet (erythema, oedema, vesicles, oozing); chronic = dry (scale, thickening, lichenification, fissures, pigment and nail change).
- **Exogenous:** irritants (wet work, detergents, solvents, friction), contact allergens, proteins (food, latex).

Endogenous: atopy/filaggrin variants, family history,.

Modifiers: cold dry weather, frequent washing/sanitiser use (sharp rise during COVID-19 — PPE, washing and stress), stress, secondary microbes (S. aureus, dermatophytes).

High-risk jobs: hairdressers, healthcare workers, cleaners, catering/food handlers, bakers, metalworkers (metalworking fluids), construction (cement), mechanics, florists/gardeners.

Water/wet work (especially hot, hard or chlorinated); soaps, detergents, cleaning agents, bleach/ammonia; antiseptics, disinfectants, hand sanitisers (alcohol more of a stinging than a barrier issue); solvents/degreasers; acids and alkalis (incl. wet cement); oils and metalworking fluids; dusts, powders, soil; foods (citrus, onion, garlic, raw meat/fish juices); plants; occlusion and friction (gloves, tools); topical retinoids/benzoyl peroxide on skin.

(Poison ivy/urushiol is a North American allergen — mention for completeness, rare in UK.)

Detailed exposure history **patch testing** (baseline + targeted series + own materials, read D2/D4 ± D7) assess **current relevance** (does the allergen exist in the patient's environment, at the site, with timing that fits?) **trial of avoidance** (improvement supports relevance; repeat open application test can help) education with allergen information sheets and product checking (INCI labels, safety data sheets).

Prevention pillars: education, protective gloves (right type, liners, short periods), skin care (emollients, soap substitutes), substitution of hazards, workplace surveillance, early reporting.

Hand eczema questions and answers

Clinical features — acute vs chronic

Causes

Common irritants (ICD)

Common allergens (ACD) on the hands

ALLERGEN GROUP	TYPICAL SOURCES
Metals — nickel, **chromate**, cobalt	Jewellery/tools/keys (nickel); **cement**, leather gloves (chromate); cement, paints (cobalt)
Rubber accelerators — thiurams, carbamates, mercaptobenzothiazole	Rubber gloves, handles, boots
Preservatives — methylisothiazolinone/MCI, formaldehyde & releasers	Hand washes, wet wipes, paints, cutting fluids
Fragrances (fragrance mixes, Myroxylon pereirae)	Soaps, creams, essential oils

ALLERGEN GROUP	TYPICAL SOURCES
(Meth)acrylates	Gel/acrylic nails, dental materials, adhesives (fingertip pulpitis)
Epoxy resins	Construction, adhesives, coatings
Hair-dye — para-phenylenediamine (PPD)	Hairdressers
Plants — sesquiterpene lactones (Compositae), primin (Primula), garlic (diallyl disulfide), tulip bulbs (tuliposide)	Gardeners, florists, chefs
Topical medicaments — neomycin, corticosteroids, lanolin	Self-treatment

Identifying allergens and correlating with symptoms

Prevention and treatment

Self-check quiz

5 questions · optional

Q1 When is hand eczema 'chronic'?

✓ Lasting more than 3 months, or relapsing at least twice a year despite adequate treatment.

Q2 'Two feet, one hand' suggests?

✓ Tinea manuum (with tinea pedis): take scrapings before steroids.

Q3 Typical first-line steroid for a hand eczema flare?

✓ A potent TCS ointment (e.g. mometasone 0.1%) once daily for 2–4 weeks, then twice-weekly maintenance.

Q4 Who needs patch testing?

✓ All chronic hand eczema (> 3 months) or suspected allergic contact dermatitis.

Q5 Painful punched-out erosions on eczematous hands?

✓ Eczema herpeticum: same-day assessment.

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Photo gallery

Chronic hand eczema 7

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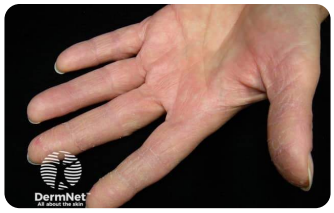

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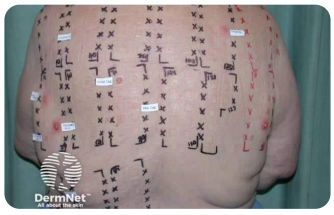
Patch testing 2



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Infected hand dermatitis 1

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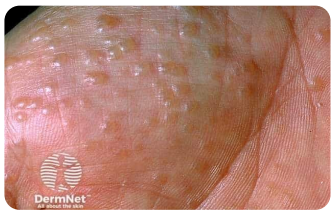
Pompholyx 3



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Differential: tinea manuum and pedis 5

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Hands, feet & skin folds Module 3 · Week 6

Keratolysis exfoliativa

A common, harmless condition of recurrent superficial peeling on the palms and fingers, starting as air-filled blisters, usually in warm weather. Reassure, use emollients and avoid irritants.

ALSO CALLED lamellar dyshidrosis · peeling palms · palm peeling · air-filled blisters · collarette

RELATED PAGES Hand eczema (chronic, occupational & pompholyx) · Tinea: corporis, cruris & pedis · Palmoplantar pustulosis & palmoplantar psoriasis · Hyperhidrosis

KEY FACTS

- **Keratolysis exfoliativa** is recurrent, painless superficial peeling of the palms (and soles), worse in warm weather.
- Air-filled blisters burst into expanding **collarettes** of scale with no inflammation.
- Benign, not contagious, not an allergy; often confused with hand eczema or tinea.
- Irritant avoidance, emollients and keratolytic creams (urea, lactic acid); topical steroids do not help.
- Scrape for fungus if one-sided or with a raised edge.

Overview

A common, **benign, non-inflammatory** disorder of recurrent superficial peeling of the **palms** (soles less often), mostly in young active adults. Synonyms: exfoliative keratolysis, **dyshidrosis lamellosa sicca**, (recurrent) focal palmar peeling, lamellar dyshidrosis. Often under-recognised and mislabelled as "hand eczema", which is why patients arrive having failed topical steroids.

Epidemiology

- **Who**
  - Young adults; often associated with sweaty hands; summer flares.

Aetiology: causes and risk factors

- Cause unknown; may run in families.
- Splitting occurs **within the stratum corneum** — corneodesmosomes (the "rivets" between corneocytes) appear to break down prematurely, so sheets of horny layer detach (Chang et al., BJD 2012 described it as a distinct peeling entity; no causative gene identified).
- **AGGRAVATING FACTORS**
  heat/summer, **palmar hyperhidrosis**, and irritants — water, soap, detergents, solvents, friction. Roughly half of patients notice summer worsening.

- Not an allergy, not infective, and not steroid-responsive inflammation.

Risk factors and associations

- Unknown; possibly a variant of mild pompholyx or an inherited keratin defect; friction, water and detergents aggravate.

Myths to address

- **"IT'S ECZEMA, USE STEROID"**
  there is no inflammation, so steroids are wasted and thin the skin.

Pathophysiology and pathology

The outer **stratum corneum splits**, forming air-filled blisters that peel.

1. **Corneal splitting**
   Cohesion within the stratum corneum fails, creating an air space.

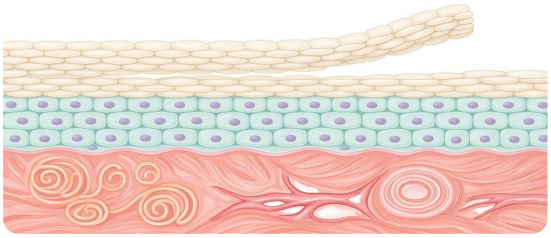

Fig. 1 · A split within the stratum corneum forms an air-filled blister.
AI-generated illustration

2. **Expanding collarettes**
   The roof peels outwards, leaving rings of scale over normal skin.

Fig. 2 · Air blister collarette peeling.

AI-generated illustration

3. **Healing**
   New stratum corneum forms beneath within 1–3 weeks.

> **HISTOPATHOLOGY**
>
> - Rarely needed: splitting within the upper stratum corneum without inflammation.

Clinical features

FEATURE	DETAIL
Earliest sign	One or more tiny **air-filled** (not fluid-filled) superficial blisters on palms/fingers
Evolution	Blisters burst **expanding collarettes of scale** round/oval peeled patches; multiple lamellae may peel

FEATURE	DETAIL
Symptoms	**Not itchy**; peeled areas can feel tender/dry and crack
Inflammation	Absent or minimal (no oedema, no vesicles with fluid, no lichenification)
Fingertips	Split can be deeper — skin feels **hard and numb** before it sheds (explains the numbness patients describe)
Course	Heals to normal skin, then **recurs every few weeks**; chronic over years; often worse in summer
Nails	Not involved (useful vs psoriasis)

Fig. 3 · KE — superficial air-filled blisters burst to leave expanding collarettes of scale on palm/fingers; no erythema, no itch.

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Diagnosis and investigations

- **Clinical** — history of recurrent non-itchy peeling + collarettes on non-inflamed palms.
- Supportive tests if doubt: **skin scrapings negative** for fungus; **patch tests negative**; biopsy (rarely needed) shows cleavage within the stratum corneum.
- **PEARL — THE NUMBNESS**
  usually explained by thick, deeply split fingertip horny layer; still ask about diabetes, B12/alcohol, carpal tunnel symptoms and vibration-tool use if numbness is persistent, sensory-distributed or present when skin is intact (diabetic neuropathy is a sensible safety-net).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Irritant hand dermatitis	Red, dry, fissured; wet work; improves with avoidance and steroid.
	Pompholyx	**Intensely itchy** fluid-filled 'tapioca' vesicles.

	CONDITION	DISTINGUISHING FEATURES
	Palmoplantar psoriasis	Sharp border, thick scale, nail pits; worse in winter.
	Early PPP	Sterile pustules brown dots; smoker.
	Tinea manuum	Often **one hand**; white scale in creases; check feet.

Management

Explain first: benign, cosmetic-nuisance condition, recurrent, not contagious and not an allergy; tends to improve with age; steroids won't help (stop them — avoids palmar atrophy and wasted courses).

Self-care / first-line (OTC/GSL/P):

- **IRRITANT AVOIDANCE**
  cotton-lined waterproof gloves for wet work; soap substitutes; dry hands thoroughly.
- **Emollient / keratolytic hand creams**, applied frequently: **urea** (e.g. 5–10% for maintenance, 10–25% for peeling — several UK OTC brands), **lactic acid** or **salicylic acid**-containing products; silicone/dimeticone barrier creams. US sources cite higher strengths (urea 20–40%, ammonium lactate 12%, salicylic acid 6%) — UK availability is limited, so use licensed UK products/specials per formulary.
- **Treat hyperhidrosis** if present (topical aluminium chloride hexahydrate at night; refer for tap-water iontophoresis if severe) — reduces summer flares.

Specialist options (refer if severe, painful fissuring, or diagnostic doubt): acitretin (teratogenic, specialist-only, Pregnancy Prevention Programme), hand **PUVA**/photochemotherapy — limited evidence, reserved for refractory cases given skin-cancer and burden trade-offs.

Treatment ladder

1 **Explain and protect**

Stop steroids: they don't help and risk atrophy. Cotton-lined gloves for wet work; soap substitutes; dry hands well.

2 **Keratolytic creams**

Urea 5–10% maintenance, 10–25% for peeling; lactic or salicylic acid creams; dimeticone barrier creams. Use licensed UK products per formulary.

3 Treat sweating

Aluminium chloride hexahydrate at night; iontophoresis if severe.

4 Specialist

Acitretin (teratogenic, specialist only) or hand PUVA for refractory cases: limited evidence.

Referral

URGENT — Peeling fingertips in a **child with fever** **5 days**: think Kawasaki disease.

ROUTINE — Painful fissuring, severe disease or diagnostic doubt: dermatology.

Red flags and when to refer

- Peeling fingertips in a **child with fever** for 5 days or more: think Kawasaki disease (urgent).
- Peeling after a sore throat and rash: scarlet fever.

Complications and prognosis

Complications

- Tender fissures and secondary irritant dermatitis.

Prognosis

- Recurrent; improves with age.

Prescriber pearls (UK pharmacist IP)

- "Peeling but not itchy, not red, worse in summer, steroid-proof" = think KE before re-issuing steroid.
- Scrape any one-handed or treatment-resistant peeling to exclude tinea before labelling KE.
- Avoid picking/peeling the lamellae — trim loose edges with clean scissors to reduce painful cracks.

Fingertip numbness that persists between episodes examine/assess for neuropathy.

Self-check quiz

4 questions · optional

Q1 Are KE blisters fluid-filled?

✓ No: tiny air-filled blisters that burst into collarettes.

Q2 Why stop the steroid?

✓ KE isn't inflammatory; steroids don't help and risk palmar atrophy.

Q3 Useful first-line cream?

✓ Urea (5–10% maintenance, 10–25% for peeling) or lactic / salicylic acid creams.

Q4 Peeling fingertips with fever for 5 days in a child?

✓ Kawasaki disease: urgent.

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dermnetnz.org

Photo gallery

Keratolysis exfoliativa 3

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Hands, feet & skin folds Module 3 · Week 6

Palmoplantar pustulosis & palmoplantar psoriasis

Sterile yellow pustules that turn brown on the palms and soles (PPP), strongly linked with smoking, and well-demarcated hyperkeratotic plaques of palmoplantar psoriasis. Both are stubborn and disabling; use potent steroids under occlusion and refer early.

ALSO CALLED PPP · pustular psoriasis palms · palmar psoriasis · plantar psoriasis · smoking pustules · sterile pustules

RELATED PAGES Psoriasis · Hand eczema (chronic, occupational & pompholyx) · Tinea: corporis, cruris & pedis · Keratolysis exfoliativa · Biologics, JAK inhibitors & conventional systemics

KEY FACTS

- **Palmoplantar pustulosis (PPP)** is a chronic, relapsing eruption of sterile yellow pustules on the palms and soles that turn brown and scale.
- Strongly linked to **smoking** (most patients smoke or have smoked); mainly women aged 40–60.
- Distinct from palmoplantar psoriasis, though some have psoriasis elsewhere.
- Stop smoking; very potent topical steroid (clobetasol) ± occlusion; refer for hand–foot PUVA, acitretin or other systemics.
- Screen for thyroid disease and coeliac disease if symptomatic; SAPHO (bone pain) is a rare association.

Overview

Palmoplantar pustulosis

- chronic relapsing **sterile** pustular eruption confined to palms/soles; now regarded as **distinct from psoriasis** (different genetics — not linked to PSORS1/HLA-C*06; IL-36/IL-17/IL-8 pathways), though ~10–25% also have plaque psoriasis. Acrodermatitis continua of Hallopeau = digit-tip variant.

Palmoplantar psoriasis

- psoriasis localised to palms/soles, alone or with plaques elsewhere; disproportionately disabling (manual tasks, walking).

Epidemiology

- **Who**
  - Women 3–4× more often; peak 40–60; uncommon (about 1 in 1,000–10,000).

Aetiology: causes and risk factors

Palmoplantar pustulosis

- adults 20–60, **women 4:1**, **70–90% current/ex-smokers** (nicotinic receptors on eccrine ducts); associations: thyroid disease, coeliac disease, diabetes, psoriatic arthritis, SAPHO/sternoclavicular osteitis; drug-induced by **TNF-α inhibitors**.

Palmoplantar psoriasis

- trauma/friction (Koebner — manual work), infection, stress, lithium, β-blockers, smoking, alcohol, obesity.

Risk factors and associations

- Smoking (nicotine receptors on eccrine sweat ducts), genetics (IL-36 pathway), focal infection; thyroid autoimmunity.

Myths to address

- **"THE PUSTULES ARE INFECTED"**
  they are sterile; antibiotics do not help.

Pathophysiology and pathology

PPP begins in the **eccrine sweat duct**, where neutrophils gather into sterile pustules.

1. **Duct inflammation**
   Nicotine and other triggers activate the acrosyringium (top of the sweat duct).
2. **IL-36 / IL-17 axis**
   Innate cytokines recruit neutrophils.
3. **Pustule evolution**
   Vesicle yellow pustule brown macule scale.

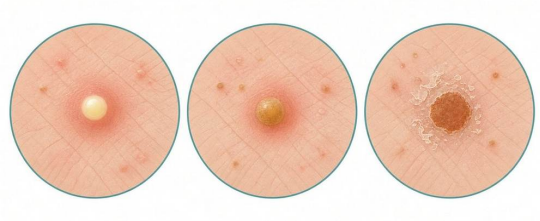

Fig. 1 · PPP lesions evolve from vesicle to pustule to brown scale.
AI-generated illustration

4 **Distribution**

Central palms, thenar and hypothenar eminences, heels and insteps.

Fig. 2 · Typical PPP sites.

AI-generated illustration

HISTOPATHOLOGY

- Intraepidermal unilocular pustule filled with neutrophils, near the eccrine duct.

Clinical features

Palmoplantar pustulosis

- crops of 1–10 mm yellow pustules on thenar/hypothenar eminences, central palms, insteps, heels dry to **brown "rusty" macules**; scale, fissures; burning/pain > itch; symmetrical.

Palmoplantar psoriasis

- (1) well-defined red scaly plaques; (2) diffuse thick hyperkeratosis without much redness (acquired keratoderma). Painful **fissures**; sharp border; symmetrical. More frequent **nail psoriasis** and **psoriatic arthritis** – screen (PEST questionnaire).

Fig. 3 · PPP – sterile yellow pustules evolving to rusty-brown macules in a scaly plaque; strongly linked with smoking.

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Types to recognise

Palm PPP

Pustules on thenar and central palm; culture sterile. Site: Palm.

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Sole PPP

Redness, scale, pustules and brown dots. Site: Instep, heel.

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Palmar psoriasis

Well-demarcated red scaly plaques. Site: Palm.

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Plantar psoriasis

Thick scale with fissuring. Site: Weight-bearing sole.

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Special groups

- **Drug-induced**
  - TNF inhibitors can cause paradoxical PPP.

Diagnosis and investigations

Palmoplantar pustulosis

- clinical; swab (sterile) and scrapings to exclude infection/tinea if unilateral.

Clinical

- Pustules + brown dots on palms / soles, symmetrical.

Swab

- Pustules are **sterile**: swab only to exclude infection.

Scrape

- Scrapings if **unilateral** or interdigital scale: exclude tinea.

Screen

- PEST for psoriatic arthritis; thyroid / coeliac if symptomatic.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Hyperkeratotic hand eczema	Hardest to separate; may need biopsy or patch tests; can coexist.
	Tinea pedis / manuum	Often one-sided; scale in creases; scrape.
	Pompholyx	Itchy deep 'tapioca' vesicles on finger sides.
	Keratolysis exfoliativa	Air-filled blisters collarettes; no inflammation.
	Generalised pustular psoriasis	Fever, sheets of pustules beyond hands and feet: emergency.

Palmoplantar psoriasis

- hyperkeratotic hand eczema (hardest — biopsy or patch testing sometimes needed; they can coexist), tinea (scrape), KE, PPP, keratoderma.

Management

Palmoplantar pustulosis

- **smoking cessation** (strongest modifiable factor; refer to stop-smoking service/NRT); emollients; **very potent TCS** (clobetasol) ± occlusion overnight; salicylic acid for scale. Refer: hand–foot PUVA, **acitretin**, methotrexate, ciclosporin; biologics less reliable than in psoriasis. Screen thyroid/coeliac if symptomatic.

Palmoplantar psoriasis

- emollients and keratolytics (salicylic acid 2–5%, urea); **potent/very potent TCS** ± vitamin D analogue (calcipotriol) or combination products; occlusion overnight; lifestyle (smoking, alcohol, weight). Refer (CKS/NICE CG153): hand–foot PUVA, acitretin, methotrexate, ciclosporin, apremilast and biologics per NICE psoriasis TAs.

Treatment ladder

1 Foundation

Stop smoking: the strongest modifiable factor; stop-smoking service / NRT. Emollients; salicylic acid 2–5% or urea for scale.

2 Topical

Very potent TCS (clobetasol) ± overnight occlusion for PPP. Psoriasis: potent / very potent TCS ± calcipotriol or combination product.

3 Specialist

Hand–foot **PUVA**, **acitretin**, methotrexate, ciclosporin. Apremilast and biologics per NICE psoriasis TAs; biologics less reliable in PPP.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Antibiotics**: Pustules are sterile: antibiotics don't help.
- **Acitretin**: Teratogenic: specialist only, pregnancy prevention.

Referral

999 / A&E	Fever with sheets of pustules beyond hands and feet: **generalised pustular psoriasis**.
URGENT	Can't work or walk: early dermatology referral.
ROUTINE	Not responding to very potent steroid and smoking cessation.

Red flags and when to refer

- Fever with sheets of pustules beyond the hands and feet: **generalised pustular psoriasis**, emergency.
- Inability to work or walk: early dermatology referral.
- Pustules on one foot only, or with interdigital scale: scrape for **tinea**.

Complications and prognosis

Complications

- Pain, fissures, difficulty walking and working; impaired quality of life.

Prognosis

- Chronic, with remission in some after stopping smoking.

Prescriber pearls (UK pharmacist IP)

- Smoking cessation is the single most effective change in PPP; offer stop-smoking support at every visit.
- Use very potent steroids under cling-film or cotton-glove occlusion at night for short periods, then step down.
- Tell patients the pustules are **sterile**: antibiotics do not help.
- Screen for psoriatic arthritis (PEST) and ask about chest-wall pain (SAPHO).

Self-check quiz

4 questions · optional

Q1 Single biggest modifiable factor in PPP?

✓ Smoking: 70–90% are current or ex-smokers; stopping helps most.

Q2 Pustules on one foot with interdigital scale?

✓ Scrape for tinea before labelling PPP.

Q3 Fever and sheets of pustules spreading off the hands?

✓ Generalised pustular psoriasis: emergency.

Q4 First-line topical for PPP?

✓ Very potent topical steroid (e.g. clobetasol) ± overnight occlusion, plus emollient.

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Photo gallery

Palmoplantar pustulosis 3

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Palmoplantar psoriasis 3

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CHAPTER 5

Nails

Paronychia and ingrowing toenail, fungal nail infection, melanonychia and nail melanoma, and the nail signs of skin and systemic disease.

1. Paronychia (acute and chronic)
2. Fungal nail infection & nail discolouration
3. Ingrowing toenail (onychocryptosis)
4. Melanonychia & nail melanoma
5. Nail signs: psoriatic nails, Beau's lines & clubbing

Nails Module 3 · Week 2

Paronychia (acute and chronic)

Acute paronychia is a painful bacterial nail-fold infection; chronic paronychia is an irritant, wet-work inflammation of the nail fold with loss of the cuticle, often colonised by Candida.

ALSO CALLED paronychia · whitlow · nail fold infection · chronic paronychia · candida paronychia · wet work · +1

RELATED PAGES Ingrowing toenail (onychocryptosis) · Fungal nail infection & nail discolouration · Hand eczema (chronic, occupational & pompholyx) · Nail signs: psoriatic nails, Beau's lines & clubbing · Herpes simplex labialis (cold sores)

KEY FACTS

- **Paronychia** is inflammation of the nail fold: **acute** (usually bacterial, hours–days) or **chronic** (over about 6 weeks).
- Acute: S. aureus or streptococci after minor trauma; one digit; pain, redness, pus.
- Chronic: **wet work** destroys the cuticle; irritant dermatitis of the fold with Candida colonisation; several fingers.
- Acute: warm soaks, drain pus; antibiotics only if spreading, unwell or high risk. Chronic: keep hands dry, cotton-lined gloves, potent steroid to the fold ± azole.
- Grouped vesicles: **herpetic whitlow**, do not incise. A single persistent 'paronychia' may be melanoma or SCC.

Overview

Paronychia = inflammation of the proximal and/or lateral nail folds. **Acute** (<~6 weeks) is usually bacterial (S. aureus, streptococci, sometimes Pseudomonas) or herpetic whitlow; **chronic** (>~6 weeks) is often related to irritant dermatitis of the nail fold plus secondary **Candida** (and sometimes Gram-negative bacteria).

Candidal nail disease usually follows chronic paronychia: yeast involves the fold first, then may spread to the nail plate (onychomycosis / onycholysis). Typical case: **candidal onychomycosis with paronychia** in a wet-work hairdresser with KOH yeast cells.

Epidemiology

- **Who**
  - Acute: any age, nail-biters, manicures, children sucking fingers.
  - Chronic: wet-work occupations (hairdressers, cleaners, caterers, healthcare), diabetes.

Aetiology: causes and risk factors

- Repeated **wet work**, cold hands, manicure trauma, cuticle picking, and irritants lift/lose the cuticle gap between fold and plate colonisation by Candida albicans (± bacteria).
- Occupations: hairdressers, cleaners, bartenders, food handlers, dairy/fishing, home wet work.
- Host factors: hand dermatitis, diabetes, immunosuppression, poor peripheral circulation.
- Candida nail-plate infection is **less common** than dermatophyte onychomycosis overall, and is relatively **more often fingernails** than toenails. Relevance of a positive Candida culture must be judged clinically (Candida can secondarily colonise dystrophic nails from psoriasis/trauma).

Risk factors and associations

- Breach of the cuticle seal between fold and nail plate.

Myths to address

- **"CHRONIC PARONYCHIA IS A FUNGAL INFECTION"**
  it is mainly irritant dermatitis of the fold; Candida is secondary.

Pathophysiology and pathology

The **cuticle** is the seal; once it is lost, irritants and microbes get under the fold.

1. **Cuticle loss**
   Trauma, picking, manicures or wet work break the cuticle seal.

2. **Acute infection**
   Bacteria enter the fold, forming pus beneath the eponychium.

Fig. 1 · Acute (single tense red fold with pus) versus chronic paronychia (several swollen folds with lost cuticles).
AI-generated illustration

3. **Chronic dermatitis**
   Repeated wetting causes irritant dermatitis; the fold lifts from the plate and Candida and Gram-negative bacteria colonise the gap.

4. **Nail plate changes**
   Matrix inflammation causes transverse ridges, discolouration and onycholysis.

5 **Restoring the barrier**

Dry hands and cotton under rubber gloves let the cuticle regrow.

Fig. 2 · Cotton gloves under waterproof gloves protect the nail folds.

AI-generated illustration

Clinical features

FEATURE	TYPICAL FINDINGS
Chronic paronychia	Swollen, tender, erythematous proximal (± lateral) folds; **loss of cuticle**; fold lifted off plate; intermittent white/yellow/green discharge
Nail plate	Thickening, yellow/white/green/black discolouration, **transverse ridging**, onycholysis starting near the fold
Sites	Fingers (thumbs common); often **several** digits over time; wet-work hands
Case clues	Hairdresser; 6 months yellow thumbnails; painful cuticles; yellow malodorous secretion; onycholysis; KOH **yeast**

Acute bacterial paronychia: rapid hours–days, usually **one** digit, marked pain, possible abscess — different tempo from chronic Candida-associated disease.

Fig. 3 · Chronic Candida paronychia — swollen proximal nail fold; hairdresser/wet-work risk.

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Types to recognise

Chronic candidal

Swollen proximal fold; wet-work hands. Site: Fingers.

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Cuticle loss

Lifted fold: a portal for yeast. Site: Nail fold.

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Candidal nail

Plate change starting near the cuticle. Site: Fingernail.

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Special groups

- **Diabetes or immunosuppression**
  - Lower threshold for antibiotics and review.

Diagnosis and investigations

- Clinical: chronicity, wet-work history, cuticle loss, multi-digit tendency, discharge.
- **KOH / microscopy ± culture** of nail clippings and/or fold swab — yeast on KOH supports Candida; culture identifies species and co-pathogens.
- Correlate lab with clinic: Candida in a dystrophic nail without fold disease may be secondary.
- Consider bacterial swab if acute flare/abscess; viral swab if grouped vesicles (herpetic whitlow).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Herpetic whitlow	Grouped vesicles, severe pain: **don't incise**.
	Dermatophyte onychomycosis	Starts distal / lateral; toenails; tinea pedis.
	Psoriatic nails	Pits, oil-drop; psoriasis elsewhere.
	Drug-related paronychia	EGFR / BRAF inhibitors, retinoids.

	CONDITION	DISTINGUISHING FEATURES
	Nail-unit SCC / melanoma	Single digit, persistent, older adult.

Management

Non-drug (essential — often determines success):

- Keep hands **dry and warm**; cotton-lined waterproof gloves for wet work; change when damp.
- Avoid cuticle cutting/picking and harsh manicures.
- Emollient barrier frequently; trim nails short.
- Occupational advice for hairdressers: PPE, dry between clients, treat hand dermatitis.

Drug therapy (align with BNF/local formulary; confirm organism when possible):

- Chronic paronychia: treat **dermatitis of the fold** (short courses of topical corticosteroid ointment to the fold are often used in specialist pathways) **plus** topical antifungal/antiseptic solutions to the nail fold for weeks–months (e.g. imidazole solutions — check BNF).
- Confirmed Candida nail/fold disease not settling: oral **itraconazole** or **fluconazole** (Candida-active). **Itraconazole** is usually preferred (broader spectrum vs Candida than terbinafine; fingernail regimens often ~6 weeks continuous or pulsed per product licence — **verify current BNF/SmPC**).
- **Terbinafine** is first-line for many **dermatophyte** nail infections but is **less reliable** for Candida — do not default to it when yeast is the culprit.
- Topical nail lacquers alone are often insufficient if matrix/proximal disease or significant fold infection.
- Abscess: drainage pathway; antibiotics only when bacterial infection is clear.

If no improvement after appropriate oral course: revisit adherence, ongoing wet exposure, alternative diagnoses (psoriasis, trauma), re-mycology, and dermatology referral.

Acute (bacterial) paronychia

- **NO ABSCESS**
  warm water (± saline) soaks 3–4 times a day for 10–15 min, keep the finger dry between soaks, rest/elevate, and give analgesia. Most cases settle without antibiotics.
- **ABSCESS (FLUCTUANT, POINTING PUS)**
  incision and drainage (or lifting the eponychium with a blade) in primary care, minor injuries or urgent care. Partial nail removal if pus has tracked under the plate. Antibiotics are often not needed after adequate drainage.
- **Antibiotics only if** there is spreading cellulitis, the patient is systemically unwell, or they are immunocompromised or diabetic: **flucloxacillin 500 mg four times daily for 5–7 days**; **clarithromycin 500 mg twice daily** if penicillin-allergic (erythromycin in pregnancy); **co-amoxiclav** if from a finger-sucking child or a bite (anaerobes, Eikenella)

- **Refer same day** for suspected **flexor tenosynovitis** (Kanavel signs: fusiform swelling, flexed posture, tenderness along the sheath, pain on passive extension), **felon** (pulp-space abscess), osteomyelitis, or spreading infection in diabetes.
- **Recurrent acute episodes** in the same fold should make you think of an underlying chronic paronychia, an ingrowing nail, HSV, or (single digit, persistent, in an older adult) **SCC of the nail unit or amelanotic melanoma**.

Ingrowing toenail is a separate problem: see Ingrowing toenail.

Fig. 7 · Wet-work protection: a thin cotton glove under a rubber glove keeps the nail folds dry. Change when damp.
AI-generated illustration

Medicines at a glance

Flucloxacillin (only if spreading, unwell, diabetic or immunocompromised)

Acute abscess

DOSE	500 mg four times daily · 5–7 days

Treatment ladder

1 **Chronic**

Keep hands **dry and warm**; **cotton-lined waterproof gloves**; no cuticle cutting; emollient barrier. Potent topical steroid to the fold ± topical azole (CKS), weeks–months.

2 **Acute, no abscess**

Warm (± saline) soaks 3–4 times a day, 10–15 min; dry between; elevate; analgesia. Most settle without antibiotics.

3

Acute abscess

Flucloxacillin (only if spreading, unwell, diabetic or immunocompromised)

4

Confirmed Candida not settling

Oral **itraconazole** or fluconazole (not terbinafine); check CYP3A4 interactions.

Referral

SAME DAY	**Kanavel signs** (flexor tenosynovitis), **felon**, spreading cellulitis, fever: surgical assessment.
URGENT	Diabetes, PAD or immunosuppression with a toe infection.
URGENT	Single nail with persistent 'paronychia', destruction or pigment: exclude **melanoma / SCC**.

Red flags and when to refer

- Spreading cellulitis, lymphangitis, fever or a **tense swollen finger pulp** (felon) or tendon-sheath signs: same-day surgical assessment.
- Grouped vesicles on the finger: **herpetic whitlow**, so do not incise.
- Diabetes, peripheral arterial disease or immunosuppression with a toe infection: low threshold for referral.
- A single nail with persistent 'paronychia', nail destruction or pigment: exclude **nail-unit melanoma or SCC**.

Complications and prognosis

Complications

- Abscess, felon, flexor tenosynovitis, osteomyelitis.
- Nail dystrophy; chronic disease lasting months.

Prognosis

- Acute: settles in days with drainage. Chronic: months, and recurs unless wet work is controlled.

Prescriber pearls (UK pharmacist IP)

Treat the **fold environment**, not only the tablet — wet-work continuation is the common reason for "failed" antifungals.

- Warn about itraconazole/fluconazole **interactions** (CYP3A4), LFTs, pregnancy/contraception issues — counsel and check interactions software.
- Nails grow slowly: cosmetic improvement lags mycological cure by months.
- Yellow-green nails: consider Pseudomonas co-colonisation.

Common misconceptions corrected

POST	STATEMENT	CORRECTION
Yours	Chronic paronychia is caused by streptococci/ Pseudomonas	**Chronic paronychia (>6 weeks)** is mainly an **irritant/eczematous** condition of the nail fold (wet work, cuticle loss) with **secondary Candida (± bacterial) colonisation**. Strep/ staph cause **acute** paronychia, and Pseudomonas is a colonist (green nail). Treat with barrier repair: dry, gloves (cotton under rubber), emollient, **potent TCS ± topical azole** (CKS)
Peer (Jason)	**Terbinafine is preferred for candidal paronychia/nail infection**	**Wrong for Candida.** Terbinafine is poorly active against yeasts and is the drug of choice for **dermatophyte** onychomycosis. Candida nail disease uses **itraconazole** pulses (or fluconazole off-label). Candidal paronychia without plate involvement uses a **topical azole ± TCS**

Self-check quiz

3 questions · optional

Q1 Grouped vesicles on a painful fingertip?

✓ Herpetic whitlow: don't incise.

Q2 Antibiotic for spreading acute paronychia?

✓ Flucloxacillin 500 mg four times daily for 5–7 days.

Q3 Main driver of chronic paronychia?

✓ Wet work and irritant damage to the cuticle, with secondary Candida.

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Photo gallery

Chronic (candidal) paronychia

5

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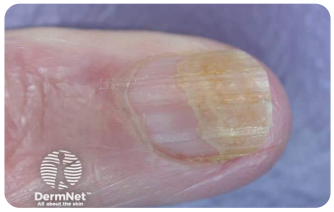

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Nails Module 3 · Week 2

Fungal nail infection & nail discolouration

Thick, discoloured, crumbly nails are often fungal but not always: trauma, psoriasis and onycholysis look the same. Confirm with clippings before committing to months of oral terbinafine.

ALSO CALLED onychomycosis | fungal nail | tinea unguium | terbinafine | amorolfine | onycholysis | +4

RELATED PAGES Tinea: corporis, cruris & pedis | Nail signs: psoriatic nails, Beau's lines & clubbing | Paronychia (acute and chronic) | Psoriasis

KEY FACTS

- Fungal infection of the nail: **dermatophytes** (over 75%, mostly T. rubrum), moulds or Candida.
- Affects about **10%** of adults, rising to 20% over 60 and higher in diabetes; **toenails** far more than fingernails.
- Thickened, discoloured (white or yellow), crumbly nail with onycholysis and subungual debris.
- **Confirm with nail clippings** (microscopy and culture) before oral treatment; about half of abnormal nails are not fungal.
- Oral **terbinafine** is first line for dermatophyte infection (CKS); amorolfine lacquer for limited distal disease.

Overview

Onychomycosis = fungal infection of the nail plate/bed. Dermatophyte infection is also called **tinea unguium**. It accounts for a large share of all nail disease presentations in primary care.

Epidemiology

- **Prevalence**
  - About **10%** overall; up to **20% over 60** and 50% over 70; rare in children.
- **Risk groups**
  - Older adults, **diabetes**, peripheral vascular disease, immunosuppression, psoriasis, athletes and communal-shower users.
  - Often coexists with **tinea pedis**.

Aetiology: causes and risk factors

Organisms & patterns

CATEGORY	TYPICAL ORGANISMS	CLINICAL CLUES
Dermatophyte (~majority)	T. rubrum, T. interdigitale, others	Toenails ≫ fingernails; often with tinea pedis/manuum; **distal lateral subungual (DLSO)** commonest
Yeast (Candida)	C. albicans ± non-albicans	Fingernails; **chronic paronychia** first; proximal change near cuticle; wet work
Non-dermatophyte moulds	Scopulariopsis, Fusarium, Aspergillus, etc.	Similar to dermatophyte; culture/PCR needed; treatment harder
Trauma / non-fungal dystrophy	N/A	Single nail; footwear/sport; negative mycology; ± subungual haematoma that grows out

Clinical subtypes (know the names): DLSO; white superficial (WSO); proximal subungual (PSO — consider immunosuppression/HIV); endonyx; total dystrophic; mixed.

Risk factors and associations

- **ORGANISMS**
  dermatophytes (T. rubrum, T. interdigitale), non-dermatophyte moulds (about 10%; Scopulariopsis, Fusarium, Aspergillus), yeasts (fingernails with chronic paronychia).
- Predisposing: nail trauma, occlusive footwear, hyperhidrosis, smoking, genetic susceptibility; **biofilm** makes cure harder.

Myths to address

> **"IT'S JUST COSMETIC"**
> it can cause pain, mobility problems and cellulitis risk in diabetes.

> **"EVERY THICK YELLOW NAIL IS FUNGAL"**
> trauma, psoriasis and ageing nails mimic it, so confirm first.

Pathophysiology and pathology

Fungi invade the **nail bed and plate keratin**, usually starting under the free edge.

1. **Entry**
   Fungi from tinea pedis or the environment invade under the free edge (distal), from the surface (white superficial) or from the cuticle (proximal).

Fig. 1 · Patterns of invasion: distal-lateral subungual, white superficial, proximal subungual and total dystrophic.
AI-generated illustration

2 **Keratin digestion**

Keratinases digest nail-bed and plate keratin; the bed responds with **subungual hyperkeratosis**, lifting the plate (onycholysis).

3 **Spread and dystrophy**

Infection spreads proximally towards the matrix, causing thick, crumbly, discoloured nails; longitudinal 'spikes' and dermatophytomas (fungal balls) resist treatment.

4 **Reservoir**

The infected nail reseeds the feet and groin, causing recurrent tinea.

HISTOPATHOLOGY

- Fungal hyphae in the nail plate on **PAS-stained nail clipping** (sensitive test), with parakeratosis and neutrophils in the bed.
- Sample correctly: clip the affected nail back as far as possible and scrape subungual debris.
- See the sampling diagram in Diagnosis.

Clinical features

Distal and lateral subungual (commonest): yellow-white thickening spreading back from the free edge, with crumbly debris under the nail and lifting (onycholysis).

WHITE SUPERFICIAL

chalky white patches on the nail surface, usually toenails.

PROXIMAL SUBUNGUAL

white change starting near the cuticle; think of immunosuppression, including HIV.

CANDIDAL

fingernails, usually after chronic paronychia, with nail-fold swelling and onycholysis.

Usually several toenails, often with scaly soles or toe webs (tinea pedis). A single dystrophic nail after injury is more often trauma.

Fig. 2 · Distal lateral subungual onychomycosis — commonest dermatophyte pattern.

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Types to recognise

DLSO

Commonest dermatophyte pattern. Site: Toenail.

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White superficial

Chalky white surface. Site: Toenail.

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Subungual haematoma

Trauma mimic; grows out. Site: Toenail.

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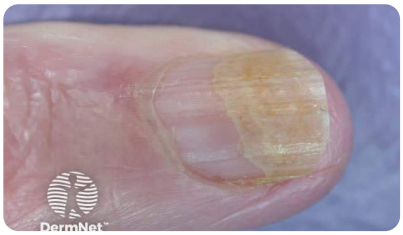
Candidal nail

Onycholysis after paronychia. Site: Fingernail.



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Special groups

- **Diabetes and peripheral arterial disease**
  - Treat more actively and refer to podiatry; thick nails cause pressure ulcers.
- **Children**
  - Rare; check parents and siblings for tinea pedis.

Diagnosis and investigations

- Examine **all nails** and feet/hands/groin for associated tinea.
- **Mycology before long systemic therapy** (NICE/CKS): clip crumbling free edge + scrape most proximal accessible subungual debris; KOH microscopy + culture (± PCR where available).
- Dermoscopy may support fungal vs traumatic onycholysis but does not replace mycology when oral therapy is planned.

Nail sampling technique

- **Clean** the nail with an alcohol wipe, and stop topical antifungals **2–4 weeks** before (and oral ones longer, if feasible).
- **DISTAL/LATERAL SUBUNGUAL PATTERN**
  clip back to the **most proximal** affected nail and collect **subungual debris** with a curette. **Superficial white** pattern: scrape the surface. **Proximal subungual** pattern: punch through the plate, or send a nail biopsy (specialist).
- Send **generous full-thickness clippings + debris** in a **folded dark paper packet** (e.g. a Dermapak), not a sealed plastic pot, because that encourages bacterial overgrowth.
- **YIELD (APPROXIMATE)**
  KOH microscopy ~50–60% sensitive. A negative result does not exclude disease; **Culture** takes up to **2–4 weeks** and has a significant false-negative rate; **Histology with PAS** stain of clippings is the **most sensitive (up to ~90–95%)** but doesn't give the species; **PCR** is quick and species-specific where the lab offers it
- **Repeat sampling** if negative but clinically suspicious, before starting 3–6 months of terbinafine/itraconazole (CKS).

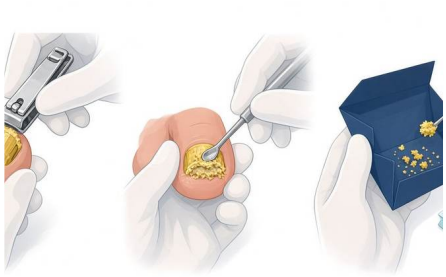

Fig. 7 · Sampling for mycology: clip back towards healthy nail, scrape the crumbly debris from underneath, and send generous clippings in a folded dark paper packet.
AI-generated illustration

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Trauma / gait dystrophy	Avoid months of needless terbinafine.
	Nail psoriasis	Pits, oil-drop patches, skin or scalp signs.
	Pseudomonas green nail	Green-black, onycholytic, wet work: dry, antiseptic soaks.

	CONDITION	DISTINGUISHING FEATURES
	Subungual melanoma / SCC	Pigment that doesn't grow out, widening band, nail-fold pigment.

Management

Management sketch for IP

- Mild distal disease (<50% plate, few nails, no matrix): consider topical lacquer (e.g. amorolfine) per BNF — expect long courses and modest cure rates.
- More extensive dermatophyte disease: oral **terbinafine** often preferred (fingernails shorter course than toenails — check BNF); alternatives itraconazole/fluconazole in selected cases.
- **CANDIDA-PREDOMINANT**
  prefer azoles (itraconazole/fluconazole), not terbinafine first-line.
- Always counsel duration, LFTs/interaction checks, and that "cure" waits for nail outgrowth.
- Relapse common if footwear/hygiene/diabetes not addressed.

Practical prescribing points

- Treatment is optional for mild, asymptomatic toenail disease in a healthy adult: discuss the long course, the cure rate (complete cure is much lower than mycological cure) and recurrence.
- **Oral terbinafine** for confirmed dermatophyte infection: fingernails 6 weeks, toenails 3–6 months (BNF). Ask about liver disease; counsel to report jaundice, dark urine, severe tiredness, rash or taste disturbance.
- **Oral itraconazole** (pulsed) for Candida or non-dermatophyte moulds, or if terbinafine is unsuitable: check interactions (CYP3A4) and heart failure history.
- **Amorolfine 5% lacquer** for superficial or distal disease involving under about half of one or two nails; file the nail first and use weekly for 6–12 months.
- Treat co-existing tinea pedis, and advise on footwear, drying between toes and not sharing nail tools.

Medicines at a glance

Amorolfine 5% lacquer

Limited distal

DOSE	File the nail; apply weekly · 6–12 months
CAUTIONS	For < about half of one or two nails; modest cure rates.

Oral terbinafine

Extensive dermatophyte

DOSE	As per BNF · Fingernails 6 weeks; toenails 3–6 months
CAUTIONS	First choice for dermatophytes.

Treatment ladder

1 **Prevent relapse**

Footwear, foot hygiene, treat tinea pedis, diabetes control.

2 **Limited distal**

Amorolfine 5% lacquer

3 **Extensive dermatophyte**

Oral terbinafine

4 **Candida / moulds**

Oral **itraconazole** (pulsed) or fluconazole; check CYP3A4 interactions and heart failure.

Referral

URGENT	Pigment that doesn't grow out, a widening band, or nail-fold pigment: **possible melanoma** (NG12).
URGENT	Diabetic foot with nail disease and skin breaks: foot-protection team.
ROUTINE	Stop terbinafine if hepatitis symptoms or severe skin reaction.

Red flags and when to refer

- Pigment that does not grow out, a widening band or pigment on the nail fold: **refer as possible melanoma** (NG12).
- Diabetic foot with nail disease and skin breaks: foot-protection team.
- Check liver function history and interactions before oral terbinafine; stop if hepatitis symptoms or severe skin reactions.

Complications and prognosis

Complications

- Pain, difficulty walking and footwear problems.
- Secondary bacterial infection and **cellulitis** (especially diabetes).
- Recurrent tinea pedis and cruris (reservoir).
- Psychosocial distress.

Prognosis

- Oral terbinafine gives the best cure rates, but nails take **6 months (fingers) to 12–18 months (toes)** to grow out.
- About **20–25% relapse** after treatment.

Prescriber pearls (UK pharmacist IP)

Get mycology **before** a long oral course: many thick nails are trauma or psoriasis.

The nail looks abnormal for months after cure, because it grows out slowly (toenails 12–18 months). Set expectations at the start.

A negative result does not exclude infection, so repeat sampling if the picture is convincing.

Self-check quiz

4 questions · optional

Q1 Commonest pattern of onychomycosis?

✓ Distal lateral subungual (DLSO), usually T. rubrum.

Q2 Oral terbinafine duration for toenails?

✓ 3–6 months (fingernails 6 weeks).

Q3 How should clippings be sent?

✓ Generous clippings + subungual debris in a folded dark paper packet.

Q4 White change starting at the cuticle?

✓ Proximal subungual onychomycosis: consider immunosuppression or HIV.

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Photo gallery

Onychomycosis patterns 3

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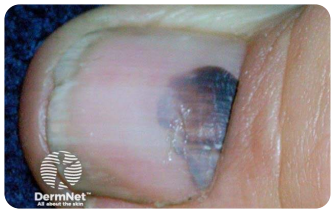
Mimic: subungual haematoma 1



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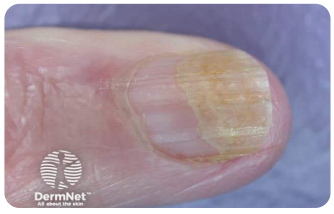
Candidal onycholysis 2



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Nails Module 3 · Week 2

Ingrowing toenail (onychocryptosis)

The edge of the nail plate (usually the big toe) pierces the lateral nail fold, causing pain, inflammation, infection and overgrowth of granulation tissue. Mostly managed conservatively or by podiatry with partial nail avulsion and phenol.

ALSO CALLED ingrowing toenail · ingrown toenail · onychocryptosis · unguis incarnatus

RELATED PAGES Paronychia (acute and chronic) · Pyogenic granuloma · Fungal nail infection & nail discolouration · Nail signs: psoriatic nails, Beau's lines & clubbing

KEY FACTS

- **Ingrowing toenail** (onychocryptosis): the nail edge pierces the lateral fold, usually of the **big toe**.
- Commonest in teenagers and young men; causes: cutting nails curved or too short, tight shoes, hyperhidrosis, involuted nails.
- Stages: inflammation infection with granulation tissue chronic fold hypertrophy.
- Cut straight across, wide shoes, soaks, lift the edge; antibiotics only for spreading cellulitis.
- Recurrent: **partial nail avulsion with phenol** by podiatry (about 95% success). Diabetes or PAD: refer early.

Overview

An **ingrowing toenail** (onychocryptosis) is penetration of the **lateral nail fold** by the edge of the nail plate, most often of the **great toe**. It is commonest in teenagers and young men.

Epidemiology

- **Who**
  - Adolescents and young adults, males more than females; also older people with thickened nails.

Aetiology: causes and risk factors

- **CAUSES**
  cutting nails curved or too short, tight shoes, hyperhidrosis, a naturally involuted ("pincer") nail. Commonest in teenagers and young men. **Retinoids** (isotretinoin) and some EGFR inhibitors cause periungual granulation tissue that looks the same.

Risk factors and associations

- Nail cutting, footwear, sweaty feet, trauma, pincer nails.
- Drugs: isotretinoin and EGFR inhibitors cause similar periungual granulation.

Myths to address

> **"CUT A V IN THE MIDDLE OF THE NAIL"**
> this does not work; cut straight across.

> **"ANTIBIOTICS FIX IT"**
> they do not remove the spike of nail.

Pathophysiology and pathology

A **spike of nail** acts as a foreign body in the nail fold.

1. **Penetration**
   A sharp nail edge pierces the lateral fold, often after cutting the corner short.
2. **Inflammation**
   The fold becomes red, swollen and painful; bacteria colonise the wound.
3. **Granulation tissue**
   Continued irritation produces moist, bleeding 'proud flesh' overhanging the nail.

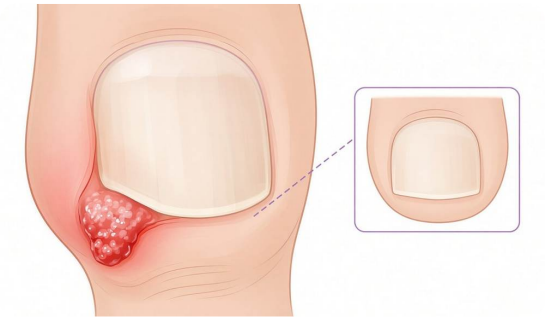

Fig. 1 · Ingrowing toenail with granulation tissue on the lateral fold; inset: nail cut straight across.
AI-generated illustration

4. **Cure**
   Removing the edge and destroying that strip of matrix (phenol) stops regrowth.

Clinical features

- **STAGES**
  inflammation infection with granulation tissue ("proud flesh") chronic hypertrophy of the fold.
- Pain, redness and swelling of the lateral fold; later discharge, pus and **granulation tissue** ("proud flesh") overhanging the nail.

Fig. 2 · Ingrowing toenail: the lateral nail edge digs into an inflamed, swollen nail fold.
© DermNet · **Source**

Fig. 3 · Granulation tissue overgrowing the nail fold in a longstanding ingrowing toenail.
© DermNet · **Source**

Fig. 4 · Ingrowing toenail with periungual inflammation of the big toe.
© DermNet · **Source**

Special groups

- **Diabetes and peripheral arterial disease**
  - High risk of infection, ulceration and amputation: early podiatry.

Diagnosis and investigations

Clinical. Consider X-ray only if a subungual exostosis is suspected; biopsy any atypical or non-healing granulation (amelanotic melanoma, SCC).

Differential diagnoses

Acute paronychia (see Paronychia): fingers more often, no nail spicule.

SUBUNGUAL EXOSTOSIS
firm bony lump under the distal nail; X-ray.

Pyogenic granuloma from retinoids or EGFR inhibitors (see Pyogenic granuloma).

AMELANOTIC MELANOMA OR SCC
non-healing, atypical granulation tissue: biopsy.

Management

CONSERVATIVE CARE
cut the nail straight across, wear wide-toe-box shoes, warm salt soaks, lift the edge (cotton wisp or dental floss), use a gutter splint. Topical antiseptic. Antibiotics only for spreading cellulitis.

DEFINITIVE
partial nail avulsion with phenol matricectomy by **podiatry**. Success is ~95% and lower recurrence than avulsion alone. Total avulsion is for recurrent or bilateral disease. Diabetes or PAD **refer early to podiatry** (foot-protection service).

Treatment ladder

1

Self-care

Cut nails **straight across**, not too short; wide-toe-box shoes. Warm salt-water soaks; lift the nail edge with a cotton wisp or dental floss; topical antiseptic.

2

Infected / granulation

Gutter splint or podiatry review. **Antibiotics only for spreading cellulitis.**

3

Recurrent or severe

Partial nail avulsion with phenol matricectomy by podiatry (~95% success); total avulsion for recurrent or bilateral disease.

Referral

URGENT	Diabetes, peripheral arterial disease or neuropathy with an ingrowing or infected toenail: **podiatry / foot-protection service**.
SAME DAY	Spreading cellulitis with systemic upset.
ROUTINE	Recurrent ingrowing toenail despite conservative care: podiatry for partial avulsion and phenol.
URGENT	Atypical or non-healing 'granulation' tissue: biopsy to exclude amelanotic melanoma or SCC.

Red flags and when to refer

- Diabetes, peripheral arterial disease or neuropathy: **refer early to podiatry** (foot-protection service).
- Spreading cellulitis or systemic upset.
- Atypical or non-healing granulation tissue: biopsy to exclude malignancy.

Complications and prognosis

Complications

- Paronychia, cellulitis, osteomyelitis (rare).

- Recurrence after simple avulsion.

Prognosis

- Mild cases settle with conservative care; phenol matricectomy rarely recurs.

Prescriber pearls (UK pharmacist IP)

- Cut toenails **straight across**, not too short.
- Partial avulsion **plus phenol** matricectomy has the lowest recurrence.

Self-check quiz

4 questions · optional

Q1 What is the definitive treatment for a recurrent ingrowing toenail?

✓ Partial nail avulsion with phenol matricectomy (podiatry).

Q2 Which patients need early podiatry referral?

✓ People with diabetes, peripheral arterial disease or neuropathy.

Q3 Which drugs cause periungual granulation tissue that mimics an ingrowing nail?

✓ Isotretinoin (retinoids) and EGFR inhibitors.

Q4 How should toenails be cut to prevent ingrowing?

✓ Straight across, not too short.

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dermnetnz.org

Photo gallery

Clinical photographs

3

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Nails Module 3 · Week 2

Melanonychia & nail melanoma

A brown or black band in the nail is usually benign (ethnic, drug or trauma-related), but a new, widening or irregular band in one nail, or pigment spreading onto the nail fold, needs urgent assessment for nail-unit melanoma.

ALSO CALLED melanonychia · longitudinal melanonychia · brown nail stripe · nail melanoma · subungual melanoma · Hutchinson sign · +1

RELATED PAGES Melanoma: recognition & referral · Nail signs: psoriatic nails, Beau's lines & clubbing · Fungal nail infection & nail discolouration · Melanocytic naevi (moles) & the changing mole

KEY FACTS

- **Longitudinal melanonychia** is a brown-black band in the nail from melanin in the nail plate.
- Common and benign in darker skin (often several nails); a **single** widening or irregular band in an adult needs assessment.
- Nail melanoma red flags (ABCDEF): Age, Band >3 mm or irregular, Change, Digit (thumb, big toe, index), Extension onto the fold (**Hutchinson sign**), Family/personal history.
- Differentials: subungual haematoma (moves out with growth), fungal melanonychia, drug-induced, naevus, lentigo.
- Suspicious band: suspected cancer pathway for **nail matrix biopsy**.

Overview

Melanonychia = brown/black nail discolouration, often a **longitudinal** band (melanonychia striata) from melanin deposited in the growing nail plate. May be diffuse. Pigment arises from:

1. **Melanocytic activation** (more melanin, normal melanocyte numbers) — racial/physiological, trauma, drugs, inflammatory disease, endocrine, infection, etc.
2. **Melanocytic hyperplasia** — lentigo, naevus, or **melanoma** of the nail unit.

Epidemiology

- **Who**
  - Up to 77% of Black adults have melanonychia in at least one nail; nail melanoma is rare and diagnosed late.

Aetiology: causes and risk factors

Who gets it / risk context

- Common in **skin of colour** (Fitzpatrick V–VI); nearly all Afro-Caribbean adults develop some nail pigmentation by mid-life; also frequent in some Asian populations. Multiple regular bands = often benign racial pattern.
- **Nail-unit melanoma** is uncommon but important: any age possible; relatively higher proportion of melanomas in people with darker skin types are acral/nail; classic teaching ages ~50–70; digits = **thumb, hallux, index** most often. Family history of melanoma increases risk. Trauma association is debated — trauma may draw attention to a lesion already present (Bob Marley's acral melanoma is the famous example).
- Up to a substantial fraction of nail melanomas are **amelanotic/hypomelanotic** — do not wait for a black streak.

Myths to address

- **"A DARK NAIL STREAK IS A BRUISE"**
  a haematoma grows out; a melanocytic band starts at the matrix and persists.

Pathophysiology and pathology

Melanonychia comes from **melanocytes in the nail matrix** depositing pigment into the growing plate.

1 **Activation**

Matrix melanocytes are activated (ethnicity, trauma, drugs, pregnancy) without increasing in number.

Fig. 1 · Pigment-producing melanocytes in the nail matrix lay down a longitudinal brown band in the growing nail plate.
AI-generated illustration

2 **Proliferation**

Increased melanocytes (lentigo, naevus, melanoma) produce a band.

3 **Band width**

The width reflects how much of the matrix is involved; widening or blurring suggests melanoma.

4 **Hutchinson sign**

Pigment on the nail fold shows spread beyond the matrix.

Clinical features

Benign vs concerning features

MORE REASSURING (OFTEN BENIGN)	MORE CONCERNING (REFER URGENTLY)
Multiple nails, stable, similar bands	**Single** nail, new or changing band
Narrow (<~3 mm), regular parallel lines, homogeneous brown	Width **3 mm**, especially widening **proximally**; irregular borders
Stable over years	Change in colour/width/speed of growth; dystrophy, splitting, bleeding, nodule
No periungual pigment	**Hutchinson's sign** — pigment on proximal/lateral nail fold/cuticle
Child with stable congenital naevus pattern (still specialist if unsure)	Adult new single-digit band without clear racial/multi-nail pattern

Hutchinson's sign: periungual extension of pigment historically linked to nail melanoma. Valuable but **not pathognomonic** — pseudo-Hutchinson's can occur (e.g. pigment showing through thin fold, Bowen disease, some benign naevi). Still, Hutchinson's + irregular dermoscopy in an adult single nail = **urgent specialist pathway**, not watchful waiting in primary care IP practice.

ABCDE / ABCDEF of nail melanoma

Both **ABCDE** (Seebacher-style teaching) and full **ABCDEF**:

LETTER	MEANING (PARAPHRASED)
A	**Age** ~50–70; higher relative concern in African, Asian, Native American ancestry (any ethnicity can be affected)
B	**Band** brown-black, width **3 mm**, blurred/irregular borders
C	**Change** — size, colour, or nail structure not improving
D	**Digit** — thumb, hallux, or index finger most often
E	**Extension** — Hutchinson's sign (pigment onto periungual skin)
F	**Family** history of melanoma (ABCDEF version)

Use as a **memory aid**, not a scoring rule that replaces clinical judgement.

Worked case (incidental toe pigment)

Adult of African descent; incidental second-toe band; football (trauma history possible); dermoscopy heterogeneous irregular brown/black pigment **extending to proximal fold and cuticle** (= Hutchinson's). Differentials include subungual haematoma and fungal melanonychia, but irregular pigment + Hutchinson's **do not reassure on trauma alone**.

Fig. 2 · Benign longitudinal melanonychia — regular brown band; commoner in skin of colour.

© DermNet · **Source**

Diagnosis and investigations

Dermoscopic clues (onychoscopy)

Benign pattern (typical): regular brown lines, even spacing/thickness/colour, parallel, sharp margins.

Suspicious pattern: irregular lines (colour, spacing, thickness, parallelism), grey-black colours, width occupying large fraction of plate, granular pigmentation, micro-Hutchinson (pigment only visible on dermoscopy), associated dystrophy. Dermoscopy helps target biopsy site but **does not replace histology** when melanoma is plausible.

Diagnosis & next step for UK IP

- History: onset, change, trauma, drugs (e.g. hydroxyurea, chemotherapies, antimalarials), endocrine, family history, occupation.
- Examine **all nails**, skin, mucosa if syndromic clues (Laugier–Hunziker, Peutz–Jeghers).
- **NEXT STEP IF SUSPICIOUS**
  urgent specialist dermatology / skin-cancer pathway referral for consideration of **nail-unit biopsy** (matrix ± bed as directed by dermatosurgeon; histopathology is gold standard).
- **IP ROLE**
  recognise red flags, photograph if helpful, **do not** attempt nail-matrix biopsy in community pharmacy/primary-care IP settings; avoid prolonged antifungal "trials" that delay cancer referral.
- Subungual haematoma usually migrates distally with growth; static/widening proximal pigment does not behave like simple blood.

Differential diagnoses

> **ETHNIC (RACIAL) MELANONYCHIA**
> several nails, often thumbs and index fingers, in darker skin types; stable, even bands.

> **DRUG-INDUCED**
> several nails after chemotherapy (hydroxyurea, cyclophosphamide), zidovudine, antimalarials or minocycline.

> **FRICTIONAL OR TRAUMATIC**

toenails rubbed by footwear; **subungual haematoma** is red-purple to black, grows out and does not reach the cuticle as a band.

Fungal melanonychia (Trichophyton rubrum var. nigricans, moulds): often wider distally, with nail-plate debris.

NAIL MATRIX NAEVUS OR LENTIGO
a single even band in a child or young adult; still needs specialist assessment if changing.

Nail-unit melanoma and pigmented SCC (Bowen's disease of the nail).

Management

Management outlook

- Benign activation/racial melanonychia: explanation, photo-monitor, safety-net for change.
- Confirmed nail melanoma: specialist surgical excision pathways (may include digit amputation depending on stage); prognosis historically worse than many cutaneous melanomas — **early referral saves lives**.

Referral

URGENT — **Single nail**, band **>3 mm** or widening, irregular colour, blurred borders, **Hutchinson sign** (pigment on the nail fold), nail dystrophy or bleeding: **urgent suspected-cancer referral** (NICE NG12).

ROUTINE — Any new band in an adult over 50, or a history of melanoma.

Red flags and when to refer

- **Single nail**, band **>3 mm** or widening, irregular colour, blurred borders, **Hutchinson sign** (pigment on the nail fold), nail dystrophy or bleeding: **urgent suspected-cancer referral** (NICE NG12).
- Any new band in an adult over 50, or a history of melanoma.

Complications and prognosis

Complications

- Late-diagnosed nail melanoma (poor prognosis); nail dystrophy after biopsy.

Prognosis

- Benign bands persist; melanoma requires amputation or wide excision of the nail unit.

Prescriber pearls (UK pharmacist IP)

Photograph the nail with a ruler and date it: change over time is the most useful sign.

Mark the top of a suspected haematoma with a small notch: if the pigment moves out with the nail, it is blood.

Pigment on the nail fold (Hutchinson sign) can be faked by a see-through cuticle (pseudo-Hutchinson sign), so let a specialist decide.

Ask about change, family history and previous melanoma; examine all nails and the skin.

A widening band, pigment spilling onto the nail fold (Hutchinson's sign) or a dark mark on the sole: see melanoma recognition (acral melanoma).

Self-check quiz

5 questions · optional

Q1 When should you think of Melanonychia & nail melanoma?

✓ A brown or black longitudinal stripe in one or more nails.

Q2 What drives it?

✓ Melanocyte activation in the nail matrix (ethnic, drug, trauma, inflammation) or proliferation (naevus, melanoma).

Q3 What is the backbone of care?

✓ Use the nail ABCDEF rule, photograph with a ruler, and refer suspicious single-digit bands on the suspected-cancer pathway.

Q4 What must you never miss?

✓ Subungual melanoma: often diagnosed late, especially in darker skin where bands are common.

Q5 Name a red flag that needs escalation.

✓ **Single nail**, band **>3 mm** or widening, irregular colour, blurred borders, **Hutchinson sign** (pigment on the nail fold), nail dystrophy or bleeding: **urgent suspected-cancer referral** (NICE NG12).

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Photo gallery

Benign melanonychia 2

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Nail-unit melanoma 3

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Nails Module 3 · Week 2

Nail signs: psoriatic nails, Beau's lines & clubbing

Common nail signs that point to skin or systemic disease: psoriatic pitting and onycholysis, Beau's lines after illness or chemotherapy, and clubbing as a red flag for lung, heart, liver or bowel disease.

ALSO CALLED nail pitting · psoriatic nail · oil drop · onycholysis · Beau's lines · clubbing · +2

RELATED PAGES Melanonychia & nail melanoma · Fungal nail infection & nail discolouration · Psoriasis · Paronychia (acute and chronic) · Osler nodes, Janeway lesions & splinter haemorrhages

KEY FACTS

- Nails record matrix and nail-bed events over months.
- **Matrix** disease: pits, ridges, Beau's lines, leukonychia. **Nail bed**: onycholysis, oil-drop patches, splinters, subungual thickening.
- **Clubbing** with chest symptoms needs a chest X-ray (lung cancer, fibrosis, bronchiectasis).
- New single-digit irregular **melanonychia** or Hutchinson's sign urgent skin cancer referral.
- Do mycology before long oral antifungals.

Overview

Nails grow slowly and record what has happened to the matrix and nail bed over months. Many nail signs are clues to skin disease (psoriasis, eczema, lichen planus) or systemic disease (illness, drugs, lung, heart, liver or bowel disease).

Aetiology: causes and risk factors

- **Matrix** disease changes the nail plate: pits, ridges, Beau's lines, leukonychia.
- **Nail-bed** disease changes how the plate attaches and its colour: onycholysis, oil-drop patches, splinter haemorrhages, subungual thickening.
- **Clubbing** reflects soft-tissue overgrowth of the fingertip, thought to be driven by platelet-derived growth factors bypassing the lung capillaries.

Myths to address

- **"WHITE SPOTS MEAN CALCIUM DEFICIENCY"**
  they are usually minor trauma.

Pathophysiology and pathology

Where the damage is in the nail unit **decides the sign**.

1. **Matrix**
   The matrix makes the plate; damage changes the plate surface (pits, Beau's lines).
2. **Nail bed**
   The bed anchors the plate; inflammation lifts it (onycholysis).
3. **Vessels**
   Splinters are bed capillary bleeds along longitudinal ridges.
4. **Clubbing**
   Fingertip soft tissue grows, possibly from platelet-derived growth factors bypassing the lung.
5. **Pigment**
   Matrix melanocytes form bands.

Fig. 1 · A longitudinal band begins in the matrix.
AI-generated illustration

Clinical features

Nail signs

Psoriatic nails (cross-link Module 4)

Matrix disease: irregular **pitting**, leukonychia, red lunula spots, Beau lines, crumbling. **Nail-bed disease: onycholysis**, oil-drop/salmon patches, subungual hyperkeratosis, splinter haemorrhages. Often coexists with skin/scalp/genital psoriasis or psoriatic arthritis — examine beyond the nail. DDx onychomycosis (mycology if treating as fungus). Management is that of psoriasis (topicals, systemic/biologic pathways via dermatology) — deepen in **Module 4**.

Beau's lines

Transverse **depressions** across the nail (often multiple nails at same level) after temporary **arrest of matrix growth** — severe systemic illness, high fever, chemotherapy, childbirth, major stressor. Grow out with the nail (fingernails ~3–4 mm/month rough guide). Usually no specific nail drug therapy — explain and date the insult. Multiple waves can mark repeated insults.

Clubbing — red-flag awareness

Clubbing: increased nail-fold soft tissue, increased nail curvature, **loss of Lovibond angle** (<165° normally preserved), often boggy nail bed. Associated with lung cancer, interstitial lung disease, bronchiectasis/abscess, cyanotic heart disease, IBD, cirrhosis, and other systemic diseases; also hypertrophic osteoarthropathy.

IP action: recognise **history/exam for cardiopulmonary/GI red flags** urgent appropriate medical referral/ investigation — **not** a dermatology cream problem. Congenital/familial clubbing exists but new clubbing in adults is sinister until proven otherwise.

Fig. 2 · Psoriatic nail pitting — matrix involvement.

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Diagnosis and investigations

- Look at **all 20 nails**, the skin, scalp and joints.
- Ask about recent illness, chemotherapy, new drugs, trauma, wet work and nail cosmetics.
- Send clippings if fungal infection is possible, because psoriasis and onychomycosis can look identical and can coexist.
- For clubbing, check Schamroth's window (lost diamond-shaped gap when the backs of two fingertips are pressed together) and look for chest, cardiac, liver or bowel signs.

Differential diagnoses

SIGN	COMMON CAUSES	THINK ALSO OF
Pitting	Psoriasis, eczema, alopecia areata	Lichen planus
Onycholysis	Psoriasis, trauma, fungal infection, wet work	Thyroid disease, drug photo-onycholysis (doxycycline)
Beau's lines	Severe illness, high fever, chemotherapy	Hand, foot and mouth disease (nail shedding weeks later)

SIGN	COMMON CAUSES	THINK ALSO OF
Clubbing	Lung cancer, interstitial lung disease, bronchiectasis	Cyanotic heart disease, endocarditis, IBD, cirrhosis, thyroid acropachy
Splinter haemorrhages	Trauma, psoriasis	Infective endocarditis (with fever or murmur)

Management

Quick reference for nails

SCENARIO	DO	AVOID
Wet-work chronic paronychia + yeast	Dryness/PPE + fold care; Candida-active azole if systemic needed	Blind long terbinafine; ignoring occupation
Suspected onychomycosis	Mycology before long oral Rx; treat associated tinea pedis	Months of oral antifungal with repeatedly negative mycology and no clinical review
New single-digit irregular melanonychia ± Hutchinson	**Urgent derm / skin cancer referral** for biopsy pathway	Reassuring as "football bruise" without follow-through; community matrix biopsy
Multi-nail stable racial melanonychia	Explain; photo diary; safety-net	Unnecessary biopsy of classic multi-nail racial pattern without change
New clubbing	Systemic work-up / medical referral	Treating as local nail dystrophy

Referral

URGENT **New clubbing**: same-week GP review for chest, cardiac, liver or bowel causes; chest X-ray is often the first test.

URGENT Nail psoriasis with joint pain or dactylitis: screen for **psoriatic arthritis** (PEST) and refer to rheumatology.

Red flags and when to refer

- **NEW CLUBBING**
  same-week GP review for chest, cardiac, liver or bowel causes; chest X-ray is often the first test.
- Nail psoriasis with joint pain or dactylitis: screen for **psoriatic arthritis** (PEST) and refer to rheumatology.

Complications and prognosis

Complications

- Missed melanoma or systemic disease.

Prognosis

- Changes grow out over 6 months (fingers) to 12–18 months (toes).

Prescriber pearls (UK pharmacist IP)

Dating a Beau's line: fingernails grow about 3 mm a month, so a line 6 mm from the cuticle marks an event about 2 months ago.

Nail psoriasis predicts psoriatic arthritis: ask about joint pain and morning stiffness at every review.

Treat nail psoriasis expectations carefully: improvement takes months, and topical treatment works best on the nail folds.

Self-check quiz

5 questions · optional

Q1 When should you think of Nail signs: psoriatic nails, Beau's lines & clubbing?

✓ Pits, oil-drop patches, lifting nails, transverse grooves, or a bulbous fingertip.

Q2 What drives it?

✓ Psoriasis in the nail matrix and bed; a temporary halt in nail growth (Beau's lines); tissue proliferation in clubbing.

Q3 What is the backbone of care?

✓ Link the nail to the whole patient: joints for psoriasis, recent illness or drugs for Beau's lines, a systemic review for clubbing.

Q4 What must you never miss?

✓ New clubbing (lung cancer, fibrosis, endocarditis, IBD, cirrhosis).

Q5 Name a red flag that needs escalation.

✓ **New clubbing**: same-week GP review for chest, cardiac, liver or bowel causes; chest X-ray is often the first test.

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Photo gallery

Psoriatic nails 2

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Beau's lines 1

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Clubbing 1

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CHAPTER 6

Acne & follicles

Disorders of the pilosebaceous unit, with NICE NG198 treatment ladders, antibiotic stewardship and the acne mimics.

1. Acne vulgaris
2. Acne mimics: rosacea, steroid eruptions and folliculitis
3. Acanthosis nigricans
4. Perioral (periorificial) dermatitis
5. Rosacea (incl. Demodex & steroid rosacea)

Acne & follicles Module 6 · Week 2

Acne vulgaris

Comedones, papules, pustules and nodules of the pilosebaceous unit. Follow NICE NG198 combinations for 12 weeks, never use antibiotic monotherapy, and refer early for scarring.

ALSO CALLED acne · spots · comedones · blackheads · whiteheads · nodulocystic acne · +3

RELATED PAGES Acne mimics: rosacea, steroid eruptions and folliculitis · Acne scars · Acne prescribing & the combined pill · Retinoids (topical & oral) · Rosacea (incl. Demodex & steroid rosacea)

KEY FACTS

- Chronic inflammatory disease of the **pilosebaceous unit**: comedones, papules, pustules, nodules.
- Affects **85–95% of adolescents**; 20–35% moderate–severe; often persists into adulthood, especially in women.
- Four drivers: **sebum**, **follicular plugging**, **C. acnes**, **inflammation**.
- UK care follows **NICE NG198**: 12-week fixed-combination topicals ± an oral tetracycline; **never antibiotic alone**.
- Refer **acne fulminans the same day**; refer scarring or distress at any severity.

Overview

Acne vulgaris is the common form of acne: a chronic inflammatory disorder of the **pilosebaceous unit** (the hair follicle and its sebaceous gland). It produces a mixed eruption of non-inflammatory lesions (**comedones**) and inflammatory lesions (**papules, pustules, nodules**).

- It favours areas with the most sebaceous follicles: **face, neck, upper chest, shoulders and back**. About half of patients have truncal acne as well as facial acne.
- Diagnosis is **clinical**. Comedones are the hallmark and help separate acne from rosacea and folliculitis.
- It matters beyond the skin: **scarring**, persistent pigment change and real **psychological distress**, often out of proportion to how the acne looks.
- Severe variants (**acne conglobata**, **acne fulminans**) are uncommon but need specialist care.

Epidemiology

- **Prevalence**
  - Most adolescents (about **85–95%**) have acne to some degree; **20–35%** develop moderate to severe disease.
  - It is one of the most common skin conditions worldwide.

- **Age**
  - Usually starts around puberty and peaks in the **mid to late teens**.
  - It can persist into, or begin in, adult life: roughly **50% of women in their 20s** and 30–40% in their 30s have some acne.
  - About **a quarter of adult women** with acne had none as teenagers (adult-onset acne).
- **Sex**
  - More common and often more severe in **males during adolescence**.
  - More common in **females in adulthood** (hormonal and PCOS-related acne).
- **Ethnicity**
  - Affects all ethnic groups. In **skin of colour**, post-inflammatory hyperpigmentation and keloid scarring are more frequent and are often the main concern.
- **Family history**
  - Strongly heritable: twin studies attribute about **80%** of the variation in acne risk to genetics.

Aetiology: causes and risk factors

Acne results from a **genetic predisposition** acted on by androgens and local factors. No single cause is found in most patients.

Causes and associated factors

- **ANDROGENS**
  drive sebum production and follicular plugging.
  - Puberty; **polycystic ovary syndrome (PCOS)**; congenital adrenal hyperplasia and other hyperandrogenic states.
  - Acne often changes in **pregnancy**.
- **Drugs**
  - Corticosteroids (systemic and topical), testosterone and **anabolic steroids**, lithium, **EGFR inhibitors**, JAK inhibitors, phenytoin, halogens (iodides, bromides).
- **Occlusion and friction** (acne mechanica)
  - Headbands, helmets, chin straps, backpacks, sports pads, face masks.
- **Cosmetics and hair products**
  - Occlusive or comedogenic creams and make-up; **pomade acne** along the hairline from greasy hair products.
- **Picking and squeezing**, humid environments and heavy sweating.

Myths to address

> **DIET**
> links with high-glycaemic foods and dairy (and whey protein supplements) have been reported, but **NICE finds insufficient evidence to recommend any specific acne diet**.
>
> - A balanced diet is reasonable; strict exclusion diets are not.

Dirt or poor hygiene does not cause acne.

- The dark colour of a blackhead is oxidised keratin and melanin, not dirt. Over-washing and scrubbing make irritation worse.

Acne is not infectious and cannot be caught.

SUNLIGHT DOES NOT CURE ACNE

any improvement is temporary, and retinoids and tetracyclines increase sun sensitivity.

NOTE

Drug-induced acneiform eruptions

A sudden, **monomorphic** papulopustular eruption without comedones after starting a drug (steroids, EGFR inhibitors) is an acneiform drug eruption rather than acne vulgaris.

Pathophysiology and pathology

Four linked processes act in a single follicle. They explain each lesion type and which drug targets which step.

1 **Androgens drive excess sebum**

At puberty, androgens (testosterone, DHEAS) enlarge the sebaceous glands and increase **sebum** output. IGF-1 also stimulates the glands, which may explain the reported link with high-glycaemic diets.

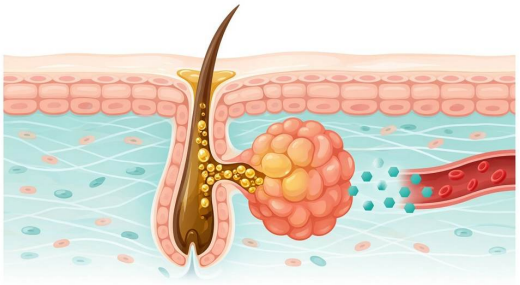

Fig. 1 · Step 1: androgens from the blood stimulate an enlarged sebaceous gland to pour excess sebum into the follicle.

AI-generated illustration

2 **Follicular hyperkeratinisation forms a microcomedone**

Keratinocytes lining the upper follicle multiply and fail to shed. Sticky keratin and sebum **plug the follicle**, forming a **microcomedone**, the precursor of every acne lesion. A closed plug is a whitehead; a plug open to the surface is a blackhead.

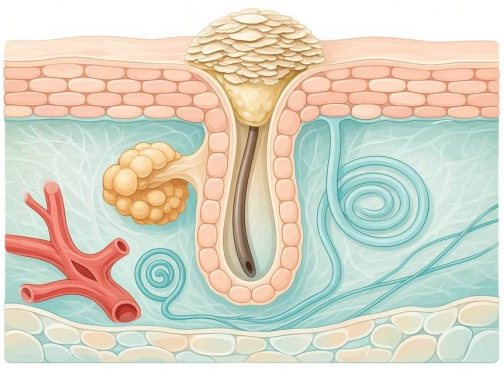

Fig. 2 · Step 2: sticky keratinocytes and sebum plug the follicle opening, forming a comedone.

AI-generated illustration

3 **Cutibacterium acnes proliferates**

The plugged, lipid-rich, low-oxygen follicle lets the commensal **C. acnes** multiply. It activates **Toll-like receptor 2** on immune cells, releasing IL-1β, IL-8, IL-12 and TNF-α; Th17 cells and IL-17 recruit neutrophils.

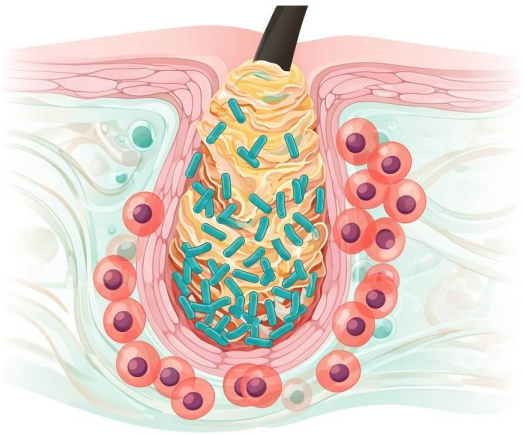

Fig. 3 · Step 3: C. acnes multiplies inside the plug and immune cells gather around the follicle wall.

AI-generated illustration

4 **Inflammation and rupture**

Inflammation turns comedones into **papules and pustules**. If the follicle wall ruptures, keratin, sebum and bacteria spill into the dermis, causing a deeper **nodule** and, when healing is disordered, **scarring** or pigment change.

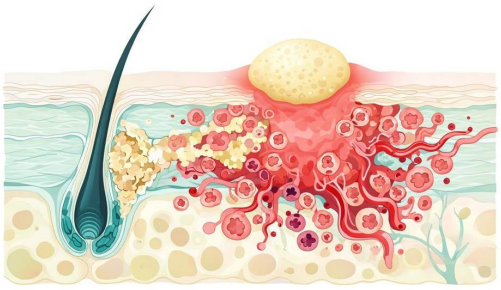

Fig. 4 · Step 4: the follicle ruptures into the dermis, producing a pustule and deeper inflammatory nodule.

AI-generated illustration

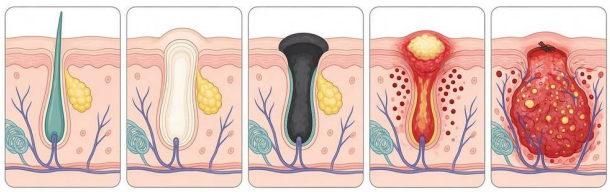

Fig. 5 · One follicle, five stages: normal closed comedone open comedone inflamed papule-pustule deep nodule.

AI-generated illustration

How treatments map onto the mechanism

> **Retinoids** (adapalene, tretinoin; isotretinoin): normalise keratinisation and unplug follicles (step 2). Isotretinoin also shrinks sebaceous glands (step 1).

> **BENZOYL PEROXIDE**
> kills C. acnes without resistance and is mildly comedolytic (steps 2–3).

> **Antibiotics** (topical clindamycin, oral tetracyclines): reduce C. acnes and are anti-inflammatory (steps 3–4).

> **AZELAIC ACID**

antibacterial, comedolytic and reduces pigmentation.

COMBINED PILL AND ANTIANDROGENS

reduce androgen-driven sebum (step 1).

HISTOPATHOLOGY

MICROCOMEDONE / COMEDONE

dilated follicular infundibulum filled with compact keratin and sebum; a narrow (closed) or wide (open) opening.

INFLAMMATORY PAPULE / PUSTULE

perifollicular lymphocytes, then neutrophils; follicular wall thinning.

RUPTURED FOLLICLE / NODULE

dermal abscess with a **foreign-body granulomatous** reaction to keratin and hair.

SCAR

dermal fibrosis (hypertrophic or keloid) or loss of dermal tissue (atrophic ice-pick, rolling, boxcar).

Biopsy is **rarely needed**; histology is mainly used when another diagnosis is suspected.

Clinical features

Symptoms

- Often few physical symptoms; inflamed papules and nodules can be **tender or painful**.
- Oily skin (**seborrhoea**) is common.
- Ask about the **psychological impact**: embarrassment, social withdrawal, low mood, bullying.
- Systemic symptoms (fever, joint pain) are **not** part of ordinary acne: think **acne fulminans**.

Lesions and morphology

Several lesion types usually occur together. Individual lesions last under 2 weeks; deep nodules can persist for months.

Open comedones (blackheads)

Dilated follicles with a dark keratin plug at the surface. Non-inflammatory.

© Raimo Suhonen · **Source**

Closed comedones (whiteheads)

Small skin-coloured or white papules with no visible opening; easiest to see with

© Raimo Suhonen · **Source**

Papules

Small (< 5 mm), tender, red bumps arising from inflamed comedones.

© Te Whatu Ora · **Source**

Pustules

Papules with a visible white or yellow centre of pus.

© DermNet · **Source**

Nodules and pseudocysts

Large, deep, painful red lumps (5 mm) or fluctuant cyst-like swellings. High risk

© DermNet · **Source**

Secondary lesions

Excoriations, red marks (**post-inflammatory erythema**), dark marks

© DermNet · **Source**

Distribution

- **Face** (most common): forehead, cheeks, nose, chin and jawline.
- **TRUNK**
  upper chest, upper back and shoulders, in about half of patients.
- Adult female acne favours the **lower face and jawline**.
- Pomade acne follows the **hairline**; acne mechanica follows the area of friction.

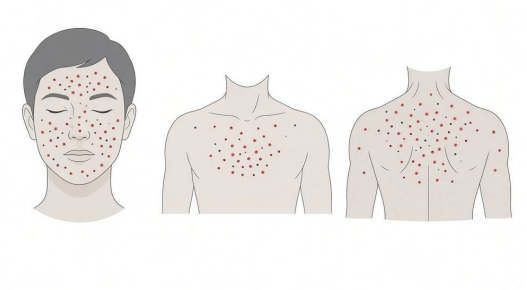

Fig. 12 · Where acne appears: face, central chest, upper back and shoulders. Truncal involvement shapes the choice of treatment.
AI-generated illustration

Variants and subtypes

Comedonal acne

Mainly open and closed comedones with few inflamed lesions. Topical

© DermNet · **Source**

Papulopustular acne

Inflamed papules and pustules with comedones; the commonest pattern.

© DermNet · **Source**

Truncal acne

Papules, pustules and nodules across the back and chest.

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Nodulocystic acne

Deep nodules and cysts with scarring. Refer for isotretinoin.

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Acne conglobata

Interconnecting nodules, abscesses and sinus tracts on the trunk, with severe

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Acne fulminans

Sudden painful **ulcerating, haemorrhagic** nodules with fever, malaise and joint pain,

© DermNet · **Source**

SAPHO-associated acne

Synovitis, acne, pustulosis, hyperostosis, osteitis: severe acne with bone or joint pain.

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Adult female acne

Inflamed papules on the lower face and jawline; ask about the menstrual cycle and PCOS.

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Neonatal and infantile acne

A separate pattern in babies; see the dedicated page.

Open page →

Special groups

- **Skin of colour**
  - Redness is less visible; **post-inflammatory hyperpigmentation** is often the main reason to seek help and can last months.
  - Pomade acne is more common; **keloid scars** are more likely on the jawline, chest and upper back.
  - Treat inflammation early; azelaic acid helps pigmentation.
- **Pregnancy**
  - Topical and oral **retinoids** and **tetracyclines** are contraindicated.
  - **Benzoyl peroxide** with or without clindamycin can be used cautiously.
- **Adult women and PCOS**
  - Look for irregular periods, **hirsutism**, acanthosis nigricans and weight gain.
- **Children**
  - Acne before age 8–9 is unusual: consider an endocrine cause and seek advice.

Fig. 21 · Skin of colour: post-inflammatory hyperpigmentation after truncal acne.

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Fig. 22 · Acne with PCOS: check for hirsutism, irregular periods and acanthosis nigricans.

© DermNet · **Source**

Fig. 23 · Acne in pregnancy: avoid retinoids and tetracyclines.

© DermNet · **Source**

Diagnosis and investigations

Diagnosis is **clinical**, from the lesion mix and distribution. **Comedones** are the key sign.

History

- Duration and course; **previous treatments**, how long each was used and adherence.
- **Contraception and pregnancy plans**; current drugs (steroids, lithium, anabolic steroids, EGFR inhibitors).
- **PCOS FEATURES**
  irregular periods, hirsutism, weight gain.
- Cosmetics, hair products and occlusion; picking.
- **Psychosocial impact**, mood and any self-harm thoughts; scarring.

Examination

- Lesion types and approximate counts; **nodules**; trunk involvement.
- **Scarring** and pigment change.
- Signs of hyperandrogenism (hirsutism, acanthosis nigricans, androgenic alopecia).

Defining severity (NICE NG198)

CATEGORY	DEFINITION
Mild to moderate	Any of: comedones; up to 34 inflammatory lesions; up to 2 nodules
Moderate to severe	35 or more inflammatory lesions, and/or 3 or more nodules

NG198 uses this two-level split to choose a regimen.

Fig. 24 · Mild: scattered papules and comedones.
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Fig. 25 · Moderate: papules and pustules on the chin with early marks.
© DermNet · **Source**

Fig. 26 · Severe: many papulopustules with early scarring.
© Raimo Suhonen · **Source**

- Other grading tools (Leeds, Investigator's Global Assessment, lesion counts) are mainly for specialist documentation and trials.
- Record scarring, pigment change and psychological impact separately: they can justify referral at **any** severity.

Investigations

- **Usually none.**

- **Hormone profile** (e.g. testosterone, SHBG, LH/FSH) if the picture is atypical or suggests hyperandrogenism or PCOS.
- **Skin swab** if an unusual pustular infection is suspected (e.g. Gram-negative folliculitis after long antibiotic use).
- Baseline bloods for isotretinoin are arranged by the specialist team.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Rosacea	Central facial flushing and telangiectasia; papules and pustules but **no comedones**; usually older adults; eye symptoms.
	Perioral (periorificial) dermatitis	Small papules around the mouth, nose and eyes with a **spared rim** next to the lip; often triggered by topical steroid.
	Bacterial folliculitis	Follicular pustules, often on the trunk, buttocks or beard area; **no comedones**; culture may help.
	Malassezia (pityrosporum) folliculitis	Itchy, **monomorphic** papules and pustules on the upper trunk; worse with heat and antibiotics.
	Milia	Tiny white keratin cysts; firm, not inflamed.
	Acneiform drug eruption	Sudden, monomorphic eruption after starting a drug (steroids, EGFR inhibitors); no comedones.
	Hidradenitis suppurativa	Recurrent nodules, abscesses and **tunnels in the flexures** (axillae, groin); double-ended comedones.

Management

Principles

- Follow **NICE NG198**: agree one **12-week** course after shared decision-making (severity, preference, adverse effects, pregnancy plans).
- Treat early to **prevent scarring**; set expectations that improvement takes **6–8 weeks** and full effect up to 12.
- **ANTIMICROBIAL STEWARDSHIP**
  never use an antibiotic alone; always pair it with BPO or a topical retinoid.
- Review at **12 weeks**; then step down to **maintenance**.
- Ask about mood and offer support at every review.

Defining severity

Use the NG198 split above (**mild to moderate** vs **moderate to severe**) to choose a regimen. Scarring, pigment change or distress can justify escalation or referral at any severity.

Conservative and self-care

- Wash **twice daily** with a syndet (non-alkaline) cleanser; avoid harsh scrubbing and over-cleansing.
- Use **non-comedogenic** moisturisers, sunscreen and make-up; remove make-up at night.
- **DO NOT PICK OR SQUEEZE**
  it causes scarring.
- OTC **benzoyl peroxide** (or adapalene where available) while waiting for a prescription.
- No specific diet is recommended (NICE).

Medical treatment: topical

Adapalene + benzoyl peroxide

Fixed-combination topical retinoid + antimicrobial

DOSE	Once daily in the evening, 12 weeks.
USE	First line at **any severity**; also maintenance.
CAUTIONS	Retinoid: **avoid in pregnancy** and when planning pregnancy. Irritation, dryness.
COUNSELLING	Start on alternate nights or as short contact; pea-sized amount to the whole area, not spot-treating. BPO **bleaches** fabric and hair. Use sunscreen.

Tretinoin + clindamycin

Fixed-combination topical retinoid + antibiotic

DOSE	Once daily in the evening, 12 weeks.
USE	First line at any severity.
CAUTIONS	Retinoid: avoid in pregnancy. Contains an antibiotic: **do not combine with an oral antibiotic**.
COUNSELLING	Irritation and photosensitivity: start gradually and use sunscreen.

Benzoyl peroxide + clindamycin

Fixed-combination antimicrobial + antibiotic

DOSE	Once daily, 12 weeks.
USE	**Mild to moderate** acne.
CAUTIONS	Contains an antibiotic: not with oral antibiotics; can be used cautiously in pregnancy.
COUNSELLING	Bleaches fabric; dryness and irritation.

Benzoyl peroxide alone

Antimicrobial, comedolytic

USE	If the patient wants to avoid a retinoid or antibiotic; OTC while awaiting review; maintenance.
CAUTIONS	Irritation; **bleaching** of towels, pillowcases and hair.
COUNSELLING	No bacterial resistance develops, so it is the preferred antibiotic partner.

Azelaic acid 15–20%

Antibacterial, comedolytic, depigmenting

DOSE	Twice daily.
USE	With an oral tetracycline for moderate to severe acne; alone for maintenance; helpful for **post-inflammatory hyperpigmentation**.
COUNSELLING	Mild stinging at first.

Medical treatment: systemic

Lymecycline 408 mg

Oral
tetracycline

DOSE	Once daily; review at 12 weeks.
USE	**Moderate to severe** acne, always **with** adapalene + BPO or azelaic acid.
CAUTIONS	**Contraindicated in pregnancy**, breastfeeding and under 12s. Never with a topical antibiotic.
COUNSELLING	Separate from antacids, iron, zinc and calcium. Photosensitivity: use sunscreen.

Doxycycline 100 mg

Oral
tetracycline

DOSE	Once daily; review at 12 weeks.
USE	As lymecycline.
CAUTIONS	Contraindicated in pregnancy and under 12s. **Photosensitivity** is more marked.
COUNSELLING	Swallow with plenty of water, sitting or standing upright (oesophageal irritation). Separate from antacids, iron and calcium.

Trimethoprim or an oral macrolide

Alternative oral antibiotics

USE

Only when tetracyclines are unsuitable, in the combinations set out in NG198.

CAUTIONS

Stewardship: same rules as tetracyclines.

Oral isotretinoin

Systemic retinoid

SPECIALIST ONLY

USE

Severe, scarring or treatment-resistant acne. Initiated **only by or under a consultant dermatologist-led team**; two independent prescribers must agree in **under-18s**.

CAUTIONS

- Highly teratogenic Pregnancy Prevention Programme for anyone who could become pregnant (highly effective contraception from 1 month before to 1 month after; supervised pregnancy test before, **monthly** during and 1 month after; **30-day** prescription limit, dispensed within **7 days**).
- **No tetracyclines** (benign intracranial hypertension); no vitamin A supplements; a progestogen-only pill alone is not adequate contraception for the PPP.
- Acne may flare in the first weeks; a severe flare can be isotretinoin-triggered acne fulminans.

COUNSELLING

- Mental health report low mood or suicidal thoughts; structured assessment at baseline and every visit.
- Sexual function possible erectile or libido problems, rarely persistent.
- Dry lips and skin (lip balm)

Fig. 27 · Isotretinoin cheilitis: near-universal lip dryness; counsel regular lip balm.
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Fig. 28 · Early flare on isotretinoin: warn patients; a severe flare needs urgent review.
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Hormonal treatment and antiandrogens

- If hormonal contraception is wanted, a **combined oral contraceptive** can be added to topical treatment; it is preferred to a progestogen-only pill, which can worsen acne.
- **Co-cyprindiol** (cyproterone acetate with ethinylestradiol) is an option in selected women (e.g. with PCOS) when other treatments fail. Discuss **VTE risk** and review at about 6 months.
- **Spironolactone** is not among the NG198 options; dermatologists sometimes use it off-label for adult women.
- Prescribing detail: acne prescribing and the combined pill.

Treatment ladder

1 **Everyone**

Skin care, non-comedogenic products, no picking; OTC benzoyl peroxide.

2 **Mild to moderate**

12 weeks of **adapalene + BPO**, **tretinoin + clindamycin**, or **BPO + clindamycin**.

3 **Moderate to severe**

12 weeks of adapalene + BPO (or azelaic acid) **plus lymecycline 408 mg or doxycycline 100 mg** daily.

4 **Maintenance**

Once clear: adapalene + BPO, or adapalene, azelaic acid or BPO alone.
Stop the oral antibiotic.

Specialist

Isotretinoin, under MHRA measures and the Pregnancy Prevention Programme.

UK GUIDANCE

Antimicrobial stewardship

- **Never** use a topical antibiotic alone, an oral antibiotic alone, or topical and oral antibiotics together.
- Review at **12 weeks**. Stop the oral antibiotic once clear and continue the topical.
- If improved but not clear, the oral antibiotic can continue for up to another 12 weeks. Use antibiotic regimens beyond 6 months only in exceptional cases, with 3-monthly review.

Psychological impact

- Acne is linked with **depression, anxiety, low self-esteem** and, in adolescents, increased **suicidal ideation**, regardless of treatment.
- Ask about mood at every visit, not only when isotretinoin is involved.
- Significant distress or a mental health disorder is a reason to **refer at any severity**.
- Signpost support (e.g. the British Skin Foundation, Changing Faces) and involve the GP or mental health services when needed.

Referral (NICE NG198)

SAME DAY	**Acne fulminans**: refer the same day to on-call dermatology.
URGENT	Low mood, self-harm or suicidal thoughts: assess and escalate (mental health), especially around isotretinoin.
ROUTINE	Diagnostic uncertainty; acne conglobata; nodulocystic acne.
ROUTINE	Mild to moderate acne that has failed **two** completed courses; moderate to severe acne that has failed a regimen including an oral antibiotic.

ROUTINE **Scarring**, persistent pigment change, or psychological distress at **any** severity.

Red flags and when to refer

SAME DAY	Sudden ulcerating, haemorrhagic nodules with **fever and joint pain** (acne fulminans), including after starting isotretinoin.
URGENT	Suicidal thoughts or severe low mood.
ROUTINE	Scarring or keloids starting at any severity; acne before age 8–9 or with virilisation (endocrine cause).
ROUTINE	Sudden monomorphic eruption after a new drug: review the drug.

Fig. 29 · Acne fulminans with ulceration and crusting: same-day dermatology.

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Complications and prognosis

Complications

- **Scarring**
  - Atrophic: **ice-pick**, rolling and boxcar scars.
  - Hypertrophic and **keloid** scars, especially on the jawline, chest and shoulders in skin of colour.
- **Post-inflammatory erythema** (red marks, fair skin) and **hyperpigmentation** (dark marks, skin of colour).
- PSYCHOLOGICAL

depression, anxiety, social withdrawal.

- Treatment-related: antibiotic resistance, irritant dermatitis, isotretinoin adverse effects.

Fig. 30 · Ice-pick atrophic scars: scarring is a referral trigger at any severity.
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Fig. 31 · Post-inflammatory hyperpigmentation in skin of colour.
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Fig. 32 · Keloid scars on the shoulders after acne.
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Scar treatment (and the timing of procedures around isotretinoin) is covered on acne scars and keloids.

Prognosis

- Most adolescent acne improves by the **early to mid-20s**, but it can persist or start in adulthood.
- Relapse after stopping treatment is common: **maintenance** treatment reduces it.
- Early effective treatment is the best way to prevent permanent scarring.

Prescriber pearls (UK pharmacist IP)

Start topicals on **alternate days** or as short-contact applications, then build up.

Apply topicals to the **whole affected area**, not just individual spots.

BPO **bleaches** fabric and hair: warn about towels and pillowcases.

Retinoids and tetracyclines cause **photosensitivity**: give sun advice.

Pregnancy: no topical or oral retinoids, no tetracyclines; BPO ± clindamycin can be used cautiously.

If hormonal contraception is wanted, a **combined pill** is preferred over a progestogen-only pill.

Combination rules: always pair an oral antibiotic with BPO and/or a topical retinoid; review at 12 weeks.

When referring for isotretinoin, document mental health history thoroughly.

In skin of colour, treat early and consider azelaic acid for pigmentation.

Depression is common with acne itself: screen regardless of the drug.

Related: retinoids, acne mimics, acne prescribing and the combined pill.

Self-check quiz

8 questions · optional

Q1 **What is the hallmark lesion that separates acne from rosacea?**

✓ Comedones (blackheads and whiteheads).

Q2 **Name the four pathogenic factors in acne.**

✓ Androgen-driven sebum, follicular hyperkeratinisation, C. acnes proliferation and inflammation.

Q3 **How does NICE NG198 define moderate to severe acne?**

✓ 35 or more inflammatory lesions and/or 3 or more nodules.

Q4 **Which first-line option suits any severity?**

✓ Fixed adapalene + BPO, or fixed tretinoin + clindamycin, once daily in the evening for 12 weeks.

Q5 **Name three antibiotic combinations to avoid.**

✓ Topical antibiotic alone; oral antibiotic alone; topical and oral antibiotics together.

Q6 Why must tetracyclines not be given with isotretinoin?

✓ Risk of benign intracranial hypertension.

Q7 Which acne patient needs same-day referral?

✓ Acne fulminans: systemically unwell with ulcerating nodules.

Q8 Is a specific acne diet recommended?

✓ No: NICE finds insufficient evidence for any specific diet.

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Photo gallery

Comedonal acne: the hallmark 3

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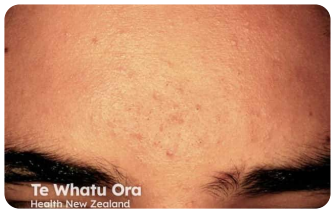
Grading strip: mild moderate severe

5



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Truncal acne

3

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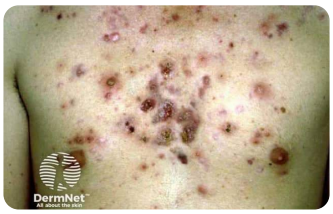

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Post-inflammatory hyperpigmentation

1

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Module 6 Week 2: red flag, acne fulminans

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Module 6 Week 2: isotretinoin side effects

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Module 6 Week 2: adult female acne, PCOS and pregnancy

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Module 6 Week 2: scarring and keloids

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AI teaching illustration

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Acne & follicles Module 6 · Week 2

Acne mimics: rosacea, steroid eruptions and folliculitis

Rosacea (no comedones), steroid-induced eruptions, perioral dermatitis, and Gram-negative, Malassezia and Pseudomonas folliculitis.

ALSO CALLED rosacea · rhinophyma · ocular rosacea · pyoderma faciale · rosacea fulminans · perioral dermatitis · +4

RELATED PAGES Acne vulgaris · Rosacea (incl. Demodex & steroid rosacea) · Perioral (periorificial) dermatitis · Pityriasis versicolor · Folliculitis

KEY FACTS

- **No comedones?** Think rosacea, perioral dermatitis, steroid acne or folliculitis rather than acne vulgaris.
- Rosacea age 30–60, flushing, central face; avoid topical steroids. **Rosacea fulminans**: explosive facial nodules in young women.
- Steroid acne sudden monomorphic papules/pustules on chest and back after oral, inhaled, topical or bleaching steroids.
- **Gram-negative folliculitis** after long-term tetracyclines: stop the tetracycline, swab, refer.
- Malassezia folliculitis itchy uniform papules on upper trunk; **hot-tub folliculitis**: Pseudomonas, settles in days.

Overview

This is a **differential page**: conditions that look like acne but need different treatment. Rosacea and perioral dermatitis have their own pages.

Not every facial papule is acne. This page covers the common **mimics** taught with Module 6 Week 2: **rosacea** (and rosacea fulminans), **steroid-induced** acne/rosacea/perioral dermatitis, and **folliculitis variants** (Gram-negative, Malassezia, Pseudomonas/hot-tub).

True acne vulgaris is covered separately: acne vulgaris.

Aetiology: causes and risk factors

ROSACEA

neurovascular dysregulation, innate immunity (cathelicidin), Demodex, UV; triggers include heat, alcohol, spicy food, hot drinks, topical steroids.

STEROID ACNE / ROSACEA / PERIORAL DERMATITIS

oral, inhaled, topical or illicit bleaching steroids.

GRAM-NEGATIVE FOLLICULITIS

long-term oral tetracyclines for acne/rosacea overgrowth of Klebsiella, Enterobacter, Proteus.

MALASSEZIA FOLLICULITIS

yeast in oily follicles, often after antibiotics or steroids.

HOT-TUB FOLLICULITIS

Pseudomonas in under-chlorinated spa water.

Myths to address

- **"More antibiotics will fix stubborn 'acne'"**: Gram-negative and Malassezia folliculitis are caused or worsened by antibiotics.

Pathophysiology and pathology

Each mimic has a **different driver** in or around the follicle.

1. **Acne**

   Comedone formation, sebum and C. acnes inflammation.

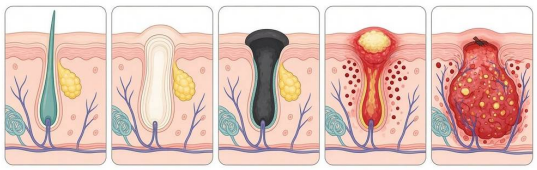

Fig. 1 · Acne starts with a comedone; mimics usually don't.

AI-generated illustration

2. **Rosacea**

   Neurovascular dysregulation, cathelicidin, Demodex and UV.

Fig. 2 · Rosacea favours the convex central face.

AI-generated illustration

3 **Steroids**

Steroids alter follicular keratinisation and local immunity, giving monomorphic lesions.

Fig. 3 · Perioral dermatitis spares a rim around the lips.

AI-generated illustration

4 **Microbial overgrowth**

Antibiotics allow Gram-negative bacteria or Malassezia yeast to fill follicles; hot tubs deliver Pseudomonas.

Clinical features

FEATURE	ACNE VULGARIS	ROSACEA
Age	Teen / young adult	30–60
Comedones	Present (hallmark)	Absent
Flushing / background erythema	Not typical	Typical, triggered
Sites	Face, chest, back	Central convex face; trunk spared
Eyes / phyma	No	Ocular rosacea; rhinophyma (mainly men)

ROSACEA FULMINANS (PYODERMA FACIALE)

explosive painful papules, pustules and nodules on a red swollen central face in young women; systemically well (unlike acne fulminans).

STEROID ACNE

sudden monomorphic papules/pustules on chest/back; few comedones.

PERIORAL DERMATITIS

papules around mouth with vermilion sparing; often after facial steroid.

GRAM-NEGATIVE FOLLICULITIS

new perinasal pustules after months of tetracycline that initially worked.

MALASSEZIA

itchy monomorphic follicular pustules on chest/back.

HOT-TUB

itchy pustules under the swimsuit line 1–4 days after spa use.

Mimics to recognise

Papulopustular rosacea

Papules and pustules on a red background; no comedones. Site: Cheeks.

© DermNet · **Source**

Rosacea fulminans

Explosive painful nodules on a swollen face; young women. Site: Central face.

© DermNet · **Source**

Steroid eruption

Monomorphic papulopustules after steroid exposure. Site: Face, chest.

© DermNet · **Source**

Perioral dermatitis

Perioral papules, clear rim at the lip border. Site: Mouth, nose.

© Dr Richard Ashton · **Source**

Gram-negative

Sudden pustular flare round the nose on long antibiotics. Site: Perinasal.

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Malassezia

Itchy, uniform small follicular pustules. Site: Chest, back.

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Hot tub

Follicular pustules after spa use. Site: Swimsuit area.

© DermNet · **Source**

Diagnosis and investigations

Clinical. Swab pustules and anterior nares (request Gram-negative culture) when acne worsens on long antibiotics. Consider ANA only if lupus is realistic. Do not diagnose rosacea if comedones dominate — rethink acne.

Clinical

- Comedones dominate it is **acne**.
- Ask about steroids, inhalers and antibiotic duration.

Swab

- Worse on long antibiotics: swab pustules and **anterior nares**; request Gram-negative culture.

Bloods

- ANA only if **lupus** is realistic.

Exposure

- Spa or hot tub in the last few days?

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Acne vulgaris	Comedones; polymorphic lesions; trunk.
	Seborrhoeic dermatitis	Greasy scale in folds.
	Lupus	Photosensitive plaques; serology.
	Acne fulminans	Systemically unwell young man; ulcerating trunk acne.

Management

PRESCRIBER TIP

Rosacea in depth

Phenotypes, flushing work-up, ocular rosacea, Demodex and steroid-induced rosacea now have their own page: rosacea.

Rosacea (NICE CKS / PCDS themes)

- Trigger diary, gentle skincare, daily sunscreen; **avoid topical steroids**.
- Persistent erythema: topical brimonidine (warn about rebound).
- Mild–moderate papulopustular: **ivermectin 1% cream** first-line, or azelaic acid 15%, or metronidazole 0.75%.
- Moderate–severe: topical ivermectin **plus** oral **doxycycline 40 mg MR** (licensed) or other tetracycline; erythromycin if tetracyclines unsuitable.
- Ocular: lid hygiene, lubricants ± oral tetracycline; **same-day ophthalmology** for pain, photophobia or reduced vision.

Steroid / perioral dermatitis

- Stop the topical/inhaled facial steroid exposure (taper systemic steroids with the original prescriber — never abruptly).
- Warn of a rebound flare. Topical metronidazole or azelaic acid; oral tetracycline 6–8 weeks if needed (erythromycin if under 12).

Perioral dermatitis now has its own page: Perioral dermatitis.

Folliculitis variants

- Gram-negative: **stop the tetracycline**; swab; specialist options include isotretinoin or culture-guided antibiotics — refer rather than endless antibiotic switches.
- Malassezia: topical or oral antifungal per local practice; stop unnecessary antibiotics/steroids.
- Hot-tub: usually self-limiting in days; antibiotics only if severe or immunocompromised.

Rosacea fulminans

- **Urgent dermatology referral.** Do not keep cycling oral antibiotics. Specialist care often uses a short steroid course then isotretinoin (pregnancy exclusions apply).

UK GUIDANCE

Check BNF

Confirm doses, pregnancy advice and interactions before prescribing.

Treatment ladder

Rosacea basics

Trigger diary, gentle skincare, daily sunscreen; **avoid topical steroids**.
Persistent erythema: brimonidine (warn about rebound).

2

Rosacea papules

Ivermectin 1% cream first line, or azelaic acid 15%, or metronidazole 0.75%. Moderate–severe: add **doxycycline 40 mg MR** (licensed); erythromycin if tetracyclines unsuitable.

3

Steroid / perioral

Stop the facial steroid; taper systemic steroids with the prescriber, never abruptly. Warn of rebound. Topical metronidazole or azelaic acid; oral tetracycline 6–8 weeks if needed (erythromycin if under 12).

4

Folliculitis variants

Gram-negative: stop the tetracycline, swab, refer (isotretinoin or culture-guided antibiotics). **Malassezia**: topical or oral antifungal; stop unnecessary antibiotics / steroids.

5

Rosacea fulminans

Urgent dermatology. Don't keep cycling oral antibiotics.

Referral

SAME DAY	Ocular pain, photophobia or reduced vision ophthalmology.
URGENT	**Rosacea fulminans** or suspected acne fulminans dermatology.
ROUTINE	Phyma needing procedures; diagnostic doubt.
ROUTINE	Acne worse on long tetracyclines: culture for Gram-negatives and **stop the antibiotic**.

Red flags and when to refer

- Ocular pain, photophobia or reduced vision same-day ophthalmology.
- Rosacea fulminans or suspected acne fulminans urgent dermatology.
- Phyma needing procedural treatment; diagnostic doubt.

- Acne worsening on long tetracyclines culture for Gram-negatives; stop the antibiotic.

Complications and prognosis

Complications

- Scarring (rosacea fulminans), ocular rosacea, prolonged mistreatment.

Prognosis

- Good once the right diagnosis is made.

Prescriber pearls (UK pharmacist IP)

- **No comedones + flushing + central face = rosacea until proven otherwise.**
- Maintenance topical therapy is often needed — rosacea is relapsing.
- Pharmacists often spot inhaled-steroid perioral dermatitis — advise rinsing/wiping the face after inhalers.
- Laser/IPL for telangiectasia is usually private, not NHS-funded.
- For the full rosacea ladder by phenotype, see rosacea.

Self-check quiz

5 questions · optional

Q1 Which single sign separates acne from its mimics?

✓ Comedones: present in acne, absent in rosacea and the folliculitis mimics.

Q2 New perinasal pustules after months of a tetracycline that first worked?

✓ Gram-negative folliculitis: swab (incl. nares), stop the tetracycline, refer.

Q3 Papules around the mouth with a clear rim at the lip border after a steroid cream?

✓ Perioral (periorificial) dermatitis: stop the steroid, warn of rebound.

Q4 Explosive painful facial nodules in a young woman who is systemically well?

✓ Rosacea fulminans (pyoderma faciale): urgent dermatology.

Q5 Itchy follicular pustules under the swimsuit 2 days after a spa?

✓ Hot-tub (Pseudomonas) folliculitis: usually self-limiting.

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Photo gallery

Rosacea and pyoderma faciale 6

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Steroid-induced eruptions 3

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Folliculitis mimics 6

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Acne & follicles Module 6 · Week 2 · Module 1 · Week 5

Acanthosis nigricans

Velvety brown thickening of the neck and axillae — usually a marker of insulin resistance; rarely a paraneoplastic red flag.

ALSO CALLED acanthosis nigricans · insulin resistance · PCOS · velvety skin · malignant acanthosis · tripe palms

RELATED PAGES Hirsutism & the Ferriman–Gallwey score · Skin signs of lung cancer & internal malignancy · Intertrigo & the flexural differential · Cushing's syndrome: skin signs

KEY FACTS

- **Acanthosis nigricans** is velvety brown-black thickening of the skin in the **neck, axillae and groin**.
- Commonest cause: **insulin resistance** (obesity, type 2 diabetes, PCOS); also drugs, endocrine disease, inherited forms.
- Sudden extensive AN with mucosal or palm involvement ('tripe palms') or weight loss: **malignant AN** (gastric adenocarcinoma).
- Check HbA1c or glucose and lipids; treat the cause (weight loss improves it).
- Topical retinoids or keratolytics for appearance.

Overview

Acanthosis nigricans (AN) is symmetrical **velvety grey-brown to black thickening** of flexural skin. In adolescents and young adults it almost always signals **insulin resistance**. Rarely, abrupt extensive AN is **paraneoplastic** (malignant AN).

Acanthosis nigricans, PCOS and insulin resistance in depth

Acanthosis nigricans (AN): symmetrical, **velvety, hyperpigmented, thickened plaques** in the **flexures**: back and sides of the neck, axillae, groin, inframammary skin, and over the knuckles/elbows. It is very common with obesity (especially in skin of colour) and is a marker of **insulin resistance**. Rarely it is a **paraneoplastic** sign.

Epidemiology

- **Who**
  - Common in obese adolescents and adults; more visible in darker skin.

Aetiology: causes and risk factors

- Obesity / insulin resistance (commonest): high insulin acts on IGF-1 receptors papillomatosis.
- Endocrine: PCOS, Cushing's, acromegaly, hypothyroidism; drug-induced (systemic steroids, nicotinic acid, some contraceptives, protease inhibitors).

- Malignant AN: tumour growth factors (notably TGF-α) — strongly linked to **gastric adenocarcinoma** and other abdominal adenocarcinomas.

Acanthosis nigricans, PCOS and insulin resistance in depth

Myths to address

- **"IT'S DIRT"**
  it does not wash off and signals insulin resistance.

Pathophysiology and pathology

High insulin acts like a **growth factor** on keratinocytes and fibroblasts.

1. **Hyperinsulinaemia**
   Insulin resistance raises circulating insulin.
2. **IGF-1 receptor stimulation**
   At high levels insulin activates IGF-1 receptors on keratinocytes.
3. **Papillomatosis**
   The epidermis thickens into velvety folds; pigment is mostly from the thick keratin.
4. **Malignant AN**
   Tumour growth factors (TGF-α) cause sudden, extensive disease.

How it is caused

> **ENDOCRINE/BENIGN AN**
> **Hyperinsulinaemia** — at high concentrations insulin binds **IGF-1 receptors** on keratinocytes and dermal fibroblasts proliferation; Also hyperandrogenism (ovarian), raised free IGF-1, and FGFR3 mutations (syndromic)

> **MALIGNANT AN**
> tumour-secreted **TGF-α** (also EGF, IGF-1) activates **EGFR** on keratinocytes. Usually **gastric adenocarcinoma** (~60%, intra-abdominal adenocarcinomas overall).

> **Why some people and not others?** It depends on the degree of hyperinsulinaemia, receptor expression/sensitivity and genetics (ethnicity, skin type), friction and maceration in flexures, and — in malignancy — the tumour's growth-factor secretion.

> **HISTOLOGY**
> **papillomatosis and hyperkeratosis** with only mild, irregular acanthosis and little real increase in melanin. The dark colour comes mostly from the thickened keratin. No inflammation.

> **OTHER ASSOCIATIONS**
> **obesity, T2DM, PCOS** — HAIR-AN syndrome (hyperandrogenism, insulin resistance, AN); Cushing's, acromegaly, hypothyroidism; **drugs:** nicotinic acid, systemic glucocorticoids, insulin injection sites, oestrogens/COC, protease inhibitors, testosterone, fusidic acid (topical); genetic/syndromic forms; acral AN in skin of colour (often isolated, benign)

HISTOPATHOLOGY

- Papillomatosis and hyperkeratosis with mild acanthosis; little increase in melanin.

Clinical features

- Neck (back/sides), axillae, groin, under breasts, knuckles; may be faint on the face.
- Skin tags often coexist; usually asymptomatic or mildly itchy.
- **MALIGNANT AN RED FLAGS**
  sudden rapid onset in an older, often non-obese adult; mucosal involvement; **tripe palms**; sudden multiple seborrhoeic keratoses (Leser–Trélat); weight loss or GI symptoms.

Acanthosis nigricans, PCOS and insulin resistance in depth

Clinical features & sites (describe the rash)

Ill-defined, symmetrical, **brown-grey velvety thickening with accentuated skin markings**, ± skin tags (acrochordons), on the **back of the neck**; usually asymptomatic or mildly itchy.

Malignant AN features: sudden onset and rapid spread; extensive, including **mucosae** (lips, oral cavity) and **palms (tripe palms –**; pruritus; weight loss; older and non-obese patients; ± Leser–Trélat sign.

Fig. 1 · Acanthosis nigricans on the neck: velvety grey-brown thickening with accentuated skin markings — commonest association is obesity/ insulin resistance; not "dirt".

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Special groups

- **Malignant acanthosis nigricans**
  - Rapid onset in older non-obese adults, mucosal involvement, tripe palms: urgent investigation for cancer.

Diagnosis and investigations

- Clinical. Biopsy only if atypical.

- Metabolic screen: BMI/waist, BP, HbA1c (or fasting glucose), lipids, LFTs; PCOS screen in females; TSH if indicated.
- Suspected malignant AN urgent GP-led suspected-cancer referral (e.g. upper GI pathway per NICE NG12).

Acanthosis nigricans, PCOS and insulin resistance in depth

(likely diagnosis)

> **CASE**
> a 22-year-old with **oligomenorrhoea for 12 months, weight gain and jawline hirsutism** that emerged after stopping a COC 18 months ago, **AN on the back of the neck**, **testosterone 2.3 nmol/L (mildly raised)**, normal prolactin/TFT/estradiol, fasting glucose 5.5 (upper limit) **PCOS with insulin resistance (HAIR-AN phenotype)**. Rotterdam criteria (2 of 3): oligo-/anovulation , clinical/biochemical hyperandrogenism , polycystic ovaries (not needed here). The COC had been masking the symptoms.

> **FURTHER TESTS (INTERNATIONAL PCOS GUIDELINE 2023 / CKS)**
> **SHBG/free androgen index**, LH/FSH, **17-OH-progesterone** (exclude NC-CAH), consider **DHEAS** (adrenal tumour if very high); pregnancy test; **75 g OGTT** (preferred) or HbA1c/fasting glucose for dysglycaemia, repeated every 1–3 years; **lipids**; BP; screen for **depression/anxiety**, **eating disorders** and **OSA**; exclude Cushing's if features are present; Pelvic USS is not needed when two criteria are already met

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Terra firma-forme dermatosis	Wipes away with 70% isopropyl alcohol
	Confluent and reticulated papillomatosis	Trunk reticulate scale; responds to tetracyclines
	Friction / PIH	No velvety thickening
	Erythrasma	Coral-red under Wood's lamp
	Addison's disease	Generalised pigment including mucosa; unwell

Acanthosis nigricans, PCOS and insulin resistance in depth

Confluent and reticulated papillomatosis (Gougerot–Carteaud — responds to minocycline/doxycycline), terra firma-forme dermatosis ("dirt" that wipes off with alcohol), erythrasma (coral-red Wood's light), tinea versicolor, post-inflammatory hyperpigmentation, Dowling–Degos disease, pellagra (Casal's necklace), Addison's (generalised), lichen simplex.

Management

- **TREAT THE CAUSE**
  sustained weight management and activity; family-based approach in adolescents; metformin only via GP/ specialist if insulin resistance/prediabetes.
- Review causative drugs with the original prescriber.
- Cosmetic adjuncts (modest evidence): urea/lactic acid keratolytics, topical retinoid or calcipotriol off-label; specialist peels/lasers.
- Colour improves slowly over months after metabolic change — scrubbing worsens PIH.

Acanthosis nigricans, PCOS and insulin resistance in depth

> **TREAT THE CAUSE**
> **Weight loss** of 5–10% improves AN, cycles and androgens; **Metformin** (off-label in PCOS, BNF) improves insulin resistance and cycle regularity, with modest AN benefit; GLP-1 RAs where eligible (NICE weight-management TAs); Stop culprit drugs

> **PCOS PLAN (CKS/INTERNATIONAL GUIDELINE 2023)**
> **COC** — cycle control, **endometrial protection** (with fewer than 4 periods a year, give cyclical progestogen or a COC to induce withdrawal bleeds and reduce endometrial hyperplasia risk), and hirsutism/acne benefit. Choose by UKMEC (BMI 35 UKMEC 3); Cosmetic hair removal; anti-androgens via a specialist with contraception (the therapeutics notes see Acanthosis nigricans and PCOS); Annual metabolic and mood checks; Fertility advice when needed (letrozole is first-line for ovulation induction — specialist)

> **SKIN (COSMETIC)**
> keratolytics (**urea 10–20%, lactic acid, salicylic acid**), **topical retinoids** (tretinoin — off-label), calcipotriol (off-label), laser (specialist/private). Gentle skin care; do not scrub.

> **COUNSEL**
> "It's not dirt and it's not caused by poor hygiene." AN is a **warning sign for diabetes** — test regularly.

IP scope

Diagnosis support, bloods/OGTT, lifestyle/metformin/COC within competence, topical keratolytics, and referral for red flags or fertility/specialist anti-androgen therapy.

Referral

URGENT — Sudden extensive AN ± tripe palms / oral pigment in an older adult urgent cancer pathway.

ROUTINE — Raised HbA1c or other metabolic comorbidity in a young person local obesity/endocrine pathways.

Red flags and when to refer

- Sudden extensive AN ± tripe palms / oral pigment in an older adult urgent cancer pathway.
- Raised HbA1c or other metabolic comorbidity in a young person local obesity/endocrine pathways.

Red flags / refer

Rapid onset, extensive or mucosal AN, tripe palms, weight loss, age >40, non-obese **urgent investigation for malignancy (NICE NG12 — upper GI: urgent OGD if appropriate)**. Hyperandrogenism with testosterone >5 nmol/L or virilisation urgent endocrinology. Hyperglycaemia.

Complications and prognosis

Complications

- Marker of diabetes and cardiovascular risk.

Prognosis

- Improves with weight loss and insulin sensitisation; malignant AN tracks the cancer.

Prescriber pearls (UK pharmacist IP)

Velvet, not dirt — the alcohol-swab test separates terra firma-forme dermatosis.

In a young person, AN is a cardiometabolic risk marker; lifestyle advice for the whole household matters most.

Use weight-stigma-sensitive language.

Often seen alongside adult female acne / PCOS — see acne vulgaris.

Sudden, widespread or mucosal acanthosis nigricans, or tripe palms, in an older adult: see skin signs of lung cancer and internal malignancy.

Acanthosis nigricans, PCOS and insulin resistance in depth

- **"Velvety neck = check glucose (and think PCOS in young women)."**
- **"Sudden, florid AN in an older thin person = look for a gastric cancer."**

Self-check quiz

5 questions · optional

Q1 When should you think of Acanthosis nigricans?

✓ Symmetrical velvety grey-brown thickening of the neck or axillae, often with skin tags, in someone with overweight or PCOS.

Q2 What drives it?

✓ High insulin / growth factors drive keratinocyte and fibroblast proliferation papillomatosis and hyperkeratosis.

Q3 What is the backbone of care?

✓ Treat the metabolic driver (weight, activity, screen HbA1c/lipids); keratolytics and retinoids are cosmetic adjuncts only.

Q4 What must you never miss?

✓ Sudden extensive AN in a slim older adult, especially with tripe palms or oral involvement urgent suspected-cancer pathway.

Q5 Name a red flag that needs escalation.

✓ Sudden extensive AN ± tripe palms / oral pigment in an older adult urgent cancer pathway.

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Photo gallery

Acanthosis nigricans

3

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Acanthosis nigricans in insulin resistance

1

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Acne & follicles Module 6 · Week 2

Perioral (periorificial) dermatitis

Small red papules and pustules around the mouth, nose or eyes with a clear rim next to the lips, often provoked by topical or inhaled steroids and heavy creams. Stop the steroid, expect a rebound, and treat with topical metronidazole or azelaic acid, or an oral tetracycline.

ALSO CALLED perioral dermatitis · periorificial dermatitis · periocular dermatitis · steroid-induced perioral dermatitis

RELATED PAGES Rosacea (incl. Demodex & steroid rosacea) · Acne mimics: rosacea, steroid eruptions and folliculitis · Acne vulgaris · Topical corticosteroids: potency, quantities & safety · Contact dermatitis

KEY FACTS

- **Perioral dermatitis** is a facial eruption of tiny papules and pustules around the mouth, nose or eyes with **sparing next to the lips**.
- Commonest in women aged 20–45 and children; often provoked by **topical, inhaled or nasal steroids** and heavy creams.
- No comedones (unlike acne); no flushing (unlike rosacea).
- Stop steroids and cosmetics, expect a rebound; topical metronidazole or azelaic acid; oral tetracycline for 6–8 weeks if needed.
- Children under 12: erythromycin rather than tetracycline.

Overview

Perioral dermatitis (periorificial dermatitis) is a common facial eruption of small papules and pustules around the mouth, nose and sometimes eyes, with a characteristic **clear zone next to the lip vermilion**. It mainly affects women aged 20–45 and children.

Epidemiology

- **Who**
  - Women 20–45 most; prepubertal children (often periocular, steroid-related).

Aetiology: causes and risk factors

- **Topical corticosteroids** on the face are the commonest trigger; also inhaled or nasal steroids, and heavy moisturisers or foundations.
- Fluorinated toothpaste and barrier disruption may contribute.

Risk factors and associations

- Steroids disrupt the barrier and the follicle; occlusive cosmetics; possibly fluorinated toothpaste and microbiome changes.

Myths to address

- **"HYDROCORTISONE WILL SETTLE IT"**
  it masks it briefly and then makes it worse.

Pathophysiology and pathology

Steroids and occlusion damage the **barrier around facial follicles**, triggering inflammation.

1. **Barrier disruption**
   Steroids and heavy creams thin the barrier and alter the follicular microbiome.
2. **Follicular inflammation**
   Innate immune activation causes small papules and pustules around follicles.
3. **Steroid dependence**
   Steroids suppress the rash; stopping causes rebound, so patients reapply.

Fig. 1 · Perioral dermatitis: papules around the mouth and nose, with a clear rim next to the lips.
AI-generated illustration

4. **Resolution**
   Stopping the trigger and anti-inflammatory treatment restores the barrier over weeks.

HISTOPATHOLOGY

- Rarely needed: perifollicular lymphohistiocytic infiltrate; granulomatous in the granulomatous variant.

Clinical features

- Monomorphic 1–2 mm red papules, papulovesicles and pustules on a red, sometimes scaly background.

- Around the mouth, nasolabial folds, chin; may involve the sides of the nose and eyelids (**periocular**).
- **VERMILION SPARING**
  a narrow clear rim around the lips.
- Burning or tightness more than itch; no comedones.

Fig. 2 · Perioral papules sparing the skin immediately next to the lips.
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Fig. 3 · Florid perioral dermatitis after potent topical steroid use on the face.
© Dr Richard Ashton · **Source**

Special groups

- **Granulomatous periorificial dermatitis**
  - Yellow-brown papules in children of African descent; slower to clear; refer if persistent.

Diagnosis and investigations

Clinical. Ask specifically about steroid creams, inhalers, nasal sprays and cosmetics.

Differential diagnoses

LOOK-ALIKE	CLUES
Rosacea	Older; flushing; central face; no clear perioral rim
Acne vulgaris	Comedones; polymorphic lesions
Seborrhoeic dermatitis	Greasy scale in nasolabial folds and brows
Contact dermatitis	Itch, eczematous, linked to a product

Management

- **Stop the facial steroid** (taper systemic or high-potency use with the prescriber; never stop inhaled asthma steroids: rinse and wipe the face after use).
- Warn of a **rebound flare** for 1–2 weeks after stopping.
- 'Zero therapy': stop all creams and cosmetics; use a gentle cleanser only.
- Topical **metronidazole** or **azelaic acid**.
- Oral tetracycline for 6–8 weeks if needed (erythromycin if under 12).

Treatment ladder

1

Stop triggers

Stop facial steroids and heavy creams; gentle cleanser only.

2

Topical

Topical metronidazole or azelaic acid.

3

Oral

Oral tetracycline 6–8 weeks if needed (erythromycin if under 12).

Referral

ROUTINE — Not responding after 8–12 weeks, granulomatous or periocular disease, diagnostic doubt.

Red flags and when to refer

- Granulomatous or periocular disease not responding, or diagnostic doubt: dermatology.

Complications and prognosis

Complications

- Steroid dependence, rebound flares, distress.

Prognosis

- Clears over 6–12 weeks with treatment; recurs if steroids are restarted.

Prescriber pearls (UK pharmacist IP)

Pharmacists often spot inhaled-steroid perioral dermatitis: advise rinsing and wiping the face after inhalers.

Hydrocortisone makes it look better briefly, then worse.

Self-check quiz

4 questions · optional

Q1 Commonest trigger of perioral dermatitis?

✓ Topical (or inhaled/nasal) corticosteroids on the face.

Q2 What is the classic sign around the lips?

✓ A clear rim next to the vermilion (vermilion sparing).

Q3 What happens after stopping the steroid?

✓ A rebound flare for 1–2 weeks.

Q4 Oral option in under-12s?

✓ Erythromycin (tetracyclines avoided).

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Photo gallery

Clinical photographs 2

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Steroid-induced perioral dermatitis

1

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Acne & follicles Module 3 · Week 1

Rosacea (incl. Demodex & steroid rosacea)

Chronic central facial redness, flushing and papulopustules without comedones. Treat by phenotype, control triggers, look for eye involvement, and never put a potent steroid on it.

ALSO CALLED rosacea | flushing | erythematotelangiectatic | papulopustular rosacea | rhinophyma | ocular rosacea | +5

RELATED PAGES Acne mimics: rosacea, steroid eruptions and folliculitis | Acne vulgaris | Topical corticosteroids: potency, quantities & safety | Discoid lupus erythematosus

KEY FACTS

- Chronic inflammatory facial disease with **flushing, persistent central redness, papules and pustules** and telangiectasia; **no comedones**.
- Affects about **5%** of adults, usually after 30; men and women about equally (phymas mainly men).
- Driven by neurovascular hyper-reactivity, innate immune activation (cathelicidin/LL-37) and Demodex.
- UK: trigger avoidance and sun protection; **ivermectin 1%**, azelaic acid or metronidazole for papules; **brimonidine** for redness; oral doxycycline for moderate–severe.
- **Ocular rosacea** with pain or reduced vision needs urgent eye review.

Overview

NOTE

Core rosacea is in the acne notes.

This section goes further.

ROSCO 2017/2019 and the NRS 2017 criteria replaced the four rigid "subtypes" with **phenotypes**, because patients often overlap (e.g. ETR + PPR + ocular).

DIAGNOSTIC (ONE IS ENOUGH)

persistent centrofacial erythema that can flare periodically, or **phymatous change**.

MAJOR (TWO ARE NEEDED IF NO DIAGNOSTIC FEATURE)

flushing/transient erythema, inflammatory papules/pustules, telangiectasia, ocular manifestations.

MINOR/SECONDARY

burning, stinging, oedema, dryness.

The older “four subtypes” teaching maps onto these phenotypes. **Teach phenotypes, and treat each feature separately.**

Demodex and demodicosis

Demodex folliculorum (in the follicle infundibulum) and D. brevis (in sebaceous glands) are normal commensal mites. **Demodicosis** is a skin disease caused by too many of them. It can be **primary** (rosacea-like, with fine scale) or **secondary** (to immunosuppression, TCS/TCI use, HIV or chemotherapy).

Steroid-induced rosacea

> **NOTE**
>
> Core is in the acne notes (steroid and drug-induced acneiform eruptions; photo K08). Withdrawal practicalities are added here.

A rosacea-like eruption (erythema, telangiectasia, papulopustules, burning) caused by **prolonged facial topical (or inhaled/nasal) corticosteroid** use. It overlaps with periorificial dermatitis and with "topical steroid withdrawal" (red skin syndrome; **MHRA Drug Safety Update, Sept 2021**).

Epidemiology

- **Prevalence**
  - About **5%** of adults worldwide; under-recognised in skin of colour.
- **Age and sex**
  - Usually presents **after 30** and becomes commoner with age; roughly equal sex distribution; **rhinophyma** is mainly in men.
- **Skin type**
  - Most common in fair skin (Celtic and Northern European descent).
- **Associations**
  - Depression, anxiety, migraine, cardiovascular disease, inflammatory bowel disease and H. pylori have been linked.

Aetiology: causes and risk factors

> **HEAT, HOT DRINKS AND SPICY FOOD (CAPSAICIN)**
>
> activate **TRPV1** on sensory nerves and vessels, so neuropeptides are released (substance P, CGRP) and vessels dilate. Pepper and chilli are classic triggers.

> **ALCOHOL, ESPECIALLY RED WINE**
>
> acetaldehyde, histamine and polyphenols cause vasodilation. Red wine is a classic trigger.

> **UV**
>
> induces MMPs, ROS and **cathelicidin** activation through **kallikrein-5**, and damages dermal matrix so telangiectasia shows. This is the case's "worse with sun".

> **COLD, WIND, EXERCISE, STRESS, MENOPAUSE**
>
> vascular reactivity through neurogenic routes.

> **TOPICAL STEROIDS AND CALCINEURIN INHIBITORS**
> local immunosuppression lets Demodex overgrow, steroids cause vasodilation, and there is rebound on withdrawal.

> **DRUGS THAT CAUSE FLUSHING**
> nicotinic acid, CCBs (nifedipine, amlodipine), nitrates, sildenafil, opioids, vancomycin, tamoxifen and aromatase inhibitors.

> **SYSTEMIC ASSOCIATIONS**
> H. pylori (conflicting evidence; **eradication is not routine**), **IBD/SIBO**, migraine, depression and anxiety, and possibly cardiovascular risk. These are associations, not reasons for routine tests.

Demodex and demodicosis

- Rosacea skin carries **roughly 6–15× more mites** (the density is higher).
- Mites and their bacterium (Bacillus oleronius) activate **TLR2**, which drives kallikrein-5 cathelicidin inflammation.
- When dying mites burst, they release antigens.
- The **response to ivermectin and permethrin** is indirect evidence that the mites play a part.
- Whether they cause rosacea or simply thrive in it is unresolved. Teach "**contributor and amplifier**".

Steroid-induced rosacea

Steroids cause vasoconstriction and then **rebound vasodilation**. They also cause skin atrophy, which makes telangiectasia visible. Local immunosuppression lets Demodex and bacteria overgrow, and the patient gets **steroid addiction** (stopping causes a flare, so they restart). Typical sources: OTC hydrocortisone, potent TCS borrowed or prescribed for "eczema", and imported skin-lightening creams (clobetasol, illegal in the UK).

Risk factors and associations

> **Genetic** susceptibility (HLA class II polymorphisms; family history).

> **TRIGGERS**
> UV light, heat and cold, exercise, hot drinks, spicy food, **alcohol**, stress, cosmetics, **topical steroids**.

> **MICROBES**
> increased Demodex folliculorum density; Bacillus oleronius; S. epidermidis.

Myths to address

> **"ROSACEA IS ADULT ACNE"**
> no comedones; acne treatments such as benzoyl peroxide often irritate.

> **"RED NOSE MEANS ALCOHOLISM"**
> rhinophyma occurs in non-drinkers; alcohol is just one trigger.

Pathophysiology and pathology

Rosacea combines an **over-reactive innate immune system** with **unstable facial blood vessels and nerves**.

1. **Triggers activate the skin**
   UV, heat, spicy food, alcohol and Demodex activate **TLR2** receptors on keratinocytes.

2 **Cathelicidin overdrive**

Raised **kallikrein-5** cleaves cathelicidin into **LL-37**, which drives inflammation, vasodilatation and new vessel growth.

3 **Neurovascular reactivity**

Sensory nerves (TRPV1 channels) release neuropeptides such as **CGRP**, causing **flushing**, burning and stinging; repeated flushing leaves **persistent erythema** and telangiectasia.

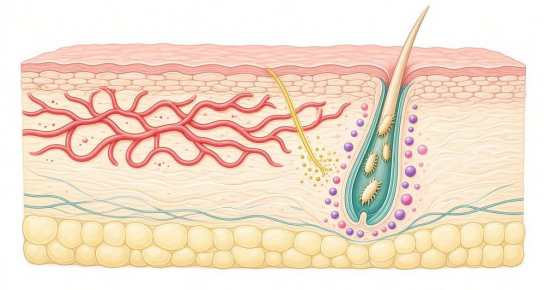

Fig. 1 · Rosacea skin: dilated tortuous capillaries, a sensory nerve releasing neuropeptides, and Demodex mites with inflammation around a follicle.

AI-generated illustration

4 **Papules and pustules**

Inflammatory cells gather around follicles, producing papules and sterile pustules (no comedones).

5 **Phyma and fibrosis**

Chronic inflammation and lymphatic leakage cause sebaceous hyperplasia and fibrosis, thickening the nose (**rhinophyma**).

Fig. 2 · Central face distribution: cheeks, nose, chin and forehead.

AI-generated illustration

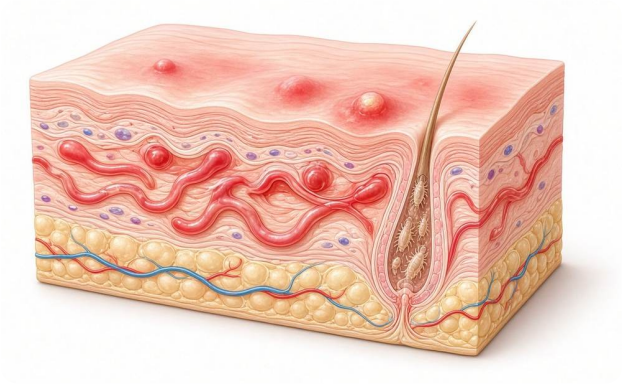

Fig. 3 · Rosacea skin: widened vessels just under the surface (telangiectasia), inflamed papules and a pustule, and Demodex mites crowding a follicle. Note: no comedone plug.

AI-generated illustration

HISTOPATHOLOGY

- Dilated dermal vessels, solar elastosis and a perivascular and perifollicular lymphohistiocytic infiltrate.
- Papulopustular: perifollicular inflammation, sometimes granulomas; increased Demodex.
- Phymatous: sebaceous hyperplasia, fibrosis and dilated follicles. Biopsy is rarely needed.

Clinical features

PHENOTYPE	MILD	MODERATE	SEVERE
Persistent erythema	Faint, central	Clear, cheeks + nose	Fiery, widespread, oedema
Flushing	Occasional	Frequent, bothersome	Daily, disabling
Papules/pustules	Few (<5)	Several (5–20)	Many (>20) ± plaques or nodules
Telangiectasia	Barely visible	Clearly visible	Prominent, widespread
Phyma	Patulous follicles, slight thickening	Contour change without nodules	Nodular, disfiguring
Ocular	Mild lid redness, grittiness	Lid-margin telangiectasia, MGD, chalazia	**Keratitis, scleritis**: same-day eye referral

Grading tools (for documentation and audit): ROSCO global severity, the Investigator's Global Assessment (IGA 0–4, used in trials and for ivermectin/brimonidine licensing), and **RosaQoL** for quality of life. Record **phenotype + severity + QoL impact** at every review.

Skin of colour: erythema is often subtle or brownish. Look for papules, telangiectasia under dermoscopy, warmth, and a history of stinging and flushing.

Demodex and demodicosis

Pointers to demodicosis rather than plain rosacea:

- **fine whitish follicular scale ("frosted", "nutmeg-grater" or "sandpaper" texture)**
- **itch or burning** rather than flushing
- **asymmetric**, sometimes unilateral
- small superficial papulopustules
- a history of **facial TCS/TCI or immunosuppression**
- **DEMODEX BLEPHARITIS**
  cylindrical "collarettes" at the lash bases

Steroid-induced rosacea

- **erythema** (often a "fire-red" colour), **telangiectasia**, papulopustules, **burning or stinging**, skin atrophy and shine
- **a flare within days of stopping the steroid**
- periocular or perioral involvement is often present

Fig. 4 · Moderate papulopustular rosacea: consider topical ivermectin plus doxycycline 40 mg MR. Review at 8–12 weeks and taper to topical maintenance.

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Phenotypes to recognise

Erythema & telangiectasia

Persistent central redness, visible vessels. Site: Cheeks, nose.

© DermNet · **Source**

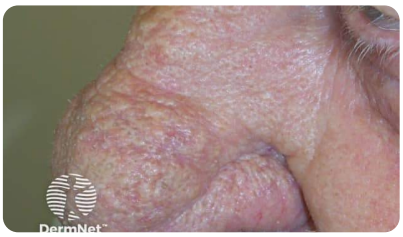
Papulopustular

Dome-shaped papules and pustules, no comedones. Site: Cheeks, chin.

© DermNet · **Source**

Phymatous

Thickened skin, contour change; mostly men. Site: Nose.



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Ocular

Gritty eyes, lid-margin vessels, blepharitis. Site: Eyelids.

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Demodicosis

Fine 'frosted' follicular scale, itch, asymmetric. Site: Cheeks.

© DermNet · **Source**

Steroid rosacea

Fiery red, burning; flares within days of stopping. Site: Face.

© DermNet · **Source**

Special groups

- **Skin of colour**
  - Redness may be hard to see; warmth, papules, burning and dryness are clues; granulomatous forms may be more common.
- **Children**
  - Rare; ocular symptoms may come first; consider perioral dermatitis.

Diagnosis and investigations

Diagnosis is **clinical**. Investigate flushing **only** if there are atypical features:

> **DIARRHOEA, WHEEZE OR PALPITATIONS**
> carcinoid (24 h urine 5-HIAA), mastocytosis (tryptase)

> **PAROXYSMAL HEADACHES, SWEATING, HYPERTENSION**

> phaeochromocytoma (plasma metanephrines)

> **PHOTOSENSITIVE, SCALY, SYSTEMIC SYMPTOMS**
> SLE or dermatomyositis (ANA, CK)

> **SWEATY, PERIMENOPAUSAL**
> menopause

> **medication-related**

Demodex and demodicosis

- **DERMOSCOPY**
  "Demodex tails" (creamy gelatinous threads poking out of follicles) and follicular plugs.
- **Standardised skin-surface biopsy (SSSB)** with cyanoacrylate glue: **>5 mites/cm²** is abnormal. This is a specialist test.
- **Microscopy** of scrapings or plucked eyelashes (optometry or ophthalmology).

Steroid-induced rosacea

Clinical, from the history. Ask specifically about **every cream, its strength and how long it has been used**, including over-the-counter and online products. Scrape for tinea if the eruption is scaly or annular (tinea incognito).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Acne vulgaris	Comedones; younger; trunk too.
	Perioral dermatitis	Perioral papules, vermilion border spared; after steroids.
	Seborrhoeic dermatitis	Greasy scale in nasolabial folds, brows; often coexists.
	Lupus	Photosensitive malar rash, systemic symptoms.
	Acne mimics	Steroid acne, Gram-negative, Malassezia folliculitis.

Core DDx is in the acne notes. New here:

- **granulomatous rosacea**. Look-alikes: sarcoidosis, lupus miliaris disseminatus faciei, granulomatous periorificial dermatitis. Biopsy if unsure.
- **demodicosis**
- **periorificial dermatitis**
- **steroid rosacea**

- **Morbihan disease** (persistent solid facial oedema)
- **seborrhoeic dermatitis**, which often coexists
- **carcinoid**
- **polycythaemia**

Demodex and demodicosis

Rosacea (often coexists), perioral or periorificial dermatitis, seborrhoeic dermatitis, folliculitis (bacterial, Malassezia, gram-negative), contact dermatitis, steroid rosacea, tinea faciei (do a scraping if scaly and asymmetric).

Steroid-induced rosacea

Rosacea, demodicosis, periorificial dermatitis, ACD (to the steroid molecule or vehicle), tinea incognito, lupus.

Management

UK doses (paraphrased from the BNF and SmPCs; confirm before prescribing):

DRUG	TYPICAL UK REGIMEN	NOTES
Ivermectin 1% cream	Once daily, up to 4 months; repeat courses allowed	CKS first line for papulopustular disease. Anti-Demodex and anti-inflammatory. Transient burning
Metronidazole 0.75% gel/cream	Twice daily, usually 6–8 weeks then review (BNF)	Pregnancy-compatible topical choice. Has 1% versions
Azelaic acid 15% gel	Twice daily (once daily if irritation)	Stings at first. Also helps post-inflammatory pigmentation
Brimonidine 3 mg/g gel (0.33%)	Once daily, pea-size amount over 5 areas	For **persistent erythema** only. Effect lasts ~12 h. **Rebound or paradoxical erythema** in ~10–20%, so test-patch a small area first. Not for flushing
Doxycycline 40 mg MR capsule	Once daily, 16 weeks then review	Licensed, sub-antimicrobial dose (less resistance). Alternatives: lymecycline 408 mg or doxycycline 100 mg (off-label). Not in pregnancy or under 12
Erythromycin or **clarithromycin**	Second-line oral if tetracyclines are unsuitable	Clarithromycin is off-label here. Check interactions (statins, QT)
Oral metronidazole	Specialist, off-label	Course mention. Not primary care
Low-dose isotretinoin	Specialist, off-label (e.g. 0.1–0.3 mg/kg/day or 10 mg/day)	Refractory PPR/phyma. The **full PPP applies**, and the MHRA rules do too
Clonidine (flushing)	**50 micrograms twice daily**, off-label	Doses quoted as "50 mg" are wrong by 1000-fold. Watch for hypotension and do not stop abruptly
Propranolol / carvedilol (flushing)	Off-label, low dose	Specialist or with caution. Contraindicated in asthma
Laser / IPL	Telangiectasia, erythema, phyma	Generally **not NHS-funded**. Phyma surgery or ablative laser via plastics/dermatology (variable funding)

Combination approach: CKS supports topical ivermectin + doxycycline 40 mg MR for moderate–severe PPR. After control, **step down to topical maintenance** (ivermectin twice weekly, or azelaic acid/metronidazole) to prevent relapse.

Demodex and demodicosis

- **Ivermectin 1% cream once daily**, which treats both rosacea and Demodex. This is the practical first choice.
- **Permethrin 5% cream** (off-label for this use; specialist-advised regimens exist), or **metronidazole** topical (weak anti-mite activity).
- **Stop facial TCS/TCI.**
- Refractory or immunosuppressed patients: **oral ivermectin** (specialist, single doses repeated).
- **DEMODEX BLEPHARITIS**
  lid hygiene and tea tree oil lid products (OTC); refer to optometry. Lotilaner eye drops are US-approved but **not UK-licensed** as of 2026 (check).

Steroid-induced rosacea

1. **Stop the facial steroid.** For **potent** steroids used for months, specialists may **taper** (e.g. step down to 1% hydrocortisone, then stop over 1–2 weeks) to soften the rebound. Do not swap to another steroid.
2. **Warn about the rebound flare.** Expect it over **1–3 weeks** and for symptoms to persist for weeks to months. This is the main reason patients restart.
3. **Cover the withdrawal** with an **oral tetracycline** (lymecycline 408 mg daily or doxycycline 40 mg MR/100 mg for 6–12 weeks) **± topical metronidazole, azelaic acid or ivermectin**. Topical **tacrolimus or pimecrolimus** is an off-label specialist option. It reduces rebound but can itself cause rosacea-like eruptions.
4. Cool compresses, a bland emollient and **"zero therapy"** for cosmetics. Use mineral sunscreen.
5. **MHRA Yellow Card** and documentation if TSW is suspected. Refer if severe or there is diagnostic doubt.

Medicines at a glance

Ivermectin 1% cream

Papules & pustules

DOSE	Once daily · Up to 4 months; repeat courses allowed
CAUTIONS	CKS first line; anti-Demodex. Transient burning.

Metronidazole 0.75% gel/cream

Papules & pustules

DOSE	Twice daily · Usually 6–8 weeks, then review
CAUTIONS	Pregnancy-compatible topical choice.

Azelaic acid 15% gel

Papules & pustules

DOSE	Twice daily (once if irritated) · Review response
CAUTIONS	Stings at first; helps pigmentation.

Brimonidine 3 mg/g gel

Persistent redness

DOSE	Once daily, pea-size over 5 areas · Effect ~12 h
CAUTIONS	Rebound / paradoxical erythema in ~10–20%: test a small area. Not for flushing.

Doxycycline 40 mg MR

Moderate–severe

DOSE	Once daily · 16 weeks, then review
CAUTIONS	Licensed. Not in pregnancy or under 12. Erythromycin if tetracyclines unsuitable.

Treatment ladder

1

Everyone

Trigger diary; gentle, fragrance-free skincare. **Daily SPF** (mineral if sensitive).

2

Papules & pustules

Ivermectin 1% cream; Metronidazole 0.75% gel/cream; Azelaic acid 15% gel

3

Persistent redness

Brimonidine 3 mg/g gel

4

Moderate–severe

Doxycycline 40 mg MR

5

Specialist

Low-dose isotretinoin (PPP applies) for refractory disease or phyma.
Laser / IPL for telangiectasia (usually private).

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Potent facial steroid**: Never without a specialist plan. OTC hydrocortisone licence excludes the face.
- **Clonidine for flushing**: 50 **micrograms** twice daily (off-label), not 50 mg.
- **Relapsing**: Maintenance topical therapy is often needed.

Referral

SAME DAY — **Eye pain, photophobia or reduced vision**: possible keratitis ophthalmology.

URGENT — Flushing with **diarrhoea, wheeze, palpitations or hypotension**: carcinoid or mastocytosis.

URGENT Photosensitive malar rash with systemic symptoms: exclude **lupus**.

ROUTINE Rash after potent topical steroids: stop or taper with support; refer if severe.

Red flags and when to refer

- **EYE PAIN, PHOTOPHOBIA OR REDUCED VISION**
  possible keratitis, so same-day ophthalmology.
- Flushing with **diarrhoea, wheeze, palpitations or hypotension**: investigate for carcinoid syndrome or mastocytosis.
- A photosensitive malar rash with systemic symptoms: exclude **lupus**.
- Facial rash after potent topical steroids: steroid rosacea, so stop or taper the steroid with support.

Complications and prognosis

Complications

- **OCULAR ROSACEA**
  blepharitis, conjunctivitis, keratitis, rarely corneal ulceration.
- **Rhinophyma** and other phymas.
- Psychological distress, low self-esteem and social avoidance.

Prognosis

- Chronic and relapsing; treatment controls rather than cures.
- Persistent erythema and telangiectasia respond best to vascular lasers or IPL (not on the NHS routinely).

Prescriber pearls (UK pharmacist IP)

Trigger diary for 2–4 weeks. Avoid only your triggers, not every listed one.

Daily SPF 30–50 mineral or tinted sunscreen. Use green-tinted camouflage for redness. Avoid **alcohol-based toners, menthol, witch hazel, exfoliants and fragrance**.

No topical steroids on the face. Check OTC hydrocortisone and imported "lightening" creams.

Set expectations: **papules respond, erythema and flushing respond poorly** to the antimicrobial-type drugs. That is why brimonidine and laser exist.

Before brimonidine, check for **MAOIs, TCAs and antihypertensives**, and avoid it in **Raynaud's or orthostatic hypotension**.

"Rosacea that fails metronidazole or azelaic acid, especially if it's scaly or itchy" think Demodex, switch to **ivermectin**, and ask about steroid creams.

Mites are on everyone. Reassure the patient that this is **not a hygiene problem** and not contagious in any practical sense.

Never prescribe a potent or very potent TCS for the face without a specialist plan. When counselling on OTC hydrocortisone, the licence **excludes the face** (with limited exceptions).

Ask about **skin-lightening products** in a non-judgemental way. Mercury and clobetasol products are illegal for sale in the UK and are a trading-standards issue.

Demodex and demodicosis

Steroid-induced rosacea

Common misconceptions and course-vs-UK differences

POST / ITEM	STATEMENT	CORRECTION (UK)
Student answer	"Clonidine orally **50 mg**" for flushing	**50 micrograms twice daily** (BNF; off-label for rosacea flushing; up to 75 micrograms BD). 50 mg would be a dangerous overdose
Student answer	Brimonidine "0.5%"	The UK gel contains **3 mg/g brimonidine (0.33% base)**, which equals 5 mg/g (0.5%) **brimonidine tartrate**. Your figure is the salt strength. Prescribe by the BNF/product description to avoid confusion
Student answer	Metronidazole, azelaic acid, ivermectin listed without regimens	Metronidazole 0.75% **twice daily**; azelaic acid 15% gel **twice daily**; ivermectin 1% **once daily up to 4 months**; doxycycline **40 mg MR once daily** (licensed)
Student answer	Oral metronidazole and isotretinoin listed as options	Both are **specialist and off-label** for rosacea. They are not IP-initiated in primary care
Course teaching	"Tetracycline 6–12 weeks"	Fine in principle. The UK licensed choice is **doxycycline 40 mg MR** (sub-antimicrobial), or lymecycline/doxycycline 100 mg off-label. Step down to topical maintenance
Course teaching	"Physical sunblock"	Agrees with UK advice (mineral SPF 30+, daily)
Course teaching	H. pylori / IBD associations	Evidence is inconsistent. **UK guidance does not recommend routine H. pylori testing** for rosacea
Student answer	Oral antibiotic with topical antibiotic, or oral alone	**NG198: never.** Always combine with BPO and/or a topical retinoid
Student answer	Clarithromycin as the alternative oral	**NG198 names erythromycin or trimethoprim**, not clarithromycin

POST / ITEM	STATEMENT	CORRECTION (UK)
Student answer	"Switch at 16 weeks"	**Review at 12 weeks.** If improved but not clear, continue up to 12 more weeks. Exceptional use beyond 6 months needs 3-monthly review
Student answer	Monotherapy-led ladder	**NG198 makes fixed-dose topical combinations first line** (adapalene/BPO, tretinoin/clindamycin, BPO/clindamycin)
Course case	Clindamycin/BPO for 8 months	Too long for an antibiotic-containing topical (resistance risk). Review at 12 weeks
Course case (student answer)	"Try another antibiotic course"	She has already failed an oral antibiotic regimen, so **refer for isotretinoin**

Self-check quiz

5 questions · optional

Q1 Which lesion is absent in rosacea but the hallmark of acne?

✓ Comedones.

Q2 First-line topical for papulopustular rosacea (CKS)?

✓ Ivermectin 1% cream once daily, up to 4 months.

Q3 What is the key warning with brimonidine gel?

✓ Rebound or paradoxical erythema in about 10–20%: test a small area first. It is for persistent redness, not flushing.

Q4 Licensed oral option for moderate–severe rosacea?

✓ Doxycycline 40 mg MR once daily for 16 weeks, then review.

Q5 Rosacea with eye pain and photophobia?

✓ Possible keratitis: same-day ophthalmology.

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Photo gallery

Rosacea phenotypes 7

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Ocular rosacea 2

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Granulomatous rosacea 1

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Demodex and demodicosis 3

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Steroid-induced rosacea 2

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Differential: perioral dermatitis 1

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CHAPTER 7

Infections — fungal

Dermatophyte and yeast infections: tinea of the body, groin and feet, pityriasis versicolor and candidal intertrigo.

1. Tinea: corporis, cruris & pedis
2. Pityriasis versicolor
3. Candidal intertrigo

Infections — fungal Adult core · deepened with Module 5 Week 1

Tinea: corporis, cruris & pedis

Dermatophyte 'ringworm' of the body, groin and feet. Look for an asymmetrical scaly active edge, examine the feet, use a topical allylamine or azole past clearance, and never a steroid alone.

ALSO CALLED ringworm · tinea corporis · tinea cruris · jock itch · tinea pedis · athlete's foot · +6

RELATED PAGES Fungal nail infection & nail discolouration · Candidal intertrigo · Annular (ring-shaped) rashes: differential · Intertrigo & the flexural differential

KEY FACTS

- **Dermatophyte** infection of keratin (stratum corneum, hair, nails), mostly Trichophyton rubrum.
- **Asymmetrical** annular plaque with a scaly **active edge** and central clearing; tinea pedis causes macerated toe webs.
- Scrape the **edge** for microscopy if uncertain, extensive or before oral treatment.
- Topical terbinafine or an imidazole **until 1–2 weeks after clearance**; treat feet and nails (the reservoir).
- A topical steroid alone causes **tinea incognito**.

Overview

FORM	PLAIN DEFINITION
Tinea corporis	'Ringworm' of the body skin
Tinea cruris	Dermatophyte infection of the groin ('jock itch')
Tinea pedis	Athlete's foot: interdigital, moccasin or vesicular

All three are caused by **dermatophytes**, most often Trichophyton rubrum. They are **contagious**, spreading by spores and fomites (towels, floors, clothing) and from animals.

Tinea manuum and pedis on the palms and soles (Module 3 Week 6)

- dermatophyte infection of the hand (tinea manuum — much less common) and foot (tinea pedis — very common). Usual organisms Trichophyton rubrum, T. interdigitale, Epidermophyton floccosum; zoophilic (Microsporum canis, T. verrucosum, T. erinacei) cause inflamed lesions.

Epidemiology

- **Who**
  - Worldwide, especially hot humid climates; tinea pedis affects up to 15–25% of adults at some point.

  - Tinea corporis is most common in **children and young adults**; tinea cruris and pedis in adult **men**.

- **Spread**
  - Skin-to-skin contact, shared towels, bedding and changing rooms; animals (Microsporum canis from cats and dogs, T. verrucosum from cattle).
  - Incubation 1–3 weeks.

Aetiology: causes and risk factors

Dermatophytes are keratin-loving fungi that invade the **stratum corneum** (and hair and nails). The infection spreads outwards in a ring, which produces the **active edge with central clearing**.

Autoinoculation is common: infection spreads from the feet and nails to the groin and body.

> **RISK FACTORS**
> occlusion, sweating, sport, shared towels and changing rooms.

> **PETS AND FARM ANIMALS**
> zoophilic species cause more inflamed, pustular lesions.

ⓘ **NOTE**

Tinea cruris risk factors (Module 5)

Hot, humid weather; sweating and sport; tight, occlusive clothing; **diabetes**; and immunosuppression. The usual organism is T. rubrum, with T. interdigitale and Epidermophyton floccosum also seen.

⚠ **CAUTION**

Why steroids make tinea worse

A topical steroid suppresses the local immune response that is holding the fungus back. The redness fades, but the dermatophyte spreads freely. The result is **tinea incognito**: a larger, less scaly, atypical rash that is easily mistaken for eczema.

Risk factors and associations

> **ORGANISMS**
> anthropophilic T. rubrum (most common), T. interdigitale, T. tonsurans; zoophilic M. canis, T. verrucosum, T. erinacei (hedgehogs) give more inflamed lesions.

> **MEDICAL RISK FACTORS**
> previous tinea, **diabetes**, immunosuppression, hyperhidrosis, obesity, ichthyosis.

> **ENVIRONMENT**
> occlusive footwear and clothing, sport, crowding, pets, contact sports (tinea gladiatorum).

> Emerging **sexually transmitted** T. mentagrophytes genotype VII and **terbinafine-resistant** T. indotineae (extensive, recalcitrant groin and body tinea, often after travel to South Asia).

Myths to address

> **"ANY RING IS RINGWORM"**
> granuloma annulare, discoid eczema and others make rings; scrape if unsure.

> **"STOP ONCE IT LOOKS CLEAR"**
> continue for 1–2 weeks after clearance or it relapses.

> **"A COMBINED STEROID CREAM IS BEST"**
> steroid-antifungal is a short bridge only (up to 7 days) for very inflamed skin.

Pathophysiology and pathology

Dermatophytes live only in **dead keratin**; the shape of the rash reflects how they spread and how the host reacts.

1. **Spores land on skin**
   Spores from people, animals, soil or fomites stick to keratinocytes, helped by warmth, moisture and minor trauma.

2. **Keratin digestion**
   Fungi release **keratinases** and proteases to invade the **stratum corneum**, hair and nail; they rarely go deeper because of serum inhibitory factors and host immunity.

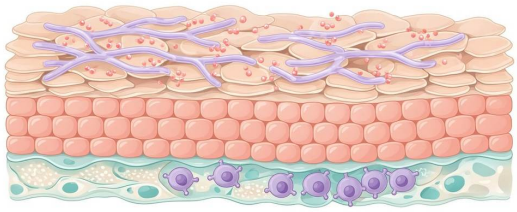

Fig. 1 · Dermatophyte hyphae spread only through the dead outer layer (stratum corneum), releasing keratin-digesting enzymes; immune cells gather beneath.
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3. **Outward spread**
   Hyphae grow outwards, while the host's inflammatory response clears the centre: an **active scaly edge with central clearing**.

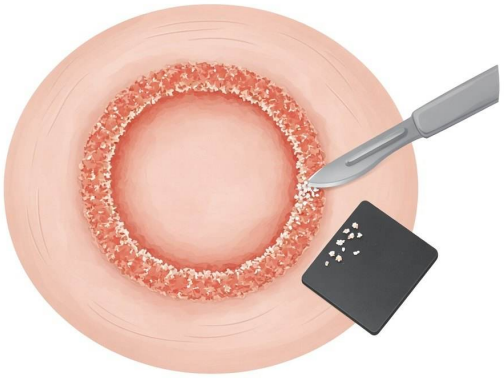

Fig. 2 · The ring grows outwards: the most active fungus is at the scaly edge, which is where to scrape.
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4 **Host response decides the look**

Strong cell-mediated immunity (often zoophilic species) gives inflamed, pustular lesions; weak responses (T. rubrum, immunosuppression, steroid use) give chronic, widespread, less inflamed disease (**tinea incognito**).

5 **Reservoirs and reinfection**

Feet and nails act as reservoirs; scratching and towels spread fungus to the groin and body (**autoinoculation**).

HISTOPATHOLOGY

- Fungal hyphae in the stratum corneum (seen with PAS or Grocott stain), sometimes the 'sandwich sign' between orthokeratotic and parakeratotic layers.
- Mild spongiosis and a superficial perivascular infiltrate; neutrophils in the stratum corneum.
- In practice, diagnosis is by **skin scrapings** (microscopy with KOH and culture or PCR), not biopsy.

Clinical features

Tinea corporis

One or more **asymmetrical annular** plaques with an active red, scaly, advancing edge and central clearing. Usually itchy.

> **TINEA INCOGNITO**
> altered by steroid, so less scaly and more widespread.

> **MAJOCCHI GRANULOMA**
> a deeper, follicular infection, often after shaving or steroid use.

> **KERION-LIKE INFLAMMATION**
> boggy and pustular, typical of zoophilic species.

Tinea cruris

Affects the inguinal folds, with or without the inner thighs. The scaly edge may be less obvious in moist folds. The **scrotum is usually spared**, whereas **Candida often involves it**.

Tinea cruris often starts on **one side or asymmetrically**. It can spread to the thighs and buttocks, and sometimes has vesicles at the active edge. The scrotum is usually spared, but this is a teaching rule rather than an absolute one.

Tinea pedis

> **INTERDIGITAL**
> maceration, peeling and fissures between the toes, especially the 4th–5th web space.

> **MOCCASIN**
> dry, hyperkeratotic scaling of the sole and side of the foot.

> **VESICOBULLOUS**
> inflamed vesicles on the instep.

Tinea manuum and pedis on the palms and soles (Module 3 Week 6)

> **CLUES**
>
> **"two feet–one hand" syndrome** (hyperkeratotic scale on one palm + both soles); **white scale accentuating creases** (vs red lichenified creases in eczema); asymmetric; spread contiguous from palm to dorsum with a raised edge; **nails** (onychomycosis); toe-web maceration; moccasin scaling of soles. Risk: manual work, hyperhidrosis, pre-existing hand eczema, animal/soil contact.

> **STEROID TRAP**
>
> topical steroid alters appearance (tinea incognito) — scrape before treating any unexplained unilateral palm rash.

Fig. 3 · Classic ringworm: annular plaque with raised scaly advancing edge and central clearing.

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Distribution

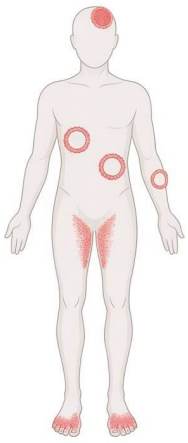

Fig. 4 · Where tinea shows up: scalp, rings on the trunk and limbs, both groin folds onto the inner thighs, and the feet. Always check the feet and nails.

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Forms to recognise

Corporis

Annular plaque, raised scaly edge, central clearing. Site: Trunk, limbs.

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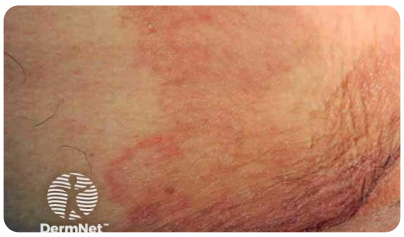
Cruris

Arcuate scaly edge spreading onto the thigh. Site: Groin.



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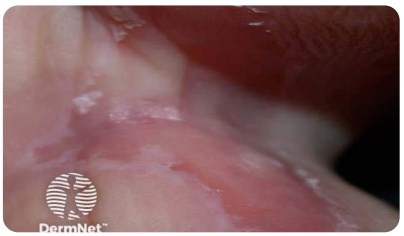
Interdigital pedis

White, macerated, peeling toe webs. Site: 4th–5th web.



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Moccasin pedis

Diffuse dry scaling of the soles. Site: Soles, sides.

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Special groups

- **Skin of colour**
  - Edges may look grey, violaceous or hyperpigmented rather than red; post-inflammatory pigmentation is common.
- **Children**
  - Tinea corporis and capitis are commonest; check pets and family members.
- **Diabetes and immunosuppression**
  - Treat tinea pedis promptly: fissured toe webs are a portal for cellulitis.

Diagnosis and investigations

- Clinical pattern plus risk factors.

- Take **skin scrapings from the active edge** for microscopy, with or without culture, if: the diagnosis is uncertain, treatment has failed, before starting oral antifungals, for extensive disease, or if immunocompromised.
- **Always examine the feet, nails and hands** when the groin or body is involved.
- Swab the folds if bacterial co-infection is suspected.

Tinea manuum and pedis on the palms and soles (Module 3 Week 6)

- **skin scrapings** (± nail clippings) for microscopy and culture.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Discoid eczema	Coin-shaped, itchy, no central clearing.
	Pityriasis rosea	Herald patch, then 'Christmas-tree' pattern.
	Granuloma annulare	Ring of papules, **no scale**.
	Candidal intertrigo	Satellites; **involves the scrotum**.
	Flexural psoriasis	Sharp salmon-pink, no scale in folds, other sites.
	Erythrasma	Brown-red; **coral-red** under Wood's light.

Management

Self-care (all forms)

- Keep the skin clean and **dry**, especially the toe webs and folds. Wear loose cotton clothing.
- Do not share towels. Wash clothing and bedding hot.
- Treat any foot or nail infection at the same time.
- Children with body or groin tinea do **not** usually need to be kept off school (CKS).

Tinea corporis/cruris: mild, localised

- Topical **terbinafine**, or an imidazole (**clotrimazole, miconazole, econazole**). Continue for **1–2 weeks after the skin looks clear**.
- If very inflamed, add a mild steroid (e.g. hydrocortisone 1%) for a **maximum of 7 days**. **Never use a steroid alone.**

Extensive or severe disease

- Adults: oral **terbinafine** first line, ideally after sending scrapings. Alternatives are itraconazole and griseofulvin.
- Children with extensive disease: refer to paediatric dermatology.

Tinea pedis

- Same topical classes. **Allylamines (terbinafine)** are often preferred for dermatophytes.
- Advise on drying the toe webs and on footwear. Treat onychomycosis through its separate pathway.
- Consider oral therapy if severe, if topicals fail, or if immunocompromised. Check LFTs and interactions for terbinafine and itraconazole.

Tinea manuum and pedis on the palms and soles (Module 3 Week 6)

- tinea pedis — topical **terbinafine 1%** (1–2 weeks) or an imidazole (clotrimazole/miconazole, ~4 weeks); foot hygiene, dry webs, change socks, treat shoes. Tinea manuum (hyperkeratotic palm) often needs **oral terbinafine** (adult 250 mg daily, typically 2–6 weeks) after mycology confirmation; check LFTs/drug interactions per BNF; treat feet/nails simultaneously to prevent re-infection. Refer if diagnosis uncertain or immunosuppressed.

Medicines at a glance

Terbinafine 1% cream

Mild, localised

DOSE As per product · Until 1–2 weeks after clear (pedis 1–2 weeks)

CAUTIONS Allylamines often preferred for dermatophytes.

Imidazole (clotrimazole, miconazole, econazole)

Mild, localised

DOSE As per product · Until 1–2 weeks after clear (pedis ~4 weeks)

CAUTIONS Very inflamed: add hydrocortisone 1% for max 7 days.

Oral terbinafine

Extensive or severe · adults

DOSE 250 mg daily (adult) · Typically 2–6 weeks (e.g. manuum)

CAUTIONS Ideally after scrapings. Check LFTs and interactions. Alternatives: itraconazole, griseofulvin.

Treatment ladder

1 **Self-care**

Keep skin and toe webs **dry**; loose cotton. No shared towels; wash clothing and bedding hot.

2 **Mild, localised**

Terbinafine 1% cream; Imidazole (clotrimazole, miconazole, econazole)

3

Extensive or severe · adults

Oral terbinafine

4

Refer

Children with extensive disease paediatric dermatology. Diagnosis uncertain, or immunosuppressed.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Steroid alone**: Never: it creates tinea incognito.
- **Combinations**: Steroid–antifungal is a short bridge only (up to 7 days).
- **Azoles**: Watch CYP interactions with oral itraconazole and fluconazole.

Referral

URGENT	**Kerion**, **Majocchi granuloma**, or suspected deep infection.
URGENT	Tinea pedis in **diabetes** or past cellulitis: fissured webs are an entry point.
ROUTINE	Immunocompromised with extensive or atypical disease; children needing oral treatment.
ROUTINE	Failure despite adherence: scrape and reconsider; occupational or institutional outbreak.

Red flags and when to refer

- Diagnostic doubt, or failure despite good adherence: take scrapings and reconsider.
- **Kerion** or **Majocchi granuloma**, or suspected deep infection.
- Immunocompromised patient with extensive or atypical disease.
- Children needing oral treatment.
- Occupational or institutional outbreak.

- Tinea pedis in a person with diabetes or a history of cellulitis: fissured toe webs are an **entry point for cellulitis**.

Complications and prognosis

Complications

- Spread to other sites and household contacts.
- **Bacterial superinfection** and cellulitis through fissured toe webs.
- **Majocchi granuloma** (follicular invasion, often after steroid use) and **kerion** (scalp).
- Id (dermatophytide) reaction: itchy vesicles at a distant site.
- Chronic dermatophytosis (T. rubrum at 4 or more sites) and resistant infection.

Prognosis

- Usually cured with correct treatment and adherence.
- Relapse is common if treatment stops too early or reservoirs (feet, nails, contacts, pets) are not treated.

Prescriber pearls (UK pharmacist IP)

Always look at the **feet** in a 'groin rash'.

A steroid–antifungal combination is a **short bridge only** (up to 7 days). Explain why steroid on its own is harmful.

Oral terbinafine: check interactions and LFTs if used for a prolonged period. Taste disturbance is uncommon but worth mentioning.

Azoles: watch for CYP interactions, especially with oral itraconazole and fluconazole.

OTC clotrimazole and terbinafine are widely available and are a good first pharmacy intervention for limited tinea.

Recurrent tinea cruris means an untreated reservoir: the feet, the nails, or another family member.

Groin rash in a person with diabetes: also think of Candida and secondary bacterial infection.

Thick, discoloured toenails with tinea pedis: see fungal nail infection and send clippings before oral treatment.

Scaly rings that fail antifungals: work through the annular rash differential.

Self-check quiz

5 questions · optional

Q1 Where do you take scrapings from in tinea corporis?

✓ The scaly active (advancing) edge.

Q2 How long to continue a topical antifungal?

✓ Until 1–2 weeks after the skin looks clear.

Q3 Groin rash involving the scrotum with satellites: tinea or Candida?

✓ Candida. Tinea cruris usually spares the scrotum.

Q4 What does a steroid alone do to tinea?

✓ Suppresses local immunity so the fungus spreads: tinea incognito (larger, less scaly, atypical).

Q5 'Two feet, one hand' suggests?

✓ Tinea manuum with tinea pedis.

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Photo gallery

Tinea corporis: scrape the scaly margin

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Tinea pedis: interdigital, moccasin, dorsal, with nails

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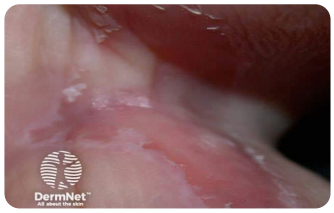

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Tinea cruris: raised arcuate border

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Module 5: tinea cruris

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Differential pairing: tinea cruris vs candida vs flexural psoriasis

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Tinea manuum and pedis (Module 3 Week 6)

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Annular differential (Module 4 Week 1)

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Infections — fungal

Pityriasis versicolor

Malassezia yeast overgrowth causing pale, pink or brown finely scaly patches on the trunk. Treat with a ketoconazole wash and warn that the colour takes months to return.

ALSO CALLED tinea versicolor · malassezia · pityrosporum · ketoconazole shampoo · pale patches

RELATED PAGES Hypopigmentation & skin of colour · Tinea: corporis, cruris & pedis · Pityriasis rosea · Folliculitis

KEY FACTS

- Superficial infection with the lipophilic yeast **Malassezia** causing scaly **hypo- or hyperpigmented** patches on the upper trunk.
- Most common in **young adults**, hot humid climates and people who sweat heavily; recurs each summer.
- **Fine scale** shows when the skin is stretched or scraped; Wood's light may give yellow fluorescence.
- First line (CKS): **ketoconazole 2% shampoo** as a short-contact wash; topical azole for small areas; oral itraconazole if extensive or resistant.
- Colour takes **months** to normalise after the yeast is cleared; recurrence is common (prophylaxis may help).

Overview

Pityriasis versicolor is a common, superficial yeast infection that causes **patchy colour change with fine scale**, usually on the trunk. It is caused by Malassezia, a yeast that is part of normal skin flora.

Because the organism is a normal skin resident, the condition is **not contagious in the usual sense**. It is an overgrowth rather than something caught.

Epidemiology

- **Who**
  - Young adults, slightly more men; up to 40% in tropical climates, 1–4% in temperate climates.
  - Malassezia is normal flora in 90–100% of adults.

Aetiology: causes and risk factors

- Lipophilic Malassezia overgrows in sebum-rich skin (chest, back, shoulders, neck).
- It produces substances that **interfere with melanin**, giving hypopigmented patches, or causes mild inflammation, giving pink or brown hyperpigmented patches.

- **PROMOTERS**
  heat, humidity, sweating, oily skin and young adulthood. Mild immunosuppression can contribute.

Risk factors and associations

- **ORGANISMS**
  M. globosa, M. sympodialis, M. furfur.
- Triggers: heat, humidity, sweating, oily skin, occlusive clothing, immunosuppression, pregnancy, genetic predisposition.

Myths to address

> **"THE WHITE PATCHES ARE VITILIGO"**
> PV patches are scaly and fade after treatment; vitiligo is smooth and depigmented.

> **"IT'S CONTAGIOUS"**
> it comes from your own skin yeast.

Pathophysiology and pathology

Normal skin yeast switches to a **hyphal form** and alters pigment in the stratum corneum.

1 **Yeast to hyphae**

Heat, humidity and sebum let Malassezia convert from yeast to the invasive **mycelial** form in the stratum corneum.

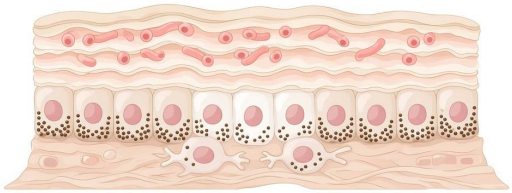

Fig. 1 · Malassezia yeasts and short hyphae in the stratum corneum; melanocytes beneath produce less pigment, so the patch looks paler.
AI-generated illustration

2 **Hypopigmentation**

The yeast produces **azelaic acid** and other dicarboxylic acids that inhibit tyrosinase and melanocytes, so patches look paler, especially after a tan.

Fig. 2 · Typical distribution: scaly pale or brown patches on the upper back, chest and upper arms.
AI-generated illustration

3 **Hyperpigmentation**

In other patients, thicker keratin and inflammation make patches pink or brown.

4 **Slow recovery**

After the yeast is killed, pigment returns only slowly over months.

HISTOPATHOLOGY

- Short hyphae and round yeasts ('spaghetti and meatballs') in the stratum corneum on KOH microscopy of scrapings.
- Biopsy is not needed.

Clinical features

- Well-demarcated **hypopigmented, hyperpigmented or pink-fawn** macules that coalesce into patches.
- **Fine, bran-like scale**, which becomes more obvious when the skin is gently scratched or stretched (the 'evoked scale' sign).
- Sites: upper trunk, shoulders, neck and upper arms.
- Usually asymptomatic or mildly itchy. Often noticed **after tanning**, because affected skin does not tan.

Fig. 3 · Hypopigmented pityriasis versicolor: coalescing pale macules with fine scale on the trunk.

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Colour variants

Hypopigmented

Coalescing pale scaly macules. Site: Upper trunk.

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On tanned skin

Small pale scaly macules; noticed after sun. Site: Shoulders.

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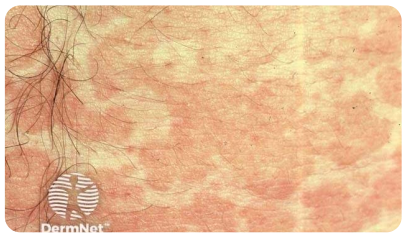
Hyperpigmented / pink

Fawn-pink scaly macules coalescing. Site: Chest, back.



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Special groups

- **Skin of colour**
  - Hypopigmentation is especially noticeable and distressing; reassure that colour returns slowly.

Diagnosis and investigations

- Usually **clinical**.
- Skin scrapings or tape-stripping for microscopy show the classic **'spaghetti and meatballs'** hyphae and spores.
- Wood's lamp may show a yellow-green fluorescence, but this is variable.

Clinical

- Usually clinical.
- Stretch the skin to **evoke scale**.

Microscopy

- Scrapings or tape strip: **'spaghetti and meatballs'** hyphae and spores.

Wood's lamp

- Yellow-green fluorescence: variable.

If no scale

- Complete depigmentation, no scale think **vitiligo**.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Vitiligo	Chalk-white, **no scale**; hands, face, orifices.
	Pityriasis alba	Children; ill-defined pale facial patches; atopy.
	Post-inflammatory hypopigmentation	Previous rash at the same sites; no scale.
	Seborrhoeic dermatitis	Greasy scale; scalp, face, central chest.
	Pityriasis rosea	Herald patch, then 'Christmas-tree'; settles in 6–8 weeks.

Management

First line (CKS)

> **EXTENSIVE**
>
> **ketoconazole 2% shampoo** used as a body wash once daily for up to **5 days**. Lather onto the affected skin, leave for 3–5 minutes, then rinse. Licensed for this use from age 12.

> **SMALL AREAS**
>
> an azole (clotrimazole, econazole, ketoconazole) or terbinafine cream twice daily for 2–3 weeks. **Clotrimazole is preferred in pregnancy.**

If topicals fail or disease is very widespread

Oral **itraconazole** or **fluconazole** in adults. Check interactions and pregnancy status. Seek specialist advice for pregnancy or children.

Preventing recurrence

Recurrence is common. Options are ketoconazole shampoo every 1–4 weeks, or a 3-day course before a sunny holiday.

Medicines at a glance

Ketoconazole 2% shampoo as body wash

First line

DOSE	Once daily; lather, leave 3–5 min, rinse · Up to 5 days
CAUTIONS	Licensed for this use from age 12.

Azole (clotrimazole, econazole, ketoconazole) or terbinafine cream

First line

DOSE	Twice daily · 2–3 weeks
CAUTIONS	Clotrimazole preferred in pregnancy.

Treatment ladder

1

First line

Ketoconazole 2% shampoo as body wash; Azole (clotrimazole, econazole, ketoconazole) or terbinafine cream

2

Topicals fail or very widespread

Oral **itraconazole** or **fluconazole** in adults: check interactions and pregnancy. Pregnancy or children: specialist advice.

3

Prevent recurrence

Ketoconazole shampoo every 1–4 weeks, or a **3-day course before a sunny holiday**.

⚠ CAUTION

Prescribing limits and pitfalls

- **Expectations**: Colour takes **weeks to months** to return: not treatment failure.
- **Ongoing scale**: Suggests the infection is still active.

Referral

ROUTINE — Diagnostic doubt: especially **no scale** or complete depigmentation (vitiligo).

ROUTINE Very extensive, recurrent or unusual: consider immunosuppression.

ROUTINE Pregnancy or children needing oral therapy: specialist advice.

Red flags and when to refer

- Diagnostic doubt, especially no scale or complete depigmentation (think vitiligo).
- Very extensive, recurrent or unusual disease: consider immunosuppression.
- Pregnancy or children needing oral therapy: seek specialist advice.

Complications and prognosis

Complications

- Persistent pigment change for months.
- Recurrence in up to 60% within a year and 80% within 2 years without prophylaxis.

Prognosis

- Responds well to antifungals; recurrence is very common, so maintenance or seasonal prophylaxis may be offered.

Prescriber pearls (UK pharmacist IP)

Manage expectations: the colour takes **weeks to months** to return after the yeast is cleared. This is **not** treatment failure.

Reassure that it is **not** caused by poor hygiene and is not usually catching.

Even sun exposure after treatment helps the colour even out. Ongoing scale suggests the infection is still active.

A prophylaxis plan before holidays prevents a lot of repeat consultations.

Self-check quiz

5 questions · optional

Q1 What organism causes pityriasis versicolor?

✓ Malassezia, a normal skin yeast, overgrowing.

Q2 How is ketoconazole shampoo used?

✓ As a body wash once daily for up to 5 days: lather, leave 3–5 minutes, rinse.

Q3 What does microscopy show?

✓ 'Spaghetti and meatballs' hyphae and spores.

Q4 Pale patches but no scale and chalk-white?

✓ Think vitiligo, not pityriasis versicolor.

Q5 Yeast cleared but still pale. Failure?

✓ No: colour takes weeks to months to return.

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Photo gallery

Hypopigmented, tanned-skin and pink/fawn variants

3

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Infections — fungal Adult core · deepened with Module 5 Week 1

Candidal intertrigo

Beefy-red, macerated skin folds with satellite pustules. Moisture control plus a topical azole or nystatin, and look for diabetes or recent antibiotics.

ALSO CALLED candida · candidiasis · thrush skin · yeast infection skin folds · intertrigo · erosio interdigitalis · +2

RELATED PAGES Intertrigo & the flexural differential · Tinea: corporis, cruris & pedis · Napkin (nappy) dermatitis · Psoriasis

KEY FACTS

- Candida (mostly C. albicans) infection of **warm, moist skin folds**.
- **Beefy red**, macerated fold with **satellite papules and pustules** at the edge; may involve the scrotum.
- Risk factors: obesity, diabetes, sweating, occlusion, incontinence, antibiotics, steroids, immunosuppression.
- Treat with **moisture control** plus a topical imidazole (e.g. clotrimazole) until 1–2 weeks after clearance; hydrocortisone 1% short term if inflamed.
- Recurrent or extensive: check **HbA1c**, review drugs and consider swabs.

Overview

Candidal intertrigo is a **yeast infection of moist skin folds**, usually caused by Candida albicans. Common sites are under the breasts, under an abdominal apron, the groin, the axillae and the finger or toe webs (erosio interdigitalis blastomycetica).

Intertrigo itself is inflammation of the skin folds from heat, moisture, friction and poor ventilation. Secondary Candida infection is common, and **satellite papules and pustules** are the key diagnostic clue.

Epidemiology

- **Who**
  - Common at both ends of life: **infants** (napkin area) and older, less mobile adults.
  - More common in hot, humid weather and in people with obesity or diabetes.

Aetiology: causes and risk factors

Candida is an **opportunist**. It overgrows where skin is wet, warm, occluded and macerated.

Risk factors

- **Diabetes** (look for it)
- Recent **antibiotics** or systemic steroids

- Immunosuppression
- Obesity and deep skin folds
- Incontinence, nappies and pads
- Wet work, for finger-web disease

Risk factors and associations

> **LOCAL**
> heat, moisture, friction, occlusive clothing, incontinence, poor hygiene of folds.

> **SYSTEMIC**
> **diabetes**, obesity, broad-spectrum antibiotics, systemic or topical steroids, immunosuppression (HIV, chemotherapy), pregnancy.

Myths to address

- **"ALL FOLD RASHES ARE THRUSH"**
  irritant intertrigo, flexural psoriasis, erythrasma and tinea cruris look similar.

Pathophysiology and pathology

Candida is a normal commensal that becomes invasive when **local moisture or host defences** change.

1. **Moist, damaged skin**
   Sweat and friction macerate the stratum corneum in folds, weakening the barrier (irritant intertrigo).

2. **Overgrowth and adhesion**
   Commensal Candida multiplies in the warm, wet fold and adheres using surface adhesins.

Fig. 1 · A macerated skin fold with Candida yeasts and hyphae invading the softened stratum corneum.
AI-generated illustration

3. **Hyphal invasion**
   Yeasts switch to **pseudohyphae and hyphae**, releasing proteases and phospholipases that invade the stratum corneum and trigger neutrophil inflammation.

4. **Satellite lesions**
   Small pustules and papules form beyond the main edge where yeasts seed new follicles: the **satellite** sign.

HISTOPATHOLOGY

- Yeasts and pseudohyphae in the stratum corneum (PAS stain) with neutrophils and subcorneal pustules.
- Diagnosis is clinical; swab or scrape for microscopy and culture if unsure or recurrent.

Clinical features

- **Beefy-red, macerated plaques** in folds, often glistening or eroded.
- **Satellite papules and pustules** beyond the main edge. This is the key sign.
- Peripheral peeling scale at the margin.
- Soreness, itch or burning.
- In the groin, Candida often **involves the scrotum**, which tinea tends to spare.

NOTE

Look elsewhere

Candidal intertrigo can coexist with oral or genital thrush, so ask and examine.

Fig. 2 · Candidal intertrigo: beefy red macerated fold with satellite papules/pustules at the edge.

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Sites to recognise

Groin fold

Macerated red fold with satellites. Site: Groin.

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Toe web

Moist, red, macerated web skin. Site: Toe webs.

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Finger web

Erosio interdigitalis blastomycetica: wet work. Site: Finger webs.

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Special groups

- **Infants**
  - Candidal napkin dermatitis involves the skin creases (irritant napkin dermatitis spares them).
- **Skin of colour**
  - Erythema may look dusky or purple; post-inflammatory pigmentation is common.

Diagnosis and investigations

- Usually clinical, based on the site and satellite lesions.
- Take a swab or scrapings for yeast if uncertain, recurrent or not responding. Swab for bacteria if secondary infection is suspected.
- **Check HbA1c or glucose** if recurrent or there are other risk features.

Clinical

- Site + satellites.
- Ask about oral or genital thrush.

Sample

- Swab or scrapings for yeast if uncertain, recurrent or not responding.
- Bacterial swab if secondary infection.

Glucose

- **HbA1c or glucose** if recurrent or other risk features.

Medicines

- Recent antibiotics, steroids, immunosuppressants?

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Simple intertrigo	Mirror-image redness, **no satellites**.
	Flexural psoriasis	Sharp salmon-pink plaque, no satellites; nails.
	Tinea cruris	Arcuate scaly edge onto the thigh; scrotum spared.
	Erythrasma	Brown-red, fine scale; coral-red under Wood's light.
	Hailey–Hailey / pemphigus vegetans	Chronic eroded recurrent fold disease: refer.

Management

Moisture control: the most important step

- Dry the folds thoroughly after washing (pat dry, or use a cool hairdryer).
- Use absorbent interface textiles or cotton separators. Choose barrier products for incontinence.
- Wear loose cotton clothing, and give weight-management advice where relevant.

Topical antifungal

- **Imidazole** (clotrimazole or miconazole) or **nystatin**. Continue until 1–2 weeks after clearance.
- For very inflamed folds, a short course of a combination with mild steroid (e.g. hydrocortisone with clotrimazole or miconazole) is sometimes used. Follow the local formulary and limit to about 7 days.

Oral

Fluconazole for severe or recalcitrant disease. Check interactions and pregnancy status.

Look for and treat the cause

Review for diabetes, recent antibiotics and immunosuppression.

Medicines at a glance

Imidazole (clotrimazole or miconazole) or nystatin

Topical antifungal

DOSE	As per product · Until 1–2 weeks after clear
CAUTIONS	OTC clotrimazole is a sensible first pharmacy step.

Treatment ladder

1 **Moisture control**

Dry folds after washing (pat dry or cool hairdryer). Interface textiles or cotton separators; barrier products for incontinence.

2 **Topical antifungal**

Imidazole (clotrimazole or miconazole) or nystatin

3 **Very inflamed**

Short combination with mild steroid (hydrocortisone + clotrimazole or miconazole), per local formulary, **about 7 days**.

4 **Severe or recalcitrant**

Oral **fluconazole**: check interactions and pregnancy status.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Find the cause**: Review diabetes, antibiotics, immunosuppression.
- **Toe webs**: Often mixed yeast, dermatophyte and bacteria: dry thoroughly plus the right antimicrobial.

Referral

URGENT — Spreading **cellulitis**, fever or systemically unwell: escalate.

ROUTINE — Recurrent or extensive: **screen for diabetes**, consider immunosuppression.

ROUTINE — No response to an adequate course: rethink (psoriasis, tinea, erythrasma) and sample.

Red flags and when to refer

- Recurrent or extensive candida: **screen for diabetes** and consider immunosuppression.
- No response to an adequate course: reconsider the diagnosis (psoriasis, tinea, erythrasma) and take samples.
- Spreading cellulitis, fever or a systemically unwell patient: escalate.

Complications and prognosis

Complications

- Secondary bacterial infection; fissuring and pain.
- Spread to other folds, nails (chronic paronychia) or mouth.
- Persistent post-inflammatory pigmentation.

Prognosis

- Responds well to treatment, but **recurs** unless moisture and risk factors are controlled.

Prescriber pearls (UK pharmacist IP)

💡 **Satellites mean Candida.** No satellites and a very sharp edge suggests flexural psoriasis.

- Treatment fails without moisture control. Make the drying advice practical and specific.
- OTC clotrimazole is a sensible first pharmacy intervention for limited disease.
- In finger-web disease, ask about wet work and rings.
- **Satellites mean Candida until proven otherwise.**
- The toe-web 'dermatophytosis complex' is often a **mixed** yeast, dermatophyte and bacterial picture. It needs thorough drying plus the right antimicrobial.
- KOH-negative fold rash, or one that keeps coming back? Work through intertrigo and the flexural differential.

Self-check quiz

5 questions · optional

Q1 What is the key diagnostic sign of candidal intertrigo?

✓ Satellite papules and pustules beyond the main edge.

Q2 Which blood test for recurrent candidal intertrigo?

✓ HbA1c or glucose: screen for diabetes.

Q3 First-line topical treatment and duration?

✓ An imidazole (clotrimazole or miconazole) or nystatin until 1–2 weeks after clearance.

Q4 Sharp salmon-pink fold plaque with no satellites?

✓ Think flexural (inverse) psoriasis.

Q5 What is the single most important step?

✓ Moisture control: keep the fold dry.

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Photo gallery

Candidal intertrigo: groin, toe web, finger web 3

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Module 5: candidal and simple intertrigo 2

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Differential pairing: candida vs tinea cruris vs flexural psoriasis 3

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CHAPTER 8

Infections — bacterial

Superficial to deep bacterial skin infections: impetigo (including MRSA), cellulitis and erysipelas, folliculitis, boils and carbuncles, scarlet fever and cutaneous tuberculosis.

1. Impetigo
2. Cellulitis & erysipelas
3. Folliculitis
4. Boils & carbuncles
5. Scarlet fever & strawberry tongue
6. Cutaneous tuberculosis (scrofuloderma & others)

Infections — bacterial Adult core · deepened with Module 5 Week 1

Impetigo

Highly contagious superficial infection with honey-coloured crusts or flaccid blisters. Follow NICE NG153: hydrogen peroxide 1% first for localised disease, oral flucloxacillin for bullous or widespread disease.

ALSO CALLED school sores · honey crust · bullous impetigo · non-bullous impetigo · hydrogen peroxide cream · fusidic acid · +3

RELATED PAGES Boils & carbuncles · Folliculitis · Eczema / dermatitis · Cellulitis & erysipelas

KEY FACTS

- Superficial, contagious bacterial skin infection, mostly Staphylococcus aureus, sometimes Streptococcus pyogenes.
- Most common in **young children**; spreads by direct contact.
- Non-bullous honey-coloured crusts around the nose and mouth; **bullous**: flaccid blisters from staphylococcal toxin.
- UK (NICE NG153): localised and well **hydrogen peroxide 1% cream**; topical antibiotic if unsuitable; oral flucloxacillin if widespread or unwell.
- Stay off school or nursery until lesions crust over or 48 h after starting antibiotics.

Overview

Impetigo is a **superficial bacterial infection of the epidermis**. It is highly contagious and most common in children. It is caused by Staphylococcus aureus, sometimes with Streptococcus pyogenes.

FORM	DESCRIPTION
Non-bullous (about 70%)	Crusted, honey-coloured lesions; often around the face and on the limbs
Bullous	Thin-roofed, flaccid bullae that rupture to leave erosions with a collarette of scale; more often systemic symptoms; common in infants

Epidemiology

- **Who**
  - Most common skin infection in children aged **2–5 years**, but any age can be affected.
  - More common in warm, humid weather and in crowded living conditions.
- **Spread**
  - Highly contagious by direct contact, towels and toys; outbreaks in nurseries, schools and sports teams.

Aetiology: causes and risk factors

A **break in the skin barrier** lets bacteria in. Common causes are eczema, scabies, insect bites, minor trauma and herpes simplex.

NON-BULLOUS
infection of the epidermis, producing an exudate that dries to a honey-coloured crust.

BULLOUS
staphylococcal **exfoliative toxins** cleave **desmoglein-1** in the upper epidermis, producing flaccid blisters. The same toxins, spread systemically, cause staphylococcal scalded skin syndrome.

Spread is by direct contact, towels and fomites, and scratching (autoinoculation, 'kissing lesions').

Risk factors and associations

- **ORGANISMS**
  S. aureus (most cases in the UK) and S. pyogenes (group A streptococcus), alone or together.
- **Primary** impetigo on intact skin, or **secondary** impetigo (impetiginisation) on damaged skin: eczema, scabies, insect bites, chickenpox, herpes simplex, trauma, burns.
- Risk factors: nasal carriage of S. aureus, immunosuppression, diabetes, poor hygiene, crowding, contact sports.

Myths to address

"IMPETIGO MEANS POOR HYGIENE"
it is very common in clean, healthy children.

"FUSIDIC ACID IS FIRST LINE"
NICE now favours hydrogen peroxide 1% cream for localised disease to reduce resistance.

Pathophysiology and pathology

Intact skin resists these bacteria; a **break in the barrier** lets them attach and multiply in the epidermis.

1. **Barrier breach and attachment**
   Minor trauma, eczema or bites expose **fibronectin** receptors; staphylococcal and streptococcal adhesins (e.g. teichoic acids) bind and colonise the surface.

2. **Non-bullous impetigo**
   Bacteria multiply under the stratum corneum, causing a neutrophilic pustule that ruptures; serum dries into **golden crusts**.

Fig. 1 · Non-bullous impetigo (crusted pustule under the stratum corneum) compared with bullous impetigo (toxin splitting the granular layer).
AI-generated illustration

3 **Bullous impetigo**

S. aureus phage group II releases **exfoliative toxins A and B**, which cleave **desmoglein 1** in the granular layer, so the upper epidermis lifts into flaccid blisters, even on intact skin.

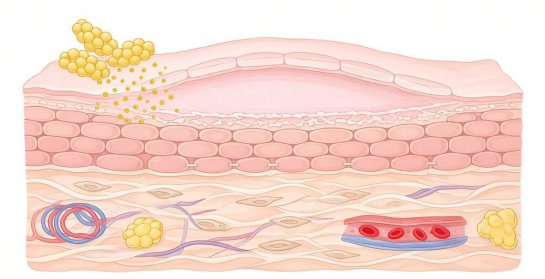

Fig. 2 · Bullous impetigo: staphylococcal toxins cut the connections between upper epidermal cells, lifting a flaccid blister.

AI-generated illustration

4 **Local spread**

Scratching and contact carry bacteria to new sites (autoinoculation) and other people.

Fig. 3 · Typical distribution around the nose and mouth.

AI-generated illustration

HISTOPATHOLOGY

- Subcorneal pustule with neutrophils and Gram-positive cocci.
- Bullous impetigo: cleavage in the **granular layer** with few inflammatory cells (the same level as staphylococcal scalded skin syndrome).
- Diagnosis is clinical; swab only if recurrent, extensive, or not responding.

Clinical features

- **Honey-coloured (golden) crusts** on a red base, often around the nose and mouth, and on the hands and limbs.
- Satellite lesions, and 'kissing lesions' where skin surfaces touch.
- **BULLOUS**
  intact or burst thin-walled blisters, shallow erosions and collarette scale. Often in the axillae, trunk or nappy area.
- Usually mildly itchy or sore. Fever and lymphadenopathy are more likely with bullous or extensive disease.

Fig. 4 · Non-bullous impetigo: golden 'honey-coloured' crusts around the mouth.
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Forms to recognise

Non-bullous

Golden ‘honey’ crusts. Site: Nose, mouth.

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Skin of colour

Crusted golden lesions; redness less visible. Site: Face.

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Bullous

Flaccid blister, collarette edge. Site: Trunk, axillae, nappy.

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Special groups

- **Skin of colour**
  - Erythema is less visible; crusting and post-inflammatory hypo- or hyperpigmentation are prominent.
- **Newborns**
  - Bullous impetigo in neonates needs prompt assessment (risk of spread and SSSS).

Diagnosis and investigations

- **Clinical.**
- Swab if: recurrent, treatment has failed, MRSA risk, or widespread disease.
- Look for and treat an **underlying dermatosis** (eczema, scabies, head lice).
- Examine for **eczema herpeticum**: monomorphic punched-out erosions on eczematous skin, often with an unwell patient.

Clinical

- Diagnosis is clinical.

Swab

- Recurrent, failed treatment, **MRSA risk**, widespread, outbreak.

Find the trigger

- Eczema, scabies, head lice.

Exclude HSV

- Monomorphic **punched-out** erosions on eczema, unwell **eczema herpeticum**.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Eczema herpeticum	Grouped vesicles punched-out erosions; painful; unwell.
	Eczema flare	Itchy, ill-defined; crusts less golden.
	Herpes simplex	Grouped vesicles at a recurrent site.
	Bullous bites	Grouped on exposed sites; tense blisters.
	Pemphigus foliaceus	Adult, recurrent superficial erosions; no antibiotic response.

Management

Aligned with **NICE NG153** (antimicrobial prescribing).

Everyone

- Hygiene advice to reduce spread: wash the lesions gently, do not share towels, and avoid scratching.
- **SCHOOL OR NURSERY**
  keep the child off until lesions are crusted and healed, or for **48 hours after starting antibiotics**, in line with UKHSA and local public health advice.

Localised non-bullous, not unwell

- **Hydrogen peroxide 1% cream** 2–3 times daily for 5 days.
- If unsuitable or ineffective: a short course of **topical antibiotic**, usually **fusidic acid** for 5 days. Reserve mupirocin for MRSA in line with the local pathway.

Widespread non-bullous, not unwell

Topical **or** oral antibiotic for 5 days (shared decision). Repeated topical antibiotic use drives resistance.

Bullous, systemically unwell, or high risk of complications

- **Oral antibiotic** for 5 days (up to 7 if severe).
- First line: **flucloxacillin**. Penicillin allergy: **clarithromycin** (erythromycin in pregnancy).

MRSA and non-response (Module 5)

- MRSA is becoming more common. **Swab** if impetigo is not responding, recurrent, part of an outbreak, or there are MRSA risk factors (recent hospital stay, care home, known carriage).
- Choose the antibiotic using the **swab sensitivities and local microbiology advice**. Flucloxacillin and fusidic acid will not reliably treat MRSA.
- Do **not** keep repeating fusidic acid courses. Resistance develops quickly.
- Recurrent impetigo: consider nasal carriage and decolonisation for the patient and household, through the local pathway.

⚠ **CAUTION**

Don't reach for a steroid–antibiotic combination by default

Use a topical antibiotic plus steroid only when you have actually diagnosed infection on inflamed (e.g. eczematous) skin, and keep the course short.

Medicines at a glance

Hydrogen peroxide 1% cream

Localised, not unwell

DOSE	2–3 times daily · 5 days
CAUTIONS	Stewardship win.

Fusidic acid (if peroxide unsuitable or fails)

Localised, not unwell

DOSE	Topical · 5 days
CAUTIONS	Don't repeat courses: resistance. Mupirocin only for MRSA via local pathway.

Flucloxacillin (oral)

Bullous, unwell or high risk

DOSE	As BNF · 5 days (up to 7 if severe)
CAUTIONS	Penicillin allergy: clarithromycin (erythromycin in pregnancy).

Treatment ladder

1 **Everyone**

Wash lesions gently; no shared towels; avoid scratching. School / nursery: off until crusted and healed, or **48 h after starting antibiotics** (UKHSA).

2 **Localised, not unwell**

Hydrogen peroxide 1% cream; Fusidic acid (if peroxide unsuitable or fails)

3 **Widespread, not unwell**

Topical **or** oral antibiotic for 5 days (shared decision).

4 **Bullous, unwell or high risk**

Flucloxacillin (oral)

5 **MRSA or not responding**

Swab; choose by **sensitivities** and local microbiology. Recurrent: consider nasal carriage and household decolonisation.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Steroid–antibiotic combos**: Only for proven infection on inflamed skin, briefly.
- **Safety-net**: Worse after 2–3 days, spreading redness or fever review.

Referral

SAME DAY	Suspected **eczema herpeticum**.
SAME DAY	**Infant with bullous impetigo**, or suspected **SSSS** (tender widespread redness, peeling).
URGENT	Cellulitis, deeper infection, or systemically unwell.
ROUTINE	Immunocompromised with widespread disease, or frequent recurrence (carriage, MRSA).

Red flags and when to refer

- Suspected **eczema herpeticum**: same-day referral.
- Signs of **cellulitis** or deeper infection, or a systemically unwell patient.
- **Infants with bullous impetigo**, or suspected staphylococcal scalded skin syndrome (tender, widespread erythema with peeling).
- Immunocompromised with widespread disease, or frequent recurrence (consider nasal carriage and MRSA).

Complications and prognosis

Complications

- **Cellulitis**, lymphangitis and bacteraemia.
- **Staphylococcal scalded skin syndrome** (young children).
- **ECTHYMA**
  deeper, punched-out ulcers that scar.
- Post-streptococcal **glomerulonephritis** (2–6 weeks later) and, rarely, scarlet fever or streptococcal toxic shock.
- Post-inflammatory pigment change; scarring is uncommon.

Prognosis

- Usually self-limiting; without treatment it heals in about 2–3 weeks, and within about 10 days with treatment.
- Recurrence suggests nasal carriage or an untreated underlying skin condition.

Prescriber pearls (UK pharmacist IP)

- Hydrogen peroxide 1% is first line for localised disease. It is a useful **stewardship** win.
- **Avoid repeated fusidic acid** courses because of resistance. Use 5 days, then stop.
- Crust does not always mean impetigo, so think about **HSV** first in eczema.
- Treat the trigger (eczema, scabies, lice) or impetigo will keep coming back.
- Safety-net: worsening after 2–3 days, spreading redness or fever means review.
- **Skin of colour:** crusts and erythema can be subtler. Palpate, and rely on the history (spread to contacts, weeping, crusting).
- New to skin terms? Describing skin lesions explains crust, erosion, bulla and vesicle.

Self-check quiz

5 questions · optional

Q1 First-line treatment for localised non-bullous impetigo (NG153)?

✓ Hydrogen peroxide 1% cream 2–3 times daily for 5 days.

Q2 Bullous or systemically unwell: what antibiotic?

✓ Oral flucloxacillin for 5 days (up to 7); clarithromycin if penicillin-allergic (erythromycin in pregnancy).

Q3 What causes the blisters in bullous impetigo?

✓ Staphylococcal exfoliative toxins cleaving desmoglein-1 in the upper epidermis.

Q4 When can a child return to school?

✓ When lesions are crusted and healed, or 48 hours after starting antibiotics.

Q5 Punched-out erosions on eczema in an unwell child?

✓ Eczema herpeticum: same-day referral.

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Photo gallery

Non-bullous impetigo: honey-coloured crusts

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Bullous impetigo

3

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Module 5: non-bullous and bullous impetigo

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Do not confuse: eczema herpeticum

1

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AI teaching illustration

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Infections — bacterial Adult core · deepened with Module 5 Week 2

Cellulitis & erysipelas

Spreading dermal infection, usually in one leg. Exclude the mimics, draw the border, give flucloxacillin for 5–7 days and safety-net hard. Bilateral red legs are rarely cellulitis.

ALSO CALLED cellulitis · erysipelas · red leg · skin and soft tissue infection · SSTI · periorbital cellulitis · +7

RELATED PAGES Red leg: differentials & referral · Venous leg ulcer · Tinea: corporis, cruris & pedis · Boils & carbuncles · Impetigo

KEY FACTS

- Acute bacterial infection of the **dermis and subcutis** (cellulitis) or **upper dermis and lymphatics** (erysipelas).
- Usually Streptococcus pyogenes (about two thirds) or Staphylococcus aureus (about one third).
- Typically **one** hot, red, swollen, tender leg; bilateral red legs are rarely cellulitis.
- UK: **NICE NG141**; flucloxacillin 500 mg–1 g QDS for 5–7 days for most adults; mark the border and review at 2–3 days.
- Emergency if **necrotising fasciitis** or sepsis is possible.

Overview

CONDITION	DEPTH	EDGE	USUAL ORGANISM
Erysipelas	Superficial dermis and lymphatics	Sharp, raised, well-demarcated	Usually streptococci
Cellulitis	Deeper dermis and subcutis	Less distinct	S. pyogenes, S. aureus

In practice the two overlap and are managed the same way in primary care.

Epidemiology

- **Common**
  - One of the commonest reasons for antibiotic prescribing in primary care and for emergency admission with skin disease.
- **Who**
  - All ages; more common in **middle-aged and older adults**.
  - Erysipelas is more common in older adults and infants.
- **Recurrence**
  - Up to about a third have a further episode, especially with lymphoedema, venous disease or untreated tinea pedis.

Aetiology: causes and risk factors

- A **breach in the skin barrier** lets bacteria in. Common entry points are **tinea pedis**, eczema, ulcers, trauma, bites, lymphoedema and previous surgery.
- Bacteria spread through the dermis and subcutaneous tissue, with or without lymphangitis.
- The host inflammatory response causes the redness, heat, swelling and systemic symptoms.
- Risk factors: obesity, venous insufficiency, lymphoedema, diabetes, and previous cellulitis.

PATIENT	USUAL ORGANISMS (MODULE 5)
Immunocompetent	S. pyogenes (about two-thirds) and S. aureus (about one-third)
Immunocompromised	A broader range, including **Gram-negatives**, so take advice

Risk factors and associations

> **PORTAL OF ENTRY**
> toe-web **tinea pedis** or fissured heels, eczema, ulcers, trauma, bites, surgical wounds, IV drug use.

> **HOST FACTORS**
> previous cellulitis, **lymphoedema**, venous insufficiency, **obesity**, diabetes, immunosuppression, chronic kidney or liver disease, pregnancy, alcohol excess.

> **ORGANISMS IN SPECIAL SETTINGS**
> Pasteurella (cat or dog bites), Pseudomonas (puncture wounds of the foot), Vibrio or Aeromonas (salt or fresh water), MRSA, Gram-negatives in the immunocompromised.

Myths to address

> **"IT WAS A SPIDER BITE"**
> documented spider bites rarely cause cellulitis; look for tinea pedis or a skin break.

> **"STILL RED ON DAY 3, SO THE ANTIBIOTIC FAILED"**
> redness takes days to settle; check the border and the patient.

> **"CELLULITIS IS CONTAGIOUS"**
> it is not usually spread person to person.

Pathophysiology and pathology

Bacteria cross a **breach in the skin**, multiply in the dermis and trigger an acute inflammatory response.

1. **A breach in the barrier**
   A fissure, cut, bite, ulcer or **macerated toe web** lets skin flora into the dermis; lymphoedema and venous stasis impair clearance.

2. **Bacteria multiply and spread**
   Streptococci release hyaluronidase, streptokinase and DNases that help them spread through tissue planes; staphylococci form local abscesses more often.

3 **Depth decides the edge**

Erysipelas involves the upper dermis and lymphatics, giving a raised, sharp edge; **cellulitis** extends into deep dermis and fat, so the edge is flat and ill-defined.

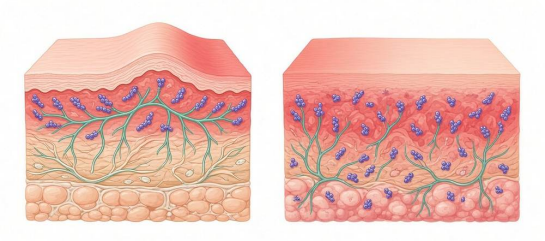

Fig. 1 · Erysipelas (left) in the upper dermis and lymphatics vs cellulitis (right) in the deep dermis and fat.

AI-generated illustration

4 **Inflammation**

Cytokines (TNF, IL-1, IL-6) cause the **redness, heat, swelling and pain**, fever and malaise; toxins can cause systemic illness.

Fig. 2 · Typical cellulitis in one leg, with the portal of entry in a fissured toe web.

AI-generated illustration

5 **Healing and damage**

Lymphatics damaged by each episode drain less well, increasing oedema and the risk of **recurrence**.

HISTOPATHOLOGY

- Dermal oedema, dilated lymphatics and a diffuse neutrophilic infiltrate extending into subcutaneous fat.
- Organisms are often scanty; biopsy is not used for diagnosis.

Clinical features

CELLULITIS

expanding redness, heat, swelling and tenderness, usually in **one** leg. May have fever, malaise, lymphadenopathy and lymphangitic streaking. Can have blisters.

ERYSIPELAS

bright red, hot, **raised, sharply demarcated** plaque, often on the face or leg. Fever and chills are common and may come before the rash.

SKIN OF COLOUR

redness is much less obvious, so rely on **swelling, warmth, pain and induration**.

- Look for **peau d'orange**, bullae, purpura, **lymphangitic streaks** and tender regional lymph nodes.
- Fever and rigors can come **before** the rash.

PRESCRIBER TIP

Milian's ear sign

Redness involving the **pinna** favours **erysipelas** over cellulitis. The ear has no deep dermis or subcutaneous fat, so deep cellulitis cannot really involve it.

Fig. 3 · Cellulitis: spreading erythema of the limb with the border marked in pen to monitor progression.

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Look for

Peau d'orange

Dimpled swollen skin: mark and reassess. Site: Leg.

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Bullous cellulitis

Still a soft-tissue infection; escalate if severe. Site: Leg.

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Facial erysipelas

Bright, well-demarcated plaque. Site: Face.

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Milian's ear sign

Pinna involved erysipelas. Site: Ear.

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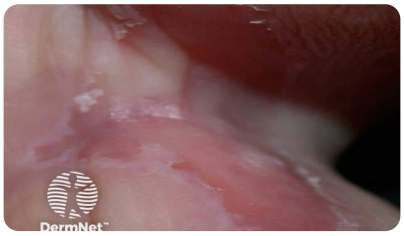
Portal of entry

Macerated toe-web tinea. Site: Toe webs.



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Special groups

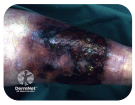
- **Skin of colour**
  - Redness may be hard to see: rely on warmth, swelling, tenderness and induration, and compare with the other leg.
- **Children**
  - Periorbital or facial cellulitis needs a low threshold for same-day assessment.
- **Pregnancy**
  - Erythromycin if penicillin allergic (NICE NG141).

Diagnosis and investigations

- **Clinical.** Exclude non-infective mimics before giving antibiotics.
- **Mark the border** with a skin pen and record the date and time. Photograph if possible, to monitor spread.

- Assess severity: observations, NEWS2 or sepsis screen, comorbidities and site (face or periorbital).
- Swab only if there is broken skin with a penetrating injury, water exposure, or infection acquired outside the UK.
- Bloods and blood cultures belong in the hospital pathway for unwell patients.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Venous eczema / lipodermatosclerosis	**Bilateral**, chronic, itchy, little pain, no fever.
	DVT	Calf swelling and tenderness, less redness; Wells, D-dimer, Doppler.
	Gout	Joint-centred, very tender, past attacks.
	Insect bite reaction	Itch > pain, central punctum, rapid onset.
	Contact dermatitis	Itchy, vesicular, matches exposure, no fever.
	**Necrotising fasciitis**	**Pain out of proportion**, dusky skin, crepitus: 999.

⚠ **RED FLAG**

Necrotising fasciitis

Severe **pain out of proportion** to the signs, rapid spread, systemic toxicity, dusky or purple skin, bullae or crepitus. This is an **emergency**: 999 or ED.

See also: Red leg: differentials & referral

Management

Aligned with **NICE NG141** and CKS.

Supportive

- Draw around the edge, elevate the limb, give analgesia and encourage fluids.
- Use emollient for dry skin. **Avoid compression** during acute cellulitis.

- **TREAT THE PORTAL OF ENTRY**
  tinea pedis, eczema, ulcers, lymphoedema.

Antibiotics (adults, oral)

SITUATION	CHOICE
First line	**Flucloxacillin** 500 mg–1 g four times daily for **5–7 days** (longer if clinically needed)
Penicillin allergy	Clarithromycin, doxycycline or erythromycin (erythromycin in pregnancy)
Near the eyes or nose (periorbital / 'danger triangle')	**Co-amoxiclav** (penicillin allergy: clarithromycin + metronidazole). Consider specialist advice
Severe, or unable to take oral	IV in hospital or the community IV service; review switch to oral by 48 hours

- **Reassess** if worse at any time, or not improving after **2–3 days**.
- Recurrent cellulitis (2 or more episodes in 12 months): a specialist may consider prophylaxis (e.g. phenoxymethylpenicillin). This is not routine in primary care.

NOTE

Recurrent cellulitis

Two or more episodes: treat the portals of entry (tinea pedis, eczema), manage oedema with compression once the acute episode has settled, and discuss prophylactic penicillin with a clinician (based on the PATCH trials).

Medicines at a glance

Flucloxacillin

First line · adults

DOSE 500 mg–1 g four times daily · 5–7 days (longer if clinically needed)

CAUTIONS Penicillin allergy: clarithromycin, doxycycline or erythromycin (pregnancy).

Co-amoxiclav

Near eyes or nose

DOSE As BNF · Per NG141

CAUTIONS Penicillin allergy: clarithromycin + metronidazole. Consider specialist advice.

Treatment ladder

1

Supportive

Elevate, analgesia, fluids; emollient for dry skin. **No compression** during acute cellulitis.

2

First line · adults

Flucloxacillin

3

Near eyes or nose

Co-amoxiclav

4

Severe or can't take oral

IV in hospital or community IV service; review switch to oral by 48 h.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Day 3 redness**: Skin takes days to settle: not automatic failure.
- **Not better at 48–72 h**: Swab open wounds, consider MRSA, **revisit the diagnosis**.
- **Recurrent (2 in 12 months)**: Treat portals, compression once settled; specialist may consider penicillin prophylaxis.

Referral

999 / A&E — **Necrotising fasciitis**: pain out of proportion, rapid spread, purple bullae, crepitus, very unwell.

999 / A&E — **Sepsis**: low BP, raised HR or RR, confusion, mottled skin.

SAME DAY — **Periorbital / orbital**: pain on eye movement, proptosis, reduced vision.

URGENT Spread beyond the border, or failure after 48–72 h oral therapy.

ROUTINE Infants, frail elderly, immunocompromised, lymphoedema: lower threshold.

Red flags and when to refer

> **NECROTISING FASCIITIS SIGNS**
> pain out of proportion, rapid spread, purple bullae, crepitus, or a very unwell patient. **Emergency.**
>
> **SEPSIS**
> low blood pressure, raised heart or breathing rate, confusion, mottled skin.
>
> **PERIORBITAL OR ORBITAL CELLULITIS**
> pain on eye movement, proptosis, reduced vision. Same-day ophthalmology or ED.
>
> Rapid spread beyond the marked border, or failure after 48–72 hours of oral therapy.
>
> Infants, frail older adults, the immunocompromised, or significant lymphoedema: lower threshold to refer.

Complications and prognosis

Complications

- **Necrotising fasciitis**, gas gangrene and streptococcal toxic shock.
- **Sepsis**; bacteraemia with seeding to bone, joints or heart valves (endocarditis).
- Abscess formation; lymphangitis.
- **Recurrent cellulitis** and secondary **lymphoedema**.
- Orbital cellulitis with visual loss or cavernous sinus thrombosis (facial infection).

Prognosis

- Most cases respond to oral antibiotics without lasting harm; skin can take weeks to look normal.
- Recurrence is common; treating portals of entry and oedema reduces it.

Prescriber pearls (UK pharmacist IP)

Bilateral red legs are not cellulitis until proven otherwise. Think venous eczema or lipodermatosclerosis.

The skin takes **days** to look normal after the bacteria are controlled. Residual redness on day 3 is not automatic failure, so avoid reflexively extending the course.

Always treat athlete's foot to reduce recurrent leg cellulitis.

On the face, have a lower threshold for review or admission.

Safety-net: rapid worsening, severe pain, purple blisters or feeling faint means same-day emergency care.

At 48–72 hours, if it is not better: swab any open wound, consider MRSA, and **revisit the diagnosis**.

Self-check quiz

5 questions · optional

Q1 Erysipelas vs cellulitis: which has a sharp raised edge and why?

✓ Erysipelas: it involves the upper dermis and lymphatics. Cellulitis is deeper, with an ill-defined edge.

Q2 First-line antibiotic and duration (NG141)?

✓ Flucloxacillin 500 mg–1 g four times daily for 5–7 days.

Q3 Which antibiotic near the eyes or nose?

✓ Co-amoxiclav (penicillin allergy: clarithromycin + metronidazole).

Q4 Pain out of proportion with rapid spread?

✓ Possible necrotising fasciitis: emergency, 999 / ED.

Q5 Both legs red and itchy, no fever?

✓ Probably not cellulitis: think venous eczema or lipodermatosclerosis.

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emedicine.medscape.com

Photo gallery

Cellulitis: mark the margin

4

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Erysipelas: sharp, raised edge, and Milian's ear sign

5

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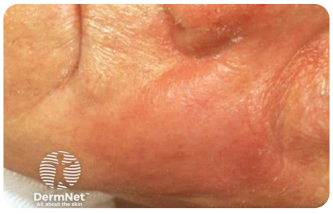

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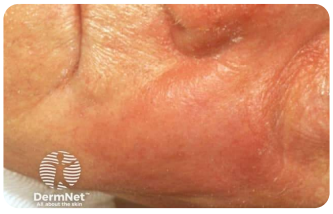

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Portal of entry: interdigital tinea pedis

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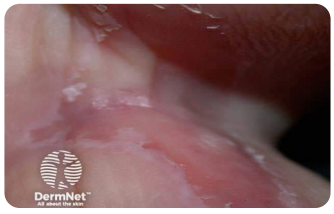

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Infections — bacterial

Folliculitis

Pustules centred on hair follicles, usually from S. aureus, shaving or occlusion. Most cases need hygiene advice and an antiseptic rather than an antibiotic, so think about other causes if it recurs.

ALSO CALLED bacterial folliculitis · hot tub folliculitis · pseudomonas folliculitis · pityrosporum folliculitis · malassezia folliculitis · shaving rash · +1

RELATED PAGES Boils & carbuncles · Acne mimics: rosacea, steroid eruptions and folliculitis · Acne vulgaris · Impetigo

KEY FACTS

- Inflammation of hair follicles: **follicular papules and pustules** centred on hairs.
- Most often bacterial (Staphylococcus aureus); also Malassezia, Pseudomonas (hot tubs), Gram-negative, viral and non-infective causes.
- Triggers: shaving, waxing, occlusion, sweating, friction, topical steroids and antibiotics.
- Mild: antiseptic washes and stop the trigger; persistent bacterial: topical or oral antibiotic per culture; Malassezia: antifungals.
- Deep or spreading infection (furuncle, cellulitis) needs review.

Overview

Folliculitis is **inflammation of the hair follicle**, seen as small pustules or papules centred on hairs. Superficial bacterial folliculitis is usually caused by S. aureus.

Deeper follicular infection becomes a **boil** (furuncle). Clusters form a **carbuncle**. See Boils & carbuncles.

Epidemiology

- **Who**
  - Common at all ages worldwide; most often adolescents and young adult men.
  - Pseudofolliculitis barbae is especially common in men with tightly curled hair.

Aetiology: causes and risk factors

- Follicular occlusion or damage from **shaving, waxing, tight clothing, oils**, sweating and friction.
- Bacterial proliferation in the follicle opening, usually S. aureus. Nasal carriage drives recurrence.
- Topical steroids and immunosuppression increase risk.

Not all folliculitis is staphylococcal

TYPE	CLUE	PATHWAY
Hot tub (Pseudomonas)	1–2 days after a hot tub or spa; under swimwear	Usually self-limiting; check spa hygiene
Pityrosporum (Malassezia)	Itchy, uniform papules on the upper back and chest; young adults; after antibiotics	**Antifungal**, not antibiotic
Gram-negative	Long-term antibiotics for acne	Specialist
Eosinophilic	Very itchy; HIV or immunosuppression	Specialist
Pseudofolliculitis barbae	Ingrown hairs in curly beard hair after close shaving	Shaving technique, beard growth

Risk factors and associations

INFECTIVE

S. aureus (commonest), coagulase-negative staphylococci, Pseudomonas (spa pools), Gram-negative (after long antibiotic use for acne), Malassezia, dermatophytes, HSV, Demodex.

NON-INFECTIVE

shaving (pseudofolliculitis), occlusion and oils, drugs (steroids, EGFR inhibitors, lithium), eosinophilic folliculitis (HIV).

HOST FACTORS

atopic eczema, diabetes, obesity, immunosuppression, nasal S. aureus carriage.

Pathophysiology and pathology

The level of the follicle that is inflamed decides how the lesion looks and how deep the risk goes.

1. **Follicle damage**

   Shaving, friction or occlusion damage the follicular opening and let organisms in.

Fig. 1 · Superficial folliculitis (top of the follicle), furuncle (deep follicle abscess) and carbuncle (linked follicles).

AI-generated illustration

2. **Superficial folliculitis**

   Organisms multiply in the upper follicle; neutrophils gather into a small pustule pierced by a hair.

3. **Deep folliculitis**

   Inflammation extends down the follicle into the dermis (sycosis, furuncle), giving tender nodules and possible scarring.

4 **Ingrowing hairs**

In pseudofolliculitis, tightly curled hairs re-enter the skin after shaving, causing a foreign-body reaction.

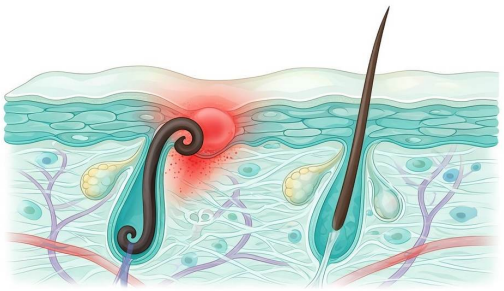

Fig. 2 · Pseudofolliculitis: a curled hair re-entering the skin next to its follicle.

AI-generated illustration

HISTOPATHOLOGY

- Neutrophils in and around the follicle; organisms on Gram or PAS stain.
- Swab pustules for culture when persistent, recurrent or after long antibiotic use.

Clinical features

- **Follicular-centred pustules**, often with a hair visible in the centre, on a red base.
- Sites: scalp, beard (sycosis barbae), trunk, buttocks, thighs and axillae.
- Itchy or tender. Systemic upset is unusual.

Fig. 3 · Superficial bacterial folliculitis: small pustules each centred on a hair follicle.

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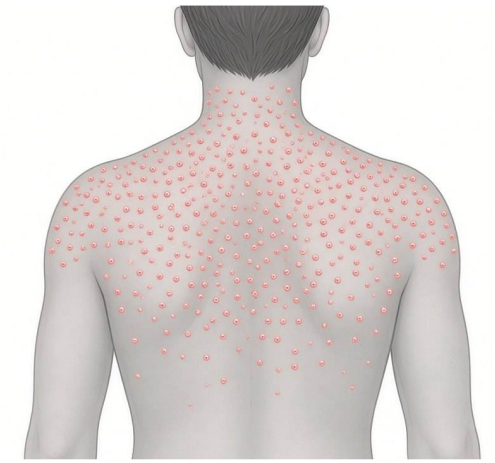

Fig. 4 · Pityrosporum (Malassezia) folliculitis: itchy, uniform small papules and pustules on the upper back and shoulders, often in young adults after antibiotics. It needs an antifungal, not an antibiotic.

AI-generated illustration

Look for

Superficial bacterial

Small pustules centred on follicles. Site: Any hair-bearing skin.

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Hair-bearing skin

Scattered inflamed follicular papules. Site: Thighs, buttocks.

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Axillary

After shaving. Site: Axillae.

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Special groups

- **Skin of colour**
  - Pseudofolliculitis barbae and keloidal scarring are common; post-inflammatory hyperpigmentation is often the main concern.

Diagnosis and investigations

- Clinical. The distribution and history (shaving, hot tub, recent antibiotics) point to the type.
- **Swab a pustule** for culture if persistent, recurrent, extensive or atypical.
- Consider a **nasal swab** for S. aureus carriage in recurrent disease.

Clinical

- Distribution + history point to the type.

Ask

- Shaving, waxing, hot tub, recent antibiotics, steroid creams?

Swab

- A pustule if persistent, recurrent, extensive or atypical.

Recurrent

- **Nasal swab** for carriage; screen for **diabetes**.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Acne vulgaris	Comedones; face, chest, back; mixed lesions.
	Pustular miliaria	Heat and occlusion; pustules not follicle-centred.
	Candida	Satellite pustules around a red moist fold.
	Keratosis pilaris	Rough follicular papules, upper arms, thighs; not pustular.
	Malassezia / Gram-negative	After antibiotics; needs a different treatment.

Management

Mild

- Stop or modify the trigger: change shaving technique (a clean sharp blade, shave with the grain, or a pause in shaving), use loose clothing, and avoid oils and occlusion.
- Use an **antiseptic wash** (e.g. chlorhexidine or a benzoyl peroxide wash).
- Often **no antibiotic** is needed.

Persistent bacterial

- Topical antiseptic or antibiotic in line with local guidance, for short courses.
- Oral **flucloxacillin** if extensive (clarithromycin if penicillin allergy).

Recurrent

Swab, consider decolonisation (via the local pathway), and screen for diabetes. Think about other causes such as Pityrosporum or Gram-negative folliculitis.

Medicines at a glance

Flucloxacillin (oral)

Extensive

DOSE	As BNF · Short course
CAUTIONS	Penicillin allergy: clarithromycin.

Treatment ladder

1 **Mild**

Change shaving technique (clean sharp blade, with the grain, or pause). Loose clothing; avoid oils and occlusion.

2 **Persistent bacterial**

Topical antiseptic or antibiotic, short course, per local guidance.

3 **Extensive**

Flucloxacillin (oral)

4 **Recurrent**

Swab; decolonisation via local pathway; screen for diabetes. Rethink: Pityrosporum (antifungal) or Gram-negative (specialist).

Referral

URGENT Progression to **boils, carbuncles or cellulitis**, or fever.

ROUTINE Recurrent despite measures: carriage, **PVL** or MRSA, diabetes, immunosuppression.

ROUTINE Widespread itchy folliculitis in immunosuppression (eosinophilic): refer.

Red flags and when to refer

- Progression to **boils, carbuncles or cellulitis**, or fever.
- Recurrent disease despite measures: consider carriage, PVL or MRSA, diabetes and immunosuppression.
- Widespread itchy folliculitis in an immunosuppressed patient (eosinophilic): refer.

Complications and prognosis

Complications

- Furuncles and carbuncles; cellulitis.
- Scarring, keloids and permanent hair loss in deep or chronic forms.
- Post-inflammatory pigmentation.

Prognosis

- Superficial folliculitis usually settles in days to weeks once the trigger is removed.
- Recurrence is common with ongoing shaving, occlusion or carriage.

Prescriber pearls (UK pharmacist IP)

Pustules on the upper back and chest after antibiotics? Think Pityrosporum folliculitis, which needs an antifungal.

A hot-tub history explains a sudden ‘swimsuit-distribution’ rash.

Most mild bacterial folliculitis settles without antibiotics, which is a useful stewardship opportunity.

Pseudofolliculitis barbae is common in curly hair. Discuss shaving technique and beard growth sensitively.

Self-check quiz

5 questions · optional

Q1 What is the usual organism in superficial bacterial folliculitis?

✓ Staphylococcus aureus.

Q2 Swimsuit-distribution pustules 1–2 days after a spa?

✓ Hot-tub (Pseudomonas) folliculitis: usually self-limiting.

Q3 Itchy uniform papules on the upper back after antibiotics?

✓ Pityrosporum (Malassezia) folliculitis: treat with an antifungal.

Q4 First step for mild folliculitis?

✓ Stop or modify the trigger and use an antiseptic wash; often no antibiotic needed.

Q5 What to check in recurrent folliculitis?

✓ Swab, nasal carriage, diabetes; consider PVL/MRSA and other types.

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Photo gallery

Bacterial folliculitis

3

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Infections — bacterial

Boils & carbuncles

Deep staphylococcal follicular abscesses. Use warm compresses for small boils, incision and drainage for large boils and all carbuncles, and antibiotics only with specific risk features. Think about PVL if recurrent.

ALSO CALLED boil · furuncle · carbuncle · skin abscess · PVL · MRSA · +2

RELATED PAGES Folliculitis · Cellulitis & erysipelas · Hidradenitis suppurativa · Epidermoid, pilar and dermoid cysts (and Gardner syndrome)

KEY FACTS

- A **boil (furuncle)** is a deep Staphylococcus aureus infection of a hair follicle forming a painful abscess; a **carbuncle** is several linked boils.
- Common in adolescents and adults; risk factors include nasal carriage, diabetes, obesity, friction and close contact.
- Small boils: warm compresses; large or fluctuant: **incision and drainage**.
- Antibiotics (e.g. flucloxacillin) only if systemically unwell, spreading cellulitis, high-risk site or comorbidity (CKS).
- Recurrent boils: swab, consider **PVL-producing S. aureus** and decolonisation.

Overview

A **boil (furuncle)** is a deep infection of a hair follicle that forms a small abscess. A **carbuncle** is a group of interconnected boils with several openings, and is more likely to cause systemic upset.

Most are caused by Staphylococcus aureus. Recurrent or severe cases raise the possibility of **Panton–Valentine leukocidin (PVL)-producing** S. aureus or **MRSA**.

Epidemiology

- **Who**
  - Common in adolescents and young adults; outbreaks in families, sports teams, prisons and the military.

Aetiology: causes and risk factors

- Deep follicular infection, usually from the patient's own S. aureus carriage (nose, axillae, perineum).
- Friction, shaving, sweating and minor trauma help bacteria enter.
- Risk factors: **diabetes**, obesity, immunosuppression, poor hygiene, close-contact sport, crowded living, and household contacts with boils.

Risk factors and associations

ORGANISM

S. aureus, including MRSA and **Panton–Valentine leukocidin (PVL)**-producing strains (recurrent, multiple, necrotic boils).

RISK FACTORS

nasal or perineal carriage, diabetes, obesity, immunosuppression, eczema, friction, shaving, poor hygiene, close contact and shared towels.

Myths to address

- **"SQUEEZE IT TO CLEAR IT"**
  squeezing can spread infection; drain only when fluctuant and by a clinician.

Pathophysiology and pathology

A boil is a **follicular abscess**: neutrophils wall off staphylococci deep in the follicle.

1. **Follicle invasion**
   S. aureus enters a damaged follicle and spreads deep into it.

Fig. 1 · From superficial folliculitis to a deep follicular abscess (boil) and a carbuncle.
AI-generated illustration

2. **Abscess formation**
   Neutrophils and necrotic tissue form pus; a red, hot, tender nodule becomes **fluctuant** with a yellow point.

3. **Carbuncle**
   Infection spreads through connected follicles in thick skin (back of neck, back, thighs), forming a larger mass with several heads.

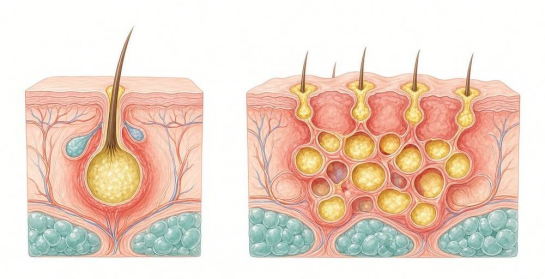

Fig. 2 · Single boil vs carbuncle with multiple draining heads.
AI-generated illustration

4 **Drainage and healing**

Pus discharges spontaneously or after incision; warm compresses help it point.

Fig. 3 · Warm compress to help a small boil come to a head.

AI-generated illustration

HISTOPATHOLOGY

- Dense neutrophilic abscess centred on a follicle with necrosis; Gram-positive cocci.
- Swab pus for culture and sensitivity (and PVL if recurrent).

Clinical features

- A **painful, red, hot nodule** that becomes fluctuant and **points** to a yellow-white head before discharging.
- Common sites: neck, face, axillae, buttocks and thighs.
- **CARBUNCLE**
  a larger, indurated area with **multiple heads**. Fever and malaise are more likely.
- Surrounding cellulitis may develop.

Fig. 4 · Furuncle (boil): tender red nodule with a central yellow pointing head of pus.

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Look for

Furuncle

Tender red nodule with a central yellow point. Site: Neck, thighs.

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With cellulitis

Spreading redness around the boil. Site: Any.

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Axillary abscess

Deep red fluctuant nodule; recurrent think HS. Site: Axillae.

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Carbuncle

Coalescent boils, multiple discharging heads. Site: Neck, back.

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Diagnosis and investigations

- Clinical.
- **Swab pus** if drained, recurrent, or PVL/MRSA is suspected.
- **RECURRENT BOILS**
  check glucose or HbA1c, ask about household and close contacts, and consider nasal carriage, PVL and MRSA risk.

Clinical

- Diagnosis is clinical.

Swab pus

- If drained, recurrent, or PVL / MRSA suspected.

Recurrent

- **Glucose or HbA1c**; nasal carriage.

Contacts

- Household, gym, team: **PVL clusters**?

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Inflamed epidermoid cyst	Long-standing lump with a punctum that flares.
	Hidradenitis suppurativa	**Recurrent** nodules, sinus tracts, scars in folds.
	Folliculitis	Superficial small pustules; no deep abscess.
	Cellulitis	Diffuse spreading redness, no collection.

Management

Small boil

- Apply **moist heat** (a warm compress) 3–4 times a day to encourage spontaneous drainage.
- **Do not squeeze.** Cover with a dressing and give hygiene advice.

Large or fluctuant boils, and all carbuncles

Incision and drainage, the same day: in primary care if skilled, otherwise via surgical or urgent care.

Antibiotics are needed if

- Fever or systemic upset
- Surrounding **cellulitis**
- **Facial** lesion, especially the central face or 'danger triangle'
- Severe pain or a large lesion
- **Diabetes** or immunosuppression

First line: **flucloxacillin**. Penicillin allergy: clarithromycin, or erythromycin in pregnancy.

Recurrent boils

Screen for diabetes, discuss PVL and MRSA, and arrange **decolonisation** (e.g. topical nasal treatment plus antiseptic body wash) through local infection control or a microbiologist. Treat household contacts if advised.

Fig. 9 · Small boils: moist heat (a warm compress) 3–4 times a day encourages spontaneous drainage. Don't squeeze, and use a separate towel.

AI-generated illustration

Medicines at a glance

Flucloxacillin

Antibiotics
if...

DOSE	As BNF · Per local guidance
CAUTIONS	Only with fever, cellulitis, facial site, large / very painful lesion, diabetes or immunosuppression. Penicillin allergy: clarithromycin (erythromycin in pregnancy).

Treatment ladder

1 **Small boil**

Moist heat (warm compress) 3–4 times a day. **Don't squeeze**; dress; hygiene advice.

2 **Large, fluctuant, or any carbuncle**

Incision and drainage the same day: primary care if skilled, otherwise surgical or urgent care.

3

Antibiotics if...

Flucloxacillin

4

Recurrent

Screen for diabetes; discuss **PVL** and MRSA. **Decolonisation** (nasal treatment + antiseptic body wash) via infection control or microbiology; household if advised.

⚠ CAUTION

Prescribing limits and pitfalls

- **Facial boils**: Don't squeeze; lower threshold for antibiotics and review.
- **Hygiene**: Own towel, hot-wash bedding, cover the lesion, wash hands.

Referral

SAME DAY	**Carbuncle** or large abscess: needs drainage.
SAME DAY	Boil on the **central face** or near the eye.
URGENT	Spreading cellulitis, fever or systemically unwell; deep infection in diabetes or immunosuppression.
URGENT	**PVL suspicion**: recurrent aggressive boils, household / team clusters, necrotising features microbiology or public health.

Red flags and when to refer

- **Carbuncle** or a large abscess: needs drainage.
- Boil on the **central face** or near the eye.
- Spreading cellulitis, fever or a systemically unwell patient.
- **PVL-S. AUREUS SUSPICION**
  recurrent, painful, aggressive boils, clusters in a household, gym or team, or necrotising features. Escalate to microbiology or public health.

- Immunocompromised or diabetic patient with a deep infection.

Complications and prognosis

Complications

- Cellulitis, lymphangitis; scarring.
- Bacteraemia, osteomyelitis, endocarditis (rare).
- Facial 'danger triangle' boils: rare cavernous sinus thrombosis.
- PVL strains: necrotising soft-tissue infection and necrotising pneumonia.

Prognosis

- Most heal within 1–2 weeks after draining.
- Recurrence is common with carriage; decolonisation helps some.

Prescriber pearls (UK pharmacist IP)

Pus needs a way out. Drainage is the treatment, and antibiotics are an add-on only with specific features.

'Recurrent boils' in the axillae or groin may be **hidradenitis suppurativa**, which follows a different pathway.

Ask about **household contacts and close-contact sport**, as PVL clusters spread this way.

Do not squeeze facial boils. Use a lower threshold for antibiotics and review.

Give practical hygiene advice: own towel, hot-wash bedding, cover the lesion, and wash hands.

Self-check quiz

5 questions · optional

Q1 **Treatment of a large fluctuant boil?**

✓ Incision and drainage, the same day.

Q2 When are antibiotics needed for a boil?

✓ Fever, cellulitis, facial site, large or very painful lesion, diabetes or immunosuppression.

Q3 First-line antibiotic?

✓ Flucloxacillin (clarithromycin if penicillin-allergic; erythromycin in pregnancy).

Q4 Recurrent painful boils in a rugby team?

✓ Suspect PVL-S. aureus: swab, escalate to microbiology/public health, decolonisation.

Q5 Recurrent 'boils' with sinus tracts in the groin?

✓ Hidradenitis suppurativa.

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Photo gallery

Boils (furuncles) 3

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Carbuncle: refer for drainage

1

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Infections — bacterial Module 5 · Week 5

Scarlet fever & strawberry tongue

A group A streptococcal infection, mostly in children: sore throat, fever, then a fine 'sandpaper' rash, flushed cheeks with a pale ring round the mouth and a strawberry tongue. Treat with 10 days of penicillin V; it is notifiable.

ALSO CALLED scarlatina · group A streptococcus · GAS · strawberry tongue · sandpaper rash · Pastia lines · +3

RELATED PAGES Cellulitis & erysipelas · Impetigo · Morbilliform drug eruption & drug urticaria · Neonatal pustular rashes

KEY FACTS

- **Scarlet fever** is a group A streptococcal (S. pyogenes) infection with a toxin-mediated rash; commonest in children aged 2–8.
- Sore throat, fever, then a **sandpaper** rash (neck, chest, flexures), flushed cheeks with **circumoral pallor**, **strawberry tongue**, Pastia lines.
- **Phenoxymethylpenicillin for 10 days**; clarithromycin or azithromycin if penicillin allergic.
- **Notifiable**; exclude from school or work until 24 hours after starting antibiotics.
- Watch for **invasive group A strep** (iGAS): fever with severe illness, spreading cellulitis, necrotising signs: same-day emergency care.

Overview

A toxin-mediated illness caused by **group A β-haemolytic Streptococcus** (Streptococcus pyogenes, GAS), usually after **streptococcal pharyngitis/tonsillitis** (occasionally a skin/wound source). Mainly children **2–10 years**; peaks in winter/spring. Notable UK surges (e.g. 2014–2018, winter 2022–23 with rise in invasive GAS). **Strawberry tongue** = inflamed, enlarged fungiform papillae on the dorsum of the tongue.

Epidemiology

- **Who**
  - Children 2–8 years in winter and spring; outbreaks in schools and nurseries.
  - UK outbreaks rose sharply in 2022–23 with more iGAS.

Aetiology: causes and risk factors

Spread by respiratory droplets and direct contact (incubation 2–5 days). GAS strains producing **streptococcal pyrogenic exotoxins (SpeA/B/C — superantigens)** cause the diffuse rash and enanthem; immunity is toxin-specific, so second episodes are possible. Complications arise by local spread (suppurative), immune mechanisms (non-suppurative), or invasion (iGAS).

Risk factors and associations

- Streptococcal **pyrogenic exotoxins** (SpeA, SpeC) act as superantigens.

Myths to address

- **"SCARLET FEVER IS A DISEASE OF THE PAST"**
  it is common again in the UK.

Pathophysiology and pathology

The rash is caused by **toxins**, not by bacteria in the skin.

1. **Pharyngeal infection**
   GAS infects the throat or tonsils (occasionally skin or wounds).

2. **Exotoxin release**
   Pyrogenic exotoxins act as **superantigens**, activating many T cells non-specifically.

3. **Rash**
   Cytokines dilate dermal capillaries, giving the fine sandpaper erythema, flushed face and strawberry tongue.

Fig. 1 · Flushed cheeks with circumoral pallor.
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4. **Tongue changes**
   White-coated then red 'strawberry' tongue.

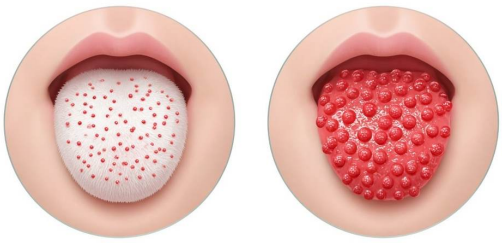

Fig. 2 · White then red strawberry tongue.

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5. **Desquamation**
   Peeling of fingertips and toes after 1–2 weeks.

HISTOPATHOLOGY

- Not needed. Throat swab for culture or rapid antigen test supports diagnosis (UKHSA).

Clinical features

- Abrupt **fever**, **sore throat** (exudative tonsillitis, tender anterior cervical nodes), headache, nausea/vomiting, abdominal pain.
- Rash after 12–48 h: **fine, pinhead, "sandpaper" erythema** starting on neck/chest, spreading to trunk/limbs; blanches; **Pastia lines** (linear petechial accentuation in skin folds — antecubital, axillae); **flushed cheeks with circumoral pallor**. On darker skin the rash may be hard to see but **feels rough**.
- **TONGUE**
  day 1–2 **"white strawberry tongue"** (white coat with red papillae poking through) day 3–5 **"red strawberry/ raspberry tongue"** (coat sheds). Palatal petechiae (Forchheimer spots).
- Rash fades in ~1 week, followed by **desquamation** of fingertips, toes, groin.
- **CAUSES OF STRAWBERRY TONGUE**
  scarlet fever; **Kawasaki disease**; **toxic shock syndrome** (staph or strep); also reported with vitamin B12/folate deficiency (glossitis), yellow fever, allergy/drug reactions and some other infections.

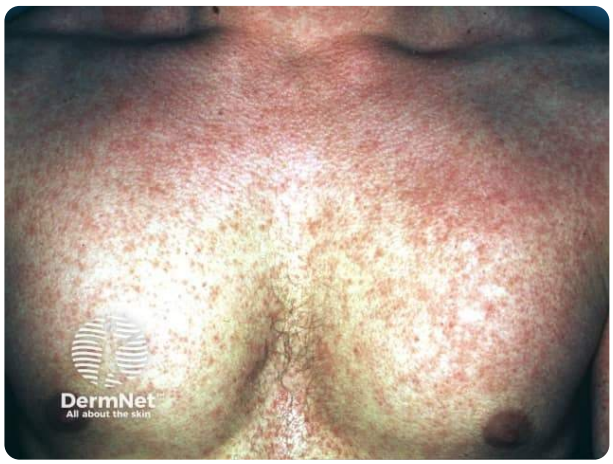

Fig. 3 · Fine punctate "sandpaper" erythema on the trunk, typically starting 12–48 h after fever/ sore throat on neck/chest and spreading — feel the roughness.

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Types to recognise

Sandpaper rash

Fine punctate erythema on trunk. Site: Trunk.

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Flexural accentuation

Pastia lines in creases. Site: Antecubital, axillae.

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Face

Flushed cheeks, circumoral pallor. Site: Face.

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Desquamation

Peeling 1–2 weeks later. Site: Fingers, toes.

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Special groups

- **Chickenpox or influenza co-infection**
  - Higher risk of iGAS: low threshold for review.

Diagnosis and investigations

Clinical in primary care (UKHSA). Throat swab for culture (or rapid antigen test where available) if uncertain or in outbreaks/closed settings. Not routinely needed before treatment. If unwell: FBC, CRP, blood cultures, ASO titre (retrospective).

Clinical

- Diagnose clinically in primary care (UKHSA).

Throat swab

- Culture or rapid antigen if uncertain or outbreaks.

If unwell

- FBC, CRP, blood cultures; ASO retrospectively.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Kawasaki disease	**Fever 5 days**, conjunctivitis, red cracked lips, swollen then peeling hands / feet; < 5 years: urgent paediatrics.
	Toxic shock syndrome	Hypotension, multi-organ, 'sunburn' erythroderma: 999.
	SSSS	Tender erythema, flaccid blisters, Nikolsky +, no tonsillitis.
	Viral exanthems	Measles (coryza, cough, Koplik), slapped cheek; no sandpaper texture.
	Glandular fever ± amoxicillin rash	Teenager, big nodes, splenomegaly.

Management

Based on UKHSA Guidelines for the public health management of scarlet fever and NICE CKS Scarlet fever:

First-line antibiotics (to speed recovery, reduce transmission and complications)

- **Phenoxymethylpenicillin for 10 days** (BNF age-banded doses; e.g. adults/children 12 years 500 mg four times daily, or 1 g twice daily per local guidance).
- Penicillin allergy: **clarithromycin** (5 days) or **azithromycin** (per BNF/local guidance); **amoxicillin** (10 days) is an alternative where liquid palatability/supply is an issue (follow UKHSA/local advice during shortages).
- Complete the full course.

Public health

- **Notifiable disease** in England & Wales — the diagnosing registered medical practitioner notifies UKHSA (pharmacist prescribers should ensure this happens via the GP/local process).
- **EXCLUSION**
  children and adults stay off school/nursery/work **until 24 hours after starting antibiotics**; untreated cases are infectious for ~2–3 weeks.
- Outbreaks (2 cases in a setting within 10 days) inform local Health Protection Team; extra caution if chickenpox co-circulating (iGAS risk).
- Hygiene: hand washing, cover coughs, don't share cutlery/towels.

Supportive / self-care

- Paracetamol/ibuprofen for fever and throat pain (avoid ibuprofen if dehydrated or chickenpox co-infection per local advice), fluids, soft foods, emollient for itch/peeling.

- NHS **Pharmacy First** (England) acute sore throat pathway uses FeverPAIN for pharyngitis; check the current pathway on how suspected scarlet fever is handled (typically outside PGD scope prescriber/GP).

Medicines at a glance

Phenoxymethylpenicillin

First line

DOSE	BNF age-banded; e.g. 12 years 500 mg four times daily (or 1 g twice daily per local guidance) · 10 days
CAUTIONS	Empty stomach; complete the course.

Treatment ladder

1 **Public health**

Notifiable: diagnosing practitioner notifies UKHSA. Off school / nursery / work **until 24 h after starting antibiotics**; untreated cases infectious ~2–3 weeks.

2 **First line**

Phenoxymethylpenicillin

3 **Penicillin allergy / supply**

Clarithromycin (5 days) or azithromycin per BNF; **amoxicillin** 10 days if liquid palatability or supply is an issue.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Chickenpox in the household**: Higher iGAS risk: lower threshold for review.
- **Safety-net**: In writing, for iGAS; dark urine or swelling 1–3 weeks later.

Referral

999 / A&E — Rapid deterioration, severe pain, spreading skin infection, breathing difficulty, sepsis: **invasive GAS**.

URGENT — Fever 5 days with red eyes, red lips, swollen hands / feet: consider **Kawasaki disease**.

ROUTINE — Dark urine or swelling 1–3 weeks later: check urine (post-strep nephritis).

Red flags and when to refer

- Rapid deterioration, severe pain, a spreading skin infection, breathing difficulty or signs of sepsis: **invasive GAS**, emergency.
- Fever for 5 days or more with red eyes, red lips, swollen hands and feet: consider **Kawasaki disease**.
- Chickenpox plus scarlet fever in the same household: higher invasive GAS risk, so lower threshold for review.

Red flags / refer — safety-net for invasive GAS

- Same-day / 999: persistent high fever despite antipyretics, drowsiness, confusion, severe muscle pain, rapidly spreading painful red skin (necrotising fasciitis), difficulty breathing, dehydration, signs of sepsis/ shock (NICE NG51).
- Quinsy (trismus, uvula deviation), otitis media, sinusitis, cervical abscess, pneumonia.
- Late: **acute rheumatic fever** (2–4 weeks; arthritis, carditis), **post-streptococcal glomerulonephritis** (1–3 weeks; dark urine, oedema, hypertension), **PANDAS** (rare), and the link to **guttate psoriasis**.
- Features of Kawasaki (fever 5 days, conjunctivitis) or TSS (hypotension) urgent.

Complications and prognosis

Complications

- Otitis media, quinsy, sinusitis, pneumonia.
- **IGAS**
  necrotising fasciitis, sepsis, toxic shock.
- Late: rheumatic fever, post-streptococcal glomerulonephritis.

Prognosis

- Recovers within a week with antibiotics.

Prescriber pearls (UK pharmacist IP)

"Sandpaper rash + sore throat + strawberry tongue + circumoral pallor" = scarlet fever until proven otherwise; on darker skin rely on texture and the tongue.

- Strawberry tongue is **not specific** — a child with 5 days of fever and red eyes needs Kawasaki excluded.
- Phenoxymethylpenicillin is best taken on an empty stomach; four-times-daily dosing needs adherence support (use bd regimen where appropriate).
- Tell parents peeling of fingers/toes 1–2 weeks later is expected.
- Safety-net clearly and in writing for iGAS symptoms; advise urine check/return if dark urine or swelling after 1–3 weeks.

Self-check quiz

4 questions · optional

Q1 First-line antibiotic and duration?

✓ Phenoxymethylpenicillin for 10 days.

Q2 When can the child return to school?

✓ 24 hours after starting antibiotics.

Q3 Is scarlet fever notifiable?

✓ Yes, to UKHSA (England & Wales).

Q4 Strawberry tongue and 6 days of fever with conjunctivitis?

✓ Consider Kawasaki disease: urgent paediatric referral.

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Photo gallery

Scarlet fever rash

6

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Strawberry tongue (Wikimedia Commons)

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Infections — bacterial Module 5 · Week 5

Cutaneous tuberculosis (scrofuloderma & others)

Skin TB is uncommon in the UK but important: a cold neck abscess or sinus over a lymph node (scrofuloderma), a slowly spreading reddish-brown plaque (lupus vulgaris), or warty or nodular forms. Refer to TB services; treatment follows NICE NG33.

ALSO CALLED scrofuloderma · lupus vulgaris · TB verrucosa cutis · erythema induratum · tuberculids · Mycobacterium tuberculosis · +1

RELATED PAGES Leprosy (Hansen disease) & lepra reactions · Cutaneous leishmaniasis · Sarcoidosis (incl. Löfgren syndrome) · The returning traveller with a skin problem

KEY FACTS

- Skin infection with Mycobacterium tuberculosis: uncommon in the UK; risk with high-incidence countries, HIV, anti-TNF, homelessness.
- Forms by route and immunity: **scrofuloderma** (from underlying node), **lupus vulgaris**, TB verrucosa cutis, chancre, tuberculids.
- Lupus vulgaris: red-brown plaques with **"apple-jelly" colour on diascopy**.
- Refer **urgently** to the TB service; NICE NG33 regimen is 2 months RHZE then 4 months RH.
- TB is **notifiable**; contacts are traced.

Overview

Skin infection by Mycobacterium tuberculosis complex (rarely M. bovis / BCG). Uncommon in the UK but seen in people from high-incidence countries, the immunosuppressed (HIV, anti-TNF, steroids), homeless people and those with prior TB contact. **Scrofuloderma** is the commonest form in many series: TB of an underlying structure (usually **cervical lymph node**, also bone/joint/epididymis) breaks through to the skin. Case 1 is a classic scrofuloderma with active pulmonary TB.

Aetiology: causes and risk factors

Classify by **route** and **host immunity**:

ROUTE	FORM	HOST IMMUNITY / BACILLARY LOAD
Exogenous — first inoculation (non-immune)	Tuberculous chancre (primary inoculation TB)	Naïve multibacillary, becomes paucibacillary as immunity develops
Exogenous — re-inoculation (sensitised)	**TB verrucosa cutis** (warty TB; "prosector's wart")	Good immunity, paucibacillary

ROUTE	FORM	HOST IMMUNITY / BACILLARY LOAD
Endogenous — **contiguous** spread	**Scrofuloderma** (node/bone skin); periorificial TB (autoinoculation at mouth/anus)	Scrofuloderma: moderate/variable, multibacillary; periorificial: poor immunity, advanced internal TB
Endogenous — haematogenous/lymphatic	**Lupus vulgaris**	Good immunity (strongly Mantoux+), paucibacillary
Haematogenous	Acute **miliary** TB of skin; tuberculous gumma (metastatic abscess)	Poor immunity (HIV, infants), multibacillary
Hypersensitivity, no bacilli in lesion	**Tuberculids**: papulonecrotic tuberculid, lichen scrofulosorum, erythema induratum (Bazin)	Strong immunity; TB focus elsewhere; PCR may detect DNA; resolve with anti-TB therapy

Mechanism in Case 1: pulmonary TB lymphatic spread to cervical nodes caseating necrosis cold abscess direct (contiguous) extension through overlying dermis nodules, ulcers, sinuses.

Myths to address

- **"SKIN TB ISN'T CONTAGIOUS"**
  associated pulmonary TB may be.

Pathophysiology and pathology

How TB looks in skin depends on **how it got there and the host's immunity**.

1. **Exogenous inoculation**
   Direct entry through a wound: chancre (non-immune) or warty TB (immune).

2. **Contiguous spread**
   Underlying node or bone TB breaks through skin: scrofuloderma.

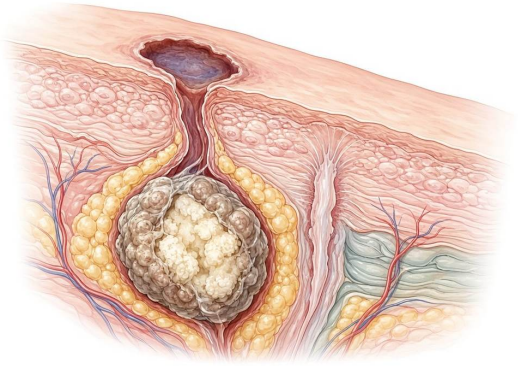

Fig. 1 · Scrofuloderma: TB node breaking through the skin.
AI-generated illustration

3. **Haematogenous spread**
   Lupus vulgaris, miliary TB, tuberculous gumma.

4. **Granuloma**
   Caseating granulomas contain the bacilli.

Clinical features

> **SCROFULODERMA**

firm, **painless**, initially skin-coloured then **dusky red/blue** subcutaneous nodules over the **neck** (submandibular, supraclavicular, anterior triangle), axilla, groin, chest wall; nodes **matted, non-tender** ("cold"). Nodules soften, discharge cheesy/watery material, form **undermined ulcers and sinus tracts**, healing slowly with **cord-like, puckered scars** and skin bridges.

Systemic: chronic cough, weight loss, night sweats, fever (Case 1).

LUPUS VULGARIS

slowly enlarging red-brown plaque on head/neck; diascopy shows **"apple-jelly" nodules**; scarring, tissue destruction; risk of SCC in long-standing lesions.

TB VERRUCOSA CUTIS

warty plaque on hands, knees, buttocks.

TUBERCULIDS

crops of necrotic papules on extensor limbs (papulonecrotic); tiny lichenoid follicular papules on trunk in children (lichen scrofulosorum); tender violaceous nodules on posterior calves of women that may ulcerate (erythema induratum).

Fig. 2 · Scrofuloderma: firm, painless, dusky-red cold nodule over the neck overlying a matted node — exactly the Case 1 picture (painless neck lesions + matted non-tender cervical nodes + pulmonary TB).

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Types to recognise

Scrofuloderma

Dusky-red cold nodule over a node. Site: Neck.

© DermNet · **Source**

Sinus tracts & scars

Linear sinuses, puckered scars. Site: Neck.

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Lupus vulgaris

Red-brown plaque; apple-jelly. Site: Face, neck.

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Warty TB

Re-inoculation; good immunity. Site: Hands.

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Erythema induratum

Tender violaceous calf nodules. Site: Calves.

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Diagnosis and investigations

Specialist-led (dermatology + TB/respiratory/ID):

- Skin/node **biopsy**: caseating granulomas; **Ziehl–Neelsen** stain for AFB; **mycobacterial culture** (gold standard, weeks; gives drug sensitivity); **TB PCR / GeneXpert** (rapid, rifampicin resistance).
- Node FNA/excision; sputum smear/culture (AFB+ in Case 1); **chest X-ray**.
- **Mantoux / IGRA** support infection but do not separate latent from active disease (may be false-negative in HIV/miliary TB).
- **HIV test** for every patient with TB (NICE NG33); also hepatitis B/C, LFTs, renal function, visual acuity (ethambutol) baselines.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Lymphoma	Rubbery or hard nodes, B symptoms: biopsy.
	Pyogenic abscess / infected cyst	Acute, hot, tender; responds to drainage.
	Deep fungal / sporotrichosis	Gardening, travel; lymphocutaneous chain.
	Lupus vulgaris mimics	Sarcoidosis, discoid lupus, leprosy.

Management

- **Not managed in primary care** — refer **urgently** to the local **TB service** (respiratory/ID) and dermatology; suspected pulmonary TB with cough needs same-day advice and infection-control (mask, avoid waiting rooms).
- Standard regimen (NICE NG33, specialist-initiated): **2 months RHZE** (rifampicin, isoniazid, pyrazinamide, ethambutol) then **4 months RH** (longer for CNS/bone, drug resistance). Directly observed therapy where adherence risk. Pyridoxine with isoniazid in at-risk patients.
- TB is a **notifiable disease** (the diagnosing clinician notifies UKHSA); **contact tracing** via the TB service.
- Surgery (excision/drainage) is adjunctive only.

Treatment ladder

1 **Refer**

Urgent TB service referral; TB is notifiable; contact tracing via the TB team.

2 **Standard regimen (specialist)**

2 months RHZE then **4 months RH** (NICE NG33); longer for CNS / bone or resistance. DOT if adherence risk; pyridoxine with isoniazid if at risk.

3 **Your role**

Interaction check for rifampicin: contraception, warfarin / DOACs, antiretrovirals, antiepileptics, methadone, steroids, azoles. Counsel: orange urine and tears (stains soft lenses); report jaundice, dark urine, vomiting; ethambutol report colour-vision change.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Contraception**: Rifampicin defeats combined hormonal contraception: non-hormonal / IUD (FSRH).
- **Don't**: Repeatedly treat a chronic neck sinus with flucloxacillin.

Referral

SAME DAY	Cough, haemoptysis, weight loss, night sweats: urgent chest assessment with infection control.
URGENT	Chronic painless neck lump or sinus, especially from a high-incidence country: **TB until proven otherwise**.
ROUTINE	Notify TB to UKHSA.

Red flags and when to refer

- Cough, weight loss, night sweats or haemoptysis: urgent chest assessment.
- Rifampicin is a strong enzyme inducer: review every interacting drug (contraceptives, anticoagulants, antiretrovirals).
- Notify TB to UKHSA.

Complications and prognosis

Complications

- Scarring, contractures, SCC in long-standing lupus vulgaris.

Prognosis

- Curable with full treatment.

Prescriber pearls (UK pharmacist IP)

💡 **Interaction checker is essential:** rifampicin is a potent enzyme inducer — reduces efficacy of combined hormonal contraception (advise non-hormonal/IUD/medroxyprogesterone with adjustment per FSRH), warfarin/DOACs, antiretrovirals, antiepileptics, methadone, steroids, azoles. Orange urine/tears (stains soft contact lenses).

Hepatotoxicity (RHZ): counsel to report jaundice, dark urine, vomiting; avoid alcohol; LFT monitoring per specialist.

Ethambutol: report any visual change (colour vision) immediately.

A chronic painless neck lump/sinus, especially with cough, weight loss, or origin from a high-incidence country, is **TB until proven otherwise** — do not repeatedly treat with flucloxacillin.

Skin lesions heal with scarring; tuberculids clear only with anti-TB treatment of the hidden focus.

Self-check quiz

4 questions · optional

Q1 **Commonest form of cutaneous TB in many series?**

✓ Scrofuloderma (contiguous spread from a node).

Q2 **Diascopy sign of lupus vulgaris?**

✓ 'Apple-jelly' nodules.

Q3 **Standard TB regimen?**

✓ 2 months rifampicin, isoniazid, pyrazinamide, ethambutol, then 4 months rifampicin + isoniazid.

Q4 **Key contraception issue on rifampicin?**

✓ Enzyme induction makes combined hormonal contraception unreliable: use non-hormonal / IUD.

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dermnetnz.org

Photo gallery

Scrofuloderma 3

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Other forms of cutaneous TB 6

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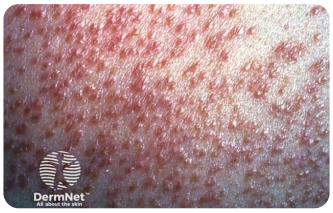
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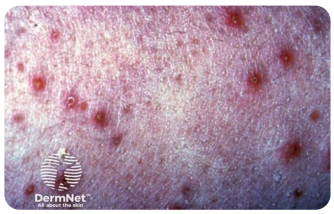
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CHAPTER 9

Infections — viral & infestations

Viral warts, shingles, cold sores and pityriasis rosea, plus scabies, head lice, Lyme disease, bite reactions and perineal dermatoses.

1. Viral warts (HPV)
2. Herpes zoster (shingles)
3. Scabies
4. Head lice (pediculosis capitis)
5. Herpes simplex labialis (cold sores)
6. Lyme disease & erythema migrans
7. Pityriasis rosea
8. Blister beetle & Paederus dermatitis
9. Papular urticaria & insect bites
10. Perineal dermatoses

Infections — viral & infestations Module 5 · Week 1

Viral warts (HPV)

Benign HPV-driven epidermal growths. Most clear on their own. Salicylic acid is the community first line, cryotherapy is clinic-based, and refer the immunosuppressed and anything atypical.

ALSO CALLED warts · verruca · verrucae · plantar wart · common wart · plane warts · +5

RELATED PAGES Perineal dermatoses · Seborrhoeic keratosis · Cutaneous squamous cell carcinoma (cSCC)

KEY FACTS

- Benign epidermal growths from **human papillomavirus (HPV)** infection of keratinocytes.
- Very common in **school-age children**; spread by direct contact, wet floors and autoinoculation.
- Common, plantar (verrucae), plane, filiform and periungual types; **thrombosed capillaries** (black dots) are typical.
- Most clear spontaneously: **half within 6 months in children, about 90% within 2 years**; treat only if painful or distressing.
- Options: **salicylic acid** paints and occlusion, cryotherapy; refer atypical, extensive, facial or immunosuppressed cases.

Overview

Viral warts (verrucae) are very common **benign** growths of the epidermis caused by cutaneous **human papillomavirus (HPV)**. This page covers **cutaneous** warts. Anogenital warts belong to sexual health and are counselled separately.

Epidemiology

- **Who**
  - Up to a third of school-age children have warts at any time; less common in adults.
  - Meat and fish handlers (HPV-7 'butcher's warts'), people with eczema and the **immunosuppressed** (transplant recipients) are at higher risk.
- **Spread**
  - Direct contact, swimming pools and changing rooms, shaving, nail biting and picking; incubation up to 12 months.

Aetiology: causes and risk factors

- HPV is a double-stranded DNA virus. Common cutaneous types are **1, 2, 3, 4, 10, 27, 29 and 57**.
- It infects **basal keratinocytes**, producing hyperkeratosis and new virus particles.

- Spread is by skin-to-skin contact, fomites (swimming pools, changing rooms), and **autoinoculation**. A line of warts along a scratch is the pseudo-Koebner phenomenon.
- Incubation takes weeks to months. The wart burden is higher with dermatitis or **immunosuppression** (transplant, HIV).

Risk factors and associations

- **HPV TYPES**
  1 (plantar myrmecia), 2 and 4 (common and plantar), 3 and 10 (plane), 7 (butchers), 27 and 57.
- Risk factors: barrier damage, moist skin, trauma, immunosuppression (e.g. azathioprine, ciclosporin, HIV).

Myths to address

> **"WARTS HAVE ROOTS"**
> they are purely epidermal; aggressive digging causes scars.

> **"CHILDREN MUST STAY OUT OF SWIMMING"**
> not necessary; covering verrucae with a plaster or sock is enough.

Pathophysiology and pathology

HPV infects the **basal layer** through tiny breaks in the skin and hijacks keratinocyte growth.

1. **Entry through micro-trauma**
   Virus reaches **basal keratinocytes** through small breaks; maceration and trauma help it enter.

2. **Viral genes drive proliferation**
   Early viral proteins (E6, E7) push infected cells to keep dividing, producing **acanthosis, papillomatosis and hyperkeratosis**.

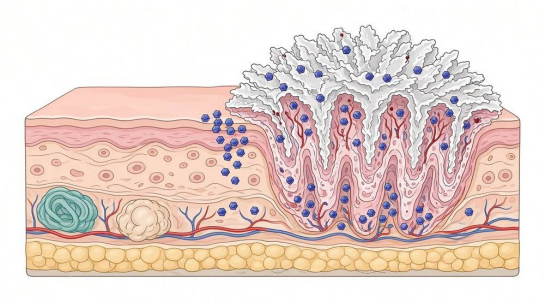

Fig. 1 · A wart in section: thickened, folded epidermis with dilated capillaries in the papillae.
AI-generated illustration

3. **Virus made at the surface**
   New virus is assembled in upper layers (**koilocytes**) and shed in skin flakes, spreading to other sites (lines of warts from scratching).

4 **Capillaries and black dots**

Dermal papillae with dilated capillaries extend up into the wart; clotted vessels show as **black dots**, unlike corns.

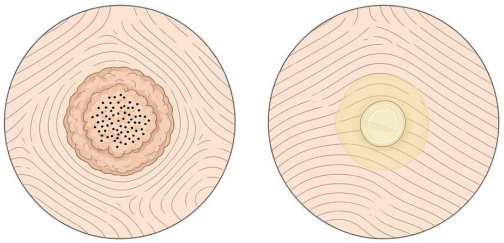

Fig. 2 · Verruca (black dots, interrupted skin lines) vs corn (central core, preserved skin lines).

AI-generated illustration

5 **Immune clearance**

Cell-mediated immunity eventually recognises the virus and clears it, which is why most warts go without treatment.

HISTOPATHOLOGY

- Papillomatosis, acanthosis and hyperkeratosis with parakeratosis over tips of papillae.
- **Koilocytes** (vacuolated keratinocytes with shrunken nuclei) and large keratohyaline granules in the upper epidermis.
- Biopsy only for atypical, persistent or bleeding lesions (exclude SCC in the immunosuppressed).

Clinical features

TYPE	FEATURES	SITES
Common (verruca vulgaris)	Rough hyperkeratotic papules; black or red dots are thrombosed capillaries	Hands, knees, around the nails
Plantar	Inward-growing and painful (myrmecia), or clustered 'mosaic' warts; **interrupt the skin lines**	Soles
Plane (plana)	Flat-topped, skin-coloured, often numerous; spread by shaving	Face, backs of hands, shins
Filiform / digitate	Thin fronds on a narrow base	Face
Periungual / subungual	Around or under the nail; risk of nail dystrophy	Nail folds

Fig. 3 · Common wart (verruca vulgaris) on hand — rough hyperkeratotic papule with capillary dots.

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Types to recognise

Common

Rough hyperkeratotic papules with capillary dots. Site: Hands, knees, nails.

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Plantar

Inward-growing and painful, or 'mosaic' clusters. Site: Soles.

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Plane

Flat-topped, skin-coloured, numerous; spread by shaving. Site: Face, hands,

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Filiform

Thin fronds on a narrow base. Site: Face.

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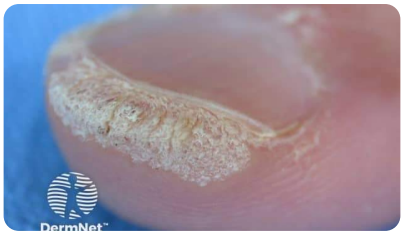
Periungual

Around or under the nail; nail dystrophy risk. Site: Nail folds.



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Special groups

- **Immunosuppressed**
  - Warts are often numerous and resistant; any growing, painful or bleeding lesion in a transplant recipient needs urgent review (SCC).
- **Children**
  - Watchful waiting is reasonable; avoid painful treatments in young children.

Diagnosis and investigations

- **Clinical.** Paring reveals pinpoint bleeding or black dots. Dermoscopy helps.
- **PLANTAR WART VS CORN**
  a wart is tender when **squeezed from the sides**. A corn is tender on **direct pressure**, and the skin lines run through it.
- **Biopsy** if SCC is a concern, especially in immunosuppressed patients.

Look

- **Clinical** diagnosis.
- Paring shows pinpoint bleeding or black dots; dermoscopy helps.

Squeeze test

> **WART**
> tender when squeezed **from the sides**.

> **CORN**
> tender on **direct pressure**; skin lines run through.

Who is at risk

- Immunosuppression (transplant, HIV), dermatitis.
- Diabetes or neuropathy changes treatment choice.

Tests

- Usually **none**.
- **Biopsy** if SCC is a concern, especially if immunosuppressed.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Corn / callus	Pressure site; skin lines continue through; no capillary dots.
	Seborrhoeic keratosis	'Stuck-on' look in an older adult.
	SCC / keratoacanthoma	Rapid growth, ulceration or tenderness; older or immunosuppressed. **Refer**.

	CONDITION	DISTINGUISHING FEATURES
	Molluscum contagiosum	Pearly, umbilicated papules (poxvirus).

Management

Many warts **resolve without treatment**: roughly half of childhood warts within 6 months, and most within 2 years. Treat if they are painful, disabling, in an immunosuppressed patient, or if the patient prefers.

First line (community)

- **Salicylic acid** paint, gel or plaster **daily**. Soak and pare (or file) first. Persist for **8–12 weeks**. Protect the surrounding skin, and avoid in diabetes or neuropathy with poor sensation.
- **Cryotherapy** (liquid nitrogen) is clinic-based and needs repeated sessions. It can blister and hurt, and may cause hypo- or hyperpigmentation (more so in skin of colour). Take care over superficial nerves.

Second line / specialist (BAD guideline themes)

Curettage or electrosurgery (scarring risk), radiofrequency cautery, CO_2 laser, intralesional bleomycin, contact immunotherapy (e.g. diphencyprone), microwave (plantar), and podophyllin or imiquimod (better evidence for anogenital than cutaneous warts).

Medicines at a glance

Salicylic acid paint, gel or plaster

First line · community

DOSE	Daily, after soaking and paring · 8–12 weeks
CAUTIONS	Protect surrounding skin. Avoid in diabetes or poor sensation.

Cryotherapy (liquid nitrogen)

Clinic

DOSE	Repeated sessions · Clinic-based
CAUTIONS	Blistering, pain, pigment change (more in skin of colour); mind superficial nerves.

Treatment ladder

Watchful waiting

Treat only if painful, disabling, immunosuppressed, or the patient prefers. Flip-flops in communal showers; no shared towels; cover verrucae for swimming.

2

First line · community

Salicylic acid paint, gel or plaster

3

Clinic

Cryotherapy (liquid nitrogen)

4

Specialist

Curettage or electrosurgery (scarring), radiofrequency, CO_2 laser. Intralesional bleomycin, contact immunotherapy (diphencyprone), microwave (plantar).

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Expectations**: No treatment 'kills' HPV instantly; recurrence is common until immunity clears it.
- **Duct tape**: Mixed evidence: optional add-on, not a cure.
- **Feet in diabetes**: No self-treatment with salicylic acid or cryotherapy.

Referral

URGENT	Rapid growth, bleeding, ulceration or atypical look: consider **SCC**.
ROUTINE	**Immunosuppressed** with extensive or recalcitrant warts dermatology.
ROUTINE	Extensive facial or periungual warts where destruction risks scarring or nail damage.
DON'T	Diabetes, neuropathy or vascular disease: no self-treatment on the feet.

Red flags and when to refer

- Rapid growth, bleeding, ulceration or an atypical appearance: consider **SCC** and refer.
- **Immunosuppressed** patient (transplant, HIV) with extensive or recalcitrant warts: refer to dermatology.
- Extensive facial or periungual warts where destructive treatment risks scarring or nail damage.
- Diabetes, peripheral neuropathy or vascular disease: avoid self-treatment with salicylic acid or cryotherapy on the feet.

Complications and prognosis

Complications

- Pain on walking (plantar warts); nail dystrophy (periungual).
- Spread to other sites and people; psychosocial distress.
- Scarring from destructive treatment.
- Rarely, malignant change (SCC) in immunosuppressed or epidermodysplasia verruciformis.

Prognosis

- Children: about 50% clear in 6 months and 90% in 2 years without treatment.
- Adults' warts are more persistent and recur more often.

Prescriber pearls (UK pharmacist IP)

No treatment 'kills' HPV instantly. Treatment removes virus-bearing skin, and recurrence is common until immunity clears that type.

Duct-tape occlusion has mixed evidence. It is an optional add-on, not a cure.

Periungual warts: avoid aggressive destruction near the **nail matrix**.

Advise flip-flops in communal showers, not sharing towels, and covering verrucae for swimming.

Self-check quiz

5 questions · optional

Q1 How do you tell a plantar wart from a corn?

✓ Wart: tender when squeezed from the sides, breaks the skin lines, black dots. Corn: tender on direct pressure, skin lines run through.

Q2 What are the black dots in a wart?

✓ Thrombosed capillaries.

Q3 Community first-line treatment and duration?

✓ Salicylic acid daily after soaking and paring, for 8–12 weeks.

Q4 Who should not self-treat warts on the feet?

✓ People with diabetes, peripheral neuropathy or vascular disease.

Q5 When is a 'wart' a red flag?

✓ Rapid growth, bleeding, ulceration or atypical look, especially if immunosuppressed: consider SCC and refer.

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emedicine.medscape.com

Photo gallery

Common, plantar, plane, filiform and periungual warts

5

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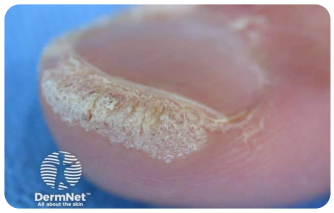

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Infections — viral & infestations Module 5 · Week 5

Herpes zoster (shingles)

Reactivation of varicella-zoster virus in one dermatome: pain, then grouped vesicles on a red base. Start an antiviral within 72 hours when indicated, protect the eye, and think about immunosuppression in severe or recurrent cases.

ALSO CALLED shingles · zoster · VZV · dermatomal rash · ophthalmic zoster · Hutchinson sign · +4

RELATED PAGES Herpes simplex labialis (cold sores) · Impetigo

KEY FACTS

- Reactivation of **varicella-zoster virus (VZV)** from a sensory ganglion, causing a painful **unilateral dermatomal** vesicular rash.
- Lifetime risk about **1 in 4**; risk rises sharply after **50** and with immunosuppression.
- Pain or tingling often precedes the rash by 2–3 days; the rash crusts in 7–10 days.
- Antiviral (e.g. **aciclovir 800 mg five times daily for 7 days**) within 72 h for most adults over 50, or with eye, immunosuppression, severe pain or rash (CKS).
- **Ophthalmic zoster** (Hutchinson's sign) and immunocompromised patients need same-day assessment; offer the **shingles vaccine** (Shingrix) to eligible groups.

Overview

Reactivation of latent **varicella-zoster virus (VZV)** from a dorsal root or cranial nerve ganglion, producing a painful, unilateral, dermatomal vesicular eruption. Lifetime risk ~1 in 4; risk rises with age and with reduced cell-mediated immunity. In a young adult (Case 2, 32M), zoster should prompt an **HIV test** — it is an HIV indicator condition (BHIVA/NICE).

Epidemiology

- **Who**
  - Anyone who has had chickenpox (or, rarely, the live varicella vaccine).
  - Incidence rises with age: most cases after 50; up to half of people reaching 85 have had it.
- **Risk factors**
  - Immunosuppression (HIV, cancer, chemotherapy, transplant, high-dose steroids, JAK inhibitors), older age, stress.
  - Recurrence is uncommon (about 1–5%).

Aetiology: causes and risk factors

After chickenpox, VZV stays latent in sensory ganglia. Waning VZV-specific T-cell immunity (age, HIV, chemotherapy, transplant, high-dose steroids, biologics/JAK inhibitors, stress) allows reactivation; virus travels down the sensory nerve **neuritis** (prodromal pain) then epidermal infection (vesicles). In HIV, zoster can occur at any CD4 count but is more frequent, more severe, recurrent, multi-dermatomal, necrotic, chronic (verrucous) or **disseminated** as CD4 falls.

Risk factors and associations

- Triggers sometimes identified: local trauma or surgery, radiotherapy, intercurrent infection.
- Zoster is **infectious** to people who have not had chickenpox (direct contact with fluid from blisters) until lesions crust.

Myths to address

> **"YOU CAN CATCH SHINGLES FROM CHICKENPOX"**
> contacts can catch **chickenpox** from shingles, not shingles itself.

> **"IT CAN'T COME BACK"**
> recurrence is uncommon but possible, especially with immunosuppression.

Pathophysiology and pathology

After chickenpox, VZV hides in **sensory ganglia**; when cell-mediated immunity falls, it reactivates and travels down the nerve.

1 **Latency after chickenpox**

VZV travels up sensory nerves from chickenpox lesions and stays **latent** in dorsal root or cranial nerve ganglia for decades.

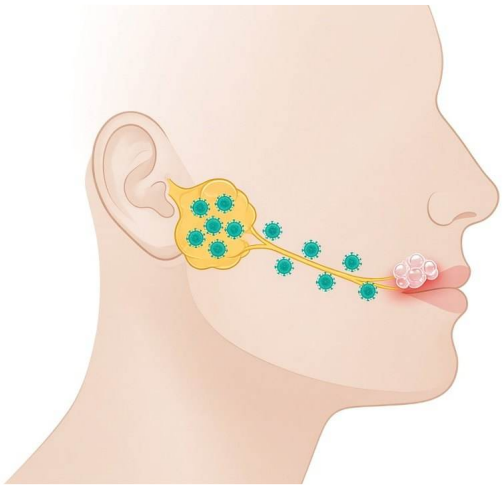

Fig. 1 · Latent virus in a sensory ganglion, travelling along the nerve to the skin when it reactivates.

AI-generated illustration

2 **Reactivation**

Waning **VZV-specific T-cell immunity** (age, immunosuppression) lets the virus multiply in the ganglion, causing inflammation and **prodromal pain**.

3 **Spread to the dermatome**

Virus moves down the sensory nerve to the skin supplied by that ganglion, producing grouped vesicles on a red base in **one dermatome**, not crossing the midline.

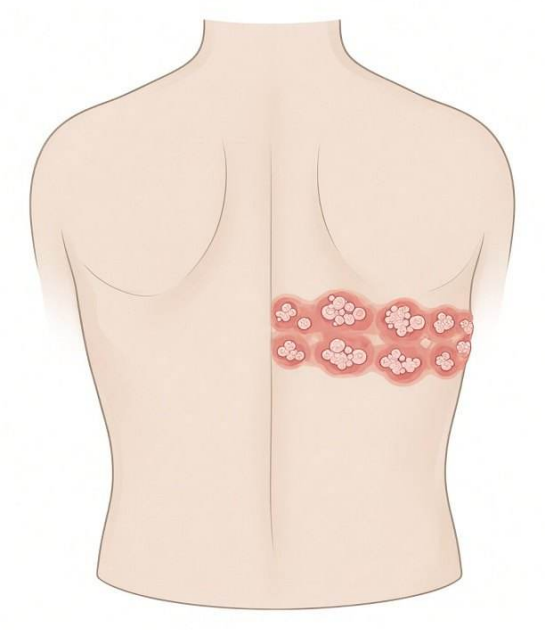

Fig. 2 · Unilateral band-like dermatomal rash on the trunk.

AI-generated illustration

4 **Nerve damage**

Ganglion and nerve inflammation can cause **post-herpetic neuralgia** and, with motor nerve involvement, weakness (e.g. Ramsay Hunt facial palsy).

5 **Healing**

Vesicles become pustules and crust in 7–10 days and heal in 2–4 weeks, sometimes with scarring and pigment change.

HISTOPATHOLOGY

- Intraepidermal vesicles with **ballooning degeneration** and multinucleated giant keratinocytes (identical to herpes simplex).
- Diagnosis is clinical; **PCR** of vesicle fluid if atypical or immunocompromised.

Clinical features

- **Prodrome** 1–5 days: burning, tingling, itching or severe pain in a dermatome (can mimic MI, pleurisy, renal colic, cholecystitis); malaise, fever.
- Rash: erythematous papules → **grouped vesicles on a red base** → pustules → crusts over 7–10 days, **unilateral, respecting the midline**, one (or two adjacent) dermatomes — Case 2: T9–T10 back.
- Commonest: **thoracic** dermatomes, then **ophthalmic (V1)**.
- Immunocompromised features: **haemorrhagic/necrotic** lesions, **>1–2 dermatomes**, **>20 vesicles outside the primary dermatome** (disseminated), slow healing (>2 weeks), recurrent episodes, chronic verrucous lesions, visceral involvement (pneumonitis, hepatitis, encephalitis).
- **HERPES ZOSTER OPHTHALMICUS (HZO)**

V1 forehead/scalp/eyelid; **Hutchinson's sign** (lesions on tip/side of nose nasociliary nerve high risk of ocular involvement).

- **RAMSAY HUNT SYNDROME**
  ear/external canal vesicles + ipsilateral facial palsy ± vertigo, tinnitus, hearing loss.

Fig. 3 · Thoracic zoster: grouped vesicles on red base in a band-like unilateral dermatome stopping at the midline — the commonest site; Case 2 was T9–T10 on the back.
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Patterns to recognise

Thoracic band

Grouped vesicles, one side. Site: Chest, back.

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Crusting / haemorrhagic

Partly haemorrhagic: check for immunosuppression. Site: Dermatome.

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Ophthalmic (V1)

Forehead and eyelid; same-day eye review. Site: Forehead, eye.

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Special groups

- **Immunocompromised**
  - Multi-dermatomal, disseminated or necrotic zoster; needs urgent specialist advice and often IV aciclovir.
- **Pregnancy**
  - Zoster in pregnancy does not cause congenital varicella; seek specialist advice before prescribing antivirals.

- **Skin of colour**
  - Erythema may be subtle; pigment change and keloid scarring after healing are more common.

Diagnosis and investigations

Usually **clinical**. Swab vesicle base for **VZV PCR** if atypical, immunocompromised, or diagnosis uncertain (PCR has replaced Tzanck smear/culture). In immunosuppression (e.g. HIV): VZV PCR, **HIV viral load + CD4 count**, FBC, U&E/LFT (renal dosing of aciclovir), syphilis/hepatitis screen as part of HIV care; ophthalmic exam if V1.

Clinical

- Usually clinical: dermatomal, unilateral, painful.

Swab

- **VZV PCR** of vesicle base if atypical, immunocompromised or unsure.

Immunosuppressed

- HIV viral load + CD4, FBC, U&E / LFT (renal dosing).

V1

- Check the **nose tip** and the eye; ophthalmic exam.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Zosteriform HSV	Recurs at the same site; smaller; less pain. PCR.
	Contact / phytophotodermatitis	Itch > pain; linear; exposure history.
	Bullous impetigo	Honey crust; not dermatomal; children.
	Insect bites	Scattered, itchy, bilateral.
	Prodrome-only phase	Cardiac, abdominal or pleuritic causes first.

Management

Based on NICE CKS Shingles themes:

Self-care / all patients

- Keep rash clean and dry; loose clothing; non-adherent dressing if needed; avoid topical antivirals/antibiotic creams; paracetamol ± NSAID for mild pain.

- **Infectious** (can cause chickenpox in non-immune contacts) until all lesions crusted (usually 5–7 days): cover lesions; avoid pregnant women without chickenpox history, neonates and immunosuppressed people.

Antiviral (immunocompetent)

- Offer oral antiviral **within 72 h** of rash onset if: age 50, **non-truncal** involvement (neck, limbs, perineum), moderate–severe pain or rash, or other risk; consider up to 1 week in those at higher risk.
- **Aciclovir 800 mg five times daily for 7 days** (first-line in many formularies); alternatives **valaciclovir 1 g three times daily for 7 days** or **famciclovir 500 mg three times daily for 7 days** (better adherence; useful in poor-adherence/renal dosing contexts). Adjust for renal impairment.
- NHS **Pharmacy First** (England) includes a shingles clinical pathway for adults — follow the current PGD inclusion/exclusion criteria (e.g. pregnancy, severe immunosuppression, ophthalmic/ear involvement are referral/exclusion points).

Immunocompromised / HIV (Case 2)

- **Offer antiviral regardless of time since onset** (new lesions may keep forming) and **seek same-day specialist advice** (HIV/ID team). Mild localised disease in stable HIV may be treated orally (often longer course, e.g. aciclovir 800 mg 5×/day, valaciclovir or famciclovir, 7–10 days until lesions crust); severe, multi-dermatomal, disseminated, ophthalmic or visceral disease **admission for IV aciclovir** (e.g. 10 mg/kg 8-hourly, renally adjusted).
- Ensure HIV care/ART is optimised (risk of IRIS-related zoster soon after ART start).

Pain / PHN

- Acute moderate–severe pain: consider short-term stronger analgesia; neuropathic agents (amitriptyline, duloxetine, gabapentin, pregabalin) for neuropathic pain/PHN per NICE CG173; capsaicin cream for localised PHN once healed; lidocaine plasters only per specialist/local policy.

Medicines at a glance

Aciclovir

Antiviral · within 72 h

DOSE	800 mg five times daily · 7 days
CAUTIONS	If 50, non-truncal, moderate–severe pain or rash, or other risk.

Valaciclovir

Antiviral · within 72 h

DOSE	1 g three times daily · 7 days
CAUTIONS	Better adherence.

Famciclovir

Antiviral · within 72 h

DOSE	500 mg three times daily · 7 days
CAUTIONS	Alternative.

Treatment ladder

1

Everyone

Clean, dry, loose clothing; non-adherent dressing; no antiviral or antibiotic creams. Paracetamol ± NSAID.

2

Antiviral · within 72 h

Aciclovir; Valaciclovir; Famciclovir

3

Pharmacy First (England)

Shingles pathway for adults: follow the PGD; pregnancy, severe immunosuppression, eye or ear involvement = refer.

4

Immunocompromised

Antiviral **regardless of timing**; **same-day specialist** advice. Severe, disseminated, ophthalmic or visceral ***IV* aciclovir** admission.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Neuropathic pain / PHN**: Amitriptyline, duloxetine, gabapentin or pregabalin (NICE CG173); capsaicin once healed.
- **Vaccination**: Shingrix (non-live) for eligible adults: check Green Book ch. 28a. Zostavax (live) contraindicated in immunosuppression.

Referral

SAME DAY	Forehead, eyelid or **nose tip** (Hutchinson), eye pain or visual change	**ophthalmology**.
SAME DAY	Facial palsy with ear vesicles (**Ramsay Hunt**)	ENT.
URGENT	Immunosuppressed, disseminated or multi-dermatomal: specialist advice, often IV aciclovir.	
ROUTINE	Under 50 with no cause, or recurrent zoster: offer an **HIV test**.	

Red flags and when to refer

- Rash on the forehead, eyelid or **tip of the nose** (Hutchinson sign), eye pain or visual change: **same-day ophthalmology**.
- Immunosuppressed, disseminated or multi-dermatomal zoster: urgent specialist advice, often IV aciclovir.
- Facial palsy with ear vesicles (Ramsay Hunt): same-day ENT.
- Zoster in an under-50 without a cause, or recurrent zoster: consider an HIV test.

Red flags / refer

> **SAME-DAY OPHTHALMOLOGY**
> V1 rash, Hutchinson's sign, red eye, visual change, photophobia.

> **SAME-DAY ENT/MEDICAL**
> Ramsay Hunt (facial palsy).

> **ADMIT/URGENT**
> severe immunosuppression, disseminated rash, systemic illness (pneumonitis, hepatitis, encephalitis, meningism), motor weakness, urinary retention (sacral zoster), neonates.

> **PREGNANCY**

> seek specialist advice; non-immune pregnant contacts need assessment for VZIG/antiviral.

Complications and prognosis

Complications

- **Post-herpetic neuralgia** (pain over 3 months): the commonest complication, more likely with age.
- **HERPES ZOSTER OPHTHALMICUS**
  keratitis, uveitis, visual loss.
- **RAMSAY HUNT SYNDROME**
  facial palsy, ear vesicles, hearing loss and vertigo.
- Secondary bacterial infection, scarring.
- Dissemination, pneumonitis, hepatitis, encephalitis or stroke (vasculopathy), especially when immunocompromised.

Prognosis

- Most immunocompetent patients recover fully in 2–4 weeks.
- PHN affects about 10–20%, more often over 60; it usually improves with time.

Prescriber pearls (UK pharmacist IP)

- Pain in a dermatome before any rash: consider zoster, but don't miss cardiac/surgical causes.
- In a patient <50 with no obvious reason for zoster, **offer an HIV test** (Case 2 learning).
- Hydrate on aciclovir; check eGFR (neurotoxicity/nephrotoxicity in renal impairment, elderly).
- Explain **postherpetic neuralgia** (pain >3 months; commoner >50, HZO, severe acute pain) — early antiviral and good analgesia may reduce burden.
- **Vaccination:** Shingrix (recombinant, non-live) is offered on the NHS programme to eligible older adults and to severely immunosuppressed adults (including people with HIV meeting criteria) — check current Green Book chapter 28a for eligibility. Zostavax (live) is contraindicated in immunosuppression.

Self-check quiz

5 questions · optional

Q1 Antiviral and dose for shingles in an immunocompetent adult?

✓ Aciclovir 800 mg five times daily for 7 days (or valaciclovir 1 g TDS / famciclovir 500 mg TDS for 7 days), ideally within 72 h.

Q2 What is Hutchinson's sign?

✓ Vesicles on the tip or side of the nose (nasociliary nerve): high risk of eye involvement, same-day ophthalmology.

Q3 Ear vesicles with facial palsy?

✓ Ramsay Hunt syndrome: same-day ENT.

Q4 Zoster in a 32-year-old with no obvious cause?

✓ Offer an HIV test: zoster is an HIV indicator condition.

Q5 How long is a patient infectious?

✓ Until all lesions have crusted (usually 5–7 days).

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Photo gallery

Dermatomal zoster

4

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Ophthalmic zoster

2

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Dermatomal distribution

1

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Infections — viral & infestations Module 5 · Week 3

Scabies

Burrows, nocturnal itch and itchy contacts. Treat the whole household with permethrin or malathion, and never miss crusted scabies.

ALSO CALLED scabies | Sarcoptes | burrows | permethrin | malathion | ivermectin | +2

RELATED PAGES Head lice (pediculosis capitis) | Papular urticaria & insect bites | Eczema / dermatitis

KEY FACTS

- Infestation with the mite **Sarcoptes scabiei** var. hominis, spread by **prolonged skin-to-skin contact**.
- Intensely **itchy, worse at night**; burrows in finger webs, wrists, genitals; household members often affected.
- Itch is a hypersensitivity reaction starting **3–6 weeks** after first infestation (1–3 days on re-infestation).
- UK: **permethrin 5% cream** (malathion 0.5% aqueous if unsuitable), applied to the **whole body**, repeated after 7 days; treat **all close contacts at the same time**.
- **Crusted scabies** (immunosuppressed, frail) is highly contagious: specialist input.

Overview

Scabies is an intensely itchy infestation with the human itch mite **Sarcoptes scabiei**. It occurs at all ages and social backgrounds. UK clusters are common in households, care homes, nurseries and student halls.

Classical scabies usually involves only about 10–15 mites per person. **Crusted (Norwegian) scabies** is a hyperinfestation with thousands to millions of mites and is highly contagious.

NOTE

Not a hygiene failure

Reassure patients: scabies is not caused by poor hygiene. Keep consultations confidential and stigma-aware.

Epidemiology

- **Global burden**
  - About **200 million** people affected at any time; common in crowded and low-resource settings.

- **In the UK**
  - Family outbreaks, schools and **care homes**; sexual spread in young adults.
  - Most common in children, adolescents and older people.

Aetiology: causes and risk factors

- Spread is mainly by **prolonged skin-to-skin contact** (household, sexual, carer–patient). Brief handshakes rarely transmit it. Fomites matter more in crusted disease.
- The fertilised female burrows into the stratum corneum and lays eggs. Larvae hatch in about 3–4 days and mature in about 10–14 days. Off the host, mites usually die within a few days.
- **Itch is delayed hypersensitivity.** At first infestation it starts about **3–6 weeks** later; on re-infestation within 1–3 days. People can transmit before they itch, so asymptomatic contacts still need treatment.
- **Crusted scabies** occurs when the host cannot mount an effective immune or scratch response (frailty, dementia, learning disability, neurological disease, immunosuppression, or potent topical steroids).

Risk factors and associations

- **TRANSMISSION**
  prolonged skin contact (family, sexual partners, carers); fomites matter mainly in **crusted scabies**.
- Risk factors: crowding, care homes, immunosuppression (HIV, steroids), cognitive impairment or reduced sensation (cannot scratch).

Myths to address

> **"SCABIES MEANS DIRT"**
> it spreads by contact, regardless of hygiene.

> **"STILL ITCHING, SO TREATMENT FAILED"**
> itch can persist for **up to 4–6 weeks** after successful treatment.

Pathophysiology and pathology

The female mite lives in the **stratum corneum**; most symptoms come from the host's **allergic reaction** to mites and their products.

1. **Mating and burrowing**
   After mating on the skin surface, the female burrows into the stratum corneum at 0.5–5 mm a day for her 4–8 week life.

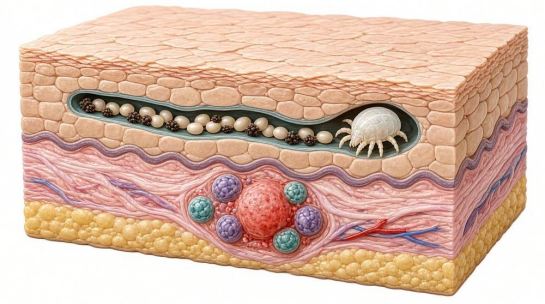

Fig. 1 · A female mite at the end of a burrow in the stratum corneum, with eggs and faeces behind her.
AI-generated illustration

2 **Eggs and the life cycle**

She lays **2–3 eggs a day**; larvae hatch in 2–5 days and become adults in about 2 weeks. A typical patient carries only **10–15 mites**.

3 **Delayed hypersensitivity**

Mites, eggs and faeces (scybala) trigger a **type IV** reaction: generalised itch and papules appear 3–6 weeks after first infestation.

4 **Crusted scabies**

With poor T-cell immunity or inability to scratch, mites multiply to **thousands or millions**, producing thick crusts and very high contagiousness.

5 **Distribution**

Mites prefer thin skin with few hair follicles: finger webs, wrists, elbows, axillae, genitals and breasts.

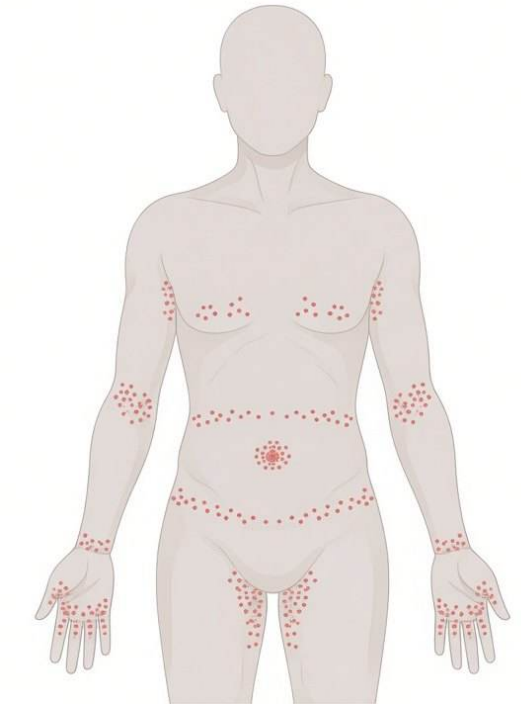

Fig. 2 · Typical sites: finger webs, wrists, elbows, axillae, waist, buttocks, genitals; head spared in adults.

AI-generated illustration

HISTOPATHOLOGY

- Mite, eggs or scybala within the stratum corneum (if biopsied), with a dermal infiltrate rich in **eosinophils**.
- Diagnosis is clinical or by **dermoscopy** (delta-wing sign) or skin scraping microscopy.

Clinical features

FEATURE	WHAT YOU SEE
Symptom	Generalised itch, often worse at night and after a warm bath; contacts itchy too
Burrows (pathognomonic)	Short wavy greyish threads (2–10 mm), sometimes with a vesicle; mite at the leading end

FEATURE	WHAT YOU SEE
Secondary eruption	Excoriated papules and eczematised patches, often away from the burrows
Nodules	Itchy red-brown nodules in axillae, groin, penis/scrotum; may persist weeks after cure
Infants	Palms, soles, face and scalp can be involved; irritability
Older adults	Few signs, atypical distribution; back involvement; care-home crusted disease

Typical sites: finger web spaces, flexor wrists, elbows, axillae, nipples, beltline, buttocks, genitalia and feet. Head and neck are usually spared in healthy adults but not in infants, older adults or immunocompromised people.

PRESCRIBER TIP

Skin of colour

Erythema is less obvious. Papules and nodules look violaceous or hyperpigmented; post-inflammatory pigmentation is common.

RED FLAG

Red flag: crusted scabies

Thick grey-yellow hyperkeratotic crusts on hands, feet, scalp or trunk; itch may be mild or absent. Refer urgently the same day. Combination oral ivermectin plus topical scabicide, isolation and outbreak control are needed.

Fig. 3 · Classic scabies burrow on the palm: short, wavy, greyish thread-like line with a vesicle — the mite sits at the leading end. Pathognomonic sign; look here first.

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Signs to recognise

Finger webs

Commonest site: papules, scale, burrows. Site: Web spaces.

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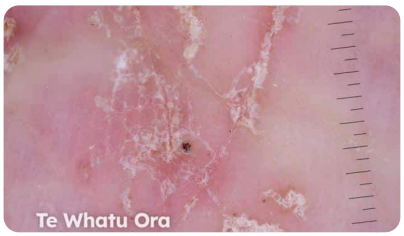
Dermoscopy

Dark ‘delta-wing jet’ at the burrow end. Site: Burrow.



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Genital papules

Itchy papules on penis or scrotum: highly suggestive. Site: Genitals.

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Nodular

Persistent itchy nodules. Site: Axilla, groin.

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Infants

Palms, soles, face and scalp. Site: Whole body.

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Crusted

Thick grey-yellow crusts; itch may be mild. Site: Hands, feet, scalp.

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Special groups

- **Infants**
  - Palms, soles, face and scalp can be involved; vesicles and nodules; treat the head and face too.
- **Older people and care homes**
  - Atypical presentation with little visible rash; crusted scabies may be missed: outbreak management with the health protection team.
- **Skin of colour**
  - Papules and nodules may look brown or violaceous; post-inflammatory hyperpigmentation is common.

Diagnosis and investigations

Diagnosis is **clinical** in most primary-care cases: nocturnal itch plus itchy contacts plus typical sites plus burrows.

- **BURROW INK TEST**

rub washable ink over a suspected burrow, wipe with alcohol; ink tracks the burrow.

- **DERMOSCOPY**
  dark triangular "delta-wing jet" (mite head/forelegs) at the end of the burrow "contrail".
- **Skin scraping** for microscopy (mites, eggs, scybala) when available.
- Offer an **STI screen** when scabies seems sexually acquired; refer to GUM for partner notification if needed.
- Refer if diagnosis uncertain, treatment fails despite correct use and simultaneous contact treatment, age under 2 months, crusted disease, or an institutional outbreak.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Atopic eczema	Flexural, atopy; no burrows; contacts not itchy.
	Insect bites	Grouped on exposed sites; pets or bed bugs.
	Dermatitis herpetiformis	Grouped vesicles on elbows, knees, buttocks; coeliac.
	Pompholyx	Deep palm vesicles; no trunk or genital papules.
	Body lice	Lice and eggs in clothing seams; homelessness.
	'Psoriasis' in a care home	Scrape it: could be **crusted scabies**.

Management

Self-care / OTC

- **Permethrin 5% cream** is a Pharmacy medicine for ages 2 months and over. Supply enough for **two applications for every person** treated at the same time.
- Do not put topical steroid on undiagnosed itch before scabicide treatment — it can mask scabies or encourage crusted disease.
- Avoid repeated excessive applications; they irritate and are a common cause of "failure" itch.

First-line treatment (NICE CKS / BNF)

OPTION	HOW TO USE	NOTES
Permethrin 5% cream (first-line)	Cool dry skin, whole body from neck down including webs, under nails, genitals and soles. Include scalp/face/neck/ears (avoid eyes/mouth) if under 2, aged 65, immunocompromised, or lesions there. Leave 8–12 h; reapply to washed hands. **Repeat after 7 days.**	OK in pregnancy and breastfeeding. ~30 g per adult application.
Malathion 0.5% aqueous liquid	Whole body, leave 24 h; repeat after 7 days. Aqueous (not alcohol) if eczema/asthma.	Alternative if permethrin allergy or intolerance; 6 months.
Oral ivermectin 200 micrograms/kg	With food; single dose repeated after 7 days.	CKS alternative (e.g. if eczema reacts to topicals). Avoid in pregnancy, breastfeeding and <15 kg. Check local formulary.

Contacts, laundry and return to school

- **Treat everyone at the same time** — household, sexual partners (past month), carers and regular sleep-over contacts — symptomatic or not.
- On day 1: hot-wash and hot-tumble bedding, towels and recent clothing, or bag unwashables for **3–4 days**. Classical scabies does not need fumigation.
- Children can return to school or nursery **after the first treatment** (UKHSA).

After treatment: post-scabetic itch

- Warn that itch and rash can last **2–4 weeks** (nodules longer). This is hypersensitivity, not failure.
- Crotamiton 10%, hydrocortisone 1% once mites are eradicated, emollients, and a night-time sedating antihistamine if sleep is poor.
- Review if itch lasts >2–4 weeks after the last application or if **new burrows** appear — check technique and contacts before re-treating.

UK GUIDANCE

Always check BNF / SmPC

Doses, age limits and pregnancy advice change. Confirm current BNF and local formulary before prescribing.

Medicines at a glance

Permethrin 5% cream

First line

DOSE Whole body from neck down incl. webs, under nails, genitals, soles; leave 8–12 h · Repeat after 7 days

CAUTIONS ~30 g per adult application. Include scalp, face, ears if < 2, 65, immunocompromised or lesions there. OK in pregnancy and breastfeeding. P medicine from 2 months.

Malathion 0.5% aqueous liquid

Alternatives

DOSE Whole body, leave 24 h · Repeat after 7 days

CAUTIONS Permethrin allergy or intolerance; 6 months.

Oral ivermectin

Alternatives

DOSE 200 micrograms/kg with food · Repeat after 7 days

CAUTIONS Avoid in pregnancy, breastfeeding, < 15 kg. Check formulary.

Treatment ladder

1

Contacts and laundry

Treat **everyone at once**: household, sexual partners (past month), carers.
Day 1: hot-wash and tumble bedding, towels, clothes, or bag for 3–4 days.

2 **First line**

Permethrin 5% cream

3 **Alternatives**

Malathion 0.5% aqueous liquid; Oral ivermectin

4 **Post-scabetic itch**

Lasts **2–4 weeks** (nodules longer): not failure. Crotamiton 10%, hydrocortisone 1% once mites cleared, emollients, night-time sedating antihistamine.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Steroid before scabicide**: Can mask scabies or encourage crusted disease.
- **Over-application**: Irritant dermatitis is a common cause of 'failure' itch.
- **Fire risk**: Paraffin products on fabrics catch fire easily.

Referral

SAME DAY	Suspected **crusted scabies**: specialist advice; isolation and outbreak control in care homes.
URGENT	Infant **under 2 months**: paediatric dermatology advice.
ROUTINE	Failure after two correct courses with all contacts treated.
ROUTINE	Significant secondary infection, systemic illness, or diagnostic doubt.

Red flags and when to refer

- Suspected **crusted scabies** same-day specialist advice; isolation and outbreak control in care homes.

- Infant **under 2 months** paediatric dermatology advice.
- Treatment failure after two correctly applied courses with all contacts treated.
- Significant secondary infection, systemic illness, or diagnostic uncertainty.

Complications and prognosis

Complications

- **Secondary bacterial infection** (impetigo, cellulitis), with post-streptococcal glomerulonephritis in endemic settings.
- **NODULAR SCABIES**
  itchy nodules that persist for months.
- Post-scabetic itch and eczema.
- Crusted scabies with institutional outbreaks.

Prognosis

- Cured with correct treatment of the patient and all contacts; reinfestation is common if contacts are missed.
- Itch settles over 2–6 weeks; nodules can take longer.

Prescriber pearls (UK pharmacist IP)

- **Treat the household, not the patient.** Agree one treatment night for everyone.
- Itch will not stop immediately — saying so prevents over-application and irritant dermatitis.
- Cut nails short; apply cream under nails; reapply to hands after every wash; mittens on infants.
- **Fire risk:** paraffin-containing products on clothing or bedding can catch fire easily — keep away from naked flames.
- Pregnancy: permethrin first choice; malathion alternative; avoid ivermectin.
- Care homes: think crusted scabies in any resident with “psoriasis” or “eczema” plus itchy staff.

Self-check quiz

5 questions · optional

Q1 How is permethrin 5% applied and repeated?

✓ Whole body from the neck down (incl. webs, under nails, genitals, soles), left 8–12 h, repeated after 7 days.

Q2 Who else must be treated, and when?

✓ All close contacts (household, sexual partners in the past month, carers) on the same night, symptomatic or not.

Q3 Why do contacts need treatment without itch?

✓ Itch is a delayed hypersensitivity starting 3–6 weeks after first infestation: they can be infested and infectious already.

Q4 Itch 2 weeks after correct treatment?

✓ Usually post-scabetic itch (hypersensitivity), not failure. Review if beyond 2–4 weeks or new burrows.

Q5 Care-home resident with thick crusts and itchy staff?

✓ Suspect crusted scabies: same-day specialist advice and outbreak control.

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Photo gallery

Classic scabies: burrows, web spaces, genital and axillary papules

10

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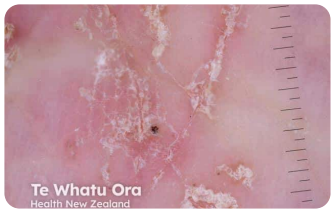

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Red flag: crusted scabies

2

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Infections — viral & infestations Module 5 · Week 3

Head lice (pediculosis capitis)

Only a live louse confirms infestation. Wet combing or dimeticone 4% are first-line; permethrin is not recommended in the UK.

ALSO CALLED head lice · nits · pediculosis · dimeticone · wet combing · bug busting

RELATED PAGES Scabies · Impetigo · Papular urticaria & insect bites

KEY FACTS

- Infestation of the scalp hair by the louse **Pediculus humanus capitis**.
- Very common in children aged **4–14**; spread by **head-to-head contact** (lice crawl, they cannot jump or fly).
- Diagnose only by finding a **live louse** with wet or dry **detection combing**; nits alone do not prove active infestation.
- Options: **wet combing** (bug busting) or a physically acting product such as **dimeticone 4%**, repeated after 7 days.
- Check all household members and treat only those with live lice; children need not miss school.

Overview

Head lice are scalp infestations with Pediculus humanus capitis, a wingless blood-feeding insect about 2–3 mm long. They are commonest in **children aged about 4–11**, especially girls. They are **not a sign of poor hygiene** and do not transmit disease in the UK.

Epidemiology

- **Who**
  - Worldwide, all social groups; commonest in primary-school children, more often **girls** (longer hair, closer contact).
- **Spread**
  - Direct head-to-head contact; sharing brushes or hats is a minor route.

Aetiology: causes and risk factors

- Spread is by **direct head-to-head contact**. Lice crawl; they cannot jump, fly or swim. Hats and brushes rarely matter because lice die within about 1–2 days off the scalp.
- Females glue eggs near the scalp. Eggs hatch in about 7–10 days; adults live about 3–4 weeks on the host.

- **Nits** are empty white egg cases that move out as hair grows. Nits far from the scalp usually mean an old infestation.
- Itch may take **4–6 weeks** to appear at first infestation; many children never itch.

Risk factors and associations

- Risk factors: female sex, more children in the family, sharing beds, combs and hats.
- Resistance to older insecticides (permethrin, malathion) is widespread, so physical or mechanical methods are preferred.

Myths to address

> **"LICE MEAN DIRTY HAIR"**
> lice prefer clean or dirty hair equally.

> **"LICE JUMP"**
> they crawl from hair to hair.

> **"NITS MEAN ACTIVE LICE"**
> empty egg cases stay glued to hair for months.

Pathophysiology and pathology

The louse lives on the scalp and feeds on blood; itch is an **allergic reaction** to its saliva.

1. **Eggs glued to hair**
   Females lay up to 10 eggs a day, cemented to the hair shaft close to the scalp; they hatch in 7–10 days.

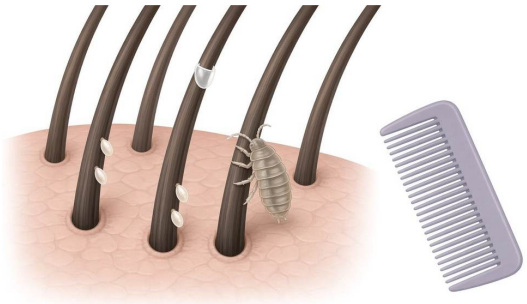

Fig. 1 · Louse and nits (eggs) glued close to the scalp on the hair shaft.
AI-generated illustration

2. **Nymphs mature**
   Nymphs become adults in about 7–10 days; adults live about 30 days and die within 1–2 days off the head.

3. **Feeding and itch**
   Lice feed on blood several times a day; sensitisation to saliva causes **itch after 4–6 weeks** of a first infestation.

4 **Detection**

Combing wet, conditioned hair with a fine-toothed comb is the most reliable way to find live lice.

Fig. 2 · Detection combing in sections from the scalp to the ends.

AI-generated illustration

HISTOPATHOLOGY

- Not relevant; diagnosis is by detection combing and seeing a live louse.

Clinical features

- Scalp itch (often absent), tickling, and excoriations — especially **behind the ears and at the nape**.
- Live grey-brown lice that move away from light; eggs/nits cemented to the shaft.
- Complications: secondary impetigo, occipital lymphadenopathy, eczematisation, sleep disturbance and school stigma.

Fig. 3 · Adult head louse (Pediculus humanus capitis) among scalp hairs — only finding a LIVE louse (ideally by detection combing) confirms active infestation.

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Signs to recognise

Adult louse

Six legs with claws for gripping hair. Site: Scalp hair.

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Eggs / nits

Cemented to the shaft near the scalp. Site: Hair shafts.

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Nape dermatitis

Itchy excoriated skin with nits. Site: Nape.

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Diagnosis and investigations

Only a live louse confirms active infestation (NICE CKS). Nits alone do not justify treatment.

- **Detection combing** with a fine-toothed plastic comb on wet conditioned hair (or dry) is more sensitive than looking alone.
- Check **all household members** and treat only those with live lice, **all on the same day**.
- Nits stay put; dandruff and hair casts brush or slide off.

Fig. 7 · Detection combing: wet, conditioned hair combed in sections from the scalp to the ends with a fine-toothed detection comb, wiping the comb on a tissue after each stroke. A live louse confirms active infestation.

AI-generated illustration

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Dandruff / seborrhoeic	Flaky scale that brushes off; no lice.
	Hair casts	Tubular casts that slide along the shaft.
	Scalp psoriasis	Thick scale beyond the hairline; nails, body plaques.
	Tinea capitis	Broken hairs, alopecia, black dots; mycology.
	Nape impetigo	Honey crust: look for lice as the driver.

Management

Treat **only when a live louse is found**. Choice depends on preference, age, asthma/eczema and previous treatment.

OPTION	HOW TO USE	NOTES
Wet combing (Bug Busting)	Conditioner + detection comb days 1, 5, 9 and 13; check day 17	Suitable for everyone including infants, pregnancy, asthma and eczema
Dimeticone 4% lotion/ spray/gel	Dry hair, leave overnight (check product), wash out; **repeat after 7 days**	Physical insecticide; CKS first-line in pregnancy, breastfeeding, young children, asthma/eczema. Flammable while on hair
Isopropyl myristate + cyclomethicone	Short contact (~10 min), comb out; repeat after 7 days	Check age and scalp licence
Malathion 0.5% aqueous liquid	Dry hair, leave 12 h; repeat after 7 days	Only chemical insecticide still recommended; resistance common; aqueous if asthma/eczema

UK GUIDANCE

Not recommended in the UK

Permethrin creme rinses (unsuitable contact time and widespread resistance), shampoos alone, herbal/essential-oil products, electric combs, and “preventive” treatment of people without live lice.

Checking cure

- Detection-comb 2–3 days after finishing treatment and again after 7 days.

- If live lice persist, check technique then switch to a **different mode of action**.
- Oral ivermectin or co-trimoxazole for resistant lice are **specialist/off-label** — refer rather than start them yourself.

Medicines at a glance

Dimeticone 4% lotion / spray / gel

Physical insecticide

DOSE	Dry hair, leave overnight (check product), wash out · Repeat after 7 days
CAUTIONS	CKS first line in pregnancy, breastfeeding, young children, asthma / eczema. Flammable while on hair.

Isopropyl myristate + cyclomethicone

Physical insecticide

DOSE	Short contact (~10 min), comb out · Repeat after 7 days
CAUTIONS	Check age and scalp licence.

Malathion 0.5% aqueous liquid

Chemical

DOSE	Dry hair, leave 12 h · Repeat after 7 days
CAUTIONS	Only chemical still recommended; resistance common; aqueous if asthma / eczema.

Treatment ladder

Option for everyone

Wet combing (Bug Busting): conditioner + detection comb on days **1, 5, 9 and 13**; check day 17. Fine for infants, pregnancy, asthma and eczema.

2

Physical insecticide

Dimeticone 4% lotion / spray / gel; Isopropyl myristate + cyclomethicone

3

Chemical

Malathion 0.5% aqueous liquid

4

Check cure

Detection-comb 2–3 days after finishing, and again after 7 days. Live lice persist: check technique, then **switch mode of action**.

5

Resistant

Oral ivermectin or co-trimoxazole are specialist / off-label: **refer**.

⚠ CAUTION

Prescribing limits and pitfalls

- **Not recommended (UK)**: Permethrin creme rinse, shampoos alone, herbal oils, electric combs, 'preventive' treatment.
- **No exclusion**: No school exclusion, boil-washing or fumigation.

Referral

URGENT	Pubic lice on a child's **eyelashes** safeguarding pathway.
ROUTINE	Infant **under 6 months**: wet combing only; seek advice if it fails.
ROUTINE	Repeated failure despite correct technique.
ROUTINE	Extensive secondary infection, severe eczematous scalp, or suspected misdiagnosis.

Red flags and when to refer

- Infant under 6 months (wet combing only; seek advice if this fails).
- Repeated failure despite correct technique.
- Extensive secondary infection or severe eczematous scalp.
- Pubic lice on a child's eyelashes safeguarding pathway.
- Suspected misdiagnosis (tinea capitis, psoriasis).

Complications and prognosis

Complications

- Scalp **excoriations and dermatitis**; secondary bacterial infection (impetigo) and enlarged neck nodes.
- Anxiety, stigma and school disruption.

Prognosis

- Curable; reinfestation from untreated contacts is common.
- Treatment failure is usually due to incorrect use, resistance or reinfestation.

Prescriber pearls (UK pharmacist IP)

No live louse, no treatment.

There is **no school exclusion** and no need to boil-wash bedding or fumigate. Inform school only as local practice.

Asthma/eczema: prefer wet combing or aqueous dimeticone; avoid alcohol-based lotions.

Itch can persist a few days after cure; leftover nits alone do not mean failure.

UK teaching point: course materials that recommend permethrin for head lice are **out of step with CKS**.

Self-check quiz

5 questions · optional

Q1 What confirms active head lice?

✓ A live louse (found by detection combing). Nits alone don't.

Q2 Wet-combing schedule?

✓ Days 1, 5, 9 and 13, with a check on day 17.

Q3 How is dimeticone 4% used?

✓ On dry hair, left on (often overnight per product), washed out, repeated after 7 days.

Q4 Is permethrin recommended for head lice in the UK?

✓ No: unsuitable contact time and widespread resistance.

Q5 Does a child need to stay off school?

✓ No school exclusion is needed.

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Photo gallery

Lice, nits and nape dermatitis

5

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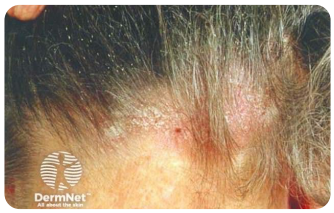

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Infections — viral & infestations Module 5 · Week 5

Herpes simplex labialis (cold sores)

Recurrent grouped vesicles on the lip border triggered by sun, illness, stress or menstruation. Self-limiting in healthy adults; know when to use oral antivirals and when it is an emergency (eczema herpeticum, eye, neonates, immunosuppression).

ALSO CALLED cold sores · HSV-1 · herpes labialis · fever blister · aciclovir cream · herpetic whitlow · +2

RELATED PAGES Herpes zoster (shingles) · Erythema multiforme · Impetigo · Eczema / dermatitis

KEY FACTS

- Recurrent **HSV-1** infection of the lip border: tingling, then grouped **vesicles**, crusting and healing in 7–10 days.
- Most adults carry HSV-1; the virus lies **latent in the trigeminal ganglion** and reactivates with sun, fever, stress, menstruation or trauma.
- Infectious from tingling until fully crusted; also spread by **asymptomatic shedding**.
- Self-care and OTC aciclovir cream for mild episodes; oral antiviral for severe, frequent or immunocompromised cases.
- Same-day assessment for **eczema herpeticum**, eye involvement, neonates and the immunocompromised.

Overview

Diagnosis confirmed from the course case: fever + recurrent perioral vesicles at the same site; the diagnosis is **recurrent herpes simplex labialis (HSV-1)**; questions covered confirmation tests and treatment of recurrences.

Recurrent vesicular eruption of the lips/perioral skin due to reactivation of latent **herpes simplex virus**, usually **HSV-1** (HSV-2 increasingly causes orolabial and HSV-1 genital disease). "Cold sores" / "fever blisters" — fever is a classic trigger. Primary infection is often childhood **herpetic gingivostomatitis**.

Epidemiology

- **Very common**
  - HSV-1 seroprevalence rises with age: most adults worldwide; under half of UK university entrants (less childhood exposure).
  - About 20–40% of infected people get recurrent cold sores.
- **Primary infection**
  - Usually in childhood, often silent or as **gingivostomatitis**; HSV-2 mainly after puberty (genital).

Aetiology: causes and risk factors

Spread by direct contact with lesions or asymptomatic viral shedding (saliva). After primary infection, HSV travels along sensory nerves to the **trigeminal ganglion** where it stays latent. **Triggers** of reactivation: fever/intercurrent infection (URTI), UV light, cold wind, menstruation, stress, fatigue, trauma (dental work, lip procedures, laser), immunosuppression. Recurrences hit the **same site** each time. Immunosuppression (HIV, transplant, chemo) larger, longer, chronic ulcerative lesions; zinc deficiency is sometimes suggested as a contributor (weak evidence).

Risk factors and associations

- **TRIGGERS FOR RECURRENCE**
  UV light, fever and colds, stress, fatigue, menstruation, lip trauma, dental work, immunosuppression.
- Transmission by kissing, sharing cups or towels; infectious for about 7–12 days per episode.

Myths to address

> **"COLD SORES ONLY SPREAD WHEN VISIBLE"**
> asymptomatic shedding can transmit HSV.

> **"ANTIVIRAL CREAM CURES IT"**
> it only shortens episodes if started at the first tingle.

Pathophysiology and pathology

HSV is a **neurotropic** DNA virus: it hides in sensory ganglia and travels along nerves to the skin.

1 **Primary infection**
Virus enters through mucosa or broken skin, replicates in keratinocytes and travels up sensory nerves.

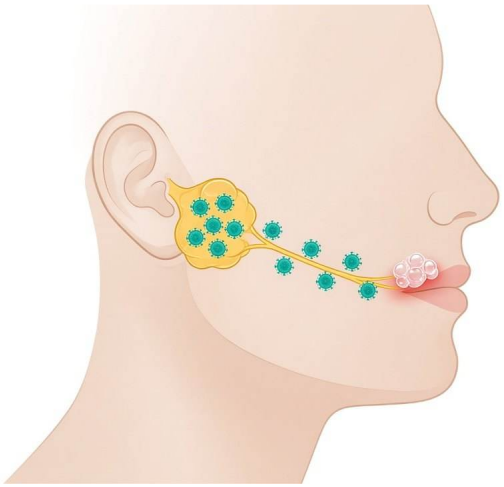

Fig. 1 · Latent HSV in the trigeminal ganglion and its path along the nerve to the lip.
AI-generated illustration

2 **Latency**
HSV persists **latent in the trigeminal ganglion** for life, invisible to the immune system.

3 **Reactivation**
Triggers (UV, fever, stress) reactivate the virus, which travels down the nerve: **prodromal tingling or burning** for 6–48 hours.

4 **Vesicles and healing**

Viral replication in keratinocytes causes ballooning and cell fusion, giving clustered vesicles that ulcerate, crust and heal in 7–10 days.

Fig. 2 · Stages of a cold sore: tingle, vesicles, ulcer, crust, healing.

AI-generated illustration

HISTOPATHOLOGY

- Intraepidermal vesicle with **ballooning degeneration**, multinucleated giant cells and nuclear moulding (Tzanck smear).
- Diagnosis is clinical; viral PCR swab if atypical, severe or immunocompromised.

Clinical features

- **Prodrome** (hours–1 day): tingling, burning, itching at the site.
- Then **grouped small vesicles on an erythematous base** at the **vermilion border**/perioral skin → pustules → painful erosions → crust → heal in **7–10 days** without scarring.
- Recurrences: a few per year typical; frequent (>6/year) in some.
- Other HSV patterns: cheek/nose, **herpetic whitlow** (finger), herpes gladiatorum (contact sports), **eczema herpeticum** (atopic skin), **HSV keratitis** (dendritic ulcer), **HSV-associated erythema multiforme** (target lesions 1–2 weeks after an outbreak).

Fig. 3 · Recurrent herpes labialis: small grouped vesicles on the vermilion border/lip, preceded by tingling — Case 4 ("fever" + similar lesions several times before = "fever blisters/cold sores").

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Patterns to recognise

Crusting stage

Crust extending onto the skin above the lip. Site: Lip, perioral.

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Skin of colour

Redness less obvious; look for grouped vesicles. Site: Lip.

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Other facial site

Recurrent HSV can appear on the cheek. Site: Cheek.

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Herpetic whitlow

Painful vesicles and swelling of a fingertip. Site: Finger.

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Eczema herpeticum

Monomorphic punched-out erosions on eczema. Site: Eczematous skin.

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Special groups

- **Atopic eczema**
  - Risk of **eczema herpeticum** (widespread punched-out erosions, fever): emergency.
- **Neonates and pregnancy**
  - Avoid kissing newborns during an episode; neonatal HSV is life-threatening.
- **Immunocompromised**
  - Large, persistent or atypical ulcers; needs oral or IV antivirals and specialist advice.

Diagnosis and investigations

Clinical in typical recurrent labialis. If atypical, severe, immunocompromised, near the eye or recurrent erythema multiforme: **viral swab for HSV PCR** (fast, most sensitive; preferred over culture/Tzanck smear). Serology (IgG) has little role in recurrent labialis.

Clinical

- Typical recurrent labialis: clinical.

Swab

- **HSV PCR** if atypical, severe, immunocompromised, near the eye, or recurrent EM.

Serology

- Little role in recurrent labialis.

Who is at risk

- Eczema, pregnancy near term, neonates nearby, immunosuppression?

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Impetigo	Honey crusts, spreading, children; no prodrome.
	Herpes zoster	Dermatomal, more painful, usually one episode.
	Aphthous ulcers	Inside the mouth; no vesicles.
	Angular cheilitis	Fissured mouth corners; dentures, iron / B12.
	Fixed drug eruption	Same site with each dose; dusky patch.
	EM / SJS	Targets, several mucosae, drug history: urgent if extensive.

Management

Based on NICE CKS Herpes simplex – oral:

Self-care / OTC (immunocompetent)

- Usually self-limiting; paracetamol/ibuprofen for pain; simple analgesic gels or creams; adequate fluids.
- **Topical aciclovir 5% cream** (OTC, five times daily for 5 days, start at the first tingle) — modest benefit (shortens episode by roughly half a day to a day); penciclovir 1% (Rx/OTC per product). Barrier/hydrocolloid cold-sore patches may help comfort and reduce spread.
- Hygiene: don't touch/pick; wash hands; don't share towels, cups, lip products; **avoid kissing**, especially **neonates** and people with eczema or immunosuppression; avoid oral sex during lesions.

- **Sunscreen lip balm (SPF 30+)** if UV is a trigger.

Oral antiviral (prescriber)

- Consider for severe or frequent episodes in immunocompetent adults, e.g. **episodic aciclovir 400 mg five times daily for 5 days** (start at prodrome) or **valaciclovir 2 g twice daily for 1 day** (cold-sore regimen in the UK SmPC; check local formulary).
- **Suppressive therapy** for frequent/severe recurrences, HSV-associated erythema multiforme, or before a known trigger: **aciclovir 400 mg twice daily**, review after 6–12 months (specialist advice often sought; off-label in some uses).
- Immunocompromised: oral antivirals (higher dose/longer) and **seek specialist advice**; chronic/large ulcers may need IV therapy.

Medicines at a glance

Aciclovir 5% cream

First tingle

DOSE	Five times daily · 5 days
CAUTIONS	Modest benefit (~½–1 day shorter). Penciclovir 1% alternative.

Aciclovir (oral)

Severe or frequent · adults

DOSE	400 mg five times daily · 5 days, from prodrome
CAUTIONS	Dose-reduce in renal impairment.

Valaciclovir (oral)

Severe or frequent · adults

DOSE	2 g twice daily · 1 day
CAUTIONS	UK SmPC cold-sore regimen; check formulary.

Aciclovir (oral)

Suppression

DOSE	400 mg twice daily · Review after 6–12 months
CAUTIONS	Frequent / severe recurrences or HSV-associated EM; specialist advice often sought.

Treatment ladder

1

Self-care · OTC

Paracetamol or ibuprofen; fluids; cold-sore patches for comfort. Don't pick; wash hands; no sharing cups or lip products.

2

First tingle

Aciclovir 5% cream

3

Severe or frequent · adults

Aciclovir (oral); Valaciclovir (oral)

4

Suppression

Aciclovir (oral)

5

Immunocompromised

Higher dose / longer oral antivirals; **specialist advice**; IV for chronic large ulcers.

⚠ CAUTION

Prescribing limits and pitfalls

- **No steroid or antibiotic creams**: Unless secondary impetigo is confirmed.
- **Timing**: Benefit falls sharply once vesicles form.

Referral

SAME DAY — Punched-out erosions spreading over eczema: **eczema herpeticum**.

SAME DAY — Red painful eye, photophobia, blurred vision, vesicles on nose tip: **ophthalmology**. Never topical steroid.

URGENT **Neonate** (< 1 month) or primary HSV late in pregnancy.

URGENT Immunosuppressed with extensive or non-healing lesions (> 2 weeks).

ROUTINE Child with gingivostomatitis not drinking, or recurrent erythema multiforme.

Red flags and when to refer

- Painful punched-out erosions spreading over eczema: **eczema herpeticum**, same-day.
- Red painful eye, photophobia or blurred vision: same-day ophthalmology (herpetic keratitis).
- Newborn contact, pregnancy near term, or immunosuppressed with extensive or non-healing lesions: urgent specialist advice.

Red flags / refer

- **Eczema herpeticum** — punched-out erosions spreading over eczema, unwell same-day assessment/oral or IV aciclovir; involve ophthalmology if periocular.
- **Eye involvement** (red painful eye, photophobia, vesicles on eyelid/nose tip) same-day ophthalmology; never give topical steroid.
- **Neonate** (<1 month) with any suspected HSV, or primary HSV in late pregnancy urgent.
- Immunocompromised, lesions >2 weeks or very extensive, systemic upset, dehydration in primary gingivostomatitis (child not drinking), recurrent erythema multiforme, diagnostic doubt GP/specialist.

Complications and prognosis

Complications

- Eczema herpeticum (Kaposi varicelliform eruption).
- Herpetic keratitis and eye involvement.
- Herpetic whitlow (finger).
- Secondary impetigo.
- **Erythema multiforme** triggered by recurrent HSV.
- Rarely encephalitis or disseminated infection.

Prognosis

- Lifelong latent infection; recurrences often become less frequent with age.
- Episodes heal without scarring in 7–10 days.

Prescriber pearls (UK pharmacist IP)

- Start antivirals at the **first tingle** — benefit falls sharply once vesicles form.
- Recurrent lesions at the same place + fever = HSV; one-off band of vesicles = zoster.
- Don't prescribe topical corticosteroid or antibiotic creams for cold sores (unless confirmed secondary impetigo).
- Patients with frequent severe recurrences: consider underlying immunosuppression (e.g. HIV) and offer testing if risk.
- Aciclovir: dose-reduce in renal impairment; keep hydrated.

Self-check quiz

5 questions · optional

Q1 Where does HSV-1 lie latent?

✓ In the trigeminal (sensory) ganglion.

Q2 OTC aciclovir cream regimen?

✓ 5% cream five times daily for 5 days, started at the first tingle.

Q3 An oral option for a severe cold sore in an adult?

✓ Aciclovir 400 mg five times daily for 5 days, or valaciclovir 2 g twice daily for 1 day.

Q4 Vesicles on the nose tip with a red painful eye?

✓ Possible herpetic eye disease: same-day ophthalmology; no topical steroid.

Q5 Which test if atypical?

✓ Viral swab for HSV PCR.

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Photo gallery

Herpes labialis 4

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Complications and variants 4

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Infections — viral & infestations Module 5 · Week 3

Lyme disease & erythema migrans

An expanding erythema migrans after a tick bite is clinical — treat with doxycycline for 21 days without serology (NICE NG95).

ALSO CALLED Lyme disease · erythema migrans · tick bite · Borrelia · Ixodes · acrodermatitis chronica atrophicans · +2

RELATED PAGES Annular (ring-shaped) rashes: differential · Papular urticaria & insect bites · Cellulitis & erysipelas · The returning traveller with a skin problem

KEY FACTS

- **Lyme disease** is infection with Borrelia burgdorferi sensu lato from an infected **Ixodes tick** bite.
- Erythema migrans an expanding red patch (often over 5 cm) appearing 1–4 weeks after the bite: diagnose clinically, no tests needed (NICE NG95).
- Doxycycline 100 mg twice daily or 200 mg once daily for 21 days; amoxicillin or azithromycin as alternatives.
- Remove ticks with fine-tipped tweezers or a tick tool; no prophylaxis or serology after an asymptomatic bite.
- Later disease: facial palsy, meningitis, carditis, arthritis, acrodermatitis chronica atrophicans: specialist.

Overview

Lyme disease is a tick-borne infection with Borrelia burgdorferi sensu lato. It is the commonest tick-borne infection in the UK, and **the skin is the commonest presenting organ**.

Erythema migrans (EM) is the hallmark of early localised disease. **Acrodermatitis chronica atrophicans (ACA)** is a late cutaneous sign (months to years later), seen almost only in European Lyme (B. afzelii). Borrelial lymphocytoma (bluish nodule on earlobe, nipple or scrotum) is a less common early European feature.

Epidemiology

- **Who**
  - UK: a few thousand cases a year; New Forest, Exmoor, South Downs, Lake District, Scottish Highlands.
  - Peak May–September; walkers, gardeners, outdoor workers.

Aetiology: causes and risk factors

- Vector: hard tick **Ixodes ricinus** (sheep/deer tick). Tiny nymphs cause most infections. **Most bites do not transmit**; risk rises after about 24–48 hours of attachment, so prompt correct removal helps (NICE NG95).
- UK risk: grassy/wooded areas, parks and gardens — higher in southern England and the Scottish Highlands, but possible anywhere. Peak spring–autumn.
- EM: dermal spread from the bite. Dissemination can cause multiple EM, neuroborreliosis, carditis and arthritis.
- ACA: chronic skin infection on acral extensors, often without a remembered bite.

Risk factors and associations

- Ixodes ricinus ticks; transmission usually needs prolonged attachment (over 24 hours).

Myths to address

> **"NO BULL'S-EYE, NO LYME"**
> most EM is uniformly red, not target-like.

> **"A NEGATIVE BLOOD TEST EXCLUDES IT"**
> serology is often negative in early EM.

Pathophysiology and pathology

Borrelia spreads **outwards in the skin** from the bite, then through the blood.

1. **Tick bite**
   An infected tick feeds for over a day; spirochaetes migrate from tick gut to saliva.
2. **Local spread**
   Borrelia spreads centrifugally in the dermis, causing the expanding erythema migrans.

Fig. 1 · Erythema migrans expanding outwards from the bite site.
AI-generated illustration

3. **Dissemination**
   Spirochaetes spread to the nervous system, heart, joints and other skin sites.
4. **Late disease**
   Persistent infection causes arthritis or acrodermatitis chronica atrophicans.

5 **Prevention**

Prompt correct tick removal reduces transmission.

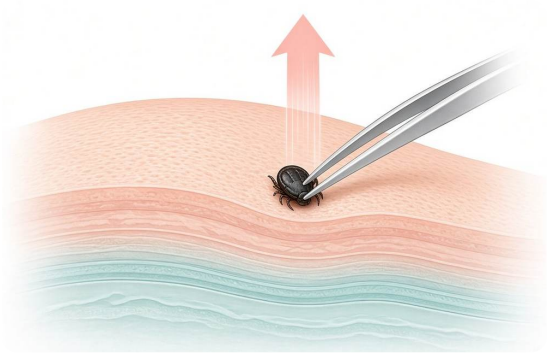

Fig. 2 · Grasp close to the skin and pull steadily up.

AI-generated illustration

HISTOPATHOLOGY

- Not needed for EM. Serology (ELISA then immunoblot) for suspected disseminated disease.

Clinical features

Erythema migrans (NICE NG95 themes)

- A **red rash that enlarges** over days; may have central clearing ("bull's-eye"), but many lesions are uniformly red or bluish.
- Usually **not** hot, itchy or painful. Appears **3 days to 3 months** after a bite (commonly 1–4 weeks).
- Often 5 cm at the bite site (legs, groin, axillae, beltline; head/neck in children). Harder to see in darker skin (bruise-like).
- May accompany flu-like non-focal symptoms.

PRESCRIBER TIP

Not EM

A tick-bite reaction that appears and fades within about 48 hours, and is hot, itchy or painful, is local hypersensitivity — not Lyme.

Acrodermatitis chronica atrophicans

INFLAMMATORY STAGE

bluish-red doughy discolouration of an extensor limb (hand, forearm, lower leg), often unilateral.

ATROPHIC STAGE

thin "cigarette-paper" skin with visible veins — easily mistaken for ageing or venous disease.

Often accompanies sensory neuropathy; mostly older women with European exposure.

Fig. 3 · Erythema (chronicum) migrans: expanding red macular patch at a tick-bite site, days–weeks after the bite — diagnose and treat clinically (NICE NG95); no blood test needed.

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Types to recognise

Erythema migrans

Expanding red patch. Site: Bite site.

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Central clearing

'Bull's-eye': not required. Site: Trunk, limbs.

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Large EM

Keeps growing over days. Site: Limbs.

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ACA atrophic

Cigarette-paper skin, visible veins. Site: Extensor limbs.

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Special groups

- **Pregnancy and children**
  - Amoxicillin in pregnancy; children under 12 per NG95 weight-based doses or specialist advice.

Diagnosis and investigations

- **EM: diagnose and treat clinically. No blood test** — early serology is often negative (NG95).
- Without EM (including suspected ACA): clinical picture plus **ELISA then immunoblot** via a UKAS-accredited lab. ACA is nearly always strongly IgG-positive.
- **Do not** diagnose or test after an asymptomatic tick bite.
- Do not rely on private "Lyme specialty" panels without checking accreditation.
- Absence of a remembered bite does **not** exclude Lyme.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Tick-bite hypersensitivity	Within hours, fades 48 h, itchy / hot, small.
	Cellulitis	Hot, tender, rapid spread, unwell.
	Tinea corporis	Scaly active edge; EM is not scaly.
	Granuloma annulare	Chronic firm papular ring; no daily expansion.

CONDITION	DISTINGUISHING FEATURES
ACA mimics	Chilblains, venous disease, morphoea, ageing skin.

Management

Prevention and tick removal

- Cover skin, use DEET-type repellent, and check skin folds/scalp after outdoor activity.
- Remove ticks with **fine-tipped tweezers or a tick tool**: grasp close to the skin and pull steadily up — do not twist, crush, burn or smother with oils (UKHSA).
- **No antibiotic prophylaxis or serology** after an asymptomatic bite. Watch the site for weeks and return if a spreading rash or flu-like illness appears.

Erythema migrans (adults / 12 years, no focal symptoms) — NICE NG95

CHOICE	REGIMEN	DURATION
First-line	Doxycycline 100 mg BD or 200 mg OD	21 days
First alternative	Amoxicillin 1 g TDS	21 days
Second alternative	Azithromycin 500 mg OD (avoid in carditis / QT risk)	17 days

- Doxycycline: take upright with water; separate from antacids/iron/calcium; sun protection; avoid in pregnancy and under 12.
- Pregnancy: usually **amoxicillin** after assessment.
- Children under 12 / under 18 with more than a single uncomplicated EM: discuss with a specialist (NG95).
- Warn about a self-limiting **Jarisch–Herxheimer** flu-like flare in the first day or two — keep taking the antibiotic.

ACA and late / focal disease

- Refer for confirmation (serology ± biopsy).
- Adult ACA: doxycycline (or amoxicillin) for **28 days** per NG95; IV ceftriaxone is a second alternative.
- Atrophy and neuropathy may not reverse — set expectations.

UK GUIDANCE

Check BNF

Confirm current doses, pregnancy advice and interactions before prescribing.

Medicines at a glance

Doxycycline

First line

DOSE	100 mg twice daily or 200 mg once daily · 21 days
CAUTIONS	Upright with water; separate from antacids / iron / calcium; sun protection; avoid in pregnancy and < 12.

Amoxicillin

Alternatives

DOSE	1 g three times daily · 21 days
CAUTIONS	Usually used in pregnancy after assessment.

Azithromycin

Alternatives

DOSE	500 mg once daily · 17 days
CAUTIONS	Avoid in carditis / QT risk.

Treatment ladder

1 **Prevent & remove**

Cover skin, repellent, check folds and scalp after outdoors. Remove ticks per UKHSA (tweezers / tick tool, pull steadily).

2 **First line**

Doxycycline

3 **Alternatives**

Amoxicillin; Azithromycin

4 **Specialist**

Children < 12 (and some < 18), focal disease, ACA: per NG95 / specialist.

⚠ CAUTION

Prescribing limits and pitfalls

- **Two courses done, still symptomatic**: No more antibiotics: specialist / reference-lab advice.

Referral

999 / A&E	Meningitis / encephalitis signs, carditis or heart block, uveitis.
SAME DAY	Facial palsy, radiculopathy, inflammatory arthritis, multiple EM with systemic upset: discuss with specialist **without delaying antibiotics**.
ROUTINE	Suspected ACA: serology and dermatology.

Red flags and when to refer

- Emergency pathways even if Lyme is possible: meningitis/encephalitis signs, uveitis, carditis or complete heart block.
- Same-day/urgent GP: facial palsy, radiculopathy, inflammatory arthritis, multiple EM with systemic upset — discuss with a specialist without delaying antibiotics.
- Persistent symptoms after two completed antibiotic courses: do not give more antibiotics; seek specialist/ reference-lab advice.

Complications and prognosis

Complications

- Neuroborreliosis, carditis (heart block), arthritis, ACA.
- Jarisch–Herxheimer reaction on starting treatment.

Prognosis

- EM treated promptly has an excellent outlook; some have lingering fatigue.

Prescriber pearls (UK pharmacist IP)

💡 **Expanding red ring after the outdoors = treat. Don't test.**

- Draw round the edge and photograph to monitor spread.
- A full **21-day** course matters for EM.
- Reinfection is possible — infection does not confer lasting immunity.
- ACA is often missed as "poor circulation" in an older person's limb — ask about European countryside exposure and test.

Self-check quiz

4 questions · optional

Q1 Blood test for typical EM?

✓ No: diagnose and treat clinically.

Q2 First-line EM regimen in adults?

✓ Doxycycline 100 mg BD or 200 mg OD for 21 days.

Q3 Antibiotics after an asymptomatic tick bite?

✓ No; advise watching the site.

Q4 Red itchy patch that appeared within hours and faded by 48 h?

✓ Tick-bite hypersensitivity, not Lyme.

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Photo gallery

Tick and erythema migrans

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Late Lyme: acrodermatitis chronica atrophicans

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Infections — viral & infestations Module 5 · Week 5 · Module 4 · Week 1

Pityriasis rosea

A self-limiting rash of young adults: a single herald patch, then oval scaly patches along skin lines on the trunk ('Christmas tree'), fading over 6–12 weeks. Reassure, but test for syphilis when the picture is atypical.

ALSO CALLED PR · herald patch · Christmas tree rash · collarette scale · HHV-6 · HHV-7 · +2

RELATED PAGES Psoriasis · Tinea: corporis, cruris & pedis · Annular (ring-shaped) rashes: differential · Pityriasis versicolor

KEY FACTS

- Self-limiting rash of young people: a **herald patch** then oval pink patches with a **collarette of scale** in a 'Christmas tree' pattern on the trunk.
- Incidence about **0.5–2%**; most common at **10–35 years**, often in spring and autumn.
- Probably triggered by reactivation of **HHV-6/7**; not contagious in practice.
- Reassure; emollient and a mild or moderate topical steroid or antihistamine for itch. Resolves in **6–12 weeks**.
- In **pregnancy** (first 15 weeks) there are reports of miscarriage: inform the obstetric team. Consider secondary syphilis if palms or soles are involved.

Overview

Diagnosis confirmed from the course case: the case shows a single initial lesion (Picture 1) followed ~2 weeks later by a generalised eruption (Picture 2) after a URTI. The diagnosis is **pityriasis rosea**; guttate psoriasis was listed as a **differential**, not the answer. The key mimic to exclude is **secondary syphilis** as the must-not-miss differential.

A common, **self-limiting** papulosquamous eruption of adolescents and young adults (10–35 years), beginning with a single **herald patch** followed days to ~2 weeks later by a **secondary eruption** of smaller oval scaly patches on the trunk and proximal limbs. Lasts about **6–12 weeks** (most 6–8). Recurrence is uncommon (~2–3%).

Epidemiology

- **Who**
  - Teenagers and young adults (10–35 years); slightly more common in women.
  - Incidence about 0.5–2%; seasonal clustering in spring and autumn.
- **Recurrence**
  - Uncommon (about 2–3%).

Aetiology: causes and risk factors

Cause not proven; strongest evidence links it to **reactivation of human herpesvirus 6 and 7 (HHV-6/HHV-7)** — supported by preceding URTI/prodrome, seasonal clustering, and low recurrence. **PR-like (pityriasiform) drug**

eruptions occur with e.g. ACE inhibitors, NSAIDs, isotretinoin, gold, some biologics/vaccines — these lack a herald patch and persist while the drug continues.

Risk factors and associations

- **HHV-6 and HHV-7** reactivation is the strongest association.
- **Drug-induced** PR-like eruptions: ACE inhibitors, NSAIDs, metronidazole, isotretinoin, imatinib and others.
- Reported after vaccines (BCG, influenza, COVID-19) and viral infections.

Myths to address

> **"IT'S RINGWORM"**
> tinea has an advancing scaly edge; PR has trailing scale just inside the edge.

> **"IT'S CATCHING"**
> it is not considered contagious.

Pathophysiology and pathology

PR is thought to be a **T-cell response to reactivated HHV-6/7**, following skin tension lines.

1. **Viral reactivation**
   HHV-6/7 reactivate (often after a mild viral illness), triggering a cell-mediated immune response in the skin.
2. **Herald patch**
   A single 2–5 cm oval pink patch appears first, often on the trunk.
3. **Secondary eruption**
   1–2 weeks later many smaller ovals appear along **cleavage (Langer) lines**, giving the Christmas-tree pattern.

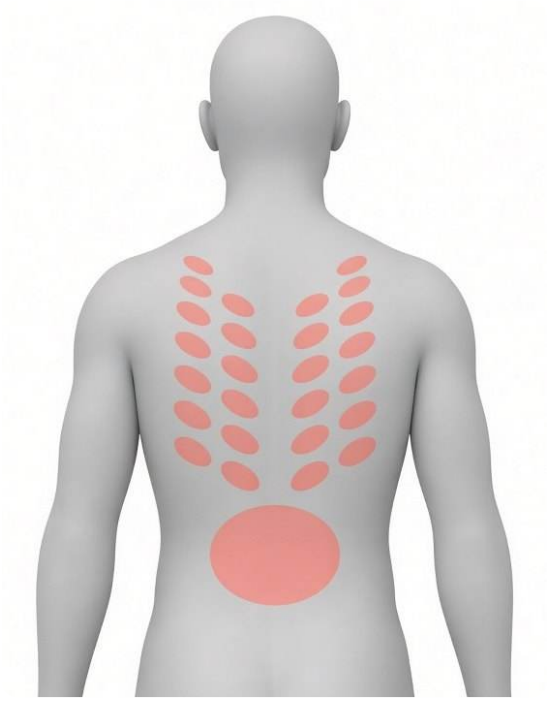

Fig. 1 · Oval patches following the ribs: the Christmas-tree pattern on the back.
AI-generated illustration

4 **Collarette scale**

Scale sits **just inside the edge**, trailing behind it, unlike tinea where scale is at the advancing edge.

Fig. 2 · Pityriasis rosea collarette (trailing scale) vs tinea (scale at the advancing edge).

AI-generated illustration

HISTOPATHOLOGY

- Mild spongiosis with focal parakeratosis (mounds), extravasated red cells and a superficial perivascular lymphocytic infiltrate.
- Biopsy only if atypical or persistent beyond 3 months.

Clinical features

- Possible prodrome: malaise, headache, sore throat, low-grade fever (URTI — Case 3).
- **HERALD PATCH**
  single, 2–5 cm, oval, salmon-pink (grey-brown/violaceous in darker skin) plaque with fine **collarette scale** inside the edge; often trunk, upper arm, neck, thigh.
- **Secondary eruption** (after ~1–2 weeks): many smaller oval macules/plaques with collarette scale, long axes along skin cleavage lines **"Christmas-tree" pattern** on the back, "V" on the chest; usually spares face, scalp, palms and soles.
- Itch: none to moderate (~25% significant).
- Variants: **inverse** (flexural — axillae/groin, commoner in children/skin of colour), papular, vesicular, purpuric, urticarial, giant, unilateral, oral lesions.
- Darker skin: more papular/facial lesions; prolonged post-inflammatory hyper/hypopigmentation.

Fig. 3 · Herald patch with the later secondary eruption of smaller ovals — Case 3 "Picture 2" (rash ~2 weeks after the first patch, post-URTI).

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Types to recognise

Herald patch

Solitary oval, collarette scale. Site: Trunk.

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Secondary eruption

Ovals along cleavage lines. Site: Back.

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Skin of colour

Grey-brown / violaceous lesions. Site: Trunk.

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Secondary syphilis

Copper-red palm / sole papules. Site: Palms, soles.

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Special groups

- **Skin of colour**
  - Lesions look grey-brown or violaceous; papular and inverse (flexural) forms are more common; pigment change can persist.
- **Pregnancy**
  - Possible link with miscarriage in the first 15 weeks: inform the obstetric team.

Diagnosis and investigations

Clinical. Tests only to exclude mimics:

- **Syphilis serology** (treponemal EIA ± RPR) if atypical, palms/soles involved, no herald patch, sexual risk, or lasting >12 weeks.
- **Skin scrapings for mycology** if the herald patch alone looks like tinea (scale at advancing edge).
- Drug history (pityriasiform drug eruption).
- Biopsy rarely (atypical/persistent).
- Pregnancy test/status relevant — see 3.6.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Guttate psoriasis	Many small drop papules, thicker silvery scale, **no herald patch**, after strep throat.
	Secondary syphilis	Palms / soles, mucous patches, nodes, malaise; positive serology.
	Tinea corporis	Active scaly edge, central clearing; KOH +.
	Discoid eczema	Coin-shaped, very itchy, crusted; limbs.
	Pityriasiform drug eruption	New drug, no herald patch, persists.

What it is

Guttate psoriasis is an acute eruption of small (<1 cm) drop-like scaly papules/plaques, typically in children, teens, and young adults — often **1–3 weeks after streptococcal pharyngitis**. Module 4 Spot Diagnosis consensus: widespread scaly eruption on trunk/limbs in a 14-year-old with no prior skin disease **guttate psoriasis**. Main clinical DDx: **pityriasis rosea**.

How it is caused / pathophysiology

- Same psoriatic immune axis; strong link to **Group A streptococcus** and HLA-C*06:02-enriched early-onset disease.
- May be first presentation of psoriasis or remain a one-off self-limited episode.
- PR (DDx): putative viral trigger; herald patch then secondary oval plaques on Langer lines.

How to diagnose

- Clinical. Ask about sore throat; consider throat swab / ASO if indicated.
- Look for herald patch and Christmas-tree pattern to favour PR.
- Biopsy rarely needed.
- Consider syphilis serology if PR-like but atypical (palms/soles, systemic, persistent).

Management for pharmacist/IP (UK)

Guttate:

- Emollients; mild–potent TCS matched to extent/site; vitamin D analogues sometimes.
- Treat proven strep infection per local antimicrobial guidance (treating strep does not always clear guttate quickly).
- Counsel possible recurrence or later plaque psoriasis.

PR (if that is the diagnosis):

- Reassure; emollient ± mild TCS for itch; UV advice individualised; refer if severe/atypical.

Prescriber pearls & counselling

- Teen + abrupt drop-like scaly rash + recent sore throat guttate until proven otherwise.
- Avoid oral steroids.
- Safety-net return if eruption becomes confluent, pustular, or erythrodermic.

Photo / reference URLs

- DermNet — Guttate psoriasis: dermnetnz.org
- DermNet — Pityriasis rosea: dermnetnz.org

Management

Based on NICE CKS Pityriasis rosea and PCDS/BAD leaflet themes:

Self-care / reassurance

- Explain benign, **non-contagious**, self-limiting (6–12 weeks); no isolation needed; scars do not occur (pigment change may persist months in darker skin).
- Emollients and soap substitutes; avoid hot showers, perfumed products, tight clothes; moderate sun exposure may speed resolution in some.

Symptomatic Rx

- Itch: emollient ± **mild–moderate topical corticosteroid** (e.g. clobetasone butyrate 0.05%) short course to itchy areas; **sedating antihistamine** at night (e.g. chlorphenamine; counsel drowsiness) or non-sedating by day.
- Evidence for **high-dose aciclovir** early in extensive/symptomatic disease is limited — specialist decision.

Guttate psoriasis (if that is the diagnosis instead)

- Throat swab only if sore throat ongoing (treat streptococcal infection per NG84); emollients; topical vitamin D analogue ± topical steroid, coal-tar; refer dermatology for **NB-UVB phototherapy** if widespread/distressing (lesions too numerous for topicals). Often clears in 3–4 months; ~⅓ later develop chronic plaque psoriasis.

Treatment ladder

1 **Reassure**

Benign, **non-contagious**, settles in 6–12 weeks; no isolation; no scarring (pigment may linger).

2 **Skin care**

Emollients, soap substitutes; avoid hot showers, perfumes, tight clothes; moderate sun may help.

3 **Itch**

Mild–moderate topical steroid (e.g. clobetasone butyrate 0.05%) short course; sedating antihistamine at night or non-sedating by day.

4 **Specialist**

High-dose aciclovir early in extensive disease: limited evidence, specialist decision.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Oral steroids**: Don't: may prolong or worsen PR.

Referral

URGENT Palm / sole lesions, mouth ulcers, nodes or sexual risk: **syphilis serology** and sexual health.

URGENT PR in the **first 15 weeks of pregnancy**: inform midwife / obstetric team.

ROUTINE Lasting > 3 months: rethink the diagnosis.

Red flags and when to refer

- Palm or sole lesions, mouth ulcers, lymphadenopathy or sexual risk: **syphilis serology** and sexual-health referral.
- Pityriasis rosea in the **first 15 weeks of pregnancy**: inform the midwife or obstetric team.
- Lasting more than 3 months: reconsider the diagnosis.

Red flags / refer

- **PREGNANCY**
  PR in pregnancy (esp. first 15 weeks) has been associated with miscarriage/premature delivery — inform/refer to midwife/obstetrician (CKS).
- Atypical features, palms/soles/mucosal involvement or sexual risk **syphilis serology / refer to sexual health (GUM)**.
- Persisting >12 weeks, diagnostic doubt, severe symptoms GP/dermatology.
- Suspected drug cause review medication with prescriber.

Complications and prognosis

Complications

- Itch; post-inflammatory hyper- or hypopigmentation lasting months.
- Pregnancy complications (rare reports).

Prognosis

- Resolves spontaneously in 6–12 weeks; recurrence is uncommon.

Prescriber pearls (UK pharmacist IP)

"One big patch first, then many small ovals along the ribs" = PR; "shower of small drops after strep throat" = guttate psoriasis.

The herald patch is often mis-treated as ringworm — antifungal failure is a clue.

Every generalised papulosquamous rash: ask about sexual history and look at palms and soles.

Do not prescribe oral steroids for PR (may prolong/ worsen).

Provide BAD/PCDS patient leaflet; review if not resolving by 3 months.

Self-check quiz

4 questions · optional

Q1 Where is the scale on a herald patch?

✓ A collarette just inside the edge (tinea: at the advancing edge).

Q2 Typical duration?

✓ 6–12 weeks (most 6–8).

Q3 When to test for syphilis?

✓ Atypical, palms / soles, no herald patch, sexual risk, or > 12 weeks.

Q4 PR at 10 weeks' pregnancy?

✓ Inform the midwife / obstetric team (miscarriage risk).

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Photo gallery

Pityriasis rosea

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Differential: guttate psoriasis

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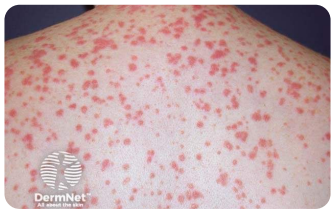
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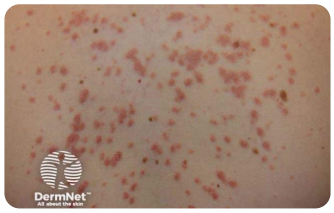
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Differential: secondary syphilis

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Infections — viral & infestations Module 5 · Week 3

Blister beetle & Paederus dermatitis

Burning linear blisters after a beetle is crushed on the skin (cantharidin or paederin). Wash promptly; do not crush beetles; refer eye involvement same day.

ALSO CALLED: blister beetle · cantharidin · Paederus · rove beetle · Nairobi fly · kissing lesions

RELATED PAGES: Contact dermatitis · Phytophotodermatitis · Papular urticaria & insect bites · The returning traveller with a skin problem

KEY FACTS

- **Irritant blistering** from **cantharidin** in blister beetles: the beetle doesn't bite; lesions follow crushing it on skin.
- **Paederus (rove beetle) dermatitis** is the look-alike: toxin paederin, **linear and "kissing" lesions**.
- Blisters appear within about a day, often on exposed skin after night-time exposure near lights.
- First aid: blow or flick beetles off, **don't crush**; wash with soap and water.
- Treat with cool compresses and a short potent topical steroid (milder on face); antibiotics only if infected.

Overview

Blister beetle dermatitis is an acute irritant contact reaction to **cantharidin** in Meloidae beetles (e.g. Lytta vesicatoria, Mylabris). The beetle does **not** bite — lesions follow crushing or brushing it on the skin, often at night near lights. It is uncommon in the UK and mainly seen in travellers or outdoor workers from endemic regions.

Paederus (rove beetle) dermatitis is the closely related look-alike taught alongside it. The toxin is **paederin**, producing linear and “kissing” lesions. Management overlaps heavily.

Aetiology: causes and risk factors

- Cantharidin breaks desmosomes acantholysis and an intraepidermal blister with little dermal inflammation (usually no scarring). The same mechanism is used therapeutically on warts and molluscum.
- Paederin from crushed Paederus beetles causes delayed linear erythema then vesicles/pustules.
- **Ingesting** cantharidin (e.g. as a purported aphrodisiac) is systemically toxic — a medical emergency, not a skin problem.

Myths to address

- **"IT'S SHINGLES"**

linear lesions without dermatomal pain and with a beetle history point to Paederus.

Pathophysiology and pathology

Beetle toxins **break the bonds between skin cells**, so the epidermis splits.

1. **Toxin contact**
   Crushing the beetle releases cantharidin (or paederin) onto skin.
2. **Desmosome breakdown**
   Cantharidin breaks desmosomes, causing acantholysis.
3. **Intraepidermal blister**
   A superficial blister forms with little dermal inflammation, so it heals without scarring.
4. **Kissing lesions**
   Toxin transfers to opposing skin folds (elbow crease, neck).

Fig. 1 · Kissing lesions in Paederus dermatitis.
AI-generated illustration

Clinical features

- Onset within hours to about 24–48 h; the patient often does not link the insect.
- Vesicles/bullae on an erythematous base — often **linear or bizarre** where the beetle was wiped; "kissing" lesions in flexures favour Paederus.
- **Burning or stinging** more than itch.
- Exposed sites: face, neck, arms, lower legs; toxin can be finger-transferred to eyes or genitals.
- Heals in about a week, sometimes with lasting pigmentation in skin of colour.

Fig. 2 · Cantharidin blisters: cantharidin is the vesicant in blister beetle (Meloidae) haemolymph — it causes intraepidermal blisters on an erythematous base by contact, the same mechanism as blister beetle dermatitis (shown here after therapeutic use).

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Lesions and beetles

Cantharidin blisters

Bullae from Meloidae. Site: Exposed skin.

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Paederus dermatitis

Linear streaks, kissing lesions. Site: Neck, flexures.

© DermNet · **Source**

Paederus beetle

Slender orange-and-black. Site: —.

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Blister beetle

Meloidae, e.g. Spanish fly.

Site: —.

Stefanie Hamm · **Source**

Diagnosis and investigations

Clinical: burning blisters on exposed skin after possible beetle contact, travel, camping or night lights. Clusters of cases support the diagnosis. Swab only if secondary infection or HSV/impetigo is realistic.

Clinical

- Burning blisters after beetle contact, travel, camping, night lights; **clusters of cases** support it.

Swab

- Only if secondary infection or HSV / impetigo is realistic.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Herpes zoster	Dermatomal, unilateral, prodromal pain.
	Bullous impetigo	Flaccid bullae, honey crust, children.
	Phytophotodermatitis	Plant juice + sun; drip pattern; marked later pigmentation.
	Bullous insect bite	Itchy, punctum, bite history.
	Paederus vs Meloidae	Paederus more linear / kissing / pustular; Meloidae more cantharidin bullae.

Management

First aid

- If a beetle lands on skin, **blow or flick it off — do not crush or swat**.
- Wash promptly with soap and water; avoid touching eyes or genitals.
- Cool compresses; leave small blisters intact; drain large tense bullae aseptically keeping the roof on.
- Simple analgesia.

Prescriber options

- Short course of **potent topical corticosteroid** on inflamed non-facial skin (mild/moderate on face or flexures).
- Antihistamine if itch; sedating at night if sleep is poor.
- Secondary infection: treat as impetigo/cellulitis per NICE — no routine prophylactic antibiotics.

Treatment ladder

1 **First aid**

Blow or flick the beetle off; don't crush or swat. Wash promptly with soap and water; keep hands off eyes and genitals.

2 **Prescriber options**

Short **potent topical steroid** on inflamed non-facial skin (**mild / moderate on face or flexures**). Antihistamine if itchy; sedating at night if sleep is poor.

3 **Infection**

Treat as impetigo / cellulitis per NICE; **no** routine prophylactic antibiotics.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Prevention**: Nets, screens, cover skin, shake bedding, lights off near open windows.

Referral

SAME DAY — Eye pain, redness, photophobia or reduced vision: irrigate and **same-day eye casualty**.

999 / A&E **Ingestion** of cantharidin: A&E / NPIS (TOXBASE).

URGENT Extensive blistering, genital involvement, severe pain or systemic illness; returned traveller with fever.

Red flags and when to refer

- Eye pain, redness, photophobia or reduced vision irrigate and **same-day eye casualty**.
- Extensive blistering, genital involvement, severe pain or systemic illness.
- Ingestion A&E / NPIS (TOXBASE).
- Returned traveller with fever — think beyond beetle dermatitis.

Complications and prognosis

Complications

- Secondary infection, eye involvement (Nairobi eye), post-inflammatory pigmentation.

Prognosis

- Heals in 1–2 weeks.

Prescriber pearls (UK pharmacist IP)

Ask "Did it **burn** or itch?" Burning linear blisters after lights outdoors beetle; itchy grouped papules bites.

Prevention: nets, screens, cover skin, shake bedding; never crush beetles on skin.

Teaching: quiz images of linear kissing lesions are usually **Paederus**, not Meloidae — but first aid is the same.

Self-check quiz

4 questions · optional

Q1 Does the blister beetle bite?

✓ No: blisters follow crushing or brushing it on skin.

Q2 Mirror-image linear lesions across the elbow crease?

✓ 'Kissing' lesions of Paederus dermatitis.

Q3 Steroid potency on the face?

✓ Mild or moderate (potent on non-facial skin).

Q4 Toxin in the eye?

✓ Irrigate and same-day eye casualty.

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Photo gallery

Cantharidin and Paederus dermatitis; the beetles 6

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Hectonichus · **Source**

Infections — viral & infestations Module 5 · Week 3

Papular urticaria & insect bites

Recurrent itchy grouped papules from flea, mite or bed-bug hypersensitivity. Treat the source (pets, home, pest control) as well as the itch.

ALSO CALLED papular urticaria · insect bite hypersensitivity · flea bites · bed bugs · Cimex · breakfast lunch dinner

RELATED PAGES Scabies · Urticaria & angioedema · Impetigo · Lyme disease & erythema migrans · Cutaneous larva migrans

KEY FACTS

- **Papular urticaria** is a persistent itchy hypersensitivity reaction to insect bites (fleas, bed bugs, mites, mosquitoes).
- Grouped or linear ('breakfast, lunch and dinner') papules, sometimes with central punctum or blisters, on exposed skin or ankles.
- Commonest in children aged 2–10; other household members may be unaffected.
- Eradicate the source (treat **all pets**, professional pest control for bed bugs); antihistamine and mild–moderate topical steroid.
- Most bites do not need antibiotics; treat only clear secondary infection.

Overview

Papular urticaria is a **chronic or recurrent eruption of intensely itchy grouped papules** (± vesicles). It is a hypersensitivity reaction to arthropod bites — most often **fleas**, mites, **bed bugs**, mosquitoes and gnats. Despite the name it is **not true urticaria** (lesions last days, not hours). It is commonest in children aged about 2–10.

Epidemiology

- **Who**
  - Children, newcomers to a household, travellers; tolerance develops with ongoing exposure.

Aetiology: causes and risk factors

- Arthropod saliva triggers type I and type IV hypersensitivity. Bites are often unnoticed; reactions arrive hours to days later.
- **Cat fleas** are the commonest fleas biting UK humans (even in dog-only homes). The home holds most of the eggs/larvae/pupae.
- **Bed bugs** hide in mattress seams and feed at night; linked to travel and second-hand furniture.
- Most children grow out of it as they desensitise.

Risk factors and associations

- Type I and IV hypersensitivity to insect saliva antigens.

Myths to address

- **"Only one child is affected, so it can't be bites"**: sensitivity varies between people.

Pathophysiology and pathology

The rash is an **allergic reaction to insect saliva**, not the bite itself.

1. **Bite**
   The insect injects saliva containing anticoagulant and enzyme proteins.
2. **Immediate reaction**
   IgE on mast cells triggers a wheal within minutes.
3. **Delayed reaction**
   T cells respond over 24–48 hours, producing a firm itchy papule that lasts days to weeks.

Fig. 1 · Clusters of bites on the ankle.

AI-generated illustration

4. **Desensitisation**
   With repeated exposure, reactions fade (tolerance).

> **HISTOPATHOLOGY**
>
> - Wedge-shaped perivascular infiltrate with eosinophils; rarely needed.

Clinical features

- Firm red itchy papules 3–10 mm, sometimes with a punctum, vesicle or blister; excoriation and later pigmentation are common.
- Grouping in clusters or lines of three ("breakfast, lunch and dinner").

- Seasonal spring–autumn for fleas/outdoor insects; bed bugs all year.

SOURCE	CLUES
Fleas	Lower legs/ankles, waistband; pets; often only one person reacts; new house can awaken pupae
Bed bugs	Face, neck, arms (exposed in bed); linear clusters; blood/faecal spots on sheets; travel
Mosquitoes/midges	Exposed outdoor sites; seasonal

Fig. 2 · Papular urticaria: grouped, very itchy firm papules — a delayed hypersensitivity to bites (often cat/dog fleas or mites); bites themselves rarely noticed.

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Types to recognise

Papular urticaria

Grouped firm itchy papules. Site: Legs, arms.

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Flea bites

Clusters at the ankles. Site: Ankles.

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Bed bug bites

Linear clusters on sleep-exposed skin. Site: Arms, neck.

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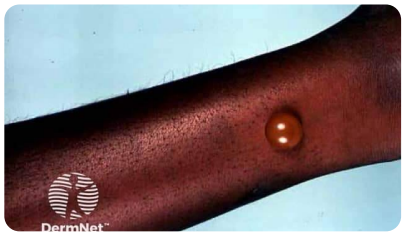
Bullous reaction

Tense blisters on bites. Site: Lower legs.



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Special groups

- **Bed bugs**
  - Linear bites on the face, neck and arms; blood spots on sheets; professional eradication needed.

Diagnosis and investigations

Clinical from morphology, grouping and history (pets, new house, travel, season). Find the source: flea-comb pets, inspect mattress seams. Investigations are rarely needed. Explain that only the sensitised person reacts.

History

- Pets, new house, travel, season, who else itches.

Find the source

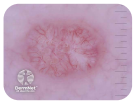
- Flea-comb pets; inspect mattress seams.

Tests

- Rarely needed.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Scabies	Web spaces, wrists, genitals, burrows; **whole household itchy**.
	Acute urticaria	Wheals fade within 24 h; no crusted papules.
	Chickenpox	Fever; crops at different stages including scalp.

	CONDITION	DISTINGUISHING FEATURES
	**Molluscum**	Pearly, umbilicated, barely itchy.
	Delusional infestation	Conviction of infestation, no insects confirmed.

Management

1. Eradicate the source

> **FLEAS**
> veterinary flea products for **all** pets; hot-wash bedding; vacuum thoroughly; household IGR spray for weeks.

> **BED BUGS**
> council pest control or BPCA/NPTA professional firm — home sprays rarely suffice.

> Outdoor: clothing and DEET-type repellent.

2. Symptom control

- Cool compresses, short nails, soap and water.
- Non-sedating antihistamine by day; short sedating antihistamine at night if sleep is poor (check age licence).
- Short mild-to-moderate **topical corticosteroid** on new itchy spots (potent short courses only on thick adult lesions).

3. Secondary infection

- Most bites do **not** need antibiotics — redness and itch can last ~10 days.
- If spreading cellulitis/impetigo: flucloxacillin first-line (NICE NG182); Pharmacy First pathway for infected insect bites (England, age 1) within PGD rules.

Treatment ladder

1

Eradicate the source

Fleas: vet flea products for **all** pets; hot-wash bedding; vacuum; household IGR spray for weeks. **Bed bugs**: council or professional (BPCA / NPTA) pest control.

2

Symptom control

Cool compresses, short nails. Non-sedating antihistamine by day; short sedating antihistamine at night (check age licence).

3

Secondary infection only

Spreading cellulitis / impetigo: **flucloxacillin** first-line (NICE NG182); Pharmacy First infected insect bite pathway (England, 1 year) within PGD.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **No routine antibiotics**: Redness and itch can last ~10 days without infection.
- **Long-term potent steroids**: Avoid.

Referral

999 / A&E	Anaphylaxis or airway compromise.
SAME DAY	Systemically unwell, spreading cellulitis, bites in mouth / throat or around eyes.
URGENT	Fever or a persisting unusual lesion after bites abroad.
ROUTINE	Atypical persistent eruption or suspected delusional infestation: GP / dermatology.

Red flags and when to refer

- Anaphylaxis or ABC compromise emergency.
- Systemically unwell, spreading cellulitis, immunocompromised with infection, bites in mouth/throat or around eyes.
- Bites abroad with fever or persistent exotic lesion.
- Atypical persistent eruption or suspected delusional infestation GP/dermatology.

Complications and prognosis

Complications

- Impetigo, scarring, post-inflammatory pigmentation, sleep disturbance.

Prognosis

- Settles when the source is removed; children grow out of it.

Prescriber pearls (UK pharmacist IP)

- **The pet may look fine — the house is the reservoir.**
- Antihistamines and steroids control symptoms; recurrences stop only when the source is gone.
- Bed bug vs flea: sleep-exposed sites and linen spots vs ankles/pets.
- Avoid long-term potent steroids and antibiotics without infection.

Self-check quiz

4 questions · optional

Q1 Why does only one family member get spots?

✓ Only sensitised people react to the bites.

Q2 Best treatment for flea bites that keep recurring?

✓ Treat all pets with vet products + vacuum + household spray.

Q3 When are antibiotics needed?

✓ Only for genuine secondary infection (flucloxacillin, NG182).

Q4 Lines of three bites on arms after a hotel stay?

✓ Bed bugs.

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Photo gallery

Papular urticaria pattern

4

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Flea vs bed-bug clues

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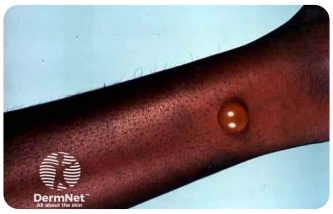

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Infections — viral & infestations Module 3 · Week 4

Perineal dermatoses

Anogenital warts and HPV counselling, groin hidradenitis first-line care (bridge to Week 5 HS), and brief DDx: molluscum, PPP, Fordyce, candida and lichen sclerosus.

ALSO CALLED anogenital warts · condyloma · genital warts · HPV · hidradenitis groin · pearly penile papules · +3

RELATED PAGES Viral warts (HPV) · Hidradenitis suppurativa · Candidal intertrigo

KEY FACTS

- Anogenital warts HPV 6/11; about 30% regress in months, so no treatment is a valid choice.
- Patient-applied: podophyllotoxin or imiquimod per SmPC (avoid podophyllotoxin in pregnancy); clinic: cryotherapy, TCA, excision.
- Recurrence is common; offer sexual health screening and HPV vaccination where eligible.
- Normal variants to recognise: **pearly penile papules**, Fordyce spots.
- Groin HS, molluscum, candidal intertrigo and lichen sclerosus are key differentials.

Overview

This page groups **perineal papules and fold conditions**: anogenital warts in depth, plus short sections on groin hidradenitis and look-alikes.

Module 3 Week 4 focuses on **perineal papules and folds**: full coverage of **anogenital warts**, a **groin hidradenitis** bridge to the Week 5 HS deep-dive, and a short differential set (molluscum, pearly penile papules, Fordyce spots, candidal intertrigo, lichen sclerosus).

Cutaneous viral warts are separate: viral warts. Candidal folds also deepen on candida / intertrigo.

Aetiology: causes and risk factors

- Anogenital warts: sexually acquired low-risk **HPV 6/11** infecting basal keratinocytes through microabrasions; incubation weeks to months.
- HS: follicular occlusion tunnels and scarring in apocrine folds; smoking and obesity are major drivers — **not** contagious infection.
- Molluscum: poxvirus; genital adult disease may be STI-context.
- PPP and Fordyce: normal anatomical variants (angiofibromas; ectopic sebaceous glands).

Myths to address

- **"WARTS MEAN A PARTNER WAS UNFAITHFUL"**

incubation can be months to years.

Pathophysiology and pathology

HPV **infects basal keratinocytes** through tiny abrasions.

1. **Entry**
   Microabrasions expose basal cells to HPV.
2. **Replication**
   Viral genes drive keratinocyte proliferation.
3. **Latency**
   Virus persists in normal-looking skin, causing recurrence.
4. **Groin HS**
   Follicular occlusion in apocrine folds forms tunnels.

Fig. 1 · Hurley stages of hidradenitis.

AI-generated illustration

Clinical features

Anogenital warts

- Flesh-coloured, pink, grey or brown verrucous papules; may coalesce into cauliflower plaques on vulva, penis, scrotum, perianal skin.
- Often asymptomatic; itch, bleeding or psychosexual distress common.

Groin hidradenitis (bridge)

- Recurrent painful nodules/abscesses, double-ended comedones, sinus tracts and rope-like scars in the inguinal folds.
- Hurley I: abscesses without tracts; II: separated tracts/scarring; III: diffuse interconnected tracts — deepen staging/escalation in Module 3 Week 5 when the dedicated HS page is published.

Brief DDx strip

ENTITY	CLUE
Molluscum	Smooth pearly papules with central umbilication
Pearly penile papules	Uniform rows on the corona — normal
Fordyce spots	Tiny yellow-white sebaceous papules — normal
Candida	Bright fold erythema + satellite pustules
Lichen sclerosus	White atrophic plaques, itch, scarring — separate pathway

Fig. 2 · Flesh-coloured cauliflower papules/ plaque — classic condyloma acuminata morphology (Case 1 pattern).

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Types to recognise

Condylomata

Cauliflower papules / plaque. Site: Genital.

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Groin HS

Nodules, comedones, scarring. Site: Inguinal fold.

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Molluscum

Umbilicated pearly papules. Site: Genital.

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Pearly penile papules

Coronal rows: normal. Site: Corona.

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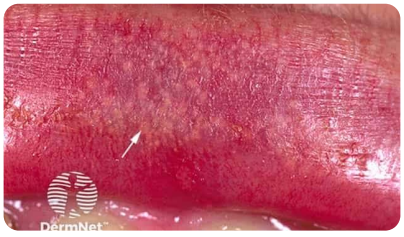
Fordyce spots

Yellow-white pinheads: normal. Site: Lip, genital.



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Lichen sclerosus

Porcelain-white atrophy. Site: Vulva, foreskin.

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Diagnosis and investigations

- Warts: usually clinical; biopsy if atypical/ulcerated/resistant. Offer STI screen and partner discussion via local sexual-health services — routine HPV typing is not needed in primary care.
- HS: clinical (typical lesions + sites + chronicity). Swabs often colonisers — do not withhold the diagnosis for a negative culture.
- Children with anogenital warts: specialist paediatric/SHS + safeguarding pathways.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Pearly penile papules / vestibular papillae	Symmetric normal variants: reassure.
	Molluscum	Umbilication.
	Anogenital SCC / HSIL	Indurated, ulcerated, evolving: urgent.
	Furuncle vs HS	Single acute boil without tracts or multi-site history.
	Candida	Bright fold erythema + satellite pustules.
	Crohn fistula disease	Gut symptoms, complex perianal disease.

Management

Anogenital warts (NICE CKS themes)

- ~30% may regress in months — **no treatment** is valid if acceptable to the patient.
- Patient-applied (external): **podophyllotoxin** or **imiquimod** per SmPC — avoid podophyllotoxin in pregnancy; teach exact application.
- Clinic: cryotherapy, TCA (often preferred in pregnancy if treatment needed), excision/electrosurgery for selected disease via SHS.
- Recurrence is common because latent HPV persists — set expectations.
- **HPV vaccine catch-up** (Gardasil 9 programme themes): discuss if eligible (routine Year 8; catch-up to 25th birthday for many; GBMSM and immunosuppressed schedules differ) — vaccine does not clear existing warts but may protect against other types.
- Pregnancy: prefer expectant care or clinic methods; many home topicals contraindicated.

Groin HS — first-line IP actions

- Smoking cessation; weight management; loose clothing; absorbent dressings; reassure not contagious.

- Antiseptic wash (e.g. chlorhexidine) + **topical clindamycin** to active areas.
- Oral doxycycline (or other tetracycline) as a prolonged anti-inflammatory course if needed — not a short "boil antibiotic".
- Hurley II–III or refractory disease dermatology (Week 5 covers combination antibiotics, hormonals, biologics, surgery).

DDx management snippets

- Molluscum: often expectant; treat for cosmesis/spread via local practice; offer SHS screen if genital adult.
- PPP/Fordyce: **reassure** — procedures only for significant distress.
- Candida: topical azole ± brief hydrocortisone; keep dry; see candidal intertrigo page.
- Lichen sclerosus: do not manage as "thrush forever" — potent steroid pathways and referral for chronic white genital disease.

Treatment ladder

1 **Warts: watch & counsel**

~30% regress in months: **no treatment** is valid. Recurrence is common (latent HPV).

2 **Warts: patient-applied**

Podophyllotoxin or **imiquimod** per SmPC; teach exact application; **avoid podophyllotoxin in pregnancy**.

3 **Warts: clinic**

Cryotherapy, TCA (often preferred in pregnancy), excision / electrosurgery via SHS.

4 **Groin HS first line**

Smoking cessation, weight, loose clothing, absorbent dressings. Antiseptic wash (e.g. chlorhexidine) + **topical clindamycin**.

5 **Specialist**

Hurley II–III / refractory HS dermatology.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Look-alikes**: PPP / Fordyce: reassure. Molluscum: often expectant. Candida: topical azole ± brief hydrocortisone.
- **Lichen sclerosus**: Not 'thrush forever': potent steroid pathway and referral.

Referral

URGENT	Ulcerated or indurated 'wart': urgent specialist.
URGENT	Anogenital warts in a child: safeguarding-aware specialist referral.
ROUTINE	Hurley II–III HS, non-healing ulcer in long-standing sinuses (SCC risk), LS with architectural change.

Red flags and when to refer

- Ulcerated/indurated "wart" urgent specialist.
- Hurley II–III HS or non-healing ulcers in long-standing sinuses (SCC risk, especially older men).
- Anogenital warts in children safeguarding-aware specialist referral.
- LS with architectural change or any suspicion of neoplasia.

Complications and prognosis

Complications

- Psychosexual distress, recurrence; rare high-risk HPV dysplasia.

Prognosis

- Many clear within a year.

Prescriber pearls (UK pharmacist IP)

Condoms reduce but do not abolish HPV transmission (skin outside the condom).

Incision and drainage alone almost always fails in HS — treat the disease, not only the abscess.

- Sexual-health interface: STI screen offer, partner pathways, vaccine catch-up.
- Always verify current Green Book / NHS HPV schedule and BNF/SmPC before advising.
- Hidradenitis suppurativa beyond the groin, with Hurley staging and the treatment ladder: see hidradenitis suppurativa.

Self-check quiz

4 questions · optional

Q1 Uniform rows of tiny papules on the corona?

✓ Pearly penile papules: normal, reassure.

Q2 Wart treatment to avoid in pregnancy?

✓ Podophyllotoxin.

Q3 Does HPV vaccine clear existing warts?

✓ No; may protect against other types.

Q4 Why does I&D alone fail in HS?

✓ It treats the abscess, not the tunnels / disease.

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Photo gallery

Anogenital warts 5

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Hidradenitis in the groin (Week 5 bridge) 5

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DDx: molluscum

2

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DDx: pearly penile papules (normal)

2

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DDx: Fordyce spots (normal; lip exemplars)

2

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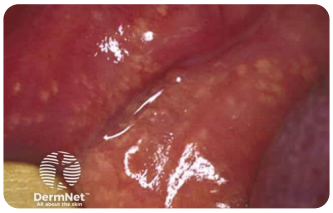

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DDx: candidal intertrigo

2

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Flag: lichen sclerosus

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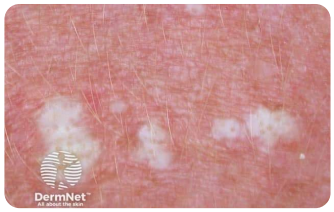

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CHAPTER 10

Tropical & travel

The returning traveller with a skin problem: cutaneous leishmaniasis, larva migrans, dengue, leprosy and mycetoma.

1. The returning traveller with a skin problem
2. Cutaneous leishmaniasis
3. Cutaneous larva migrans
4. Dengue: skin signs & safety
5. Leprosy (Hansen disease) & lepra reactions
6. Mycetoma (Madura foot)

Tropical & travel Module 5 · Week 4

The returning traveller with a skin problem

A structured approach to skin problems after travel or migration: where, when, what exposure, and is there fever? Fever after the tropics means same-day malaria exclusion.

ALSO CALLED travel rash · tropical dermatology · migrant health · fever and rash · NaTHNaC · travel history

RELATED PAGES Cutaneous larva migrans · Cutaneous leishmaniasis · Dengue: skin signs & safety · Leprosy (Hansen disease) & lepra reactions · Mycetoma (Madura foot)

KEY FACTS

- Most skin problems after travel are **common**: bite reactions, impetigo, sunburn, tinea, pityriasis versicolor.
- Take a full travel history: regions, dates, activities, bites, water and animal contact, sexual contact.
- Incubation guides diagnosis: dengue ~4–10 days, CLM days–weeks, leishmaniasis 2 weeks–6+ months, leprosy years.
- **Fever + rash in a traveller** same-day assessment; consider dengue before NSAIDs.
- Refer exotic or uncertain cases early to ID/tropical medicine; remember notifiable diseases.

Overview

Core approach

Most skin problems after travel are common things (insect-bite reactions, impetigo/cellulitis from scratched bites, sunburn, tinea, pityriasis versicolor, contact/phyto-dermatitis). Tropical-specific diagnoses are less common but matter because they are missed (PCDS).

TRAVEL HISTORY CHECKLIST

countries and regions (urban vs rural/forest/beach), dates (incubation windows: dengue ~4–10 days; CLM days–weeks; leishmaniasis 2 weeks–6+ months; leprosy years; mycetoma months–years), activities (barefoot on beaches/soil, jungle trekking, farming, military service), insect bites, freshwater/animal contact, sexual contact, prior residence (migrants, VFR travellers — "visiting friends and relatives"), pre-travel vaccines/prophylaxis, and illness or treatment while abroad.

Fever + rash in a returning traveller = same-day medical assessment. Malaria must be excluded urgently in anyone with fever after travel to a malarious area (usually within the past year, especially within 3 months), even if they took prophylaxis. Consider dengue, chikungunya, rickettsial infection (look for an eschar), meningococcal disease, measles, acute HIV, typhoid.

WHERE TO REFER IN THE UK

local Infectious Diseases (ID) or tropical medicine service; national centres include the **Hospital for Tropical Diseases (HTD), UCLH, London** (GP e-Referral; 24-hour on-call tropical medicine registrar via UCLH switchboard; emergency walk-in for unwell returnees within 6 months, weekday clinic hours) and the Liverpool School of Tropical Medicine/Royal Liverpool tropical service. Travel-health resources: **NaTHNaC / TravelHealthPro** country pages; **UKHSA Rare and Imported Pathogens Laboratory (RIPL)** for arbovirus testing.

NOTIFICATION (ENGLAND)

leprosy is a statutorily **notifiable disease** for clinicians; dengue virus is a **notifiable organism** reported by the laboratory (clinicians do not need to notify suspected dengue).

Aetiology: causes and risk factors

COMMON UK CONDITIONS ACQUIRED ABROAD

insect-bite reactions, impetigo and cellulitis from scratched bites, sunburn and phototoxic reactions, tinea, pityriasis versicolor, contact and phytophotodermatitis.

EXPOSURE-SPECIFIC INFECTIONS

sandflies (leishmaniasis), mosquitoes (dengue, chikungunya, Zika), beaches and soil (larva migrans, tungiasis, mycetoma), ticks (rickettsial eschar, Lyme disease), fresh water (schistosome cercarial dermatitis), sexual contact (syphilis, HIV).

Myths to address

- **"IT'S JUST A BITE"**
  a sore that won't heal months after travel may be leishmaniasis.

Pathophysiology and pathology

Tropical skin disease follows **how the organism gets in**.

1. **Vector bites**
   Sandflies (leishmaniasis), mosquitoes (dengue) inject pathogens.

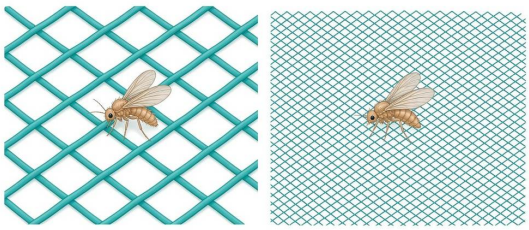

Fig. 1 · Fine mesh stops sandflies.

AI-generated illustration

2 **Skin penetration**

Hookworm larvae burrow through bare skin.

Fig. 2 · Larva migrans track.

AI-generated illustration

3 **Inoculation injury**

Thorns introduce mycetoma organisms.

4 **Chronic granuloma**

Some infections form slow granulomatous ulcers.

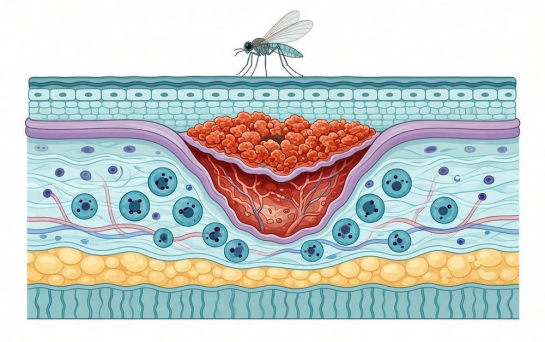

Fig. 3 · Leishmaniasis ulcer.

AI-generated illustration

Clinical features

PRESENTATION	THINK OF
Moving, itchy serpiginous track	Cutaneous larva migrans
Painless ulcer with a rolled edge, weeks to months after travel	Cutaneous leishmaniasis
Fever, aches and a flushing or ‘white islands’ rash	Dengue, chikungunya, measles; exclude malaria

PRESENTATION	THINK OF
Numb pale or red patches, thickened nerves	Leprosy
Swollen foot with discharging sinuses	Mycetoma
Black eschar at a bite site with fever	Rickettsial infection (tick typhus)
Boil-like lump with a central pore and movement	Myiasis (botfly larva)

Fig. 4 · Hallmark: thin (2–3 mm), raised, serpiginous, intensely itchy track on the sole after barefoot beach/soil contact.

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Patterns to link

Larva migrans

Serpiginous track. Site: Feet.

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Leishmaniasis

Volcano ulcer. Site: Exposed skin.

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Dengue

White islands in a sea of red. Site: Limbs.

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Leprosy

Anaesthetic plaques. Site: Any.

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Diagnosis and investigations

- Ask where (including rural, beach or forest), when (incubation windows), what they did, bites, water and animal contact, sexual contact, and treatment taken abroad.
- Examine the whole skin, lymph nodes and mucosa.
- **Fever = same-day malaria film or rapid test**, plus FBC and specialist advice.
- Use the national tropical-medicine services for advice on biopsy, culture or PCR; take photographs to share.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Scabies	Web spaces, household itch.
	Impetigo	Honey crusts on scratched bites.
	Tinea	Scaly active edge.
	Drug eruption	New antimalarial / antibiotic.
	Urticaria	Wheals < 24 h.
	Contact / phytophotodermatitis	Plant juice + sun; drip marks.

Most rashes after travel are the same conditions seen at home. Keep scabies, impetigo, tinea, drug eruptions, urticaria and contact dermatitis on the list before deciding it is a tropical disease.

Management

- Treat common conditions as usual, with review dates.
- Refer exotic or uncertain cases early to infectious diseases or tropical medicine.
- Do not give NSAIDs or aspirin to a febrile traveller until dengue has been considered.
- Remember notifiable diseases and give prevention advice for the next trip (insect repellent with DEET 50%, covering up, footwear on beaches).

Treatment ladder

1 **Common conditions**

Treat as usual, with review dates.

2 **Exotic / uncertain**

Refer early to ID / tropical medicine (e.g. HTD e-Referral, on-call tropical registrar).

3 **Safety**

No NSAIDs or aspirin in a febrile traveller until dengue considered. **No steroid creams on undiagnosed tropical ulcers; no systemic steroids** if Strongyloides or TB possible without specialist input.

4 **Public health**

Remember notifiable diseases; next-trip advice: DEET 50%, cover up, footwear on beaches.

Referral

SAME DAY — **Fever after travel to a malaria area**: same-day malaria testing, whatever the rash.

999 / A&E — Dengue warning signs (abdominal pain, vomiting, bleeding, lethargy).

URGENT — Nasal or oral ulceration after leishmaniasis exposure.

Red flags and when to refer

- **FEVER AFTER TRAVEL TO A MALARIA AREA**
  same-day malaria testing, whatever the rash.
- Dengue warning signs (abdominal pain, vomiting, bleeding, lethargy): emergency.
- Nasal or oral ulceration after leishmaniasis exposure: specialist referral.

Complications and prognosis

Complications

- Missed diagnosis and delayed treatment.

Prognosis

- Depends on cause.

Prescriber pearls (UK pharmacist IP)

Cross-cutting prescriber pearls

1. **"Where have you been, and when?"** Take the travel and migration history for every unexplained or non-healing skin lesion. Incubation runs from days (dengue, CLM) to months (leishmaniasis) to years (leprosy, mycetoma).
2. **Fever after travel = same-day assessment.** Exclude malaria first. In suspected dengue, give **paracetamol and no NSAIDs or aspirin**.
3. **Painless is a warning sign, not a comfort.** Painless ulcers suggest leishmaniasis, leprosy (neuropathic) or Buruli ulcer. Painless swelling with sinuses suggests mycetoma.
4. **Test sensation** in any persistent hypopigmented or red patch in someone from a leprosy-endemic country. **Palpate the nerves.**
5. **Do not put steroid creams on undiagnosed tropical ulcers, and do not give systemic steroids** to someone who may have Strongyloides (larva currens) or TB without specialist input.
6. Know your UK specialist routes HTD (UCLH) e-Referral, the 24-hour on-call tropical registrar via UCLH switchboard, the HTD Leishmaniasis MDT, the national leprosy Consultant Advisors (UKHSA memorandum), and local ID. **Leprosy is clinician-notifiable in England.**
7. Teratogens in tropical dermatology miltefosine, thalidomide (ENL), albendazole/ivermectin (avoid in pregnancy), and live Qdenga (contraindicated in pregnancy). Always check pregnancy and contraception status. **Rifampicin defeats hormonal contraception.**
8. **Pre-travel prevention is the pharmacist's strongest lever:** footwear (CLM, mycetoma); day-time Aedes bite avoidance and DEET 50% (dengue); dusk/dawn sandfly avoidance and fine-mesh nets (leishmaniasis); Qdenga only for previously infected travellers

Self-check quiz

4 questions · optional

Q1 Febrile returnee with a rash: first test?

✓ Same-day malaria film / rapid test.

Q2 Painless rolled-edge ulcer 3 months after Belize?

✓ Cutaneous leishmaniasis.

Q3 Why avoid systemic steroids without specialist input?

✓ Strongyloides hyperinfection or TB.

Q4 Boil-like lump with a central pore and movement?

✓ Myiasis.

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Photo gallery

Common travel-related presentations 4

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Tropical & travel Module 5 · Week 4

Cutaneous leishmaniasis

A slowly enlarging, painless papule that ulcerates with a raised edge, weeks to months after sandfly bites in the Middle East, Mediterranean, Africa, Asia or Latin America. Refer to tropical medicine; Latin American species can spread to the nose and mouth.

ALSO CALLED leishmaniasis · sandfly · oriental sore · Baghdad boil · New World leishmaniasis · mucocutaneous leishmaniasis · +1

RELATED PAGES The returning traveller with a skin problem · Cutaneous tuberculosis (scrofuloderma & others) · Leprosy (Hansen disease) & lepra reactions · Mycetoma (Madura foot)

KEY FACTS

- Protozoal infection spread by **sandfly bites**; endemic in 90+ countries including the Mediterranean, Middle East and Latin America.
- A painless papule at the bite enlarges over weeks into a **crusted ulcer with a raised edge ("volcano")**.
- **New World (Viannia) species** can cause destructive **mucocutaneous** disease later (about 5% after L. braziliensis).
- Diagnose by biopsy/scraping with PCR (specialist).
- Primary care: **recognise and refer** (in England, HTD Leishmaniasis MDT); don't start empirical treatment.

Overview

Leishmaniasis is a protozoal infection caused by more than 20 Leishmania species and spread by the bite of infected female **phlebotomine sandflies**. It is endemic in 90+ countries (Latin America, the Mediterranean basin including southern Europe, the Middle East, Central/South Asia and East Africa). WHO lists it as a neglected tropical disease. Clinical forms:

- **Cutaneous leishmaniasis (CL)** is the most common form. It is a chronic sore at the bite site. WHO estimates roughly 600,000 to 1 million new cases a year.
- **Mucocutaneous leishmaniasis (MCL, "espundia")** is a destructive infection of the nose, mouth and throat. It occurs only with New World Viannia species (L. braziliensis, L. guyanensis, L. panamensis) and can follow the skin lesion by months or years. Lifetime risk after L. braziliensis CL is around 5%.
- **Diffuse CL** is rare and anergic, with many non-ulcerating nodules (L. aethiopica, L. amazonensis).
- **Leishmaniasis recidivans** means new lesions keep appearing around an old scar (L. tropica).
- **Visceral leishmaniasis (kala-azar)** causes fever, weight loss, splenomegaly and pancytopenia, and is fatal without treatment. **Post-kala-azar dermal leishmaniasis (PKDL)** may follow it.

Old World vs New World:

	OLD WORLD CL	NEW WORLD (AMERICAN) CL
Regions	Middle East, North Africa, Mediterranean, Central/South Asia, East Africa	Central and South America (incl. Belize, Brazil, Peru, Bolivia, Colombia)
Species	L. major, L. tropica, L. infantum, L. aethiopica	L. mexicana, L. amazonensis, L. (Viannia) braziliensis/ guyanensis/panamensis
Old names	Oriental sore, Baghdad/Delhi boil, Aleppo button	Chiclero ulcer, uta, forest yaws, pian bois
Course	Often heals on its own over months–1 year, leaving a scar	**All treated as complex** because of the mucosal risk

Aetiology: causes and risk factors

- Sandflies are tiny (1.5–3 mm), silent and mostly bite at **dusk and dawn**. They breed in wall cracks, animal burrows and leaf litter. The bite usually goes unnoticed and lands on exposed skin. Risk is highest in rural or forest settings and when sleeping outdoors or near the ground.
- The infected sandfly injects **promastigotes** into the dermis. Macrophages take them up, they change into **amastigotes**, and these multiply inside the cells. This produces a granulomatous dermal infiltrate that breaks down into an ulcer.
- The **host's cell-mediated immunity** shapes the disease. A good Th1 response gives a localised, self-limiting sore. A poor response (anergy, HIV, immunosuppression) gives diffuse or disseminated disease. An over-exuberant response is linked to mucosal and recidivans forms.
- Some regions have an animal reservoir (rodents, dogs, sloths). L. tropica and L. donovani can pass human to human via sandflies.
- Transmission is by **vector bite**. Spread by blood transfusion, needle-sharing and congenital routes is rare. CL is **not** spread by casual contact.

Myths to address

- **"A NON-HEALING TRAVEL SORE IS JUST INFECTED"** antibiotics don't cure leishmaniasis.

Pathophysiology and pathology

Leishmania parasites **live inside macrophages** in the dermis.

1. **Sandfly bite**
   Promastigotes are injected into the skin.
2. **Macrophage infection**
   They transform into amastigotes and multiply inside macrophages.

3 **Granulomatous ulcer**

The immune response forms a granuloma that ulcerates in the centre.

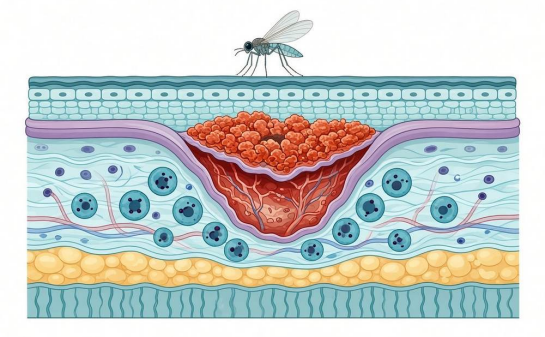

Fig. 1 · Volcano-like ulcer with parasite-laden macrophages.

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4 **Prevention**

Fine mesh nets and repellents stop sandfly bites.

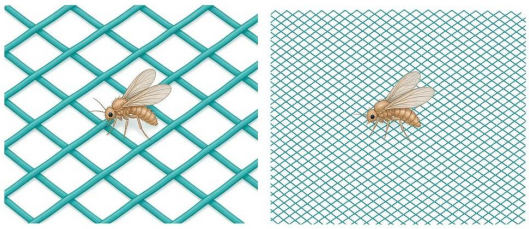

Fig. 2 · Sandflies pass through standard nets; fine mesh is needed.

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Clinical features

- Incubation is usually **2 weeks to 6 months** and sometimes longer, so ask about travel well before the lesion appeared.
- The lesion starts as a **painless red papule** at the bite site. It slowly grows into a nodule or plaque, often 1–5 cm, then usually **ulcerates centrally**. The result is the classic **"volcano" ulcer** with a **raised, rolled, indurated (often violaceous) border** and either a **dry adherent crust** or a moist, granulating or purulent base.
- It sits on **exposed sites**: face, ears, forearms, hands and lower legs. There may be one lesion or several (several bites).
- It is usually **painless**, and the patient is otherwise well (as in Case 1). Pain usually means secondary bacterial infection.
- **Sporotrichoid spread** (nodules in a line along the lymphatics) and regional lymphadenopathy can occur, especially with Viannia species.
- **HEALING**

  spontaneous healing over 2 months to more than a year leaves an **atrophic, cribriform, "cigarette-paper" scar**.
- **Mucosal red flags** (New World, sometimes years later) are nasal stuffiness, nosebleeds, crusting, septal perforation, lip/palate ulcers and hoarseness.
- **SKIN OF COLOUR**

  expect hyperpigmented or violaceous borders, with post-inflammatory hyper- or hypopigmentation and scarring.

Fig. 3 · Classic "volcano" lesion: painless ulcer with raised rolled edge and adherent crust on an exposed limb — mirrors Case 1 forearm sore.

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Types to recognise

Volcano ulcer

Painless, raised rolled edge. Site: Exposed skin.

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Early papule

Weeks–months after the bite. Site: Bite site.

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New World ulcer

Clean granulating base. Site: Limbs.

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Facial crusted

Facial lesions: priority referral. Site: Face.

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Mucosal (espundia)

Destructive nose / mouth disease. Site: Nose, mouth.

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Diagnosis and investigations

- **Clinical suspicion** comes from the travel/residence history plus a chronic, non-healing, painless ulcer or nodule on exposed skin (DermNet, PCDS).
- **CONFIRMATION (SPECIALIST)**
  a **punch or incisional biopsy from the active raised edge**. Send part **unfixed in sterile saline** for **PCR** (most sensitive, and identifies the **species**, which decides treatment) and culture, and part in formalin for histology (amastigotes inside macrophages; Giemsa stain). Slit-skin smears or scrapings from the edge can also be used. UK specialist parasitology labs (e.g. HTD/UKHSA-linked) want the **travel history on the request form**.
- **Species ID matters** because New World Viannia infection needs systemic treatment and ENT/mucosal examination.
- In New World cases, check for mucosal involvement with **ENT/nasal examination**. If visceral disease is suspected (fever, splenomegaly, cytopenias), do FBC, LFTs and serology.
- Serology is not useful for CL. It is used for visceral disease.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Infected insect bite	Acute, itchy / painful, resolves; no rolled edge.
	Ecthyma / bacterial ulcer	Painful, purulent, responds to antibiotics.
	BCC / SCC	Older, sun-exposed, pearly or keratotic edge.
	Atypical mycobacteria	Fish tank (M. marinum), Buruli: undermined ulcer.
	Lupus vulgaris	Apple-jelly nodules; chronic plaque.
	Sporotrichosis / deep fungal	Gardening / thorn; sporotrichoid nodules.
	Leprosy	Anaesthetic plaques, thickened nerves.

Management

Primary-care role: recognise and refer. Do not start empirical treatment.

Refer every suspected CL to an ID/tropical medicine service. In England, the **HTD Leishmaniasis MDT** (UCLH) accepts referrals and advice requests, or use your local ID unit. **New World exposure, facial/joint/genital lesions, more than 4 cm or multiple lesions, lymphatic spread, immunosuppression or any mucosal symptoms** are priority referrals.

SAME-DAY/URGENT REFERRAL

mucosal destruction, fever, weight loss or splenomegaly (possible visceral disease), immunosuppression (HIV, biologics, transplant).

INTERIM CARE

gentle cleansing and a simple non-adherent dressing. Treat **clear secondary bacterial infection** per NICE impetigo/cellulitis guidance (local spreading erythema, pain, pus). Antibiotics will not cure CL itself. **Avoid potent topical steroids** on an undiagnosed ulcer.

Specialist treatment options (for awareness and counselling): **Simple Old World CL:** watchful waiting, or local therapy such as cryotherapy, heat therapy, intralesional antimonials (sodium stibogluconate/meglumine antimoniate), topical paromomycin, or photodynamic therapy; **Complex or all New World CL: systemic therapy**, such as **liposomal amphotericin B**, **oral miltefosine** (increasingly used because it is oral), parenteral **sodium stibogluconate (Pentostam)**, or sometimes azoles or pentamidine depending on the species

MEDICINES-SAFETY POINTS (IF CO-MANAGING)

Miltefosine is **teratogenic**. Women of child-bearing potential need a pregnancy test and effective contraception during treatment and for months afterwards (check the SmPC/local protocol). Take with food. GI upset is common. It can be supplied via named-patient/specialist routes in the UK; **Antimonials** carry risks of QT prolongation and arrhythmia, pancreatitis, raised liver enzymes and myalgia. Patients need ECG and blood monitoring under specialist care; **Liposomal amphotericin** can cause nephrotoxicity, hypokalaemia and infusion reactions

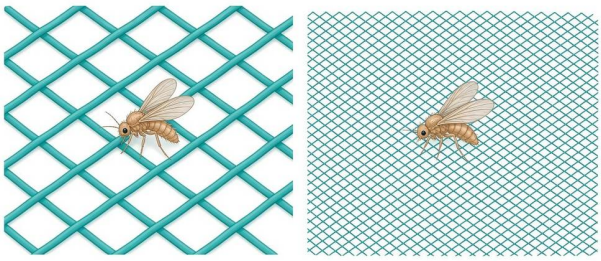

Fig. 9 · Sandflies are small enough to pass through standard mosquito nets (left); fine-mesh nets (right) keep them out.

AI-generated illustration

Treatment ladder

1

Refer

Every suspected CL **ID / tropical medicine** (e.g. HTD Leishmaniasis MDT, UCLH). **Priority**: New World exposure, facial / joint / genital, > 4 cm or multiple, lymphatic spread, immunosuppression, mucosal symptoms.

2

Interim care

Gentle cleansing, simple non-adherent dressing. Treat **clear secondary infection** per NICE (antibiotics won't cure CL).

3

Specialist treatment

Species-guided: local or systemic therapy; New World Viannia needs systemic treatment.

⚠ CAUTION

Prescribing limits and pitfalls

- **Prevention (no vaccine)**: Avoid dusk / dawn outdoors; long sleeves; DEET; permethrin clothing; **fine-mesh nets**; fans / AC; sleep off the ground.

Referral

SAME DAY	Mucosal destruction, fever, weight loss or splenomegaly (possible visceral disease), immunosuppression.
URGENT	Nasal stuffiness, bleeding or oral / nasal ulcers months–years after New World CL: **mucosal leishmaniasis**.
ROUTINE	Any suspected CL: tropical medicine referral.

Red flags and when to refer

- Nasal stuffiness, bleeding or oral or nasal ulceration months or years later: **mucosal leishmaniasis**, specialist referral.
- Immunosuppression or HIV: specialist advice early.

Complications and prognosis

Complications

- Scarring, mucosal destruction (New World), secondary infection.

Prognosis

- Old World lesions often self-heal over months with scarring.

Prescriber pearls (UK pharmacist IP)

A **"non-healing insect bite" for more than 4–6 weeks after travel** should trigger the question about leishmaniasis. The **Mediterranean** (Spain, Portugal, Italy, Greece, Cyprus, Malta, Turkey) and **Belize/ Central America** are common UK holiday and military sources.

Healing of the skin lesion does not exclude New World mucosal disease. Advise patients to report nasal or oral symptoms even years later.

Lesions are **not contagious person to person**. Scarring is common, so set expectations and note the psychological impact of facial scars.

Prevention (no vaccine or chemoprophylaxis): avoid outdoor activity at dusk and dawn; wear long sleeves and trousers; use a DEET-based repellent on exposed skin; use permethrin-treated clothing and **fine-mesh bed nets** (sandflies pass through standard mosquito nets); fans and air-conditioning help; sleep off the ground.

Self-check quiz

4 questions · optional

Q1 Vector?

✓ Female phlebotomine sandflies.

Q2 Classic lesion?

✓ Painless ‘volcano’ ulcer with a raised rolled indurated border.

Q3 Where to biopsy and why send unfixed tissue?

✓ Active edge; PCR identifies the species, which guides treatment.

Q4 Which species risk mucosal disease?

✓ New World Viannia (e.g. L. braziliensis).

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Photo gallery

Cutaneous leishmaniasis 6

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Mucocutaneous disease 1

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Laboratory 1

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Tropical & travel Module 5 · Week 4

Cutaneous larva migrans

An intensely itchy, red, snake-like track that moves a few millimetres to centimetres a day, usually on the feet after walking barefoot on a tropical beach. Treat with oral ivermectin or albendazole.

ALSO CALLED CLM · creeping eruption · hookworm rash · beach rash · serpiginous track · ivermectin · +1

RELATED PAGES The returning traveller with a skin problem · Scabies · Tinea: corporis, cruris & pedis · Cutaneous leishmaniasis

KEY FACTS

- **Animal hookworm larvae** wander in the epidermis, leaving an intensely itchy, **serpiginous track**.
- Common in travellers returning from tropical beaches; usual source dog and cat hookworms.
- The track moves a few millimetres to centimetres a day, usually on feet, buttocks or hands.
- Self-limiting (2–8 weeks), but treatment gives rapid relief: **ivermectin 200 micrograms/kg single dose** (off-label; prescriber).
- Don't freeze the track: the larva is ahead of it.

Overview

Cutaneous larva migrans (CLM), also called hookworm-related CLM or **"creeping eruption"**, is a self-limiting skin infestation. **Animal hookworm larvae** (mostly from dogs and cats) burrow into the epidermis and wander there, leaving an intensely itchy, serpiginous track. It is one of the most common dermatoses in travellers returning from tropical beaches (Caribbean, Central/South America, South-East Asia, Africa). Occasional UK-acquired cases have been reported in gardeners and runners.

Aetiology: causes and risk factors

- Usual causes are **Ancylostoma braziliense** (dog/cat hookworm in the Americas and Caribbean) and A. caninum, plus Uncinaria stenocephala (European dog hookworm) and Bunostomum phlebotomum (cattle).
- **TRANSMISSION**
  infected animals pass eggs in their faeces onto **warm, moist, sandy soil or beaches**. The eggs hatch into infective filariform larvae, which **penetrate human skin** through follicles, cracks or intact skin when someone walks **barefoot**, sits or lies on sand, or plays in sandpits.
- Humans are an **accidental "dead-end" host**. The larvae lack the enzymes to cross the basement membrane, so they stay in the epidermis and **cannot complete their life cycle**. They die within weeks to months. The track is an inflammatory/allergic reaction to the moving larva and its secretions.
- Not spread person to person.

Myths to address

- **"CRYOTHERAPY KILLS THE WORM"**
  painful and ineffective.

Pathophysiology and pathology

Humans are a **dead-end host**: the larva can't get through the basement membrane.

1. **Eggs to larvae**
   Infected animals pass eggs onto warm sand, where they hatch.
2. **Skin penetration**
   Infective larvae burrow into bare skin.
3. **Trapped in the epidermis**
   Lacking the enzymes to cross the basement membrane, they wander.

Fig. 1 · Serpiginous track on the foot.

AI-generated illustration

4. **Itch and death**
   The host reaction causes itch; the larva dies within weeks.

Clinical features

- There may be tingling or prickling at the entry site within hours. The track appears **days to a few weeks** later. The larvae can lie dormant first.

- The track is **2–3 mm wide, raised, serpiginous** ("snake-like") and **red or skin-coloured** (on darker skin it can look **hypochromic or pigmented**, as in Case 2). It is **intensely itchy** and moves **a few mm to a few cm a day**. The larva usually lies 1–2 cm **beyond** the visible end of the track.
- **SITES**
  feet (soles, toe webs, dorsum) most often, then **hands** (sand play, as in Case 2), buttocks, knees and trunk (sunbathing).
- **VARIANTS**
  several tracks or loops, **vesicles and bullae**, and **hookworm folliculitis** (itchy follicular papulopustules, often on the buttocks). Look for eczematised or impetiginised skin from scratching.
- The patient is systemically well. Rarely there is eosinophilia, or (with heavy infestation) Löffler-type pulmonary infiltrates.

Fig. 2 · Hallmark: thin (2–3 mm), raised, serpiginous, intensely itchy track on the sole after barefoot beach/soil contact.
© DermNet (Dr Ian Coulson) · **Source**

Types to recognise

Classic track

Thin, raised, serpiginous. Site: Foot.

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Multiple loops

Advances daily. Site: Foot.

© Dr Richard Ashton · **Source**

Hand

Sand play in children. Site: Hand.

© Dr Lawrence Meyerson · **Source**

Bullous

Don't mistake for impetigo.
Site: Foot.

© Dr Richard Ashton · **Source**

Diagnosis and investigations

- The diagnosis is **clinical**: the typical migrating serpiginous itchy track plus barefoot exposure in a warm-climate setting.
- No routine tests are needed. A biopsy is usually unhelpful because the larva is ahead of the track. Dermoscopy may show translucent segmental structures (the larval body) beyond the track.
- Check for eosinophilia if the picture is atypical. Think of **Strongyloides serology** if the eruption is fast-moving, perianal/buttock or recurrent.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Larva currens (Strongyloides)	Fast (**5–10 cm / hour**), urticarial band, perianal / buttocks / trunk, recurs for years; **hyperinfection with steroids**.
	Tinea	Annular scaly edge that doesn't creep daily; KOH +.
	Phytophotodermatitis / contact	Fixed linear streaks, no migration.
	Jellyfish sting	Immediate painful whip marks after swimming.
	Scabies burrows	Short (< 1 cm) burrows in web spaces; household itch.

Management

Primary care can manage most cases. It is self-limiting (usually 2–8 weeks), but treatment brings rapid relief and lowers the risk of infection.

> **SELF-CARE / OTC (INTERIM)**
> reassure (not contagious, the larva will die). Relieve itch with a **non-sedating oral antihistamine** (or sedating at night for older children/adults per licence) and emollients. A **mild–moderate topical steroid** helps eczematous change (only once CLM is confidently diagnosed and **not** larva currens). Cut nails short and avoid scratching. Treat secondary impetigo per NICE NG153. **Do not** attempt cryotherapy: the larva is ahead of the track, it is painful and it does not work.

> **Specific treatment (prescriber, off-label in UK; check local formulary and ID advice): First line: ivermectin 200 micrograms/kg orally as a single dose** (rounded to the nearest 3 mg tablet). Repeat the next day or after 1–2 weeks if it fails. In the UK this is usually **unlicensed/imported (named-patient)**, but it is listed for CLM on several NHS formularies (e.g. SE London JMF) and in NHS Tayside guidance. Avoid in pregnancy and breastfeeding. Safety in **children under 15 kg** is not established; **Alternative: albendazole 400 mg once daily for 3 days** (up to 5–7 days for extensive disease or folliculitis). It is unlicensed for CLM in the UK and can be used in children over about 2 years. Take with fatty food. **Avoid in pregnancy.**; **Topical option** for single early tracks, or when oral drugs are unsuitable (young children, pregnancy, on specialist advice): **10–15% thiabendazole or albendazole ointment**, applied three times daily for 1–2 weeks. These are extemporaneous or specialist products in the UK. Topical ivermectin 1% cream (licensed for rosacea) has been used off-label; Mebendazole is poorly absorbed and works less well. The UK OTC threadworm pack (single 100 mg dose) is **not** adequate for CLM, so do not sell it as a "cure"

> **CASE 2 (A 6-YEAR-OLD)**
> if he weighs at least 15 kg, single-dose ivermectin or albendazole 400 mg daily for 3 days are both reasonable, prescribed off-label by a competent prescriber or via GP/paediatrics. Add an antihistamine. Itch usually settles within 24–48 hours, and tracks fade within about a week.

> **REFER / SEEK ADVICE**
> diagnostic doubt; features of **larva currens** (fast, urticarial, perianal, recurrent, especially if steroids or immunosuppression are planned); widespread disease, folliculitis or treatment failure; pregnancy or a child under 15 kg; systemic symptoms (wheeze, eosinophilia).

Medicines at a glance

Ivermectin

First line (off-label)

DOSE	200 micrograms/kg orally, single dose (nearest 3 mg tablet) · Once; repeat next day or after 1–2 weeks if it fails
CAUTIONS	Usually unlicensed / imported; empty stomach; avoid in pregnancy / breastfeeding; < 15 kg not established.

Albendazole

Alternative

DOSE	400 mg once daily · 3 days (5–7 for extensive / folliculitis)
CAUTIONS	Unlicensed for CLM; with fatty food; avoid in pregnancy; children > ~2 years.

Treatment ladder

1 **Self-care / interim**

Reassure: **not contagious**; larva will die. Non-sedating antihistamine (sedating at night per licence), emollients, short nails.

2 **First line (off-label)**

Ivermectin

3 **Alternative**

Albendazole

4 **Topical (specialist)**

10–15% thiabendazole or albendazole ointment three times daily 1–2 weeks for single early tracks / when oral drugs unsuitable.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Cryotherapy**: Don't: larva is ahead of the track.
- **OTC threadworm mebendazole**: Single 100 mg dose is **not** adequate for CLM.

Referral

URGENT — Features of **larva currens**, especially if steroids or immunosuppression are planned.

ROUTINE — Diagnostic doubt, widespread disease / folliculitis, treatment failure, pregnancy, child < 15 kg, wheeze or eosinophilia.

SAME DAY — Spreading cellulitis from scratching.

Red flags and when to refer

- Rapidly moving perianal or buttock tracks (larva currens) or a patient on steroids: think Strongyloides hyperinfection.
- Spreading cellulitis from scratching.

Complications and prognosis

Complications

- Secondary impetigo, eczematisation; rarely Löffler syndrome.

Prognosis

- Excellent with treatment.

Prescriber pearls (UK pharmacist IP)

💡 **Prevention: avoid walking barefoot** on beaches and soil where dogs and cats roam. Wear shoes or sandals, lie on a mat or sunbed rather than directly on sand, and choose beaches that are tide-washed or animal-free. Supervise children in sandpits. Dry beach towels and clothing indoors rather than on sand.

💡

Ivermectin tablets: give on an empty stomach with water (follow the product information). Counsel on mild transient GI upset and dizziness.

The two "larva" names: larva **migrans** creeps slowly from an animal hookworm and is a dead end in humans. Larva **currens** "runs" fast and comes from the human Strongyloides, which can persist for decades and **disseminate if steroids are given**.

Self-check quiz

4 questions · optional

Q1 Why does freezing the track fail?

✓ The larva is 1–2 cm beyond the visible end.

Q2 First-line oral treatment?

✓ Ivermectin 200 micrograms/kg single dose (off-label).

Q3 Fast urticarial perianal track recurring for years?

✓ Larva currens (Strongyloides): serology before any steroids.

Q4 Is CLM contagious?

✓ No.

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Photo gallery

Cutaneous larva migrans 6

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Tropical & travel Module 5 · Week 4

Dengue: skin signs & safety

A mosquito-borne viral fever with severe aches, retro-orbital headache and a rash that classically shows 'islands of white in a sea of red'. Exclude malaria, avoid NSAIDs and aspirin, and watch for warning signs as the fever settles.

ALSO CALLED dengue fever | breakbone fever | islands of white in a sea of red | petechiae | tourniquet test | NSAIDs dengue | +1

RELATED PAGES The returning traveller with a skin problem | Morbilliform drug eruption & drug urticaria | Purpura: the differential diagnosis

KEY FACTS

- The most common imported arboviral infection in the UK (UKHSA: 904 cases in 2024).
- Fever, headache, retro-orbital pain, myalgia; then a rash with **"white islands in a sea of red"** during recovery.
- Critical phase days 3–7 warning signs (abdominal pain, vomiting, bleeding, lethargy, rising haematocrit with falling platelets).
- Supportive care: **paracetamol only**; avoid aspirin, ibuprofen and NSAIDs (bleeding risk).
- Any fever in a returning traveller same-day medical assessment.

Overview

Dengue is an acute mosquito-borne viral illness and the most common travel-associated arboviral infection seen in the UK. UKHSA counted **904 confirmed/probable imported cases in England, Wales and NI in 2024** (a record) and **344 in 2025**. Most were acquired in South and South-East Asia (India, Thailand, Indonesia), with large outbreaks also in Latin America and the Caribbean. Sporadic local transmission now occurs in southern Europe (France, Italy, Spain, Croatia). Most infections are mild or asymptomatic. A minority progress to **severe dengue** (plasma leakage, shock, haemorrhage, organ impairment), which is more likely with a **second infection with a different serotype**.

WHO 2009 classification: **dengue without warning signs** **dengue with warning signs** **severe dengue**. The older term "dengue haemorrhagic fever" has been replaced.

Aetiology: causes and risk factors

AETIOLOGY

dengue virus is a **flavivirus** (same family as yellow fever and Zika). It is a single-stranded RNA virus with **four serotypes (DENV-1 to DENV-4)**.

VECTOR

Aedes aegypti (and Ae. albopictus). These are **day-biting** mosquitoes, most active in the early morning and late afternoon. They breed in **small collections of clean standing water** (containers, tyres, plant pots), so risk is high

in **urban** areas. Ae. albopictus is found intermittently in parts of Europe, but not established in the UK, so UK onward transmission risk is negligible.

Incubation is **4–10 days** (range about 3–14). A patient can infect feeding mosquitoes during the viraemic phase.

IMMUNITY

infection gives lifelong immunity to that serotype and only brief cross-protection to the others. **Antibody-dependent enhancement** at a later infection with a different serotype is thought to drive more severe disease.

SKIN MECHANISMS

capillary dilatation causes the early flush. An immune-mediated viral exanthem causes the morbilliform rash. **Thrombocytopenia, capillary fragility and endothelial leak** cause petechiae and purpura.

Myths to address

- **"DENGUE IS ALWAYS MILD"**
  second infection with a different serotype raises the risk of severe dengue.

Pathophysiology and pathology

Severe dengue is due to **leaky capillaries**, often worse on a second infection.

1. **Mosquito bite**
   Aedes mosquitoes transmit the virus, mainly by day.
2. **Viraemia**
   Virus infects immune cells, causing fever and pain.
3. **Antibody enhancement**
   Non-neutralising antibodies from a past serotype can increase viral uptake.
4. **Plasma leakage**
   Cytokines and endothelial damage cause leakage around defervescence.

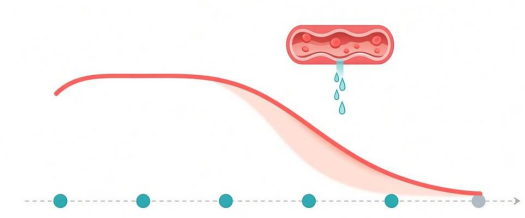

Fig. 1 · The critical phase follows defervescence.
AI-generated illustration

5 **Recovery rash**

A confluent rash with spared islands appears.

Fig. 2 · White islands in a sea of red.

AI-generated illustration

Clinical features

Illness phases:

1. Febrile phase (days 1–3/7) abrupt high fever, severe **frontal/retro-orbital headache**, **myalgia, arthralgia and back pain ("break-bone fever")**, nausea and vomiting, and a positive tourniquet test.
2. Critical phase (around defervescence, days 3–7) plasma leakage can develop **as the fever settles**. This is the danger window.
3. Recovery phase fluid reabsorbs, and there may be bradycardia and the convalescent rash.

Skin manifestations. About half of patients get a rash, and it is often missed on darker skin.

> **EARLY (FIRST 24–48 H)**
> transient **flushing/blanching erythema** of the face, neck and chest.

> **DAYS 3–6**
> a **maculopapular or morbilliform exanthem**. It often starts on the dorsum of the hands and feet, spreads to the limbs and trunk, and may be itchy.

> **CONVALESCENT RASH**
> confluent erythema with round pale **"islands of sparing", the classic "white islands in a sea of red"**, the characteristic sign. It is often **itchy** (palms/soles may be involved) and can be followed by fine desquamation.

> **HAEMORRHAGIC SIGNS**
> **petechiae** (especially on the **lower legs**, feet and at pressure sites), purpura/ecchymoses, conjunctival injection or bleeding, gum bleeding and nosebleeds. A **positive tourniquet test** shows capillary fragility.

> **CASE 3 FINAL QUESTION (LESIONS ON THE LEGS)**

> these are **petechiae/purpura**. They signify **thrombocytopenia and increased vascular (capillary) fragility or leakage**. They can occur in uncomplicated dengue, but they are a prompt to check the **platelet count and haematocrit** and to look for **warning signs of severe dengue**.

Warning signs (refer to hospital same day):

- abdominal pain or tenderness
- persistent vomiting
- clinical fluid accumulation (effusions, ascites)
- **mucosal bleeding**
- lethargy or restlessness
- liver enlargement over 2 cm
- **rising haematocrit with a rapidly falling platelet count**

Severe dengue:

- shock or respiratory distress from plasma leakage
- severe bleeding
- severe organ impairment (AST/ALT 1000, altered consciousness, heart or kidney failure)

Fig. 3 · Recovery-phase rash: confluent erythema with islands of sparing ("white islands in a sea of red").

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Types to recognise

White islands

Confluent erythema with sparing. Site: Legs.

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Truncal exanthem

Confluent morbilliform rash. Site: Trunk.

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Darker skin

Erythema subtle. Site: Trunk.

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Diagnosis and investigations

HISTORY

fever plus travel to an endemic area within the past 2 weeks (dengue's incubation fits returns within about 14 days). **Always exclude malaria first** if the patient travelled to a malarious area.

BLOODS

FBC: **leucopenia, thrombocytopenia, rising haematocrit**; LFTs: raised transaminases; U&E, CRP (usually low or modest)

Specific tests (in the UK, via the local lab or the **UKHSA Rare and Imported Pathogens Laboratory, RIPL**): **Days 1–5/7: RT-PCR** (viral RNA) and/or **NS1 antigen**; **After about day 5: dengue IgM** (single positive = probable); Paired IgG rise or IgM seroconversion confirms

Serology **cross-reacts** with other flaviviruses (Zika, yellow fever/Japanese encephalitis vaccines), so give the full travel and vaccine history.

NOTIFIABILITY

dengue virus is a laboratory-notifiable organism in England. Clinicians do not need to notify suspected cases.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Malaria	Exclude urgently; usually no rash.
	Chikungunya	Same vector; **severe persistent polyarthralgia**.
	Zika	Milder fever, itchy rash, conjunctivitis; pregnancy implications.
	Rickettsial infection	**Eschar**, regional nodes; responds to doxycycline.

	CONDITION	DISTINGUISHING FEATURES
	Meningococcal septicaemia	Non-blanching purpura + unwell: treat immediately.
	Measles	Coryza, cough, conjunctivitis, Koplik spots.
	Acute HIV seroconversion	Morbilliform rash, pharyngitis, nodes, oral ulcers.

Management

Treatment: there is **no specific antiviral**. Care is **supportive** and focuses on fluids and early detection of warning signs. Early supportive care brings case fatality in severe dengue below 1%.

> **Any fever in a returning traveller same-day medical assessment** (GP, urgent care, or A&E/HTD walk-in if unwell). This is not a pharmacy-only problem.

> **ANALGESIA/ANTIPYRETIC**
> **paracetamol only** (standard doses; caution with liver involvement).

> **Avoid aspirin, ibuprofen and other NSAIDs** (bleeding risk with thrombocytopenia), and **do not sell them OTC** to a febrile returnee. Avoid IM injections.

> **HYDRATION**
> oral fluids and oral rehydration salts, rest, and monitoring of urine output.

> **Outpatient care** (only if senior clinician-assessed and no warning signs) includes daily review or bloods (FBC/haematocrit) through the critical phase (days 3–7, especially as the fever falls). Give written "come back immediately if..." safety-netting: abdominal pain; persistent vomiting; bleeding gums or nose; black stools; drowsiness or restlessness; cold, clammy skin; passing little urine

> **Admit** with warning signs, severe dengue, pregnancy, infants or older adults, significant comorbidity (diabetes, renal/cardiac disease), or poor social support. Hospital care includes careful IV crystalloid and haematocrit-guided fluid management.

> **ITCH IN THE CONVALESCENT RASH**
> emollient, menthol in aqueous cream, or a non-sedating antihistamine. It settles over days.

> **MEDICATION REVIEW**
> pause or discuss anticoagulants and antiplatelets with the treating team if platelets are low. Check for herbal products (e.g. papaya leaf) that patients may take.

Treatment ladder

1

Same-day assessment

Any fever in a returning traveller same-day GP / urgent care / A&E (HTD walk-in if unwell): **not a pharmacy-only problem**.

2

Supportive

Paracetamol only (standard doses; caution with liver involvement). Oral fluids / ORS, rest, monitor urine output.

3

Prevention

Bite avoidance **by day**: DEET 50% (adults, children > 2 months, pregnancy), sunscreen first then repellent; long sleeves, permethrin clothing; screens / AC; nets for daytime naps; remove standing water. **Qdenga** (live, 2 doses 3 months apart): only 4 years **with previous dengue** travelling to risk areas (Green Book 15a).

⚠ CAUTION

Prescribing limits and pitfalls

- **NSAIDs and aspirin**: Avoid; don't sell OTC to a febrile returnee. Avoid IM injections.

Referral

999 / A&E — **Warning signs**: abdominal pain, persistent vomiting, mucosal bleeding, lethargy / restlessness, fluid accumulation, rising haematocrit with falling platelets.

SAME DAY — Fever after the tropics: **exclude malaria the same day**.

Red flags and when to refer

- **WARNING SIGNS**
  abdominal pain, persistent vomiting, mucosal bleeding, lethargy or restlessness, fluid accumulation, rising haematocrit with falling platelets: emergency admission.
- Never give **NSAIDs or aspirin** in suspected dengue.
- Fever after the tropics: exclude **malaria** the same day.

Complications and prognosis

Complications

- Shock, haemorrhage, organ impairment.

Prognosis

- Case fatality in severe dengue under 1% with good supportive care.

Prescriber pearls (UK pharmacist IP)

Bite avoidance is the main prevention. Aedes bite **during the day**, so a malaria-style "dusk to dawn" message is not enough: Use **DEET-based repellent (50% recommended by NaTHNaC for adults, children over 2 months, and pregnancy/breastfeeding)**. Alternatives are icaridin (picaridin), eucalyptus citriodora oil (PMD) or IR3535. Apply **sunscreen first, repellent second**, and reapply; Wear long sleeves and trousers, and **permethrin-treated clothing**; Stay in screened or air-conditioned rooms. Use nets for **daytime naps** (infants); Remove standing water around accommodation

Vaccine (UK, Green Book chapter 15a, Oct 2024): Qdenga® (TAK-003) is a **live attenuated** vaccine given as **2 doses 3 months apart**; It is recommended in the UK **only for people aged 4 years and over with a previous dengue infection** (confirmed or likely) who are travelling to risk or outbreak areas, or who are exposed through work. It is **not recommended for dengue-naive travellers** (JCVI precaution); Consider delaying for 1 year after a lab-confirmed infection; Contraindicated in **pregnancy, breastfeeding and immunosuppression**, and after previous anaphylaxis to the vaccine; NICE CKS and TravelHealthPro carry the eligibility algorithm. Testing for past infection goes via UKHSA RIPL

A second dengue infection carries a higher risk of severe disease. Previously infected travellers deserve a careful pre-travel consultation.

Dengue cannot be caught person to person in the UK (no established vector), apart from rare blood/organ/vertical transmission.

Self-check quiz

4 questions · optional

Q1 Analgesic of choice?

✓ Paracetamol; avoid NSAIDs and aspirin.

Q2 When is the critical phase?

✓ Around defervescence, days 3–7.

Q3 Classic convalescent rash?

✓ 'White islands in a sea of red'.

Q4 First test in a febrile returnee from a malarious area?

✓ Same-day malaria film / RDT.

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Photo gallery

Dengue rash 3

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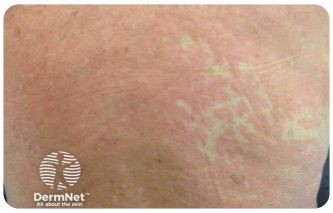

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Tropical & travel Module 5 · Week 4

Leprosy (Hansen disease) & lepra reactions

A chronic mycobacterial infection of skin and nerves: pale or red patches with reduced sensation, thickened nerves and, in multibacillary disease, nodules and facial infiltration. Curable with WHO multidrug therapy; reactions are emergencies for the nerves.

ALSO CALLED Hansen disease · leprosy · Mycobacterium leprae · numb patch · lepra reaction · erythema nodosum leprosum · +2

RELATED PAGES Cutaneous tuberculosis (scrofuloderma & others) · The returning traveller with a skin problem · Hypopigmentation & skin of colour · Cutaneous leishmaniasis

KEY FACTS

- Chronic infection of **skin and peripheral nerves** by Mycobacterium leprae; curable with **multidrug therapy**, but nerve damage is permanent.
- WHO: about 200,000 new cases a year (India, Brazil, Indonesia); UK about 10–15 cases a year, all acquired abroad.
- Cardinal signs: **hypopigmented or red patch with loss of sensation**, thickened peripheral nerves, acid-fast bacilli on smear.
- Spectrum from tuberculoid (few lesions, strong immunity) to lepromatous (many lesions, weak immunity).
- Refer every suspected case to a national leprosy centre; **reactions** (new nerve pain, weakness, eye symptoms) are urgent. Notifiable.

Overview

Leprosy is a chronic granulomatous infection of the **skin and peripheral nerves** (also the nasal mucosa, eyes and testes). It is caused by **Mycobacterium leprae** (and M. lepromatosis). It is **curable with multidrug therapy (MDT)**, but nerve damage already present is permanent, so **early diagnosis prevents disability**. WHO registers about **200,000 new cases a year**, mostly in **India, Brazil and Indonesia**, with others in Nigeria, Bangladesh, Nepal, Ethiopia, the DR Congo, Myanmar and the Pacific.

UK: around 10–15 notifications a year, all acquired abroad, mostly from the South-East Asia, Americas and Africa regions. Recent UKHSA data show 255 cases from 2000–2025 and no UK transmission. It is **statutorily notifiable** in England.

Classification:

> **WHO (OPERATIONAL)**
>
> **paucibacillary (PB)** = 1–5 skin lesions and a negative smear. **Multibacillary (MB)** = more than 5 lesions, or nerve involvement (pure neuritis or any lesions with neuritis), or a positive slit-skin smear.

> **RIDLEY–JOPLING SPECTRUM (IMMUNOLOGICAL)**
> TT – BT – BB – BL – LL, plus indeterminate and pure neural forms.

Aetiology: causes and risk factors

> M. leprae is an **obligate intracellular acid-fast bacillus**. It divides extremely slowly (about every 12–14 days) and prefers **cooler body sites** (skin, superficial nerves, nose, earlobes, testes). Incubation averages around 5 years (range from months to over 20 years).

> **TRANSMISSION**
> mainly **respiratory droplets from the nasal secretions of untreated MB patients**, after **prolonged close (household) contact**. Most exposed people (around 95%) are naturally resistant. **Armadillos** are a zoonotic source in the southern USA and Brazil. Patients become **non-infectious within days of starting rifampicin-containing MDT**.

> **IMMUNE SPECTRUM**
> **Tuberculoid pole (TT/BT):** strong Th1 cell-mediated immunity, few bacilli, few well-defined lesions, early and severe nerve damage near the lesions; **Lepromatous pole (LL/BL):** weak cell-mediated immunity, a Th2/ antibody response, huge numbers of bacilli, widespread symmetrical infiltration, and slower but extensive nerve involvement (glove and stocking)

> **NERVE DAMAGE**
> bacilli invade **Schwann cells**. This leads to sensory loss (temperature first, then light touch, pain, pressure), then autonomic loss (dry skin, anhidrosis), then motor loss (claw hand, foot drop, lagophthalmos).

> **Lepra reactions** are immune flare-ups and cause most of the disability. They occur in **30–50%** of patients, before, **during** or after MDT: **Type 1 (reversal) reaction:** delayed-type hypersensitivity in **borderline** disease (BT/ BB/BL). Existing lesions become red, swollen and tender, with **acute neuritis**; **Type 2 reaction = erythema nodosum leprosum (ENL):** an **immune-complex** (type III) systemic vasculitis/panniculitis in **LL/BL** (high bacillary load). Antigens from dying bacilli form antibody complexes that deposit in vessels and tissues, with high TNF-α. This is why Case 4 has **recurrent bouts of fever, joint pain and painful red nodules** on the legs and abdomen

Myths to address

- **"Leprosy is highly contagious and makes fingers fall off"**: it is hard to catch; digit loss follows numbness and injury.

Pathophysiology and pathology

M. leprae **lives in Schwann cells**, and the host's immunity shapes the disease.

1. **Spread**
   Droplets from untreated patients after prolonged close contact; long incubation (years).
2. **Nerve invasion**
   Bacilli infect Schwann cells in cool peripheral nerves.
3. **Immune spectrum**
   Strong Th1 response few granulomatous lesions; weak response many bacilli.

4 **Sensory loss**

Nerve damage causes numb patches and injuries.

Fig. 1 · Testing sensation in a skin patch.

AI-generated illustration

Clinical features

Cardinal signs (WHO). Leprosy is diagnosed if 1 is present:

1. **Definite loss of sensation in a pale (hypopigmented) or reddish skin patch**
2. **Thickened/enlarged peripheral nerve** with sensory loss and/or weakness of the muscles it supplies
3. **Acid-fast bacilli on slit-skin smear**

Spectrum:

TYPE	SKIN	NERVES	BACILLI
Indeterminate	1 or a few faint hypopigmented macules, ill-defined	Minimal sensory change	Few/none
TT (tuberculoid)	1–3 **sharply defined** hypopigmented or red plaques with raised edge; **dry, hairless, anaesthetic, anhidrotic**	Thickened nerve near lesion, early damage	None (smear –)
BT	More (5–20), larger, asymmetrical lesions; satellite lesions	Asymmetrical, prone to type 1 reaction	Few
BB	Many varied lesions; "**Swiss-cheese**"/inverted-saucer annular lesions	Several nerves	Moderate
BL	Many, fairly symmetrical macules/papules/plaques/nodules; sensation partly preserved	Widespread, glove-and-stocking	Many
LL (lepromatous)	Widespread, ill-defined, shiny symmetrical macules **diffuse infiltration, nodules (lepromas)**; **leonine facies**, **madarosis** (loss of lateral eyebrows/lashes), **thickened earlobes**, nasal stuffiness/epistaxis/collapse, oedematous legs, ichthyotic skin	Symmetrical glove-and-stocking sensory loss; nerves thickened	Very many (smear ++++)
Histoid (LL variant)	Shiny dome-shaped skin-coloured to coppery papules/nodules	–	Very many; may signal resistance/relapse
Pure neural	No skin lesions; nerve thickening/tenderness, sensory loss, trophic ulcers	–	–

Nerves to palpate: ulnar (elbow), median (wrist), radial cutaneous (wrist), **common peroneal / lateral popliteal** (fibular neck), posterior tibial (medial malleolus), great auricular (neck) and supra-orbital.

Case 4: findings in favour of leprosy:

- **Infiltrated facial lesions** for 2 years (LL infiltration, early leonine facies)

- **Glove-and-stocking loss of temperature and touch** (temperature lost first)
- **Thickened, tender ulnar and lateral popliteal nerves**
- **Painless ulcer after a thorn prick** (sensory neuropathy trophic ulcer)
- **"Slipping footwear"** (insensitive feet ± early foot drop from common peroneal palsy)
- **Recurrent fever + arthralgia + crops of painful nodules = ENL (type 2 reaction)**, typical of LL/BL
- He is a **farmer**, likely from an endemic rural setting

Other organs in LL:

> **EYES**
> corneal anaesthesia, lagophthalmos, keratitis, iritis, blindness
>
> **TESTES**
> orchitis, atrophy, gynaecomastia, infertility
>
> **NOSE**
> septal perforation, saddle nose
>
> **BONES**
> resorption of digits
>
> **KIDNEY**
> amyloidosis, glomerulonephritis

ENL (type 2):

- **Crops of tender red subcutaneous nodules/plaques** on the **face, extensor arms, thighs, trunk and abdomen**. The distribution is wider than classic EN. Nodules may pustulate or ulcerate (necrotic ENL).
- **Fever, malaise, arthralgia/arthritis**, with neuritis, iritis/uveitis, orchitis, dactylitis, lymphadenitis, hepatitis or nephritis.
- It is often recurrent or chronic, as in Case 4 (every 15 days or so).

Type 1 reaction:

- Existing patches suddenly become red, swollen and tender, with new lesions, oedema of the hands, feet and face, and **acute painful neuritis with new weakness or numbness**.
- **This is an emergency because of the risk of permanent nerve damage.**

Fig. 2 · Contrast pole: tuberculoid (PB) — few sharply defined anaesthetic plaques; test sensation in any persistent patch.

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Types to recognise

Tuberculoid

Few defined anaesthetic plaques. Site: Any.

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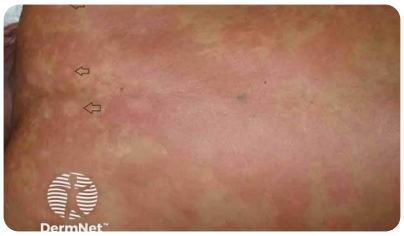
Borderline

'Swiss-cheese' annular plaques. Site: Trunk, limbs.



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Lepromatous

Diffuse facial infiltration. Site: Face.

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Earlobe

Thickening: LL site, smear site. Site: Ears.

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Type 1 reaction

Plaques become red, swollen. Site: Existing lesions.

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Trophic ulcer

Painless, from sensory loss. Site: Feet, hands.

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Diagnosis and investigations

UKHSA advice: **discuss every suspected case with a Consultant Advisor in Leprosy before smears or biopsies are taken**, and refer to a national leprosy centre.

Investigations:

> **CLINICAL EXAMINATION**
> sensory testing of lesions (cotton wool for touch, warm/cold tubes for temperature, monofilaments), nerve palpation, and a voluntary muscle test.

> **Slit-skin smear (SSS)** from the **earlobes**, forehead and lesions. Stain with Ziehl–Neelsen/Fite. Report the **bacterial index (BI)** and **morphological index**. The smear is positive in MB and negative in TT.

> **Skin biopsy** (deep incisional, from the active edge, down to fat) for histology. This shows granulomas, nerve involvement and AFB on **Fite–Faraco stain**, and helps classify on the Ridley–Jopling scale. Lepromatous disease shows foamy macrophages (Virchow/lepra cells) packed with bacilli and globi, with a **Grenz zone**.

> **M. leprae* PCR** on tissue or smear is highly specific. It can also detect drug-resistance mutations (rifampicin rpoB, dapsone folP1, fluoroquinolone gyrA).

> **NERVE STUDIES**
> nerve conduction, ultrasound or nerve biopsy (pure neural leprosy).

> **LEPROMIN (MITSUDA) TEST**
> not diagnostic. It helps **classify** (positive in TT, negative in LL).

> **SEROLOGY (ANTI-PGL-I)**
> positive in most MB disease, but limited for diagnosis.

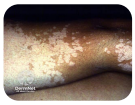
> **BEFORE MDT**
> FBC, LFTs, U&E, **G6PD** (dapsone), and screening for HIV, TB (before steroids and rifampicin), hepatitis B/C, Strongyloides (before prolonged steroids) and pregnancy.

"Leonine facies":

> **MEANING**
> a lion-like face from diffuse infiltration and thickening of the facial skin, with deepened furrows on the forehead and cheeks, thickened earlobes and nose, and often loss of eyebrows.

> **CAUSES**
> **Lepromatous leprosy** (classic); **Cutaneous T-cell lymphoma** (mycosis fungoides / Sézary syndrome), leukaemia cutis, other lymphomas; **Diffuse cutaneous leishmaniasis** / PKDL; **Sarcoidosis**; **Scleromyxoedema** / lichen myxoedematosus; **Amyloidosis**, lipoid proteinosis; **Actinic reticuloid / chronic actinic dermatitis**; **Mastocytosis**; **Paget's disease of bone** / fibrous dysplasia ("leontiasis ossea", where the cause is bony); **Severe rosacea/ rhinophyma**

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	**Pityriasis versicolor**	Fine scale, KOH +, **normal sensation**.

	CONDITION	DISTINGUISHING FEATURES
	Vitiligo	Depigmented (not hypo-), not anaesthetic.
	Tinea / granuloma annulare / sarcoid	Red annular plaques, **sensation normal**.
	Lupus vulgaris	Apple-jelly nodules.
	Erythema nodosum (vs ENL)	Tender shin nodules; strep, sarcoid, IBD, drugs.
	Cutaneous lymphoma	Infiltrated plaques; biopsy.

Management

Primary-care role: suspect, refer, support adherence, and spot reactions and adverse drug effects.

Refer every suspected case to a **national leprosy centre** via a Consultant Advisor in Leprosy (e.g. **HTD, UCLH, London**; see the UKHSA Memorandum on leprosy 2023, Appendix 5). The diagnosing doctor **notifies the case to the local authority proper officer / UKHSA health protection team** (England), handled confidentially. Contact management goes through the CCDC and UKHSA TB Unit.

URGENT/SAME-DAY REFERRAL

a **reaction** (type 1 with new nerve pain or weakness; ENL with fever), **acute eye symptoms** (red painful eye, reduced vision, cannot close the eye), or a **new foot drop or claw hand**.

WHO MDT 2018. Three drugs for all patients (adult doses, specialist-supervised): Rifampicin 600 mg once a month (supervised); **Clofazimine 300 mg once a month (supervised) + 50 mg daily**; **Dapsone 100 mg daily**; **Duration: PB 6 months; MB 12 months.** Case 4 = **MB, 12 months**; **Rifampicin-resistant leprosy:** clofazimine + at least two of clarithromycin, minocycline or a fluoroquinolone (e.g. ofloxacin/moxifloxacin) for 6 months, then clofazimine + one of these for 18 months; UK centres may use modified regimens (e.g. daily rifampicin, or longer courses for high bacterial index)

MANAGEMENT OF ENL

Continue MDT. Reactions are **not** a reason to stop; **Mild ENL:** rest, **NSAIDs/aspirin** for fever and arthralgia (with GI protection); **Moderate–severe ENL (or neuritis, iritis, orchitis): oral prednisolone** (e.g. starting around 40–60 mg/day, then tapered over weeks–months by the specialist); **Clofazimine at higher anti-inflammatory doses** (e.g. up to 300 mg/day for limited periods): a steroid-sparing option for chronic or recurrent ENL. Its effect is slow (4–6 weeks); **Thalidomide** (100–400 mg/day, then tapered) is very effective for recurrent ENL. It is **highly teratogenic**: in the UK it is supplied under a **Pregnancy Prevention Programme**, with neuropathy and VTE risks. It is used when steroids fail or cause problems; **Refractory ENL:** methotrexate, ciclosporin, **TNF-α inhibitors** (specialist, case-based); Treat iritis urgently with ophthalmology (topical steroid + mydriatic)

TYPE 1 REACTION

prednisolone (typically starting around 40 mg/day and tapering slowly over about 3–6 months under specialist supervision) to prevent permanent nerve damage. Splint the affected limb at rest.

PREVENTION OF DISABILITY (POD)

Daily **self-examination of hands, feet and eyes**; Soak and oil dry skin; Protective **footwear** (microcellular rubber, well-fitting) and gloves for cooking or hot objects; Prompt dressing and rest of any wound; never walk on an ulcer; Eye protection (glasses; blink exercises; lubricants for lagophthalmos); Physiotherapy, splints, and reconstructive surgery later

CONTACTS

examine household contacts. WHO recommends **single-dose rifampicin post-exposure prophylaxis (SDR-PEP)** for eligible contacts (adults 600 mg once) **after excluding leprosy and TB**. This reduces risk by around 50–60%. UK use is per the UKHSA/Consultant Advisor. BCG gives partial protection.

Medicines at a glance

Rifampicin

WHO MDT 2018 (specialist) — SPECIALIST ONLY

DOSE	600 mg once a month (supervised)

Clofazimine

WHO MDT 2018 (specialist) — SPECIALIST ONLY

DOSE	300 mg once a month (supervised) + 50 mg daily

Dapsone

WHO MDT 2018 (specialist) — SPECIALIST ONLY

DOSE	100 mg daily
CAUTIONS	Check G6PD first.

Treatment ladder

1

Refer & notify

Refer every suspected case to a **national leprosy centre** (e.g. HTD, UCLH). Notify to UKHSA (confidential); contacts via CCDC / UKHSA.

2

WHO MDT 2018 (specialist)

Rifampicin; Clofazimine; Dapsone

3

Reactions (specialist)

Continue MDT. Type 1: prednisolone (~40 mg/day tapered over 3–6 months) + splint. ENL: NSAIDs if mild; prednisolone (~40–60 mg/day) if moderate–severe; clofazimine high-dose; **thalidomide** under a Pregnancy Prevention Programme.

4

Your safety checks

Rifampicin: enzyme induction (contraception IUD / condoms per FSRH; warfarin, DOACs, ARVs, azoles, ciclosporin, methadone, steroids); orange urine and tears. **Dapsone**: haemolysis, methaemoglobinaemia, **hypersensitivity syndrome** in first 4–6 weeks; FBC, LFT.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Prevent disability**: Daily self-examination of numb hands, feet and eyes; protective footwear.

Referral

SAME DAY	Reaction: new **nerve pain or weakness**, ENL with fever, **new foot drop or claw hand**.
SAME DAY	Eye pain, redness, reduced vision or can't close the eye: **same-day ophthalmology**.
URGENT	Any suspected leprosy: Consultant Advisor / national leprosy centre.

Red flags and when to refer

- New **nerve pain, numbness or weakness**, or painful nodules with fever on treatment: lepra reaction, **urgent** specialist contact.
- Eye pain or redness: same-day ophthalmology.
- Thalidomide for ENL is specialist-only with a strict pregnancy prevention programme.

Complications and prognosis

Complications

- Disability, ulcers, eye damage, stigma; leprosy reactions.

Prognosis

- Cured with MDT (6–12 months); nerve damage persists.

Prescriber pearls (UK pharmacist IP)

Rifampicin (even monthly) induces enzymes, which matters around the dose day: It reduces combined/progestogen-only pill, implant and emergency contraception efficacy. Advise a copper IUD/LNG-IUS or condoms, and follow FSRH guidance; It interacts with warfarin, DOACs, antiretrovirals, azoles, ciclosporin, methadone and steroids (a higher prednisolone dose may be needed); It colours urine, tears and contact lenses **orange-red**

Dapsone: Check G6PD before starting. It causes dose-related haemolysis and methaemoglobinaemia (breathlessness, cyanosis); **Dapsone hypersensitivity syndrome** usually appears in the first 4–6 weeks: fever, rash, lymphadenopathy, hepatitis. Stop and seek urgent help. HLA-B*13:01 is a risk marker in some Asian populations; Monitor FBC and LFTs; agranulocytosis has been reported

Clofazimine: Red-brown to grey-black skin pigmentation (most marked in lesions and on dark skin), which fades **slowly over months to years after stopping**. Warn patients in advance, as it adds to stigma; Dry, ichthyotic skin (use emollients), red-coloured sweat, tears and urine; GI upset and, at high doses, enteropathy (crystal deposition), so take with food. QT prolongation

Steroid courses for reactions are often long. Give bone protection, GI protection where indicated, glucose and BP checks, steroid card and sick-day rules. Screen for and treat Strongyloides and latent TB first.

Adherence: the course is long (6–12 months) and skin lesions fade slowly. Explain that **reactions are not treatment failure**, and that **numbness may not recover** even when the infection is cured.

Stigma: Use person-first language ("person affected by leprosy", not "leper"); Emphasise that leprosy is **curable**, that people are **not infectious once on MDT**, and that casual contact (touch, sharing meals, sex) does not spread it; Maintain **strict confidentiality** (notification is handled in confidence); Signpost **LEPRA** and **The Leprosy Mission England & Wales** for patient support

(incl. stigma)

Self-check quiz

4 questions · optional

Q1 Three WHO cardinal signs?

✓ Anaesthetic patch, thickened nerve with loss, AFB on slit-skin smear.

Q2 MDT duration for MB disease?

✓ 12 months (PB 6 months).

Q3 Check before dapsone?

✓ G6PD.

Q4 New nerve pain on MDT?

✓ Type 1 reaction: urgent specialist (steroids); continue MDT.

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Photo gallery

Lepromatous and borderline leprosy

5

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Tuberculoid and borderline

2

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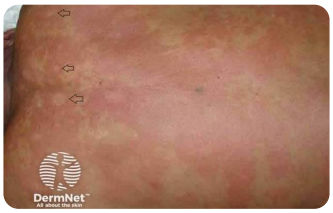

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Reactions and complications

3

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Tropical & travel Module 5 · Week 4

Mycetoma (Madura foot)

A slowly progressive, painless swelling of the foot (or other site) with sinuses discharging grains, after soil organisms enter through a minor wound. Refer to tropical medicine early; fungal and bacterial forms are treated very differently.

ALSO CALLED Madura foot · eumycetoma · actinomycetoma · discharging sinuses · grains · Nocardia

RELATED PAGES The returning traveller with a skin problem · Cutaneous tuberculosis (scrofuloderma & others) · Cutaneous leishmaniasis

KEY FACTS

- Chronic granulomatous infection of skin, fascia and bone from soil organisms entering through minor injury.
- Triad: **painless firm swelling, multiple sinuses, discharge with grains**.
- Two forms: **eumycetoma** (fungal, e.g. Madurella) and **actinomycetoma** (bacterial, Nocardia, Actinomadura).
- Usually the foot in rural workers in the tropical "mycetoma belt"; WHO neglected tropical disease since 2016.
- Refer to ID/tropical medicine; **don't incise or partially excise** in primary care.

Overview

Mycetoma is a chronic, slowly progressive, granulomatous infection of the **skin, subcutaneous tissue, fascia and eventually bone**. It is caused by organisms from soil or plants introduced through minor injury. Its hallmark is a **triad**:

1. **painless firm subcutaneous swelling**
2. **multiple sinuses**
3. **discharge containing grains** (colonies of the organism)

There are two forms:

- **Eumycetoma** is **fungal** (e.g. Madurella mycetomatis, the most common cause worldwide).
- **Actinomycetoma** is **bacterial**, from filamentous aerobic actinomycetes (Nocardia, Actinomadura, Streptomyces).

WHO recognised mycetoma as a **neglected tropical disease in 2016** (WHA69.21). It is endemic in the **"mycetoma belt"** between latitudes about 15°S and 30°N: **Sudan** (the highest burden), Senegal, Mauritania, Chad, Niger, Somalia, Yemen, **India**, **Mexico** (mainly actinomycetoma, Nocardia brasiliensis), Venezuela and Brazil. In the UK it is seen only in **migrants or long-term residents from endemic areas**, often years after exposure.

Aetiology: causes and risk factors

- **Inoculation** happens through thorn pricks, splinters or minor cuts, typically on the **feet of barefoot rural workers** such as farmers, herders and labourers (Case 5 is a 52-year-old farmer). Most patients are men aged 20–40, though any age can be affected.
- There is **no person-to-person spread** and no known vector or animal reservoir. Incubation runs from months to years.
- The organism forms **grains** (compact colonies in a matrix). These provoke suppurative and granulomatous inflammation with **abscesses that track to the skin as sinuses**, and also spread along fascial planes into **muscle and bone** (osteomyelitis, periosteal reaction, lytic lesions).
- **Actinomycetoma** tends to be more inflammatory, spreads faster and **invades bone earlier**. **Eumycetoma** is slower, better encapsulated and more localised.
- **GRAIN COLOUR HINTS AT THE CAUSE**
  black = eumycetoma (Madurella, Trematosphaeria, Falciformispora); **white/pale** = eumycetoma (Scedosporium, Fusarium, Acremonium) or actinomycetoma (Nocardia, Actinomadura madurae); **red/pink** = Actinomadura pelletieri; **yellow/brown** = Streptomyces somaliensis

Myths to address

- **"IT'S RECURRENT CELLULITIS"**
  repeated antibiotics miss the diagnosis.

Pathophysiology and pathology

Organisms form **grains** that the body walls off but cannot clear.

1. **Inoculation**
   Thorns or splinters introduce organisms.
2. **Grain formation**
   Organisms grow as compact colonies (grains) in tissue.
3. **Granulomas and sinuses**
   Inflammation forms abscesses that drain grains through sinuses.

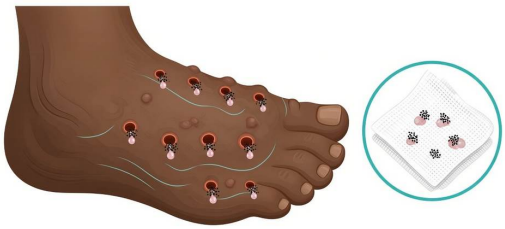

Fig. 1 · Mycetoma foot with sinuses discharging grains.
AI-generated illustration

4. **Bone spread**
   Infection invades fascia and bone.

Clinical features

SITE

about **70% on the foot** (forefoot, dorsum, sole), then hand, leg and knee. The **back, shoulder, head and neck** can be affected in carriers of loads or sacks (often Nocardia). The chest wall and scalp are rare sites but dangerous.

COURSE

A small, painless, firm subcutaneous nodule, easily dismissed as a "boil" or "cyst"; Gradual **swelling**, then multiple nodules that soften and **open as sinuses**; Discharge of **serous, serosanguinous or purulent fluid containing grains**; Sinuses heal and new ones open (old sinuses scar); The skin becomes **hyperpigmented**, shiny and stretched

Pain is mild or absent until bone involvement or secondary bacterial infection. Lymph nodes are usually not involved, except sometimes with actinomycetoma.

The patient is typically systemically well (fever is rare unless there is secondary infection), which helps explain the late presentation.

LATE FEATURES

massive swelling and deformity, bone destruction, joint ankylosis, disuse atrophy, chronic lymphoedema, disability, sometimes **amputation**, and rarely death from cranial, spinal or thoracic spread or sepsis.

Fig. 2 · Triad: painless firm swelling + multiple sinuses + discharge with grains — "Madura foot" (Case 5 farmer).

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Types to recognise

Classic triad

Swelling + sinuses + grains. Site: Foot.

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Advanced

Gross forefoot swelling. Site: Foot.

© DermNet (courtesy Dr Sriyani Samaraweera) · **Source**

Actinomycetoma

Numerous crusted sinuses. Site: Foot, limb.

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Spread up the leg

Chronic indurated limb. Site: Leg.

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Diagnosis and investigations

Clinical triad + endemic exposure. Look for grains in the discharge (shake gauze in saline; dermoscopy may show grains).

GRAINS AND TISSUE

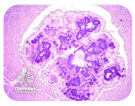
collect by **deep surgical biopsy or FNA** (not superficial swabs) for: **direct microscopy** (KOH: **fine filaments under 1 µm = actinomycete; broad septate hyphae 2–5 µm = fungus**); **Gram/PAS/Grocott stains**; **culture** (slow, up to 6 weeks; tell the lab); **histology** (grains in abscesses surrounded by neutrophils, then granuloma; Splendore–Hoeppli material); **PCR/sequencing** for species ID, now standard at reference centres

IMAGING

Plain X-ray: soft-tissue mass, periosteal reaction, lytic "moth-eaten" bone, cortical thinning; **Ultrasound:** grains as bright hyper-reflective echoes in cavities (helps tell eumycetoma from actinomycetoma); **MRI:** extent and bone involvement, with the characteristic **"dot-in-circle" sign**; **CT:** bony involvement

> **BASELINE BEFORE LONG ANTIMICROBIAL COURSES**
> FBC, LFTs, U&E, glucose and HIV status.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Chronic osteomyelitis	Sinus over bone, prior fracture, metalwork, diabetic foot; no grains.
	Scrofuloderma / NTM	Over nodes or bone; AFB; undermined sinuses.
	Botryomycosis	Grain-forming S. aureus; standard antibiotics work.
	**Actinomycosis**	Cervicofacial / dental; 'sulphur granules'.
	Hidradenitis	Flexural tunnels, comedones.
	Podoconiosis / lymphoedema	Mossy foot, no sinuses with grains.

Infective mimics of mycetoma:

Non-infective mimics:

- **Podoconiosis** (non-filarial "mossy foot" from irritant volcanic soil) or **lymphatic filariasis/lymphoedema**: swelling **without** sinuses or grains
- **Kaposi sarcoma**, **soft-tissue/bone sarcoma**, **SCC in a chronic sinus (Marjolin ulcer)**, foreign-body granuloma

Causes of discharging sinuses in general:

> **SKIN FOLLICULAR/INFLAMMATORY**
> **hidradenitis suppurativa**, **acne conglobata**, **dissecting cellulitis of the scalp** (the follicular occlusion tetrad, with pilonidal sinus), pyoderma vegetans, **Crohn's disease** (perianal/cutaneous fistulae)

> **DENTAL (ODONTOGENIC)**
> chronic dental abscess draining on the chin or jaw skin, often misdiagnosed as a skin lesion

> **INFECTIVE**
> osteomyelitis, mycetoma, actinomycosis, botryomycosis, scrofuloderma/TB, NTM, chronic furuncles/carbuncles

> **OTHER**
> foreign body, post-surgical (suture/mesh), congenital (preauricular sinus, branchial cleft, thyroglossal duct), malignancy (carcinoma breaking down)

Management

Primary-care role: recognise, avoid inappropriate repeated antibiotic courses or incision, and refer.

> **Refer every suspected case** to **Infectious Diseases / tropical medicine** (e.g. **HTD London**, which runs a fungal infections service, or a regional ID unit), working jointly with **orthopaedic/plastic surgery** and **dermatology**. Mark it urgent if there is spread toward the trunk or scalp, rapid progression, or secondary sepsis.

> **Do not** perform simple incision and drainage or partial excision in primary care. Inadequate excision causes recurrence and spread. Do not label the condition "recurrent cellulitis/abscess".

> **INTERIM**
> treat **acute secondary bacterial infection** per NICE cellulitis guidance, keep wounds clean and dressed, give simple analgesia and limb elevation, check tetanus status, and offer footwear advice.

> **SPECIALIST TREATMENT**
> **Actinomycetoma (bacterial):** usually curable **medically**. Prolonged combination antibiotics for months, e.g. **co-trimoxazole + amikacin** cycles (the "Welsh regimen"). Alternatives include co-amoxiclav, dapsone, rifampicin, doxycycline/minocycline, linezolid or carbapenems depending on species and severity. Surgery is rarely needed; **Eumycetoma (fungal): prolonged oral itraconazole** (e.g. 200 mg twice daily for at least 6–12 months, often longer). This shrinks and encapsulates the lesion, which is then followed by **wide surgical excision** and continued antifungal. Alternatives include voriconazole, posaconazole and terbinafine (ketoconazole is no longer used systemically in the UK because of hepatotoxicity). Recurrence is common and cure rates are modest. Advanced disease may need **amputation**. Fosravuconazole is under evaluation (DNDi/Sudan trial)

Treatment ladder

1 **Refer**

Every suspected case **ID / tropical medicine** (e.g. HTD) with orthopaedics / plastics and dermatology. **Urgent** if spreading toward trunk or scalp, rapid progression or sepsis.

2 **Interim**

Treat acute secondary infection per NICE cellulitis guidance; clean dressings; analgesia; elevation; **tetanus check**; footwear advice.

3 **Specialist**

Actinomycetoma: usually medically curable with prolonged combination antibiotics. **Eumycetoma**: long antifungals (e.g. itraconazole) + surgery.

Co-management checks

Itraconazole: capsules need food and acid (avoid PPIs / antacids); strong CYP3A4 inhibitor (simvastatin, some DOACs, CCBs, colchicine…); LFTs; avoid in heart failure. **Co-trimoxazole** for months: FBC, potassium, renal function, rash; avoid with methotrexate; warfarin / phenytoin interactions.

⚠ CAUTION

Prescribing limits and pitfalls

- **Don't**: Incise, partially excise, or label as 'recurrent cellulitis / abscess'.

Referral

URGENT	Bone pain, deformity, rapid spread: urgent specialist imaging and referral.
ROUTINE	Any suspected mycetoma: ID / tropical medicine.

Red flags and when to refer

- Bone pain, deformity or rapid spread: urgent specialist imaging and referral.
- Do not treat repeatedly with short courses of antibiotics without a diagnosis.

Complications and prognosis

Complications

- Bone destruction, deformity, amputation, secondary infection.

Prognosis

- Actinomycetoma responds to antibiotics; eumycetoma often needs surgery plus long antifungals.

Prescriber pearls (UK pharmacist IP)

The farmer with a foot that keeps "discharging" despite antibiotics: think mycetoma, chronic osteomyelitis or TB. Ask where they grew up and worked, and whether they went barefoot.

Itraconazole (if co-managing): **Capsules** need food and stomach acid, so avoid PPIs/H2 blockers/ antacids around the dose or use the oral solution (empty stomach). Formulations are not interchangeable; It is a **strong CYP3A4 inhibitor**. Many interactions, some contraindicated:

simvastatin, some DOACs, certain calcium-channel blockers, colchicine in renal/hepatic impairment, some antiarrhythmics, ergot alkaloids, midazolam; Monitor **LFTs**. It is a **negative inotrope**, so avoid in heart failure. Take TDM levels if available

Co-trimoxazole for months: FBC (folate antagonism, marrow), **potassium** and renal function, rash (SJS/TEN warning); Avoid with methotrexate; interacts with warfarin and phenytoin; Folic acid supplementation per specialist

Amikacin: therapeutic drug monitoring, renal function, and **audiometry** (ototoxicity is cumulative over cycles).

Adherence: therapy lasts months to years. Explain that **stopping early = relapse**, and that swelling and scarring may persist after cure. Arrange physiotherapy.

Prevention: wear **shoes** and protective clothing during farming and herding, clean and dress minor wounds promptly, and seek early care for any slowly enlarging painless lump on the foot.

Self-check quiz

4 questions · optional

Q1 **Mycetoma triad?**

✓ Painless swelling, multiple sinuses, discharge with grains.

Q2 **Black grains suggest?**

✓ Eumycetoma (fungal).

Q3 **Which form is usually medically curable?**

✓ Actinomycetoma (bacterial).

Q4 **Why not incise in primary care?**

✓ Inadequate excision causes recurrence and spread.

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Photo gallery

Mycetoma 5

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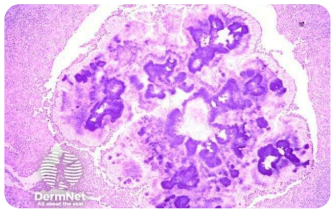

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CHAPTER 11

Ulcers & red leg

Venous, arterial and neuropathic ulcers, pyoderma gangrenosum, and the red-leg differential, so you don't treat every red leg as cellulitis.

1. Venous leg ulcer
2. Pyoderma gangrenosum
3. Arterial (ischaemic) leg ulcer
4. Neuropathic and diabetic foot ulcers
5. Red leg: differentials & referral

Ulcers & red leg Module 5 · Week 2

Venous leg ulcer

The commonest chronic leg ulcer: a shallow, sloping, exudative ulcer in the gaiter area. Compression after ABPI is the cornerstone. Colonised does not mean infected.

ALSO CALLED: venous ulcer | stasis ulcer | gravitational ulcer | leg ulcer | compression | ABPI | +3

RELATED PAGES: Arterial (ischaemic) leg ulcer | Neuropathic and diabetic foot ulcers | Red leg: differentials & referral | Cellulitis & erysipelas | Pyoderma gangrenosum | Wound aftercare: sutures, activity, infection & scars

KEY FACTS

- **Venous leg ulcers** are the commonest chronic leg ulcers (about 70%), caused by **venous hypertension**.
- Gaiter area, especially over the **medial malleolus**; shallow, sloping-edged, exudative ulcer with haemosiderin, venous eczema and lipodermatosclerosis.
- Always check **ABPI** before high compression to exclude significant arterial disease.
- **Compression** is the cornerstone; plus elevation, walking, wound care and treatment of venous eczema.
- Refer if not healing after about 2 weeks of optimal care, atypical, mixed arterial disease or for venous intervention (NICE CG168).

Overview

A venous leg ulcer is full-thickness skin loss caused by **chronic venous insufficiency** (venous hypertension). It is the **most common** chronic leg ulcer. Other names are stasis ulcer and gravitational ulcer. Avoid calling it a 'varicose ulcer' if there are no varicose veins.

Epidemiology

- **Who**
  - Older adults, more women; prevalence rises steeply after 65.
  - Risk factors: varicose veins, previous **DVT**, obesity, immobility, multiple pregnancies, leg trauma.
- **Burden**
  - Often chronic and recurrent: many recur within a year without ongoing compression.

Aetiology: causes and risk factors

- **Valve incompetence**, previous **DVT**, obesity, immobility and calf-muscle pump failure cause **venous hypertension**.

- This leads to oedema, haemosiderin deposition, inflammation and lipodermatosclerosis, with poor oxygen and nutrient delivery.
- The skin then breaks down after **minor trauma** and fails to heal.

Risk factors and associations

- Calf-muscle pump failure (immobility, ankle stiffness, arthritis).
- Post-thrombotic syndrome after DVT.

Myths to address

> **"SWAB EVERY ULCER"**
> all chronic ulcers grow bacteria; swab only if clinically infected.

> **"COMPRESSION IS UNSAFE IN THE ELDERLY"**
> it is safe once ABPI shows adequate arterial supply.

Pathophysiology and pathology

Persistently high pressure in the leg veins damages the skin from the inside.

1. **Valve failure**
   Incompetent valves in superficial, perforating or deep veins (often after DVT) let blood **reflux** downwards.
2. **Venous hypertension**
   Walking normally lowers ankle venous pressure; with reflux or pump failure it stays high (**ambulatory venous hypertension**).
3. **Capillary leak**
   High pressure stretches capillaries; fluid, red cells (later **haemosiderin**) and fibrinogen leak into the tissues, causing oedema and brown staining.
4. **Inflammation and fibrosis**
   White cells trapped in capillaries release enzymes and cytokines; the dermis and fat scar (**lipodermatosclerosis**) and oxygen delivery falls.
5. **Ulceration**
   Minor trauma breaks the fragile skin, which cannot heal while venous pressure stays high: compression reverses this.

HISTOPATHOLOGY

- Not needed routinely. **Biopsy** a non-healing or atypical ulcer (rolled edge, exuberant tissue) to exclude SCC (Marjolin ulcer) or BCC.

Clinical features

- **Gaiter area**, most commonly over the **medial malleolus**.
- **Shallow** ulcer with **sloping** edges, often **heavily exudative**.

- Surrounding skin: varicose veins, brown **haemosiderin** staining, **venous eczema**, atrophie blanche, and **lipodermatosclerosis** (the 'inverted champagne bottle' leg).
- **Dependent oedema**, worse when the leg is down.
- Pain is usually milder than with arterial ulcers, unless the ulcer is infected.

Fig. 1 · Venous ulcer near medial malleolus / gaiter — most common venous ulcer site.
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Special groups

- **Mixed arterial–venous disease**
  - ABPI below 0.8 needs reduced or no compression and vascular referral; diabetes can give a falsely high ABPI.

Diagnosis and investigations

- Clinical, with signs of venous disease.
- **ABPI** (ankle–brachial pressure index) **before any high compression**, to exclude significant arterial disease.
- Venous duplex if intervention is planned.
- Swab **only** if there are clinical signs of infection.

ABPI

- (ankle–brachial pressure index) **before any high compression**, to exclude significant arterial disease.

Key points

- Clinical, with signs of venous disease.
- Venous duplex if intervention is planned.
- Swab **only** if there are clinical signs of infection.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Arterial ulcer	Punched-out, distal (toes, pressure points), claudication or rest pain, cold foot, weak pulses, **low ABPI**
	Neuropathic ulcer	Plantar pressure points, **painless**, callus, diabetes with neuropathy
	Pyoderma gangrenosum	Rapid, very painful, **violaceous undermined edge**; pathergy
	Vasculitic or malignant ulcer	Atypical site or edge, rolled or raised margin, non-healing. **Biopsy or refer**
	Pressure ulcer	Over a bony prominence in an immobile patient

See also: arterial ulcers and diabetic foot ulcers · Pyoderma gangrenosum

Management

Based on NICE CG168 and local leg ulcer and tissue viability pathways.

- **Compression** (multilayer bandaging or hosiery) is the cornerstone, **once ABPI is satisfactory**.
- Leg elevation, walking and ankle exercises, weight management, and emollients.
- Treat **venous eczema** with short courses of mild to potent topical steroid around (not in) the ulcer.
- Wound care: moist healing and debridement by tissue viability. Do not use topical antibiotics routinely.
- **INFECTION**
  give systemic antibiotics **only** for spreading cellulitis or systemic signs. Colonisation is not infection.
- Pentoxifylline is sometimes used as an adjunct, following specialist or local policy.

Referral

DON'T — **Low or abnormal ABPI**, absent pulses, rest pain or a cold foot: arterial disease, so **do not compress**. Refer to vascular.

ROUTINE — Spreading redness, fever or systemic upset: cellulitis or sepsis.

ROUTINE Rolled, raised or irregular edge, bleeding, or failure to heal: biopsy to exclude **SCC or BCC** (Marjolin ulcer).

ROUTINE Rapidly enlarging, very painful ulcer with a purple edge: suspect **pyoderma gangrenosum**.

Red flags and when to refer

- **Low or abnormal ABPI**, absent pulses, rest pain or a cold foot: arterial disease, so **do not compress**. Refer to vascular.
- Spreading redness, fever or systemic upset: cellulitis or sepsis.
- Rolled, raised or irregular edge, bleeding, or failure to heal: biopsy to exclude **SCC or BCC** (Marjolin ulcer).
- Rapidly enlarging, very painful ulcer with a purple edge: suspect **pyoderma gangrenosum**.

Refer

To vascular surgery, tissue viability or dermatology if: not healing after about 2 weeks of optimal care, atypical appearance, mixed arterial disease, or for venous intervention (NICE: refer for venous assessment).

Complications and prognosis

Complications

- Cellulitis; contact allergy to dressings and topical treatments (common in leg ulcer patients).
- Chronic pain, reduced mobility and isolation.
- Malignant change in long-standing ulcers (rare).

Prognosis

- Most heal within months with good compression; recurrence is common, so lifelong compression hosiery is advised after healing.

Prescriber pearls (UK pharmacist IP)

Never start high compression without a vascular assessment or ABPI.

Explain **lifelong compression stockings** after healing, because recurrence is common.

Good analgesia and exudate control improve adherence to compression.

Topical steroids on surrounding venous eczema help. Watch for contact allergy to dressings and topical products, which is common in leg ulcer patients.

Self-check quiz

5 questions · optional

Q1 When should you think of Venous leg ulcer?

✓ A shallow, sloping, wet ulcer over the medial malleolus with haemosiderin staining, venous eczema, varicose veins and oedema that is worse at the end of the day.

Q2 What drives it?

✓ Venous hypertension from incompetent valves, previous DVT or calf-pump failure, causing oedema, inflammation and skin breakdown.

Q3 What is the backbone of care?

✓ ABPI, then multilayer compression, elevation, walking, emollient, and lifelong stockings after healing.

Q4 What must you never miss?

✓ Arterial disease before compressing, and a non-healing or atypical ulcer (malignancy, PG, vasculitis).

Q5 Name a red flag that needs escalation.

✓ **Low or abnormal ABPI**, absent pulses, rest pain or a cold foot: arterial disease, so **do not compress**. Refer to vascular.

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Photo gallery

Venous ulcers: gaiter area, shallow and sloping

3

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Venous stasis changes and lipodermatosclerosis

1

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Ulcers & red leg Module 5 · Week 2 · Module 1 · Week 5

Pyoderma gangrenosum

A rare, very painful, rapidly enlarging neutrophilic ulcer with a purple undermined edge. It is not an infection. Avoid debridement and refer urgently to dermatology.

ALSO CALLED PG | neutrophilic dermatosis | pathergy | violaceous undermined ulcer | IBD skin ulcer

RELATED PAGES Venous leg ulcer | Arterial (ischaemic) leg ulcer | Approach to cutaneous vasculitis | Erythema nodosum (incl. drug-induced) | Biologics, JAK inhibitors & conventional systemics

KEY FACTS

- **Pyoderma gangrenosum (PG)** is a rare **neutrophilic dermatosis**: a rapidly enlarging, very painful ulcer with a **violaceous, undermined edge**.
- Not an infection and not gangrene; often starts as a pustule after minor trauma (**pathergy**).
- About half have an associated disease: **IBD**, inflammatory arthritis, haematological disorders (MGUS, myeloma, leukaemia).
- Diagnosis of exclusion: biopsy the edge, swab, exclude vasculitis, infection and malignancy.
- Avoid debridement and surgery; urgent dermatology for potent topical/systemic steroids or ciclosporin, biologics.

Overview

Pyoderma gangrenosum (PG) is a rare **neutrophilic**, autoinflammatory ulcerating skin disease. Despite the name, it is **not an infection**. It causes a rapidly enlarging, very painful ulcer with a **violaceous, undermined** border, and shows **pathergy**: it worsens after trauma or surgery.

Epidemiology

- **Who**
  - Rare (about 1 in 100,000 per year); adults 20–50; slightly more women.

Aetiology: causes and risk factors

Neutrophil dysfunction and autoinflammation. About **half are idiopathic**. Associations include:

- Inflammatory bowel disease (ulcerative colitis more than Crohn's)
- Rheumatoid arthritis
- Haematological malignancy and monoclonal gammopathy

- Behçet disease and PAPA syndrome
- Drugs, e.g. cocaine contaminated with levamisole

Myths to address

- **"DEBRIDE IT LIKE ANY ULCER"**
  surgery worsens PG through pathergy.

Pathophysiology and pathology

PG is driven by **overactive neutrophils** in otherwise sterile tissue.

1. **Immune dysregulation**
   Innate immune overactivity (IL-1, IL-8, IL-17, TNF-α) recruits neutrophils.
2. **Pathergy**
   Minor trauma triggers an exaggerated neutrophilic response at the site.
3. **Destructive ulcer**
   Neutrophils destroy dermis and fat, undermining the edge.

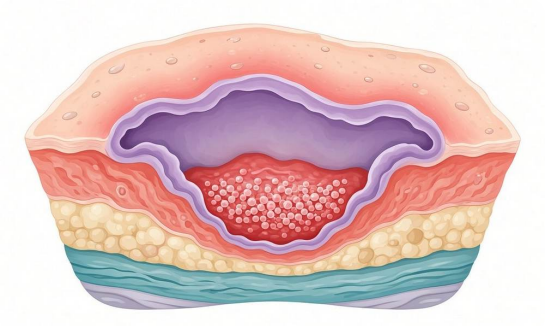

Fig. 1 · Pyoderma gangrenosum: a deep ulcer packed with neutrophils, with an overhanging violaceous (undermined) edge.
AI-generated illustration

4. **Cribriform healing**
   Ulcers heal with thin, sieve-like (cribriform) scars.

Mechanism, pathergy and associations in more depth

> **NEUTROPHIL DYSREGULATION AND AUTOINFLAMMATION**
> overexpression of IL-1β, IL-8, IL-17, IL-23 and TNF-α drives neutrophils into the dermis, which then destroy tissue. Innate-immune gene variants (e.g. PSTPIP1 in **PAPA** syndrome) show the mechanism.

> **PATHERGY**
> minor trauma (a graze, a needle, surgery, a stoma, debridement) triggers or worsens lesions in ~25–30%. **The case: a car-door graze pustule within days painful ulcer.**

> **Associations to screen for:**

GROUP	EXAMPLES
Gut	**Crohn's disease, ulcerative colitis** (PG activity often, but not always, parallels bowel activity); peristomal PG after IBD surgery

GROUP	EXAMPLES
Joints	**Rheumatoid arthritis**, seronegative spondyloarthritis, psoriatic arthritis
Blood	**Myeloid dyscrasias**: AML, **myelodysplasia**, myeloproliferative neoplasms; **monoclonal gammopathy (especially IgA MGUS)**, myeloma
Other	Hidradenitis suppurativa (PASH/PAPASH syndromes), Behçet's, granulomatosis with polyangiitis, autoimmune hepatitis, solid tumours (rare)
Drugs	**Levamisole-adulterated cocaine**, isotretinoin, propylthiouracil, sunitinib and other kinase inhibitors, rituximab, G-CSF

- Azathioprine (the patient's Crohn's treatment) does **not** protect against PG; immunosuppression also raises the chance that an ulcer is **infected or atypical infection**, which must be excluded.

HISTOPATHOLOGY

- Dense neutrophilic infiltrate without vasculitis or organisms; findings are supportive, not specific (Delphi/PARACELSUS criteria).

Clinical features

- Starts as a pustule, papule or haemorrhagic blister, often after a **minor injury**, then becomes a deep ulcer within days.
- Classic **purple, overhanging (undermined) edge** with a necrotic or purulent base.
- **Severe pain**, out of proportion to the size.
- Legs most often; also around stomas in IBD, and at surgical sites (pathergy).
- Heals with **cribriform** (sieve-like) scars.

Variants and clinical clues (Module 1 Week 5)

CLASSIC ULCERATIVE PG

starts as a **tender papule, pustule or haemorrhagic bulla**, often at a site of minor injury, which **breaks down within days** into an ulcer with: a **violaceous (purple-blue), undermined, ragged edge** and surrounding erythema; a **necrotic, purulent or haemorrhagic base**; **pain out of proportion** to the size (often 7/10); **rapid centrifugal growth** (centimetres in days to weeks)

SITES

pretibial skin (commonest); also trunk, head and neck, **peristomal skin**, post-surgical wounds (breast surgery, caesarean section), genitals; rarely mucosa and internal organs (sterile neutrophilic abscesses in lung, spleen).

VARIANTS

Bullous — superficial haemorrhagic blisters on the arms or face; **strong link with myeloid leukaemia/MDS** (check the blood film); **Pustular** — sterile pustules in active IBD; **Vegetative (superficial granulomatous)** — slow, less painful, rarely associated with systemic disease; **Peristomal** — around an ileostomy/colostomy; often misread as an irritant or allergic stoma reaction; **Postoperative** — wound "dehiscence" or "infection" that fails to respond to antibiotics and worsens with debridement

Healing leaves **atrophic cribriform (sieve-like) scars** — a useful clue to past PG.

> **SYSTEMIC**
> low-grade fever, malaise, arthralgia and raised CRP/ESR are common and **do not by themselves mean infection**.
> **The case: T 37.5 °C, HR 95, high CRP and ESR.**

Fig. 2 · PG — violaceous undermined border; painful; urgent dermatology.
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Special groups

- **Peristomal PG**
  - Around stomas in IBD; involve the stoma team.

Diagnosis and investigations

- Clinical, after **excluding** infection, vasculitis and malignancy. Biopsy of the edge supports the diagnosis (neutrophilic infiltrate; Delphi/Maverakis criteria).
- Swabs only to look for secondary infection.
- **Seek dermatology urgently.**

Diagnostic criteria (Su, Delphi, PARACELSUS) and the work-up

There is no single diagnostic test. Diagnosis combines the clinical picture, **exclusion of mimics**, histology and the treatment response.

- **Su criteria (2004)** — the course answer: **both major** criteria (rapidly progressive painful necrolytic ulcer with an irregular violaceous undermined border; other causes excluded) **plus** **2 minor** (pathergy or cribriform scarring; associated systemic disease; compatible histology; rapid response to systemic steroids).
- **Delphi consensus criteria (Maverakis et al., JAMA Dermatology 2018)** — now widely used in UK specialist practice: **1 major criterion — biopsy of the ulcer edge showing a neutrophilic infiltrate** — plus **4 of 8 minor criteria**: exclusion of infection (including histology stains and tissue cultures); pathergy; personal history of IBD or inflammatory arthritis; papule, pustule or vesicle that ulcerates within 4 days; peripheral erythema, undermining border and tenderness at the ulcer site; multiple ulcers, at least one on the anterior lower leg; cribriform or "wrinkled paper" scars at healed sites; ulcer shrinking within 1 month of starting immunosuppression

- **PARACELSUS score (2019)** — a weighted score from German wound clinics (rapid progression, assessment of differentials, violaceous border as the main items); a total of **10 points** suggests PG. Useful as a teaching checklist.
- **BIOPSY (SPECIALIST)**
  a **deep incisional or punch biopsy from the inflamed edge** (including subcutis) for **H&E** plus **tissue culture (bacteria, mycobacteria, fungi)** and special stains. Histology shows a **sterile dermal neutrophilic infiltrate** — supportive but non-specific. Its main job is to **exclude the mimics**. Pathergy makes biopsy a small risk; dermatology usually does it with steroid cover in mind.
- **PRIMARY-CARE WORK-UP WHILE REFERRING**
  Swab the ulcer (bacterial culture) — a positive swab often reflects colonisation, not the cause; **FBC and blood film** (myeloid disorders; **azathioprine-induced cytopenias**), **U&E, LFT, CRP/ESR**; **serum protein electrophoresis and serum free light chains** (paraprotein); **ANCA, ANA, rheumatoid factor/anti-CCP, antiphospholipid antibodies, cryoglobulins** (vasculitis and vasculopathy mimics), hepatitis B/C and HIV serology as indicated; **coagulation screen**; HbA1c; **ABPI** (vascular contribution and before any compression); **faecal calprotectin** and bowel history if IBD is not yet known; photograph and **measure the ulcer** (length × width, edge marked) to document speed of spread

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Infection — S. aureus/ streptococcal ecthyma, necrotising soft-tissue infection, atypical mycobacteria (M. marinum, M. chelonae, M. ulcerans), deep fungal infection, herpes simplex (immunosuppressed), leishmaniasis, synergistic gangrene	Responds to the right antimicrobial; tissue culture/PCR positive; atypical mycobacterial ulcers usually evolve **over weeks to months**; on azathioprine, infection must be excluded first
	Martorell (hypertensive ischaemic) ulcer	Older patient with **long-standing hypertension ± diabetes**; very painful necrotic ulcer on the **lateral or posterior lower leg / Achilles area**; subcutaneous arteriolosclerosis on deep biopsy. **Debridement and skin grafting help Martorell but worsen PG** — so the distinction matters
	Vasculitis and vasculopathy — GPA, PAN, cryoglobulinaemia, rheumatoid vasculitis, antiphospholipid syndrome, livedoid vasculopathy, calciphylaxis	Palpable purpura, livedo racemosa, nodules, renal involvement; ANCA, cryoglobulins, APL antibodies; calciphylaxis in renal failure
	Venous, arterial and neuropathic ulcers	Site, ABPI, varicose veins and haemosiderin, neuropathy; less painful and slower (venous/neuropathic)
	Malignancy — SCC (including Marjolin ulcer in a chronic wound), BCC, lymphoma	Rolled or heaped edge, slower growth, no inflammatory violaceous border; biopsy

	CONDITION	DISTINGUISHING FEATURES
	Other	Factitious ulcers (bizarre shapes, inaccessible sites spared), spider bite (loxoscelism — not UK), drug-induced (hydroxycarbamide leg ulcers, levamisole-cocaine vasculopathy), Sweet syndrome (plaques, not ulcers), Behçet's

Misdiagnosis is common. Consider venous or arterial ulcers, vasculitis, antiphospholipid syndrome ulcers, infection (ecthyma, atypical mycobacteria, deep fungal), factitious ulcers and ulcerated tumours.

Management

REFER

Recognise and refer: do not debride

Do not debride or excise active PG, because pathergy makes it worse. Do not assume antibiotics will cure it. Refer **urgently** to dermatology.

Specialist treatment (awareness, Module 5)

- **Oral corticosteroids** first line for rapidly progressive disease. **Ciclosporin** is an alternative first-line option.
- Later lines: dapsone, tetracyclines, and biologics (e.g. **infliximab**, particularly with IBD).
- Potent topical or intralesional steroid, or topical tacrolimus, for small or early lesions.
- Gentle, non-adherent wound care, good **analgesia**, and treatment of the associated disease (e.g. IBD).

Steroid safety: the IP role

- Monitor **blood glucose and blood pressure**.
- BONE HEALTH
  the Module 5 notes flag prednisolone **above 7.5 mg daily for 6 months or more** as an osteoporosis risk needing bone assessment (DEXA). UK guidance (NOGG) advises assessing bone health in **anyone expected to take oral steroids for 3 months or more**, so follow local policy and consider bone protection early.
- Counsel on steroid cards, infection risk and not stopping abruptly.

UK management in more depth: STOP GAP, IBD and medicines safety

There is no NICE guideline or BAD guideline specific to PG; UK practice follows the **UK STOP GAP trials**, BAD and PCDS clinical guidance, the BSG IBD guideline (2019) and specialist expert consensus. **PG is specialist-managed; the IP role is recognition, avoiding harm, symptom control and medicines safety.**

1. Recognise and refer. **Urgent dermatology referral** (same week; **same day** if rapidly spreading, very painful, multiple ulcers, peristomal, or systemically unwell). **Tell the IBD/rheumatology team** already looking after the patient; **Admit / same-day medical review** if signs of sepsis (NEWS2), necrotising infection cannot be excluded, or there is severe pain or rapid spread
2. Do no harm. **Do not debride, incise or excise** active PG, and avoid sharp debridement by wound clinics — **pathergy enlarges the ulcer**; Avoid repeated courses of antibiotics for a "non-healing infection" without a

diagnosis — escalate instead. Antibiotics are appropriate only for **proven secondary infection** (e.g. cellulitis spreading from the ulcer); Compression only after an ABPI and specialist advice (painful, and trauma can provoke pathergy)

3. Wound care and pain. Moist, **non-adherent** dressings (e.g. silicone or paraffin gauze, hydrogel or foam); antimicrobial dressings (silver, iodine) if heavily colonised; protect the surrounding skin. Involve the **tissue viability / district nursing** team; **Analgesia is a priority:** paracetamol ± weak or strong opioids (PG pain is often severe); **avoid NSAIDs in IBD** where possible (flare risk); review at dressing changes. Peristomal PG: **stoma nurse** for appliance fit
4. **Topical and local treatment** (small or early lesions; specialist advice) — **UK STOP GAP topical cohort (Thomas et al., J Am Acad Dermatol 2016; a prospective cohort, not an RCT)**: **very potent topical corticosteroid** (clobetasol propionate 0.05% ointment) to the ulcer edge/base, or; **topical tacrolimus** (0.03% or 0.1% ointment — off-label); overall **about 44% healed on topical treatment alone by 6 months** (median 145 days); **smaller ulcers healed faster**; **intralesional triamcinolone** into the active edge (specialist)
5. **Systemic treatment (specialist-initiated)** — **UK STOP GAP RCT (Ormerod et al., BMJ 2015)**: **Prednisolone 0.75 mg/kg/day** (maximum 75 mg) **or ciclosporin 4 mg/kg/day** (maximum 400 mg) gave **similar healing (about half healed by 6 months)**; serious adverse effects (especially infections) were somewhat commoner with prednisolone. Choice depends on comorbidity (diabetes, osteoporosis, hypertension, renal function, infection risk); **With IBD: anti-TNF therapy — infliximab** (the only biologic with RCT evidence in PG; BSG 2019 and ECCO) **or adalimumab** — is often preferred, treating gut and skin together. Ustekinumab, JAK inhibitors and IL-1/IL-23 blockers are emerging options (off-label); Others/steroid-sparing: **dapsone** (G6PD first), **doxycycline or minocycline** (mild disease, anti-neutrophil effect), mycophenolate, methotrexate, IVIG, tacrolimus (oral); **Treat the underlying disease** (IBD, arthritis, haematological disorder) and **stop a culprit drug** (e.g. cocaine; review isotretinoin)
6. Pharmacist medicines-safety checklist **Steroids:** blood glucose and BP, mood/sleep; **bone protection** (FRAX; calcium/vitamin D ± bisphosphonate for those on 3 months of prednisolone at higher risk — NICE CKS Corticosteroids – oral; Royal Osteoporosis Society 2022); consider gastroprotection; **Steroid Emergency Card** and sick-day rules when on prednisolone **5 mg/day for 4 weeks** (NatPSA 2020); never stop abruptly; **Ciclosporin:** baseline and regular **BP, creatinine/eGFR (stop or reduce if creatinine rises > 25–30% above baseline), potassium, magnesium, LFT, lipids**; **major CYP3A4/P-gp interactions** (clarithromycin, azole antifungals, diltiazem, grapefruit, St John's wort, rifampicin, carbamazepine); avoid NSAIDs/nephrotoxins; sun protection; **Azathioprine** (already prescribed): TPMT checked at baseline; shared-care FBC/LFT monitoring; **allopurinol/febuxostat interaction** (dangerous myelosuppression); any new infection or cytopenia changes the picture; **Before biologics:** TB screen (IGRA ± CXR), hepatitis B/C, HIV, VZV status; inactivated vaccines (influenza, pneumococcal, COVID-19, Shingrix where eligible); **avoid live vaccines**; Counsel about **pathergy**: protect the legs, avoid injections into affected areas, and plan elective surgery (including stoma surgery) with dermatology input and the disease quiet

IP scope

Recognition and urgent referral; ulcer measurement and photography; swabs and the baseline bloods above; wound-care and dressing advice with the nursing team; analgesia; stopping or questioning antibiotic cycling; medicines reconciliation, interaction checks and shared-care monitoring of steroids, ciclosporin, azathioprine and biologics; steroid card and bone protection; vaccination advice. Diagnosis (biopsy) and systemic or biologic treatment are specialist.

Worked case: 38-year-old woman with Crohn's on azathioprine

- **QA = 4, pyoderma gangrenosum.** A minor graze (pathergy) pustule within days **rapidly enlarging, very painful ulcer** in a woman with **Crohn's disease**, low-grade fever and raised CRP/ESR: 1. Atypical mycobacterial infection: evolves over weeks to months; it should still be excluded (tissue culture) because she is on azathioprine; 2. S. aureus infection: would not usually produce a violaceous undermined ulcer this painful and fast, and would respond to antibiotics; swab anyway; 3. Martorell ulcer: an ischaemic ulcer of long-standing **hypertension ± diabetes**, usually older patients, lateral/posterior leg; her BP is 125/80; 5. SCC: would not reach this size in 4 weeks, is rarely this painful and lacks an inflammatory violaceous border
- **QB = 2, diagnosis based on clinical and laboratory criteria.** There is no confirmatory test; PG is a diagnosis of exclusion using the **Su criteria** (the course answer); current UK specialist practice uses the **Delphi consensus criteria**: 4/5. Skin biopsy for histology (and tissue culture): an essential part of the work-up — mainly to **exclude infection, vasculitis and malignancy** — but histology is non-specific, so biopsy alone does not confirm PG (and a biopsy of the edge is the major Delphi criterion; see the course-vs-UK table); 1. ABPI: assesses arterial disease and safety of compression; no diagnostic role in PG; 3. Nerve conduction studies/EMG: for neuropathic ulcers
- **QC = 1, myeloid blood dyscrasias** (AML, myelodysplasia, myeloproliferative disorders; also IgA monoclonal gammopathy/myeloma). Other well-described associations are **rheumatoid and other inflammatory arthritis** and **IBD**. Eosinophilic asthma, penicillin antibiotics, pregnancy and ischaemic heart disease are not recognised associations.
- **UK ADDITIONS FOR THIS PATIENT**
  exclude infection first (swab, FBC — azathioprine can cause neutropenia — CRP, tissue culture via biopsy); **urgent dermatology referral and tell her IBD team**; no debridement; analgesia (avoid NSAIDs in Crohn's); non-adherent dressings; check her Crohn's activity. Expect **prednisolone or ciclosporin** (STOP GAP), given active IBD, an **anti-TNF (infliximab/adalimumab)**; steroid safety, interaction and vaccination checks.

Referral

ROUTINE	A painful ulcer with a **purple undermined edge** that expands over days.
ROUTINE	An ulcer getting worse after debridement or surgery (pathergy).
ROUTINE	Peristomal ulcer in IBD.
ROUTINE	Systemic upset, or rapid spread despite antibiotics.

Red flags and when to refer

- A painful ulcer with a **purple undermined edge** that expands over days.
- An ulcer getting worse after debridement or surgery (pathergy).
- Peristomal ulcer in IBD.

- Systemic upset, or rapid spread despite antibiotics.

Red flags / refer

Rapidly enlarging, extremely painful ulcer with a violaceous undermined edge (urgent dermatology); fever with spreading cellulitis, crepitus, pain out of proportion or systemic toxicity (exclude **necrotising fasciitis** — same-day/999); an ulcer worsening after debridement or surgery (pathergy); bullous or haemorrhagic PG or abnormal blood film (**urgent haematology** — myeloid malignancy); peristomal ulcer in IBD; new PG with bowel symptoms (undiagnosed IBD — urgent GI referral); a patient on immunosuppressants with a new ulcer (infection first).

Complications and prognosis

Complications

- Severe pain, infection, scarring; recurrence; complications of immunosuppression.

Prognosis

- Healing takes months; relapses common.

Prescriber pearls (UK pharmacist IP)

Pathergy counselling: protect the skin, and plan procedures only when the disease is quiet, with specialist input.

In the community, escalate early. Do not trial antibiotic after antibiotic.

Ask about bowel symptoms and joint pain. PG can be the first sign of IBD.

"**Graze pustule fast, agonising, purple-edged ulcer in someone with IBD = PG until proved otherwise — and never debride it.**"

PG is a diagnosis of exclusion: **infection first** (especially on azathioprine or a biologic), then vasculitis, vascular disease and cancer.

Pain out of proportion and an ulcer that **gets bigger after debridement** are the two most useful bedside clues.

Cribriform scars on the legs at a later consultation = past PG: ask about the bowel, joints and blood disorders.

Steroids and ciclosporin heal PG equally (STOP GAP): the choice is a **medicines-safety decision** — a natural place for pharmacist input.

Pyoderma gangrenosum is one of the skin signs of systemic disease (IBD, arthritis, blood disorders): see also skin signs of systemic disease.

Self-check quiz

5 questions · optional

Q1 When should you think of Pyoderma gangrenosum?

✓ A pustule or small wound that rapidly becomes an exquisitely painful ulcer with a violaceous, overhanging edge, often in a patient with IBD or rheumatoid arthritis.

Q2 What drives it?

✓ Neutrophilic autoinflammation, not infection. Trauma makes it worse (pathergy).

Q3 What is the backbone of care?

✓ Recognise it and refer urgently. Specialist treatment with oral steroids or ciclosporin. No wide debridement.

Q4 What must you never miss?

✓ Misdiagnosis as an infected ulcer, with repeated antibiotics and surgical debridement that make it worse.

Q5 Name a red flag that needs escalation.

✓ A painful ulcer with a **purple undermined edge** that expands over days.

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Photo gallery

Pyoderma gangrenosum: violaceous undermined edge

4

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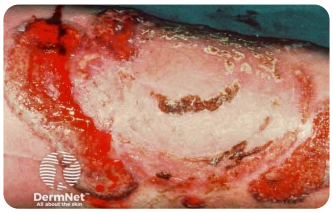

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Multiple ulcers, pathergy after surgery and cribriform scarring

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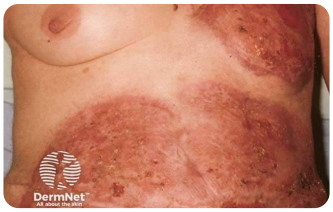

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Ulcers & red leg Module 5 · Week 2

Arterial (ischaemic) leg ulcer

Punched-out, painful, distal ulcers from peripheral arterial disease. Never compress an ischaemic limb; refer urgently to vascular, same day for critical limb ischaemia.

ALSO CALLED arterial ulcer · ischaemic ulcer · peripheral arterial disease · PAD · critical limb ischaemia

RELATED PAGES Neuropathic and diabetic foot ulcers · Venous leg ulcer · Pyoderma gangrenosum · Red leg: differentials & referral

KEY FACTS

- **Arterial ulcers** are caused by **peripheral arterial disease** and account for about 10–20% of leg ulcers (alone or mixed).
- Punched-out, deep, painful ulcers on the toes, heel, lateral malleolus or shin; cold, hairless foot with weak pulses.
- Pain worse on elevation and at night, eased by hanging the leg down.
- Measure ABPI (toe pressures in diabetes); **never compress** an ischaemic limb.
- Urgent vascular referral; same day for critical limb ischaemia. Stop smoking; antiplatelet and statin.

Overview

An **arterial ulcer** is skin loss caused by tissue **ischaemia** from peripheral arterial disease. It accounts for about 10–20% of leg ulcers, alone or mixed with venous disease.

Epidemiology

- **Who**
  - Older smokers, people with diabetes, CKD, hypertension and hyperlipidaemia.

Aetiology: causes and risk factors

Atherosclerosis (or embolism) reduces flow, so poorly perfused sites become necrotic after minor trauma. Risk factors: smoking, diabetes, hypertension, hyperlipidaemia, CKD and age.

Myths to address

- **"ALL LEG ULCERS NEED COMPRESSION"**
  compression can cause gangrene in arterial disease.

Pathophysiology and pathology

Too little arterial blood reaches the skin, so it **dies from ischaemia**.

1. **Atherosclerosis**
   Plaques narrow the arteries of the leg.
2. **Reduced perfusion**
   Distal skin receives too little oxygen, especially at pressure points and on elevation.
3. **Necrosis**
   Minor trauma causes tissue death that cannot heal: a punched-out ulcer.
4. **Critical ischaemia**
   Rest pain, gangrene and tissue loss signal limb-threatening disease.

HISTOPATHOLOGY

- Not used; diagnosis is clinical and vascular (ABPI, duplex, angiography).

Clinical features

- Site: toes, heel, bony prominences, lateral malleolus, shin.
- **Punched-out**, deep, dry or necrotic base.
- Pain often **severe**, worse on elevation and at night (eased by hanging the leg out of bed).
- Foot: cold, pale or mottled, shiny, hairless, weak or absent pulses; dependent rubor; **claudication**.

Special groups

- **Mixed ulcers**
  - ABPI 0.5–0.8: reduced compression only under specialist guidance; below 0.5: no compression, urgent vascular.

Diagnosis and investigations

- Examine pulses and measure **ABPI** (or TBPI in diabetes, where calcified vessels give falsely high ABPI).
- Urgent vascular assessment (duplex, angiography) for critical limb ischaemia.

Differential diagnoses

Venous leg ulcer (gaiter area, shallow, exudative), diabetic foot ulcer, pressure ulcer, pyoderma gangrenosum, vasculitis, malignancy.

Management

- **Do not compress** ischaemic limbs.

- Urgent **vascular referral**, especially with rest pain, gangrene or critical ischaemia.
- Risk factors: stop smoking, antiplatelet and statin therapy (NICE CG147 and vascular advice); optimise blood pressure and diabetes.
- Analgesia, and wound care directed by tissue viability or vascular.

Treatment ladder

1 **Risk factors**

Stop smoking; antiplatelet and statin per NICE CG147; optimise BP and diabetes.

2 **Wound care**

Analgesia; dry, protective dressings guided by tissue viability.

3 **Vascular**

Urgent vascular assessment for revascularisation.

⚠ CAUTION

Prescribing limits and pitfalls

- **Compression**: Contraindicated in significant arterial disease (low ABPI).

Referral

SAME DAY	Rest pain, gangrene, tissue loss or a cold pulseless foot.
URGENT	Any suspected arterial ulcer: vascular surgery.

Red flags and when to refer

- CRITICAL LIMB ISCHAEMIA
  rest pain, gangrene, tissue loss, or a cold pulseless foot: **same-day vascular**.
- Acute limb ischaemia (pain, pallor, pulselessness, paraesthesia, paralysis, perishingly cold): emergency.

Complications and prognosis

Complications

- Infection, gangrene, amputation; high cardiovascular mortality.

Prognosis

- Healing depends on revascularisation.

Prescriber pearls (UK pharmacist IP)

Any leg ulcer may be mixed. Always assess the arterial supply before compression.

Painful ulcer worse on elevation suggests arterial.

Self-check quiz

4 questions · optional

Q1 Where do arterial ulcers occur?

✓ Toes, heel, bony prominences, lateral malleolus.

Q2 Why can ABPI mislead in diabetes?

✓ Calcified vessels give falsely high readings; use toe pressures (TBPI).

Q3 What eases ischaemic rest pain?

✓ Hanging the leg down (dependency).

Q4 What must you never do?

✓ Apply compression to an ischaemic limb.

References

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Photo gallery

Ischaemic ulcers

1

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Ulcers & red leg Module 5 · Week 2

Neuropathic and diabetic foot ulcers

Painless ulcers on plantar pressure points surrounded by callus, from loss of protective sensation. Any diabetic foot ulcer needs referral to the foot care service within 1 working day (NICE NG19).

ALSO CALLED neuropathic ulcer · diabetic foot ulcer · diabetic foot · NG19 · osteomyelitis · Charcot foot · +1

RELATED PAGES Arterial (ischaemic) leg ulcer · Venous leg ulcer · Cellulitis & erysipelas · Leprosy (Hansen disease) & lepra reactions

KEY FACTS

- **Neuropathic (diabetic foot) ulcers** form on insensate pressure points: plantar metatarsal heads, heel and toe tips.
- Painless, surrounded by **callus**, with warm feet and good pulses unless neuro-ischaemic.
- Test with a **10 g monofilament**; probe to bone and X-ray/MRI for osteomyelitis.
- Any diabetic foot ulcer: refer to the multidisciplinary foot service **within 1 working day** (NICE NG19).
- Same-day admission for spreading infection, sepsis, crepitus, wet gangrene or suspected Charcot foot.

Overview

A **neuropathic ulcer** follows loss of **protective sensation**, with or without deformity and ischaemia ('neuro-ischaemic' foot). Most are **diabetic foot ulcers**, a leading cause of amputation.

Epidemiology

- **Who**
  - Up to a quarter of people with diabetes develop a foot ulcer in their lifetime; most amputations in diabetes follow an ulcer.

Aetiology: causes and risk factors

Peripheral neuropathy (usually diabetes; also alcohol, B12 deficiency, leprosy) means repetitive trauma goes unnoticed. Deep ulcers can reach bone, with a risk of **osteomyelitis**.

Myths to address

- **"IT DOESN'T HURT, SO IT'S NOT SERIOUS"**
  absence of pain is the problem.

Pathophysiology and pathology

Without **protective sensation**, repeated pressure goes unnoticed and breaks down the skin.

1. **Neuropathy**

   Sensory loss removes pain warning; motor neuropathy causes clawing and abnormal pressure; autonomic neuropathy dries and cracks skin.

2. **Callus**

   High plantar pressure builds thick callus, which raises pressure further.

3. **Ulceration**

   Tissue breaks down beneath the callus and the ulcer deepens towards bone.

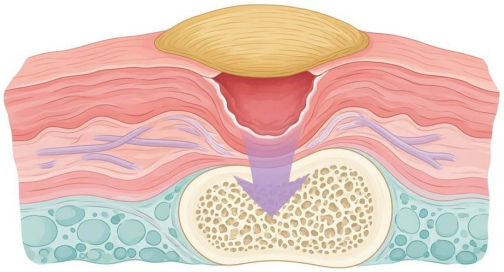

Fig. 1 · Neuropathic plantar ulcer: callus over a deep ulcer above the metatarsal head, with damaged nerve fibres and concentrated pressure.

AI-generated illustration

4. **Infection and ischaemia**

   Poor perfusion and impaired immunity let infection spread to bone (osteomyelitis).

> **HISTOPATHOLOGY**
>
> - Bone biopsy or deep tissue samples guide treatment of osteomyelitis.

Clinical features

- Site: **plantar** metatarsal heads, heel, tips of toes, under bunions.
- Deep, punched-out, surrounded by **callus**; may reach bone.
- Often **painless**; warm foot with good pulses (unless neuro-ischaemic).
- Clawed toes, Charcot deformity, dry fissured skin.

Special groups

- **Charcot foot**
  - Hot, swollen, red foot in neuropathy, often painless: immobilise and refer the same day.

Diagnosis and investigations

- Test sensation with a **10 g monofilament** and vibration.

- Pulses and ABPI/TBPI (neuro-ischaemic foot).
- **Probe to bone**, plus X-ray or MRI if osteomyelitis is suspected.
- Deep tissue samples rather than surface swabs if infected.

Differential diagnoses

Arterial ulcer, pressure ulcer (immobile or paraplegic patients), venous ulcer, malignancy (non-healing).

Management

- **ANY DIABETIC FOOT ULCER**
  refer urgently to the **multidisciplinary foot care service** or foot protection service (within 1 working day, NICE NG19).
- **Offloading**, glycaemic optimisation, debridement, and assessment for infection, which may need admission and IV antibiotics.
- Diabetic foot infection: follow NICE NG19 and local antimicrobial guidance.

Treatment ladder

1 **Prevention**

Daily foot checks, footwear, podiatry; glycaemic control.

2 **Any ulcer**

Multidisciplinary foot care service within 1 working day; offloading, debridement.

3 **Infected**

Antibiotics per NICE NG19 and local guidance; admit if severe.

Referral

SAME DAY	Spreading infection, fever, crepitus, wet necrosis or suspected Charcot foot.
URGENT	Any diabetic foot ulcer (within 1 working day).

Red flags and when to refer

- Diabetic foot with **spreading redness, fever, crepitus or wet necrosis**: **ED / same-day admission**.

- Probe to bone, or suspected osteomyelitis or **Charcot foot** (hot, swollen foot in neuropathy).
- **Never ignore a 'small' plantar ulcer in diabetes.**

Complications and prognosis

Complications

- Osteomyelitis, sepsis, amputation; high mortality.

Prognosis

- Many heal with offloading; recurrence is common, so lifelong foot protection is needed.

Prescriber pearls (UK pharmacist IP)

Teach **daily foot checks**, sensible footwear and podiatry follow-up; never walk barefoot with neuropathy.

Painless ulcer under a callus suggests neuropathic.

Self-check quiz

4 questions · optional

Q1 How is protective sensation tested?

✓ 10 g monofilament (plus vibration).

Q2 What does a positive probe-to-bone test suggest?

✓ Osteomyelitis.

Q3 How quickly should a diabetic foot ulcer be referred?

✓ Within 1 working day to the foot care service (NG19).

Q4 What is a hot swollen foot in neuropathy until proven otherwise?

✓ Charcot neuroarthropathy.

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Photo gallery

Neuropathic plantar / pressure ulcers

4

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Ulcers & red leg Module 5 · Week 2

Red leg: differentials & referral

A structured approach to the red leg: is it really cellulitis? Bilateral, chronic, itchy redness is usually venous. Plus a one-table guide to when to refer urgently.

ALSO CALLED red leg · bilateral cellulitis · pseudocellulitis · cellulitis mimics · stasis dermatitis · lipodermatosclerosis · +2

RELATED PAGES Cellulitis & erysipelas · Venous leg ulcer · Contact dermatitis · Erythema nodosum (incl. drug-induced) · Eczema / dermatitis

KEY FACTS

- The **red leg** is a common diagnostic trap: most bilateral red legs are **not cellulitis**.
- Cellulitis is usually **unilateral**, acute, hot, tender, spreading, with fever or raised inflammatory markers.
- Mimics: venous (stasis) eczema, lipodermatosclerosis, contact dermatitis, DVT, haematoma, gout, erythema nodosum, vasculitis.
- Bilateral red legs: treat the cause (compression, emollient, steroid for eczema); avoid unnecessary antibiotics.
- Same-day: sepsis, rapidly spreading, severe pain out of proportion (necrotising fasciitis), suspected DVT.

Overview

'Red leg' is a presentation, not a diagnosis. Up to a third of patients admitted with 'cellulitis' turn out to have a **mimic** (pseudocellulitis). This page brings together the Module 5 red-leg teaching and the referral triggers for leg ulcers.

Epidemiology

- **Who**
  - Up to a third of patients admitted with 'cellulitis' have a mimic, often stasis dermatitis.

Aetiology: causes and risk factors

CLUE	POINTS TOWARDS
Unilateral, acute, fever or malaise, spreading	Cellulitis or erysipelas
Bilateral, chronic, itchy, scaly	Venous (stasis) eczema
Painful, woody, red gaiter skin, often bilateral	Acute lipodermatosclerosis

CLUE	POINTS TOWARDS
Calf swelling and tenderness, less skin redness	DVT
Joint-centred, exquisitely tender	Gout or septic arthritis
Long history of swelling	Lymphoedema (with or without a flare)
Pain out of proportion, rapid spread, toxic	**Necrotising fasciitis**

Myths to address

- **"BOTH LEGS RED MEANS DOUBLE CELLULITIS"**
  bilateral cellulitis is rare.

Pathophysiology and pathology

Different processes make legs red: **infection, venous hypertension, allergy, thrombosis or inflammation**.

1. **Venous hypertension**
   Stasis causes oedema, inflammation and eczema (bilateral, itchy, chronic).
2. **Infection**
   Bacteria enter via breaks (tinea, ulcers) and spread through dermis and fat (unilateral, acute).
3. **Contact allergy**
   Dressings and topical medicines cause itchy, sharply bordered eczema.
4. **Thrombosis**
   DVT causes swollen, tender calf with warmth: Wells score and D-dimer.

Clinical features

Compare the photos below. Venous eczema is scaly and itchy on **both** legs. Lipodermatosclerosis is indurated and tapering. True cellulitis is hot, tender, **one-sided** and expanding.

Fig. 1 · Venous (stasis/gravitational) eczema: lower-leg eczema with haemosiderin staining over varicose veins — key leg-rash differential; treat venous hypertension (compression after ABPI).

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Special groups

- **Necrotising fasciitis**
  - Pain out of proportion, rapid spread, systemic toxicity, crepitus: emergency surgery.

Diagnosis and investigations

- Ask: **one leg or two?** Acute or chronic? Fever or rigors? Itch or pain? Is there a portal of entry?
- Mark the border and record observations (NEWS2).
- Wells score with D-dimer or Doppler if DVT is possible.
- Check pulses or ABPI if there is an ulcer, or before compression.

Key points

- Ask: **one leg or two?** Acute or chronic? Fever or rigors? Itch or pain? Is there a portal of entry?
- Mark the border and record observations (NEWS2).
- Wells score with D-dimer or Doppler if DVT is possible.
- Check pulses or ABPI if there is an ulcer, or before compression.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Venous eczema / stasis dermatitis	Bilateral or chronic, itchy, scale, haemosiderin, no fever
	Acute lipodermatosclerosis	Painful, woody, red skin above the ankle, often misdiagnosed as cellulitis

	CONDITION	DISTINGUISHING FEATURES
	DVT	Calf pain and swelling, less intense redness
	Lymphoedema flare	Chronic swelling history
	Gout / septic arthritis	Joint-centred
	Contact dermatitis (e.g. to dressings)	Itchy, vesicular, matches the area covered
	Necrotising fasciitis	**Severe pain out of proportion**, rapid progression, systemic toxicity. **ED**
	Pyoderma gangrenosum / vasculitis	Ulcer edge features, purpura, systemic disease

Management

Referral

ROUTINE	Pain out of proportion, purple bullae, crepitus, or a rapidly spreading area: **necrotising fasciitis**.
ROUTINE	Sepsis signs: low blood pressure, tachycardia, confusion, mottling.
ROUTINE	A cold, pulseless or ischaemic limb.
ROUTINE	A diabetic foot that is hot, red or ulcerated.

Red flags and when to refer

- Pain out of proportion, purple bullae, crepitus, or a rapidly spreading area: **necrotising fasciitis**.
- Sepsis signs: low blood pressure, tachycardia, confusion, mottling.
- A cold, pulseless or ischaemic limb.

- A diabetic foot that is hot, red or ulcerated.

Referral summary (Module 5)

SITUATION	ACTION
Suspected necrotising infection or sepsis	**Immediate ED (999)**
Facial or orbital cellulitis, or rapidly spreading	**Urgent, same day**
Critical limb ischaemia, rest pain or gangrene	**Urgent vascular**
Diabetic foot ulcer	**Urgent diabetic foot pathway (NG19)**
Suspected PG (painful violaceous undermined ulcer)	**Urgent dermatology**; avoid debridement
Leg ulcer not improving, or atypical edge	Tissue viability plus dermatology or vascular
Red leg with no fever, bilateral, chronic and itchy	Rethink: venous eczema. Emollient, topical steroid and compression, **no antibiotic**
Recurrent cellulitis	Treat portals of entry; discuss prophylaxis with a clinician

Cellulitis treatment itself: see Cellulitis & erysipelas.

Complications and prognosis

Complications

- Unnecessary antibiotics and admissions; missed DVT or necrotising infection.

Prognosis

- Depends on cause; stasis dermatitis is chronic and needs compression.

Prescriber pearls (UK pharmacist IP)

Bilateral ‘cellulitis’ is rarely infection.

Venous eczema responds to emollient plus a topical steroid and compression (after ABPI). Antibiotics do nothing for it.

Recurrent ‘cellulitis’ in the same leg may be lipodermatosclerosis. Look at the whole leg history.

Self-check quiz

5 questions · optional

Q1 When should you think of Red leg: differentials & referral?

✓ Any patient presenting with 'cellulitis', especially if both legs are involved, it is chronic or itchy, or there is no fever.

Q2 What drives it?

✓ Most bilateral red legs are venous (stasis dermatitis, lipodermatosclerosis) or oedema, not infection.

Q3 What is the backbone of care?

✓ Ask: one leg or two, acute or chronic, fever or not, itch or pain? Treat the actual cause.

Q4 What must you never miss?

✓ Necrotising fasciitis, sepsis, DVT, critical ischaemia and diabetic foot infection.

Q5 Name a red flag that needs escalation.

✓ Pain out of proportion, purple bullae, crepitus, or a rapidly spreading area: **necrotising fasciitis**.

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Photo gallery

True infection: cellulitis

2

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Mimics: venous eczema and lipodermatosclerosis

3

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CHAPTER 12

Photodermatoses

Rashes driven by light: polymorphic light eruption, chronic actinic dermatitis, drug-induced photosensitivity and porphyria cutanea tarda, with a sun-protection script.

1. Polymorphic light eruption
2. Chronic actinic dermatitis
3. Drug-induced photosensitivity
4. Porphyria cutanea tarda (& pseudoporphyria)

Photodermatoses Module 5 · Week 6

Polymorphic light eruption

The commonest photodermatosis: itchy papules or vesicles on sun-exposed skin hours after the first strong sun of spring or a holiday. Sun protection, short potent steroid bursts and, for severe cases, specialist 'hardening' phototherapy.

ALSO CALLED PMLE · PLE · sun allergy · prickly heat (lay term) · spring rash · holiday rash · +2

RELATED PAGES Drug-induced photosensitivity · Chronic actinic dermatitis · Discoid lupus erythematosus · Phytophotodermatitis · Urticaria & angioedema

KEY FACTS

- **Polymorphic light eruption (PLE)** is the commonest photodermatosis: itchy papules, vesicles or plaques hours to days after the first strong sun of spring or a holiday.
- Affects exposed skin that is usually covered in winter (V of neck, arms), sparing the face often; settles over days without sun.
- Tends to improve over the summer ('hardening').
- Prevention: sun protection, gradual exposure; potent topical steroid for 5–7 days for flares.
- Refer if severe for prophylactic phototherapy (hardening); exclude lupus (check ANA/Ro if suspicious).

Overview

The commonest **idiopathic (primary) photodermatosis**: a recurrent, itchy, non-scarring eruption on sun-exposed skin appearing hours to a couple of days after the first significant sun exposure of spring/summer or of a sunny holiday. Also called polymorphous light eruption or "prickly heat"/"sun allergy" by the public (not true prickly heat — that is miliaria). Affects roughly 10–20% of people in northern Europe; more common in women, onset usually in the teens to 30s; all skin types.

Epidemiology

- **Who**
  - Up to 1 in 5 people in northern Europe; women more; onset in the teens to 30s.

Aetiology: causes and risk factors

A delayed-type (cell-mediated) immune reaction to a UV-induced skin antigen. Normally UV exposure suppresses skin immunity; in PMLE that **UV-induced immunosuppression fails**, so the newly photo-induced antigen provokes inflammation. **UVA is the main trigger in most patients** (UVB alone in a minority, both in some) — which is why a

high-SPF but weak-UVA sunscreen fails (Case 1). Skin "hardens" with repeated exposure, so it is typically worse early in the season and improves through summer. Family history in a proportion.

Risk factors and associations

- Delayed-type hypersensitivity to a UV-induced skin antigen, with failure of normal UV immunosuppression.
- Mainly UVA; can occur through window glass.

Myths to address

- **"IT'S AN ALLERGY TO SUNSCREEN"**
  PLE is a reaction to light itself, although sunscreen allergy can occur.

Pathophysiology and pathology

Normally UV **dampens skin immunity**; in PLE this fails and the immune system attacks UV-altered skin.

1. **UV antigen**
   UV (mainly UVA) alters skin proteins, creating new antigens.
2. **Failed suppression**
   Normal UV-induced immunosuppression (Langerhans cell migration, regulatory T cells) does not occur.
3. **Delayed hypersensitivity**
   T cells react hours to days later, producing itchy papules and vesicles.
4. **Hardening**
   Repeated gradual exposure restores tolerance, so PLE eases as summer goes on.

HISTOPATHOLOGY

- Papillary dermal oedema with a perivascular lymphocytic infiltrate; spongiosis. Biopsy if lupus is suspected.

Clinical features

- **Itch or burning** often starting within hours (Case 1: 30 minutes); rash within hours to 1–3 days.
- Morphology varies between patients but is **usually consistent in one individual**: small papules (commonest), papulovesicles, plaques, targetoid/EM-like lesions, rarely purpuric.
- Sites: **V of neck, upper chest, outer arms/forearms, dorsal hands, legs**; face often relatively spared (habitually exposed, hardened).
- Resolves over days–2 weeks without scarring **if further sun is avoided**; can spread and persist with continued exposure (Case 1 extension to trunk).
- Variant: **juvenile spring eruption** (papulovesicles on helices of ears in boys).

Fig. 1 · PMLE — itchy papulovesicles on exposed forearm hours–days after first strong sun of holiday/spring.

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Special groups

- **Skin of colour**
  - Pinpoint papular variant common; may look skin-coloured or hypopigmented.

Diagnosis and investigations

Clinical: timing (first strong sun), photodistribution, recurrent seasonal pattern, normal health. In **severe, atypical or treatment-resistant** cases specialists consider: **ANA, anti-Ro/La antibodies** (exclude lupus — SCLE can be triggered by sun and by drugs), **porphyrin screen** (exclude erythropoietic protoporphyria/other porphyrias), skin biopsy (dermal oedema, perivascular lymphocytes — non-specific but helps exclude lupus), and **phototesting/ provocation testing** in a photobiology unit. Always take a full drug history to exclude drug-induced photosensitivity.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Sunburn	Uniform painful erythema; no papules; dose-related; within hours, peaks 24 h
	Drug-induced photosensitivity	New photosensitiser (doxycycline, thiazide, NSAID…); "exaggerated sunburn" or eczema; ask about drugs
	Cutaneous lupus (SCLE/DLE)	Persistent annular/psoriasiform or scarring plaques; ANA/Ro positive; lasts weeks–months
	Solar urticaria	Wheals within **minutes** of exposure, fade within hours
	Actinic prurigo	Childhood onset, excoriated papules/nodules incl. covered sites, cheilitis, persists into winter

	CONDITION	DISTINGUISHING FEATURES
	Erythema multiforme	Targets on palms/soles; HSV/drug trigger; not photodistributed
	Miliaria (true prickly heat)	Heat/sweat, covered/occluded sites, tiny vesicles
	Airborne contact dermatitis / CAD	Eczematous, chronic; shaded sites involved (airborne) or older man, lichenified (CAD)

Management

Self-care / OTC

- Avoid further sun until settled; cool showers/compresses, emollient.
- **Oral non-sedating antihistamine** for itch (helps itch, not the rash) — hydroxyzine in Case 1 had limited effect.
- OTC hydrocortisone 1% is usually too weak for established PMLE on limbs/trunk.

First-line Rx

- **Short course (5–7 days, up to ~2 weeks) of a potent topical corticosteroid** (e.g. betamethasone valerate 0.1%, mometasone 0.1%) to the trunk/limbs; avoid potent on face — use moderate for 5–7 days if needed.
- Severe extensive flares: a **short course of oral prednisolone** may be used (GP/specialist decision; e.g. a few days at the start of a holiday is a recognised specialist strategy for predictable attacks).

Prevention (the key intervention)

- Graded, gradual sun exposure in spring ("natural hardening").
- **Broad-spectrum SPF 30–50+ with high UVA protection (UK 4–5 star UVA or EU "UVA" circle logo)**, applied generously 15–30 min before going out and re-applied every 2 h and after swimming.
- Clothing, hat, shade, avoid 11:00–15:00; UVA passes through glass (car side windows, conservatory).
- Severe recurrent: **referral for prophylactic phototherapy ("hardening") with narrowband UVB or PUVA in spring**; hydroxychloroquine or immunosuppressants are rarely used (specialist).

Sun-protection counselling script

SPF is not the only measure of sunscreen quality — choose by UVA protection too.

PARAMETER	WHAT TO ADVISE
SPF (UVB/sunburn protection)	SPF **30 minimum; SPF 50+** for photodermatoses
UVA protection	**UK star rating 4–5 stars** (Boots scale) or **EU "UVA" circle logo** (UVA-PF ⅓ of SPF). Critical — UVA drives PMLE (~75% UVA-induced), drug phototoxicity, phytophotodermatitis, CAD
Visible light (PCT, some CAD/solar urticaria)	Chemical filters inadequate — **mineral (titanium dioxide/zinc oxide) tinted/iron-oxide sunscreens**; Dundee cream via specialists

PARAMETER	WHAT TO ADVISE
Broad spectrum / photostable	Look for "broad spectrum"; photostable filters
Water resistance	Water-resistant if swimming/sweating, but **re-apply after towelling**
Formulation/skin type	Fragrance-free and minimal-filter for CAD / photoallergy (avoid benzophenone/oxybenzone, octocrylene if allergic — especially with topical ketoprofen history); sprays/gels for hairy skin
Amount	~**2 mg/cm²** — about **6 teaspoons (~35 ml)** for an adult whole body; most people apply a third to half of this
Timing	Apply 15–30 min before going out; **re-apply every 2 hours** and after swimming/sweating
Shade/timing	Avoid **11:00–15:00** in UK summer (sun strongest); UV index 3 needs protection; cloud doesn't block UVA
Glass	**UVA passes through window glass** — car side windows, conservatories, office windows; UV-filter window film helps (PCT, CAD, drug photosensitivity)
Clothing	Tightly woven/UPF 50+ clothing, long sleeves, collars, **wide-brimmed hat**, UV-protective sunglasses, gloves for driving (PCT/CAD)
Sunbeds	**Avoid entirely** (UVA-rich; pseudoporphyria; skin cancer)
Vitamin D	Strict photoprotection **10 micrograms (400 units) colecalciferol daily** (UK advice), or check levels
Expiry/storage	Check expiry (usually 12 months after opening); store cool
Prescribing	Sunscreens are **ACBS-approved** on FP10 for genetic disorders/photodermatoses (e.g. PMLE, PCT, CAD, porphyrias) — endorse "ACBS"; otherwise self-care purchase

Treatment ladder

1

Self-care / OTC

Avoid further sun until settled; cool showers/compresses, emollient. **Oral non-sedating antihistamine** for itch (helps itch, not the rash) — hydroxyzine in Case 1 had limited effect.

2

First-line Rx

Short course (5–7 days, up to ~2 weeks) of a potent topical corticosteroid (e.g. betamethasone valerate 0.1%, mometasone 0.1%) to the trunk/limbs; avoid potent on face — use moderate for 5–7 days if needed. Severe extensive flares: a **short course of oral prednisolone** may be used (GP/specialist decision; e.g. a few days at the start of a holiday is a recognised specialist strategy for predictable attacks).

3

Red flags / refer

Diagnostic doubt (rule out lupus/porphyria), scarring, persistent >2–3 weeks without sun, systemic features, failure of potent topical steroid, severe recurrent PMLE affecting quality of life (phototherapy referral).

Red flags and when to refer

Photodermatoses: red flags and when to refer

SITUATION	ACTION
Widespread blistering, mucosal involvement, fever/ unwell after new drug	**Same-day/ED** — exclude SJS/TEN, severe drug reaction
Extensive phytophotodermatitis/hogweed burns, eye involvement, hand/face/genital blisters	Same-day urgent care / burns or eye service
Blisters + fragility + scars/milia on dorsal hands (± alcohol, hep C, oestrogen, iron)	Refer dermatology; urine/plasma porphyrins, LFT, ferritin, hep C/ HIV; stop iron/oestrogen review/alcohol
Acute porphyria features (abdominal pain, neuropathy, confusion, hyponatraemia)	**Urgent** medical assessment; check porphyria-safe drug lists
Chronic lichenified photo-distributed eczema in older adult	Dermatology (phototesting, patch/photopatch testing)
Erythroderma, lymphadenopathy, infiltrated plaques	**Urgent** dermatology (exclude CTCL/Sézary)
PMLE not responding to potent topical steroid, scarring, persisting >3 weeks without sun, lupus features	GP/dermatology — ANA/Ro, porphyrins, consider phototherapy
Photosensitivity on essential drug (amiodarone, voriconazole, oncology, HIV)	Don't stop — photoprotect; liaise with specialist; dermatology skin cancer surveillance for voriconazole/long-term HCTZ with prior NMSC
Persistent photosensitivity after culprit drug stopped	Dermatology (persistent light reaction/CAD)
New severe photosensitivity in a young adult with no drug cause	Consider HIV, lupus, porphyria — GP bloods/dermatology
Child with bizarre streaks/handprint marks	Careful history (plants/limes); safeguarding if history inconsistent
Suspected ADR	MHRA Yellow Card

Red flags / refer

Diagnostic doubt (rule out lupus/porphyria), scarring, persistent >2–3 weeks without sun, systemic features, failure of potent topical steroid, severe recurrent PMLE affecting quality of life (phototherapy referral).

Complications and prognosis

Complications

- Distress and holiday disruption; rare link with later lupus or autoimmune thyroid disease.

Prognosis

- Recurs each spring for years; often improves with age.

Prescriber pearls (UK pharmacist IP)

- "It's not an allergy to the sun cream — it's a reaction to UVA the cream didn't block." Check the **UVA rating**, not just SPF.
- Expect recurrence each spring/holiday but often milder later in summer; tends to improve over years.
- Vitamin D: if strict sun avoidance, advise 10 micrograms (400 units) daily supplement (UK public health advice).
- Always run a quick drug check — a new doxycycline or thiazide changes the diagnosis.

Self-check quiz

4 questions · optional

Q1 When should you think of Polymorphic light eruption?

✓ An itchy rash on the V of the neck, forearms and backs of hands hours to a day after early-season sun, settling within days without further exposure.

Q2 What drives it?

✓ A delayed immune reaction to a UV-induced skin antigen, with a failure of normal UV immunosuppression; it improves through the summer ('hardening').

Q3 What is the backbone of care?

✓ Broad-spectrum high-UVA sunscreen, clothing and gradual exposure; potent topical steroid for flares; refer for phototesting or hardening if severe.

Q4 What must you never miss?

✓ Lupus (persistent rash, systemic symptoms, positive ANA) and drug photosensitivity.

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Photo gallery

Polymorphic light eruption 5

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Photodermatoses Module 5 · Week 6

Chronic actinic dermatitis

A persistent, severe, eczematous photodermatosis, usually in older men, affecting the face, neck and backs of the hands with sharp cut-offs at clothing. Often linked to contact allergy (Compositae, fragrance) and needs specialist phototesting.

ALSO CALLED CAD · actinic reticuloid · photosensitive eczema · older man eczema face · Compositae allergy

RELATED PAGES Polymorphic light eruption · Contact dermatitis · Eczema / dermatitis · Photoageing

KEY FACTS

- **Chronic severe eczema of sun-exposed skin** due to abnormal sensitivity to UVB, often UVA and sometimes visible light.
- Typically **older fair-skinned men (60+)** in the UK; also younger people with darker skin and atopic eczema.
- Sharp cut-off at collar and sleeve lines, sparing behind ears, eyelid folds and under the chin.
- Up to about three-quarters have positive patch tests, especially to **Compositae**, fragrance and sunscreen filters.
- Refer for phototesting and patch testing; strict photoprotection, allergen avoidance, emollients and topical steroids.

Overview

A **chronic, persistent, often severe eczema of sun-exposed skin** caused by abnormal sensitivity to UVB, often UVA and sometimes visible light. Typically affects **older men (60+), usually fair-skinned**, in the UK; also seen in younger people with darker skin types and in people with longstanding atopic eczema. Older synonyms: persistent light reaction, photosensitive eczema, actinic reticuloid (severe pseudolymphomatous end of the spectrum).

Aetiology: causes and risk factors

Thought to be a **delayed-type hypersensitivity (allergic contact-type) reaction to an endogenous skin antigen photo-modified by UV** — i.e. "photo-allergy to one's own skin". Strong links with **contact allergy**: up to ~3/4 have positive patch tests, especially to **Compositae (sesquiterpene lactones — chrysanthemums, daisies, dandelions)**, fragrance, colophony, rubber chemicals and **sunscreen filters**. May follow a photoallergic drug reaction (persistent light reaction) or airborne contact dermatitis. Cumulative sun damage and outdoor work/gardening/hobbies in older men are risk factors. Some drugs (thiazides, certain antibiotics) can aggravate it.

Myths to address

- **"SUNSCREEN WILL SOLVE IT"**
  sunscreen allergy is common and visible light may be involved.

Pathophysiology and pathology

CAD behaves like an **allergic contact eczema to the patient's own UV-altered skin**.

1. **UV photo-modification**
   UV alters an endogenous skin protein, creating a new antigen.
2. **T-cell sensitisation**
   A delayed-type hypersensitivity develops, as in allergic contact dermatitis.
3. **Lower threshold**
   With time, smaller UV doses (and visible light) trigger eczema.
4. **Contact allergy overlap**
   Co-existing Compositae, fragrance and sunscreen allergies keep it going.

HISTOPATHOLOGY

- Spongiotic dermatitis; in severe cases dense dermal lymphocytes mimicking lymphoma (actinic reticuloid).

Clinical features

- **Itchy, red, thickened, lichenified eczematous plaques** on face (forehead, cheeks, nose), scalp in bald men, ears, **back and sides of neck, V of chest, dorsal hands/forearms**.
- **Sharp cut-off** at collar and wrist/cuff; **sparing** of upper eyelids, skin creases, behind/under ears, under chin, finger webs (DermNet gallery).
- Can extend to covered sites and progress to **erythroderma** in severe cases.
- Present year-round but worse in summer; patients may say even indoor light/fluorescent light triggers flares.
- Darker skin: hyperpigmentation and lichenification more than redness.
- Severe: infiltrated papules/plaques with atypical lymphocytes (actinic reticuloid) — mimics cutaneous T-cell lymphoma (Sézary syndrome).

Fig. 1 · CAD — lichenified eczema in light-exposed skin with sharp cut-off at collar.
© DermNet · **Source**

Diagnosis and investigations

Clinical suspicion + specialist confirmation:

- **Monochromator phototesting** (photobiology unit): reduced minimal erythema dose to UVB ± UVA ± visible — confirms diagnosis and guides photoprotection.
- **Patch testing and photopatch testing** (Compositae mix, fragrance, sunscreen agents, plant extracts).
- Biopsy: chronic spongiotic dermatitis ± atypical lymphoid infiltrate; exclude lymphoma (Sézary cells, T-cell receptor gene studies if erythrodermic).
- Bloods: ANA/ENA (exclude lupus), porphyrins; HIV in younger patients with new severe photosensitivity.
- Drug history (exclude drug photosensitivity).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Airborne contact dermatitis (e.g. Compositae, sawdust)	Involves **shaded** sites too (upper eyelids, under chin, behind ears); patch tests; may coexist with CAD
	Photoallergic drug reaction	Temporal link to drug; resolves after stopping (persistent light reaction = CAD-like evolution)
	Seborrhoeic / atopic eczema	Flexural/scalp/nasolabial involvement; not light-cut-off
	Cutaneous lupus	Plaques with scarring/atrophy; ANA/Ro
	Cutaneous T-cell lymphoma / Sézary	Erythroderma, lymphadenopathy, atypical cells — biopsy

	CONDITION	DISTINGUISHING FEATURES
	PMLE	Younger, intermittent, clears within weeks, not lichenified
	Rosacea	Central face, papulopustules, telangiectasia, flushing; no eczema

Management

Refer to dermatology (phototesting/patch testing are specialist). While waiting and long-term:

- **Rigorous photoprotection** (UVB, UVA ± visible): broad-spectrum SPF 50+ high UVA, **fragrance-free**, ideally **mineral/tinted** if visible-light sensitive or sunscreen-allergic; wide-brimmed hats, long sleeves, gloves, UV-protective window film; avoid sun 10:00–16:00.
- **Allergen avoidance** per patch tests (e.g. Compositae plants in garden, fragrance).
- **Emollients + soap substitutes.**
- **Topical corticosteroids** — potent to body/hands; moderate short courses to face; **topical calcineurin inhibitors** (tacrolimus 0.1%/pimecrolimus — off-label for CAD, specialist-guided) as steroid-sparing on the face. Counsel: TCIs can cause transient burning; strict sun protection while using them.
- Flares: short oral prednisolone (prescriber decision).
- Specialist systemic: **azathioprine**, ciclosporin, mycophenolate, methotrexate, dupilumab (case reports); low-dose PUVA/narrowband UVB under steroid cover (hardening) in selected patients.
- **Vitamin D supplementation** (10 micrograms daily, or check levels) because of strict sun avoidance.
- Review **aggravating drugs** (thiazides, etc.).

Referral

SAME DAY — Erythroderma or rapidly spreading eczema: same-day dermatology.

ROUTINE — Persistent photosensitive eczema in an older adult: refer for **phototesting and patch testing**.

Red flags and when to refer

- Erythroderma or rapidly spreading eczema: same-day dermatology.
- Persistent photosensitive eczema in an older adult: refer for **phototesting and patch testing**.

Red flags / refer

Suspected CAD routine dermatology (photobiology). **Urgent** if erythrodermic, systemically unwell, lymphadenopathy or atypical infiltrated plaques (exclude lymphoma).

Complications and prognosis

Complications

- Erythroderma, severe impact on quality of life, social isolation.

Prognosis

- Chronic; some improve over years.

Prescriber pearls (UK pharmacist IP)

- "Eczema in an older man that is worst on the face, neck and backs of hands — and spares behind the ears" = think CAD.
- Patients may be sensitised to their **own sunscreen** — choose fragrance-free, minimal-filter or physical products and ask about reactions.
- Gardening is a double hit (UV + Compositae): gloves, long sleeves, avoid chrysanthemums/daisies if allergic.
- Severe disease can be life-limiting (house-bound): signpost BAD leaflet and support; consider mental health.
- Many improve over years with strict avoidance.

Self-check quiz

5 questions · optional

Q1 When should you think of Chronic actinic dermatitis?

✓ Thickened, itchy eczema on the face, scalp, neck and backs of hands in an older man, sparing the upper eyelids, behind the ears and under the chin.

Q2 What drives it?

✓ Extreme sensitivity to UVB, UVA and sometimes visible light, behaving like contact allergy to an endogenous light-altered antigen; contact allergy is common alongside.

Q3 What is the backbone of care?

✓ Strict photoprotection (including visible light), emollients, potent topical steroid or tacrolimus; refer for phototesting and patch testing; specialist immunosuppression.

Q4 What must you never miss?

✓ Cutaneous T-cell lymphoma and drug photosensitivity as mimics; erythroderma.

Q5 Name a red flag that needs escalation.

✓ Erythroderma or rapidly spreading eczema: same-day dermatology.

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Photo gallery

Chronic actinic dermatitis 5

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Photodermatoses Module 5 · Week 6

Drug-induced photosensitivity

Many common drugs make the skin burn or react in sunlight. Phototoxic reactions look like exaggerated sunburn; photoallergic ones look like eczema. A medication review is the key intervention.

ALSO CALLED phototoxic · photoallergic · doxycycline sunburn · thiazide photosensitivity · amiodarone · NSAID photosensitivity · +2

RELATED PAGES Polymorphic light eruption · Phytophotodermatitis · Chronic actinic dermatitis · Drug-induced pigmentation (amiodarone, minocycline) · Approach to a suspected drug eruption

KEY FACTS

- **Drug photosensitivity** is a rash on sun-exposed skin caused by a drug plus UV (mostly UVA).
- **Phototoxic** (commonest, dose-related, 'exaggerated sunburn') versus **photoallergic** (eczematous, immune).
- Common culprits: tetracyclines (doxycycline), thiazides, amiodarone, voriconazole, NSAIDs, quinolones, phenothiazines.
- Sharp cut-off at clothing lines; spares under the chin, behind the ears and skin folds.
- Stop or switch the culprit with the prescriber (never stop amiodarone, voriconazole or antiretrovirals without specialist input); strict UVA protection.

Overview

An abnormal skin reaction to UV (mostly **UVA, 320–400 nm**) or visible light caused by a **systemic or topical drug** (or chemical). Two mechanisms: **phototoxic** (common; non-immune) and **photoallergic** (uncommon; immune). Some drugs do both. A core pharmacist topic because most cases follow a medicine started in the previous days–months, and many are preventable with counselling at the point of supply.

An abnormal skin reaction to light (usually **UVA**, 320–400 nm) caused by a systemic or topical drug.

Epidemiology

- **Who**
  - Common in older people on multiple drugs; holiday or summer onset.

Aetiology: causes and risk factors

- **Dose- and light-dose-dependent**; can happen to **anyone, on first exposure**, no sensitisation needed.

- Requires **prior sensitisation**, only affects a minority, not strictly dose-related; may cross-react with chemically related drugs and may **persist** after drug withdrawal (persistent light reaction CAD). Topical agents (ketoprofen gel, sunscreen filters, fragrances, phenothiazine creams) are commoner causes than oral drugs.

Risk factors and associations

- Photosensitising drug in the skin plus UVA (passes through window glass).

Myths to address

- **"SUNSCREEN FIXES IT"**
  many sunscreens protect poorly against UVA; clothing and avoiding sun matter.

Pathophysiology and pathology

A drug in the skin **absorbs UV energy** and damages cells directly or creates an allergen.

1. **Drug in skin**
   The drug or its metabolite reaches the dermis and epidermis.
2. **Phototoxic**
   Absorbed UVA excites the drug, releasing free radicals that damage cells: sunburn-like reaction within hours.
3. **Photoallergic**
   UV converts the drug into a hapten; T cells become sensitised: eczema days later, may spread beyond exposed sites.
4. **Pigmentation**
   Some drugs (amiodarone, minocycline) leave slate-grey or blue pigment in exposed skin.

How it is caused

- Some drugs do both (**thiazides**, quinine, NSAIDs, phenothiazines).
- Thiazides/sulfonamides can also cause **photo-distributed lichenoid eruptions**, **SCLE** and **pseudoporphyria**.
- **Hydrochlorothiazide** carries a dose-dependent increased risk of **non-melanoma skin cancer** (MHRA DSU Nov 2018). The evidence applies to HCTZ — bendroflumethiazide data are limited — but photoprotection advice applies to all thiazides.
- **COMMON UK PHOTOSENSITISERS**
  thiazides (bendroflumethiazide, HCTZ, indapamide), **doxycycline/tetracyclines**, **quinolones**, sulfonamides (co-trimoxazole), **amiodarone**, NSAIDs (piroxicam, naproxen, ketoprofen gel — photoallergic), phenothiazines (chlorpromazine), **voriconazole** (SCC risk), St John's wort, retinoids, methotrexate (UV recall), quinine, hydroxychloroquine, furosemide (pseudoporphyria), vemurafenib.

HISTOPATHOLOGY

- Phototoxic: necrotic keratinocytes like sunburn. Photoallergic: spongiotic dermatitis. Phototesting confirms in specialist units.

Clinical features

FEATURE	PHOTOTOXIC	PHOTOALLERGIC
Frequency	Common	Uncommon
Prior sensitisation	**Not needed** — first exposure	**Needed**
Onset after sun	Minutes–hours (24 h)	24–72 h
Appearance	**Exaggerated sunburn**: erythema, oedema, blistering, then hyperpigmentation/desquamation; may cause photo-onycholysis (tetracyclines, psoralens), slate-grey pigmentation (amiodarone), pseudoporphyria	**Eczematous**: papules, vesicles, scaling, lichenification
Symptom	**Burning, stinging, pain**	**Itch**
Distribution	**Sharply limited to exposed skin**, clear cut-off (watch strap, collar, T-shirt)	Exposed skin **and can spread to covered sites**
Dose dependence	Yes (drug + UV dose)	No (small amounts enough)
Cross-reactivity	No	Possible
Resolution	Quickly after stopping drug/UV	Can persist (persistent light reaction)
Examples	**Tetracyclines (doxycycline)**, fluoroquinolones, NSAIDs (oral), amiodarone, thiazides, voriconazole, psoralens, chlorpromazine, St John's wort	Topical **ketoprofen**, sunscreen filters (benzophenones/oxybenzone, octocrylene), fragrance (musk ambrette), sulfonamides, phenothiazines, quinine/quinidine, thiazides (both)

(Three discriminators: sensitisation, distribution, burning vs itching.)

Other drug-related photo-patterns: **drug-induced subacute cutaneous lupus** (thiazides, terbinafine, PPIs, calcium-channel blockers, checkpoint inhibitors), **drug-induced pellagra** (isoniazid), **lichenoid photo-eruption** (thiazides, quinine), **methotrexate UV-recall** of a recent sunburn.

- Itchy or burning erythema on the **face, V of the neck, nape, backs of hands and forearms**.
- There is a sharp cut-off at clothing, with **sparing of the upper eyelids, under the chin (submental), behind the ears, nasolabial folds and finger webs**.

Fig. 1 · Phototoxic — exaggerated sunburn within hours; doxycycline is the commonest UK primary-care culprit.

© DermNet · **Source**

Special groups

- **Photo-onycholysis**
  - Tetracyclines and psoralens can lift the nails from the bed after sun exposure.

Diagnosis and investigations

Clinical + **temporal drug history** (including OTC, topical, herbal — St John's wort — and supplements, and recent dose increases). Specialist: **phototesting on and off the drug**, **photopatch testing** for suspected photoallergens, biopsy if lupus/lichenoid suspected; ANA/Ro if lupus-like.

- Clinical: distribution + drug history + sun exposure.
- **Investigations** mainly exclude differentials: **ANA, anti-Ro/La** (SCLE — note her **omeprazole**, the commonest drug cause of SCLE, and thiazides also cause SCLE); porphyrin screen if blistering or fragility (PCT, pseudoporphyria); biopsy if uncertain

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Ordinary sunburn	Proportionate to exposure; no new drug
	PMLE	Papular, recurrent seasonal, no drug link
	Phytophotodermatitis	Bizarre streaks/drips where plant/juice touched; no drug
	Airborne/allergic contact dermatitis	Shaded sites involved; contact history

	CONDITION	DISTINGUISHING FEATURES
	CAD	Chronic, older men, persists; patch/phototests
	Drug-induced lupus (SCLE)	Annular/psoriasiform plaques; Ro positive
	Pseudoporphyria	Blisters + fragility dorsal hands (naproxen etc.)

PLE (spring, no drug, papulovesicular), **SCLE** (annular/psoriasiform, Ro+, drugs: PPIs, thiazides, terbinafine, calcium-channel blockers), **chronic actinic dermatitis** (older men, lichenified), airborne contact dermatitis (Compositae — involves the shaded areas too), dermatomyositis (heliotrope, Gottron's — the skin signs of systemic disease notes), rosacea, seborrhoeic dermatitis, pellagra.

Management

Identify and stop/switch the culprit where possible, in discussion with the original prescriber (e.g. doxycycline lymecycline is also a tetracycline, so consider a non-tetracycline per indication; thiazide alternative antihypertensive). **Never stop amiodarone, voriconazole or antiretroviral/oncology drugs without specialist input** — manage with photoprotection instead; amiodarone photosensitivity can persist for months after stopping.

ACUTE PHOTOTOXIC REACTION

cool compresses, emollients, analgesia (paracetamol; avoid if NSAID is the culprit), **potent topical corticosteroid** short course for inflammation on body; burn-type dressings for blisters; oral antihistamine for itch.

PREVENTION

counsel at supply, especially before holidays: broad-spectrum high-UVA sunscreen, cover up, avoid sunbeds; consider evening dosing only where evidence supports it (e.g. some advise doxycycline with evening meal — mixed evidence; protection remains essential).

Report suspected ADRs via **MHRA Yellow Card** (serious, novel, or black-triangle drugs).

Photosensitiser drug checklist for medication review

Paraphrased from DermNet and BNF photosensitivity side-effect listings; **always check the current BNF monograph and SmPC** for an individual drug. = commonest/most important in UK primary care.

CLASS	EXAMPLES (UK-RELEVANT)	MAIN TYPE / NOTES
Tetracyclines	**doxycycline**, demeclocycline (strongest), lymecycline, minocycline (less), oxytetracycline	Phototoxic; photo-onycholysis; pseudoporphyria. Counsel every supply (acne, rosacea, malaria prophylaxis)
Fluoroquinolones	ciprofloxacin, levofloxacin, ofloxacin, moxifloxacin; nalidixic acid	Phototoxic; pseudoporphyria (nalidixic acid)
Sulfonamides/ antifolates	co-trimoxazole, trimethoprim (occasional), sulfasalazine, dapsone	Photoallergic/phototoxic

CLASS	EXAMPLES (UK-RELEVANT)	MAIN TYPE / NOTES
Antifungals	**voriconazole** (also SCC/melanoma risk), griseofulvin, itraconazole/ ketoconazole (rare), terbinafine (drug-induced SCLE)	Phototoxic; skin cancer with voriconazole
NSAIDs	**topical ketoprofen** (photoallergy — avoid sun on site during + 2 weeks after), piroxicam, naproxen (pseudoporphyria), ibuprofen (rare), diclofenac, celecoxib	Both types
Diuretics	**thiazides/related** (bendroflumethiazide, hydrochlorothiazide — MHRA NMSC risk, indapamide, chlortalidone), furosemide, bumetanide (pseudoporphyria)	Phototoxic/photoallergic; lichenoid; SCLE
Cardiovascular	**amiodarone** (phototoxic, blue-grey pigmentation; persists months), dronedarone, quinidine; calcium-channel blockers (amlodipine, nifedipine, diltiazem — photo-distributed telangiectasia/ pigmentation), ACE inhibitors (rare), statins (rare)	Do not stop amiodarone without cardiology
Alpha-blockers	tamsulosin (DermNet gallery case)	Photosensitive eczema/purpura (rare)
Psychotropics	chlorpromazine and other phenothiazines (incl. topical promethazine), tricyclic antidepressants, St John's wort (hypericin)	Phototoxic + photoallergic; check herbal use
Retinoids	isotretinoin, acitretin, alitretinoin; topical tretinoin/adapalene	Increased burning/fragility (stratum corneum thinning)
Antimalarials	quinine, chloroquine, hydroxychloroquine (paradoxically treats some photodermatoses), mefloquine (rare)	Photoallergic/lichenoid
Oncology/ immunology	methotrexate (UV recall), fluorouracil (systemic/topical), vemurafenib (marked UVA phototoxicity), BRAF/MEK & checkpoint inhibitors, hydroxycarbamide	Phototoxic; specialist-managed
Antidiabetics	sulfonylureas (glibenclamide, gliclazide — rare)	Rare
Antiretrovirals/TB	efavirenz; isoniazid (pellagra-like photosensitivity), pyrazinamide	Photo-distributed eruptions
PPIs	omeprazole, lansoprazole, esomeprazole	Drug-induced SCLE (photo-aggravated)
Psoralens	methoxsalen (PUVA), bergamot oil/citrus essential oils	Phototoxic — therapeutic in PUVA
Topicals/cosmetics	sunscreen filters (oxybenzone, octocrylene), fragrances, coal tar, benzoyl peroxide (irritant)	Photoallergic (filters/fragrance), phototoxic (tar)

1. **Stop the probable culprit — bendroflumethiazide** (the classic phototoxic agent here). Choose an alternative antihypertensive per **NICE NG136**: She already takes an ARB (losartan) for T2DM/HTN; Add a **CCB (amlodipine)**. Avoid indapamide, a sulfonamide-type thiazide-like diuretic that also photosensitises; Check BP after the switch
2. Symptomatic potent topical steroid (body/hands) and moderate/mild steroid on the face for 1–2 weeks; emollient; antihistamine for itch; cool compresses.
3. Review other contributors omeprazole (SCLE — ask about annular lesions, check Ro; deprescribe if no ongoing indication); the HbA1c of 78 means diabetes needs intensifying (NICE NG28 — SGLT2 inhibitor/GLP-1 RA). Sitagliptin is linked to **bullous pemphigoid** rather than photosensitivity — keep that in mind if blisters appear.

4. **Yellow Card**; document "photosensitivity — bendroflumethiazide".

IP scope

Stopping/switching antihypertensives within competence, topical steroids, photoprotection advice, Yellow Card.

Referral

SAME DAY Blistering, extensive or systemic reaction: same-day assessment.

ROUTINE Long-term voriconazole with photodamage: monitor for skin cancer.

Red flags and when to refer

- Blistering, extensive or systemic reaction: same-day assessment.
- Long-term voriconazole with photodamage: monitor for skin cancer.

Red flags / refer

Blistering over large area, mucosal involvement or systemic illness (consider SJS/TEN, not photosensitivity) **same-day**; persistent eruption after stopping drug (persistent light reaction/CAD) dermatology; lupus-like eruption GP bloods/dermatology; **voriconazole long-term + photodamage** dermatology for skin cancer surveillance.

Red flags / refer

Blistering/erosions, widespread or persistent eruption after stopping the drug (persistent light reaction/chronic actinic dermatitis), suspected SCLE/lupus, diagnostic doubt dermatology (photobiology).

Complications and prognosis

Complications

- Blistering, pigmentation, persistent light reaction (chronic actinic dermatitis).

Prognosis

- Phototoxic reactions clear when the drug stops (amiodarone may take months); photoallergy can persist.

Prescriber pearls (UK pharmacist IP)

Always ask **"any new tablets, creams or herbal remedies in the last 3 months?"** for any photo-distributed rash.

Doxycycline (acne, rosacea, malaria prophylaxis, Lyme, chest infections) — the commonest UK primary-care culprit; give the sun advice **with every supply**, especially malaria prophylaxis for sunny destinations.

Topical ketoprofen gel — BNF/MHRA advice: avoid exposure of treated area to sunlight/UV (incl. sunbeds) **during treatment and for 2 weeks after**, wash hands, avoid octocrylene-containing sunscreens/fragrance co-sensitisers.

- **Hydrochlorothiazide** — MHRA (2018) Drug Safety Update: cumulative-dose-related increased risk of **non-melanoma skin cancer**; advise sun protection, skin checks, and review in those with prior skin cancer. (Bendroflumethiazide is the usual UK thiazide; photosensitivity also listed.)
- **Voriconazole** — phototoxicity and **squamous cell carcinoma/melanoma** risk with long-term use (MHRA warning): strict photoprotection; dermatology review.
- **Fluoroquinolones** — photosensitivity listed; MHRA restrictions on use for other safety reasons; avoid strong sun/sunbeds during and for a few days after.
- **Methotrexate** — sunburn recall; advise sun protection.
- **Isotretinoin/acitretin** — increased sunburn/skin fragility (retinoid effect, not classic phototoxicity).
- Check the **BNF monograph side-effects / "Patient and carer advice"** for photosensitivity statements — the definitive UK source for counselling labels.
- **BNF cautionary label 11** applies to doxycycline, demeclocycline, chlorpromazine and others. Use the same wording verbally for thiazides and amiodarone.
- Doxycycline photosensitivity is dose-related — useful when choosing lymecycline vs doxycycline for summer acne courses.
- Thiazide users with lots of sun exposure or previous skin cancer: discuss NMSC risk (MHRA 2018) and **skin checks**.

Self-check quiz

5 questions · optional

Q1 **When should you think of Drug-induced photosensitivity?**

✓ Exaggerated sunburn, blistering, eczema or pigment change on sun-exposed skin after starting a drug (doxycycline, thiazides, amiodarone, NSAIDs, quinine, voriconazole).

Q2 What drives it?

✓ Phototoxic: the drug absorbs UVA and directly damages skin (dose-dependent). Photoallergic: UV turns the drug into an allergen (type IV).

Q3 What is the backbone of care?

✓ Review and stop or switch the culprit where possible, counsel UVA protection, treat with cool compresses and potent topical steroid; Yellow Card.

Q4 What must you never miss?

✓ Voriconazole-related SCC risk, and lupus or pellagra as differentials.

Q5 Name a red flag that needs escalation.

✓ Blistering, extensive or systemic reaction: same-day assessment.

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Photo gallery

Drug-induced photosensitivity

7

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Drug photosensitivity

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Photodermatoses Module 5 · Week 6 · Module 4 · Week 6

Porphyria cutanea tarda (& pseudoporphyria)

Fragile skin, blisters, erosions, scars and milia on the backs of the hands, with facial hair growth and dark urine. Look for alcohol, hepatitis C, haemochromatosis, oestrogens and HIV; pseudoporphyria from drugs looks the same.

ALSO CALLED PCT · porphyria · blisters back of hands · skin fragility sun · hypertrichosis · dark urine · +4

RELATED PAGES Milia · Skin fragility & dermatoporosis · Drug-induced photosensitivity · Epidermolysis bullosa (EB) · Bullous pemphigoid (incl. MMP & pemphigoid gestationis)

KEY FACTS

- **Porphyria cutanea tarda (PCT)** is the commonest porphyria: blisters, erosions, **skin fragility**, scars and **milia** on the backs of the hands.
- Also facial hypertrichosis and hyperpigmentation; urine may be dark.
- Triggers: **alcohol, oestrogens, hepatitis C, HIV, iron overload (haemochromatosis)**, smoking.
- Confirm: plasma and urine porphyrins; check ferritin, LFT, hepatitis C, HFE genotype.
- Specialist: venesection (iron reduction) or low-dose hydroxychloroquine; remove triggers; visible-light protection.

Overview

The commonest porphyria worldwide: a **blistering photosensitivity with skin fragility** caused by reduced activity of the hepatic haem-pathway enzyme **uroporphyrinogen decarboxylase (UROD)**. Usually presents in middle-aged adults (more often men). **Sporadic/acquired (type 1, ~75–80%)** or **familial (type 2, inherited heterozygous UROD mutation)** — the latter still needs acquired triggers to become clinically active. A **non-acute** porphyria: no neurovisceral attacks.

Epidemiology

- **Who**
  - Adults over 40; about 1 in 10,000–25,000; familial form in about a fifth.

Aetiology: causes and risk factors

When hepatic UROD activity falls below ~20–25% of normal, **uroporphyrin and heptacarboxyl porphyrins accumulate in the liver**, spill into plasma and urine, and deposit in skin. Porphyrins absorb strongly in the **Soret**

band (~400–410 nm, visible violet light), generating reactive oxygen species damage at the dermo-epidermal junction fragility and subepidermal blisters. **Triggers / associations (mostly hepatic iron and oxidative stress):**

- **Alcohol** (Case 2: ~40 units/week) and **smoking** (Case 2: 10/day)
- **Iron overload** — including **HFE haemochromatosis** mutations
- **Hepatitis C** (also hepatitis B), **HIV**
- **Oestrogens** (COC, HRT, prostate cancer therapy)
- Chronic kidney disease/haemodialysis; occasionally hepatocellular carcinoma ("hepatic porphyrinoma"); polyhalogenated aromatic hydrocarbons (historical)

Liver is the internal organ most affected: steatosis, siderosis, fibrosis/cirrhosis and raised risk of hepatocellular carcinoma.

Myths to address

- **"SUNSCREEN PROTECTS"**
  ordinary sunscreens do not block the visible light (around 400 nm) that excites porphyrins.

Pathophysiology and pathology

Reduced activity of **uroporphyrinogen decarboxylase** in the liver lets **light-sensitive porphyrins** build up in the skin.

1. **Enzyme inhibition**
   Iron and oxidative stress (alcohol, hepatitis C, oestrogen) inhibit hepatic UROD.
2. **Porphyrin accumulation**
   Uroporphyrins accumulate in the liver, blood and skin.
3. **Phototoxicity**
   Visible light (about 400–410 nm) excites porphyrins, generating reactive oxygen that damages the dermo-epidermal junction.
4. **Fragility and blisters**
   Minor trauma causes subepidermal blisters that heal with scars and milia.

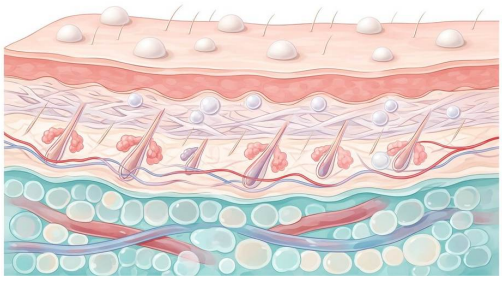

Fig. 1 · Milia form as blisters heal: tiny keratin cysts just beneath the epidermis.
AI-generated illustration

HISTOPATHOLOGY

- Cell-poor subepidermal blister with 'festooning' of dermal papillae; caterpillar bodies.

Clinical features

- **Tense vesicles and bullae (some haemorrhagic)** on **dorsal hands and fingers**, forearms, face, and scalp/ears in bald men — often after minor knocks.
- **SKIN FRAGILITY**
  erosions and crusts from trivial trauma; slow healing over weeks.
- **Milia**, atrophic and white/pink scars on dorsal hands.
- **Hypertrichosis** of temples/cheeks, mottled hyper/hypopigmentation, sclerodermoid thickening (late).
- **Dark, red-brown or "tea"/port-wine urine**; fluoresces pink-red under Wood's lamp.
- Worse in summer (Case 2) — but patients often don't link it to sun, because the trigger is visible light through windows and damage is delayed.

Fig. 2 · PCT — tense blisters/erosions on dorsal hands after minor trauma; check urine/plasma porphyrins.
© DermNet · **Source**

Special groups

- **Pseudoporphyria**
  - Same skin signs with normal porphyrins: naproxen, tetracyclines, furosemide, sunbeds, dialysis.

Diagnosis and investigations

- **Most relevant first test: urine (and/or plasma) porphyrins** — raised uroporphyrin and heptacarboxylic porphyrin; faecal isocoproporphyrin raised. Plasma fluorescence emission peak ~619–620 nm distinguishes PCT from variegate porphyria (~626 nm) — this matters because variegate porphyria can cause acute attacks. ALA and porphobilinogen normal.
- **ASSOCIATED-CAUSE WORK-UP**
  LFTs, **ferritin + transferrin saturation**, **HFE genotyping**, **hepatitis C and B, HIV serology**, glucose/HbA1c, renal function, consider liver ultrasound (± HCC surveillance if cirrhotic), ANA if diagnostic overlap.
- Skin biopsy (subepidermal blister with "festooning" of dermal papillae; immunofluorescence) when needed to exclude other bullous disease.
- Sent via specialist porphyria laboratory; UK National Acute Porphyria Service / specialist dermatology.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Pseudoporphyria	Same clinical picture but **normal porphyrins**; drugs — **naproxen** (and other NSAIDs), tetracyclines, **furosemide**, bumetanide, nalidixic acid, isotretinoin, dapsone, voriconazole; **sunbeds**; haemodialysis. Hypertrichosis and dark urine absent. Stop the drug/sunbed.
	Variegate porphyria / hereditary coproporphyria	Same skin changes **plus risk of acute neurovisceral attacks** — porphyrin pattern differs; dangerous drug list applies
	Epidermolysis bullosa acquisita	Trauma-induced blisters/milia, but not photo-related; biopsy/ immunofluorescence
	Bullous pemphigoid	Elderly, itchy urticated plaques + tense blisters trunk/flexures
	Bullous drug eruption / fixed drug eruption	Drug timing; recurring same site
	Phytophotodermatitis / phototoxic drug	Acute single episode; streaks; no fragility/milia
	Erythropoietic protoporphyria	Childhood, immediate burning pain in sun, little blistering

Pseudoporphyria and EBA

> **NOTE**
>
> **Full PCT coverage**
>
> (pathophysiology, urine porphyrin testing, venesection, low-dose hydroxychloroquine, visible-light sunscreen) is in the photodermatoses notes. Only the case-specific points and the mimics are covered here.

What it is

PCT is the **commonest porphyria**. It comes from reduced **hepatic uroporphyrinogen decarboxylase (UROD)** activity, which is acquired (type 1, ~80%) or familial (type 2). Uroporphyrins build up and cause **blistering photosensitivity on sun-exposed skin**.

How it is caused / pathophysiology

Porphyrins absorb visible light (Soret band ~400–410 nm), which damages the dermal–epidermal junction, so the skin becomes fragile and blisters below the epidermis. **Triggers that the history in Case 3 points to:**

- **alcohol** (dependence)
- **hepatitis C** and **HIV** (homelessness and possible injecting drug use, which may also explain the **forearm scars**)
- **iron overload / HFE haemochromatosis**
- oestrogens
- smoking

Clinical features & typical sites

- **fragile skin, tense and haemorrhagic blisters and erosions on the backs of the hands** after minor trauma (Case 3; photo D01, and the photodermatoses notes B01–B06)
- healing with **scarring, milia** and dyspigmentation
- **facial hypertrichosis** (upper cheeks and temples)
- photo-distributed hyperpigmentation or "tanned" face
- morphoea-like sclerodermoid plaques later on
- **dark or red-brown urine**

How to diagnose

Urine porphyrins (a random urine sample **protected from light**; it may glow pink under Wood's lamp), plasma porphyrins with a **plasma fluorescence emission scan (peak ~619 nm)** to separate PCT from variegate porphyria, and stool porphyrins. Add LFT, **ferritin/transferrin saturation, HFE genotype**, **HCV, HBV and HIV serology**, glucose and liver ultrasound (HCC risk). Skin biopsy with DIF (perivascular IgG/C3) is not usually needed. Specialist support: **UK Porphyria Medicines Information Service / National Acute Porphyria Service**, Cardiff and King's.

Key differentials

Differentials for Case 3:

> **PSEUDOPORPHYRIA**
> looks the same but porphyrins are **normal**. Drug causes are **naproxen** and other NSAIDs, tetracyclines, nalidixic acid, furosemide, retinoids, and **sunbeds**. Also seen in dialysis. Stop the drug (**Module 5 exam Q2**).

> **VARIEGATE PORPHYRIA / HEREDITARY COPROPORPHYRIA**
> the skin looks the same, but there is a risk of **acute neurovisceral attacks** with porphyrinogenic drugs. This is why the plasma scan matters before choosing drugs.

> **EBA**
> trauma-induced acral blisters, scarring and milia; porphyrins normal; linear IgG on the dermal floor of salt-split skin.

> **BP**, **bullous impetigo**, **friction or trauma blisters** (rough sleeping), **bullous drug eruption**, **bullosis diabeticorum**, **burns** (cigarettes in intoxication), and **erythropoietic protoporphyria** (painful, burning, no blisters, children).

Management for pharmacist/IP (UK)

- Refer to dermatology or hepatology (routine to urgent). Treat the drivers: **alcohol cessation** (local alcohol services, thiamine), **HCV direct-acting antivirals** (which often cure PCT), and stop oestrogens. Specialists then use **venesection** (to ferritin ~20 µg/L) or **low-dose hydroxychloroquine 100–200 mg twice weekly** (not daily,

which causes hepatotoxicity), with eye screening per the RCOphth schedule. **Visible-light protection:** reflectant tinted sunscreen such as zinc/titanium with iron oxides; standard UV sunscreen is not enough.

- **SOCIAL CONTEXT FOR CASE 3**
  homelessness makes venesection and follow-up hard. Arrange a GP registration route, link with the hepatology outreach or HCV peer-support team, and give **gloves and hand protection**.

Prescriber pearls & counselling

- **Blisters on the backs of the hands plus hypertrichosis or dark urine: check porphyrins and do a medication review (NSAIDs, so think pseudoporphyria).**
- The porphyria drug-safety database (Welsh Medicines Information / NAPS) matters for **acute** porphyrias, not for PCT. Until the plasma scan excludes variegate porphyria, prescribe cautiously.

Photo / reference URLs

- See the the photodermatoses notes URL list. DermNet: dermnetnz.org · dermnetnz.org
- UK Porphyria Medicines Information Service / NAPS: wmic.wales.nhs.uk

Management

Refer — PCT diagnosis and treatment are specialist (dermatology ± hepatology/porphyria service). The pharmacist/IP role:

- **Recognise** the pattern (blisters + fragility + scars/milia on dorsal hands; alcohol/smoking/oestrogen/hep C) and **refer** with suspicion stated; topical steroids don't help (Case 2).
- **REMOVE TRIGGERS**
  stop alcohol and smoking (offer NHS stop-smoking support and alcohol brief intervention), **review oestrogens** (COC/HRT — discuss alternatives with prescriber), **stop iron supplements and iron-containing multivitamins**.
- Specialist treatments (know, don't initiate): **venesection** (reduce ferritin), **low-dose hydroxychloroquine** (e.g. 100 mg–200 mg **twice weekly**, not standard daily antimalarial doses, which can cause hepatotoxicity), treatment of hepatitis C with direct-acting antivirals often induces remission. Chloroquine sometimes used at low dose.
- **VISIBLE-LIGHT PHOTOPROTECTION**
  conventional chemical UV filters don't cover 400–410 nm — recommend **reflectant sunscreens containing titanium dioxide/zinc oxide, ideally tinted (iron-oxide)**, e.g. Dundee cream (via specialists) or tinted mineral SPF 50; gloves, long sleeves, hats; **window film** for car/home.
- Wound care for erosions (non-adherent dressings, watch for infection); protect hands from trauma (gloves for DIY/gardening).

Referral

URGENT — Signs of chronic liver disease or a known liver mass: urgent review.

DON'T — Ordinary sunscreens do not block visible light: use opaque or tinted (iron oxide) protection.

DON'T Do not use standard-dose hydroxychloroquine: only low-dose under specialist advice (risk of hepatotoxicity).

Red flags and when to refer

- Signs of chronic liver disease or a known liver mass: urgent review.
- Ordinary sunscreens do not block visible light: use opaque or tinted (iron oxide) protection.
- Do not use standard-dose hydroxychloroquine: only low-dose under specialist advice (risk of hepatotoxicity).

Red flags / refer

All suspected PCT routine-soon dermatology with urine porphyrins requested in primary care if local pathway allows. Urgent if abdominal pain, neuropathy, psychiatric features or hyponatraemia (possible acute porphyria — check drug safety via the Welsh Medicines Advice/UK Porphyria Medicines Information Service database). Signs of decompensated liver disease urgent.

Complications and prognosis

Complications

- Scarring, cirrhosis, hepatocellular carcinoma (in liver disease).

Prognosis

- Remits with iron depletion or hydroxychloroquine; relapse if triggers return.

Prescriber pearls (UK pharmacist IP)

- **Blisters on the back of the hands in a drinker = check porphyrins**; "worse in summer" but light through glass counts.
- **Pseudoporphyria** is the pharmacist's catch: look for **naproxen**, doxycycline, furosemide or sunbed use — normal porphyrins, resolves (slowly, months) after withdrawal.
- Before supplying drugs to anyone with a porphyria diagnosis, check porphyria drug safety lists (BNF "Acute porphyrias" section and UK Porphyria Medicines Information Service) — strictly relevant to acute porphyrias, but confirm the type.
- Ask about **urine colour**.
- Long-term: alcohol abstinence reduces relapse; screen for HCC if cirrhosis.

Self-check quiz

5 questions · optional

Q1 When should you think of Porphyria cutanea tarda (& pseudoporphyria)?

✓ Blisters and erosions on the backs of the hands after minor knocks, healing with scars and milia; facial hypertrichosis; reddish-brown urine.

Q2 What drives it?

✓ Reduced hepatic uroporphyrinogen decarboxylase activity (iron, alcohol, hepatitis C, oestrogen) causes porphyrin build-up that reacts with visible light.

Q3 What is the backbone of care?

✓ Refer for porphyrin testing; stop alcohol and oestrogens; screen liver, iron, hepatitis B/C and HIV; specialist venesection or low-dose hydroxychloroquine; visible-light protection.

Q4 What must you never miss?

✓ Underlying liver disease or hepatocellular carcinoma; haemochromatosis; pseudoporphyria from naproxen or other drugs.

Q5 Name a red flag that needs escalation.

✓ Signs of chronic liver disease or a known liver mass: urgent review.

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Photo gallery

Porphyria cutanea tarda

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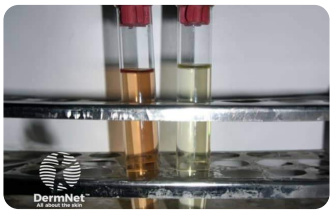
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Pseudoporphyria and EBA

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NSAID pseudoporphyria (mimic)

1

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CHAPTER 13

Pigment & colour change

Hypo- and hyperpigmentation. Progressive macular hypomelanosis in full, plus melasma, lichenoid pigmentation and vitiligo (Module 6 Week 3).

1. Progressive macular hypomelanosis (PMH)
2. Melasma & facial melanosis
3. Contact & lichenoid pigmentation
4. Phytophotodermatitis
5. Cutaneous amyloidosis (macular and lichen)
6. Hypopigmentation & skin of colour

Pigment & colour change Module 5 · Week 1

Progressive macular hypomelanosis (PMH)

Ill-defined, non-scaly pale patches merging on the trunk of young adults. Think of it when 'pityriasis versicolor' has no scale and does not respond to antifungals.

ALSO CALLED PMH | pale patches trunk | hypopigmentation | Cutibacterium acnes | Wood's lamp red fluorescence

RELATED PAGES Pityriasis versicolor | Hypopigmentation & skin of colour | Acne vulgaris

KEY FACTS

- Benign, ill-defined, **non-scaly pale macules** merging on the trunk; no itch or prior inflammation.
- Common in young adults, especially with darker skin; slowly progressive, may improve after mid-life.
- Wood's lamp often shows **follicular red fluorescence** (Cutibacterium acnes).
- Differentials: pityriasis versicolor (fine scale, yeast), PIH, vitiligo (chalk-white), leprosy (numb).
- Options with variable evidence: benzoyl peroxide ± clindamycin, NB-UVB, oral tetracycline; relapse is common.

Overview

PMH is an acquired, **benign** condition of ill-defined, **non-scaly** hypopigmented macules. They often merge into patches on the trunk. It is **asymptomatic**, with no itch, pain or preceding inflammation. It progresses slowly over years and may stabilise or improve after mid-life.

Aetiology: causes and risk factors

The cause is not fully known. The leading hypothesis is **Cutibacterium (Propionibacterium) acnes** in the hair follicles producing a depigmenting factor. This is supported by **follicular red fluorescence** under a Wood's lamp in many, but not all, patients. Genetic factors may contribute.

Melanin is reduced while melanocyte numbers are normal, and melanosome packaging is altered on electron microscopy.

Myths to address

- **"IT'S FUNGAL, SO ANTIFUNGALS WILL FIX IT"**
  it doesn't scale and doesn't respond.

Pathophysiology and pathology

A follicular bacterium is thought to **reduce melanin production** in nearby skin.

1. **C. acnes in follicles**
   Bacteria in follicles may produce a depigmenting factor.
2. **Less melanin**
   Melanocyte numbers stay normal but melanin falls.
3. **Altered melanosomes**
   Melanosome packaging changes on electron microscopy.

Clinical features

- Round or oval, **poorly bordered** pale macules that merge on the back and trunk, with or without the proximal limbs and neck.
- **No scale.** This is the key difference from pityriasis versicolor.
- **No preceding rash.** This is the key difference from post-inflammatory hypopigmentation.
- More noticeable, and more often reported, in **darker skin**. Typically teens and young adults, possibly more common in women.

Fig. 1 · Progressive macular hypomelanosis — ill-defined non-scaly hypopigmented macules on trunk.

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Diagnosis and investigations

- Clinical.
- **WOOD'S LAMP**
  follicular **coral-red** fluorescence supports C. acnes.
- **Skin scrapings (KOH) are negative**, unlike pityriasis versicolor.
- Dermoscopy shows ill-defined white macules with a faint pigment network. Biopsy is rarely needed.

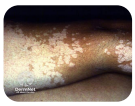
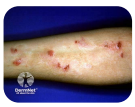
Wood's lamp

- follicular **coral-red** fluorescence supports C. acnes.

Key points

- Clinical.
- **Skin scrapings (KOH) are negative**, unlike pityriasis versicolor.
- Dermoscopy shows ill-defined white macules with a faint pigment network. Biopsy is rarely needed.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	**Pityriasis versicolor**	**Fine scale**; 'spaghetti and meatballs' on KOH; often yellow-green under Wood's lamp
	**Post-inflammatory hypopigmentation**	History of eczema, psoriasis or another rash at the same sites
	Vitiligo	Sharp, **chalk-white** patches, often on the hands, face and around orifices; bright white under Wood's lamp
	Pityriasis alba	Faces of atopic children; mild scale
	Hypopigmented mycosis fungoides	Persistent, atypical, progressive patches in an adult. **Refer**
	Ash-leaf macules (tuberous sclerosis)	Congenital, leaf-shaped; look for other features of tuberous sclerosis

Management

- **Reassure** that it is benign. Explain the C. acnes theory and the slow course.
- Options with **variable evidence**: topical **benzoyl peroxide ± clindamycin**, **narrowband UVB** (specialist), oral **tetracyclines**, or combinations. Isotretinoin appears only in case reports.
- Relapse after stopping treatment is common.
- Refer to dermatology if the diagnosis is uncertain, or if the patient wants treatment for cosmetic reasons.

Referral

ROUTINE — Atypical, persistent or progressive patches in an adult: exclude **hypopigmented mycosis fungoides**.

ROUTINE Congenital or ash-leaf-shaped macules with other signs: consider **tuberous sclerosis**.

ROUTINE 'Pityriasis versicolor' that does not respond to antifungals: re-check for scale and do KOH.

Red flags and when to refer

- Atypical, persistent or progressive patches in an adult: exclude **hypopigmented mycosis fungoides**.
- Congenital or ash-leaf-shaped macules with other signs: consider **tuberous sclerosis**.
- 'Pityriasis versicolor' that does not respond to antifungals: re-check for scale and do KOH.

Complications and prognosis

Complications

- Cosmetic concern.

Prognosis

- Slowly progressive; may stabilise.

Prescriber pearls (UK pharmacist IP)

Do not keep treating it as 'fungus'. If there is scale, take a **KOH scraping first**.

Photoprotection still helps even out skin tone.

Manage expectations: response is **slow and often incomplete**.

Self-check quiz

5 questions · optional

Q1 **When should you think of Progressive macular hypomelanosis (PMH)?**

✓ Poorly defined, non-scaly, symptom-free pale macules merging on the back and trunk of a teenager or young adult, often with darker skin.

Q2 What drives it?

✓ Probably Cutibacterium acnes in the follicles producing a depigmenting factor. Melanin is reduced but melanocyte numbers are normal.

Q3 What is the backbone of care?

✓ Reassure. Options with variable evidence: benzoyl peroxide ± clindamycin, narrowband UVB, oral tetracycline.

Q4 What must you never miss?

✓ Persistent, atypical, progressive patches in an adult may be hypopigmented mycosis fungoides. Refer.

Q5 Name a red flag that needs escalation.

✓ Atypical, persistent or progressive patches in an adult: exclude **hypopigmented mycosis fungoides**.

References

1. DermNet: Progressive macular hypomelanosis — https://dermnetnz.org/topics/progressive-macular-hypomelanosis
dermnetnz.org

Photo gallery

PMH: ill-defined, non-scaly, confluent pale macules 2

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Compare: pityriasis versicolor (fine scale) 2

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Pigment & colour change Module 6 · Week 3

Melasma & facial melanosis

Acquired brown facial patches driven by hormones, UV and visible light. Strict tinted sunscreen is the foundation; avoid long-term hydroquinone (ochronosis risk).

ALSO CALLED melasma · chloasma · facial melanosis · exogenous ochronosis · hydroquinone · drug photosensitivity

RELATED PAGES Drug-induced pigmentation (amiodarone, minocycline) · Solar lentigines · Photoageing · Hypopigmentation & skin of colour · Contact & lichenoid pigmentation

KEY FACTS

- **Melasma** is acquired, symmetrical brown patches on the face (cheeks, forehead, upper lip, chin), driven by **UV, visible light and hormones**.
- Commonest in women with Fitzpatrick III–V skin; triggered by pregnancy ('mask of pregnancy') and the combined pill.
- Wood's light or dermoscopy suggests epidermal versus dermal pigment; dermal pigment responds poorly.
- Foundation: daily **tinted** broad-spectrum SPF 30–50 (iron oxides block visible light); azelaic acid 15–20%; avoid unregulated bleaching creams.
- Specialist: triple-combination creams, tranexamic acid, peels; relapse is common. Consider drug-induced pigmentation and exogenous ochronosis.

Overview

Melasma is an acquired disorder of facial hyperpigmentation — symmetrical brown to grey-brown patches on sun-exposed face. It is commoner in women of reproductive age and in darker skin phototypes, but men are affected too.

This page also covers **exogenous ochronosis** (hydroquinone-related blue-black pigment) and a brief note on **drug photosensitivity/pigmentation**, which sit in the same facial-melanosis teaching week.

Epidemiology

- **Who**
  - About 90% are women; onset in the 20s–40s; common in Latin American, East and South Asian, Middle Eastern and African skin.
  - Family history in about half.

Aetiology: causes and risk factors

Genetic predisposition plus **UV and visible light**, heat, pregnancy and hormonal contraception, thyroid disease, and cosmetics/irritation.

Melanocytes overproduce melanin; pigment may sit in epidermis, dermis, or both (Wood's lamp / clinical pattern).

EXOGENOUS OCHRONOSIS
prolonged topical hydroquinone (often unregulated bleaching creams) dermal ochre pigment; can be permanent.

DRUG PHOTOSENSITIVITY
tetracyclines, thiazides, amiodarone, NSAIDs, quinolones, sulfonylureas and others cause photo-distributed eruptions or slate pigment.

Risk factors and associations

- Sun (UVA, UVB and **visible light**), oestrogen and progesterone, heat, genetics; possibly thyroid disease.
- Irritant cosmetics, phototoxic products.

Myths to address

"SPF 50 ALONE IS ENOUGH"
ordinary sunscreens do not block visible light; use a tinted one.

"STRONGER BLEACH WORKS FASTER"
high-strength hydroquinone or mercury/steroid creams cause ochronosis and harm.

Pathophysiology and pathology

In melasma, **overactive melanocytes** and a damaged dermis keep producing pigment.

1. **Triggers**
   UV and visible light, heat and hormones stimulate melanocytes and keratinocytes.

2. **Melanocyte overactivity**
   Melanocytes enlarge and transfer more melanin to keratinocytes (epidermal melasma).

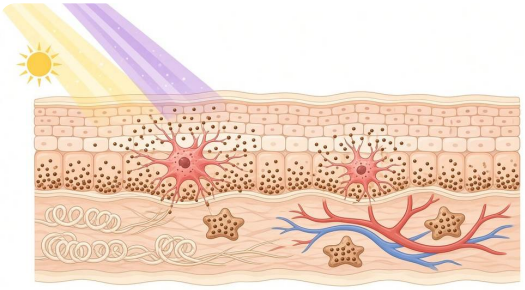

Fig. 1 · Melasma: light and hormones stimulate enlarged melanocytes to pass extra pigment into the epidermis; some pigment drops into dermal melanophages.
AI-generated illustration

3. **Dermal pigment**
   Pigment falls into the dermis and is taken up by melanophages; this dermal component is hard to treat.

4 Dermal signalling

Solar elastosis, increased blood vessels and a damaged basement membrane keep stimulating melanocytes, explaining relapse.

HISTOPATHOLOGY

- Increased epidermal melanin, prominent melanocytes, dermal melanophages, solar elastosis and vessels. Biopsy rarely needed.

Clinical features

- Centro-facial, malar or mandibular patterns; upper lip common.
- Usually asymptomatic; psychological impact can be high.
- Exogenous ochronosis: speckled blue-black "caviar" papules on treated sites.
- Drug photo-eruptions: sharp cut-off at clothing, sparing under chin and upper eyelids.

PRESCRIBER TIP

Skin of colour

Melasma is commoner and often mixed epidermal–dermal. Judge severity on colour contrast and area, not redness.

Fig. 2 · Melasma: symmetrical light-to-dark brown macular patches on sun-exposed face — centrofacial pattern (forehead, cheeks, upper lip, nose, chin) is the commonest.

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Special groups

- **Pregnancy**
  - Often fades after delivery; azelaic acid is the safer option; avoid retinoids and hydroquinone.

Diagnosis and investigations

- Clinical. Ask about hormones, pregnancy plans, drugs, cosmetics and skin-lightening (hydroquinone, steroids, mercury).
- Wood's lamp may help map epidermal vs dermal pigment but is imperfect in darker skin.
- Biopsy if exogenous ochronosis, lichenoid pigment or malignancy is suspected.
- Drug timeline for photo-distributed pigment.

Key points

- Clinical. Ask about hormones, pregnancy plans, drugs, cosmetics and skin-lightening (hydroquinone, steroids, mercury).
- Wood's lamp may help map epidermal vs dermal pigment but is imperfect in darker skin.
- Biopsy if exogenous ochronosis, lichenoid pigment or malignancy is suspected.
- Drug timeline for photo-distributed pigment.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Post-inflammatory hyperpigmentation	Prior inflammatory disease or injury
	Pigmented contact dermatitis / Riehl	Cosmetic allergen history; patch testing
	Lichen planus pigmentosus	Slate-grey; may need biopsy
	Exogenous ochronosis	Long hydroquinone use; caviar papules
	Drug pigmentation	Drug history; photo distribution

Management

Foundation for everyone

- Daily **tinted** broad-spectrum SPF 30–50 (iron oxides block visible light); reapply outdoors; hat and shade; reduce heat exposure (saunas, hot kitchens).
- Stop irritating acids/scrubs and unregulated bleaching products.
- Camouflage cosmetics if wanted.

UK primary-care lightening options

- **Azelaic acid** 15–20% (also useful in pregnancy as a safer option).

- Topical retinoid (adapalene/tretinoin) off-label for pigment — avoid in pregnancy.
- Specialist: short Kligman-type triple combinations, tranexamic acid, chemical peels, carefully chosen lasers — slow response; dermal pigment is hardest.

Exogenous ochronosis & drug pigment

- Stop hydroquinone and illegal bleaching creams immediately; strict tinted sunscreen; refer — lasers are specialist and results incomplete.
- Drug photo-reactions: medication review with the prescriber, UVA-strong sunscreen, short topical steroid for acute dermatitis; Yellow Card if appropriate.

⚠ **CAUTION**

Hydroquinone caution

Long-term or high-strength HQ (especially from unregulated creams) risks exogenous ochronosis. Prefer time-limited specialist courses only.

Referral

ROUTINE	Blue-black speckled pigment after bleaching creams stop HQ, refer.
ROUTINE	Diagnostic doubt with lichenoid or contact pigment dermatology (biopsy ± patch testing).
ROUTINE	Sudden photo-distributed pigment with systemic drug — review the culprit with the original prescriber.

Red flags and when to refer

- Blue-black speckled pigment after bleaching creams stop HQ, refer.
- Diagnostic doubt with lichenoid or contact pigment dermatology (biopsy ± patch testing).
- Sudden photo-distributed pigment with systemic drug — review the culprit with the original prescriber.

Complications and prognosis

Complications

- Psychological impact; post-inflammatory hyperpigmentation from irritating treatments; exogenous ochronosis from hydroquinone.

Prognosis

- Chronic and relapsing, worse in summer; strict photoprotection is lifelong.

Prescriber pearls (UK pharmacist IP)

- The single most important advice is **strict photoprotection including visible light** (tinted sunscreen), not another cream.
- Melasma in pregnancy: sunscreen, camouflage and azelaic acid; defer stronger agents.
- Ask specifically about imported "fairness" creams — they may hide steroids, mercury or HQ.
- Set expectations: months of improvement, and relapse is common without ongoing sun protection.

Self-check quiz

5 questions · optional

Q1 When should you think of Melasma & facial melanosis?

✓ Symmetrical brown patches on cheeks, forehead or upper lip in an adult (often after pregnancy or hormonal contraception), worse with sun and heat.

Q2 What drives it?

✓ Melanocyte hyperactivity with epidermal ± dermal pigment; UV and visible light, hormones, heat and inflammation drive it.

Q3 What is the backbone of care?

✓ Daily tinted broad-spectrum SPF 30–50, trigger avoidance, and UK-available lighteners (azelaic acid, retinoid). Refer for specialist combinations.

Q4 What must you never miss?

✓ Exogenous ochronosis from prolonged hydroquinone or illegal bleaching creams — stop HQ and refer; do not escalate lighteners.

Q5 Name a red flag that needs escalation.

✓ Blue-black speckled pigment after bleaching creams stop HQ, refer.

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Photo gallery

Melasma 6

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Exogenous ochronosis and hydroquinone pigmentation 4

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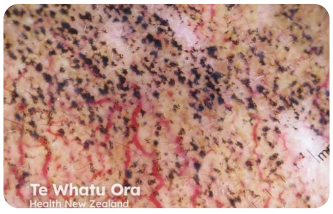

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Photo-induced drug eruptions

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Pigment & colour change Module 6 · Week 3

Contact & lichenoid pigmentation

Grey-brown facial or flexural pigment from contact allergens (Riehl/PCD) or lichenoid interface dermatitis (LPP, ashy dermatosis). Allergen avoidance and patch testing are key.

ALSO CALLED Riehl melanosis · pigmented contact dermatitis · lichen planus pigmentosus · ashy dermatosis · erythema dyschromicum perstans · patch testing

RELATED PAGES Melasma & facial melanosis · Contact dermatitis · Lichen planus · Drug-induced pigmentation (amiodarone, minocycline) · Hypopigmentation & skin of colour

KEY FACTS

- Pigmented contact dermatitis (Riehl melanosis) grey-brown facial and neck pigment with little visible eczema, mostly in **Fitzpatrick IV–VI** skin.
- Triggers: fragrances, essential oils, **PPD hair dye**, kumkum/bindi dyes, textile dyes, preservatives.
- Related: **lichen planus pigmentosus** (check the hairline for frontal fibrosing alopecia) and **ashy dermatosis**.
- Key step: **patch (± photopatch) testing**; stop non-essential cosmetics and fragrance.
- Pigment is dermal, so it fades slowly over many months; avoid potent facial steroids.

Overview

Pigmented contact dermatitis (PCD) / Riehl melanosis presents mainly as grey-brown to slate facial and neck pigment with little visible eczema, typically in Fitzpatrick IV–VI skin, often linked to cosmetics ("pigmented cosmetic dermatitis").

Lichen planus pigmentosus (LPP) and **erythema dyschromicum perstans (ashy dermatosis)** are related acquired dermal macular hyperpigmentation patterns with lichenoid histology. Overlap is common; specialist diagnosis matters.

Aetiology: causes and risk factors

- PCD: repeated low-dose allergens (fragrances, essential oils, PPD hair dye, kumkum/bindi dyes, textile dyes, preservatives) interface dermatitis dermal pigment.
- LPP: T-cell lichenoid attack; triggers may include UV, hair dyes, oils; association with frontal fibrosing alopecia — check the hairline.
- Ashy dermatosis: idiopathic ash-grey ovals on trunk/proximal limbs.

Myths to address

- **"IT'S MELASMA"**
  grey-blue tones, neck involvement and cosmetic history suggest contact or lichenoid pigmentation.

Pathophysiology and pathology

Low-grade allergy **damages the basal layer** and pigment **drops into the dermis**.

1. **Repeated allergen**
   Low-dose allergens trigger a T-cell reaction.
2. **Interface damage**
   Basal keratinocytes and melanocytes are injured (lichenoid/interface dermatitis).
3. **Pigment incontinence**
   Melanin falls into the dermis and is eaten by melanophages.

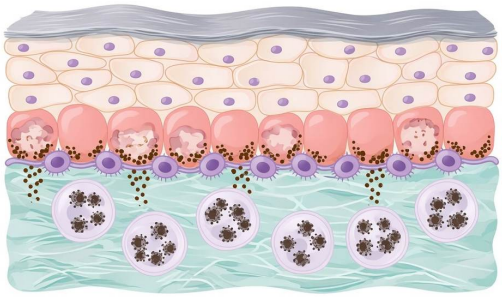

Fig. 1 · Pigment incontinence: lymphocytes damage the basal layer and melanin drops into dermal macrophages, giving a slate-grey tone.
AI-generated illustration

4. **Slate colour**
   Deep pigment looks grey-blue (Tyndall effect) and fades slowly.

HISTOPATHOLOGY

- Interface dermatitis with dermal melanophages.

Clinical features

- Gradual asymptomatic pigment; brief red/itchy flares after product use.
- Forehead, temples, hairline, lateral cheeks, neck; may spare nose and perioral skin in cosmetic patterns.
- LPP: face, neck, flexures (LPP inversus); slate-grey macules.
- Ashy dermatosis: oval ash-grey patches on trunk.

Fig. 2 · Patch testing: allergen chambers applied to the upper back, removed at 48 h, read at 48 h and ~96 h — the key investigation for suspected pigmented contact dermatitis (refer to a contact dermatitis clinic).

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Diagnosis and investigations

- Detailed cosmetic/fragrance/hair-dye/textile history.
- **Patch testing** (British baseline + cosmetic/fragrance/hair-dye + patient's own products) at a contact clinic; readings day 2 and 3–4 (± day 7).
- **Photopatch testing** when sunscreen filters, NSAIDs or photoallergens are suspected.
- Biopsy when LPP / ashy dermatosis / ochronosis need separating.

Patch testing

- (British baseline + cosmetic/fragrance/hair-dye + patient's own products) at a contact clinic; readings day 2 and 3–4 (± day 7).

Photopatch testing

- when sunscreen filters, NSAIDs or photoallergens are suspected.

Key points

- Detailed cosmetic/fragrance/hair-dye/textile history.
- Biopsy when LPP / ashy dermatosis / ochronosis need separating.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Melasma	Brown, malar/centrofacial; no allergen story

	CONDITION	DISTINGUISHING FEATURES
	Exogenous ochronosis	Hydroquinone; caviar papules
	Drug pigmentation	Drug timeline
	Berloque dermatitis	Bergamot perfume streaks + sun
	Acanthosis nigricans (flexural)	Velvety thickening

Management

- **Allergen avoidance** first: stop non-essential cosmetics, fragrances and essential oils; bland fragrance-free cleanser, emollient and mineral sunscreen.
- Refer for patch (± photopatch) testing — the key next step.
- Short mild/moderate topical steroid or topical calcineurin inhibitor for itchy flares only (avoid potent facial steroids).
- Specialist lightening and lasers are slow; pigment may take many months to fade after avoidance.

Referral

ROUTINE	Ongoing potent facial steroid from borrowed/imported creams.
ROUTINE	Diagnostic uncertainty — refer before escalating lighteners.
ROUTINE	Frontal fibrosing alopecia signs with facial pigment — dermatology.

Red flags and when to refer

- Ongoing potent facial steroid from borrowed/imported creams.
- Diagnostic uncertainty — refer before escalating lighteners.
- Frontal fibrosing alopecia signs with facial pigment — dermatology.

Complications and prognosis

Complications

- Persistent pigmentation, psychological distress.

Prognosis

- Slow improvement after allergen avoidance.

Prescriber pearls (UK pharmacist IP)

- Ask about borrowed steroid creams and imported cosmetics.
- Antihistamines do not block patch-test readings (type IV).
- Pharmacists can help patients find fragrance-free, patch-test–compatible products.
- Cross-link atopic/contact eczema pathways: eczema / dermatitis.

Self-check quiz

5 questions · optional

Q1 When should you think of Contact & lichenoid pigmentation?

✓ Slow grey-brown reticulate facial/neck pigment in darker skin, sometimes with brief itchy flares after cosmetics or fragrances.

Q2 What drives it?

✓ Type IV hypersensitivity or lichenoid basal-layer attack pigment incontinence (dermal melanophages).

Q3 What is the backbone of care?

✓ Stop fragrances/essential oils/non-essential cosmetics; fragrance-free regimen; refer for patch (± photopatch) testing.

Q4 What must you never miss?

✓ Do not keep borrowing potent facial steroids — they mask dermatitis and damage the face.

Q5 Name a red flag that needs escalation.

✓ Ongoing potent facial steroid from borrowed/imported creams.

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Photo gallery

Contact and lichenoid pigmentation 12

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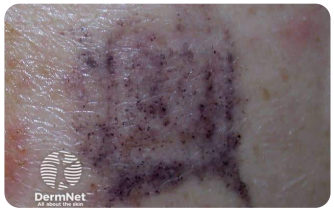

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Pigment & colour change Module 6 · Week 3

Phytophotodermatitis

Plant furocoumarins (psoralens) on the skin plus UVA cause a non-immune phototoxic burn in bizarre streaks, drips or hand-prints, followed by pigmentation that lasts months.

ALSO CALLED phytophotodermatitis · phototoxic plant dermatitis · berloque dermatitis · giant hogweed burn · lime burn · margarita dermatitis · +1

RELATED PAGES Cutaneous amyloidosis (macular and lichen) · Drug-induced photosensitivity · Polymorphic light eruption · Blister beetle & Paederus dermatitis

KEY FACTS

- A **non-immune phototoxic** reaction: plant **furocoumarins** (psoralens) on the skin plus **UVA** light.
- Anyone can be affected with enough sap and sun; no previous sensitisation is needed.
- Burning redness and **blisters** appear about **24 h** after exposure and peak at **48–72 h**, in **streaks, drips or hand-prints**.
- **Hyperpigmentation** follows and can last months to years; it is often why people present.
- Manage like a superficial burn: wash sap off, keep out of sun for 48 h, short topical steroid, strict sun protection. Think **safeguarding** with odd marks in children.

Overview

Phytophotodermatitis is a **phototoxic** skin reaction that happens when light-sensitising chemicals in some plants (**furocoumarins**, also called psoralens) get on the skin and are then exposed to **ultraviolet A (UVA)**. It is not an allergy: the sap and light damage the skin directly, so it can happen to anyone at first exposure.

- The burn has **bizarre, geometric shapes** (streaks, drips, splashes, finger- and hand-prints) that trace where the juice or sap touched.
- It heals with **brown pigmentation** that can persist for months.
- Classic UK sources: **giant hogweed**, wild parsnip and rue (strimming, walking), and **limes** (cocktails and cooking on holiday).
- **Berloque dermatitis** is a variant from bergamot-containing perfume on the neck.

Epidemiology

- **Who**
  - Any age, sex or skin type; incidence is unknown because mild cases are not reported.

- **When**
  - Mostly **summer** and sunny holidays, when people handle plants or citrus outdoors.
- **At-risk groups**
  - Gardeners, farm and grounds workers, bartenders and chefs (limes), children playing outdoors, hikers.
  - Fair skin may burn more readily, while darker skin shows more marked and longer-lasting pigmentation.

Aetiology: causes and risk factors

Two things are needed: a **furocoumarin on the skin** and enough **UVA**. Moisture (sweat, water) and heat increase absorption and the reaction.

Common plant sources

FAMILY	EXAMPLES
Apiaceae (Umbelliferae)	**Giant hogweed**, cow parsley, wild and cultivated **parsnip**, celery, carrot, fennel, dill, parsley
Rutaceae	**Lime**, lemon, bergamot orange, bitter orange, grapefruit, **rue**
Moraceae	**Fig** (leaves and sap)
Other	Some St John's wort and other garden plants

> **ACTIVITIES**
> strimming or mowing weeds, clearing riverbanks, picking wild plants, squeezing limes or making drinks in the sun.

> **BERLOQUE DERMATITIS**
> perfume containing bergamot oil (5-methoxypsoralen) applied to the neck, then sun.

Myths to address

> **"IT MUST BE AN ALLERGY"**
> no; phototoxicity needs no sensitisation and anyone can react.

> **"IT WAS A SUNBURN"**
> plain sunburn is uniform over exposed skin; the streaks and drips point to a plant or citrus.

> **"THE MARKS ARE BRUISES OR SCARS"**
> they are pigment and fade slowly without scarring.

Pathophysiology and pathology

1 **Sap meets light**

Furocoumarins in plant sap or citrus juice penetrate the epidermis. **UVA (320–400 nm, peak effect around 335 nm)** activates them; a cloudy day still delivers enough UVA.

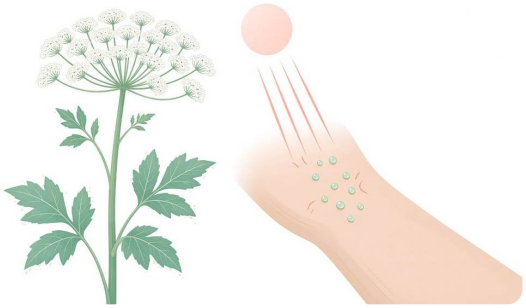

Fig. 1 · Step 1: furocoumarin-rich sap (e.g. hogweed) on skin, then UVA exposure.

AI-generated illustration

2 **Phototoxic damage to DNA and membranes**

Activated psoralens act in two ways: **type I** reactions bind directly to **DNA**, forming cross-links between strands; **type II** reactions generate reactive oxygen species that damage cell membranes. No immune memory is involved.

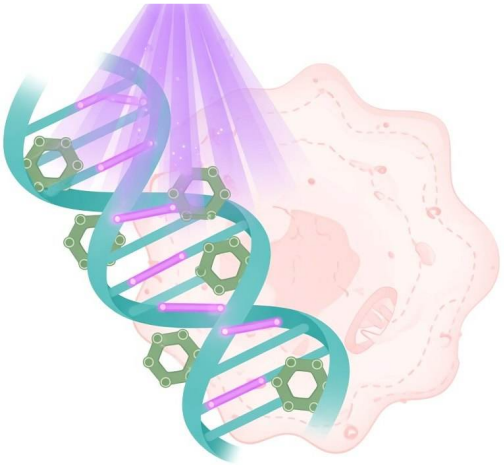

Fig. 2 · Step 2: UVA-activated psoralen molecules slot between DNA strands and cross-link them.

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3 **Keratinocyte death and blistering**

Damaged keratinocytes die ("sunburn cells"), with dermal oedema and inflammation. Severe damage separates the epidermis, producing **vesicles and bullae** after about 24 h, peaking at 48–72 h.

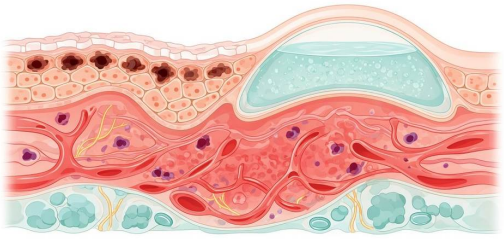

Fig. 3 · Step 3: necrotic keratinocytes and an intraepidermal blister above an inflamed dermis.

AI-generated illustration

4 **Pigment follows**

Psoralen–UVA stimulates **melanocytes** to make more melanin and melanin drops into the dermis where **melanophages** hold it, so brown or grey-brown pigment appears 1–2 weeks later and lasts months.

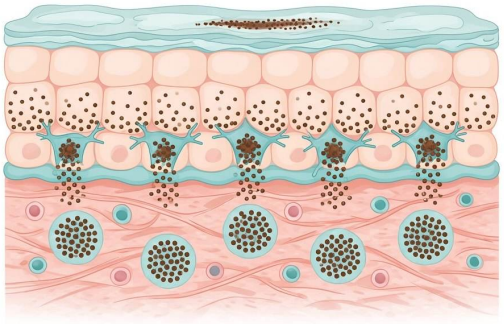

Fig. 4 · Step 4: active melanocytes and pigment-laden melanophages in the upper dermis cause the long-lasting brown marks.

AI-generated illustration

Fig. 5 · The shape gives it away: drips, streaks and hand-prints that follow where juice or sap touched the skin.

AI-generated illustration

HISTOPATHOLOGY

- Acute: necrotic keratinocytes, spongiosis, intraepidermal or subepidermal **vesicles**, dermal oedema and a mild perivascular infiltrate (like an acute sunburn).
- Late: **pigment incontinence** with melanophages in the upper dermis and increased basal melanin.
- Biopsy is rarely needed; history and shape are diagnostic.

Clinical features

Symptoms

- **Burning or stinging pain** more than itch, starting about 24 h after sun exposure.
- Usually no systemic symptoms, except with extensive hogweed burns.
- Later, **brown marks** with no memory of the inflammatory phase: often the presenting complaint.

Lesions and morphology

Streaky erythema

Sharply outlined **linear streaks** of redness and oedema where sap brushed

© DermNet · **Source**

Vesicles and bullae

Tense blisters, sometimes large, resembling a partial-thickness burn.

© Dr Richard Ashton · **Source**

Drips and hand-prints

Drip, splash and finger-print shapes from citrus juice.

© DermNet · **Source**

Hyperpigmentation

Brown to grey-brown streaks that persist for months after the burn settles.

© Dr Richard Ashton · **Source**

Distribution

- **Exposed skin** that touched the plant or juice: hands, forearms, legs and ankles (strimming), around the mouth and chin (citrus), neck (perfume).
- Shapes follow contact, **not** the whole sun-exposed area (unlike drug photosensitivity).

Fig. 10 · Linear redness and blisters after strimming.

© DermNet · **Source**

Fig. 11 · Streaks where juice brushed the skin: an outside cause.

© DermNet · **Source**

Variants

Lime ("margarita") dermatitis

Drips and hand-prints on hands, wrists, face and chest after squeezing limes in the

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Giant hogweed burn

Severe blistering on arms and legs after contact with sap on riverbanks and

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Bullous phytophotodermatitis

Large blisters, usually on the lower legs and ankles.

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Pigmentation-only presentation

Streaky brown marks without remembered redness.

© Te Whatu Ora · **Source**

Berloque dermatitis

Pendant-shaped streaks on the neck from bergamot perfume plus sun.

Special groups

- **Children**
  - Hand-print or splash patterns can look like **bruising or non-accidental injury**; take a careful plant and sun history and follow safeguarding procedures if the story does not fit.
- **Skin of colour**
  - Redness may be subtle; the **hyperpigmentation** is often marked and lasts longer.

Fig. 16 · Residual pigmentation for months; atypical patterns in children need a careful history.
© DermNet · **Source**

Diagnosis and investigations

Diagnosis is **clinical**, from the **bizarre shape** plus an **exposure history**.

History

- Plants handled, strimming or gardening, walks through long vegetation, citrus drinks or cooking, perfumes.
- Sun exposure in the previous 1–2 days; timing of symptoms.
- Drugs (to exclude drug photosensitivity); previous similar episodes.

Examination

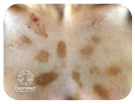
- Shape and site of lesions; blister size and extent; eye involvement; signs of infection.

Investigations

- **Usually none.**
- Swab if secondary infection is suspected.
- Patch or photopatch testing only if allergic contact dermatitis is a possibility.
- Biopsy rarely.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Allergic contact dermatitis (plants)	Very **itchy** eczema; needs sensitisation; does not need sun; spreads beyond the contact.

	CONDITION	DISTINGUISHING FEATURES
	Drug photosensitivity	**Uniform** redness over all sun-exposed skin with a drug history (e.g. doxycycline, thiazides).
	Sunburn	Uniform over exposed sites; no streaks.
	Blister beetle dermatitis	Linear blisters after crushing a beetle; no sun needed.
	Polymorphic light eruption	Itchy papules on sun-exposed skin hours after sun; recurrent each spring.
	Non-accidental injury	Inconsistent or changing history, injuries in unusual places, other concerns: **safeguarding** assessment.
	**Burns or artefact**	History of thermal or chemical injury; patterns may be self-inflicted.
	Fixed drug eruption	Round dusky patches that recur at the same site with each dose.

Management

Principles

- Treat the acute phase like a **superficial burn**; then protect the pigmented skin from the sun.
- Explain the cause so the patient can avoid it; most cases settle without scarring.

Immediate care after contact (e.g. hogweed sap)

- **Wash off sap promptly with soap and water**, cover the skin and keep it **out of sunlight for at least 48 h**.
- If sap gets in the eyes: rinse and seek **urgent eye assessment**.

Conservative and self-care

- Cool compresses, emollients and simple analgesia.
- Leave blisters intact or use non-adherent dressings; keep clean (topical antiseptic only if needed).
- **OTC hydrocortisone 1%** for mild small areas (not on broken skin).

Medical treatment

Potent topical corticosteroid

Short course

DOSE	Short course of **5–7 days** to inflamed **intact** skin on the limbs or trunk; **moderate** potency on the face.
CAUTIONS	Not on broken or infected skin.
COUNSELLING	Apply thinly; stop when redness settles.

Oral antihistamine

Symptom relief

USE	If itchy.

Secondary infection

Antibiotic per NICE guidance

USE	Infected blisters: treat per NICE impetigo / secondary bacterial infection guidance.
COUNSELLING	Link: impetigo.

Pigmentation phase

- Reassure: it **fades slowly over months**.
- **Strict sun protection** (even limited UV can darken it again): clothing, broad-spectrum high-SPF sunscreen.
- Avoid bleaching agents unless advised by a specialist.

Prevention

- Learn to recognise **giant hogweed**; wear long sleeves, trousers and gloves when strimming or clearing plants.
- Wash skin after gardening; avoid squeezing limes or applying citrus or perfume to skin before sun.

Treatment ladder

1

Everyone

Wash off sap, 48 h out of sun, cool compresses, analgesia, emollient.

2

Inflamed intact skin

Potent topical steroid 5–7 days (moderate on face); antihistamine if itchy.

3

Pigmentation

Reassurance and strict sun protection for months.

4

Burns-level injury

Large or blistering burns, face, genitals or hands: same-day urgent care or burns unit.

Referral

SAME DAY	Large or extensive blistering, burns on the **face, genitals or hands**, severe pain or infection: same-day GP, urgent care or **burns unit**.
SAME DAY	Sap in the eye: urgent eye assessment.
URGENT	**Safeguarding** concern in a child: follow local procedures.
ROUTINE	Diagnostic doubt or persistent distressing pigmentation.

Red flags and when to refer

SAME DAY	Extensive hogweed burns, eye exposure, or blistering of the face, hands or genitals.
URGENT	Unexplained marks in a child where the history does not fit: safeguarding.

ROUTINE Signs of infection: spreading redness, pus, fever.

Complications and prognosis

- **Post-inflammatory hyperpigmentation** for months, sometimes years.
- Secondary bacterial infection of blisters.
- Rarely scarring after severe bullous burns.
- Misdiagnosis as abuse or burns causing distress.

Prognosis

- The acute reaction settles in 1–2 weeks; pigment fades gradually over months with sun protection.
- Recurs only with re-exposure.

Prescriber pearls (UK pharmacist IP)

- Pigmentation, not the acute burn, is often why patients attend.
- Berloque = perfume + sun streaks.
- Tell gardeners about **giant hogweed**: sap plus sun is a classic UK summer presentation.
- Ask about holidays, cocktails and cooking with limes.
- Check for photosensitising drugs before blaming the plant.
- Other light-related rashes: polymorphic light eruption, drug-induced photosensitivity and chronic actinic dermatitis.

Self-check quiz

6 questions · optional

Q1 What two things cause phytophotodermatitis?

✓ Plant furocoumarins (psoralens) on the skin plus UVA light.

Q2 Is it allergic?

✓ No: it is phototoxic and needs no previous sensitisation.

Q3 When does it appear and peak?

✓ About 24 h after exposure, peaking at 48–72 h.

Q4 What clue separates it from drug photosensitivity?

✓ Bizarre streaks, drips and hand-prints that follow contact, not uniform sun-exposed redness.

Q5 Immediate advice after hogweed sap contact?

✓ Wash with soap and water, cover and keep out of sunlight for at least 48 h.

Q6 Why mention safeguarding?

✓ Hand-prints and splashes in children can mimic non-accidental injury.

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Photo gallery

Phytophotodermatitis 6

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Phytophotodermatitis (Module 5 Week 6) 5

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Pigment & colour change Module 6 · Week 3

Cutaneous amyloidosis (macular and lichen)

Localised keratin-derived amyloid in the papillary dermis: rippled grey-brown patches on the upper back (macular) or very itchy firm papules on the shins (lichen), driven by chronic friction and scratching.

ALSO CALLED macular amyloidosis · lichen amyloidosis · primary localised cutaneous amyloidosis · PLCA · friction amyloidosis · amyloid

RELATED PAGES Phytophotodermatitis · Lichen planus · Contact & lichenoid pigmentation · Acanthosis nigricans

KEY FACTS

- Primary localised cutaneous amyloidosis amyloid made from **damaged keratinocytes** builds up in the papillary dermis; it is **not** systemic amyloidosis.
- Two common forms: **macular** (rippled grey-brown patches, upper back) and **lichen** (very itchy firm papules, shins); they coexist in about a quarter of cases.
- Most common in people of **Asian, Middle Eastern and South American** descent with skin types IV–V.
- Driven by chronic **friction and scratching** (nylon scrubs, brushes, notalgia paraesthetica); about 10% are familial.
- Clinical diagnosis ± biopsy (**Congo red, apple-green birefringence**). Stop friction, potent steroid ± occlusion, treat itch; refer if refractory.

Overview

Cutaneous amyloidosis is a group of conditions in which **amyloid** (abnormally folded protein) is deposited in the skin. In the common **primary localised** forms, the amyloid comes from **keratin** released by damaged keratinocytes, and it stays in the skin.

> **MACULAR AMYLOIDOSIS**
> itchy or symptomless, grey-brown, **rippled** (reticulate) pigmentation, usually between the shoulder blades.

> **LICHEN AMYLOIDOSIS**
> intensely **itchy**, firm, brown, scaly papules on the shins, often in rows, merging into plaques.

> **Biphasic** amyloidosis: both forms together.

> Rare forms (nodular amyloidosis, which can link to systemic disease) and **systemic** amyloidosis are different conditions needing specialist assessment.

Epidemiology

- **Ethnicity and skin type**
  - Most common in people from **Asia, the Middle East and South America**, and in **skin types IV–V**.

- **Age and sex**
  - Usually adults; macular amyloidosis is more common in **women**; lichen amyloidosis is more common in **men** aged 50–60.
- **Family history**
  - Up to **10%** are familial (autosomal dominant, **OSMR** or **IL31RA** genes, part of the itch pathway).
- **Overlap**
  - Macular and lichen amyloidosis **coexist in about 25%**.

Aetiology: causes and risk factors

Chronic **friction and scratching** damage keratinocytes; their keratin is processed into amyloid in the upper dermis.

Causes and associations

- **Friction**
  - Nylon towels, bath brushes, loofahs, back scrubbers and rough clothing ("friction amyloidosis").
- **Chronic itch**
  - **Notalgia paraesthetica** (itchy patch below the shoulder blade) often leads to macular amyloidosis.
  - **Atopic dermatitis** and lichen simplex are linked with lichen amyloidosis.
- **Genetic**
  - Familial forms (OSMR, IL31RA); rarely **multiple endocrine neoplasia type 2A** (RET mutations) with interscapular amyloidosis.
- Sun exposure and darker skin types are risk factors.
- Occasionally associated with autoimmune conditions.

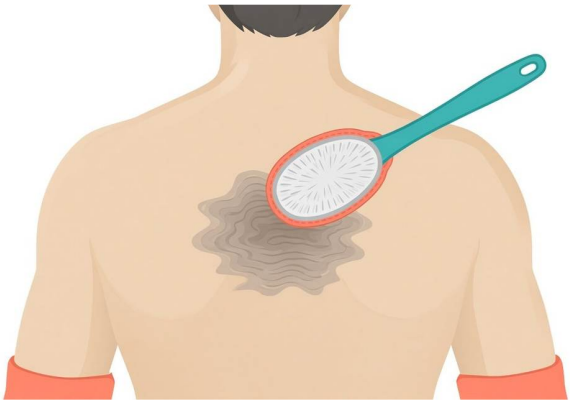

Fig. 1 · Repeated scrubbing of the upper back with a brush or nylon towel is a common trigger of macular amyloidosis.

AI-generated illustration

Myths to address

> **"IT'S DIRT, SO SCRUB HARDER"**
> scrubbing is the cause and makes it worse.

> **"AMYLOID IN THE SKIN MEANS ORGAN DISEASE"**
> localised cutaneous amyloidosis is not systemic amyloidosis.

Pathophysiology and pathology

1 **Friction and scratching injure keratinocytes**

Repeated rubbing or scratching damages basal keratinocytes, which undergo apoptosis.

2 **Keratin becomes amyloid**

Keratin filaments from dying cells form **colloid (keratin) bodies** that drop into the papillary dermis, where macrophages and fibroblasts modify them into **amyloid**.

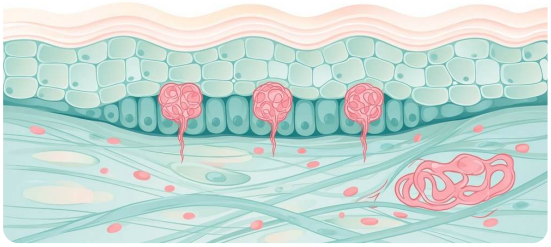

Fig. 2 · Step 2: damaged basal keratinocytes shed keratin bodies into the papillary dermis.

AI-generated illustration

3 **Deposits and pigment build up**

Small globules of amyloid collect in the **dermal papillae** with melanin-laden macrophages, giving the grey-brown colour. In lichen amyloidosis the epidermis thickens over them, forming papules.

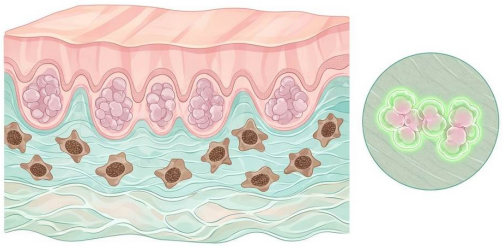

Fig. 3 · Step 3: amyloid globules in the dermal papillae with pigment-laden macrophages; Congo-red staining shows apple-green birefringence (inset).

AI-generated illustration

4 **Itch keeps the cycle going**

Amyloid and the thickened skin itch, so scratching continues and more amyloid forms. Faults in the **OSMR / IL-31** itch pathway explain familial cases.

HISTOPATHOLOGY

- Small globules of pink amorphous **amyloid** in the **dermal papillae**, just beneath the epidermis.
- **Pigment incontinence** with melanophages.
- Lichen amyloidosis: acanthosis and hyperkeratosis over the deposits.
- **Congo red** stains amyloid, with **apple-green birefringence** in polarised light; crystal violet and thioflavin T also help; immunostaining shows keratin origin.
- Deposits can be sparse, so multiple sections or a deeper biopsy may be needed.

Clinical features

Symptoms

- **ITCH**
  mild or absent in macular amyloidosis, often **severe and persistent** in lichen amyloidosis.
- Cosmetic concern about pigmentation.

Lesions and morphology

Rippled grey-brown macules

Small dusky macules merge into a **rippled or reticulate** pattern.

© DermNet · **Source**

Interscapular patch

Ill-defined pigmentation between the shoulder blades.

© DermNet · **Source**

Firm itchy papules

Tan-brown, scaly, dome-shaped papules, often in rows, on the shins.

© DermNet · **Source**

Confluent plaques

Papules merge into thickened, ridged plaques.

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Distribution

MACULAR

interscapular back, upper chest, shoulders and outer upper arms.

LICHEN

shins, then thighs, forearms and the backs of the hands.

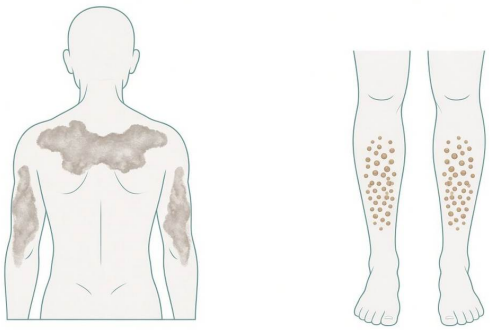

Fig. 8 · Macular amyloidosis on the upper back, shoulders and upper arms; lichen amyloidosis on the shins.

AI-generated illustration

Variants and subtypes

Macular amyloidosis

Rippled grey-brown patches, upper back; women; friction.

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Notalgia-linked macular amyloidosis

Pigmented patch below one shoulder blade from chronic scratching of notalgia

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Lichen amyloidosis

Intensely itchy papules on the shins; atopy.

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Biphasic

Both forms together in about a quarter.

Nodular amyloidosis

Waxy nodules; light-chain amyloid; may relate to plasma cell disorders: specialist work-up.

Special groups

- **Skin of colour**
  - The most affected group; pigmentation is often the main concern and can be confused with post-inflammatory hyperpigmentation.

- **Families**
  - Ask about relatives; consider MEN2A in familial interscapular amyloidosis (specialist).

Diagnosis and investigations

Diagnosis is usually **clinical** from the rippled pattern, site and history of friction or itch.

History

- Scrubbing habits, brushes, towels; chronic itch; atopy; family history.
- Systemic symptoms (to exclude systemic amyloidosis): tiredness, weight loss, purpura, enlarged tongue, swelling, frothy urine.

Investigations

- **DERMOSCOPY**
  a central white or brown hub with surrounding pigment.
- **Skin biopsy** with Congo red when the diagnosis is uncertain.
- If systemic or nodular amyloidosis is suspected: specialist investigations (serum and urine protein electrophoresis, free light chains).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Post-inflammatory hyperpigmentation	Follows a known rash; no ripple; no papules.
	Lichen planus pigmentosus	Grey-brown patches on the face, neck and flexures; may follow lichen planus.
	Hypertrophic lichen planus	Thick violaceous plaques on the shins; Wickham striae; biopsy.
	Lichen simplex chronicus / prurigo	Single lichenified plaque or nodules from scratching; no amyloid on biopsy.
	Acanthosis nigricans	Velvety pigmentation in the flexures and neck; insulin resistance.
	Pretibial myxoedema	Waxy plaques on shins with thyroid eye disease.
	Drug-induced pigmentation	Drug history (e.g. minocycline, amiodarone).

Management

Principles

- Break the **itch–scratch–friction** cycle; set realistic expectations: pigment improves slowly and recurs if friction restarts.
- Treatment is often difficult; many options are specialist and evidence is limited.

Conservative and self-care

- **STOP SCRUBBING**
  no loofahs, brushes or nylon towels; soft, loose clothing.
- Regular **emollients**; keep nails short.
- Sun protection for pigmentation.

Medical treatment

Potent topical corticosteroid

Anti-itch, anti-inflammatory

USE Itch and lichen amyloidosis; occlusion improves effect.

CAUTIONS Limit courses and monitor for atrophy, especially under occlusion.

Topical calcineurin inhibitor

Steroid-sparing (off-label)

USE Itch where long-term steroid is unwanted.

Sedating antihistamine at night

Sleep

USE To help sleep when itch is severe; does not treat the itch mechanism.

Specialist options

Dermatology **SPECIALIST ONLY**

USE Intralesional steroid, topical capsaicin or DMSO, phototherapy (NB-UVB, PUVA), oral acitretin, laser or dermabrasion for lichen amyloidosis.

Treatment ladder

1

Everyone

Stop friction, emollients, explanation.

2

Itchy or papular

Potent topical steroid ± occlusion, or calcineurin inhibitor; night sedating antihistamine if sleep is affected.

3

Refractory

Refer: phototherapy, retinoids, procedures.

Referral

URGENT — Features of **systemic amyloidosis** or **nodular** amyloidosis.

ROUTINE — Severe or refractory itch, diagnostic doubt (biopsy), or for specialist treatments.

Red flags and when to refer

URGENT — Purpura (especially around the eyes), enlarged tongue, nerve symptoms, heart or kidney symptoms: possible **systemic** amyloidosis.

ROUTINE — Waxy nodules (nodular amyloidosis): screen for plasma cell disease.

Complications and prognosis

- Persistent itch, sleep disturbance and lichenification.
- Long-lasting **cosmetic** pigmentation and distress.
- Steroid side effects with long courses under occlusion.

Prognosis

- **Chronic** and benign; it does not progress to systemic amyloidosis.
- Improves slowly once friction and scratching stop; tends to recur.

Prescriber pearls (UK pharmacist IP)

- Ask about **back scrubbers, brushes and nylon towels**: stopping them is the key treatment.
- Rippled interscapular pigmentation in skin of colour is often macular amyloidosis, not dirt or PIH.
- An itchy patch under one shoulder blade suggests **notalgia paraesthetica**.
- Potent steroid under occlusion helps lichen amyloidosis, but time-limit courses.
- Localised cutaneous amyloid is **not** systemic amyloidosis; refer if systemic features appear.
- Related: phytophotodermatitis (the other half of the Module 6 · Week 3 topic).

Self-check quiz

6 questions · optional

Q1 Where does the amyloid in macular and lichen amyloidosis come from?

✓ Keratin from damaged keratinocytes.

Q2 Typical site of macular amyloidosis?

✓ Between the shoulder blades (interscapular back).

Q3 Typical site of lichen amyloidosis?

✓ Shins.

Q4 Which stain confirms amyloid?

✓ Congo red, with apple-green birefringence under polarised light.

Q5 Key self-care advice?

✓ Stop scrubbing with brushes, loofahs or nylon towels.

Q6 Does it progress to systemic amyloidosis?

✓ No; systemic features need separate specialist assessment.

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Photo gallery

Cutaneous amyloidosis 5

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Pigment & colour change Module 6 · Week 3

Hypopigmentation & skin of colour

PIH and physiological pigment in skin of colour, plus piebaldism vs vitiligo and the harms of unlicensed bleaching creams.

ALSO CALLED PIH | post-inflammatory hyperpigmentation | piebaldism | vitiligo | pityriasis alba | bleaching cream | +1

RELATED PAGES Melasma & facial melanosis | Skin layers, Fitzpatrick skin types & dermoscopy | Progressive macular hypomelanosis (PMH) | Contact & lichenoid pigmentation | Leprosy (Hansen disease) & lepra reactions

KEY FACTS

- In skin of colour, **post-inflammatory hyperpigmentation** and flexural darkening are everyday problems; true **depigmentation** (vitiligo, piebaldism) is separate.
- Treat the driver of PIH (acne, eczema, shaving friction); daily tinted broad-spectrum sunscreen; azelaic acid.
- Warn about **unlicensed bleaching creams** (mercury, potent steroids, high-strength hydroquinone).
- Piebaldism is congenital and stable (white forelock); vitiligo is acquired and autoimmune.
- Pale patches: think pityriasis alba, PIH (hypo), pityriasis versicolor, PMH, vitiligo, leprosy (numb).

Overview

In skin of colour, **post-inflammatory hyperpigmentation (PIH)** and constitutional/flexural darkening are everyday presentations. True **depigmentation** (vitiligo, piebaldism) needs a different pathway. This page also flags harm from **unlicensed bleaching and steroid creams**.

Aetiology: causes and risk factors

- PIH: any inflammation (acne, eczema, bites, procedures, friction from shaving) leaves excess epidermal ± dermal melanin.
- Constitutional pigment: periorbital dark circles, maturational and flexural darkening.
- Piebaldism: congenital KIT-pathway melanocyte absence (stable white forelock + patches).
- Vitiligo: acquired autoimmune melanocyte loss; associated thyroid and other autoimmunity.

Myths to address

- **"BLEACHING CREAMS ARE SAFE COSMETICS"**
  illegal products cause steroid damage, ochronosis and mercury toxicity.

Pathophysiology and pathology

Pigment change after inflammation depends on **where the melanin ends up**.

1. **Epidermal PIH**
   Inflamed melanocytes over-produce pigment in the epidermis: brown, fades in months.
2. **Dermal PIH**
   Basal damage drops pigment into the dermis: grey-blue, slow to fade.

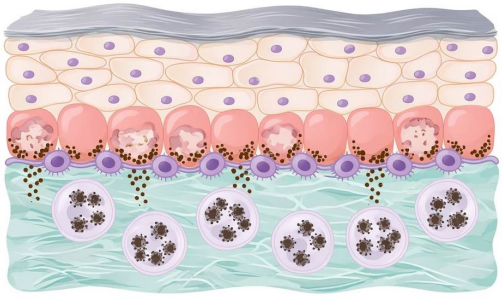

Fig. 1 · Pigment incontinence makes deeper, greyer pigment.
AI-generated illustration

3. **Hypopigmentation**
   Inflammation may reduce melanin transfer, leaving pale areas.
4. **Depigmentation**
   Absent melanocytes (vitiligo, piebaldism) give chalk-white skin.

Clinical features

PATTERN	CLUES
PIH	Prior inflammation/injury; brown to grey; fades slowly
Pityriasis alba	Children; ill-defined slightly scaly pale patches; hypo- not de-pigmented
Piebaldism	Present at birth; white forelock; stable islands of normal skin inside patches
Vitiligo	Acquired milky-white patches; Wood's lamp enhances; may spread
Bleaching harm	Atrophy, striae, telangiectasia, ochronosis, infection

Fig. 2 · Non-segmental vitiligo at the wrists — acquired, symmetrical, milky-white under Wood's lamp; can progress; ask about thyroid/ autoimmune history.

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Types to recognise

PIH

Prior inflammation/injury; brown to grey; fades slowly

Pityriasis alba

Children; ill-defined slightly scaly pale patches; hypo- not de-pigmented

Piebaldism

Present at birth; white forelock; stable islands of normal skin inside patches

Vitiligo

Acquired milky-white patches; Wood's lamp enhances; may spread

Bleaching harm

Atrophy, striae, telangiectasia, ochronosis, infection

Diagnosis and investigations

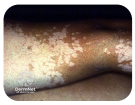
- History of inflammation, procedures, shaving, cosmetics and bleaching products.
- Wood's lamp: chalk-white sharp edges favour vitiligo/piebaldism over pityriasis alba.
- Vitiligo: consider TSH ± antibodies if indicated; piebaldism: genetics/hearing if Waardenburg features.
- Cross-check fungal differentials (pityriasis versicolor) when scale is present.

Key points

- History of inflammation, procedures, shaving, cosmetics and bleaching products.
- Wood's lamp: chalk-white sharp edges favour vitiligo/piebaldism over pityriasis alba.
- Vitiligo: consider TSH ± antibodies if indicated; piebaldism: genetics/hearing if Waardenburg features.

- Cross-check fungal differentials (pityriasis versicolor) when scale is present.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	**Pityriasis versicolor**	Fine scale, KOH positive
	Naevus depigmentosus	Stable segmental pale patch from early life
	Idiopathic guttate hypomelanosis	Small white shin/arm macules in older adults
	Chemical leukoderma	Phenols/monobenzone contact
	Acanthosis nigricans	Velvety flexural thickening — see acanthosis nigricans

Management

PIH and flexural darkening

- Treat the driver (acne, eczema, bites). Stop harsh scrubs, lemon juice and bleaching creams.
- Hair removal in darker skin: prefer clippers/trimming or accredited long-pulsed Nd:YAG; avoid aggressive IPL settings.
- Daily tinted broad-spectrum sunscreen; azelaic acid; topical retinoid off-label; time-limited specialist HQ only.
- Vitamin D advice for people with darker skin who avoid sun (UK public-health framing).

Vitiligo / piebaldism

- Piebaldism: reassurance that it is stable; cosmetic options; genetics if needed — no “repigmenting cream” expectation.
- Vitiligo: potent topical steroid or calcineurin inhibitor on limited areas; phototherapy/JAK inhibitors are specialist; camouflage and psychological support.
- Refer early for extensive, facial, or rapidly progressive vitiligo.

⚠ **RED FLAG**

Illegal bleaching / steroid creams

Advise stopping immediately. Counsel on skin atrophy, infection, adrenal suppression and ochronosis. Signpost licensed pathways and Trading Standards/MHRA concerns for supply.

Referral

ROUTINE	Rapidly progressive vitiligo on the face — early dermatology.
ROUTINE	White forelock plus deafness/heterochromia Waardenburg work-up.
ROUTINE	Systemic steroid side-effects from bleaching creams — medical review.

Red flags and when to refer

- Rapidly progressive vitiligo on the face — early dermatology.
- White forelock plus deafness/heterochromia Waardenburg work-up.
- Systemic steroid side-effects from bleaching creams — medical review.

Complications and prognosis

Complications

- Psychological impact; harm from bleaching products.

Prognosis

- PIH fades over months to years if the trigger stops.

Prescriber pearls (UK pharmacist IP)

- Set timelines: epidermal PIH months; dermal longer.
- Ask what people already put on their skin — many worsen PIH with home acids and bleaching.
- Milky-white + Wood's lamp pityriasis alba.
- See also progressive macular hypomelanosis and pityriasis versicolor.

Self-check quiz

5 questions · optional

Q1 When should you think of Hypopigmentation & skin of colour?

✓ Dark marks after acne/eczema/friction, or milky-white patches — decide hyperpigmentation vs true depigmentation (vitiligo/piebaldism).

Q2 What drives it?

✓ Inflammation drives melanin excess (PIH) or autoimmune/congenital melanocyte loss (vitiligo/piebaldism).

Q3 What is the backbone of care?

✓ Treat the cause, stop friction and illegal bleaching, daily tinted SPF, azelaic acid/retinoid for PIH; refer vitiligo/piebaldism pathways.

Q4 What must you never miss?

✓ Unlicensed steroid/mercury/hydroquinone "fairness" creams — stop and counsel on harm; Waardenburg features with white forelock need hearing assessment.

Q5 Name a red flag that needs escalation.

✓ Rapidly progressive vitiligo on the face — early dermatology.

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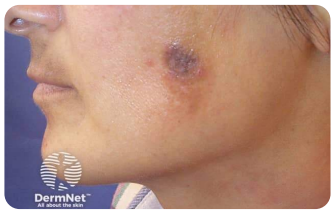
Photo gallery

PIH and physiological pigmentation 8



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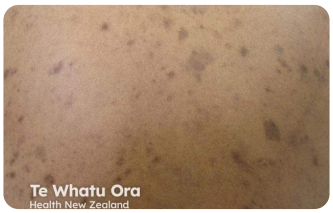

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Piebaldism 4

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Vitiligo 5

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Pityriasis alba and hypopigmentation

2

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Harm from unlicensed bleaching / steroid creams

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CHAPTER 14

Childhood & neonatal

Module 6 Week 1: neonatal pustular rashes, infant acne, nappy rash, vascular and pigmented birthmarks, epidermolysis bullosa and aplasia cutis.

1. Neonatal pustular rashes
2. Neonatal & infantile acne
3. Napkin (nappy) dermatitis
4. Infantile haemangioma (strawberry mark)
5. Pigmented & epidermal birthmarks
6. Port-wine stain (capillary malformation)
7. Epidermolysis bullosa (EB)
8. Aplasia cutis congenita

Childhood & neonatal Module 6 · Week 1

Neonatal pustular rashes

Erythema toxicum, TNPM and miliaria are benign and need reassurance. Neonatal HSV is the one you must never ‘wait and see’ on. Includes a one-table differential.

ALSO CALLED erythema toxicum neonatorum · ETN · toxic erythema of the newborn · transient neonatal pustular melanosis · TNPM · miliaria · +7

RELATED PAGES Neonatal & infantile acne · Milia · Napkin (nappy) dermatitis · Infantile haemangioma (strawberry mark) · Herpes simplex labialis (cold sores)

KEY FACTS

- Most neonatal rashes are **benign and transient**: erythema toxicum neonatorum (ETN), transient neonatal pustular melanosis (TNPM), miliaria, milia and neonatal acne.
- ETN: blotchy red macules with central pustules in days 1–3; spares palms and soles; eosinophils.
- TNPM: pustules at birth that rupture to leave collarettes and brown macules, more in darker skin.
- Miliaria: blocked sweat ducts from overheating; clear vesicles (crystallina) or red papules (rubra).
- Red flags: unwell baby, vesicles (**HSV**: emergency aciclovir), purpura, blistering, widespread pustules with fever.

Overview

RASH	IN ONE LINE
Erythema toxicum neonatorum (ETN)	Common, benign, self-limiting rash in up to about half of **full-term** babies; uncommon in premature babies
Transient neonatal pustular melanosis (TNPM)	Uncommon benign pustular rash **present at birth**, more common in darker skin; leaves pigmented macules
Miliaria (heat rash)	Blocked or inflamed **eccrine sweat ducts**; common with immature ducts, heat and overbundling

Epidemiology

- **Who**
  - ETN affects up to half of term babies; TNPM about 5% of Black infants; miliaria in hot, overwrapped babies.

Aetiology: causes and risk factors

> **ETN**
> the exact cause is unknown (theories include a reaction to mechanical or thermal stimuli, or a transient immune response). Pustules are **sterile** and **eosinophil-rich**. It is not infection and not poor hygiene.

> **TNPM**
> cause unknown, possibly a 'precocious' form of ETN. Pustules are sterile and **neutrophil-rich**.

> **MILIARIA**
> the level of sweat-duct blockage determines the type (below).

MILIARIA TYPE	LEVEL OF BLOCK	CLINICAL GIST
Crystallina	Stratum corneum	Clear 'dewdrop' vesicles, no inflammation
Rubra	Mid-epidermis	Itchy red papules or papulovesicles
Pustulosa	Variant of rubra	Pustules
Profunda	Dermo-epidermal junction or deeper	Rare; flesh-coloured deep papules (mainly adults)

Myths to address

- **"SPOTS IN A NEWBORN MEAN INFECTION"**
  most are harmless, but vesicles or an unwell baby need urgent review.

Pathophysiology and pathology

Newborn skin is **adapting** to air, bacteria and temperature regulation.

1. **Immune adaptation**
   ETN reflects an innate immune response to colonising bacteria at hair follicles (eosinophils).

2. **Sweat duct immaturity**
   Immature ducts block easily, causing miliaria when overheated.

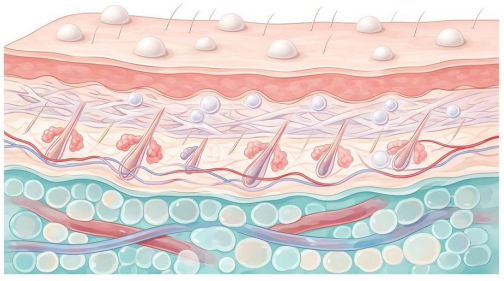

Fig. 1 · Milia: tiny keratin cysts common on newborn noses and cheeks.
AI-generated illustration

3. **Maternal hormones**
   Maternal androgens stimulate sebaceous glands (neonatal acne, sebaceous hyperplasia).

Clinical features

Erythema toxicum neonatorum

- Onset in the **first few days** of life (up to about 2 weeks).
- Blotchy red macules, papules and small white or yellow pustules (a 'flea-bitten' look). Lesions wax and wane, each lasting under 24 hours.
- Face, trunk and proximal limbs. **Palms and soles spared.**
- **The baby is well.**

TNPM

- **Present at birth.**
- Pustules on a **non-red** base rupture to leave a collarette of scale and **hyperpigmented macules**, which fade over weeks to months.
- Can involve the forehead, neck, trunk, limbs, **palms and soles**.

Miliaria

> **CRYSTALLINA**
> 1–2 mm clear fragile vesicles on the head, neck and upper trunk, often in the first week.

> **RUBRA**
> itchy red non-follicular papules on the trunk and in **skin folds**.

> Triggered by heat, fever, synthetic or tight clothing, and occlusion.

Fig. 2 · ETN close-up: fleeting erythematous macules and tiny pustules that wax and wane over days; infant otherwise well — no treatment needed.

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Special groups

- **Neonatal HSV**
  - Grouped vesicles, especially on the scalp or presenting part, with or without fever: emergency IV aciclovir.

Diagnosis and investigations

- **Clinical** in a well baby with typical morphology.
- If uncertain: a smear of pustule contents (eosinophils in ETN, neutrophils in TNPM), Gram stain and culture, and **HSV PCR** where HSV is possible.
- Biopsy is rarely needed.

Clinical

- in a well baby with typical morphology.

Key points

- If uncertain: a smear of pustule contents (eosinophils in ETN, neutrophils in TNPM), Gram stain and culture, and **HSV PCR** where HSV is possible.
- Biopsy is rarely needed.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	ETN	Days 1–14
	TNPM	At birth
	Miliaria	Heat or occlusion
	Neonatal acne / cephalic pustulosis	Weeks
	Neonatal HSV	Days to weeks
	Bacterial (staphylococcal)	Variable
	Candida (congenital)	Birth or early

Management

ETN and TNPM

- **REASSURE**
  benign and self-limiting. ETN usually clears within days to about 2 weeks and may briefly recur. TNPM pigment fades over weeks to months.

- **No specific treatment.** Do not pick the pustules. Use gentle cleansing and avoid unnecessary antiseptics.
- Do **not** treat as infection if the diagnosis is clear and the baby is well.

Miliaria

- **COOL THE BABY**
  fewer layers, a cooler room, breathable cotton, and prompt removal of wet clothing.
- Cool baths or compresses. For itch, short-term calamine or a light emollient. A short course of mild topical steroid is sometimes used for miliaria rubra on suitable sites.
- Usually clears within 1–2 days once cool.

Referral

999 / A&E	**Any neonate with vesicles who is febrile, lethargic, poorly feeding, or has eye involvement:** same-day emergency pathway. **Never 'wait and see' for possible HSV.**
ROUTINE	Clustered or grouped vesicles, mucosal or eye involvement.
ROUTINE	A premature or unwell infant, heavy involvement of the palms and soles (outside typical TNPM), or diagnostic doubt.
DON'T	Fever plus rash: do **not** assume miliaria.

Red flags and when to refer

- **Any neonate with vesicles who is febrile, lethargic, poorly feeding, or has eye involvement:** same-day emergency pathway. **Never 'wait and see' for possible HSV.**
- Clustered or grouped vesicles, mucosal or eye involvement.
- A premature or unwell infant, heavy involvement of the palms and soles (outside typical TNPM), or diagnostic doubt.
- Fever plus rash: do **not** assume miliaria.

Complications and prognosis

Complications

- Rarely, misdiagnosis of infection; miliaria pustulosa can become infected.

Prognosis

- ETN and TNPM resolve within days to weeks (TNPM pigment within months).

Prescriber pearls (UK pharmacist IP)

- Parents often fear allergy or sepsis. Explain that ETN pustules are sterile and the baby should stay well.
- Document 'well neonate with typical ETN' carefully, and keep a low threshold to investigate anything atypical.
- Post-TNPM pigmented macules can be mistaken for bruises or Mongolian spots. The history of preceding pustules helps.
- **Skin of colour:** ETN erythema is easily missed, so look and feel for papules and pustules.
- Overbundling advice is the main 'prescription' for miliaria.

Self-check quiz

5 questions · optional

Q1 When should you think of Neonatal pustular rashes?

✓ A well term baby with blotchy red macules and small pustules (ETN), pustules present at birth on a non-red base (TNPM), or dewdrop vesicles or red fold papules with overheating (miliaria).

Q2 What drives it?

✓ ETN: sterile eosinophilic pustules. TNPM: sterile neutrophilic pustules. Miliaria: blocked eccrine sweat ducts.

Q3 What is the backbone of care?

✓ Reassure, give no specific treatment, cool the baby for miliaria, and document that the baby is well.

Q4 What must you never miss?

✓ Neonatal HSV: grouped vesicles, or an unwell, febrile or poorly feeding baby. Same-day emergency pathway.

Q5 **Name a red flag that needs escalation.**

✓ **Any neonate with vesicles who is febrile, lethargic, poorly feeding, or has eye involvement:** same-day emergency pathway. **Never 'wait and see' for possible HSV.**

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Photo gallery

Erythema toxicum neonatorum 3

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Transient neonatal pustular melanosis 3

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Miliaria 2

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Red flag: neonatal herpes simplex

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AI teaching illustration

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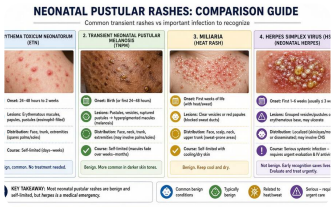

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Childhood & neonatal Module 6 · Week 1

Neonatal & infantile acne

Neonatal acne (birth to 6 weeks) is common and harmless. Infantile acne (6 weeks to about 16 months) can scar and may need treatment. Mid-childhood acne needs an endocrine review.

ALSO CALLED neonatal acne · infantile acne · baby acne · neonatal cephalic pustulosis · Malassezia · mid-childhood acne

RELATED PAGES Neonatal pustular rashes · Acne vulgaris · Milia · Napkin (nappy) dermatitis

KEY FACTS

- **Neonatal acne** (cephalic pustulosis) appears in the first weeks: papules and pustules on the cheeks; **no comedones**; settles in weeks.
- **Infantile acne** starts at 3–6 months with **comedones**, papules and sometimes nodules; may last 1–2 years and scar.
- Infantile acne needs examination for signs of virilisation or early puberty (refer if present).
- Neonatal: reassure, gentle washing; infantile: topical benzoyl peroxide or retinoid; erythromycin if moderate (tetracyclines contraindicated).
- Severe or scarring infantile acne: refer (may need isotretinoin).

Overview

TYPE	AGE	FEATURES
Neonatal acne	Birth to about 6 weeks	Common (about 20% of newborns); facial papules and pustules ± comedones; usually no scarring
Infantile acne	About 6 weeks to 12–16 months	Rarer; comedones, papules, pustules, occasional nodules; **may scar**; boys more than girls
Neonatal cephalic pustulosis	First weeks	Facial pustules **without true comedones**, often linked to Malassezia

Epidemiology

- **Who**
  - Neonatal acne affects up to 20% of newborns; infantile acne is less common, more in boys.

Aetiology: causes and risk factors

Overactive sebaceous glands respond to **maternal and neonatal androgens**, with follicular Cutibacterium colonisation and the same plugging and inflammation cascade as later acne. Androgen drive falls after infancy

and rises again at adrenarche, which is why **mid-childhood acne (ages 1–6) is very rare** and suggests an endocrine cause.

Myths to address

- **"Baby acne means the baby will have bad teenage acne"**: neonatal acne does not; infantile acne may predict later acne.

Pathophysiology and pathology

Maternal and infant androgens briefly stimulate the baby's sebaceous glands.

1. **Hormonal stimulus**
   Maternal androgens (neonatal) and transient infant adrenal and testicular androgens (infantile) enlarge sebaceous glands.

2. **Malassezia (neonatal)**
   Neonatal cephalic pustulosis may be a reaction to Malassezia yeasts.

Fig. 1 · Neonatal acne: papules and pustules on the cheeks, no comedones.
AI-generated illustration

3. **Comedones (infantile)**
   Follicular plugging creates true comedones and inflammatory lesions.

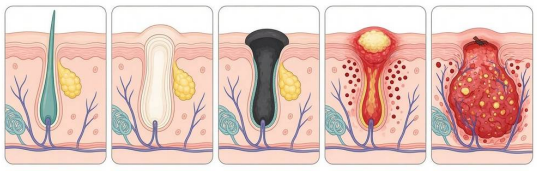

Fig. 2 · Acne follicle stages from microcomedone to inflamed papule.
AI-generated illustration

Clinical features

> **NEONATAL**
> comedones on the scalp, chest and back; inflammatory lesions on the cheeks, chin and forehead.

> **INFANTILE**

> mainly the cheeks; can be more inflammatory or nodular.

Fig. 3 · Infantile acne (weeks–months of age): comedones, papules and pustules on cheeks — may scar; distinguish from neonatal cephalic pustulosis.

© DermNet · **Source**

Types to recognise

Cephalic pustulosis / neonatal acne

Facial papules and pustules in the first weeks. Site: Face.

© DermNet · **Source**

Infantile acne

Comedones, papules, pustules over weeks–months. Site: Cheeks.

© DermNet · **Source**

Special groups

- **Red flags in infantile acne**
  - Growth acceleration, body odour, pubic hair, clitoromegaly or testicular enlargement: paediatric endocrinology.

Diagnosis and investigations

Clinical. If the age is atypical or the disease severe, look for accelerated growth, early sexual development or hirsutism. Refer mid-childhood or severe unresponsive acne to endocrinology.

Clinical

- Age + comedones confirm the type.

Atypical age or severe

- Look for fast growth, early puberty signs, hirsutism, body odour.

Refer

- Mid-childhood or severe unresponsive acne endocrinology.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Cephalic pustulosis	No or few comedones; may respond to topical antifungal.
	ETN / miliaria	Timing and distribution; not comedonal.
	Periorificial dermatitis	Around mouth and eyes; often steroid-related.
	Folliculitis	Follicular; culture.

Management

Neonatal acne

Usually **observe**. Gentle cleansing; avoid greasy, comedogenic products. Most cases clear without scarring.

Infantile acne (if treatment needed)

> **MILD**
>
> benzoyl peroxide ± topical retinoid (adapalene or tretinoin) as tolerated. Check the age licence.

> **MODERATE**
>
> a topical regimen ± an age-appropriate oral **erythromycin**-class antibiotic. **Avoid tetracyclines under 12.**

> **SEVERE, NODULAR OR SCARRING**
>
> refer to dermatology. Specialists do use isotretinoin in infants when indicated.

Treatment ladder

Neonatal

Observe; gentle cleansing; avoid greasy, comedogenic products.

2

Infantile, mild

Benzoyl peroxide ± topical retinoid (adapalene or tretinoin) as tolerated; check age licence.

3

Infantile, moderate

Topical regimen ± age-appropriate oral **erythromycin**-class antibiotic.
Avoid tetracyclines under 12.

4

Severe / nodular / scarring

Dermatology: specialists may use isotretinoin in infants.

Referral

URGENT — **Mid-childhood acne (1–6 years)** or signs of hyperandrogenism (fast growth, body odour, pubic hair, clitoromegaly).

ROUTINE — Nodular or scarring disease, or treatment failure.

Red flags and when to refer

- **Mid-childhood acne (age 1–6).**
- Suspected hyperandrogenism (accelerated growth, body odour, pubic hair, clitoromegaly).
- Nodular or scarring disease, or treatment failure.

Complications and prognosis

Complications

- Scarring (infantile nodular acne); parental anxiety.

Prognosis

- Neonatal: resolves by 3 months. Infantile: usually by 2 years.

Prescriber pearls (UK pharmacist IP)

Reassure parents that neonatal acne is not caused by poor hygiene.

Do not use adult OTC acne kits on infants without checking age licences.

Persistent facial pustules in a neonate: consider cephalic pustulosis rather than true acne.

Self-check quiz

4 questions · optional

Q1 Neonatal vs infantile acne: age ranges?

✓ Neonatal: birth to ~6 weeks. Infantile: ~6 weeks to 12–16 months.

Q2 Which oral antibiotic class to avoid under 12?

✓ Tetracyclines.

Q3 Acne in a 4-year-old?

✓ Very rare: look for hyperandrogenism and refer to endocrinology.

Q4 Pustules without comedones in a 3-week-old?

✓ Neonatal cephalic pustulosis (Malassezia-linked).

References

1. DermNet: Acne in children – https://dermnetnz.org/topics/acne-in-children dermnetnz.org

Photo gallery

Neonatal cephalic pustulosis and infantile acne 2

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Childhood & neonatal Module 6 · Week 1

Napkin (nappy) dermatitis

Irritant rash on the convex skin with the folds spared, or Candida with fold involvement and satellites. Use barrier ointment at every change, with short-course hydrocortisone and/or an azole.

ALSO CALLED nappy rash · diaper rash · napkin dermatitis · candidal nappy rash · irritant napkin dermatitis · barrier cream · +1

RELATED PAGES Candidal intertrigo · Eczema / dermatitis · Psoriasis · Neonatal pustular rashes

KEY FACTS

- Inflammation of the skin under a nappy, most often **irritant** contact dermatitis from urine, faeces and occlusion.
- Peaks at **9–12 months** (range 3–15 months); much less common with modern breathable disposables.
- Irritant: convex surfaces affected, **skin creases spared**; Candida: creases involved with **satellite** papules.
- Frequent changes, nappy-free time, gentle cleansing and a **barrier preparation**; topical imidazole if candidal; hydrocortisone 1% short term if inflamed.
- Refer if atypical, nodular, ulcerated or not responding (e.g. psoriasis, Langerhans cell histiocytosis, zinc deficiency).

Overview

Inflammation in the nappy area, most often **irritant contact dermatitis** from occlusion, urine and faecal enzymes. Secondary **Candida** is common. Peak age is about **3–15 months**. It also affects older children and adults with incontinence.

Epidemiology

- **Who**
  - Up to a third of babies in nappies at any time; peak 9–12 months.
  - Also affects incontinent older children and adults (incontinence-associated dermatitis).

Aetiology: causes and risk factors

- Occlusion and urine cause over-hydration and maceration.
- Faecal enzymes and bile salts damage the barrier, and ammonia raises the pH.
- Friction, and an atopic or seborrhoeic tendency.
- Candida albicans thrives on damaged, moist skin.

- It is not usually an allergy to the nappy, and it is not tinea.

Risk factors and associations

- Prolonged contact with urine and faeces; **diarrhoea** (faecal enzymes) and antibiotics (Candida) are common triggers.
- Infrequent changes, tight nappies, harsh wipes and soaps.

Myths to address

- **"LEAVE THE CREAM THICK AND NEVER WIPE IT OFF"**
  a barrier layer is fine, but clean skin gently at each change.

Pathophysiology and pathology

Wet, occluded skin with a **high pH** lets faecal enzymes break down the barrier.

1. **Overhydration**
   Occlusion keeps the stratum corneum wet, making it fragile and more permeable.
2. **Alkaline pH and enzymes**
   Bacteria convert urine urea to **ammonia**, raising pH, which activates faecal **proteases and lipases** that digest the skin barrier.
3. **Irritant dermatitis**
   Friction and enzymes inflame the convex skin in contact with the nappy; creases are protected.

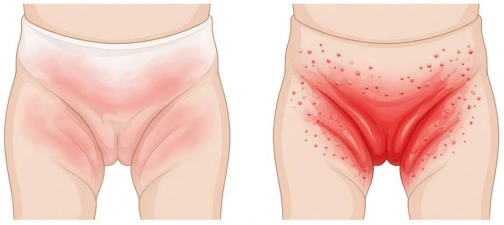

Fig. 1 · Irritant napkin dermatitis (convex surfaces, creases spared) vs candidal (creases involved, satellite spots).
AI-generated illustration

4. **Secondary Candida**
   After about 3 days, Candida from the gut often colonises damaged skin, spreading into the folds.

HISTOPATHOLOGY

- Not needed; diagnosis is clinical. Swab satellite pustules if unsure or recurrent.

Clinical features

PATTERN	FEATURES
Irritant napkin dermatitis	Redness ± scale or erosions on **convex surfaces** in contact with the nappy; **folds often spared**
Candidal	Bright red, shiny plaques **involving the folds**, with **satellite** papules and pustules

PATTERN	FEATURES
Impetigo	Irregular blisters, pustules or honey crust
Seborrhoeic	Salmon-pink patches in folds ± cradle cap
Psoriasis ('napkin psoriasis')	Persistent, well-demarcated, shiny plaques
Atopic eczema	Itchy rash elsewhere; nappy area often relatively spared

Special groups

- **Skin of colour**
  - Redness is less obvious; look for scaling, erosion and post-inflammatory pigmentation.

Diagnosis and investigations

Clinical. Swab if severe, atypical or not responding. In refractory cases, consider rarities such as **zinc deficiency (acrodermatitis enteropathica)** or **Langerhans cell histiocytosis**, which need specialist input.

Clinical

- Diagnosis is clinical.
- Check the **folds** and look for satellites.

Swab

- If severe, atypical or not responding.

Think rare

- Refractory: **zinc deficiency** (acrodermatitis enteropathica) or **Langerhans cell histiocytosis**: specialist.

Safeguarding

- Injury pattern that does not fit dermatitis: document and escalate locally.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Candida	Folds involved; satellite papules and pustules.
	Impetigo	Blisters, pustules, honey crust.
	Infantile seborrhoeic	Salmon-pink fold patches, cradle cap.
	Napkin psoriasis	Persistent, sharply demarcated shiny plaques.

	CONDITION	DISTINGUISHING FEATURES
	Perianal streptococcal	Bright perianal redness, painful defecation.

See the table under Clinical features. Also consider **perianal streptococcal disease**, and **safeguarding** if the injury pattern does not fit dermatitis. Document carefully and escalate through local safeguarding.

Management

Aligned with NICE CKS nappy rash and PCDS. Re-check the live CKS page and the BNF for Children.

Barrier and self-care

- Change nappies frequently, with nappy-free time when practical.
- Cleanse gently with water or a non-soap cleanser, and pat dry.
- Apply a **thick barrier ointment** (petrolatum ± zinc oxide) at **every change**.
- Use modern absorbent disposables. With cloth nappies, use liners and avoid plastic pants. Avoid fragranced wipes.

Topical prescribing

- Inflamed irritant rash: **hydrocortisone 1%**, thin layer, for a short course (usually up to 7 days).
- Suspected Candida: a topical **imidazole** (clotrimazole, miconazole, econazole), continued after the skin looks clear (often 7–14 days in total).
- Combination steroid and antifungal products: short courses only. **No potent steroids on nappy skin.**
- Secondary bacterial infection: follow local guidance (oral flucloxacillin if needed; topical fusidic acid only for very localised areas).

Medicines at a glance

Hydrocortisone 1%

Inflamed irritant rash

DOSE	Thin layer · Short course, usually up to 7 days
CAUTIONS	No potent steroids on nappy skin.

Topical imidazole (clotrimazole, miconazole, econazole)

Suspected Candida

DOSE	As per product · Continue after clear; often 7–14 days in total
CAUTIONS	Combination steroid–antifungal: short courses only.

Treatment ladder

1 **Barrier and self-care**

Frequent changes; nappy-free time. Water or non-soap cleanser; pat dry; no fragranced wipes.

2 **Inflamed irritant rash**

Hydrocortisone 1%

3 **Suspected Candida**

Topical imidazole (clotrimazole, miconazole, econazole)

4 **Secondary bacterial infection**

Follow local guidance: oral flucloxacillin if needed. Topical fusidic acid only for very localised areas.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Potent steroids**: Not for the nappy area in primary care.
- **How much barrier?**: Parents often use too little: show them.

Referral

SAME DAY	Infant **under 1 month** with significant rash, or a **systemically unwell** infant.
URGENT	Bruising or a pattern suggesting **non-accidental injury**: safeguarding.
ROUTINE	Severe or **ulcerated** rash, or no improvement with good care and treatment.
ROUTINE	Diagnostic doubt, or suspected nutritional deficiency or rare disease.

Red flags and when to refer

- Infant **under 1 month** with significant rash, or a **systemically unwell** infant.
- Severe or **ulcerated** rash, or failure to improve with good barrier care and appropriate treatment.
- Diagnostic uncertainty, or suspected nutritional deficiency or rare disease.
- Bruising or a pattern suggesting non-accidental injury.

Complications and prognosis

Complications

- Secondary Candida or bacterial (impetigo) infection.
- Erosive **Jacquet** dermatitis and granuloma gluteale infantum (steroid misuse).
- Distress and sleep disturbance.

Prognosis

- Usually settles within days with good nappy care; recurs with diarrhoea or infrequent changes.

Prescriber pearls (UK pharmacist IP)

Folds involved with satellites: treat Candida. Folds spared: irritant pattern.

Parents often use too little barrier ointment, so show them how much.

A short course of mild hydrocortisone is appropriate. Potent steroids are not for the nappy area in primary care.

Breastfed babies' stools are often less alkaline, which may reduce the irritant load (supportive counselling, not a treatment claim).

Self-check quiz

5 questions · optional

Q1 **Folds spared or involved: which suggests Candida?**

✓ Involved folds with satellite papules/pustules suggest Candida; spared folds suggest irritant.

Q2 What goes on at every nappy change?

✓ A thick barrier ointment (petrolatum ± zinc oxide).

Q3 Steroid choice and duration for inflamed irritant nappy rash?

✓ Hydrocortisone 1%, thin layer, short course (usually up to 7 days). No potent steroids.

Q4 Rash refractory to good care: which rarities?

✓ Zinc deficiency (acrodermatitis enteropathica) and Langerhans cell histiocytosis: specialist input.

Q5 Which infants need same-day assessment?

✓ Under 1 month with significant rash, or systemically unwell.

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Photo gallery

Irritant, candidal and severe erosive napkin dermatitis 3

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Compare: infant atopic eczema (nappy area spared)

2

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Childhood & neonatal Module 6 · Week 1

Infantile haemangioma (strawberry mark)

A benign vascular tumour that appears in the first weeks, grows fast for a few months, then slowly involutes. Refer high-risk sites early (ideally by about 4 weeks), while propranolol works best.

ALSO CALLED infantile haemangioma · strawberry naevus · strawberry mark · haemangioma · PHACE · propranolol · +1

RELATED PAGES Port-wine stain (capillary malformation) · Pyogenic granuloma · Pigmented & epidermal birthmarks · Neonatal pustular rashes

KEY FACTS

- **Infantile haemangioma** is the commonest tumour of infancy (about 4–10% of infants): absent or faint at birth, then rapid growth.
- Most growth happens in the **first 1–3 months**; involution over years, often leaving fibrofatty tissue or telangiectasia.
- High-risk sites: **periocular, airway/beard, lip, nasal tip, large segmental facial, perineal**, ulcerated, or 5 or more lesions.
- Refer high-risk lesions early, ideally by about **4 weeks** of age; **propranolol** is first-line and specialist-initiated.
- Five or more skin haemangiomas: liver ultrasound. Large facial segmental lesions: think PHACE.

Overview

An **infantile haemangioma (IH)** is a benign vascular **tumour** of infancy. It is absent or a faint mark at birth, **grows rapidly** (mostly in the first 1–3 months, largely complete by about 5 months), plateaus, then **involutes** over years. About **4–10%** of infants are affected.

Epidemiology

- **Who**
  - More in girls, premature and low-birth-weight babies, and twins.
  - Head and neck in about 60%.

Aetiology: causes and risk factors

- Clonal endothelial proliferation, **GLUT-1 positive** (unlike malformations and congenital haemangiomas).
- Risk factors: prematurity, low birth weight, female sex, multiple pregnancy, placental anomalies.

Myths to address

- **"WAIT AND IT WILL GO AWAY"**
  most do, but high-risk ones need treatment in the early growth window.

Pathophysiology and pathology

A haemangioma is a **benign proliferation of immature endothelial cells** that later switches off.

1. **Trigger**
   Tissue hypoxia (possibly placental) induces HIF-1α and VEGF in a progenitor cell population.
2. **Proliferation**
   GLUT-1–positive endothelial cells proliferate rapidly in the first months.
3. **Plateau**
   Growth slows by about 5 months.
4. **Involution**
   Endothelial cells undergo apoptosis and are replaced by fibrofatty tissue over years.

Fig. 1 · Natural history: appearance, rapid growth, slow involution.
AI-generated illustration

5. **Propranolol effect**
   Beta-blockade causes vasoconstriction, reduces VEGF signalling and speeds involution.

HISTOPATHOLOGY

- Lobules of capillaries lined by GLUT-1–positive endothelium (GLUT-1 negative in congenital haemangiomas and malformations).

Clinical features

TYPE	APPEARANCE
Superficial	Bright red, raised, well-defined ('strawberry')
Deep	Bluish, rubbery swelling under the skin
Mixed	Both

Haemangiomas can occur anywhere, most often on the head and neck. **5 or more** skin haemangiomas raise concern for **liver haemangiomas**. White surface change can herald **ulceration**.

Fig. 2 · Natural history of infantile haemangioma: appearance (0–2 weeks), rapid growth (1–6 months), slow involution, and completion (5–7+ years).

AI-generated illustration

Special groups

- **PHACE syndrome**
  - Posterior fossa anomalies, Haemangioma (large facial segmental), Arterial, Cardiac (coarctation) and Eye anomalies: needs MRI/MRA, echo and eye review.
- **LUMBAR syndrome**
  - Lumbosacral segmental haemangioma with spinal and urogenital anomalies: spinal imaging.

Diagnosis and investigations

- Usually **clinical**.
- Ultrasound if deep, or uncertain against a malformation or soft-tissue mass. MRI for complex facial or segmental lesions.
- Abdominal ultrasound if there are 5 or more skin lesions.

Differential diagnoses

LOOK-ALIKE	DISTINGUISHING CLUES
Port-wine stain (see Port-wine stain)	Present at birth, flat, **grows with the child**, no rapid tumour growth
Congenital haemangioma	Fully formed at birth; different natural history
Venous or lymphatic malformation	Soft, compressible, persistent; imaging differs
Pyogenic granuloma	Later onset; often a pedunculated bleeding papule

Management

Uncomplicated small haemangioma

Active observation with **photo documentation**. Reassure parents about involution.

Specialist treatment

- Oral **propranolol** is first line for problematic IH. It is started and supervised by a specialist, with heart rate, BP and glucose monitoring. Feed regularly, and **hold doses if the baby is unwell or not feeding**.
- Topical **timolol** for selected small, thin, superficial lesions (off-label; NICE ESUOM47).
- Wound care, analgesia and antibiotics for ulceration; laser or surgery for selected residual changes.

Treatment ladder

1 **Uncomplicated**

Active observation with photographs; reassure.

2 **Small superficial (specialist)**

Topical timolol (off-label; NICE ESUOM47).

3 **Problematic / high-risk**

Oral propranolol, specialist-initiated with monitoring; hold doses if not feeding or unwell.

Referral

URGENT	Periocular, airway/beard, lip, nasal tip, segmental facial, perineal, ulcerated or 5 lesions: refer ideally by about 4 weeks of age.
SAME DAY	Stridor with a beard-area haemangioma.

Red flags and when to refer

- High-risk haemangioma sites (listed above): **do not wait to see if it goes away**.
- Stridor or noisy breathing with a beard-area haemangioma: **urgent**.
- Ulcerated or bleeding haemangioma.

Refer promptly (ideally by about 4 weeks of age if high risk)

- **Periocular** (risk to vision and amblyopia)
- **Beard or airway** distribution (risk of stridor)
- **Lip or perioral** (feeding)
- Nasal tip, large facial, or **segmental** (PHACE risk with large facial segmental lesions)

- **Perineal or flexural** (ulceration)
- **Ulcerated, bleeding or rapidly enlarging**
- **5 or more** skin lesions
- Breast (girls) or large scalp lesions

Complications and prognosis

Complications

- Ulceration (painful, scarring), bleeding, amblyopia, airway obstruction, feeding problems, disfigurement; heart failure with large liver haemangiomas.

Prognosis

- Most involute; residual skin changes are common in large or deep lesions.

Prescriber pearls (UK pharmacist IP)

The rapid growth window is early, so **late referral misses the easiest window for treatment**.

IP role: recognise high-risk sites, explain that beta-blockers are specialist-supervised, and support dressings for ulceration under the specialist plan.

BAD patient leaflet and the Birthmark Support Group are useful for families.

Self-check quiz

4 questions · optional

Q1 When does most haemangioma growth occur?

✓ In the first 1–3 months; largely complete by about 5 months.

Q2 Which marker distinguishes IH from malformations?

✓ GLUT-1.

Q3 What does 5 skin haemangiomas prompt?

✓ Abdominal (liver) ultrasound.

Q4 First-line systemic treatment?

✓ Oral propranolol, started by a specialist.

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nice.org.uk

Photo gallery

Infantile haemangioma: proliferating, involuting, high-risk sites 4

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Childhood & neonatal Module 6 · Week 1

Pigmented & epidermal birthmarks

Naevus of Ota, achromic and anaemic naevi, Becker naevus, Mongolian spots and linear epidermal naevi. Mostly benign and needing reassurance, with key exceptions: the eye in Ota, tuberous sclerosis, and documenting Mongolian spots for safeguarding.

ALSO CALLED naevus of Ota · nevus of Ota · achromic naevus · naevus depigmentosus · naevus anaemicus · nevus anemicus · +7

RELATED PAGES Congenital melanocytic naevi (CMN) · Port-wine stain (capillary malformation) · Sebaceous naevus (naevus sebaceous) · Hypopigmentation & skin of colour · Infantile haemangioma (strawberry mark)

KEY FACTS

- **Congenital dermal melanocytosis** (Mongolian spot): blue-grey patches on the sacrum in darker-skinned babies; **document at birth** (not bruising).
- Naevus of Ota blue-grey patch around the eye (V1/V2): check for **glaucoma**; laser is effective.
- Becker naevus brown hairy patch on the shoulder appearing in adolescent males.
- Naevus depigmentosus / anaemicus stable pale patches; ash-leaf macules may signal **tuberous sclerosis**.
- Linear epidermal naevus warty streaks along Blaschko lines; extensive forms may be syndromic.

Overview

This page brings together several **pigmented and epidermal birthmarks**, each in its own short section.

BIRTHMARK	COLOUR / LOOK	TYPICAL SITE	KEY POINT
Naevus of Ota	Blue-grey or brown-violet, mottled	Forehead, temple, around the eye (V1/V2) ± sclera	Check the eye: **ophthalmology** (glaucoma)
Achromic naevus (naevus depigmentosus)	Off-white, stable, serrated edge	Trunk; may be segmental	Not vitiligo; consider ash-leaf macules
Naevus anaemicus	Pale; **does not redden** when rubbed	Trunk	Vasoconstriction, not melanin loss; NF1 association
Becker naevus	Large brown patch, **hairy** after puberty	Shoulder or upper trunk; men about 5:1	Rare Becker naevus syndrome (breast or pectoral hypoplasia)
Mongolian spot	Flat blue-grey	Lower back and buttocks	**Document** to avoid confusion with bruising; most fade by about age 4

BIRTHMARK	COLOUR / LOOK	TYPICAL SITE	KEY POINT
Linear verrucous epidermal naevus	Linear, warty, along Blaschko lines	Trunk or limbs, usually one side	Refer if extensive or syndromic

Aetiology: causes and risk factors

> **Dermal melanocytoses** (Ota, Mongolian spot): melanocytes arrested in the dermis during embryonic migration (GNAQ/GNA11 mutations described).

> **ACHROMIC NAEVUS**
> a mosaic developmental pigment defect (melanocytes present, but melanin production or transfer reduced). It is not autoimmune.

> **NAEVUS ANAEMICUS**
> localised α-adrenergic hypersensitivity of dermal vessels.

> **BECKER NAEVUS**
> a hamartoma of epidermis, melanocytes and hair follicles, triggered by androgens around puberty.

> **EPIDERMAL NAEVUS**
> postzygotic keratinocyte mosaicism (FGFR3, PIK3CA, HRAS).

Myths to address

- **"BLUE PATCHES ON A BABY ARE BRUISES"**
  Mongolian spots are common; record them to avoid safeguarding confusion.

Clinical features

See the table under 'What it is'. Extra points:

- Naevus of Ota: present at birth in about half; more common in East Asian women. Rarely, melanoma develops within dermal melanocytosis.
- Naevus anaemicus: **diascopy** (pressing the surrounding skin) makes the contrast disappear.
- Epidermal naevus: starts flat and tan, then becomes warty, and may extend through childhood. **ILVEN** is an itchy inflammatory variant.

Fig. 1 · Mongolian spot (lumbosacral dermal melanocytosis): blue-grey sacral/buttock patch present at birth — usually fades by early childhood; document to avoid bruise misinterpretation.

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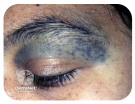
Diagnosis and investigations

Clinical, with Wood's lamp and dermoscopy where helpful. Achromic naevus shows a faint residual network, unlike the 'dead white' of vitiligo. Biopsy is rarely needed.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	**Naevus of Ota**	Mongolian spot, blue naevus, melasma, drug pigmentation, bruise
	Achromic naevus	Vitiligo, ash-leaf macules (tuberous sclerosis), pityriasis alba, post-inflammatory hypopigmentation, naevus anaemicus, piebaldism
	Becker naevus	Café-au-lait macule, congenital melanocytic naevus, plexiform neurofibroma, smooth muscle hamartoma
	Mongolian spot	**Bruising** (acute, tender, colour changes over time), naevus of Ota/Ito
	Epidermal naevus	Lichen striatus, linear lichen planus, linear porokeratosis, sebaceous naevus (scalp)

Management

> **Reassure.** Most need no medical treatment. Offer **camouflage** cosmetics.

> NAEVUS OF OTA

specialist laser (Q-switched Nd:YAG or picosecond) over several sessions. **Ophthalmology** if the eye is involved. Any changing nodule needs dermatology.

ACHROMIC NAEVUS

refer if there is doubt, especially with multiple ash-leaf-like macules (tuberous sclerosis work-up).

BECKER NAEVUS

sun protection reduces contrast. Hair removal (shaving, waxing, or specialist laser). Examine for breast or limb asymmetry if extensive.

MONGOLIAN SPOT

no treatment. **Document site, size and colour** in the red book and notes.

EPIDERMAL NAEVUS

emollients. Specialist options include keratolytics, calcipotriol, laser or surgery (recurrence possible). Refer if extensive, syndromic, a scalp sebaceous naevus, or ILVEN. Do not treat as warts with destructive OTC products.

Referral

ROUTINE	Naevus of Ota involving the **sclera or eye**: glaucoma screening.
ROUTINE	**Multiple** hypopigmented ash-leaf macules, seizures or developmental delay: tuberous sclerosis.
ROUTINE	Extensive epidermal naevus with eye, neurological or skeletal features (epidermal naevus syndrome).
ROUTINE	Extrasacral or atypical blue-grey marks without documentation: safeguarding questions. Document early.

Red flags and when to refer

- Naevus of Ota involving the **sclera or eye**: glaucoma screening.
- **Multiple** hypopigmented ash-leaf macules, seizures or developmental delay: tuberous sclerosis.
- Extensive epidermal naevus with eye, neurological or skeletal features (epidermal naevus syndrome).
- Extrasacral or atypical blue-grey marks without documentation: safeguarding questions. Document early.

Complications and prognosis

Complications

- Glaucoma (Ota); tuberous sclerosis associations; psychosocial impact.

Prognosis

- Mongolian spots fade over childhood; most others persist.

Prescriber pearls (UK pharmacist IP)

- Mottled blue-grey pigment around the eye is not 'just a birthmark' if the sclera is involved.
- Clear documentation of Mongolian spots protects families from later misunderstanding.
- Naevus anaemicus does not redden on rubbing, but vitiligo and achromic naevus do.
- Moles present at birth, melanoma risk by size and surveillance: see congenital melanocytic naevi.

Self-check quiz

5 questions · optional

Q1 When should you think of Pigmented & epidermal birthmarks?

✓ A congenital or early-childhood patch: blue-grey (dermal melanocytosis), pale (achromic or anaemic), brown and hairy (Becker), or linear and warty (epidermal naevus).

Q2 What drives it?

✓ Developmental pigment or mosaic abnormalities: dermal melanocytes, reduced melanin, vasoconstriction, or keratinocyte mosaicism.

Q3 What is the backbone of care?

✓ Reassure, offer camouflage, document carefully, and refer selectively for specialist laser or surgery.

Q4 What must you never miss?

✓ Ocular Ota (glaucoma), multiple ash-leaf macules (tuberous sclerosis), extensive or syndromic epidermal naevus, and misreading a Mongolian spot as a bruise.

Q5 Name a red flag that needs escalation.

✓ Naevus of Ota involving the **sclera or eye**: glaucoma screening.

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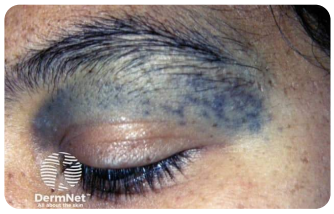
pcds.org.uk

Photo gallery

Naevus of Ota 2



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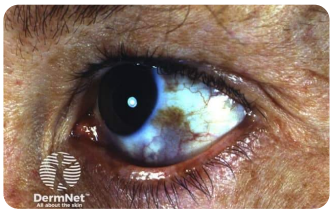

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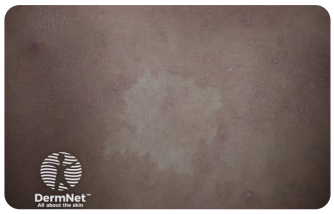
Pale birthmarks: achromic naevus and naevus anaemicus 3



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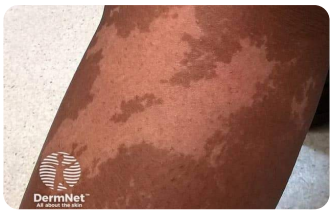

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Becker naevus

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Mongolian spot (dermal melanocytosis)

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Epidermal naevus

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Childhood & neonatal Module 6 · Week 1

Port-wine stain (capillary malformation)

A flat red or purple capillary malformation present at birth that grows with the child and never involutes. Facial V1 stains need assessment for Sturge–Weber syndrome; pulsed dye laser is specialist.

ALSO CALLED port-wine stain · capillary malformation · naevus flammeus · Sturge-Weber · Klippel-Trénaunay · vascular birthmark

RELATED PAGES Infantile haemangioma (strawberry mark) · Pigmented & epidermal birthmarks · Neonatal pustular rashes · Intense pulsed light (IPL)

KEY FACTS

- **Port-wine stain** is a capillary **malformation**, present at birth, that grows with the child and never involutes.
- Caused by a **somatic GNAQ** mutation; about 0.3% of newborns.
- Darkens and thickens with age (cobblestoning, overgrowth).
- Facial **V1** distribution: assess for **Sturge–Weber** (glaucoma, seizures); limb stain with overgrowth: consider Klippel–Trénaunay.
- **Pulsed dye laser** (specialist) lightens it, best started early; camouflage for all.

Overview

A **port-wine stain (PWS)** is a **capillary malformation**: present at birth as a flat red or purple patch, it **grows in proportion with the child**, may darken and thicken with age, and **never involutes**. It affects about **0.3%** of newborns.

Epidemiology

- **Who**
  - Equal sexes; any site, face commonest.

Aetiology: causes and risk factors

Dilated, malformed dermal capillaries, usually from a **somatic mosaic GNAQ** mutation; reduced nerve supply to the vessels lets them dilate progressively.

Myths to address

- **"IT'S A STORK MARK THAT WILL FADE"**
  salmon patches fade; port-wine stains do not.

Pathophysiology and pathology

A port-wine stain is a patch of **malformed, progressively widening capillaries**.

1. **Mosaic mutation**
   A GNAQ mutation in a developing cell gives a patch of abnormal vessels.

2. **Ectatic capillaries**
   Dermal capillaries are dilated and increase in calibre with age.

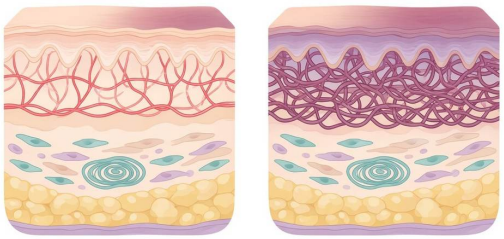

Fig. 1 · Normal fine dermal capillaries (left) versus the widened, crowded capillaries of a port-wine stain (right).
AI-generated illustration

3. **Reduced neural control**
   Poor sympathetic innervation of the vessels allows progressive dilation.

4. **Thickening**
   Over decades vessels enlarge further, causing darker colour, blebs and soft-tissue overgrowth.

HISTOPATHOLOGY

- Dilated capillaries and venules in the papillary and reticular dermis without endothelial proliferation.

Clinical features

- Flat, well-demarcated pink, red or purple patch, often unilateral and following a dermatome-like segment.
- With age: darker colour, thickening and **cobblestone** nodules (vascular blebs), soft-tissue overgrowth.
- Blanches partly on pressure; does not grow out of proportion.

Special groups

- **Sturge–Weber syndrome**
  - V1 facial PWS with leptomeningeal angiomatosis and glaucoma: seizures, hemiparesis, learning difficulties.

Diagnosis and investigations

Clinical. MRI brain and eye review for facial V1 stains (Sturge–Weber); limb imaging and leg-length check for limb stains with overgrowth.

Differential diagnoses

LOOK-ALIKE	DISTINGUISHING CLUES
Salmon patch (naevus simplex)	Pale pink, midline (nape, glabella, eyelids); fades in the first years
Infantile haemangioma (see Infantile haemangioma)	Absent or faint at birth, then rapid growth and involution
Venous malformation	Blue, compressible, swells when dependent

Management

- Reassure for small, uncomplicated stains; camouflage cosmetics.
- **Pulsed dye laser** is specialist, and earlier childhood referral is often preferred.
- **Facial PWS in the V1 distribution: consider Sturge–Weber** (neurology and ophthalmology for glaucoma and seizures).
- **Limb PWS with overgrowth: consider Klippel–Trénaunay** and related syndromes.

Treatment ladder

1 **Reassure**

Explain permanence; camouflage.

2 **Laser**

Pulsed dye laser via specialist; earlier childhood treatment often preferred.

Referral

URGENT — Facial V1 stain: ophthalmology (glaucoma) and neurology (Sturge–Weber).

ROUTINE — Limb stain with overgrowth or asymmetry.

Red flags and when to refer

- Facial segmental port-wine stain with eye or neurological symptoms (seizures, glaucoma).
- Limb asymmetry or overgrowth.

Complications and prognosis

Complications

- Glaucoma, seizures (Sturge–Weber); limb overgrowth; bleeding from blebs; psychosocial impact.

Prognosis

- Permanent; laser lightens but rarely clears.

Prescriber pearls (UK pharmacist IP)

A malformation is not a tumour: be clear with parents whether it is a PWS or a haemangioma.

PWS does not fade; salmon patches do.

Self-check quiz

4 questions · optional

Q1 Does a port-wine stain fade?

✓ No; it grows with the child and may darken and thicken.

Q2 Which mutation is usually responsible?

✓ Somatic mosaic GNAQ.

Q3 Which syndrome must be considered with a V1 facial stain?

✓ Sturge–Weber syndrome.

Q4 How is a salmon patch different?

✓ Pale, midline and fades in early childhood.

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Photo gallery

Port-wine stain (capillary malformation)

2

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Childhood & neonatal Module 6 · Week 1

Epidermolysis bullosa (EB)

Inherited skin fragility in which minimal friction causes blisters. Recognise it, handle the baby gently, and refer urgently to a specialist EB centre. Primary care supports under the specialist plan.

ALSO CALLED EB | EB simplex | junctional EB | dystrophic EB | RDEB | Kindler EB | +5

RELATED PAGES Approach to blistering diseases | Bullous pemphigoid (incl. MMP & pemphigoid gestationis) | Aplasia cutis congenita | Milia

KEY FACTS

- **Inherited** skin fragility: blisters after **minimal trauma** due to defective adhesion proteins.
- Types by split level: **EB simplex** (keratins 5/14, intraepidermal), **junctional** (laminin-332, lamina lucida), **dystrophic** (type VII collagen, below lamina densa, scarring), **Kindler**.
- Differs from **EB acquisita**, an autoimmune adult disease.
- Refer urgently any baby with unexplained blistering to the EB team.
- Handle gently: no adhesive tapes, lift on a soft pad, silicone dressings; don't lance blisters unless trained.

Overview

A group of **inherited mechanobullous disorders**: skin (and often mucosa) blisters after **minimal trauma**, because adhesion molecules are defective. This is different from **EB acquisita**, which is autoimmune and usually affects adults.

TYPE	LEVEL OF SPLIT	KEY MOLECULES (EXAMPLES)	GIST
EB simplex (EBS)	Within the epidermis	Keratins 5/14	Often milder, localised, acral
Junctional EB (JEB)	Lamina lucida	Laminin-332, integrins	Severe forms are life-limiting
Dystrophic EB (DEB)	Below the lamina densa	**Type VII collagen (COL7A1)**	Scarring; recessive DEB is severe
Kindler EB	Mixed levels	Kindlin-1	Photosensitivity, poikiloderma

Aetiology: causes and risk factors

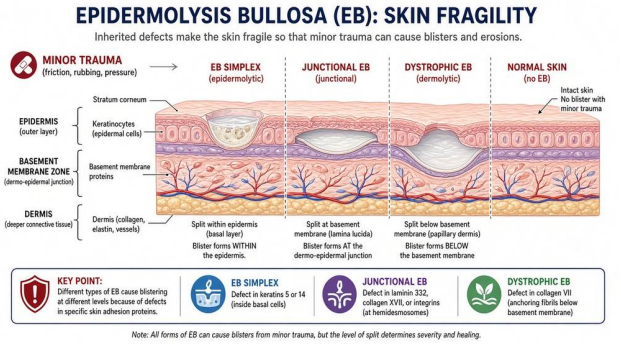

Fig. 1 · Skin fragility in EB: the level of the split in EB simplex, junctional and dystrophic EB compared with normal skin.

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Loss of structural integrity at a specific skin level means minor shear causes a blister. Inheritance is autosomal dominant or recessive depending on subtype. **Key Module 6 links: DEB is caused by type VII collagen (COL7A1) defects, and chronic RDEB wounds carry a high risk of cutaneous SCC.**

Myths to address

- **"EB IS CONTAGIOUS"**
  it is genetic.

Pathophysiology and pathology

The **level of the missing glue** decides EB type and severity.

1. **Keratin defect (EBS)**
   Basal keratinocytes break under friction.
2. **Lamina lucida defect (JEB)**
   Laminin-332 or integrin loss separates epidermis from basement membrane.
3. **Anchoring fibril defect (DEB)**
   Type VII collagen loss splits the skin beneath the basement membrane, healing with scars and milia.

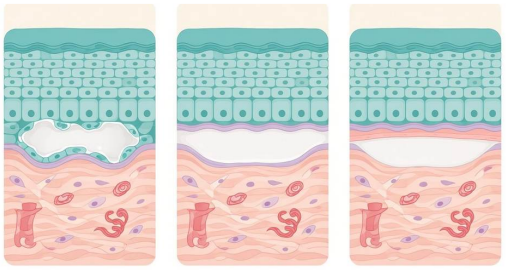

Fig. 2 · EB types split at different levels: within the basal cells, in the basement membrane, or beneath it.

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4 **Fragile skin**

Everyday friction tears skin.

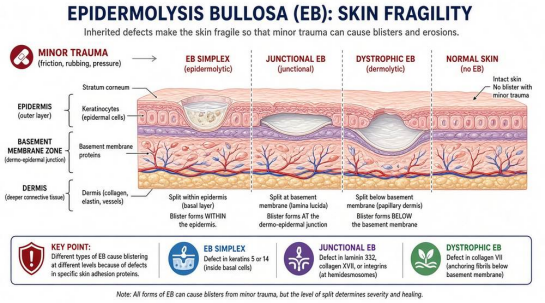

Fig. 3 · Skin fragility in EB.

AI-generated illustration

Clinical features

- Blisters and erosions at **friction sites** (hands, feet, belt line, nappy area). They may be present at birth.
- Nail dystrophy, milia and scarring (especially dystrophic types). Mitten deformities in severe RDEB.
- **MUCOSA**
  mouth, oesophagus (dysphagia), larynx (hoarse cry in severe JEB).
- Localised EBS often presents when the child starts **crawling or walking**.
- A neonate with unexplained blistering, with or without congenital absence of skin: **consider EB until proven otherwise**.

Fig. 4 · Epidermolysis bullosa in a neonate: fragile skin with blisters/erosions from minimal trauma — urgent specialist/EB service referral.

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Diagnosis and investigations

Urgent recognition and referral to a **specialist EB centre** (UK: e.g. GOSH and other DEBRA-linked centres). Subtyping uses immunofluorescence mapping, electron microscopy and genetic testing. Primary care does **not** attempt to subtype.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Friction or sucking blisters	Localised; well baby; no family history
	SSSS / bullous impetigo	Signs of infection; microbiology
	HSV / varicella	Viral morphology; PCR
	Incontinentia pigmenti	Blaschko-line stages; usually female
	Autoimmune neonatal blistering (maternal antibodies)	Maternal history; transient
	Mastocytosis, other bullous genodermatoses	Specialist

Management

> **NOTE**
>
> **A specialist multidisciplinary disease**
>
> The IP and pharmacist role is recognition, safe interim handling, timely referral, and medicines support under the EB team's plan.

Immediate

- **Refer urgently** any baby with unexplained fragility or blistering to paediatric dermatology or the EB team. For GOSH, contact the dermatology registrar via switchboard.
- **HANDLE GENTLY**
  no adhesive tapes, lift on a soft pad rather than under the arms, keep the environment cool, and reduce friction.
- Do **not** lance blisters unless trained in EB technique. Specialist nurses teach sterile piercing and silicone dressings.

Ongoing primary-care support

> **DRESSINGS**
> soft silicone or low-adherent, with no ordinary adhesives. Emollients and paraffin to lubricate.

> Pain, constipation, nutrition, dental and eye care through the MDT.

Infection: colonisation is common. Discuss significant infection with the EB team.

MEDICINES OPTIMISATION

liquid formulations if there are oesophageal strictures, avoid rectal routes where possible, and check interactions.

Awareness: **Oleogel-S10** (birch triterpenes) is a specialist wound therapy for selected JEB and DEB wounds.

RDEB

surveillance for **SCC**. Any changing chronic wound needs urgent dermatology review.

Referral

999 / A&E	Respiratory distress or hoarseness (airway involvement in JEB).
ROUTINE	Dehydration, failure to thrive, or sepsis.
ROUTINE	A new, enlarging or ulcerating chronic wound in RDEB (**SCC**).

Red flags and when to refer

- Respiratory distress or hoarseness (airway involvement in JEB).
- Dehydration, failure to thrive, or sepsis.
- A new, enlarging or ulcerating chronic wound in RDEB (**SCC**).

Complications and prognosis

Complications

- Infection, malnutrition, anaemia, strictures, fusion of fingers (DEB), aggressive SCC (recessive DEB).

Prognosis

- EBS often improves with age; severe JEB and recessive DEB are life-limiting.

Prescriber pearls (UK pharmacist IP)

- Give parents **DEBRA UK** support details early.
- Genetic counselling for future pregnancies is a specialist role.

Never blame ‘rough handling’ alone when true fragility is present.

Even mild localised EBS benefits from a correct diagnosis (footwear and blister care education).

Self-check quiz

5 questions · optional

Q1 When should you think of Epidermolysis bullosa (EB)?

✓ A neonate or infant with blisters and erosions from minimal friction (hands, feet, nappy area), ± absent skin at birth, nail dystrophy or mouth blisters.

Q2 What drives it?

✓ Inherited defects in adhesion proteins; the level of split defines the type (simplex, junctional, dystrophic, Kindler).

Q3 What is the backbone of care?

✓ Urgent referral to an EB centre, gentle handling, no adhesives, and soft silicone dressings under specialist guidance.

Q4 What must you never miss?

✓ Airway involvement (hoarse cry in JEB), sepsis, dehydration or failure to thrive, and SCC in chronic RDEB wounds.

Q5 Name a red flag that needs escalation.

✓ Respiratory distress or hoarseness (airway involvement in JEB).

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Photo gallery

Epidermolysis bullosa

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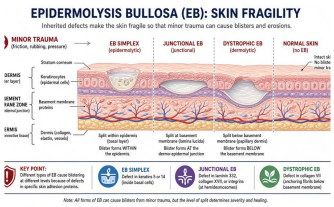

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Childhood & neonatal Module 6 · Week 1

Aplasia cutis congenita

Congenital focal absence of skin, usually on the vertex scalp. Small superficial lesions heal with gentle care. The hair collar sign, large size or a midline position means imaging and urgent specialist care.

ALSO CALLED aplasia cutis congenita · ACC · scalp defect newborn · absent skin · hair collar sign · Adams-Oliver · +2

RELATED PAGES Epidermolysis bullosa (EB) · Sebaceous naevus (naevus sebaceous) · Scarring alopecia overview

KEY FACTS

- Congenital **focal absence of skin**, most often a **solitary scalp vertex** defect; depth ranges from epidermis to bone or dura.
- Appears at birth as an ulcer, erosion, membrane or blister and heals to a **hairless scar**.
- **Hair collar sign** (ring of dark coarse hair) suggests underlying neural tissue: imaging and specialist review.
- Associations: teratogens (**carbimazole/methimazole**, misoprostol, valproate), Adams–Oliver syndrome, Bart syndrome with EB.
- Small superficial lesions heal with gentle care; large, deep or midline lesions need urgent specialist input.

Overview

Congenital **focal absence of skin**. Depth varies from epidermis only through to bone or dura. Most often it is a **solitary scalp vertex** defect. It may be isolated or part of a syndrome (e.g. **Adams–Oliver**, or **Bart syndrome** when limb ACC occurs with EB).

Aetiology: causes and risk factors

- Genetic factors (e.g. BMS1 and other loci).
- **TERATOGENS**
  methimazole/carbimazole, misoprostol, valproate, cocaine.
- Intrauterine infection (varicella, HSV), vascular disruption, amniotic bands, fetus papyraceus.
- Tension at the hair whorl during rapid fetal brain growth.

Myths to address

- **"It's a birth injury from forceps or a scalp clip"**: lesions are present before delivery.

Pathophysiology and pathology

Aplasia cutis results from **failed skin development or loss in utero**.

1. **Genetic factors**
   Mutations (e.g. BMS1) affect skin development.
2. **Tension at the hair whorl**
   Rapid fetal brain growth stretches vertex skin.
3. **Vascular disruption**
   Placental infarcts, amniotic bands or twin demise reduce blood supply.
4. **Healing**
   The defect heals in utero or after birth to a membranous, hairless scar.

Clinical features

- Well-demarcated **ulcer, erosion, membrane or bullous lesion** on the vertex scalp at birth, healing to an **alopecic scar**.
- Less often on the trunk or limbs.
- **HAIR COLLAR SIGN**
  a ring of darker, longer, coarser hair around a scalp lesion. It marks possible **cranial dysraphism or ectopic neural tissue** (encephalocele or meningocele spectrum).
- Large midline defects, vascular stains or neurological signs increase the risk of underlying skull or CNS involvement.

Fig. 1 · Aplasia cutis congenita: congenital absence of skin, classically a scalp vertex defect — assess depth, underlying skull, and associated anomalies.

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Diagnosis and investigations

Clinical at birth. Document size and depth. **Imaging** (ultrasound ± MRI) for large, midline, deep or hair-collar lesions. Exclude birth-trauma mimics (forceps, scalp electrode) from the history and appearance. Genetic or syndrome work-up if there are other anomalies.

Differential diagnoses

Birth trauma, herpes erosions, EB with congenital absence of skin, sebaceous naevus, heterotopic brain tissue, and sinus pericranii.

Management

> **SMALL, SUPERFICIAL LESIONS**
> usually heal with conservative care. Cleanse gently, apply a bland ointment (± a topical antibacterial if infected), and protect from trauma.

> **LARGE, DEEP, MULTIPLE, HAIR-COLLAR OR MIDLINE LESIONS**
> urgent paediatric, plastic surgery or neurosurgery input. Grafting or careful dressings may be needed to prevent **sagittal sinus haemorrhage** and infection.

> **COUNSELLING**
> a scar and bald patch may remain. The teratogen history matters for future pregnancy counselling (specialist or obstetric).

> **PHARMACIST ANGLE**
> know the carbimazole/methimazole association. **Do not stop** maternal antithyroid treatment without obstetric or endocrine advice; flag the association to specialists.

Referral

ROUTINE	**Hair collar sign: image before assuming it is 'just a scar'.**
ROUTINE	Large, deep or midline defects, or exposed dura or bone.
ROUTINE	ACC with blistering (think EB) or limb defects (think Adams–Oliver).
ROUTINE	Bleeding or signs of infection.

Red flags and when to refer

- **Hair collar sign: image before assuming it is 'just a scar'.**
- Large, deep or midline defects, or exposed dura or bone.

- ACC with blistering (think EB) or limb defects (think Adams–Oliver).
- Bleeding or signs of infection.

Complications and prognosis

Complications

- Infection, haemorrhage (sagittal sinus with deep defects), cosmetic alopecia.

Prognosis

- Small lesions heal well; a bald patch remains.

Prescriber pearls (UK pharmacist IP)

Never probe deep scalp defects.

Ask about maternal medicines in pregnancy, especially antithyroid drugs.

Most small defects heal well, so reassure while arranging appropriate follow-up.

Self-check quiz

5 questions · optional

Q1 When should you think of Aplasia cutis congenita?

✓ A well-demarcated erosion, ulcer or shiny membrane on the vertex scalp at birth, healing to a hairless scar.

Q2 What drives it?

✓ Varied causes: genetic, teratogens (carbimazole or methimazole), intrauterine infection, vascular disruption, amniotic bands.

Q3 What is the backbone of care?

✓ Small and superficial: gentle cleansing, bland ointment and protection from trauma.

Q4 What must you never miss?

✓ Hair collar sign (possible neural tube defect), a large midline defect (sagittal sinus), and ACC with blistering (EB) or limb defects (Adams–Oliver).

Q5 Name a red flag that needs escalation.

✓ **Hair collar sign: image before assuming it is 'just a scar'.**

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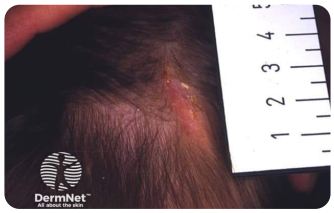
Photo gallery

Aplasia cutis congenita: at birth and healed 2



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CHAPTER 15

Ageing skin & older adults

Itch and xerosis, skin fragility and dermatoporosis, actinic keratoses and photoageing in older adults.

1. Itch & xerosis in older adults
2. Skin fragility & dermatoporosis
3. Actinic keratoses (solar keratoses)
4. Photoageing

Ageing skin & older adults Module 6 · Week 4

Itch & xerosis in older adults

Asteatotic eczema, venous and discoid eczema, lichen simplex and prurigo — with scabies and bullous pemphigoid as don't-miss itch differentials.

ALSO CALLED pruritus · xerosis · asteatotic eczema · eczema craquelé · winter itch · venous eczema · +2

RELATED PAGES Eczema / dermatitis · Emollients & soap substitutes · Skin fragility & dermatoporosis · Bullous pemphigoid (incl. MMP & pemphigoid gestationis) · Scabies

KEY FACTS

- **Xerosis** (dry skin) is the commonest cause of itch in older adults; asteatotic eczema (eczema craquelé) is its inflamed form, especially on the shins.
- Look for a cause of itch without rash: iron deficiency, renal or liver disease, thyroid disease, haematological malignancy, drugs.
- Cornerstone: emollients **liberally and often** (2–3 times a day and after washing), soap substitutes, short lukewarm showers.
- Short course of a moderately potent steroid for asteatotic flares; warn about **fire risk** of emollient-soaked fabric and slippery floors.
- Do not forget scabies, bullous pemphigoid (prodrome) and drug reactions in itchy older people.

Overview

Xerosis (dry skin) is the commonest skin problem and the commonest cause of itch in older adults. **Asteatotic eczema** (eczema craquelé) is the inflammatory end — polygonal "crazy-paving" red fissures, typically on the shins.

This page also covers related elderly itch patterns: venous (stasis) eczema, discoid eczema, lichen simplex / nodular prurigo, and the red-flag differentials of scabies and pre-bullous pemphigoid.

Epidemiology

- **Who**
  - Most people over 65 have some xerosis; worse in winter, with central heating and frequent washing.

Aetiology: causes and risk factors

- Ageing epidermis: fewer lipids and NMF, slower repair, higher water loss; less sebum/sweat.
- Environment: low humidity, central heating, hot showers, soaps and antiseptic washes.
- Drugs: diuretics, **statins**, retinoids, some antiandrogens, chemotherapy/EGFR inhibitors.

- Systemic: hypothyroidism, CKD, diabetes, malnutrition, HIV, malignancy (acquired ichthyosis).

Risk factors and associations

- Ageing skin: less lipid and filaggrin, slower barrier repair, reduced sweat and sebum.
- Diuretics, statins and other drugs; hypothyroidism; low humidity.

Myths to address

> **"ITCHING IN OLDER PEOPLE IS JUST AGE"**
> always examine for scabies, pemphigoid and systemic causes.

> **"ANTIHISTAMINES HELP DRY-SKIN ITCH"**
> sedating ones cause falls and confusion and do little for xerotic itch.

Pathophysiology and pathology

Ageing skin loses its **lipid barrier**, lets water escape and becomes itchy and cracked.

1. **Barrier lipid loss**
   Ceramides, cholesterol and natural moisturising factor fall with age.

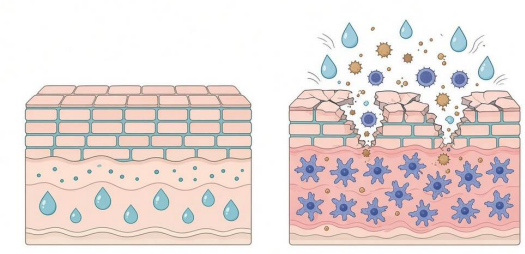

Fig. 1 · A leaky skin barrier lets water out and irritants in.
AI-generated illustration

2. **Water loss and scaling**
   Transepidermal water loss rises; corneocytes shrink and scale.

3 **Cracking**

Fine fissures form a 'crazy-paving' pattern on the shins (eczema craquelé).

Fig. 2 · Asteatotic eczema: crazy-paving pattern on the shins.

AI-generated illustration

4 **Itch–scratch cycle**

Nerve fibres in dry skin become sensitised; scratching damages the barrier further.

5 **Repair**

Regular emollient restores lipids and breaks the cycle.

Fig. 3 · A long-handled applicator helps older people apply emollient to the legs.

AI-generated illustration

Clinical features

- Dull rough scale, tight feel, nocturnal itch; winter flares.
- Asteatotic eczema: shallow red fissures in a cracked-porcelain network on shins (± thighs, arms).

- Venous eczema: medial gaiter area with haemosiderin, oedema, varicose veins — see also venous leg ulcer / red leg.
- Discoid eczema: coin-shaped weeping plaques.
- Lichen simplex / prurigo: thick scratched plaques or nodules at reachable sites.

PRESCRIBER TIP

Skin of colour

Erythema may look violaceous or grey-brown; judge on scale, thickening and symptoms.

Fig. 4 · Asteatotic eczema (eczema craquelé): dry, fissured 'crazy-paving' pattern on the shins of an older adult — worse in winter/low humidity; emollient + soap substitute are the cornerstone.

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Patterns to recognise

Eczema craquelé

Shallow red fissures between plates. Site: Shins.

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Winter itch

Excoriations, little primary rash. Site: Back, limbs.

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Venous eczema

Haemosiderin, oedema, varicose veins. Site: Gaiter area.

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Discoid eczema

Coin-shaped itchy plaques. Site: Limbs.

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Nodular prurigo

Excoriated firm nodules at reachable sites. Site: Limbs, back.

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Crusted scabies

Thick crusts; care-home outbreaks. Site: Hands, body.

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Special groups

- **Itch without rash (generalised pruritus)**
  - Check FBC, ferritin, U&E, LFT, TFT, HbA1c; consider chest X-ray if there are systemic symptoms.

Diagnosis and investigations

- Clinical: season, bathing habits, products, drugs, care-home contacts.
- Itch >> rash, new generalised dryness, or resistance systemic screen and “don’t miss” list below.
- Swab only if infected; patch test if products seem to worsen it.

History

- Season, bathing, products, drugs, care-home contacts.

Itch >> rash?

- New generalised itch or resistant: systemic screen and don't-miss list.

Tests

- Swab only if infected; patch test if products seem to worsen it.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Scabies	Burrows, night itch, itchy contacts or care home.
	Pre-bullous pemphigoid	Urticated plaques, intense itch, older adult: refer.
	Contact dermatitis	Follows medicaments or emollient ingredients.
	Tinea	Annular active edge; scrape.
	Systemic itch	Little rash; weight loss, night sweats, nodes.

Management

Self-care (cornerstone)

- Emollients **liberally and often** (2–3× daily and after washing). Greasier ointments for legs; urea 5–10% for scale.
- Soap substitutes; short lukewarm showers; pat dry; no bubble bath.
- Humidify rooms; cotton clothes.
- **FIRE RISK**
  paraffin (and other) emollients on fabrics — no smoking/naked flames; wash clothing often (MHRA).
- **FALLS RISK**
  slippery bathroom floors — non-slip mats; apply after stepping out.

Prescribing

- Asteatotic flare: short **moderately potent** topical steroid (e.g. clobetasone butyrate) 1–2 weeks plus ongoing emollient; potent short-term for thick leg plaques.
- Infected eczema: treat per NICE guidance — avoid routine topical antibiotics.
- Drug review with the GP (diuretic/statin) — do not stop cardiovascular drugs unilaterally.
- Venous eczema: emollient + steroid + compression after ABPI assessment pathways.
- Lichen simplex/prurigo: potent steroid ± occlusion; address itch driver; refer if refractory.

Fig. 11 · Help older people reach their legs: a long-handled applicator, a pump dispenser, and a non-slip mat because emollients make floors slippery.

AI-generated illustration

Treatment ladder

1

Self-care

Emollients **liberally and often** (2–3 times daily, after washing); greasier on legs; urea 5–10% for scale. Soap substitutes; short lukewarm showers; pat dry; humidify; cotton clothes.

2

Flare

Asteatotic flare: short **moderately potent** steroid (e.g. clobetasone butyrate) 1–2 weeks + emollient. Thick leg plaques: potent, short term.

3

Other patterns

Venous eczema: emollient + steroid + compression after ABPI. Lichen simplex / prurigo: potent steroid ± occlusion; refer if refractory.

4

Drug review

Review diuretics / statins **with the GP**: don't stop cardiovascular drugs unilaterally.

⚠ CAUTION

Prescribing limits and pitfalls

- **Fire risk**: Emollients on fabrics: no smoking or naked flames; wash clothing often (MHRA).
- **Falls**: Slippery floors: non-slip mats; apply after stepping out.

Referral

URGENT	Intense itch with **urticated plaques**: consider pemphigoid; refer.
URGENT	Generalised itch with weight loss, night sweats or nodes: systemic work-up.
SAME DAY	Suspected **scabies outbreak** in a care home.
ROUTINE	Non-healing 'eczema': patch test or rethink the diagnosis.

Red flags and when to refer

- Generalised itch with little rash, weight loss, night sweats or lymphadenopathy systemic work-up.
- Suspected scabies outbreak in a care home.
- New intense itch with urticated plaques in an older adult consider pemphigoid; refer.
- Non-healing "eczema" patch testing or alternative diagnosis.

Complications and prognosis

Complications

- Infected eczema; skin tears and dermatoporosis.
- Sleep disturbance, depression.
- Falls from slippery emollients; burns from emollient-soaked fabrics.

Prognosis

- Chronic and seasonal; controlled with daily skin care.

Prescriber pearls (UK pharmacist IP)

- "Steroid settles it but it always comes back" usually means **no maintenance emollient**.
- Help older people reach their legs — carers, pumps, long-handled applicators.
- Teach fingertip units; steroid treats the flare, emollient prevents the next one.

Self-check quiz

4 questions · optional

Q1 Commonest cause of itch in older adults?

✓ Xerosis (dry skin).

Q2 Two drug classes that dry the skin?

✓ Diuretics and statins (also retinoids, antiandrogens, EGFR inhibitors).

Q3 Itch with urticated plaques in an 80-year-old?

✓ Consider pre-bullous pemphigoid: refer.

Q4 Steroid for an asteatotic flare?

✓ Short moderately potent course (e.g. clobetasone butyrate) 1–2 weeks with ongoing emollient.

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Photo gallery

Xerosis and asteatotic eczema

6

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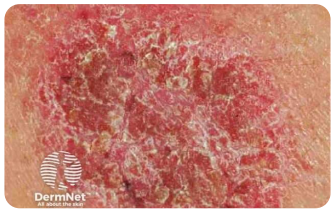

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Venous, discoid eczema and lichen simplex

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Prurigo and neuropathic itch

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Don’t miss: scabies and bullous pemphigoid

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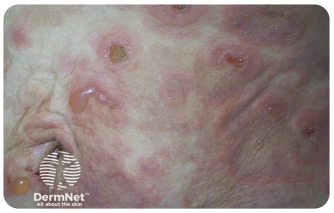

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Ageing skin & older adults Module 6 · Week 4

Skin fragility & dermatoporosis

Actinic purpura, thin skin, skin tears and stellate pseudoscars — protect fragile skin and review anticoagulants/steroids carefully.

ALSO CALLED senile purpura · actinic purpura · Bateman purpura · dermatoporosis · skin tears · stellate pseudoscars

RELATED PAGES Photoageing · Purpura: the differential diagnosis · Topical corticosteroids: potency, quantities & safety · Itch & xerosis in older adults · Wound aftercare: sutures, activity, infection & scars

KEY FACTS

- Actinic (senile) purpura benign purple patches on thin sun-damaged forearms and hands after minor trauma.
- Dermatoporosis chronic skin insufficiency with atrophy, purpura, stellate pseudoscars and **skin tears**.
- Aggravators: anticoagulants, antiplatelets, corticosteroids (topical, oral, inhaled), SSRIs, fish oil, ginkgo.
- Platelets and clotting are usually normal.
- Protect skin (sleeves, padding, careful transfers); **never stop anticoagulants** just for purpura without review.

Overview

Actinic (senile/Bateman) purpura is benign purple-red ecchymoses on thin sun-damaged forearms and hands after minimal trauma. **Dermatoporosis** describes chronic cutaneous insufficiency — atrophy, purpura, stellate pseudoscars and **skin tears**.

Aetiology: causes and risk factors

- Chronic UV + intrinsic ageing dermal collagen/elastin loss; vessels rupture with minor shear.
- Aggravators: anticoagulants, antiplatelets, topical/oral/inhaled **corticosteroids**, SSRIs, fish oil/ginkgo/high-dose vitamin E.
- Platelets and clotting are usually normal in classic actinic purpura.

Myths to address

- **"BRUISING LIKE THIS MEANS A BLOOD DISORDER"**
  classic actinic purpura has normal clotting; check if bruises are elsewhere.

Pathophysiology and pathology

Sun and age **thin the dermis**, so vessels lose support.

1. **Collagen loss**
   UV and ageing reduce dermal collagen and elastin.
2. **Unsupported vessels**
   Small vessels rupture with minor shear.
3. **Haemosiderin**
   Purpura fades to brown staining.
4. **Skin tears**
   The thin epidermis and dermis separate easily.

Clinical features

- Irregular dark-red/purple patches 1–5 cm on extensor forearms and dorsal hands; fade to brown haemosiderin.
- Thin translucent wrinkled skin; stellate pseudoscars; skin tears from shear/adhesive dressings.
- Not tender; **not palpable** (palpable = vasculitis).

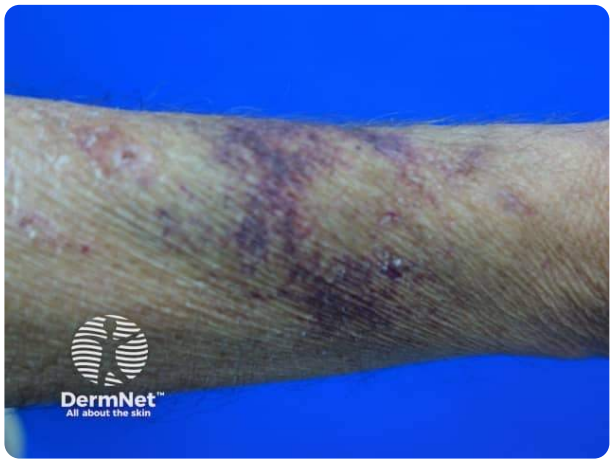

Fig. 1 · Actinic (senile/Bateman) purpura: irregular dark-red/purple ecchymoses on thin, sun-damaged forearms after trivial trauma — check anticoagulant/antiplatelet/steroid use.

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Diagnosis and investigations

Clinical at typical sites. Investigate FBC/clotting/LFTs (± INR) if purpura is on covered sites, palpable, with mucosal bleeding, systemic symptoms, new anticoagulant, or unexpected INR rise. Consider elder abuse if history/pattern mismatch.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Steroid purpura	Long-term steroid exposure
	Thrombocytopenia / coagulopathy	Petechiae, mucosa, abnormal bloods
	Vasculitis	Palpable purpura, legs, systemic features
	Scurvy	Perifollicular haemorrhage, corkscrew hairs
	Non-accidental injury	Grip marks, multiple ages of bruise

Management

- Reassure: benign but marks fragility.
- Protect: long sleeves, tubular forearm protectors, pad furniture, careful carer transfers; avoid harsh adhesives.
- Sun protection and daily emollient.
- Medication review with the GP — **never stop an anticoagulant solely for actinic purpura** without discussion.
- Skin tears: gently approximate, non-adherent dressings, tetanus status; refer complex wounds.

Referral

ROUTINE	Palpable purpura or systemic vasculitis signs.
ROUTINE	Unusual distribution / safeguarding concern.
ROUTINE	Large unexplained bruises with rising INR.

Red flags and when to refer

- Palpable purpura or systemic vasculitis signs.
- Unusual distribution / safeguarding concern.
- Large unexplained bruises with rising INR.

Complications and prognosis

Complications

- Skin tears, infection, delayed healing.

Prognosis

- Chronic; protection reduces injuries.

Prescriber pearls (UK pharmacist IP)

- Actinic purpura is a counselling and protection problem more than a cream problem.
- Inhaled as well as topical steroids can contribute — ask.
- Document skin-tear classification if using local wound pathways.

Self-check quiz

5 questions · optional

Q1 When should you think of Skin fragility & dermatoporosis?

✓ Painless purple patches on thin sun-damaged forearms after minimal trauma in an older person.

Q2 What drives it?

✓ UV and age destroy dermal support around vessels easy shear bleeding; advanced form is dermatoporosis with skin tears.

Q3 What is the backbone of care?

✓ Reassure, protect (sleeves, pad furniture), daily emollient and sun protection; medication review with the prescriber.

Q4 What must you never miss?

✓ Palpable purpura, non-exposed-site bruising, mucosal bleeding or safeguarding patterns — investigate / escalate.

Q5 Name a red flag that needs escalation.

✓ Palpable purpura or systemic vasculitis signs.

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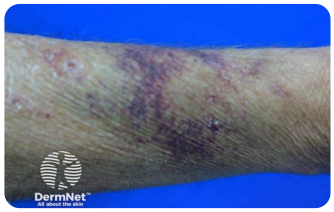
Photo gallery

Purpura, thin skin and skin tears 7



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Ageing skin & older adults Module 6 · Week 4

Actinic keratoses (solar keratoses)

Rough, scaly UV-induced dysplastic patches on sun-damaged skin: a marker of field cancerisation with a small risk of SCC. Sun protection, then lesion- or field-directed treatment, and when to refer.

ALSO CALLED actinic keratosis · solar keratosis · AK · actinic cheilitis · field change · field cancerisation · +2

RELATED PAGES Cutaneous squamous cell carcinoma (cSCC) · Seborrhoeic keratosis · Photoageing · Solar lentigines

KEY FACTS

- **Actinic (solar) keratoses** are UV-induced patches of dysplastic keratinocytes on chronically sun-exposed skin.
- Very common: up to **25% of people over 60** in northern England and Ireland; more in fair skin, outdoor workers and the immunosuppressed.
- Rough, **gritty** scale on a pink base, often easier to feel than see; face, bald scalp, ears, forearms and backs of hands.
- A single AK rarely becomes SCC, but people with **many AKs** have a 10–15% lifetime SCC risk (field cancerisation).
- Sun protection for all; cryotherapy for few lesions; **field treatment** (5-FU, imiquimod, diclofenac 3%, PDT). Tender, thick or growing exclude SCC.

Overview

Actinic keratoses (AKs) are common UV-induced dysplastic lesions on chronically sun-exposed skin of older adults. They mark **field cancerisation** and a small risk of progression to squamous cell carcinoma.

Benign **seborrhoeic keratoses** are covered on their own page: seborrhoeic keratosis.

Epidemiology

- **Who**
  - Older adults with fair skin (Fitzpatrick I–II), chronic sun exposure and photoageing.
  - Organ transplant recipients and other immunosuppressed people have many more, and more aggressive, lesions.
- **Natural history**
  - Individual AKs can regress (up to a quarter a year), persist or, rarely, progress to SCC.

Aetiology: causes and risk factors

- AK: cumulative UVB/UVA; fair skin, outdoor work, immunosuppression (transplant) raise burden and SCC risk.

- Actinic cheilitis is AK of the lip vermilion.

Risk factors and associations

- Cumulative **UVB** (and UVA) exposure: outdoor work, sunbeds, living in sunny climates.
- Immunosuppression (transplant drugs), older age, previous skin cancer; HPV types may contribute.

Myths to address

> **"IT'S JUST DRY SKIN"**
> a persistent gritty patch on sun-damaged skin is an AK until shown otherwise.

> **"TREAT EACH SPOT"**
> once cancer is excluded, treating the whole field matters more.

Pathophysiology and pathology

UV damage to keratinocyte DNA creates a **field** of genetically abnormal cells; the AK is the visible tip.

1. **UV DNA damage**
   UVB forms pyrimidine dimers; mutations in **TP53** let damaged keratinocytes survive and divide.

2. **Clonal expansion**
   Mutated clones spread along the basal layer, producing **dysplasia** of the lower epidermis.

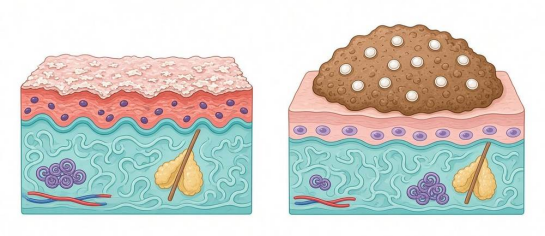

Fig. 1 · Actinic keratosis (left: rough scale over dysplastic epidermis) compared with a benign seborrhoeic keratosis (right).
AI-generated illustration

3. **Abnormal keratinisation**
   Dysplastic cells make abnormal keratin, giving adherent, rough **parakeratotic** scale.

4. **Field cancerisation**
   Surrounding sun-damaged skin carries similar mutations, so new AKs and SCCs arise across the field.

Fig. 2 · Multiple AKs across a sun-damaged bald scalp.
AI-generated illustration

5. **Progression**
   If dysplasia becomes full-thickness (Bowen's disease) or invades the dermis, it has become SCC.

HISTOPATHOLOGY

- Atypical keratinocytes in the **basal layer** (partial-thickness dysplasia), with parakeratosis that spares the follicular openings (alternating pink and blue).
- Solar elastosis in the dermis.
- Biopsy if thick, tender, ulcerated or not responding (to exclude SCC).

Clinical features

- AK: rough sandpaper scale on erythematous base; face, bald scalp, ears, forearms, hands; often multiple.
- Hypertrophic AK and cutaneous horn need careful SCC exclusion.

Fig. 3 · Actinic keratosis on the nasal bridge: pink base with adherent hyperkeratotic scale — often better felt (sandpaper) than seen.

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Special groups

- **Immunosuppressed (transplant)**
  - Many AKs and high SCC risk: lower threshold for specialist referral and regular skin checks.
- **Lips (actinic cheilitis)**
  - Blurred, scaly vermilion border of the lower lip; higher risk of SCC: lower threshold for specialist review.

Diagnosis and investigations

Clinical ± dermoscopy. Biopsy or 2WW/urgent skin cancer referral if SCC or melanoma cannot be excluded. Never destroy an undiagnosed pigmented lesion with cryotherapy or laser.

Clinical

- Fingertip + dermoscopy.

Doubt

- Biopsy or **suspected cancer referral** if SCC or melanoma can't be excluded.

Never

- Destroy an undiagnosed pigmented lesion with cryo or laser.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	SCC	Indurated, tender, enlarging, ulcerated.
	Bowen's disease	Persistent scaly plaque.
	SK	Stuck-on waxy vs gritty flat scale.
	Solar lentigo	Flat even brown macule.
	Lentigo maligna	Irregular facial pigment: urgent pathway.

Management

Everyone with AKs

- Daily broad-spectrum SPF 30–50, hat, behaviour change; skin self-check education.
- Emollient to reduce scale.

Lesion- and field-directed options (local formulary / CKS themes)

- Few discrete AKs: **cryotherapy** (light freeze) in trained hands.
- Field change: topical **5-fluorouracil**, **imiquimod**, **diclofenac 3% gel**, or photodynamic therapy via dermatology — expect an inflammatory treatment phase; counsel clearly.
- Actinic cheilitis: lower threshold for specialist review.
- Immunosuppressed: lower threshold for dermatology.

Treatment ladder

Everyone

Daily broad-spectrum **SPF 30–50**, hat, behaviour change; self-check education. Emollient to reduce scale.

2 **Few discrete AKs**

Cryotherapy (light freeze) in trained hands.

3 **Field change**

Topical **5-fluorouracil**, **imiquimod**, **diclofenac 3% gel**, or PDT via dermatology. Counsel: an **inflammatory phase** is expected.

4 **Lower threshold for dermatology**

Actinic cheilitis; immunosuppressed patients.

5 **SKs**

Reassure when typical; cryo / curettage only if symptomatic or doubtful (often not NHS-funded if cosmetic).

Referral

URGENT	Possible SCC or melanoma: suspected cancer referral (NG12).
URGENT	Rapidly enlarging tender nodule in an AK field.
URGENT	Lip AK with induration or ulceration.

Red flags and when to refer

- Possible SCC or melanoma urgent suspected-cancer referral (NG12).
- Rapidly enlarging tender nodule in an AK field.
- Lip AK with induration or ulceration.

Complications and prognosis

Complications

- Progression to **cutaneous SCC**.
- Cutaneous horn (may hide SCC).

- Inflammatory reactions to field treatments; cosmetic concern.

Prognosis

- Chronic: new lesions keep appearing in sun-damaged skin.
- Treatment clears most lesions; ongoing sun protection and surveillance are needed.

Prescriber pearls (UK pharmacist IP)

- Feel with a fingertip — AKs are gritty.
- Treating the field matters more than chasing every spot once cancer is excluded.
- Warn that 5-FU/imiquimod look worse before they look better.
- Tender, thickened or growing AK, or an AK on the lip or ear: see cutaneous squamous cell carcinoma for when it may have become SCC.

Self-check quiz

5 questions · optional

Q1 Why treat a field of AKs rather than single spots?

✓ The whole sun-damaged area carries mutated cells (field cancerisation); treating it reduces new AKs and SCCs.

Q2 Which AK features suggest SCC?

✓ Tenderness, thickening, induration, rapid growth, ulceration or bleeding.

Q3 Name three field treatments.

✓ Topical 5-fluorouracil, imiquimod and diclofenac 3% gel; photodynamic therapy via dermatology.

Q4 What should patients expect from 5-FU or imiquimod?

✓ An inflammatory phase: redness, crusting and soreness before healing.

Q5 What is actinic cheilitis?

✓ Actinic keratosis of the lip vermilion; higher risk of SCC.

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Photo gallery

Actinic keratoses 6

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Ageing skin & older adults Module 6 · Week 4

Photoageing

UV-driven dermatoheliosis: solar elastosis, Favre–Racouchot, poikiloderma of Civatte, IGH and related patterns — prevention first.

ALSO CALLED photoageing · dermatoheliosis · solar elastosis · Favre-Racouchot · cutis rhomboidalis nuchae · poikiloderma of Civatte · +2

RELATED PAGES Solar lentigines · Actinic keratoses (solar keratoses) · Skin fragility & dermatoporosis · Retinoids (topical & oral) · Facial ageing & neuromodulators

KEY FACTS

- **Photoageing** is premature skin ageing from chronic UV (and visible/infrared) exposure, on top of intrinsic ageing.
- Wrinkles, coarse texture, mottled pigment, solar lentigines, telangiectasia, yellowish thickening (**solar elastosis**).
- Marks cumulative sun damage and higher risk of AK, BCC, SCC and melanoma: examine the skin.
- Prevention: sun protection and stopping smoking; topical **retinoids** have the best evidence for treatment.
- Procedures (peels, lasers, injectables) are cosmetic and not NHS-funded.

Overview

Photoageing (dermatoheliosis) is the cumulative UV contribution to ageing skin, distinct from intrinsic chronological ageing. Patterns include solar elastosis, cutis rhomboidalis nuchae, Favre–Racouchot (nodular elastosis with cysts and comedones), poikiloderma of Civatte, idiopathic guttate hypomelanosis (IGH) and senile comedones.

Epidemiology

- **Who**
  - Fair skin, outdoor workers, sunbed users, sunny climates; smoking adds damage.

Aetiology: causes and risk factors

- UVA/UVB matrix metalloproteinases, DNA damage, solar elastosis.
- Smoking and pollution amplify extrinsic ageing.

- Favre–Racouchot: severe elastosis + follicular plugging on the malar/periorbital skin, mainly in older men with heavy sun/smoking history.

Risk factors and associations

- UVA penetrates the dermis (collagen and elastin damage); UVB damages the epidermis and DNA.
- Smoking, pollution, infrared radiation.

Myths to address

> **"A TAN PROTECTS THE SKIN"**
> a tan is a sign of DNA damage.

> **"ANTI-AGEING CREAMS REVERSE WRINKLES"**
> only retinoids and sun protection have good evidence.

Pathophysiology and pathology

UV turns on **collagen-destroying enzymes** while slowing new collagen formation.

1. **Free radicals**
   UV generates reactive oxygen species in skin cells.
2. **Matrix breakdown**
   Signalling (AP-1) switches on matrix metalloproteinases that digest collagen.
3. **Reduced repair**
   New collagen synthesis falls; fragmented collagen weakens the dermis: wrinkles.
4. **Solar elastosis**
   Abnormal elastic material builds up in the upper dermis, causing yellow, thickened skin.
5. **Pigment and vessels**
   Uneven melanocyte activity (lentigines) and dilated vessels (telangiectasia) appear.

HISTOPATHOLOGY

- Solar elastosis (blue-grey amorphous elastic material in the papillary dermis), reduced collagen, atrophic or uneven epidermis.

Clinical features

- Atrophy or thickened leathery skin, deep wrinkles, blotchy pigment, telangiectasia.
- Cutis rhomboidalis nuchae: deep crossed furrows on the nape.
- Poikiloderma of Civatte: red-brown reticulate pigment and atrophy on the sides of the neck, sparing shaded under-chin.
- IGH: small porcelain-white macules on sun-exposed limbs.
- Senile comedones / Favre–Racouchot: open blackheads in elastotic cheek skin.

Fig. 1 · Atrophic photoageing: thin, telangiectatic, smoother skin with lentigines and keratoses — higher keratinocyte cancer risk.

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Special groups

- **Skin of colour**
  - Photoageing appears later and is mainly pigmentary (uneven tone, dermatosis papulosa nigra) rather than wrinkles.

Diagnosis and investigations

Clinical. Use photoageing as a prompt to examine for skin cancer and AK field change. Biopsy atypical pigmented or ulcerated lesions.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Intrinsic ageing alone	Less elastosis on covered sites
	Rosacea / steroid damage	Central face papules or atrophy history
	Vitiligo vs IGH	Larger milky patches; Wood's lamp; progression
	Acne comedones	Younger; inflammatory lesions

Management

- **PREVENTION**
  daily SPF 30–50, shade, clothing, no sunbeds; stop smoking.

- Emollients; treat xerosis as on the itch/xerosis page.
- Topical retinoids (POM) can modestly improve texture — irritation and photosensitivity counselling; avoid in pregnancy.
- Extraction/retinoids for Favre–Racouchot comedones; procedures (laser, peels, IPL) are usually private — counselling only here.
- Evidence for oral collagen/HA supplements is limited — do not oversell; prioritise sunscreen and lifestyle.
- Cross-link cosmetic counselling: facial ageing & neuromodulators, solar lentigines.

Referral

URGENT — Any changing, bleeding or irregular lesion in a photoaged field urgent skin-cancer pathway.

ROUTINE — Unilateral or atypical "poikiloderma" — reconsider contact, connective tissue disease or mycosis fungoides (rare).

Red flags and when to refer

- Any changing, bleeding or irregular lesion in a photoaged field urgent skin-cancer pathway.
- Unilateral or atypical "poikiloderma" — reconsider contact, connective tissue disease or mycosis fungoides (rare).

Complications and prognosis

Complications

- Actinic keratoses and skin cancers; skin fragility, purpura and skin tears.

Prognosis

- Partly preventable and partly reversible with photoprotection and retinoids.

Prescriber pearls (UK pharmacist IP)

Photoageing is a **risk marker**, not only a cosmetic complaint.

Under-chin sparing is a useful Civatte clue.

Supplements are not a substitute for SPF.

Sun-damaged skin is where keratinocyte cancers arise: see basal cell carcinoma and squamous cell carcinoma.

Self-check quiz

5 questions · optional

Q1 When should you think of Photoageing?

✓ Leathery, wrinkled, unevenly pigmented sun-exposed skin with open comedones on the cheeks or rhomboid nuchal furrows.

Q2 What drives it?

✓ Chronic UV degrades collagen/elastin (solar elastosis) and dysregulates pigment and follicular keratinisation.

Q3 What is the backbone of care?

✓ Lifelong photoprotection, smoking cessation, emollients; retinoids/procedures are adjunctive and usually private.

Q4 What must you never miss?

✓ Photoaged fields hide AK, BCC, SCC and lentigo maligna — examine the whole field, not just the wrinkle.

Q5 Name a red flag that needs escalation.

✓ Any changing, bleeding or irregular lesion in a photoaged field urgent skin-cancer pathway.

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Photo gallery

Photoageing and benign lesions

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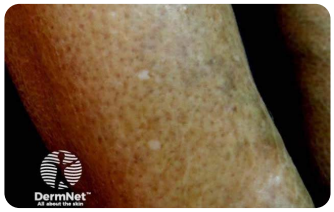

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CHAPTER 16

Hair & scalp

Pattern hair loss (incl. PRP), alopecia areata, telogen effluvium, trichotillomania and scarring alopecia: name the pattern and spot scarring early.

1. Androgenetic alopecia (male & female)
2. Alopecia areata
3. Telogen effluvium
4. Trichotillomania
5. Scarring alopecia overview

Hair & scalp Module 6 · Week 6

Androgenetic alopecia (male & female)

Male pattern: recession plus vertex. Female pattern: a widened parting with the frontal line kept. Topical minoxidil for both sexes, finasteride 1 mg for men, and an honest view of PRP and off-label options.

ALSO CALLED AGA · male pattern hair loss · MPHL · female pattern hair loss · FPHL · hair loss · +7

RELATED PAGES Telogen effluvium · Alopecia areata · Hirsutism & the Ferriman–Gallwey score · Scarring alopecia overview

KEY FACTS

- **Androgenetic alopecia** (male- and female-pattern hair loss) is the commonest cause of hair loss: genetic sensitivity of follicles to androgens.
- Men: temporal recession and vertex thinning (Hamilton–Norwood); women: diffuse crown thinning with a **preserved frontal hairline** and widened parting (Ludwig).
- Hallmark: **miniaturisation** (hairs of varying thickness).
- Topical **minoxidil** 2% or 5% for both sexes (continuous, about 6 months to judge); **finasteride 1 mg** in men.
- In women with signs of hyperandrogenism (acne, hirsutism, irregular periods) check for PCOS or another cause.

Overview

Androgenetic alopecia (AGA) is genetically predisposed, androgen-mediated, progressive **follicular miniaturisation**.

	MALE PATTERN (MPHL)	FEMALE PATTERN (FPHL)
Pattern	Frontotemporal recession plus vertex (Norwood–Hamilton)	Mid-frontal and crown thinning with a **widened parting** ('Christmas tree'); **frontal hairline usually preserved** (Ludwig/Sinclair)
Androgens	Normal	Usually **normal**. Check for PCOS if there are acne, hirsutism or irregular periods

Epidemiology

- **Who**
  - About half of men by 50 and up to 40% of women by 70.
  - Strong family history; onset after puberty.

Aetiology: causes and risk factors

- **DHT** (from testosterone via 5α-reductase) shortens anagen and miniaturises follicles in a genetically susceptible scalp.
- Polygenic inheritance from **both** sides of the family.
- Women with true hyperandrogenism may show a male-type pattern with acne, hirsutism and irregular periods (consider PCOS).

Risk factors and associations

- Polygenic inheritance (androgen receptor gene on the X chromosome and others).
- **Dihydrotestosterone (DHT)** formed by 5α-reductase in the scalp.

Myths to address

> **"HATS OR FREQUENT WASHING CAUSE BALDNESS"**
> they do not.

> **"MINOXIDIL CAN BE STOPPED ONCE HAIR RETURNS"**
> stopping loses the gains within months.

Pathophysiology and pathology

DHT gradually **shrinks** genetically susceptible scalp follicles.

1. **Androgen activation**
   5α-reductase type II converts testosterone to **DHT** in the follicle; finasteride blocks this step.
2. **Shorter anagen**
   DHT shortens each growth phase, so hairs grow for less time and are shorter.
3. **Miniaturisation**
   Each cycle produces a thinner, shorter, less pigmented hair until terminal hairs become vellus-like.
4. **Visible thinning**
   More follicles in telogen and fewer terminal hairs give see-through scalp in a patterned distribution.

HISTOPATHOLOGY

- Rarely needed: reduced terminal-to-vellus ratio, increased telogen hairs, no scarring. Dermoscopy shows variable hair diameter (anisotrichosis).

Clinical features

- Gradual, **non-scarring** thinning with miniaturised hairs and **variable hair diameter**.
- The pull test is usually **negative** (positive in telogen effluvium).
- Psychological distress is common in both sexes.

Fig. 1 · Male pattern AGA — patterned thinning (frontotemporal/vertex).
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Special groups

- **Women**
  - Usually normal androgen levels; test only if there are features of hyperandrogenism.
  - Spironolactone is an off-label specialist option; contraception needed.

Diagnosis and investigations

- Clinical ± dermoscopy (miniaturisation, hair diameter diversity).
- **Bloods** if there is sudden shedding, signs of scarring, or features of hyperandrogenism or systemic disease in women: **ferritin, TFTs**, and hormones as indicated.
- Biopsy is rarely needed.

Bloods

- if there is sudden shedding, signs of scarring, or features of hyperandrogenism or systemic disease in women: **ferritin, TFTs**, and hormones as indicated.

Key points

- Clinical ± dermoscopy (miniaturisation, hair diameter diversity).
- Biopsy is rarely needed.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Telogen effluvium	Acute diffuse shedding 2–3 months after a trigger; **positive pull test**
	Alopecia areata	Patchy, with exclamation-mark hairs; may be diffuse

	CONDITION	DISTINGUISHING FEATURES
	Traction / trichotillomania	Trauma pattern; broken hairs of different lengths
	Scarring alopecias	Loss of follicle openings, redness, scale, pain. **Refer**
	Nutritional / thyroid	Systemic features; blood tests

Management

Check doses in the current BNF and SmPC. Off-label agents follow local dermatology advice.

Established options

- **Topical minoxidil** 2% or 5% for **both sexes**. It must be used continuously. An **initial shed** is possible in the first weeks. Facial hair can appear if it runs onto the face. Allow about 6 months to judge the effect.
- **Oral finasteride 1 mg** (men), a type II 5α-reductase inhibitor. Counsel on **sexual and mood side effects** (MHRA warning on psychiatric effects and suicidal thoughts). It is **teratogenic to a male fetus**: contraindicated in pregnancy, and women who are or may become pregnant must not handle crushed or broken tablets. Not licensed for women of childbearing potential.

Other options: overview, caution and referral

OPTION	NOTES
Topical finasteride	Emerging; systemic absorption possible. Specialist
Dutasteride	Off-label in men. Specialist
Spironolactone (FPHL)	Off-label anti-androgen, especially with hyperandrogenism. Monitor potassium and BP; contraception needed. Specialist or an experienced IP within competence
Low-dose oral minoxidil	Off-label, with growing evidence. Side effects: hypertrichosis, fluid retention, tachycardia. **Specialist initiation**, not a casual community start
PRP (platelet-rich plasma)	Autologous concentrated platelets injected into the scalp. **Mixed evidence quality**, non-standard protocols, private cost. An adjunct, not a replacement
LLLT, transplant, micropigmentation, camouflage	Variable evidence. Set expectations

Referral

ROUTINE Signs of **scarring alopecia** (lost follicle openings, inflammation, pain or itch): refer.

ROUTINE Rapid or patchy loss, or a positive pull test: look for another diagnosis.

ROUTINE Women with acne, hirsutism and irregular periods: assess for **PCOS** or hyperandrogenism.

ROUTINE Mood change or sexual dysfunction on finasteride: review and stop if needed (MHRA).

Red flags and when to refer

- Signs of **scarring alopecia** (lost follicle openings, inflammation, pain or itch): refer.
- Rapid or patchy loss, or a positive pull test: look for another diagnosis.
- Women with acne, hirsutism and irregular periods: assess for **PCOS** or hyperandrogenism.
- Mood change or sexual dysfunction on finasteride: review and stop if needed (MHRA).

Complications and prognosis

Complications

- Psychological distress; sunburn and actinic damage on the bald scalp.

Prognosis

- Slowly progressive; treatment stabilises or partly reverses loss while it continues.

Prescriber pearls (UK pharmacist IP)

- **Early treatment preserves more hair** than late rescue.
- Stopping treatment means **gains reverse** over months, so explain this before starting.
- Women: exclude scarring alopecia and reversible causes (iron, thyroid) before long-term cosmetic treatment.
- Finasteride counselling must cover psychiatric and sexual side effects and pregnancy safety for partners.
- PRP: avoid overselling it. Document informed expectations. Platelet disorders and anticoagulation need specialist judgement.

Self-check quiz

5 questions · optional

Q1 When should you think of Androgenetic alopecia (male & female)?

✓ Gradual, non-scarring thinning: frontotemporal recession and vertex in men; widening central parting or crown in women.

Q2 What drives it?

✓ DHT (via 5α-reductase) miniaturises genetically susceptible follicles and shortens anagen.

Q3 What is the backbone of care?

✓ Topical minoxidil (both sexes) and oral finasteride 1 mg (men), started early and continued. Gains reverse on stopping.

Q4 What must you never miss?

✓ Scarring alopecia (lost follicle openings, inflammation), hyperandrogenism in women, and telogen effluvium triggers (iron, thyroid).

Q5 Name a red flag that needs escalation.

✓ Signs of **scarring alopecia** (lost follicle openings, inflammation, pain or itch): refer.

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Photo gallery

Male pattern hair loss

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Female pattern hair loss

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Hair & scalp Module 3 · Week 3

Alopecia areata

Autoimmune non-scarring bald patches with exclamation-mark hairs. Many limited patches regrow; refer extensive, childhood or distressed disease.

ALSO CALLED alopecia areata · patchy hair loss · exclamation mark hairs · alopecia totalis · alopecia universalis · ophiasis

RELATED PAGES Androgenetic alopecia (male & female) · Telogen effluvium · Trichotillomania · Scarring alopecia overview · Tinea: corporis, cruris & pedis

KEY FACTS

- **Alopecia areata** is autoimmune, **non-scarring** hair loss in round or oval patches with normal-looking skin.
- **Exclamation-mark hairs** at the edge; can affect beard, brows, lashes and nails (pitting).
- Severe forms: **ophiasis** (band along the hairline), totalis (whole scalp) and universalis (all body hair).
- About 80% of small, short-duration patches regrow within a year, but relapse is common.
- Adults: potent or very potent scalp steroid for about 3 months (off-label); refer children, extensive or distressing disease (JAK inhibitors are specialist).

Overview

Alopecia areata (AA) is a common autoimmune **non-scarring** hair loss causing sudden smooth bald patches on any hair-bearing site. Lifetime risk is about 2%. Spectrum: patchy AA, ophiasis, diffuse AA, **alopecia totalis** (all scalp) and **universalis** (all body hair).

Epidemiology

- **Who**
  - Lifetime risk about 2%; any age, often starting in childhood or young adulthood.
  - Associated with atopy, thyroid disease, vitiligo and Down syndrome.

Aetiology: causes and risk factors

- Collapse of hair-follicle immune privilege peribulbar T-cell attack; IFN-γ/IL-15 via JAK–STAT (rationale for specialist JAK inhibitors).
- Genetic predisposition (~1 in 5 family history). Triggers sometimes cited: stress, illness — often none found.

- Associations: autoimmune thyroid disease, vitiligo, type 1 diabetes, atopy, Down syndrome; higher anxiety/depression rates.

Risk factors and associations

- Loss of the hair follicle's **immune privilege** with CD8+ T-cell attack; genetic susceptibility (HLA, ULBP3).
- Triggers sometimes reported: stress, illness.

Myths to address

> **"STRESS ALONE CAUSES IT"**
> it is autoimmune; stress may trigger flares.

> **"HAIR LOSS IS PERMANENT"**
> follicles are not destroyed, so regrowth is possible even after years.

Pathophysiology and pathology

Immune cells attack **growing (anagen) hair bulbs** but spare the stem cells, so hair can regrow.

1. **Immune privilege collapse**
   Anagen bulbs normally hide their antigens (low MHC class I); in AA this privilege fails.
2. **T-cell attack**
   Interferon-γ and **JAK–STAT** signalling recruit CD8+ T cells around the bulb ('swarm of bees'): the target of JAK inhibitors.
3. **Dystrophic hairs**
   Damaged hairs narrow near the scalp and break: **exclamation-mark hairs**; follicles enter early catagen and telogen.
4. **Regrowth**
   Stem cells in the bulge survive, so regrowth often starts as fine white hairs that later repigment.

HISTOPATHOLOGY

- Peribulbar lymphocytic infiltrate around anagen follicles; increased catagen/telogen hairs; no scarring. Biopsy only if the diagnosis is uncertain.

Clinical features

- Round/oval smooth skin-coloured patches; **no scale, erythema or scarring**; ostia preserved.
- **Exclamation-mark hairs** at active margins; pull test positive at the edge.
- Grey/white hairs often spared; regrowth may start as fine white vellus hairs.
- Nail **pitting** (thimble pits); also trachyonychia.
- May involve brows, lashes and beard — avoid potent topical steroids on the face.

Fig. 1 · Classic AA: a well-defined, round, smooth, skin-coloured bald patch with NO scale, redness or scarring — follicular openings preserved.

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Special groups

- **Children**
  - Refer; use lower-potency steroids, and consider the psychosocial impact and school support.
- **Nails**
  - Fine regular pitting or trachyonychia in 10–20%, more with severe disease.

Diagnosis and investigations

- Clinical: extent, speed, prior episodes, autoimmune/atopy history, psychosocial impact.
- Trichoscopy: exclamation marks, yellow dots (ostia present), black dots, tapered/regrowing hairs.
- Mycology if scale or childhood doubt (exclude tinea). Syphilis serology if moth-eaten. ANA if lupus possible.
- Thyroid: consider TSH ± TPO especially in totalis/universalis or symptoms; check TSH is in range if already hypothyroid (coexistent telogen effluvium).
- Routine iron/zinc/vitamin D supplementation is not evidence-based for AA (BAD).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Trichotillomania	Irregular patches; hairs of different lengths; pull test negative — see trichotillomania
	Tinea capitis	Scale, black dots, nodes; children; mycology
	Telogen effluvium	Diffuse shed; trigger 2–3 months earlier — see telogen effluvium

	CONDITION	DISTINGUISHING FEATURES
	Pattern hair loss	Gradual miniaturisation — see androgenetic alopecia
	Scarring alopecia	Lost ostia, shiny scalp, symptoms — see scarring alopecia

Management

Primary care (CKS / BAD themes)

- Explain unpredictability; ~80% of limited short-duration patches regrow within a year, but relapses are common. Extensive/AT/AU: full recovery ~1 in 10.
- Sun protection on bald patches; signpost Alopecia UK / BAD leaflets; screen psychological distress.
- **No treatment / watchful waiting** if already regrowing or small non-distressing patches.
- Adults: potent or very potent **scalp** steroid once daily for about **3 months** (off-label); reassess 3–6 months. Never potent steroid on face/brows.
- Camouflage, fibres, brows/lashes cosmetics; NHS wigs need a dermatologist's prescription.

Referral

ROUTINE	Childhood scaly patch	treat/investigate as tinea.
URGENT	Lost follicular openings / symptoms of scarring	urgent derm.
ROUTINE	Significant distress, school withdrawal, self-harm risk	mental-health pathways.

Red flags and when to refer

- Childhood scaly patch treat/investigate as tinea.
- Lost follicular openings / symptoms of scarring urgent derm.
- Significant distress, school withdrawal, self-harm risk mental-health pathways.

Refer

- Child, pregnancy/breastfeeding, diagnostic doubt, severe/rapid progression (ophiasis, AT/AU), no response at 3–6 months, major psychosocial impact, facial-site treatment wanted, or wig prescription needed.
- Specialist options (counsel only): intralesional steroid, contact immunotherapy, JAK inhibitors, systemic agents — do not initiate without training/pathway.

Complications and prognosis

Complications

- Anxiety, depression and social withdrawal.
- Loss of eyelashes and brows (eye irritation).
- Associated autoimmune disease.

Prognosis

- Unpredictable. Worse outlook: onset in childhood, extensive or ophiasis pattern, nail disease, atopy, long duration.

Prescriber pearls (UK pharmacist IP)

- Yellow dots on trichoscopy prove follicles are still there — hope for regrowth.
- Regrowth takes months and may be white at first — set expectations.
- Check TSH is optimised if the patient is already on thyroxine.

Self-check quiz

5 questions · optional

Q1 When should you think of Alopecia areata?

✓ Sudden smooth round bald patch without scale or scarring; exclamation-mark hairs at the active edge; nails may be pitted.

Q2 What drives it?

✓ CD8+ T-cell attack on the anagen bulb (JAK–STAT); stem-cell bulge spared non-scarring and potentially reversible.

Q3 What is the backbone of care?

✓ Explain and support; watchful waiting or potent scalp steroid for 3 months in adults; camouflage/ wigs; refer severe, childhood or non-responding disease.

Q4 What must you never miss?

✓ Scaly childhood patch = tinea until proven otherwise. Loss of follicular ostia = scarring alopecia — urgent dermatology.

Q5 Name a red flag that needs escalation.

✓ Childhood scaly patch treat/investigate as tinea.

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Photo gallery

Patchy alopecia areata 3

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Clinching signs: exclamation hairs, trichoscopy, nails 4

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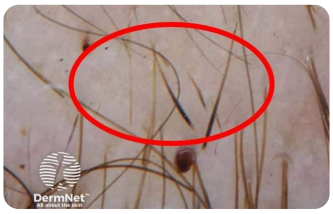

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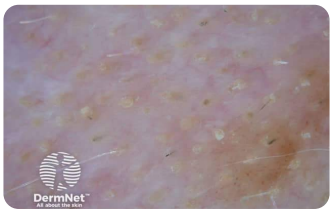

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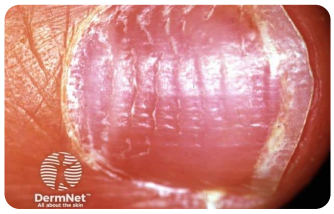

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Regrowth and poliosis 2

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Children and facial sites 4

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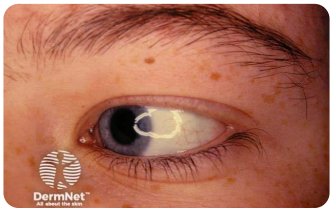

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Severity spectrum 3

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Hair & scalp Module 3 · Week 3

Telogen effluvium

Diffuse shedding of club hairs 2–3 months after a trigger. Reassure, find reversible causes, and cross-link pattern hair loss rather than duplicating AGA photo sets.

ALSO CALLED telogen effluvium | diffuse hair shedding | club hairs | postpartum hair loss | TE

RELATED PAGES Androgenetic alopecia (male & female) | Alopecia areata | Trichotillomania

KEY FACTS

- **Telogen effluvium** is diffuse shedding about **2–3 months after a trigger**, when many follicles shift into telogen together.
- Triggers: childbirth, febrile illness (including COVID-19), surgery, crash diets, iron deficiency, thyroid disease, new drugs.
- Positive **hair-pull test** (more than 4–6 club hairs) in the acute phase; no scarring, no miniaturisation.
- Check ferritin, thyroid function and full blood count if the trigger is unclear.
- Reassure: regrowth over **6–12 months** once the trigger has gone; chronic TE lasts longer than 6 months.

Overview

Telogen effluvium (TE) is non-scarring **diffuse** shedding of telogen (club) hairs. Acute TE follows a trigger by about **2–3 months**. Chronic TE lasts beyond 6 months.

Androgenetic (pattern) hair loss is the other common diffuse-looking thinning of mid-life — full UK IP detail and Module 6 Week 6 photos live on androgenetic alopecia. This page only sketches the DDx so Week 3 teaching stays joined up without duplicating that photo set.

Epidemiology

- **Who**
  - Common at any age, more often reported by women.
  - Postpartum shedding affects many women 2–4 months after delivery.

Aetiology: causes and risk factors

- Classic triggers: high fever, COVID/flu-like illness, major surgery, postpartum (including when stopping breastfeeding), crash diets, haemorrhage, new drugs (retinoids, anticoagulants, antithyroid, beta-blockers, some anticonvulsants — check BNF), severe stress.
- Iron deficiency, thyroid disease and other metabolic insults.

- Chronic TE: ongoing trigger or idiopathic; overlap with female pattern loss is common.

Risk factors and associations

- Physiological stress, hormonal change, nutritional deficiency or drugs (retinoids, anticoagulants, beta-blockers, stopping the combined pill).

Myths to address

> **"SHEDDING MEANS I'LL GO BALD"**
> follicles are alive and will regrow.

> **"HAIR SUPPLEMENTS CURE IT"**
> only replace a proven deficiency.

Pathophysiology and pathology

A shock **synchronises** hair cycles so many hairs rest and then fall together.

1. **Normal cycle**
   Normally about 85–90% of scalp hairs are growing (anagen) and 10–15% resting (telogen) at any time.
2. **Trigger**
   A physiological shock pushes many anagen follicles into early **telogen** at once.
3. **Delayed shedding**
   Telogen lasts about 3 months, so club hairs shed in large numbers 2–3 months after the trigger.
4. **Recovery**
   New anagen hairs push out the old ones; short regrowing hairs at the hairline are a good sign.

HISTOPATHOLOGY

- Biopsy only if unclear: normal follicle numbers with an increased proportion of telogen follicles; no miniaturisation or inflammation.

Clinical features

- Diffuse thinning; parted hair looks thinner; **no discrete bald patch**.
- Increased daily shed; club hairs with a white bulb on the brush (photo teaching: D10).
- Pull test: many club hairs come out easily when active.
- Scalp skin looks normal; ostia preserved.

Brief AGA contrast (cross-link)

- Pattern loss is gradual miniaturisation over years (male temples/vertex; female crown with widened part, frontal hairline usually kept).
- Management (minoxidil, male finasteride MHRA mood warnings, etc.) is covered on the androgenetic alopecia page, with its own photo gallery.

Fig. 1 · TE: diffuse thinning all over, often 2–3 months after a trigger (illness, childbirth, crash diet, major stress, new drug); positive pull test diffusely; no patches.

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Special groups

- **Postpartum**
  - Shedding peaks 3–4 months after delivery and usually settles by 12 months.

Diagnosis and investigations

- History timed to a trigger 2–3 months earlier.
- Examine for patches, scale, inflammation, pattern miniaturisation.
- Bloods when indicated: FBC, ferritin, TFT (± others guided by history).
- Child with a scaly patch mycology (tinea), not TE.

Key points

- History timed to a trigger 2–3 months earlier.
- Examine for patches, scale, inflammation, pattern miniaturisation.
- Bloods when indicated: FBC, ferritin, TFT (± others guided by history).
- Child with a scaly patch mycology (tinea), not TE.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Androgenetic alopecia	Patterned miniaturisation over years — AGA page
	Diffuse AA	Exclamation hairs, yellow dots, rapid patches
	Tinea capitis	Scale, black dots, children

	CONDITION	DISTINGUISHING FEATURES
	Traction alopecia	Marginal loss matching hairstyle (D11)
	Scarring alopecia	Lost ostia — scarring overview

Management

- Reassure: acute TE usually regrows over **6–12 months** once the trigger has passed; shedding can look alarming but follicles are alive.
- Correct reversible causes (iron if deficient, thyroid replacement optimisation, review culprit drugs with the original prescriber).
- Gentle hair care; avoid crash diets; camouflage if wanted.
- Do not promise “hair vitamins” as evidence-based cure-alls.
- If patterned miniaturisation coexists, manage as on the AGA page.
- Refer if diagnostic doubt, progressive patterned loss with virilisation, scarring signs, or severe distress.

Referral

ROUTINE	Virilisation or sudden severe FPHL endocrine/tumour pathways.
ROUTINE	Patchy, scaly or shiny bald areas — not TE.
URGENT	Kerion — urgent antifungal pathway.

Red flags and when to refer

- Virilisation or sudden severe FPHL endocrine/tumour pathways.
- Patchy, scaly or shiny bald areas — not TE.
- Kerion — urgent antifungal pathway.

Complications and prognosis

Complications

- Anxiety; may unmask underlying androgenetic alopecia.

Prognosis

- Acute TE resolves in most within 6–12 months; chronic TE fluctuates over years but rarely causes baldness.

Prescriber pearls (UK pharmacist IP)

- Always ask “what happened 2–3 months ago?”
- Postpartum shedding is physiological — counsel antenatally when possible.
- Teaching grid: TE (D09) vs FPHL (D06) vs extensive AA — different time courses and trichoscopy.

Self-check quiz

5 questions · optional

Q1 When should you think of Telogen effluvium?

✓ Sudden increase in hairs in the brush or shower, diffuse thinning, no bald patches — often 2–3 months after illness, childbirth or a new drug.

Q2 What drives it?

✓ A stressor synchronises many anagen follicles into telogen; club hairs shed together weeks later.

Q3 What is the backbone of care?

✓ Identify the trigger, check thyroid/ferritin when indicated, reassure that acute TE usually settles in 6–12 months; treat coexistent pattern loss separately.

Q4 What must you never miss?

✓ Patchy loss, scale, or lost ostia are not TE — rethink AA, tinea or scarring alopecia.

Q5 Name a red flag that needs escalation.

✓ Virilisation or sudden severe FPHL endocrine/tumour pathways.

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Photo gallery

Telogen effluvium

2

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Must-not-miss in children: tinea / kerion

3

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DDx thumbnails: pattern hair loss (see AGA page for full gallery)

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Traction alopecia

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Hair & scalp Module 3 · Week 3

Trichotillomania

Habitual hair pulling causing irregular patches of broken hairs. Habit-reversal therapy is first-line; watch for trichophagia and trichobezoar.

ALSO CALLED trichotillomania · hair pulling · hair-pulling disorder · trichophagia · trichobezoar

RELATED PAGES Alopecia areata · Tinea: corporis, cruris & pedis · Scarring alopecia overview · Telogen effluvium

KEY FACTS

- **Trichotillomania** is recurrent hair pulling causing hair loss: a **body-focused repetitive behaviour** (OCD-related disorder).
- Irregular patches with hairs of **different lengths**, broken hairs, often on the side of the dominant hand; scalp, brows, lashes.
- Many deny pulling; dermoscopy shows broken hairs, coiled hairs and 'flame' hairs.
- First-line: **habit reversal therapy** (CBT); treat associated anxiety or depression.
- Ask about **trichophagia** (eating hair): risk of trichobezoar.

Overview

Trichotillomania (hair-pulling disorder) is a body-focused repetitive behaviour causing noticeable hair loss. It is not scarring in early disease. It is commoner in children and adolescents and often under-disclosed.

Epidemiology

- **Who**
  - 1–2% lifetime; children (often self-limiting) and young women; associated with anxiety, OCD and depression.

Aetiology: causes and risk factors

- Classed with obsessive–compulsive related disorders; anxiety, boredom or sensory urge often precede pulling.
- **Automatic** pulling (unaware, while reading/screens) vs **focused** pulling (conscious tension relief).
- May coexist with skin picking or nail biting.

Myths to address

- **"IT'S JUST A BAD HABIT"**
  it is a recognised mental health condition that responds to specific therapy.

Pathophysiology and pathology

Repeated pulling **mechanically breaks** hairs, which regrow normally if pulling stops.

1. **Urge and relief**
   Tension or boredom builds; pulling gives relief or pleasure, reinforcing the behaviour.
2. **Mechanical damage**
   Hairs fracture at different lengths; follicles bleed or become distorted.
3. **Regrowth**
   Follicles are not destroyed early on, so hair regrows if pulling stops; chronic pulling can scar.

> **HISTOPATHOLOGY**
>
> - Rarely needed: trichomalacia, pigment casts, empty follicles and increased catagen hairs without inflammation.

Clinical features

- Irregular, angular or tonsure-like patches; hairs of **unequal lengths**; broken hairs firmly anchored.
- Common sites: vertex, frontotemporal (dominant-hand side), brows and lashes.
- Trichoscopy: black dots, broken hairs, **V-sign**, flame hairs — contrast with AA exclamation marks.
- Chronic severe pulling can eventually scar; secondary infection/crust possible.

Fig. 1 · Irregular, angular patch with hairs broken at many different lengths — not a smooth round patch; pull test negative.

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Special groups

- **Children**
  - Early-onset pulling often settles; support parents without punishment.

Diagnosis and investigations

Clinical pattern ± behaviour history (ask privately, without blame). Pull test negative. Exclude tinea if scaly. Differentiate from alopecia areata and scarring alopecia.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Alopecia areata	Smooth patch; exclamation hairs; positive edge pull test
	Tinea capitis	Scale; mycology
	Traction alopecia	Hairstyle distribution; fringe sign
	Scarring alopecia	Lost ostia; symptoms

Management

- Validate and reduce shame; involve parents/carers sensitively in children.
- **Habit-reversal training / CBT** via CAMHS, psychology or specialist services is first-line.
- Stimulus control (fidget tools, gloves at night, shorter nails); treat secondary infection if present.
- Medicines are not first-line in primary care; any drug therapy is specialist.
- Ask about **chewing/swallowing hair** every time.

Referral

URGENT — Abdominal pain, early satiety, vomiting, weight loss, halitosis, epigastric mass → urgent surgical review (trichobezoar / Rapunzel syndrome).

URGENT — Self-harm or suicidal ideation → urgent mental-health assessment.

ROUTINE — Diagnostic doubt or progressive scarring → dermatology.

Red flags and when to refer

- Abdominal pain, early satiety, vomiting, weight loss, halitosis, epigastric mass urgent surgical review (trichobezoar / Rapunzel syndrome).
- Self-harm or suicidal ideation urgent mental-health assessment.
- Diagnostic doubt or progressive scarring dermatology.

Complications and prognosis

Complications

- Permanent alopecia, infection, shame and social isolation; trichobezoar.

Prognosis

- Chronic and fluctuating in adults; good outcomes with habit reversal therapy.

Prescriber pearls (UK pharmacist IP)

- Different hair lengths in one patch is the bedside clue.
- Eyebrow/eyelash pulling in a child is still trichotillomania until proven otherwise — also consider AA and hypothyroidism for lateral brows.
- Steroids will not fix the behaviour.

Self-check quiz

5 questions · optional

Q1 When should you think of Trichotillomania?

✓ Irregular or geometric bald patch with hairs of many different lengths; pull test negative; often on the dominant-hand side.

Q2 What drives it?

✓ Body-focused repetitive behaviour (OCRD spectrum) — automatic or focused pulling; broken shafts, not autoimmune bulb attack.

Q3 What is the backbone of care?

✓ Non-judgemental recognition, habit-reversal / CBT referral, wound care if excoriated; do not treat as alopecia areata with steroids alone.

Q4 What must you never miss?

✓ Trichophagia with abdominal pain, vomiting or mass urgent surgical assessment for trichobezoar.

Q5 Name a red flag that needs escalation.

✓ Abdominal pain, early satiety, vomiting, weight loss, halitosis, epigastric mass urgent surgical review (trichobezoar / Rapunzel syndrome).

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Photo gallery

Pattern recognition 4

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Behaviour and complications 2

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Trichoscopy vs alopecia areata

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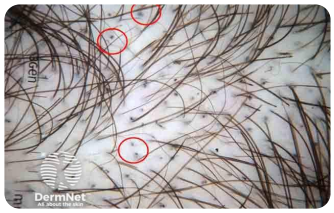

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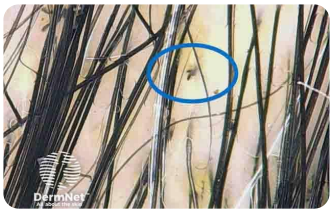

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Hair & scalp Module 3 · Week 3

Scarring alopecia overview

When follicular openings are lost, hair loss is permanent. Recognise LPP, FFA, DLE, FD and CCCA and refer early — primary care holds the line while waiting.

ALSO CALLED cicatricial alopecia · lichen planopilaris · frontal fibrosing alopecia · discoid lupus · folliculitis decalvans · CCCA · +1

RELATED PAGES Alopecia areata · Androgenetic alopecia (male & female) · Telogen effluvium · Lichen planus · Discoid lupus erythematosus

KEY FACTS

- **Scarring (cicatricial) alopecia** permanently destroys follicles: loss of follicular openings is the key sign.
- Primary lymphocytic: **lichen planopilaris, frontal fibrosing alopecia (FFA), DLE, CCCA**; neutrophilic: **folliculitis decalvans**.
- FFA is rising, mainly in post-menopausal women; CCCA mostly affects women of African descent.
- **Refer early** (PCDS): hair lost is not regained; send photos.
- Don't delay referral for blood tests.

Overview

Scarring (cicatricial) alopecia permanently destroys follicles. Primary types target the follicle (lymphocytic: LPP, FFA, DLE, CCCA; neutrophilic: folliculitis decalvans). Secondary scarring follows burns, trauma, tumours, kerion or long-standing traction.

Contrast with non-scarring loss (alopecia areata, telogen effluvium, androgenetic alopecia) where ostia remain.

Aetiology: causes and risk factors

- Lymphocytic attack (LPP/FFA/DLE/CCCA) or neutrophilic folliculitis (FD, often S. aureus).
- FFA incidence is rising, mainly in post-menopausal women.
- CCCA predominantly affects women of African descent; hair practices contribute alongside genetics.

Myths to address

- **"IT WILL GROW BACK LIKE OTHER HAIR LOSS"**
  scarring loss is permanent; early treatment saves the rest.

Pathophysiology and pathology

Inflammation attacks the **follicle stem-cell bulge**, so the follicle can't regrow.

1. **Bulge attack**
   Lymphocytes or neutrophils target the upper follicle, where stem cells live.
2. **Stem cell loss**
   Without stem cells the follicle can't cycle.
3. **Fibrosis**
   The follicle is replaced by a fibrous tract and its opening disappears.

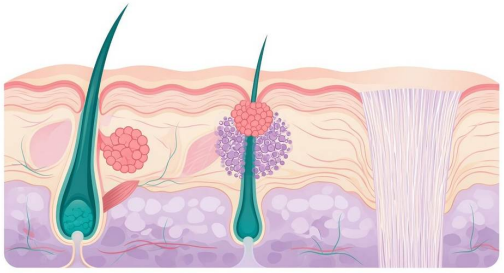

Fig. 1 · Inflammation around the upper follicle destroys it, leaving a fibrous tract.
AI-generated illustration

4. **Shiny scalp**
   The skin loses openings and becomes smooth.

Clinical features

CLUE	NON-SCARRING	SCARRING
Follicular ostia	Present (yellow/black dots)	Absent — shiny polished scalp
Edge signs	Exclamation hairs / miniaturisation	Perifollicular erythema, scale, plugs, pustules, tufting
Symptoms	Often none	Itch, pain, burning, tenderness
Prognosis	Regrowth possible	Permanent — treat to halt spread

> **LPP**
> white patches with spiky perifollicular scale; check mouth/skin LP.

> **FFA**
> frontal/temporal recession, lonely hairs, eyebrow loss, "unsunned" forehead band.

> **DLE**
> red scaly plugged plaques dyspigmented scars; ears/face; ask lupus systemic features.

> **FD**
> pustules, crusts, doll's-hair tufting at the vertex — swab.

> **CCCA**
> crown spreading centrifugally.

Fig. 2 · The core sign of ANY scarring alopecia: loss of follicular openings (ostia) on close/ dermoscopic inspection.

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Diagnosis and investigations

Clinical + trichoscopy (absent ostia). Specialist biopsy from the **active inflamed margin**. Primary care: good photos (clinical + dermoscopy if available), swab pustules, mycology if scale.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Alopecia areata	Ostia preserved; yellow dots
	Kerion	Boggy child scalp infection — can scar; start antifungal pathway; do not incise
	Skin cancer on bald scalp	Ulcerated/bleeding nodule NG12
	End-stage traction / trichotillomania	History; may finally scar

Management

- **Urgent dermatology / Advice & Guidance** with photos (PCDS: scarring alopecia needs early referral).
- While waiting: swab, mycology if indicated; baseline FBC, ferritin, U&E/LFT, TFT, vitamin D; ANA if lupus suspected; syphilis serology if indicated.
- Do not delay referral for blood results.
- Dermatologist may suggest potent topical steroid via A&G for symptoms.

- Specialist ladders (awareness only): hydroxychloroquine, doxycycline, anti-androgens (FFA), systemic immunosuppressants, culture-guided antibiotics for FD — not routine IP initiation.
- Signpost Alopecia UK and camouflage.

Referral

ROUTINE — Any loss of ostia, symptomatic inflamed edge, FFA pattern, CCCA pattern, lupus features, kerion, or uncertain "AA that will not regrow".

ROUTINE — Ulcerated/bleeding lesion on bald scalp suspected-cancer pathway.

Red flags and when to refer

- Any loss of ostia, symptomatic inflamed edge, FFA pattern, CCCA pattern, lupus features, kerion, or uncertain "AA that will not regrow".
- Ulcerated/bleeding lesion on bald scalp suspected-cancer pathway.

Complications and prognosis

Complications

- Permanent hair loss, distress.

Prognosis

- Treatment slows or stops progression; lost hair doesn't return.

Prescriber pearls (UK pharmacist IP)

- Side-by-side teaching: yellow dots (AA) vs absent ostia (scar).
- FFA is often dismissed as "just ageing" — check brows and the frontal band.
- Once scarred, the goal is halt progression, not regrowth.

Self-check quiz

5 questions · optional

Q1 When should you think of Scarring alopecia overview?

✓ Shiny bald patch without follicular openings, often with itch, burning, perifollicular scale or pustules at the active edge.

Q2 What drives it?

✓ Inflammation destroys the stem-cell bulge fibrosis replaces the follicle.

Q3 What is the backbone of care?

✓ Photograph, swab/mycology if pus or scale, baseline bloods, Advice & Guidance / urgent dermatology — do not watch and wait for "AA to regrow".

Q4 What must you never miss?

✓ Scarring alopecia is irreversible once established. Also refer non-healing nodules on bald scalp via suspected-cancer pathways.

Q5 Name a red flag that needs escalation.

✓ Any loss of ostia, symptomatic inflamed edge, FFA pattern, CCCA pattern, lupus features, kerion, or uncertain "AA that will not regrow".

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Photo gallery

Discoid lupus erythematosus

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Lichen planopilaris

3

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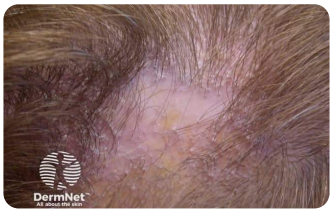

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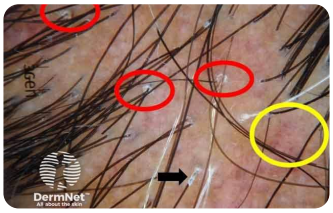

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Frontal fibrosing alopecia

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Folliculitis decalvans & CCCA

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CHAPTER 17

Cosmetic & aesthetic

Acne scars and keloids, peels, needling and lasers, and counselling-level aesthetic topics (neuromodulators, solar lentigines, IPL).

1. Acne scars
2. Keloids and hypertrophic scars
3. Cosmetic procedures: peels, needling & lasers
4. Facial ageing & neuromodulators
5. Solar lentigines
6. Intense pulsed light (IPL)
7. Eyelash enhancement (bimatoprost)

Cosmetic & aesthetic Module 6 · Week 6

Acne scars

Atrophic acne scars (ice-pick, boxcar, rolling), how to tell scar from pigment, and which specialist modalities suit which scar. Prevention is treating active acne early; scarring is a NICE referral trigger.

ALSO CALLED acne scarring · ice-pick scars · boxcar scars · rolling scars · atrophic scars · post-inflammatory hyperpigmentation · +2

RELATED PAGES Acne vulgaris · Keloids and hypertrophic scars · Cosmetic procedures: peels, needling & lasers · Retinoids (topical & oral)

KEY FACTS

- **Atrophic acne scars** follow collagen loss after inflammatory acne: **ice-pick**, **boxcar** and **rolling**.
- Ice-pick scars are the commonest (about 60–70%).
- Separate true scars from **post-inflammatory hyperpigmentation or erythema** (flat colour change, which fades).
- Prevention matters most: treat acne early; scarring is a referral trigger under NICE NG198.
- Treatments (TCA CROSS, subcision, lasers, microneedling) are specialist or cosmetic; aim for improvement, not removal.

Overview

NOTE

Scope: counselling and recognition

For the pharmacist/IP, this page covers recognising scar types, setting expectations, safety and referral. It is **not** a procedure how-to.

Acne scarring is fibrous remodelling after inflammatory acne. **Atrophic** scars follow collagen destruction; raised scars are covered on Keloids and hypertrophic scars. Scars are usually permanent but may soften with time or combined treatment. The psychological impact is often high, even when scarring looks 'mild'.

Epidemiology

- **Who**
  - Some scarring affects many people with moderate to severe acne; risk rises with delayed treatment and picking.

Aetiology: causes and risk factors

- Inflammatory nodules and cysts cause net **collagen destruction**, giving **atrophic** scars.
- Risk rises with acne severity and duration, delay in treatment, picking (acne excoriée) and smoking.
- **Post-inflammatory erythema and hyper- or hypopigmentation** are colour change, not true scarring, but patients often call both 'scars'.

Myths to address

- **"SCARS WILL FADE COMPLETELY"**
  atrophic scars are permanent; colour marks fade.

Pathophysiology and pathology

Deep inflammation **destroys collagen** faster than it can be rebuilt.

1. **Deep inflammation**
   Ruptured follicles release contents into the dermis, causing intense inflammation.
2. **Collagen destruction**
   Enzymes (MMPs) break down dermal collagen.
3. **Imperfect repair**
   Healing lays down too little collagen, leaving a depression.

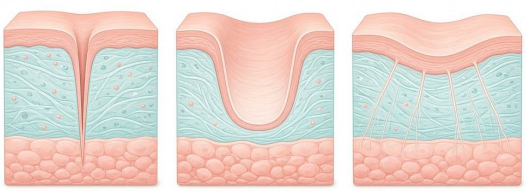

Fig. 1 · Atrophic scar types: ice-pick, boxcar and rolling.
AI-generated illustration

4. **Tethering**
   Fibrous bands pull the surface down in rolling scars (released by subcision).

Clinical features

TYPE	APPEARANCE	TYPICAL SITES
Ice-pick	Narrow, deep, V-shaped; deeper than wide (about 60–70% of atrophic scars)	Cheeks
Boxcar	Round or oval depressions with sharp vertical edges	Cheeks, temples
Rolling	Wider, sloping edges, tethered by dermal bands	Cheeks

TYPE	APPEARANCE	TYPICAL SITES
Perifollicular elastolysis	Small, pale, follicular atrophic scars	Back, chest
PIH / PIE (not scar)	Flat brown or red marks; no change in skin contour	Anywhere acne occurred

Special groups

- **Skin of colour**
  - PIH is very common and often more distressing than scars; higher procedure risk.

Diagnosis and investigations

Clinical: history of acne plus scar morphology. Document **Fitzpatrick skin type**, whether acne is still active, and the patient's goals.

Differential diagnoses

LOOK-ALIKE	DISTINGUISHING CLUES
PIH or erythema alone	Flat colour change, with no depression
Chickenpox scars	History of varicella; scattered
Traumatic or surgical scar	History of injury

Management

Prevent

Treat active acne **early and adequately** (see Acne vulgaris). **Scarring is a referral trigger** under NICE NG198. Treat active acne first, before any scar procedures.

Specialist modalities (overview)

SCAR TYPE	OPTIONS (SPECIALIST OR COSMETIC)
Ice-pick	**TCA CROSS** (chemical reconstruction of skin scars), punch excision or grafting
Boxcar	Resurfacing (fractional lasers, peels), punch techniques, microneedling
Rolling	**Subcision**, dermal fillers or fat, microneedling, fractional laser

Combined, multi-session approaches are usual. Refer to dermatology, plastic surgery or a reputable aesthetic clinic, and explain that the aim is **improvement, not removal**.

Treatment ladder

1. **Prevent**

   Treat active acne **early and adequately**; scarring = refer (NG198).

2 **Specialist: atrophic**

Ice-pick: TCA CROSS, punch excision / grafting. Boxcar: fractional laser, peels, punch, microneedling.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Expectations**: Improvement, not removal.
- **Isotretinoin**: Check status before procedures.
- **Skin of colour**: Higher PIH risk: experienced operator only.

Referral

ROUTINE Scarring or significant distress at any acne severity (NG198).

ROUTINE Low mood or body dysmorphic concerns: mental-health support.

Red flags and when to refer

- Scarring or significant distress at any acne severity: refer (NG198).
- Low mood, anxiety or body dysmorphic concerns: signpost or refer for mental health support.

Complications and prognosis

Complications

- Psychological distress, low self-esteem, depression.

Prognosis

- Permanent but improvable with combined treatment.

Prescriber pearls (UK pharmacist IP)

Separate scar from pigment. The treatments are different.

Check isotretinoin status before procedures: see Cosmetic procedures.

Self-check quiz

4 questions · optional

Q1 Which atrophic scar type is commonest?

✓ Ice-pick (about 60–70%).

Q2 What treatment suits rolling scars?

✓ Subcision (± fillers, microneedling, laser).

Q3 How do you tell PIH from a scar?

✓ PIH is flat colour change with no contour change.

Q4 What does NICE say about scarring?

✓ Scarring is a referral trigger at any acne severity.

References

1. DermNet: Acne scarring — https://dermnetnz.org/topics/acne-scarring
   dermnetnz.org
2. NICE NG198: Acne (referral for scarring) — https://www.nice.org.uk/guidance/ng198
   nice.org.uk

Photo gallery

Atrophic acne scars: ice-pick, rolling, boxcar 5

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Pigment, not scar: PIH

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Cosmetic & aesthetic Module 6 · Week 6

Keloids and hypertrophic scars

Raised scars from excess collagen: hypertrophic scars stay within the wound and often settle; keloids spread beyond it, persist and recur after excision. Silicone and steroid first; procedures are specialist.

ALSO CALLED keloid | hypertrophic scar | raised scar | silicone gel | intralesional steroid

RELATED PAGES Acne scars | Dermatofibroma | Dermatofibrosarcoma protuberans (DFSP) | Skin surgery: consent, complications & danger zones

KEY FACTS

- **Hypertrophic scars** stay within the wound and often settle; **keloids** extend beyond it, persist and recur.
- Keloids are much commoner in skin of colour and families; sites: earlobes, jawline, chest, shoulders, upper back.
- First line: **silicone gel or sheet** for months; pressure; potent topical steroid under occlusion.
- Specialist: intralesional steroid, cryotherapy, laser, excision with adjuvant therapy.
- Biopsy an atypical or rapidly growing 'keloid' to exclude DFSP.

Overview

A **hypertrophic scar** is a raised scar that stays **within** the original wound and often improves over 1–2 years. A **keloid** is firm scar tissue that grows **beyond** the original wound border, persists and often recurs after excision.

Epidemiology

- **Who**
  - Keloids peak at 10–30 years; up to 15–20× more common in people of African descent.

Aetiology: causes and risk factors

- Excess collagen production with reduced breakdown during wound healing.
- Triggers: acne, piercing, surgery (especially under tension), burns, vaccination, minor trauma.
- **Keloids** are much more common in **skin of colour** and run in families.

Myths to address

- **"CUT IT OUT AND IT'S GONE"**
  keloids recur in most cases after simple excision.

Pathophysiology and pathology

Raised scars come from **too much collagen** and not enough breakdown.

1. **Wound**
   Injury starts normal healing: inflammation, proliferation, remodelling.
2. **Overactive fibroblasts**
   TGF-β signalling keeps fibroblasts producing collagen.
3. **Hypertrophic vs keloid**
   Hypertrophic scars stay in the wound; keloids invade surrounding skin with thick, glassy collagen.

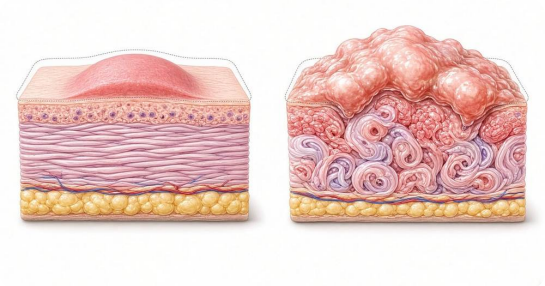

Fig. 1 · Hypertrophic scar (left) within the wound; keloid (right) extends beyond it.
AI-generated illustration

4. **Tension**
   Wounds under tension (chest, shoulders) are most prone.

> **HISTOPATHOLOGY**
> - Keloid: thick hyalinised (glassy) collagen bundles; hypertrophic scar: nodules of collagen parallel to the surface.

Clinical features

	HYPERTROPHIC SCAR	KELOID
Border	Stays within the wound	Extends **beyond** the wound
Onset	Weeks after injury	Months, may keep growing
Course	Often flattens over 1–2 years	Persists; **high recurrence** after excision
Sites	Anywhere, under tension	Earlobes, jawline, chest, shoulders, upper back

Keloids may itch or hurt.

Diagnosis and investigations

Clinical. Biopsy only for an atypical, rapidly growing or ulcerated 'keloid' (exclude dermatofibrosarcoma protuberans or other tumour).

Differential diagnoses

LOOK-ALIKE	DISTINGUISHING CLUES
Dermatofibroma	Small firm papule, dimple sign
DFSP	Slowly enlarging plaque without a clear injury: biopsy
Sarcoidosis in a scar	Purple papules in old scars; systemic signs

Management

- Prevention: avoid elective procedures (piercing, cosmetic surgery) on keloid-prone sites in people with a keloid history.
- **Silicone gel or sheet** (OTC) for months, not weeks; pressure (earlobe pressure clips).
- Potent topical steroid under occlusion for early raised scars.
- Specialist: **intralesional steroid**, cryotherapy, pulsed dye laser, careful excision with adjuvant therapy (**high recurrence** after excision alone).

Explain that the aim is **improvement, not removal**.

Treatment ladder

1

Early raised scar

Silicone gel or sheet for months; pressure for earlobes.

2

Persistent

Potent topical steroid under occlusion.

3

Specialist

Intralesional steroid, cryotherapy, pulsed dye laser, excision with adjuvant (high recurrence).

Referral

URGENT — Rapidly growing or atypical ‘keloid’: exclude a tumour.

ROUTINE — Symptomatic or disfiguring keloids.

Red flags and when to refer

- A rapidly growing or atypical ‘keloid’: exclude a tumour.
- Keloid tendency or skin of colour before elective procedures: experienced operator only.

Complications and prognosis

Complications

- Itch, pain, restricted movement, disfigurement.

Prognosis

- Hypertrophic scars often improve over 1–2 years; keloids persist.

Prescriber pearls (UK pharmacist IP)

Keloids extend beyond the wound; hypertrophic scars do not.

OTC silicone gel is a reasonable first step for early raised scars: months, not weeks.

Self-check quiz

4 questions · optional

Q1 Key difference between keloid and hypertrophic scar?

✓ Keloids extend beyond the original wound; hypertrophic scars do not.

Q2 Which skin types are most keloid-prone?

✓ Skin of colour (Fitzpatrick IV–VI), with family history.

Q3 First-line OTC treatment?

✓ Silicone gel or sheet for months.

Q4 Why avoid excision alone?

✓ Keloids recur in most cases without adjuvant therapy.

References

1. DermNet: Keloid and hypertrophic scar — https://dermnetnz.org/topics/keloid-and-hypertrophic-scar
dermnetnz.org

Photo gallery

Keloid and hypertrophic scars 4

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Cosmetic & aesthetic Module 6 · Week 6

Cosmetic procedures: peels, needling & lasers

A counselling guide to peel depths, microneedling and laser classes, and an evidence-based view on procedures during or after isotretinoin (the old blanket 6-month wait is outdated).

ALSO CALLED chemical peel · microneedling · skin needling · collagen induction · laser · fractional laser · +6

RELATED PAGES Intense pulsed light (IPL) · Acne scars · Facial ageing & neuromodulators · Photoageing · Retinoids (topical & oral)

KEY FACTS

- Chemical peels controlled chemical injury to epidermis ± dermis (superficial, medium, deep).
- Microneedling fine needles cause micro-injuries that trigger new collagen.
- Lasers **selective photothermolysis**, a wavelength targets water, haemoglobin or melanin.
- Darker skin types have a higher risk of **post-inflammatory pigmentation** and scarring: choose operators carefully.
- Isotretinoin: document it and **let the operating clinician decide** on timing; HSV history may need antiviral cover.

Overview

Counselling and recognition only: what procedures are, expectations, safety and referral. Not a procedure how-to.

NOTE

Scope: counselling and recognition

For the pharmacist/IP, this page covers recognising types, setting expectations, safety and referral. It is **not** a procedure how-to (no injection technique, laser settings or peel recipes).

PROCEDURE	WHAT IT IS
Chemical peel	Controlled chemical injury to the epidermis ± dermis to resurface skin (photoageing, pigment, fine lines, acne, superficial scars)
Microneedling	Collagen induction therapy: fine needles (professional about 0.5–3 mm; home devices much shorter) create micro-injuries that trigger new collagen

PROCEDURE	WHAT IT IS
Lasers	Selective photothermolysis: the wavelength targets a chromophore (water, haemoglobin or melanin)

Aetiology: causes and risk factors

Each procedure injures the skin in a controlled way, triggering inflammation, re-epithelialisation and collagen remodelling. **'Frosting'** (whitening) with some peel agents, such as TCA, reflects protein coagulation and indicates depth. This is for recognition only.

Microneedling microchannels close within minutes and the epidermis is largely preserved, so it often suits darker skin better than ablative options.

Myths to address

- **"Home microneedling is the same as clinic treatment"**: home devices are shorter and carry infection risk.

How it works

Most resurfacing procedures **injure the skin in a controlled way** so it repairs with new collagen.

1. **Controlled injury**
   Peels, needles and ablative lasers damage set depths of skin.
2. **Wound healing**
   Inflammation, then fibroblasts lay down new collagen over months.
3. **Selective targets**
   Lasers heat one chromophore (melanin, haemoglobin, water) and spare the rest.
4. **Depth = downtime**
   Deeper injury gives more effect, more downtime and more risk.

Clinical features

Peel depth

DEPTH	EXAMPLE AGENTS	DOWNTIME (APPROX.)	COMMON AIMS
Superficial ('lunchtime')	Glycolic (AHA), salicylic, Jessner	Hours to 2 days	Dullness, mild pigment, acne adjunct
Medium	TCA about 20–35%	About a week of crusting and swelling	Texture, freckling, AKs, fine lines, some scars
Deep	Phenol–croton oil (rare now)	Prolonged; **systemic (cardiac) toxicity risk**	Deep wrinkles. Specialist only

Laser classes

CLASS	EXAMPLES	USE (OVERVIEW)
Ablative	CO_2, Er:YAG	Deep wrinkles and scars; more downtime and risk
Non-ablative	Nd:YAG, infrared	Milder, more sessions
Fractional (ablative or non-ablative)	Microscopic columns with skin bridges left intact	Atrophic acne scars, photoageing; faster healing
Vascular	**PDL**, KTP	Telangiectasia, redness, **pink hypertrophic scars**, some vascular birthmarks
Pigment / melasma	Various	Specialist; melasma often relapses and laser can **worsen PIH**

Fig. 1 · Boxcar / atrophic scarring pattern with sharper edges than rolling scars.

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Diagnosis and investigations

There is no disease to diagnose. Instead, the **pre-procedure assessment** covers: Fitzpatrick type, **HSV history**, **isotretinoin status**, keloid tendency, pregnancy, sun exposure, active acne or infection, and realistic goals.

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
Laser hair removal; fractional lasers; pigment lasers; fractional RF microneedling; superficial and medium peels; microdermabrasion; dermaroller; biopsies; superficial excisions	**Deep** phenol peels; fully ablative resurfacing; mechanical dermabrasion

Isotretinoin and procedures: the evidence

The old rule was to delay elective procedures for **about 6 months after isotretinoin**, because of theoretical impaired healing and scarring. A task force review (Mysore et al., Standard Guidelines of Care), cited in Module 6, found **insufficient evidence** to support a blanket delay for many procedures.

GENERALLY CONSIDERED ACCEPTABLE (IN EXPERIENCED HANDS)	STILL A HIGHER-RISK DISCUSSION
Laser hair removal; fractional lasers; pigment lasers; fractional RF microneedling; **superficial and medium peels**; microdermabrasion; dermaroller; biopsies; superficial excisions	**Deep** phenol peels; fully ablative resurfacing; mechanical dermabrasion

Management

> **Do not** perform medium or deep peels or procedures without the appropriate training, governance and insurance.

> **ISOTRETINOIN**
> do **not** clear patients for procedures yourself. Document isotretinoin use and **refer the decision to the operating clinician**. Some UK centres remain conservative. Continue standard isotretinoin safety (pregnancy prevention programme, blood tests, mucocutaneous effects).

> **AFTERCARE COUNSELLING**
> gentle cleansing, **SPF**, moisturiser, no picking, and expect temporary redness, peeling or swelling.

> **Antiviral prophylaxis** may be advised by the operator if there is a history of HSV.

> **HOME MICRONEEDLING ROLLERS**
> warn about infection and poor hygiene or quality.

Adverse effects to recognise

- Infection (bacterial, candida, **HSV reactivation**)
- **PIH or hypopigmentation** (especially darker skin), scarring, burns, blistering
- Acne flare, persistent redness, granulomatous reactions (products driven into the skin with needling)
- Paradoxical hypertrichosis (hair lasers)

Referral

ROUTINE	Pain, spreading redness, pus or fever after a procedure: infection.
ROUTINE	Grouped vesicles after a facial procedure: **HSV**, needing prompt antivirals.
ROUTINE	Blistering, burns or dyspigmentation: refer back to the operator or dermatology.
ROUTINE	Deep peels: systemic toxicity risk. Specialist settings only.

Red flags and when to refer

- Pain, spreading redness, pus or fever after a procedure: infection.
- Grouped vesicles after a facial procedure: **HSV**, needing prompt antivirals.
- Blistering, burns or dyspigmentation: refer back to the operator or dermatology.
- Deep peels: systemic toxicity risk. Specialist settings only.

Complications and prognosis

Complications

- PIH, scarring, infection, HSV reactivation, burns, eye injury (lasers).

Prognosis

- Results build over 3–6 months; repeat sessions are common.

Prescriber pearls (UK pharmacist IP)

More ablative is not always better. Match downtime and risk to the goal and skin type.

Darker skin: start cautiously, as there is a higher risk of dyspigmentation and keloid.

Pregnancy: most elective peels and lasers are deferred.

Advise patients to check the provider's **qualifications and insurance**.

Patient phrase: "Many modern guidelines no longer insist on a blanket 6-month wait for milder procedures, but your dermatologist or cosmetic clinician decides case by case."

Self-check quiz

5 questions · optional

Q1 **When should you think of Cosmetic procedures: peels, needling & lasers?**

✓ A patient asking about peels, needling or laser for scars, pigment or ageing, or a patient on isotretinoin asking 'can I have this done?'

Q2 What drives it?

✓ Controlled injury (chemical, mechanical or thermal) leading to re-epithelialisation and collagen remodelling.

Q3 What is the backbone of care?

✓ Match depth and risk to the goal and skin type, refer to trained operators, and counsel on SPF and aftercare.

Q4 What must you never miss?

✓ PIH and keloid risk in skin of colour, HSV reactivation, and unregulated providers. Do not 'clear' isotretinoin patients yourself.

Q5 Name a red flag that needs escalation.

✓ Pain, spreading redness, pus or fever after a procedure: infection.

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1. DermNet: Chemical peels — https://dermnetnz.org/topics/chemical-peels
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2. DermNet: Skin needling — https://dermnetnz.org/topics/skin-needling
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4. Mysore V et al. Standard guidelines of care: procedures in patients on isotretinoin (Indian J Dermatol Venereol Leprol) — https://pubmed.ncbi.nlm.nih.gov/?term=Mysore+isotretinoin+procedures+standard+guidelines
pubmed.ncbi.nlm.nih.gov

Photo gallery

Common targets: atrophic acne scars 2

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Procedure risk in skin of colour: PIH and keloid

2

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Cosmetic & aesthetic Module 6 · Week 5

Facial ageing & neuromodulators

Counselling-level guide to intrinsic vs extrinsic ageing, dynamic vs static lines, and botulinum toxin / filler safety — no injection protocols.

ALSO CALLED wrinkles | facial ageing | dynamic lines | static lines | botulinum toxin | Botox | +4

RELATED PAGES Cosmetic procedures: peels, needling & lasers | Photoageing | Hyperhidrosis

KEY FACTS

- Ageing is **intrinsic** (time, genes) and **extrinsic** (UV, smoking): volume loss, bone remodelling, elastosis, wrinkles.
- **Dynamic lines** come from muscle movement; **static lines** stay at rest.
- Botulinum toxin softens dynamic lines; it **doesn't lift sagging skin or replace volume**.
- Toxin is **prescription-only**; GPhC: prescribers must assess **face to face**. Illegal for under-18s cosmetically (2021 Act).
- Filler red flags: **vascular occlusion** (pain, blanching, mottling, visual change) is an emergency.

Overview

Counselling level only: what patients ask about toxin and fillers, UK regulation and red flags. No injection technique, doses or units.

Facial ageing mixes **intrinsic** (genetics, time) and **extrinsic** (UV, smoking, pollution) change: volume loss, bone remodelling, elastosis and wrinkles. **Dynamic lines** appear with expression; **static lines** remain at rest.

This page is **counselling-level only** for community pharmacist IPs: what patients ask about neuromodulators and fillers, UK regulation, expected effects and red flags. It does **not** teach injection technique, doses, units or reconstitution.

Aetiology: causes and risk factors

- Dynamic lines: repeated frontalis, corrugator, orbicularis oculi and other facial muscle activity.
- Static lines and folds: dermal thinning, fat-pad descent, bone resorption, gravity.
- Botulinum toxin type A temporarily blocks acetylcholine release at the neuromuscular junction reduced muscle contraction for about 3–4 months.
- HA fillers restore volume by placing hyaluronic-acid gel in soft tissue — different products and planes (specialist detail).

Myths to address

- **"FILLERS AND TOXIN ARE THE SAME"**

toxin relaxes muscle; fillers add volume.

How it works

Botulinum toxin **stops nerves signalling to muscle** for a few months.

1. **Nerve terminal**
   Toxin is taken up into the motor nerve terminal.
2. **SNAP-25 cleavage**
   It cuts SNAP-25, so acetylcholine vesicles can't release.
3. **Muscle relaxes**
   The injected muscle weakens and dynamic lines soften.
4. **Recovery**
   New nerve sprouts restore function over about 3–4 months.

Clinical features

- Glabellar frown lines, forehead lines, crow's feet are the common aesthetic requests.
- Expected toxin timeline: onset over days, peak around 2 weeks, duration about 3–4 months.
- Common local effects: bruising, short-lived headache, asymmetric smile if diffusion occurs.
- Filler: immediate volume; swelling and bruising common early.

Fig. 1 · Dynamic lines appear only with expression (here a smiling teenager) — muscle-driven; neuromodulator territory when they become bothersome.

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Diagnosis and investigations

Aesthetic assessment belongs to a trained face-to-face prescriber. Community role: screen for skin-cancer lesions in photoaged faces before anyone treats pigment, and triage complications.

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
New facial asymmetry after toxin	Diffusion / brow or lid ptosis — usually temporary
Acute unilateral facial weakness with forehead spared	Stroke — 999 FAST
Painful livedo or blanching after filler	Vascular occlusion — emergency
Descending paralysis, dysphagia after unlicensed injections	Iatrogenic botulism — 999 / UKHSA

Management

Before anyone injects (counselling)

- Daily SPF, stop smoking, realistic expectations — toxin softens dynamic lines; it does not lift sagging skin or replace volume.
- Botulinum toxin for cosmetic use is a **prescription-only medicine**. GPhC guidance: pharmacist prescribers must **examine face to face**; remote prescribing of these medicines is not appropriate.
- Illegal under-18 cosmetic toxin/fillers (Botulinum Toxin and Cosmetic Fillers (Children) Act 2021).
- Advise licensed UK product via a regulated, insured practitioner — never buy toxin online for self-injection.

Aftercare themes (practitioner-led detail)

- Follow the treating clinician's aftercare (often avoid rubbing, lying flat or vigorous exercise for the first hours).
- Review at about 2 weeks rather than unplanned "top-ups".
- OTC "Botox-like" peptide creams have weak evidence — do not oversell.

Filler adverse-effect awareness (no procedure detail)

> Early: swelling, bruising, lumps. Delayed: nodules, biofilm infection — refer back to the injector/dermatology.

> **VASCULAR OCCLUSION**
> severe pain, livedo, blanching soon after injection **emergency** (injector/A&E; hyaluronidase is a specialist treatment).

> **VISUAL LOSS / OPHTHALMOPLEGIA AFTER FILLER**
> sight-threatening emergency — call 999; do not delay.

⚠ **RED FLAG**

Unlicensed toxin / botulism

Descending weakness, dysarthria, dysphagia or breathing difficulty after aesthetic injections 999. UKHSA may advise on antitoxin; notify health protection; Yellow Card. Check UK-licensed product and legitimate supply chain.

ⓘ **NOTE**

Therapeutic facial palsy toxin

Specialist facial-palsy teams use toxin for synkinesis and asymmetry alongside physiotherapy. Acute Bell's palsy pathways (steroid within 72 h, eye protection) are separate — forehead-sparing weakness is stroke until proven otherwise.

Referral

999 / A&E	Visual symptoms or severe pain after filler emergency.
999 / A&E	Bilateral descending paralysis after unlicensed injections 999 / UKHSA botulism pathway.
URGENT	Drooping eyelid with corneal exposure risk urgent eye care.
999 / A&E	Stroke pattern facial weakness 999.

Red flags and when to refer

- Visual symptoms or severe pain after filler emergency.
- Bilateral descending paralysis after unlicensed injections 999 / UKHSA botulism pathway.
- Drooping eyelid with corneal exposure risk urgent eye care.
- Stroke pattern facial weakness 999.

Complications and prognosis

Complications

- Bruising, headache, eyelid ptosis, asymmetry; filler: lumps, infection, vascular occlusion, blindness (rare).

Prognosis

- Effects wear off; repeat treatment needed.

Prescriber pearls (UK pharmacist IP)

- This site intentionally omits units, dilution and injection maps — those belong in accredited aesthetic training, not a teaching website.
- Photoaged faces need a skin-cancer check before cosmetic energy devices — see photoageing and solar lentigines.
- Document advice and signpost legitimate practitioners; do not prescribe toxin remotely.

Self-check quiz

5 questions · optional

Q1 When should you think of Facial ageing & neuromodulators?

✓ Someone asking about frown lines, crow's feet or "preventative Botox", or worried after a recent aesthetic treatment.

Q2 What drives it?

✓ Intrinsic ageing plus UV/smoking drive volume loss and elastosis; dynamic lines from repeated muscle movement; static lines persist at rest.

Q3 What is the backbone of care?

✓ Photoprotection and expectations first; botulinum toxin is POM and face-to-face prescribing only; recognise ptosis, infection and filler emergencies.

Q4 What must you never miss?

✓ Filler vascular occlusion (pain, livedo, visual loss) emergency pathway. Suspected iatrogenic botulism after unlicensed toxin 999 / UKHSA.

Q5 Name a red flag that needs escalation.

✓ Visual symptoms or severe pain after filler emergency.

References

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Photo gallery

Facial ageing 8

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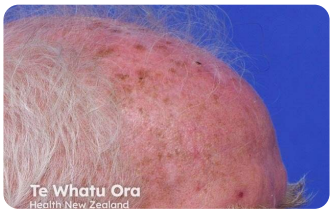

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Botulinum toxin: before / after and adverse effects

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Cosmetic & aesthetic Module 6 · Week 5

Solar lentigines

Benign sun-induced lentigines versus freckles, melasma and seborrhoeic keratoses — with lentigo maligna as the red-flag differential.

ALSO CALLED: solar lentigo · age spots · liver spots · sun spots · ink-spot lentigo

RELATED PAGES: Seborrhoeic keratosis · Melanoma: recognition & referral · Photoageing · Melasma & facial melanosis · Actinic keratoses (solar keratoses)

KEY FACTS

- **Solar lentigines** ('liver spots', 'age spots') are flat, well-defined brown macules on sun-exposed skin.
- Face, backs of hands, forearms and shoulders in middle-aged and older adults; a marker of sun damage.
- Dermoscopy: fine reticular pattern, 'moth-eaten' edges, fingerprint pattern.
- Differentiate from **lentigo maligna** (irregular, enlarging, multiple colours on the face) and pigmented AK or SK.
- Benign: sun protection; cryotherapy or laser are cosmetic options.

Overview

Solar lentigines (“age spots”, “liver spots”) are benign flat tan-to-dark-brown macules on chronically sun-exposed skin. They mark cumulative UV damage and skin-cancer risk but are not themselves premalignant.

Epidemiology

- **Who**
  - Over 90% of fair-skinned people over 60; increase with age and sun exposure.

Aetiology: causes and risk factors

UV-driven localised melanocyte hyperactivity. Risk rises with fair skin, outdoor work, sunburn history, sunbeds and age.

Risk factors and associations

- Chronic UV exposure; PUVA and sunbeds produce similar lentigines.

Myths to address

- **"THEY'RE HARMLESS SO NEVER NEED CHECKING"**
  a changing, irregular facial lentigo may be lentigo maligna.

Pathophysiology and pathology

UV causes a **local increase in melanocytes** and pigment along the basal layer.

1. **UV injury**
   Repeated UV exposure stimulates basal melanocytes.
2. **Melanocyte increase**
   Melanocyte numbers increase modestly along the basal layer (no nests).
3. **Elongated rete ridges**
   Rete ridges lengthen and are heavily pigmented, creating a flat brown patch.

HISTOPATHOLOGY

- Elongated, club-shaped, pigmented rete ridges with a slight increase in single basal melanocytes; no atypia.

Clinical features

- Uniform light-to-dark brown macules with a regular border; face, dorsal hands, forearms, shoulders.
- Do **not** fade in winter (unlike freckles).
- May coexist with AK field change and photoageing.

Fig. 1 · Solar lentigo on the nose: evenly pigmented flat macule with regular border — sun-induced, benign.

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Special groups

- **Lentigo maligna warning**
  - Asymmetric pigmentation, multiple colours, grey dots, enlargement: refer via the suspected cancer pathway.

Diagnosis and investigations

Clinical ± dermoscopy (fingerprint/moth-eaten border patterns). Biopsy or refer asymmetric, irregular, enlarging, ugly-duckling or grey-regression lesions — especially on the face.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Freckles (ephelides)	Childhood onset; fade in winter
	Melasma	Symmetrical patches; hormones; worsened by heat/IPL — see melasma
	Seborrhoeic keratosis	Stuck-on, waxy, raised
	Lentigo maligna	Irregular colours, slow enlargement on older face NG12 referral
	Pigmented AK	Rough surface

Management

- Daily broad-spectrum SPF 30–50 (± tinted); hats/gloves for driving and gardening — without this, treated lesions return.
- Examine the whole photodamaged field for AK/BCC/SCC/lentigo maligna.
- Modest topicals: retinoids, azelaic acid; short specialist hydroquinone courses only (ochronosis risk).
- Procedural (usually private): light cryotherapy (hypopigmentation risk, especially in darker skin), pigment laser for discrete spots, IPL for diffuse pigment+redness — only after diagnosis is secure.
- Melasma is a poor IPL indication and can worsen.

Referral

URGENT Any lesion suspicious for lentigo maligna or melanoma urgent suspected-cancer pathway.

DON'T Do not treat undiagnosed pigment with cryotherapy, IPL or laser.

Red flags and when to refer

- Any lesion suspicious for lentigo maligna or melanoma urgent suspected-cancer pathway.

- Do not treat undiagnosed pigment with cryotherapy, IPL or laser.

Complications and prognosis

Complications

- Cosmetic concern; may evolve into a flat seborrhoeic keratosis (reticulated SK).

Prognosis

- Persist and increase; fade slightly with strict sun protection.

Prescriber pearls (UK pharmacist IP)

- Solar lentigines are a UV dosimeter — use the visit for skin-cancer education.
- Hand spots are a classic patient question; same rules as the face.
- See also photoageing and intense pulsed light.
- Flat brown marks that become raised and warty are usually seborrhoeic keratoses; a lentigo that darkens or spreads unevenly on the face may be lentigo maligna.

Self-check quiz

5 questions · optional

Q1 When should you think of Solar lentigines?

✓ Flat, evenly brown macules on the face or backs of the hands of an older outdoor worker that do not fade in winter.

Q2 What drives it?

✓ Chronic UV increases local melanocyte number and activity along elongated rete ridges.

Q3 What is the backbone of care?

✓ Photoprotection forever; topical retinoids/azelaic acid for modest fading; cryotherapy/IPL/laser only after cancer exclusion — usually private.

Q4 What must you never miss?

✓ Irregular, multicoloured, enlarging facial pigment lentigo maligna — urgent suspected-cancer referral. Never laser an undiagnosed pigmented lesion.

Q5 Name a red flag that needs escalation.

✓ Any lesion suspicious for lentigo maligna or melanoma urgent suspected-cancer pathway.

References

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Photo gallery

Solar lentigines 7

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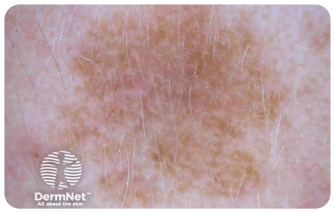

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Red flag: lentigo maligna 2

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Look-alikes: freckles, melasma, seborrhoeic keratosis

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Cosmetic & aesthetic Module 6 · Week 5

Intense pulsed light (IPL)

Counselling on intense pulsed light for redness, pigment and hair: who is a good candidate, what to expect, and the burn and pigment risks, especially in tanned or darker skin.

ALSO CALLED IPL · intense pulsed light · photofacial · photorejuvenation · hypopigmentation after IPL

RELATED PAGES Solar lentigines · Photoageing · Rosacea (incl. Demodex & steroid rosacea) · Cosmetic procedures: peels, needling & lasers · Eyelash enhancement (bimatoprost)

KEY FACTS

- **IPL** uses filtered broad-spectrum flashlamp light to target **melanin** (lentigines, hair) and **haemoglobin** (redness, telangiectasia).
- Usually 3–6 sessions; treated pigment darkens ('coffee grounds') then flakes over 1–2 weeks.
- Risks: burns, blistering, hypo- or hyperpigmentation, scarring, eye injury; highest in tanned or darker skin.
- Never treat undiagnosed pigmented lesions; caution with photosensitising drugs, recent isotretinoin, HSV.
- Melasma is a poor candidate; cosmetic and not NHS-funded.

Overview

Intense pulsed light (IPL) is a non-laser flashlamp delivering broad-spectrum light through filters to treat vascular redness, solar lentigines and unwanted dark hair, with modest texture benefit.

Aetiology: causes and risk factors

- Selective photothermolysis: melanin and oxyhaemoglobin absorb the light and heat up; dermal heating may remodel collagen modestly.
- Darker or tanned skin absorbs more epidermal energy, giving a higher risk of burns and pigment change.

Myths to address

- **"IPL IS A LASER"**
  it is non-coherent broad-spectrum light, less selective than lasers.

Pathophysiology and pathology

IPL heats **coloured targets** in the skin, which absorb the light more than their surroundings.

1. **Light absorption**
   Melanin and oxyhaemoglobin absorb specific wavelengths.
2. **Selective heating**
   Short pulses heat the target faster than heat spreads (selective photothermolysis).
3. **Clearance**
   Damaged pigment is shed; heated vessels collapse and are reabsorbed.
4. **Collateral risk**
   Epidermal melanin also absorbs energy, so tanned or darker skin burns more easily.

Clinical features

- Sessions often 3–6, weeks apart; treated pigment darkens like "coffee grounds" then flakes over 1–2 weeks.
- Adverse effects: burns, blistering, **hypo-/hyperpigmentation** (sometimes grid footprints), scarring, paradoxical hair growth, eye injury.

Diagnosis and investigations

Before IPL: confirm the diagnosis of every pigmented spot (no undiagnosed lesions), Fitzpatrick type, recent tan, photosensitising drugs and HSV history. Melasma is generally a poor IPL candidate.

Differential diagnoses

REQUEST	BETTER FRAMED AS
Diffuse redness + lentigines	IPL photofacial after a skin cancer check
Discrete stubborn lentigo	Often a pigment laser rather than IPL
Melasma	Sunscreen + medical therapy; avoid heat devices first-line (see Melasma)

Management

IPL counselling (UK)

- Check practitioner training, insurance, patch test and (in Wales) HIW registration for Class 3B/4 lasers/IPL in independent clinics.
- Contraindications/cautions: undiagnosed pigmented lesions, active infection/HSV, recent tan, photosensitivity disorders, many photosensitising drugs; extra caution in skin types IV–VI and on isotretinoin (many delay treatment).
- Aftercare: SPF, gentle skincare, no picking; seek review for blistering or infection.
- See also solar lentigines and photoageing.

Treatment ladder

1 **Select**

Diagnose every pigmented lesion first; check skin type, tan, drugs, HSV.

2 **Treat**

3–6 sessions by a trained, registered operator; patch test.

3 **Aftercare**

SPF, no picking; review blistering or infection.

Referral

SAME DAY — Eye pain or visual change after IPL: ophthalmology.

URGENT — Undiagnosed changing pigmented lesion: suspected cancer pathway.

Red flags and when to refer

- Eye pain or visual change during/after IPL: **urgent ophthalmology**.
- Blistering burns or striped hypopigmentation: stop further treatment; specialist wound/pigment advice.
- Undiagnosed changing pigmented lesion: cancel IPL; refer.

Complications and prognosis

Complications

- Burns, pigment change (grid pattern), scarring, eye injury, paradoxical hair growth.

Prognosis

- Results fade with new sun damage; maintenance sessions are common.

Prescriber pearls (UK pharmacist IP)

IPL is versatile but unforgiving on tanned or darker skin without expert settings.

Coffee-ground darkening is expected; picking prolongs marks.

Self-check quiz

4 questions · optional

Q1 What are IPL's target chromophores?

✓ Melanin and oxyhaemoglobin (and hair melanin).

Q2 Who is at higher risk of IPL burns and pigment change?

✓ Tanned or darker skin (types IV–VI).

Q3 Is melasma a good IPL indication?

✓ Generally no; heat can worsen it.

Q4 What is the 'coffee grounds' effect?

✓ Expected darkening of treated pigment that flakes off over 1–2 weeks.

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Photo gallery

IPL effects and adverse effects 4

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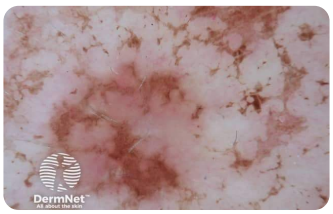

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Cosmetic & aesthetic Module 6 · Week 5

Eyelash enhancement (bimatoprost)

Counselling on bimatoprost (a prostamide) for longer, darker lashes: how it works, permanent iris darkening, reversible eyelid pigmentation and when to exclude a disease cause of lash loss.

ALSO CALLED bimatoprost | Latisse | eyelash growth | eyelash serum | hypotrichosis of the eyelashes | madarosis

RELATED PAGES Alopecia areata | Trichotillomania | Intense pulsed light (IPL) | Cosmetic procedures: peels, needling & lasers

KEY FACTS

- **Bimatoprost** is a prostamide (glaucoma drug) that lengthens, thickens and darkens eyelashes when applied to the upper lid margin.
- Works by **prolonging anagen**; results build over weeks to months and reverse on stopping.
- Warn about **permanent iris darkening** (hazel/green eyes), reversible eyelid pigmentation, irritation and redness.
- Exclude a disease cause of lash loss (alopecia areata, thyroid disease, blepharitis, trichotillomania).
- Cosmetic serums vary in content; prescription products need proper counselling.

Overview

Eyelash enhancement options include cosmetics, lifts and **bimatoprost** (a prostamide also used in glaucoma drops) applied to the upper lid margin to lengthen lashes.

Aetiology: causes and risk factors

Bimatoprost increases eyelash length, thickness and darkness by **extending anagen** and stimulating follicle melanogenesis; periocular skin and iris melanocytes can also respond.

Myths to address

- **"NATURAL SERUMS ARE SAFER"**
  many contain undeclared prostaglandin analogues.

Pathophysiology and pathology

Bimatoprost acts on **prostamide receptors** in the lash follicle to keep it growing.

1. **Receptor binding**
   Bimatoprost binds prostamide/FP receptors in the dermal papilla.

2. **Longer anagen**
   Follicles stay in the growth phase longer and more follicles enter anagen.
3. **Thicker, darker lashes**
   Follicle size and melanogenesis increase.
4. **Melanocyte side effects**
   Iris and eyelid melanocytes increase pigment production.

Clinical features

- Longer, darker lashes over weeks to months.
- Side effects: periocular hyperpigmentation, **iris colour change** (especially hazel/green eyes), itch, redness; periorbital fat atrophy (glaucoma-dose context).

Diagnosis and investigations

Before use: exclude a disease cause of sparse lashes (**madarosis**): alopecia areata, thyroid disease, blepharitis, trichotillomania, chemotherapy.

Differential diagnoses

CAUSE OF SPARSE LASHES	CLUE
Alopecia areata	Patchy loss elsewhere; exclamation-mark hairs
Trichotillomania	Broken lashes of different lengths
Blepharitis	Crusted, red lid margins
Thyroid disease	Loss of the outer brows; systemic symptoms

Management

Bimatoprost eyelash counselling

- Explain cosmetic versus glaucoma origins; apply only as directed to the upper lid margin, not into the eye for cosmetic use.
- Warn about **permanent iris darkening**, reversible periocular skin darkening, possible hair growth on unintended skin if the product migrates, and ocular irritation.
- Avoid with active ocular inflammation unless an ophthalmologist agrees; remove contact lenses as per product advice.
- Effects reverse over weeks to months if stopped.
- Check the current SmPC/BNF; this page does not cover off-label periocular protocols.

Medicines at a glance

Bimatoprost (eyelash)

Bimatoprost

DOSE	as per SmPC/BNF · continuous
CAUTIONS	Permanent iris darkening possible; periocular pigment reverses.

Treatment ladder

1 **Exclude disease**

Check for alopecia areata, blepharitis, thyroid disease, trichotillomania.

2 **Bimatoprost**

Bimatoprost (eyelash)

Referral

ROUTINE	Lash loss with scalp or brow loss, or lid disease: GP/dermatology/ophthalmology.
URGENT	Eye pain, redness or visual change on treatment.

Red flags and when to refer

- Eye pain, redness or visual change: stop and seek ophthalmology review.
- Patients on glaucoma drops: discuss with their ophthalmologist (additive effect on intraocular pressure).

Complications and prognosis

Complications

- Iris pigmentation (permanent), periocular pigmentation, conjunctival hyperaemia, periorbital fat atrophy.

Prognosis

- Lashes return to baseline within weeks to months after stopping.

Prescriber pearls (UK pharmacist IP)

Eyelash serums sold cosmetically vary widely; prescription bimatoprost needs proper counselling on iris pigment.

Self-check quiz

4 questions · optional

Q1 How does bimatoprost lengthen lashes?

✓ It prolongs the anagen (growth) phase.

Q2 Which side effect is permanent?

✓ Iris darkening.

Q3 Which effect reverses on stopping?

✓ Periocular skin darkening and the lash growth.

Q4 What is madarosis?

✓ Loss of eyelashes or eyebrows, which may signal disease.

References

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Photo gallery

Bimatoprost eyelash growth

2

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CHAPTER 18

Therapeutics

Prescribing tools: topical steroids and fingertip units, emollients, vitamin D analogues, tar and phototherapy, retinoids, acne pills, hirsutism and the biologics interface.

1. Topical corticosteroids: potency, quantities & safety
2. Emollients & soap substitutes
3. Vitamin D analogues, coal tar, dithranol & phototherapy
4. Retinoids (topical & oral)
5. Acne prescribing & the combined pill
6. Hirsutism & the Ferriman–Gallwey score
7. Biologics, JAK inhibitors & conventional systemics

Therapeutics Module 1 · Week 2

Topical corticosteroids: potency, quantities & safety

How topical steroids work, the UK potency ladder, matching potency to age, site and severity, fingertip-unit quantities, adverse effects and when not to use them.

ALSO CALLED TCS | steroid cream | potency ladder | fingertip unit | FTU | steroid atrophy | +4

RELATED PAGES Eczema / dermatitis | Emollients & soap substitutes | Psoriasis | Perioral (periorificial) dermatitis | Rosacea (incl. Demodex & steroid rosacea)

KEY FACTS

- Potency ladder (UK): **mild** (hydrocortisone 1%), **moderate** (clobetasone butyrate 0.05%), **potent** (betamethasone valerate 0.1%, mometasone 0.1%), **very potent** (clobetasol propionate 0.05%).
- Match potency to **site and severity**: mild on face and folds, potent on body, very potent on palms and soles for short courses.
- Dose with **fingertip units**: 1 FTU 0.5 g covers two adult palms (about 2% body surface).
- Usually once or twice daily; for eczema, treat flares fully then step down or use **weekend (proactive) therapy**.
- Side effects come from misuse: atrophy, striae, perioral dermatitis, steroid rosacea, **topical steroid withdrawal**; under-treatment from **steroid phobia** is commoner.

Overview

Synthetic analogues of cortisol applied to the skin for their **anti-inflammatory, immunosuppressive, anti-proliferative and vasoconstrictive** effects. They are the mainstay for eczema/dermatitis flares. They are also used in psoriasis (with vitamin D), lichen planus, lichen sclerosus, discoid lupus, insect-bite reactions and many other inflammatory dermatoses.

Systemic glucocorticoids covered on Case 1 (approximate anti-inflammatory potency vs hydrocortisone = 1): cortisone 0.8; prednisolone 4; methylprednisolone 5; dexamethasone and betamethasone ~25–30. Topical potency is a separate scale (see Topical corticosteroids). It depends on the molecule, its esterification (more lipophilic better skin penetration), the concentration and the vehicle.

Aetiology: causes and risk factors

Myths to address

- **"Steroid creams thin the skin, so use as little as possible"**: correct potency for a short course is safe; under-treatment prolongs flares.

How it works (mechanism of action)

Corticosteroids **switch off inflammatory genes** in skin cells.

1. **Receptor binding**

   Steroids cross the cell membrane and bind the glucocorticoid receptor.

2. **Gene regulation**

   The complex suppresses NF-κB and AP-1 (inflammatory genes) and induces anti-inflammatory proteins.

3. **Effects**

   Anti-inflammatory, immunosuppressive, vasoconstrictive and antiproliferative effects.

4. **Absorption by site**

   Thin skin (eyelids, genitals, folds) absorbs far more than palms and soles.

Fig. 1 · Relative absorption varies widely by body site.

AI-generated illustration

5 **Atrophy with misuse**

Prolonged potent use thins epidermis and dermal collagen.

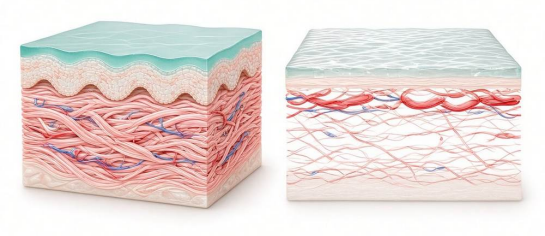

Fig. 2 · Steroid atrophy: thinned epidermis and dermis.

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6 **Dosing**

Fingertip units guide quantity.

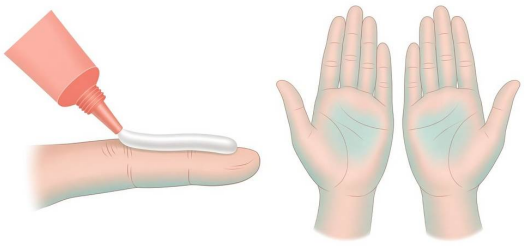

Fig. 3 · One fingertip unit.

AI-generated illustration

Clinical features

How it works

GENOMIC (CLASSICAL) PATHWAY

the lipophilic steroid crosses the cell membrane and binds the cytoplasmic **glucocorticoid receptor (GR)**, which is released from heat-shock proteins. The complex then moves into the nucleus: **Transactivation:** the complex binds glucocorticoid response elements and switches on anti-inflammatory genes. Examples: **annexin-1 (lipocortin-1)**, which inhibits phospholipase A2 less arachidonic acid fewer prostaglandins and leukotrienes; also MKP-1, IκBα and IL-10; **Transrepression:** the GR "tethers" to the pro-inflammatory transcription factors **NF-κB and AP-1** and blocks them. This switches off cytokines (IL-1, IL-2, IL-6, TNF-α), chemokines, adhesion molecules and COX-2, and probably drives most of the anti-inflammatory effect

NON-GENOMIC (RAPID) PATHWAYS

mostly, but not entirely. Rapid effects (seconds to minutes) occur through (a) **membrane-bound GR**, (b) cytosolic GR signalling that needs no gene transcription, and (c) at high concentrations, **non-specific physicochemical membrane interactions** that need no receptor at all. Vasoconstriction (the basis of the potency assay) appears within hours and is partly non-genomic.

CELL EFFECTS THAT MATTER IN SKIN

inhibition of Langerhans cells, T cells, mast-cell mediator release and eosinophils; an **anti-proliferative** effect on keratinocytes (useful in psoriasis); and **inhibition of fibroblast collagen and hyaluronic-acid synthesis**. That last effect causes atrophy, striae and purpura; course teaching singled it out.

POTENCY TESTING

the McKenzie–Stoughton **vasoconstrictor (skin-blanching) assay** tracks clinical potency and underlies the BNF classes.

Clinical uses & where it fits — the UK potency ladder

UK practice uses **four BNF classes**. The US uses seven (I = super-potent VII = least potent). The two systems do **not** line up product by product, so use UK classes for UK prescribing and teaching.

UK CLASS (BNF)	EXAMPLES (UK BRAND)	TYPICAL USE
Mild	Hydrocortisone 0.5–2.5% (1% most used)	Face, flexures, infants; mild eczema
Moderate	Clobetasone butyrate 0.05% (Eumovate); betamethasone valerate 0.025% (Betnovate-RD); alclometasone 0.05%; fluocinolone acetonide 0.00625% (Synalar 1 in 4)	Moderate eczema on the body; short courses on face/flexures in adults; flares in children
Potent	Betamethasone valerate 0.1% (Betnovate); **mometasone furoate 0.1% (Elocon)**; fluticasone propionate 0.05% (Cutivate); betamethasone dipropionate 0.05% (Diprosone); hydrocortisone butyrate 0.1% (Locoid); fluocinolone acetonide 0.025% (Synalar)	Severe eczema on trunk/limbs; psoriasis (with vitamin D); lichen planus; hand eczema
Very potent	**Clobetasol propionate 0.05% (Dermovate)**; diflucortolone valerate 0.3% (Nerisone Forte)	Short courses for thick/lichenified plaques, palms/soles, lichen sclerosus, discoid lupus, scalp; in children usually only with specialist advice

Combination products take the potency of their steroid. Examples: steroid + antimicrobial (Fucibet, Fucidin H, Daktacort, Trimovate, Timodine, Canesten HC) and steroid + salicylic acid (Diprosalic). The course point: use them **only when infection and inflammation genuinely coexist, and only briefly** (e.g. fusidic acid for 1–2 weeks because of resistance), then continue the plain steroid. If there is no response, **recheck the diagnosis**.

The vehicle matters:

- **Ointments** are greasy and occlusive, with the fewest preservatives. Best for dry or lichenified skin and for eczema.
- **Creams** suit weeping or hair-bearing skin and are more acceptable to patients, but their preservatives can cause contact allergy.
- **Lotions, scalp applications, foams and mousses** are for hairy areas.
- **Occlusion** (wet wraps, hydrocolloid, steroid tape such as fludroxycortide) raises effective potency.

Regional absorption (approximate, forearm = 1): soles ~0.1, palms ~0.8, scalp ~3.5, face ~7–13, eyelids/genitals up to ~40. Thin, flexural and infant skin absorbs more, which is why the face, flexures and infant get mild potency.

How to choose / assess

> **Match potency to severity, site and age.** Use the lowest potency that works, for the shortest time. In practice, under-treatment from steroid phobia is commoner than harm: NICE CG57 (children): mild for mild eczema, moderate for moderate, potent for severe; **Do not use potent steroids on the face/neck, or very potent steroids in children, without specialist advice.**; On the face/neck, use moderate potency for only 3–5 days in severe flares; Apply once or twice daily; once daily is usually enough for potent preparations

> **Step down** once the eczema is controlled. For frequently relapsing eczema, consider **proactive "weekend" therapy** (twice weekly to the usual flare sites) (NICE CG57 / CKS).

> **ASSESS BEFORE EVERY RE-ISSUE**

how often repeats are requested, the amount used (FTU arithmetic, see Topical corticosteroids), sites, signs of atrophy, alternative diagnoses (scabies, tinea, contact allergy), infection, and emollient adherence.

SIGNS OF STEROID OVERUSE

Epidermal/dermal atrophy: shiny, thin, translucent skin with fine "cigarette-paper" wrinkling, visible veins and loss of skin markings; Telangiectasia (especially on the face), **steroid rosacea**, and **perioral/periorificial dermatitis**; **Striae** (irreversible; flexures, inner thighs, axillae; adolescents in growth spurts), **purpura/easy bruising**, stellate pseudoscars; Hypopigmentation (more visible in skin of colour), hypertrichosis, **steroid acne/ folliculitis**; Delayed wound healing and ulceration. Infections masked or changed by the steroid (**tinea incognita**, crusted scabies, spread of HSV); Allergic contact dermatitis to the steroid or a preservative. Suspect it when eczema gets worse on treatment; Systemic effects (large areas, occlusion, very potent steroids, infants): HPA-axis suppression, iatrogenic Cushing's, growth suppression, hyperglycaemia. Periocular use glaucoma, cataract; Grading scales exist (e.g. atrophy and telangiectasia scored 0–3 in trials — the "objective gradings" a classmate posted). In practice, record the sign, the site and the steroid history

Fig. 4 · Steroid telangiectasia on facial skin after prolonged steroid use — face and eyelids are the highest-risk sites; use mild potency short-term or a TCI; periocular steroid also risks glaucoma/cataract.

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Adverse effects to recognise

Telangiectasia

Facial vessels after prolonged use. Site: Face.

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Purpura

Bruising on thinned skin. Site: Forearms.

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Steroid rosacea

Rebound flare when stopped. Site: Face.

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Tinea incognita

Less scale, spreads on steroid. Site: Anywhere.

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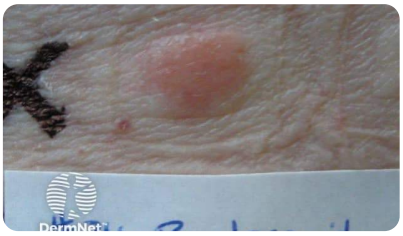
Red sleeve (TSW)

Confluent redness, sharp cut-off. Site: Forearm.

© DermNet · **Source**

Steroid allergy

Positive budesonide patch test. Site: Patch test.



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Diagnosis and investigations

1 FTU

- Ointment from a 5 mm nozzle, **crease to fingertip 0.5 g**, covers **two adult palms**.

Adult FTUs

- Face & neck 2.5 · arm 3 · hand 1 · leg 6 · foot 2 · front trunk 7 · back 7. Whole body 40 FTU (20 g).

BNF weekly (bd)

- Face 15–30 g · hands 15–30 g · arms 30–60 g · legs 100 g · trunk 100 g · groins 15–30 g (recheck BNF).

Before re-issue

- Repeats, amount used, sites, atrophy, other diagnoses, infection, emollient use.

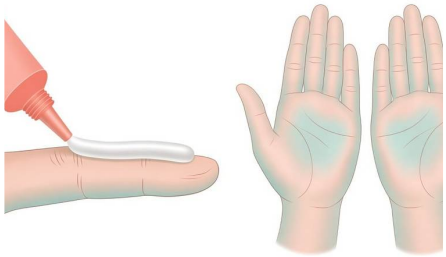

Fig. 11 · One fingertip unit: ointment squeezed from the last finger crease to the tip of an adult index finger, about 0.5 g. It covers the area of two adult palms with fingers together.
AI-generated illustration

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Untreated bacterial infection	Spreads: treat infection first.
	Fungal infection	**Tinea incognita**: scrape; antifungal.
	Viral (HSV, zoster)	Dissemination, e.g. eczema herpeticum.
	Scabies	Masks signs crusted scabies.
	Rosacea / perioral dermatitis	Steroids cause and rebound them.
	Acne, ulcers, broken skin	Steroid acne; impaired healing.

Key comparisons / differentials — where TCS are contraindicated or need caution

Any eight with reasons answers the course question. The BNF wording: contraindicated in untreated bacterial, fungal or viral skin lesions, acne, rosacea and perioral dermatitis. It adds cautions for widespread psoriasis, infants and long-term use.

TCS vs topical calcineurin inhibitors (TCIs) — Case 1 alternative:

Management

Fingertip units (FTU) and quantities

> **1 FTU** is the ointment squeezed from a tube with a **5 mm nozzle**, from the **distal skin crease to the tip of an adult index finger**. It weighs about **0.5 g** and covers **two adult palms with fingers together** (~2% BSA). For children, still measure on an adult finger; fewer FTUs are needed per site.

> Adult FTUs per application (Long & Finlay 1991; BAD/NES charts): face & neck **2.5**; one arm **3**; one hand (both sides) **1**; one leg **6**; one foot **2**; front of trunk **7**; back incl. buttocks **7**. The whole adult body takes about 40 FTU (20 g) per application.

> **BNF SUITABLE QUANTITIES (ADULT, TWICE DAILY FOR 1 WEEK)**
> face & neck 15–30 g; both hands 15–30 g; scalp 15–30 g; both arms 30–60 g; both legs 100 g; trunk 100 g; groins & genitalia 15–30 g. These are rounded teaching values — recheck the current BNF table.

> **PRACTICAL PRESCRIBING**
> state the **name, potency, site, frequency, duration and number of FTUs** on the label/instructions, not "use as directed". MHRA asks that packaging and counselling make the **potency** clear. Apply the emollient first and wait ~15–30 minutes before the steroid (or always use the same order). The gap and the quantity matter more than the order.

MHRA Drug Safety Update (Sept 2021): topical steroid withdrawal (TSW) reactions

- Rare but potentially severe reactions after **long-term, continuous or inappropriate use** of topical steroids, mainly **moderate-to-high potency**, typically **on stopping**.
- Features: redness that may look like a flare but **spreads beyond the treated area**; burning or stinging worse than the original condition; peeling; oedema; oozing. The **"red sleeve" sign** (forearm erythema sparing the palms) and facial/eyelid redness are described.
- **ADVICE TO PRESCRIBERS AND PHARMACISTS**
  Use the **lowest potency for the shortest time**; Review patients on long-term use, and **reduce gradually rather than stopping abruptly** after prolonged use; If TSW is suspected, seek specialist advice rather than restarting the same potent steroid as if it were a "flare"; Report via **Yellow Card**, and counsel on potency and duration
- TSW has to be told apart from a rebound flare, contact allergy and the underlying disease, which often needs dermatology.

Tachyphylaxis

- A reduced response with continuous use. In experimental models the vasoconstriction fades within days, but how much this matters clinically is debated. Poor adherence and a wrong diagnosis are commoner reasons that "the steroid stopped working".
- **WHAT TO DO**
  Check adherence, quantity used and the diagnosis (tinea, scabies, contact allergy, infection); Use **intermittent or pulsed regimens** (e.g. potent once daily, then weekends only) or short rest periods; Step up briefly, then down; Or switch to a **steroid-sparing agent** (TCI; vitamin D in psoriasis; phototherapy)

Steroid-phobic patients (Sophie's and Siddharth's experience)

- Explore the patient's beliefs (online TSW content is common). Explain potency, FTU and duration, and that **untreated eczema** causes more lichenification and infection than correctly used TCS. Offer written plans, TCIs for face/flexures, and follow-up.

IP scope

- Starting and continuing mild–potent TCS for common dermatoses, within competence and the local formulary, is routine IP practice.
- Very potent steroids, steroids in infants, genital lichen sclerosus regimens and TCIs may be within competence after training, but follow local formulary/shared-care rules.
- TCIs in under-2s and any long-term facial/eyelid plan are specialist decisions.

Worked cases

CASE 1

moderate–severe eczema with steroid atrophy

SIGNS OF OVERUSE

thin, shiny, translucent skin; fine wrinkling; telangiectasia; purpura/bruising; striae; hypopigmentation; perioral dermatitis, steroid rosacea or acne; infection modified by the steroid (tinea incognita); delayed healing; TSW on stopping.

WHAT TO ADVISE

potency and site rules; FTU quantity; **duration and step-down**; stinging on application; signs of infection and allergy; **do not stop abruptly after long-term use** (MHRA TSW); periocular risks (glaucoma/cataract); a steroid-sparing plan; emollients; fire risk from paraffin-soaked fabrics.

ALTERNATIVE

a **topical calcineurin inhibitor** (tacrolimus 0.03%/0.1% ointment or pimecrolimus 1% cream). It blocks calcineurin/IL-2 T-cell activation without collagen loss, so there is **no atrophy** — suitable for face, eyelids, flexures and long-term proactive twice-weekly use. Counsel about initial burning and sun protection. Alongside it: generous emollients, and referral for **phototherapy** or systemic therapy if still uncontrolled.

MECHANISM, TYPES, POTENCY

see Topical corticosteroids (four UK classes; vasoconstrictor assay).

All via GR? Mostly, via cytosolic GR (genomic transactivation/transrepression). There are also **non-genomic** effects via membrane GR, cytosolic GR signalling, at high concentration, non-specific membrane effects (see Topical corticosteroids).

QUESTIONS 6–8

see Vitamin D analogues, coal tar (vitamin D analogues; coal tar — Pott 1775, scrotal SCC; cohort data reassuring) and (phototherapy/PUVA). **Tachyphylaxis:** see Topical corticosteroids.

CASE 2

retinoids and steroid contraindications

CLASS/MOA/USES

see Retinoids. **Topical retinoids in pregnancy:** low absorption and no clear signal, but **contraindicated in pregnancy and when planning** (EMA 2018/MHRA). Inadvertent exposure stop, reassure, contact **UKTIS**.

8 CONTRAINDICATIONS

see Topical corticosteroids (untreated bacterial/fungal/viral infection, scabies, rosacea, perioral dermatitis, acne, ulcers/wounds; caution in widespread psoriasis, infants and allergy).

COMBINATION CREAMS

use briefly for genuinely infected eczema, then switch to the plain steroid; recheck the diagnosis if it does not settle.

CASE 3

19-year-old with mild–moderate acne

QUESTION 1

skin care + a **12-week fixed-combination topical** (adapalene/BPO, tretinoin/clindamycin or BPO/clindamycin) per NICE NG198. Review at 12 weeks; maintain with a topical retinoid; screen mood; refer if failing or scarring. No antibiotic monotherapy; no minocycline.

QUESTION 2

yes. COCs reduce inflammatory and comedonal acne (SHBG, ovarian androgen) and match oral antibiotics at 6 months. The evidence is best for mild–moderate and hormonally influenced lower-face acne; benefit takes 3–6 months.

QUESTION 3

if she wants contraception, start with a **low-VTE (levonorgestrel) COC**, checked against UKMEC. Move to drospirenone/desogestrel options if needed. Use **co-cyprindiol** only for moderate–severe acne after topicals + an oral antibiotic have failed, with VTE counselling and a stop plan (MHRA 2013). Mechanism as above, plus androgen-receptor blockade by cyproterone/drospirenone.

ACNE FULMINANS

systemic, ulcerating, haemorrhagic acne. It needs **dermatology within 24 h** for urgent oral steroids to prevent scarring and manage the systemic illness; it is often triggered by isotretinoin.

CASE 4

25-year-old with localised facial hirsutism

QUESTION 1

most likely **idiopathic/peripheral hirsutism** (normal androgens, regular cycles). Check SHBG/FAI, consider HbA1c, ± mid-luteal progesterone to confirm ovulation. Then: **weight loss**; **cosmetic methods, laser or electrolysis**; eflornithine only if still obtainable (now discontinued in UK dm+d); a **COC** if distress persists; anti-androgens only via a specialist and with contraception; psychological support; red flags for referral (see Hirsutism case and the Ferriman–Gallwey score).

QUESTION 2

mFG = 9 areas × 0–4, total 36. The traditional cut-off is 8 (4–6 in some ethnic groups). Use it to track response, not as the only basis for decisions (see Hirsutism case and the Ferriman–Gallwey score).

CASE 5

biologics

See Biologics, JAK inhibitors and conventional systemics — the primary-care interface: targets (TNF, IL-12/23, IL-23, IL-17; IL-4Rα/IL-13; JAK); NICE eligibility (psoriasis: PASI 10 and DLQI >10 after a conventional systemic; AE: after systemic failure). The **primary-care role**: record the drug, check interactions, no live vaccines, recognise infections and AEs, and do shared care where it applies.

Practice case: 6-year-old with AE not responding to 1% hydrocortisone

QA = 1, aqueous cream (SLS irritation and barrier damage; MHRA 2013).

QB = 4, 250–500 g/week (NICE CG57).

QC = 3

5 mm nozzle, from the distal crease to the tip of an **adult** index finger (0.5 g; covers two adult palms).

EXTRA TEACHING

a 6-year-old failing hydrocortisone 1% on the body probably needs a **moderate** steroid (e.g. clobetasone butyrate 0.05%) for 7–14 days during flares, adequate emollient quantities, and review for infection or an alternative diagnosis. Give fire-safety advice.

Treatment ladder

Prescribe clearly

State **name, potency, site, frequency, duration and FTUs**, not ‘as directed’. Emollient first, wait 15–30 min (or keep the same order).

2 **Treat the flare**

Mild for mild, moderate for moderate, potent for severe (NICE CG57); **no potent steroids on the face** without advice.

3 **Step down / proactive**

Once controlled, step down; for frequent relapse, **weekend therapy** twice weekly to flare sites.

4 **Steroid-sparing**

TCI (tacrolimus, pimecrolimus) for face, eyelids, flexures: no atrophy. Vitamin D analogues in psoriasis.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **TSW (MHRA 2021)**: After long-term moderate–potent use, on stopping: **reduce gradually**, seek specialist advice.
- **Adrenal suppression**: Large amounts of potent steroid, children, occlusion.
- **Skin lightening**: Never supply potent steroids for lightening: illegal use causes atrophy, striae, Cushing's.
- **Fire**: Paraffin-based preparations on fabrics (MHRA 2018).

Referral

URGENT	Children needing **very potent** steroid, or potent on face / flexures: specialist advice.
ROUTINE	Suspected **TSW**, steroid **contact allergy** (patch test), widespread atrophy or striae.
ROUTINE	Adult eczema not controlled after 4 weeks of correct potent TCS + emollients.

Red flags and when to refer

- Rash worse or spreading on a steroid with scale at the edge: scrape for **tinea incognito** before going on.

- Facial redness, burning and rebound after stopping a potent steroid: think **steroid rosacea / topical steroid withdrawal** (MHRA 2021).
- Large quantities of potent or very potent steroid in a child or under occlusion: risk of **adrenal suppression**, so review and seek advice.

Red flags / refer

- Suspected **TSW**, **contact allergy** to a steroid (patch testing), or widespread atrophy/striae.
- Adult eczema not controlled after 4 weeks of correct potent TCS + emollients, or children meeting the NICE CG57 referral criteria; facial eczema needing repeated moderate/potent steroid; diagnostic doubt.
- **Eczema herpeticum** (same day), erythroderma, suspected HPA suppression or Cushingoid features.
- Children who need **very potent** steroids, or potent steroids on the face/flexures.

Complications and prognosis

Complications

- Atrophy, striae, telangiectasia, acne, rosacea, perioral dermatitis, infection masking (tinea incognito), HPA suppression (rare, large amounts).

Prognosis

- Safe and effective when the right potency is used for the right time.

Prescriber pearls (UK pharmacist IP)

- "**Right potency, right place, right amount, right duration** — then step down."
- Write the FTU count on the prescription (e.g. "apply 2.5 FTU to face once daily for 5 days then stop").
- Inflamed skin may **sting** briefly with creams/TCIs. Persistent burning or worsening needs review (allergy, TSW, infection).
- Eyelids: mild steroid for a few days only; TCI preferred for recurrent eyelid eczema. Consider ophthalmology if periocular steroid has been used long-term (IOP).
- Skin of colour: atrophy and hypopigmentation are more conspicuous — warn patients.
- **Never** sell or supply potent steroids (e.g. clobetasol creams) for skin lightening. Illegal cosmetic use causes severe atrophy, striae and Cushing's (relevant in community pharmacy).

Paraffin-based steroid preparations soaked into dressings/clothing are a **fire hazard** (the same MHRA 2018 warning as for emollients).

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK POSITION (2026)
Steroid potency classes	A student answer (and some students) used the **US 7-class** list — e.g. betamethasone valerate 0.1% cream "mild", mometasone "moderate", hydrocortisone butyrate "mild"	UK BNF uses **4 classes**, and all three are **potent**; clobetasol is very potent. Teach and prescribe by **BNF class**
Contraindications	Course materials varied	BNF: contraindicated in untreated bacterial/fungal/viral lesions, acne, rosacea and perioral dermatitis; cautions in widespread psoriasis, infants, long-term/large-area use. MHRA TSW 2021 adds stopping gradually after long-term use
Topical retinoids in pregnancy	Course teaching: "limited data — avoid"	Agrees. EMA/MHRA: **contraindicated in pregnancy and in women planning pregnancy**; UKTIS for exposures
Isotretinoin regulation	Course (Sept 2023) predates the MHRA Oct 2023 rules	MHRA **Oct 2023**: mental health and sexual function assessment, new forms, two prescribers for under-18s. **Jan 2026**: two-prescriber rule removed, new AoR form, BAD audit and video. PPP unchanged
COC in acne	"Yasmin, Dianette and dienogest are used for acne in the UK"; "co-cyprindiol 2 mg + EE 35–50 mcg"; newer progestogens carry "2–4×" the VTE risk	Only **co-cyprindiol** is **licensed for acne** in the UK, and it contains EE **35 micrograms** only. Drospirenone/dienogest pills are contraceptives (acne benefit is off-label). FSRH/EMA: desogestrel/gestodene/drospirenone **1.5–2×** LNG (9–12 vs 5–7 per 10,000); cyproterone 1.5–2×. NICE NG198: consider a COC alongside topicals; co-cyprindiol not first-line
Spironolactone dose	"50–100 mg **twice** daily"	Usual acne dose is **50–100 mg once daily** (up to 200 mg/day under specialist guidance), off-label, with contraception
Acne first-line	Some student answers suggested "oral doxycycline 14 days"	NICE NG198: oral tetracycline only with a topical retinoid/BPO, review at 12 weeks, maximum ~6 months
Eflornithine	Students suggested eflornithine for the hirsutism case	**Vaniqa is now discontinued/not available in the UK (dm+d)** and "not recommended" in several ICB formularies — check availability
Hirsutism referral	Course discussion: "UK hirsutism is managed by endocrine/private clinics"	CKS: primary care can assess, exclude red flags and start a COC; refer for red flags or failed treatment. Laser is mostly self-funded
Oral steroids in psoriasis	Course teaching: avoid (pustular flare)	Agrees with UK practice (NICE CG153/BAD)
Short oral steroid in eczema	Course teaching: acceptable as a bridge	UK: occasionally, short and specialist-guided; avoid repeated courses; steroid emergency card if prolonged
Biologics in primary care	Course teaching: GPs may not initiate	Correct: **hospital-only (red)** in the UK. Primary care records the drug, watches for AEs and avoids live vaccines; methotrexate is via shared care

TOPIC	COURSE TEACHING	UK POSITION (2026)
Coal tar	"Contraindicated in pregnancy/ breastfeeding"	Manufacturers advise caution (avoid in the first trimester and over large areas); not an absolute contraindication in UK SmPCs — check the product

Self-check quiz

4 questions · optional

Q1 How much is one FTU and what does it cover?

✓ About 0.5 g; two adult palms (~2% BSA).

Q2 Potency of betamethasone valerate 0.1%?

✓ Potent (BNF), not mild.

Q3 Rash spreading on a steroid with scale at the edge?

✓ Scrape for tinea incognita.

Q4 What does MHRA advise after long-term potent use?

✓ Reduce gradually, not abruptly; seek specialist advice if TSW suspected.

References

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Photo gallery

Adverse effects of topical steroids

7

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Topical steroid withdrawal

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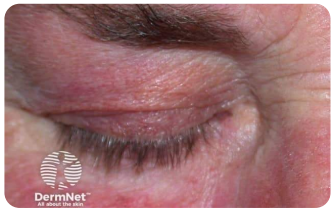
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Fingertip unit and vehicles

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Therapeutics Module 1 · Week 2

Emollients & soap substitutes

Why emollients are the backbone of dry-skin and eczema care, how to choose between lotions, creams, gels and ointments, how much to prescribe, and the safety messages (fire risk, aqueous cream).

ALSO CALLED moisturiser · leave-on emollient · soap substitute · aqueous cream · paraffin fire risk · emollient quantity

RELATED PAGES Eczema / dermatitis · Itch & xerosis in older adults · Topical corticosteroids: potency, quantities & safety · Psoriasis · Hand eczema (chronic, occupational & pompholyx)

KEY FACTS

- **Emollients** restore the skin barrier: the foundation of treatment for eczema, psoriasis, ichthyosis and dry skin.
- Types run from lotions (light, more water) through creams and gels to **ointments** (greasiest, most occlusive, fewest preservatives).
- Use **liberally and often**: adults with eczema may need about 250–500 g a week; apply in the direction of hair growth.
- Use as a **soap substitute**; leave 15–30 minutes between emollient and topical steroid.
- Fire risk paraffin-based (and other) emollients dried on fabric are flammable; warn about naked flames and smoking (MHRA).

Overview

Moisturisers that soften and hydrate the stratum corneum.

- **Occlusives** (paraffins, petrolatum, dimeticone) reduce transepidermal water loss.
- **Humectants** (urea, glycerol, lactic acid) draw water in.
- **Emollient lipids** (ceramides, oils) fill gaps in the barrier.

Formulations range from light lotions/gels creams greasy ointments, which work best and contain the least preservative. They are used as **leave-on** treatments, as **soap substitutes**, and (now rarely) as bath additives.

Aetiology: causes and risk factors

Myths to address

> **"EMOLLIENTS ARE JUST MOISTURISERS, OPTIONAL"**
> regular use reduces flares and steroid use.

> **"AQUEOUS CREAM IS A GOOD EMOLLIENT"**
> it contains sodium lauryl sulfate, an irritant; avoid as a leave-on.

How it works (mechanism of action)

Emollients work by **occluding, humectant attraction of water and replacing lipids**.

1. **Occlusion**

   Oily layers (paraffin, petrolatum) slow transepidermal water loss.

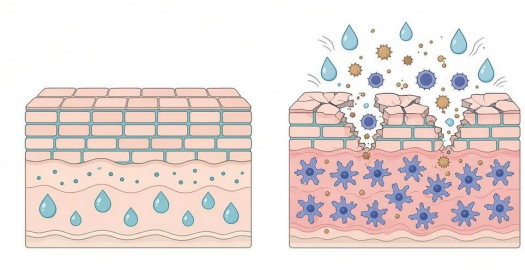

Fig. 1 · A leaky barrier loses water and lets irritants in; emollients seal it.

AI-generated illustration

2. **Humectants**

   Urea, glycerol and lactic acid draw water into the stratum corneum.

3. **Lipid replacement**

   Ceramides and fatty acids refill gaps between corneocytes.

4. **Choice by greasiness**

   More oil = more occlusion but less cosmetic acceptability.

Fig. 2 · Spectrum from light lotions to greasy ointments.

AI-generated illustration

5. **Technique**

   Smooth on downwards in the direction of hair growth to avoid folliculitis.

Fig. 3 · Apply in long downward strokes.

AI-generated illustration

Clinical features

How it works

Emollients restore the lipid "mortar" between corneocytes. They reduce water loss, irritant/allergen penetration and itch, and have a **steroid-sparing** effect (fewer flares, less steroid needed). In atopic eczema, filaggrin loss and soap/surfactant exposure drive barrier failure (see the eczema notes see Vitamin D analogues, coal tar).

Clinical uses & where it fits

Step 0 of every eczema, psoriasis and ichthyosis plan. Also xerosis in older people, hand dermatitis and dryness after isotretinoin.

How to choose / assess

- **"The best emollient is the one the patient will use."** Offer 2–4 to try (sample packs or self-selection, photo C03). Consider dryness, site, time of day, school/work, preference and **allergens/irritants**: avoid **fragrance**, and check for **lanolin** (wool alcohols), **SLS**, **methylisothiazolinone** and preservatives in creams.
- **QUANTITY CHECK**
  NICE CG57 — **250–500 g/week** for children; about **500 g/week** for adults with widespread dryness. A patient collecting one 500 g tub every 3 months is under-using.
- Prescribe enough for **home and nursery/school** (NICE CG57).

Key comparisons — the options

PRODUCT	LEAVE-ON?	TEACHING POINT
Aqueous cream BP	**No** — contraindicated as a leave-on	Contains **sodium lauryl sulfate (SLS)** stinging, burning, redness and measurable **barrier damage** (Danby 2011; MHRA 2013). At most a wash product; most formularies have dropped it
Aveeno cream (colloidal oatmeal)	Yes	Oat protein — rare contact allergy; check the formulary (many ICBs restrict it)
Cetraben cream	Yes	Paraffin-based cream
Diprobase cream	Yes	Lighter paraffin-based cream
Epaderm ointment	Yes	Greasy ointment, also a soap substitute; **high fire risk** when soaked into fabric
Emulsifying ointment	Yes (and as a wash)	Older versions contained SLS — switch if it stings

Fig. 4 · Emollients from light to greasy — "the right one is the one the patient will use"; children need 250–500 g/week. Avoid aqueous cream as leave-on.

Open access image (DermNet) · **Source**

Fig. 5 · Emollients range from light lotions to very greasy ointments — the right one is the one the patient will use.

Open access image · **Source**

Diagnosis and investigations

Quantity

- **250–500 g / week** for children; **~500 g / week** for adults with widespread dryness.

Under-use check

- One 500 g tub every 3 months = under-using.

Allergy?

- Worse with a product: suspect contact allergy; patch test.

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
Aqueous cream BP	**Not** a leave-on: SLS stings and damages the barrier (MHRA 2013).
Colloidal oatmeal cream	Leave-on; rare oat allergy; check formulary.
Paraffin-based creams	Leave-on; lighter, daytime.
Greasy ointments	Leave-on and soap substitute; **high fire risk** on fabric.
Emulsifying ointment	Leave-on and wash; older versions had SLS: switch if it stings.

Management

> **Prescribe generously** (250–500 g/week for a child; ~500 g for an adult), in **pump dispensers** to reduce contamination, with a separate tub for the soap substitute.

> **HOW TO APPLY**

smooth on in the direction of hair growth (reduces folliculitis), often (at least 2–4 times daily and after washing), including when the skin is clear. Leave a gap between emollient and steroid.

MHRA SAFETY

Fire risk with **both paraffin-containing and paraffin-free** emollients that dry onto clothing, bedding or dressings (DSU Dec 2018; reinforced 2020). Advise no smoking or naked flames; washing fabrics at high temperature may not remove all residue; tell carers and care homes; **Slips** in the bath/shower with emollient bath additives. Bath additives add no benefit (BATHE trial), and NICE CG57 (updated 2023) no longer advises offering them routinely

Aqueous cream (MHRA DSU 2013): may irritate the skin, particularly in children with eczema, probably because of SLS.

IP scope

- Fully within community/primary-care IP and OTC practice. Follow ICB self-care/formulary policies; many ICBs expect OTC purchase for mild dry skin.

Fig. 6 · Smooth emollient on in long strokes along the direction of hair growth; rubbing hard against it can block follicles and cause folliculitis.

AI-generated illustration

Treatment ladder

1 **Prescribe**

Pump dispensers; separate tub for soap substitute. Generous quantities (as above).

2 **Apply**

Smooth on **in the direction of hair growth** (less folliculitis). At least **2–4 times daily** and after washing, even when clear.

3 **Counsel safety**

Fire risk with paraffin and paraffin-free emollients on clothing, bedding, dressings: no smoking or naked flames; tell carers. **Slips** with bath additives; BATHE trial: no added benefit.

⚠ CAUTION

Prescribing limits and pitfalls

- **Fire**: Residue on fabric burns easily; washing may not remove it.
- **Aqueous cream**: Wash product at most.

Referral

ROUTINE — Uncontrolled despite adequate emollient + TCS: refer per NICE CG57.

ROUTINE — Suspected emollient contact allergy: patch testing.

SAME DAY — Erythroderma or infected eczema: assess promptly.

Red flags and when to refer

- Warn about **fire risk**: dressings, clothing and bedding soaked in any emollient (paraffin or paraffin-free) can catch fire easily (MHRA).

Red flags / refer

- Eczema still uncontrolled despite adequate emollient + TCS (see NICE CG57 referral criteria); suspected emollient contact allergy (patch testing); erythroderma; infection.

Complications and prognosis

Complications

- Folliculitis, irritant reactions (preservatives, lanolin), slipping and falls, fire risk.

Prognosis

- Ongoing, lifelong in chronic conditions.

Prescriber pearls (UK pharmacist IP)

"Emollients are the treatment, not an extra."

Show the FTU and a handful of emollient. For a child with widespread eczema, a 500 g tub should last about a week.

Fire-safety counselling is an MHRA expectation — record that you gave it.

Self-check quiz

4 questions · optional

Q1 Weekly emollient for a child with eczema (NICE)?

✓ 250–500 g a week.

Q2 Why not aqueous cream as a leave-on?

✓ Contains SLS: stinging and barrier damage (MHRA 2013).

Q3 Which direction to apply?

✓ In the direction of hair growth, to reduce folliculitis.

Q4 Fire risk applies to which emollients?

✓ Both paraffin-containing and paraffin-free products soaked into fabric.

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Photo gallery

Clinical photographs 1

Open access image · **Source**

Emollients from light to greasy 1

Open access image (DermNet) · **Source**

Therapeutics Module 1 · Week 2

Vitamin D analogues, coal tar, dithranol & phototherapy

The non-steroid topical options for psoriasis and the hospital light treatments: how they work, where they fit on the NICE CG153 ladder, how to counsel, and their safety issues.

ALSO CALLED calcipotriol · calcitriol · Dovobet · Enstilar · coal tar · dithranol · +4

RELATED PAGES Psoriasis · Biologics, JAK inhibitors & conventional systemics · Eczema / dermatitis · Topical corticosteroids: potency, quantities & safety · Hypopigmentation & skin of colour

KEY FACTS

- **Vitamin D analogues** (calcipotriol, calcitriol) are first-line topicals for plaque psoriasis, often combined with a potent steroid.
- **Coal tar** is an older, effective but messy option for scalp and plaque psoriasis.
- **Narrowband UVB** (NB-UVB, 311 nm) for psoriasis, eczema, vitiligo; **PUVA** (psoralen + UVA) for thicker or resistant disease.
- Phototherapy is usually 2–3 times a week for 6–10 weeks; cumulative dose increases skin cancer risk (especially PUVA: SCC).
- Calcipotriol: max about 100 g a week in adults (hypercalcaemia risk with more); irritating on face and folds.

Overview

VITAMIN D ANALOGUES (UK)

Calcipotriol 50 micrograms/g ointment or scalp solution (Dovonex), and combined with betamethasone dipropionate 0.05% (**Dovobet** gel/ointment, **Enstilar** foam, generics); **Calcitriol** 3 micrograms/g ointment (Silkis) — gentler, used on face/flexures; **Tacalcitol** 4 micrograms/g (Curatoderm) — once daily

COAL TAR

crude coal tar or refined extracts in creams, ointments, scalp shampoos and bath products (e.g. Exorex lotion, Sebco, Capasal, Polytar, Cocois). Specialist units compound crude tar.

Dithranol (anthralin): short-contact therapy for chronic plaques in specialist/day units. It stains purple-brown and burns normal skin.

Phototherapy and PUVA

Controlled ultraviolet exposure, delivered in **hospital dermatology units** and some home-phototherapy schemes. Main types:

MODALITY	WAVELENGTH	MAIN USES
Narrowband UVB (NB-UVB, TL-01)	**311–313 nm**	First-line phototherapy: psoriasis, atopic eczema, PLE desensitisation, vitiligo, pruritus
Broadband UVB	290–320 nm	Largely replaced by NB-UVB
PUVA = psoralen + UVA	UVA **320–400 nm**	Thick/resistant psoriasis, palmoplantar pustulosis, hand eczema (hand-and-foot PUVA), mycosis fungoides, some vitiligo
UVA1	340–400 nm	Morphoea, scleroderma, some eczema (specialist)
Targeted UVB / excimer 308 nm	308 nm	Localised psoriasis/vitiligo
Photodynamic therapy (PDT)	Visible light + photosensitiser	Actinic keratoses, superficial BCC, Bowen's

Psoralens are furocoumarins. **8-methoxypsoralen (methoxsalen, 8-MOP)** is taken orally ~2 h before UVA, or used as **bath/soak** (topical) PUVA. 5-MOP, where available, causes fewer GI effects.

Aetiology: causes and risk factors

Myths to address

- **"SUNBEDS ARE LIKE PHOTOTHERAPY"**
  sunbeds emit mainly UVA, are unregulated and raise melanoma risk.

How it works (mechanism of action)

These treatments slow **keratinocyte overgrowth** and **suppress skin T cells**.

1. **Vitamin D receptor**
   Calcipotriol binds the VDR, normalising keratinocyte proliferation and differentiation.
2. **Tar**
   Coal tar activates the aryl hydrocarbon receptor, reducing inflammation and scale.
3. **UVB**
   UVB penetrates the epidermis, causing T-cell apoptosis and immunosuppression.

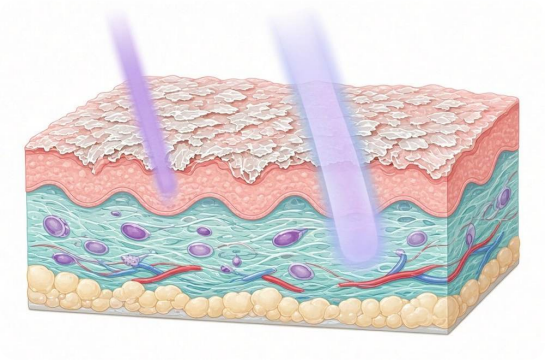

Fig. 1 · UVB acts in the epidermis; UVA reaches the dermis.
AI-generated illustration

4. **PUVA**
   Psoralen intercalates in DNA; UVA cross-links it, stopping cell division.

5 **Delivery**

Treatment is given in controlled whole-body cabinets.

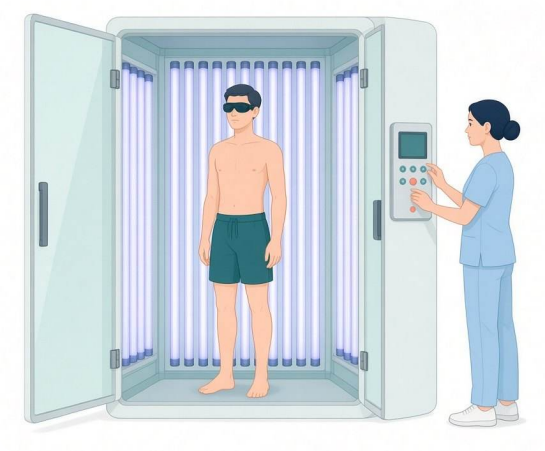

Fig. 2 · Narrowband UVB cabinet.

AI-generated illustration

Clinical features

How it works

VITAMIN D ANALOGUES

They bind the nuclear **vitamin D receptor (VDR)**, which pairs with RXR to regulate genes; This **slows keratinocyte proliferation** (e.g. via the cyclin-dependent kinase inhibitors p21/p27) and **promotes normal differentiation** (cornified-envelope proteins); They are also **immunomodulatory**: they lower Th1/Th17 cytokines such as IL-17 and IL-8, and promote regulatory responses; Clinically they flatten and de-scale plaques but do less for erythema than steroids

COAL TAR

A complex mixture (polycyclic aromatic hydrocarbons, phenols). It activates the **aryl hydrocarbon receptor (AhR)**, which normalises keratinocyte differentiation, restores barrier proteins (filaggrin) and damps Th2/Th17 inflammation; It is also anti-proliferative (reduces DNA synthesis), antipruritic and antifungal (Malassezia); It is phototoxic with UVA; the historic Goeckerman regimen combined it with UVB

DITHRANOL

generates free radicals, giving an anti-proliferative effect; it stains and irritates.

Clinical uses & where it fits

PLAQUE PSORIASIS (NICE CG153), TRUNK/LIMBS

First-line: **potent corticosteroid once daily + vitamin D once daily**, applied separately (one in the morning, the other in the evening), for **up to 4 weeks**; No improvement after **8 weeks**: vitamin D **twice daily** alone; Still poor after 8–12 weeks: potent steroid twice daily for up to 4 weeks, or coal tar once/twice daily, or the **combined calcipotriol + betamethasone dipropionate** product once daily for up to 4 weeks. NICE places the combined product where the separate regimen fails or adherence is the problem

SCALP PSORIASIS

potent steroid for up to 4 weeks first, then the combined product, then vitamin D alone.

FACE/FLEXURES/GENITALS

mild–moderate steroid short-term, or calcitriol / a TCI (off-label).

COAL TAR

chronic plaque and scalp psoriasis, seborrhoeic dermatitis (shampoos), and some specialist use in lichenified eczema.

How to choose / assess

- Thin plaques, or patients who struggle with adherence combined once-daily gel/foam.
- Thick scale descale first (salicylic acid; coal tar/salicylic acid/sulfur scalp ointments such as Cocois).
- Flexures avoid calcipotriol irritation (use calcitriol).
- Large BSA watch the vitamin D maximum doses.

Key comparisons

	VITAMIN D ANALOGUE	POTENT TCS	COAL TAR
Speed	Weeks (4–8)	Days	Weeks
Atrophy	None	Yes	None
Irritation	Common (face/flexures)	Rare	Folliculitis common; stains, odour
Safety	**Hypercalcaemia** if the maximum is exceeded	HPA suppression, rebound	Phototoxicity; theoretical carcinogenicity
Pregnancy	Avoid unless essential (limited data)	Mild–potent acceptable short-term	Avoid in first trimester (manufacturers)

Phototherapy and PUVA

How it works

- UV light is **immunosuppressive** and anti-proliferative: It induces **T-cell apoptosis** and depletes Langerhans cells; It shifts cytokines (IL-12/IL-23/IL-17 and TNF-α; IL-10 and regulatory T cells); It forms DNA photoproducts that slow keratinocyte turnover
- **Psoralens** intercalate into DNA. On UVA activation they form **mono-adducts and interstrand cross-links**, which block DNA replication and proliferation, and they also generate reactive oxygen species.
- UVA penetrates to the dermis, deeper than UVB.

Clinical uses & where it fits

NICE CG153: offer **NB-UVB** for plaque or guttate psoriasis that topical treatment alone cannot control. Offer **PUVA** for palmoplantar pustulosis and when NB-UVB fails.

How to choose / assess

COURSE LOGISTICS

2–3 visits a week for ~6–10 weeks (NB-UVB ~20–30 exposures). Dose is set by **skin type or minimal erythema dose (MED) testing**. Patients must be able to attend regularly.

EXCLUDE

photosensitising drugs or conditions (lupus, porphyria, xeroderma pigmentosum); previous skin cancer or high cumulative UV; inability to stand, or claustrophobia; pregnancy (oral PUVA)

Key comparisons

	NB-UVB	PUVA
Efficacy (psoriasis)	Good	Slightly higher, longer remissions
Pregnancy/ children	**Can be used** (folate may be depleted — check)	Oral PUVA **avoided** in pregnancy and children
Acute AE	Erythema/burn, itch, PLE flare	Nausea (8-MOP), phototoxic burn (peaks at 48–72 h), **eye damage** (cataract)
Long-term risk	Small; skin ageing; NMSC risk thought low	**Dose-dependent SCC** (much higher after >150–200 treatments), BCC; **melanoma** reported after very high exposure; lentigines
UK caps	Lifetime counts recorded; review after >500 exposures (BAD/BPG)	**Lifetime ~150–200 treatments** as a guide (BAD/BPG); skin-cancer surveillance after high exposure

Fig. 3 · Natal cleft/buttock psoriasis before calcipotriol. Note: calcipotriol can irritate flexures — calcitriol is better tolerated there.

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Diagnosis and investigations

Ladder (NICE CG153)

- Potent TCS od + vitamin D od, separately, **up to 4 weeks**.
- No better at **8 weeks**: vitamin D bd alone.
- Still poor at 8–12 weeks: potent TCS bd 4 weeks, coal tar, or combined product.

Max doses (BNF, recheck)

- Calcipotriol **100 g / week**.
- Combination: **15 g / day, 100 g / week, 30% BSA**, 4-week courses.
- Calcitriol **30 g / day**, 35% BSA.

Avoid

- Hypercalcaemia, severe renal impairment; caution in erythrodermic or pustular psoriasis.

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
Potent steroid alone	Faster on redness; atrophy and rebound with long use.
Combined calcipotriol + betamethasone	One daily application; **contains a steroid**: count it.
Coal tar	Messy, odour, stains; very low cancer risk at therapeutic doses.
Dithranol	Effective; stains purple-brown; specialist short-contact.
Biologics / systemics	When topicals and light fail.

Management

> **MAXIMUM DOSES (ADULTS, BNF — RECHECK)**
>
> Calcipotriol: **100 g/week** (less if the scalp solution is also used); **Calcipotriol + betamethasone** products: **max 15 g/day and 100 g/week, 30% BSA, 4-week courses**. The foam can be used longer as twice-weekly maintenance per its SmPC; Calcitriol: **max 30 g/day** and 35% BSA; Children: specialist use, with lower age-based maxima

> **AVOID IN**
>
> hypercalcaemia and severe renal impairment (calcium handling). Use with caution in generalised pustular or erythrodermic psoriasis (absorption, hypercalcaemia).

> **COUNSEL**
>
> wash hands after use; keep calcipotriol off the face; irritation usually settles; with phototherapy, apply after UV exposure, not before.

> **COAL TAR SAFETY (COURSE TEACHING QUESTION)**
>
> **Percivall Pott (1775)** described **scrotal squamous cell carcinoma** ("chimney sweeps' cancer") from soot — the first recognised occupational cancer; Large cohort data (e.g. Roelofzen et al. 2010, ~13,000 psoriasis/eczema patients) found **no increase in overall cancer risk** from therapeutic coal tar, so the risk at dermatological concentrations is considered very low; Coal tar is used less now mainly because of mess, odour, staining and EU/UK reformulation of products, not because of cancer data; Avoid on inflamed/pustular psoriasis and broken skin, avoid concentrated products on face/genitals, and avoid sun after application (phototoxicity)

IP scope

Starting vitamin D analogues, combined calcipotriol/betamethasone and coal-tar preparations for chronic plaque/scalp psoriasis is within primary-care IP competence. Dithranol is specialist.

Phototherapy and PUVA

> **ORAL PSORALEN**
>
> Wear **UVA-protective wrap-around glasses for 24 h** after each dose (some units say 12 h); Avoid the sun on treatment days, and take the dose with food to reduce nausea; Avoid eating large amounts of furocoumarin-rich foods (celery, figs, parsnips, limes)

> **LONG-TERM**
>
> lifelong **skin-cancer surveillance** after high PUVA exposure, with self-examination. Any new keratotic or ulcerated lesion goes on the NICE NG12 pathway.

> **WITH CALCIPOTRIOL OR ACITRETIN**
> units may combine them with light (e.g. re-PUVA with acitretin). Apply topical vitamin D after UV exposure, not before.

IP scope

Referral and counselling only; phototherapy prescription and dosing are specialist.

Treatment ladder

1 **Counsel**

Wash hands after; keep calcipotriol **off the face**; irritation usually settles. With phototherapy, apply vitamin D **after** UV, not before.

2 **First line**

Steroid in the morning, vitamin D at night (or the reverse) for up to 4 weeks, or one combined product daily.

3 **Step up**

Vitamin D bd; coal tar; combined product; calcitriol for face and flexures.

4 **Specialist**

NB-UVB (often safest step-up in **pregnancy**), PUVA, dithranol, systemics.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Hypercalcaemia**: Over weekly maximum or symptoms: check calcium, stop.
- **PUVA eyes & cancer**: UVA-blocking eye protection ~24 h after oral psoralen; lifelong skin checks.
- **Tar**: Avoid inflamed / pustular skin, face, genitals; avoid sun after (phototoxic).

Referral

SAME DAY Erythrodermic or generalised pustular psoriasis: urgent dermatology.

ROUTINE Not controlled on topicals at 8–12 weeks: refer for phototherapy or systemics.

Red flags and when to refer

- More than the BNF weekly maximum of calcipotriol, or symptoms of hypercalcaemia: check calcium and stop.
- Patients on PUVA need UVA-blocking eye protection for about 24 hours after oral psoralen, and lifelong skin-cancer checks.

Red flags / refer

- Erythrodermic or generalised pustular psoriasis (same day).
- Psoriasis not responding to two topical steps, extensive (>10% BSA) or high-impact disease (NICE CG153 criteria for referral, phototherapy or systemic therapy).
- Suspected hypercalcaemia (thirst, confusion) after heavy vitamin D use.

Red flags / refer

Complications and prognosis

Complications

- Irritation (vitamin D, tar), hypercalcaemia (excess calcipotriol), burning, photoageing, skin cancer (PUVA > UVB).

Prognosis

- Good clearance in many; remissions last months.

Prescriber pearls (UK pharmacist IP)

- "Steroid in the morning, vitamin D at night" (or the other way round) for the first 4 weeks, or one combined product once daily.
- Combined products **contain a steroid** — count them in the steroid budget and use breaks.
- Tar shampoo: lather, leave 5–10 minutes, rinse. It can discolour blond, grey or dyed hair.
- "Phototherapy is a prescription of light" — the dose, course length and lifetime total are recorded like a drug's.
- NB-UVB is often the **safest step-up in pregnancy** for psoriasis/eczema.

Phototherapy and PUVA

Self-check quiz

4 questions · optional

Q1 NICE first-line topical for trunk plaque psoriasis?

✓ Potent steroid once daily + vitamin D analogue once daily, applied separately, up to 4 weeks.

Q2 Weekly calcipotriol maximum?

✓ 100 g a week (adult, BNF).

Q3 Which vitamin D analogue for face and flexures?

✓ Calcitriol (gentler).

Q4 Advice after oral psoralen for PUVA?

✓ UVA-blocking eye protection for about 24 hours.

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Photo gallery

Vitamin D analogues and coal tar 3

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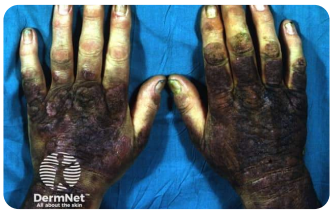

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Phototherapy 2

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Therapeutics Module 1 · Week 2

Retinoids (topical & oral)

Topical retinoids for acne and photoageing, and the oral retinoids (isotretinoin, acitretin, alitretinoin): mechanism, uses, adverse effects, MHRA pregnancy-prevention rules and what the primary-care prescriber monitors.

ALSO CALLED adapalene · tretinoin · isotretinoin · acitretin · alitretinoin · pregnancy prevention programme · +1

RELATED PAGES Acne vulgaris · Photoageing · Psoriasis · Acne prescribing & the combined pill · Cosmetic procedures: peels, needling & lasers

KEY FACTS

- **Topical retinoids** (adapalene, tretinoin, tazarotene) treat comedonal and inflammatory acne and photoageing.
- **Isotretinoin** (oral) for severe or scarring acne: dermatology-initiated, with the **Pregnancy Prevention Programme** and mood monitoring (MHRA).
- **Acitretin** for psoriasis and keratinisation disorders: avoid pregnancy for **3 years** after stopping.
- Topical retinoids are contraindicated in pregnancy and in women planning pregnancy (UK).
- Start low, apply thinly (pea-sized amount for the face), expect irritation for 2–4 weeks; use sunscreen.

Overview

Natural and synthetic compounds related to **vitamin A (retinol)** that act on retinoid receptors.

GENERATION (CLASSIC GROUPING)	AGENTS	UK USE
1st (non-aromatic)	**Tretinoin** (all-trans retinoic acid), **isotretinoin** (13-cis), **alitretinoin** (9-cis)	Topical tretinoin with clindamycin (Treclin); **oral isotretinoin** (severe acne); **oral alitretinoin** (severe chronic hand eczema, NICE TA177)
2nd (mono-aromatic)	Etretinate (withdrawn), **acitretin**	**Oral acitretin** — severe/pustular/palmoplantar psoriasis, keratinisation disorders (Darier's, ichthyoses), chemoprevention in transplant recipients
3rd (poly-aromatic, "arotinoids")	**Adapalene**, tazarotene (UK topical discontinued), bexarotene	**Adapalene** 0.1% (Differin) and with BPO (**Epiduo** 0.1%/2.5% and 0.3%/2.5%); oral **bexarotene** for CTCL (specialist)
4th	**Trifarotene** (RAR-γ selective)	Topical (Aklief) for facial and truncal acne — formulary status varies

Over-the-counter cosmetic **retinol/retinaldehyde** are weaker precursors that the skin converts.

Aetiology: causes and risk factors

Myths to address

- **"RETINOIDS THIN THE SKIN"**
  they thin the stratum corneum but thicken the epidermis and increase collagen.

How it works (mechanism of action)

Retinoids act on **nuclear retinoic acid receptors** to normalise how skin cells grow and shed.

1. **Receptors**
   Retinoids bind RAR and RXR nuclear receptors and alter gene expression.
2. **Normal keratinisation**
   Reduce follicular plugging and comedones.
3. **Anti-inflammatory**
   Reduce TLR-2 expression and inflammatory signalling.
4. **Sebum reduction (isotretinoin)**
   Oral isotretinoin shrinks sebaceous glands dramatically.

Fig. 1 · Isotretinoin shrinks sebaceous glands.
AI-generated illustration

5. **Application**
   A pea-sized amount covers the whole face.

Fig. 2 · Pea-sized amount for the whole face.
AI-generated illustration

Clinical features

How it works

- Retinoids bind the nuclear **retinoic acid receptors (RAR-α, -β, -γ)** and **retinoid X receptors (RXR)**. Receptor dimers bind retinoic-acid response elements and regulate genes for **keratinocyte proliferation and differentiation**. RAR-γ predominates in the epidermis, which is why adapalene and trifarotene are skin-selective.
- **TOPICAL RETINOIDS IN ACNE**
  They normalise follicular keratinisation: **comedolytic**, and anti-comedogenic because they prevent the microcomedone; They are anti-inflammatory (down-regulate **TLR-2** and AP-1) and help other agents penetrate; In photoageing they stimulate collagen and even out pigment
- **Oral isotretinoin** is the only drug that acts on **all four acne mechanisms**: profound **sebaceous-gland shrinkage and sebum suppression** (largely through intracellular isomerisation and sebocyte apoptosis, not simply direct receptor binding); normalised keratinisation; fewer C. acnes (a knock-on effect of sebum loss); anti-inflammatory effects
- **Acitretin** is anti-proliferative and normalises differentiation in psoriasis. It is not immunosuppressive, so it is useful when infection or malignancy risk rules out immunosuppressants.

Clinical uses & where it fits

> **TOPICAL**
> acne (first-line, NICE NG198); photoageing (tretinoin — cosmetic, not NHS); some ichthyoses; off-label melasma combinations.

> **ISOTRETINOIN**
> severe acne (nodular/conglobate), acne at risk of **permanent scarring**, or moderate acne that has failed adequate standard therapy including an oral antibiotic. **Specialist-initiated only** (consultant-led team).

> **ACITRETIN, ALITRETINOIN, BEXAROTENE**
> specialist-initiated (hospital-only prescribing).

How to choose / assess

- Topical retinoids suit **comedonal and mixed acne**. Choose adapalene for tolerability, or adapalene/BPO for inflammatory lesions. Avoid in pregnancy or when planning pregnancy.
- Before an oral retinoid, assess **whether the patient could become pregnant**, mental health, lipids/liver, and whether a tetracycline is being taken.

Key comparisons

	TOPICAL RETINOID	ISOTRETINOIN	ACITRETIN	ALITRETINOIN
Teratogenic	Low systemic absorption; **avoid in pregnancy/when planning**	**Yes — PPP**; contraception until **1 month** after	**Yes — PPP**; contraception until **3 YEARS** after (etretinate depot)	**Yes — PPP**; until 1 month after
Alcohol	—	Caution	**Avoid during and for 2 months after** (forms etretinate)	—

	TOPICAL RETINOID	ISOTRETINOIN	ACITRETIN	ALITRETINOIN
Blood donation	–	Not during or for 1 month after	Not during or for 3 years after	Not during or for 1 month after
Key AE	Irritation, dryness, photosensitivity	Mucocutaneous dryness, **mood/ psychiatric effects**, **sexual dysfunction**, lipids/LFTs, myalgia/ CK, IIH, poor night vision	Mucocutaneous dryness, hair loss, nail fragility, lipids/LFTs, hyperostosis	Headache (IIH), lipids, TFTs (central hypothyroidism)

Fig. 3 · Sunburn on isotretinoin — photosensitivity and skin fragility; counsel SPF 30+ / UVA, avoid sunbeds; no waxing/laser/ dermabrasion during and for 6 months after.

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Adverse effects to recognise

Sunburn on isotretinoin

Photosensitivity and fragility. Site: Face.

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Impetigo

Dry cracked skin, S. aureus. Site: Face.

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Retinoid dermatitis

Dryness and peeling: start slowly. Site: Face.

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Acitretin nails

Fragility, paronychia. Site: Nails.

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Hair shedding

Dose-related telogen effluvium (acitretin). Site: Scalp.

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Diagnosis and investigations

Before oral retinoid

- Could they become pregnant? Mental health, lipids / liver, any tetracycline?

PPP contraception

- **1 highly effective user-independent method** (LARC, sterilisation) or **two complementary** methods, from 1 month before.

PPP pregnancy tests

- Supervised test before, **ideally monthly** during, **1 month after** (acitretin: periodically for 3 years).

Monitor

- Isotretinoin: mood, sexual function, lipids, LFTs. Acitretin: lipids, LFTs. Alitretinoin: lipids, TFTs, headache.

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
Isotretinoin	Dryness, **mood / psychiatric**, sexual dysfunction, lipids, LFTs, myalgia, IIH, night vision.
Acitretin	Dryness, hair loss, nail fragility, lipids / LFTs; **no alcohol** (forms etretinate); no blood donation.
Alitretinoin	Headache (IIH), lipids, central hypothyroidism.
Tetracycline + retinoid	**Raised intracranial pressure**: don't combine.
Topical in pregnancy	Contraindicated (EMA 2018); ask and record.

Management

Pregnancy and topical retinoids

> Systemic absorption from topical tretinoin/adapalene is very low. Observational data have **not shown a clear rise in malformations**, though isolated case reports exist.

> Since the **EMA 2018 retinoid review** (reflected in MHRA materials and SmPCs), the EU/UK position is that **topical retinoids are contraindicated in pregnancy and in women planning pregnancy**. No formal PPP is needed for topicals, but **ask about pregnancy intention and contraception**, and record the advice.

> **INADVERTENT EXPOSURE TO A TOPICAL**
> stop the product, reassure that the risk appears very low, and seek advice from **UKTIS** (UK Teratology Information Service, 0344 892 0909).

> **EXPOSURE TO AN ORAL RETINOID**
> urgent specialist/obstetric referral and a Yellow Card report.

Oral retinoid Pregnancy Prevention Programme (PPP) — MHRA (2019 materials, still current)

> Applies to **isotretinoin, acitretin and alitretinoin** in anyone who **could become pregnant**.

> **CONTRACEPTION**
> **at least one highly effective, user-independent method** (LARC — implant, IUD/IUS — or sterilisation) **or two complementary user-dependent methods** (e.g. combined pill + condom), started at least 1 month before treatment.

> **PREGNANCY TESTING**
> a **medically supervised test** before starting, **ideally monthly** during treatment, and **1 month after stopping** (acitretin: periodically for 3 years).

> **SUPPLY**
> prescriptions within the PPP are limited to **30 days' supply** and must be dispensed within **7 days** of the prescription date.

> **PAPERWORK**
> an annual **acknowledgement of risk** form, a patient reminder card and a **pharmacist checklist** at dispensing.

Isotretinoin — MHRA rules (Oct 2023, updated Jan 2026)

> **31 OCT 2023 (CHM ISOTRETINOIN EXPERT WORKING GROUP)**
> Counselling on possible **psychiatric and sexual side effects**, including reports of sexual dysfunction persisting after stopping; **Assessment of mental health and sexual function before starting** and at every follow-up (patient-reported measures, e.g. PHQ-9/GAD-7); An **in-person review about 1 month** after starting; A new Acknowledgement of Risk Form, Patient Reminder Card and Pharmacist Checklist; The **Lead Prescriber** must have expertise in systemic retinoids (consultant-led); For **under-18s, agreement of two independent prescribers**

> **OCTOBER 2025 CLARIFICATIONS**
> follow-ups after the first may be **remote**; supervised pregnancy testing may be done remotely with oversight; by the third appointment the sexual-function questions can be brief.

> **UPDATE, 22 JAN 2026**
> The **two-prescriber requirement for under-18s was removed**; It was replaced by an updated two-page Acknowledgement of Risk Form: the prescriber confirms there is no other appropriate effective treatment, and

the patient confirms they know they can ask for a second opinion; A **BAD-run national clinical audit** starts in 2026, and patients should watch a **BAD patient information video** before starting; All other measures remain

WHAT THIS MEANS FOR A PRIMARY-CARE IP

You **do not initiate** isotretinoin; you **refer** (NICE NG198 criteria); **Pre-counsel**: MHRA asks the referrer to give initial information on the risks, including mental health and sexual function; **Check and record contraception** and **screen mood**; Avoid interacting prescriptions: **no tetracyclines** (IIH) and no high-dose vitamin A supplements; Report suspected ADRs via **Yellow Card**; If a patient on isotretinoin presents with low mood or suicidal thoughts stop the drug, arrange urgent mental-health review, and inform the dermatologist

MONITORING (BNF/BAD)

LFTs and fasting lipids before starting, after ~1 month, then 3-monthly (or per the specialist protocol); pregnancy tests as above. Typical dose 0.5–1 mg/kg/day to a cumulative 120–150 mg/kg; lower-dose regimens are common.

COUNSELLING LIST

Dry lips, skin and eyes (lip balm, emollient, lubricant eye drops; contact lenses may become uncomfortable), nosebleeds; Photosensitivity and **sunburn**; Skin fragility: **no waxing, laser or dermabrasion during treatment and for 6 months after**; An early **acne flare**; Myalgia (cut back strenuous exercise); Headache or visual disturbance (**IIH** — urgent); reduced night vision (driving); **Hair thinning**: telogen effluvium that usually recovers after treatment, though persistent cases are reported; Impetigo/S. aureus colonisation; Mood and sexual changes; No blood donation during treatment or for 1 month after

Acitretin and alitretinoin (hospital-only)

- **ACITRETIN**
  PPP with contraception continuing for **3 years** after stopping; **No alcohol** during treatment and for 2 months after, because alcohol converts acitretin to etretinate, which has a very long half-life; Avoid tetracyclines and methotrexate (hepatotoxicity); monitor LFTs and lipids; Nail fragility and hair loss
- **Alitretinoin** (NICE TA177): severe chronic hand eczema unresponsive to potent TCS; PPP; check lipids and TFTs; headache.

IP scope

Topical retinoids: initiate within competence (NICE NG198). Oral retinoids: **specialist-only** initiation and supply (hospital/homecare). Your role is referral, counselling, interaction and contraception checks, and safety-netting.

Fig. 9 · A pea-sized amount covers the whole face: dot it on, then spread thinly at night on dry skin.

AI-generated illustration

Treatment ladder

1

Topical technique

Pea-sized for the whole face, at night, on dry skin; alternate nights at first. Moisturiser and daily SPF; expect 'purging' 2–4 weeks, benefit 8–12 weeks.

2

Pregnancy and topicals

Contraindicated in pregnancy / planning; no formal PPP, but **ask about pregnancy and contraception** and record. Inadvertent exposure: stop, reassure risk looks very low, advice from **UKTIS**.

3

Oral retinoid support

Support PPP checks, mood questions, and bloods per specialist plan. Oral retinoid exposure in pregnancy: urgent specialist / obstetric referral and Yellow Card.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Pregnancy**: Oral retinoids are highly teratogenic: PPP applies to anyone who could become pregnant.
- **Mood**: New low mood or suicidal thoughts on isotretinoin: stop, same-day referral.

Referral

SAME DAY	**Low mood, anxiety or suicidal thoughts** on isotretinoin: stop, same-day referral.
SAME DAY	Possible **pregnancy** on an oral retinoid: stop, urgent specialist advice.
URGENT	Headache, visual disturbance, vomiting on retinoid + tetracycline: possible **raised ICP**.

Red flags and when to refer

- Any possibility of pregnancy on an oral retinoid: stop and seek urgent specialist advice.
- New low mood, anxiety or suicidal thoughts on isotretinoin: stop and refer the same day.

- Headache, visual disturbance or vomiting on a retinoid plus a tetracycline: possible **raised intracranial pressure**.

Red flags / refer

- Pregnancy on an oral retinoid urgent: stop the drug, obstetric/fetal medicine referral, UKTIS, Yellow Card.
- **Mood change, suicidal ideation or psychosis** on isotretinoin stop, urgent mental-health assessment.
- Severe headache with visual symptoms (IIH), severe abdominal pain (pancreatitis with very high triglycerides), jaundice.
- **Acne fulminans** (see Acne therapeutics case and the combined oral contraceptive) dermatology the same day or within 24 h.

Complications and prognosis

Complications

- Irritation, dryness, photosensitivity; teratogenicity; isotretinoin: dry lips and eyes, raised lipids and liver enzymes, mood changes.

Prognosis

- Topical: improvement over 8–12 weeks. Isotretinoin: long remission in most.

Prescriber pearls (UK pharmacist IP)

Topical: a **pea-sized amount for the whole face**, at night, on dry skin. Start on alternate nights, use a moisturiser and daily SPF. Expect "purging" for 2–4 weeks and benefit by 8–12 weeks. Avoid if pregnant or trying to conceive.

Ask every woman on a topical retinoid "Could you be pregnant, or are you planning to be?" — and document the answer.

Isotretinoin memory aid: "contraception, mood, sex, sun, skin procedures, no tetracyclines".

Self-check quiz

4 questions · optional

Q1 **Why not combine isotretinoin with doxycycline?**

✓ Risk of raised intracranial pressure.

Q2 How long to avoid pregnancy after acitretin?

✓ 3 years after stopping.

Q3 PPP contraception requirement?

✓ At least one highly effective user-independent method, or two complementary user-dependent methods.

Q4 How much topical retinoid for the face?

✓ A pea-sized amount for the whole face, at night.

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Photo gallery

Retinoid adverse effects 5

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Therapeutics Module 1 · Week 2

Acne prescribing & the combined pill

The NICE NG198 12-week first-line combinations for acne, how to choose and counsel, antibiotic stewardship, and how combined oral contraceptives (including co-cyprindiol) fit in, with their VTE risk.

ALSO CALLED NICE NG198 · benzoyl peroxide · adapalene · clindamycin · lymecycline · doxycycline · +3

RELATED PAGES Acne vulgaris · Retinoids (topical & oral) · Hirsutism & the Ferriman–Gallwey score · Acne scars

KEY FACTS

- NICE NG198: first-line acne treatment is a **12-week fixed-combination topical** (adapalene + benzoyl peroxide, tretinoin + clindamycin, or BPO + clindamycin).
- Add an oral **tetracycline (lymecycline or doxycycline)** with a topical for moderate–severe acne; never use oral and topical antibiotics together.
- The **combined oral contraceptive (COC)** is an alternative or add-on for women; choose using **UKMEC** (VTE, migraine with aura, smoking over 35).
- Co-cyprindiol has higher VTE risk: use only if other options fail and stop 3 months after acne control.
- Review at 12 weeks; refer scarring, severe or persistent acne for isotretinoin.

Overview

(the case)

A **19-year-old woman** with numerous **comedones and pustules** on the cheeks, jawline, perioral and nasolabial areas, unresponsive to cleansers. Periods are regular; she does not smoke or drink and has had no treatment yet.

- Under **NICE NG198** definitions this is **mild-to-moderate acne**: any number of non-inflammatory lesions, **up to 34 inflammatory lesions and up to 2 nodules**. The lesion count, scarring or psychological impact could move her higher (course teaching: **psychological impact can raise the grade**).
- Moderate-to-severe acne = **35 inflammatory lesions and/or 3 nodules**.

The **lower-face/jawline** distribution suggests an androgen ("hormonal") component. Regular cycles and no hirsutism make PCOS less likely, but still ask about hirsutism, weight, cycle changes and family history.

Aetiology: causes and risk factors

Myths to address

- **"THE PILL CAUSES ACNE"**
  progestogen-only methods can worsen acne; combined pills usually improve it.

How it works (mechanism of action)

The COC reduces acne by **lowering free androgen**, which drives sebum production.

1. **Ovarian suppression**
   Ethinylestradiol and progestogen suppress ovulation and ovarian androgen output.
2. **Raised SHBG**
   Oestrogen increases sex hormone–binding globulin, reducing free testosterone.
3. **Less sebum**
   Lower androgen activity shrinks sebaceous output and comedone formation.

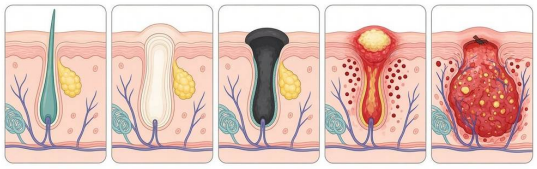

Fig. 1 · Acne develops from microcomedone to inflamed lesion.
AI-generated illustration

4. **Anti-androgenic progestogens**
   Cyproterone, drospirenone and others block androgen receptors.

Fig. 2 · Hormonal pattern acne along the jawline.
AI-generated illustration

How it works (acne pathophysiology drug targets)

Four mechanisms:

1. **Androgen-driven sebum** production.
2. **Follicular hyperkeratinisation** (the microcomedone).
3. Cutibacterium acnes proliferation/biofilm.
4. **Inflammation** (TLR-2, IL-1).

DRUG	MECHANISMS TARGETED
Topical retinoids	2, 4
BPO	3, 2 — no resistance
Topical/oral antibiotics	3, 4 — but resistance
Azelaic acid	2, 3, 4
Hormonal therapy (COC, spironolactone, co-cyprindiol)	1

DRUG	MECHANISMS TARGETED
Isotretinoin	All four

Clinical features

Clinical uses & where it fits — NICE NG198 first-line (12-week courses)

SEVERITY	OFFER ONE OF (FIXED COMBINATIONS PREFERRED)
Mild–moderate	(a) **adapalene 0.1% + BPO 2.5%** gel (Epiduo); (b) **tretinoin 0.025% + clindamycin 1%** gel (Treclin); (c) **BPO 3–5% + clindamycin 1%** gel (Duac)
Moderate–severe	(a) **adapalene 0.3% + BPO 2.5%**; (b) tretinoin + clindamycin; (c) adapalene/BPO **+ oral lymecycline or doxycycline**; (d) **azelaic acid 15–20% + oral lymecycline/doxycycline**
Alternatives	BPO alone if the above are contraindicated; **COC** in women (with topicals); oral erythromycin/trimethoprim if tetracyclines are contraindicated (specialist advice)

> **NG198 RULES**
> **No** topical or oral antibiotic **monotherapy**, and **no** topical + oral antibiotic together; Oral antibiotics **only with a topical retinoid or BPO**; Review at **12 weeks**; continue oral antibiotics beyond 6 months only in exceptional cases; Once clear, **maintain** with a topical retinoid ± BPO (fixed combination)

> **Minocycline is not recommended** (lupus-like syndrome, autoimmune hepatitis, blue-grey pigmentation — see.

> **TETRACYCLINES**
> **Avoid in pregnancy, breastfeeding and under-12s.**; Doxycycline causes photosensitivity; take it upright with water and use sun protection; Doses: lymecycline 408 mg once daily; doxycycline 100 mg once daily (BNF — recheck)

How to choose / assess

For this case:

- Start **adapalene 0.1%/BPO 2.5% gel once daily at night**: it covers comedones and pustules with no antibiotic resistance.
- Use tretinoin/clindamycin instead if BPO irritation or bleaching is a concern.
- If she could become pregnant, NICE says to discuss that **topical retinoids should not be used in pregnancy**; **BPO/clindamycin** is the option without a retinoid.
- Discuss her **contraceptive needs** at the same visit. If she wants contraception anyway, a COC can be added for its acne benefit.

Key comparisons — COCs in acne

Does the COC have merit? Yes.

> **HOW IT WORKS**
> oestrogen hepatic **SHBG** (less free testosterone) and suppresses **LH-driven ovarian androgen** production. Progestogens with anti-androgenic activity (cyproterone, drospirenone, dienogest) also **block the androgen receptor** and reduce 5α-reductase activity.

> **EVIDENCE**

Systematic reviews (Cochrane; Arowojolu 2012) show every COC studied reduces inflammatory and non-inflammatory lesions vs placebo, with **few consistent differences between progestogens**; A meta-analysis (Koo 2014, which you cited) found COCs **comparable to oral antibiotics by 6 months**; antibiotics work faster at 3 months

IN PRACTICE

benefit takes **3–6 months**. COCs help inflammatory and comedonal acne and acne with hyperandrogenism (PCOS). They are not monotherapy for severe/nodular acne.

OPTION (UK)	LICENSED FOR ACNE?	VTE RISK PER 10,000 WOMEN/YEAR (FSRH/EMA)	NOTES
Non-user	–	~2	Pregnancy itself ~29
Levonorgestrel / norethisterone / norgestimate COC (e.g. Microgynon 30, Rigevidon, Cilest)	No (contraceptive)	**~5–7**	Reasonable first COC. The progestogen is androgenic in theory, but acne still improves via the oestrogen/SHBG effect
Desogestrel / gestodene COC (e.g. Marvelon, Femodene)	No	~9–12	Less androgenic
Drospirenone/EE (Yasmin; 24/4 Eloine/Yaz-type)	**Not in the UK** (anti-androgenic progestogen; Yaz-type pills have an acne indication in the US)	~9–12	Also anti-mineralocorticoid – check potassium if on ACE-i/ARB/spironolactone
Co-cyprindiol (cyproterone 2 mg + EE **35 micrograms**, Dianette/Clairette)	**Yes** – moderate–severe acne related to androgen sensitivity and/or hirsutism, **after topical therapy and oral antibiotics have failed**	**~1.5–2× LNG** (EMA 2013)	MHRA 2013: use only for the licensed indication. It is also a contraceptive, so **do not combine with another hormonal contraceptive**. **Stop 3–4 cycles after the condition has resolved**; review regularly
Dienogest + estradiol valerate (Qlaira)	No	~8–11 (limited data)	–

UKMEC (FSRH)

before any COC, check: BMI (35 = UKMEC 3) and BP; **migraine with aura (UKMEC 4)**; smoking at age 35; personal/family VTE, thrombophilia, immobility, postpartum, breastfeeding; liver disease, breast cancer, SLE with antiphospholipid antibodies, multiple CV risk factors

Choice for this 19-year-old (if she wants hormonal contraception): Start a **second-generation (levonorgestrel) COC** alongside the topical regimen; If acne is still a problem after 6 months, consider a third-generation or drospirenone pill; If acne is moderate–severe after topical + antibiotic failure, consider **co-cyprindiol**, with explicit VTE counselling and a stop date; NICE NG198 also says to **consider a COC in addition to topicals** in women, and not to use co-cyprindiol first-line

COMMON QUESTION: BREAKTHROUGH BLEEDING (BTB)

BTB is common in the first **3 cycles**; Check adherence, interacting drugs (enzyme inducers, St John's wort), vomiting/diarrhoea and smoking. **Exclude STI (chlamydia) and pregnancy**, and check cervical screening status; If it persists beyond 3 months, FSRH suggests trying a pill with a different progestogen or a higher oestrogen dose

(30–35 micrograms); Tailored regimens (shortened or no pill-free interval) can increase early BTB; The observation that COCs cause less BTB than progestogen-only pills fits the FSRH data

SPIRONOLACTONE

Off-label anti-androgen for persistent lower-face acne in women, typically **50–100 mg once daily** (up to 200 mg/day under specialist guidance); Needs **effective contraception** (risk of feminising a male fetus); Check U&E/ potassium at baseline. Routine monitoring is often unnecessary in young healthy women — follow the local/ specialist protocol; Side effects: menstrual irregularity, breast tenderness. Not used in men (gynaecomastia). In the UK it is usually started on **dermatology advice**

Fig. 3 · Retinoid dermatitis (dryness/peeling) from a topical retinoid — start alternate nights, pea-sized amount, moisturiser; topical retinoids are avoided in pregnancy.

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Diagnosis and investigations

Mild–moderate

- Adapalene 0.1% + BPO 2.5%; tretinoin 0.025% + clindamycin 1%; or BPO 3–5% + clindamycin 1%.

Moderate–severe

- Adapalene 0.3% + BPO 2.5%; tretinoin + clindamycin; adapalene/BPO **+ oral lymecycline or doxycycline**; or azelaic acid 15–20% + oral tetracycline.

NG198 rules

- No antibiotic monotherapy; **no topical + oral antibiotic** together; oral only with retinoid or BPO; review at 12 weeks.

Before a COC

- **UKMEC**

  BMI (35 = UKMEC 3), BP, **migraine with aura (UKMEC 4)**, VTE history, thrombophilia, postpartum.

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
Tetracyclines	Avoid in pregnancy, breastfeeding, under-12s; doxycycline photosensitivity.
Minocycline	Not recommended: lupus-like syndrome, hepatitis, pigmentation.
Co-cyprindiol	Not first-line; highest VTE risk; stop plan (review at 3 months).
Drospirenone COC	Anti-androgenic; check potassium with ACE-i / ARB / spironolactone.
Spironolactone	Off-label, 50–100 mg daily for persistent lower-face acne in women; per formulary.

Management

1. Skin care non-comedogenic products, a gentle pH-neutral cleanser twice daily, no picking or squeezing, oil-free sunscreen. Diet: no strong evidence for elimination diets (NICE).
2. **12-week fixed-combination topical** (as above), applied thinly to the whole affected area, not just the spots. If irritated, start on alternate nights or wash off after 1 h, then build up to daily.
3. **Discuss contraception**, if wanted, choose a COC using UKMEC (above).
4. Review at 12 weeks Clear stop antibiotics and **maintain** with a topical retinoid ± BPO; Improving continue; Not improved switch to another first-line option; Moderate–severe acne that has failed two courses, including an oral antibiotic **refer for isotretinoin**
5. **Screen psychologically** (mood, self-harm) at every review — acne carries a high psychosocial burden.

IP scope

All NICE NG198 primary-care options (topicals, oral tetracyclines, and COCs within contraceptive competence) are within IP scope if you are competent. For **co-cyprindiol** and spironolactone in acne, follow the local formulary (often specialist advice). Isotretinoin: refer.

Fig. 4 · A pea-sized amount dotted on forehead, cheeks, nose and chin, then spread thinly over the whole face, avoiding eyes and lips.

AI-generated illustration

Medicines at a glance

Lymecycline (if oral added)

12-week topical

DOSE	408 mg once daily · Review at 12 weeks
CAUTIONS	With a topical retinoid or BPO; never with a topical antibiotic.

Treatment ladder

1 **Skin care**

Non-comedogenic products, gentle cleanser twice daily, no picking, oil-free sunscreen. No strong evidence for elimination diets (NICE).

2 **12-week topical**

Lymecycline (if oral added)

3 **COC option**

Consider a COC **in addition to topicals** (NICE); choose with UKMEC. Still a problem after 6 months: third-generation or drospirenone pill; co-cyprindiol for moderate–severe with explicit VTE counselling and stop date.

4 **Review at 12 weeks**

Clear stop antibiotic, **maintain** with retinoid ± BPO. Not improved switch option; failed two courses incl. oral antibiotic **refer for isotretinoin**.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Acne fulminans**: Systemic features: dermatologist within 24 hours.
- **Mood**: Screen mood and self-harm at every review.
- **BPO bleaches**: Towels, clothes, hair.

Referral

SAME DAY	**Acne fulminans** (fever, arthralgia, ulcerating acne): seen by dermatology **within 24 hours**.
SAME DAY	Calf swelling, chest pain or breathlessness on a COC: possible **VTE**.
ROUTINE	Scarring, nodulocystic, rapidly worsening acne or severe distress: refer for isotretinoin.

Red flags and when to refer

- Scarring, nodulocystic or rapidly worsening acne, or severe psychological distress: refer for isotretinoin assessment.
- Calf swelling, chest pain or breathlessness on a combined pill: possible **VTE**, same-day assessment.
- Co-cyprindiol: review at 3 months and stop once acne is controlled (MHRA).

Red flags / refer (NICE NG198)

- **Acne fulminans** **urgent referral, to be seen by a dermatologist within 24 hours** (the course "why?"): It is sudden, severe, ulcerating and haemorrhagic acne with **systemic features**: fever, malaise, arthralgia/bone pain (osteitis), raised inflammatory markers, leucocytosis; It is often triggered by **starting isotretinoin**, or by testosterone/anabolic steroids; It needs **oral corticosteroids** quickly (± pausing or reducing isotretinoin) to prevent severe scarring
- **REFER TO CONSULTANT DERMATOLOGY FOR**
  diagnostic uncertainty; **acne conglobata** or **nodulocystic acne**; mild–moderate acne not responding to two 12-week courses; moderate–severe acne not responding to a course that included an oral antibiotic; **scarring** or persistent PIH; **psychological distress** or a mental health disorder (also refer to mental health services for suicidal ideation); a suspected underlying endocrine disorder (PCOS with abnormal results GP/endocrinology)

Complications and prognosis

Complications

- VTE (higher with co-cyprindiol, drospirenone, desogestrel than levonorgestrel), breakthrough bleeding, mood change.

Prognosis

- Benefit takes 3–6 months.

Prescriber pearls (UK pharmacist IP)

- "Twelve weeks before judging" — show her a calendar.
- BPO **bleaches** towels, clothes and hair; retinoids **dry the skin and sensitise it to sun**.
- On doxycycline: photosensitivity; oesophagitis (take upright with water); keep indigestion remedies and iron 2–3 h apart.
- Women on a COC should know the **VTE warning symptoms** (calf pain/swelling, chest pain, breathlessness) and the **stroke symptoms**.

Self-check quiz

4 questions · optional

Q1 NICE moderate–severe threshold?

✓ 35 inflammatory lesions and/or 3 nodules.

Q2 Can you combine oral and topical antibiotics?

✓ No: oral antibiotics only with a topical retinoid or BPO.

Q3 UKMEC category for migraine with aura on a COC?

✓ UKMEC 4 (unacceptable risk).

Q4 When to refer for isotretinoin?

✓ Moderate–severe acne failing two courses including an oral antibiotic, or scarring / distress.

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Therapeutics Module 1 · Week 2

Hirsutism & the Ferriman–Gallwey score

Male-pattern terminal hair growth in women: scoring with the modified Ferriman–Gallwey scale, causes (mostly PCOS or idiopathic), the tests that matter, and treatment from hair removal and the pill to specialist anti-androgens.

ALSO CALLED excess hair · mFG score · PCOS · eflornithine · Vaniqa · androgen excess · +1

RELATED PAGES Acne vulgaris · Androgenetic alopecia (male & female) · Acanthosis nigricans · Acne prescribing & the combined pill · Intense pulsed light (IPL)

KEY FACTS

- **Hirsutism** is excess terminal hair in women in a **male pattern** (upper lip, chin, chest, abdomen, back).
- Commonest cause: **PCOS**; also idiopathic hirsutism, congenital adrenal hyperplasia, drugs; rarely androgen-secreting tumours.
- Red flags: **rapid onset**, virilisation (deep voice, clitoromegaly), very high testosterone: urgent endocrine referral.
- Check testosterone (± SHBG, LH/FSH, prolactin, 17-OH progesterone); weight loss; hair removal; COC; eflornithine for the face.
- Effects of hormonal treatment take 6–12 months.

Overview

Hirsutism is excess **terminal** hair in a woman, in an **androgen-dependent (male-pattern)** distribution: upper lip, chin, jawline/sideburns, chest, abdomen, back, thighs. It differs from **hypertrichosis**, which is excess hair in a non-androgen pattern (e.g. drug-induced by ciclosporin, minoxidil or phenytoin, or paraneoplastic lanugo).

Epidemiology

- **Who**
  - 5–10% of women of reproductive age; varies by ethnicity.

Aetiology: causes and risk factors

Myths to address

- **"SHAVING MAKES HAIR THICKER"**
  it does not.

Pathophysiology and pathology

Androgens convert fine **vellus hairs into thick terminal hairs** in sensitive follicles.

1. **Androgen excess or sensitivity**
   Higher androgens (PCOS) or more sensitive follicles (idiopathic).
2. **5α-reductase**
   Follicles convert testosterone to DHT.
3. **Vellus to terminal**
   DHT enlarges follicles, producing coarse pigmented hairs.

Fig. 1 · Androgens transform fine vellus hairs into coarse terminal hairs.
AI-generated illustration

4. **Treatment**
   COCs raise SHBG and lower androgens; laser targets hair melanin.

Fig. 2 · Laser hair removal targets the pigmented hair follicle.
AI-generated illustration

How it is caused

The cause is increased **androgen production** (ovary, adrenal) and/or increased **follicle sensitivity**: 5α-reductase in the pilosebaceous unit converts testosterone to DHT.

- **PCOS** (~70–80% of hirsutism); **idiopathic hirsutism** (normal androgens and ovulatory cycles, ~5–15%); idiopathic hyperandrogenaemia.
- Less common causes: **non-classic congenital adrenal hyperplasia** (21-hydroxylase deficiency — check 17-OH-progesterone); **Cushing's**, acromegaly, hyperprolactinaemia, thyroid disease; **androgen-secreting ovarian/adrenal tumours** (rapid onset, virilisation, testosterone often >5 nmol/L); drugs (anabolic steroids, testosterone, danazol, valproate); menopause
- Normal hair density varies widely with ethnicity and family. Mediterranean, Middle Eastern and South Asian women often have more terminal hair; East Asian women less.

Clinical features

The case: **dark terminal hair limited to the jawline and sideburns**, FG 2 in that area, **regular periods**, distress, **BMI 28**, **testosterone 1.2 nmol/L (<2)**, **androstenedione 6.5 nmol/L (<10)**. No virilisation, acne or AN is recorded.

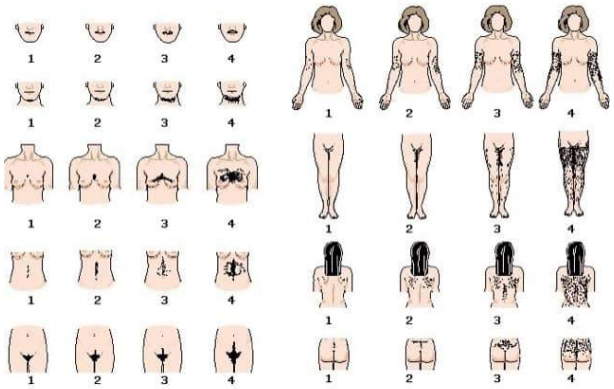


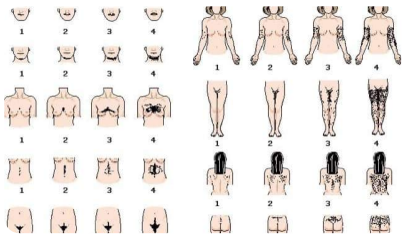
Fig. 3 · Modified Ferriman–Gallwey visual scale — 9 androgen-sensitive sites scored 0–4 (max 36); 8 usually defines hirsutism in white/black women (lower cut-offs in East Asian women). Case score of 2 at jaw/sideburns = localised/mild.

Open access image (DermNet) · **Source**

Types to recognise

mFG visual scale

9 androgen-sensitive sites. Site: Chart.



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Facial hair

Measure impact. Site: Chin, lip.

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Jawline hair

Common distressing site. Site: Jawline.

© DermNet · **Source**

Post-removal folliculitis

Shaving / waxing complication. Site: Face.

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Special groups

- **Hypertrichosis**
  - Generalised non-androgenic hair growth (drugs such as minoxidil, ciclosporin; anorexia): different investigations.

Diagnosis and investigations

SCORING

9 androgen-sensitive areas — upper lip, chin, chest, upper abdomen, lower abdomen, upper arms, thighs, upper back, lower back. Each scores **0 (no terminal hair) to 4 (frankly virile)**, for a **total of 0–36**.

CUT-OFFS

8 is the traditional threshold in white and black women; Lower cut-offs (**4–6**) are used in some populations, e.g. East Asian women. The 2018/2023 International PCOS guideline accepts **4–6** depending on ethnicity; Also recognise "**patient-important hirsutism**": localised hair that distresses the patient even with a low total, as here

LIMITATIONS

subjective, with inter-observer variation; underestimates hair after hair removal; no extra weighting for the face (the most distressing site); does not capture distress, so pair it with a QoL measure; Its main use is to **monitor response**

WORK-UP (UK — CKS HIRSUTISM / ENDOCRINE SOCIETY)

History: onset and pace, cycles, fertility, weight change, drugs (including supplements and anabolic steroids), family history; **Examination:** virilisation (clitoromegaly, deep voice, male-pattern balding, increased muscle bulk), acne, AN, Cushingoid signs, BP, BMI; **Bloods:**; **Total testosterone** (early follicular, morning) and **SHBG/free androgen index**; As indicated: **LH/FSH, prolactin, TSH, 17-OH-progesterone**; If PCOS is likely: HbA1c/OGTT and lipids; Pregnancy test; **Pelvic USS** only if PCOS criteria are uncertain. It is not needed when 2 criteria are already met, and is not used within 8 years of menarche

INTERPRETING THE CASE

Normal androgens, regular cycles and mild localised hair are most consistent with **idiopathic (peripheral) hirsutism** or a constitutional/ethnic hair pattern; PCOS is **not excluded**: regular cycles can still be anovulatory

> (check mid-luteal progesterone), and ~20–30% of women with PCOS have a normal total testosterone (check SHBG/FAI); Her BMI of 28 justifies an insulin-resistance work-up (HbA1c)

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
Idiopathic	Slow, peripubertal; regular cycles; normal androgens.
PCOS	Oligo-/anovulation, mild T, SHBG, AN, obesity, acne.
NC-CAH	17-OHP; family history.
Androgen-secreting tumour	**Rapid** onset, virilisation, T > 5 nmol/L, DHEAS.
Cushing's	Striae, myopathy, hypertension.

Management

Therapeutic strategy for the case: treat the patient's goals, layer treatments, and set expectations — hormonal effects take 6–12 months, and a hair cycle is about 6 months.

1. Lifestyle **weight reduction** (even 5% improves SHBG, androgens and insulin resistance) and exercise; address any PCOS/metabolic risk.
2. **Cosmetic/physical methods** (first-line; mostly self-funded in the UK): Shaving — it does not make hair thicker; Plucking, threading or waxing — risk of folliculitis and PIH; Depilatory creams (irritant) and bleaching; **Laser/IPL** — best for dark hair on light skin; **Nd:YAG** for darker skin; needs multiple sessions. NHS funding is very limited and ICB-specific, often only for severe cases or gender services; **Electrolysis** — permanent but slow; works on any hair colour
3. Topical eflornithine 11.5% cream (Vaniqa) An irreversible ornithine-decarboxylase inhibitor that slows facial hair growth. Apply twice daily; effect by 6–8 weeks; stop if no benefit at 4 months. The effect is lost about 8 weeks after stopping. Works best combined with laser; **2026 status:** NHS dm+d lists **Vaniqa as discontinued/not available** in the UK, and several ICBs (e.g. Pan Mersey, BSW) class it as **not recommended**. Check local availability before suggesting it
4. **Systemic hormonal therapy** (if distress persists despite 6 months of cosmetic measures): A **COC** is the first-line systemic option (ovarian androgen, SHBG); any low-VTE COC such as a levonorgestrel pill is a reasonable start; **Co-cyprindiol** is licensed for **severe hirsutism** but carries ~1.5–2× the VTE risk (MHRA 2013); stop 3–4 cycles after control; **Anti-androgens** (spironolactone 50–200 mg/day, higher-dose cyproterone acetate, finasteride) are **off-label and specialist-advised**, need **reliable contraception** (feminisation of a male fetus), and are usually added to a COC after 6 months of inadequate response; **Higher-dose cyproterone (10 mg/day)** carries a dose-related **meningioma** risk (MHRA 2020) — specialist only
5. **Psychological support** — hirsutism has a major impact on self-.

IP scope

Assessment, bloods, lifestyle and cosmetic advice, and **COC prescribing within contraceptive competence and UKMEC** are within IP scope. Anti-androgens, co-cyprindiol for hirsutism (formulary-dependent) and laser referral criteria follow local specialist pathways.

Fig. 8 · Laser or IPL hair removal works best on dark hair and light skin (Nd:YAG for darker skin) and needs several sessions; eye protection is essential. NHS funding is limited.

AI-generated illustration

Treatment ladder

1 **Lifestyle**

Weight loss (even 5% improves SHBG and androgens), exercise; PCOS / metabolic risk.

2 **Physical removal**

Shaving (doesn't thicken hair), plucking, threading, waxing (folliculitis, PIH), creams, bleaching. Laser / IPL (dark hair, light skin; Nd:YAG for darker skin), electrolysis. NHS funding limited.

3 **COC**

First-line systemic if distress persists after 6 months: ovarian androgen, SHBG; a low-VTE levonorgestrel pill is a reasonable start. Eflornithine cream discontinued in the UK: check availability.

4 **Anti-androgens**

Spironolactone 50–200 mg/day, cyproterone, finasteride: **off-label, specialist-advised**, with reliable contraception.

⚠ CAUTION

Prescribing limits and pitfalls

- **Teratogenic**: Anti-androgens can feminise a male fetus: effective contraception essential.

Referral

URGENT — Rapid onset over months, virilisation or very high testosterone: **urgent endocrine** referral.

ROUTINE — Suspected NC-CAH, Cushing's or other endocrine cause.

Red flags and when to refer

- Rapid onset over months, virilisation or a very high testosterone: **urgent** endocrine referral to exclude an androgen-secreting tumour.
- Anti-androgens (spironolactone, cyproterone, finasteride) are teratogenic to a male fetus: effective contraception is essential.

Red flags / refer

- **Rapid onset or progression**, **virilisation**, **testosterone >5 nmol/L** (or >2× ULN), raised DHEAS, or a pelvic/abdominal mass **urgent endocrinology/gynaecology** for a suspected androgen-secreting tumour (NICE NG12 if there is a mass).
- Features of **Cushing's**, acromegaly or hyperprolactinaemia; suspected NC-CAH; new hirsutism after the menopause.
- Distress not responding to primary-care measures endocrinology/dermatology per the local pathway.

Complications and prognosis

Complications

- Psychological distress; folliculitis and PIH from hair removal.

Prognosis

- Chronic; controlled with combined treatment.

Prescriber pearls (UK pharmacist IP)

"Hormonal treatment stops **new** hair coarsening; it doesn't remove existing hair" — so combine it with physical removal.

Shaving is safe and effective, and does not make regrowth thicker or darker.

Ask about **drugs and supplements**: testosterone gel transferred by contact with a partner, anabolic steroids, valproate; minoxidil causes hypertrichosis.

Self-check quiz

4 questions · optional

Q1 Commonest cause of hirsutism?

✓ PCOS (about 70–80%).

Q2 mFG scoring?

✓ 9 areas scored 0–4, total 0–36; 8 traditional cut-off.

Q3 Red flag for a tumour?

✓ Rapid onset with virilisation or testosterone > 5 nmol/L.

Q4 First-line systemic therapy?

✓ A combined oral contraceptive, after 6 months of cosmetic measures.

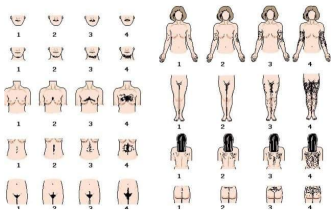
References

1. DermNet: Hirsutism — https://dermnetnz.org/topics/hirsutism
dermnetnz.org
2. CKS Hirsutism: Endocrine Society hirsutism guideline 2018 International evidence-based PCOS guideline 2023 — https://cks.nice.org.uk/topics/hirsutism/
cks.nice.org.uk

Photo gallery

Scoring and patterns

4



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Therapeutics Module 1 · Week 2

Biologics, JAK inhibitors & conventional systemics

The primary-care interface for patients on systemic treatment for psoriasis and eczema: what each drug targets, the NICE criteria, monitoring, vaccines, infections, interactions and when to call the specialist team.

ALSO CALLED methotrexate · ciclosporin · adalimumab · dupilumab · ustekinumab · JAK inhibitor · +3

RELATED PAGES Psoriasis · Eczema / dermatitis · Hidradenitis suppurativa · Vitamin D analogues, coal tar, dithranol & phototherapy · Urticaria & angioedema

KEY FACTS

- Conventional systemics: **methotrexate** (weekly), ciclosporin, acitretin, azathioprine, mycophenolate; biologics target specific cytokines.
- Psoriasis biologics: TNF-α (adalimumab), IL-17 (secukinumab, ixekizumab, bimekizumab), IL-23 (guselkumab, risankizumab); eczema: dupilumab, tralokinumab, JAK inhibitors.
- Methotrexate is **once weekly**: prescribing errors (daily dosing) are fatal; folic acid on another day; monitor FBC, LFT, U&E.
- Screen before biologics: TB (IGRA), hepatitis B/C, HIV; avoid live vaccines.
- Infection on immunosuppressants needs early review; record specialist drugs in the GP record.

Overview

CLASS	PSORIASIS (UK LICENSED, NICE-APPRAISED)	ATOPIC ECZEMA
TNF-α inhibitors	Adalimumab (biosimilars), etanercept, infliximab, certolizumab pegol	–
IL-12/23 (p40)	Ustekinumab (biosimilars)	–
IL-23 (p19)	Guselkumab, risankizumab, tildrakizumab	–
IL-17A / IL-17A+F / IL-17RA	Secukinumab, ixekizumab / **bimekizumab** / brodalumab	–
IL-4Rα (IL-4 + IL-13)	–	**Dupilumab** (NICE TA534; further TAs for adolescents/children)
IL-13	–	**Tralokinumab**, **lebrikizumab**
IL-31RA	–	Nemolizumab (prurigo nodularis/AD – check current NICE status)

CLASS	PSORIASIS (UK LICENSED, NICE-APPRAISED)	ATOPIC ECZEMA
Oral JAK/TYK2 inhibitors	Deucravacitinib (TYK2)	**Abrocitinib, upadacitinib, baricitinib**
Oral small molecules	Apremilast (PDE-4), dimethyl fumarate (Skilarence)	–
Conventional systemics	**Methotrexate**, ciclosporin, acitretin, NB-UVB/PUVA	**Ciclosporin** (licensed); methotrexate, azathioprine, mycophenolate (off-label)

Aetiology: causes and risk factors

Myths to address

- **"BIOLOGICS ARE JUST STRONGER STEROIDS"**
  they block one immune pathway, with different risks.

How it works (mechanism of action)

Systemic drugs and biologics **interrupt the immune circuits** that drive chronic skin inflammation.

1. **IL-23/Th17 axis**
   Dendritic cell IL-23 sustains Th17 cells that release IL-17, driving psoriasis.

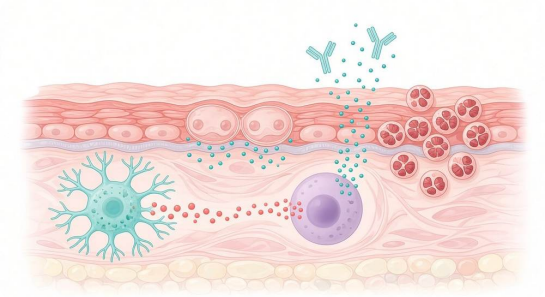

Fig. 1 · The IL-23/Th17 axis targeted by psoriasis biologics.
AI-generated illustration

2. **Th2 axis**
   IL-4 and IL-13 drive atopic eczema; dupilumab blocks the IL-4 receptor α.

3. **JAK–STAT**
   Many cytokine receptors signal through JAKs; oral JAK inhibitors block several at once.

4. **Antimetabolites**
   Methotrexate inhibits folate pathways in rapidly dividing immune cells.

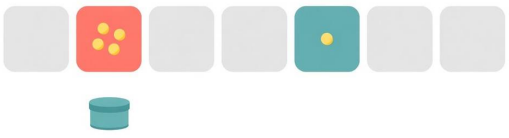

Fig. 2 · Methotrexate is taken once a week.
AI-generated illustration

Clinical features

How it works (target disease)

- **Psoriasis** is driven by the **IL-23/Th17 axis**: Dendritic cells release IL-23, which drives Th17 cells to make **IL-17A/F** and IL-22. These cause keratinocyte hyperproliferation and recruit neutrophils; TNF-α amplifies the loop; Blocking IL-23 (p19) or IL-17 gives PASI 90 responses in most patients; IL-17 blockade works fastest
- **Atopic eczema** is **Th2-dominant**: **IL-4 and IL-13** impair barrier proteins (filaggrin, loricrin) and drive IgE and itch; **IL-31** is a key itch cytokine; Dupilumab blocks the shared **IL-4Rα**, so it blocks both IL-4 and IL-13 signalling. Tralokinumab and lebrikizumab neutralise IL-13; JAK inhibitors block intracellular signalling for many cytokines (IL-4, IL-13, IL-31, IFN-γ), giving rapid itch relief

Clinical uses & where it fits (NICE)

> **PSORIASIS BIOLOGICS**
> Typically for **severe disease — PASI 10 and DLQI >10** — that has not responded to, is intolerant of, or is contraindicated for **conventional systemic therapy** (methotrexate, ciclosporin, acitretin) and phototherapy. Some TAs use lower DLQI thresholds; check each one; Choice depends on comorbidity:; **psoriatic arthritis** TNF, IL-17 or IL-23 options; **IBD** avoid IL-17 (may trigger or worsen IBD); TNF or IL-23 preferred; **demyelination or heart failure** avoid TNF; **pregnancy** certolizumab; also consider TB and hepatitis risk

> **ECZEMA BIOLOGICS/JAKI**
> moderate–severe AE (e.g. EASI 16, DLQI 10) after at least one conventional systemic has failed or is unsuitable.

> **WHERE PRIMARY CARE FITS**
> These are **hospital-initiated and hospital/homecare-supplied ("red") drugs**. GPs and IPs **do not prescribe** them; They must still be **recorded on the GP record as "hospital-prescribed"**, so that interactions, vaccines and infection risk are visible

How to choose / assess (what you do in primary care)

> **SPECIALIST SCREEN BEFORE STARTING**
> TB (IGRA ± CXR), hepatitis B/C, HIV, VZV serology, pregnancy, malignancy and demyelination history, vaccines up to date (give **live vaccines 4 weeks before** starting).

> **DURING TREATMENT**
> infections, vaccines, surgery planning, pregnancy, new symptoms.

Key comparisons — adverse effects you will meet

AGENT/CLASS	AE TO RECOGNISE IN PRIMARY CARE
All biologics	Injection-site reactions; infections (URTI); **live vaccines contraindicated** during therapy and for a period after (check SmPC); hold for major surgery per specialist
TNF-i	Serious infection, **TB reactivation**, paradoxical (palmoplantar pustular) psoriasis, demyelination, worsening heart failure, lupus-like syndrome
IL-17	**Mucocutaneous candidiasis**, **new or worsening IBD**, eczematous reactions
IL-23 / IL-12/23	Generally well tolerated; infections; headache

AGENT/CLASS	AE TO RECOGNISE IN PRIMARY CARE
Dupilumab	**Conjunctivitis/blepharitis/keratitis** (ocular surface disease — photo G01; ~10–25%), facial/neck erythema ("dupilumab facial redness"), injection-site reactions, eosinophilia, new psoriasiform rash, arthralgia, oral HSV
Tralokinumab/ lebrikizumab	Conjunctivitis (less often), URTI
JAK inhibitors	**MHRA (DSU Apr 2023):** increased risk of **major adverse CV events, malignancy (incl. NMSC and lymphoma), VTE, serious infections and mortality**. Use in people **65, current or past long-time smokers, or those with CV/ malignancy/VTE risk factors only if there is no suitable alternative**. Also acne, herpes zoster, raised lipids/CK, cytopenias
Apremilast	Diarrhoea, nausea, headache, weight loss, **low mood/suicidality**

Fig. 3 · Dupilumab-associated ocular surface disease (conjunctivitis/blepharitis) 8 weeks after starting — the commonest dupilumab AE primary care sees; lubricants, refer to ophthalmology if painful/visual change, tell the dermatology team.

© DermNet · **Source**

Adverse effects to recognise

Dupilumab eye

Conjunctivitis, blepharitis, keratitis (~10–25%). Site: Eyes.

© DermNet · **Source**

Severe plaque psoriasis

Biologic eligibility: PASI 10 and DLQI > 10. Site: Trunk, limbs.

© Raimo Suhonen · **Source**

Diagnosis and investigations

Psoriasis (NICE)

- **PASI 10 and DLQI > 10** after conventional systemic and phototherapy failed or unsuitable.

Eczema (NICE)

- Moderate–severe (e.g. EASI 16, DLQI 10) after 1 conventional systemic.

Pre-start screen

- TB (IGRA ± CXR), hepatitis B/C, HIV, VZV, pregnancy; **live vaccines 4 weeks before**.

Differential diagnoses

CONDITION	DISTINGUISHING FEATURES
TNF inhibitors	TB reactivation, serious infection, paradoxical PPP, demyelination, heart failure.
IL-17 inhibitors	**Candidiasis**, new or worse **IBD**: avoid in IBD.
Dupilumab	Ocular surface disease, facial redness, eosinophilia, oral HSV.
JAK inhibitors	MHRA 2023: **MACE, malignancy, VTE, serious infection**; caution 65, smokers, CV risk.
Ciclosporin	**BP and creatinine**: reduce or stop if creatinine rises > 25–30%.

Management

Patients on biologics/JAKi — primary-care checklist

> **Record** the drug (hospital-only flag), the dose interval and the homecare provider; check interactions with every new prescription.

> **INFECTION**
> Lower threshold for review and antibiotics; **Hold the next dose during a significant infection** and contact the specialist team; Fever or serious infection same-day assessment (presentations may be atypical)

> **VACCINES**
> Inactivated vaccines (influenza, COVID-19, pneumococcal) are encouraged; **No live vaccines** (e.g. MMR, yellow fever, live intranasal flu in children); For shingles, use the **non-live Shingrix**, which immunosuppressed adults are eligible for under the NHS programme, not the live Zostavax

> **PREGNANCY PLANNING**
> specialist advice. Many biologics can be continued; JAKi are contraindicated.

> **Skin-cancer checks** (JAKi, TNF-i, previous PUVA or ciclosporin).

Methotrexate safety (IP level — the conventional systemic you are most likely to co-manage)

> **Once WEEKLY.** MHRA DSU (2020) followed fatal overdoses from inadvertent **daily** dosing: The prescriber must **record the day of the week** on the prescription and in the record; The patient gets clear counselling and the **patient card/booklet**; Pharmacies dispense **2.5 mg tablets only** (avoid 10 mg tablets in the same patient) and check understanding at every supply; Never write "as directed"

Folic acid 5 mg once weekly on a different day (some regimens use up to 6 days a week, never on the methotrexate day). It reduces GI, mucosal and hepatic toxicity.

MONITORING (BSR/BAD SHARED CARE — FOLLOW THE LOCAL PROTOCOL)

Baseline: FBC, U&E/creatinine, LFTs (ALT/albumin), ± hepatitis B/C, HIV, VZV, CXR. In psoriasis, add PIIINP or transient elastography/FIB-4 (liver fibrosis risk with metabolic syndrome or alcohol); Then **every 2 weeks until on a stable dose for 6 weeks, then monthly for 3 months, then every 3 months**; Test more often after a dose change, in renal impairment and in older people

Stop and seek advice if (check local thresholds): WCC <3.5 × 10^9/L, neutrophils <1.6, platelets <140; ALT/AST >100 IU/L (or >2× ULN), or an unexplained fall in albumin; falling eGFR; **new cough or breathlessness (pneumonitis)**; mouth ulcers, sore throat, fever or bruising (marrow suppression); rash

KEY INTERACTIONS

Trimethoprim and co-trimoxazole: contraindicated (pancytopenia); other antifolates; **NSAIDs** reduce renal clearance. Occasional low-dose use is often accepted in RA; psoriasis protocols may avoid them — follow the specialist; **PPIs** (with high-dose MTX) and penicillins reduce clearance; Live vaccines; Alcohol (keep within limits, or avoid in psoriasis — hepatotoxicity); acitretin (hepatotoxicity); leflunomide

PREGNANCY/FERTILITY (UK SMPC/BNF)

contraindicated in pregnancy: **Women:** effective contraception during treatment and for **at least 6 months** after the last dose; **Men:** contraception during treatment and for **at least 3 months** after (some SmPCs say 6 months — check the product); no semen donation

DOSE IN SKIN DISEASE

typically 7.5–25 mg once weekly, oral or subcutaneous; sometimes a 2.5–5 mg test dose first. Specialist-initiated.

Ciclosporin (eczema/psoriasis; specialist-initiated, short courses)

- Monitor BP and creatinine; stop or reduce if creatinine rises >25–30% from baseline.
- Many CYP3A4 interactions (clarithromycin, azoles, grapefruit, St John's wort).
- **Avoid phototherapy** and sun (skin cancer).
- Gingival hyperplasia, hypertrichosis, tremor; hyperkalaemia (avoid potassium-sparing drugs).

Oral corticosteroids — course Case 5 comment

PSORIASIS

avoid systemic steroids. Withdrawal can trigger **generalised pustular or erythrodermic psoriasis** (photo A06 shows localised pustular psoriasis; see the the psoriasis notes).

SEVERE ECZEMA

A **short course** (e.g. prednisolone ~0.5 mg/kg, up to 30–40 mg, for 1–2 weeks, tapered as needed — specialist/GP judgement) can "buy time" while phototherapy or systemics are arranged; Rebound is common; avoid repeated courses; If used for 3 weeks or repeatedly, consider HPA suppression, bone health and the **NHS Steroid Emergency Card** (NatPSA 2020)

IP scope

No initiation of biologics or JAKi (hospital-only). Methotrexate and ciclosporin only under **shared care**, if competent and locally agreed: continuing supply, monitoring and acting on results. Every IP should flag hospital-only drugs on the record and manage interactions.

Fig. 6 · Methotrexate is taken on one chosen day each week; folic acid is taken on a different day, never the methotrexate day. Write the day on the prescription.
AI-generated illustration

Treatment ladder

1 **Record**

Flag the hospital-only drug, dose interval and homecare provider. Check interactions with **every** new prescription.

2 **Infection**

Lower threshold to review and treat; **hold the next dose** in significant infection and contact the team. Fever or serious infection: same-day assessment.

3 **Vaccines**

Inactivated (flu, COVID-19, pneumococcal) encouraged; **no live vaccines**. Shingles: non-live **Shingrix**, not Zostavax.

4 **Methotrexate (shared care)**

Once WEEKLY; record the **day of the week** (MHRA 2020); patient card.
Folic acid 5 mg once weekly on a different day.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Trimethoprim / co-trimoxazole**: Never with methotrexate.
- **Pregnancy**: Specialist advice; JAK inhibitors contraindicated.
- **Skin cancer**: Checks on JAKi, TNF-i, previous PUVA or ciclosporin.

Referral

SAME DAY	Fever, sore throat, **mouth ulcers or bruising** on methotrexate: same-day FBC; withhold until reviewed.
SAME DAY	Serious infection or **TB symptoms** on a biologic or JAKi: urgent assessment, contact team.
URGENT	Cough or breathlessness on MTX (pneumonitis); chest pain or VTE symptoms on JAKi.
URGENT	New IBD symptoms on IL-17 inhibitor; eye pain or visual loss on dupilumab.

Red flags and when to refer

- Fever, sore throat, mouth ulcers or bruising on methotrexate: same-day FBC and stop until reviewed.
- Never co-prescribe **trimethoprim or co-trimoxazole** with methotrexate.
- Signs of serious infection or TB on a biologic or JAK inhibitor: urgent assessment and contact the specialist team.

Red flags / refer

On a biologic, JAKi, MTX or ciclosporin:

- fever or sepsis; TB symptoms
- new neurological symptoms (TNF-i — demyelination)
- jaundice; unexplained bruising or bleeding
- persistent cough or breathlessness (MTX pneumonitis; infections on JAKi)
- new IBD symptoms on an IL-17 inhibitor
- eye pain or visual loss on dupilumab
- chest pain or VTE symptoms on a JAKi
- pregnancy.

Complications and prognosis

Complications

- Infection (including TB reactivation), cytopenias, liver toxicity, kidney toxicity (ciclosporin), conjunctivitis (dupilumab), thrombosis risk (JAK inhibitors).

Prognosis

- Most patients achieve major improvement; treatment is long-term.

Prescriber pearls (UK pharmacist IP)

- "A red eye on dupilumab is ocular surface disease until proven otherwise, not just viral conjunctivitis — tell the dermatology team."
- "IL-17 inhibitor + diarrhoea or bloody stool = IBD until proven otherwise."
- "Methotrexate: one day a week, write the day down, no trimethoprim."

Self-check quiz

4 questions · optional

Q1 Which antibiotic must never be co-prescribed with methotrexate?

✓ Trimethoprim (and co-trimoxazole).

Q2 Shingles vaccine for a patient on a JAK inhibitor?

✓ The non-live recombinant vaccine (Shingrix), not live Zostavax.

Q3 NICE psoriasis biologic threshold?

✓ PASI 10 and DLQI > 10 after conventional systemic failure or unsuitability.

Q4 Bloody diarrhoea on secukinumab?

✓ Suspect new or worsening IBD: assess and contact the team.

References

1. DermNet: Dupilumab — https://dermnetnz.org/topics/dupilumab
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gov.uk

Photo gallery

Biologic adverse effects and psoriasis

2

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CHAPTER 19

Drug eruptions

When a medicine causes the rash: morbilliform eruptions, DRESS and AGEP, ACE-inhibitor angioedema, erythema nodosum, pigmentation, fixed drug eruption and how to report.

1. Approach to a suspected drug eruption
2. Morbilliform drug eruption & drug urticaria
3. DRESS (drug reaction with eosinophilia and systemic symptoms)
4. ACE-inhibitor (bradykinin) angioedema
5. AGEP (acute generalised exanthematous pustulosis)
6. Erythema nodosum (incl. drug-induced)
7. Drug-induced pigmentation (amiodarone, minocycline)
8. Fixed drug eruption

Drug eruptions Module 1 · Week 3

Approach to a suspected drug eruption

How to work out whether a rash is drug-induced: the drug timeline, morphology, severity markers (SCARs), which drug to stop, documenting allergy properly and reporting on the Yellow Card scheme.

ALSO CALLED drug rash · adverse drug reaction · drug allergy · Yellow Card · Naranjo · SCAR · +1

RELATED PAGES Morbilliform drug eruption & drug urticaria · SJS/TEN and drug-induced blistering · DRESS (drug reaction with eosinophilia and systemic symptoms) · AGEP (acute generalised exanthematous pustulosis) · Fixed drug eruption

KEY FACTS

- Drug reactions are common: first **classify the morphology** (maculopapular, urticarial, blistering, pustular, fixed, photosensitive, lichenoid).
- Build a **drug timeline**: everything started in the last 8 weeks, including OTC and herbal products.
- Latency clues: urticaria/anaphylaxis minutes–hours; AGEP hours–2 days; MDE 4–14 days; SJS/TEN 4–28 days; DRESS 2–8 weeks.
- Red flags: **mucosal involvement, skin pain, blisters, facial oedema, fever, organ dysfunction**: same-day referral.
- Stop the likely culprit, document the reaction precisely, Yellow Card serious reactions, and refer for allergy testing when it matters.

Overview

Cutaneous ADRs affect ~2–3% of hospital inpatients. Most (~90%) are benign **exanthems, urticaria or FDE**. About **2% are SCARs** (SJS/TEN, DRESS, AGEP), which are potentially fatal. Antibiotics (aminopenicillins, sulfonamides), anticonvulsants (carbamazepine, phenytoin, lamotrigine), allopurinol, NSAIDs (oxicams) and contrast media dominate the culprit lists.

Aetiology: causes and risk factors

Myths to address

- **"ANY RASH ON AMOXICILLIN = PENICILLIN ALLERGY"**
  many are viral or non-allergic; mislabelled allergy harms future care.

Pathophysiology and pathology

Drug reactions follow **different immune pathways**, which explain their timing and appearance.

1. **Type I (IgE)**
   Mast cell degranulation within minutes: urticaria, angioedema, anaphylaxis.

2. **Type IV T-cell reactions**
   Delayed days to weeks: maculopapular (IVb/c), SJS/TEN (IVc), AGEP (IVd), DRESS (IVb).

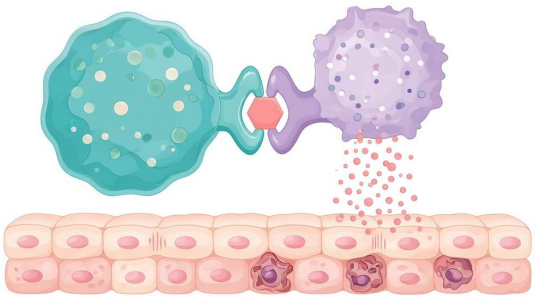

Fig. 1 · Delayed drug reaction: a drug held by an antigen-presenting cell activates a T cell, whose cytokines damage keratinocytes.
AI-generated illustration

3. **p-i concept**
   Some drugs bind HLA or T-cell receptors directly; HLA alleles predict risk (HLA-B*58:01, B*15:02, B*57:01).

4. **Non-immune**
   Phototoxicity, pigmentation and pharmacological effects (NSAID-exacerbated urticaria).

How it is caused

> **IMMUNE-MEDIATED (GELL & COOMBS, REVISED)**
> **Type I** (IgE): urticaria, angioedema, anaphylaxis; minutes to hours; Type II: drug-induced immune thrombocytopenia/pemphigus; Type III (immune complex): serum-sickness-like reaction, vasculitis (the vasculitis notes); **Type IV** (T-cell, delayed), subdivided:; IVa (Th1) — eczematous; **IVb** (Th2/eosinophils) — **DRESS**, maculopapular; **IVc** (cytotoxic CD8 T cells, granulysin) — **SJS/TEN, FDE**; **IVd** (neutrophils, IL-8) — **AGEP**

> **NON-IMMUNE**
> pharmacological (bradykinin — ACE-i angioedema; COX-1 inhibition — NSAID urticaria/angioedema); **phototoxicity**; direct mast-cell activation (opioids, vancomycin "red man" — MRGPRX2); deposition (amiodarone, minocycline pigmentation); overdose/cumulative toxicity

> **HOST FACTORS**
> HLA alleles (see Stevens–Johnson syndrome / toxic epidermal necrolysis), viral infection (EBV + amoxicillin; HIV sulfonamide reactions; HHV-6 reactivation in DRESS), older age, renal impairment, autoimmune disease.

Clinical features

Clinical features & sites

Pattern **likely culprit**, as asked in Case 2 for the 76-year-old woman:

PATTERN	TYPICAL TIMING AFTER STARTING THE DRUG	USUAL SUSPECTS
Urticarial / angioedema	Minutes–hours (IgE); days (serum-sickness-like)	Penicillins/cephalosporins, NSAIDs/aspirin, opioids, contrast; ACE-i (angioedema, weeks–years)
Morbilliform (exanthem)	**4–14 days** (sooner on re-exposure)	Aminopenicillins, cephalosporins, sulfonamides (co-trimoxazole), anticonvulsants, allopurinol
Photosensitive (face, V of neck, backs of hands, sparing behind ears/under chin)	Days–weeks of sun exposure	**Thiazides**, **tetracyclines (doxycycline)**, **amiodarone**, quinolones, NSAIDs (piroxicam), voriconazole, phenothiazines, PPIs/terbinafine (SCLE)
Blistering	Hours–days (FDE), 4–28 days (SJS/TEN), weeks–months (pseudoporphyria, drug-induced BP)	SJS/TEN: allopurinol, carbamazepine, lamotrigine, phenytoin, sulfonamides, oxicam NSAIDs, nevirapine. FDE: NSAIDs, co-trimoxazole, tetracyclines, paracetamol. **DPP-4 inhibitors (gliptins) bullous pemphigoid**. Naproxen pseudoporphyria. Furosemide/ ACE-i/PD-1 inhibitors BP
Pustular (AGEP)	**<48 h** (often hours)	Aminopenicillins, macrolides, pristinamycin, hydroxychloroquine, diltiazem, terbinafine
Fever + rash + organ involvement (DRESS)	**2–8 weeks**	Allopurinol, carbamazepine, phenytoin, lamotrigine, sulfonamides, dapsone, minocycline, vancomycin
Lichenoid	Months	Thiazides, β-blockers, ACE-i, antimalarials, gold, PD-1 inhibitors
Vasculitic purpura	1–3 weeks	Antibiotics, NSAIDs, allopurinol (see the vasculitis notes)

Fig. 2 · Morbilliform (exanthematous) drug eruption — the commonest pattern; symmetrical truncal macules/papules 4–14 days after a new drug (aminopenicillins, sulfonamides, anticonvulsants, allopurinol). Check for SCAR red flags.

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Types to recognise

Urticarial / angioedema

Minutes–hours (IgE); days (serum-sickness-like) Site: Penicillins/cephalosporins, NSAIDs/aspirin, opioids, contrast; ACE-i (angioedema, weeks–years).

Morbilliform (exanthem)

4–14 days (sooner on re-exposure) Site: Aminopenicillins, cephalosporins, sulfonamides (co-trimoxazole), anticonvulsants, allopurinol.

Photosensitive (face, V of neck, backs of hands, sparing behind ears/under chin)

Days–weeks of sun exposure Site: **Thiazides**, **tetracyclines (doxycycline)**, **amiodarone**, quinolones, NSAIDs (piroxicam), voriconazole, phenothiazines, PPIs/terbinafine (SCLE).

Blistering

Hours–days (FDE), 4–28 days (SJS/TEN), weeks–months (pseudoporphyria, drug-induced BP) Site: SJS/TEN: allopurinol, carbamazepine, lamotrigine, phenytoin, sulfonamides, oxicam NSAIDs, nevirapine. FDE: NSAIDs, co-trimoxazole, tetracyclines, paracetamol. **DPP-4 inhibitors (gliptins)** **bullous pemphigoid**, Naproxen, pseudoporphyria. Furosemide/ACE-i/PD-1 inhibitors BP.

Pustular (AGEP)

<48 h (often hours) Site: Aminopenicillins, macrolides, pristinamycin, hydroxychloroquine, diltiazem, terbinafine.

Fever + rash + organ involvement (DRESS)

2–8 weeks Site: Allopurinol, carbamazepine, phenytoin, lamotrigine, sulfonamides, dapsone, minocycline, vancomycin.

Lichenoid

Months Site: Thiazides, β-blockers, ACE-i, antimalarials, gold, PD-1 inhibitors.

Vasculitic purpura

1–3 weeks Site: Antibiotics, NSAIDs, allopurinol (see the vasculitis notes).

Diagnosis and investigations

1. **Drug chart/timeline**. List every drug, with start/stop/dose-change dates, for the previous **8 weeks** (up to 3 months for DRESS). Include **OTC, herbal, supplements, recreational drugs, contrast, vaccines and depot/long-acting drugs**. Plot the rash onset against each one. A drug started >8 weeks before the rash (and taken without problem) is less likely; one started after onset can be excluded.
2. Previous exposure a previous reaction to the same or a cross-reacting drug; re-exposure shortens the latency.
3. Morphology and extent pattern (above), **mucosal involvement**, skin pain, Nikolsky sign, blisters/detachment (% BSA), pustules, facial oedema, lymphadenopathy, purpura.
4. Systemic features fever, lymphadenopathy, organ dysfunction.
5. Investigations **FBC with differential** (eosinophilia DRESS; lymphopenia SJS/TEN; atypical lymphocytes; neutrophilia AGEP); U&E and LFTs (DRESS hepatitis/nephritis); CRP, glucose and bicarbonate (SCORTEN); Urinalysis (vasculitis); CK; TFTs (DRESS late effect); Viral serology/PCR (EBV, HHV-6, Mycoplasma in EM/SJS-like mucositis); **Skin biopsy** (H&E ± direct immunofluorescence — confirms SJS/TEN, AGEP, vasculitis; excludes AIBD/SSSS); **Mast-cell tryptase** within 1–4 h of suspected anaphylaxis; Later, by the **allergy service**: skin prick/intradermal tests, patch tests (FDE lesional patch, DRESS/AGEP delayed patch), lymphocyte transformation tests, and oral **drug provocation** for low-risk penicillin allergy. **Never rechallenge after SJS/TEN or DRESS.**

6. Causality tools **Naranjo** — a general ADR probability scale (10 questions; 9 definite, 5–8 probable, 1–4 possible, 0 doubtful); **ALDEN** — specific to SJS/TEN (see Stevens–Johnson syndrome / toxic epidermal necrolysis); **RegiSCAR** — DRESS scoring

Differential diagnoses

Viral exanthem (EBV, parvovirus, measles, COVID-19), Mycoplasma-associated mucositis (RIME), graft-versus-host disease, staphylococcal scalded skin syndrome (SSSS — superficial split, children/renal failure), autoimmune bullous disease, Kawasaki disease, scarlet fever, toxic shock syndrome, acute cutaneous lupus, pustular psoriasis (vs AGEP), cellulitis (vs FDE/EN).

Management

- **Stop the probable culprit** (and any non-essential drugs). Choose a structurally unrelated alternative (e.g. levetiracetam rather than another aromatic anticonvulsant).
- **BENIGN EXANTHEM**
  emollient, moderate–potent topical steroid, sedating or non-sedating antihistamine for itch. The rash may worsen for 1–3 days after stopping, then fades with desquamation over 1–2 weeks.
- **Record the reaction** (NICE CG183 Drug allergy): drug name, reaction description, date, severity, and whether it is an allergy or an intolerance. **Tell the patient in writing**, and make sure the record is shared (GP record, hospital discharge letter, community pharmacy PMR).
- **Report via the MHRA Yellow Card scheme** (https://yellowcard.mhra.gov.uk): all suspected serious reactions; any reaction to ▼ black triangle drugs; and any reaction in children. SJS/TEN, DRESS and AGEP should always be reported.
- **PENICILLIN ALLERGY DE-LABELLING**
  ~90% of labels are inaccurate. Refer suitable patients to allergy or local de-labelling pathways (BSACI 2022). A "rash on amoxicillin during glandular fever" is not true allergy.

IP scope

IPs can stop and substitute drugs, treat benign exanthems/urticaria symptomatically, document allergies and submit Yellow Cards. SCARs need emergency admission (dermatology/burns/ITU). Allergy testing and drug provocation are specialist.

Worked cases

CASE 1

memorable drug reaction

There is no single answer. Strong answers covered: **timing** relative to a new drug, **morphology** (pattern recognition, see Approach to a suspected drug eruption), **resolution on withdrawal**, recurrence on re-exposure, **skin of colour** (erythema looks violaceous/grey, and PIH dominates), and patient education (name the drug, avoid it, carry an alert, read OTC labels).

CASE 2

76-year-old woman on a medical ward, multiple drugs

HISTORY

a **dated drug chart** for 8 weeks: new drugs, dose changes, OTC/herbal, contrast, IV antibiotics, and drugs given in other settings (ED, theatre); previous drug reactions and tolerance; onset and evolution of the rash; itch vs pain, fever, mucosal/eye symptoms; viral illness; renal/hepatic function (drug accumulation)

EXAMINATION

morphology and distribution (urticarial / morbilliform / photo-distributed / blistering / pustular / purpuric), mucosae, % BSA, Nikolsky sign, facial oedema, lymph nodes, temperature, organ signs.

INVESTIGATIONS

FBC + differential (eosinophils, lymphocytes, neutrophils), U&E, LFT, CRP, urinalysis, ± tryptase, viral PCR, **skin biopsy (H&E + DIF)**. Score causality with **Naranjo** (general) or **ALDEN** (SJS/TEN). Later testing by the **allergy service** (patch tests, specific IgE, provocation).

LIKELY CANDIDATES (TYPICAL INPATIENT DRUGS)

Urticarial: β-lactam antibiotics, NSAIDs/aspirin, opioids (morphine, codeine — pseudo-allergic), contrast media, vancomycin, ACE-i (angioedema); **Photosensitive:** thiazides/loop diuretics (pseudoporphyria — furosemide), amiodarone, doxycycline, quinolones (ciprofloxacin), NSAIDs, phenothiazines, PPIs (SCLE); **Blistering:** in an older patient think **drug-induced bullous pemphigoid** (gliptins, loop diuretics, spironolactone, PD-1 inhibitors), **SJS/TEN** (allopurinol, anticonvulsants, co-trimoxazole, oxicams), **FDE/GBFDE**, pseudoporphyria (furosemide, naproxen) and linear IgA disease (**vancomycin**)

CASE 2

EM vs SJS vs TEN

See the table in and the classification in Stevens–Johnson syndrome / toxic epidermal necrolysis:

EM

HSV-triggered, typical targets, acral, mild mucosa, good prognosis; treat HSV (aciclovir prophylaxis if recurrent).

SJS (<10%) / OVERLAP / TEN (>30%)

drug-induced CD8/granulysin necrosis, atypical targets, severe mucositis. Stop the drug, admit, supportive care, ± ciclosporin/IVIg.

Case 3

- **Case 1: ACE-inhibitor (ramipril)-induced, bradykinin-mediated angioedema**, possibly triggered by minor surgery. No useful response to antihistamine or steroid. **Investigations:** clinical; C4/C1-INH if atypical, tryptase if anaphylaxis is possible. **Management:** airway assessment, stop ramipril permanently (class allergy), consider an

ARB with caution or a CCB, avoid gliptins/sacubitril, Yellow Card, explain that recurrences can happen for weeks (see ACE-inhibitor (bradykinin-mediated) angioedema).

- **Case 2: erythema nodosum**, most likely **COC-induced**. Exclude infection (strep), sarcoid/TB (CXR), IBD and pregnancy, and **exclude DVT/thrombophlebitis** given the unilateral leg + COC + tranexamic acid; cellulitis is a valid differential. **Treat:** stop the COC (alternative contraception/cycle control), rest, elevation, NSAID analgesia; review; biopsy if atypical (see Drug-induced erythema nodosum).
- **Case 3: Question 1** — main causes of photosensitivity (see Drug-induced pigmentation — amiodarone); **Question 2** — the visible abnormality is **blue-grey (slate) pigmentation of sun-exposed facial skin** ± phototoxic erythema; **Question 3** — **amiodarone-induced phototoxicity and pigmentation** (UVA-absorbing drug ROS; lipofuscin-like drug–lipid deposits in dermal macrophages; long half-life); **Question 4** — cardiology liaison (dose reduction/ alternative), strict UVA photoprotection, monitoring (TFT/LFT/eyes/lung), counselling that the pigment fades slowly, Yellow Card (see Drug-induced pigmentation — amiodarone)

Case 4

- **Case 1: drug-induced photosensitivity (phototoxic) from bendroflumethiazide**. Investigate clinically ± ANA/Ro (SCLE — omeprazole), porphyrins if blistering; phototesting/photopatch rarely. **Treat:** stop the thiazide amlodipine; topical steroid, emollient, antihistamine; strict photoprotection and education; review diabetes (HbA1c 78) and the PPI indication (see Drug-induced photosensitivity — thiazide case).
- **Case 2: fixed drug eruption from diclofenac** (taken during two consecutive periods), misdiagnosed as a cold sore. Stop/avoid diclofenac (± related NSAIDs), alternative dysmenorrhoea treatment, topical steroid, document and report (see Fixed drug eruption).

Practice case: 56-year-old man, 60% detachment

- **QA = TEN** (>30% detachment + severe mucositis + systemic illness). course follow-up "does TEN involve >10%?" — >10% is the SCORTEN item and the lower bound for overlap. **TEN is >30%**; SJS <10%; overlap 10–30%.
- **QB = 5–28 days** (typical latency from starting the culprit; the ALDEN "suggestive" window). BAD quotes 4–28 days.
- **QC = carbamazepine.** Worked **ALDEN**, with index day = fever onset 5 days ago:

DRUG	DELAY	PRESENT	PRE/ RECHALLENGE	DECHALLENGE	NOTORIETY	OTHER CAUSE	SCORE
Carbamazepine (started 20 d ago 15 d before index)	**+3**	0	0	0	**+3**	0	**6 — very probable**
Ibuprofen (4 d ago, after index)	–3	0	0 (or –2 if previously tolerated)	0	+1	–1	**–3 — very unlikely**
Flucloxacillin (1 d ago, after index)	–3	0	0	0	+1	–1	**–3 — very unlikely**

DRUG	DELAY	PRESENT	PRE/ RECHALLENGE	DECHALLENGE	NOTORIETY	OTHER CAUSE	SCORE
Lamotrigine (stopped 20 d ago; t½ ~1–1.5 d)	–1	–3	0	0	+3	–1	**–2 — very unlikely** (unless still present during a cross-taper — check the regimen)
Amlodipine (long-term)	–1	0	–2 (tolerated for years)	–2 if continued	–1	–1	**–5 — very unlikely**

SCORTEN FROM THE DATA GIVEN

age 40 (+1) and BSA >10% (+1) = **at least 2 (12% predicted mortality)**. HR, urea, glucose and bicarbonate are needed, and the WCC of 16 and ALT of 150 suggest systemic involvement. Recalculate within 24 h and on day 3.

SUPPORTING RESULTS

lymphopenia (0.3) is typical of SJS/TEN; **normal eosinophils** argue against DRESS; raised ALT occurs in TEN; anaemia (110).

MANAGEMENT

Stop carbamazepine (and the non-essential ibuprofen/flucloxacillin unless infection is proven); Switch to a **non-aromatic anticonvulsant** (e.g. **levetiracetam**, or valproate if appropriate — **not** phenytoin/oxcarbazepine/phenobarbital/lamotrigine, all of which cross-react or carry high risk); Transfer to a burns/ITU centre; ophthalmology within 24 h; supportive care; ± ciclosporin/MIg; HLA-B*15:02 / A*31:01 if of relevant ancestry (for family counselling); **Yellow Card**; lifelong alert

Referral

SAME DAY — Skin pain, blisters, skin detachment or mucosal erosions: possible **SJS/TEN**, stop the drug and admit the same day.

ROUTINE — Fever, facial swelling, lymph nodes and abnormal bloods 2–8 weeks after a new drug: possible **DRESS**.

999 / A&E — Swelling of lips, tongue or throat, or breathing difficulty: **999**.

Red flags and when to refer

- Skin pain, blisters, skin detachment or mucosal erosions: possible **SJS/TEN**, stop the drug and admit the same day.
- Fever, facial swelling, lymph nodes and abnormal bloods 2–8 weeks after a new drug: possible **DRESS**.
- Swelling of lips, tongue or throat, or breathing difficulty: **999**.

Red flags / refer

Mucosal involvement (2 sites); skin pain or burning; dusky/targetoid lesions or blisters; **Nikolsky sign**; **fever**; **facial oedema**; lymphadenopathy; abnormal LFTs/renal function; eosinophilia; widespread pustules; signs of anaphylaxis (call 999, IM adrenaline 500 micrograms in adults).

Complications and prognosis

Complications

- SJS/TEN, DRESS, anaphylaxis; mislabelled allergies.

Prognosis

- Most simple drug rashes resolve in 1–2 weeks after stopping the drug.

Prescriber pearls (UK pharmacist IP)

"**Any rash on any drug** — take a dated drug history, then think SCAR red flags."

Give the patient the **drug name, the reaction and the date** in writing; advise wearing a medical alert if severe; check OTC products (ibuprofen) and combination products contain none of the culprit.

Write the reaction, not just "allergy", in the record (e.g. "amoxicillin — maculopapular rash day 7, 2026, no mucosal involvement").

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK POSITION (2026)
SJS/TEN thresholds	course explanation: SJS <10%, TEN >30%. Some student answers said "TEN >10%"	Agrees with BAD 2016 / Bastuji-Garin: SJS <10%, **overlap 10–30%**, TEN >30%. ">10%" is the SCORTEN item, not the TEN definition
Latency	QB answer 5–28 days	BAD 2016 says **4–28 days**; ALDEN "suggestive" 5–28 days — both acceptable
Causality	Course teaching: Naranjo + allergy team	UK: Naranjo is generic; for SJS/TEN use **ALDEN** (BAD 2016). Allergy-service involvement per **NICE CG183**. Add **Yellow Card** reporting (not emphasised in the course)
SJS/TEN management	Student answers mentioned topical steroids/ antihistamines for a TEN case	UK: **same-day admission to a burns/ITU or specialist dermatology centre**, supportive care, ophthalmology within 24 h; systemic immunomodulation (ciclosporin/MIg) is a specialist decision; topical steroids/antihistamines alone are not management
HLA screening	Not discussed	MHRA/BNF/SmPC: **HLA-B*15:02 testing before carbamazepine** in Han Chinese/Thai/SE Asian ancestry; allopurinol low-start/titrate (HLA-B*58:01)

TOPIC	COURSE TEACHING	UK POSITION (2026)
ACE-i angioedema	Course case and student answers: stop ACE-i; ARB alternative	Agrees; UK adds airway-first triage (999), **class reaction — avoid all ACE-i**, ARB **with caution** (low recurrence risk), avoid **sacubitril/valsartan and gliptins**; C4/C1-INH only if atypical
EN case	Course teaching: drug-induced EN (COC); DDx cellulitis	Agrees; UK practice would also **exclude DVT/superficial thrombophlebitis** (unilateral, COC + tranexamic acid) and do a CXR/ASO, and notes the caution about combining TXA with a COC
Thiazide photosensitivity	Course teaching: phototoxic commoner; biopsy/photopatch rarely	Agrees; UK extras: **omeprazole/thiazide SCLE** DDx (check Ro), **HCTZ NMSC MHRA 2018**, choose a non-photosensitising replacement (amlodipine rather than indapamide)
FDE case	Diclofenac FDE; stop drug	Agrees; UK adds documentation of the NSAID class risk, OTC checks, paracetamol/hormonal alternatives for dysmenorrhoea, Yellow Card
Amiodarone	Students: stop amiodarone	UK: **don't stop without cardiology**; photoprotection for months after stopping (BNF); routine TFT/LFT monitoring

Self-check quiz

5 questions · optional

Q1 When should you think of Approach to a suspected drug eruption?

✓ Any new rash in someone who started, changed or restarted a medicine in the last 8 weeks.

Q2 What drives it?

✓ Immune reactions (type I to IV) or non-immune effects (phototoxicity, deposition, pharmacology) of a drug or its metabolites.

Q3 What is the backbone of care?

✓ Build a drug timeline, name the morphology, check for SCAR red flags, stop the most likely culprit, record the reaction precisely (NICE CG183) and report.

Q4 What must you never miss?

✓ SJS/TEN, DRESS, AGEP, anaphylaxis and angioedema; vasculitis with organ involvement.

Q5 Name a red flag that needs escalation.

✓ Skin pain, blisters, skin detachment or mucosal erosions: possible **SJS/TEN**, stop the drug and admit the same day.

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Drug eruptions Module 1 · Week 3

Morbilliform drug eruption & drug urticaria

The commonest drug rash: symmetrical pink macules and papules 4–14 days after a new drug, and drug-induced urticaria. How to tell them from SCARs and viral rashes, what to stop, and how to label the allergy.

ALSO CALLED maculopapular drug rash · exanthematous eruption · amoxicillin rash · drug urticaria · penicillin allergy label

RELATED PAGES Approach to a suspected drug eruption · DRESS (drug reaction with eosinophilia and systemic symptoms) · SJS/TEN and drug-induced blistering · Urticaria & angioedema · Scarlet fever & strawberry tongue

KEY FACTS

- **Morbilliform (maculopapular) drug eruption** is the commonest drug rash: measles-like pink macules and papules starting on the trunk.
- Begins 4–14 days after starting a drug (sooner on re-exposure); itchy, symmetrical; mucosa spared.
- Common culprits: aminopenicillins, cephalosporins, sulfonamides, anticonvulsants, allopurinol.
- Stop the drug; emollient plus moderate or potent topical steroid for 5–7 days; antihistamine for itch.
- Red flags for **DRESS** (fever, facial oedema, lymph nodes, eosinophilia, abnormal liver tests) or **SJS/TEN** (pain, blisters, mucosa): urgent referral.

Overview

MORBILLIFORM/MACULOPAPULAR EXANTHEM (MPE)

the commonest cADR (~75–95%). Symmetrical, itchy, pink macules and papules starting on the trunk.

DRUG URTICARIA/ANGIOEDEMA

wheals lasting <24 h ± angioedema, in the minutes to hours after a dose.

Epidemiology

- **Who**
  - About 1–3% of hospital inpatients; amoxicillin rash in EBV infection is classic.

Aetiology: causes and risk factors

Risk factors and associations

- T-cell mediated (type IV) hypersensitivity; viral infection (EBV, HIV) increases risk.

Myths to address

- **"An amoxicillin rash in glandular fever means lifelong penicillin allergy"**: often it does not; refer for allergy testing if unsure.

Pathophysiology and pathology

Drug-specific **T cells** recognise the drug bound to proteins in the skin.

1. **Sensitisation**
   Over 1–2 weeks, the drug (or a drug–protein complex) activates T cells.
2. **Skin infiltration**
   Activated T cells home to the dermis and release cytokines.
3. **Rash**
   Dermal vessel dilation and mild interface damage cause pink macules and papules.
4. **Resolution**
   Clears 1–2 weeks after stopping, often with fine peeling.

How it is caused

- MPE is a delayed **type IVb/Vc** T-cell reaction; latency is typically 4–14 days.
- Urticaria is **IgE-mediated** (β-lactams), **pseudo-allergic** (NSAIDs via COX-1 inhibition → leukotriene shift; opioids, vancomycin, contrast → direct mast-cell degranulation), or part of a serum-sickness-like reaction (cefaclor, amoxicillin in children; 1–3 weeks).
- The **amoxicillin–EBV rash** occurs in most patients with infectious mononucleosis given aminopenicillins. It is usually not a true allergy.

> **HISTOPATHOLOGY**
>
> - Superficial perivascular lymphocytes with eosinophils, mild interface change.

Clinical features

Clinical features & sites

> **MPE**
>
> begins on the trunk/flexures and spreads to the limbs, symmetrically. It often spares the face and does not involve mucosae. Mild fever is possible. It resolves in 1–2 weeks after stopping, often with fine desquamation.

> **URTICARIA**
>
> wheals with a flare, individual lesions lasting <24 h; ± angioedema of lips, eyelids or genitals.

Fig. 1 · Morbilliform (exanthematous) drug eruption — the commonest pattern; symmetrical truncal macules/papules 4–14 days after a new drug (aminopenicillins, sulfonamides, anticonvulsants, allopurinol). Check for SCAR red flags.
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Special groups

- **Viral exanthem**
  - Similar appearance; fever, contacts and timing help distinguish.

Diagnosis and investigations

Clinical diagnosis plus the drug timeline. For MPE, **FBC/LFTs** help exclude early DRESS (eosinophilia, transaminitis). With urticaria, check for anaphylaxis. Refer to allergy for possible β-lactam allergy testing/de-labelling.

Differential diagnoses

Viral exanthem; early **DRESS** (facial oedema, fever, eosinophilia, longer latency); early **SJS/TEN** (skin pain, dusky lesions, mucositis); urticarial vasculitis (the vasculitis notes: wheals >24 h, bruising); acute GVHD; scarlet fever.

Management

- Stop the drug (or "treat through" only if essential and benign, in hospital with specialist agreement).
- MPE: emollient + moderate/potent TCS twice daily for 5–7 days, and an antihistamine for itch.
- Urticaria: a **non-sedating antihistamine** (e.g. cetirizine 10 mg; up to 4× the licensed dose off-label for chronic urticaria — BSACI). A short oral prednisolone course only for severe acute urticaria. Anaphylaxis adrenaline.
- **NSAID-EXACERBATED URTICARIA/ANGIOEDEMA**
  avoid all COX-1 inhibitors. Paracetamol is usually tolerated (below 1 g); a selective COX-2 inhibitor may be tolerated after specialist challenge.
- Document, Yellow Card if serious, and arrange allergy referral if the label matters (e.g. penicillin).

IP scope

Within IP scope (symptomatic treatment, stopping/switching drugs, documentation, referral to allergy).

Referral

ROUTINE — Rash with fever, facial swelling, eosinophilia or abnormal LFTs: treat as possible **DRESS**.

999 / A&E — Urticaria with breathing, throat or circulation symptoms: anaphylaxis, IM adrenaline and **999**.

Red flags and when to refer

- Rash with fever, facial swelling, eosinophilia or abnormal LFTs: treat as possible **DRESS**.
- Urticaria with breathing, throat or circulation symptoms: anaphylaxis, IM adrenaline and **999**.

Red flags / refer

Mucosal involvement, skin pain, blistering, fever >38.5 °C with facial oedema/lymphadenopathy, eosinophilia >1.5 × 10^9/L, deranged LFTs treat as a possible SCAR. Anaphylaxis 999.

Complications and prognosis

Complications

- Progression to DRESS or SJS/TEN (rare); misdiagnosis leading to unnecessary allergy labels.

Prognosis

- Resolves in 1–2 weeks after stopping the drug.

Prescriber pearls (UK pharmacist IP)

The rash may worsen for a couple of days after stopping — that does not mean the stop failed.

"Amoxicillin rash in glandular fever" is not usually penicillin allergy — but label carefully and refer for de-labelling rather than ignoring it.

Self-check quiz

5 questions · optional

Q1 When should you think of Morbilliform drug eruption & drug urticaria?

✓ A symmetrical, itchy measles-like rash on the trunk 1–2 weeks after an antibiotic, anticonvulsant or allopurinol.

Q2 What drives it?

✓ Delayed T-cell reaction (morbilliform) or immediate IgE/mast-cell reaction (urticaria) to the drug.

Q3 What is the backbone of care?

✓ Stop the culprit if possible; emollient, antihistamine and a moderate steroid; review in 48 h; record the reaction type precisely.

Q4 What must you never miss?

✓ Fever, facial oedema, mucosal or skin pain, blisters or organ involvement: it is not a simple exanthem.

Q5 Name a red flag that needs escalation.

✓ Rash with fever, facial swelling, eosinophilia or abnormal LFTs: treat as possible **DRESS**.

References

1. DermNet: Morbilliform drug reaction — https://dermnetnz.org/topics/morbilliform-drug-reaction
dermnetnz.org

Photo gallery

Morbilliform drug eruption 2

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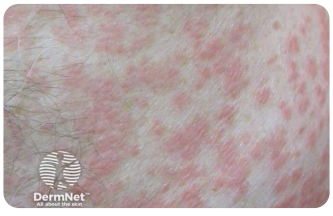

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Drug urticaria

1

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Drug eruptions Module 1 · Week 3

DRESS (drug reaction with eosinophilia and systemic symptoms)

A delayed (2–8 weeks), multi-organ drug reaction: widespread rash, fever, facial oedema, lymphadenopathy, eosinophilia and liver, kidney, lung or heart involvement. Same-day hospital assessment.

ALSO CALLED DRESS · DIHS · drug hypersensitivity syndrome · drug reaction with eosinophilia · RegiSCAR · allopurinol · +1

RELATED PAGES AGEP (acute generalised exanthematous pustulosis) · Morbilliform drug eruption & drug urticaria · SJS/TEN and drug-induced blistering · Approach to a suspected drug eruption

KEY FACTS

- **DRESS** is a severe, delayed drug reaction: rash, fever, **facial oedema**, lymphadenopathy, eosinophilia and organ involvement.
- Latency **2–8 weeks**; culprits: allopurinol, carbamazepine and other aromatic anticonvulsants, lamotrigine, sulfonamides/dapsone, minocycline, vancomycin.
- The liver is the commonest organ involved; also kidney, lung, heart, thyroid. Mortality about 2–10%.
- Stop the drug and arrange **same-day hospital assessment**; systemic steroids tapered slowly over 6–12 weeks.
- Follow up thyroid function and glucose for 1–2 years (late autoimmunity).

Overview

DRESS (drug reaction with eosinophilia and systemic symptoms; also called DIHS) is a delayed, multi-organ reaction with fever, rash and internal organ involvement. Mortality is about **2–10%**.

Epidemiology

- **Who**
  - About 1 in 1,000 to 1 in 10,000 exposures to high-risk drugs; adults more than children.
  - HLA risk alleles: HLA-B*58:01 (allopurinol), HLA-A*31:01 (carbamazepine in Europeans).

Aetiology: causes and risk factors

- Type IVb (Th2 / eosinophil) reaction with **sequential herpesvirus reactivation** (HHV-6, then EBV/CMV), which explains the relapsing course.
- Culprits: **allopurinol**, **carbamazepine**, phenytoin, lamotrigine, phenobarbital, **sulfonamides/dapsone**, minocycline, vancomycin, abacavir/nevirapine, strontium.

- Latency **2–8 weeks** (the longest of the severe drug reactions).

Myths to address

- **"IT'S JUST A DRUG RASH"**
  facial swelling, fever and abnormal blood tests mean a systemic emergency.

Pathophysiology and pathology

DRESS combines a **drug-specific T-cell attack** with **herpesvirus reactivation**.

1. **Drug presentation**
   The drug or a metabolite binds HLA molecules and activates T cells (often in genetically susceptible people).
2. **Th2 and eosinophils**
   IL-5 and other Th2 cytokines drive eosinophilia and tissue infiltration.
3. **Viral reactivation**
   HHV-6, then EBV and CMV reactivate, sustaining inflammation and relapses.
4. **Organ damage**
   Infiltrating T cells and eosinophils injure the liver, kidney, lung and heart.
5. **Late autoimmunity**
   Immune dysregulation after recovery can cause thyroiditis or type 1 diabetes.

HISTOPATHOLOGY

- Interface or spongiotic dermatitis with dense perivascular lymphocytes and eosinophils; not specific. RegiSCAR scoring combines clinical and lab features.

Clinical features

- Morbilliform rash that becomes infiltrated and oedematous, often **over 50% of body surface**; may become exfoliative.
- **Facial oedema** (key sign); fever above 38 °C; **lymphadenopathy**.
- Blood: eosinophilia (over 1.5×10^9/L), atypical lymphocytes.
- **ORGAN INVOLVEMENT**
  liver (commonest), kidney, lung, heart (myocarditis), thyroid.
- Course over weeks to months with relapses; **late autoimmune disease** (thyroiditis, type 1 diabetes).
- **RegiSCAR** score grades the likelihood of DRESS.

Fig. 1 · Widespread morbilliform eruption of DRESS — check for fever, facial oedema, nodes, eosinophils and liver function.

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Special groups

- **Allopurinol in CKD**
  - Highest risk with recent start, high dose, renal impairment and HLA-B*58:01 (common in Han Chinese, Thai and Korean people).

Diagnosis and investigations

FBC with differential and film, LFT, U&E, urinalysis, troponin/ECG if symptomatic, TFT, lipase, HHV-6 PCR and skin biopsy. Patch testing later can help identify the culprit.

Differential diagnoses

Viral exanthem, lymphoma (especially angioimmunoblastic), hypereosinophilic syndrome, SJS/TEN, Kawasaki disease, simple morbilliform drug eruption.

Management

- **Stop the culprit**, document, Yellow Card, and avoid for life (including cross-reacting drugs).
- **Same-day hospital assessment** if fever, facial oedema, eosinophilia or abnormal LFTs.
- Mild disease without significant organ involvement: potent topical steroids.
- With organ involvement: **systemic corticosteroids** (e.g. prednisolone 0.5–1 mg/kg/day) **tapered slowly over 6–12 weeks**: rapid tapers cause relapse.
- Monitor organs for months, and check **TFT and glucose** for 1–2 years.

IP scope

Recognise, stop, refer, document, Yellow Card. After discharge: support the specialist steroid taper (steroid card if prolonged) and monitor TFT/glucose per the specialist letter.

Medicines at a glance

Prednisolone (specialist)

Organ involvement	SPECIALIST ONLY
DOSE	0.5–1 mg/kg/day · taper over 6–12 weeks
CAUTIONS	Rapid tapers cause relapse.

Treatment ladder

1 **Stop and record**

Stop the culprit; record the allergy; Yellow Card.

2 **Mild, no organ involvement**

Potent topical steroid under specialist care; monitor bloods.

3 **Organ involvement**

Prednisolone (specialist)

Referral

SAME DAY Rash + fever + facial oedema / lymph nodes 2–8 weeks after a new high-risk drug.

ROUTINE Follow-up TFT and glucose for 1–2 years (late autoimmunity).

Red flags and when to refer

- Fever + rash + facial oedema or lymphadenopathy **2–8 weeks** after a new anticonvulsant, allopurinol or sulfonamide: **same-day admission** (hepatitis, nephritis and myocarditis can follow).

- Allopurinol (HLA-B*58:01), carbamazepine (HLA-B*15:02) and lamotrigine are high-risk starts: counsel to report a rash early.

Complications and prognosis

Complications

- Fulminant hepatitis, interstitial nephritis, pneumonitis, myocarditis; death.
- Autoimmune thyroiditis, type 1 diabetes, alopecia areata after recovery.

Prognosis

- Recovery over weeks to months; relapses during steroid taper are common.

Prescriber pearls (UK pharmacist IP)

DRESS has the **longest latency** of the severe drug reactions (2–8 weeks).

"**Allopurinol + CKD + recent start + rash** = DRESS/SJS risk": dose adjustment and HLA-B*58:01 awareness.

Self-check quiz

4 questions · optional

Q1 What is the typical latency of DRESS?

✓ 2–8 weeks after starting the drug.

Q2 Name three common culprits.

✓ Allopurinol, carbamazepine (and other aromatic anticonvulsants), sulfonamides/dapsone, minocycline.

Q3 Which organ is most often involved?

✓ The liver.

Q4 Why taper steroids slowly?

✓ Rapid tapers cause relapse (linked to herpesvirus reactivation).

References

1. DermNet: Drug hypersensitivity syndrome (DRESS) — https://dermnetnz.org/topics/drug-hypersensitivity-syndrome
dermnetnz.org

Photo gallery

Clinical photographs 1

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DRESS 1

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Drug eruptions Module 1 · Week 3

ACE-inhibitor (bradykinin) angioedema

Deep swelling of the lips, tongue or face without wheals or itch in a patient on an ACE inhibitor, which can start years into treatment and does not respond to antihistamines.

ALSO CALLED ramipril angioedema · lisinopril · bradykinin · swelling without wheals · ARB · sacubitril · +1

RELATED PAGES Urticaria & angioedema · Approach to a suspected drug eruption · Morbilliform drug eruption & drug urticaria

KEY FACTS

- Non-pitting swelling of lips, tongue, face, larynx or gut **without wheals or itch**, caused by ACE inhibitors.
- Affects about 0.1–0.7% of users; can start **from the first dose to years later**.
- It is **bradykinin-mediated**: antihistamines and steroids don't work, and adrenaline often doesn't.
- Risk factors: **black ethnicity**, female sex, age over 65, smoking, gliptins, mTOR inhibitors.
- Airway first (999). Stop the ACE inhibitor **permanently** and record a class reaction; an ARB may be considered with caution.

Overview

Episodic, non-pitting swelling of the deep dermis/submucosa (lips, tongue, face, larynx, gut) **without wheals or itch**, caused by ACE inhibitors. It affects ~0.1–0.7% of users and accounts for up to a third of angioedema presentations to emergency departments. It can start **at any time — from the first dose to years later** (often within the first months).

Aetiology: causes and risk factors

Myths to address

- **"It can't be the ramipril, they've taken it for years"**: onset can be years after starting.
- **"No hives, so not an allergy... so it's fine to continue"**: it is not allergic but it can be fatal.

Pathophysiology and pathology

ACE inhibitors stop the breakdown of **bradykinin**, which leaks fluid into deep tissues.

1. **ACE blocked**
   ACE (kininase II) normally degrades bradykinin; ACE inhibitors block it.
2. **Bradykinin builds up**
   Bradykinin and substance P accumulate.

3 **B2 receptors**

B2 receptor activation causes vasodilatation and leaky vessels.

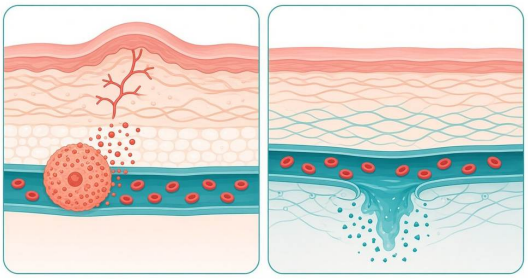

Fig. 1 · Histamine versus bradykinin angioedema.

AI-generated illustration

4 **Deep swelling**

Fluid collects in the deep dermis and submucosa: no wheals, no itch.

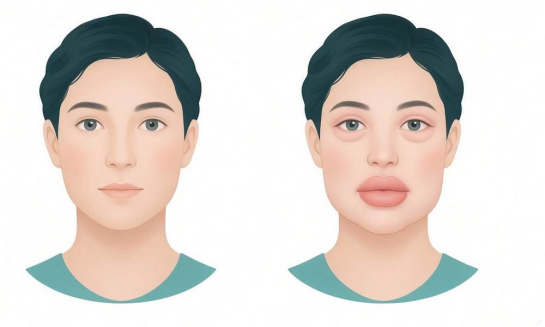

Fig. 2 · Facial and lip angioedema.

AI-generated illustration

How it is caused

ACE (kininase II) normally breaks down **bradykinin**. ACE inhibition bradykinin (and substance P) accumulate **B2 receptors** vasodilatation and increased vascular permeability. It is **not histamine- or IgE-mediated**, which is why antihistamines, steroids and often adrenaline don't work.

RISK FACTORS

black ethnicity (3–5×), female sex, age >65, smoking, previous idiopathic angioedema, seasonal allergies, and **co-prescription of DPP-4 inhibitors (gliptins)**, mTOR inhibitors (sirolimus, everolimus), **sacubitril/valsartan** (neprilysin inhibition — contraindicated within 36 h of an ACE-i) or alteplase.

TRIGGERS

minor trauma/surgery, dental work and intubation can precipitate attacks. That fits this case (swelling a week after skin-tag removal), though the procedure is a trigger, not the cause.

Clinical features

Face, periorbital tissue, lips, **tongue**, floor of mouth and larynx; recurrent intestinal angioedema (abdominal pain). Swelling develops over hours and lasts **24–72 h**. Attacks may recur for weeks after stopping the ACE-i (the effect wanes over ~1–3 months), so the episodes that follow are not "treatment failure".

Fig. 3 · ACE-inhibitor angioedema of the lip — bradykinin-mediated, no wheals/itch, poor response to antihistamine/steroid. Stop ACE-i permanently; airway first.

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Fig. 4 · Severe lip angioedema without wheals — typical of ACE-inhibitor (bradykinin) angioedema.

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Types to recognise

Lip angioedema

Bradykinin, no wheals. Site: Lips.

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Diagnosis and investigations

- Clinical: angioedema with **no urticaria**, on an ACE-i, not responding to antihistamine/steroid.
- **INVESTIGATIONS**
  Usually none are needed to diagnose it; **C4** (+ C1-inhibitor level and function) if there is no ACE-i, a family history, onset before 40, or recurrence after stopping — to exclude **hereditary/acquired C1-inhibitor deficiency** (C4 low in both). In acquired deficiency, look for lymphoproliferative disease and C1q; **Mast-cell tryptase** if anaphylaxis is possible; FBC/TFT if chronic urticaria is in the differential

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Histamine angioedema	Wheals / itch, minutes, < 24 h, responds to antihistamine / adrenaline.
	Hereditary / acquired C1-INH deficiency	Low C4; family history; acquired look for lymphoproliferative disease.
	Contact dermatitis of eyelids / face	Itch, eczema, allergen.
	Cellulitis / erysipelas	Hot, tender, fever, one-sided.
	Orofacial granulomatosis / dermatomyositis	Persistent lip swelling; periorbital heliotrope.

Also: contact dermatitis of the eyelids/face, cellulitis/erysipelas, SVC obstruction, hypothyroid myxoedema, Melkersson–Rosenthal syndrome / orofacial granulomatosis, dermatomyositis (periorbital oedema — the skin signs of systemic disease notes).

Management

1. **Airway first.** Tongue, floor-of-mouth or laryngeal swelling, voice change, stridor or dysphagia **999 / ED**. Adrenaline is given if anaphylaxis cannot be excluded. In hospital, airway management is the priority; off-label icatibant, C1-INH concentrate or fresh frozen plasma are sometimes used (evidence is mixed — RCTs of icatibant in ACE-i angioedema were negative).
2. **Stop ramipril permanently** and record "**ACE inhibitor — angioedema (class reaction): avoid all ACE inhibitors**". Do not rechallenge. Yellow Card.
3. **Alternative for hypertension/T2DM renoprotection (NICE NG136/NG28):** An **ARB** can be considered **with caution**. Recurrence on ARBs is uncommon (~2–10% in studies), so discuss with the patient, consider specialist advice, and safety-net; Or use a **calcium-channel blocker** (amlodipine), with an SGLT2 inhibitor for renal protection where indicated
4. **Avoid** sacubitril/valsartan, and think twice before adding a **gliptin** (DPP-4 inhibitors also reduce bradykinin/substance-P breakdown).
5. **Explain** that swelling may recur for several weeks after stopping. Stop cetirizine/prednisolone if they gave no benefit (avoid steroid harm in new diabetes — hyperglycaemia).
6. **Metformin and simvastatin** are not implicated and can continue.

IP scope

Recognise, stop and switch the ACE-i (within competence for hypertension/diabetes), document the class allergy, Yellow Card, safety-net. Immunology referral for suspected HAE/AAE.

Treatment ladder

1 **Stop & document**

Stop the ACE-i permanently: record “ACE inhibitor: angioedema (class reaction), avoid all ACE inhibitors”. No rechallenge. **Yellow Card**. Explain swelling may recur for several weeks.

2 **Alternative (NICE NG136 / NG28)**

ARB with caution (recurrence ~2–10%): discuss, consider specialist advice, safety-net. Or a **calcium-channel blocker** (amlodipine) ± SGLT2 inhibitor for renal protection.

Avoid

Sacubitril / valsartan; think twice before a **gliptin**. Stop antihistamine / prednisolone that gave no benefit (steroid harm, e.g. hyperglycaemia).

Hospital

Airway management; adrenaline if anaphylaxis can't be excluded; off-label icatibant / C1-INH / FFP (mixed evidence).

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Metformin, statins**: Not implicated; continue.

Referral

999 / A&E — Tongue / throat swelling, voice change, stridor, difficulty swallowing: **999**.

ROUTINE — Recurrent angioedema without ACE-i or with family history; abdominal attacks; > 3 months after stopping: **immunology** (C4 / C1-INH).

Red flags and when to refer

- Tongue or throat swelling, voice change, stridor or difficulty swallowing: **999** now.
- Stop the ACE inhibitor **permanently** even if the swelling has settled, and document it as an adverse reaction.

Red flags / refer

Airway involvement (999); recurrent angioedema without an ACE-i, or with a family history (immunology — C4/C1-INH); abdominal attacks; angioedema persisting >3 months after stopping.

Complications and prognosis

Complications

- Airway obstruction and death; abdominal angioedema mimicking surgical abdomen.

Prognosis

- Attacks stop after the drug is withdrawn, but can continue for weeks to months.

Prescriber pearls (UK pharmacist IP)

- "Angioedema **without wheals** on an ACE-i = bradykinin. Antihistamines won't fix it — **stopping the ACE-i** will (eventually)."
- Check every ACE-i patient on the medication review for previous facial or tongue swelling — patients often don't link it to a long-standing tablet.
- The risk persists for years on treatment, so this is not just a "first-month" issue.

Self-check quiz

4 questions · optional

Q1 Mechanism?

✓ ACE inhibition blocks bradykinin breakdown (not histamine / IgE).

Q2 Swelling recurs 2 weeks after stopping ramipril: treatment failure?

✓ No: attacks can recur for weeks–months.

Q3 Which combination drug is contraindicated within 36 h of an ACE-i?

✓ Sacubitril / valsartan.

Q4 Test if no ACE-i and family history?

✓ C4 (± C1-INH level and function).

References

1. DermNet: Angioedema — https://dermnetnz.org/topics/angioedema
dermnetnz.org

Photo gallery

Clinical photographs 1

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ACE-inhibitor angioedema 1

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Drug eruptions Module 1 · Week 3

AGEP (acute generalised exanthematous pustulosis)

Sudden sheets of tiny sterile non-follicular pustules on red, swollen skin within hours to 2 days of a drug (often an antibiotic), with fever. Usually settles in 1–2 weeks once the drug is stopped.

ALSO CALLED AGEP · acute generalised exanthematous pustulosis · pustular drug eruption · toxic pustuloderma

RELATED PAGES DRESS (drug reaction with eosinophilia and systemic symptoms) · Palmoplantar pustulosis & palmoplantar psoriasis · Psoriasis · Approach to a suspected drug eruption · Folliculitis

KEY FACTS

- **AGEP** is a pustular drug reaction: hundreds of tiny **non-follicular** sterile pustules on red, swollen skin, with fever.
- Starts in the flexures and face **within hours to 2 days** of an antibiotic (aminopenicillins, macrolides) or other drug (hydroxychloroquine, diltiazem, terbinafine).
- Neutrophilia is common; organ involvement is mild and uncommon.
- Stop the drug, potent topical steroid and supportive care; settles in **1–2 weeks** with peeling.
- Main differential: generalised pustular psoriasis (history of psoriasis, slower course).

Overview

AGEP (acute generalised exanthematous pustulosis) is an abrupt eruption of **sterile, non-follicular pinhead pustules** on oedematous erythema. It is usually self-limiting; mortality is under 5%.

Epidemiology

- **Who**
  - Rare (about 1–5 per million per year); any age; more women.

Aetiology: causes and risk factors

- Type IVd reaction: drug-specific T cells release **IL-8 (CXCL8)**, recruiting neutrophils into the epidermis.
- Culprits: **aminopenicillins**, macrolides, cephalosporins, **pristinamycin**, **hydroxychloroquine**, **diltiazem**, terbinafine.
- Latency **under 48 hours** for antibiotics; up to 1–2 weeks for some drugs.
- IL36RN variants overlap with generalised pustular psoriasis.

Myths to address

- **"PUSTULES MEAN INFECTION"**

AGEP pustules are sterile; more antibiotics can make it worse.

Pathophysiology and pathology

Drug-specific T cells summon **neutrophils** into the top of the epidermis.

1. **T-cell activation**
   Drug-specific CD4+ and CD8+ T cells migrate into the epidermis.
2. **IL-8 release**
   T cells and keratinocytes release **IL-8 (CXCL8)** and IL-17.
3. **Subcorneal pustules**
   Neutrophils gather just under the stratum corneum, forming pinhead pustules.

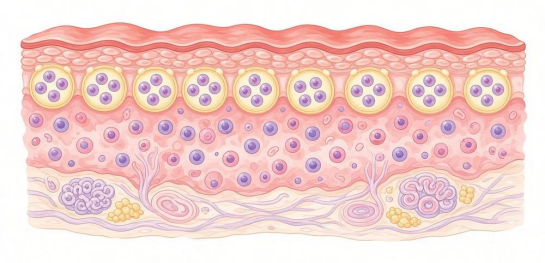

Fig. 1 · AGEP: neutrophil-filled pustules collect just under the stratum corneum (subcorneal), not around hair follicles, over swollen inflamed skin.
AI-generated illustration

4. **Desquamation**
   As pustules dry, the surface peels in collarettes over 1–2 weeks.

HISTOPATHOLOGY

- Subcorneal or intraepidermal spongiform pustules, papillary oedema, eosinophils, sometimes necrotic keratinocytes.

Clinical features

- Pinhead pustules on erythema, starting in the **flexures and face** and becoming generalised; burning or itch.
- Fever above 38 °C; neutrophilia; mild organ involvement in up to 20%.
- Resolves in 1–2 weeks with **collarette desquamation**.

Diagnosis and investigations

Clinical with history; FBC (neutrophilia), U&E, LFT. Biopsy shows subcorneal or intraepidermal spongiform pustules (to distinguish it from pustular psoriasis). The EuroSCAR score helps. Patch testing later.

Differential diagnoses

Generalised pustular psoriasis (history of psoriasis, slower onset, longer course), subcorneal pustular dermatosis, pustular DRESS, folliculitis, SJS/TEN (early AGEP can show Nikolsky-like sloughing).

Management

- **Stop the drug**, document, Yellow Card, avoid for life.
- Potent topical steroid; antipyretics; supportive care. Systemic steroids are rarely needed.
- Usually managed in dermatology, with admission if unwell.

Treatment ladder

1 **Stop and record**

Stop the culprit; record allergy; Yellow Card.

2 **Symptom control**

Potent topical steroid, emollients, antipyretics.

3 **Unwell**

Same-day dermatology or admission.

Referral

SAME DAY Widespread pustules with fever within days of a new drug.

Red flags and when to refer

- Widespread pustules within days of an antibiotic: **same-day dermatology**.
- Mucosal erosions, skin sloughing or systemic organ dysfunction: hospital.

Complications and prognosis

Complications

- Superinfection; rarely liver, kidney or lung involvement; fluid loss in frail patients.

Prognosis

- Excellent once the drug is stopped; recurs on re-exposure.

Prescriber pearls (UK pharmacist IP)

AGEP has the **shortest latency** of the severe drug reactions (hours to 2 days).

Pustules in AGEP are **non-follicular** (unlike folliculitis).

Self-check quiz

4 questions · optional

Q1 What is the typical latency of AGEP after an antibiotic?

✓ Under 48 hours.

Q2 Which cytokine drives the neutrophils in AGEP?

✓ IL-8 (CXCL8).

Q3 Main differential?

✓ Generalised pustular psoriasis.

Q4 How does AGEP resolve?

✓ Over 1–2 weeks with collarette desquamation after stopping the drug.

References

1. DermNet: AGEP — https://dermnetnz.org/topics/acute-generalised-exanthematous-pustulosis
dermnetnz.org

Photo gallery

AGEP 2

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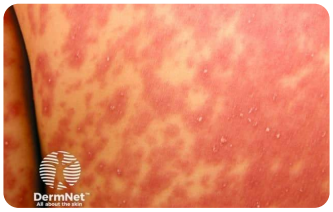

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Drug eruptions Module 1 · Week 3

Erythema nodosum (incl. drug-induced)

Tender red nodules on the shins that bruise as they fade. Causes include streptococcal infection, sarcoidosis, IBD, pregnancy and drugs such as the combined pill; the work-up looks for the trigger.

ALSO CALLED EN · tender shin nodules · septal panniculitis · COC · streptococcal · sarcoidosis · +1

RELATED PAGES Sarcoidosis (incl. Löfgren syndrome) · Cellulitis & erysipelas · Approach to cutaneous vasculitis · Cutaneous tuberculosis (scrofuloderma & others) · Pyoderma gangrenosum

KEY FACTS

- **Erythema nodosum (EN)** is a **septal panniculitis**: tender red nodules on the shins that bruise as they heal.
- Often with fever, malaise and arthralgia; lasts 3–6 weeks; does not ulcerate or scar.
- Causes: streptococcal throat, sarcoidosis, IBD, pregnancy, drugs (combined pill), TB; about half idiopathic.
- Tests: FBC, CRP, ASO titre, chest X-ray (bilateral hilar lymphadenopathy), throat swab; others guided by history.
- Treat the cause; rest, elevation, compression and NSAIDs.

Overview

Erythema nodosum (EN) is the commonest **septal panniculitis**: a hypersensitivity reaction producing tender red nodules, usually on the shins. It is most common in women aged 20–45.

Epidemiology

- **Who**
  - Women 3–6× more often; age 20–45; commonest panniculitis.

Aetiology: causes and risk factors

Risk factors and associations

- Delayed hypersensitivity to infection (streptococcus, TB, Yersinia), sarcoidosis, IBD, pregnancy, drugs, malignancy (lymphoma).

Myths to address

- **"IT'S A BRUISE OR CELLULITIS"**
  bilateral tender nodules plus systemic symptoms point to EN.

Pathophysiology and pathology

EN is an **immune reaction in the fibrous septa** between fat lobules.

1. **Antigen exposure**
   An infection, drug or systemic disease provides the antigen.
2. **Immune complex / T-cell reaction**
   Immune reactions target the septal blood vessels of the subcutaneous fat.
3. **Septal panniculitis**
   Inflammatory cells thicken the septa while lobules are spared.

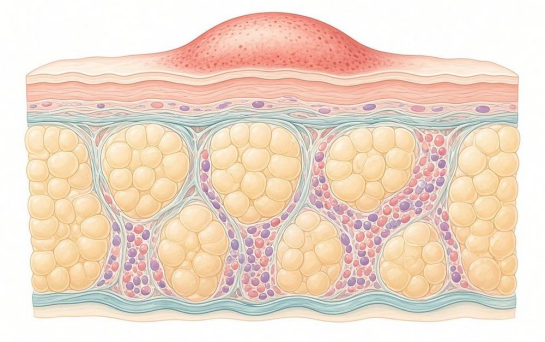

Fig. 1 · Erythema nodosum: inflammation thickens the fibrous septa between fat lobules (septal panniculitis), causing a tender red nodule.
AI-generated illustration

4. **Resolution**
   Haemoglobin breakdown changes colour from red to purple to yellow-green, like a bruise.

How it is caused

A delayed hypersensitivity response to many antigens:

- **idiopathic** (~30–50%)
- **streptococcal infection** (the commonest identifiable cause, especially in children)
- **sarcoidosis** (Löfgren syndrome — the skin signs of systemic disease notes)
- **drugs** — **oestrogen-containing contraceptives/HRT**, sulfonamides, penicillins/amoxicillin, bromides, minocycline, gold, and newer agents (BRAF inhibitors, immune checkpoint inhibitors, isotretinoin rarely)
- **pregnancy**
- **IBD** (Crohn's/UC)
- TB; Yersinia, Salmonella, Campylobacter; Chlamydia, Mycoplasma; Behçet's; lymphoma (rare).

HISTOPATHOLOGY

- Septal panniculitis without vasculitis; Miescher radial granulomas. Biopsy only if atypical (deep incisional).

Clinical features

- Crops of **tender, warm, ill-defined red nodules 1–5 cm** on the **anterior shins**, usually **bilateral**. Less often on the thighs, forearms or trunk.
- They evolve through bruise-like colours ("**erythema contusiformis**", photo E02) and resolve in **3–6 weeks without ulceration or scarring**.
- Prodrome: fever, malaise, arthralgia (ankles), sometimes before the nodules.
- **Unilateral EN is uncommon but described**. A single unilateral lesion makes it essential to exclude the differentials below.

Fig. 2 · Erythema nodosum — tender, ill-defined red nodules on the shins (septal panniculitis). Oestrogen-containing COC is a recognised trigger.
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Special groups

- **Löfgren syndrome**
  - EN + bilateral hilar lymphadenopathy + fever + arthritis: acute sarcoidosis with a good prognosis.

Diagnosis and investigations

- Clinical diagnosis; look for the cause: **FBC, ESR/CRP, U&E/LFT**; **ASO/anti-DNase B** titre ± throat swab; **CXR** (bilateral hilar lymphadenopathy — sarcoid/TB/lymphoma) ± serum ACE/calcium; **IGRA**/Mantoux if TB risk; **pregnancy test**; stool culture if diarrhoea; faecal calprotectin if bowel symptoms; urinalysis
- **Deep incisional/punch biopsy** (to fat) if atypical — unilateral, ulcerating or persistent >6–8 weeks.
- **CASE-SPECIFIC**
  she has a **unilateral tender shin lesion** and takes a **COC + tranexamic acid** (both prothrombotic). Apply **Wells/D-dimer and venous Doppler** if there is calf swelling, cord-like tenderness or leg oedema, to exclude **DVT/superficial thrombophlebitis**. The course discussion did not mention this.

Differential diagnoses

	EN	CELLULITIS	SUPERFICIAL THROMBOPHLEBITIS / DVT	ERYTHEMA INDURATUM (NODULAR VASCULITIS)	OTHER PANNICULITIS
Laterality	Usually bilateral	**Unilateral**	Unilateral	Bilateral	Variable
Site	Anterior shins	Any, spreading	Along a vein (cord) / calf	**Posterior calves**	Pancreatic (ulcerates), lupus profundus (face/arms)
Features	Tender nodules, no ulcer	Spreading erythema, fever, portal of entry, raised WCC	Tender cord; calf swelling	Ulcerates, scars; TB association	Atrophy/scarring
Course	3–6 weeks, resolves	Responds to antibiotics	—	Chronic	—

Also: insect-bite reactions, cutaneous PAN (nodules + livedo — the vasculitis notes), Sweet syndrome, lipodermatosclerosis.

Management

1. **Treat the cause.** In this case, **consider stopping the COC**, which is the most likely drug trigger. **Tranexamic acid** is a less established EN trigger but adds thrombotic risk with the COC. **Metformin** is not a recognised cause: **Contraception and cycle control for PCOS:** a progestogen-only method (desogestrel POP, implant, **LNG-IUS 52 mg** — also helps dysmenorrhoea and heavy bleeding), or cyclical progestogen to protect the endometrium. Weight loss and metformin help cycle regularity; **Dysmenorrhoea:** an NSAID such as mefenamic acid or naproxen (both effective for dysmenorrhoea and EN pain); LNG-IUS; Note the **FSRH/BNF caution** about combining tranexamic acid with a COC — prescribe TXA only for heavy menstrual bleeding, at the licensed dose, and not in people at VTE risk
2. Symptomatic rest, **leg elevation**, compression hosiery (if arterial supply is adequate), cool compresses; **NSAIDs** (ibuprofen/naproxen) for pain if not contraindicated; paracetamol.
3. Specialist options potassium iodide (rarely used in the UK); colchicine; systemic corticosteroids only after infection (TB, strep) and malignancy are excluded.
4. **Review in 2–4 weeks.** Persistent or recurrent EN dermatology/rheumatology; biopsy if atypical.

IP scope

Investigation, contraceptive switch (within competence and UKMEC), NSAID analgesia, and referral. Thrombosis assessment through local DVT pathways.

Referral

ROUTINE — EN with cough, breathlessness, weight loss or night sweats: CXR to look for **sarcoidosis, TB or lymphoma**.

ROUTINE — Ulcerating or one-sided nodules: not typical EN, so think again (erythema induratum, vasculitis, cellulitis).

Red flags and when to refer

- EN with cough, breathlessness, weight loss or night sweats: CXR to look for **sarcoidosis, TB or lymphoma**.
- Ulcerating or one-sided nodules: not typical EN, so think again (erythema induratum, vasculitis, cellulitis).

Red flags / refer

Unilateral swelling with features of DVT (same-day DVT pathway); fever + spreading erythema (cellulitis); BHL on CXR or respiratory symptoms (sarcoid/TB/lymphoma); bowel symptoms (IBD); ulceration (not EN); pregnancy; lesions lasting >6–8 weeks.

Complications and prognosis

Complications

- Pain, arthralgia; complications depend on the cause.

Prognosis

- Resolves in 3–6 weeks; recurrence possible if the cause persists.

Prescriber pearls (UK pharmacist IP)

- "EN is a **sign, not a diagnosis**" — the job is finding the trigger.
- It heals without scarring, but **new crops can appear for several weeks** after the trigger is removed.
- In any woman on a COC + TXA with a **tender, red, unilateral leg**, think **clot** before panniculitis.

Self-check quiz

5 questions · optional

Q1 When should you think of Erythema nodosum (incl. drug-induced)?

✓ Painful, warm, ill-defined red nodules on both shins, often with fever or aching joints.

Q2 What drives it?

✓ A hypersensitivity reaction in the fat septa (septal panniculitis) to an infection, drug, inflammatory disease or pregnancy.

Q3 What is the backbone of care?

✓ Find the trigger (throat swab/ASO, CXR, FBC, CRP, pregnancy test, drug review); rest, elevation, NSAID if suitable; stop a culprit COC.

Q4 What must you never miss?

✓ Sarcoidosis with hilar nodes; TB; IBD flare; cellulitis or DVT mistaken for EN.

Q5 Name a red flag that needs escalation.

✓ EN with cough, breathlessness, weight loss or night sweats: CXR to look for **sarcoidosis, TB or lymphoma**.

References

1. DermNet: Erythema nodosum — https://dermnetnz.org/topics/erythema-nodosum
dermnetnz.org
2. Medscape: Erythema nodosum (incl. drug-induced) — https://emedicine.medscape.com/article/1081633-overview
emedicine.medscape.com

Photo gallery

Erythema nodosum 2

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Drug eruptions Module 1 · Week 3

Drug-induced pigmentation (amiodarone, minocycline)

Blue-grey, slate or brown discolouration caused by drugs such as amiodarone and minocycline, usually on sun-exposed skin or scars. Recognise it, review the drug and set expectations about slow fading.

ALSO CALLED blue-grey skin · slate-grey · amiodarone · minocycline · antimalarial · drug hyperpigmentation

RELATED PAGES Fixed drug eruption · Melasma & facial melanosis · Drug-induced photosensitivity · Contact & lichenoid pigmentation · Addison's disease: skin signs

KEY FACTS

- Drugs cause about 10–20% of acquired hyperpigmentation; colours and sites are clues.
- Minocycline blue-grey on scars, shins and sclera; **amiodarone**: slate-grey on sun-exposed face; **hydroxychloroquine**: grey-blue shins and palate.
- Chemotherapy (bleomycin flagellate streaks, 5-FU), zidovudine (nails), clofazimine (red-brown), heavy metals (argyria).
- Fixed drug eruption leaves round grey-brown patches at the same sites.
- Stop or switch the drug with the prescriber; pigmentation fades slowly (months–years) or may be permanent; photoprotection.

Overview

- **Amiodarone** causes two linked skin problems: **Phototoxicity** – burning/erythema within hours of sun exposure (UVA); up to 30–75% of patients; **Blue-grey (slate) pigmentation** of sun-exposed skin (~1–10%), usually after **months of treatment and high cumulative doses** (>40–160 g)
- The case: a 50-year-old man in a sunny country, 3 months after starting amiodarone (dose recently increased), with increased sun sensitivity and a facial photo showing blue-grey discoloration.

Aetiology: causes and risk factors

Myths to address

- **"IT'LL GO AS SOON AS THE DRUG STOPS"**
  dermal pigment can persist for years.

Pathophysiology and pathology

Drugs pigment skin by **increasing melanin, depositing themselves, or forming drug–iron–melanin complexes**.

1. **Increased melanin**
   Some drugs stimulate melanocytes (oral contraceptives in melasma, chemotherapy).

2 **Drug deposition**

Drug or metabolites deposit in the dermis (amiodarone lipofuscin, gold, silver).

3 **Complexes**

Drug, iron and melanin bind together in dermal macrophages (minocycline).

4 **Post-inflammatory**

Inflammatory reactions leave melanin in the dermis (fixed drug eruption).

How it is caused

- Amiodarone and its metabolite **desethylamiodarone** absorb **UVA (300–400 nm)** and generate reactive oxygen species phototoxic injury.
- Both are **lipophilic**, with an extremely **long half-life (~50 days, range 20–100)**. They accumulate in dermal macrophages as **lipofuscin-like granules with drug–lipid complexes**, and melanin rises blue-grey colour.

Clinical features

Burning, erythema and prickling minutes to hours after sun, even through glass (UVA). Later, **blue-grey/violaceous** colour of the **face (nose, cheeks), ears and backs of hands**. Also look for other amiodarone toxicity: thyroid (hypo-/hyperthyroidism), liver, lung (pneumonitis/fibrosis), eyes (corneal microdeposits, optic neuropathy), peripheral neuropathy.

Fig. 1 · Blue-grey (slate) pigmentation of sun-exposed facial skin after long-term amiodarone. Fades over months–years after dose reduction/stop.

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Diagnosis and investigations

Clinical (history + distribution). Biopsy only if doubtful (yellow-brown granules in macrophages; Fontana-Masson). Check **TFT, LFT**, CXR ± PFT, and arrange ophthalmology review per the amiodarone monitoring schedule (BNF: TFT/LFT every 6 months).

Differential diagnoses

Causes of photosensitivity:

> **EXOGENOUS**
> systemic drugs — phototoxic or photoallergic (see Drug-induced photosensitivity — thiazide case); topical agents — fragrances, sunscreens (photoallergic), **phytophotodermatitis** (psoralens in limes, figs, celery, giant hogweed), tar

> **ENDOGENOUS/METABOLIC**
> **porphyrias** (PCT, EPP), pellagra (niacin deficiency).

> **IDIOPATHIC PHOTODERMATOSES**
> **polymorphic light eruption (PLE)**, actinic prurigo, **chronic actinic dermatitis**, solar urticaria, hydroa vacciniforme.

> **GENETIC**
> xeroderma pigmentosum, albinism, Bloom, Cockayne and Rothmund–Thomson syndromes.

Blue-grey pigmentation DDx: minocycline, antimalarials (hydroxychloroquine — shins/palate), phenothiazines (chlorpromazine), tricyclics (imipramine), argyria (silver), chrysiasis (gold), ochronosis (hydroquinone), Addison's (brown, not blue — the skin signs of systemic disease notes), haemochromatosis, cyanosis, methaemoglobinaemia (dapsone).

Management

1. **Do not stop amiodarone yourself.** Discuss with the **cardiologist** (dose reduction or an alternative antiarrhythmic, rate/rhythm strategy, ablation).
2. The BNF advises avoiding sun exposure **during treatment and for several months after stopping** (long half-life).
3. **Pigmentation** fades very slowly over months to years after stopping or reducing the dose, and may be permanent. Camouflage makeup; Q-switched/picosecond laser (specialist, private) for persistent cases.
4. **Review other photosensitisers** (thiazides, doxycycline, quinolones, NSAIDs).
5. Monitoring TFT, LFT (6-monthly), annual eye check, CXR baseline. Yellow Card.

IP scope

Counselling, photoprotection, monitoring, liaison with cardiology, Yellow Card. Antiarrhythmic changes are specialist/shared-care decisions.

Referral

ROUTINE — New pigmentation on amiodarone: check **thyroid and liver function** and ask about breathlessness (amiodarone toxicity).

ROUTINE — Generalised pigmentation with tiredness, weight loss or low blood pressure: exclude **Addison's disease**.

Red flags and when to refer

- New pigmentation on amiodarone: check **thyroid and liver function** and ask about breathlessness (amiodarone toxicity).
- Generalised pigmentation with tiredness, weight loss or low blood pressure: exclude **Addison's disease**.

Red flags / refer

Blistering phototoxicity; new breathlessness/cough (pneumonitis); visual change (optic neuropathy); thyroid symptoms; jaundice.

Complications and prognosis

Complications

- Cosmetic distress; marker of cumulative dose (amiodarone).

Prognosis

- Variable; laser may help some (minocycline).

Prescriber pearls (UK pharmacist IP)

- "Amiodarone sensitivity **goes through window glass**" — UVA, so driving and office windows count.
- Effects persist for months after stopping (half-life ~50 days), and the pigment for longer.
- Counsel at the first supply — the BNF lists phototoxicity among the amiodarone counselling points.

Self-check quiz

5 questions · optional

Q1 When should you think of Drug-induced pigmentation (amiodarone, minocycline)?

✓ Blue-grey or slate discolouration of the face or sun-exposed skin in someone on long-term amiodarone or minocycline.

Q2 What drives it?

✓ Drug or drug–melanin complexes deposited in the dermis, often worsened by UV light.

Q3 What is the backbone of care?

✓ Medication review with the prescriber (amiodarone is often essential); sun protection; explain fading takes months to years.

Q4 What must you never miss?

✓ Other amiodarone toxicity (thyroid, liver, lung, eyes); Addison's disease or haemochromatosis as non-drug causes.

Q5 Name a red flag that needs escalation.

✓ New pigmentation on amiodarone: check **thyroid and liver function** and ask about breathlessness (amiodarone toxicity).

References

1. DermNet: Drug induced pigmentation — https://dermnetnz.org/topics/drug-induced-pigmentation
dermnetnz.org
2. Medscape: Drug-induced pigmentation (amiodarone, minocycline) — https://emedicine.medscape.com/article/1069686-overview
emedicine.medscape.com

Photo gallery

Amiodarone and minocycline pigmentation 2

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Drug eruptions Module 1 · Week 3

Fixed drug eruption

A round dusky patch or blister that comes back in exactly the same place each time a drug is taken, leaving brown-grey pigmentation. Common triggers are NSAIDs, paracetamol, antibiotics and laxatives.

ALSO CALLED FDE | recurring rash same place | NSAID | co-trimoxazole | paracetamol | lip FDE | +1

RELATED PAGES Approach to a suspected drug eruption | Erythema multiforme | SJS/TEN and drug-induced blistering | Herpes simplex labialis (cold sores) | Drug-induced pigmentation (amiodarone, minocycline)

KEY FACTS

- **Fixed drug eruption (FDE)** recurs at the **same site** each time the culprit drug is taken.
- Round dusky red-purple patch, sometimes blistering, within hours; lips, genitals, hands, face; leaves grey-brown pigmentation.
- Culprits: NSAIDs, paracetamol, co-trimoxazole, tetracyclines, laxatives (phenolphthalein historically).
- Stop and avoid the drug; record the allergy; potent topical steroid for a few days.
- Generalised bullous FDE can mimic SJS/TEN: urgent assessment.

Overview

A drug reaction that **recurs at exactly the same site(s)** each time the drug is taken. Usually there are one or a few round, dusky-violet plaques, leaving **slate-brown/grey pigmentation**.

Epidemiology

- **Who**
  - Any age; often over-the-counter painkillers; recurrent episodes before diagnosis.

Aetiology: causes and risk factors

Risk factors and associations

- Intermittent use of a drug, often OTC.

Myths to address

- **"IT'S AN INFECTION LIKE HERPES"**
  recurrence at the same site after a drug points to FDE (genital FDE is often mistaken for HSV).

Pathophysiology and pathology

FDE is caused by **memory T cells that stay in the skin** at the original site.

1. **First reaction**
   The drug triggers CD8+ T cells that attack keratinocytes at one site.
2. **Resident memory**
   Some CD8+ T cells remain in the epidermis of that patch long-term.

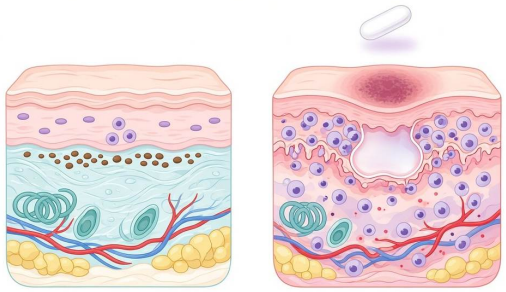

Fig. 1 · Fixed drug eruption: memory T cells resting at the site (left) are reactivated by the drug, causing interface damage, a blister and a dusky patch (right).
AI-generated illustration

3. **Re-exposure**
   Each dose reactivates them within hours, causing a dusky patch, sometimes with blistering.
4. **Pigment**
   Damaged basal cells drop melanin into the dermis, leaving persistent brown-grey pigment.

How it is caused

Intraepidermal CD8+ tissue-resident memory T cells persist at the original site. On re-exposure they activate within hours and release IFN-γ and granzymes localised keratinocyte death (type Mc, like a "mini-TEN"). Melanin dropping into the dermis gives the persistent pigment.

Common culprits: NSAIDs (diclofenac, ibuprofen, naproxen, **mefenamic acid** — note the menstrual link), **co-trimoxazole/sulfonamides**, **tetracyclines**, **paracetamol**, metronidazole, fluconazole, barbiturates, carbamazepine, **sildenafil** (genital), antihistamines (cetirizine, hydroxyzine), quinine (in tonic water), phenolphthalein (old laxatives) and food dyes.

HISTOPATHOLOGY

- Interface dermatitis with necrotic keratinocytes and pigment incontinence; eosinophils.

Clinical features

- **Lips**, **genitals** (glans penis), hands, feet, face; the trunk and limbs less often.
- Round/oval, sharply demarcated, **dusky red-violet plaques**, often with a darker/targetoid or bullous centre. Burning or itch appears **30 min–8 h** (up to 48 h) after a dose.

- Each recurrence may add **new sites**. Lesions resolve over 1–3 weeks with **post-inflammatory hyperpigmentation**, which is especially marked in skin of colour.
- **GENERALISED BULLOUS FDE (GBFDE)**
  multiple large blistering plaques (>10% BSA) in older patients. It mimics SJS/TEN, carries significant mortality (~20%) and **needs admission**.
- **THE CASE**
  a 26-year-old with a **pigmented patch on the upper lip**, after **diclofenac during two consecutive periods**. Erythema/blistering recurring at the same site, then pigmentation **FDE (diclofenac)**. It had been treated as a **cold sore**.

Fig. 2 · Pigmented fixed drug eruption on the lip.

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Special groups

- **Skin of colour**
  - Pigmentation is marked and long-lasting.

Diagnosis and investigations

- History of **recurrence at the same site** after intermittent drug use (menstrual NSAIDs, weekend antibiotics, "headache tablets") — ask specifically about intermittent OTC products.
- Biopsy if doubtful (interface dermatitis, necrotic keratinocytes, dermal melanophages).
- **Lesional patch test** (drug applied to the old site) or supervised **oral provocation** in specialist settings. Not for GBFDE.

Differential diagnoses

	FDE	HERPES LABIALIS (THE MISDIAGNOSIS)	EM	INSECT BITE	PIH / MELASMA
Recurs at same site	**Yes, after the drug**	Yes, after triggers (sun, stress, menses)	No	No	—

	FDE	HERPES LABIALIS (THE MISDIAGNOSIS)	EM	INSECT BITE	PIH / MELASMA
Prodrome	Burning hours after the drug	Tingling 24 h before	HSV preceding	–	–
Lesion	Dusky plaque ± bulla, **then slate pigmentation**	**Grouped vesicles** crust, no lasting pigment	Typical targets, acral	Papule with punctum	Flat brown, no inflammation
Test	Patch/provocation	**HSV PCR swab**	Clinical	–	–

Also: lichen planus pigmentosus, Bowen's disease (fixed, but persistent), bullous pemphigoid, SJS/TEN (vs GBFDE).

Management

1. **Stop and avoid diclofenac.** Record "**diclofenac – fixed drug eruption (lip)**", and inform her that structurally related NSAIDs (acetic-acid group: aceclofenac, indometacin, etodolac) may cross-react. Some patients tolerate structurally unrelated NSAIDs, but cross-reactivity is unpredictable – specialist advice before a trial.
2. Dysmenorrhoea alternatives **paracetamol**, local heat, a hormonal method (COC/LNG-IUS/desogestrel) that suppresses ovulation/pain, TENS. Consider gynaecology review if dysmenorrhoea is severe (endometriosis).
3. Acute lesion potent topical steroid for a few days (a moderate steroid on the lip, short course), emollient/lip balm; for erosions, non-adherent care. The pigmentation fades slowly (months), and more slowly with repeated exposure – sun protection helps.
4. **Check OTC purchases** (diclofenac gel, combination cold remedies, mefenamic acid) and tell her community pharmacy. Yellow Card (FDE is reportable, especially if bullous).

IP scope

Fully within IP scope (recognition, stopping the drug, alternative analgesia/contraception, topical steroid, documentation, Yellow Card). Provocation testing is specialist.

Referral

SAME DAY — Many or widespread blistering patches, mucosal involvement or an unwell patient: possible **generalised bullous FDE**, same-day assessment.

ROUTINE — A 'cold sore' that always appears after the same tablet: ask about over-the-counter NSAIDs and paracetamol combinations.

Red flags and when to refer

- Many or widespread blistering patches, mucosal involvement or an unwell patient: possible **generalised bullous FDE**, same-day assessment.

- A 'cold sore' that always appears after the same tablet: ask about over-the-counter NSAIDs and paracetamol combinations.

Red flags / refer

Multiple/generalised blistering plaques (GBFDE same-day admission); mucosal erosions beyond the lip; uncertain diagnosis (genital lesions — exclude STI/HSV).

Complications and prognosis

Complications

- Persistent pigmentation; generalised bullous FDE (rare, serious).

Prognosis

- Does not recur if the drug is avoided; pigment fades slowly.

Prescriber pearls (UK pharmacist IP)

- "**Same place, same drug, every time.**" Ask what she took in the 24 h before each flare.
- A "cold sore" that leaves a **dark patch** and no vesicles is FDE until proven otherwise.
- Menstrual NSAIDs are a classic hidden culprit; so are the "weekend antibiotics" for UTIs (co-trimoxazole/trimethoprim).

Self-check quiz

5 questions · optional

Q1 When should you think of Fixed drug eruption?

✓ A round, dusky red-brown patch on the lip, genitals, hands or face that recurs at the same site, often mistaken for a cold sore.

Q2 What drives it?

✓ Memory T cells sitting in the skin at that site are reactivated by the drug within hours.

Q3 What is the backbone of care?

✓ Find the intermittent drug (ask about period pain relief, cold remedies, laxatives); stop and avoid it; potent topical steroid; document.

Q4 What must you never miss?

✓ Generalised bullous FDE, which can resemble SJS/TEN and needs same-day assessment.

Q5 Name a red flag that needs escalation.

✓ Many or widespread blistering patches, mucosal involvement or an unwell patient: possible **generalised bullous FDE**, same-day assessment.

References

1. DermNet: Fixed drug eruption – https://dermnetnz.org/topics/fixed-drug-eruption
   dermnetnz.org
2. Medscape: Fixed drug eruption – https://emedicine.medscape.com/article/1336702-overview
   emedicine.medscape.com

Photo gallery

Fixed drug eruption 4

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CHAPTER 20

Vasculitis & purpura

Palpable purpura, urticarial vasculitis, polyarteritis nodosa, livedo, IgA vasculitis, endocarditis signs and the purpura differential, with the same-day checks.

1. Approach to cutaneous vasculitis
2. Cutaneous small-vessel (leukocytoclastic) vasculitis
3. Urticarial vasculitis & HUVS
4. Polyarteritis nodosa (systemic & cutaneous)
5. Livedo reticularis vs livedo racemosa
6. IgA vasculitis (Henoch–Schönlein purpura)
7. Osler nodes, Janeway lesions & splinter haemorrhages
8. Purpura: the differential diagnosis

Vasculitis & purpura Module 1 · Week 4

Approach to cutaneous vasculitis

How to recognise vasculitis in the skin, link the lesion to the vessel size involved, look for organ involvement the same day, and choose the tests and referral speed.

ALSO CALLED palpable purpura · vessel size · ANCA · Chapel Hill · urine dipstick · leukocytoclastic

RELATED PAGES Cutaneous small-vessel (leukocytoclastic) vasculitis · IgA vasculitis (Henoch–Schönlein purpura) · Urticarial vasculitis & HUVS · Polyarteritis nodosa (systemic & cutaneous) · Livedo reticularis vs livedo racemosa

KEY FACTS

- **Vasculitis** is classified by vessel size: **small** (palpable purpura), **medium** (nodules, livedo racemosa, ulcers), **large** (rarely skin).
- Always ask: is it skin-limited or systemic? Check **urine dipstick and BP** in every patient.
- Biopsy an early lesion (<48 h) with direct IF; deep biopsy for medium-vessel disease.
- Screen: FBC, U&E, LFT, CRP, ANA, ANCA, complement, hepatitis B/C, HIV, cryoglobulins, immunoglobulins.
- Same-day: unwell, renal involvement, haemoptysis, neuropathy, gut ischaemia, or child with possible meningococcal disease.

Overview

Vasculitis is inflammation of blood-vessel walls causing damage (haemorrhage, ischaemia). The skin is involved in many systemic vasculitides, and can be the only organ affected (**single-organ cutaneous small-vessel vasculitis**, the commonest form seen in dermatology).

Chapel Hill Consensus Conference (CHCC 2012) nomenclature:

CATEGORY	EXAMPLES
Large vessel	Giant-cell arteritis, Takayasu arteritis
Medium vessel	**Polyarteritis nodosa**, Kawasaki disease
Small vessel — ANCA-associated	Microscopic polyangiitis (MPA), **granulomatosis with polyangiitis (GPA)**, eosinophilic GPA (EGPA)
Small vessel — immune complex	Anti-GBM disease, **cryoglobulinaemic vasculitis**, **IgA vasculitis (HSP)**, **hypocomplementaemic urticarial vasculitis (anti-C1q vasculitis)**
Variable vessel	Behçet's disease, Cogan syndrome
Single-organ	**Cutaneous leukocytoclastic angiitis**, cutaneous arteritis (= cutaneous PAN)

CATEGORY	EXAMPLES
Associated with systemic disease	Lupus, rheumatoid, sarcoid vasculitis
Associated with a probable aetiology	**HCV-associated cryoglobulinaemic vasculitis**, **HBV-associated PAN**, **drug-associated** immune-complex/ANCA vasculitis, cancer-associated vasculitis

Aetiology: causes and risk factors

Myths to address

- **"SKIN VASCULITIS IS JUST A RASH"**
  it can be the first sign of kidney or lung disease.

Pathophysiology and pathology

Vasculitis is **inflammation of vessel walls**; the size of vessel decides what you see.

1. **Small vessels**
   Immune complexes in post-capillary venules: palpable purpura, wheals.

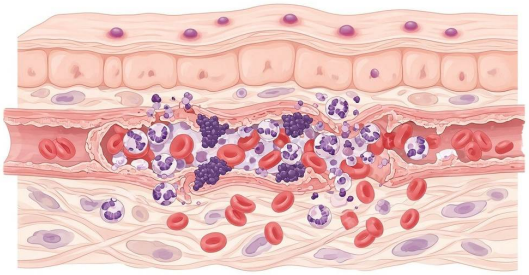

Fig. 1 · Small-vessel vasculitis with neutrophils and red-cell leakage.
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2. **Medium vessels**
   Arteries at the dermo-subcutaneous junction: nodules, livedo racemosa, ulcers, digital infarcts.

3. **ANCA vasculitis**
   ANCA activate neutrophils against small–medium vessels (GPA, MPA, EGPA).

4. **Systemic spread**
   The same process in kidney, lung, nerve and gut causes organ damage.

How it is caused

- **Immune-complex deposition** (antigen from a drug, infection, autoantigen or tumour complexes deposit in post-capillary venules complement activation neutrophil influx **leukocytoclasis** (nuclear dust), fibrinoid necrosis, red-cell extravasation = purpura).
- **ANCA-mediated** neutrophil activation (MPO/PR3).
- **Cell-mediated/granulomatous** inflammation.
- **Direct infection** of the vessel wall (septic vasculitis — meningococcus, gonococcus, endocarditis).
- **COMMON CAUSES OF CUTANEOUS SMALL-VESSEL VASCULITIS**

idiopathic (~45–55%), **infection** (~15–20% — streptococcal URTI, hepatitis B/C, HIV, endocarditis), **drugs** (~10–20% — β-lactams, sulfonamides, NSAIDs, allopurinol, thiazides, propylthiouracil/hydralazine (ANCA), TNF inhibitors, levamisole-contaminated cocaine), **connective tissue disease** (~15–20% — SLE, RA, Sjögren's), **malignancy** (<5% — haematological especially).

Clinical features

Clinical features & sites — lesion vessel size

VESSEL SIZE	SKIN LESIONS
Small (post-capillary venules, dermal)	**Palpable purpura**, petechiae, **urticarial plaques lasting >24 h**, haemorrhagic vesicles/bullae, pustules, superficial ulcers, splinter haemorrhages
Medium (muscular arteries at the dermal–subcutaneous junction)	**Subcutaneous nodules**, **livedo racemosa**, **deep punched-out ulcers**, digital infarcts/gangrene, papulonecrotic lesions
Mixed (ANCA, cryo, CTD)	Any of the above

Distribution: dependent areas — **lower legs, ankles, feet, buttocks**; under tight clothing; in bed-bound patients the back and buttocks. Symptoms: burning/stinging or pain more than itch. Look for **systemic symptoms** (fever, arthralgia, abdominal pain, haematuria, cough/haemoptysis, sinusitis, neuropathy).

Fig. 2 · Palpable purpura with haemorrhagic papules on the ankle — "palpable purpura is vasculitis until proven otherwise". Dipstick the urine and check BP today.

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Diagnosis and investigations

1. Confirm it is purpura **non-blanching on diascopy** (press a glass slide). Is it palpable (inflammatory vasculitis) or flat (petechiae/ecchymoses platelet/coagulation or capillaritis)?
2. **Skin biopsy** — the most useful test: **H&E:** punch from a **fresh lesion 24–48 h old** (LCV, vessel size); **Direct immunofluorescence (DIF):** a second biopsy from a lesion **<24 h old** (or perilesional), sent in **Michel's medium/saline, not formalin**. IgA IgA vasculitis; IgM/C3 cryoglobulins/immune complex; lupus band HUVS/SLE; For

suspected medium-vessel disease (nodules, livedo, ulcers), take a **deep incisional or elliptical biopsy to subcutaneous fat**, from a nodule or the **white centre of livedo**. A 4 mm punch often misses it

3. Baseline screen for cause and organ involvement **Urinalysis (dipstick for blood and protein — the key bedside test)**; urine ACR/PCR; urine microscopy (red-cell casts); **Blood pressure.**; **FBC** (platelets — exclude thrombocytopenia; eosinophilia — EGPA/drug), **U&E/eGFR**, LFTs, **CRP/ESR**, clotting; **Immunology:** ANA, ENA, dsDNA, **ANCA (MPO/PR3)**, **C3/C4** (± C1q, anti-C1q), rheumatoid factor, **cryoglobulins** (sample kept at **37 °C** in transport), IgA level; **Infection: hepatitis B and C**, HIV, ASO/anti-DNase B, throat swab, **blood cultures** (endocarditis), ± IGRA (before immunosuppression); **Paraprotein:** serum protein electrophoresis, immunofixation, serum free light chains; As indicated: **CXR**, ECG/echo, faecal occult blood/calprotectin, CT angiography (PAN), nerve conduction studies
4. Severity assessment BVAS (Birmingham Vasculitis Activity Score) in specialist care; Five-Factor Score for PAN/ANCA (see Polyarteritis nodosa — systemic, HBV-associated and cutaneous).

Differential diagnoses

NON-VASCULITIC PURPURA

thrombocytopenia (ITP, leukaemia, drugs), coagulopathy/anticoagulants, **DIC/meningococcal sepsis**, senile/actinic and **steroid purpura**, scurvy (perifollicular), capillaritis (pigmented purpuric dermatoses), trauma/non-accidental injury in children, raised intrathoracic pressure (petechiae above the nipple line after vomiting/coughing).

VASCULOPATHIES/OCCLUSION (NON-INFLAMMATORY)

antiphospholipid syndrome, cholesterol emboli, calciphylaxis, livedoid vasculopathy, warfarin necrosis, cryoglobulin/cryofibrinogen occlusion, septic emboli.

Also: pigmented purpuric dermatoses, stasis dermatitis, erythema multiforme/urticaria (blanching).

Management

TREAT THE CAUSE

stop a suspected drug (record + Yellow Card); treat infection.

SKIN-LIMITED, MILD DISEASE

rest, **leg elevation**, **compression hosiery** if arterial supply is adequate, emollient, potent TCS for symptoms, antihistamine; NSAIDs only if not the culprit and renal function is normal. Most self-limiting episodes resolve in **3–4 weeks**.

CHRONIC/RECURRENT SKIN-LIMITED DISEASE (SPECIALIST)

colchicine, **dapsone** (G6PD check; FBC/methaemoglobin monitoring), hydroxychloroquine; short prednisolone courses for necrotic/ulcerated disease; azathioprine/methotrexate/MMF for steroid-sparing.

SYSTEMIC VASCULITIS

specialist (rheumatology/renal) — glucocorticoids + cyclophosphamide/rituximab (ANCA), etc.

MONITORING

clinical (old lesions healing, new crops) + ESR/CRP; **urinalysis and BP** at each review — renal involvement can appear weeks later.

IP scope

Recognition, urinalysis/BP, first-line bloods, stopping culprit drugs, symptomatic care and correct-urgency referral are within IP scope. Biopsy, immunosuppression and dapsone/colchicine regimens are specialist-led (shared care at most).

Worked cases

Case 1

LESIONS SUGGESTING VASCULITIS

palpable purpura, petechiae, urticarial plaques lasting >24 h, haemorrhagic vesicles/bullae, pustules, necrosis/ulcers, nodules, livedo racemosa, digital infarcts, splinter haemorrhages (see Approach to cutaneous vasculitis).

HOW TO DIAGNOSE

clinical suspicion **biopsy (H&E 24–48 h lesion + DIF <24 h; deep incisional for medium vessels)** screen for cause and organ involvement (urinalysis, BP, FBC, U&E, LFT, CRP/ESR, ANA/ENA/dsDNA, ANCA, C3/C4, cryoglobulins, RF, hepatitis B/C, HIV, ASO, SPEP, blood cultures, CXR) (see Approach to cutaneous vasculitis).

OSLER/JANEWAY

Osler = tender immunological finger-pad nodules; Janeway = painless septic-embolic palm/sole macules. **Not pathognomonic** (SLE, gonococcaemia, typhoid, haemolytic anaemia, marantic endocarditis). **Splinters:** commonly traumatic/psoriatic; no validated threshold — new, multiple, proximal or multi-nail splinters plus fever/murmur raise concern (see Osler nodes, Janeway lesions and splinter haemorrhages).

CASE 2

36-year-old woman

DIAGNOSIS

hypocomplementaemic urticarial vasculitis (HUVS / anti-C1q vasculitis), with **lupus overlap** to exclude (ANA+, photosensitivity, arthralgia, low C3/C4, renal impairment): Strictly, Schwartz criteria require **6 months** of urticaria; she has had 3 months. Label it "hypocomplementaemic urticarial vasculitis, probable HUVS"; The **creatinine of 2.3 mg/dL (~203 µmol/L)** indicates renal involvement urgent nephrology

CONFIRMATORY TESTS

skin biopsy with **DIF** (immunoreactants around vessels ± lupus band); **C1q level and anti-C1q antibodies**; CH50; **dsDNA, ENA**, antiphospholipid antibodies; **urinalysis, uPCR, urine microscopy ± renal biopsy**; cryoglobulins; hepatitis B/C; SPEP; CXR/PFTs; ophthalmology review (see Urticarial vasculitis and hypocomplementaemic urticarial vasculitis syndrome).

Case 3

TASK 1

UV vs urticaria table; investigations (see Urticarial vasculitis and hypocomplementaemic urticarial vasculitis syndrome).

3A

symmetrical non-blanching **palpable purpuric papules** (some confluent/haemorrhagic) on the lower legs.

3B

cutaneous leukocytoclastic (small-vessel) vasculitis, drug-induced (NSAID).

3C

hydrostatic pressure, venous stasis and slow flow in dependent venules immune-complex deposition (hence rest/elevation/hosiery).

3D

see Cutaneous small-vessel (leukocytoclastic) vasculitis.

3E

urinalysis.

3F

see Cutaneous small-vessel (leukocytoclastic) vasculitis.

3G

stop the NSAID, alternative analgesia, rest/elevation/compression, potent TCS/antihistamine; dapsone/colchicine/short prednisolone if persistent or necrotic; monitor clinically + ESR/CRP + urine/BP (see Cutaneous small-vessel (leukocytoclastic) vasculitis).

CASE 4

30-year-old woman, HBsAg+

DIAGNOSIS

HBV-associated systemic polyarteritis nodosa.: ACR criteria met: weight loss >4 kg, livedo, myalgia/leg tenderness, neuropathy, diastolic BP >90, raised creatinine (2.7 mg/dL 239 µmol/L), **HBsAg+** = **7 of 10**; Predictive items: HBsAg+, neuropathy positive; ANCA/cryo/asthma/ENT/GN negative

SUPPORTING

nodules along arteries, necrotic ulcers, anaemia, thrombocytosis, leucocytosis. Renal impairment with **hypertension** reflects **renal-artery vasculopathy (renovascular hypertension), not GN**.

INVESTIGATIONS STILL NEEDED

CT/MR or catheter angiography (renal/mesenteric microaneurysms), deep skin/muscle/nerve biopsy, HBeAg/HBV DNA, LFT, nerve conduction studies, urinalysis (bland sediment expected), echo/ECG.

DDX

ANCA-associated vasculitis (MPA), cryoglobulinaemic vasculitis, SLE vasculitis, infective endocarditis, APS, cholesterol emboli, lymphoma, DADA2.

TREATMENT

antiviral (entecavir/tenofovir) + short-course steroids + plasma exchange (avoid prolonged immunosuppression in HBV-PAN); BP control; specialist MDT (rheumatology, hepatology, renal); contact tracing/vaccination (see Polyarteritis nodosa — systemic, HBV-associated and cutaneous).

Case 5

Diagnosis of PAN; cutaneous vs systemic PAN; causes of livedo and vessel involvement; treatment (see Polyarteritis nodosa — systemic, HBV-associated and cutaneous and Livedo reticularis vs livedo racemosa).

Practice case: 7-year-old boy

> **QA = 4, HENOCH–SCHÖNLEIN PURPURA (IGA VASCULITIS)**
> purpura + arthralgia + abdominal pain + haematuria/proteinuria after a URTI; normal platelets.

> **QB = 4, LCV WITH PERIVASCULAR IGA AND C3 ON DIF.**
> The course explanation attributes "necrotising vasculitis of medium-sized arteries with negative DIF" to **GPA** as well as PAN. The correct match is: option 1 = **PAN**; option 2 (**small- and medium-vessel vasculitis with necrosis and granulomas**) = **GPA**; option 3 = giant-cell arteritis; option 5 (IgM) = cryoglobulinaemia

> **QC = 2, group A β-haemolytic streptococcus.**

> **UK ADDITIONS**
> exclude meningococcal sepsis first. The urinalysis already shows blood + protein, so **check BP and the urine protein:creatinine ratio, and discuss with paediatrics** (CKS). Then a structured urine/BP follow-up for 6 months.

Referral

SAME DAY	Blood or protein on urine dipstick, or raised BP: same-day discussion with renal or medical team.
999 / A&E	Non-blanching rash in an unwell or feverish patient: **suspected meningococcal disease**, 999 (NICE NG240).
URGENT	Haemoptysis, breathlessness, neuropathy or abdominal pain with purpura: systemic vasculitis, urgent referral.

Red flags and when to refer

- Blood or protein on urine dipstick, or raised BP: same-day discussion with renal or medical team.
- Non-blanching rash in an unwell or feverish patient: **suspected meningococcal disease**, 999 (NICE NG240).
- Haemoptysis, breathlessness, neuropathy or abdominal pain with purpura: systemic vasculitis, urgent referral.

Red flags / refer

> **SAME DAY (999/ED)**
> purpura in an **unwell or febrile** patient — especially a child — **meningococcal sepsis/DIC** (NICE NG240); suspected **endocarditis**; rapidly spreading necrosis; haemoptysis; acute kidney injury; GI bleeding/severe abdominal pain; mononeuritis multiplex.

> **URGENT (DAYS)**
> palpable purpura with **haematuria/proteinuria**, raised creatinine or **hypertension** (renal); systemic symptoms; positive ANCA; nodules/ulcers/livedo racemosa (medium-vessel disease); persistent >4 weeks or recurrent.

> **ROUTINE DERMATOLOGY**

> skin-limited, well patient, normal urine/bloods, cause unclear.

Complications and prognosis

Complications

- Renal failure, pulmonary haemorrhage, neuropathy, gut infarction, ulcers.

Prognosis

- Skin-limited forms usually settle; systemic vasculitis needs immunosuppression.

Prescriber pearls (UK pharmacist IP)

"**Glass test first.**" Non-blanching + palpable = vasculitis work-up; non-blanching + unwell = 999.

Always **dip the urine** and **check BP** — then repeat at follow-up.

Ask about **every new drug in the last 1–3 weeks**, including OTC ibuprofen and recreational cocaine (levamisole).

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK POSITION (2026)
course explanation	Course teaching: "necrotising vasculitis of medium-sized arteries with negative DIF" for **GPA** (and PAN)	That option fits **PAN**. GPA = **necrotising small/medium-vessel vasculitis with granulomas** (option 2)
HSP management	course explanation: monitor urine and BP during active disease	NICE CKS: **exclude meningococcal disease** and leukaemia/ITP first; a **structured urinalysis + BP schedule for 6 months**; same-day referral criteria (abdominal, testicular, renal, unwell); paracetamol first, NSAIDs only if no renal involvement
HSP epidemiology	course explanation: ~75% under 8; more severe in adults	Agrees; teach that adults get HSP too (biopsy with DIF — higher nephritis risk)
HBV-PAN treatment	A student answer: prednisolone 1 mg/kg tapered over 6 months ± cyclophosphamide for 12 months, after antivirals	For **HBV-associated PAN**: **antivirals + short steroid course + plasma exchange**, avoiding prolonged steroids/cyclophosphamide (HBV flare risk). Long steroid ± cyclophosphamide applies to severe **non-HBV** PAN (ACR/VF 2021; FFS-guided)
PAN renal disease	A student answer: "high risk of renal involvement"	Correct, but in PAN it is **vascular (renovascular hypertension, infarcts)**; **glomerulonephritis argues against PAN** (a negative predictive item)
HUVS diagnosis	Students: HUVS	Agrees, but the 3-month history does not meet the Schwartz **6-month** criterion, and the SLE overlap must be formally assessed; urgent renal referral for the raised creatinine

TOPIC	COURSE TEACHING	UK POSITION (2026)
LCV bedside test	Course teaching: urinalysis	Agrees; UK practice adds **BP** at the same visit and repeated urinalysis/BP at follow-up
LCV treatment	Course teaching: bed rest, hosiery, dapsone	Agrees; dapsone requires **G6PD testing** and FBC/LFT/methaemoglobin monitoring (BNF); check arterial supply before compression
Endocarditis signs	Students: Osler/Janeway "pathognomonic" in some student answers	Not pathognomonic; **Duke–ISCVID 2023** minor criteria; **blood cultures before antibiotics**; NICE CG64: no routine antibiotic prophylaxis
Units	Course labs in mg/dL	UK: creatinine µmol/L (× 88.4), urea mmol/L (BUN mg/dL × 0.357)

Self-check quiz

5 questions · optional

Q1 When should you think of Approach to cutaneous vasculitis?

✓ Palpable purpura, haemorrhagic blisters, nodules, livedo or ulcers, especially on the lower legs.

Q2 What drives it?

✓ Inflammation of vessel walls from immune complexes, ANCA or infection; small vessels give purpura, medium vessels give nodules, livedo racemosa and ulcers.

Q3 What is the backbone of care?

✓ Urine dipstick and BP today; FBC, U&E, LFT, CRP, ANCA, complement, hepatitis B/C, cryoglobulins as indicated; biopsy within 48 h; stop culprit drugs.

Q4 What must you never miss?

✓ Renal involvement, systemic ANCA vasculitis, endocarditis and meningococcal sepsis.

Q5 Name a red flag that needs escalation.

✓ Blood or protein on urine dipstick, or raised BP: same-day discussion with renal or medical team.

References

1. DermNet: Cutaneous small vessel vasculitis — https://dermnetnz.org/topics/cutaneous-small-vessel-vasculitis

dermnetnz.org

Vasculitis & purpura Module 1 · Week 4

Cutaneous small-vessel (leukocytoclastic) vasculitis

Crops of palpable purpura on the lower legs, often 1–3 weeks after a drug (such as an NSAID) or infection. Usually skin-limited, but every patient needs a urine and BP check.

ALSO CALLED CSVV · hypersensitivity vasculitis · NSAID vasculitis · palpable purpura · drug-induced vasculitis

RELATED PAGES IgA vasculitis (Henoch–Schönlein purpura) · Urticarial vasculitis & HUVS · Approach to cutaneous vasculitis · Purpura: the differential diagnosis · Polyarteritis nodosa (systemic & cutaneous)

KEY FACTS

- **Cutaneous small-vessel (leukocytoclastic) vasculitis** presents as **palpable purpura** on the lower legs, often in crops.
- Causes: drugs, infection (strep, hepatitis B/C), autoimmune disease, malignancy; up to half idiopathic.
- Look for systemic disease: urine dipstick and BP, FBC, U&E, LFT, CRP, ANA, ANCA, complement, hepatitis serology, cryoglobulins.
- Biopsy an early lesion (<48 h) for histology and direct IF.
- Stop culprit drug, rest, elevation, compression; refer if systemic or recurrent.

Overview

Leukocytoclastic vasculitis (LCV) is the histological term for neutrophil-rich small-vessel vasculitis of dermal post-capillary venules. Clinically, when it is **single-organ**, it is called **cutaneous small-vessel vasculitis (CSVV)**, **hypersensitivity vasculitis** or cutaneous leukocytoclastic angiitis (CHCC). Case 3 is **drug-induced LCV after an NSAID**.

Epidemiology

- **Who**
  - Any age; adults most often.

Aetiology: causes and risk factors

Myths to address

- **"PURPURA IS ALWAYS LOW PLATELETS"**
  palpable purpura with normal platelets suggests vasculitis.

Pathophysiology and pathology

Immune complexes trigger **neutrophils to destroy small vessels**.

1. **Antigen**
   A drug, infection or autoantigen forms immune complexes.
2. **Deposition**
   Complexes lodge in post-capillary venules, especially in dependent legs.
3. **Neutrophil damage**
   Complement attracts neutrophils that fragment (leukocytoclasis) and damage vessels.

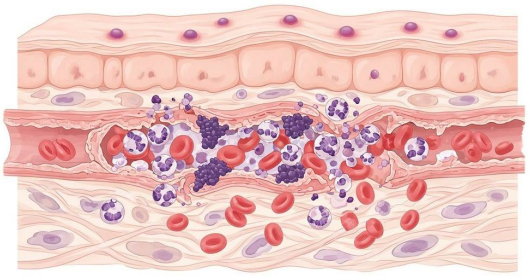

Fig. 1 · Immune complexes and fragmenting neutrophils damage the vessel wall; red cells leak out.
AI-generated illustration

4. **Purpura**
   Red cells leak, causing raised purpura that does not blanch.

How it is caused

Immune complexes (drug–antibody) form and deposit in venules, activating complement and attracting neutrophils.

- Drug-induced LCV usually starts **7–21 days** after a new drug (sooner on re-exposure; this case says "a few days").
- **WHY THE LEGS?**
  gravity/hydrostatic pressure and **venous stasis** in dependent limbs slower flow and turbulence in post-capillary venules; so immune complexes deposit more readily; plus cooler peripheral temperature; and tight footwear/socks — the same reason it appears on the buttocks/back in people lying in bed; That is why **bed rest, elevation and support hosiery** help

HISTOPATHOLOGY

- Fibrinoid necrosis of small vessels, neutrophils with nuclear dust, red cell extravasation.

Clinical features

Clinical features & sites

Symmetrical, non-blanching, palpable purpuric papules and macules (1–3 mm), some coalescing, on the **lower legs/ankles/dorsal feet**, appearing in crops; ± haemorrhagic vesicles, pustules, small necrotic centres or shallow ulcers; ± mild oedema. Burning/itch. Well otherwise.

Fig. 2 · Palpable purpura with haemorrhagic papules on the ankle — "palpable purpura is vasculitis until proven otherwise". Dipstick the urine and check BP today.

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Diagnosis and investigations

- **3e — the single bedside test: urinalysis** (dipstick for **blood and protein**). It tells you whether the kidneys are involved (systemic vasculitis/IgA nephritis), which changes urgency and management. Measure BP at the same visit.
- **3F — OTHER INVESTIGATIONS**
  FBC (platelet count — exclude thrombocytopenic purpura; eosinophilia), U&E/eGFR, LFT, CRP/ESR, clotting; **urine ACR/PCR** and microscopy; **ANA, ANCA, C3/C4, RF, cryoglobulins**; **hepatitis B/C, HIV**, ASO/throat swab; **SPEP**/immunofixation; **IgA**; blood cultures if febrile; faecal occult blood if abdominal pain; CXR if respiratory symptoms; **skin biopsy for H&E (LCV) + DIF** (IgA deposition IgA vasculitis even in adults, with a higher nephritis risk)

Differential diagnoses

IgA vasculitis (HSP — adults, abdominal pain, arthritis, haematuria), urticarial vasculitis, cryoglobulinaemic vasculitis (HCV), ANCA vasculitis (MPA/GPA — renal/lung), vasculitis secondary to CTD (SLE, RA, Sjögren's), septic vasculitis/emboli (endocarditis, meningococcaemia, gonococcaemia), thrombocytopenic purpura (ITP, drug-induced — **NSAIDs rarely cause thrombocytopenia too**), pigmented purpuric dermatosis, stasis/capillaritis, **exercise-induced vasculitis** ("golfer's vasculitis" — after long walks in heat), scurvy, Waldenström hypergammaglobulinaemic purpura.

Management

1. **Stop the NSAID.** Record "NSAID (name) — cutaneous vasculitis" and **Yellow Card**. Advise against OTC NSAIDs (ibuprofen, naproxen, aspirin in analgesic doses) until specialist advice clarifies cross-reactivity.
2. Back pain analgesia without NSAIDs (NICE NG59) exercise/physiotherapy, heat, and — if needed — a short course of a **weak opioid ± paracetamol** (paracetamol alone is not recommended for low back pain in NG59, but may be used as an adjunct). Avoid gabapentinoids for non-specific back pain.
3. General measures rest, **leg elevation**, **graduated compression hosiery** (check ABPI/pulses if arterial disease is possible), emollient, potent TCS on symptomatic lesions, antihistamine for itch. Expect resolution in **2–4 weeks**, sometimes with residual pigmentation.
4. Ulcerating/necrotic or persistent (>4 weeks) disease specialist — **short oral prednisolone course** (e.g. 0.5 mg/kg tapering over 2–4 weeks), **dapsone** (course teaching; 50–100 mg/day after G6PD check; FBC/LFT/methaemoglobin monitoring), **colchicine** 500 micrograms once–twice daily; immunosuppressants for chronic disease.
5. Monitoring clinical (old lesions healing, new crops) + **ESR/CRP**; repeat **urinalysis and BP** at each review (e.g. weekly while active).

IP scope

Within IP scope: stop the NSAID, alternative analgesia, urinalysis/bloods, compression and topical care, Yellow Card, and referral. Systemic steroids/dapsone are usually specialist-advised.

Referral

SAME DAY — Haematuria, proteinuria or rising creatinine: same-day renal advice.

URGENT — Necrosis, ulceration or purpura above the waist: consider systemic vasculitis and refer urgently.

Red flags and when to refer

- Haematuria, proteinuria or rising creatinine: same-day renal advice.
- Necrosis, ulceration or purpura above the waist: consider systemic vasculitis and refer urgently.

Red flags / refer

Haematuria/proteinuria, raised creatinine, hypertension, systemic symptoms, abdominal pain/GI bleed, haemoptysis, neuropathy, necrosis/ulceration, persistence >4 weeks, recurrence, or positive ANCA/cryoglobulins urgent specialist (rheumatology/renal/dermatology). Fever + purpura + unwell same day.

Complications and prognosis

Complications

- Ulceration; systemic vasculitis (kidney, gut, nerves).

Prognosis

- Single episodes settle in weeks; chronic forms recur.

Prescriber pearls (UK pharmacist IP)

"**Drug, legs, palpable purpura, well patient stop the drug, dip the urine, elevate.**"

Warn that **new crops can appear for 1–2 weeks** after stopping the drug.

Self-check quiz

5 questions · optional

Q1 When should you think of Cutaneous small-vessel (leukocytoclastic) vasculitis?

✓ Symmetrical palpable purpura, sometimes with blisters or small ulcers, on the shins and ankles after a new drug or infection.

Q2 What drives it?

✓ Immune complexes lodge in small skin vessels, recruit neutrophils and break down (leukocytoclasis).

Q3 What is the backbone of care?

✓ Stop the trigger; rest, elevation and compression if no arterial disease; analgesia (not the culprit NSAID); dipstick and bloods; refer if systemic or persistent.

Q4 What must you never miss?

✓ Kidney involvement; IgA vasculitis in children; vasculitis that is part of lupus, ANCA disease or cryoglobulinaemia.

Q5 Name a red flag that needs escalation.

✓ Haematuria, proteinuria or rising creatinine: same-day renal advice.

References

1. DermNet: Cutaneous small vessel vasculitis – https://dermnetnz.org/topics/cutaneous-small-vessel-vasculitis
dermnetnz.org

2. Medscape: Cutaneous small-vessel (leukocytoclastic) vasculitis — https://emedicine.medscape.com/article/1083719-overview
emedicine.medscape.com

Photo gallery

Small-vessel vasculitis

6

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Vasculitis & purpura Module 1 · Week 4

Urticarial vasculitis & HUVS

Wheal-like plaques that last longer than 24 hours, burn rather than itch and leave bruising. Low complement (HUVS) points to systemic disease such as lupus.

ALSO CALLED UV | hypocomplementaemic urticarial vasculitis | HUVS | wheals lasting more than 24 hours | bruising wheals

RELATED PAGES Urticaria & angioedema | Cutaneous small-vessel (leukocytoclastic) vasculitis | Discoid lupus erythematosus | Approach to cutaneous vasculitis

KEY FACTS

- **Urticarial vasculitis** looks like urticaria, but wheals last **over 24 hours**, burn more than itch, and leave **bruising**.
- Normocomplementaemic (often skin-limited) or **hypocomplementaemic** (linked to lupus, systemic disease).
- Check C3, C4, C1q, ANA, urine, ESR/CRP; biopsy confirms vasculitis.
- Antihistamines help less than in urticaria; specialist options include dapsone, colchicine, hydroxychloroquine.
- Hypocomplementaemic UV: watch for lung (COPD), kidney and eye disease.

Overview

Urticarial vasculitis (UV): a small-vessel (leukocytoclastic) vasculitis that presents as **urticaria-like lesions**. Two forms:

- **Normocomplementaemic UV** (~70–80%): usually skin-limited, often idiopathic, and milder.
- **Hypocomplementaemic UV** (~20–30%): associated with systemic disease. Its severe form is **HUVS** (McDuffie syndrome; CHCC 2012: **anti-C1q vasculitis**), with **arthritis, angioedema, ocular inflammation, COPD, glomerulonephritis** and a strong overlap with **SLE**.

Epidemiology

- **Who**
  - Uncommon; middle-aged women most; about 5% of chronic 'urticaria' cases.

Aetiology: causes and risk factors

Myths to address

- **"ALL WHEALS ARE URTICARIA"**
  wheals lasting over a day with bruising need biopsy.

Pathophysiology and pathology

Immune complexes cause **vasculitis** in vessels beneath a wheal.

1. **Immune complexes**
   Complexes activate complement (low C3/C4/C1q in hypocomplementaemic disease).
2. **Mast cell and neutrophil activation**
   C5a degranulates mast cells (wheal) and attracts neutrophils (vasculitis).
3. **Persistent lesion**
   Vessel damage makes wheals last over 24 hours and leave purpura.

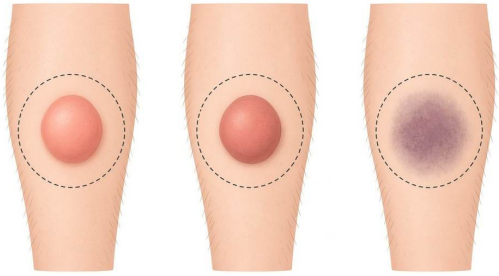

Fig. 1 · Circling a wheal with a pen shows how long each lasts.
AI-generated illustration

How it is caused

Immune-complex deposition in post-capillary venules with complement consumption (low C3/C4/C1q). **Anti-C1q autoantibodies** in HUVS. **Causes**:

- **drugs** (ACE-i, penicillins, sulfonamides, fluoxetine, cimetidine, diltiazem, thiazides, potassium iodide, NSAIDs, glatiramer; also TNF inhibitors)
- **CTD** (SLE, Sjögren's)
- **paraproteins** (IgM – **Schnitzler syndrome** – and IgA gammopathy), mixed cryoglobulinaemia
- **malignancy** (haematological, solid)
- **infection** (hepatitis B/C, EBV, Lyme)
- idiopathic.

HISTOPATHOLOGY

- Leukocytoclastic vasculitis beneath an urticarial lesion; DIF may show immunoglobulin and complement in vessels.

Clinical features

Clinical features & sites

FEATURE	URTICARIAL VASCULITIS	ORDINARY (CHRONIC SPONTANEOUS) URTICARIA
Lesion duration	**>24 h (often 1–3 days)**	<24 h (each wheal)
Symptom	**Burning, pain, tenderness** ± itch	Itch
Resolution	**Leaves purpura/bruising or brown pigment**	Leaves normal skin
Systemic	Fever, arthralgia, abdominal pain, angioedema, lymphadenopathy, eye/kidney/lung involvement (especially HUVS)	Rarely (angioedema only)
Bloods	ESR ; **C3/C4/C1q low in HUV**; ANA ±	Usually normal (± thyroid autoantibodies)
Antihistamine response	**Poor**	Good
Biopsy	**LCV** (fibrinoid necrosis, nuclear dust, RBC extravasation); DIF Ig/C3 around vessels ± lupus band at the DEJ	Dermal oedema, sparse infiltrate, no vasculitis

Fig. 2 · Urticarial vasculitis on the leg — wheal-like plaques lasting >24 h, burning/tender rather than itchy, leaving bruising/pigment.

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Types to recognise

Leg plaques

Wheal-like, lasting > 24 h. Site: Legs.

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Residual purpura

Dusky staining after. Site: Legs.

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Diagnosis and investigations

Biopsy a lesion (H&E + DIF, ideally a lesion <48 h old) — essential, because the clinical overlap with urticaria is large.

UNDERLYING CAUSE AND ORGAN SCREEN

C3, C4, C1q, anti-C1q, CH50; **ANA, dsDNA, ENA (Ro/La/Sm)**; ESR/CRP, FBC, U&E/eGFR, LFT; **urinalysis + uPCR** (± renal biopsy); **hepatitis B/C**, HIV, EBV; **SPEP/immunofixation** (IgM paraprotein — Schnitzler), **cryoglobulins**; ASO; CXR and **PFTs** (COPD/emphysema in HUVS, especially smokers); **ophthalmology** (uveitis/episcleritis/conjunctivitis); a cancer screen if indicated

HUVS CRITERIA (SCHWARTZ 1982)

Major (both required): chronic urticarial exanthem for **6 months**, and **hypocomplementaemia**; **Minor (2):** dermal venulitis on biopsy; arthralgia/arthritis; uveitis/episcleritis; glomerulonephritis; recurrent abdominal pain; positive C1q precipitin (anti-C1q); **Exclusions:** cryoglobulinaemia, high-titre anti-dsDNA, HBV, C1-INH deficiency

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Chronic spontaneous urticaria	Wheals < 24 h, itch, normal skin after, antihistamine-responsive.
	SLE	Photosensitivity, ANA +, low complement, nephritis; ~50% of HUVS meet SLE criteria.
	Schnitzler syndrome	IgM paraprotein, fever, bone pain.

	CONDITION	DISTINGUISHING FEATURES
	Leukocytoclastic vasculitis	Palpable purpura on legs.
	Urticarial bullous pemphigoid	Older, itchy, later tense blisters.
	Erythema multiforme	Targets, acral, mucosa.

Chronic spontaneous urticaria; **SLE** (photosensitivity, ANA+, low complement and nephritis also fit — HUVS and SLE overlap, and ~50% of HUVS patients meet SLE criteria); Schnitzler syndrome (IgM paraprotein, fever, bone pain); cryoglobulinaemic vasculitis; serum-sickness-like reaction; Still's disease; neutrophilic urticarial dermatosis; urticarial bullous pemphigoid; erythema multiforme.

Management

- **Refer** to dermatology ± rheumatology/renal (urgently if renal involvement — the case has a creatinine of ~203 µmol/L).
- Stop any suspect drug (ACE-i, NSAIDs, etc.). Yellow Card.
- **SYMPTOMATIC**
  non-sedating antihistamines (help itch/angioedema, but not the vasculitis), NSAIDs only if renal function allows.
- **SPECIALIST LADDER**
  dapsone, **colchicine**, **hydroxychloroquine** (first-line for mild/normocomplementaemic disease) oral corticosteroids for flares steroid-sparing **mycophenolate, azathioprine, methotrexate** **rituximab** (HUVS/SLE), omalizumab (case reports, normocomplementaemic UV), anakinra/canakinumab (Schnitzler). Cyclophosphamide/MMF for nephritis.
- **Shared-care monitoring** (dapsone — FBC, LFT, methaemoglobin; HCQ — annual eye screening after 5 years, or from 1 year if risk factors, RCOphth 2020; MMF — FBC/LFT; contraception).

IP scope

Recognition (do not keep up-titrating antihistamines for "urticaria" that bruises), urinalysis, bloods, drug review, referral, and shared-care monitoring.

Treatment ladder

Your role

Refer dermatology ± rheumatology / renal (urgent if renal involvement).
Stop suspect drugs (ACE-i, NSAIDs…); Yellow Card.

2

Symptomatic

Non-sedating antihistamines (itch / angioedema, not vasculitis); NSAIDs only if renal function allows.

3

Specialist ladder

Dapsone, colchicine, hydroxychloroquine (mild / normocomplementaemic) oral steroids for flares **mycophenolate, azathioprine, methotrexate** **rituximab** (HUVS / SLE); omalizumab (case reports); anakinra / canakinumab (Schnitzler); cyclophosphamide / MMF for nephritis.

4

Shared-care monitoring

Dapsone: FBC, LFT, methaemoglobin. HCQ: annual eye screening after 5 years (from 1 year if risk factors, RCOphth 2020). MMF: FBC / LFT; contraception.

Referral

URGENT — Wheals with **joint pain, fever, abdominal pain, breathlessness or abnormal urine**: suspected systemic (hypocomplementaemic) disease.

ROUTINE — Wheals > 24 h, burning or bruising: dermatology for biopsy.

Red flags and when to refer

- Wheals with joint pain, fever, abdominal pain, breathlessness or abnormal urine: suspect **systemic (hypocomplementaemic) disease**, refer urgently.

Red flags / refer

Wheals lasting >24 h, painful, leaving bruising; systemic symptoms; abnormal urine, creatinine or complement; angioedema with laryngeal symptoms (999); eye pain/redness (same-day ophthalmology); breathlessness.

Complications and prognosis

Complications

- Angioedema, arthritis, glomerulonephritis, COPD, uveitis (hypocomplementaemic).

Prognosis

- Variable; may last years.

Prescriber pearls (UK pharmacist IP)

"**Wheals that last more than a day, burn, and leave a bruise = biopsy.**"

Ask patients to **circle a wheal with a pen** and photograph it daily — it documents duration (cheap, effective).

Self-check quiz

4 questions · optional

Q1 Key duration clue?

✓ Individual lesions last > 24 hours.

Q2 What do lesions leave behind?

✓ Purpura / bruising or brown pigment.

Q3 Test distinguishing HUVS?

✓ Complement: low C3, C4, C1q; anti-C1q.

Q4 Do antihistamines treat the vasculitis?

✓ No: only itch / angioedema.

References

1. DermNet: Urticarial vasculitis — https://dermnetnz.org/topics/urticarial-vasculitis
dermnetnz.org
2. Medscape: Urticarial vasculitis & HUVS — https://emedicine.medscape.com/article/1085087-overview
emedicine.medscape.com

Photo gallery

Urticarial vasculitis

2

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Vasculitis & purpura Module 1 · Week 4

Polyarteritis nodosa (systemic & cutaneous)

Medium-vessel vasculitis giving tender nodules, livedo racemosa and punched-out ulcers on the legs. Cutaneous PAN is skin-limited; systemic PAN (sometimes hepatitis-B related) affects nerves, gut, kidneys and testes.

ALSO CALLED PAN | cutaneous PAN | medium-vessel vasculitis | hepatitis B | tender nodules | livedo racemosa

RELATED PAGES Livedo reticularis vs livedo racemosa | Cutaneous small-vessel (leukocytoclastic) vasculitis | Approach to cutaneous vasculitis | Erythema nodosum (incl. drug-induced)

KEY FACTS

- **Polyarteritis nodosa (PAN)** is a necrotising vasculitis of **medium-sized arteries**.
- Skin: tender subcutaneous nodules, **livedo racemosa**, ulcers, digital ischaemia; cutaneous PAN may be skin-limited.
- Systemic PAN: hypertension, renal infarcts, mononeuritis multiplex, abdominal pain, weight loss; **ANCA negative**.
- Associated with hepatitis B; deep incisional biopsy of a nodule needed.
- Specialist treatment: steroids ± cyclophosphamide; antivirals for HBV-PAN.

Overview

PAN is a **necrotising arteritis of medium (and small) muscular arteries**, **without glomerulonephritis** or vasculitis of arterioles/capillaries/venules, and **not associated with ANCA** (CHCC 2012). It is rare (~2–30 per million), with a peak at 40–60 years. Forms:

- **systemic PAN** — idiopathic, or **HBV-associated** (now rare in vaccinated populations), HCV, HIV, hairy-cell leukaemia, **DADA2** (adenosine deaminase-2 deficiency, especially childhood or young-onset with strokes)
- **cutaneous (single-organ) PAN** — skin ± local muscle/nerve/joint, a chronic relapsing course, rarely becoming systemic.

Epidemiology

- **Who**
  - Rare; adults 40–60; children with cutaneous PAN after streptococcal infection.

Aetiology: causes and risk factors

Myths to address

- **"NORMAL ANCA EXCLUDES VASCULITIS"**

PAN is ANCA-negative.

Pathophysiology and pathology

PAN inflames and destroys the **walls of medium arteries**, causing aneurysms and infarcts.

1. **Trigger**
   Hepatitis B immune complexes or unknown triggers.
2. **Transmural necrosis**
   Fibrinoid necrosis of the full arterial wall at the dermo-subcutaneous junction.
3. **Aneurysms and thrombosis**
   Weakened walls balloon (microaneurysms); thrombosis causes infarction.
4. **Livedo and ulcers**
   Patchy arterial blockage gives broken livedo, nodules and ulcers.

How it is caused

Immune complexes (e.g. HBsAg–antibody in HBV-PAN, typically within 6–12 months of infection) or other triggers cause **segmental transmural fibrinoid necrosis** at arterial branch points **microaneurysms**, thrombosis, stenosis **infarction** of the organs supplied:

- **renal arteries** **renovascular hypertension, infarcts (not GN)**
- vasa nervorum **mononeuritis multiplex**
- mesenteric abdominal pain/bowel infarction
- testes orchitis
- coronary arteries
- skin nodules, livedo racemosa, ulcers, digital gangrene.

HISTOPATHOLOGY

- Necrotising arteritis of medium vessels in deep dermis/subcutis; deep biopsy required.

Clinical features

Clinical features & sites

SYSTEMIC

fever, **weight loss**, myalgia, arthralgia; **mononeuritis multiplex/polyneuropathy** (foot/wrist drop, patchy numbness); **hypertension**; renal impairment; abdominal pain after meals (mesenteric angina), GI bleed; **testicular pain**; MI/heart failure; stroke (DADA2).

SKIN (~25–60%)

tender subcutaneous nodules (0.5–2 cm) along arteries of the lower legs/thighs; **livedo racemosa** (feet, legs); **punched-out ulcers** (D02 — sensitive); digital ischaemia; "starburst" livedo around nodules.

CUTANEOUS PAN

recurrent painful nodules ± livedo/ulcers on the lower legs, with local myalgia/arthralgia/neuropathy in the same limb. Triggers: **streptococcal infection** (children), IBD, HBV/HCV, **minocycline** (drug-induced cPAN — link to the drug eruption notes/acne). ANCA negative; usually no internal organ involvement.

Fig. 1 · Cutaneous PAN — tender subcutaneous nodules along the leg with livedo (medium-vessel vasculitis at the dermal–subcutaneous junction).

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Diagnosis and investigations

No single blood test. Diagnosis combines clinical features, exclusion of mimics, **histology** and **angiography**: **ESR/CRP** raised; normocytic anaemia; leucocytosis; thrombocytosis; **ANCA negative** (positive ANCA points to MPA/GPA/EGPA); **Hepatitis B (HBsAg, HBeAg, HBV DNA) and C**, HIV, cryoglobulins, complement, ANA, APL antibodies, blood cultures; **Urinalysis** (no active sediment — red-cell casts suggest GN, so not PAN); **ADA2 activity/genetics** in young patients

BIOPSY

deep **incisional skin biopsy of a nodule to subcutis** (medium artery with fibrinoid necrosis; DIF non-specific), or **muscle/sural nerve biopsy** guided by symptoms/EMG.

IMAGING

CT/MR angiography or catheter angiography — **microaneurysms** and stenoses of renal/mesenteric/hepatic arteries ("arteriographic abnormality").

ACR 1990 CLASSIFICATION CRITERIA (3 OF 10)

weight loss 4 kg; livedo reticularis; testicular pain/tenderness; myalgias/weakness/leg tenderness; mono- or polyneuropathy; diastolic BP >90 mmHg; raised urea/creatinine (BUN >40 mg/dL or creatinine >1.5 mg/dL); HBV markers; arteriographic abnormality; biopsy showing neutrophils in a small/medium artery wall. These are **classification** criteria, not diagnostic.

French Vasculitis Study Group predictive items (Henegar 2008; course list): positive — **HBsAg+, arteriographic abnormalities, mono/polyneuropathy**; negative — **ANCA, asthma, ENT signs, glomerulopathy, cryoglobulinaemia**.

Cutaneous vs systemic PAN:

	CUTANEOUS PAN	SYSTEMIC PAN
Organs	Skin ± same-limb muscle/nerve/joint	Kidney (renovascular), nerves (mononeuritis multiplex), gut, heart, testes
Systemic symptoms	Mild/none (low-grade fever)	Fever, weight loss, myalgia
BP / renal	Normal	**Hypertension**, raised creatinine
Angiography	Normal	**Microaneurysms**
Associations	Strep, IBD, minocycline, HBV/HCV	HBV, HCV, hairy-cell leukaemia, DADA2
Course	Chronic relapsing, benign	Potentially fatal untreated; relapses ~10–20%
Treatment	NSAIDs, colchicine, dapsone, topical/intralesional/oral steroids, methotrexate, IVIg; penicillin if strep	Glucocorticoids ± cyclophosphamide; antivirals for HBV-PAN

- **Severity — revised Five-Factor Score (FFS 2011)** for PAN: age >65, cardiac insufficiency, GI involvement, renal insufficiency (creatinine 150 µmol/L). Each = 1 point; 1 point = worse prognosis cyclophosphamide added.

Differential diagnoses

ANCA-associated vasculitis (MPA — GN and lung haemorrhage; GPA; EGPA — asthma, eosinophilia), cryoglobulinaemic vasculitis (HCV), SLE/RA vasculitis, **infective endocarditis** (emboli, aneurysms — blood cultures), **antiphospholipid syndrome**, cholesterol emboli, segmental arterial mediolysis, fibromuscular dysplasia, Buerger's disease, calciphylaxis, lymphoma, EN/panniculitis (the drug eruption notes see Livedo reticularis vs livedo racemosa for nodules), DADA2.

Management

Systemic PAN urgent rheumatology/renal admission. BP 160/100 with raised creatinine, neuropathy and necrotic ulcers is organ-threatening.

NON-HBV SYSTEMIC PAN (ACR/VF 2021; EULAR)

Induction: **high-dose glucocorticoids** (e.g. IV methylprednisolone pulses then prednisolone ~1 mg/kg/day, max ~60 mg, tapering over months); **Add cyclophosphamide** for severe disease (FFS 1, neuropathy, renal, GI, cardiac involvement); Maintenance: azathioprine or methotrexate (total ~18 months); Supportive: **BP control** (ACE-i/ARB for renin-mediated hypertension, with care if renal artery stenosis); PJP prophylaxis (co-trimoxazole) on cyclophosphamide; bone protection; gastroprotection; Steroid Emergency Card; vaccines (no live); fertility counselling before cyclophosphamide

HBV-ASSOCIATED PAN (THE CASE) — DIFFERENT STRATEGY

Antiviral therapy (entecavir or tenofovir; lamivudine historically) via hepatology; A **short course of glucocorticoids** (e.g. ~2 weeks, to control life-threatening inflammation) and **plasma exchange** to clear immune complexes, until HBeAg seroconversion; **Avoid prolonged steroids and cyclophosphamide**, which boost HBV replication and risk fulminant hepatitis (Guillevin regimen); Notify/contact-trace and vaccinate household and sexual contacts (UKHSA); This differs from the "prednisolone 6 months + cyclophosphamide 12 months" approach in a student answer, which suits severe **non-HBV** PAN

CUTANEOUS PAN

> NSAIDs/colchicine/dapsone short steroid courses methotrexate/azathioprine; treat strep (penicillin prophylaxis in children with recurrent strep-triggered cPAN); stop minocycline.

IP scope

Recognition, bloods (incl. hepatitis serology, urinalysis, BP), referral, and post-discharge support: shared-care monitoring of azathioprine/methotrexate (FBC/LFT/U&E schedules), PJP prophylaxis checks, steroid card, vaccines (inactivated only), BP management. Induction therapy is specialist.

Referral

URGENT — Foot drop or new numbness, severe abdominal pain or testicular pain with nodules or livedo: **urgent** referral for systemic PAN.

Red flags and when to refer

- Foot drop or new numbness, severe abdominal pain or testicular pain with nodules or livedo: **urgent** referral for systemic PAN.

Red flags / refer

Hypertension + renal impairment; mononeuritis multiplex (foot/wrist drop); post-prandial abdominal pain/GI bleeding; testicular pain; chest pain; digital ischaemia/gangrene; nodules + livedo racemosa + ulcers; HBsAg positive with vasculitic features urgent/same day.

Complications and prognosis

Complications

- Renal failure, hypertension, bowel infarction, stroke, neuropathy.

Prognosis

- Cutaneous PAN is chronic but benign; systemic PAN is serious but often remits with treatment.

Prescriber pearls (UK pharmacist IP)

- **"Nodules + livedo + neuropathy + hypertension, ANCA negative think PAN, and check hepatitis B."**
- PAN **spares glomeruli**. Haematuria with red-cell casts points to ANCA vasculitis.
- Hepatitis B serology **before** starting steroids/immunosuppressants for any vasculitis (reactivation risk) — this applies to rituximab especially.

Self-check quiz

5 questions · optional

Q1 When should you think of Polyarteritis nodosa (systemic & cutaneous)?

✓ Tender subcutaneous nodules along vessels, livedo racemosa and painful ulcers on the lower legs.

Q2 What drives it?

✓ Necrotising inflammation of medium-sized arteries; hepatitis B, streptococcal infection and some drugs are triggers.

Q3 What is the backbone of care?

✓ BP, urine, renal function, hepatitis B/C serology; deep (incisional) biopsy; refer to rheumatology/ dermatology; specialist immunosuppression.

Q4 What must you never miss?

✓ Systemic PAN: mononeuritis multiplex, abdominal pain, testicular pain, hypertension, renal infarcts.

Q5 Name a red flag that needs escalation.

✓ Foot drop or new numbness, severe abdominal pain or testicular pain with nodules or livedo: **urgent** referral for systemic PAN.

References

1. DermNet: Cutaneous polyarteritis nodosa — https://dermnetnz.org/topics/cutaneous-polyarteritis-nodosa
dermnetnz.org
2. Medscape: Polyarteritis nodosa (systemic & cutaneous) — https://emedicine.medscape.com/article/330717-overview
emedicine.medscape.com

Photo gallery

Cutaneous polyarteritis nodosa

2

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Vasculitis & purpura Module 1 · Week 4

Livedo reticularis vs livedo racemosa

A violaceous net-like pattern on the skin. A regular, complete net that fades on warming is usually physiological; a broken, irregular, persistent net (racemosa) points to vessel disease such as antiphospholipid syndrome or vasculitis.

ALSO CALLED mottled skin · net-like rash · cutis marmorata · antiphospholipid · Sneddon syndrome · livedoid vasculopathy

RELATED PAGES Polyarteritis nodosa (systemic & cutaneous) · Approach to cutaneous vasculitis · Purpura: the differential diagnosis · Osler nodes, Janeway lesions & splinter haemorrhages

KEY FACTS

- **Livedo reticularis** is a violaceous, net-like (reticulate) mottling of the skin from sluggish blood flow.
- Physiological (cold, complete regular rings, fades on warming) versus pathological (**livedo racemosa**: broken, irregular, persistent).
- Pathological causes: antiphospholipid syndrome, vasculitis (PAN), cholesterol emboli, cryoglobulins, calciphylaxis, drugs (amantadine).
- Investigate persistent or broken livedo: FBC, clotting, antiphospholipid antibodies, ANA, cryoglobulins, renal function.
- Livedo plus stroke: Sneddon syndrome (antiphospholipid).

Overview

A **violaceous, net-like mottling** of the skin, caused by reduced blood flow/deoxygenation at the edges of the cone-shaped skin segments supplied by single ascending dermal arterioles.

> **LIVEDO RETICULARIS**
> a **regular, complete, fine** net. Often physiological.

> **LIVEDO RACEMOSA**
> **irregular, broken, branching** rings ("lightning bolt") that **persist when warm**. Always pathological.

> **CUTIS MARMORATA**
> physiological cold-induced mottling (infants, young women) that **disappears on rewarming**.

Epidemiology

- **Who**
  - Physiological cutis marmorata common in infants and young women.

Aetiology: causes and risk factors

Myths to address

- **"MOTTLED SKIN IS ALWAYS HARMLESS"**
  persistent, broken livedo may signal clotting or vasculitis.

Pathophysiology and pathology

Livedo appears when blood **pools in the venous network** at the edge of each skin perfusion cone.

1. **Perfusion cones**
   Each arteriole supplies a cone of skin; the edges are drained by a venous plexus.
2. **Slow flow**
   Cold, spasm or blockage slows flow so deoxygenated blood pools at the cone edges: a violet net.
3. **Racemosa**
   Patchy arteriolar blockage (thrombosis, vasculitis) breaks the rings into irregular shapes.

How it is caused

Each cone's centre is supplied by an ascending arteriole at the **dermal–subcutaneous junction**. The periphery (venous plexus) is least oxygenated. Anything that **slows arteriolar inflow** (spasm, occlusion, vasculitis), **raises blood viscosity** or **impairs venous outflow** dilates and deoxygenates the peripheral venous plexus visible blue-violet net.

MECHANISM	CAUSES
Physiological / vasospasm	Cold (cutis marmorata), Raynaud's
Vessel-wall disease (vasculitis)	**PAN / cutaneous PAN**, cryoglobulinaemic vasculitis, SLE/RA vasculitis, ANCA vasculitis
Intraluminal occlusion (non-inflammatory)	**Antiphospholipid syndrome** (± **Sneddon syndrome** — livedo racemosa + strokes), **cholesterol emboli** (after angiography/vascular surgery/anticoagulation — painful toes, eosinophilia, AKI), livedoid vasculopathy, **calciphylaxis** (renal failure), cryoglobulins/cryofibrinogen/cold agglutinins, thrombocythaemia/ polycythaemia, DIC, hyperoxaluria, septic emboli
Hyperviscosity	Myeloma, Waldenström's, polycythaemia
Drugs	**Amantadine** (classic), catecholamines (noradrenaline, adrenaline), interferon, minocycline, bismuth, quinidine, cocaine/levamisole; **heparin/warfarin** (necrosis)
Other	Intravascular lymphoma, hypothyroidism, CNS lesions, erythema ab igne (heat), endocarditis

Clinical features

Clinical features & sites

Legs, feet, buttocks, arms; pattern, persistence on warming, pain, ulceration (atrophie blanche/livedoid vasculopathy, calciphylaxis), nodules (PAN), and neurological symptoms (Sneddon).

Fig. 1 · Livedo reticularis — regular, complete violaceous net. Physiological if it fades on warming.

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Diagnosis and investigations

Warm the limb (does it fade?). FBC, clotting, **APL antibodies (lupus anticoagulant, anticardiolipin, anti-β2GPI)**, ANA, ANCA, cryoglobulins, complement, hepatitis B/C, SPEP, U&E (calciphylaxis/cholesterol emboli), eosinophils, lipids. **Deep biopsy from the white centre** (or several biopsies) if pathological; MRI brain if neurological symptoms.

Differential diagnoses

Erythema ab igne (reticulate pigmentation from heat — laptops, hot-water bottles), reticulate eruptions (confluent and reticulated papillomatosis, poikiloderma), port-wine stain/cutis marmorata telangiectatica congenita.

Management

- Physiological: warmth, reassurance.
- Pathological: treat the cause (anticoagulation in APS; stop amantadine/vasopressors if possible; PAN treatment).
- **LIVEDOID VASCULOPATHY**
  compression, antiplatelet/anticoagulant (e.g. rivaroxaban — specialist), pain control.

IP scope

Recognition, drug review (amantadine, vasopressors), bloods and referral.

Referral

ROUTINE — New livedo racemosa with stroke, TIA, miscarriage or clots: test for **antiphospholipid syndrome**.

URGENT — Painful purple toes and livedo after an arterial procedure: **cholesterol emboli**, urgent referral.

Red flags and when to refer

- New livedo racemosa with stroke, TIA, miscarriage or clots: test for **antiphospholipid syndrome**.
- Painful purple toes and livedo after an arterial procedure: **cholesterol emboli**, urgent referral.

Red flags / refer

Livedo **racemosa**, livedo with **ulcers/nodules/purpura**, livedo + neurological symptoms (stroke/TIA — Sneddon), new livedo after an arterial procedure (cholesterol emboli), in renal failure (calciphylaxis — urgent), or in pregnancy loss (APS).

Complications and prognosis

Complications

- Depends on cause: ulceration, stroke, renal failure.

Prognosis

- Physiological livedo is benign; pathological livedo follows its cause.

Prescriber pearls (UK pharmacist IP)

"Regular, complete and fades with warmth = reticularis (usually benign). Broken and persistent = racemosa — find the cause."

Ask about **amantadine** in Parkinson's patients with leg mottling and ankle oedema.

Self-check quiz

5 questions · optional

Q1 **When should you think of Livedo reticularis vs livedo racemosa?**

✓ A purple-blue lace-like pattern on the legs, arms or trunk.

Q2 **What drives it?**

✓ Slowed blood flow in the venous plexus around each skin 'cone': physiological in cold, or due to vessel blockage, hyperviscosity or vasculitis.

Q3 What is the backbone of care?

✓ Decide: complete and fading (reassure) or broken and fixed (investigate: FBC, clotting, antiphospholipid antibodies, ANA, cryoglobulins, lipids).

Q4 What must you never miss?

✓ Antiphospholipid syndrome, Sneddon syndrome (stroke), cholesterol emboli after angiography, PAN, calciphylaxis.

Q5 Name a red flag that needs escalation.

✓ New livedo racemosa with stroke, TIA, miscarriage or clots: test for **antiphospholipid syndrome**.

References

1. DermNet: Livedo reticularis – https://dermnetnz.org/topics/livedo-reticularis
dermnetnz.org

Photo gallery

Livedo reticularis and racemosa 3

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Livedoid vasculopathy 1

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Vasculitis & purpura Module 1 · Week 4

IgA vasculitis (Henoch–Schönlein purpura)

The commonest childhood vasculitis: palpable purpura on the legs and buttocks, joint pain, abdominal pain and sometimes kidney involvement, usually after a sore throat or cold.

ALSO CALLED HSP | Henoch–Schönlein | purpura in children | IgA nephropathy | abdominal pain purpura

RELATED PAGES Cutaneous small-vessel (leukocytoclastic) vasculitis | Purpura: the differential diagnosis | Approach to cutaneous vasculitis | Urticarial vasculitis & HUVS

KEY FACTS

- **IgA vasculitis** (Henoch–Schönlein purpura) is the commonest childhood vasculitis: **palpable purpura** on the buttocks and legs.
- Tetrad: purpura, arthralgia/arthritis, abdominal pain (intussusception risk) and **renal involvement** (haematuria, proteinuria).
- Often follows an upper respiratory infection; ages 3–10.
- First exclude **meningococcal sepsis** in an unwell child with a non-blanching rash.
- Supportive care; monitor **urine and BP** for 6–12 months; adults have more renal disease.

Overview

The **commonest childhood vasculitis** (~20 per 100,000 children/year). A systemic **IgA-immune-complex small-vessel vasculitis** affecting skin, joints, gut and kidneys. **~90% of cases are in children** (peak 3–10 years; ~75% under 8; slightly more boys), but it **does occur in adults**, where renal disease is more often severe.

Epidemiology

- **Who**
  - About 20 per 100,000 children per year; autumn and winter; adults less often but more severely.

Aetiology: causes and risk factors

Myths to address

- **"THE RASH SETTLED, SO FOLLOW-UP CAN STOP"**
  nephritis can appear weeks to months later.

Pathophysiology and pathology

Abnormal **IgA1** forms immune complexes that lodge in small vessels.

1. **Abnormal IgA1**
   Galactose-deficient IgA1 is produced after a mucosal infection.
2. **Immune complexes**
   IgA complexes deposit in small vessel walls in skin, gut, joints and kidney.
3. **Leukocytoclastic vasculitis**
   Neutrophils damage vessels and red cells leak: palpable purpura.

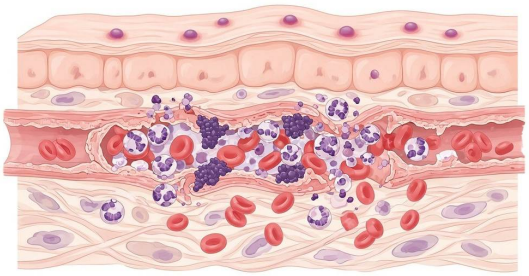

Fig. 1 · Small-vessel vasculitis: immune complexes on the vessel wall attract neutrophils that fragment and damage it, letting red cells leak out (palpable purpura).
AI-generated illustration

4. **Nephritis**
   Mesangial IgA deposits cause haematuria and proteinuria.

How it is caused

Galactose-deficient IgA1 forms immune complexes (with anti-glycan IgG) that deposit in vessel walls and the glomerular mesangium. It often follows an **upper respiratory infection** — **group A β-haemolytic streptococcus** is the most reported trigger (positive throat swabs in 10–30%, raised ASO in 20–50%). Other triggers: parainfluenza, adenovirus, Mycoplasma, parvovirus, drugs, vaccines, malignancy in adults.

HISTOPATHOLOGY

- Leukocytoclastic vasculitis with IgA on direct IF of a new lesion.

Clinical features

Clinical features & sites

- **Palpable purpura** (100%, required): symmetrical on the **buttocks, legs, ankles and extensor arms**, ± urticarial and bullous/necrotic lesions; scrotal oedema; oedema of the scalp/face in toddlers.
- **Arthralgia/arthritis** (~75%): knees and ankles, transient and non-erosive.
- **Abdominal pain** (~65%) ± GI bleeding (~30%), **intussusception** (ileo-ileal — 1–5%), pancreatitis.

- **Renal** (~30–50%): microscopic/macroscopic **haematuria**, proteinuria, nephritic/nephrotic syndrome, hypertension, raised creatinine. Usually within **4–6 weeks** (up to 6 months). The main long-term risk: ~1–2% of children progress to CKD; higher if nephrotic/nephritic at onset.
- Rare: orchitis/testicular torsion mimic, CNS involvement, pulmonary haemorrhage.

Fig. 2 · IgA vasculitis (HSP) — palpable purpura on the ankle/foot, the classic dependent distribution.

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Diagnosis and investigations

> **EULAR/PRINTO/PRES CLASSIFICATION (CHILDREN)**
>
> **purpura or petechiae, lower-limb predominant (mandatory)** + 1 of: abdominal pain; histology with IgA deposition; arthritis/arthralgia; renal involvement (proteinuria >0.3 g/24 h or uACR >30 mg/mmol, or haematuria/red-cell casts).

> **BLOODS**
>
> **FBC + film (platelets normal — excludes ITP/leukaemia)**, clotting, U&E/creatinine, albumin, CRP; ASO ± throat swab; IgA (raised in ~50%; not diagnostic).

> **Urinalysis + BP** at diagnosis; **urine ACR/PCR** if proteinuria.

> **Biopsy** only if atypical (adults, uncertain diagnosis): LCV with **perivascular IgA and C3 on DIF**. Renal biopsy if significant proteinuria or impaired function.

Differential diagnoses

Meningococcal septicaemia (unwell, fever, rapidly spreading non-blanching rash — must be excluded first), **ITP** (low platelets, flat petechiae/bruising, well), **acute leukaemia** (pallor, bone pain, lymphadenopathy, abnormal FBC), **acute haemorrhagic oedema of infancy**, non-accidental injury, cryoglobulinaemia, GPA, SLE, post-streptococcal GN, HUS (diarrhoea, anaemia, low platelets, AKI), papular-purpuric gloves-and-socks syndrome (parvovirus), scurvy, drug vasculitis.

Management

- **Exclude sepsis first.** A child with non-blanching rash who is unwell or febrile **NICE NG240**: treat as **meningococcal disease** — 999. Give **IM/IV benzylpenicillin** in primary care if it does not delay transfer.

- **MOST CHILDREN ARE MANAGED AT HOME**
  rest, **paracetamol** for joint pain. **NSAIDs** may be used for arthralgia if there is **no renal involvement** and no GI bleeding (avoid in renal disease). Elevate the legs for purpura.
- **Oral corticosteroids** may help **severe abdominal pain/arthritis/orchitis** (specialist advice). They do **not prevent nephritis** and are not given routinely.
- **Renal monitoring** (core primary-care task — the course mentions it only briefly): **urinalysis and BP** at presentation and then regularly for **at least 6 months** — typically **weekly for the first month, then every 2 weeks to month 3, monthly to month 6, then at 12 months** (CKS/UK paediatric nephrology practice — check the current CKS schedule). Any proteinuria or hypertension paediatric/renal referral.
- **Recurrence** occurs in ~1/3, usually within 4 months.
- **PREGNANCY**
  women with a history of HSP nephritis need BP/urine monitoring in pregnancy (pre-eclampsia risk).

IP scope

Within IP scope for a well child or adult with typical HSP and normal urine/BP: analgesia, safety-netting, arranging the urinalysis/BP schedule and recognising when to escalate. Paediatric IPs: follow local competence frameworks for children.

Referral

ROUTINE	Child unwell, febrile or with spreading non-blanching rash: treat as **meningococcal disease** until proven otherwise.
SAME DAY	Severe abdominal pain or blood in stool: possible **intussusception**, same-day paediatric review.
ROUTINE	Proteinuria, haematuria or high BP: paediatric nephrology advice; keep checking urine for 6–12 months.

Red flags and when to refer

- Child unwell, febrile or with spreading non-blanching rash: treat as **meningococcal disease** until proven otherwise.
- Severe abdominal pain or blood in stool: possible **intussusception**, same-day paediatric review.
- Proteinuria, haematuria or high BP: paediatric nephrology advice; keep checking urine for 6–12 months.

Red flags / refer

Unwell/febrile child or **atypical rash** (exclude meningococcal sepsis/leukaemia); **severe abdominal pain**, bloody stools, signs of **intussusception**; **testicular pain/swelling** (exclude torsion); **renal involvement** — macroscopic haematuria, proteinuria 2+ or uPCR >100 mg/mmol (nephrotic range — CKS), oedema, **hypertension**, raised creatinine; adults (higher renal risk — refer for assessment); diagnostic uncertainty; low platelets.

Complications and prognosis

Complications

- Nephritis and chronic kidney disease, intussusception, orchitis.

Prognosis

- Most children recover in 4–6 weeks; recurrence in a third.

Prescriber pearls (UK pharmacist IP)

"**HSP = purpura with a NORMAL platelet count.**" If platelets are low, it is not HSP.

Give the family a **urine dipstick plan** and a written list of red flags (abdominal pain, blood in urine/stools, swollen testicles, headache, reduced urine output).

The rash can recur with activity/standing — reassure.

Self-check quiz

5 questions · optional

Q1 When should you think of IgA vasculitis (Henoch–Schönlein purpura)?

✓ A child (often 3–10 years) with palpable purpura on the legs and buttocks after a cold or sore throat, with sore joints or tummy pain.

Q2 What drives it?

✓ IgA-containing immune complexes in small vessels of the skin, gut, joints and kidneys.

Q3 What is the backbone of care?

✓ Exclude sepsis; BP and urine dipstick; FBC and clotting if unsure; simple analgesia; urine and BP monitoring for 6–12 months.

Q4 What must you never miss?

✓ Meningococcal sepsis, leukaemia or ITP (check platelets), intussusception, nephritis.

Q5 Name a red flag that needs escalation.

✓ Child unwell, febrile or with spreading non-blanching rash: treat as **meningococcal disease** until proven otherwise.

References

1. DermNet: Henoch schoenlein purpura — https://dermnetnz.org/topics/henoch-schoenlein-purpura
   dermnetnz.org
2. cks.nice.org.uk — https://cks.nice.org.uk/topics/henoch-schonlein-purpura/
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3. nice.org.uk: ng240 — https://www.nice.org.uk/guidance/ng240
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4. Medscape: IgA vasculitis (Henoch–Schönlein purpura) — https://emedicine.medscape.com/article/984105-overview
   emedicine.medscape.com

Photo gallery

IgA vasculitis 3

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Vasculitis & purpura Module 1 · Week 4

Osler nodes, Janeway lesions & splinter haemorrhages

The fingertip and palm signs of infective endocarditis, what they mean, why none of them is diagnostic alone, and why fever plus any of them needs same-day admission.

ALSO CALLED infective endocarditis · Duke criteria · splinter haemorrhages · Osler nodes · Janeway lesions · septic emboli

RELATED PAGES Approach to cutaneous vasculitis · Nail signs: psoriatic nails, Beau's lines & clubbing · Purpura: the differential diagnosis · Cutaneous small-vessel (leukocytoclastic) vasculitis

KEY FACTS

- Classic peripheral signs of **infective endocarditis**: Osler nodes, Janeway lesions, splinter haemorrhages, petechiae, Roth spots.
- **Osler nodes** are **tender** nodules on finger and toe pads; **Janeway lesions** are **painless** macules on palms and soles.
- Each sign is now seen in under 10–15% of cases because IE is diagnosed earlier.
- Fever + new murmur or IE risk + embolic signs **same-day hospital assessment**.
- **Don't start oral antibiotics before blood cultures**; routine dental prophylaxis is not recommended (NICE CG64).

Overview

Classic peripheral signs of **infective endocarditis (IE)**. They are now uncommon (each is seen in <10–15% of modern IE cases) because IE is diagnosed earlier.

> **OSLER NODES**
> **tender**, pea-sized red-purple nodules on the **pads of fingers and toes** (and thenar/hypothenar eminences), lasting hours to days.

> **JANEWAY LESIONS**
> **painless**, flat or slightly raised haemorrhagic macules on the **palms and soles**.

> **SPLINTER HAEMORRHAGES**
> thin, longitudinal, red-brown/black lines in the nail bed, running in the direction of nail growth.

Aetiology: causes and risk factors

Myths to address

- **"SPLINTER HAEMORRHAGES MEAN ENDOCARDITIS"**
  most are from trauma; distal, few and in manual workers is typical.

Pathophysiology and pathology

Skin signs come from **infected emboli and immune complexes** shed from the valve.

1. **Vegetation**
   Bacteria colonise a damaged or prosthetic valve, forming a vegetation.

2. **Septic microemboli**
   Fragments lodge in small skin vessels: Janeway lesions, splinters, petechiae.

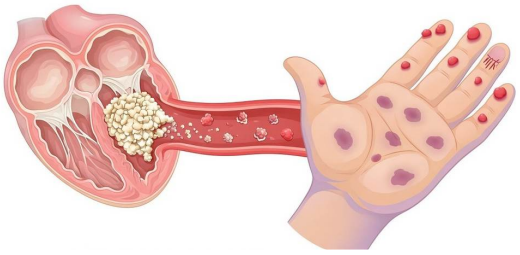

Fig. 1 · Fragments of a valve vegetation travel to small vessels in the fingertips and palms.
AI-generated illustration

3. **Immune complexes**
   Circulating immune complexes cause vasculitis: Osler nodes, Roth spots.

4. **Systemic spread**
   Emboli also reach brain, kidney and spleen.

How it is caused

> **OSLER NODES**
> traditionally **immunological** (immune-complex vasculitis); histology also shows microemboli. They count as an **immunological phenomenon** in the Duke criteria.

> **JANEWAY LESIONS**
> **septic microemboli** with dermal microabscesses (a **vascular phenomenon**), so organisms may be cultured from them.

> **SPLINTERS**
> rupture of longitudinally oriented nail-bed capillaries (trauma) or microemboli/vasculitis.

Clinical features

Clinical features & sites

The Osler/Janeway distinction is tender vs painless. Look also for **Roth spots** (retinal haemorrhages with pale centres — fundoscopy), conjunctival petechiae, clubbing (subacute), new murmur, fever, splenomegaly, haematuria, and embolic events (stroke, limb ischaemia).

Fig. 2 · Osler nodes — tender red-purple nodules on finger pads/dorsum (immunological phenomenon, usually subacute endocarditis). Not pathognomonic (also SLE, gonococcaemia).

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Diagnosis and investigations

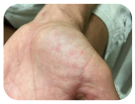
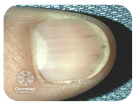
ENDOCARDITIS

2023 Duke–ISCVID / ESC 2023 criteria: **Major:** typical organisms in 2 blood cultures; imaging evidence (echo/PET-CT vegetation, abscess, new valvular regurgitation); **Minor:** predisposition (prosthetic valve, previous IE, IV drug use, congenital heart disease, cardiac devices), **fever 38 °C**, **vascular phenomena** (major arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial haemorrhage, conjunctival haemorrhages, **Janeway lesions**), **immunological phenomena** (glomerulonephritis, **Osler nodes**, Roth spots, positive rheumatoid factor), microbiological evidence not meeting the major criterion

TESTS (HOSPITAL)

3 sets of blood cultures **before antibiotics**, echocardiography (TTE TOE), FBC, CRP, urinalysis.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Osler nodes	SLE, **gonococcaemia**, typhoid, haemolytic anaemia, marantic (non-bacterial thrombotic) endocarditis, infected radial-artery lines (distal emboli), vasculitis
	**Janeway lesions**	SLE, gonococcaemia, typhoid, other septicaemia, haemolytic anaemia
	**Splinter haemorrhages**	**Trauma (commonest — manual workers, distal splinters)**, **psoriasis**, **onychomycosis**, eczema, haemodialysis, vasculitis/connective tissue disease, **antiphospholipid syndrome**, trichinosis, high altitude, mitral stenosis, scurvy, drugs (tetracyclines, VEGF-inhibitors such as sunitinib/sorafenib), and some normal elderly people

No. None of the three is specific to IE.

SIGN	OTHER CAUSES
Osler nodes	SLE, **gonococcaemia**, typhoid, haemolytic anaemia, marantic (non-bacterial thrombotic) endocarditis, infected radial-artery lines (distal emboli), vasculitis
Janeway lesions	SLE, gonococcaemia, typhoid, other septicaemia, haemolytic anaemia
Splinter haemorrhages	**Trauma (commonest — manual workers, distal splinters)**, **psoriasis**, **onychomycosis**, eczema, haemodialysis, vasculitis/connective tissue disease, **antiphospholipid syndrome**, trichinosis, high altitude, mitral stenosis, scurvy, drugs (tetracyclines, VEGF-inhibitors such as sunitinib/sorafenib), and some normal elderly people

How many splinters matter? There is **no validated number**. Splinters are more suspicious when they are **new**, **multiple** (several in one nail or across **several nails** — some texts suggest 4), **proximal** (rather than distal — trauma), **painful**, and **accompanied by fever, a murmur or IE risk factors**. Context, not count, makes the diagnosis.

Management

> **Suspected IE** (fever + new murmur or IE risk + embolic/vascular signs) **same-day hospital assessment**. **Do not start oral antibiotics** before blood cultures — this sterilises cultures and delays diagnosis (a common primary-care pitfall).

> **IE PROPHYLAXIS (NICE CG64, UPDATED 2016)**
> antibiotic prophylaxis for dental/other procedures is **not routinely recommended**. Advise high-risk patients on good **dental hygiene**, to avoid body piercing/tattooing, and on **symptoms to report** (persistent fever, night sweats, weight loss).

> **PEOPLE WHO INJECT DRUGS**
> harm reduction (needle-exchange, PWID services), awareness of IE (often right-sided, S. aureus).

IP scope

Recognition and urgent referral; no antibiotic initiation for suspected IE in primary care; counselling of at-risk patients.

Referral

SAME DAY — Fever with a new murmur, splinters or Osler/Janeway lesions: **same-day admission**; do not give antibiotics before blood cultures unless septic.

Red flags and when to refer

- Fever with a new murmur, splinters or Osler/Janeway lesions: **same-day admission**; do not give antibiotics before blood cultures unless septic.

Red flags / refer

Any febrile patient with new splinters/Osler/Janeway lesions, a murmur, a prosthetic valve, IV drug use or embolic events same day. Isolated distal splinters in a well manual worker reassure and review.

Complications and prognosis

Complications

- Heart failure, stroke, abscesses, kidney injury.

Prognosis

- Depends on organism and speed of treatment.

Prescriber pearls (UK pharmacist IP)

"**Osler = Ouch** (tender); **Janeway = painless**."

"Fever + murmur + splinters = **blood cultures before antibiotics**."

Self-check quiz

5 questions · optional

Q1 When should you think of Osler nodes, Janeway lesions & splinter haemorrhages?

✓ Tender pulp nodules (Osler), painless palm or sole macules (Janeway) or splinters, especially with fever or a murmur.

Q2 What drives it?

✓ Immune-complex vasculitis (Osler) and septic micro-emboli (Janeway) from infected heart valves.

Q3 What is the backbone of care?

✓ Fever + signs = same-day admission for blood cultures before antibiotics and echocardiography (modified Duke criteria).

Q4 What must you never miss?

✓ Endocarditis in injecting drug users, people with prosthetic valves or devices; starting antibiotics before cultures.

Q5 Name a red flag that needs escalation.

✓ Fever with a new murmur, splinters or Osler/Janeway lesions: **same-day admission**; do not give antibiotics before blood cultures unless septic.

References

1. DermNet: Osler nodes and janeway lesions – https://dermnetnz.org/topics/osler-nodes-and-janeway-lesions
dermnetnz.org
2. Medscape: Osler nodes, Janeway lesions & splinter haemorrhages – https://emedicine.medscape.com/article/216650-overview
emedicine.medscape.com

Photo gallery

Osler nodes 2

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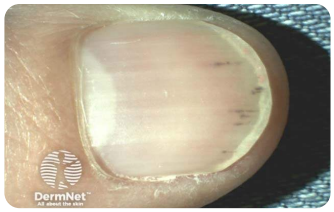
Splinter haemorrhages 3



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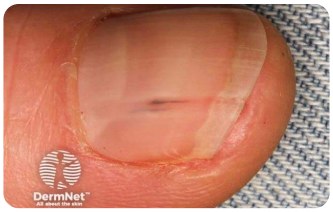

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Septic emboli (Janeway pattern) 1

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Vasculitis & purpura Module 1 · Week 4

Purpura: the differential diagnosis

Not every purple spot is vasculitis. Sort purpura into flat (platelets, clotting, fragile vessels) and palpable (vasculitis), and look for the dangerous causes: sepsis, leukaemia, DIC and systemic vasculitis.

ALSO CALLED non-blanching rash · petechiae · ITP · DIC · cryoglobulinaemia · acute haemorrhagic oedema · +1

RELATED PAGES IgA vasculitis (Henoch–Schönlein purpura) · Cutaneous small-vessel (leukocytoclastic) vasculitis · Skin fragility & dermatoporosis · Approach to cutaneous vasculitis · Dengue: skin signs & safety

KEY FACTS

- **Purpura** is bleeding into the skin that does **not blanch**: petechiae (<2 mm), purpura (2–10 mm), ecchymoses (>1 cm).
- **Non-blanching rash in an unwell or febrile child: meningococcal sepsis until proven otherwise** (NICE NG240: benzylpenicillin, 999).
- Flat purpura: platelets, clotting, vessel fragility (steroids, age, scurvy); palpable purpura: vasculitis.
- Tests: FBC and film, clotting, U&E, LFT, CRP, blood cultures if febrile; urine and BP for vasculitis.
- Consider leukaemia (petechiae, pallor, bone pain), ITP, anticoagulants and non-accidental injury in children.

Overview

Short comparison cards for the four other options and the emergency that must always be excluded.

Aetiology: causes and risk factors

How they are caused / 8.3 Clinical features

CONDITION	MECHANISM	WHO / FEATURES
Acute haemorrhagic oedema of infancy (AHEI)	Immune-complex LCV (often post-infection/vaccine/antibiotic)	**Infants 4 months–2 years**: large **rosette/target/medallion purpuric plaques** on face, ears and limbs with oedema and fever, but **the child looks well**. Usually skin-only; resolves in 1–3 weeks
Cryoglobulinaemic vasculitis	Cold-precipitating immunoglobulins (type II/III mixed — **hepatitis C**, Sjögren's, lymphoma; type I — myeloma/ Waldenström's occlusion)	Adults; palpable purpura on legs, **acral necrosis/ulcers, livedo**, Raynaud's, arthralgia, neuropathy, **membranoproliferative GN**, low C4; DIF IgM (option 5)

CONDITION	MECHANISM	WHO / FEATURES
Granulomatosis with polyangiitis (GPA)	**PR3-ANCA (c-ANCA)** necrotising granulomatous small/medium-vessel vasculitis	Rare in children; sinusitis, crusting, epistaxis, saddle nose, lung nodules/haemorrhage, **GN**; skin: palpable purpura, ulcers, nodules, **oral ulcers/"strawberry gums"**. Biopsy: **necrotising vasculitis with granulomas** (option 2)
ITP	Autoantibody platelet destruction (often post-viral in children)	**Well child, flat petechiae and bruising, mucosal bleeding; platelets <100 (often <20) × 10^9/L**; otherwise normal FBC/film. Most children recover in 6 months
Meningococcal sepsis / DIC	N. meningitidis endotoxin DIC and septic vasculitis	**Unwell, febrile child or young adult**; rapidly spreading **non-blanching petechiae purpura fulminans**; cold peripheries, leg pain, abnormal colour, tachycardia, ± neck stiffness — a **999 emergency**

Myths to address

- **"THE GLASS TEST IS NEGATIVE SO IT'S FINE"**
  early meningococcal rash may blanch; judge the whole child.

Pathophysiology and pathology

Purpura means **red cells outside vessels**: from low platelets, clotting defects, fragile or inflamed vessels.

1. **Platelet problem**
   Low or dysfunctional platelets cause petechiae and mucosal bleeding.
2. **Coagulation problem**
   Clotting factor deficiency causes larger bruises and deep bleeds.
3. **Vessel fragility**
   Thin dermis (age, steroids) or weak collagen (scurvy) lets vessels tear: senile purpura.
4. **Vasculitis**
   Inflamed vessels leak red cells, causing raised palpable purpura.

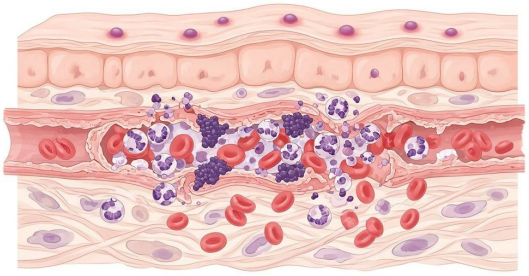

Fig. 1 · Vasculitis: neutrophils damage small vessels and red cells leak into the dermis.
AI-generated illustration

Clinical features

Fig. 2 · Acute haemorrhagic oedema of infancy — large rosette/target purpuric plaques with oedema in a well infant (4 months–2 years).

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Diagnosis and investigations

FBC + film (platelets), clotting/D-dimer/fibrinogen (DIC), blood cultures/meningococcal PCR, ANCA, cryoglobulins (37 °C), HCV, C3/C4, urinalysis.

Differential diagnoses

Also: non-accidental injury, Henoch–Schönlein purpura (see IgA vasculitis), leukaemia, haemophilia/von Willebrand, scurvy, HUS, drug-induced thrombocytopenia (heparin, quinine, valproate, co-trimoxazole).

Management

NICE NG240: in suspected meningococcal disease, transfer immediately to hospital; give **parenteral benzylpenicillin** (or ceftriaxone/cefotaxime) at the earliest opportunity **if it will not delay transfer**. Doses: benzylpenicillin IM/IV — **adults and children 10 years 1.2 g; 1–9 years 600 mg; <1 year 300 mg** (check BNF/BNFC). Many community IP settings will not hold it — call 999.

IP scope

Recognition and emergency referral.

Referral

999 / A&E — Unwell with a non-blanching rash: **999**, benzylpenicillin if there will be a delay (NICE NG240).

SAME DAY — Petechiae with pallor, lymph nodes, bone pain or bruising: **same-day FBC and film** (leukaemia, ITP).

Red flags and when to refer

- Unwell with a non-blanching rash: **999**, benzylpenicillin if there will be a delay (NICE NG240).
- Petechiae with pallor, lymph nodes, bone pain or bruising: **same-day FBC and film** (leukaemia, ITP).

Red flags / refer

Any non-blanching rash in an unwell or febrile patient 999. Purpura + low platelets same-day haematology. Purpura + sinus/lung/renal symptoms or positive ANCA urgent rheumatology/renal.

Complications and prognosis

Complications

- Depends on cause: sepsis, haemorrhage, organ damage.

Prognosis

- Depends on cause.

Prescriber pearls (UK pharmacist IP)

Teach parents the **glass/tumbler test**, but stress that an **unwell child needs urgent help even if the rash blanches**.

Self-check quiz

5 questions · optional

Q1 When should you think of Purpura: the differential diagnosis?

✓ Non-blanching red or purple spots or bruises, in a child or an adult.

Q2 What drives it?

✓ Low or faulty platelets, clotting problems, fragile vessels (age, steroids, scurvy) or vessel inflammation.

Q3 What is the backbone of care?

✓ Glass test; is the patient unwell? FBC and film, clotting, U&E, LFT, CRP, urine; same-day if unwell or platelets low.

Q4 What must you never miss?

✓ Meningococcal sepsis, leukaemia, ITP, DIC, cryoglobulinaemic or ANCA vasculitis, non-accidental injury.

Q5 Name a red flag that needs escalation.

✓ Unwell with a non-blanching rash: **999**, benzylpenicillin if there will be a delay (NICE NG240).

References

1. DermNet: Acute haemorrhagic oedema of infancy – https://dermnetnz.org/topics/acute-haemorrhagic-oedema-of-infancy
dermnetnz.org

Photo gallery

Purpura differentials 4

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CHAPTER 21

Skin signs of systemic disease

When the skin points inside: dermatomyositis, lung cancer signs, Cushing's, hyperhidrosis, Addison's, eruptive xanthomas, acanthosis nigricans, sarcoidosis, and the worked cases (including pyoderma gangrenosum).

1. Dermatomyositis
2. Skin signs of lung cancer & internal malignancy
3. Cushing's syndrome: skin signs
4. Hyperhidrosis
5. Addison's disease: skin signs
6. Eruptive xanthomas & skin signs of lipids
7. Sarcoidosis (incl. Löfgren syndrome)

Skin signs of systemic disease Module 1 · Week 5

Dermatomyositis

An autoimmune disease of skin and muscle: violet eyelids, Gottron papules over the knuckles, a photo-distributed rash and proximal weakness. In adults it can herald a cancer, so every new case needs a cancer screen.

ALSO CALLED heliotrope rash · Gottron papules · shawl sign · V-sign · myositis · anti-TIF1-γ · +2

RELATED PAGES Discoid lupus erythematosus · Skin signs of lung cancer & internal malignancy · Photoageing · Polymorphic light eruption

KEY FACTS

- **Dermatomyositis** is an autoimmune disease of skin and muscle: **heliotrope** eyelids, **Gottron papules**, shawl and V signs, nailfold changes.
- Proximal muscle weakness (may be absent: amyopathic DM); check CK, myositis antibodies, ANA.
- Adults: strongly linked to **cancer** (ovary, lung, breast, GI, nasopharynx): screen at diagnosis and for 3 years.
- Anti-MDA5 disease: rapidly progressive interstitial lung disease and skin ulcers.
- Specialist treatment: photoprotection, steroids, steroid-sparing agents (methotrexate, MMF), IVIG, hydroxychloroquine for skin.

Overview

An idiopathic **inflammatory myopathy** with characteristic skin signs. Incidence ~1 per 100,000. Peaks in childhood (juvenile DM, 5–14 years) and at 40–60 years (F>M). Around **20% are clinically amyopathic** (skin only — CADM). In adults it is a **paraneoplastic** marker in a significant minority.

Epidemiology

- **Who**
  - Two peaks: children 5–15 and adults 40–60; women twice as often.

Aetiology: causes and risk factors

Myths to address

- **"NORMAL CK EXCLUDES DERMATOMYOSITIS"**
  amyopathic disease has typical skin signs with normal muscles.

Pathophysiology and pathology

DM is an **interferon-driven attack** on small blood vessels in skin and muscle.

1. **Autoantibodies**
   Myositis-specific antibodies (Mi-2, TIF1-γ, NXP2, MDA5) mark subtypes and cancer risk.
2. **Type I interferon**
   Interferon signatures drive inflammation and complement activation.
3. **Capillary damage**
   Complement destroys capillaries (nailfold dilated loops, muscle perifascicular atrophy).
4. **Photosensitive interface dermatitis**
   UV-exposed skin shows violaceous erythema (shawl, V sign).

How it is caused

- Complement-mediated **microangiopathy** (MAC deposition in capillaries perifascicular atrophy), with type I interferon signatures, in genetically predisposed people.
- Triggers: malignancy, UV, drugs (hydroxyurea, statins — more often necrotising myopathy; TNF-i; checkpoint inhibitors).
- **Myositis-specific antibodies (MSA) predict phenotype:**

ANTIBODY	PHENOTYPE
Anti-TIF1-γ	**Strongest cancer association** (adults >40; ~50–75% of cancer-associated DM); extensive skin disease, mild myositis
Anti-NXP2	Cancer (adults), calcinosis (juvenile), oedema
Anti-SAE	Skin first, then myositis; dysphagia; some cancer risk
Anti-MDA5	**Rapidly progressive ILD**, skin ulcers, palmar papules, arthritis; little myositis (CADM); high mortality
Anti-Mi-2	Classic DM, florid skin, good steroid response, low cancer risk
Anti-synthetase (Jo-1 etc.)	ILD, **mechanic's hands**, arthritis, Raynaud's, fever

HISTOPATHOLOGY

- Interface dermatitis with dermal mucin (like lupus); muscle biopsy: perifascicular atrophy.

Clinical features

HELIOTROPE RASH

violaceous upper-eyelid erythema ± periorbital oedema.

GOTTRON PAPULES

violaceous flat-topped papules over the **MCP/PIP/DIP joints** (pathognomonic, A02). **Gottron sign:** macular erythema over the knuckles, **elbows, knees**.

NAILFOLD

dilated capillary loops, **ragged cuticles** (Samitz sign), periungual erythema.

Others: **itchy, scaly scalp** (psoriasiform/seb-derm-like), **mechanic's hands** (antisynthetase), calcinosis (juvenile), flagellate erythema, panniculitis, ulcers (MDA5), oedema.

Itch is prominent — a useful contrast with lupus.

MUSCLE

symmetrical **proximal weakness** (rising from a chair, climbing stairs, combing hair), myalgia, **dysphagia** (pharyngeal), respiratory muscle weakness.

SYSTEMIC

ILD (cough, breathlessness), arthritis, Raynaud's, myocarditis/arrhythmia.

Fig. 1 · Gottron papules — violaceous flat-topped papules over MCP/IP joints (pathognomonic). Contrast SLE, which favours the skin between the knuckles.

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Special groups

- **Juvenile DM**
  - Calcinosis and vasculopathy; no cancer link.

Diagnosis and investigations

BLOODS

CK (may be normal in CADM), aldolase, LDH, ALT/AST (raised from muscle — don't misread as liver disease), ESR/CRP, **ANA and myositis antibody panel (MSA/MAA)**, TFT, FBC, U&E.

SPECIALIST TESTS

EMG, **MRI of thighs** (oedema guides biopsy), **muscle ± skin biopsy** (interface dermatitis with mucin, similar to lupus).

ILD SCREEN

HRCT, **PFTs incl. TLCO**.

> **HEART**
> ECG ± echo/troponin.

> **Swallow assessment.**

> **EULAR/ACR 2017 classification criteria.**

> **CANCER SCREENING**
> Adult DM carries a **~3–7× increased cancer risk** (~15–25% of adults with DM). Highest in the 3 years around diagnosis (especially the first year), in those >40, with anti-TIF1-γ/NXP2, rapid onset, dysphagia, cutaneous necrosis, or a poor treatment response; Typical cancers: **ovarian**, **lung**, pancreatic, stomach, colorectal, breast, **non-Hodgkin lymphoma**; **nasopharyngeal** in South-East Asian patients; **IMACS 2023 international guideline** stratifies risk:; **Basic screen for all:** history/exam (incl. breast, pelvic, rectal/prostate, lymph nodes), FBC, LFT, ESR/CRP, SPEP/free light chains, urinalysis, CXR, and **age-appropriate national screening** (cervical, breast, bowel); **Enhanced screen for moderate/high risk: CT chest/abdomen/pelvis**, ± mammography, pelvic/transvaginal USS + CA-125, PSA, upper/lower endoscopy, **FDG-PET/CT**; Repeat annually for **3 years** in high-risk patients; Juvenile DM is **not** cancer-associated

Differential diagnoses

SLE/SCLE (spares knuckles — lupus favours skin between the joints; nasolabial folds spared; less itch), psoriasis (scalp/elbows), seborrhoeic dermatitis, contact/photo-dermatitis (eyelids), rosacea, **drug-induced DM-like eruptions** (hydroxyurea), lichen planus, polymorphic light eruption. For weakness: statin myopathy/**immune-mediated necrotising myopathy (anti-HMGCR)**, hypothyroid myopathy, inclusion-body myositis, steroid myopathy, polymyalgia rheumatica (pain, stiffness, not weakness; normal CK).

Management

> **Refer** to dermatology + rheumatology (urgently if weakness, dysphagia, breathlessness or a high CK). Same-day if there is respiratory compromise or difficulty swallowing.

> **SKIN**
> **strict photoprotection** (broad-spectrum high-UVA sunscreen); potent TCS or **topical tacrolimus** (face); antihistamines for itch; **hydroxychloroquine** — note that drug eruptions to HCQ are more common in DM than in lupus; RCOphth retinal screening applies

> **SYSTEMIC (SPECIALIST)**
> high-dose oral prednisolone (~1 mg/kg) ± IV methylprednisolone; steroid-sparing **methotrexate**, azathioprine or **mycophenolate**; **IVIg** (licensed in the UK for DM since 2021 — Octagam 10%); rituximab; JAK inhibitors (emerging, off-label); Anti-MDA5 ILD combination immunosuppression urgently

> **PHARMACIST CHECKLIST**
> **Steroid Emergency Card**, bone protection (calcium/vitamin D ± bisphosphonate — NOGG), gastroprotection as needed; PJP prophylaxis on combination therapy (specialist); inactivated vaccines (flu, COVID-19, pneumococcal; Shingrix); no live vaccines; methotrexate safety (the therapeutics notes see Dermatitis herpetiformis); **review statins** if weakness develops (statin-associated IMNM); support cancer screening appointments

IP scope

Recognition, CK/bloods, photoprotection and topical care, safe-prescribing support (steroids, MTX, HCQ shared care), and urgent referral. Diagnosis and immunosuppression are specialist.

Worked cases

CASE 1

dermatomyositis, lung cancer and the striae case

SKIN SIGNS OF DERMATOMYOSITIS

heliotrope rash (± periorbital oedema); **Gottron papules** (pathognomonic) and Gottron sign over knuckles, elbows and knees; photodistributed poikiloderma — **shawl, V and holster signs**; facial erythema that **involves the nasolabial folds**; **ragged cuticles and dilated nailfold capillaries**; itchy scaly scalp; mechanic's hands (antisynthetase); calcinosis (juvenile); ulcers (anti-MDA5). Itch is prominent (see Dermatomyositis).

ASSOCIATION WITH UNDERLYING DISEASE

Malignancy in adults — 3–7× risk (ovary, lung, pancreas, stomach, colorectal, breast, NHL; nasopharynx in South-East Asian patients), highest within 3 years of diagnosis, with **anti-TIF1-γ or anti-NXP2**; screen using the **IMACS 2023** risk-stratified approach (see Dermatomyositis); **Interstitial lung disease** (anti-MDA5, antisynthetase antibodies), dysphagia, myocarditis; overlap with other connective tissue diseases; Juvenile DM is not cancer-associated

SKIN SIGNS OF LUNG CANCER

clubbing ± HPOA; cutaneous metastases (chest wall, scalp, umbilicus; carcinoma erysipeloides); SVC obstruction; Horner's syndrome (Pancoast); dermatomyositis; **tripe palms** and malignant AN; ectopic ACTH pigmentation; erythema gyratum repens, acquired hypertrichosis lanuginosa, Leser–Trélat, Trousseau's sign, Bazex syndrome; EGFR-inhibitor and checkpoint-inhibitor rashes. Know the **NICE NG12 CXR triggers** (see Skin signs of lung cancer).

40-year-old woman with abdominal striae, amenorrhoea, weight gain, BP 142/92: Most probable diagnosis: Cushing's syndrome — the striae are wide, purple ("livid"), with menstrual disturbance, central weight gain and hypertension. **Exclude exogenous glucocorticoids first** (all routes, skin-lighteners, ritonavir/cobicistat + inhaled or nasal steroid); **Bedside/clinical tests:** BP; **proximal myopathy** (rise from a chair or squat without the arms); thin skin and bruising; width and colour of striae; facial plethora, fat pads; capillary glucose/urine dipstick; weight and waist; visual fields; **Single primary-care investigation:** the course teaches **24-h urinary free cortisol** (or late-night salivary cortisol where available). UK practice most often uses the **1 mg overnight dexamethasone suppression test** (cortisol < 50 nmol/L = normal suppression), and many pathways send suspected Cushing's straight to endocrinology. Any of the three is an acceptable first test; avoid the DST in women on oestrogen-containing contraception (see Cushing's syndrome — the striae case; see Where course teaching differs from UK guidance)

CASE 2

41-year-old man with hyperhidrosis

- **FURTHER HISTORY**
  onset, pattern (**focal vs generalised, day vs night, symmetry**), triggers, family history; **drugs** (SSRIs/SNRIs, TCAs, opioids, insulin/sulfonylureas, anabolic steroids or testosterone withdrawal, PPIs); **alcohol and recreational drugs — specifically ask about cocaine** (common and often not asked); **hypogonadism** (erectile dysfunction, low libido, **reduced shaving**, fatigue, gynaecomastia); **acromegaly** (**ring or shoe size change**, coarse features, headache); **OSA** (snoring, apnoeas, sleepiness); phaeochromocytoma (paroxysms of headache, palpitations, pallor), carcinoid (flushing, diarrhoea), infection or lymphoma (fever, **night sweats**, weight loss, lymph nodes, TB/HIV risk), hypoglycaemia, anxiety, GORD; impact (HDSS, DLQI)
- **Is primary hyperhidrosis likely? No.** Primary focal hyperhidrosis needs 6 months of focal sweating plus 2 of: bilateral and symmetrical; impairs daily activity; 1 episode a week; **onset before 25**; positive family history; **no sweating during sleep**. New sweating at 41, at most times of day, with obesity, points to a **secondary cause** (see Hyperhidrosis — the 41-year-old man).

	PRIMARY (FOCAL)	SECONDARY
Onset	Childhood–adolescence (< 25)	Any age, often adult
Distribution	Focal, symmetrical (axillae, palms, soles, face)	Generalised, or asymmetrical/unilateral
Sleep	No night sweats	Often at night
Family history	Positive in 30–50%	Usually negative
Cause	Central sympathetic overactivity	Drugs, endocrine, infection, malignancy, neurological, obesity

DIFFERENTIAL

obesity-related sweating and **obesity-related functional hypogonadism** (the course answer), OSA, drugs/alcohol/cocaine, thyrotoxicosis (excluded by normal TFT), hypoglycaemia on treatment, acromegaly, phaeochromocytoma, carcinoid, lymphoma/TB/HIV, anxiety, primary focal hyperhidrosis (unlikely).

FURTHER INVESTIGATIONS

9 am fasting total testosterone (repeat if low) with **SHBG, LH/FSH and prolactin**; fasting glucose and lipids; urine drug screen with consent; metanephrines if paroxysmal or hypertensive; **IGF-1** if acromegalic; ESR/CRP, HIV test and CXR if systemic features; 24-h urinary 5-HIAA if flushing; sleep study if OSA likely (see Hyperhidrosis — the 41-year-old man).

CASE 3

Addison's disease and the 40-year-old man with new diabetes

ADDISON'S — SKIN FEATURES

generalised bronze hyperpigmentation, most marked in **palmar creases, knuckles, elbows, knees, scars, nipples, gums and buccal mucosa**; darkening of naevi; vitiligo (autoimmune overlap); loss of axillary/pubic hair in women. On black or brown skin the change may be subtle — ask and look inside the mouth (NICE NG243).

SCREENING TEST IN PRIMARY CARE

an **8–9 am serum cortisol** with **U&E** (low sodium, high potassium), glucose and FBC (NICE NG243): **< 150 nmol/L refer to endocrinology** (and treat as crisis if unwell); **150–300 nmol/L** uncertain, repeat or seek advice; **> 300 nmol/L** AI very unlikely. **Never delay hydrocortisone 100 mg IM/IV in suspected adrenal crisis** (see Addison's disease).

Why does pigmentation occur? Low cortisol removes negative feedback more **POMC** cleaved into **ACTH and α-MSH**, which stimulate **MC1R** on melanocytes more melanin. Secondary (pituitary) adrenal insufficiency therefore causes **pallor, not pigmentation**.

ONE INVESTIGATION

a **fasting lipid profile** — expect a **turbid/milky (lipaemic) sample** with **very high triglycerides**. The diagnosis is **eruptive xanthomas** from hypertriglyceridaemia driven by uncontrolled type 2 diabetes (± obesity, alcohol, drugs, familial). It matters because of the **high risk of acute pancreatitis**: **TG > 20 mmol/L urgent specialist referral** (NICE NG238).

DIFFERENTIALS

molluscum contagiosum, dermatitis herpetiformis, folliculitis, acne/keloidal papules, papular granuloma annulare, xanthoma disseminatum/histiocytoses, papular sarcoidosis, lichen planus, insect bites. A shave biopsy (foamy macrophages ± extracellular lipid) is needed only if uncertain. "Nappy rash with cleansers and emollients" is not a diagnosis for persistent papules in an adult — re-examine and test.

CASE 4

acanthosis nigricans with PCOS, and dermatitis herpetiformis

DESCRIBE THE RASH

ill-defined, symmetrical, **velvety brown-grey thickening with accentuated skin markings** on the back of the neck, ± skin tags.

LIKELY DIAGNOSIS

acanthosis nigricans with **polycystic ovary syndrome and insulin resistance** (HAIR-AN phenotype). Rotterdam criteria: oligo-ovulation + clinical and biochemical hyperandrogenism (jawline hirsutism; testosterone 2.3 nmol/L, mildly raised). The COC had masked the symptoms. Normal TFT and prolactin help exclude other causes; check **SHBG/free androgen index, 17-OH-progesterone** (NC-CAH) and a **75 g OGTT or HbA1c** (see Acanthosis nigricans and PCOS).

ASSOCIATIONS OF AN

obesity and **insulin resistance** (pre-diabetes, type 2 diabetes, metabolic syndrome), PCOS, other endocrinopathies (Cushing's, acromegaly, hypothyroidism), familial/syndromic forms, **drugs** (systemic glucocorticoids, nicotinic acid, insulin injection sites, oral contraceptives, protease inhibitors), and **malignancy** (see Acanthosis nigricans and PCOS).

HISTOLOGY AND MALIGNANCY LINK

papillomatosis and hyperkeratosis with only mild acanthosis, little or no extra melanin — the brown colour comes from the thickened keratin. **Why some people get it: hyperinsulinaemia** activates **IGF-1 receptors** on keratinocytes and fibroblasts (endocrine AN, mild, gradual); in **malignant AN** tumour-derived growth factors (TGF-α acting on EGFR) cause **sudden, extensive, often mucosal or palmar ("tripe palms") disease** in older, non-obese people — most often **gastric adenocarcinoma**; endocrine AN is mild, malignant AN is severe and rapid. Advise about diabetes risk and weight (see Acanthosis nigricans and PCOS).

VISIBLE ABNORMALITIES

grouped small papules and vesicles on the extensor elbows, mostly **excoriated, eroded and crusted** (intact blisters are rare because of the intense itch); symmetrical, also knees and sacrum (see Dermatitis herpetiformis).

CLINICAL DIAGNOSIS

dermatitis herpetiformis, the skin form of **coeliac disease** (bloating, malaise and weight loss support it).

CONFIRM AND FURTHER TESTS

Skin biopsy — **perilesional** skin for **DIF (granular IgA in the dermal papillae)**, sent in **Michel's medium or saline, not formalin**, ± lesional biopsy **in formalin** for H&E; usually by dermatology; **tTG-IgA with total IgA** while the patient is **eating gluten** (more than one meal a day for 6 weeks; NICE NG20); IgG-based tests if IgA-deficient; EMA if equivocal; FBC, ferritin, folate, B12, LFT, calcium, vitamin D, TFT, HbA1c; **G6PD before dapsone**; bone density if indicated; Duodenal biopsy is usually not needed when skin DIF confirms DH (BSG), but refer to gastroenterology for significant GI symptoms or red flags (see Dermatitis herpetiformis)

CASE 5

40-year-old man with breathlessness, weight loss, night sweats and painful shins

DESCRIBE THE RASH

bilateral, **tender, warm, ill-defined red-violaceous nodules and plaques on the shins** = **erythema nodosum** (see Sarcoidosis — Löfgren syndrome).

CXR

bilateral hilar lymphadenopathy (Scadding stage I) ± right paratracheal nodes.

LIKELY DIAGNOSIS AND WHY

acute sarcoidosis, Löfgren-type — EN + BHL, with **hypercalcaemia (2.7 mmol/L)** from granuloma 1-α-hydroxylase activity and normocytic anaemia. **But weight loss, night sweats and anaemia mean TB and lymphoma must be excluded**: IGRA, sputum AFB/culture, HIV, LDH, film, HRCT and tissue (EBUS-TBNA) — classic Löfgren without B symptoms can skip tissue confirmation, this case cannot (see Sarcoidosis — Löfgren syndrome).

RELATED QUESTION — HOW AN EN NODULE EVOLVES

bright red, raised, tender nodule (**1–5 cm**) flattens over 1–2 weeks and turns **dusky violaceous, then yellow-green like a bruise** (erythema contusiformis) fades **without ulceration or scarring** over **3–6 weeks** (occasionally up to 8); crops recur; post-inflammatory pigmentation may linger in darker skin.

RELATED QUESTION — CAUSES OF EN

idiopathic (~30–50%); **infection** (streptococcal pharyngitis — the commonest identifiable cause; TB; Yersinia, Salmonella, Campylobacter; Chlamydia, Mycoplasma; EBV; hepatitis B; fungal infections such as coccidioidomycosis); **sarcoidosis**; **IBD**; **pregnancy and oestrogen-containing contraception**; **drugs** (sulfonamides, penicillins, oral contraceptives, bromides, vemurafenib/BRAF inhibitors); Behçet's; lymphoma and leukaemia (rare). First tests: throat swab/ASO titre, FBC, CRP/ESR, CXR, IGRA, pregnancy test, stool culture if diarrhoea, faecal calprotectin.

RELATED QUESTION — DIFFERENTIALS OF EN

cellulitis (unilateral, spreading, fever); superficial thrombophlebitis and DVT; **erythema induratum (nodular vasculitis)** — calves, ulcerates, linked to TB; other panniculitides (lupus profundus, pancreatic panniculitis, α1-antitrypsin deficiency, cold panniculitis); cutaneous polyarteritis nodosa (painful nodules with livedo); Sweet syndrome; insect bites; bruises/trauma; subcutaneous sarcoid; lymphoma. See the drug eruption notes for the full table.

Practice case: 38-year-old woman with Crohn's on azathioprine

- **QA = 4, pyoderma gangrenosum.** A minor graze (pathergy) pustule within days **rapidly enlarging, very painful ulcer** in a woman with **Crohn's disease**, low-grade fever and raised CRP/ESR (see Pyoderma gangrenosum): 1. Atypical mycobacterial infection: evolves over weeks to months; it should still be excluded (tissue culture) because she is on azathioprine; 2. S. aureus infection: would not usually produce a violaceous undermined ulcer this painful and fast, and would respond to antibiotics; swab anyway; 3. Martorell ulcer: an ischaemic ulcer of long-standing **hypertension ± diabetes**, usually older patients, lateral/posterior leg; her BP is

125/80; 5. SCC: would not reach this size in 4 weeks, is rarely this painful and lacks an inflammatory violaceous border (see Pyoderma gangrenosum)

- **QB = 2, diagnosis based on clinical and laboratory criteria.** There is no confirmatory test; PG is a diagnosis of exclusion using the **Su criteria** (the course answer); current UK specialist practice uses the **Delphi consensus criteria** (see Pyoderma gangrenosum): 4/5. Skin biopsy for histology (and tissue culture): an essential part of the work-up — mainly to **exclude infection, vasculitis and malignancy** — but histology is non-specific, so biopsy alone does not confirm PG (and a biopsy of the edge is the major Delphi criterion; see Where course teaching differs from UK guidance); 1. ABPI: assesses arterial disease and safety of compression; no diagnostic role in PG; 3. Nerve conduction studies/EMG: for neuropathic ulcers
- **QC = 1, myeloid blood dyscrasias** (AML, myelodysplasia, myeloproliferative disorders; also IgA monoclonal gammopathy/myeloma). Other well-described associations are **rheumatoid and other inflammatory arthritis** and **IBD**. Eosinophilic asthma, penicillin antibiotics, pregnancy and ischaemic heart disease are not recognised associations (see Pyoderma gangrenosum).
- **UK ADDITIONS FOR THIS PATIENT**
  exclude infection first (swab, FBC — azathioprine can cause neutropenia — CRP, tissue culture via biopsy); **urgent dermatology referral and tell her IBD team**; no debridement; analgesia (avoid NSAIDs in Crohn's); non-adherent dressings; check her Crohn's activity. Expect **prednisolone or ciclosporin** (STOP GAP), given active IBD, an **anti-TNF (infliximab/adalimumab)**; steroid safety, interaction and vaccination checks.

Week summary

The course summary lists the week's themes: Cushing's, dermatomyositis, sarcoid and dermatitis herpetiformis (diagnosis, differentials, investigations and associations); Addison's features and screening; eruptive xanthomas and their pancreatitis risk; and pyoderma gangrenosum, a painful ulcerating disease whose commonest association is IBD. The common thread: **name the skin sign, think of the internal disease, do the first test, and refer at the right speed.**

Referral

SAME DAY — Difficulty swallowing, breathlessness or rapidly worsening weakness: **same-day** assessment.

ROUTINE — New dermatomyositis in an adult: screen for **cancer** (NICE NG12 symptoms plus specialist-led screening).

Red flags and when to refer

- Difficulty swallowing, breathlessness or rapidly worsening weakness: **same-day** assessment.
- New dermatomyositis in an adult: screen for **cancer** (NICE NG12 symptoms plus specialist-led screening).

Red flags / refer

Proximal weakness, **dysphagia/aspiration**, breathlessness/cough (ILD), very high CK, chest pain/palpitations (myocarditis), skin ulceration (MDA5), constitutional symptoms or other cancer red flags (NG12 — urgent cancer pathway).

Complications and prognosis

Complications

- Cancer, interstitial lung disease, dysphagia, aspiration, calcinosis, cardiac involvement.

Prognosis

- Chronic; survival depends on cancer and lung disease.

Prescriber pearls (UK pharmacist IP)

"**Purple eyelids + knuckle papules + itch = dermatomyositis — check CK and look for cancer.**"

A DM rash with "raised ALT" may be muscle, not liver — check CK before stopping a statin for "LFTs".

Itchy scalp in a middle-aged woman with eyelid redness: think DM, not just seborrhoeic dermatitis.

The worked practice case on the painful leg ulcer in Crohn's disease is on the pyoderma gangrenosum page, with diagnostic criteria and UK treatment.

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK POSITION (2026)
Cushing's first test	Course teaching: 24-h urinary free cortisol, or late-night salivary cortisol where available	All three tests are valid (Endocrine Society); UK pathways most often use the **1 mg overnight dexamethasone suppression test** (cortisol < 50 nmol/L normal) or refer straight to endocrinology. **Exclude exogenous steroids first** (incl. ritonavir/cobicistat + inhaled/nasal steroids). Avoid the DST in women on oral oestrogen (false positive)
Addison's screen	Course teaching: "8 am cortisol"; a student answer: 9 am cortisol < 100 nmol/L suggests Addison's	**NICE NG243 (2024):** 8–9 am serum cortisol with U&E; **< 150 nmol/L = refer** (consider treating); 150–300 uncertain; > 300 very unlikely. Do not wait for tests in suspected crisis: **hydrocortisone 100 mg IM/IV and 999**
Addison's pigmentation	Course: generalised darkening of the skin	NG243 adds that pigmentation **may not be visible on black or brown skin** — ask, and check the buccal mucosa and scars
Estradiol units	450 nmol/L (range 150–800)	**pmol/L** — UK labs report estradiol in pmol/L; nmol/L values would be 1,000-fold too high
Female testosterone	2.3 nmol/L flagged against "< 2"	Reference ranges vary by lab and assay; interpret with **SHBG/free androgen index**. A total testosterone **5 nmol/L**, rapidly worsening hirsutism or virilisation **urgent endocrinology referral** to exclude an androgen-secreting tumour, NC-CAH or Cushing's (NICE draft PMOS/PCOS guideline 2026; local pathways)

TOPIC	COURSE TEACHING	UK POSITION (2026)
DH biopsy medium	Course teaching: "skin biopsy for histology in formalin with DIF (requires special media)"	Two samples: **lesional skin in formalin** (H&E) and **perilesional skin in Michel's medium or saline for DIF — never formalin**
Coeliac serology before testing	Course teaching: "prep to the test"	**NICE NG20:** the patient must eat gluten in **more than one meal every day for at least 6 weeks** before serology; always measure **total IgA** with tTG-IgA
Eruptive xanthoma investigation	Course teaching: lipids (turbid sample, very high TG); GP in the case: "nappy rash", cleansers and emollients	Agree with lipids; add **NICE NG238**: **TG > 20 mmol/L urgent specialist referral**; 10–20 mmol/L repeat fasting within 5–14 days. Treat glycaemia urgently; review alcohol and drugs (thiazides, β-blockers, oestrogens, retinoids, antipsychotics)
Hyperhidrosis and hypogonadism	Course teaching: check testosterone — obesity-related hypogonadism	Agree; repeat a **9 am fasting** sample with SHBG and LH/FSH before diagnosing; **weight loss is first-line** for obesity-related functional hypogonadism and testosterone replacement is a specialist decision (Society for Endocrinology / BSSM). Treat sweating per **NICE CKS *Hyperhidrosis** (aluminium chloride; iontophoresis; specialist botulinum toxin or anticholinergics)
EN nodule size and course	Student answers: nodules "2–5 mm" resolving over 8 weeks	EN nodules are **1–5 cm**; they usually fade in **3–6 weeks** (sometimes up to 8) through bruise-like colour change, without scarring
Sarcoid diagnosis	Student answers: EN + BHL + hypercalcaemia = sarcoid	Agree it fits Löfgren, but **weight loss, night sweats and anaemia demand exclusion of TB and lymphoma** (BTS 2021); serum ACE is supportive only. **Stop vitamin D/calcium supplements** if hypercalcaemic
PG — what confirms it	Answer 2: clinical and laboratory criteria (Su 2004); some students chose 5 (biopsy + tissue culture)	Agree there is no single confirmatory test. Current consensus (**Delphi 2018**) makes **biopsy of the ulcer edge the major criterion**, with 4 minor criteria including **exclusion of infection by histology and tissue culture** — so biopsy + culture is expected in practice, even though it does not confirm PG on its own
PG treatment	Course: rapid response to systemic steroids (a minor criterion)	UK STOP GAP RCT: **prednisolone 0.75 mg/kg/day and ciclosporin 4 mg/kg/day are equally effective**; very potent topical steroid or topical tacrolimus for smaller ulcers; anti-TNF (infliximab) when IBD coexists. **Never debride active PG**
Martorell ulcer distractor	Course teaching: excluded because there is no history of hypertension	Correct; teach also that Martorell responds to **debridement and grafting**, the opposite of PG — a practical reason the distinction matters

Self-check quiz

5 questions · optional

Q1 When should you think of Dermatomyositis?

✓ Violaceous eyelids, scaly papules over the knuckles, a red V or shawl on the chest and back, ragged cuticles, with or without proximal muscle weakness.

Q2 What drives it?

✓ Autoimmune damage to small vessels of skin and muscle; myositis-specific antibodies predict cancer (TIF1-γ, NXP-2) or lung disease (MDA5).

Q3 What is the backbone of care?

✓ CK, ANA and myositis antibodies, urgent rheumatology/dermatology referral; age-appropriate cancer screening; strict photoprotection.

Q4 What must you never miss?

✓ Underlying malignancy (ovary, lung, gut, breast); interstitial lung disease; swallowing and respiratory muscle weakness.

Q5 Name a red flag that needs escalation.

✓ Difficulty swallowing, breathlessness or rapidly worsening weakness: **same-day** assessment.

References

1. DermNet: Adult onset dermatomyositis — https://dermnetnz.org/topics/adult-onset-dermatomyositis
   dermnetnz.org
2. Medscape: Dermatomyositis — https://emedicine.medscape.com/article/332783-overview
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Photo gallery

Dermatomyositis 6

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Skin signs of systemic disease Module 1 · Week 5

Skin signs of lung cancer & internal malignancy

Clubbing, tripe palms, sudden acanthosis nigricans, eruptive seborrhoeic keratoses, skin metastases and other paraneoplastic clues that should prompt an urgent cancer search.

ALSO CALLED paraneoplastic · clubbing · tripe palms · carcinoma erysipeloides · skin metastases · Leser–Trélat · +1

RELATED PAGES Nail signs: psoriatic nails, Beau's lines & clubbing · Dermatomyositis · Acanthosis nigricans · Melanoma immunotherapy & targeted therapy: side-effects

KEY FACTS

- Lung cancer is the commonest cause of UK cancer death; skin signs may be the first clue.
- Paraneoplastic clubbing/HPOA, dermatomyositis, acanthosis/tripe palms, ectopic ACTH pigmentation.
- Local SVC obstruction (facial swelling, distended chest veins), Horner's syndrome; **direct**: skin metastases (chest wall, scalp).
- Treatment-related EGFR-inhibitor acneiform rash, checkpoint-inhibitor rashes.
- Use smoking history as a red-flag multiplier; refer per NICE NG12. SVC obstruction is same-day.

Overview

Lung cancer is the commonest cause of UK cancer death. Its skin signs are **direct** (metastases), **local** (SVC obstruction, Horner's syndrome), **paraneoplastic** (clubbing/HPOA, dermatomyositis, acanthosis/tripe palms, ectopic ACTH) or **treatment-related** (EGFR-inhibitor acneiform rash, checkpoint-inhibitor rashes, radiotherapy).

Aetiology: causes and risk factors

Myths to address

- **"An acne-like rash on cancer treatment means stop the drug"**: EGFR rashes are managed with oncology protocols and often signal response.

Pathophysiology and pathology

Tumours affect skin through **growth factors, hormones, immunity and spread**.

1. **Growth factors**
   VEGF and PDGF (from platelet clumps bypassing the lung) drive clubbing and HPOA.

2 **Ectopic hormones**

Small-cell tumours make ACTH: pigmentation, hypokalaemia.

3 **Immune cross-reaction**

Anti-tumour immunity cross-reacts with skin and muscle (dermatomyositis).

4 **Spread and obstruction**

Metastases and SVC compression.

How it is caused

Tumour-derived growth factors (TGF-α, EGF, VEGF, PDGF in clubbing/HPOA), ectopic hormones (ACTH — small-cell lung cancer), immune cross-reactivity (DM), haematogenous/lymphatic spread, mechanical obstruction (SVC).

Clinical features

Clinical features & sites

SIGN	NOTES
Finger clubbing ± hypertrophic pulmonary osteoarthropathy (HPOA)	Loss of the nail-fold angle, Schamroth window lost; HPOA = painful wrists/ankles with periostitis. **NSCLC** especially (adenocarcinoma). Also bronchiectasis, ILD, cyanotic heart disease, IBD, cirrhosis, endocarditis
Cutaneous metastases	Firm, painless pink-purple or skin-coloured nodules on the **chest wall, scalp, abdomen/umbilicus**; **carcinoma erysipeloides** (inflammatory, cellulitis-like plaque — photo B01)
SVC obstruction	Facial/neck oedema and plethora, dilated chest-wall veins, worse lying down — an oncological emergency
Horner's syndrome	Pancoast (apical) tumour: ptosis, miosis, anhidrosis, ± shoulder/arm pain (T1)
Dermatomyositis	see Dermatomyositis
Acanthosis nigricans (malignant) / tripe palms	Velvety ridged palms; tripe palms alone lung cancer more likely; with AN gastric (see Acanthosis nigricans and PCOS)
Ectopic ACTH (small-cell)	Rapid **hyperpigmentation**, hypokalaemia, myopathy, oedema, hypertension; usually **no time for classic striae**
Others	**Erythema gyratum repens** (rare, wood-grain concentric bands — strongly associated with lung cancer), **acquired hypertrichosis lanuginosa**, sign of **Leser–Trélat** (sudden eruptive seborrhoeic keratoses), **Trousseau's migratory thrombophlebitis**, **Bazex syndrome** (acrokeratosis paraneoplastica — upper aerodigestive SCC, psoriasiform acral plaques), pruritus, Sweet syndrome, paraneoplastic pemphigus, palmar hyperkeratosis
Treatment-related	**EGFR inhibitors** (erlotinib, gefitinib, osimertinib): acneiform folliculitis, paronychia, xerosis — doxycycline/lymecycline + topical steroid; **checkpoint inhibitors**: lichenoid/eczematous rashes, vitiligo, bullous pemphigoid, SJS (rare)

Fig. 1 · Tripe palms — velvety, ridged palmar thickening; strongly paraneoplastic (lung cancer when alone; gastric when with AN).
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Diagnosis and investigations

- **Refer on a suspected cancer pathway** if CXR findings suggest lung cancer, or if aged **40 with unexplained haemoptysis**.
- **Offer an urgent CXR (within 2 weeks)** to people **40** with **two or more** unexplained symptoms — or **one** if they have **ever smoked**: cough, fatigue, shortness of breath, chest pain, weight loss, appetite loss.
- **Consider an urgent CXR** in people 40 with: persistent or recurrent chest infection; **finger clubbing**; supraclavicular or persistent cervical lymphadenopathy; chest signs consistent with lung cancer; **thrombocytosis**.
- Skin metastasis biopsy (usually via a dermatology urgent pathway) + CT.

Differential diagnoses

Clubbing (above); carcinoma erysipeloides vs **cellulitis** (afebrile, no response to antibiotics, firm); metastases vs cysts/lipomas/DFSP; SVC obstruction vs angioedema/heart failure.

Management

Recognise and refer per NG12. Use **smoking history as a red-flag multiplier**. Smoking cessation support (NCSCT, varenicline/NRT). Manage EGFR-inhibitor rashes with oncology protocols (pre-emptive doxycycline, emollients, SPF, mild steroid).

IP scope

Recognition, CXR request (where the IP can order it locally), suspected-cancer referral per local IP rights, smoking cessation, supportive dermatology for oncology drug rashes.

Referral

URGENT — New finger clubbing, haemoptysis, weight loss or persistent cough in a smoker over 40: **urgent CXR / suspected cancer pathway** (NICE NG12).

URGENT — 'Cellulitis' on the chest wall after breast cancer that does not respond to antibiotics: think **carcinoma erysipeloides**, urgent referral.

Red flags and when to refer

- New finger clubbing, haemoptysis, weight loss or persistent cough in a smoker over 40: **urgent CXR / suspected cancer pathway** (NICE NG12).
- 'Cellulitis' on the chest wall after breast cancer that does not respond to antibiotics: think **carcinoma erysipeloides**, urgent referral.

Red flags / refer

SVC obstruction (same day); haemoptysis 40 (suspected cancer pathway); new clubbing with chest symptoms; cutaneous nodules in a smoker; rapid pigmentation + hypokalaemia (ectopic ACTH).

Complications and prognosis

Complications

- Late diagnosis; treatment interruptions.

Prognosis

- Depends on cancer stage.

Prescriber pearls (UK pharmacist IP)

"**Clubbing in a smoker over 40 = chest X-ray.**"

An "**antibiotic-resistant cellulitis**" on the chest wall of a cancer patient may be carcinoma erysipeloides.

Self-check quiz

5 questions · optional

Q1 When should you think of Skin signs of lung cancer & internal malignancy?

✓ New clubbing, velvety ridged palms, sudden widespread acanthosis nigricans or seborrhoeic keratoses, or a firm red 'cellulitis' that does not respond.

Q2 What drives it?

✓ Tumour-derived growth factors and cytokines, or direct spread of cancer to the skin.

Q3 What is the backbone of care?

✓ Take a smoking and symptom history; CXR and NICE NG12 urgent pathways; don't keep treating a non-responding 'cellulitis'.

Q4 What must you never miss?

✓ Lung, gastric and breast cancer; carcinoma erysipeloides mistaken for infection.

Q5 Name a red flag that needs escalation.

✓ New finger clubbing, haemoptysis, weight loss or persistent cough in a smoker over 40: **urgent CXR / suspected cancer pathway** (NICE NG12).

References

1. DermNet: Skin metastasis — https://dermnetnz.org/topics/skin-metastasis
dermnetnz.org
2. NICE NG12 Suspected cancer: recognition and referral: NICE NG122 lung cancer — https://www.nice.org.uk/guidance/ng12
nice.org.uk

Photo gallery

Paraneoplastic and metastatic signs

2

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Skin signs of systemic disease Module 1 · Week 5

Cushing's syndrome: skin signs

Wide purple striae, thin bruising skin, a round red face and central weight gain from too much cortisol, most often from steroid medicines (oral, inhaled, topical or injected).

ALSO CALLED striae · moon face · steroid excess · iatrogenic Cushing's · purple striae · buffalo hump

RELATED PAGES Addison's disease: skin signs · Topical corticosteroids: potency, quantities & safety · Hirsutism & the Ferriman–Gallwey score · Acanthosis nigricans

KEY FACTS

- Glucocorticoid excess: **iatrogenic** (steroids) is far commoner than endogenous Cushing's.
- Skin signs: **thin skin, wide purple striae, easy bruising**, acne, hirsutism, plethora, poor wound healing, fungal infections.
- Body habitus: moon face, dorsocervical and supraclavicular fat pads, proximal myopathy.
- Iatrogenic: **never stop steroids abruptly**; taper, Steroid Emergency Card and sick-day rules.
- Endogenous: refer to endocrinology, urgently with hypokalaemia, severe hypertension or rapid progression.

Overview

The clinical state caused by prolonged **glucocorticoid excess**.

- **Exogenous (iatrogenic)** — by far the commonest: oral, inhaled, injected, **potent topical** steroids, and unlicensed skin-lightening creams (link to the therapeutics notes see Dermatomyositis).
- **Endogenous** (~2–3 per million/year): **ACTH-dependent** (~80%): **pituitary adenoma = Cushing's disease** (~70%) or **ectopic ACTH** (small-cell lung cancer, carcinoid); **ACTH-independent:** adrenal adenoma/carcinoma, bilateral adrenal hyperplasia

Aetiology: causes and risk factors

Myths to address

- **"STRIAE ALWAYS MEAN CUSHING'S"**
  pale narrow striae are common; wide (>1 cm) purple striae are the concern.

Pathophysiology and pathology

Excess cortisol **breaks down collagen and protein**, thinning skin and vessels.

1. **Catabolism**
   Cortisol inhibits fibroblasts and collagen synthesis.
2. **Thin skin and striae**
   Dermis thins; stretching tears it into wide purple striae.
3. **Fragile vessels**
   Poor dermal support causes bruising.
4. **Androgen and immune effects**
   Adrenal androgens cause acne and hirsutism; immunosuppression allows fungal infection.

How it is caused

Cortisol is catabolic to protein **thin skin, striae, bruising, proximal myopathy**, osteoporosis. It causes insulin resistance diabetes and central fat redistribution (moon face, dorsocervical and supraclavicular fat pads). Mineralocorticoid effect **hypertension**, hypokalaemia. Gonadotrophin suppression and adrenal androgens **amenorrhoea/oligomenorrhoea**, hirsutism, acne. Immunosuppression infections, poor healing.

Clinical features

Clinical features & sites

- **Most discriminating features** (Endocrine Society): **easy bruising**, **facial plethora**, **proximal myopathy**, and **wide (>1 cm), purple/"livid" striae** on the abdomen, flanks, axillae and thighs.
- Others: moon face, dorsocervical "buffalo hump", supraclavicular fat pads, central obesity with thin limbs, thin skin with purpura, acne, hirsutism, acanthosis nigricans, fungal infections (tinea versicolor), poor wound healing, oedema, depression/psychosis, insomnia, kidney stones, osteoporotic fractures. **Children: weight gain with slowed growth.**
- **THE CASE**
  a 40-year-old woman with **abdominal striae, amenorrhoea, weight gain and BP 142/92** most likely **Cushing's syndrome** (after excluding exogenous steroids). Why: wide purple striae + menstrual disturbance + hypertension + weight gain is a combination of discriminatory and common features; at 40 (not pregnant, no recent large weight change) striae should not be dismissed as physiological.

Fig. 1 · Wide purple (livid) striae — the Cushing's clue; exogenous (incl. potent topical/inhaled) steroids are the commonest cause.

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Diagnosis and investigations

FIRST, EXCLUDE EXOGENOUS GLUCOCORTICOIDS

prescribed, OTC (hydrocortisone creams), inhaled/nasal, joint injections, **skin-lightening products**, herbal remedies adulterated with dexamethasone. Ritonavir/cobicistat + inhaled/nasal fluticasone or budesonide (CYP3A4) iatrogenic Cushing's — a key pharmacist check.

BEDSIDE/CLINICAL TESTS

BP; **capillary glucose**/urinalysis for glycosuria; weight, BMI and waist; **Proximal muscle strength** — rise from a chair or squat without using the arms; climb a step; **Skin thickness** (skin fold on the back of the hand <2 mm — Liddle's sign); look for bruises, striae width and colour, plethora, fat pads; Visual fields (pituitary macroadenoma — bitemporal hemianopia), mood, and signs of infection

SINGLE PRIMARY-CARE INVESTIGATION

Course teaching suggests **24-h urinary free cortisol** (2 collections), or **late-night salivary cortisol** (2) where available; **UK practice:** the **1 mg overnight dexamethasone suppression test** (dexamethasone 1 mg at 23:00, serum cortisol at 08:00–09:00; normal suppression **<50 nmol/L**) is the most commonly used first-line screen. Many ICB pathways advise **referring to endocrinology for testing** rather than testing in primary care — follow the local pathway; Pitfalls: oestrogen-containing COCs raise CBG (false-positive DST — stop the COC 6 weeks before or use UFC/salivary cortisol); enzyme inducers (carbamazepine, phenytoin, rifampicin) speed dexamethasone clearance (false positive); pseudo-Cushing's states (alcohol, depression, severe obesity, poorly controlled diabetes)

SECONDARY CARE

ACTH (dependent vs independent), high-dose DST/CRH test, pituitary MRI, inferior petrosal sinus sampling, adrenal CT, K^+, HbA1c, DEXA.

Differential diagnoses

Physiological striae (pregnancy, adolescence, rapid weight gain/bodybuilding — usually narrower, pink then silvery-white), **iatrogenic Cushing's** (topical/inhaled/oral steroids), metabolic syndrome/obesity, **PCOS** (oligomenorrhoea,

hirsutism, acne, AN — no myopathy/bruising), pseudo-Cushing's (alcohol, depression), hypothyroidism, Marfan/Ehlers–Danlos (striae).

Management

> **Refer to endocrinology** (urgently if hypokalaemia, severe hypertension, rapid progression, psychosis or suspected ectopic ACTH/adrenal carcinoma).

> **IATROGENIC CUSHING'S**
> **Do not stop steroids abruptly** — the HPA axis is suppressed. Taper to the lowest effective dose; use steroid-sparing therapy; **Steroid Emergency Card** and sick-day advice (NatPSA 2020; NICE NG243); Bone protection; glucose and BP monitoring; Review inhaled/nasal steroid + CYP3A4 inhibitor combinations

> **MEANWHILE, MANAGE THE COMORBIDITIES**
> BP (NICE NG136), diabetes, VTE risk (Cushing's is prothrombotic), osteoporosis, infection, mood.

> **DEFINITIVE TREATMENT (SPECIALIST)**
> transsphenoidal surgery, adrenalectomy, or medical therapy (metyrapone, osilodrostat, ketoconazole — interaction-heavy; levoketoconazole), radiotherapy.

IP scope

History and medicine review (crucial — iatrogenic causes), bedside assessment, bloods/first-line screen per local pathway, steroid-taper planning and emergency cards, comorbidity management, and referral.

Referral

999 / A&E — Never stop long-term systemic steroids suddenly: taper, issue a **steroid emergency card** and sick-day rules.

URGENT — Cushingoid features without steroid exposure, especially with rapid onset, weakness or low potassium: **urgent endocrine** referral.

Red flags and when to refer

- Never stop long-term systemic steroids suddenly: taper, issue a **steroid emergency card** and sick-day rules.
- Cushingoid features without steroid exposure, especially with rapid onset, weakness or low potassium: **urgent endocrine** referral.

Red flags / refer

Rapid onset with hypokalaemia, hyperpigmentation, muscle wasting (ectopic ACTH — urgent); virilisation (adrenal carcinoma); visual field loss; severe hypertension, psychosis or fractures.

Complications and prognosis

Complications

- Diabetes, hypertension, osteoporosis, VTE, infection, mood disorder.

Prognosis

- Skin signs improve as cortisol normalises.

Prescriber pearls (UK pharmacist IP)

- "**Before you test for Cushing's, take the steroid history — every route.**"
- "**Wide, purple, and with weakness or bruising** — not just stretch marks."
- Fluticasone (inhaled/nasal) + ritonavir = a classic, preventable cause.

Self-check quiz

5 questions · optional

Q1 When should you think of Cushing's syndrome: skin signs?

✓ Wide purple striae, easy bruising, thin skin, moon face, acne and central weight gain, often in someone on steroids.

Q2 What drives it?

✓ Excess glucocorticoid breaks down collagen and redistributes fat; usually iatrogenic, sometimes pituitary, adrenal or ectopic ACTH.

Q3 What is the backbone of care?

✓ Total steroid review (all routes); if not iatrogenic, overnight dexamethasone suppression test or late-night salivary cortisol and endocrine referral; never stop long-term steroids abruptly.

Q4 What must you never miss?

✓ Adrenal crisis from stopping steroids suddenly; hypertension, diabetes, osteoporosis; ectopic ACTH from lung cancer.

Q5 Name a red flag that needs escalation.

✓ Never stop long-term systemic steroids suddenly: taper, issue a **steroid emergency card** and sick-day rules.

References

1. DermNet: Cushing syndrome — https://dermnetnz.org/topics/cushing-syndrome
dermnetnz.org

Photo gallery

Cushing's syndrome 3

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Skin signs of systemic disease Module 1 · Week 5

Hyperhidrosis

Sweating beyond what is needed to control temperature. Primary focal hyperhidrosis (armpits, palms, soles, face) starts young and is symmetrical; generalised or night sweating needs a search for a cause.

ALSO CALLED excessive sweating · axillary hyperhidrosis · palmar sweating · aluminium chloride · iontophoresis · botulinum toxin · +1

RELATED PAGES Keratolysis exfoliativa · Hand eczema (chronic, occupational & pompholyx) · Tinea: corporis, cruris & pedis · Facial ageing & neuromodulators

KEY FACTS

- **Hyperhidrosis** is sweating beyond what is needed for temperature control; **primary focal** (axillae, palms, soles, face) or **secondary generalised**.
- Primary: starts in childhood or adolescence, bilateral and symmetrical, absent during sleep, family history.
- Secondary: generalised or night sweats, later onset: look for drugs (SSRIs), endocrine disease, infection, malignancy.
- Aluminium chloride hexahydrate 20% at night; then iontophoresis (palms/soles), botulinum toxin, oral anticholinergics (specialist).
- Huge quality-of-life impact; ask about anxiety and social avoidance.

Overview

Sweating beyond what is needed for thermoregulation.

> **PRIMARY FOCAL HYPERHIDROSIS**
> idiopathic; axillae, palms, soles, craniofacial; affects ~1–3% of people.

> **SECONDARY HYPERHIDROSIS**
> generalised (or focal from neurological lesions), due to a medical condition or drug.

Epidemiology

- **Who**
  - About 1–3% of people; onset in childhood or adolescence for primary focal hyperhidrosis.

Aetiology: causes and risk factors

Myths to address

- **"IT'S JUST NERVES"**

anxiety worsens it, but primary hyperhidrosis is a physiological overactivity of sweat glands.

Pathophysiology and pathology

Sweat glands are normal; the **nerves driving them are overactive**.

1. **Central drive**
   Hypothalamic and emotional centres send excess sympathetic signals.
2. **Cholinergic nerves**
   Sympathetic fibres release acetylcholine onto eccrine glands.

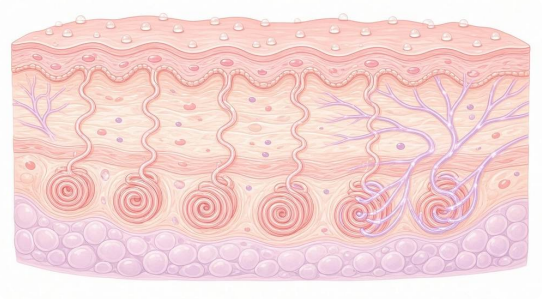

Fig. 1 · Overactive sympathetic nerve fibres (cholinergic) wrap around eccrine sweat glands, driving excess sweat to the surface.
AI-generated illustration

3. **Eccrine output**
   Glands secrete large volumes onto palms, soles and axillae.
4. **Treatment targets**
   Aluminium salts plug ducts; botulinum toxin blocks acetylcholine; anticholinergics block receptors.

How it is caused

Primary: overactivity of the sympathetic cholinergic supply to normal eccrine glands (central dysregulation; family history in ~30–50%). Secondary: increased metabolic rate (thyrotoxicosis, phaeochromocytoma), hormonal change (**hypogonadism**, menopause), infection/malignancy (cytokines), drugs, **obesity** (insulation, higher metabolic demand), **obstructive sleep apnoea**, autonomic/neurological disease.

HISTOPATHOLOGY

- Not needed; sweat glands are histologically normal.

Clinical features

Clinical features & sites

The case: a **41-year-old man**, sweating **"at the drop of a hat", at most times of day**, **forehead and axillae**, **deteriorating over 1 year**, potent antiperspirants ineffective, **BMI 34**; FBC/LFT/TFT/HbA1c normal.

	PRIMARY FOCAL	SECONDARY
Onset	**<25 years** (childhood/adolescence)	**Adult onset**, often recent

	PRIMARY FOCAL	SECONDARY
Distribution	Focal, **bilateral, symmetrical** (axillae, palms, soles, face)	**Generalised**, or asymmetric/unilateral (neurological)
Night sweats	**Stop during sleep**	**May occur during sleep**
Family history	Often positive	Usually negative
Associated symptoms	None	Weight change, fever, flushing, palpitations, sexual dysfunction, snoring

Primary focal criteria (Hornberger 2004): focal, visible, excessive sweating for **6 months** without an apparent cause, plus **2** of: bilateral and relatively symmetrical; impairs daily activities; **1 episode/week**; **onset <25 years**; **family history**; **stops during sleep**. This man meets few of them (onset at 40, 1-year history), so **secondary hyperhidrosis must be excluded** — the answer to the course question.

Fig. 2 · Axillary hyperhidrosis — visible sweat patches; new onset at 40 with forehead + axillae + all-day sweating = look for a secondary cause.

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Special groups

- **Secondary causes to exclude**
  - Hyperthyroidism, diabetes, menopause, phaeochromocytoma, TB, lymphoma, drugs (SSRIs, opioids).

Diagnosis and investigations

- **FURTHER HISTORY**

  Drugs: SSRIs/SNRIs (sertraline, venlafaxine, duloxetine), TCAs, opioids (incl. methadone), insulin/sulfonylureas (hypoglycaemia), pilocarpine/cholinesterase inhibitors, antipyretics, GnRH analogues, tamoxifen/aromatase inhibitors, **testosterone withdrawal/anabolic steroids**, PPIs, prostaglandin analogues; **Alcohol** (and withdrawal); **recreational drugs — cocaine, amphetamines, MDMA** (course teaching: often not asked); caffeine; nicotine; **Hypogonadism symptoms:** erectile dysfunction, low libido, **reduced shaving frequency**, fatigue, low mood, gynaecomastia; **Acromegaly:** ring/shoe size change, coarse features, headaches, snoring; **OSA:** snoring, witnessed apnoeas, daytime sleepiness (STOP-BANG, Epworth); **Phaeochromocytoma:** paroxysmal headache, palpitations, pallor, hypertension; **Infection/malignancy:** fever, **night sweats**, weight loss, lymph nodes, TB

exposure/travel, HIV risk; **Carcinoid:** flushing, diarrhoea, wheeze; GORD (nocturnal sweats); anxiety/social phobia; heart failure; neurological (Parkinson's, spinal cord injury, stroke); hypoglycaemic episodes; **Impact:** HDSS (Hyperhidrosis Disease Severity Scale, 1–4), DLQI, mood

- **Investigations** beyond those done: **9 am fasting total testosterone** (repeat if low) with **SHBG, LH/FSH** and **prolactin**; **fasting glucose/lipids** (metabolic); **urine drug screen** (with consent); **plasma free or 24-h urinary metanephrines** if paroxysmal/hypertensive; **IGF-1** if acromegalic features; **ESR/CRP, HIV test, CXR** (TB/lymphoma) if systemic features; **24-h urinary 5-HIAA** if flushing/diarrhoea; **sleep study** if OSA likely; BP, ECG; the starch–iodine (Minor) test to map focal areas (specialist)

Differential diagnoses

Primary focal hyperhidrosis; obesity-related sweating; **male hypogonadism** (obesity-related functional hypogonadism); OSA; drugs/alcohol/cocaine; thyrotoxicosis (TFT normal); diabetes/hypoglycaemia (HbA1c normal, but ask about sulfonylureas/insulin); phaeochromocytoma; acromegaly; carcinoid; lymphoma/TB/HIV/endocarditis; anxiety; gustatory sweating (Frey syndrome, diabetes); compensatory hyperhidrosis (after sympathectomy).

Management

- **Treat the cause.** Weight loss (NICE obesity pathways; GLP-1/GIP agonists where eligible), alcohol/cocaine cessation support, CPAP for OSA, endocrine referral for hypogonadism (testosterone replacement is specialist-guided and usually reserved for confirmed organic hypogonadism or persistent symptoms after weight loss), drug switches (e.g. a different SSRI, or a dose reduction).
- **SYMPTOMATIC (CKS/BAD STEPWISE)**
  Lifestyle: loose natural fibres, avoid triggers (spicy food, alcohol, caffeine), dress shields; **Aluminium chloride hexahydrate 20%** (e.g. Driclor) applied **at night to dry skin**, washed off in the morning; nightly until controlled, then 1–2×/week. If it irritates, use less often or apply hydrocortisone 1%; do not use on broken or recently shaved skin; **Tap-water iontophoresis** (palms/soles; also axillae) — secondary care or home devices; **Botulinum toxin A** intradermal injections (licensed for severe axillary hyperhidrosis; lasts 4–9 months); **Oral anticholinergics: propantheline bromide** 15 mg three times daily (licensed for hyperhidrosis); **oxybutynin** or **glycopyrronium** (off-label, specialist). Anticholinergic burden matters — dry mouth, constipation, urinary retention, glaucoma, cognitive effects in older people; **Microwave thermolysis** (miraDry — NICE IPG601) and **endoscopic thoracic sympathectomy** (NICE IPG487 — compensatory sweating is common) for severe refractory disease

IP scope

History/drug review, first-line tests, aluminium chloride and propantheline prescribing within competence, deprescribing/switching culprits, lifestyle support, and referral for endocrine/specialist treatments.

Referral

ROUTINE — Generalised or night sweats with weight loss, fever or lymph nodes: look for **TB, lymphoma or other systemic disease**.

ROUTINE One-sided or new-onset sweating in later life: consider a neurological cause.

Red flags and when to refer

- Generalised or night sweats with weight loss, fever or lymph nodes: look for **TB, lymphoma or other systemic disease**.
- One-sided or new-onset sweating in later life: consider a neurological cause.

Red flags / refer

Night sweats, weight loss, fever, lymphadenopathy (lymphoma/TB — urgent); paroxysmal headache/palpitations/hypertension (phaeochromocytoma); unilateral or asymmetric sweating (neurological — e.g. Pancoast tumour, stroke); new onset after 40 without an explanation; acromegalic features; symptoms of hypogonadism.

Complications and prognosis

Complications

- Maceration, pitted keratolysis, fungal infection, eczema; social and occupational disability.

Prognosis

- Primary hyperhidrosis often persists for years but can improve with age.

Prescriber pearls (UK pharmacist IP)

"**New sweating at 40 is secondary until proven otherwise**" — ask about drugs, cocaine, alcohol, sleep, sex and weight.

Aluminium chloride works only on **dry skin at night**, when sweat glands are least active.

Many SSRIs and venlafaxine cause dose-related sweating. Switching or reducing the dose often helps; the evidence for adding an anticholinergic is limited.

Self-check quiz

5 questions · optional

Q1 **When should you think of Hyperhidrosis?**

✓ Excessive, symmetrical sweating of the armpits, palms, soles or face that interferes with daily life.

Q2 What drives it?

✓ Overactive cholinergic sympathetic signalling to normal eccrine glands (primary), or a systemic cause (secondary).

Q3 What is the backbone of care?

✓ Exclude secondary causes; aluminium chloride at night; tap-water iontophoresis for hands and feet; specialist botulinum toxin or oral anticholinergics.

Q4 What must you never miss?

✓ Secondary causes: thyrotoxicosis, infection (TB), lymphoma, menopause, drugs (SSRIs), phaeochromocytoma.

Q5 Name a red flag that needs escalation.

✓ Generalised or night sweats with weight loss, fever or lymph nodes: look for **TB, lymphoma or other systemic disease**.

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Photo gallery

Hyperhidrosis 2

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Skin signs of systemic disease Module 1 · Week 5

Addison's disease: skin signs

Generalised bronzing, darker palmar creases, scars and gums, with tiredness, weight loss, salt craving and low blood pressure. Spotting it early can prevent a fatal adrenal crisis.

ALSO CALLED primary adrenal insufficiency · hyperpigmentation · buccal pigmentation · palmar crease pigmentation · adrenal crisis

RELATED PAGES Cushing's syndrome: skin signs · Drug-induced pigmentation (amiodarone, minocycline) · Melasma & facial melanosis

KEY FACTS

- Primary adrenal insufficiency failure of the adrenal cortex, with low cortisol and aldosterone. About 100–140 per million in the UK.
- 70–90% is **autoimmune** in the UK; other causes include TB, metastases, haemorrhage and drugs.
- Skin clue: **hyperpigmentation** of creases, scars, gums, knuckles and pressure areas; vitiligo may coexist.
- **Adrenal crisis** (vomiting, hypotension, collapse) 999 and **hydrocortisone 100 mg IM/IV immediately** (NICE NG243).
- Every patient needs an **NHS Steroid Emergency Card**, injection kit and **sick-day rules**.

Overview

Primary adrenal insufficiency: failure of the adrenal cortex **cortisol and aldosterone deficiency** (and adrenal androgens). About 100–140 per million in the UK.

- In the UK ~70–90% is **autoimmune** (21-hydroxylase antibodies), often with other autoimmune diseases (thyroid, T1DM, vitiligo, coeliac, pernicious anaemia, premature ovarian insufficiency — autoimmune polyendocrine syndromes).
- Other causes: TB, adrenal metastases/haemorrhage (anticoagulants, meningococcal — Waterhouse–Friderichsen), HIV/CMV, congenital adrenal hyperplasia, adrenoleukodystrophy (young men), and drugs (ketoconazole, etomidate, checkpoint inhibitors, mitotane).
- **Secondary/tertiary adrenal insufficiency** (pituitary disease or **suppression by exogenous glucocorticoids**) is far commoner in practice.

Aetiology: causes and risk factors

Myths to address

- **"Pigmentation happens in all adrenal insufficiency"**: only primary (high ACTH); secondary causes are pale.

Pathophysiology and pathology

Pigmentation comes from **excess ACTH**, which also stimulates melanocytes.

1. **Adrenal failure**
   Autoimmune destruction stops cortisol production.
2. **Loss of feedback**
   Low cortisol removes negative feedback, so the pituitary makes more POMC.
3. **POMC cleavage**
   POMC is cut into ACTH and α-MSH.
4. **Melanocyte stimulation**
   ACTH and α-MSH stimulate MC1R on melanocytes, increasing melanin, most visibly in creases and mucosa.

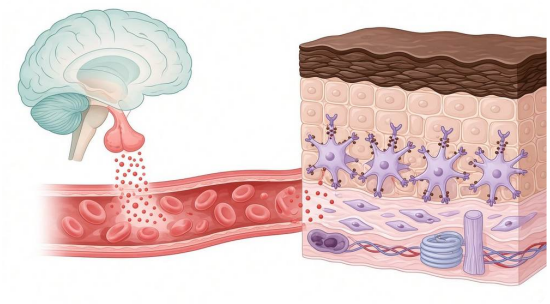

Fig. 1 · Excess ACTH and MSH from the pituitary drive melanocytes to make more pigment.
AI-generated illustration

How it is caused — why the hyperpigmentation?

Low cortisol removes negative feedback the pituitary makes more **pro-opiomelanocortin (POMC)**, which is cleaved into **ACTH** and **α-MSH**. Both bind the **melanocortin-1 receptor (MC1R)** on melanocytes more eumelanin **hyperpigmentation**. This occurs only in **primary** AI. In secondary AI, ACTH is low and patients look **pale**.

Clinical features

Clinical features & sites

> **SKIN AND MUCOSA**
> **generalised bronzing**, accentuated in **sun-exposed skin**, **palmar creases**, **knuckles**, **friction/pressure sites** (belt line, elbows, knees, bra strap), **recent scars**, nipples, axillae, perineum; **buccal mucosa, gums, tongue and lips** (blue-black patches); longitudinal nail bands; darkening of hair and naevi; **vitiligo** (~10–20%, autoimmune); loss of axillary/pubic hair in women (adrenal androgen loss)

> **NICE NG243**
> hyperpigmentation **may not be visible on black or brown skin** — ask whether the patient has noticed a change and **check the buccal mucosa and scars**.

> **SYSTEMIC**
> fatigue, weakness, **weight loss**, anorexia, nausea/vomiting, abdominal pain, diarrhoea, **salt craving**, **postural dizziness/hypotension**, myalgia/arthralgia, low mood; **hyponatraemia, hyperkalaemia**, hypoglycaemia (especially in children), raised urea, eosinophilia, mild anaemia, raised TSH.

> **ADRENAL CRISIS**

> vomiting, abdominal pain, confusion, hypotension/shock — often triggered by intercurrent illness; can be fatal (the press cases).

Fig. 2 · Generalised tanned hyperpigmentation in Addison's disease (ACTH drives melanogenesis).

James Heilman, MD · CC BY-SA 4.0 · **Source**

Diagnosis and investigations

- **NICE NG243 (2024): offer an 8 am–9 am serum cortisol** to people with suspected AI (course teaching: 8 am cortisol). Interpretation for adults (modern immunoassays):

8–9 AM SERUM CORTISOL	MEANING (ADULTS 16)	ACTION
<150 nmol/L	May have adrenal insufficiency	**Refer to endocrinology**; consider starting treatment; if acutely unwell treat as adrenal crisis
150–300 nmol/L	Uncertain	Consider repeating; if it stays in this range, seek endocrinology advice/ referral
>300 nmol/L	AI very unlikely	—

- Measure **U&E** (Na^+, K^+), glucose, FBC, and **ACTH** with the cortisol sample if possible (local lab rules — EDTA, on ice). Check TFT and **renin/aldosterone**.
- **Don't test** people on oral glucocorticoids at physiological doses or above. **Inhaled, injected or topical steroids can lower the 9 am cortisol.** Wait 4 weeks after IM/intra-articular steroid injections.
- **CONFIRMATION (SECONDARY CARE)**
  short Synacthen (ACTH stimulation) test, ACTH, renin, **21-hydroxylase antibodies**, adrenal imaging if antibody-negative, very-long-chain fatty acids in young men.

Differential diagnoses

(hyperpigmentation)

Haemochromatosis (bronze diabetes), **drugs** (minocycline, amiodarone, antimalarials, chemotherapy — busulfan, bleomycin; zidovudine), **ectopic ACTH/Nelson syndrome**, **PCT**, malabsorption/B12 or folate deficiency, pregnancy/

melasma, **chronic kidney disease**, primary biliary cholangitis, acanthosis nigricans, post-inflammatory hyperpigmentation, constitutional/racial pigmentation, **arsenic**, hyperthyroidism, POEMS.

Management

Suspected adrenal crisis (vomiting, hypotension, collapse, confusion) **999**. Give **hydrocortisone 100 mg IM/IV immediately** — do not wait for tests (NG243; adults; children by age). Then IV fluids and hydrocortisone 200 mg/24 h infusion or 50 mg IM/IV 6-hourly in hospital.

MAINTENANCE (SPECIALIST-INITIATED)

hydrocortisone ~15–25 mg/day in 2–3 divided doses (largest dose on waking), or prednisolone in selected patients; **fludrocortisone** 50–300 micrograms daily for mineralocorticoid replacement; ± DHEA in women (specialist)

PHARMACIST/IP CORE TASKS (NG243 + NATPSA 2020)

Ensure the patient has an **NHS Steroid Emergency Card**, medical alert ID and a **hydrocortisone emergency injection kit** (100 mg IM + needles/syringes + instructions), and that patient and carers are **trained** to use it; **Sick-day rules:** during significant illness (fever, infection needing antibiotics, injury), take **at least 40 mg oral hydrocortisone daily in 2–4 divided doses** (or 10 mg prednisolone) until recovered. If vomiting within 30 minutes of a dose, repeat at double the dose. If vomiting recurs **IM hydrocortisone and go to ED**; **Interactions: enzyme inducers** (rifampicin, carbamazepine, phenytoin, St John's wort) increase hydrocortisone requirement; CYP3A4 inhibitors increase exposure; Plan steroid cover for surgery/procedures (Woodcock 2020) and dental extractions; Recognise **iatrogenic adrenal suppression** in people on long-term glucocorticoids by any route: emergency card if oral prednisolone 5 mg/day (or equivalent) for 4 weeks, or high-dose inhaled/topical (NatPSA criteria); Annual review: BP (lying/standing), electrolytes, weight, over-/under-replacement signs; screen for other autoimmune diseases (TFT, B12, HbA1c, coeliac serology)

IP scope

Suspect, test (9 am cortisol, U&E), refer, and — vitally — **steroid safety** (cards, sick-day rules, emergency kits, interactions, never stopping steroids abruptly). Emergency IM hydrocortisone can be given by trained staff under emergency provisions.

Referral

999 / A&E — Vomiting, abdominal pain, confusion or collapse with suspected adrenal insufficiency: **adrenal crisis**, hydrocortisone 100 mg IM and 999.

SAME DAY — Unexplained pigmentation with low sodium or high potassium: same-day 9am (or random) cortisol and endocrine advice.

Red flags and when to refer

- Vomiting, abdominal pain, confusion or collapse with suspected adrenal insufficiency: **adrenal crisis**, hydrocortisone 100 mg IM and 999.

- Unexplained pigmentation with low sodium or high potassium: same-day 9am (or random) cortisol and endocrine advice.

Red flags / refer

Unexplained hyperpigmentation + fatigue/weight loss/postural hypotension; hyponatraemia/hyperkalaemia; a 9 am cortisol <150 nmol/L; vomiting or collapse in a person on steroid replacement or long-term glucocorticoids (crisis — 999 + IM hydrocortisone).

Complications and prognosis

Complications

- Adrenal crisis (fatal if untreated), hyponatraemia, hyperkalaemia, hypoglycaemia.

Prognosis

- Normal life expectancy with replacement and crisis prevention.

Prescriber pearls (UK pharmacist IP)

"**Pigmented palmar creases and gums + salt craving + postural dizziness = check a 9 am cortisol today.**"

"**Sick-day rules save lives**" — rehearse them at every supply.

Pale patient with AI features on long-term steroids = **secondary/tertiary AI** — no pigmentation expected.

Self-check quiz

5 questions · optional

Q1 **When should you think of Addison's disease: skin signs?**

✓ Gradual tanning not explained by sun, pigmented creases, scars and buccal mucosa, with fatigue, weight loss and dizziness.

Q2 **What drives it?**

✓ Autoimmune destruction of the adrenal cortex; low cortisol drives high ACTH and MSH, which stimulate melanocytes.

Q3 What is the backbone of care?

✓ U&E (low sodium, high potassium), 9am cortisol and urgent endocrine referral; steroid emergency card and sick-day rules once diagnosed.

Q4 What must you never miss?

✓ Adrenal crisis (vomiting, hypotension, collapse): hydrocortisone 100 mg IM/IV and 999.

Q5 Name a red flag that needs escalation.

✓ Vomiting, abdominal pain, confusion or collapse with suspected adrenal insufficiency: **adrenal crisis**, hydrocortisone 100 mg IM and 999.

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Photo gallery

Clinical photographs 1

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Skin signs of systemic disease Module 1 · Week 5

Eruptive xanthomas & skin signs of lipids

Crops of yellow-orange papules with a red halo on the buttocks and limbs, signalling very high triglycerides, often with new or poorly controlled diabetes. Pancreatitis risk makes it urgent.

ALSO CALLED xanthoma · hypertriglyceridaemia · xanthelasma · tendon xanthoma · familial hypercholesterolaemia · pancreatitis

RELATED PAGES Acanthosis nigricans · Cushing's syndrome: skin signs · Retinoids (topical & oral)

KEY FACTS

- Crops of 1–5 mm **yellow papules with a red halo** on buttocks, extensor limbs and shoulders.
- Marker of **severe hypertriglyceridaemia** (usually TG over 10–20 mmol/L).
- Common drivers: **uncontrolled diabetes**, obesity, alcohol, hypothyroidism, nephrotic syndrome, drugs (oestrogens, retinoids, antipsychotics, steroids).
- Risk of **acute pancreatitis**: check glucose, HbA1c, fasting lipids, TFT, U&E, LFT, urine protein.
- Treat the cause; papules fade over weeks as triglycerides fall.

Overview

Crops of small (1–5 mm) **yellow papules with a red halo**, appearing over weeks, on the **buttocks**, extensor arms/thighs and shoulders. They are the skin marker of **severe hypertriglyceridaemia** (usually TG **>10–20 mmol/L**, chylomicronaemia).

Aetiology: causes and risk factors

Myths to address

- **"THEY'RE JUST SPOTS"**
  they mean dangerously high triglycerides.

Pathophysiology and pathology

Lipid **leaks from blood into the dermis** and macrophages eat it.

1. **Chylomicronaemia**
   Lipoprotein lipase activity falls (e.g. insulin deficiency); triglyceride-rich particles build up.
2. **Leak**
   Particles cross dermal capillaries.

3 **Foam cells**

Macrophages take up the lipid and become foam cells.

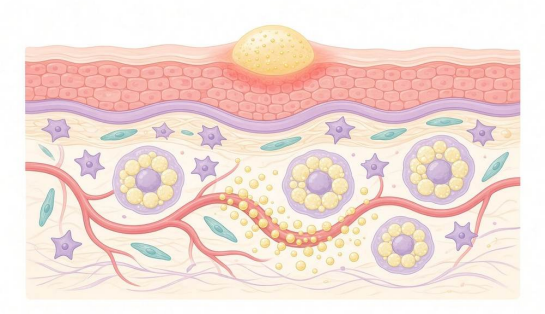

Fig. 1 · Lipid particles leave dermal capillaries and are engulfed by macrophages, forming foam cells.

AI-generated illustration

4 **Resolution**

Lowering triglycerides clears the papules.

How it is caused

Excess circulating **chylomicrons/VLDL** leak through dermal capillaries and are taken up by macrophages (foam cells), with **extracellular lipid** deposits early on.

> **SECONDARY CAUSES**
>
> **uncontrolled diabetes** (insulin deficiency less lipoprotein lipase activity — this case: fasting glucose 18.1); **obesity**/metabolic syndrome (BMI 35); **alcohol**; hypothyroidism, nephrotic syndrome, CKD, pregnancy; **drugs:** oral oestrogens, tamoxifen, **retinoids (isotretinoin, acitretin)**, corticosteroids, thiazides, β-blockers, **ciclosporin/ tacrolimus/sirolimus**, protease inhibitors, **atypical antipsychotics** (olanzapine, clozapine), propofol, L-asparaginase

> **PRIMARY**
>
> familial chylomicronaemia syndrome (LPL deficiency), multifactorial chylomicronaemia, familial hypertriglyceridaemia, dysbetalipoproteinaemia.

Clinical features

Clinical features & sites

Clusters of firm, non-tender or slightly itchy yellow-orange papules with erythematous bases; Koebnerisation. Larger coalescing (tubero-eruptive) lesions with persistent disease. Look also for **lipaemia retinalis** (creamy retinal vessels), hepatosplenomegaly, abdominal pain (**pancreatitis**). Other xanthoma types point to other lipid disorders: tendon (FH), tuberous (elbows/knees — dysbetalipoproteinaemia/FH), palmar crease (dysbetalipoproteinaemia), xanthelasma (half have normal lipids).

Fig. 2 · Eruptive xanthomas — crops of yellow papules with a red halo. Request lipids today (TG often >10–20 mmol/L).

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Diagnosis and investigations

- **Fasting lipid profile.** The sample is **turbid/milky ("lipaemic")** with **very high TG**. Cholesterol is also raised.
- **HbA1c**, glucose, U&E/eGFR, LFT (fatty liver, alcohol), **TSH**, urine ACR/PCR, amylase/lipase if abdominal pain.
- **Shave or punch biopsy** only if uncertain (foamy macrophages ± extracellular lipid).
- **NICE NG238 (2023)**
  TG >20 mmol/L (not caused by excess alcohol or poor glycaemic control) **urgent specialist referral**; **TG 10–20 mmol/L** **repeat fasting TG within 5–14 days**, review and treat secondary causes; **seek specialist advice if still >10**; In practice, a TG >20 with xanthomas and new diabetes needs **urgent** management whatever the cause, because of pancreatitis risk

Differential diagnoses

Molluscum contagiosum (umbilicated, pearly), **dermatitis herpetiformis** (itchy grouped excoriated papules/vesicles, same sites), folliculitis/acne, **papular granuloma annulare**, xanthoma disseminatum/non-Langerhans histiocytosis, papular sarcoid, milia, insect bites, juvenile xanthogranuloma.

Management

1. **Treat the diabetes promptly** (NICE NG28). A fasting glucose of 18.1 with a new diagnosis needs HbA1c and ketones. With symptomatic hyperglycaemia or very high TG, **insulin** (or a sulfonylurea) achieves rapid control and rapidly lowers TG. Then metformin ± SGLT2 inhibitor/GLP-1 RA for long-term control and weight.
2. **Stop alcohol** completely while TG is very high. **Review culprit drugs** (oestrogens, thiazides, β-blockers, retinoids, antipsychotics, steroids).
3. Dietitian very-low-fat diet (<10–15% energy from fat) for chylomicronaemia; avoid simple sugars; weight loss.
4. Lipid-lowering **Fenofibrate** (specialist-advised) when TG remains >10 mmol/L despite control of secondary causes, to reduce **pancreatitis** risk; **Statin** for CVD risk — **atorvastatin 20 mg** for primary prevention in T2DM if QRISK3 10%; **Icosapent ethyl** (NICE TA805) for secondary prevention with raised TG on statins

5. **Xanthomas regress** within weeks to months of TG control — a useful motivator.
6. Safety-net for pancreatitis severe epigastric pain radiating to the back, vomiting ED.

IP scope

Within diabetes/lipid IP competence: start diabetes therapy (incl. insulin if trained), statin, stop alcohol/culprit drugs, arrange repeat lipids, refer urgently per NG238. Fibrate/specialist therapies as advised.

Referral

SAME DAY — Triglycerides above 20 mmol/L: **urgent specialist referral** (NICE NG238); abdominal pain raises concern for **acute pancreatitis**, same-day assessment.

ROUTINE — Tendon xanthomas or xanthelasma in a young person: check for **familial hypercholesterolaemia** (NICE CG71).

Red flags and when to refer

- Triglycerides above 20 mmol/L: **urgent specialist referral** (NICE NG238); abdominal pain raises concern for **acute pancreatitis**, same-day assessment.
- Tendon xanthomas or xanthelasma in a young person: check for **familial hypercholesterolaemia** (NICE CG71).

Red flags / refer

Abdominal pain (pancreatitis — same day); TG >20 mmol/L (urgent lipid clinic); TG >10 persistently; suspected familial chylomicronaemia (onset in childhood, recurrent pancreatitis); DKA/HHS features (glucose 18 + vomiting/ketones).

Complications and prognosis

Complications

- Acute pancreatitis, cardiovascular disease.

Prognosis

- Papules clear within weeks to months of control.

Prescriber pearls (UK pharmacist IP)

"**Yellow papules on the buttocks in new diabetes = check triglycerides today — pancreatitis risk.**"

Before starting **isotretinoin, acitretin or oral oestrogen**, check fasting lipids if there are risk factors (family history, obesity, diabetes, alcohol).

Self-check quiz

5 questions · optional

Q1 When should you think of Eruptive xanthomas & skin signs of lipids?

✓ Sudden crops of small yellow papules with a red rim on the buttocks, elbows or knees, often in poorly controlled diabetes.

Q2 What drives it?

✓ Very high triglycerides leak into the dermis and are taken up by macrophages.

Q3 What is the backbone of care?

✓ Fasting lipids, HbA1c, glucose, TSH, LFT, renal function, alcohol and drug review; urgent lipid-lowering and diabetes control.

Q4 What must you never miss?

✓ Acute pancreatitis (triglycerides >10–20 mmol/L); familial hypercholesterolaemia with tendon xanthomas.

Q5 Name a red flag that needs escalation.

✓ Triglycerides above 20 mmol/L: **urgent specialist referral** (NICE NG238); abdominal pain raises concern for **acute pancreatitis**, same-day assessment.

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Photo gallery

Eruptive xanthomas 3

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Skin signs of systemic disease Module 1 · Week 5

Sarcoidosis (incl. Löfgren syndrome)

A multisystem granulomatous disease. Specific skin lesions (papules, plaques, lupus pernio, scar sarcoid) and non-specific erythema nodosum; Löfgren syndrome (EN, hilar nodes, ankle arthritis, fever) usually settles.

ALSO CALLED Löfgren · lupus pernio · scar sarcoid · granuloma · bilateral hilar lymphadenopathy · BHL · +1

RELATED PAGES Erythema nodosum (incl. drug-induced) · Discoid lupus erythematosus · Cutaneous tuberculosis (scrofuloderma & others) · Annular (ring-shaped) rashes: differential · Skin signs of lung cancer & internal malignancy

KEY FACTS

- **Sarcoidosis** is a multisystem **non-caseating granulomatous** disease; the skin is involved in about a quarter.
- Specific lesions: red-brown papules and plaques (face, scars, tattoos), **lupus pernio** (nose, cheeks); non-specific: **erythema nodosum**.
- Apple-jelly colour on diascopy; infiltration of old scars and tattoos is a clue.
- Investigate: biopsy, chest X-ray, calcium, ACE, U&E, LFT, ECG, eye review.
- Mild skin disease: potent or intralesional steroid; extensive or systemic: specialist (hydroxychloroquine, methotrexate, TNF inhibitors).

Overview

A multisystem **granulomatous** disease of unknown cause, with **non-caseating epithelioid granulomas**. It affects lungs/mediastinal nodes (~90%), skin (~25–35%), eyes (~25%), liver, heart, nerves, kidneys and bones. Peak onset 25–45 years; commoner in Afro-Caribbean and Scandinavian people.

- **LÖFGREN SYNDROME**
  acute sarcoidosis with **erythema nodosum + bilateral hilar lymphadenopathy (BHL)** ± fever and **ankle periarthritis**. It has a **good prognosis (>80–90% resolve within 2 years)**.

Epidemiology

- **Who**
  - Adults 25–45; more common and severe in people of African descent; Scandinavian peak.

Aetiology: causes and risk factors

Myths to address

- **"SKIN SARCOID IS JUST COSMETIC"**
  lupus pernio is linked with chronic lung and upper airway disease.

Pathophysiology and pathology

Sarcoidosis is an **over-exuberant granuloma response** to an unknown antigen.

1. **Antigen**
   An unknown antigen (possibly mycobacterial or environmental) is presented to CD4+ T cells.
2. **Th1/Th17 response**
   Interferon-γ and TNF-α activate macrophages.
3. **Granuloma formation**
   Epithelioid macrophages and giant cells form tight 'naked' granulomas in the dermis.
4. **Organ involvement**
   The same process in lungs, eyes, heart and lymph nodes causes systemic disease.

How it is caused

An exaggerated Th1/Th17 response to an unidentified antigen (mycobacterial/propionibacterial antigens, environmental dusts are proposed) in genetically susceptible people (HLA-DRB1*03 linked with Löfgren). It forms granulomas.

> **HYPERCALCAEMIA**
> granuloma macrophages express **1-α-hydroxylase** convert 25-OH vitamin D to **calcitriol** independently of PTH gut calcium absorption. Worse with sun or vitamin D supplements.

> **ANAEMIA**
> chronic disease, marrow/splenic involvement.

> **EN** is a reactive (non-specific) septal panniculitis, **not a granuloma**. "Specific" lesions (papules, plaques, lupus pernio, scar sarcoid) contain granulomas.

HISTOPATHOLOGY

- Non-caseating 'naked' epithelioid granulomas with few lymphocytes; stains for mycobacteria and fungi negative.

Clinical features

Clinical features & sites

> **RASH**

bilateral tender, warm, red-violaceous, ill-defined nodules/plaques on the shins (EN, photo G01), with fever, malaise and arthralgia.

HOW AN EN NODULE EVOLVES CLINICALLY

bright red, raised, tender nodule → flattens and turns **dusky violaceous/blue** → **yellow-green bruise-like** colour ("erythema contusiformis") over 1–2 weeks → fades **without ulceration, atrophy or scarring** in **3–6 weeks**; new crops may appear.

SPECIFIC CUTANEOUS SARCOID

red-brown papules/plaques (face, lips, nose — "**apple-jelly**" on diascopy, G05); **lupus pernio** (violaceous nose/cheeks/ears — chronic disease, upper-airway/lung fibrosis, bone cysts; G03); **scar/tattoo sarcoid**; subcutaneous (Darier–Roussy) nodules; annular, hypopigmented, ichthyosiform, ulcerative forms; scarring alopecia; nail dystrophy

SYSTEMIC

cough, breathlessness, chest pain; uveitis (red/painful eye, blurred vision — can be asymptomatic); parotid enlargement; facial palsy (Heerfordt); arrhythmia/heart block (cardiac sarcoid — sudden death); hepatosplenomegaly; renal stones; fatigue (very common).

THE CASE

breathlessness, **weight loss 3 kg, night sweats**, EN, **anaemia (Hb 10.8)**, **calcium 2.7 mmol/L**, CXR showing **BHL**.

Fig. 1 · Lupus pernio — violaceous nasal/cheek plaques; chronic sarcoid with upper respiratory tract and lung involvement.

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Special groups

- **Löfgren syndrome**
  - Erythema nodosum, bilateral hilar lymphadenopathy, fever and ankle arthritis: good prognosis.

Diagnosis and investigations

- **CXR: bilateral hilar lymphadenopathy** ± right paratracheal nodes and parenchymal changes. Scadding stages: 0 normal; **I BHL only**; II BHL + infiltrates; III infiltrates only; IV fibrosis.
- **Diagnosis: acute sarcoidosis, Löfgren-type** (EN + BHL; hypercalcaemia supports sarcoid): BUT **weight loss + night sweats + anaemia** mean **TB and lymphoma must be excluded** — both cause BHL, systemic symptoms and EN

- **INVESTIGATIONS (BTS CLINICAL STATEMENT 2021 / ERS)**
  TB: IGRA, sputum AFB/culture, HIV test; **Bloods:** FBC + film, LDH (lymphoma), **calcium (adjusted) + 24-h urine calcium**, U&E/eGFR, LFT, **serum ACE** (supportive only — low sensitivity and specificity; **ACE inhibitors lower it**, so interpret with the drug history), vitamin D (25-OH and 1,25-OH), immunoglobulins; **Lungs: HRCT chest**; **PFTs with TLCO**; **ECG for all** (heart block) ± Holter, echo, cardiac MRI if symptomatic or abnormal ECG; **Ophthalmology review for all** (uveitis); **Tissue: EBUS-TBNA** of mediastinal nodes or biopsy of a **specific skin lesion** (non-caseating granulomas; stain and culture for AFB/fungi). In a **classic Löfgren presentation** without atypical features, tissue confirmation can be omitted — **not here**, given the B symptoms; Urinalysis

Differential diagnoses

TB (caseating granulomas, IGRA), **lymphoma** (B symptoms, high LDH, nodes elsewhere), metastatic cancer, histoplasmosis/fungal infection, **berylliosis** (occupational), hypersensitivity pneumonitis, drug-induced sarcoid-like reactions (**checkpoint inhibitors, TNF inhibitors, interferon**), granulomatosis with polyangiitis, IgG4 disease. Skin: granuloma annulare, lupus vulgaris (cutaneous TB), leishmaniasis, foreign-body granulomas, lupus, rosacea, lymphoma. **EN differentials**: see the drug eruption notes see Eruptive xanthomas — new diabetes (cellulitis, thrombophlebitis/DVT, erythema induratum, other panniculitis, cutaneous PAN, insect bites, Sweet syndrome, bruising).

Management

> **Refer to respiratory (sarcoid clinic). Admit** if hypoxic, unwell, severe hypercalcaemia (3.0 mmol/L or symptomatic), cardiac or neurological involvement.

> **LÖFGREN SYNDROME**
> usually self-limiting — **NSAIDs** for EN/arthralgia (if renal function/GI risk allow), rest, leg elevation. A **short oral corticosteroid course** if severe.

> **HYPERCALCAEMIA**
> hydration; **stop vitamin D/calcium supplements**; avoid excess sun; review thiazides (raise calcium) and lithium; glucocorticoids treat sarcoid hypercalcaemia (specialist); monitor U&E (nephrocalcinosis/stones)

> **SYSTEMIC/ORGAN-THREATENING SARCOID (SPECIALIST; ERS 2021)**
> prednisolone (e.g. 20–40 mg/day, tapering), methotrexate (steroid-sparing), azathioprine/leflunomide/mycophenolate, **infliximab/adalimumab** for refractory disease and lupus pernio.

> **CUTANEOUS SARCOID**
> potent/very potent TCS or **intralesional triamcinolone**, **hydroxychloroquine**, **doxycycline/minocycline**, methotrexate; TNF inhibitors for disfiguring lupus pernio; JAK inhibitors (emerging, off-label). Camouflage.

> **PHARMACIST CHECKLIST**
> steroid card, bone protection (**caution with vitamin D/calcium supplements in hypercalcaemic sarcoid** — specialist advice), PJP prophylaxis if heavily immunosuppressed, vaccines (inactivated), TB screening before TNF inhibitors, MTX safety, HCQ eye screening, ECG awareness (QT, conduction).

IP scope

Recognition, bloods (calcium, U&E, LFT), CXR request (where the IP has rights), NSAID analgesia, stopping vitamin D/calcium supplements, referral, and long-term medicines support. Specialist diagnosis and immunosuppression.

Referral

URGENT — Palpitations, syncope or heart block: possible **cardiac sarcoid**, urgent assessment.

SAME DAY — Red painful eye or blurred vision: same-day ophthalmology (uveitis).

Red flags and when to refer

- Palpitations, syncope or heart block: possible **cardiac sarcoid**, urgent assessment.
- Red painful eye or blurred vision: same-day ophthalmology (uveitis).

Red flags / refer

Hypercalcaemia (thirst, polyuria, confusion), palpitations/syncope/heart block (cardiac sarcoid — urgent), visual symptoms/red eye (same-day ophthalmology), facial palsy/neurology, worsening breathlessness/hypoxia, B symptoms suggesting lymphoma/TB, lupus pernio (chronic disease).

Complications and prognosis

Complications

- Lung fibrosis, uveitis, cardiac arrhythmia, hypercalcaemia, renal stones; disfigurement from lupus pernio.

Prognosis

- Löfgren usually resolves; lupus pernio and plaques tend to be chronic.

Prescriber pearls (UK pharmacist IP)

- "**EN + BHL = Löfgren — but B symptoms or anaemia mean exclude TB and lymphoma.**"
- In patients with sarcoid, **check before recommending OTC vitamin D or calcium**.
- Every patient with sarcoid needs an **eye check and an ECG**.

Self-check quiz

5 questions · optional

Q1 When should you think of Sarcoidosis (incl. Löfgren syndrome)?

✓ Red-brown papules or plaques around the nose, lips or old scars and tattoos, or a young adult with erythema nodosum, sore ankles and fever.

Q2 What drives it?

✓ Non-caseating granulomas from an exaggerated immune response to an unknown trigger.

Q3 What is the backbone of care?

✓ CXR, calcium, renal and liver function, ECG, eye review; biopsy skin lesions; respiratory or dermatology referral; Löfgren usually needs NSAIDs only.

Q4 What must you never miss?

✓ Cardiac sarcoid (palpitations, syncope), eye inflammation, hypercalcaemia, lupus pernio with nasal involvement.

Q5 Name a red flag that needs escalation.

✓ Palpitations, syncope or heart block: possible **cardiac sarcoid**, urgent assessment.

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Photo gallery

Löfgren syndrome 2

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Specific skin sarcoid 3

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CHAPTER 22

Lumps, lesions & skin cancer

Benign lumps and lesions, BCC, SCC and cutaneous lymphoma, moles and naevi, melanoma recognition, staging and treatment, NICE NG12 referral routes and teledermatology.

1. Dermatofibroma
2. Dermatofibrosarcoma protuberans (DFSP)
3. Sebaceous naevus (naevus sebaceous)
4. Epidermoid, pilar and dermoid cysts (and Gardner syndrome)
5. Lipoma & the sarcoma lump pathway
6. Milia
7. Pyogenic granuloma
8. Angiokeratomas & Fabry disease
9. Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more
10. Chondrodermatitis nodularis helicis & ear nodules
11. Seborrhoeic keratosis
12. Clear cell acanthoma
13. Basal cell carcinoma (BCC)
14. Cutaneous squamous cell carcinoma (cSCC)
15. Mycosis fungoides & cutaneous lymphoma
16. Melanocytic naevi (moles) & the changing mole
17. Congenital melanocytic naevi (CMN)
18. Atypical naevus syndrome & familial melanoma
19. Halo naevus (Sutton naevus)
20. Blue naevus
21. Melanoma: recognition & referral
22. Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment
23. Spitz and Reed naevi
24. Melanoma immunotherapy & targeted therapy: side-effects
25. Teledermatology & lesion photography

Lumps, lesions & skin cancer Module 2 · Week 1

Dermatofibroma

A common firm button-like papule, usually on the legs, that dimples when pinched. Dermoscopy shows a central white patch with a fine peripheral network. Pigmented forms can mimic a mole or melanoma.

ALSO CALLED benign fibrous histiocytoma · dimple sign · pinch sign · pigmented dermatofibroma · leg lump after insect bite

RELATED PAGES Dermatofibrosarcoma protuberans (DFSP) · Melanocytic naevi (moles) & the changing mole · Melanoma: recognition & referral · Lipoma & the sarcoma lump pathway

KEY FACTS

- **Dermatofibroma** (benign fibrous histiocytoma) is a common firm dermal papule, usually on the **legs**.
- Firm 'button' with a brown rim; the **dimple sign**: it puckers inwards when pinched.
- Dermoscopy: central white scar-like patch with a fine peripheral pigment network.
- Benign: reassure. Refer if growing, large, atypical or pigmented in an unusual way (exclude melanoma and DFSP).
- Excision leaves a scar often worse than the lesion.

Overview

Dermatofibroma (DF) is a very common benign fibrohistiocytic proliferation in the dermis. It is seen mostly in adults aged 20–50 and is more common in women. It is a reactive or neoplastic nodule; which of the two is still debated.

Pigmented dermatofibroma

A dermatofibroma (core: the skin lumps notes) with marked overlying epidermal hyperpigmentation. It is **commoner and darker in skin of colour**, so it is often brought in as a "mole".

Epidemiology

- **Who**
  - Young to middle-aged adults, more women; lower legs commonest.

Aetiology: causes and risk factors

- Often follows minor trauma such as an insect bite, folliculitis or a shaving nick, but many patients recall none.
- Dermal fibroblasts and histiocytes proliferate and trap collagen bundles at the periphery. The overlying epidermis is induced to thicken and hyperpigment, which is why the surface is often brown.

- Multiple (eruptive) DFs (15, or many appearing quickly) are associated with altered immunity: SLE, HIV, immunosuppressive drugs and pregnancy.
- Histology: Factor XIIIa positive, CD34 negative. This is the reverse of DFSP (see Dermatofibrosarcoma protuberans), and the stain pair is a favourite exam question.

Pigmented dermatofibroma

Dermal fibrohistiocytic proliferation induces the overlying epidermis to thicken (acanthosis) and to increase basal melanin ("dirty fingers" rete ridges). Haemosiderin deposits make the aneurysmal/haemosiderotic variant blue-black.

Risk factors and associations

- Possibly a reaction to minor trauma such as an insect bite or ingrown hair; multiple lesions in immunosuppression (HIV, lupus).

Myths to address

- **"IT'S A MOLE"**
  it is a fibrous lesion, which explains its firmness and the dimple sign.

Pathophysiology and pathology

A dermatofibroma is a **fibrous reaction** in the dermis that tethers the skin above.

1. **Injury**
   Minor trauma triggers fibroblasts and histiocytes to proliferate in the dermis.
2. **Fibrosis**
   Collagen and cells form a firm nodule that tethers the overlying epidermis.

Fig. 1 · The dimple sign: pinching pushes the dermatofibroma downwards.
AI-generated illustration

3. **Epidermal response**
   Overlying epidermis thickens and becomes pigmented, causing the brown rim.

HISTOPATHOLOGY

- Spindle cells and histiocytes trapping collagen bundles at the periphery ('collagen balls'); overlying acanthosis with basal pigmentation; CD34 negative (unlike DFSP).

Clinical features

- Firm to hard papule or nodule, 3–10 mm, that feels like a "button" tethered to the skin but mobile over deeper tissue.
- Colour varies (skin-coloured, pink, brown, or a darker rim with a paler centre), especially in darker skin (see the benign lesion notes, pigmented DF).
- **Lower legs** are the classic site, then arms and trunk. Usually solitary and asymptomatic. It may itch or be tender, and it is often nicked when shaving.
- **DIMPLE (FITZPATRICK / PINCH) SIGN**
  pinching the lesion from the sides makes it dimple inward.
- Variants to know: cellular, aneurysmal/haemosiderotic (can enlarge quickly and look blue-black), atypical ("pseudosarcomatous"), and deep or subcutaneous. **Cellular, aneurysmal and atypical variants recur more often after excision, and rare cases metastasise.** This is the course Case 1 answer.

Pigmented dermatofibroma

- A firm 3–10 mm papule, brown to dark brown, often **darker at the rim with a paler centre**. Sometimes pink-brown.
- Limbs (arms in Case 1, legs classically). Slow growth over years. A positive dimple sign.

Fig. 2 · Dimple (Fitzpatrick) sign: lateral pinch makes the lesion dip below the surface because it is tethered to the dermis.

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Types to recognise

Dimple sign

Lateral pinch dip. Site: Leg.

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Typical papule

Firm pink-brown dome. Site: Leg.

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Pigmented DF

Darker rim. Site: Arm.

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Dermoscopy

Peripheral network, white centre. Site: Dermoscopy.

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Special groups

- **DFSP warning**
  - A slowly enlarging firm plaque over 2–3 cm, especially on the trunk, may be dermatofibrosarcoma protuberans: refer.

Diagnosis and investigations

- Clinical: typical site, firmness, and the dimple sign.
- Dermoscopy: a delicate peripheral pigment network with a **central white scar-like patch**. The commonest pattern, but not universal.
- Biopsy is needed only for atypical lesions: >1 cm, rapid growth, ulceration, bleeding, recurrence after removal, unusual site (face, scalp, acral), or doubt between DF and a melanocytic lesion.

Pigmented dermatofibroma

- Pinch test. Dermoscopy: **delicate peripheral pigment network + central white scar-like area**. Some show a bland structureless or homogeneous pattern, and those are less reassuring, so refer if unsure.
- **Immunohistochemistry teaching table** (course teaching extension; useful for reading histology reports):

STAIN	POSITIVE IN	USE
Factor XIIIa	Dermatofibroma (dermal dendrocytes)	DF vs DFSP
CD34	DFSP (also some vascular tumours, spindle cell lipoma)	DFSP vs DF
S100	Melanocytes and melanoma, Langerhans cells, nerve sheath (neurofibroma, schwannoma)	Broad melanocytic/neural marker (sensitive, not specific)
SOX10, Melan-A, HMB-45	Melanocytic lesions	More specific melanocytic markers. HMB-45 is patchy in naevi, diffuse in melanoma
BerEP4	BCC (positive)	BCC vs SCC (usually negative)
CD1a, langerin	Langerhans cells	Langerhans cell histiocytosis
CK20	Merkel cell carcinoma (paranuclear "dot")	Merkel vs small-cell lung metastasis (TTF-1 positive, CK20 negative)
CK7	Paget's / extramammary Paget's, adnexal tumours	Paget's
CD30	Lymphomatoid papulosis, anaplastic large cell lymphoma	(see the skin cancer notes MF)

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	DFSP	Growing plaque > 2–3 cm, trunk; CD34 +.
	Melanoma	Changing pigmented lesion, atypical dermoscopy.
	Naevus	Soft, no dimple.
	Hypertrophic scar / keloid	Injury history.
	Epidermoid cyst	Punctum, mobile.

Pigmented dermatofibroma

Melanocytic naevus; **melanoma** (EFG, change, irregular dermoscopy); pigmented BCC (pearly, ovoid nests, arborising vessels); SK (stuck-on, milia-like cysts); post-inflammatory hyperpigmentation over a prurigo nodule; blue naevus.

Management

- **TYPICAL DF**
  reassure and give the BAD/DermNet leaflet. **No biopsy or removal is needed.-** Removal for cosmetic reasons only is generally not funded on the NHS.* NHS England's *Evidence-Based Interventions programme and local "procedures of limited clinical value" policies restrict benign lesion removal to set criteria: diagnostic doubt, recurrent bleeding/infection, functional impairment or significant pain. Excision leaves a scar often larger and more visible than the DF, and incomplete removal (DF extends deeper than it looks) leads to recurrence.
- Cryotherapy or shave can flatten a DF but often leaves a white mark and recurrence. These are specialist or GP-surgeon decisions, not IP.
- **Refer (routine dermatology)** if diagnosis is uncertain, a variant is suspected, or the lesion is >1 cm and growing.
- **Refer urgently** (two-week-wait / suspected cancer pathway) if it may be melanoma: NG12 weighted 7-point checklist 3, or features of nodular melanoma. Refer urgently (sarcoma or dermatology as per local pathway) if DFSP is suspected (see Dermatofibrosarcoma protuberans).

Pigmented dermatofibroma

- Classic DF features plus the dimple sign: reassure; leaflet.
- **Any doubt about a melanocytic lesion in darker skin: refer** (routine dermatology/teledermatology with dermoscopic). If the 7-point score is 3 (e.g. change in size + irregular colour), use the **suspected cancer pathway**.
- Removal on cosmetic grounds is not routinely NHS-funded (the skin lumps notes see Pigmented dermatofibroma).

Worked cases

CASE 1

42-year-old woman, firm dome-shaped skin-coloured papule on the leg for 8 months

> **DIAGNOSIS**
> dermatofibroma.
>
> **BEDSIDE TEST**
> dimple (pinch/Fitzpatrick) sign. Dermoscopy: peripheral network with a central white patch.
>
> **SUBTYPE WITH RECURRENCE/METASTATIC POTENTIAL**
> cellular (also aneurysmal and atypical) fibrous histiocytoma.
>
> **MANAGEMENT (UK TEACHING ANSWER)**
> reassure; no biopsy or removal for a typical lesion. Removal only for symptoms or diagnostic doubt, within local funding criteria. Refer if atypical (>1 cm, growing, ulcerated, recurrent) or if the 7-point checklist is 3. The course "excise with a 3–5 mm margin" is a reasonable specialist approach if excision is indicated, not a routine plan (see Where course teaching differs from UK guidance).

CASE 2

5-year-old boy, congenital hairless scalp patch, now thicker and yellow-orange

DIAGNOSIS

sebaceous naevus of Jadassohn (pubertal/hormonal phase approaching; HRAS/KRAS mosaicism).

COUNSEL

risk of secondary neoplasms in adult life, mostly benign (trichoblastoma, syringocystadenoma papilliferum); BCC rare (<1%). Watch for any new lump, ulcer or bleeding.

MANAGEMENT

refer routinely to paediatric dermatology. Observation or elective complete excision (narrow margin, timing shared with the family; earlier for the face). Biopsy any nodule. Look for syndromic features (Schimmelpenning) if the lesion is extensive.

DIFFERENTIAL COURSE TEACHING RAISED

linear morphoea en coup de sabre (acquired, indurated, progressive, which needs urgent treatment).

CASE 3

38-year-old man with a forehead nodule, punctum and odorous contents; 56-year-old woman with multiple painful lipomas

DIAGNOSIS 1

epidermoid (infundibular) cyst. **Syndrome association:** Gardner syndrome (FAP, APC): multiple early-onset cysts, osteomas, dental anomalies, desmoids, and colorectal polyposis.

MANAGEMENT

reassure. Treat inflammation without routine antibiotics. Elective complete excision of the cyst wall once quiet if criteria are met. Image midline dermoids first.

DIAGNOSIS 2

Dercum's disease (adiposis dolorosa): painful lipomas, obesity, female, depression/fatigue. Symptomatic, multidisciplinary care; screen mood. Use NG12 sarcoma red flags for any enlarging lump.

Treatment ladder

Reassure

Explain the pinch test and benign nature; shaving guard if it catches.

2 **Removal**

NHS **Evidence-Based Interventions**: only for diagnostic doubt, recurrent bleeding / infection, function or significant pain. Excision scar often bigger than the DF; recurrence if incomplete; cryo / shave leaves white mark (specialist / GP-surgeon decision).

3 **Refer**

Routine dermatology if uncertain or symptomatic; eruptive DFs on immunosuppressants: review drugs.

Referral

URGENT Changing pigmented lesion with atypical dermoscopy: suspected cancer pathway.

URGENT Growing beyond 2–3 cm, irregular, nodular or on the trunk: **DFSP**.

Red flags and when to refer

- Growing beyond 2–3 cm, irregular, nodular or on the trunk: think **DFSP** and refer.
- Changing pigmented lesion with atypical dermoscopy: refer on a **suspected cancer pathway** (NICE NG12).

Complications and prognosis

Complications

- Itch or catching on razors; ulceration (rare).

Prognosis

- Persists indefinitely; may flatten over years.

Prescriber pearls (UK pharmacist IP)

Teach the patient the pinch test and explain that it is "a little knot of scar-like tissue". Shaving over it causes bleeding; a shaving guard helps.

Many eruptive DFs in someone on immunosuppressants (e.g. methotrexate, biologics) or with possible SLE/HIV: check the drug list and consider whether investigations are warranted (GP/specialist).

A **painful DF** that the patient wants removed is a worked case in the skin surgery notes surgery case. See the skin surgery notes for the anticoagulant, latex and local anaesthetic issues.

Five years of slow growth fits DF, but **"growth" is still a 7-point major feature (2 points)**. Document the dimple sign and size, and photograph with a ruler for comparison.

In skin of colour, do not dismiss acral or nail pigmentation as "ethnic". See the melanoma notes (acral melanoma) and Hutchinson's sign.

Pigmented dermatofibroma

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK GUIDANCE / PRACTICE	TEACHING POINT
Dermatofibroma management	Excise with a 3–5 mm margin	Typical DF: reassure. NHS commissioning (EBI, local PLCV policies) does not fund cosmetic removal. Excision only for diagnostic doubt or symptoms	Excision scar is often worse than the DF, and recurrence follows incomplete excision
Sebaceous naevus	Excise in late childhood (earlier if facial)	UK practice increasingly favours observation, given the low malignancy risk. Excision is an elective shared decision	Present both options
Sebaceous naevus	Topical sirolimus, laser	Not standard; unlicensed, specialist-only, weak evidence	Not IP options
"Infected" cysts	(Not discussed)	Most are inflamed, not infected. Antibiotics only for cellulitis (NICE NG141, stewardship)	–
Lipoma red flags	Clinical description only	NICE NG12: urgent direct-access ultrasound within 2 weeks for an unexplained lump increasing in size	Use the NG12 route explicitly

Self-check quiz

4 questions · optional

Q1 Dimple sign?

✓ Lateral pinch makes the papule dip inward.

Q2 Classic dermoscopy?

✓ Peripheral fine network + central white patch.

Q3 Stain pattern of DF?

✓ Factor XIIIa positive, CD34 negative.

Q4 'DF' growing to 4 cm on the chest?

✓ Refer to exclude DFSP.

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Photo gallery

Dermatofibroma 4

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Pigmented dermatofibroma 5

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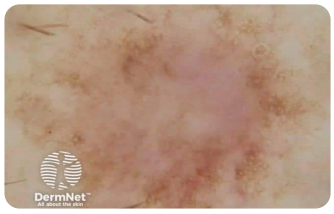

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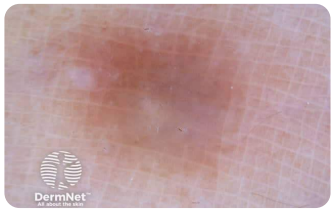

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Lumps, lesions & skin cancer Module 2 · Week 1

Dermatofibrosarcoma protuberans (DFSP)

A rare, slow-growing skin sarcoma that starts as a firm plaque, often on the trunk, and is easily mistaken for a scar, keloid or dermatofibroma. It rarely spreads but recurs locally unless widely excised (often Mohs).

ALSO CALLED DFSP | slow-growing sarcoma | trunk plaque | COL1A1-PDGFB | imatinib

RELATED PAGES Dermatofibroma | Keloids and hypertrophic scars | Lipoma & the sarcoma lump pathway | Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more

KEY FACTS

- **DFSP** is a rare, slow-growing, locally aggressive sarcoma of the dermis; metastasis is rare.
- Firm, skin-coloured or red-brown **plaque that slowly enlarges** over years, then develops nodules; trunk and proximal limbs.
- Often mistaken for a keloid, dermatofibroma or scar: biopsy any enlarging firm plaque.
- Driven by a **COL1A1–PDGFB** fusion; CD34-positive.
- Treatment: **Mohs** or wide excision; imatinib for unresectable disease; long-term follow-up for recurrence.

Overview

A rare, locally aggressive, low-grade **cutaneous sarcoma**. It infiltrates widely beneath clinically normal skin ("tentacles") and recurs if under-excised. Metastasis is rare (<5%) and mostly follows fibrosarcomatous transformation or repeated recurrence.

Epidemiology

- **Who**
  - Adults 20–50; about 1 per million per year; any ethnicity.

Aetiology: causes and risk factors

- Over 90% carry the translocation **t(17;22)** (often as a ring chromosome), fusing **COL1A1–PDGFB**. This drives autocrine PDGF receptor stimulation and is the rationale for **imatinib** in unresectable or metastatic disease.
- Histology: storiform spindle cells infiltrating fat in a "honeycomb" pattern; **CD34 positive, Factor XIIIa negative** (contrast DF).

Myths to address

- **"IT'S JUST A KELOID"**

a keloid follows injury; DFSP enlarges without one.

Pathophysiology and pathology

A gene fusion makes tumour cells **produce their own growth factor** (PDGF-B).

1. **Gene fusion**
   t(17;22) creates COL1A1–PDGFB.
2. **Autocrine loop**
   PDGF-B stimulates its own receptor, driving proliferation (imatinib target).
3. **Infiltrative growth**
   Tentacle-like extensions spread far beyond the visible edge: hence Mohs.

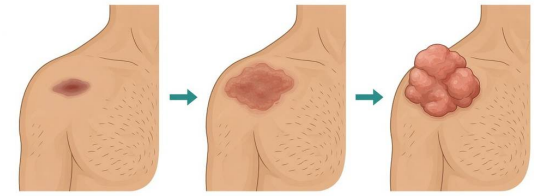

Fig. 1 · DFSP: slow plaque, then nodules, with deep tentacle-like extension.
AI-generated illustration

HISTOPATHOLOGY

- Storiform spindle cells infiltrating fat in a honeycomb pattern; CD34+, factor XIIIa–.

Clinical features

- Adults aged 20–50; trunk (especially chest and shoulder), proximal limbs, head and neck.
- Starts as a slowly enlarging, firm, skin-coloured, pink-violet or brown **indurated plaque**, often mistaken for a scar, keloid, morphoea or DF for years, the same lesion a year apart). Later it becomes **protuberant and multinodular**.
- Variants: pigmented (Bednar tumour), atrophic (looks like morphoea), and childhood/congenital forms. The the skin cancer notes the course summary notes DFSP can **mimic a keloidal BCC**.

Fig. 2 · DFSP early: ill-defined, firm pink-brown plaque on the trunk, easily dismissed as a scar or DF.

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Types to recognise

Early DFSP

Ill-defined firm plaque. Site: Trunk.

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A year later

Enlarging, nodular. Site: Trunk.

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Diagnosis and investigations

- Deep incisional or punch biopsy that includes subcutis (shave biopsies miss it). CD34 immunohistochemistry; FISH/RT-PCR for COL1A1–PDGFB when needed.
- MRI for large or recurrent tumours to define deep extent. Chest imaging only for fibrosarcomatous or metastatic disease (specialist).

Deep biopsy

- Incisional / punch **including subcutis** (shave misses it).

Pathology

- CD34; FISH / RT-PCR for COL1A1–PDGFB.

Imaging

- MRI for large / recurrent tumours; chest imaging only if fibrosarcomatous / metastatic.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Dermatofibroma	Smaller, dimple sign, Factor XIIIa +.
	Keloid / hypertrophic scar	Injury history: beware a 'keloid' without one.
	Morphoea	Ivory sclerotic plaque, lilac edge.
	Lipoma / epidermoid cyst	Soft / punctum.
	Other sarcoma / metastasis	Rapid, deep.

DF (smaller, dimple sign), keloid/hypertrophic scar (history of injury, but beware a "keloid" that appeared without one), morphoea, lipoma, epidermoid cyst, metastasis, other sarcomas, and in children vascular malformation.

Management

- **IP role: suspicion and referral only.** Any "scar", "keloid" or "DF" that keeps enlarging, is >2–3 cm, or is nodular on the trunk goes to dermatology urgently (the local skin cancer or sarcoma pathway). NG12's sarcoma recommendation is a reasonable primary-care route where dermatology access is slow.
- Specialist treatment: **surgery with complete peripheral and deep margin assessment** (Mohs or "slow Mohs"/ staged excision with en-face margins) gives lower recurrence rates than conventional wide local excision with 2–3 cm margins, and spares tissue. European consensus (EORTC/EADO 2015, updated 2022) prefers margin-controlled surgery. Radiotherapy is for positive margins that cannot be re-excised. **Imatinib** is for unresectable or metastatic PDGFB-rearranged disease.
- Follow-up: clinical, with recurrence most likely within 3–5 years.

Treatment ladder

1

Refer

Enlarging 'scar', 'keloid' or 'DF', > 2–3 cm, or nodular on the trunk **urgent** dermatology (skin cancer or sarcoma pathway; NG12 sarcoma route if access is slow).

2

Specialist surgery

Margin-controlled surgery (Mohs / slow Mohs) preferred (EORTC / EADO) over WLE with 2–3 cm margins. Radiotherapy for positive margins that can't be re-excised.

3

Systemic

Imatinib for unresectable / metastatic disease.

1

If on imatinib

CYP3A4 interactions; warfarin; caution with high-dose paracetamol.

Referral

URGENT A 'dermatofibroma' or 'keloid' that keeps growing, is > ~3 cm or becomes nodular: exclude **DFSP**.

Red flags and when to refer

- A 'dermatofibroma' or 'keloid' that keeps growing, is larger than about 3 cm or becomes nodular: refer to exclude **DFSP**.

Complications and prognosis

Complications

- Local recurrence; fibrosarcomatous transformation (higher metastatic risk).

Prognosis

- Excellent after clear-margin surgery.

Prescriber pearls (UK pharmacist IP)

"A keloid without an injury is not a keloid until proved otherwise."

Imatinib interactions (CYP3A4; warfarin; paracetamol hepatotoxicity at high doses) are worth knowing if a patient on it presents to primary care.

Self-check quiz

4 questions · optional

Q1 Driver fusion?

✓ COL1A1–PDGFB (t(17;22)).

Q2 Targeted drug and why?

✓ Imatinib: blocks PDGF receptor signalling.

Q3 Preferred surgery?

✓ Margin-controlled (Mohs / slow Mohs).

Q4 Stain pattern?

✓ CD34 positive, Factor XIIIa negative.

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Photo gallery

DFSP over one year 2

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Lumps, lesions & skin cancer Module 2 · Week 1

Sebaceous naevus (naevus sebaceous)

A yellow-orange hairless plaque on the scalp or face present from birth, which thickens at puberty. Benign tumours can develop within it in adulthood; true BCC is rare, so removal is now case by case.

ALSO CALLED naevus sebaceous of Jadassohn · hairless scalp patch · organoid naevus · epidermal naevus

RELATED PAGES Pigmented & epidermal birthmarks · Congenital melanocytic naevi (CMN) · Basal cell carcinoma (BCC) · Aplasia cutis congenita

KEY FACTS

- **Naevus sebaceus** is a congenital hamartoma: a yellow-orange hairless plaque on the **scalp** or face.
- Flat at birth, becomes warty and raised at puberty under hormonal influence.
- Caused by post-zygotic **HRAS/KRAS** mosaicism.
- Benign tumours (trichoblastoma, syringocystadenoma) can develop in adulthood; true BCC is rare.
- Observe; excision is optional (cosmetic or new nodule). Extensive lesions: think Schimmelpenning syndrome.

Overview

A congenital **hamartoma** of the folliculosebaceous-apocrine unit, present at birth in about 0.3% of newborns. It is mostly on the scalp, then the face.

Aetiology: causes and risk factors

- **Post-zygotic (mosaic) HRAS (most) or KRAS mutations.** This is why it follows Blaschko's lines and why the same mutation underlies the syndromic forms.
- It changes with hormones in **three stages**: infancy (smooth, waxy, yellow-orange, hairless); puberty (sebaceous glands enlarge, so it becomes thick, verrucous and pebbly, as in the boy in Case 2); adulthood (secondary adnexal tumours may develop).

Myths to address

- **"It must be removed in childhood to prevent cancer"**: malignant change is rare; removal is a choice.

Pathophysiology and pathology

A mosaic **RAS mutation** creates a patch of malformed hair follicles and sebaceous glands.

1. **Mosaicism**
   A post-zygotic HRAS or KRAS mutation affects one patch of skin.
2. **Hamartoma**
   Follicles are rudimentary (no hair) and sebaceous and apocrine glands are abnormal.

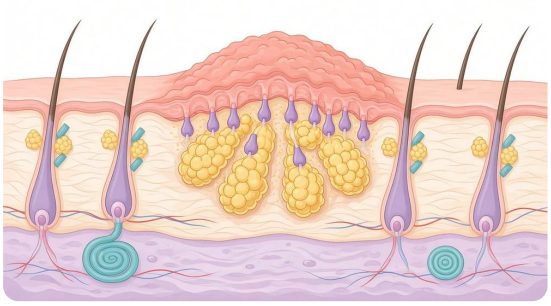

Fig. 1 · Sebaceous naevus after puberty: papillomatous surface, enlarged sebaceous glands, immature hairless follicles and an ectopic apocrine gland.
AI-generated illustration

3. **Puberty**
   Androgens enlarge the sebaceous component, so the plaque thickens.
4. **Secondary tumours**
   Benign adnexal tumours may arise within it in adulthood.

> **HISTOPATHOLOGY**
>
> - Papillomatous epidermis, abnormal immature follicles, enlarged sebaceous glands, ectopic apocrine glands.

Clinical features

Well-demarcated, hairless, yellow-orange or pink plaque, usually oval on the scalp or linear on the face/neck child; skin of colour).

In adults: rough, warty, sometimes with papules or nodules inside it.

SECONDARY TUMOURS

they occur in perhaps 10–20% of lesions, mostly in adults, and **most are benign**. Trichoblastoma is commonest, then syringocystadenoma papilliferum, trichilemmoma, sebaceoma and apocrine tumours. **BCC is rare (<1% in modern series)**, and older reports that called trichoblastomas "BCC" exaggerated the risk. SCC, sebaceous carcinoma and apocrine carcinoma are very rare.

SYNDROMIC FORMS

Schimmelpenning (linear naevus sebaceous) syndrome: an extensive, often facial, linear naevus with CNS (seizures, developmental delay, hemimegalencephaly), eye (colobomas, choristomas) and skeletal (rickets from

> hypophosphataemia, FGF23) involvement. Phacomatosis pigmentokeratotica: sebaceous naevus plus a speckled lentiginous naevus (HRAS).

Fig. 2 · Hairless yellow-orange scalp plaque in a child. Thickens and becomes verrucous at puberty.

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Diagnosis and investigations

- Clinical in most cases. Dermoscopy shows yellow globules and a cerebriform/papillomatous pattern.
- Biopsy any **new nodule, ulcer, bleeding or rapid change** within a sebaceous naevus.
- An extensive or facial lesion with neurological, ocular or skeletal signs needs paediatric referral for syndrome work-up.

Key points

- Clinical in most cases. Dermoscopy shows yellow globules and a cerebriform/papillomatous pattern.
- Biopsy any **new nodule, ulcer, bleeding or rapid change** within a sebaceous naevus.
- An extensive or facial lesion with neurological, ocular or skeletal signs needs paediatric referral for syndrome work-up.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Aplasia cutis congenita	Membranous or scarred, atrophic, sometimes a "hair collar"; not yellow or waxy
	Epidermal (verrucous) naevus	Warty, brown, not yellow; often off the scalp
	Juvenile xanthogranuloma	Dome-shaped orange papule, not a plaque

	CONDITION	DISTINGUISHING FEATURES
	Solitary mastocytoma	Darier sign positive
	Linear morphoea en coup de sabre (see Other lumps and syndromes)	Acquired (not congenital), indurated, ivory, with hair loss and a groove; **progressive**
	Alopecia areata / tinea capitis	Acquired, with normal skin texture or scale

Management

- Explain the natural history and the low (mainly benign) tumour risk. **Observation is reasonable**, and many UK paediatric dermatologists no longer advise routine prophylactic excision.
- Elective **excision** (the course answer, often in late childhood or adolescence under local anaesthetic, earlier for facial lesions or under GA) is offered for cosmetic, practical (comb trauma, bleeding) or anxiety reasons. It is a shared decision with paediatric dermatology or plastics.
- **REFER**
  routine to dermatology for confirmation and discussion. **Sooner** for a new nodule, ulceration or bleeding, which needs biopsy. **Paediatrics/neurology** for extensive or syndromic presentations.
- Not IP options: laser, PDT, **topical sirolimus** (case reports only; unlicensed; does not remove tumour risk; specialist-only).

Referral

ROUTINE — A new lump, crust, ulcer or bleeding within a sebaceous naevus: refer for **biopsy**.

Red flags and when to refer

- A new lump, crust, ulcer or bleeding within a sebaceous naevus: refer for **biopsy**.

Complications and prognosis

Complications

- Secondary tumours (mostly benign), alopecia patch, cosmetic concern.

Prognosis

- Benign lifelong lesion.

Prescriber pearls (UK pharmacist IP)

Parents often worry about "cancer". Give the honest figures (secondary tumours are mostly benign, BCC under 1%) and teach them to look for **any lump growing within the patch**.

Teenagers notice the change at puberty. Reassure them it is expected, not malignant.

Self-check quiz

5 questions · optional

Q1 When should you think of Sebaceous naevus (naevus sebaceous)?

✓ A child with a smooth, yellow-orange, hairless plaque on the scalp, or an adult with a warty plaque there.

Q2 What drives it?

✓ A hamartoma of epidermis, hair follicles and sebaceous glands (mosaic HRAS/KRAS mutations); hormones thicken it at puberty.

Q3 What is the backbone of care?

✓ Recognise, photograph and monitor; discuss elective excision; refer any new nodule, ulcer or bleeding within it.

Q4 What must you never miss?

✓ A new lump or crusted area within the naevus (trichoblastoma, rarely BCC); extensive lesions with eye, brain or bone involvement (syndromic).

Q5 Name a red flag that needs escalation.

✓ A new lump, crust, ulcer or bleeding within a sebaceous naevus: refer for **biopsy**.

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dermnetnz.org
2. BAD PIL naevus sebaceous — https://www.bad.org.uk/pils/
bad.org.uk

Photo gallery

Sebaceous naevus 4

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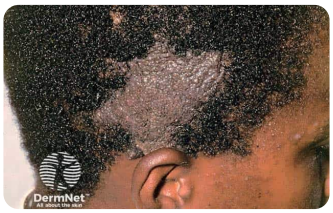
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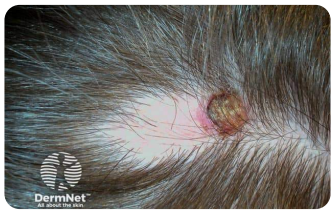
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Lumps, lesions & skin cancer Module 2 · Week 1

Epidermoid, pilar and dermoid cysts (and Gardner syndrome)

Smooth, mobile lumps under the skin. Epidermoid cysts have a central punctum; pilar cysts sit on the scalp. Recognise the inflamed cyst, when to excise, and the rare syndromic clue (multiple facial cysts in Gardner syndrome).

ALSO CALLED sebaceous cyst · epidermal cyst · pilar cyst · trichilemmal cyst · dermoid cyst · punctum · +2

RELATED PAGES Milia · Lipoma & the sarcoma lump pathway · Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more · Boils & carbuncles · Hidradenitis suppurativa

KEY FACTS

- **Epidermoid cysts** are the commonest skin cyst: a mobile dermal nodule with a **central punctum**, filled with keratin ('sebaceous cyst' is a misnomer).
- Pilar (trichilemmal) cysts about 90% on the scalp, smooth, no punctum, often multiple and inherited.
- Most 'infected' cysts are **inflamed** (ruptured keratin), not infected: antibiotics only for spreading cellulitis or systemic features.
- Excise electively once inflammation settles (6–8 weeks), removing the whole wall; cosmetic removal is usually not NHS-funded.
- Midline dermoids need imaging before surgery; multiple cysts in a child or young adult suggest **Gardner syndrome** (APC).

Overview

EPIDERMOID (EPIDERMAL INCLUSION) CYST
the commonest cutaneous cyst. Its wall resembles epidermis/infundibulum and it is filled with keratin (the "cheesy", foul-smelling contents in Case 3). "Sebaceous cyst" is a misnomer.

PILAR (TRICHILEMMAL) CYST
the scalp in ~90%. Firm, smooth, no punctum, often multiple and familial (autosomal dominant). It shells out easily.

DERMOID CYST
congenital, along embryonic fusion lines. The classic site is the **lateral eyebrow** (external angular dermoid); midline nasal or scalp dermoids may connect intracranially.

MILIA
tiny superficial keratin cysts: see Milia.

VELLUS HAIR CYSTS / STEATOCYSTOMA

> small multiple cysts on the chest; steatocystoma multiplex.

Epidemiology

- **Who**
  - Epidermoid cysts: young and middle-aged adults, more men; face, neck, trunk, scrotum.
  - Pilar cysts: middle-aged women, autosomal dominant tendency.

Aetiology: causes and risk factors

Most epidermoid cysts arise from occluded hair follicle infundibula. Others come from implantation of epidermis after trauma (palms/soles), acne, or HPV (plantar). **Rupture** releases keratin into the dermis and causes a sterile foreign-body inflammatory reaction. Most "infected cysts" are actually inflamed, not infected.

Risk factors and associations

- Occluded follicular infundibulum; implantation of epidermis after injury; acne.
- Drugs: isotretinoin, ciclosporin, BRAF inhibitors.

Myths to address

> **"IT'S A BLOCKED SEBACEOUS GLAND"**
> the cyst is lined by epidermis and contains keratin, not sebum.

> **"SQUEEZE IT OUT"**
> squeezing ruptures the wall and causes inflammation and scarring.

Pathophysiology and pathology

A cyst is a sac of **epidermis turned inside out**, shedding keratin into its centre.

1. **Trapped epidermis**
   Follicular infundibulum (or implanted epidermis) forms a closed sac in the dermis.

2. **Keratin build-up**
   The lining keeps making keratin, which collects as soft 'cheesy' material; the punctum is the blocked opening.

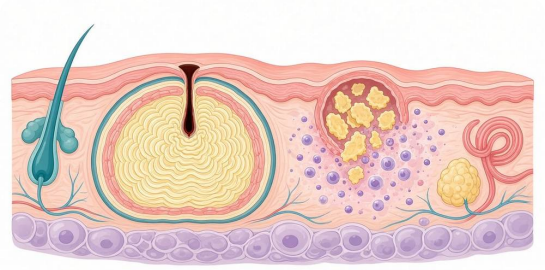

Fig. 1 · Epidermoid cyst with a central punctum (left) and a ruptured cyst spilling keratin into the dermis with inflammation (right).
AI-generated illustration

3. **Rupture and inflammation**
   Keratin released into the dermis causes a sterile **foreign-body** reaction: red, hot and tender without infection.

4 **Recurrence**

Any wall left behind regrows, so excision must remove the whole sac.

HISTOPATHOLOGY

- Epidermoid: stratified squamous lining with a granular layer and laminated keratin. Pilar: no granular layer, compact keratin (trichilemmal keratinisation).

Clinical features

- Mobile, dome-shaped dermal nodule, 0.5–5 cm, on the face, neck, trunk or scrotum, with a **central punctum** in many.
- Inflamed or ruptured cysts are red, tender and fluctuant. Recurrence follows incomplete wall removal.
- **Gardner syndrome** (a familial adenomatous polyposis variant, **APC** gene): **multiple epidermoid cysts**, often atypical sites (face, scalp, limbs) and **appearing before puberty**; **osteomas** of the skull or jaw; dental anomalies (supernumerary or unerupted teeth); desmoid tumours; CHRPE (retinal pigment lesions); and colorectal polyposis with near-100% colorectal cancer risk if untreated.

Fig. 2 · Palpating an epidermoid cyst accentuates the central punctum.

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Special groups

- **Gardner syndrome**
  - Multiple epidermoid cysts before puberty, osteomas, dental anomalies, desmoids and colorectal polyposis: refer to genetics or colorectal services.

Diagnosis and investigations

- Clinical. Ultrasound if unsure (helps distinguish lipoma, lymph node or sarcoma).
- **Dermoid cysts in the midline (nasal bridge, scalp), or any lesion with a sinus or hair, need imaging (MRI) before surgery** because of possible intracranial extension. This was the course point.

- Gardner: refer to genetics or colorectal services. Triggers are multiple cysts in a child or young adult, especially with osteomas, dental anomalies, or a family history of polyposis or young colorectal cancer.

Differential diagnoses

Lipoma (soft, no punctum, lobulated); pilomatricoma (rock-hard, child, see Other lumps and syndromes); dermoid; abscess / hidradenitis (flexural, recurrent, sinus tracts); lymph node; BCC with a cystic component; metastasis; on the scalp, pilar cyst vs proliferating trichilemmal tumour (rapidly growing, ulcerating, elderly woman biopsy).

Management

ASYMPTOMATIC CYSTS

reassure. **Cosmetic removal is usually not NHS-funded** (EBI/local policy). Squeezing worsens inflammation and scarring.

INFLAMED (NOT INFECTED) CYST

warm compresses; analgesia. A fluctuant cyst may need incision and drainage, which can be done in some GP settings. **Antibiotics only if there is spreading cellulitis or systemic features** (choose per NICE NG141 cellulitis / local formulary). Avoid routine antibiotics for a "red cyst" (antimicrobial stewardship).

Elective excision once the inflammation settles (6–8 weeks) removes the entire wall to prevent recurrence. options include standard elliptical excision, or minimal-incision / punch techniques with lower scar burden but slightly higher recurrence. These are clinician-surgeon decisions (knowledge level only; see the skin surgery notes).

REFER

suspected dermoid near the midline or orbit (plastics/paediatric surgery, imaging first); suspected Gardner (genetics/colorectal); rapidly enlarging scalp "cyst" in an older person (biopsy); any lump with sarcoma features (see Lipoma, angiolipoma and Dercum's disease; the sarcoma lump pathway).

Referral

ROUTINE — Midline lump on the nose, scalp or spine in a child: **imaging before any surgery**.

ROUTINE — Multiple cysts in childhood with osteomas or a family history of bowel cancer: consider **Gardner syndrome (FAP)** and refer.

Red flags and when to refer

- Midline lump on the nose, scalp or spine in a child: **imaging before any surgery**.
- Multiple cysts in childhood with osteomas or a family history of bowel cancer: consider **Gardner syndrome (FAP)** and refer.

Complications and prognosis

Complications

- Inflammation, rupture and abscess; scarring.
- Recurrence after incomplete excision.
- Malignant change is very rare; a rapidly growing scalp 'cyst' in an older person needs biopsy (proliferating trichilemmal tumour).

Prognosis

- Benign; persists unless excised.

Prescriber pearls (UK pharmacist IP)

- Explain that the smell comes from keratin breakdown, not poor hygiene.
- Recurrence after "popping" or drainage is expected because the wall remains.
- Isotretinoin, ciclosporin and BRAF inhibitors can provoke milia or epidermoid cysts. Look at the drug history if cysts erupt.

Self-check quiz

5 questions · optional

Q1 When should you think of Epidermoid, pilar and dermoid cysts (and Gardner syndrome)?

✓ A smooth, round, mobile lump with a central pore (epidermoid), or several firm scalp lumps (pilar).

Q2 What drives it?

✓ Keratin-filled sacs lined by epidermis or hair-root sheath; dermoid cysts are embryonic, often at the outer eyebrow.

Q3 What is the backbone of care?

✓ Reassure; leave alone or excise the whole wall when not inflamed; inflamed cysts usually are not infected (antibiotics only for cellulitis).

Q4 What must you never miss?

✓ Recurrence after incomplete removal; midline nasal or scalp lumps in children (intracranial connection); multiple cysts with osteomas in Gardner syndrome (FAP).

Q5 Name a red flag that needs escalation.

✓ Midline lump on the nose, scalp or spine in a child: **imaging before any surgery**.

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Photo gallery

Epidermoid cysts 2

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Gardner syndrome 2

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Pilar cyst 1

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Lumps, lesions & skin cancer Module 2 · Week 1

Lipoma & the sarcoma lump pathway

Soft, rubbery, mobile fatty lumps under the skin. Nearly all are benign; the job is to know the features (large, deep, growing, painful) that need an ultrasound or sarcoma referral.

ALSO CALLED fatty lump · angiolipoma · Dercum's disease · soft tissue sarcoma · lump larger than 5 cm

RELATED PAGES Epidermoid, pilar and dermoid cysts (and Gardner syndrome) · Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more · Dermatofibroma · Dermatofibrosarcoma protuberans (DFSP)

KEY FACTS

- **Lipomas** are benign tumours of mature fat cells: soft, rubbery, mobile, painless lumps.
- Commonest soft-tissue tumour; trunk, neck, shoulders and proximal limbs; usually under 5 cm.
- Ultrasound if uncertain. **Sarcoma** features: over 5 cm, deep to fascia, growing, painful or recurrent: urgent imaging or referral (NICE suspected cancer).
- Reassure; cosmetic removal is usually not NHS-funded.
- Multiple painful lipomas: angiolipomas or **Dercum's disease**; multiple lipomas can be familial.

Overview

LIPOMA
a benign tumour of mature fat. The commonest soft-tissue tumour; age 40–60.

ANGIOLIPOMA
a vascular variant. Small (<2 cm), often **multiple and tender**, on the forearms and trunk of young adults (often men). Familial in about 5%.

DERCUM'S DISEASE (ADIPOSIS DOLOROSA)
a rare chronic condition with **multiple painful lipomas or painful fatty tissue**, usually in obese women aged 35–50. It is often accompanied by fatigue, depression, sleep and cognitive problems, and joint pain. It is a diagnosis of exclusion.

Related syndromes: familial multiple lipomatosis; Madelung disease (benign symmetric lipomatosis, linked to alcohol); PTEN hamartoma (see Other lumps and syndromes); Gardner (see Epidermoid, pilar and dermoid cysts; milia; Gardner syndrome).

Epidemiology

- **Who**
  - Adults 40–60; about 1 in 100 people; more with obesity.

Aetiology: causes and risk factors

Lipomas show clonal rearrangements (often HMGA2 at 12q13–15). The cause of Dercum's pain is unclear; proposed mechanisms include nerve compression, inflammatory and neuroendocrine factors.

Risk factors and associations

- Unknown; chromosomal rearrangements (12q13–15, HMGA2) in many; familial multiple lipomatosis; trauma possibly.

Myths to address

- **"LIPOMAS TURN INTO CANCER"**
  they almost never do; the concern is a sarcoma mistaken for a lipoma.

Pathophysiology and pathology

A lipoma is a **capsule of normal-looking fat cells** growing slowly beneath the skin.

1. **Clonal fat cells**
   Mature adipocytes proliferate as a lobulated mass with a thin fibrous capsule.
2. **Slow growth**
   Growth is slow and then plateaus; the capsule makes it mobile and well defined.
3. **Variants**
   Angiolipoma (vessels, tender), spindle-cell lipoma (back of neck in older men), deep lipomas within muscle.

HISTOPATHOLOGY

- Mature adipocytes with a thin capsule; angiolipomas contain small vessels with microthrombi. Atypical lipomatous tumour shows atypical stromal cells (MDM2 amplification).

Clinical features

- Soft, rubbery, mobile, lobulated subcutaneous mass with normal overlying skin. Trunk, shoulders, neck and arms. Painless (except angiolipoma or Dercum).
- **RED FLAGS FOR SARCOMA (REFER)**
  size **>5 cm**, **increasing in size**, **deep to fascia** (fixed, doesn't move with the skin), **painful**, recurrence after excision. Thigh, buttock and retroperitoneal sites are higher risk, as is any lump that is hard or irregular.

Fig. 1 · Soft, mobile subcutaneous lipoma on the back. Multiple painful lipomas in an older woman with obesity and low mood suggest Dercum disease.

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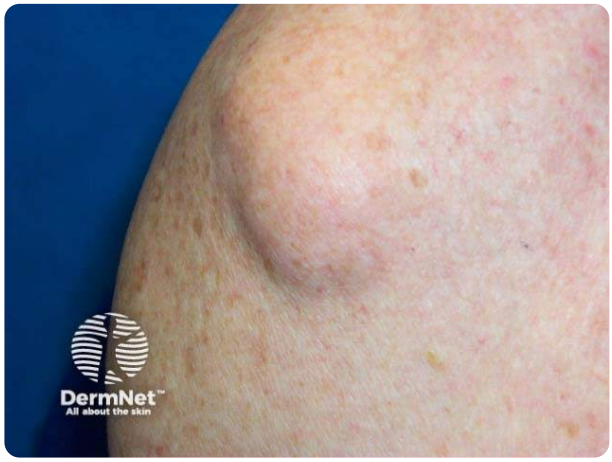

Fig. 2 · Soft, mobile subcutaneous lipoma on the shoulder.

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Special groups

- **Sarcoma red flags**
  - Size over 5 cm, growth, pain, deep location, recurrence after excision: urgent ultrasound or sarcoma pathway.

Diagnosis and investigations

Clinical for small, typical superficial lesions. **Ultrasound** is first-line imaging. MRI for deep or large lesions. Biopsy in a sarcoma centre, never by excision in primary care for a suspicious lump (an "unplanned excision" (the "whoops" procedure) compromises sarcoma outcomes).

Differential diagnoses

Epidermoid cyst (punctum), liposarcoma and other soft-tissue sarcoma, lymph node, hernia, haematoma, neurofibroma (buttonhole sign), angiolipoma, lipodystrophy injection-site lipohypertrophy (insulin sites, so check injection technique).

Management

> **TYPICAL SMALL LIPOMA**
> reassure; removal for cosmetic reasons is usually not NHS-funded.

> **NICE NG12 (SARCOMA)**
> consider an **urgent direct-access ultrasound (performed within 2 weeks) for adults with an unexplained lump that is increasing in size**. Refer on the suspected cancer pathway if the ultrasound suggests sarcoma or is uncertain but clinical concern persists. In children/young people, an unexplained lump increasing in size warrants a **very urgent ultrasound (within 48 hours)**.

> If the IP does not have direct-access imaging rights, the route is via the GP the same week, with the red flags documented.

> **DERCUM'S DISEASE**
> a GP/specialist (dermatology, pain clinic) diagnosis. Management is symptomatic: analgesia, weight management, psychological support; specialist options include lidocaine infusions, liposuction and excision of the worst lesions. Screen for and support **depression**. Chronic pain medicines need careful IP review: avoid escalating opioids, and consider neuropathic agents per NICE NG193 (chronic primary pain) principles and local formulary.

> **LIPOHYPERTROPHY FROM INSULIN/GLP-1 INJECTIONS**
> counsel on rotating sites. It also impairs absorption and glycaemic control.

Referral

URGENT — Lump that is increasing in size or larger than 5 cm, deep or painful: **urgent direct-access ultrasound** (within 2 weeks) or a suspected sarcoma pathway (NICE NG12).

Red flags and when to refer

- Lump that is increasing in size or larger than 5 cm, deep or painful: **urgent direct-access ultrasound** (within 2 weeks) or a suspected sarcoma pathway (NICE NG12).

Complications and prognosis

Complications

- Pressure on nerves; cosmetic concern; recurrence after excision (rare).

Prognosis

- Benign; persists unless excised.

Prescriber pearls (UK pharmacist IP)

"Is it bigger than a golf ball, growing, deep or painful?" Then it needs a scan.

Angiolipoma tenderness is common and benign, but multiple painful lumps plus systemic symptoms should prompt thinking of Dercum.

Self-check quiz

5 questions · optional

Q1 When should you think of Lipoma & the sarcoma lump pathway?

✓ A soft, painless, mobile, slowly growing lump on the trunk, neck or limbs.

Q2 What drives it?

✓ Benign overgrowth of mature fat cells; angiolipomas are tender; Dercum's disease gives many painful lipomas.

Q3 What is the backbone of care?

✓ Reassure if small and typical; ultrasound or refer if uncertain; excision only for symptoms or diagnosis.

Q4 What must you never miss?

✓ Soft-tissue sarcoma: larger than 5 cm, deep to fascia, growing, painful or recurrent.

Q5 Name a red flag that needs escalation.

✓ Lump that is increasing in size or larger than 5 cm, deep or painful: **urgent direct-access ultrasound** (within 2 weeks) or a suspected sarcoma pathway (NICE NG12).

References

1. DermNet: Lipoma – https://dermnetnz.org/topics/lipoma
dermnetnz.org
2. DermNet: Angiolipoma – https://dermnetnz.org/topics/angiolipoma
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3. NICE NG12 sarcoma – https://www.nice.org.uk/guidance/ng12

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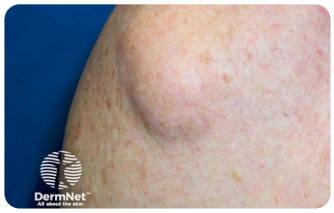
4. Medscape: Lipoma & the sarcoma lump pathway — https://emedicine.medscape.com/article/1057855-overview
emedicine.medscape.com

Photo gallery

Clinical photographs 1



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Lipoma 1

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Lumps, lesions & skin cancer Module 2 · Week 1

Milia

Milia are tiny (1–2 mm) white keratin cysts just under the epidermis. Primary milia are common in newborns and on the face; secondary milia follow blistering, trauma, dermabrasion or potent topical steroids. Harmless; usually left alone.

ALSO CALLED milia · milium · milk spots · milia en plaque

RELATED PAGES Epidermoid, pilar and dermoid cysts (and Gardner syndrome) · Neonatal pustular rashes · Porphyria cutanea tarda (& pseudoporphyria) · Epidermolysis bullosa (EB) · Acne vulgaris

KEY FACTS

- **Milia** are 1–2 mm white **keratin cysts** just under the epidermis.
- Primary very common in newborns (nose, cheeks) and on adult eyelids and cheeks.
- Secondary after blistering, burns, dermabrasion, or potent topical steroids.
- Diagnosis is clinical; look for a cause of secondary milia (e.g. PCT on the backs of the hands).
- Reassure; troublesome milia can be de-roofed and expressed; topical retinoids for multiple lesions.

Overview

Milia are 1–2 mm superficial **keratin cysts** (tiny epidermoid cysts). **Primary** milia arise spontaneously (up to half of newborns; also adults on the eyelids and cheeks). **Secondary** milia follow blistering, trauma, burns, dermabrasion or potent topical steroids.

Epidemiology

- **Who**
  - Up to half of newborns have milia; common at all ages.
  - Milia en plaque: rare plaque of milia, usually behind the ear in middle-aged women.

Aetiology: causes and risk factors

Primary milia form from the lower **infundibulum of vellus hairs**; secondary milia form when damaged sweat ducts, follicles or epidermis become trapped during healing.

Risk factors and associations

- Primary: retention of keratin in the lower infundibulum of vellus hairs.
- Secondary: damage to follicles or sweat ducts during blistering or trauma; steroid-induced skin atrophy.

Myths to address

- **"THEY'RE WHITEHEADS"**
  milia are closed cysts without an opening; they do not respond to acne cleansers.

Pathophysiology and pathology

A milium is a **miniature epidermoid cyst** trapped just under the surface.

1. **Keratin retention**
   Keratin made by a tiny follicle or sweat duct cannot escape to the surface.

2. **Microcyst**
   A sac lined by epidermis fills with layered keratin, seen as a pearly white dot.

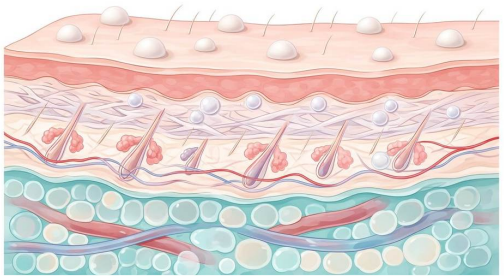

Fig. 1 · Milia: tiny keratin cysts in the upper dermis just beneath the epidermis, seen as white dome papules.
AI-generated illustration

3. **Secondary milia**
   When blisters or wounds heal, fragments of epithelium are trapped and form new cysts.

HISTOPATHOLOGY

- Small cyst with a stratified squamous lining and laminated keratin, like a tiny epidermoid cyst.

Clinical features

- Firm, pearly **white or yellow** dome papules, 1–2 mm.
- Face (eyelids, cheeks, nose) in primary milia; at sites of previous blisters or scars in secondary milia (e.g. hands in porphyria cutanea tarda or epidermolysis bullosa).
- Neonatal milia: nose and cheeks; resolve within weeks.

Fig. 2 · Periocular milia: tiny firm white keratin cysts around the eyes.
© DermNet · **Source**

Fig. 3 · Neonatal milia on the nose and cheeks; they clear without treatment.
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Fig. 4 · Secondary milia after blistering (here pemphigoid) — a clue to a past subepidermal blister.
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Special groups

- **Newborns**
  - Often with Epstein pearls in the mouth; reassure, resolves in weeks.

Diagnosis and investigations

Clinical. Dermoscopy shows a white-yellow globule. Look for a cause of secondary milia (blistering disease, PCT, steroid use).

Differential diagnoses

> **Comedones** (acne): central keratin plug in a follicle, other acne lesions.

> **SYRINGOMA**
> skin-coloured papules on the lower eyelids.

> **SEBACEOUS HYPERPLASIA**
> yellow umbilicated papules with crown vessels.

> **XANTHELASMA**
> yellow plaques on the eyelids.

> **MOLLUSCUM CONTAGIOSUM**
> umbilicated, pearly papules, often in children.

Management

- **Reassure**; neonatal milia resolve without treatment.
- If cosmetically troublesome: de-roofing with a sterile needle and gentle expression (de-roofing and extraction).
- Topical retinoids may help multiple milia; avoid potent steroids that provoke them.

Treatment ladder

1 **Reassure**

Neonatal milia resolve within weeks; many adult milia persist but are harmless.

2 **Cosmetic removal**

De-roof with a sterile needle or blade and express the keratin (trained clinician).

3 **Multiple / secondary**

Topical retinoid; treat the underlying blistering disease; avoid potent topical steroids on the face.

Referral

ROUTINE — Milia with skin fragility, blisters and scarring on the backs of the hands: investigate for **porphyria cutanea tarda** or epidermolysis bullosa.

ROUTINE — Milia en plaque or widespread milia: dermatology.

Red flags and when to refer

- Milia on scarred, fragile skin of the backs of the hands: think **porphyria cutanea tarda** or epidermolysis bullosa acquisita.

Complications and prognosis

Complications

- Cosmetic concern only; scarring if picked.

Prognosis

- Neonatal milia clear; adult and secondary milia may persist.

Prescriber pearls (UK pharmacist IP)

- Milia are **cysts**, not blocked sebaceous glands; squeezing without de-roofing does not work.

Self-check quiz

4 questions · optional

Q1 What are milia?

✓ Tiny superficial keratin cysts (miniature epidermoid cysts), 1–2 mm, white.

Q2 Name three causes of secondary milia.

✓ Blistering diseases (PCT, EB, pemphigoid), burns or dermabrasion, potent topical steroids.

Q3 How are troublesome milia removed?

✓ De-roofing with a sterile needle or blade and gentle expression.

Q4 Do neonatal milia need treatment?

✓ No; they resolve on their own within weeks.

References

1. DermNet: Milia — https://dermnetnz.org/topics/milia
dermnetnz.org

Photo gallery

Clinical photographs 3

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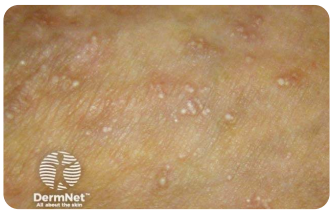

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Secondary milia 1

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Lumps, lesions & skin cancer Module 2 · Week 1

Pyogenic granuloma

A fast-growing, bright-red, friable lump that bleeds easily, often on a finger, lip or gum after minor injury or in pregnancy. Treat by curettage and cautery with histology, because amelanotic melanoma can look the same.

ALSO CALLED lobular capillary haemangioma · bleeding red lump · granuloma gravidarum · curettage and cautery

RELATED PAGES Melanoma: recognition & referral · Ingrowing toenail (onychocryptosis) · Retinoids (topical & oral) · Cutaneous squamous cell carcinoma (cSCC)

KEY FACTS

- **Pyogenic granuloma** (lobular capillary haemangioma) is a fast-growing, **bleeding**, red, moist papule: neither pyogenic nor a granuloma.
- Grows over weeks, often after minor trauma; fingers, lips, face, gums (pregnancy epulis).
- Triggers: pregnancy, isotretinoin, EGFR inhibitors, other drugs.
- Curettage and cautery with **histology** is the usual treatment; send all removed tissue.
- Amelanotic melanoma can look identical: a bleeding nodule in an adult needs histology.

Overview

A common, benign, rapidly growing vascular tumour of skin and mucosa. The name is a misnomer: it is neither pyogenic nor a granuloma.

Epidemiology

- **Who**
  - Children and young adults; pregnant women (gingiva).

Aetiology: causes and risk factors

Reactive proliferation of lobules of capillaries, often after minor trauma. Hormonal (pregnancy, the "granuloma gravidarum" on the gums) and **drug triggers**, which a pharmacist should know:

- systemic retinoids (isotretinoin, acitretin)
- EGFR inhibitors (cetuximab, erlotinib, gefitinib; around the nail folds)
- BRAF/MEK inhibitors
- some antiretrovirals (indinavir)
- ciclosporin
- capecitabine and taxanes

- the combined oral contraceptive

Risk factors and associations

- Minor injury, hormones (pregnancy), drugs (retinoids, EGFR inhibitors, protease inhibitors).

Myths to address

- **"IT'S AN INFECTION"**
  it is a benign vascular proliferation; antibiotics do not help.

Pathophysiology and pathology

Pyogenic granuloma is an **exuberant capillary overgrowth** at a site of injury.

1. **Trigger**
   Minor injury or hormones release angiogenic factors (VEGF).

2. **Lobular capillary proliferation**
   Capillaries proliferate in lobules, forming a red, friable nodule that bleeds easily.

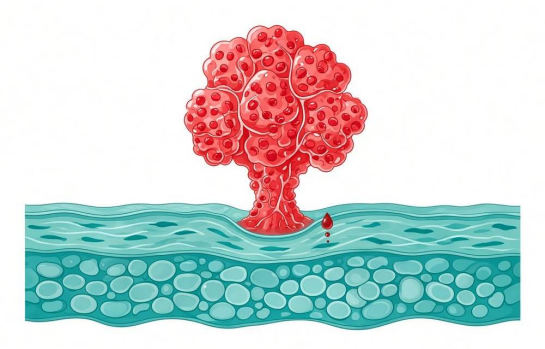

Fig. 1 · Pyogenic granuloma: lobules of capillaries with a collarette of epidermis.
AI-generated illustration

3. **Collarette**
   Epidermis grows around the base forming a collar; the surface often ulcerates.

> **HISTOPATHOLOGY**
>
> - Lobular capillary proliferation in an oedematous stroma with an epidermal collarette; ulceration common.

Clinical features

- A bright red, glistening, friable papule or polyp, 5–10 mm, developing over **weeks** (the 4-week series is in). It often has a collarette of scale.
- **Bleeds profusely with minor trauma.** Fingers, lips, gums, face, eyelid, and nail folds.

Fig. 2 · Pyogenic granuloma, new lesion on the finger.

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Types to recognise

Lip PG

Rapidly growing red papule. Site: Lip.

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Eyelid

Pedunculated. Site: Eyelid.

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Week 4

Pedunculated, friable. Site: Finger.

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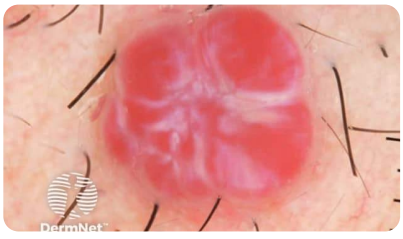
Dermoscopy

Red clods, white lines. Site: Dermoscopy.



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Special groups

- **Pregnancy**
  - Gingival lesions often regress after delivery; treat if bleeding or troublesome.

Diagnosis and investigations

- Clinical plus dermoscopy (red homogeneous areas, white "rail" lines, collarette).
- **Histology is mandatory after removal, because amelanotic melanoma can look identical** (see ,).

Clinical

- + dermoscopy.

Histology

- **Mandatory** after removal: amelanotic melanoma can look identical.

Drug history

- Retinoids, EGFR / BRAF inhibitors, ciclosporin, pill.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Amelanotic nodular melanoma	Adult, non-traumatised site, recurrence, atypical vessels.
	SCC / keratoacanthoma	Keratin, induration.
	BCC	Pearly, arborising vessels.
	Kaposi sarcoma / bacillary angiomatosis	HIV context; violaceous.
	Cherry angioma	Older, stable, small.
	Ingrown-nail granulation	Periungual, nail spike.

Amelanotic nodular melanoma (the critical one; consider in any adult with a "PG" on a non-traumatised site, recurrence, or atypical vessels), SCC/keratoacanthoma, BCC, Kaposi sarcoma, bacillary angiomatosis (HIV), cherry angioma (older, stable), infantile haemangioma (infants), metastasis (kidney, thyroid), and periungual granulation tissue from an ingrown nail.

Management

- **BLEEDING FIRST AID**
  firm pressure for 10–15 minutes; a non-adherent dressing.
- **Refer for removal with histology.** Curettage and cautery (shave/curette plus base cautery) is the usual UK primary-care or dermatology approach. It is discussed at knowledge level in the skin surgery notes.
- Silver nitrate or cryotherapy alone gives no specimen and recurs more. Only use them where the diagnosis is not in doubt (e.g. a small pregnancy gum lesion, managed by a dentist).
- **DRUG-INDUCED PG**
  discuss with the prescribing specialist before stopping a cancer drug. Topical timolol or potent steroid are sometimes used by specialists for small periungual lesions.
- Pregnancy PGs may regress after delivery; manage conservatively unless bleeding.

Why curettage and cautery

- A **friable, bleeding vascular lesion** that would bleed during shave alone. **C&C removes it and seals the base**, and the curettings go for **histology** (amelanotic melanoma, SCC and Kaposi are the "must-not-miss" mimics; the skin lumps notes see Procedures at knowledge level).
- Alternatives: shave + cautery; excision for recurrent lesions (with closure, as course teaching described); silver nitrate or cryotherapy for very small lesions where the diagnosis is certain (no specimen).
- **POSTERIOR NECK**
  superficial procedures are safe. A deeper excision would need awareness of the spinal accessory nerve (see Consent, complications and anatomical danger zones).

Treatment ladder

1 **First aid**

Firm pressure 10–15 minutes; non-adherent dressing.

2 **Remove**

Refer for **curettage and cautery** (shave / curette + base cautery) **with histology**.

3 **Drug-induced**

Discuss with the prescribing specialist before stopping a cancer drug; specialists may use topical timolol or potent steroid for small periungual lesions. Isotretinoin nail-fold granulation: **Yellow Card** + liaise with prescriber.

Pregnancy

Gum lesions may resolve after birth (dentist-managed).

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Silver nitrate / cryo alone**: No specimen, more recurrence: only if diagnosis isn't in doubt.

Referral

URGENT — Rapidly growing red nodule in an adult, especially **sole, nail unit or > 50**: histology; suspected cancer pathway if atypical (amelanotic melanoma).

ROUTINE — Typical PG: refer for curettage and cautery with histology.

Red flags and when to refer

- A rapidly growing red nodule in an adult, especially on the sole, nail unit or in someone over 50: send tissue for histology; refer on a **suspected cancer pathway** if atypical (amelanotic melanoma).

Complications and prognosis

Complications

- Recurrent bleeding; recurrence after treatment (satellite lesions).

Prognosis

- Rarely resolves untreated; treatment is usually curative.

Prescriber pearls (UK pharmacist IP)

Recurrence (up to 15%) and satellite lesions happen. Return if it recurs, and do not assume it is benign.

On isotretinoin? PG-like granulation tissue at the nail folds is a known adverse effect. Report it (Yellow Card) and liaise with the prescriber.

Self-check quiz

4 questions · optional

Q1 Why always send histology?

✓ Amelanotic melanoma can look identical.

Q2 Usual treatment?

✓ Curettage and cautery of the base, with histology.

Q3 Drug classes that trigger PG?

✓ Systemic retinoids, EGFR / BRAF inhibitors, ciclosporin, some antiretrovirals.

Q4 Bleeding first aid?

✓ Firm pressure for 10–15 minutes.

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1. DermNet: Pyogenic granuloma — https://dermnetnz.org/topics/pyogenic-granuloma
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2. BAD PIL — https://www.bad.org.uk/pils/
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3. Medscape: Pyogenic granuloma — https://emedicine.medscape.com/article/1077040-overview
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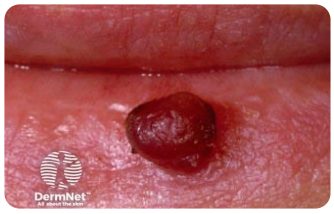
Photo gallery

Pyogenic granuloma 2



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Growth over four weeks

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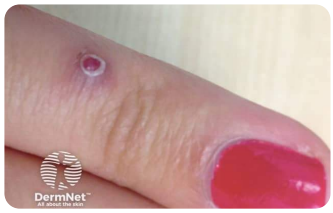

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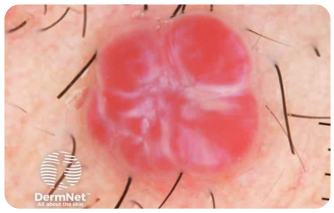

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Lumps, lesions & skin cancer Module 2 · Week 1

Angiokeratomas & Fabry disease

Dark red to black warty papules from dilated vessels near the surface. Most are harmless (scrotum, vulva); a solitary dark one can mimic melanoma; clusters in the bathing-trunk area of a young person suggest Fabry disease.

ALSO CALLED angiokeratoma of Fordyce · angiokeratoma circumscriptum · angiokeratoma corporis diffusum · Fabry disease · dark red papules

RELATED PAGES Melanoma: recognition & referral · Pyogenic granuloma · Port-wine stain (capillary malformation)

KEY FACTS

- Angiokeratomas dark red to black papules from dilated superficial capillaries with overlying hyperkeratosis.
- Variants: solitary (legs; can mimic melanoma), **Fordyce** (scrotum/vulva), Mibelli (fingers/toes), circumscriptum (congenital).
- **Multiple angiokeratomas in a bathing-trunk area** suggest **Fabry disease** (X-linked α-galactosidase A deficiency).
- Fabry clues: neuropathic pain in hands and feet, reduced sweating, proteinuria, early stroke or cardiac disease.
- Refer uncertain dark lesions; refer suspected Fabry for testing at a national lysosomal storage centre.

Overview

Benign vascular lesions: dilated superficial dermal capillaries with overlying hyperkeratosis.

Aetiology: causes and risk factors

Ectatic papillary dermal vessels with reactive epidermal thickening. **Angiokeratoma corporis diffusum** (multiple lesions) is a marker of lysosomal storage disease. The classic one is **Fabry disease**: X-linked, **GLA** gene, α-galactosidase A deficiency, with globotriaosylceramide (Gb3) accumulating in vessels, kidney, heart and nerves. Others are fucosidosis and sialidosis.

Myths to address

- **"A BLACK PAPULE THAT BLEEDS MUST BE MELANOMA"**
  angiokeratoma is a common mimic; dermoscopy helps, but refer if uncertain.

Pathophysiology and pathology

Angiokeratomas are **ectatic papillary vessels** with a thickened epidermis on top.

1. **Vessel dilatation**
   Superficial dermal capillaries dilate (pressure, trauma or storage material).

2. **Epidermal reaction**
   The overlying epidermis thickens and becomes hyperkeratotic.

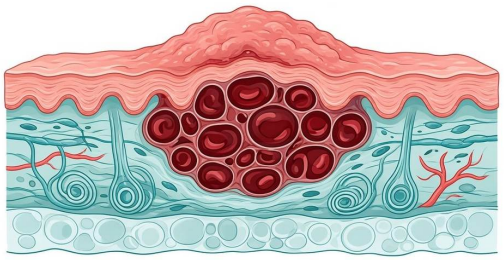

Fig. 1 · Angiokeratoma: dilated vessels hugged by a thickened epidermis.
AI-generated illustration

3. **Fabry disease**
   Gb3 accumulates in vessel walls, kidney, heart and nerves.

Clinical features

VARIANT	WHO / WHERE	CLUE
Solitary / multiple angiokeratoma	Adults, legs	Dark red-black papule. **Can mimic melanoma**
Angiokeratoma of Fordyce	Older men (scrotum), women (vulva)	Multiple small purple papules; may bleed
Angiokeratoma of Mibelli	Children/teens, dorsal fingers and toes	Associated with chilblains/ acrocyanosis
Angiokeratoma circumscriptum	Congenital, a limb	Plaque of coalescent papules
Angiokeratoma corporis diffusum	"Bathing-trunk" distribution (lower trunk, buttocks, groin, umbilicus)	Suggests **Fabry**

Fabry features: acroparaesthesia (burning hands/feet triggered by heat, exercise, fever) starting in childhood; **hypohidrosis** and heat intolerance; cornea verticillata; proteinuria and later renal failure; left ventricular hypertrophy and arrhythmias; stroke at a young age. Heterozygous women are often affected too.

Fig. 2 · Angiokeratoma circumscriptum: a congenital band of dark red warty papules on a limb (one of the five variants).

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Types to recognise

Fordyce

Scrotal purple papules. Site: Scrotum.

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Solitary dark

Melanoma mimic. Site: Leg.

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Circumscriptum

Congenital plaque. Site: Limb.

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Multiple (Fabry?)

Bathing-trunk cluster. Site: Lower trunk.

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Diagnosis and investigations

- Dermoscopy: red-blue lacunae, whitish veil, and haemorrhagic crusts.
- Biopsy if a dark lesion could be melanoma.
- **FABRY**
  α-Gal A enzyme activity in males; GLA gene testing (needed in females); plasma lyso-Gb3. Urine protein, eGFR, ECG/echo.

Dermoscopy

- Red-blue **lacunae**, whitish veil, haemorrhagic crusts.

Biopsy

- If a dark lesion could be melanoma.

Fabry tests

- α-Gal A activity (males); **GLA gene test** (needed in females); plasma lyso-Gb3; urine protein, eGFR, ECG / echo.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Melanoma / pigmented BCC	Dark solitary lesion.
	Cherry angioma	Bright red, dome, older.
	Venous lake	Lip, blanches.
	Kaposi sarcoma	Violaceous plaques; HIV.
	Pyogenic granuloma	Fast, bleeds.
	Viral wart	Hands; capillary dots.

Melanoma or pigmented BCC (dark solitary lesion), cherry angioma, haemangioma, venous lake, Kaposi sarcoma, warts (on the hands), pyogenic granuloma, and lymphangioma circumscriptum.

Management

- Benign isolated angiokeratomas: reassure. Treatment is cosmetic (laser, cautery, excision; specialist/not NHS-funded).
- **Uncertain dark lesion refer** (dermoscopy/biopsy). Use NG12 urgency if the 7-point score is 3.
- **Multiple angiokeratomas in a bathing-trunk distribution, or with neuropathic pain, sweating problems, proteinuria or early cardiac/stroke history refer** to dermatology/GP for **Fabry testing**. Diagnosis and treatment are at a **national lysosomal storage disorder centre** (enzyme replacement with agalsidase alfa/beta, pegunigalsidase alfa; or oral migalastat for amenable mutations). Cascade-screen family members.
- Neuropathic pain in Fabry: specialist-guided (carbamazepine, gabapentinoids). Avoid triggers.

Treatment ladder

1

Benign

Reassure isolated angiokeratomas; treatment is cosmetic (laser, cautery, excision; not NHS-funded).

2

Doubt

Uncertain dark lesion refer (dermoscopy / biopsy); NG12 urgency if 7-point 3.

3

Fabry

Bathing-trunk angiokeratomas, neuropathic pain, sweating problems, proteinuria or early cardiac / stroke history **Fabry testing**. National lysosomal storage centre: enzyme replacement (agalsidase alfa / beta, pegunigalsidase alfa) or oral **migalastat** (amenable mutations); cascade-screen family.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Migalastat**: Every other day; no food 2 h before and after.

Referral

ROUTINE — Many angiokeratomas in a child / young adult with burning hand / foot pain, reduced sweating or proteinuria: test for **Fabry**.

URGENT Solitary changing dark papule you can't confidently call an angiokeratoma: **possible melanoma**.

Red flags and when to refer

- Many angiokeratomas in a child or young adult with burning hand/foot pain, reduced sweating or proteinuria: test for **Fabry disease**.
- Solitary changing dark papule: if you cannot confidently call it an angiokeratoma, refer as a **possible melanoma**.

Complications and prognosis

Complications

- Bleeding; in Fabry, renal failure, cardiomyopathy and stroke.

Prognosis

- Benign; Fabry is treatable with enzyme replacement or migalastat.

Prescriber pearls (UK pharmacist IP)

Fordyce angiokeratomas bleeding on the scrotum are benign, but **always ask about other symptoms** (the patient is often embarrassed).

Migalastat is taken every other day, with no food 2 h before and after.

Self-check quiz

4 questions · optional

Q1 Inheritance and enzyme in Fabry?

✓ X-linked; α-galactosidase A (GLA).

Q2 Distribution suggesting Fabry?

✓ Bathing-trunk (umbilicus to knees).

Q3 Test needed in females?

✓ GLA gene testing (enzyme may be normal).

Q4 Migalastat food rule?

✓ No food 2 hours before and after; every other day.

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dermnetnz.org
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emedicine.medscape.com

Photo gallery

Angiokeratomas 4

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Lumps, lesions & skin cancer Module 2 · Week 1

Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more

A quick guide to lumps you will meet less often: neurofibromas and NF1, pilomatricoma in children, digital myxoid cysts, sebaceous hyperplasia and other look-alikes.

ALSO CALLED neurofibroma · NF1 · plexiform neurofibroma · pilomatricoma · digital myxoid cyst · sebaceous hyperplasia · +1

RELATED PAGES Epidermoid, pilar and dermoid cysts (and Gardner syndrome) · Lipoma & the sarcoma lump pathway · Dermatofibroma · Dermatofibrosarcoma protuberans (DFSP) · Pigmented & epidermal birthmarks

KEY FACTS

- Lumps here: **neurofibroma** (soft, 'buttonhole' sign; many = NF1), **pilomatricoma** (rock-hard, children), **digital myxoid cyst** (near the nail, osteoarthritis), sebaceous hyperplasia, connective tissue naevi and others.
- Six or more café-au-lait macules 5 mm (prepubertal) or 15 mm (adults) plus neurofibromas or freckling: **NF1**.
- Rapid growth, pain, size over 5 cm or deep location: think sarcoma, refer urgently.
- Most are benign: reassure, excise if symptomatic or uncertain.
- Use ultrasound when the depth or nature is unclear.

Overview

This page is a **compendium of less common lumps** (each covered in a short section below), rather than one disease.

Aetiology: causes and risk factors

Myths to address

- **"A HARD LUMP IS ALWAYS CANCER"**
  pilomatricoma is rock-hard but benign.

Clinical features

Lump by lump

Each card is a compressed 8-heading entry.

Neurofibroma and plexiform neurofibroma

> **WHAT/CAUSE**

a benign peripheral nerve sheath tumour. A solitary sporadic lesion is common; multiple lesions suggest **NF1** (NF1 gene, neurofibromin, RAS-MAPK, autosomal dominant, about 50% de novo).

CLINICAL

soft, skin-coloured to pinkish papule or pedunculated nodule that **invaginates on pressure (buttonhole sign)**. **Plexiform neurofibroma** is congenital or early, diffuse, "bag of worms", with overlying hyperpigmentation or hypertrichosis, and it can deform tissue.

NF1 CRITERIA (REVISED 2021)

two or more of the following: 6 café-au-lait macules (>5 mm before puberty, >15 mm after); 2 neurofibromas or 1 plexiform; axillary/inguinal freckling; optic pathway glioma; 2 Lisch nodules or choroidal abnormalities; distinctive bony lesion; pathogenic NF1 variant; an affected parent

RED FLAGS

a plexiform neurofibroma that becomes **painful, rapidly growing, hard, or causes neurological deficit** suggests **malignant peripheral nerve sheath tumour (MPNST; lifetime risk about 8–13% in NF1)**. Urgent referral to an NF1 specialist centre or sarcoma service. NF1 also raises breast cancer risk (screening from 40 in the UK).

TREATMENT

surgery or laser for symptomatic cutaneous lesions. **Selumetinib** (MEK inhibitor) for symptomatic inoperable plexiform neurofibromas in children 3 years (NICE HST20, 2022), specialist-led.

IP ROLE

recognise the NF1 pattern and refer to the national NF1 complex service or genetics. Selumetinib interactions (CYP3A4/2C19 inhibitors; vitamin E content of capsules with anticoagulants).

DermNet: dermnetnz.org

Pilomatricoma (pilomatrixoma)

WHAT/CAUSE

a benign tumour of hair matrix cells. **CTNNB1 (β-catenin)** activating mutations. Histology shows basaloid cells maturing into **ghost (shadow) cells** with calcification.

CLINICAL

children and teens (a second peak in older adults); head, neck and upper limbs. A **rock-hard**, irregular, angulated nodule, sometimes bluish. The "tent sign" (stretching the skin shows facets). It may become inflamed or extrude chalky material.

DIAGNOSIS/DDX

clinical ± ultrasound (calcified). DDx: epidermoid or dermoid cyst, calcinosis cutis. **Multiple pilomatricomas** may indicate myotonic dystrophy, Gardner, Turner, Rubinstein–Taybi or Sotos.

MANAGEMENT

routine referral for **surgical excision** (it does not regress; recurrence is rare after complete removal). Pilomatrix carcinoma is very rare, in adults.

DermNet: dermnetnz.org

Digital myxoid (mucous) cyst

WHAT/CAUSE

a ganglion-like cyst of the distal finger (sometimes toe), usually over or near the **DIP joint**, connected to the joint in many cases. Associated with **osteoarthritis** and Heberden's nodes.

CLINICAL

> a translucent dome-shaped papule between the DIP joint and the nail fold. A **longitudinal nail groove** comes from pressure on the matrix. Clear, sticky jelly if punctured.

> **DDX**
> wart, giant cell tumour of tendon sheath, fibrokeratoma, SCC/Bowen's of the nail fold.

> **MANAGEMENT**
> reassure; many resolve. **Advise patients not to puncture the cyst themselves** because of infection and **septic arthritis** risk. Options (GP/hand surgery/dermatology) include repeated sterile needling, steroid injection, cryotherapy, sclerotherapy, or surgery ligating the joint connection (lowest recurrence). For a red, hot, painful joint, same-day review.

> DermNet: dermnetnz.org

Sebaceous hyperplasia

- Yellowish, 2–4 mm, **umbilicated** papules on the forehead and cheeks of middle-aged and older adults. Common in transplant recipients on **ciclosporin**.
- Dermoscopy: yellow lobules with **"crown" vessels** that do not cross the centre (vs BCC arborising vessels).
- **Key DDx: BCC** (pearly, translucent, ulcerating, solitary, enlarging; refer if in doubt). Multiple sebaceous adenomas (not hyperplasia) suggest **Muir–Torre syndrome** (Lynch).
- Treatment is cosmetic only (not NHS-funded). Low-dose isotretinoin is used privately (specialist; pregnancy prevention programme applies).
- DermNet: dermnetnz.org

Connective tissue naevi

Hamartomas of collagen (collagenoma) or elastin (elastoma): skin-coloured to yellowish plaques or papules, often on the lower back.

> **Shagreen patch** (lumbosacral, orange-peel) is a major criterion for **tuberous sclerosis complex** (with hypomelanotic macules, facial angiofibromas, periungual fibromas, forehead plaque). Refer for TSC assessment.

> **BUSCHKE–OLLENDORFF SYNDROME**
> elastomas plus osteopoikilosis (LEMD3).

> **PROTEUS SYNDROME**
> cerebriform plantar collagenoma.

> **MANAGEMENT**
> benign. Refer to look for the syndrome when there are associated signs.

> DermNet: dermnetnz.org

Aplasia cutis congenita

- Congenital absence of skin, most often a **solitary vertex scalp defect**. It may be a healed atrophic, hairless scar, a membranous "bullous" form, or an ulcer at birth.
- **"Hair collar sign"** (a ring of long dark hair around the defect) suggests neural tube or ectopic neural tissue: **imaging before any procedure**.
- Associations: Adams–Oliver syndrome (limb defects); Bart syndrome (with epidermolysis bullosa); trisomy 13; fetus papyraceus.
- **PHARMACIST PEARL**

maternal **carbimazole/methimazole** exposure in early pregnancy is linked to scalp aplasia cutis (part of carbimazole embryopathy, MHRA 2019 advice on contraception in women of childbearing potential). Misoprostol and valproate have also been reported.

- **MANAGEMENT**
  gentle wound care (moist dressings) for small defects; most heal with a scar. Large or deep defects need specialist care (risk of haemorrhage or infection of the sagittal sinus). Scar revision later. More in the infant skin pages notes W1.
- DermNet: dermnetnz.org

Linear morphoea en coup de sabre

- Linear localised scleroderma of the frontoparietal scalp/forehead: an **indurated, ivory or hyperpigmented band with hair loss** that becomes a depressed groove, like a sabre cut. It usually starts in childhood shows linear morphoea elsewhere; no captioned en coup de sabre image was available on DermNet).
- Overlaps with **Parry–Romberg syndrome** (progressive hemifacial atrophy). It may have **neurological** (seizures, headaches, MRI changes) and **ocular** (uveitis) involvement.
- **KEY DIFFERENCE FROM SEBACEOUS NAEVUS**
  it is acquired, progressive, indurated rather than waxy-yellow, and causes tissue loss.
- **MANAGEMENT**
  prompt referral (paediatric dermatology/rheumatology). Active disease is treated with systemic **methotrexate ± corticosteroids** to prevent permanent atrophy (BAD/ PRES consensus). Ophthalmology screening; neurology if there are symptoms.
- DermNet: dermnetnz.org

Birt–Hogg–Dubé syndrome

- Autosomal dominant; **FLCN** (folliculin).
- Multiple small skin-coloured papules on the face, neck and upper trunk from about age 20–40 (**fibrofolliculomas**, trichodiscomas, acrochordons).
- **Lung cysts spontaneous pneumothorax**, and **renal tumours** (chromophobe/hybrid oncocytic; bilateral, multifocal).
- **ACTION**
  refer to clinical genetics. Renal **MRI surveillance** (UK practice: every 1–2 years from about age 20). Advise against smoking. Seek urgent review for sudden breathlessness or chest pain.
- DermNet: dermnetnz.org

PTEN hamartoma tumour syndrome: Bannayan–Riley–Ruvalcaba

- BRRS and Cowden syndrome are a spectrum caused by **PTEN** mutations.
- **BRRS** (usually diagnosed in childhood): macrocephaly, multiple **lipomas**, haemangiomas or vascular malformations, **pigmented macules on the glans penis**, intestinal hamartomatous polyps, and developmental delay.
- **COWDEN**
  facial trichilemmomas, acral keratoses, oral papillomatosis, macrocephaly. Raised risk of **breast, thyroid, endometrial, kidney and colorectal cancers**.
- **ACTION**

genetics referral and cancer surveillance. Melanoma risk is also modestly raised (see the melanoma notes).

- DermNet: dermnetnz.org

Eponymous signs

SIGN	ELICIT	POSITIVE IN
Dimple / Fitzpatrick	Lateral pinch → central depression	Dermatofibroma
Buttonhole	Press → nodule invaginates	Neurofibroma
Darier	Rub lesion → wheal/flare within minutes	Mastocytosis (urticaria pigmentosa, mastocytoma)
Nikolsky	Lateral pressure → epidermis shears off	Pemphigus vulgaris, SSSS, TEN (not BP)
Asboe-Hansen	Pressure on bulla → it spreads laterally	Pemphigus
Leser–Trélat	Sudden eruption of many SKs	Possible internal malignancy (often GI adenocarcinoma). Contentious; see the benign lesion notes
Tent sign	Stretch skin → angulated facets	Pilomatricoma

Fig. 1 · Soft, skin-coloured neurofibroma with the 'buttonhole' sign (invaginates on pressure). Solitary lesions are common and not necessarily NF1.

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Management

Referral

URGENT — Rapid growth or pain in a plexiform neurofibroma: refer urgently (malignant peripheral nerve sheath tumour).

ROUTINE Six or more café-au-lait macules in a child: refer to assess for **NF1**.

Red flags and when to refer

- Rapid growth or pain in a plexiform neurofibroma: refer urgently (malignant peripheral nerve sheath tumour).
- Six or more café-au-lait macules in a child: refer to assess for **NF1**.

Complications and prognosis

Complications

- Malignant peripheral nerve sheath tumour in plexiform neurofibromas (NF1); recurrence after excision.

Prognosis

- Most are benign and persist.

Self-check quiz

5 questions · optional

Q1 When should you think of Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more?

✓ A soft 'button-hole' papule (neurofibroma), a hard chalky lump on a child's face (pilomatricoma), a cyst by the nail (myxoid), yellow umbilicated papules on the face (sebaceous hyperplasia).

Q2 What drives it?

✓ Different tissues of origin: nerve sheath, hair matrix, joint-related mucin, enlarged sebaceous glands.

Q3 What is the backbone of care?

✓ Recognise and reassure where typical; refer when uncertain, growing or syndromic.

Q4 What must you never miss?

✓ NF1 (6 café-au-lait macules, axillary freckling), plexiform neurofibroma change, BCC mimicking sebaceous hyperplasia.

Q5 Name a red flag that needs escalation.

✓ Rapid growth or pain in a plexiform neurofibroma: refer urgently (malignant peripheral nerve sheath tumour).

Photo gallery

Neurofibromas 2

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Other lumps 4

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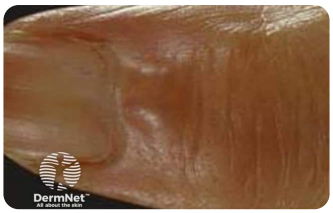

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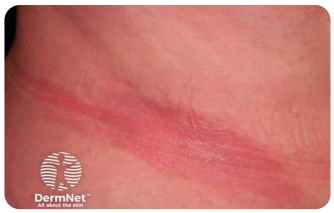

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Lumps, lesions & skin cancer Module 2 · Week 2

Chondrodermatitis nodularis helicis & ear nodules

A painful, tender nodule on the rim of the ear that stops people sleeping on that side. Relieve pressure first; know the ear-nodule differentials (BCC, SCC, keratoacanthoma, gouty tophus, AK).

ALSO CALLED CNH · painful ear nodule · Winkler's disease · pressure-relief pillow · gouty tophus · keratoacanthoma

RELATED PAGES Cutaneous squamous cell carcinoma (cSCC) · Basal cell carcinoma (BCC) · Actinic keratoses (solar keratoses)

KEY FACTS

- **Chondrodermatitis nodularis helicis (CNH)** is a painful nodule on the ear rim (helix or antihelix) from pressure.
- Typically older men; painful when lying on that side at night; central crust or scale.
- Must be distinguished from **SCC** and BCC: biopsy if atypical.
- Pressure relief is key: a pillow with a hole or ear protector; potent or intralesional steroid; surgery if persistent.
- Recurs if pressure continues.

Overview

A benign, **painful**, inflamed nodule of the ear cartilage and overlying skin. It is commonest on the **helix in older men** and on the **antihelix in women**.

Ear nodule differentials

The pinna is a **high-risk site for skin cancer**: it is sun-exposed and thin-skinned, and spread to the parotid and cervical nodes happens early. BAD 2020 classifies **ear cSCC as high-risk by site**.

Aetiology: causes and risk factors

Pressure-induced ischaemia. Ear cartilage has little subcutaneous padding and a poor blood supply. Sustained pressure (sleeping on one side, headsets, phones, hats), cold or sun damage causes perichondrial inflammation and focal cartilage necrosis that extrudes through the skin. **One-sided sleeping is typical, e.g. after a stroke, or with arthritis or CPAP masks.** In Case 2 with a previous right-sided stroke, ask which side she lies on.

Ear nodule differentials

Chronic UV (helix, especially in men with short hair), immunosuppression, and for gout, urate deposition in cool tissue.

Myths to address

- **"AN EAR NODULE IS JUST A WART"**
  painful helical nodules are usually CNH, but SCC must be excluded.

Pathophysiology and pathology

CNH is **pressure ischaemia** of thin skin over cartilage.

1. **Pressure**
   Sleeping on the ear compresses skin against cartilage with a poor blood supply.
2. **Ischaemia**
   Skin and perichondrium are damaged; cartilage degenerates.
3. **Transepidermal elimination**
   Degenerate collagen is pushed out through a central ulcer or crust.
4. **Relief**
   Removing pressure lets it heal.

Fig. 1 · A cut-out pillow or ear protector relieves pressure.
AI-generated illustration

HISTOPATHOLOGY

- Central ulcer with degenerate collagen and perichondrial fibrosis; acanthotic edges.

Clinical features

- A firm, round, 4–10 mm nodule with a **central crust, scale or keratin plug**, sometimes ulcerated.
- **EXQUISITELY TENDER**
  it wakes the patient at night when they lie on it, and **light squeezing causes disproportionate pain** (the course bedside test). Usually on the side the patient sleeps on.

Ear nodule differentials

DIAGNOSIS	CLUES	PHOTO
CNH	Very tender, central crust, the side slept on	
cSCC	Indurated, keratotic or ulcerated, growing over weeks to months, bleeds. Often painless or mildly tender. ± palpable pre-auricular/cervical node	
Keratoacanthoma	Rapid growth over **weeks**, a dome with a **central keratin crater**, may involute	
BCC	Pearly, rolled edge, telangiectasia, slowly growing; often posterior ear or sulcus	

DIAGNOSIS	CLUES	PHOTO
Actinic keratosis	Rough, flat-to-slightly-raised scale; not nodular or tender	see Worked cases
Gouty tophi	Firm, **chalky white-yellow** papules on the helix/antihelix, may discharge, ± a history of gout, diuretics (e.g. thiazides)	
Weathering nodules / elastotic nodules	Small, white, painless, bilateral helical-rim papules	–
Cutaneous horn	Keratin projection; the **base** may be SCC	

Fig. 2 · Tender crusted nodule on the helical rim. Exquisite tenderness on light squeezing is the bedside test; ask about sleeping side (post-stroke immobility).

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Types to recognise

Helical CNH

Tender crusted rim nodule. Site: Helix.

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With erythema

Upper helix. Site: Helix.

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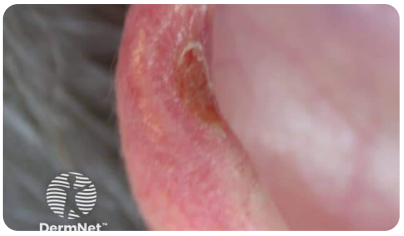
Antihelix CNH

Commoner in women. Site: Antihelix.



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Gouty tophi

Chalky white-yellow. Site: Ear.

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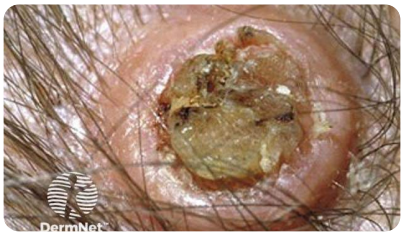
Keratoacanthoma

Rapid dome with crater. Site: Ear.



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Diagnosis and investigations

- Clinical: site + pain + central crust + sleeping history.
- Dermoscopy: central white-yellow keratin/ulcer with a radiating whitish "starburst" and peripheral dotted or linear vessels. It is **non-specific** and **cannot reliably exclude SCC/BCC**.
- **BIOPSY WHEN**
  atypical features (rapid growth, bleeding, induration, a size >1 cm, a rolled pearly edge, no pain); no improvement after **6–8 weeks of strict offloading**; immunosuppression; or patient anxiety.

Ear nodule differentials

Pain pattern, growth tempo, and palpation of regional nodes. Biopsy anything unexplained. Serum urate (when not in an acute flare) if tophi are suspected.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	cSCC	Indurated, keratotic / ulcerated, growing; often painless ± node.
	BCC	Pearly rolled edge.
	Keratoacanthoma	Rapid weeks, central keratin crater.
	AK	Rough scale, flat.

	CONDITION	DISTINGUISHING FEATURES
	Gouty tophi	Chalky, multiple, gout history.
	Chilblain / relapsing polychondritis	Cold, bilateral / whole ear red sparing lobe.

See Ear nodule differentials (SCC, BCC, keratoacanthoma, AK, gouty tophi, also cutaneous horn, wart, elastotic nodules of the ear, weathering nodules, chilblain).

Ear nodule differentials

As above. Also chilblain (cold, bilateral, purple), relapsing polychondritis (the whole ear red, sparing the lobe), and discoid lupus.

Management

Step 1: pressure relief, for everyone (first-line; the course answer)

- A **pressure-relieving device**: a CNH prosthesis/"ear protector" made of foam with a hole for the ear (sold OTC and online), a pillow with a hole, or a travel pillow; sleep on the other side. Remove headphone and phone pressure.
- Resolution or major improvement occurs in a large proportion of patients within weeks to months. Recurrence is common if offloading stops.
- Analgesia as needed. Emollient/protective dressing over the crust.

Step 2: second-line (GP, GPwSI or dermatology)

OPTION	NOTES
Intralesional triamcinolone	Specialist/trained GP. Risk of atrophy
Potent topical steroid ± occlusion	Low-quality evidence. IP-prescribable (e.g. betamethasone valerate 0.1%) for a short trial **only once SCC/BCC has been reasonably excluded**
Topical nitroglycerin (GTN) 2% gel	Off-label. Small studies show benefit via vasodilatation. Headache and hypotension. **Check PDE-5 inhibitor co-prescription** (contraindicated). Not routinely available as a dermatology product in the UK
Cryotherapy	Painful, variable results
Surgical excision of the abnormal cartilage ± skin	For refractory cases. Recurrence 10–30%

Refer: NG12 1.7.4 says to consider a suspected cancer pathway referral if SCC is suspected (a growing, indurated, bleeding, non-tender or atypical ear nodule). Refer routinely if CNH is not settling after 6–8 weeks of offloading.

Ear nodule differentials

- **SUSPECTED SCC OR KERATOACANTHOMA**

NG12 1.7.4, consider a **suspected cancer pathway referral**. BAD guidance is to **treat KA as SCC** (excise) because they cannot be reliably distinguished clinically.

- Suspected BCC: NG12 1.7.5, routine referral, unless the site or size makes delay important (1.7.6). The ear is a high-risk "H-zone" site.
- Gouty tophi: a GP/IP gout pathway. **Urate-lowering therapy (allopurinol, treat to target <360 µmol/L, or <300 if tophaceous)**, per NICE NG219 (gout, 2022). Review diuretics.
- AK: cross-reference the older skin pages notes see Worked cases (field therapy).

Treatment ladder

1 **Pressure relief**

Foam ear protector (OTC / online), pillow with a hole or travel pillow; sleep on the other side; remove headset / phone pressure. At least a month; recurrence common if offloading stops.

2 **Second line**

Potent topical steroid; intralesional triamcinolone (trained; atrophy risk) via GP / GPwSI / dermatology.

3 **Refer**

Failing or atypical: biopsy / surgery (cartilage excision).

⚠ CAUTION

Prescribing limits and pitfalls

- **Aspirin**: Don't stop for minor skin surgery (BSDS 2023).

Referral

URGENT — Rapidly growing, ulcerating or bleeding ear nodule, or with a nearby node: **suspected SCC** (NG12).

ROUTINE — No improvement after 6–8 weeks of strict offloading: biopsy.

Red flags and when to refer

- Rapidly growing, ulcerating or bleeding ear nodule, or an ear nodule with a nearby lymph node: **suspected SCC**, suspected-cancer referral (NICE NG12).

Complications and prognosis

Complications

- Persistent pain and sleep disturbance; recurrence.

Prognosis

- Good with pressure relief; recurrence common.

Prescriber pearls (UK pharmacist IP)

Case 2 medicines: aspirin is relevant only if a biopsy or excision is done. **Do not stop aspirin for minor skin surgery** (BSDS 2023; see the skin surgery notes). Losartan and simvastatin are not causative.

Explain the mechanism ("a pressure sore of the ear cartilage"). Adherence to offloading for at least a month is the key to success.

Painless ear nodule plus rapid growth means think SCC/KA. **Very painful at night** means think CNH. But both can coexist, which is why non-resolving CNH gets a biopsy.

Check transplant or immunosuppressant history (azathioprine, ciclosporin, tacrolimus): cSCC risk rises up to 65–100× in organ transplant recipients.

Ear nodule differentials

Self-check quiz

4 questions · optional

Q1 Typical history?

✓ Painful ear nodule waking them when lying on that side.

Q2 First-line treatment?

✓ Pressure relief (ear protector / pillow with a hole) for 4–6+ weeks.

Q3 When to biopsy?

✓ Atypical features or no improvement after 6–8 weeks.

Q4 Painless rapidly growing crater on the ear?

✓ SCC / keratoacanthoma: suspected cancer pathway.

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Photo gallery

Chondrodermatitis nodularis helicis 3

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Ear nodule differentials 2

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Lumps, lesions & skin cancer Module 2 · Week 2

Seborrhoeic keratosis

The commonest benign 'stuck-on' warty lesion of later life. Learn the dermoscopy clues (milia-like cysts, comedo openings, fissures), when an SK needs a second look, and the sign of eruptive SKs.

ALSO CALLED seborrhoeic wart · basal cell papilloma · stuck-on lesion · Leser–Trélat sign · SK dermoscopy · milia-like cysts

RELATED PAGES Actinic keratoses (solar keratoses) · Melanoma: recognition & referral · Solar lentigines · Melanocytic naevi (moles) & the changing mole

KEY FACTS

- Very common **benign** epidermal tumour: waxy, warty, **'stuck-on'** papules and plaques, tan to black.
- Over **90% of adults over 60** have at least one; usually start in the 30s–40s.
- Not premalignant; caused by clonal mutations (e.g. **FGFR3, PIK3CA**) in keratinocytes, not sebum or viruses.
- Dermoscopy: **milia-like cysts**, comedo-like openings, 'brain-like' ridges and sharp edges.
- Reassure if typical; refer anything atypical (melanoma mimic). Sudden eruption of many SKs (**Leser–Trélat**) may signal internal malignancy.

Overview

Core SK content (definition, cause, sites, basic management) is in the older skin pages notes and is not repeated here. This section adds dermoscopy, "the SK that changes", irritated SK and Leser–Trélat.

A benign epidermal tumour that is almost universal with age, often multiple, and the commonest "changing lesion" seen in primary care.

Hutchinson and Albright signs

Eponymous signs often mixed up in exams.

Epidemiology

- **Who**
  - Extremely common in older adults of all ethnic groups; uncommon under 20.
  - Numbers increase with age; a strong family tendency in some people.
- **Variant in skin of colour**
  - **DERMATOSIS PAPULOSA NIGRA**
    many small dark papules on the face and neck, especially in people of African or Asian descent.

Aetiology: causes and risk factors

Activating **FGFR3/PIK3CA** mutations in keratinocytes; it is a clonal benign neoplasm, not a pre-cancer. Lesions thicken, darken, catch on clothing, become **irritated/inflamed** and itch. That is exactly the "changing, itchy" story in Case 3, and the same story fits melanoma.

Risk factors and associations

- Clonal somatic mutations in **FGFR3, PIK3CA, RAS, AKT1, EGFR**.
- Sun exposure may contribute (more on exposed sites); friction in body folds.
- Eruptive SKs may follow inflammatory dermatoses or sunburn; **Leser–Trélat sign** with internal cancer is rare.

Myths to address

> **"IT'S FROM GREASY SKIN"**
> the name is misleading; SKs are not related to sebum or sebaceous glands.

> **"THEY ARE WARTS AND SPREAD"**
> they are not viral or contagious.

Pathophysiology and pathology

An SK is a **benign clonal overgrowth** of epidermal keratinocytes that sits on top of the skin.

1. **Clonal mutation**
   Keratinocytes acquire activating mutations (often FGFR3 or PIK3CA) that drive orderly but excessive growth.

2. **Upward growth**
   The thickened epidermis grows **outwards above the skin surface**, so the lesion looks stuck on and can almost be lifted off.

Fig. 1 · A seborrhoeic keratosis sitting on the skin surface, with horn cysts within it.
AI-generated illustration

3. **Horn cysts and pigment**
   Keratin collects in small cysts (milia-like cysts on dermoscopy); melanin transferred from melanocytes darkens many lesions.

4. **Irritation**
   Friction or trauma can inflame an SK, making it red, crusted or bleeding, which can mimic cancer.

HISTOPATHOLOGY

- Acanthosis of basaloid keratinocytes with **horn pseudocysts**, hyperkeratosis and papillomatosis; the base is flat (the 'string sign').
- Variants: acanthotic, hyperkeratotic, reticulated, clonal, irritated. Biopsy if any doubt.

Clinical features

"Stuck-on", waxy, verrucous, fissured papules and plaques, tan to black, on the trunk, face and scalp.

IRRITATED SK

red rim, crust and itch. This mimics SCC or melanoma.

SIGN OF LESER–TRÉLAT

sudden eruption of many SKs (often with acanthosis nigricans or itch). It has been linked to internal malignancy (gastric/colorectal adenocarcinoma, lymphoma), but **the evidence is weak and contested**, because SKs are so common. Look for systemic symptoms rather than labelling everyone. Eruptive SKs also occur with **drugs** (e.g. cytarabine, docetaxel, BRAF inhibitors) and after inflammatory dermatoses (erythroderma).

Hutchinson and Albright signs

How they arise / the signs

SIGN	MEANING
Hutchinson's sign (nail)	**Periungual pigment spreading onto the proximal or lateral nail fold** from a longitudinal melanonychia. This suggests **subungual (acral) melanoma**. Cross-reference the nail disorders notes (melanonychia) and the melanoma notes
Pseudo-Hutchinson's sign	Pigment visible through a transparent cuticle (benign naevus, ethnic pigmentation, Laugier–Hunziker, drugs such as zidovudine, minocycline, hydroxyurea, chemotherapy)
Hutchinson's sign (herpes zoster)	Vesicles on the **tip/side of the nose** (nasociliary branch of V1) predict ophthalmic involvement, so same-day ophthalmology
Hutchinson's triad	Congenital syphilis: notched incisors, interstitial keratitis, deafness
Albright's sign (pseudohypoparathyroidism / Albright hereditary osteodystrophy)	Short 4th/5th metacarpals ("knuckle-knuckle-dimple-dimple" when making a fist); osteoma cutis
McCune–Albright syndrome	Large unilateral **"coast of Maine"** café-au-lait macules (jagged), polyostotic fibrous dysplasia, precocious puberty (GNAS mosaicism). NF1 CALMs are smooth, like the "coast of California"

Fig. 2 · Classic waxy 'stuck-on' papule with a warty surface. This is the reassuring look.
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Types to recognise

Classic stuck-on

Waxy warty papule. Site: Trunk.

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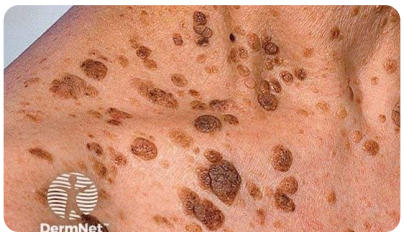
Many SKs

Older back. Site: Back.



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Irritated SK

Red, crusted. Site: Trunk.

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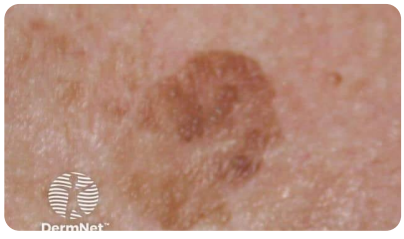
Flat tan SK

Like a solar lentigo. Site: Face.



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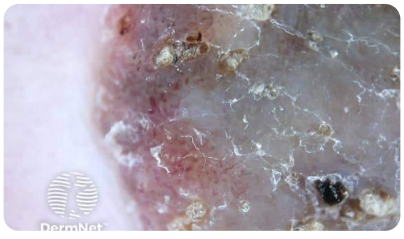
Milia-like cysts

Dermoscopy. Site: Dermoscopy.



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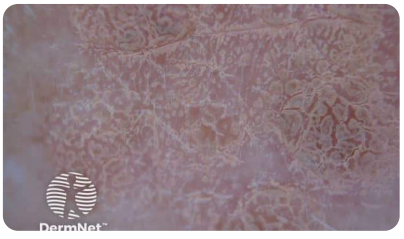
Brain-like ridges

Dermoscopy. Site: Dermoscopy.



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Special groups

- **Skin of colour**
  - Dermatosis papulosa nigra is common and benign; treat only for cosmetic reasons, with care to avoid pigment change.

Diagnosis and investigations

Key features:

- **Milia-like cysts** (white/yellow "stars" or round bodies)
- **Comedo-like openings** (keratin-filled crypts)
- **Fissures and ridges** ("brain-like"/cerebriform, "fat fingers")
- **Sharp, "moth-eaten" border**
- **Hairpin vessels** with a white halo, in irritated or thick lesions
- **CAVEAT**
  some SKs show multiple colours or blue-black areas, and **"collision" lesions** (SK beside a melanoma or BCC) exist. **Melanoma can mimic SK**, especially in men's backs, like Case 3 sailor.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Melanoma	Evolving asymmetry, atypical network, blue-white veil; no cysts or openings.
	Pigmented BCC	Pearly, leaf-like areas, arborising vessels.
	Solar lentigo	Flat, fingerprint pattern.
	Viral wart	Thrombosed capillaries; younger.
	SCC / Bowen's	Glomerular vessels, tender.
	Clear cell acanthoma	'String of pearls' vessels.

Clinical relevance, diagnosis and differentials

A **single nail band that is new, widening (>3 mm), dark, irregular, or in an adult aged over 50, especially of the thumb or great toe, or with Hutchinson's sign** needs urgent dermatology (suspected cancer pathway). Skin of colour often has multiple benign bands. A single changing band is the concern.

Management

- **Apply the NICE NG12 weighted 7-point checklist to any changing lesion:** Major features (2 points each): change in size, irregular shape, irregular colour; Minor features (1 point each): largest diameter 7 mm, inflammation, oozing, change in sensation (e.g. itch); Case 3: "changing" (2) + **itchy** (1) = **3 suspected cancer pathway**. This is unless the lesion is confidently a classic SK on dermoscopy by a trained clinician (see Where course teaching differs from UK guidance)
- Even when the score is below 3, refer if there is **any** doubt. NG12 1.7.2 says dermoscopy suggesting melanoma means a suspected cancer referral.
- **CONFIDENT SK**
  reassure; leaflet. Itch: emollient ± a short course of a mild/moderate topical steroid for irritation.
- Removal for cosmetic reasons is usually **not NHS-funded**. Cryotherapy, curettage or shave are used where symptomatic, with histology for curettage/shave if any doubt. **Never destroy (cryo) a lesion that might be melanoma.**
- Topical **hydrogen peroxide 40%** (licensed in the US, not marketed in the UK) and other cosmetic options are not IP options.

Hutchinson and Albright signs

Do not label nail pigmentation as fungal or traumatic without evidence. Haematomas grow out distally (mark the nail and re-check in 4–6 weeks), whereas melanoma bands remain at the matrix. Refer on the suspected cancer pathway when Hutchinson's sign is present.

Worked cases

CASE 1

50-year-old Asian woman, 5 mm pigmented arm papule, slow growth over 5 years

> **DIAGNOSIS**
> (pigmented) dermatofibroma.
>
> **BEDSIDE SIGN**
> dimple / Fitzpatrick (pinch) sign. Dermoscopy: peripheral network with a central white patch.
>
> **IMMUNOSTAIN**
> **Factor XIIIa positive** (CD34 negative). DFSP is the reverse.
>
> **IP PLAN**
> reassure if classic. Refer if dermoscopy is atypical or the 7-point score is 3 (growth = 2 points, so one more feature tips it).

CASE 2

68-year-old woman, tender scaly 7×8 mm pinna nodule, post-stroke, losartan/aspirin/simvastatin

BEDSIDE TEST

gentle squeeze, which causes disproportionate tenderness. Also ask about the sleeping side.

DIAGNOSIS

chondrodermatitis nodularis helicis.

FIRST-LINE TREATMENT

pressure relief (CNH foam protector/pillow with a hole; avoid sleeping on that side).

SECOND-LINE

intralesional or potent topical steroid, topical GTN (off-label), cryotherapy, cartilage excision.

BIOPSY IF

not resolving after 6–8 weeks of offloading, atypical features, or SCC/BCC suspected (NG12 1.7.4/1.7.5).

Do **not** stop aspirin for a biopsy.

CASE 3

55-year-old sailor, itchy back lesion changing over 6 months

BEDSIDE TEST

dermoscopy. Milia-like cysts, comedo-like openings, fissures/ridges, sharp border, and hairpin vessels mean SK.

DIAGNOSIS

seborrhoeic keratosis. **Management:** reassure and discharge.

UK IP VERSION

NG12 7-point score = change (2) + itch/sensation change (1) = **3**, which means the suspected cancer pathway unless a trained clinician has confidently made the dermoscopic SK diagnosis. An IP without dermoscopy competence should **refer**, not discharge. Also give sun-safety advice and a full skin check.

Practice case

- **CLEAR CELL ACANTHOMA**
  red, moist, collarette-rimmed leg papule; **string-of-pearls** vessels on dermoscopy.
- Reassure when typical. Cryotherapy or curettage only if symptomatic or diagnostic doubt; usually private/cosmetic on the NHS unless catching or bleeding.
- Ugly-duckling or atypical refer, do not freeze.

Treatment ladder

1

Confident SK

Reassure; leaflet; itch: emollient ± short mild steroid.

2

Doubt

Score 3, dermoscopy suggesting melanoma (NG12 1.7.2), or **any doubt**: refer.

3

Removal

Cryo / curettage only if symptomatic (catching, bleeding, inflamed); cosmetic removal usually not NHS-funded.

Referral

URGENT — An 'SK' that is changing, has atypical dermoscopy, or stands out from the patient's other SKs: **possible melanoma**.

ROUTINE — Sudden eruption of many SKs with itch: consider internal malignancy (**Leser–Trélat**).

Red flags and when to refer

- An 'SK' that is changing, has atypical dermoscopy or stands out from the patient's other SKs: refer as a **possible melanoma**.
- Sudden eruption of many SKs with itch: consider internal malignancy (**Leser–Trélat**).

Complications and prognosis

Complications

- Irritation, itch, catching on clothing and bleeding.
- **MISDIAGNOSIS**
  melanoma, pigmented BCC or SCC can look like an SK.
- Collision tumours (a cancer arising next to an SK) are rare.

Prognosis

- Benign and permanent; new lesions keep appearing with age.
- Removal (cryotherapy, curettage, shave) is usually cosmetic and not funded on the NHS unless symptomatic.

Prescriber pearls (UK pharmacist IP)

Case 3 sailor has lifelong UV exposure, so he is at higher risk for melanoma, BCC and SCC. Give sun-protection advice and do a **full skin check** (other lesions, scalp, ears).

Teach the "ugly duckling" concept and ask about change using photos (patients can photograph lesions monthly).

Ask about drugs that cause melanonychia (hydroxycarbamide, zidovudine, minocycline, antimalarials, cytotoxics). Multiple nails and a matching start date make it drug-related.

Hutchinson and Albright signs

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK GUIDANCE / PRACTICE	TEACHING POINT
Changing itchy lesion	Dermoscopy, then reassure and discharge	NG12 7-point checklist: change + itch = 3, so suspected cancer pathway unless confident. NG12 1.7.2 applies when dermoscopy suggests melanoma	**Reassurance is only safe after competent dermoscopy.** IPs without that training refer (teledermatology with dermoscopic images where available)
CNH management	Pressure relief only	PCDS/BAD: offloading is first-line; steroid, GTN, cryo and surgery are second-line; biopsy non-resolving or atypical lesions	A student list was valid as second-line (see Chondrodermatitis nodularis helicis)
Topical GTN	(A student answer)	Off-label; small trials; contraindicated with PDE-5 inhibitors	Specialist advice before using
Leser–Trélat	Presented as a marker of malignancy	Weak, contested evidence; eruptive SKs also follow drugs and inflammation	Look for systemic symptoms; don't over-investigate
Aspirin and skin biopsy	(Not discussed)	BSDS 2023: continue antiplatelets for skin surgery	–

Self-check quiz

4 questions · optional

Q1 Two dermoscopic SK clues?

✓ Milia-like cysts and comedo-like openings.

Q2 Changing itchy pigmented lesion: 7-point score?

✓ 3: suspected cancer pathway unless confidently SK.

Q3 Leser–Trélat sign?

✓ Sudden eruption of many SKs (weak malignancy link).

Q4 Is SK related to HPV or sun?

✓ No.

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Photo gallery

Seborrhoeic keratoses 7

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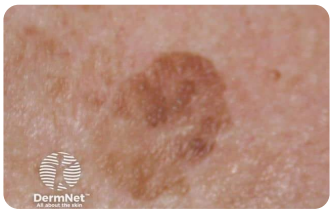

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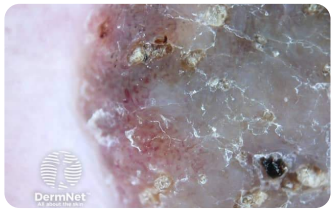
Dermoscopy of seborrhoeic keratosis

6



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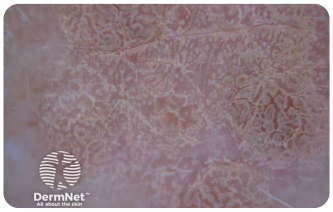

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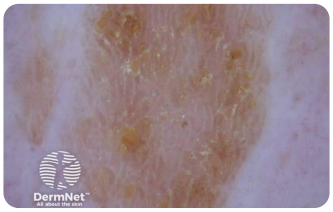

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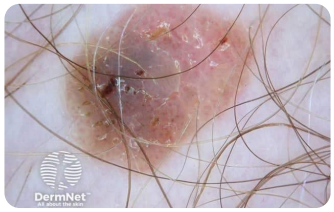

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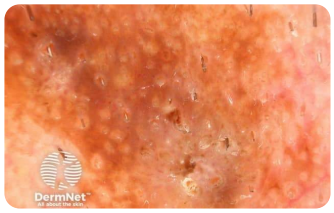

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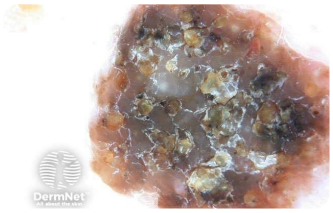

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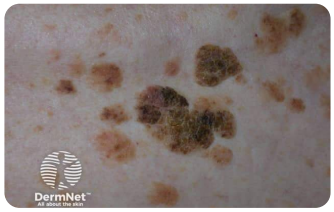
Seborrhoeic keratosis vs actinic keratosis (Module 6 Week 4)

2



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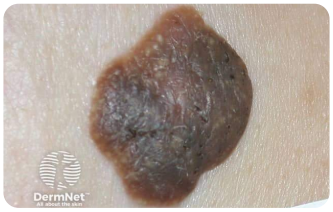

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Lumps, lesions & skin cancer Module 2 · Week 2

Clear cell acanthoma

An uncommon benign red, moist dome-shaped papule on the lower leg with a scaly collar. Dermoscopy shows dotted vessels in a ‘string of pearls’ pattern.

ALSO CALLED CCA · Degos acanthoma · string of pearls vessels · red dome papule on the leg

RELATED PAGES Pyogenic granuloma · Melanoma: recognition & referral · Basal cell carcinoma (BCC) · Seborrhoeic keratosis

KEY FACTS

- **Clear cell acanthoma** is a rare benign epidermal tumour: a moist red dome-shaped papule, usually on the leg.
- Middle-aged and older adults; slowly growing, often under 2 cm.
- Dermoscopy is characteristic: dotted vessels in a **'string of pearls'** network.
- Differentials: pyogenic granuloma, amelanotic melanoma, BCC, Bowen's disease: biopsy if unsure.
- Excision, curettage or cryotherapy if troublesome.

Overview

A rare benign epidermal tumour (Degos acanthoma) made of pale, glycogen-rich keratinocytes. Some regard it as a reactive inflammatory dermatosis rather than a true neoplasm.

Aetiology: causes and risk factors

Unknown. It often occurs on legs with venous stasis or chronic inflammation. Histology shows sharply demarcated psoriasiform acanthosis of PAS-positive pale cells, with neutrophils in the epidermis.

Myths to address

- **"RED PAPULES ON THE LEG ARE ALWAYS HARMLESS"**
  amelanotic melanoma can look similar.

Pathophysiology and pathology

A localised patch of **glycogen-rich pale keratinocytes** thickens the epidermis.

1. **Pale keratinocytes**
   Keratinocytes accumulate glycogen and look clear.
2. **Psoriasiform thickening**
   The epidermis thickens with dilated capillaries in dermal papillae.

3 **Vessel pattern**

Capillaries line up in a reticular necklace pattern.

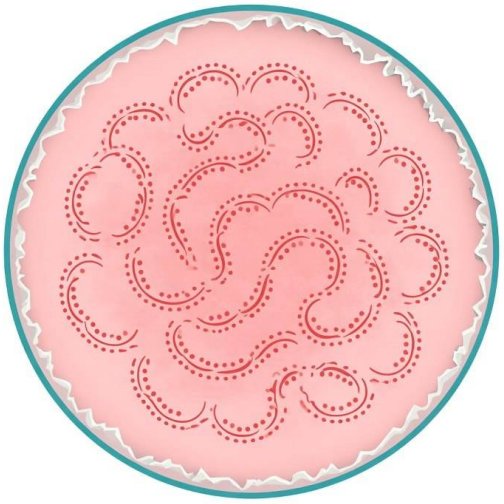

Fig. 1 · Dermoscopy: dotted vessels in a 'string of pearls'.

AI-generated illustration

HISTOPATHOLOGY

- Sharply demarcated psoriasiform acanthosis of pale, PAS-positive keratinocytes, with neutrophils.

Clinical features

A solitary, slowly growing, **moist red-brown dome-shaped papule/nodule (5–20 mm)** on the **lower leg** of a middle-aged/older adult, often with a peripheral collarette of scale. It can bleed. Multiple forms are rare.

Fig. 2 · Clear cell acanthoma: solitary moist red dome-shaped papule, usually on the lower leg of a middle-aged or older adult.

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Types to recognise

Moist red dome

Solitary, lower leg. Site: Shin.

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Collarette

Scaly rim on hairy leg. Site: Leg.

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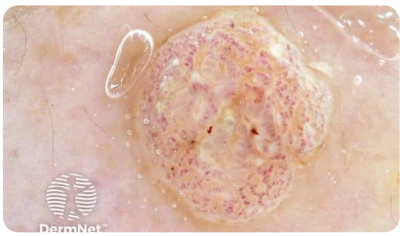
Vessels close-up

Dotted / glomerular. Site: Dermoscopy.



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Diagnosis and investigations

Dermoscopy: dotted/glomerular vessels in a **"string of pearls"** (necklace/reticular) arrangement. This is highly characteristic. Confirm with biopsy if there is doubt.

Dermoscopy

- 'String of pearls' vessels.

Biopsy

- If any doubt.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Amelanotic melanoma	Pink lesion you can't name confidently.
	Bowen's disease	Glomerular vessels in **clusters**, not strings.
	Psoriasis plaque	Usually multiple, symmetrical.
	Pyogenic granuloma	Faster, bleeds.
	BCC	Pearly, arborising vessels.

	CONDITION	DISTINGUISHING FEATURES
	Irritated SK	Cysts, openings.

Amelanotic melanoma; Bowen's disease (glomerular vessels in clusters, not strings); psoriasis plaque; pyogenic granuloma; BCC; dermatofibroma; irritated SK; eccrine poroma.

Management

Refer routinely to confirm (the differential includes amelanotic melanoma and Bowen's). Treatment (specialist/GP surgeon): excision, curettage, cryotherapy. Recurrence is uncommon.

Treatment ladder

1

Refer

Routine referral to confirm (differential includes amelanotic melanoma and Bowen's).

2

Specialist / GP surgeon

Excision, curettage or cryotherapy; recurrence uncommon.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Don't escalate steroid**: A solitary red scaly leg papule not responding to emollient / steroid isn't necessarily eczema.

Referral

URGENT — Pink lesion you **cannot confidently name**: exclude **amelanotic melanoma**.

ROUTINE — Typical CCA: refer to confirm.

Red flags and when to refer

- Pink lesion you cannot confidently name: refer, as **amelanotic melanoma** must be excluded.

Complications and prognosis

Complications

- None significant.

Prognosis

- Benign; persists.

Prescriber pearls (UK pharmacist IP)

A red, scaly leg papule that does not respond to emollient/steroid is not necessarily eczema or psoriasis. Look for a solitary lesion and **refer it for dermoscopy/biopsy** rather than escalating steroid potency.

Self-check quiz

4 questions · optional

Q1 Classic site?

✓ Lower leg (shin) of a middle-aged / older adult.

Q2 Pathognomonic dermoscopy?

✓ Dotted vessels in a ‘string of pearls’ pattern.

Q3 Critical differential?

✓ Amelanotic melanoma.

Q4 Bowen’s vessel pattern?

✓ Glomerular vessels in clusters.

References

1. DermNet: Clear cell acanthoma — https://dermnetnz.org/topics/clear-cell-acanthoma
dermnetnz.org

Photo gallery

Clear cell acanthoma

3

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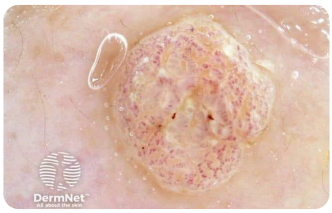

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Lumps, lesions & skin cancer Module 2 · Week 3

Basal cell carcinoma (BCC)

The commonest skin cancer: a slowly growing pearly papule or non-healing sore, usually on the face. Learn the subtypes, the dermoscopy clues, the NICE NG12 referral speed, and the treatment ladder (surgery first-line for nodular BCC, BAD 2021).

ALSO CALLED BCC | rodent ulcer | pearly papule | nodular BCC | superficial BCC | morphoeic BCC | +3

RELATED PAGES Cutaneous squamous cell carcinoma (cSCC) | Melanoma: recognition & referral | Actinic keratoses (solar keratoses) | Skin surgery procedures: excision, curettage, cryotherapy & Mohs

KEY FACTS

- The **commonest cancer in the UK**: a slow-growing, locally invasive tumour of basal keratinocytes that almost never metastasises.
- Mostly older, fair-skinned adults with sun damage; **80%** on the head and neck.
- Classic nodular BCC: **pearly** papule with rolled edge and **arborising telangiectasia**, often with central ulceration (rodent ulcer).
- Caused by **hedgehog pathway** mutations (PTCH1, SMO), usually from UV exposure.
- UK (NICE NG12): routine (not 2WW) referral for most suspected BCC; urgent if site or size threatens function. Surgery is the mainstay.

Overview

The commonest human cancer. It is a slow-growing, locally invasive malignancy of basal keratinocyte-like cells. **Metastasis is very rare (<0.1%)**, but neglected lesions destroy tissue (nose, eye, ear.

Epidemiology

- **Incidence**
  - The commonest cancer in white populations; well over 100,000 new cases a year in the UK and rising.
  - Lifetime risk about 1 in 5 in some fair-skinned populations.
- **Who**
  - Usually over 60, more men; rare in people with dark skin but does occur (more often pigmented).

Aetiology: causes and risk factors

- **Intermittent intense and cumulative UV**, with a latency of decades (course teaching gave 20–50 years). Fair skin (Fitzpatrick I–II), age, male sex.

- Other risks: previous BCC (about 40% get another within 3 years), radiotherapy, arsenic, immunosuppression (transplant), and **hydrochlorothiazide** (MHRA 2018: dose-dependent increased NMSC risk, mainly SCC).
- **HEDGEHOG PATHWAY**
  most BCCs have loss of **PTCH1** or activating **SMO** mutations, which is why hedgehog inhibitors (vismodegib, sonidegib) work in advanced disease.
- **GORLIN (NAEVOID BCC) SYNDROME**
  autosomal dominant PTCH1 (or SUFU) mutation. **Multiple BCCs from youth**, **palmar/plantar pits**, odontogenic keratocysts of the jaw, calcified falx, bifid ribs, macrocephaly, frontal bossing, and medulloblastoma in childhood (SUFU). **Avoid radiotherapy** (it induces many new BCCs) and minimise CT/X-ray.
- Histology: nests of basaloid cells with **peripheral palisading**, **retraction artefact** (clefting) and mucinous stroma.

Risk factors and associations

- **Intermittent intense UV** and childhood sunburn; fair skin, red or blond hair, freckling.
- Previous BCC (about 40% get another within 3 years), sunbeds, radiotherapy, arsenic.
- Immunosuppression, hydrochlorothiazide (possible), syndromes: **Gorlin (basal cell naevus) syndrome**, xeroderma pigmentosum.

Myths to address

> **"SKIN CANCER ALWAYS SPREADS"**
> BCC rarely metastasises but can destroy local tissue if neglected.

> **"IT'S A SPOT THAT KEEPS BLEEDING AND HEALING"**
> a non-healing or recurrently bleeding spot needs review.

Pathophysiology and pathology

BCC grows from basal-layer cells when the **hedgehog signalling pathway** is switched on permanently.

1. **UV mutation**
   UV damages **PTCH1** (a hedgehog-pathway brake) or activates **SMO**; TP53 mutations often coexist.
2. **Uncontrolled hedgehog signalling**
   GLI transcription factors stay on, so basaloid cells keep proliferating.

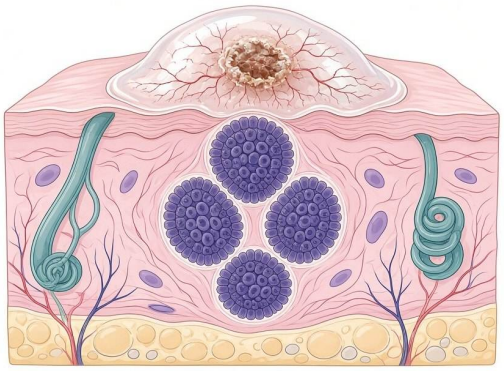

Fig. 1 · BCC in section: nests of basaloid cells growing from the basal layer into the dermis, with dilated vessels on the surface.
AI-generated illustration

3 **Local invasion**

Tumour nests depend on surrounding stroma, which is why BCC invades locally but rarely spreads by blood or lymph.

4 **Growth patterns**

Nodular, superficial, infiltrative and **morphoeic** patterns differ in how far they extend beyond what is visible.

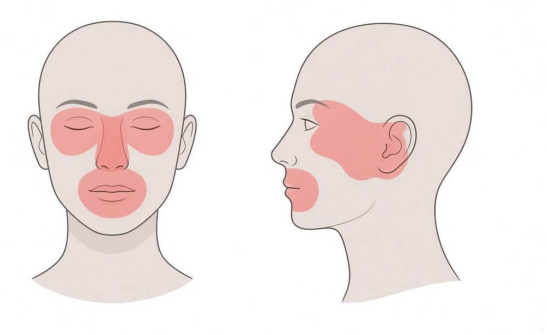

Fig. 2 · High-risk 'H-zone' of the face (eyes, nose, lips, ears), where BCCs are harder to treat.

AI-generated illustration

5 **Ulceration**

Central necrosis gives the classic **rodent ulcer**, with a rolled pearly edge.

HISTOPATHOLOGY

- Nests of **basaloid cells** with peripheral **palisading**, retraction artefact and mucinous stroma.
- Subtype (nodular, superficial, infiltrative, morphoeic, micronodular) guides treatment and recurrence risk.

Clinical features

About 80–85% occur on the head and neck. Main subtypes (the course "5"):

SUBTYPE	LOOK	PHOTO
Nodular (commonest)	Pearly/translucent papule-nodule, rolled edge, telangiectasia, central ulcer ("rodent ulcer")	
Superficial	Pink, scaly, thin plaque with a "thread-like" pearly edge; trunk and limbs; younger patients	
Morphoeic / infiltrative	Ill-defined, scar-like, waxy, indurated plaque; subclinical spread; high-risk	
Pigmented	Brown-blue-grey; commoner in skin of colour; melanoma DDx	,
Basosquamous	Mixed features; behaves more like SCC (metastatic potential)	–
Micronodular	Small nests; high-risk histology	–
Fibroepithelioma of Pinkus	Pink pedunculated lower-back papule	–

Fig. 3 · Typical ulcerated nodular BCC: pearly rolled edge, broken telangiectasia, central ulcer/crust ('rodent ulcer'). These are NG12 1.7.5 features (non-urgent referral).

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Subtypes to recognise

Nodular

Commonest. Pearly nodule, rolled edge, telangiectasia, central ulcer. Site: Face.

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Superficial

Pink, scaly thin plaque with a thread-like pearly edge. Site: Trunk, limbs; younger.

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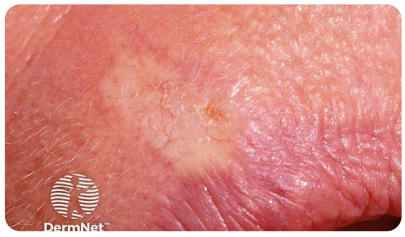
Morphoeic

Scar-like, waxy, ill-defined; spreads unseen. High-risk. Site: Face.



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Pigmented

Brown-blue-grey; commoner in skin of colour; melanoma DDx. Site: Face.

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Special groups

- **Skin of colour**
  - Less common; often **pigmented** and mistaken for a naevus or melanoma.
- **Young patients or many BCCs**
  - Consider Gorlin syndrome (jaw cysts, palmar pits) and refer.

Diagnosis and investigations

- Clinical + **dermoscopy: arborising (tree-like) vessels in sharp focus**; blue-grey ovoid nests/globules; spoke-wheel and leaf-like structures; ulceration; shiny white structures.
- The skin cancer team confirms with **punch or excision biopsy**. Histology determines the subtype and risk.

History

- Slow growth; bleeds, crusts, never fully heals.
- UV exposure, fair skin, previous BCC (~40% get another in 3 years).
- Immunosuppression, radiotherapy, **hydrochlorothiazide** (MHRA).

Examine

- Site and size (H-zone?), edge definition.
- Whole-skin check for other lesions.

Dermoscopy

- **Arborising vessels** in sharp focus.
- Blue-grey ovoid nests; spoke-wheel and leaf-like areas; shiny white structures.

Tests

- Skin cancer team: **punch or excision biopsy**.
- Histology: basaloid nests, peripheral palisading, retraction clefts.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Sebaceous hyperplasia	Yellowish umbilicated facial papules, often multiple.
	Intradermal naevus	Soft, symmetrical; vessels out of focus.
	Amelanotic melanoma	New, growing pink lesion: do not reassure.
	SCC / keratoacanthoma	Firm, indurated, keratotic, faster growth.

	CONDITION	DISTINGUISHING FEATURES
	Bowen's disease	Persistent, well-demarcated scaly red plaque (vs superficial BCC).
	Trigeminal trophic syndrome	Crescent ulcer of the nasal ala sparing the tip; numbness, picking. Biopsy.

Also: eczema or psoriasis plaque (vs superficial BCC), morphoea or scar (vs morphoeic), DFSP, molluscum, clear cell acanthoma.

Sebaceous hyperplasia (the skin lumps notes see Where course teaching differs from UK guidance), intradermal naevus (soft, symmetrical, vessels out of focus; see the naevi notes), amelanotic melanoma, SCC/KA, Bowen's (for superficial BCC), eczema/psoriasis plaque (for superficial BCC), morphoea or scar (for morphoeic), DFSP (the skin lumps notes see Trigeminal trophic syndrome (TTS) and Case 1 care-home context), **trigeminal trophic syndrome** (see Trigeminal trophic syndrome (TTS) and Case 1 care-home context), molluscum, CCA.

What it is

A rare cause of **self-inflicted ulceration of the nasal ala** after damage to the trigeminal nerve or its central pathways.

How it is caused / pathophysiology

Trigeminal **sensory loss or paraesthesia** (after a brainstem/lateral medullary stroke, trigeminal ablation, herpes zoster, acoustic neuroma surgery, or vertebrobasilar ischaemia) leads to unconscious rubbing, scratching and picking. The result is a chronic, non-healing ulcer. **Vascular dementia with brainstem infarction, plus picking behaviour, fits this picture.**

Clinical features & typical sites

A **crescent-shaped ulcer of the nasal ala** that often **spares the nasal tip** (the tip is supplied by the external nasal branch, which is often spared). It is unilateral and progressive, and can destroy the ala. The area is numb or tingly.

How to diagnose

History of trigeminal injury, sensory testing, and **biopsy to exclude BCC/SCC** (and infection: granulomatous disease, mycobacterial or fungal infection; vasculitis/GPA).

Key differentials

BCC (main), SCC, granulomatosis with polyangiitis, lupus vulgaris, leishmaniasis, NK/T-cell lymphoma, pyoderma gangrenosum, and dermatitis artefacta.

Management for pharmacist/IP (UK)

- **Refer for biopsy** (the course answer). The IP role is to **arrange the referral** and **not to treat the ulcer as infected or as eczema** without a diagnosis.
- TTS treatment (specialist): **behavioural protection** (occlusive dressings, nail care, mittens or protective devices in dementia), treat paraesthesia (carbamazepine, gabapentin or amitriptyline at low dose, with careful weighing of **anticholinergic burden and falls risk in an 85-year-old on donepezil**), and reconstructive surgery once the picking stops.
- **CAPACITY AND CONSENT (MCA 2005)**

assess capacity for each decision (biopsy, surgery). If capacity is lacking, decide in the person's **best interests**, consulting family or an LPA for health and welfare (or an IMCA if no one is appropriate). Weigh the burden of travel and procedures against benefit. For a frail resident with a slow-growing BCC, options include surgery under local anaesthetic, radiotherapy, or active monitoring with palliative wound care. This is an MDT decision.

- **Medicines for Case 1: continue aspirin for biopsy or skin surgery** (BSDS 2023). Donepezil plus beta-blockers or other bradycardic drugs carries a bradycardia/syncope risk, which matters if sedation is considered. No interaction issue with local anaesthetic at standard doses.

Prescriber pearls & counselling

Care-home staff should report non-healing lesions on the face for more than 4 weeks. An "infected scab" on the nose of an older person that won't heal is a BCC until proved otherwise.

Photo / reference URLs

- DermNet: dermnetnz.org
- Mental Capacity Act 2005 Code of Practice (gov.uk).

Management

Recognition and referral (IP role)

> **NICE NG12 1.7.5**
> suspected BCC means a **routine (non-urgent) referral**, usually to dermatology.

> **NG12 1.7.6**
> consider the **suspected cancer pathway** only if delay could have an unacceptable impact because of **site or size**, e.g. near the eye or nose, or a large or rapidly growing lesion. Case 1 nasal ulcer qualifies for discussion.

> **WHO EXCISES**
> the NICE Improving Outcomes Guidance for skin tumours (and its 2010 update) allows suitably trained, accredited primary-care clinicians to excise **only low-risk BCCs** (e.g. a small nodular BCC on the trunk or limbs in a non-immunosuppressed patient). Anything high-risk goes to a specialist skin cancer team. **IPs do not biopsy, curette, freeze or apply topical treatment to an undiagnosed suspected BCC.**

Specialist treatment ladder (BAD 2021; for teaching)

OPTION	MAIN INDICATION	NOTES
Standard surgical excision	**First-line for most BCCs, including nodular BCC on the face**	Peripheral margin **4 mm** for low-risk (95% clearance); **5 mm** for primary high-risk
Mohs micrographic surgery	**Primary high-risk BCC** (consider); **offer first-line for recurrent BCC with 1 other high-risk factor**; high-risk sites (H-zone) where tissue sparing matters; morphoeic/ill-defined	Complete margin control; lowest recurrence; tissue-sparing. Photo
Curettage & cautery	Small, well-defined, low-risk nodular/superficial BCC on low-risk sites	Not for high-risk sites or recurrent lesions
Cryosurgery	Low-risk superficial/small nodular	No histological margins; recurrence
Topical imiquimod 5%	**Superficial BCC** (licensed: 5×/week for 6 weeks)	Local inflammation is expected; flu-like symptoms

OPTION	MAIN INDICATION	NOTES
Topical fluorouracil 5%	Superficial BCC (off-label for BCC)	Twice daily for several weeks
Photodynamic therapy (PDT)	**Superficial** BCC and **thin nodular** BCC (<2 mm) at low-risk sites	Good cosmesis; higher recurrence than surgery for nodular BCC
Radiotherapy	Patients unfit or unwilling for surgery; adjuvant	**Avoid in Gorlin** and in the young
Hedgehog inhibitors (vismodegib, sonidegib)	Locally advanced/metastatic, MDT-led	Teratogenic (strict pregnancy prevention); muscle cramps, alopecia, dysgeusia. NICE did not recommend vismodegib

BAD 2021 high-risk features (determine margins/Mohs):

- Site and size: area H (central face, eyelids, nose, lips, ears, temple), area M lesions >10 mm, area L lesions >20 mm.
- Poorly defined borders.
- Recurrence.
- Immunosuppression.
- Prior radiotherapy at the site.
- Histology: infiltrative, micronodular, morphoeic or basosquamous; perineural invasion; depth >6 mm or beyond the subcutis.

Prognosis: excellent for treated primary low-risk BCC (5-year recurrence under 5% after surgery). Recurrence is higher for high-risk subtypes and sites. Mortality is negligible except in neglected or metastatic disease. **About 40% develop another BCC**, so self-examination and sun protection matter.

The BCC treatment ladder

See the skin cancer notes:

- surgery (standard excision 4 mm / 5 mm margins) Mohs for high-risk/recurrent
- C&C or cryo for small low-risk lesions
- imiquimod, 5-FU or PDT for superficial BCC
- radiotherapy if unfit
- hedgehog inhibitors for advanced disease.

NICE NG12 quick table

SUSPICION	NG12 / UK ROUTE	TIMING
BCC, typical, low-risk site	1.7.5: routine referral (or accredited primary-care excision under IOG for low-risk lesions)	Routine
BCC where site/size makes delay matter (eye, nose, large)	1.7.6: consider the suspected cancer pathway	2 weeks
cSCC (including KA)	1.7.4: consider the suspected cancer pathway	2 weeks
Lip/oral ulcer >3 weeks	NG12 oral cancer: suspected cancer pathway (or urgent dental)	2 weeks

SUSPICION	NG12 / UK ROUTE	TIMING
Non-resolving "eczema/psoriasis" in an adult, atypical sites	Routine dermatology with a note on treatment failure (biopsy). Urgent if tumours, erythroderma, nodes or B symptoms	Routine urgent
Make the referral	**Within 1 working day of deciding** (NG12 general principle)	–

Worked cases

CASE 1

85-year-old care-home resident, enlarging picked nasal ulcer, vascular dementia

DIAGNOSIS

nodular BCC (ulcerated "rodent ulcer").

KEY DIFFERENTIAL

trigeminal trophic syndrome. Also SCC and GPA.

INVESTIGATION

biopsy (dermatology). **Histology:** nests of basaloid cells with **peripheral palisading** and retraction clefts.

REFERRAL

NG12 1.7.5 says routine, but nose = H-zone, so consider the suspected cancer pathway (1.7.6) if it is large or close to vital structures.

Capacity / best-interests decisions under the MCA. Continue aspirin for any procedure.

TREATMENT (SPECIALIST)

surgical excision (or Mohs for a high-risk facial BCC) is first-line; radiotherapy if unfit. **Not PDT** (it is for superficial/thin nodular BCC) (see Where course teaching differs from UK guidance).

CASE 2

BCC related questions

AETIOLOGY

UV, PTCH1/SMO, Gorlin, immunosuppression, radiation, arsenic.

SURGERY

standard excision, 4 mm (low-risk) / 5 mm (high-risk) margins; Mohs for high-risk/recurrent/ H-zone.

VARIANTS

nodular, superficial, morphoeic/infiltrative, micronodular, pigmented, basosquamous.

PROGNOSIS

excellent; recurrence is the issue, not mortality.

CASE 3

68-year-old outdoor worker, indurated bleeding lower-lip ulcer for 6 weeks

DIAGNOSIS

SCC of the lip (on actinic cheilitis).

WHY WORRYING

the lip is a high-risk site with a higher metastatic rate.

TREATMENT

surgical excision (6 mm margin for high-risk per BAD 2020), with nodal assessment.

REFERRAL

NG12 1.7.4 suspected cancer pathway, plus a lip ulcer >3 weeks meets the oral cancer criteria.

CASE 4

SCC related questions

PRECURSORS

AK, actinic cheilitis, Bowen's, KA, horn, chronic wounds (Marjolin), lichen sclerosus.

INVESTIGATIONS

biopsy, node examination, ultrasound ± FNA, CT/MRI for very high-risk.

MANAGEMENT

excision, Mohs (selected), RT, cemiplimab for advanced disease, follow-up and photoprotection.

CASE 5

62-year-old mechanic, steroid-unresponsive rash for 12 months

DIAGNOSIS

mycosis fungoides. Rule out occupational contact dermatitis (patch test) and drug-induced pseudolymphoma.

PLAN

refer for (repeat) biopsies with TCR clonality, then staging. Skin-directed therapy (potent topical steroid, nbUVB/PUVA phototherapy, chlormethine) for early stage.

HISTOLOGY

epidermotropism with **Pautrier microabscesses**.

STAGE

cannot be assigned from the vignette. Tumours = IIB, erythroderma = III, blood involvement = IV.

Medicines at a glance

Standard surgical excision

First line · most BCCs

DOSE 4 mm margin (low-risk); 5 mm (high-risk) · 95% clearance

CAUTIONS Including nodular BCC on the face.

Mohs micrographic surgery

High-risk or recurrent

DOSE Complete margin control · Lowest recurrence

CAUTIONS H-zone, morphoeic, recurrent with another high-risk factor.

Imiquimod 5%

Superficial / small low-risk

DOSE 5 × per week · 6 weeks

CAUTIONS Superficial BCC (licensed). Expect inflammation.

Photodynamic therapy

Superficial / small low-risk

DOSE Superficial or thin (< 2 mm) nodular · Low-risk sites

CAUTIONS Good cosmesis; higher recurrence.

Curettage & cautery · cryotherapy · 5-FU

Superficial / small low-risk

DOSE Small, well-defined, low-risk · Not high-risk sites

Treatment ladder

First line · most BCCs

Standard surgical excision

2

High-risk or recurrent

Mohs micrographic surgery

3

Superficial / small low-risk

Imiquimod 5%; Photodynamic therapy; Curettage & cautery · cryotherapy · 5-FU

4

Unfit for surgery / advanced

Radiotherapy if unfit or unwilling (avoid in Gorlin and the young).
Hedgehog inhibitors (vismodegib, sonidegib): MDT-led; teratogenic.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **High-risk features**: H-zone; M-zone > 10 mm, L-zone > 20 mm; poorly defined; recurrent; immunosuppressed; prior RT; aggressive histology; perineural; depth > 6 mm.
- **Prognosis**: 5-year recurrence < 5% after surgery for low-risk primary BCC.
- **Primary-care excision**: Only low-risk BCCs by accredited clinicians (NICE IOG).

Referral

URGENT — Near the **eye, nose or ear**, large, fast-growing or recurrent: consider the **suspected cancer pathway** (NG12 1.7.6).

ROUTINE — Typical BCC at a low-risk site: routine dermatology referral (NG12 1.7.5).

ROUTINE — Multiple BCCs young, palmar pits or jaw cysts: think **Gorlin syndrome**.

DON'T — Never treat an **undiagnosed** pigmented or nodular lesion: refer for diagnosis first.

Red flags and when to refer

- BCC near the eye, nose or ear, large, rapidly growing or recurrent: consider the **suspected cancer pathway** because delay could matter (NG12 1.7.6).
- Do not freeze, curette or apply imiquimod to an **undiagnosed** pigmented or nodular lesion: refer for diagnosis first.
- Multiple BCCs in a young person, palmar pits or jaw cysts: think **Gorlin syndrome** and refer.

Complications and prognosis

Complications

- Local tissue destruction (eye, nose, ear) if neglected.
- Recurrence, especially infiltrative/morphoeic BCC and H-zone sites.
- Further BCCs and other skin cancers.
- Metastasis is very rare (under 0.1%).

Prognosis

- Excellent: over 95% cure with standard excision.
- Annual skin self-checks and sun protection are needed because new BCCs are common.

Prescriber pearls (UK pharmacist IP)

Medicines review for keratinocyte cancer patients: Hydrochlorothiazide: MHRA advises informing patients of the NMSC risk and considering alternatives in those with previous skin cancer; Long-term voriconazole: phototoxicity and SCC; Azathioprine and other transplant immunosuppressants

Imiquimod counselling: apply at bedtime, wash off after about 8 h; expect redness, crusting and erosion (it means the drug is working); stop and seek advice if there is severe reaction or flu-like illness.

Sun protection: SPF 30+ with high UVA rating, shade, hats (NICE NG34). Advise a self-check every month.

Hedgehog inhibitors: pregnancy prevention for women (and condoms for men on vismodegib); cramps, taste loss and weight loss are common.

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING OR STUDENT ANSWER	UK GUIDANCE	TEACHING POINT
Nodular BCC on the face: PDT and "avoid surgery"	PDT first; avoid facial surgery	BAD 2021: **surgery (standard excision or Mohs) is first-line** for nodular BCC, especially facial/high-risk. PDT is for **superficial** and **thin (<2 mm) nodular** BCC at low-risk sites, with higher recurrence	Gentle correction: facial site favours margin-controlled surgery (Mohs), it doesn't preclude surgery
BCC urgency	(Not specified)	NG12 1.7.5 routine; 1.7.6 suspected cancer pathway if site/size makes delay matter	Nose = consider urgent
SCC	Excision first-line	BAD 2020 margins 4/6/10 mm by risk; NG12 1.7.4 suspected cancer pathway	Aligned
SIGN 140	You cited SIGN 140	Still valid (Scotland, 2014), but BAD 2020 is the current UK-wide reference	Cite both
MF "advanced disease"	Course teaching	No staging data given. A 12-month steroid-resistant rash advanced. Advanced = stage IIB	Stage before labelling; important for counselling
MF treatment	Phototherapy	BAD/UKCLG 2019: skin-directed therapy (steroid, phototherapy, chlormethine, RT) for early stage	Aligned
Nicotinamide	(Not discussed)	Benefit in immunocompetent high-risk patients (ONTRAC); no benefit in transplant recipients (2023 RCT)	Specialist advice

Self-check quiz

5 questions · optional

Q1 What is the NICE NG12 referral speed for a typical BCC?

✓ Routine referral (1.7.5); consider the suspected cancer pathway only if site or size makes delay matter (1.7.6).

Q2 Classic dermoscopy clue?

✓ Arborising (tree-like) vessels in sharp focus; also blue-grey ovoid nests.

Q3 First-line treatment for most BCCs, including nodular BCC on the face?

✓ Standard surgical excision (4 mm margin low-risk, 5 mm high-risk); Mohs for high-risk or recurrent.

Q4 Which mimic causes a crescent ulcer of the nasal ala that spares the tip?

✓ Trigeminal trophic syndrome. Biopsy to exclude BCC.

Q5 What must an IP never do with a suspected BCC?

✓ Freeze, curette or apply imiquimod to an undiagnosed lesion. Refer for diagnosis first.

References

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emedicine.medscape.com

Photo gallery

Nodular BCC 5

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Other BCC subtypes 3

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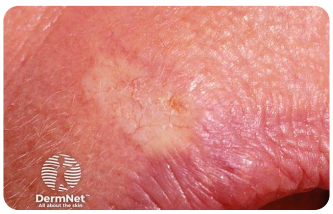

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Dermoscopy and histology

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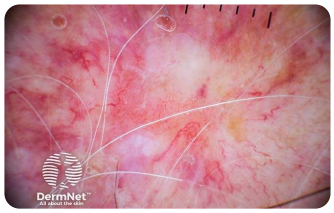

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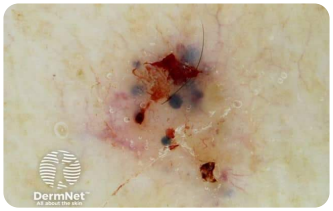

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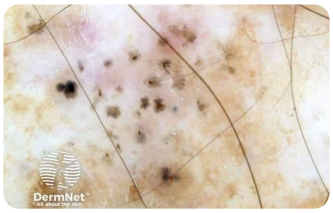

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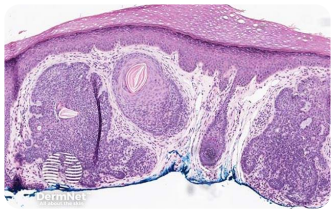

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Trigeminal trophic syndrome (mimic)

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Gorlin syndrome

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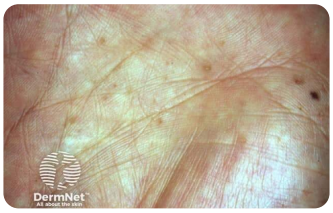

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Lumps, lesions & skin cancer Module 2 · Week 3

Cutaneous squamous cell carcinoma (cSCC)

A growing, tender, keratotic or ulcerated nodule on sun-damaged skin that can spread to lymph nodes, especially on the lip and ear. Know the precursors (AK, Bowen's, actinic cheilitis) and refer on the suspected-cancer pathway.

ALSO CALLED SCC | cSCC | lip SCC | keratoacanthoma | cutaneous horn | Marjolin ulcer | +3

RELATED PAGES Actinic keratoses (solar keratoses) | Basal cell carcinoma (BCC) | Melanoma: recognition & referral | Skin surgery procedures: excision, curettage, cryotherapy & Mohs

KEY FACTS

- Malignant tumour of **epidermal keratinocytes**; the second commonest skin cancer.
- Risk factors: cumulative **UV**, older age, fair skin, **AKs**, organ-transplant immunosuppression (65–250× risk), smoking, chronic wounds and scars.
- Tender, **keratotic, crusted or ulcerated** nodule on sun-exposed skin that grows over weeks to months.
- Can metastasise (about 2–5%): **lip, ear**, large, deep, perineural and immunosuppressed are high risk.
- UK (NICE NG12): **suspected cancer (2WW) referral**; surgery is the mainstay.

Overview

A malignant tumour of keratinocytes. It is the second commonest skin cancer, with real potential to metastasise (overall about 2–5%; higher for high-risk tumours). **Lip and ear SCCs metastasise more often** (lip around 10–15% in older series).

SCC precursors and related lesions

Lesions that sit on the UV keratinocyte dysplasia SCC pathway, plus clinical forms that conceal SCC.

Epidemiology

- **Incidence**
  - Rising; about 1 in 4 non-melanoma skin cancers in the UK.
  - Mostly over 60, more men; outdoor workers.
- **High-risk groups**
  - Organ transplant recipients develop SCCs far more often and more aggressively.
  - People with chronic lymphocytic leukaemia or HIV.

Aetiology: causes and risk factors

- **Cumulative UV** (outdoor workers, as in Case 3), older age, fair skin.
- **IMMUNOSUPPRESSION**
  organ transplant recipients have 65–250× the risk; CLL; HIV.
- **DRUGS**
  azathioprine, voriconazole, hydrochlorothiazide, BRAF inhibitors (vemurafenib monotherapy: KA-type SCCs).
- **HPV.**
- Chronic wounds, scars and burns (**Marjolin ulcer**,).
- Radiation, arsenic, smoking (lip), lichen sclerosus/planus (genital and oral).
- Genetic: xeroderma pigmentosum, recessive dystrophic epidermolysis bullosa, albinism.
- Path: UV-signature TP53 mutations progress through AK or Bowen's to invasive SCC (see SCC precursors and related lesions).

SCC precursors and related lesions

PRECURSOR	PROGRESSION RISK	NOTES
Actinic keratosis	Low per lesion (~0.1%/year; up to ~10% of patients with many AKs over a decade)	**Core in the older skin pages notes** (field therapy, PCDS AK pathway)
Actinic cheilitis	Higher than AK; lip SCC often arises on it	. Biopsy any induration or ulcer. Treatments: 5-FU, imiquimod (off-label), PDT, laser, vermilionectomy
Bowen's disease (SCC in situ)	About 3–5% to invasive SCC (higher genital/perianal)	. BAD 2014 guideline (Morton): 5-FU, imiquimod (off-label), cryo, PDT, curettage, excision; on the lower leg, PDT/5-FU is preferred over cryo (poor healing)
Keratoacanthoma	Treat as SCC	See the benign lesion notes see Cutaneous squamous cell carcinoma (cSCC), including lip SCC
Cutaneous horn	20–30% have SCC/AK at the base	; excise for histology
Chronic ulcers/scars/burns	Marjolin ulcer	; biopsy non-healing ulcers with rolled edges or exuberant granulation
Lichen sclerosus (vulval/penile)	~3–5% vulval SCC	Long-term potent steroid reduces risk
HPV-related (genital/periungual)	Bowenoid papulosis, periungual SCC	—
Porokeratosis (disseminated superficial actinic)	Small risk	—

Risk factors and associations

- **Cumulative UVB**, sunbeds and PUVA; photosensitising drugs (voriconazole, hydrochlorothiazide).
- Precursors: AKs, Bowen's disease (SCC in situ), actinic cheilitis.
- **CHRONIC INFLAMMATION AND SCARS**

burns, leg ulcers (Marjolin ulcer), lichen sclerosus, discoid lupus, epidermolysis bullosa.

- HPV (especially in immunosuppression and genital/periungual SCC), arsenic, ionising radiation, smoking.

Myths to address

- **"IT'S A WART THAT WON'T GO"**
  a fast-growing, tender keratotic lump on sun-damaged skin needs urgent assessment.

Pathophysiology and pathology

SCC develops step by step from **UV-damaged keratinocytes** that eventually breach the basement membrane.

1 **UV DNA damage**

UVB causes **TP53** mutations (UV signature) and then NOTCH and CDKN2A changes.

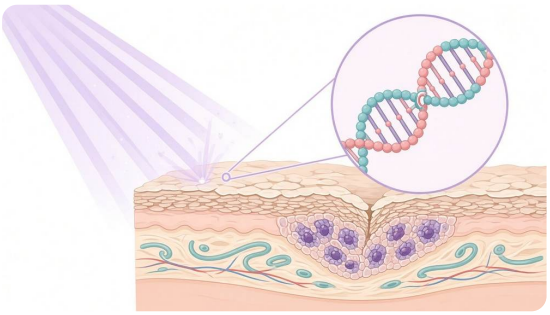

Fig. 1 · UVB damages keratinocyte DNA (inset: fused adjacent bases); mutated cells crowd together and begin to invade beneath rough scale.
AI-generated illustration

2 **Dysplasia**

Abnormal cells spread in the epidermis: partial thickness (AK) then full thickness (Bowen's disease, SCC in situ).

3 **Invasion**

Cells break through the basement membrane into the dermis as nests and strands of keratinising cells.

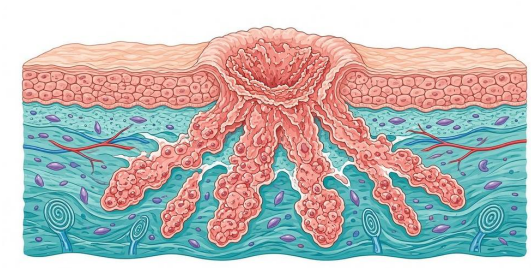

Fig. 2 · SCC invading through the basement membrane into the dermis, with keratin pearls.
AI-generated illustration

4 **Immune escape**

Immunosuppression lets mutated clones grow unchecked: why transplant recipients get many aggressive SCCs.

5 **Spread**

Some tumours invade nerves or spread to regional lymph nodes, especially on high-risk sites.

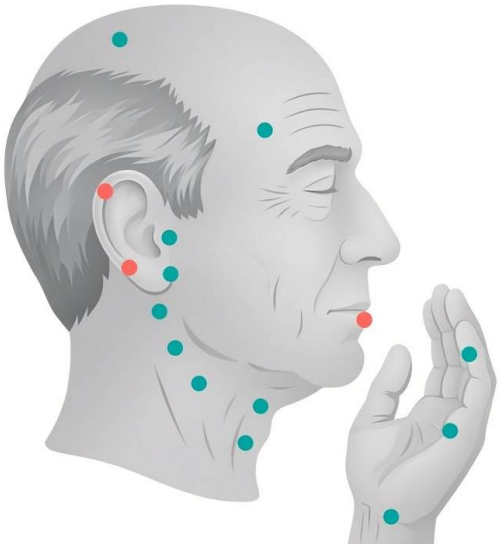

Fig. 3 · High-risk sites: lip, ear, temple, scalp and the backs of the hands.

AI-generated illustration

HISTOPATHOLOGY

- Atypical keratinocytes invading the dermis with **keratin pearls** and intercellular bridges.
- Grade (well to poorly differentiated), depth, perineural invasion and margins determine risk and follow-up.

Clinical features

- An indurated, **tender** keratotic nodule or plaque, or an **ulcer with hard everted edges**, growing over weeks to months, often bleeding.
- Sun-exposed sites: scalp, ears, lower lip, the back of the hands, and the lower legs in women.
- **LIP SCC**
  usually the **lower lip vermilion** (more UV), often on a background of **actinic cheilitis**. Smoking or pipe use. Persistent ulcer or crust, induration.
- Variants: well-differentiated/keratinising, cutaneous horn, verrucous carcinoma/carcinoma cuniculatum (plantar, slow, locally destructive;), keratoacanthoma-type (the benign lesion notes see Cutaneous squamous cell carcinoma (cSCC), including lip SCC), poorly differentiated, acantholytic, desmoplastic.

SCC precursors and related lesions

Signs that an AK or Bowen's has become SCC (refer under NG12 1.7.4):

- **induration/thickening**
- **tenderness/pain**
- **rapid growth**
- **bleeding**
- **ulceration**

- diameter >1 cm
- failure to respond to treatment.

Fig. 4 · Ulcerated, firm lower-lip lesion in a smoker = SCC. Induration is the first sign of malignancy. The lip is a high-risk site, so make a suspected cancer pathway referral (NG12 1.7.4).

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Types to recognise

Lip SCC

Indurated ulcer, lower lip, smoker. Site: Lower lip.

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Ear SCC

Bleeding ulcerated; high risk. Site: Ear.

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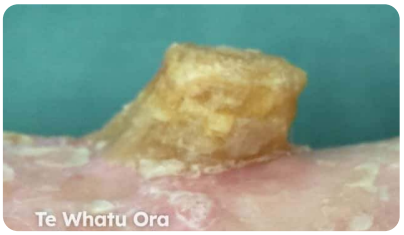
Cutaneous horn

Keratin projection. Site: Face, hands.



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Marjolin ulcer

SCC in a chronic leg ulcer. Site: Leg.

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Actinic cheilitis

Scaly blurred vermilion. Site: Lower lip.

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Bowen's disease

SCC in situ: scaly plaque. Site: Finger, leg.

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Special groups

- **Transplant recipients**
  - Annual or more frequent skin surveillance; rapid referral of any new or growing lesion.
- **Skin of colour**
  - SCC is the commonest skin cancer in people with dark skin, often arising in **scars, chronic ulcers or sun-protected sites** and presenting late.

Diagnosis and investigations

Clinical + palpation of regional lymph nodes (pre-auricular, parotid, cervical, submental/submandibular for the lip;).

BIOPSY

excision biopsy is preferred where feasible. Otherwise an incisional/punch biopsy deep enough to assess depth.

PATHOLOGY REPORT ITEMS

differentiation, thickness/depth, perineural and lymphovascular invasion, margins.

STAGING

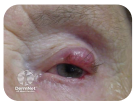
the AJCC 8th edition (head and neck cSCC) or UICC 8. The BAD 2020 risk categories (low, high, very high) guide management.

Imaging (specialist): ultrasound ± FNA of clinically suspicious nodes; CT/MRI for very high-risk tumours or perineural symptoms. **Consider SLNB only in trials** (not standard UK practice).

SCC precursors and related lesions

Clinical + dermoscopy (Bowen's: glomerular vessels in clusters + scale). Biopsy if there is doubt or atypia.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Keratoacanthoma	Rapid crateriform nodule; treat as SCC.
	BCC	Pearly, telangiectasia, slow.
	Hypertrophic AK / Bowen's	Flat scaly; no induration.
	Amelanotic melanoma	Pink nodule; EFG.
	**Merkel cell carcinoma**	Rapid red-violet nodule in the elderly.
	Lip: herpes / aphthous / chancre	Herpes heals; chancre: syphilis risk.

KA, BCC, AK (hypertrophic), Bowen's, irritated SK, viral wart, amelanotic melanoma, Merkel cell carcinoma (rapid red-violet nodule in the elderly), chancre (syphilis; lip), herpes labialis (it heals), aphthous ulcer, traumatic ulcer, and for the lip discoid lupus or lichen planus.

SCC precursors and related lesions

For Bowen's: psoriasis, eczema, superficial BCC, tinea (scrape), and extramammary Paget's (genital).

Management

Recognition and referral

> **NICE NG12 1.7.4**
>
> **consider a suspected cancer pathway referral** (appointment within 2 weeks) for suspected cSCC. Case 3 (a 6-week indurated bleeding lip ulcer in an outdoor worker) should be referred on that pathway. **Make the referral within 1 working day of deciding.**

> **Any lip or oral ulcer that persists for more than 3 weeks** also meets NG12 head-and-neck/oral cancer criteria (suspected cancer pathway or urgent dental referral).

> **DO NOT**
>
> freeze, curette or treat a suspected SCC in primary care unless you are an accredited skin-surgery practitioner working to the IOG.

Specialist management (BAD 2020; teaching)

RISK GROUP	PERIPHERAL CLINICAL MARGIN (EXCISION)	NOTES
Low risk	**4 mm**	Aim for 1 mm histological clearance
High risk	**6 mm**	Lip and ear are high-risk by site
Very high risk	**10 mm**	Consider adjuvant radiotherapy; MDT

- **Surgery is first-line** (the course answer). Lip: wedge or V-excision ± vermilionectomy for associated actinic cheilitis (maxillofacial/plastics).
- Mohs: selected high-risk facial tumours.
- **RADIOTHERAPY**
  primary treatment if surgery is unsuitable; adjuvant for close/positive margins, perineural invasion or nodal disease.
- Nodal disease: neck dissection ± parotidectomy, then adjuvant RT.
- Advanced/metastatic: **cemiplimab** (anti-PD-1; NICE TA802, 2022).
- **FOLLOW-UP**
  BAD suggests clinical review for high-risk tumours (e.g. 3–6-monthly for 2 years), plus education about self-examination of the scar and nodes.
- **IMMUNOSUPPRESSED/TRANSPLANT PATIENTS**
  joint care with the transplant team. Specialists may switch to mTOR inhibitors (sirolimus) or use acitretin chemoprevention. Oral **nicotinamide 500 mg twice daily** reduced new keratinocyte cancers in immunocompetent high-risk patients (ONTRAC 2015), but **showed no benefit in a 2023 trial in transplant recipients**; it is a specialist recommendation.

SIGN 140 / BAD high-risk features:

- Site: lip, ear, non-sun-exposed sites, or within scars or chronic wounds.
- Diameter >20 mm.
- Thickness >4 mm or invasion beyond the subcutaneous fat.
- Poor differentiation.
- Perineural or lymphovascular invasion.
- Recurrence.
- Immunosuppression.

SCC precursors and related lesions

- AK: IP-manageable field therapy per the older skin pages notes see Worked cases (5-FU, diclofenac 3%, imiquimod, tirbanibulin; check the BNF/formulary).
- **BOWEN'S DISEASE**
  treatment is usually a confirmed diagnosis + GP/dermatology plan. An IP may prescribe topical 5-FU 5% **within a documented specialist or GP plan after diagnosis** (e.g. once or twice daily for 3–4 weeks; local protocol).
- Actinic cheilitis: refer for diagnosis. Topical options are specialist-initiated (lip reactions can be severe).

Treatment ladder

1

Refer

NG12 1.7.4: suspected cancer pathway (2 weeks); refer **within 1 working day**. **Lip or oral ulcer > 3 weeks**: head-and-neck / oral pathway or urgent dental.

2

Specialist surgery (BAD 2020)

Peripheral margins: **low risk 4 mm; high risk 6 mm; very high risk 10 mm** (± adjuvant radiotherapy, MDT). Advanced: **cemiplimab** (TA802).

3

Precursors (IP within a plan)

AK field therapy: 5-FU, diclofenac 3%, imiquimod, tirbanibulin. Bowen's: topical **5-FU 5%** after confirmed diagnosis within a documented plan.

4

Prevention

Review **hydrochlorothiazide** (MHRA) and long-term **voriconazole**; transplant photoprotection; smoking cessation; SPF lip balm. Transplant: joint care (mTOR switch, acitretin); nicotinamide is specialist-only.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Don't**: Freeze, curette or treat suspected SCC in primary care unless accredited.
- **5-FU and brivudine**: Never together (fatal DPD inhibition); keep 5-FU off eyes / mucosa.

Referral

URGENT	Suspected cSCC: **suspected cancer pathway** (NG12 1.7.4).
URGENT	Lip or mouth ulcer > 3 weeks: head-and-neck pathway.
URGENT	New growth or non-healing area in a long-standing ulcer or scar: biopsy (Marjolin).

Red flags and when to refer

- Suspected cSCC: **suspected cancer pathway** referral (NICE NG12 1.7.4).
- Any lip or mouth ulcer persisting more than 3 weeks: suspected head-and-neck cancer pathway.
- A non-healing area or new growth in a long-standing ulcer or scar: biopsy to exclude **SCC (Marjolin ulcer)**.

Complications and prognosis

Complications

- Local recurrence and tissue destruction.
- **Metastasis** to lymph nodes (about 2–5%), higher with high-risk features.
- Perineural spread (pain, numbness, cranial nerve palsies).

Prognosis

- Most low-risk SCCs are cured by excision.
- Five-year survival falls markedly once nodes are involved; high-risk patients need specialist follow-up.

Prescriber pearls (UK pharmacist IP)

- **Review hydrochlorothiazide** (MHRA) and **voriconazole** (photosensitivity; consider an alternative azole for long-term use in high-risk patients), and remind transplant patients of photoprotection.
- Smoking cessation: lip/oral SCC risk and wound healing.
- Lip sun protection: an SPF lip balm.
- Teach patients to report a new lump in the scar or the neck.
- 5-FU: expect redness, crusting and soreness. Do not apply near the eyes or mucosa; wash hands after.
- Topical 5-FU is minimally absorbed, but **never use it with brivudine** (fatal DPD inhibition), and take care over large areas in patients with known DPD deficiency.
- Mark lesions on a body map for follow-up.

SCC precursors and related lesions

Self-check quiz

4 questions · optional

Q1 Which sites carry higher metastatic risk?

✓ Lip and ear (and scars / chronic wounds).

Q2 Referral route?

✓ Suspected cancer pathway (NG12 1.7.4), within 1 working day.

Q3 Excision margin for a low-risk SCC?

✓ 4 mm (high risk 6 mm, very high risk 10 mm).

Q4 Drug never to combine with topical 5-FU?

✓ Brivudine.

References

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Photo gallery

Cutaneous SCC 6

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Precursors 2

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Lumps, lesions & skin cancer Module 2 · Week 3

Mycosis fungoides & cutaneous lymphoma

The commonest cutaneous lymphoma: fixed, slowly changing eczema- or psoriasis-like patches in sun-protected sites that do not settle with treatment. Most patients have early-stage disease with a normal life expectancy.

ALSO CALLED MF · CTCL · cutaneous T-cell lymphoma · Sézary syndrome · parapsoriasis · patch-stage · +1

RELATED PAGES Eczema / dermatitis · Psoriasis · Hypopigmentation & skin of colour · Biologics, JAK inhibitors & conventional systemics

KEY FACTS

- **Mycosis fungoides (MF)** is the commonest cutaneous T-cell lymphoma: slowly evolving patches, plaques and sometimes tumours.
- Patches in **sun-protected 'bathing trunk'** areas (buttocks, hips, breasts), varying shape, mild itch; often misdiagnosed as eczema or psoriasis for years.
- Repeated biopsies (off steroids) with T-cell receptor gene studies may be needed.
- Early stage: skin-directed therapy (potent steroids, phototherapy, radiotherapy); most have a normal life expectancy.
- Sézary syndrome erythroderma, lymphadenopathy, circulating malignant T cells: specialist.

Overview

MF is the commonest primary cutaneous T-cell lymphoma (CTCL), about half of all primary cutaneous lymphomas. It is usually indolent over years to decades, with a minority progressing. **Sézary syndrome** is the leukaemic, erythrodermic variant.

Epidemiology

- **Who**
  - Adults 50–70; men twice as often; about 0.5 per 100,000 per year.

Aetiology: causes and risk factors

Clonal skin-homing CD4+ memory T-cells (CD4+, often CD7- and CD26-). Cause unknown; links to chronic antigen stimulation have been proposed. **Pautrier microabscesses** (intraepidermal collections of atypical lymphocytes) and **epidermotropism** without spongiosis are the histological hallmark, but they are absent in many early biopsies.

Myths to address

- **"LYMPHOMA ALWAYS SPREADS QUICKLY"**

early MF is indolent; most never progress.

Pathophysiology and pathology

MF is a **clonal expansion of skin-homing CD4+ T cells** that infiltrate the epidermis.

1. **Skin-homing clone**
   A clone of memory CD4+ T cells carries skin-homing receptors (CLA, CCR4).
2. **Epidermotropism**
   Malignant cells invade the epidermis, forming Pautrier microabscesses.
3. **Patch plaque tumour**
   Over years, deeper infiltration produces plaques and tumours.

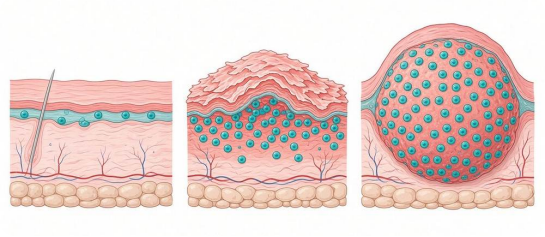

Fig. 1 · Mycosis fungoides evolves from patch to plaque to tumour.
AI-generated illustration

4. **Systemic spread**
   Rarely, cells reach nodes and blood (advanced stage, Sézary).

> **HISTOPATHOLOGY**
>
> - Atypical cerebriform lymphocytes in the epidermis (epidermotropism), Pautrier microabscesses; CD4+, loss of CD7; clonal TCR rearrangement.

Clinical features

PATCH STAGE

persistent, variably shaped, often **finely wrinkled ("cigarette-paper") pink-brown patches in sun-protected "bathing-trunk" sites** (buttocks, hips, breasts, inner thighs). Itchy or asymptomatic, **steroid-resistant or relapsing**.

PLAQUE STAGE

infiltrated, scaly, annular plaques.

TUMOUR STAGE

nodules that may ulcerate.

Erythroderma, poikiloderma (atrophy, mottled pigmentation, telangiectasia), and variants (folliculotropic, hypopigmented in skin of colour and children, granulomatous slack skin, pagetoid reticulosis).

Fig. 2 · Patches and plaques of mycosis fungoides on the trunk. A fixed, 'eczema-like' rash in sun-protected sites that fails steroids for months needs biopsy.

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Types to recognise

Patches & plaques

Trunk / hips. Site: Trunk.

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Plaque stage

Sharply defined, variable. Site: Trunk.

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Tumour stage

Ulcerated nodule. Site: Any.

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Sézary syndrome

Erythroderma. Site: Whole skin.

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Diagnosis and investigations

- **Multiple biopsies** (ideally after a 2–4-week washout of topical steroids) with immunohistochemistry and **TCR gene rearrangement** (clonality). Diagnosis often takes years.
- **ISCL/EORTC algorithm for early MF**. Diagnosis needs **4 points**: **Clinical:** 2 points for the basic criterion (persistent and/or progressive patches/plaques) plus 2 of the additional criteria (non-sun-exposed distribution; size or shape variation; poikiloderma), or 1 point for the basic criterion plus 1 additional; **Histopathology:** 2 points for a superficial lymphoid infiltrate plus both epidermotropism without spongiosis and lymphoid atypia, or 1 point for the infiltrate plus one of those; **Molecular:** clonal TCR rearrangement = 1 point; **Immunopathology:** CD2/3/5 <50% of T-cells, CD7 <10%, or epidermal/dermal discordance = 1 point
- **Staging (ISCL/EORTC TNMB)**:

STAGE	SKIN (T)	NOTES
IA	T1: patches/plaques **<10% BSA**	Life expectancy similar to the general population
IB	T2: **10% BSA**	
IIA	T1–2 with dermatopathic (N1–2) nodes	
IIB	**T3: 1 tumour (1 cm)**	"Advanced" begins at IIB
III	T4: erythroderma (80% BSA)	
IVA–IVB	Blood B2 (Sézary), N3 nodes, or visceral (M1)	

- Staging investigations (specialist): FBC + blood film/flow cytometry for Sézary cells, LDH, LFTs, HTLV-1 serology, and CT (or PET-CT) for stage IIA or as indicated. Biopsy suspicious nodes.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Chronic eczema / contact dermatitis	Patch test (e.g. mechanic).
	Psoriasis	Silvery scale, extensors.
	Tinea corporis	Scrape.
	Parapsoriasis	Small / large plaque (large can progress).
	Drug pseudolymphoma	Anticonvulsants, ACE-i / ARBs, **amlodipine**; dupilumab unmasking.
	Lichen planus / cutaneous lupus	Biopsy distinguishes.

Chronic eczema (including occupational contact dermatitis, relevant to a mechanic; patch test), psoriasis, tinea corporis (scrape), parapsoriasis (small- and large-plaque, where large-plaque can progress), pityriasis lichenoides, **drug-induced pseudolymphoma / MF-like drug eruptions** (reported with, among others, anticonvulsants, ACE inhibitors/ARBs, **amlodipine** and other calcium-channel blockers, and dupilumab-associated CTCL unmasking), lichen planus, cutaneous lupus.

Management

IP role

- A **rash not responding to an appropriate potent topical steroid over several weeks, especially in sun-protected sites, should be referred for biopsy**. Do not repeat or escalate steroids indefinitely, as happened for 12 months in Case 5. Refer **routinely to urgently depending on stage clues** (tumours, erythroderma, lymphadenopathy, systemic symptoms: urgent). Primary cutaneous lymphoma is managed in **specialist supra-network skin lymphoma MDTs**.
- Review drug triggers. Stopping a suspected culprit (e.g. amlodipine) and reviewing after 6–12 weeks may be appropriate before or alongside biopsy. This is a joint decision with the GP/dermatologist.
- Case 5 medicines: **canagliflozin (SGLT2 inhibitor)** has no link to MF but carries genital mycotic infection and DKA risk (check the patient's wellbeing). Metformin: none. Simvastatin + amlodipine: the BNF/MHRA cap simvastatin at **20 mg** with amlodipine (CYP3A4 interaction), so check the dose.

Specialist treatment (BAD/UKCLG Gilson 2019)

- **Early (IA–IIA): skin-directed therapy.** Potent/very potent topical steroid; **phototherapy** (narrowband UVB for patches, PUVA for thicker plaques; the course answer); topical **chlormethine gel** (Ledaga®); localised radiotherapy; topical bexarotene (not UK-licensed); and for many IA patients, "expectant" skin-directed care.

- **ADVANCED (IIB–IV)**
  Total-skin electron beam therapy; Systemic **bexarotene** (hypertriglyceridaemia and central hypothyroidism; needs a lipid-lowering agent and levothyroxine; **avoid gemfibrozil**); Low-dose **methotrexate**; Interferon-α; **Mogamulizumab** (anti-CCR4; NICE TA754: Sézary after 1 systemic therapy, MF stage IIB+ after 2); **Brentuximab vedotin** for CD30+ (NICE TA577); Extracorporeal photopheresis (Sézary); Chemotherapy; Allogeneic stem-cell transplant in selected patients
- Quality of life: **itch** (emollients, antihistamines, gabapentin/pregabalin, mirtazapine, aprepitant for refractory itch), and work and psychological impact.

Other primary cutaneous lymphomas

GROUP	EXAMPLES	BEHAVIOUR
CTCL	MF (and variants), **Sézary syndrome**	Sézary is aggressive
CD30+ lymphoproliferative disorders	**Lymphomatoid papulosis** (recurrent self-healing papules), **primary cutaneous anaplastic large-cell lymphoma**	Excellent prognosis
Others	Subcutaneous panniculitis-like TCL; primary cutaneous γδ TCL (aggressive); adult T-cell leukaemia/lymphoma (**HTLV-1**); extranodal NK/T-cell (nasal type)	Variable
CBCL	Primary cutaneous **marginal zone** and **follicle centre** lymphoma (indolent: head/trunk nodules); **diffuse large B-cell lymphoma, leg type** (older women, red nodules on the leg; aggressive)	Indolent vs aggressive

Treatment ladder

1 **Refer**

Rash not responding to an appropriate **potent topical steroid over several weeks**, esp. sun-protected sites biopsy. **Don't escalate indefinitely.** Routine urgent by stage clues; managed by **supra-network skin lymphoma MDTs**.

2 **Drug review**

Consider stopping a suspected culprit (e.g. amlodipine) and reviewing at 6–12 weeks, jointly with GP / dermatology.

3 **Specialist (early)**

Skin-directed: topical steroids, phototherapy; advanced disease: systemic therapy.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Dupilumab**: Biopsy atypical adult-onset ‘eczema’ before biologics: CTCL can be unmasked.

Referral

ROUTINE	Fixed patches or plaques in sun-protected sites failing treatment for months: biopsy.
URGENT	**Erythroderma with lymph nodes**, intense itch, weight loss: **Sézary syndrome**.
URGENT	Tumours or ulcerated nodules in known or suspected MF.

Red flags and when to refer

- Fixed patches or plaques in sun-protected sites that fail to respond to treatment over months: refer for **biopsy** (MF).
- Erythroderma with lymph nodes, intense itch and weight loss: **urgent** referral (Sézary syndrome).

Complications and prognosis

Complications

- Itch, infection (Staph), progression, large-cell transformation.

Prognosis

- Stage IA: normal life expectancy; advanced stages much poorer.

Prescriber pearls (UK pharmacist IP)

- Explain that early MF is usually a **chronic, indolent** condition (stage IA has near-normal life expectancy). Avoid the "advanced cancer" framing without staging.
- Cumulative UV raises skin cancer risk.
-

Dupilumab caveat: CTCL has been unmasked or accelerated in patients misdiagnosed with atopic dermatitis. In an atypical adult-onset "eczema" that is failing treatment, biopsy before biologics.

Self-check quiz

4 questions · optional

Q1 Classic distribution?

✓ Sun-protected 'bathing-trunk' sites.

Q2 Leukaemic form with erythroderma?

✓ Sézary syndrome.

Q3 Biopsy tip?

✓ Multiple biopsies after a 2–4-week steroid washout.

Q4 Why biopsy before dupilumab in atypical adult eczema?

✓ CTCL can be unmasked or accelerated.

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Photo gallery

Mycosis fungoides 3

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Sézary syndrome and histology

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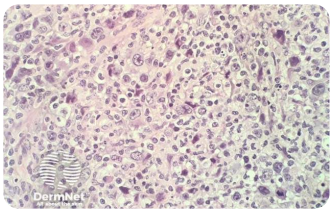

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Lumps, lesions & skin cancer Module 2 · Week 4

Melanocytic naevi (moles) & the changing mole

Common moles, how they evolve through life, benign dermoscopy patterns, and how to assess a changing mole with the NICE weighted 7-point checklist. Includes hormones, pregnancy, sunbeds, apps and mole mapping.

ALSO CALLED mole · junctional naevus · compound naevus · dermal naevus · changing mole · dermoscopy of naevi · +3

RELATED PAGES Melanoma: recognition & referral · Atypical naevus syndrome & familial melanoma · Congenital melanocytic naevi (CMN) · Halo naevus (Sutton naevus) · Blue naevus · Spitz and Reed naevi · Seborrhoeic keratosis

KEY FACTS

- **Melanocytic naevi (moles)** are benign clonal proliferations of melanocytes; most adults have 10–40.
- They evolve over life: **junctional** (flat) **compound** (raised) **intradermal** (soft, skin-coloured), and many fade after middle age.
- Common naevi are symmetrical, one or two colours, under about 6 mm, and match the person's other moles ('signature' naevi).
- Dermoscopy patterns (reticular, globular, homogeneous) help; the **ugly duckling** and changing lesion need assessment.
- Reassure typical naevi; refer any lesion meeting the NICE 7-point checklist or with clinical concern via the **suspected cancer pathway**.

Overview

Benign clonal proliferations of melanocytes ("moles"). They appear from early childhood and peak in number in the 20s–30s (a typical white adult has 10–40). Many involute after 50.

Dermoscopy of naevi

Dermoscopy (epiluminescence microscopy) shows structures in the epidermis and papillary dermis. In trained hands it raises sensitivity for melanoma by roughly 10–30% over the naked eye. In untrained hands it may not improve accuracy.

Hormones, pregnancy and sunbeds

Common patient questions about whether contraception, pregnancy, HRT or sunbeds change moles or melanoma risk.

Epidemiology

- **Who**
  - Naevi appear in childhood, peak in number in the 20s–30s and decline after 50.

  - More naevi with fair skin, childhood sun exposure and family history; acral naevi more common in darker skin.
- **Melanoma risk**
  - More than about 100 naevi, or several atypical naevi, is a major melanoma risk factor; most melanomas still arise de novo, not in a pre-existing mole.

Aetiology: causes and risk factors

- Most acquired naevi carry **BRAF V600E**, and congenital naevi carry **NRAS** mutations. After growth they enter **oncogene-induced senescence**, which is why most never become melanoma.
- **LIFE CYCLE**
  junctional (nests at the dermo-epidermal junction; flat, brown) **compound** (junctional + dermal; raised centre) **dermal (intradermal)** (dermal only; soft, skin-coloured to light-brown dome, often hair-bearing; adults).
- Drivers of naevus number: genetics (MC1R, family), **childhood sunburn and intermittent UV**, immunosuppression, BRAF inhibitors (eruptive naevi), and pregnancy/hormones (darkening rather than true growth in most cases).

Dermoscopy of naevi

Hormones, pregnancy and sunbeds

Risk factors and associations

- Somatic **BRAF V600E** (common naevi) or **NRAS** (congenital naevi) mutations start a clone, which then stops growing (oncogene-induced senescence).
- Sun exposure in childhood and genetics (number of naevi is strongly inherited).

Myths to address

> **"MOLES TURN INTO CANCER"**
> individual moles rarely do; the number of moles signals risk.

> **"HAIRS MEAN IT'S SAFE"**
> hair does not exclude melanoma.

Pathophysiology and pathology

A naevus is a clone of melanocytes that grows, then **switches itself off**.

1. **Founder mutation**
   A melanocyte acquires a **BRAF** or **NRAS** mutation and proliferates as nests at the dermo-epidermal junction (junctional naevus).
2. **Senescence**
   After a growth phase, the clone enters **oncogene-induced senescence** (p16) and stops enlarging: a stable mole.

3 **Maturation into the dermis**

Nests drop into the dermis (compound, then intradermal naevus); deeper cells become smaller and less pigmented.

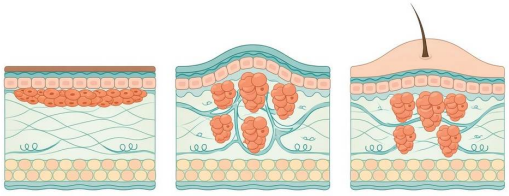

Fig. 1 · Life cycle of a naevus: junctional compound intradermal, then fading.

AI-generated illustration

4 **Involution**

Many naevi fade or become fibrous in older age; a new or changing pigmented lesion in an adult needs more caution.

How it works / pathophysiology

Pigment depth predicts colour: black is in the stratum corneum or epidermis; brown is at the dermo-epidermal junction; blue-grey is in the dermis (Tyndall effect). Network = pigmented rete ridges; globules = nests.

Evidence / pathophysiology

COCP

large meta-analyses show **no meaningful increase in melanoma risk**. Melanocytes express oestrogen receptors, and darkening of naevi or melasma can occur.

PREGNANCY

naevi may darken or enlarge on the abdomen and breasts (skin stretch). **Changing naevi in pregnancy still need assessment**, and pregnancy-associated melanoma is not more aggressive stage-for-stage, but diagnosis is often delayed.

HRT

observational studies show at most a small association. **No requirement to stop HRT** because of moles. The course "stop HRT if lesions increase" is not standard UK advice (see Where course teaching differs from UK guidance). Review the overall benefit-risk with the patient. For **people with a history of melanoma**, HRT is not contraindicated in most guidance (individualise with the oncology team).

SUNBEDS

IARC Group 1 carcinogen. Ever-use raises melanoma risk by ~20%, and **first use before 35 raises it by ~60–75%** (Boniol 2012; Colantonio 2014). Use also accelerates photoageing, and the risk rises with the number of sessions. Case 1 (3×/week for 18 months, about 230 sessions) represents substantial exposure.

HISTOPATHOLOGY

- Junctional: nests at the dermo-epidermal junction; compound: junctional plus dermal nests; intradermal: dermal nests only, showing **maturation** with depth.
- No atypia, symmetry and orderly maturation separate naevi from melanoma.

Clinical features

TYPE	LOOK	PHOTO
Junctional	Flat, uniformly brown, round	
Compound	Slightly raised, brown, may have a flat rim	
Dermal (intradermal)	Soft, dome-shaped, skin-coloured or pink-tan; may have a few fine vessels; face, trunk	. Case 1 diagnosis
Miescher naevus	Face; dome-shaped, firm, often hairy	
Unna naevus	Trunk/neck; papillomatous, "raspberry-like", soft	
Common mole	Symmetrical, regular border, one or two colours	

Benign change: a slow gradual elevation of a naevus with age (junctional dermal), lightening, and evenly spread darkening in pregnancy or adolescence.

Dermoscopy of naevi

Benign naevus patterns

> **RETICULAR**
> a regular network that fades at the periphery (junctional/compound; trunk).

> **GLOBULAR**
> regular brown globules (compound/dermal, children and young adults; the **Case 1** pattern).

> **Homogeneous** (blue = blue naevus; tan = dermal naevus).

> **STARBURST**
> symmetrical peripheral streaks, typical of a pigmented Spitz/Reed naevus (in adults it needs excision because it mimics melanoma).

> **Mixed reticular-globular** with symmetry.

> Dermal naevus vessels: **comma vessels, out of focus** ("hazy", because they are deeper).

Atypical / melanoma-specific features

Atypical network, irregular streaks, irregular dots/globules (especially peripheral), blue-white veil, regression structures (white scar-like, peppering), negative network, atypical vessels (dotted, polymorphous), and more than one pattern asymmetrically.

Naevus vs BCC (the course Case 1 point): **BCC arborising vessels are bright red and sharply in focus** on the surface, and there is **no pigment network**. Dermal naevus vessels are comma-shaped and blurred, with residual globules or homogeneous tan.

Hormones, pregnancy and sunbeds

Law and regulation

- **SUNBEDS (REGULATION) ACT 2010**
  under-18s are banned from using sunbeds in commercial premises. Scotland (Public Health etc. (Scotland) Act 2008 and the 2009 regulations) and Northern Ireland (Sunbeds Act (NI) 2011) have equivalent bans, plus supervision and information requirements. At 19, Case 1 is legally entitled to use them, so counselling is the lever.

Mole mapping, apps and AI

What they are

> **MOLE MAPPING**
> total-body photography ± sequential digital dermoscopy (specialist or private). Most useful for high-risk people (AMS, FAMMM), and it reduces unnecessary excisions.

> **SMARTPHONE APPS AND AI**
> many consumer apps are unregulated or have poor sensitivity. Under UK MDR 2002 and MHRA rules, apps that diagnose are medical devices. Some UKCA/CE-marked AI tools, e.g. **DERM (Skin Analytics)**, are used within NHS teledermatology pathways with dermatologist oversight. NICE early value assessment HTE24, now HealthTech guidance **HTG746**, supports **conditional use of DERM** for triaging adults already referred on the urgent suspected skin cancer pathway, with evidence generation over 3 years. People with black or brown skin (Fitzpatrick V–VI) need healthcare-professional review.

Limitations

False reassurance is the main harm. Amelanotic and nodular melanomas are often missed by both apps and ABCDE.

Other naevus types

NAEVUS	KEY FEATURES	MANAGEMENT
Recurrent naevus ("pseudomelanoma")	Pigment returning in a scar after a shave or incomplete excision	Refer **with the previous histology**; it mimics melanoma on dermoscopy
Naevus spilus (speckled lentiginous)	Tan patch with darker speckles	Monitor; small melanoma risk
Acral naevus	Parallel-**furrow** pattern on dermoscopy	Parallel-**ridge** pattern = melanoma (see melanoma)

Special naevi have their own pages: halo naevus, blue naevus and Spitz and Reed naevi.

Fig. 2 · Compound naevus: slightly raised centre with a flat pigmented rim (junctional plus dermal components).

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Types to recognise

Junctional

Flat, even brown. Site: Trunk, limbs.

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Compound

Raised centre. Site: Any.

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Dermal

Soft pale dome. Site: Face, trunk.

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Unna naevus

Papillomatous. Site: Trunk, neck.

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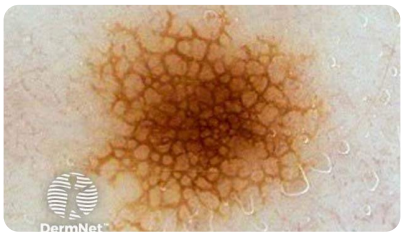
Reticular dermoscopy

Regular fading network. Site: Dermoscopy.



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Atypical dermoscopy

Irregular network. Site: Dermoscopy.

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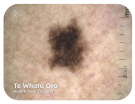
Special groups

- **Pregnancy**
  - Naevi may darken slightly, but a clearly changing mole needs the same assessment as at any other time.
- **Children**
  - New naevi are normal; childhood melanoma is rare and often amelanotic or nodular (use modified ABCD criteria).

Diagnosis and investigations

- History of change (who noticed it, over what time, photos?), the **NG12 weighted 7-point checklist**, the **ABCDE** rule, and the **"ugly duckling" sign**. See the describing skin lesions notes.
- **Dermoscopy** (trained clinician; see Dermoscopy of naevi; naevus vs BCC; biopsy rules).
- **Excision biopsy** (2 mm margin, full thickness) if melanoma cannot be excluded.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	**Melanoma**	Change, asymmetry, multiple colours; EFG nodule.
	Seborrhoeic keratosis	Stuck-on, waxy.
	Dermatofibroma	Firm, dimples on pinching.
	BCC	Pink pearly, for a pink dermal naevus.
	Neurofibroma	Soft, buttonhole sign.

Melanoma (SSM or nodular), SK, dermatofibroma, BCC (pigmented or nodular, for a pink dermal naevus), neurofibroma (soft, buttonhole sign), and molluscum.

Dermoscopy of naevi

See Acquired melanocytic naevi and the melanoma notes.

Diagnosis / differentials

See Acquired melanocytic naevi and Dermoscopy of naevi; naevus vs BCC; biopsy rules. Melasma, pregnancy-related hyperpigmentation and new SKs are non-naevus differentials.

Management

- **APPLY THE NG12 WEIGHTED 7-POINT CHECKLIST**
  Major features (2 points each): change in size, irregular shape, irregular colour; Minor features (1 point each): diameter 7 mm, inflammation, oozing, change in sensation; **3 points suspected cancer pathway** (NG12 1.7.1); **Below 3: still refer if there is any concern** (the score is a minimum, not a ceiling)
- **Case 1:** change in size = 2, which is below 3. But the history is a growing mole in a heavy sunbed user. If the IP cannot confidently diagnose a dermal naevus with dermoscopy, **refer** (routine dermatology or teledermatology with dermoscopic) and **photograph with a ruler**.
- **Cosmetic removal** of benign naevi is generally not NHS-funded. Shave removal of a dermal naevus for cosmetic reasons is private and done by a dermatologist or surgeon who sends the specimen for histology.

Dermoscopy of naevi

- **Use dermoscopy only if trained and competent.** Otherwise send good dermoscopic images via teledermatology (many ICSs run "photo-triage" for 2WW skin referrals) or refer.
- **BIOPSY RULES FOR SUSPECTED MELANOMA**

excision biopsy with a 2 mm clinical margin and a cuff of fat, oriented along the limb's long axis (to preserve options for wide local excision and SLNB). **Do not shave, punch or incise** a suspected melanoma or a naevus of uncertain nature in primary care, because partial biopsy risks sampling error and makes Breslow thickness impossible to measure. Exceptions (specialist only): large facial lentigo maligna or acral/nail lesions, where incisional biopsy is accepted.

- The NICE IOG says that **lesions suspected of melanoma should be referred, not excised, in primary care**.

Hormones, pregnancy and sunbeds

> **COCP**
> continue. No need to change the method because of moles. Melasma on the COCP: consider a progestogen-only or non-hormonal method if it is troublesome (UKMEC).

> **Sunbed counselling (NICE NG34 and NICE PH32 on skin cancer prevention):** explain the risk with concrete numbers. Use brief-intervention techniques (relate to appearance; photoageing and "self-tan alternatives"); **fake tan / DHA products are safe alternatives but do not protect** against UV. Sun protection: shade 11:00–15:00, clothing, SPF 30+ with 4–5-star UVA, reapply.

> **VITAMIN D**
> heavy UV avoidance should not lead to deficiency. Consider 10 micrograms (400 IU) daily, especially October–March (NICE NG34/PH56).

Mole mapping, apps and AI

- Recommend **monthly self-examination** with dated, rather than relying on an unregulated app. **A reassuring app result never overrides clinical concern.** Conversely, an app "flag" alone justifies professional review, not panic (the melanoma notes Case 1).
- **LIFESTYLE**
  sun protection, no sunbeds, stop smoking (general health; not a melanoma risk factor), vitamin D. **Diet and supplements do not prevent melanoma**, although nicotinamide does prevent keratinocyte cancer in high-risk patients (the skin cancer notes). Evidence for nicotinamide in melanoma prevention is lacking.

Worked cases

CASE 1

19-year-old woman, long-standing back mole growing over 6 months; sunbeds 3×/week for 18 months; COCP

> **DIAGNOSIS (COURSE TEACHING, FROM THE LOST)**
> benign **dermal (intradermal) naevus**. Dermoscopy: globular/homogeneous pattern with blurred (out-of-focus) comma vessels, **not** the sharp arborising vessels of BCC.

> **MANAGEMENT**
> reassurance + **sunbed and sun-protection counselling** + self-monitoring with; continue the COCP. **Never partially biopsy** a naevus.

> **UK IP CAVEAT**
> NG12 score = 2 (change in size), below the threshold. But without competent dermoscopy, **refer** (teledermatology with dermoscopic, or routine dermatology) rather than reassure. The under-18 sunbed ban doesn't apply; counselling does.

CASE 2

5-year-old, 30 cm trunk CMN becoming darker, rougher, larger

DIAGNOSIS

large **giant** (projected adult size >40 cm) congenital melanocytic naevus.

MELANOMA RISK

roughly 2–5% lifetime overall for large/giant (up to 10–15% with satellites/multiple CMN). The course 5–10% sits in this range. **Small/medium CMN: <1%.**

PLAN

paediatric dermatology for **lifelong surveillance**; urgent review of any nodule; MRI if multiple CMN or neurological signs; MDT decisions about surgery (which does not reduce melanoma risk); psychological support. Roughness and hairiness are benign maturation.

CASE 3

30-year-old PE teacher, mother had melanoma, blistering childhood sunburn

DIAGNOSIS

atypical naevus syndrome. **FAMMM** if 2 affected relatives (or a CDKN2A mutation).

RISK

substantially raised. >100 naevi is significant; course teaching gave a lifetime risk of ~15%. CDKN2A carriers carry much higher risk.

PLAN

dermatology for dermoscopy + total-body photography + **digital follow-up every 6–12 months**; consider genetics (ask about pancreatic cancer); self-examination; sun protection at work; no sunbeds.

Treatment ladder

1 **7-point checklist**

Major (2): change in size, irregular shape, irregular colour. Minor (1): 7 mm, inflammation, oozing, sensation change. **3 suspected cancer pathway**; **< 3: still refer if any concern** (a minimum, not a ceiling).

2 **Uncertain**

Can't confidently call it benign with dermoscopy refer (routine / teledermatology) with dermoscopic images and a ruler.

3 Cosmetic

Cosmetic removal generally not NHS-funded; specimens always to histology.

4 Prevention

No sunbeds (**illegal < 18**); SPF ~2 mg/cm² (6–8 teaspoons whole body), reapply; monthly self-checks with photos.

Referral

URGENT	7-point score **3** or strong suspicion: suspected cancer pathway.
ROUTINE	Ugly duckling or a **new mole after 40**: low threshold to refer.
ROUTINE	New / eruptive naevi on immunosuppression or BRAF inhibitors that look atypical.

Red flags and when to refer

- NICE weighted 7-point checklist score of **3 or more**, or strong suspicion: **suspected cancer pathway** referral (NG12).
- A mole that stands out from the patient's others (ugly duckling), or a new mole after 40: low threshold to refer.
- Under-18s cannot legally use sunbeds in the UK (Sunbeds (Regulation) Act 2010); advise everyone to avoid them.

Complications and prognosis

Complications

- Irritation, trauma or inflammation (folliculitis within a mole).
- Rarely melanoma arising in a naevus.
- Halo phenomenon (depigmented ring).

Prognosis

- Benign and stable; most change slowly with age. Removal is only for diagnosis or symptoms; cosmetic removal is not NHS-funded.

Prescriber pearls (UK pharmacist IP)

- Teach monthly self-examination using a partner or mirror, and photographs on the phone.
- Immunosuppressed patients (transplant, biologics, methotrexate) and those on **BRAF inhibitors** can develop new or eruptive naevi. Refer lesions that look new or atypical.
- Patients ask "can't you just shave it off?" Explain that removing a whole mole is necessary for an accurate diagnosis, and that cosmetic shaves may leave pigment that regrows (recurrent naevus, which then mimics melanoma.
- "A tan is your skin's damage response." People with many moles, red hair or freckles, a family history, or a history of sunburn should never use sunbeds.
- Check that sunscreen is applied in an adequate amount (about 2 mg/cm²: roughly 6–8 teaspoons for the whole adult body) and reapplied. Most people apply a quarter to a half of this.

Dermoscopy of naevi

Hormones, pregnancy and sunbeds

Mole mapping, apps and AI

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK GUIDANCE / EVIDENCE	TEACHING POINT
Case 1 reassurance	Reassure (benign dermal naevus)	NG12 change = 2 points; refer if any concern. Reassure only after competent dermoscopy	IP without dermoscopy refers
HRT and naevi	"Stop HRT if lesions increase"	No UK guideline recommends stopping HRT because of naevi; evidence of association is weak	Individualise; review change on its own merits
CMN risk	5–10% applied to small/medium CMN	Small/medium **<1%**; large/giant ~2–5% (up to 10–15% in highest-risk)	Risk depends on size and satellites
CMN laser		Not recommended routinely; deep cells remain; surveillance harder	Specialist only
Surveillance interval in AMS	6–12 months	UK high-risk clinics typically use 3–12 months, risk-tailored (sequential digital dermoscopy at 3–6 months for single lesions)	Broadly aligned
Lifetime risk in FAMMM	~15%	AMS without a mutation: several-fold relative risk. **CDKN2A carriers: about 30–70% by 80**	Distinguish phenotype from genotype

Self-check quiz

4 questions · optional

Q1 Referral threshold on the 7-point checklist?

✓ 3 or more (but refer below 3 if concerned).

Q2 Mole life cycle?

✓ Junctional compound dermal.

Q3 Mutation in most acquired naevi?

✓ BRAF V600E.

Q4 Can under-18s use sunbeds in the UK?

✓ No (Sunbeds (Regulation) Act 2010).

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Photo gallery

Common naevi

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Dermoscopy of naevi

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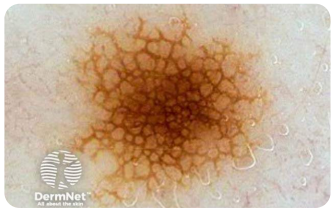

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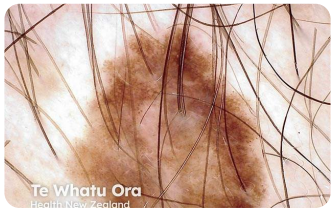

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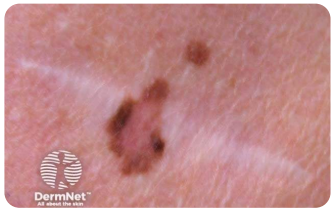
Recurrent naevus in a scar

1



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Lumps, lesions & skin cancer Module 2 · Week 4

Congenital melanocytic naevi (CMN)

Moles present at birth. Melanoma risk depends on size: small and medium CMN carry a low (<1%) lifetime risk, large and giant ones about 2–5%, higher with many satellites. Lifelong surveillance and support matter more than surgery.

ALSO CALLED CMN · birthmark mole · giant congenital naevus · projected adult size · satellite naevi · neurocutaneous melanosis

RELATED PAGES Melanocytic naevi (moles) & the changing mole · Atypical naevus syndrome & familial melanoma · Melanoma: recognition & referral · Pigmented & epidermal birthmarks · Sebaceous naevus (naevus sebaceous)

KEY FACTS

- **Congenital melanocytic naevi (CMN)** are present at birth or appear in the first months; about 1% of newborns.
- Classified by projected adult size: small (<1.5 cm), medium, large (20–40 cm), giant (>40 cm).
- Melanoma risk is low for small and medium CMN, higher for **large/giant** (a few per cent, often in childhood).
- Multiple CMN or large CMN: MRI for **neurocutaneous melanosis** (NRAS mutations).
- Monitor with photographs; refer new nodules, ulceration or rapid change; support families (Caring Matters Now).

Overview

A melanocytic naevus present at birth (or appearing in the first year, "tardive"). About 1% of newborns have one; large/giant ones occur in about 1 in 20,000.

Epidemiology

- **Who**
  - About 1% have a small CMN; large or giant CMN about 1 in 20,000 births.

Aetiology: causes and risk factors

Post-zygotic NRAS mutation in neural-crest melanoblasts (BRAF in some small CMN). The earlier the mutation, the larger the naevus and the more likely satellites and neurocutaneous melanosis (CNS melanocytes).

Myths to address

- **"ALL CONGENITAL MOLES SHOULD BE REMOVED"**
  most small CMN never cause problems; surgery has its own risks.

Pathophysiology and pathology

CMN arise from a **mutation in an embryonic melanocyte precursor**, mostly NRAS.

1. **Embryonic mutation**
   A post-zygotic NRAS (or BRAF) mutation in a migrating melanocyte precursor.
2. **Clonal expansion**
   Descendant melanocytes populate a patch of skin, sometimes deep into fat and fascia.

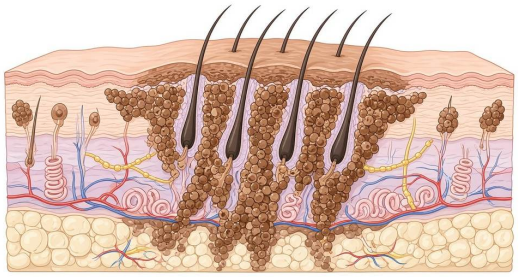

Fig. 1 · Congenital naevus cells extend deep into the dermis and fat, wrapping around follicles, glands, vessels and nerves.
AI-generated illustration

3. **Size and depth**
   Earlier mutation gives larger, deeper naevi and sometimes CNS involvement.
4. **Evolution**
   CMN become more raised, hairy and textured with age; some lighten.

HISTOPATHOLOGY

- Naevus cells extending deep into dermis and around adnexa, vessels and nerves.

Clinical features

- **SIZE BY PROJECTED ADULT SIZE (PAS)**
  small <1.5 cm; medium 1.5–20 cm (M1 1.5–10 cm, M2 10–20 cm); **large 20–40 cm**; **giant >40 cm**
- Head lesions roughly double, and trunk and limb lesions roughly triple, from infancy to adulthood. **A 30 cm trunk CMN at age 5 projects to >40 cm, so it is a giant CMN** (Case 2.
- Features: brown to black, often hairy, with varied texture. **They become rougher, lumpy, hairier or lighter with time**. These are benign maturation changes (the course point), but a new **firm nodule, ulceration, rapid growth or pain** needs urgent assessment. Also proliferative nodules (benign) occur in infancy.
- **Satellite naevi** and **multiple CMN** (2 at birth) are markers for **neurocutaneous melanosis/CNS involvement** (seizures, developmental delay, hydrocephalus).

Fig. 2 · Medium CMN (1.5–20 cm projected adult size): textured, darker, with some hair. Melanoma risk is low (around 1% or less), so monitor and photograph.

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Types to recognise

Giant CMN

> 40 cm PAS, satellites. Site: Trunk.

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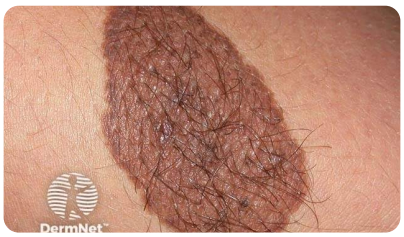
Medium CMN

1.5–20 cm PAS. Site: Any.



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Hairy CMN

Coarse hair: maturation. Site: Any.

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Irregular outline

Monitor with photos. Site: Limb.

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Diagnosis and investigations

Clinical, with serial photographs. Biopsy of new nodules. **MRI of brain and spine** (ideally in the first 6 months of life, before myelination, under natural sleep) in children with **2 CMN at birth, of any size** (the current UK expert approach; Kinsler). It is not routine for a single CMN without neurological signs.

Photos

- Serial photographs.

Biopsy

- Any new nodule.

MRI

- Brain + spine, ideally < 6 months (natural sleep), if **2 CMN at birth, any size**; not routine for a single CMN without neurology.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Café-au-lait macule	Flat, no hair, uniform.
	Becker naevus	Acquired, adolescent shoulder, hairy.
	Dermal melanocytosis	Blue-grey sacral ('Mongolian spot').

	CONDITION	DISTINGUISHING FEATURES
	Naevus spilus	Tan patch with darker speckles.
	Plexiform neurofibroma	Soft 'bag of worms' with pigment.

Café-au-lait macule (flat, no hair, uniform), Becker naevus (acquired, adolescent shoulder, hairy), dermal melanocytosis ("Mongolian spot", blue-grey, sacral), speckled lentiginous naevus (naevus spilus), smooth muscle hamartoma, and plexiform neurofibroma with overlying pigment.

Management

> **MELANOMA RISK BY SIZE (CORRECTED; RELATED QUESTION)**
>
> **Small and medium CMN: lifetime melanoma risk is low (<1%)**, similar to or slightly above the general population, and melanoma usually arises after puberty at the junctional edge; **Large/giant CMN: around 2–5% overall (up to 10–15% in the highest-risk group: >40 cm PAS with many satellites/multiple CMN)**. Melanoma often arises in childhood and **deep** in the naevus or in the CNS, which is why surface excision does not remove the risk; The course "5–10%" belongs to the large/giant group, **not** to small/medium CMN

> **Refer** (routine to paediatric dermatology; urgent for any new nodule, ulcer or rapid change) for **lifelong surveillance** and MDT input (dermatology, plastics, neurology, psychology).

> **PROPHYLACTIC EXCISION**
>
> no evidence that it reduces melanoma risk in large/giant CMN (the course point). Surgery is for function, cosmesis or symptoms, as an MDT decision. **Laser** lightens the surface but leaves deep naevus cells and can make surveillance harder. It is not recommended routinely (specialist only).

> **SUPPORT**
>
> itch (emollients; antihistamines at night if needed); dry, fragile skin; psychological impact and bullying. Point families to Caring Matters Now (UK charity for CMN).

> Sun protection, and **teach parents what change to report**.

Treatment ladder

1

Explain risk

Small / medium: lifetime melanoma risk < 1% (usually after puberty, at the edge). **Large / giant: ~2–5%** (up to 10–15% if > 40 cm with many satellites); often **childhood and deep / CNS**.

2

Refer

Paediatric dermatology (routine; **urgent** for nodule, ulcer, rapid change) for lifelong MDT surveillance (dermatology, plastics, neurology, psychology).

3 **Support**

Teach families what change to report; photograph regularly; sun protection; psychological support.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Prophylactic excision**: No evidence it reduces melanoma risk in large / giant CMN.

Referral

URGENT	New firm nodule, ulceration, rapid growth or pain within a CMN.
SAME DAY	Scalp or multiple CMN with headaches, vomiting or seizures: same-day paediatrics.
ROUTINE	2 CMN at birth or large / giant with satellites: MRI and specialist surveillance.

Red flags and when to refer

- A new firm nodule, ulceration, rapid growth or pain within a CMN: **urgent** referral.
- Two or more CMN at birth, or a large/giant CMN with satellites: refer for MRI and specialist surveillance.

Complications and prognosis

Complications

- Melanoma (large/giant), neurocutaneous melanosis, psychosocial impact; proliferative nodules in infancy (usually benign).

Prognosis

- Lifelong; most remain benign.

Prescriber pearls (UK pharmacist IP)

The rough, hairy, raised changes the parent describes are expected. A **hard lump** is not.

If scalp CMN or multiple CMN plus headaches, vomiting or seizures, arrange **same-day** paediatric review.

Self-check quiz

4 questions · optional

Q1 Giant CMN threshold?

✓ > 40 cm projected adult size.

Q2 A 30 cm trunk CMN at age 5: category?

✓ Giant (trunk lesions ~triple).

Q3 Who needs MRI?

✓ Infants with 2 CMN at birth (any size).

Q4 Does prophylactic excision remove the risk?

✓ No.

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1. DermNet: Congenital melanocytic naevi — https://dermnetnz.org/topics/congenital-melanocytic-naevi
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emedicine.medscape.com

Photo gallery

Congenital melanocytic naevi 4

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Lumps, lesions & skin cancer Module 2 · Week 4

Atypical naevus syndrome & familial melanoma

Many large, irregular moles with a personal or family history of melanoma. Know who to refer for surveillance and genetics (CDKN2A), and what to teach about self-examination and sun protection.

ALSO CALLED dysplastic naevus syndrome · FAMMM · CDKN2A · multiple atypical moles · familial melanoma · surveillance

RELATED PAGES Melanocytic naevi (moles) & the changing mole · Melanoma: recognition & referral · Congenital melanocytic naevi (CMN) · Teledermatology & lesion photography

KEY FACTS

- **Atypical (dysplastic) naevi** are moles that are larger (often >5 mm), irregular or variegated, but benign.
- Atypical mole syndrome many naevi (often >100) including several atypical ones: a strong melanoma risk factor.
- Familial melanoma 2 or more first-degree relatives with melanoma; CDKN2A, CDK4, BAP1 mutations; pancreatic cancer link with CDKN2A.
- Manage with self-examination, total-body photography and dermoscopy surveillance, sun protection.
- Refer any changing or 'ugly duckling' lesion via the suspected cancer pathway; genetics referral for familial clusters.

Overview

ATYPICAL (DYSPLASTIC) NAEVI

clinically "funny-looking" naevi (5 mm, irregular or ill-defined border, variable colour, flat component. Histologically they show architectural disorder ± cytological atypia.

ATYPICAL MOLE SYNDROME (AMS) PHENOTYPE

many naevi (often >100, or >50 in younger people), several atypical naevi, and naevi in unusual sites (scalp, buttocks, feet, iris).

FAMMM (familial atypical multiple mole melanoma syndrome): AMS phenotype + **melanoma in 1–2 first- or second-degree relatives**. Germline **CDKN2A** (p16) in about 20–40% of melanoma-prone families; CDK4 is rarer. FAMMM is covered in a worked case in the melanoma notes too.

Inherited melanoma and syndromes

About 5–10% of melanomas occur in people with a family history; a smaller proportion carry a **high-penetrance germline mutation**. **FAMMM** = many (often atypical) naevi + melanoma in relatives (the naevi notes see AJCC 8th edition staging and prognosis).

Epidemiology

- **Who**
  - Atypical naevi in about 5–10% of fair-skinned adults; inherited atypical mole syndrome is rarer.

Aetiology: causes and risk factors

Inherited susceptibility (CDKN2A, CDK4, MC1R variants, MITF E318K, BAP1, POT1, TERT) **plus UV**, especially intermittent intense exposure and **blistering childhood sunburn**.

Inherited melanoma and syndromes

Myths to address

- **"ATYPICAL MOLES SHOULD ALL BE REMOVED"**
  removal does not prevent melanoma; surveillance does detect it early.

Pathophysiology and pathology

Atypical naevi reflect a **susceptible melanocyte field** with more naevi and more mutations.

1. **Genetic susceptibility**
   Inherited variants (CDKN2A, MC1R, naevus-count genes) increase melanocyte proliferation.
2. **Many naevi**
   UV and genetics produce numerous naevi, some architecturally disordered.
3. **Ugly duckling sign**
   A lesion that differs from a person's 'signature' naevi needs assessment.

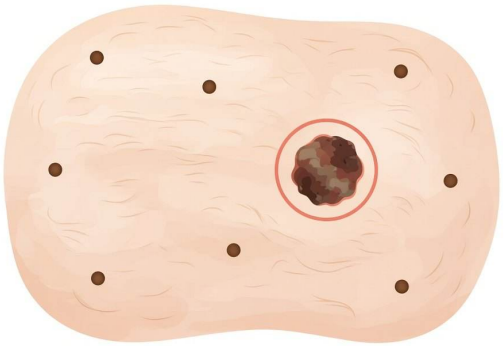

Fig. 1 · The ugly duckling: the lesion that looks different from the rest.
AI-generated illustration

4 **Surveillance**

Total-body photography and sequential dermoscopy detect change early.

Fig. 2 · Mole mapping: baseline photographs for comparison.

AI-generated illustration

How it is caused / genes

GENE / SYNDROME	PENETRANCE / FEATURES	OTHER CANCERS / NOTES
CDKN2A (p16INK4a / p14ARF)	Commonest high-penetrance gene (~20–40% of melanoma-prone families). Early-onset, multiple primaries	**Pancreatic cancer** (lifetime ~15–25% in some families; e.g. the p16-Leiden variant). UK pancreatic surveillance through EUROPAC/specialist programmes
CDK4	Rare; similar phenotype	–
BAP1 tumour predisposition	Cutaneous and **uveal** melanoma, BAP1-inactivated melanocytic tumours	Mesothelioma, renal cancer
POT1, TERT, ACD, TERF2IP	Telomere genes	Gliomas (POT1)
MC1R variants	Red hair, freckles, fair skin (**Irish ancestry**, Case 2)	Moderate penetrance; multiplies CDKN2A risk
MITF E318K	Moderate risk	Renal
Xeroderma pigmentosum	Defective nucleotide excision repair; extreme photosensitivity; melanoma risk >2000× before age 20	SCC/BCC in childhood; ocular; neurological. National XP service (Guy's)
Werner syndrome (WRN)	Progeroid; **acral/mucosal melanoma**	Sarcomas, thyroid
Cowden / PTEN hamartoma	Modestly raised melanoma risk	Breast, thyroid, endometrial, renal, colorectal (the skin lumps notes)
Noonan syndrome with multiple lentigines (LEOPARD; **PTPN11**)	Multiple lentigines; **not** a major melanoma predisposition	Cardiac (HCM), deafness
Li–Fraumeni, hereditary retinoblastoma (RB1), BRCA2	Small increases	–
c-KIT	Somatic (not germline) in acral/mucosal/CSD melanoma	Imatinib-sensitive in selected advanced cases

HISTOPATHOLOGY

- Architectural disorder, bridging nests, lamellar fibroplasia, variable cytological atypia; graded by pathologists.

Clinical features

Many naevi on the trunk and limbs, of varied sizes and colours. Look for the **"ugly duckling"**.

Inherited melanoma and syndromes

Clinical features

Early age at onset (<40), multiple primary melanomas, several affected relatives, pancreatic cancer in the family, and the AMS phenotype.

Fig. 3 · Back with numerous naevi of varying size, shape and colour. With melanoma in a first-degree relative and childhood burns this is high risk, so refer for baseline dermoscopy/ photography (mole mapping).

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Types to recognise

Many naevi

Varied size, shape, colour. Site: Back.

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Atypical naevi

Mark for comparison. Site: Back, flank.

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Irregular naevus

Variable pigment. Site: Trunk.

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Eccentric dark area

Dysplastic naevus. Site: Trunk.

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Diagnosis and investigations

Full-skin examination, a naevus count, family history (melanoma, **pancreatic cancer**, which suggests CDKN2A), personal history, and dermoscopy. Total-body photography plus sequential digital dermoscopy (specialist).

Inherited melanoma and syndromes

Detailed three-generation pedigree. Refer to **clinical genetics** for germline testing (NHS National Genomic Test Directory criteria). A common UK rule of thumb: **3 melanomas** in an individual or family (or **2 in a high-incidence country** in some guidelines), or **melanoma + pancreatic cancer**, or melanoma + mesothelioma/uveal melanoma (BAP1).

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Common naevi, high count	Uniform, no atypia.
	Melanoma	Individual changing / outlier lesion.
	Solar lentigines	Flat, sun-exposed, older.
	Sporadic multiple melanoma	High UV, fair skin, no family pattern.

Common naevi in a high-naevus-count person; melanoma (individual lesion); solar lentigines (flat, sun-exposed, older).

Inherited melanoma and syndromes

Multiple sporadic melanomas from high UV exposure in a fair-skinned person; a common naevus phenotype.

Management

> **Lifetime melanoma risk.** Each of the following raises it: Having >100 naevi (relative risk ~7); Atypical naevi; **One first-degree relative with melanoma** (about doubles the risk); **FAMMM with CDKN2A** (cumulative risk of roughly 30–70% by age 80, depending on country). The course "~15%" is a reasonable mid-range figure for AMS without a known mutation

> **Refer** (routine) to dermatology for assessment: a person with an **AMS phenotype**, **2 first-degree relatives with melanoma**, or a personal history of melanoma. Refer to **clinical genetics** for 3 melanomas in the family or individual, melanoma + pancreatic cancer in the family, or multiple primary melanomas (UK cancer-genetics criteria).

> **SURVEILLANCE (SPECIALIST)**
> total-body photography ("mole mapping") and **sequential digital dermoscopy every 3–12 months** (course teaching said 6–12) in high-risk clinics. Excise only lesions that change. **Do not prophylactically excise** atypical naevi.

> **NICE NG14**
> give people at increased risk **information on self-examination, sun protection and what changes to report**, and provide psychological support.

> **PRIMARY PREVENTION**
> sun protection, no sunbeds, vitamin D advice (see Hormones, pregnancy and sunbeds). **Children of FAMMM families** should have sun protection from birth and be examined from adolescence.

Inherited melanoma and syndromes

- Refer the 17-year-old to **dermatology** (total-body photography plus sequential digital dermoscopy) and to **clinical genetics** (father and grandfather with early melanoma plus other family cancers).

- Counsel: strict sun protection, **no sunbeds** (the under-18 ban applies to him in commercial premises), monthly self-examination, and vitamin D.
- **PANCREATIC CANCER**
  CDKN2A carriers may be offered surveillance (MRI/EUS) in research or specialist programmes, usually from 40–50 or 10 years before the youngest case. **Smoking cessation is critical** because smoking multiplies the pancreatic risk.

Treatment ladder

1 **Refer**

Dermatology (routine): AMS phenotype, **2 first-degree relatives with melanoma**, or personal melanoma (NG14). **Clinical genetics**: criteria above.

2 **Specialist surveillance**

Total-body photography + **sequential digital dermoscopy every 3–12 months**; CDKN2A families: pancreatic surveillance programmes (e.g. EUROPAC).

3 **Prevention**

Shade, UV clothing, hat, **SPF 30–50 reapplied every 2 h**; no sunbeds; no smoking. Outdoor workers: employer duties (HSE).

4 **Self-checks**

Monthly **whole-skin** checks with partner / photos: most AMS melanomas arise de novo.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Genetic testing & insurance**: UK Code: insurers can't require predictive results (except very large policies / Huntington's).

Referral

ROUTINE — Multiple atypical naevi + family history of melanoma: **dermatology surveillance**.

ROUTINE Melanoma and pancreatic cancer in the same family: **clinical genetics** (CDKN2A).

URGENT Any changing outlier lesion: suspected cancer pathway.

Red flags and when to refer

- Multiple atypical naevi with a family history of melanoma: refer to a **dermatology surveillance** service (NICE NG14).
- Melanoma and pancreatic cancer in the same family: refer to **clinical genetics** (CDKN2A).

Complications and prognosis

Complications

- Melanoma (lifetime risk greatly increased in familial disease).

Prognosis

- Lifelong surveillance.

Prescriber pearls (UK pharmacist IP)

The PE teacher has an outdoor occupation: shade during outdoor lessons, long-sleeved UV tops, a hat, SPF 30–50 reapplied every 2 h and after sweating. The employer has duties (HSE outdoor workers' advice).

Explain that **most melanomas in AMS arise de novo, not from existing moles**. That is why whole-skin self-checks matter more than watching one mole.

Genetic testing affects family members and insurance (UK Code on Genetic Testing and Insurance: insurers can't require predictive test results except for very large policies and Huntington's). Arrange genetic counselling before testing.

Inherited melanoma and syndromes

Self-check quiz

4 questions · optional

Q1 Cancer linked to CDKN2A besides melanoma?

✓ Pancreatic adenocarcinoma.

Q2 BAP1 is linked with which melanoma?

✓ Uveal (plus mesothelioma, renal cancer).

Q3 Do most AMS melanomas arise in existing moles?

✓ No: most arise de novo.

Q4 Surveillance tools?

✓ Total-body photography + sequential digital dermoscopy.

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1. DermNet: Atypical melanocytic naevus — https://dermnetnz.org/topics/atypical-melanocytic-naevus
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emedicine.medscape.com

Photo gallery

Atypical naevi 5

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Lumps, lesions & skin cancer Module 2 · Week 4

Halo naevus (Sutton naevus)

A mole surrounded by a symmetrical white ring as the immune system clears it; common in children and teenagers and linked with vitiligo. A new halo in an adult over 40, or an atypical centre, needs assessment for melanoma.

ALSO CALLED halo naevus · Sutton naevus · leukoderma acquisitum centrifugum

RELATED PAGES Melanocytic naevi (moles) & the changing mole · Blue naevus · Spitz and Reed naevi · Melanoma: recognition & referral · Hypopigmentation & skin of colour

KEY FACTS

- A **halo naevus** is a mole with a symmetrical white ring caused by immune destruction of melanocytes.
- Common in children and teenagers (about 1%); often on the back; may be multiple.
- Associated with **vitiligo**; the centre may disappear and the ring later repigment.
- Reassure if the central naevus is regular; sun-protect the white skin.
- Refer a **new halo in an adult over 40** or any atypical centre (melanoma can provoke halos).

Overview

A **halo naevus** is a melanocytic naevus surrounded by a symmetrical **depigmented ring**, caused by immune destruction of the naevus. It is common in children and teenagers and may be multiple.

Aetiology: causes and risk factors

Cytotoxic T cells recognise naevus melanocytes and destroy them and the surrounding normal melanocytes; there is an **association with vitiligo**.

Myths to address

- **"THE RING MEANS CANCER"**
  in young people it is a sign of benign immune clearance.

Pathophysiology and pathology

The immune system **clears the naevus** and the melanocytes around it.

1. **Immune recognition**
   CD8+ T cells recognise antigens on naevus melanocytes.

2 **Destruction**

Naevus cells and surrounding normal melanocytes are destroyed, creating the white halo.

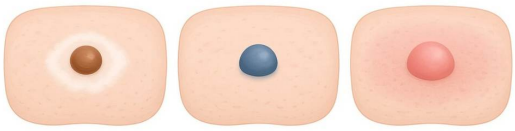

Fig. 1 · Halo naevus (white ring), blue naevus and Spitz naevus.

AI-generated illustration

3 **Regression**

The central naevus fades and may vanish.

4 **Repigmentation**

Melanocytes repopulate the ring over months to years in many.

HISTOPATHOLOGY

- Naevus with a dense lymphocytic infiltrate; loss of melanocytes in the halo.

Clinical features

- Central brown naevus with a round or oval white ring; over months the centre fades and may disappear.
- The white area later repigments in many.
- Back is the commonest site.

Diagnosis and investigations

Clinical with dermoscopy: a regular globular or homogeneous centre with a white halo. An asymmetric or atypical centre, or onset over 40, needs specialist assessment.

Differential diagnoses

Melanoma with regression or a halo (adults, atypical centre), vitiligo, post-inflammatory hypopigmentation.

Management

- Reassure if the central naevus is benign and symmetrical.
- **Refer an adult (over 40) with a new halo, or any atypical central lesion** (melanoma can provoke halos).
- Sun-protect the white ring (it burns easily).
- Examine the skin for vitiligo and other atypical naevi.

Treatment ladder

1

Typical, stable

Reassure; photograph; sun protection.

2

Atypical or adult

Dermatology via the suspected cancer pathway; excision for histology.

Referral

URGENT — New halo over 40 or atypical centre.

Red flags and when to refer

- Atypical central lesion or new halo in an adult: **suspected cancer pathway**.

Complications and prognosis

Complications

- Sunburn of the white ring; association with vitiligo.

Prognosis

- Benign; often regresses fully.

Prescriber pearls (UK pharmacist IP)

Halo naevi are benign in children; the halo is a sign of immune clearance, not of cancer.

Self-check quiz

3 questions · optional

Q1 **What causes the white ring?**

✓ T-cell destruction of the naevus and surrounding melanocytes.

Q2 Which condition is associated?

✓ Vitiligo.

Q3 When should a halo naevus be referred?

✓ New halo in an adult over 40, or an atypical central lesion.

References

1. DermNet: Halo naevus — https://dermnetnz.org/topics/halo-naevus
dermnetnz.org

Photo gallery

Halo naevus 2

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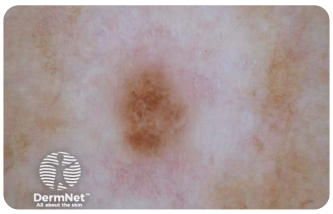

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Lumps, lesions & skin cancer Module 2 · Week 4

Blue naevus

A steel-blue, firm, dome-shaped papule from pigment-producing melanocytes deep in the dermis. Long-standing and stable lesions are benign; new, enlarging or large lesions in adults need assessment.

ALSO CALLED blue naevus · blue nevus · cellular blue naevus · Tyndall effect

RELATED PAGES Melanocytic naevi (moles) & the changing mole · Halo naevus (Sutton naevus) · Spitz and Reed naevi · Melanoma: recognition & referral · Pigmented & epidermal birthmarks

KEY FACTS

- A **blue naevus** is a benign collection of **dermal melanocytes**: a steel-blue, firm, dome-shaped papule.
- Blue because deep pigment scatters back blue light (**Tyndall effect**).
- Head and neck, backs of hands and feet, buttocks; usually under 1 cm.
- Reassure if long-standing and stable; dermoscopy shows homogeneous blue pigment.
- Refer new, enlarging or large (>1 cm) lesions in adults: nodular or metastatic melanoma can look blue.

Overview

A **blue naevus** is a benign collection of pigment-producing **dermal melanocytes**. It appears as a steel-blue, firm, dome-shaped papule, usually under 1 cm, often appearing in childhood or adolescence.

Aetiology: causes and risk factors

Melanocytes arrested in the dermis during their embryonic migration; GNAQ/GNA11 mutations. The colour is blue because deep pigment scatters back mainly blue light (**Tyndall effect**).

Myths to address

- **"BLUE MOLES ARE BRUISES OR TATTOOS"**
  a persisting blue papule without trauma is often a blue naevus, but must be stable.

Pathophysiology and pathology

Melanocytes that **stopped in the dermis** during development produce pigment deep in the skin.

1. **Arrested migration**
   Melanocytes travelling from the neural crest to the epidermis stop in the dermis.
2. **GNAQ/GNA11 mutation**
   A mutation drives a small clonal proliferation.

3 **Tyndall effect**

Deep pigment scatters blue light back to the eye.

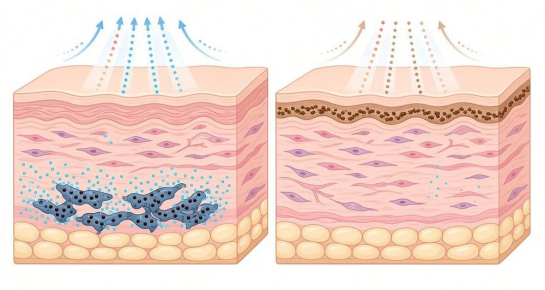

Fig. 1 · Deep dermal pigment looks blue (Tyndall effect); superficial pigment looks brown.

AI-generated illustration

HISTOPATHOLOGY

- Dendritic, heavily pigmented melanocytes and melanophages in the dermis; cellular blue naevus has nodules of spindle cells.

Clinical features

- Uniform slate- or steel-blue dome papule.
- Head and neck, backs of hands and feet, buttocks.
- **CELLULAR BLUE NAEVUS**
  larger (over 1 cm), often on the buttocks or scalp.

Fig. 2 · Blue naevus: a small, uniform slate-blue papule.

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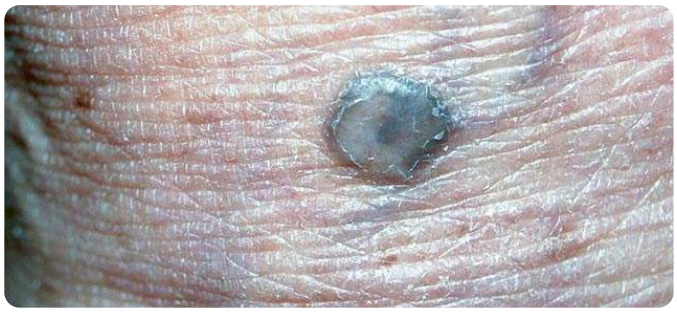

Fig. 3 · Blue naevus on the hand.

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Diagnosis and investigations

Clinical with dermoscopy (structureless homogeneous blue pigmentation). Excise if new in an adult, changing or large.

Differential diagnoses

Nodular or metastatic melanoma (can look blue), traumatic tattoo, venous lake, pigmented BCC, angiokeratoma.

Management

- Reassure if long-standing and stable.
- Refer if new in an adult, enlarging, or over 1 cm (cellular blue naevus; melanoma differential).

Treatment ladder

1 **Typical, stable**

Reassure; photograph; sun protection.

2 **Atypical or adult**

Dermatology via the suspected cancer pathway; excision for histology.

Referral

URGENT New, enlarging or >1 cm blue lesion in an adult.

Red flags and when to refer

- New or changing blue papule in an adult, or a history of melanoma (metastasis looks blue): **suspected cancer pathway**.

Complications and prognosis

Complications

- Rarely, malignant blue naevus (melanoma arising in a blue naevus).

Prognosis

- Stable for life in almost all.

Prescriber pearls (UK pharmacist IP)

Blue = deep pigment (Tyndall effect); brown = pigment near the surface.

Self-check quiz

3 questions · optional

Q1 Why does a blue naevus look blue?

✓ Deep dermal pigment scatters blue light (Tyndall effect).

Q2 Which dangerous lesion can look blue?

✓ Nodular or metastatic melanoma.

Q3 Size threshold for concern?

✓ Over 1 cm (cellular blue naevus) or enlarging.

References

1. DermNet: Blue naevus — https://dermnetnz.org/topics/blue-naevus
dermnetnz.org

Photo gallery

Clinical photographs 2

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Blue naevus 1

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Lumps, lesions & skin cancer Module 2 · Week 5

Melanoma: recognition & referral

How to spot melanoma: ABCDE, the NICE weighted 7-point checklist, the ugly-duckling sign and EFG for nodular melanoma, the clinical subtypes (including acral and amelanotic), dermoscopy and AI, and fast referral.

ALSO CALLED malignant melanoma · ABCDE · 7-point checklist · ugly duckling · EFG · nodular melanoma · +4

RELATED PAGES Melanocytic naevi (moles) & the changing mole · Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment · Atypical naevus syndrome & familial melanoma · Melanonychia & nail melanoma · Seborrhoeic keratosis

KEY FACTS

- Malignant tumour of **melanocytes**; the UK's fifth commonest cancer (about 17,500 cases a year).
- Risk: **UV and sunburn**, sunbeds, many or atypical naevi, fair skin, family history, previous melanoma, immunosuppression.
- Recognise with the **ABCDE** rule, the NICE **7-point checklist** and the **ugly duckling** sign; nodular melanoma may be EFG (elevated, firm, growing).
- Refer any suspicious pigmented lesion on the **suspected cancer pathway** (NICE NG12); never destroy an undiagnosed pigmented lesion.
- Prognosis depends mostly on **Breslow thickness** at diagnosis.

Overview

Melanoma is a malignant tumour of melanocytes. In the UK it is the **5th commonest cancer** (about 17,500–20,000 new cases a year), with rising incidence, especially in people over 65. It is one of the commonest cancers in young adults (15–34).

Clinical subtypes

Subtypes describe the growth pattern. **Breslow thickness, not subtype, drives stage and prognosis.**

Epidemiology

- **Incidence**
  - About **17,500** new cases a year in the UK; incidence has more than doubled since the 1990s.
  - One of the commonest cancers in **15–34-year-olds**; median age at diagnosis in the mid-60s.
- **Sex and site**
  - Men: trunk (back) most common; women: legs.
  - Acral lentiginous melanoma (palms, soles, nails) is the commonest type in people with dark skin.

Aetiology: causes and risk factors

- **Intermittent intense UV and sunburn** (especially in childhood), sunbeds, fair skin (MC1R), many or atypical naevi, family history, previous melanoma, and immunosuppression.
- **DRIVER MUTATIONS**
  BRAF V600 in ~40–50% (intermittently sun-exposed skin). This is the target for BRAF/MEK inhibitors; NRAS ~20%; NF1 (chronically sun-damaged skin, desmoplastic); **KIT** in acral/mucosal/CSD melanomas (the course c-KIT)
- Growth phases: radial (in situ / thin), then vertical (nodular, invasive). Breslow thickness tracks the vertical phase.

Risk factors and associations

> **UV EXPOSURE**
> intermittent intense exposure and **sunburn** (especially in childhood), sunbed use.

> **HOST**
> many naevi (over 100), atypical naevi, fair skin type I–II, red hair, freckling.

> **GENETIC**
> family history (CDKN2A, CDK4, BAP1, MC1R variants), previous melanoma, immunosuppression, giant congenital naevus.

Myths to address

> **"MELANOMA IS ALWAYS DARK AND IRREGULAR"**
> amelanotic and nodular melanomas can be pink and symmetrical.

> **"DARK SKIN DOESN'T GET MELANOMA"**
> acral and nail melanomas occur and are often diagnosed late.

Pathophysiology and pathology

Melanoma arises when **melanocytes** accumulate mutations that switch on growth and switch off brakes.

1. **Driver mutations**
   UV-signature mutations activate the **MAPK pathway** (BRAF V600E in about half, NRAS) and inactivate tumour suppressors (CDKN2A, PTEN, TP53).

2. **Radial growth**
   Most melanomas first spread sideways within the epidermis (superficial spreading, lentigo maligna), when cure by excision is very likely.

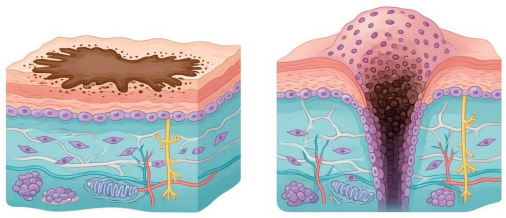

Fig. 1 · Superficial spreading melanoma (radial growth along the epidermis) vs nodular melanoma (early vertical growth into the dermis).
AI-generated illustration

3 **Vertical growth**

Cells invade the dermis; deeper tumours (thicker **Breslow**) have greater metastatic risk. Nodular melanoma enters this phase early.

4 **Immune escape**

Tumours evade T cells (e.g. via PD-L1), the basis for immunotherapy.

5 **Metastasis**

Spread to lymph nodes, skin, lung, liver and brain.

HISTOPATHOLOGY

- Atypical melanocytes in nests and single cells at all levels of the epidermis (**pagetoid spread**), with asymmetry and poor circumscription.
- Report includes **Breslow thickness**, ulceration, mitoses and margins, which guide AJCC staging and treatment.

Clinical features

Clinical features: the recognition tools

TOOL	CONTENT	STRENGTH / WEAKNESS
NICE NG12 weighted 7-point checklist	**Major (2 points each):** change in size, irregular shape, irregular colour. **Minor (1 point each):** largest diameter 7 mm, inflammation, oozing, change in sensation. **3 suspected cancer pathway** (NG12 1.7.1)	The UK primary-care standard; emphasises change
ABCDE	Asymmetry, Border, Colour, Diameter >6 mm, Evolving	Good for SSM; **misses nodular and amelanotic melanoma**
EFG	**Elevated, Firm, Growing** progressively for >1 month	**Nodular melanoma**. NG12 1.7.3: consider the suspected cancer pathway for possible nodular melanoma
Ugly duckling	A lesion that looks different from the person's other naevi	Useful with many naevi
Dermoscopy (trained)	NG12 1.7.2: dermoscopy suggesting melanoma suspected cancer pathway	Training-dependent

Case 1 shows why EFG matters: a nodule growing for 12 months scores **change in size (2) + possibly diameter 7 mm (1)**, so the threshold is met, and nodular melanoma needs urgent referral even when it is symmetrical or uniformly coloured.

Clinical subtypes

Features and typical sites

SUBTYPE	SHARE (WHITE POPULATIONS)	WHO/WHERE	LOOK	PHOTOS
Superficial spreading (SSM)	~60–70%	Young/middle-aged; trunk (men), legs (women)	Flat slightly raised, irregular, multicoloured. A **nodule** within it means vertical growth	
Nodular	~15–30%	Middle-aged/older; trunk, head and neck	Rapid (weeks to months) dome-shaped nodule, black-blue or **pink (amelanotic)**, ulcerates and bleeds. Fails ABCD; **use EFG**. Often thick at diagnosis	
Lentigo maligna / LMM	~10–15%	Older adults, **chronically sun-damaged face**	LM = melanoma in situ: a slowly enlarging irregular brown macule over years. A **nodule or darker raised area** = invasive LMM	
Acral lentiginous (ALM)	~2–3% in white people, but the **commonest subtype in Black and Asian people**	Soles, palms, nail unit	Irregular brown-black macule on the sole; **parallel-ridge** dermoscopy; Hutchinson's sign in the nail. Often diagnosed late	
Amelanotic	~2–8%	Any subtype	Pink or red; mimics PG, BCC, eczema, Bowen's	,
Desmoplastic	Rare	Older; head and neck	Scar-like firm plaque; neurotropic	–
Mucosal	~1%	Oral, nasal, anogenital, vulval	Late; poor prognosis; KIT	–
Uveal	(not cutaneous)	Eye	GNAQ/GNA11; BAP1; liver metastases	–
Childhood/ adolescent	Rare	Spitzoid or amelanotic, often nodular	"Bleeding pink bump" in a child	–

Fig. 2 · Superficial spreading melanoma with multiple notches and irregular pigment. Suspected cancer pathway or teledermatology with dermoscopic images, then excision biopsy (2 mm margins).

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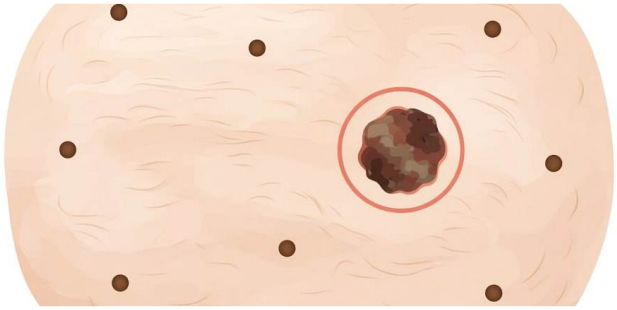

Fig. 3 · Ugly duckling sign: the mole that looks different from all the person's other moles deserves a closer look.

AI-generated illustration

Fig. 4 · Check soles and nails: an irregular dark heel macule, and a nail streak with pigment spreading onto the nail fold (Hutchinson's sign), suggest acral melanoma.

AI-generated illustration

Types to recognise

Superficial spreading

Notched, irregular colour. Site: Trunk, legs.

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Nodular (EFG)

Elevated, firm, growing. Site: Trunk, head.

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Amelanotic

Pink smooth nodule. Site: Any.

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Lentigo maligna

Slowly enlarging facial macule. Site: Face.

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Acral lentiginous

Heel, palm, nail unit. Site: Soles.

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Evolving

Larger and darker over time. Site: Any.

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Special groups

- **Skin of colour**
  - Look at **palms, soles and nails**: acral lentiginous melanoma and nail melanoma (Hutchinson sign) present late.
- **Pregnancy and children**
  - Naevi can darken in pregnancy, but a changing lesion still needs assessment; melanoma is rare in children (Spitz-like lesions need specialist review).

Diagnosis and investigations

> **Diagnostic accuracy by eye** rises with experience. The course figures (consultant ~80%, registrar 50–60%, GP ~40%) are in line with older studies, but published estimates vary widely.

Dermoscopy improves accuracy in trained users: meta-analyses show sensitivity rising from about 70% to about 90%, with an odds ratio of ~15 for diagnostic accuracy versus the naked eye. **Untrained use may not help.**

DERMOSCOPIC MELANOMA FEATURES

atypical network; irregular streaks / pseudopods; irregular dots and globules; **blue-white veil**; regression structures; negative network; shiny white lines; atypical (polymorphous/dotted) vessels; acral **parallel-ridge** pattern; facial LM: asymmetric follicular openings, rhomboid structures, annular-granular pattern; Algorithms: **3-point checklist** (asymmetry, atypical network, blue-white structures; 2 = refer), 7-point dermoscopic checklist, ABCD rule (Stolz), CASH, TADA, "chaos and clues"

TRAINING AND SIMULATORS

PCDS dermoscopy courses, the BAD/BDNG, International Dermoscopy Society (dermoscopy-ids.org) online tutorials, DermNet's dermoscopy course, and image-based simulators (e.g. "Dermoscopedia", quiz banks). The evidence supports structured training with regular practice and feedback.

AI AND ALGORITHMS

NICE HTG746 (formerly HTE24) supports conditional NHS use of **DERM** (Skin Analytics) to triage lesions **already referred** on the urgent skin cancer pathway, with dermatologist oversight. Images of black or brown skin (Fitzpatrick V–VI) still need human review. Consumer apps are mostly unregulated. **An app result never overrides clinical concern** in either direction.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Seborrhoeic keratosis	Stuck-on, waxy, milia-like cysts.
	Dysplastic naevus	Atypical but stable; many moles.
	Haemangioma / pyogenic granuloma	Red, bleeds; but amelanotic melanoma mimics PG.
	Pigmented BCC	Pearly, arborising vessels.
	Subungual haematoma	Trauma, grows out; no fold pigment.
	Solar lentigo	Uniform flat brown on sun-damaged skin.

SK (the benign lesion notes see Biopsy, Breslow thickness and the histopathology report;), atypical naevus, pigmented BCC, DF, angiokeratoma (the skin lumps notes see Immunotherapy and targeted therapy: what the IP must know), haemangioma/thrombosed angioma, pyogenic granuloma (amelanotic nodular melanoma mimic;), subungual haematoma (nail), acral naevus, lentigo/flat SK (vs LM), and metastasis.

Diagnosis and differentials

As in Recognition, diagnostic accuracy, dermoscopy and AI. LM vs solar lentigo/flat SK (dermoscopy, biopsy). ALM vs acral naevus, haematoma (talon noir, "black heel"), wart, and fungal (tinea nigra).

Management

- **Suspected cancer pathway referral (2WW)** if: the 7-point score is 3; dermoscopy suggests melanoma; or; a nodule fits EFG (nodular melanoma)
- **Make the referral within 1 working day** of deciding. Include dermoscopic/clinical images if a teledermatology triage route exists (see Teledermatology).
- Document size (ruler), duration, change, the 7-point score, and safety-netting.
- **DO NOT**
  freeze, shave, laser or "watch and wait" on a lesion meeting criteria; or reassure because an app is negative.
- Previous **repeated reassurance** is a classic **diagnostic-delay pattern**. When a patient re-presents with persistent concern, take it seriously, and consider reporting it as a learning event.

Clinical subtypes

- **Check soles, palms and nails** in every skin of colour presentation, and **do not label a pigmented sole lesion as a "bruise" or "verruca"** without follow-up. Subungual: Hutchinson's sign suspected cancer pathway (the nail disorders notes).
- LM on the face of an older person: an elderly red flag (the older skin pages notes quick reference), so use the suspected cancer pathway.

Worked cases

CASE 1

54-year-old, nodule >12 months, repeatedly reassured; an app flagged it

DIAGNOSIS

nodular melanoma (EFG: elevated, firm, growing).

ACTION

NG12 suspected cancer pathway (1.7.1 score 3 and/or 1.7.3 nodular melanoma), referral within 1 working day.

Excision biopsy with a 2 mm margin (the course "narrow margin") by the skin cancer team, then WLE per Breslow.

DIAGNOSTIC ACCURACY

consultants ~80%, registrars 50–60%, GPs ~40% (course teaching; study-dependent); dermoscopy adds about 10–20% in trained hands.

LEARNING POINTS

repeated reassurance = diagnostic delay; an app is not a diagnostic tool, but a flag deserves clinical review.

CASE 2

17-year-old of Irish ancestry, >50 atypical moles, father and grandfather with melanoma in their 30s, other family cancers

DIAGNOSIS

FAMMM (familial atypical multiple mole melanoma) syndrome.

GENE

CDKN2A (p16). Also consider CDK4, BAP1 and MC1R modifiers.

ASSOCIATED CANCER

pancreatic adenocarcinoma.

PLAN

dermatology surveillance (total-body photography plus digital dermoscopy), clinical genetics, sun protection, no sunbeds (under-18 law), self-examination, no smoking.

CASE 3

30-year-old, lesion changing over 2 years, teledermatology images

PATHWAY

2WW with images excision biopsy (2 mm) WLE.

PATHOLOGY

Breslow **1.4 mm** (depth, not 1.5 cm diameter), **ulcerated** **T2b**.

STAGE

IIA if node-negative; SLNB decides (if positive IIIA or higher).

5-YEAR MSS

~93–94% (AJCC 8), not ~80%. Mitoses >5/mm² are prognostic only, not staging.

WLE

2 cm margin (stage II, NG14).

SLNB

consider/discuss (Breslow >1 mm).

IMAGING

not required at IIA (consider at IIB).

ADJUVANT

none at IIA (pembrolizumab TA837 is for IIB/IIC).

FOLLOW-UP

3-monthly for 3 years, then 6-monthly to 5 years. Self-examination; vitamin D measured at diagnosis.

Treatment ladder

1

Refer

Suspected cancer pathway (NG12 1.7.1 score 3; 1.7.3 nodular) **within 1 working day**. Include images if teledermatology triage exists; document the score and safety-net.

2

Specialist

Excision biopsy with a 2 mm margin by the skin cancer team WLE per Breslow.

3

Prevention & family

Sun protection, no sunbeds; full-skin self-checks; first-degree relatives ~2× risk. Many atypical moles + family melanoma: consider **FAMMM / CDKN2A** (pancreatic cancer link) genetics.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Don't**: Freeze, shave, laser or 'watch and wait' a lesion meeting criteria; reassure because an app is negative.
- **Pigmented 'verruca' or 'bruise' on the sole**: Refer if not resolving; not more destructive treatment.

Referral

URGENT	7-point score **3** or dermoscopy suggesting melanoma: suspected cancer pathway.
URGENT	**Firm growing nodule > 1 month** (EFG), even if pink or symmetrical.
URGENT	New growing dark mark on sole, palm or nail; Hutchinson's sign.

Red flags and when to refer

- NICE weighted 7-point checklist score **3 or more**, or dermoscopy suggesting melanoma: **suspected cancer pathway** referral (NG12 1.7.1–1.7.3).
- A firm, growing nodule over a month (EFG), even if pink or symmetrical: suspected **nodular melanoma**.

- A new dark mark on the sole, palm or nail that is growing: suspected **acral melanoma** (check the nail fold for Hutchinson's sign).

Complications and prognosis

Complications

- Local recurrence, in-transit metastases.
- Lymph node and distant metastasis.
- Treatment toxicity (immunotherapy immune-related adverse events).

Prognosis

- Five-year survival is over **95%** for thin (stage I) melanoma, falling with thickness and spread.
- Further primary melanomas occur in about 5–10%: lifelong skin self-checks.

Prescriber pearls (UK pharmacist IP)

"Any spot that is raised, firm and growing, of any colour, needs to be seen within two weeks."

Sun-protection counselling and full-skin self-checks for family members (first-degree relatives have about twice the risk).

A "verruca" that doesn't respond to salicylic acid or cryo and is pigmented or bleeding needs referral, not more destructive treatment.

Clinical subtypes

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING OR STUDENT ANSWER	UK GUIDANCE / EVIDENCE	TEACHING POINT
Case 3 survival	"Stage IIA, ~80% 5-year survival"	AJCC 8: T2b N0 = IIA, **~93–94% 5-year MSS**	80% is AJCC 7/Balch 2009-era data
Mitoses in staging	Mitoses >5 used to support the stage	AJCC 8 removed mitotic rate from T1 staging; it is prognostic only. **Clark level is not used**	Report it, but don't stage on it
Ulceration	"Not ulcerated"	The case states ulceration T2b (not T2a/ IB)	Ulceration upstages
Breslow	Read "1.5 cm" as Breslow	Breslow is a **depth in mm** (1.4 mm)	Size thickness
SLNB	"Consider SLNB"	NG14 2022: consider for >1.0 mm (and 0.8–1.0 mm with risk features); not for IA	Aligned

TOPIC	COURSE TEACHING OR STUDENT ANSWER	UK GUIDANCE / EVIDENCE	TEACHING POINT
WLE margin	Two-stage excision	NG14: stage II **2 cm** (1 cm if 2 cm would cause unacceptable morbidity)	Add the margin
Adjuvant at IIA	(Not stated)	None. TA837 pembrolizumab covers **IIB/IIC**; stage III options TA684/TA766/TA544	–
Completion lymphadenectomy	(Not stated)	No longer routine (MSLT-II)	–
Accuracy figures	Consultant 80%, GP 40%	Wide variation in studies; dermoscopy benefit depends on training	Use as illustrative
App use	App flagged it	Unregulated apps are not recommended. NICE HTG746 covers AI within NHS triage only	Never let an app reassure

Self-check quiz

4 questions · optional

Q1 NICE 7-point referral threshold?

✓ Score of 3 or more.

Q2 Which melanoma does ABCDE miss?

✓ Nodular and amelanotic: use EFG.

Q3 Commonest subtype in Black and Asian people?

✓ Acral lentiginous melanoma.

Q4 Should primary care excise a suspected melanoma?

✓ No: refer; specialist excision biopsy with a 2 mm margin.

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Photo gallery

Nodular and amelanotic melanoma

7

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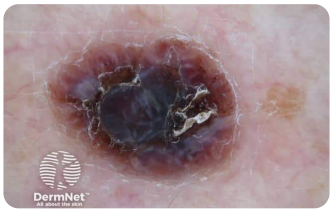

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ABCDE in practice

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Superficial spreading melanoma and the ugly duckling

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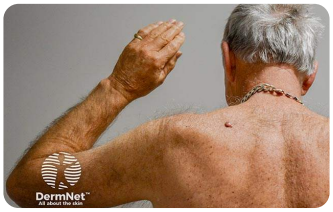

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Lentigo maligna, acral and amelanotic melanoma

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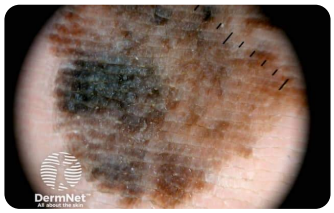

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Lumps, lesions & skin cancer Module 2 · Week 5

Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment

What happens after referral: excision biopsy, the histopathology report (Breslow, ulceration), AJCC 8th edition staging with current 5-year survival figures, NICE NG14 excision margins, sentinel node biopsy, adjuvant treatment and follow-up.

ALSO CALLED Breslow thickness · AJCC 8 · stage IIA · sentinel lymph node biopsy · SLNB · wide local excision · +2

RELATED PAGES Melanoma: recognition & referral · Melanoma immunotherapy & targeted therapy: side-effects · Atypical naevus syndrome & familial melanoma · Skin surgery procedures: excision, curettage, cryotherapy & Mohs

KEY FACTS

- Melanoma outcome depends mostly on **Breslow thickness**, **ulceration** and nodal status (AJCC 8th edition).
- Excision biopsy with a 2 mm margin first; then **wide local excision** (1 cm for <1 mm; 1–2 cm for 1–2 mm; 2 cm for >2 mm thickness).
- Sentinel lymph node biopsy offered from stage IB (pT1b) upwards (NICE NG14).
- Adjuvant immunotherapy or BRAF/MEK inhibitors for resected stage III (and some IIB/C); systemic therapy for stage IV.
- Follow-up by stage; teach self-examination of skin and lymph nodes.

Overview

Breslow thickness is the depth of invasion in millimetres. It is measured from the top of the granular layer (or the base of an ulcer) to the deepest invasive melanoma cell, and it is the single strongest prognostic factor in localised melanoma. **It is a depth in mm, never a lesion diameter in cm**.

Aetiology: causes and risk factors

Myths to address

- **"REMOVING THE MOLE MEANS IT'S ALL GONE"**
  thicker melanomas need wide excision and staging.

Pathophysiology and pathology

The deeper a melanoma grows, the more likely it has **reached lymphatics and blood vessels**.

1. **Radial growth**
   Early melanoma spreads within the epidermis: excellent prognosis.

2 **Vertical growth**

Invasion into the dermis gives access to lymphatics and vessels.

Fig. 1 · Breslow thickness: measured from the granular layer to the deepest tumour cell.

AI-generated illustration

3 **Ulceration and mitoses**

Ulceration and high mitotic rate mark aggressive biology.

4 **Margins**

Wide local excision removes in-transit cells around the scar.

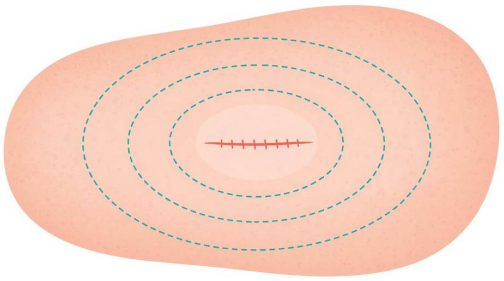

Fig. 2 · Wide local excision margins by Breslow thickness.

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The histopathology report (RCPath dataset, 2019 / updates)

ITEM	WHY IT MATTERS
Subtype; in situ vs invasive	Classification
Breslow thickness (to 0.1 mm)	T category, SLNB eligibility, margins
Ulceration (present/absent; extent)	Upstages (T "b")
Mitotic rate (/mm²)	Prognostic; **no longer a staging criterion in AJCC 8** (removed from T1 subcategorisation). Still reported
Microsatellites	N1c/N2c/N3c (stage III)
Lymphovascular and perineural invasion	Risk; NG14 SLNB considerations
Regression, tumour-infiltrating lymphocytes (TILs)	Prognostic
Growth phase, associated naevus, Clark level	**Clark level is not used in AJCC 8**
Peripheral and deep margins	Re-excision planning

ITEM	WHY IT MATTERS
pT stage (AJCC 8)	Stage
BRAF/molecular	Requested for stage III–IV or adjuvant decisions

Clinical features

Biopsy principles

- **Excision biopsy with a 2 mm clinical margin**, full thickness including subcutaneous fat, oriented to allow later WLE/SLNB (along the long axis of limbs).
- Done by the skin cancer team/specialist. In primary care, **refer rather than excise** suspected melanoma (NICE IOG).
- **Avoid shave, punch or incisional biopsy** except for specialist situations (large facial LM, acral/nail).

Fig. 3 · Persistent blue tattoo at a sentinel node biopsy site (patent blue dye). It is harmless; reassure. SLNB is offered from Breslow over 1.0 mm (or 0.8–1.0 mm with risk factors) under NG14 2022.

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Types to recognise

In-transit metastases

Lumps between scar and nodes. Site: Limb.

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Nodal disease

New regional node lump. Site: Groin, axilla.

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Cutaneous metastases

Blue-grey nodules. Site: Skin.

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SLNB tattoo

Persistent blue dye mark. Site: Node site.

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Diagnosis and investigations

Histology. Immunohistochemistry: SOX10, Melan-A, HMB-45, PRAME, Ki-67. FISH/NGS for ambiguous spitzoid lesions.

AJCC 8th edition staging and survival

T category (primary tumour)

T	BRESLOW	A	B
Tis	in situ	–	–
T1	1.0 mm	T1a: <0.8 mm, no ulceration	T1b: <0.8 mm with ulceration, or 0.8–1.0 mm ± ulceration
T2	>1.0–2.0 mm	no ulceration	**with ulceration**

T	BRESLOW	A	B
T3	>2.0–4.0 mm	no ulceration	with ulceration
T4	>4.0 mm	no ulceration	with ulceration

N = regional nodes, in-transit, satellite or microsatellite metastases (N1–N3, a/b/c). M = distant (M1a skin/soft tissue/non-regional node; M1b lung; M1c other visceral; M1d CNS; with LDH suffix).

Pathological stage groups and approximate 5-year melanoma-specific survival (AJCC 8; Gershenwald 2017)

STAGE	INCLUDES	~5-YR MSS
0	Tis	~100%
IA	T1a, T1b N0	~99%
IB	T2a N0	~97%
IIA	**T2b N0**, T3a N0	**~94%** (T2b 93%)
IIB	T3b, T4a N0	~87%
IIC	T4b N0	~82%
IIIA	T1–T2a with N1a/N2a	~93%
IIIB	—	~83%
IIIC	—	~69%
IIID	T4b with N3	~32%
IV	M1	Historically <20%. Now ~50% at 5 years or more with combination immunotherapy in trials (CheckMate 067: ~49% 10-year MSS with ipi+nivo)

Case 3 worked example:

- Breslow **1.4 mm with ulceration = T2b**.
- If N0 (clinically and after SLNB), this is **stage IIA**, with ~93–94% 5-year MSS. The course "~80%" reflects older AJCC 7-era data, and it is closer to stage IIC today.
- **Mitoses >5/mm² do not alter the stage in AJCC 8** (they are prognostic only).
- If SLNB is positive stage III (e.g. T2b N1a = **IIIA**).
- Until SLNB is done, the stage is clinical ("cIIA").

Reference URLs

- Gershenwald JE et al. Melanoma staging: evidence-based changes in the AJCC 8th edition. CA Cancer J Clin 2017;67:472–92.
- Keung EZ, Gershenwald JE. AJCC 8 update. Expert Rev Anticancer Ther 2018.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Dysplastic naevus	Histological look-alike.
	Spitz / atypical Spitz	Young, pink; FISH / NGS.
	MELTUMP	Uncertain malignant potential: specialist plan.
	Recurrent / halo naevus	Re-grown pigment, halo.

(histological)

Dysplastic naevus, Spitz naevus/atypical Spitz tumour, MELTUMP (melanocytic tumour of uncertain malignant potential), recurrent naevus, and halo naevus.

Management

Know what the letter means, so you can help patients understand it. "Breslow 0.6 mm, no ulceration" = stage IA, an excellent prognosis. Encourage patients to ask their **clinical nurse specialist (CNS)** keyworker any questions.

NICE NG14 management

Wide local excision (WLE) margins (NG14 2022)

STAGE	CLINICAL PERIPHERAL MARGIN (MEASURED FROM THE BIOPSY SCAR/LESION EDGE)
0 (in situ)	**At least 0.5 cm** (LM: consider imiquimod or radiotherapy if surgery would be disfiguring or unsuitable; specialist)
I	**1 cm**
II	**2 cm** (consider 1 cm if a 2 cm margin would cause unacceptable morbidity/disfigurement, e.g. face)
III	As stage II, plus nodal management

So Case 3 (stage IIA) needs WLE with a **2 cm margin** (two-stage surgery, the course answer).

Sentinel lymph node biopsy (SLNB)

> **WHAT**
>
> lymphoscintigraphy plus blue dye, a persistent blue tattoo) identifies the first draining node(s). These are excised for histology at the time of WLE.

> **NG14 2022**
>
> **Do not offer** SLNB (or imaging) for **stage IA**; **Consider SLNB** for **Breslow 0.8–1.0 mm** with 1 of: ulceration, lymphovascular invasion, or mitotic rate 2/mm²; **Consider (offer to discuss)** SLNB for **Breslow >1.0 mm**; **Discuss the pros and cons:** SLNB is a **staging** procedure (it determines eligibility for adjuvant therapy). It **does not**

> **improve melanoma-specific survival** (MSLT-I). Complications include seroma, infection, lymphoedema and nerve injury

> **Completion lymph node dissection** after a positive SLNB is **no longer routine** (MSLT-II, DeCOG-SLT showed no survival benefit). **Nodal ultrasound surveillance** is used instead.

> Case 3: **consider SLNB** (Breslow >1 mm). The course answer was "consider SLNB" .

Imaging

- Stage I–IIA: no routine imaging.
- **Stage IIB: consider** contrast-enhanced CT (CE-CT) staging.
- **Stage IIC–IV: offer** CE-CT (and **MRI brain** for stage IV, and for III where relevant; children may have whole-body MRI).
- **Measure vitamin D at diagnosis** and give advice on correction/supplementation.

Adjuvant systemic therapy (after complete resection)

SETTING	NICE-RECOMMENDED OPTIONS	TA
Stage IIB/IIC	**Pembrolizumab** (age 12)	**TA837**
Stage IIB/IIC	Nivolumab: **no NICE TA identified** in this session (licensed by MHRA; check for updates)	–
Stage III (node-positive, completely resected)	**Nivolumab**, **pembrolizumab**, **dabrafenib + trametinib** if BRAF V600	–
Resected stage IV	Nivolumab	–
Stage IIA	**None.** Adjuvant therapy is not recommended at IIA	–

Unresectable stage III / stage IV

- **BRAF testing** for all patients with stage III–IV disease.
- **Immunotherapy** (checkpoint inhibitors): **Ipilimumab + nivolumab** (anti-CTLA-4 + anti-PD-1; TA400): highest response and highest toxicity; **Pembrolizumab** or **nivolumab** monotherapy; **Nivolumab + relatlimab** (anti-LAG-3; NICE-recommended for untreated advanced melanoma in people aged 12; check the current TA)
- **Targeted therapy** (BRAF V600): **dabrafenib + trametinib**, **encorafenib + binimetinib**, vemurafenib + cobimetinib.
- **Talimogene laherparepvec (T-VEC)** intralesional oncolytic virus for unresectable injectable stage IIIB–IVM1a without bone/brain/lung/visceral disease.
- Other: radiotherapy and stereotactic radiosurgery (brain metastases); TIL therapy (lifileucel; licensed in the US; not routine in the UK at the time of writing); trials.
- In-transit metastases: surgery, T-VEC, isolated limb perfusion/infusion, electrochemotherapy, systemic therapy.

Follow-up

> **STAGE IA**
> 2–4 follow-up visits in the first year, then discharge with self-examination education.

> **STAGE IB–IIB (AND IIC WITHOUT IMAGING)**
> every **3 months for 3 years**, then **every 6 months to 5 years**.

> **STAGE IIC–III**

imaging surveillance (CT or whole-body MRI) per NG14 schedules (e.g. 6-monthly for 3 years, then 12-monthly to 5 years) in addition to clinical review.

Teach self-examination of the scar, the in-transit route, regional nodes and the whole skin.

Psychological support and a CNS keyworker. Fertility and pregnancy advice where relevant.

Prescriber pearls & counselling

A new lump near the scar or in the groin/axilla/neck after melanoma means **urgent contact with the skin cancer CNS/MDT**. Do not wait for a routine appointment.

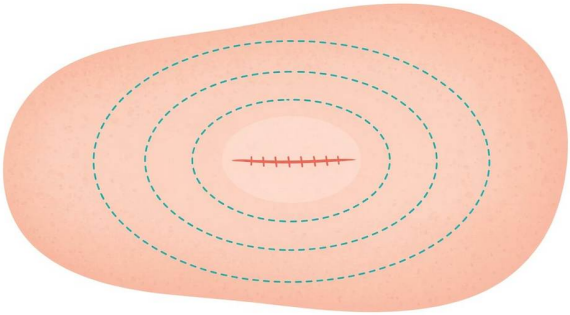

Fig. 8 · Wide local excision is planned around the biopsy scar, with margin size set by stage (0.5, 1 or 2 cm), oriented along the limb's long axis.

AI-generated illustration

Treatment ladder

1 **Explain**

Breslow = depth, not size. "0.6 mm, no ulceration" = stage IA, excellent prognosis. Link patients with their **CNS keyworker**.

2 **WLE margins**

In situ 0.5 cm; **stage I 1 cm**; **stage II 2 cm** (1 cm if 2 cm would disfigure); III as II + nodes.

3 **SLNB**

Consider if Breslow > 0.8 mm with risk features / > 1 mm; staging, **not survival**. Completion dissection no longer routine (MSLT-II): nodal ultrasound surveillance.

Adjuvant (resected)

IIB / IIC: pembrolizumab (TA837). **III**: nivolumab, pembrolizumab, or **dabrafenib + trametinib** (BRAF V600). Resected M: nivolumab. **IIA: none**.

Advanced

BRAF test all III–M. **Ipilimumab + nivolumab** (TA400), PD-1 monotherapy, nivolumab + relatlimab; BRAF / MEK combos; T-VEC; radiotherapy / SRS.

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Follow-up**: **IA**: 2–4 visits in year 1 then discharge. **IB–IIB**: 3-monthly for 3 years, then 6-monthly to 5 years. **IIC–III**: plus imaging surveillance. Teach self-examination.

Referral

URGENT — New lumps in or near the scar, between scar and nodes, or new nodes: **recurrence or in-transit disease**.

ROUTINE — Counsel with AJCC 8 (Gershenwald 2017) survival figures.

Red flags and when to refer

- New lumps in or near the scar, in the limb between the scar and nodes, or new nodes: **urgent** review for **recurrence or in-transit disease**.
- Survival figures change with each AJCC edition: use AJCC 8 (Gershenwald 2017) when counselling.

Reference URLs

- NICE NG14 (updated July 2022): nice.org.uk
- NICE TA837, TA684, TA766, TA544, TA400, TA366, TA384, TA396, TA562, TA410: nice.org.uk
- Morton DL et al. MSLT-I. NEJM 2014. Faries MB et al. MSLT-II. NEJM 2017.

Complications and prognosis

Complications

- Local recurrence, in-transit and nodal metastases, distant spread (lung, liver, brain).

Prognosis

- Five-year survival over 95% for thin melanoma (<1 mm); much lower with nodal or distant disease.

Prescriber pearls (UK pharmacist IP)

Patients often confuse lesion size with thickness. Explain: "Breslow is how deep it went, measured under the microscope."

Self-check quiz

4 questions · optional

Q1 Breslow 1.4 mm, ulcerated, node-negative: stage?

✓ IIA (T2b N0), ~93–94% 5-year MSS.

Q2 WLE margin for stage II?

✓ 2 cm (1 cm if 2 cm would cause unacceptable morbidity).

Q3 Adjuvant therapy at stage IIA?

✓ None; pembrolizumab is for IIB / IIC.

Q4 Does SLNB improve survival?

✓ No: it stages (MSLT-I).

References

1. DermNet: Melanoma — https://dermnetnz.org/topics/melanoma
dermnetnz.org
2. Medscape: Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment — https://emedicine.medscape.com/article/1295718-overview
emedicine.medscape.com

Photo gallery

Spread and staging 4

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Lumps, lesions & skin cancer Module 2 · Week 4

Spitz and Reed naevi

Benign naevi of large epithelioid or spindle melanocytes that grow quickly in children and can mimic melanoma. Children under 12 with typical lesions may be monitored by specialists; adolescents and adults usually need excision.

ALSO CALLED Spitz naevus · Reed naevus · pigmented spindle cell naevus · atypical Spitz tumour · spitzoid melanoma

RELATED PAGES Melanocytic naevi (moles) & the changing mole · Halo naevus (Sutton naevus) · Blue naevus · Melanoma: recognition & referral · Pyogenic granuloma

KEY FACTS

- **Spitz naevi** are benign naevi of large epithelioid/spindle melanocytes that **grow quickly in children**.
- Pink-red or pigmented dome; dermoscopy starburst pattern or dotted vessels. **Reed naevus**: dark symmetrical starburst, often thigh of young women.
- They can be clinically and histologically hard to tell from melanoma (spitzoid melanoma).
- Children under 12 with typical lesions may be monitored by specialists; **adolescents and adults: excision**.
- Atypical Spitz tumours need specialist histopathology review.

Overview

A **Spitz naevus** is a benign naevus made of large epithelioid and/or spindle melanocytes, typically a fast-growing pink or pigmented dome in a **child**. A **Reed naevus** (pigmented spindle cell naevus) is a dark, symmetrical variant, often on the thigh of young women.

Epidemiology

- **Who**
  - Mostly children and young adults; face and limbs.

Aetiology: causes and risk factors

Driven by HRAS mutations or kinase fusions (ALK, ROS1, NTRK and others); growth is followed by stabilisation. **Atypical Spitz tumours** lie between benign Spitz naevi and spitzoid melanoma.

Myths to address

- **"FAST GROWTH ALWAYS MEANS MELANOMA"**

in young children it is typical of Spitz naevi, but needs specialist assessment.

Pathophysiology and pathology

Spitz naevi grow from a **single driver event** (HRAS mutation or kinase fusion), then stop.

1. **Driver**
   An HRAS mutation or a kinase fusion (ALK, ROS1, NTRK, BRAF, MET) starts growth.
2. **Rapid growth**
   Large epithelioid and spindle melanocytes proliferate over months.
3. **Stabilisation**
   Senescence halts growth; many involute or become fibrotic over years.
4. **Spectrum**
   Additional mutations push lesions towards atypical Spitz tumour and spitzoid melanoma.

HISTOPATHOLOGY

- Symmetrical, well-circumscribed proliferation of large epithelioid/spindle melanocytes with Kamino bodies; maturation with depth.

Clinical features

SPITZ

pink-red or pigmented, dome-shaped, smooth, rapidly growing over months in a child; face and limbs. Dermoscopy: starburst or dotted vessels.

REED

black, symmetrical, starburst pattern; thigh in young women.

Diagnosis and investigations

Dermoscopy plus age and growth pattern. Excision for histology in adolescents and adults, and in any atypical lesion; specialist histopathology review.

Differential diagnoses

Spitzoid melanoma (adults, asymmetry), pyogenic granuloma, juvenile xanthogranuloma, haemangioma, dermatofibroma, nodular melanoma.

Management

- Children under 12 with a typical lesion: specialist may monitor with dermoscopy.
- **Adolescents/adults: excision** (atypical Spitz tumour and spitzoid melanoma differential).
- Reed naevi are usually excised for histology.

Treatment ladder

1

Typical, stable

Reassure; photograph; sun protection.

2

Atypical or adult

Dermatology via the suspected cancer pathway; excision for histology.

Referral

URGENT — Spitz-like lesion in an adolescent or adult; any atypical features.

Red flags and when to refer

- A Spitz-like lesion in an adult, or any atypical, ulcerated or rapidly growing lesion: **suspected cancer pathway**.

Complications and prognosis

Complications

- Diagnostic uncertainty; rarely malignant behaviour (atypical Spitz tumour).

Prognosis

- Benign in most children; may regress.

Prescriber pearls (UK pharmacist IP)

Rapid growth worries parents but is typical of Spitz naevi in children; age decides how worried to be.

Self-check quiz

3 questions · optional

Q1 **Typical patient with a Spitz naevus?**

✓ A child with a fast-growing pink dome.

Q2 What is a Reed naevus?

✓ A pigmented spindle-cell naevus: dark symmetrical starburst, often on a young woman's thigh.

Q3 Management in adults?

✓ Excision (spitzoid melanoma differential).

References

1. DermNet: Spitz naevus — https://dermnetnz.org/topics/spitz-naevus
dermnetnz.org

Photo gallery

Spitz and Reed naevi 2

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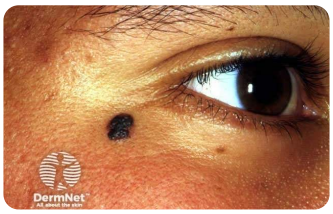

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Lumps, lesions & skin cancer Module 2 · Week 5

Melanoma immunotherapy & targeted therapy: side-effects

What a primary-care prescriber must know about patients on checkpoint inhibitors or BRAF/MEK inhibitors: immune-related side-effects (colitis, pneumonitis, hypophysitis, diabetes, skin), interactions and when to ring the acute oncology hotline.

ALSO CALLED checkpoint inhibitor · pembrolizumab · nivolumab · ipilimumab · irAE · immune-related adverse events · +3

RELATED PAGES Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment · Melanoma: recognition & referral · Bullous pemphigoid (incl. MMP & pemphigoid gestationis) · Lichen planus · Approach to a suspected drug eruption

KEY FACTS

- **Checkpoint inhibitors** (anti-PD-1: nivolumab, pembrolizumab; anti-CTLA-4: ipilimumab) release the brakes on T cells.
- Used for advanced and adjuvant melanoma; combination ipilimumab + nivolumab is most effective and most toxic.
- **Immune-related adverse events (irAEs)** can affect any organ: skin (commonest), colitis, hepatitis, pneumonitis, thyroid, pituitary, adrenals, myocarditis.
- Any new symptom on immunotherapy: contact the **oncology acute line**; do not wait. Diarrhoea, breathlessness and headache can be life-threatening.
- BRAF/MEK inhibitors (dabrafenib + trametinib, encorafenib + binimetinib) for BRAF-mutant disease: fever, photosensitivity, rashes.

Overview

- **Checkpoint inhibitors** (anti-PD-1: pembrolizumab, nivolumab; anti-CTLA-4: ipilimumab; anti-LAG-3: relatlimab) release T-cell "brakes".
- **BRAF/MEK inhibitors** block MAPK signalling in BRAF-mutant tumours.

Aetiology: causes and risk factors

Myths to address

- **"A RASH ON IMMUNOTHERAPY MEANS STOP TREATMENT"**
  most skin irAEs are mild and managed with topical steroids; oncology decides.

Pathophysiology and pathology

Checkpoint inhibitors **remove inhibitory signals** so T cells attack tumour, and sometimes normal tissue.

1. **Checkpoints**
   CTLA-4 and PD-1 normally dampen T-cell activation to prevent autoimmunity.
2. **Release**
   Blocking them reactivates tumour-specific T cells.

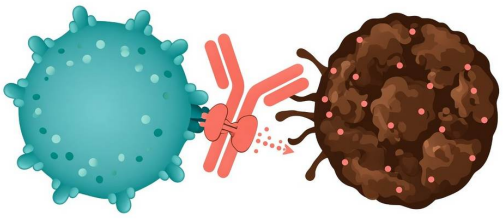

Fig. 1 · Checkpoint inhibitors release the brakes on T cells.
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3. **Off-target immunity**
   Autoreactive T cells cause organ inflammation (irAEs).

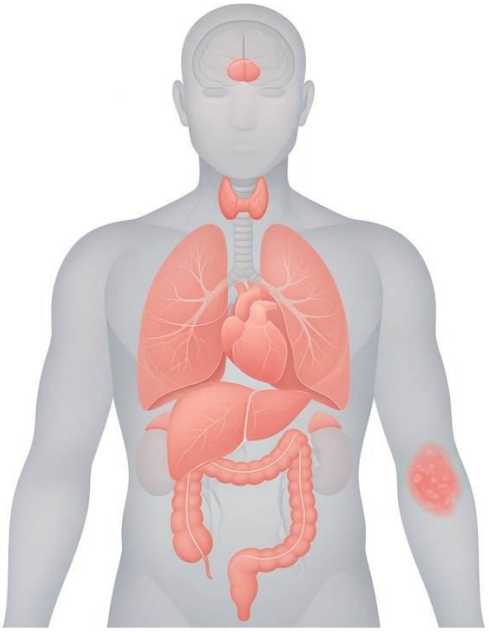

Fig. 2 · Organs affected by immune-related adverse events.
AI-generated illustration

4. **Skin irAEs**
   Itch, maculopapular rash, lichenoid and psoriasiform eruptions, vitiligo, bullous pemphigoid, rarely SJS/TEN.

Clinical features

Immune-related adverse events (irAEs): mechanism and timing

These come from autoimmune inflammation of any organ. They can start **weeks to months after the first dose, and even after stopping**. Combination ipi+nivo causes grade 3–4 irAEs in more than half of patients.

Clinical features: irAE table

SYSTEM	WATCH FOR	PRIMARY-CARE ACTION
Gut (colitis)	Diarrhoea (4 stools above baseline), blood, abdominal pain	**Same-day acute oncology contact.** Don't just give loperamide and wait
Liver (hepatitis)	Raised ALT/AST, jaundice	Urgent bloods/oncology
Lung (pneumonitis)	New cough, breathlessness, hypoxia	Same-day assessment. Don't assume a chest infection
Endocrine	**Hypophysitis** (headache, fatigue, hyponatraemia); **thyroiditis** (hyper hypothyroid); **adrenal insufficiency**; **new type 1 diabetes / DKA**	Check TFTs, glucose, cortisol, sodium. Endocrinopathies are often permanent (lifelong levothyroxine, hydrocortisone, insulin)
Skin	Rash, itch, **vitiligo** (associated with response), lichenoid, bullous pemphigoid; rarely SJS/TEN	Emollient + topical steroid for grade 1–2. Urgent for blistering/mucosal
Kidney	Rising creatinine (nephritis)	Review NSAIDs/PPIs
Heart, nerve, muscle	**Myocarditis** (chest pain, troponin), myasthenia, myositis	Emergency
Joints	Arthralgia/inflammatory arthritis	Rheumatology

Fig. 3 · Eczematous dermatitis induced by pembrolizumab (anti-PD-1).
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Fig. 4 · Lichenoid dermatitis induced by nivolumab.
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Fig. 5 · Psoriasis triggered by pembrolizumab.
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Diagnosis and investigations

Use a low threshold. Patients carry an **immunotherapy alert card** and a **24-hour acute oncology hotline** (UKONS triage tool). Always ask "are you on cancer treatment?"

Ask

- **"Are you on cancer treatment?"**; alert card + **24-hour hotline** (UKONS triage).

Bloods

- TFTs, glucose, cortisol, sodium, LFT, U&E as directed.

Think also

- Infection (e.g. C. difficile), other drug reactions, progression.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Infection	C. difficile, chest infection: but don't assume.
	Drug eruption	Other new drugs.
	SJS / TEN	Blistering, mucosal erosions: emergency.
	Bullous pemphigoid	Tense blisters on immunotherapy.
	Vitiligo	Depigmentation on anti-PD-1.

Infection (e.g. C. difficile in a patient with diarrhoea), drug reactions, and disease progression.

Management

Never start immunosuppression independently. Contact the acute oncology team, who will grade the event and prescribe high-dose corticosteroids/other immunosuppressants (per ESMO/UKONS).

STEROID STEWARDSHIP

patients on high-dose steroids for irAEs need **PPI/gastroprotection, glucose monitoring, bone protection if prolonged, PJP prophylaxis (co-trimoxazole) if 20 mg prednisolone for 4 weeks**, a **steroid emergency card**, and "sick-day" rules for adrenal insufficiency.

VACCINES

inactivated vaccines (influenza, COVID-19, pneumococcal) are fine. Live vaccines need specialist advice.

BRAF/MEK INHIBITORS

Pyrexia (dabrafenib + trametinib) needs interrupting treatment per the protocol and contacting the oncology team; **Photosensitivity** (vemurafenib): sun protection; New SCC/KA and new naevi (BRAF monotherapy); **QT prolongation** (vemurafenib, encorafenib): check interacting drugs; **CYP3A4 interactions**: dabrafenib is an inducer, which lowers levels of hormonal contraceptives (use a non-hormonal method; contraception is needed during treatment), warfarin and others. Use the BNF/SmPC interaction checker; **LVEF reduction and ocular toxicity** (MEK inhibitors: serous retinopathy); **Hand–foot skin reaction.**

PREGNANCY

effective contraception during treatment and for months afterwards (per SmPC; e.g. 4 months after pembrolizumab).

Treatment ladder

1

Contact

Never start immunosuppression independently: acute oncology grades and prescribes (ESMO / UKONS). Skin grade 1–2: emollient + topical steroid; urgent if blistering / mucosal.

2

Steroid stewardship

High-dose steroids: **PPI**, **glucose monitoring**, bone protection if prolonged, **PJP prophylaxis (co-trimoxazole)** if 20 mg prednisolone 4 weeks, **steroid emergency card**, sick-day rules.

3

BRAF / MEK

Fever on dabrafenib + trametinib: interrupt per protocol and call oncology. **Dabrafenib induces CYP3A4**: hormonal contraception unreliable (non-hormonal method); warfarin and others: interaction checker.

4

Vaccines & pregnancy

Inactivated vaccines fine; live vaccines: specialist advice. Effective contraception during and for months after (e.g. 4 months after pembrolizumab).

⚠ **CAUTION**

Prescribing limits and pitfalls

- **Loperamide-and-wait**: Not for diarrhoea on immunotherapy.

Referral

999 / A&E	Chest pain / suspected myocarditis, DKA, adrenal crisis, SJS / TEN.
SAME DAY	**Diarrhoea, breathlessness, severe headache, chest pain, new thirst / polyuria** on immunotherapy: acute oncology.
SAME DAY	Fever on dabrafenib + trametinib: protocol + oncology team.

Red flags and when to refer

- Diarrhoea, breathlessness, severe headache, chest pain or new thirst/polyuria in a patient on immunotherapy: **same-day acute oncology** contact.
- Fever on dabrafenib + trametinib: follow the protocol (interrupt) and contact the oncology team.

Complications and prognosis

Complications

- Colitis, hepatitis, pneumonitis, endocrinopathies (often permanent), myocarditis, SJS/TEN.

Prognosis

- Durable responses in about half with combination therapy; vitiligo on treatment is associated with better response.

Prescriber pearls (UK pharmacist IP)

Explain that side-effects can be serious but are usually treatable if caught early: "ring the hotline, not the GP's routine line".

Vitiligo during anti-PD-1 therapy is common and often correlates with better response. Protect the depigmented skin from sun.

Self-check quiz

4 questions · optional

Q1 Diarrhoea × 5 on nivolumab: action?

✓ Same-day acute oncology contact; don't just give loperamide.

Q2 PJP prophylaxis threshold?

✓ 20 mg prednisolone for 4 weeks.

Q3 Contraception issue with dabrafenib?

✓ CYP3A4 induction lowers hormonal levels: non-hormonal method.

Q4 Fever on dabrafenib + trametinib?

✓ Interrupt per protocol and contact oncology.

References

1. UKONS 24-hour triage tool v3, 2023 — https://www.ukons.org/ukons.org

Photo gallery

Clinical photographs 3

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Lumps, lesions & skin cancer Module 2 · Week 5

Teledermatology & lesion photography

How store-and-forward teledermatology works in the UK, the three images every lesion referral needs (regional, close-up with scale, dermoscopic), what it does well, and its pitfalls.

ALSO CALLED advice and guidance · dermoscopic images · image standards · store-and-forward · PCDS image quality · digital referral

RELATED PAGES Describing skin lesions: the language of dermatology · Melanoma: recognition & referral · Skin layers, Fitzpatrick skin types & dermoscopy

KEY FACTS

- UK teledermatology is mostly **store-and-forward**: advice and guidance, or images on all **suspected-cancer skin referrals**.
- Image standard: **three views**: regional (where), close-up with ruler, and **dermoscopic**.
- Good images let specialists triage, reassure or fast-track and avoid wasted appointments.
- Pitfalls: poor focus, glare, no scale, wrong lesion, missing history; images can't be palpated.
- Consent, secure systems and record-keeping are required.

Overview

Remote assessment of skin lesions from. In the UK it is mostly **store-and-forward** (asynchronous) **advice-and-guidance** or **2WW triage**, using dermoscopic + close-up + regional photographs.

Aetiology: causes and risk factors

Myths to address

- **"PHOTOS REPLACE EXAMINATION"**
  they support triage; full-skin checks still matter.

How it works

Teledermatology works by **sending the right images and history** to the right specialist.

1 **Capture**

Trained staff take standard regional, close-up and dermoscopic images.

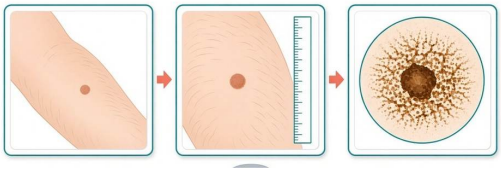

Fig. 1 · Three standard views.

AI-generated illustration

2 **Forward**

Images and history go through a secure NHS system.

3 **Triage**

Specialist decides: reassure, treat in primary care, routine or urgent clinic.

4 **AI-assisted triage**

Some services add AI risk scoring (NICE HTG746).

Clinical features

How it works

NHS England's **A teledermatology roadmap** (2023) describes models:

- primary-care advice and guidance
- referral triage (images attached to all 2WW skin referrals)
- community "image capture hubs" with trained medical photographers
- AI-assisted triage (HTG746; see Recognition, diagnostic accuracy, dermoscopy and AI).

Image standards (BAD/PCDS/Primary Care Dermatology quality standards)

- **THREE VIEWS**
  a regional/whole-site image (anatomical location), a close-up with a **ruler/scale**, and a **dermoscopic image** (polarised, in focus, the whole lesion in view).
- Good lighting, no flash glare, a plain background, a labelled patient ID, consent, and secure NHS-approved systems (**no personal phones/WhatsApp** without an approved app and policy). Include history (duration, change, symptoms, 7-point score, risk factors, Fitzpatrick type).

How it performs

For lesion triage, accuracy is comparable to face-to-face assessment for many lesions when dermoscopic images are of good quality. It reduces unnecessary face-to-face appointments and speeds up the management of cancers. Image quality is the main failure mode. Darker skin types are under-represented in training.

Limitations / pitfalls

Nodular/amelanotic melanoma can be under-called on. Teledermatology must not delay a 2WW-meeting referral; where a pathway uses, they **accompany** the urgent referral rather than replace it.

Fig. 2 · Dermoscopic skin examination — the image that travels with a teledermatology referral.
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Fig. 3 · Clinical photography for a teledermatology referral: overview, close-up and dermoscopy.
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Diagnosis and investigations

History

- Duration, **change**, symptoms, risk factors (sunburn, sunbeds, family history, immunosuppression), 7-point score.

Quality

- Check focus and framing before sending.

Tricky sites

- Scalp, genitals, nails: may need face-to-face.

Differential diagnoses

	CONDITION	DISTINGUISHING FEATURES
	Melanoma pathway	Score 3: 2WW with images.
	Changing naevus	Dermoscopy essential.
	SCC	Palpate nodes; images can't.
	BCC	Routine pathway usually.

Management

- Learn the local teledermatology pathway and image capture standards. An IP in a PCN may capture images if trained.
- **Case 3:** a lesion changing over 2 years (change = 2, plus any other feature) 2WW referral with. The specialist arranges excision biopsy within the pathway.

Treatment ladder

1 **Learn**

Local teledermatology pathway and image capture standards. IPs in a PCN may capture images if trained.

2 **Refer**

Changing lesion (change = 2 + any other) 2WW **with images**; specialist arranges excision biopsy.

3 **Consent**

Explain photos may be reviewed first and they may be booked straight for surgery; consent for storage.

⚠ CAUTION

Prescribing limits and pitfalls

- **Poor images**: Refer face to face rather than delay.

Referral

URGENT Teledermatology must **not delay** a suspected cancer pathway: poor images refer face to face.

Red flags and when to refer

- Teledermatology must not delay a **suspected cancer pathway** referral: if images are poor, refer face to face.

Complications and prognosis

Complications

- Missed diagnosis from poor images; information governance breaches.

Prognosis

- Not applicable.

Prescriber pearls (UK pharmacist IP)

Tell the patient that photos may be reviewed by a specialist first and they may be booked straight for surgery. Get consent for image storage.

Self-check quiz

4 questions · optional

Q1 Three standard views?

✓ Regional, close-up with ruler, dermoscopic.

Q2 Dermoscopic image requirements?

✓ Polarised, in focus, whole lesion in view.

Q3 Images poor on a suspected melanoma?

✓ Refer face to face; don't delay.

Q4 What must the patient consent to?

✓ Image capture and storage, and specialist review.

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dermnetnz.org

Photo gallery

Clinical photographs 2

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CHAPTER 23

Skin surgery

Minor skin surgery at knowledge level: pre-operative checks, anticoagulants, local anaesthetic doses, allergy, consent and complications, procedures and aftercare.

1. Skin surgery: pre-operative checks, anticoagulants & cardiac devices
2. Local anaesthetics & allergy in skin surgery
3. Skin surgery: consent, complications & danger zones
4. Skin surgery procedures: excision, curettage, cryotherapy & Mohs
5. Wound aftercare: sutures, activity, infection & scars

Skin surgery Module 2 · Week 6

Skin surgery: pre-operative checks, anticoagulants & cardiac devices

The medication and patient checks before minor skin surgery: allergies, anticoagulants and antiplatelets (BSDS 2023: continue most), diabetes, immunosuppression, cardiac devices, and why NICE CG64 does not recommend endocarditis prophylaxis.

ALSO CALLED pre-op assessment · warfarin skin surgery · DOAC skin surgery · BSDS 2023 · endocarditis prophylaxis · pacemaker diathermy · +1

RELATED PAGES Skin surgery: consent, complications & danger zones · Local anaesthetics & allergy in skin surgery · Skin surgery procedures: excision, curettage, cryotherapy & Mohs · Wound aftercare: sutures, activity, infection & scars

KEY FACTS

- Pre-op check: is the procedure **indicated, funded, safe and wanted**?
- Antithrombotics (BSDS 2023) usually **continue**; thrombotic harm from stopping outweighs bleeding harm.
- Write a clear **antithrombotic plan** (doses to take or omit, restart time).
- Post-op analgesia: **paracetamol first**; avoid NSAIDs in anticoagulated patients. Many antibiotics raise INR with warfarin.
- Check allergies (latex, chlorhexidine, LA, plasters), pacemakers (diathermy), keloid history, smoking, steroids.

Overview

A structured check that the procedure is **indicated, funded, safe and wanted**. It happens before any skin surgery, however minor.

Antithrombotics (BSDS 2023)

Guidance on whether to **continue, adjust or (rarely) pause** antithrombotics for dermatological surgery. The guiding principle is that **thrombotic harm from stopping usually outweighs bleeding harm from continuing**.

Aetiology: causes and risk factors

Myths to address

- **"STOP ASPIRIN A WEEK BEFORE EVERY EXCISION"**
  this is usually unnecessary and risky.

Why it matters

Most complications are **predictable from patient factors** checked before surgery.

1. **Bleeding**
   Antithrombotics and high-vascularity sites.
2. **Infection**
   Diabetes, immunosuppression, lower-leg and groin sites.
3. **Poor healing**
   Smoking, steroids, oedema and the lower leg.
4. **Scarring**
   Keloid tendency and high-tension sites.
5. **Allergy**
   Latex, chlorhexidine, dressings.

Clinical features

Why it matters

Most complications of minor skin surgery are predictable from patient factors: bleeding (antithrombotics), infection (site, diabetes), poor healing (smoking, steroids, the lower leg), bad scars (keloid tendency, site), and allergy.

Checklist (teaching version)

DOMAIN	ASK / CHECK	WHY
Indication	Diagnosis? Is removal needed (symptomatic, diagnostic doubt) or cosmetic?	NHS EBI/local policies restrict removal of benign lesions. Case 1 (**painful** DF) may qualify on symptoms
Medical history	Cardiac disease (recent MI/ACS, stents, valves, **pacemaker/ICD**), bleeding disorders, diabetes, immunosuppression, epilepsy, pregnancy, blood-borne viruses	Haemostasis method, infection risk, timing
Medicines	**Anticoagulants/antiplatelets** (see Antithrombotics and skin surgery), **β-blockers**, TCAs/MAOIs, steroids, immunosuppressants, isotretinoin (old advice to delay surgery is no longer strict; individualise), SSRIs (bleeding), supplements (fish oil, ginkgo, garlic, vitamin E)	Bleeding, LA safety, healing
Allergies	**Latex**, **LA** (true allergy rare), **chlorhexidine**, iodine/povidone, **adhesive dressings**, antibiotics, **parabens/sulfites**	see Allergy: latex, chlorhexidine, LA/parabens; anaphylaxis
Scarring history	Keloids or hypertrophic scars (self or family), skin of colour, site (chest, shoulder, earlobe, upper back)	Consent; site choice
Lifestyle	**Smoking** (healing), occupation and sport, travel, ability to keep the wound dry	Aftercare planning

DOMAIN	ASK / CHECK	WHY
Capacity and consent	MCA 2005 where relevant (the skin cancer notes see Antithrombotics and skin surgery)	–
Weight	Needed for weight-based LA dosing; **use ideal body weight in obesity**	see Local anaesthetics: doses, adrenaline, toxicity, alternatives and topical agents

How to assess

History, medication reconciliation (including OTC and herbal products), allergy history with the reaction type, and examination of the site and regional nodes (if the lesion could be malignant).

"Defer or redirect" situations

SITUATION	ACTION
Suspected melanoma	Refer to the skin cancer team. **No primary-care excision** (IOG)
Suspected high-risk BCC/SCC	Refer (the skin cancer notes see Procedures at knowledge level)
Lesion purely cosmetic	Usually not NHS-funded
Recent ACS, stent within 12 months on DAPT, unstable cardiac disease	Discuss timing. **Never stop DAPT without cardiology input**
INR above the threshold (see Antithrombotics and skin surgery)	Postpone; liaise with the anticoagulation service
Pregnancy	Defer elective procedures if possible. Lidocaine is generally considered acceptable for necessary procedures

Antithrombotics (BSDS 2023)

Evidence

Stopping warfarin or antiplatelets for skin surgery has been associated with strokes, emboli and deaths. Bleeding complications on continued therapy are mostly minor and manageable (pressure, haemostasis, dressings).

Recommendations (teaching summary; check the full BSDS document)

DRUG	BSDS 2023 APPROACH (SUMMARY)
Aspirin, clopidogrel and other antiplatelets (including DAPT)	**Continue.** Stopping DAPT after a recent stent risks stent thrombosis
Warfarin	**Continue.** Check the **INR before surgery (on the day or within ~72 h)**. **Proceed if INR <3.5** for most procedures. For higher-bleeding-risk procedures (large flaps/grafts), aim for the **lower end of the patient's therapeutic range**, liaising with the anticoagulation clinic
DOACs (apixaban, rivaroxaban, edoxaban, dabigatran)	**Continue for most procedures.** For **high-bleeding-risk procedures in patients at lower thrombotic risk**, consider omitting the DOAC **24 h before** (**48 h if eGFR <50**, especially dabigatran) and restarting **~24 h after** once haemostasis is secure. Timing relative to a once-daily dose (e.g. schedule surgery before the next dose) is a practical alternative
LMWH	Individualise. Often omit the dose on the morning of surgery (specialist/anticoagulation team)

DRUG	BSDS 2023 APPROACH (SUMMARY)
Highest thrombotic risk (do not interrupt without specialist advice)	**Valvular AF / mechanical valves**, **VTE within 3 months**, **CHA_2DS_2-VASc 5**, recent stroke/TIA, recent coronary stent
Platelet count	Generally **>50 × 10^9/L** for most skin procedures

Case 1 (warfarin for AF; small excision of an arm DF): **continue warfarin; check the INR (on the day, per course teaching, or within 72 h per local policy); proceed if <3.5**. Meticulous haemostasis; pressure dressing.

How to check

INR from the anticoagulation clinic or point-of-care testing. Renal function (eGFR/CrCl) for DOAC timing. Recent FBC if there is a bleeding history.

Pitfalls

- "Bridging" with LMWH for minor skin surgery is **not** needed and increases bleeding.
- Patients often stop aspirin themselves "to be safe". Tell them **not to**.

Antibiotic prophylaxis and cardiac devices

Infective endocarditis

- **NICE CG64 (UPDATED 2016)**
  antibiotic prophylaxis against infective endocarditis is not recommended routinely for people at risk (including those with prosthetic valves or previous endocarditis) undergoing dental, GI, GU, respiratory or other interventional procedures. This includes skin surgery.
- If the procedure is at a site of **established infection**, treat the infection with an antibiotic that covers organisms causing endocarditis (CG64 principle).
- The **SDCEP/ESC** positions differ for dentistry in high-risk patients (ESC 2023 still recommends prophylaxis for high-risk dental procedures), but **NICE CG64 is the England standard**. Individual cardiology advice can be sought for the highest-risk patients.

Surgical site infection prophylaxis

- **Clean skin surgery** (most minor procedures): **no routine antibiotic prophylaxis** (NICE NG125: do not use routinely for clean, non-prosthetic, uncomplicated surgery).
- Specialists may consider prophylaxis in **selected higher-risk situations**: procedures **below the knee**, groin, **wedge excisions of the lip or ear**, nasal flaps and grafts, and **inflamed or ulcerated lesions with clinical infection**. The decision should be individual, ideally a **single pre-operative dose** rather than a week-long course.
- The course **"1 week of prophylaxis for ulcerated lesions"** is **not standard UK practice** (stewardship; adverse effects; C. difficile). Treat established infection instead (see Aftercare: sutures, return to work and sport, infection and scars).

Pacemakers and ICDs

- Electrosurgery current can interfere with devices: inhibited pacing, inappropriate shocks, or reprogramming.
- **PREFERRED**
  electrocautery (heated wire; no current through the patient) or **bipolar diathermy** (current confined between the forceps tips), in **short bursts (<5 s)**, at the **lowest power**, **away from the device** (>15 cm).

- **Avoid monopolar** diathermy near the device. With an **ICD**, follow the **BSDS cardiac devices flowchart**: contact the cardiac physiology team about **suspending tachytherapy** if monopolar diathermy is essential, and have monitoring and a defibrillator available.
- Ask about devices at pre-assessment (see Pre-operative assessment: patient factors).

Other cardiac points

- **Recent MI** (course teaching: "defer for 6 months"): elective procedures are usually deferred after ACS, but a short procedure under LA is low risk. Timing is individual, and **cancer surgery should not be delayed** unnecessarily. **Never stop DAPT** after a stent without cardiology input.
- **UNCONTROLLED HYPERTENSION**
  check BP before surgery (adrenaline; bleeding).

Fig. 1 · Elliptical excision wound before closure.
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Fig. 2 · The same wound after primary closure with sutures.
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Fig. 3 · Healed linear scar after elliptical excision.
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Management

Lead the **medicines and allergy** part of the assessment: produce a clear pre-op medicines plan, including the antithrombotic plan, which β-blocker the patient takes, and the allergy list with reaction type.

Antithrombotics (BSDS 2023)

> Write a **clear peri-operative antithrombotic plan** in the record and give it to the patient (date of surgery, doses to take or omit, restart time).

> **POST-OPERATIVE ANALGESIA**
> **paracetamol first-line**. Avoid NSAIDs (and aspirin as an analgesic) in anticoagulated patients.

> **POST-OPERATIVE ANTIBIOTICS IN WARFARIN PATIENTS**
> many antibiotics raise the INR (clarithromycin, erythromycin, metronidazole, co-trimoxazole, fluoroquinolones, doxycycline) or lower it (rifampicin; flucloxacillin can reduce it). **Check the INR within 3–5 days** if starting an interacting antibiotic, and inform the anticoagulation service.

> DOAC interactions: clarithromycin raises levels (P-gp/CYP3A4); rifampicin lowers them.

Antibiotic prophylaxis and cardiac devices

Answer the "do I need antibiotics, I have a heart valve?" question with **CG64**: no routine prophylaxis. Advise good oral hygiene and prompt treatment of infections, and tell the patient to seek medical advice for symptoms of endocarditis (fever, night sweats, new murmur, weight loss).

Worked cases

CASE 1

54-year-old, painful arm DF; warfarin for AF; latex allergy; 60 kg

> **Is removal indicated?** Yes. It is **symptomatic (painful)**, which meets most commissioning criteria. An elliptical excision along RSTL with histology (the DF diagnosis is confirmed and variants excluded).

> **WARFARIN**
> **continue**; check the INR on the day (or within 72 h, per local policy); **proceed if INR <3.5** (BSDS 2023); meticulous haemostasis, a pressure dressing, and paracetamol for pain (no NSAIDs); check the INR if an interacting antibiotic is started

> **LATEX ALLERGY**
> first on the list, a latex-free room and kit, an anaphylaxis kit ready (IM adrenaline 500 micrograms of 1 mg/ml), and allergy documented. Check the chlorhexidine allergy status too.

> **CONSENT (MATERIAL RISKS)**
> bleeding/haematoma (higher on warfarin); infection; scar (3× lesion length; hypertrophic/keloid on the arm/shoulder); wound dehiscence; numbness/nerve injury; recurrence; histology result

Alternatives: leave it, or analgesia.

> **LA MAXIMUM (LIDOCAINE WITH ADRENALINE, TEACHING)**
> 7 mg/kg × 60 kg = **420 mg = 21 ml of 2%** (the course "~20 ml"), within the BNF absolute maximum of **500 mg**. **Plain** lidocaine: BNF maximum **200 mg** (3 mg/kg = 180 mg = 9 ml of 2%). In practice, a few ml suffices.

> **ADRENALINE**
> acceptable unless the patient is on a **non-selective β-blocker** such as propranolol or sotalol. Rate control in AF is often with bisoprolol, which is cardioselective and lower risk, so check which β-blocker they take. Use caution in Raynaud's.

CASE 2

21-year-old builder/footballer, 9 mm bleeding PG on the posterior neck

DIAGNOSIS

pyogenic granuloma (lobular capillary haemangioma). Exclude amelanotic melanoma, so **histology is mandatory**.

TREATMENT

curettage and cautery (shave/curette and cauterise the base) under LA, with the specimen to histology. Excision with layered closure is reserved for recurrence (deep absorbable plus surface nylon, as course teaching described).

AFTERCARE

keep dry and covered for 24–48 h; petroleum jelly; **no football, heavy lifting or contact sport for ~2 weeks**; **1–2 weeks of light duties** (fit note with adjustments); neck sutures (if used) out at 7–10 days; return if bleeding or infection; recurrence is possible (up to 15%)

ANATOMY

posterior triangle, so the spinal accessory nerve is superficial. Superficial C&C is safe; deeper excision needs care.

CASE 3

see the melanoma teledermatology case

See the melanoma notes. In brief: Breslow 1.4 mm ulcerated = T2b, stage IIA if N0, ~93–94% 5-year MSS; WLE 2 cm; consider SLNB; no adjuvant at IIA.

Referral

DON'T — Do not stop anticoagulants without specialist advice in **mechanical valves, recent VTE (3 months), recent stroke/TIA or recent stent**.

ROUTINE — Warfarin: check the INR before surgery; proceed for most procedures if **INR <3.5** (BSDS 2023).

Red flags and when to refer

- Do not stop anticoagulants without specialist advice in **mechanical valves, recent VTE (3 months), recent stroke/TIA or recent stent**.
- Warfarin: check the INR before surgery; proceed for most procedures if **INR <3.5** (BSDS 2023).

Complications and prognosis

Complications

- Unplanned bleeding, infection, wrong-site surgery.

Prognosis

- Not applicable.

Prescriber pearls (UK pharmacist IP)

Ask "**Do you or your family form thick or raised scars?**" Keloid-prone patients need to be told before surgery on high-risk sites.

Bleeding first aid after surgery: **firm continuous pressure for 20 minutes by the clock** with a clean pad, sitting upright, without peeking. If bleeding persists, or there is a rapidly expanding, tense, painful swelling (haematoma), seek same-day review.

Patients with valves may have been told "antibiotics before the dentist" in the past (pre-2008 practice). Explain the change and its basis (resistance and anaphylaxis risk outweighed unproven benefit).

Antithrombotics (BSDS 2023)

Antibiotic prophylaxis and cardiac devices

Where course teaching differs from UK guidance

TOPIC	COURSE TEACHING	UK GUIDANCE / PRACTICE	TEACHING POINT
DOACs	Stop 1–2 days before surgery	**BSDS 2023: continue for most procedures.** Omit 24 h (48 h if eGFR <50) only for high-bleeding-risk procedures in lower-thrombotic-risk patients; restart ~24 h after	Most skin surgery proceeds on DOACs
Warfarin INR	Check on the day, proceed if <3.5	BSDS: INR <3.5 (check on the day or within 72 h)	Aligned
Antibiotic prophylaxis for ulcerated lesions	1 week	Not routine (NICE NG125, stewardship). Consider a single dose in selected high-risk sites/situations. Treat established infection	—
Endocarditis prophylaxis (valves)		**NICE CG64: not recommended routinely**	ESC/SDCEP differ for dental procedures in high-risk patients
Defer surgery 6 months after MI	6 months	Individualise. Elective procedures are deferred after ACS, but minor LA surgery is low risk. Never stop DAPT without cardiology input	Don't delay cancer surgery unnecessarily

TOPIC	COURSE TEACHING	UK GUIDANCE / PRACTICE	TEACHING POINT
Adrenaline in ring/ digital blocks	Avoid	Evidence supports dilute adrenaline as safe in healthy digits. Avoid in Raynaud's/PVD. **Follow local policy**	Changing practice
Lidocaine maximum	7 mg/kg with adrenaline (20 ml 2% for 60 kg)	**BNF: 200 mg plain; 500 mg with adrenaline (adult)**; ideal body weight in obesity; reduce in frail, hepatic or cardiac patients	Know both conventions; never exceed the BNF absolute maximum
Adrenaline 1:80,000	Stated	A dental cartridge strength. 1:200,000 is more usual for skin surgery	Concentration maximum dose
Anaphylaxis	IM adrenaline 500 micrograms	RCUK 2021: repeat after 5 min; antihistamines/ steroids not first-line	Aligned + update
Skin prep		Alcohol preps must dry before diathermy (fire); avoid chlorhexidine near eyes/ears	—
Operative technique in course discussion	Detailed technique	**this page keeps it at knowledge level**	Skills need supervised training/credentialing

Self-check quiz

5 questions · optional

Q1 When should you think of Skin surgery: pre-operative checks, anticoagulants & cardiac devices?

✓ A patient booked for an excision, curettage or biopsy, especially on anticoagulants, antiplatelets or with a pacemaker or valve.

Q2 What drives it?

✓ Bleeding, infection, poor healing and device interference are mostly predictable from the medicine list and history.

Q3 What is the backbone of care?

✓ Continue antiplatelets and warfarin (INR under 3.5) and, for most procedures, DOACs (BSDS 2023); no routine endocarditis prophylaxis (NICE CG64); bipolar diathermy or cautery near devices.

Q4 What must you never miss?

✓ Stopping anticoagulants in high thrombotic-risk patients; stopping DAPT after a recent stent; unchecked INR.

Q5 Name a red flag that needs escalation.

✓ Do not stop anticoagulants without specialist advice in **mechanical valves, recent VTE (3 months), recent stroke/TIA or recent stent**.

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Photo gallery

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Skin surgery Module 2 · Week 6

Local anaesthetics & allergy in skin surgery

Lidocaine maximum doses (BNF: 200 mg plain, 500 mg with adrenaline), how to calculate them, adrenaline cautions, local anaesthetic systemic toxicity, topical anaesthetics and allergy to latex, chlorhexidine or local anaesthetic.

ALSO CALLED lidocaine maximum dose · lidocaine with adrenaline · LAST · Intralipid · EMLA · Ametop · +3

RELATED PAGES Skin surgery: pre-operative checks, anticoagulants & cardiac devices · Skin surgery procedures: excision, curettage, cryotherapy & Mohs · Skin surgery: consent, complications & danger zones · Wound aftercare: sutures, activity, infection & scars

KEY FACTS

- Amides (lidocaine, prilocaine, bupivacaine) and esters (tetracaine, benzocaine) **block voltage-gated sodium channels**.
- **Adrenaline** gives vasoconstriction: less bleeding, longer action, higher maximum dose.
- BNF adult maximum lidocaine (infiltration): **200 mg plain**, **500 mg with adrenaline**; **1% = 10 mg/ml**. Use ideal body weight; reduce in frail, liver or heart failure.
- Toxicity: perioral tingling, metallic taste, dizziness, then seizures and arrhythmia: stop, call for help, lipid emulsion.
- Allergy to amides is rare; latex and **chlorhexidine** are commoner perioperative allergens.

Overview

Amide local anaesthetics (lidocaine, prilocaine, bupivacaine, levobupivacaine, articaine, mepivacaine) and esters (tetracaine/amethocaine, benzocaine, chloroprocaine) block voltage-gated sodium channels in nerves. **Adrenaline** is added for vasoconstriction: less bleeding, longer action, and a higher maximum dose.

Allergy: latex, chlorhexidine and local anaesthetic

- **Latex allergy** is either **type I** (IgE-mediated: urticaria, rhinitis, asthma, **anaphylaxis**) or **type IV** (allergic contact dermatitis to rubber accelerators: thiurams, carbamates, MBT). Irritant hand dermatitis from gloves is not an allergy.
- **Chlorhexidine** is an increasingly recognised cause of **perioperative anaphylaxis** (MHRA 2012 alert). It is found in skin preps, catheters, lubricants and wipes.

Aetiology: causes and risk factors

Allergy: latex, chlorhexidine and local anaesthetic

Risk factors

- Healthcare workers.
- **Spina bifida and multiple childhood surgeries.**
- Atopy.
- **LATEX–FRUIT SYNDROME**
  cross-reaction with **banana, avocado, kiwi, chestnut** and others.

Myths to address

- **"NEVER USE ADRENALINE ON FINGERS"**
  modern evidence supports its use in digits in healthy patients (follow local policy).

How it works

Local anaesthetics **block sodium entry**, so nerves can't fire.

1. **Diffusion**
   The un-ionised drug crosses the nerve membrane.
2. **Channel block**
   Inside, the ionised form blocks voltage-gated sodium channels.

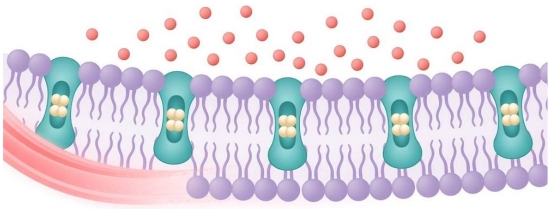

Fig. 1 · Local anaesthetic molecules plug sodium channels from inside the nerve, so the impulse stops.
AI-generated illustration

3. **Fibre order**
   Small pain fibres block first, then touch and motor.
4. **Adrenaline**
   Vasoconstriction slows removal, prolonging action and reducing bleeding.
5. **Inflamed tissue**
   Acidic infected tissue ionises the drug, so it works poorly.

Clinical features

Lidocaine maximum dose (BNF): the key Case 1 calculation

Useful conversion: 1% = 10 mg/ml; 2% = 20 mg/ml.

	BNF ADULT MAXIMUM (INFILTRATION)	WEIGHT-BASED CONVENTION (COMMON TEACHING)
Plain lidocaine	**200 mg**	~3 mg/kg (some SmPCs: 4.5 mg/kg, maximum 300 mg)
Lidocaine with adrenaline	**500 mg**	~7 mg/kg
Children (BNFC)	**Up to 3 mg/kg** (check the BNFC entry and age band)	—

- **Use ideal body weight in obesity. Reduce the dose in frail elderly people, in hepatic impairment, in heart failure** (reduced clearance), and in highly vascular sites (absorption).
- **Case 1, worked (teaching only):** 60 kg × 7 mg/kg = **420 mg** with adrenaline. This is below the BNF absolute ceiling of 500 mg. At 2% (20 mg/ml), 420 mg is **21 ml** (course teaching rounded to "20 ml"). **Plain** lidocaine at 3 mg/kg is 180 mg = **9 ml of 2%** (and under the 200 mg BNF cap). For a small DF excision, the volume actually used is typically a **few millilitres**, far below the maximum.
- **NOTE ON 1:80,000 ADRENALINE**
  lidocaine 2% with adrenaline 1:80,000 is the **dental cartridge** formulation. For skin surgery in UK practice, **1% or 2% with 1:200,000 adrenaline** is more usual. The maximum-dose principle is the same.

Adrenaline: cautions

SITUATION	CONCERN	UK TEACHING POSITION
Non-selective β-blockers (propranolol, nadolol, sotalol, timolol eye drops in large doses)	Unopposed α-stimulation, which can cause **severe hypertension with reflex bradycardia**	**Real interaction.** Use the lowest effective dose and concentration (or plain LA). Cardioselective β-blockers (bisoprolol, atenolol) are lower risk
Digital / ring blocks, penis, ear lobe, nasal tip	Historical fear of ischaemic necrosis	**Modern evidence (including large series and Cochrane review) suggests dilute adrenaline is safe in digits** in healthy patients. Phentolamine can reverse it. **Follow local policy.** Avoid in **Raynaud's, peripheral vascular disease, scleroderma and vasospasm**
Uncontrolled hypertension, ischaemic heart disease, arrhythmia, hyperthyroidism, phaeochromocytoma	Sympathomimetic effects	Minimise or use plain LA
TCAs, MAOIs, cocaine use	Potentiated pressor response (TCAs > MAOIs for exogenous adrenaline)	Caution
Sulfite allergy	Adrenaline-containing preparations contain sodium metabisulfite	Use plain LA

Local anaesthetic systemic toxicity (LAST): recognise and respond

> **EARLY (CNS)**
> perioral numbness or tingling, **metallic taste**, tinnitus, light-headedness, visual disturbance, slurred speech, agitation or drowsiness.

LATE

muscle twitching **seizures** coma; **cardiovascular:** bradycardia, arrhythmias, hypotension, **cardiac arrest** (bupivacaine is most cardiotoxic).

RESPONSE (ASSOCIATION OF ANAESTHETISTS LAST GUIDELINE)

stop injecting; call for help (999 in community settings); airway, 100% oxygen, ABC; treat seizures (benzodiazepine); CPR if arrested; **IV 20% lipid emulsion** (Intralipid®) per the guideline, where available. Settings using larger LA volumes should know where it is kept

PREVENTION

calculate the maximum dose before starting, **aspirate before injecting**, inject slowly, use the lowest effective volume, and keep patients under observation.

Distinguish from vasovagal (pallor, sweating, bradycardia, quick recovery when lying flat; the commonest event) and **adrenaline effect** (palpitations, tremor, anxiety, tachycardia).

Alternatives

AGENT	ONSET / DURATION	ADULT MAXIMUM (BNF / SMPC SUMMARY; CHECK CURRENT)	NOTES
Lidocaine	Fast / 1–2 h (longer with adrenaline)	200 mg plain; 500 mg with adrenaline	Workhorse
Prilocaine	Fast / medium	**400 mg** (plain or with felypressin)	Least toxic amide. **Methaemoglobinaemia** at high doses (avoid in infants, G6PD deficiency, and with other oxidising drugs)
Bupivacaine	Slow / long (4–8 h)	**150 mg** (with or without adrenaline; BNF); 2 mg/kg is a common convention	Most cardiotoxic. Used for post-op analgesia in larger procedures
Levobupivacaine	Slow / long	**150 mg** single dose	Less cardiotoxic S-enantiomer
Articaine	Fast / medium	Up to **7 mg/kg** (dental)	Dental mainly
Mepivacaine	Fast / medium	Per SmPC	Dental
Tetracaine (ester)	Topical	–	Higher allergy potential (PABA)

True LA allergy is rare. Most "allergies" are vasovagal, adrenaline effects or anxiety. **Ester allergy (PABA)** does not cross-react with amides. **Methylparaben** (the course point) is a preservative in **multi-dose vials** and can cause reactions in paraben-sensitive patients, so use **single-dose, preservative-free ampoules**. Refer suspected LA allergy for allergy testing before elective surgery.

Topical anaesthesia

PRODUCT	CONTENT	APPLICATION	NOTES
EMLA® 5% cream	Lidocaine 2.5% + prilocaine 2.5%	Under occlusive dressing for **60 min** (up to 5 h)	**Methaemoglobinaemia risk** in infants under 3 months and with oxidising drugs (dapsone, sulfonamides, nitrates). Blanching then erythema is normal. Max area/dose varies by age (BNF/BNFC)

PRODUCT	CONTENT	APPLICATION	NOTES
Ametop® 4% gel	Tetracaine	Under occlusion for **30–45 min** (remove by 60 min)	Vasodilatation helps cannulation. Ester, so allergy is possible. Not for infants under 1 month
LMX4®	Lidocaine 4% liposomal	30 min	Prescribe/OTC

Topicals reduce needle pain. They **do not** provide surgical anaesthesia for an excision.

Allergy: latex, chlorhexidine and local anaesthetic

Clinical features

Type I reactions within minutes of contact: urticaria, angioedema, wheeze, hypotension. Type IV: eczema at the contact site 24–72 h later.

Fig. 2 · Positive latex skin prick test (type I allergy, a wheal). Confirms immediate-type latex allergy, so the patient needs a latex-free environment.

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Diagnosis and investigations

Allergy: latex, chlorhexidine and local anaesthetic

History. Specific IgE to latex (k82) and components (Hev b 1, 3, 5, 6.01); skin prick test; patch testing for type IV.

Differential diagnoses

Allergy: latex, chlorhexidine and local anaesthetic

Irritant glove dermatitis; reaction to chlorhexidine, LA (rare), antibiotics, or adhesive dressings (acrylates, colophony); vasovagal syncope.

Management

Allergy: latex, chlorhexidine and local anaesthetic

Latex-allergic patient:

- **First on the list** (the course answer) so latex particles in the room are minimised. A **latex-free environment**: nitrile or neoprene gloves, latex-free tourniquets, BP cuffs, catheters, dressings and drug vial stoppers (check), and a sign on the door.
- **Prepared anaphylaxis kit.**
- Document the allergy clearly (type and reaction) in the record and referral.

Anaphylaxis (Resuscitation Council UK 2021):

- **IM adrenaline 1 mg/ml (1:1000): adults and children >12 years 500 micrograms (0.5 ml)**, into the anterolateral thigh. **Repeat after 5 minutes** if there is no improvement.
- Lie flat (or sit up if breathless), legs raised; call 999; high-flow oxygen; IV fluids if trained.
- **Antihistamines and steroids are not first-line** (RCUK 2021 de-emphasised them for the emergency phase).
- Refractory anaphylaxis (two IM doses without improvement) needs specialist help urgently.
- **Mast cell tryptase** samples (as soon as possible, at 1–2 h and at 24 h or later) and **referral to an allergy clinic** (NICE CG134).

Chlorhexidine: for an allergic patient use an alternative prep, e.g. **povidone-iodine** if not allergic, or alcohol alone.

Referral

999 / A&E — Perioral tingling, metallic taste, confusion, seizures or arrhythmia after injection: **local anaesthetic systemic toxicity**, stop, 999 and lipid emulsion per the AAGBI guideline.

DON'T — Never exceed the **BNF maximum**: lidocaine 200 mg plain or 500 mg with adrenaline (adult).

Red flags and when to refer

- Perioral tingling, metallic taste, confusion, seizures or arrhythmia after injection: **local anaesthetic systemic toxicity**, stop, 999 and lipid emulsion per the AAGBI guideline.
- Never exceed the **BNF maximum**: lidocaine 200 mg plain or 500 mg with adrenaline (adult).

Complications and prognosis

Complications

- Systemic toxicity, vasovagal faint, allergy (rare), haematoma, nerve injury.

Prognosis

- Not applicable.

Prescriber pearls (UK pharmacist IP)

- Warm, buffered lidocaine (with 8.4% sodium bicarbonate, 1:10) injected slowly through a fine needle reduces pain (practitioner technique; local policy).
- Tell the patient the numbness lasts 1–3 h (longer with bupivacaine). Avoid hot drinks or biting the lip after facial or oral blocks.
- **Adrenaline auto-injector** patients: check they carry **two** devices in date (MHRA 2017 advice), and that they know latex is hidden in balloons, condoms, rubber gloves and some dental and medical products.
- Latex-fruit syndrome foods may cause oral allergy or anaphylaxis.

Allergy: latex, chlorhexidine and local anaesthetic

Self-check quiz

5 questions · optional

Q1 When should you think of Local anaesthetics & allergy in skin surgery?

✓ Planning or checking a local anaesthetic dose, or a patient reporting an 'allergy' to local anaesthetic, latex or skin prep.

Q2 What drives it?

✓ Sodium-channel blockade stops nerve conduction; adrenaline prolongs action and reduces bleeding; overdose affects brain and heart.

Q3 What is the backbone of care?

✓ 1% = 10 mg/ml; BNF maximum 200 mg plain and 500 mg with adrenaline (weight-based 3 and 7 mg/kg as a cross-check); reduce in frailty, liver disease and heart failure; latex-free room and alternative prep for allergy.

Q4 What must you never miss?

✓ Local anaesthetic systemic toxicity (perioral tingling, seizures, arrhythmia): stop, call for help, AAGBI/ Intralipid protocol; chlorhexidine anaphylaxis.

Q5 Name a red flag that needs escalation.

✓ Perioral tingling, metallic taste, confusion, seizures or arrhythmia after injection: **local anaesthetic systemic toxicity**, stop, 999 and lipid emulsion per the AAGBI guideline.

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Photo gallery

Latex allergy 2

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Skin surgery Module 2 · Week 6

Skin surgery: consent, complications & danger zones

What to discuss when consenting for minor skin surgery (scar, infection, bleeding, nerve injury, recurrence), the anatomical danger zones of the face and neck, and safe practice: room, prep, specimen labelling and the histology fail-safe.

ALSO CALLED consent · dehiscence · haematoma · wound infection · keloid · temporal branch facial nerve · +3

RELATED PAGES Skin surgery: pre-operative checks, anticoagulants & cardiac devices · Wound aftercare: sutures, activity, infection & scars · Keloids and hypertrophic scars · Local anaesthetics & allergy in skin surgery

KEY FACTS

- Valid consent is **informed, voluntary and given with capacity**.
- **Montgomery (2015)** and GMC (2020): discuss **material risks** and reasonable alternatives, including **doing nothing**.
- Common complications: bleeding, infection, wound dehiscence, stretched or hypertrophic scars, keloids, nerve injury, recurrence.
- Danger zones temporal branch of the facial nerve, marginal mandibular branch, spinal accessory nerve (posterior triangle).
- Track every specimen: refer unexpected malignant histology within NG12 timeframes.

Overview

Valid consent in the UK is **informed, voluntary and given by a person with capacity**. Under **Montgomery v Lanarkshire* (2015)** and the **GMC Decision making and consent (2020)**, clinicians must discuss **material risks**: those a reasonable person in the patient's position would consider significant, or that this particular patient would. They must also discuss reasonable alternatives, including **doing nothing**.

Aetiology: causes and risk factors

Myths to address

- **"A SIGNED FORM IS CONSENT"**
  consent is the discussion; the form records it.

How it works

Complications are mostly **predictable from patient, site and technique**.

1. **Bleeding**
   Vascular sites and antithrombotics.

2 **Infection**

Contamination, site (groin, lower leg) and host factors (diabetes).

3 **Tension**

High-tension sites (back, chest, shoulders) stretch scars and dehisce.

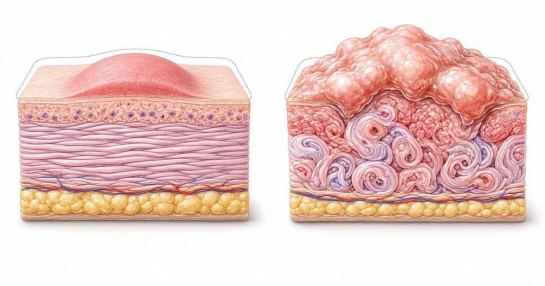

Fig. 1 · Hypertrophic scar versus keloid.

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4 **Anatomy**

Superficial nerves at danger zones are easily cut.

Clinical features

Why complications happen

Bleeding (vascular site, antithrombotics), infection (contamination, site, host), tension (dehiscence, stretched scar), genetics (keloid), and anatomy (superficial nerves).

Complications to discuss

COMPLICATION	NOTES	PHOTO
Bleeding / haematoma	Higher on antithrombotics. Pressure first aid (see Antithrombotics and skin surgery). An expanding haematoma needs review	–
Infection	~1–3% overall; higher on the **lower leg, groin, axilla**, with diabetes, smoking, immunosuppression	–
Wound dehiscence	Tension, infection, haematoma, early exertion (sport and lifting in Case 2), smoking	
Scar (always)	A line scar about **three times the lesion length** for an ellipse; **hypertrophic** (within the wound), **keloid** (beyond it), **stretched/atrophic** (back, shoulders), pigment change, "dog ears"	,
Nerve damage	Numbness around the scar is common and often improves. **Motor nerve injury** is rare but serious (danger zones, below)	–
Pain	Usually mild; paracetamol	–
Recurrence / incomplete excision	DF, cysts, lipomas, BCC; may need further surgery	
Allergy / LA reactions		–
Histology surprise	The diagnosis may differ from the clinical one	–

Anatomical danger zones

Knowledge for consent and referral decisions. These are **areas where motor nerves or key structures are superficial**:

ZONE	STRUCTURE AT RISK	CONSEQUENCE
Temple, over the zygomatic arch	**Temporal (frontal) branch of the facial nerve** (lies just under the SMAS/temporoparietal fascia)	Brow droop; unable to raise the eyebrow
Along the mandible, anterior to masseter	**Marginal mandibular branch of the facial nerve**	Asymmetric smile (lower lip weakness)
Posterior triangle of the neck (around Erb's point, mid-sternocleidomastoid)	**Spinal accessory nerve (CN XI)** (very superficial)	Trapezius weakness, shoulder droop, pain. **Relevant to Case 2 (posterior neck)**: superficial C&C of a PG is safe, but deeper excisions need care
Great auricular nerve (behind the ear, crossing SCM)	Sensory	Numb earlobe
Cheek (a line from the tragus to the mid-upper lip)	**Parotid duct** and buccal branches	Sialocele/fistula; weakness
Supraorbital and supratrochlear foramina	Sensory	Forehead/scalp numbness
Lateral fibular head	Common peroneal nerve	Foot drop
Digits	Digital nerves and vessels (lateral)	Numbness; ischaemia
Eyelid margin, lip vermilion, nasal ala, ear helix	Free margins	Distortion (ectropion, notching): refer to a specialist

Room, skin preparation and specimen handling

Room and PPE

- Minor skin surgery in primary care takes place in a clean treatment room meeting **CQC and local infection prevention standards**: washable surfaces, good lighting, an adjustable couch, a sharps bin, clinical waste, resuscitation equipment (anaphylaxis kit, AED access), and hand hygiene facilities.
- **Aseptic non-touch technique (ANTT)**, sterile gloves for excisions, eye protection (splash/plume), and surgical masks.
- **Surgical smoke** from electrosurgery contains particulates and may carry viable HPV. **Use smoke evacuation** for diathermy (NHS/HSE guidance) and FFP masks where indicated.

Skin preparation

- **Chlorhexidine 0.5% in 70% alcohol** (or 2% chlorhexidine in alcohol) or **povidone-iodine** (aqueous or alcoholic).
- **Avoid chlorhexidine near the eyes** (keratitis) and **middle ear** (ototoxicity). Use aqueous povidone-iodine there.
- **FIRE RISK**
  alcohol-based preps must dry completely before electrosurgery or diathermy. Pooled alcohol in hair or drapes has caused burns (NPSA/MHRA alerts). Wait about 3 minutes, more on hairy skin.
- Hair removal (if needed) by **clipping, not shaving** (NICE NG125).

Specimen handling

- **Every excised specimen goes for histology**, as NICE IOG and good practice require. Do not discard specimens, even "obvious" SKs or PGs.

- **Label the pot at the time** with **two patient identifiers**, the **site** (left/right, precise location) and the date. Use the **correct fixative** (10% neutral-buffered formalin, with about 10× the specimen volume). Use an **orientation suture** (e.g. "suture at 12 o'clock" or "long suture = superior") for malignancy excisions, so involved margins can be located.
- The **request form** must carry clinical details: the provisional diagnosis, previous histology, immunosuppression, and the question being asked.
- **FAIL-SAFE**
  keep a **minor surgery log** of all specimens, then check that every result has been **received, read, communicated to the patient and acted on**. Unexpected malignancy needs referral to the skin cancer MDT. This is a frequent root cause of patient-safety incidents.

How to audit

Annual audit of histology concordance (clinical vs histological diagnosis), complete-excision rates for malignancies, infection rates, and patient satisfaction (IOG requirement for accredited GPs).

Pitfalls

Unlabelled or mislabelled pots, missing request forms, results filed without action, and specimens lost in transport.

Fig. 2 · Surgical wound dehiscence on the neck. Causes are tension, infection, haematoma or early activity.

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Differential diagnoses

(post-op)

Normal healing redness vs infection (spreading erythema, pain, pus, fever); haematoma vs seroma; contact dermatitis from dressings or antiseptics (itchy, square-edged rash) vs infection.

Management

- Support consent with **written information** (BAD/BSDS/PCDS leaflets) and **time to decide**. Document the material risks discussed and the alternatives (e.g. for a painful DF: leave alone, analgesia, or excision).
- For complicated sites (eyelid margin, nose, lips, ears, danger zones) or high-risk patients, refer to dermatological surgery or plastics.

Room, skin preparation and specimen handling

IPs can own the **results-tracking fail-safe** in a practice, and can check that **unexpected malignant histology** is referred within NG12 timeframes.

Referral

DON'T — Every specimen needs a **fail-safe** system so a malignant result is never missed.

ROUTINE — Surgery over the temple, jawline or posterior triangle of the neck is a **danger zone**: specialist only.

Red flags and when to refer

- Every specimen needs a **fail-safe** system so a malignant result is never missed.
- Surgery over the temple, jawline or posterior triangle of the neck is a **danger zone**: specialist only.

Complications and prognosis

Complications

- As above.

Prognosis

- Most minor skin surgery heals without problems.

Prescriber pearls (UK pharmacist IP)

"Every cut leaves a scar. The aim is a neat scar in a natural line, but we cannot guarantee how your skin heals."

Use the photos of the wound journey and complications to make the consent conversation concrete.

Tell patients when to expect results (often 2–6 weeks) and to **chase if they hear nothing**. "No news is not good news."

Room, skin preparation and specimen handling

Self-check quiz

5 questions · optional

Q1 When should you think of Skin surgery: consent, complications & danger zones?

✓ Consenting, or answering a patient's questions about the risks of minor skin surgery.

Q2 What drives it?

✓ Tension, site, infection, bleeding and nerve anatomy drive most complications.

Q3 What is the backbone of care?

✓ Montgomery consent covering material risks; know the danger zones (temporal and marginal mandibular branches of the facial nerve, accessory nerve); label specimens at the bedside; run a histology fail-safe log.

Q4 What must you never miss?

✓ Nerve injury in danger zones; lost or unlabelled specimens; missed malignant histology without a fail-safe.

Q5 Name a red flag that needs escalation.

✓ Every specimen needs a **fail-safe** system so a malignant result is never missed.

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Photo gallery

Wound dehiscence

2

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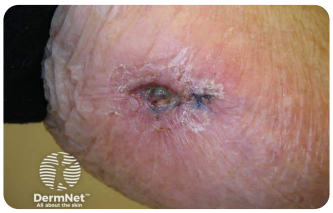

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Skin surgery Module 2 · Week 6

Skin surgery procedures: excision, curettage, cryotherapy & Mohs

What each procedure is for, at knowledge level: elliptical excision along relaxed skin tension lines, shave and punch biopsy, curettage and cautery, cryotherapy (and its blisters), Mohs surgery, flaps and grafts.

ALSO CALLED elliptical excision · shave · punch biopsy · curettage and cautery · cryotherapy · Mohs micrographic surgery · +4

RELATED PAGES Basal cell carcinoma (BCC) · Pyogenic granuloma · Melanoma: recognition & referral · Skin surgery: consent, complications & danger zones · Wound aftercare: sutures, activity, infection & scars

KEY FACTS

- Shave excision raised lesions level with skin; **not for suspected melanoma** (Breslow lost).
- Curettage and cautery pyogenic granuloma, SK, small low-risk BCC; no margin check.
- Punch biopsy full-thickness core for diagnosis; **incisional / excisional** ellipses with margins.
- Orient ellipses along **relaxed skin tension lines (RSTL)** for the best scar.
- Suspected melanoma: **excise whole** with a narrow margin, via the specialist team.

Overview

Concepts only: to understand referrals, consent and aftercare. Performing these needs supervised training.

Concepts only, for understanding referrals, consent and aftercare. Performing these needs supervised training and credentialing.

Aetiology: causes and risk factors

Myths to address

- **"FREEZE IT AND SEE"**
  cryotherapy destroys tissue and gives no histology.

How it works

Each method **trades tissue for histology and scar**.

1. **Surface methods**
   Shave, curette and cryotherapy remove or destroy superficial tissue.
2. **Full-thickness methods**
   Punch and ellipse excisions include dermis and fat for diagnosis and margins.

3 **Scar direction**

Cuts along RSTL heal with finer scars.

4 **Margins**

Standard BCC excision margins 4 mm, 5 mm for high risk; Mohs for high-risk sites.

Clinical features

Overview table

PROCEDURE	WHAT IT DOES	TYPICAL USES	SPECIMEN?	KEY LIMITATIONS
Shave excision/ biopsy	Removes a raised lesion level with the skin	SK, dermal naevus (cosmetic, private), skin tags, some BCC biopsies	Yes	**Not for suspected melanoma** (Breslow lost); naevi may recur
Curettage & cautery (C&C)	Scrapes off friable tissue, then cauterises the base	**Pyogenic granuloma**, SK, small low-risk BCC (trunk/limbs), viral warts	Yes (fragmented)	No margin assessment. Not for high-risk or facial BCC or for melanoma
Punch biopsy	Cylindrical core (2–8 mm)	Diagnosing rashes and tumours (BCC/SCC subtype, MF)	Yes	Samples only part; **not for melanoma diagnosis** (except specialist situations)
Elliptical excision	Full-thickness removal with margin, closed in a line	Suspected malignancy (with margin), symptomatic benign lesions, cysts	Yes (with margins)	Scar about 3× the lesion length; oriented along **relaxed skin tension lines (RSTL)**
Cryotherapy	Liquid nitrogen (–196 °C) freezes and destroys tissue	**AK, viral warts, SK**, some superficial BCC (specialist)	**No**	**Never on an undiagnosed or pigmented lesion that might be melanoma**
Electrosurgery	Heat or current: electrocautery (heated wire, no current through the patient), electrodesiccation/ fulguration, electrosection, **bipolar diathermy** (current between forceps tips)	Haemostasis, C&C, small lesions	Variable	Pacemakers/ICDs (see Antibiotic prophylaxis and cardiac devices); burns; fire risk with alcohol prep (see Room, PPE, skin preparation, specimen handling and the fail-safe); smoke plume
Secondary intention	Leaves the wound to heal by granulation	**Concave sites** (inner canthus, nasal alar groove, conchal bowl, temple), after C&C or shave, Mohs defects	–	Slower (weeks); dressing burden; contraction near free margins
Undermining	Freeing the skin edges in a tissue plane to reduce tension before closure	Larger ellipses	–	Haematoma risk; danger zones
Flaps / grafts	Moving adjacent skin (flap) or transferring skin from elsewhere (graft) to close defects	Larger facial defects after cancer excision	–	Specialist

PROCEDURE	WHAT IT DOES	TYPICAL USES	SPECIMEN?	KEY LIMITATIONS
Mohs micrographic surgery	Staged excision with **complete (100%) peripheral and deep margin examination** on frozen sections, repeated until clear	High-risk/recurrent BCC (BAD 2021), some cSCC, DFSP, LM (specialist)	Yes	Specialist centres; time-consuming

Langer's lines vs RSTL

- **Langer's lines** were mapped on cadavers (collagen orientation). Surgeons now prefer **relaxed skin tension lines (RSTL)** (Borges), seen as natural wrinkle lines in a living person and found by pinching the skin.
- Placing scars **parallel to RSTL** gives narrower, less visible scars. In practice: horizontal on the forehead, radial around the mouth and eyes, transverse in the neck creases, and along the long axis of limbs for melanoma excision biopsies (lymphatic drainage).

Cryotherapy: (knowledge level)

> **EXPECTED REACTIONS**
> redness and swelling within hours; **blister** (sometimes **haemorrhagic**) within 24–48 h; periorbital oedema after facial treatment; crusting and healing over 1–3 weeks

> **ADVERSE EFFECTS**
> **hypopigmentation** (avoid or use caution in skin of colour); hyperpigmentation; scarring; nail dystrophy; **nerve damage** over superficial nerves (sides of fingers); tendon damage; **recurrence or missed cancer** (recurrent BCC after cryotherapy)

> **CAUTIONS**
> cold urticaria, cryoglobulinaemia, Raynaud's, poor circulation (lower legs in elderly people heal poorly), and **undiagnosed lesions**.

> **NHS**
> commissioning for warts and SK is restricted in many areas (EBI/local policies). Self-care alternatives for warts: salicylic acid, OTC cryogen products (dimethyl ether/propane). These are less effective but suitable for many patients.

Pyogenic granuloma: why C&C

- A **friable, bleeding vascular lesion** that would bleed during shave alone. **C&C removes it and seals the base**, and the curettings go for **histology** (amelanotic melanoma, SCC and Kaposi are the "must-not-miss" mimics; the skin lumps notes see Procedures at knowledge level).
- Alternatives: shave + cautery; excision for recurrent lesions (with closure, as course teaching described); silver nitrate or cryotherapy for very small lesions where the diagnosis is certain (no specimen).
- **POSTERIOR NECK**
  superficial procedures are safe. A deeper excision would need awareness of the spinal accessory nerve (see Consent, complications and anatomical danger zones).

Fig. 1 · Ellipse marked along relaxed skin tension lines, which gives a finer scar.
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Management

BCC treatment ladder

See the skin cancer notes:

- surgery (standard excision 4 mm / 5 mm margins) Mohs for high-risk/recurrent
- C&C or cryo for small low-risk lesions
- imiquimod, 5-FU or PDT for superficial BCC
- radiotherapy if unfit
- hedgehog inhibitors for advanced disease.

Use the table to explain to patients **why** a clinician chose a method, and why some lesions (suspected melanoma) must be excised whole by the specialist team rather than shaved or frozen.

Referral

DON'T — Never use cryotherapy on a lesion without a **confident diagnosis**: there is no histology and cancers can recur beneath the scar.

Red flags and when to refer

- Never use cryotherapy on a lesion without a **confident diagnosis**: there is no histology and cancers can recur beneath the scar.

Complications and prognosis

Complications

- Recurrence (shave, C&C), scarring, infection, bleeding.

Prognosis

- Not applicable.

Prescriber pearls (UK pharmacist IP)

After cryotherapy: **leave blisters intact**, or drain with a sterile needle if tense and painful, keeping the roof. Use simple dressings and paracetamol. Seek help for signs of infection.

After C&C or shave: the wound is moist and heals over 2–3 weeks; petrolatum and a simple dressing.

Self-check quiz

5 questions · optional

Q1 When should you think of Skin surgery procedures: excision, curettage, cryotherapy & Mohs?

✓ Explaining a planned procedure to a patient, or recognising expected after-effects (cryotherapy blisters, flap scars).

Q2 What drives it?

✓ Each technique trades tissue sampling, margin control, cosmesis and recurrence differently.

Q3 What is the backbone of care?

✓ Excision for diagnosis and most cancers (3:1 ellipse along RSTLs); C&C for small low-risk lesions; cryotherapy only for confidently diagnosed benign or low-risk lesions; Mohs for high-risk or recurrent facial BCC.

Q4 What must you never miss?

✓ Freezing an undiagnosed lesion (no histology, recurrence); cryotherapy blisters and pigment loss in skin of colour.

Q5 Name a red flag that needs escalation.

✓ Never use cryotherapy on a lesion without a **confident diagnosis**: there is no histology and cancers can recur beneath the scar.

References

1. DermNet: Curettage and cautery — https://dermnetnz.org/topics/curettage-and-cautery
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2. DermNet: Cryotherapy — https://dermnetnz.org/topics/cryotherapy
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3. DermNet: Electrosurgery — https://dermnetnz.org/topics/electrosurgery
dermnetnz.org
4. DermNet: Mohs micrographic surgery — https://dermnetnz.org/topics/mohs-micrographic-surgery
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5. DermNet: Skin tension lines — https://dermnetnz.org/topics/skin-tension-lines
dermnetnz.org
6. BAD PIL: cryotherapy; Mohs micrographic surgery — https://www.bad.org.uk/pils/
bad.org.uk

Photo gallery

Excision and healing

4

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Cryotherapy after-effects

4

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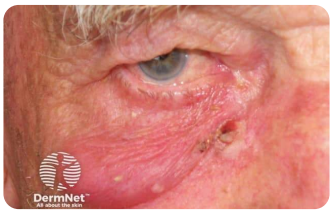

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Mohs, flaps and grafts

3

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Skin surgery Module 2 · Week 6

Wound aftercare: sutures, activity, infection & scars

Aftercare advice following minor skin surgery: dressing and washing, typical suture removal times by site, return to work and sport, recognising and treating wound infection, and managing hypertrophic and keloid scars.

ALSO CALLED suture removal | post-op wound infection | return to work | hypertrophic scar | keloid | silicone gel | +1

RELATED PAGES Skin surgery: pre-operative checks, anticoagulants & cardiac devices | Skin surgery: consent, complications & danger zones | Keloids and hypertrophic scars | Local anaesthetics & allergy in skin surgery

KEY FACTS

- Keep the dressing on and **dry for 24–48 h**, then wash gently; **plain petroleum jelly**, no routine topical antibiotics.
- Suture removal (typical): face 5–7 days, scalp and neck 7–10, trunk and arm 10–14, back, leg and joints 14 (up to 21).
- Avoid strenuous exercise and stretching the wound for 1–2 weeks.
- Infection: increasing pain, redness spreading **>1–2 cm**, pus, fever; **flucloxacillin** first line (check interactions for alternatives).
- Scars mature over 12–18 months; massage, silicone and sun protection help.

Aetiology: causes and risk factors

Myths to address

- **"ANTIBIOTIC OINTMENT HELPS HEALING"**
  it causes contact allergy and resistance; petroleum jelly is as good.

How wounds heal

Wounds heal in **overlapping phases** over months.

1. **Haemostasis**
   Clot forms within minutes.
2. **Inflammation**
   Neutrophils and macrophages clean the wound (days 1–4); mild redness is normal.
3. **Proliferation**
   Fibroblasts lay down collagen and new vessels; epithelium covers the wound (days 4–21).

4 **Remodelling**

Collagen reorganises and strengthens over 12–18 months.

Clinical features

Wound care

Keep the dressing on and **dry for 24–48 h**, then gently wash and pat dry. A thin layer of **plain petroleum jelly** keeps the wound moist (no routine topical antibiotics: they cause allergy (neomycin, bacitracin) and resistance). For scalp and posterior neck, keep the hair clean.

Suture removal (typical; local protocols vary)

SITE	DAYS
Face, eyelids	5–7 (3–5 for eyelids)
Scalp	7–10
Neck	7–10
Trunk, upper limb	10–14
Back, lower limb, over joints	14 (up to 21)

Absorbable deep sutures (e.g. poliglecaprone; the course Monocryl) dissolve over weeks. Surface nylon (Ethilon) is removed at the times above.

Activity

- **Avoid strenuous exercise, heavy lifting, contact sport and swimming for ~1–2 weeks** (until sutures are out and the wound is sound) to prevent **dehiscence and stretched scars**. course teaching said no football for 2 weeks.
- **WORK**
  a construction worker may need **1–2 weeks of modified or light duties**. The fit note (from any eligible healthcare professional, including pharmacists since July 2022) can advise "may be fit for work with adjustments".
- Protect the wound from dust and dirt at work (a waterproof dressing).

Scars

- Scars mature over 12–18 months.
- **Massage** with emollient once healed (from about 2–3 weeks).
- **Sun protection (SPF 30+) for at least a year** to avoid pigmentation.
- **Silicone gel or sheets** for hypertrophic-scar-prone patients (OTC/prescribable).
- **KELOIDS**
  specialist intralesional triamcinolone (hypopigmentation and atrophy are possible), pressure, cryotherapy, or radiotherapy after excision. **Avoid elective surgery on high-risk sites** in keloid formers.

Fig. 1 · Hypertrophic scar: raised but confined to the wound margins, and it often improves over months.

© DermNet · **Source**

Management

Infection (recognise and treat)

> **SIGNS**
>
> increasing pain, redness spreading **>1–2 cm** from the edge, swelling; **pus**, wound breakdown; fever

> Mild stitch-line redness in the first 48 h is normal.

> **MANAGEMENT (NICE NG125/NG141 PRINCIPLES, LOCAL FORMULARY)**
>
> swab if there is discharge; **flucloxacillin** first-line for S. aureus/streptococci; **penicillin allergy: clarithromycin or doxycycline**, but check **warfarin/DOAC interactions** and **statins** (clarithromycin with simvastatin is contraindicated; hold the statin); remove sutures if there is an abscess; same-day review if systemically unwell

> **MRSA** history needs a microbiology-guided choice.

The IP can prescribe analgesia (paracetamol), treat confirmed wound infection, advise on dressings and silicone, issue fit notes, and **check the INR** after interacting antibiotics. Review isotretinoin/acitretin users' wound healing.

Referral

SAME DAY — Spreading redness, fever, pus or increasing pain after surgery: assess for **wound infection or abscess** the same day.

Red flags and when to refer

- Spreading redness, fever, pus or increasing pain after surgery: assess for **wound infection or abscess** the same day.

Complications and prognosis

Complications

- Infection, dehiscence, haematoma, hypertrophic scar or keloid.

Prognosis

- Most heal well.

Prescriber pearls (UK pharmacist IP)

Give written aftercare with a **named contact for bleeding/infection** and the **date for suture removal and results**.

Self-check quiz

5 questions · optional

Q1 When should you think of Wound aftercare: sutures, activity, infection & scars?

✓ A patient a few days to weeks after minor skin surgery with a question about the wound, stitches, activity or scar.

Q2 What drives it?

✓ Healing goes through inflammation, proliferation and remodelling; tension and infection disrupt it; some people form keloids.

Q3 What is the backbone of care?

✓ Keep dry 24–48 h then wash; sutures out at 5–7 days on the face, 10–14 on the trunk and arms and about 14 on the back and legs (local protocols vary); avoid strenuous activity for 1–2 weeks; flucloxacillin for wound infection (clarithromycin or doxycycline if penicillin-allergic, checking interactions).

Q4 What must you never miss?

✓ Spreading cellulitis, abscess, dehiscence; keloids in people with skin of colour or on the chest, shoulders and earlobes.

Q5 Name a red flag that needs escalation.

✓ Spreading redness, fever, pus or increasing pain after surgery: assess for **wound infection or abscess** the same day.

References

1. DermNet: Wound closure — https://dermnetnz.org/topics/wound-closure
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2. DermNet: Suturing techniques — https://dermnetnz.org/topics/suturing-techniques
dermnetnz.org

Photo gallery

Scars and keloids 4

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APPENDIX A

Exam review

The module 2, 4 and 5 end-of-module exams with the recommended UK answer and why the distractors are wrong.

Exam review

Exam review: Module 2 · skin lesions, tumours & skin surgery

All 30 questions from the Module 2 end-of-module exam, with the recommended UK answer, why the distractors are wrong and links to the matching condition pages. Themes: Vascular vs malignant lookalikes, adnexal tumours, AK vs Bowen's, anatomy and electrosurgery.

End-of-module exam · Dec 2023

ALSO COVERS exam · SBA · revision · module 2 exam · single best answer

About this review

Revision key for the DermUSW end-of-module exam for Module 2 (Dec 2023): 30 single-best-answer questions. **Saved** is the answer ticked before submission (not a graded course key). **Recommended** is the teaching answer on current UK guidance (NICE / CKS, BNF, BAD, BASHH, Resuscitation Council UK, UKHSA).

NOTE

Key

- ✓ saved and recommended agree
- ✗ the saved answer is probably wrong
- ⚠ debatable or image-dependent; the reasoning is given and any course-vs-UK difference is flagged

✓ AGREE	✗ LIKELY WRONG	⚠ DEBATABLE
22	5 (Q12, Q19, Q25, Q30, Q8)	3 (Q3, Q5, Q22)

Main themes of errors: Vascular vs malignant lookalikes, adnexal tumours, AK vs Bowen's, anatomy, electrosurgery.

Each question links to the matching page in Lumps, lesions & skin cancer or Skin surgery, where one exists.

Questions & answers

1 **Young woman, rapidly growing bleeding papule after a scratch; granulation-like capillaries with epithelial collarette**

SAVED
Lobular capillary haemangioma

RECOMMENDED
Lobular capillary haemangioma (pyogenic granuloma)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Rapid growth after trauma, friable, collarette on histology. Amelanotic melanoma is the important DDx, so **always send for histology**. Bacillary angiomatosis needs HIV/Bartonella. Glomus is painful and subungual. KS is spindle cells and HHV-8

READ MORE
Pyogenic granuloma

2 **Warty papules on the backs of hands/feet since childhood, "church-spire" hyperkeratosis, palmar pits**

SAVED
Acrokeratosis verruciformis

RECOMMENDED
Acrokeratosis verruciformis of Hopf

WHY, AND WHY THE DISTRACTORS ARE WRONG
Linked to Darier disease (ATP2A2), which explains the palmar pits. EDV is HPV-driven flat warts with malignant potential. Stucco keratoses appear in older people. AK is sun-damaged older skin. Viral warts have no palmar pits and are not lifelong

READ MORE
—

3 **58-year-old farmer, nodular flesh-coloured neck lesion with telangiectatic rolled edge; next step to establish the diagnosis**

SAVED
Shave biopsy

RECOMMENDED
Excision biopsy (likely course key)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Looks like nodular BCC. UK practice: diagnostic **excision with a 4 mm margin** for a well-defined primary BCC (BAD BCC guideline, NICE CSG8) gives diagnosis and cure together. If unsure, a **punch/incisional** biopsy is used. **Shave biopsy** misses depth/subtype for nodular lesions. Tru-cut is for deep masses, triamcinolone is for keloid, and microneedling is cosmetic

READ MORE
Basal cell carcinoma (BCC)

4 **32-year-old, 6 weeks pregnant, nodular BCC**

SAVED
Postpone until after the birth

RECOMMENDED
Postpone excision until after delivery

WHY, AND WHY THE DISTRACTORS ARE WRONG
BCC is slow-growing and rarely metastasises, so elective surgery can wait. Local anaesthetic is safe if surgery is needed, but there is no urgency. Cryotherapy is unsuitable for nodular BCC. Radiotherapy in pregnancy is inappropriate

READ MORE
Basal cell carcinoma (BCC)

5 **60-year-old man, "moles of concern" on his back (image)**

SAVED
Junctional naevus

RECOMMENDED
Atypical (dysplastic) naevus is more likely

WHY, AND WHY THE DISTRACTORS ARE WRONG
New or flat junctional naevi are unusual at 60, because naevi mature to compound or intradermal with age. Irregular, multiple lesions in an older adult fit atypical naevi. Use the image: if one lesion meets the ABCDE/7-point checklist, think melanoma and **refer on the suspected-cancer pathway (NICE NG12)**

READ MORE
Atypical naevus syndrome & familial melanoma

6 **Bluish-black plaque since childhood, new satellite, blue-white veil, atypia, S100/Melan-A positive**

SAVED

Malignant blue naevus

RECOMMENDED

Malignant blue naevus

WHY, AND WHY THE DISTRACTORS ARE WRONG

A melanoma arising in or resembling a blue naevus: change, a satellite and atypia. LMM is facial and chronic sun damage. Pigmented BCC is S100-negative. KS is a vascular tumour

READ MORE

Halo naevus, blue naevus, Spitz and Reed naevi, Melanoma: recognition & referral

7 **Nodular melanoma: Breslow 4 mm, ulcerated, mitoses 1/mm²; strongest predictor of survival**

SAVED

Breslow

RECOMMENDED

Breslow thickness

WHY, AND WHY THE DISTRACTORS ARE WRONG

AJCC 8 T-stage is based on Breslow + ulceration, and Breslow is the strongest single predictor. Clark level is no longer used for staging. Mitotic rate was dropped from AJCC 8 T1 staging

READ MORE

Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment

8 **Pacemaker (recent battery, regular review), BCC on abdomen >5 cm away; electrosurgery**

SAVED

Pre/post checks + **monopolar**

RECOMMENDED

Bipolar short bursts + monitoring + arrest trolley + **inform cardiac physiology**. Option 4 is the likely course key; option 1 (bipolar + pre/post checks) is acceptable and more cautious

WHY, AND WHY THE DISTRACTORS ARE WRONG

Bipolar current stays between the forceps tips, so pacemaker interference is minimal. Monopolar current runs through the patient and can cause oversensing or inhibition. With a stable, recently checked **pacemaker** (not an ICD), routine pre/post checks aren't needed if there are no clinical concerns. ICDs need deactivation. Options 2, 3 and 5 all use monopolar

READ MORE

Skin surgery: pre-operative checks, anticoagulants & cardiac devices

9 **Renal transplant, multiple yellowish facial papules**

SAVED
Sebaceous hyperplasia

RECOMMENDED
Sebaceous hyperplasia

WHY, AND WHY THE DISTRACTORS ARE WRONG
Ciclosporin is a classic driver. Umbilicated yellow papules with "crown" vessels on dermoscopy. Transplant patients also get many SCCs and BCCs, so examine everything. Fibrofolliculomas suggest Birt-Hogg-Dubé. Syringomas are periorbital. Trichoepitheliomas are skin-coloured and central-facial

READ MORE
Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more

10 **Gardner syndrome, punctum, malodorous expressible contents**

SAVED
Epidermoid cyst

RECOMMENDED
Epidermoid cyst

WHY, AND WHY THE DISTRACTORS ARE WRONG
Multiple epidermoid cysts in FAP/Gardner (APC), so check colonoscopy surveillance. A punctum + keratin contents is diagnostic. Dermoid cysts are congenital lines of fusion. Milia are tiny. Hidrocystomas are translucent and periocular

READ MORE
Epidermoid, pilar and dermoid cysts

Milia

11 **12-year-old boy, hard calcified angulated cheek papule**

SAVED
Pilomatricoma

RECOMMENDED
Pilomatricoma

WHY, AND WHY THE DISTRACTORS ARE WRONG
Children, head and neck, rock-hard, "tent sign", calcification. Pilomatrical carcinoma is rare and occurs in adults. Dermatofibroma is on the legs with a dimple sign

READ MORE
Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more

12 **68-year-old, multiple painless red papules increasing over years (image), on SGLT2i etc.**

SAVED

Amelanotic melanoma

RECOMMENDED

Campbell de Morgan spots (cherry angiomas)

WHY, AND WHY THE DISTRACTORS ARE WRONG

Multiple, small, bright red, increasing with age, benign. Reassure. Amelanotic melanoma is usually a single, changing, often nodular lesion. HHT gives mucosal/digital telangiectasias with epistaxis. Pyogenic granuloma is solitary and bleeding. No drug causes this

READ MORE

Angiokeratomas & Fabry disease

13 **62-year-old street cleaner, firm indurated papillomatous lesion with central crust on the back of the hand for 2 months**

SAVED

Excision

RECOMMENDED

Surgical excision

WHY, AND WHY THE DISTRACTORS ARE WRONG

Likely cutaneous SCC (or KA, which is treated as SCC): excise with a 4–6 mm margin (BAD cSCC 2020). Curettage alone, PDT and imiquimod are not for invasive SCC. Radiotherapy is for people unfit for surgery

READ MORE

Cutaneous squamous cell carcinoma (cSCC)

14 **72-year-old woman, gradually enlarging pigmented macule on the side of the face (image)**

SAVED

Lentigo maligna

RECOMMENDED

Lentigo maligna

WHY, AND WHY THE DISTRACTORS ARE WRONG

Irregular, slowly enlarging facial pigmentation in chronically sun-damaged skin. Melanoma in situ, so refer (NG12). Solar lentigo is uniform and stable. SK is stuck-on/waxy. Nodular melanoma is raised

READ MORE

Melanoma: recognition & referral, Solar lentigines

15 **Dorsal distal-phalanx cyst with nail groove, clear jelly**

SAVED
Digital myxoid cyst

RECOMMENDED
Digital (mucous) myxoid cyst

WHY, AND WHY THE DISTRACTORS ARE WRONG
Linked to DIP osteoarthritis. Pressure on the matrix causes a longitudinal groove. Glomus is painful, blue and subungual. Subungual exostosis is hard and under the nail

READ MORE
Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more, Nail signs: psoriatic nails, Beau's lines & clubbing

16 **Lifetime malignant transformation of a single benign intradermal naevus**

SAVED
<1 in 2000

RECOMMENDED
Smaller than 1 in 2000

WHY, AND WHY THE DISTRACTORS ARE WRONG
The per-naevus risk is tiny (estimates ~1 in 200,000/year). Total naevus count matters more than any single mole

READ MORE
Melanocytic naevi (moles) & the changing mole

17 **70-year-old, 1 year of itchy worsening rash unresponsive to emollients and TCS (image), on ramipril/amlodipine**

SAVED
Skin biopsy

RECOMMENDED
Skin biopsy

WHY, AND WHY THE DISTRACTORS ARE WRONG
Persistent "eczema" that fails treatment in older age: exclude **mycosis fungoides/CTCL** (and drug eruption), which needs multiple biopsies. Oral steroids would mask it. Porphyrin screen and PDT are irrelevant

READ MORE
Mycosis fungoides & cutaneous lymphoma

18 **38-year-old woman, longstanding mole, now itchy; dermoscopy image**

SAVED
Blue naevus

RECOMMENDED
Blue naevus (as the dermoscopy image showed homogeneous steel-blue pigment)

WHY, AND WHY THE DISTRACTORS ARE WRONG
A homogeneous blue structureless pattern is typical. Itch and a family member with melanoma warrant documented review, not alarm

READ MORE
Halo naevus, blue naevus, Spitz and Reed naevi

19 **Painful solitary calf nodule for 4 years; eosinophilic tumour, Masson trichrome positive**

SAVED
Eccrine spiradenoma

RECOMMENDED
Leiomyoma

WHY, AND WHY THE DISTRACTORS ARE WRONG
Masson trichrome stains smooth muscle red. Pilar leiomyomas are painful (cold, touch). Painful skin tumours mnemonic "**ANGEL**" (Angiolipoma, Neuroma, Glomus, Eccrine spiradenoma, Leiomyoma). Of these, only leiomyoma fits the trichrome. Multiple leiomyomas raise the possibility of HLRCC (fumarate hydratase), so consider genetics

READ MORE
—

20 **20-year-old, previously pigmented mole now changed (white halo)**

SAVED
Halo naevus

RECOMMENDED
Halo naevus

WHY, AND WHY THE DISTRACTORS ARE WRONG
A symmetrical depigmented rim around a regressing naevus, common in young people (vitiligo link). Refer if the central naevus is atypical or the patient is older

READ MORE
Halo naevus, blue naevus, Spitz and Reed naevi

21 **22-year-old, rapidly growing exophytic bleeding hand lesion over 3 weeks**

SAVED
Curettage and cautery

RECOMMENDED
Curettage and cautery (send for histology)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Pyogenic granuloma. C&C removes it and gives tissue for histology. Steroids, antibiotics and reassurance are inappropriate

READ MORE
Pyogenic granuloma, Skin surgery procedures: excision, curettage, cryotherapy & Mohs

22 **60-year-old runner, firm slightly raised forehead lump of 6 weeks (image)**

SAVED
BCC

RECOMMENDED
BCC vs **keratoacanthoma/SCC**, depending on the image

WHY, AND WHY THE DISTRACTORS ARE WRONG
Six weeks of growth and firmness favour **KA (manage as well-differentiated SCC)**. A pearly rolled edge with telangiectasia would favour BCC. Either way the UK route is **excision and histology**, not watchful waiting. AK is flat/scaly. LM is pigmented

READ MORE
Cutaneous squamous cell carcinoma (cSCC), Basal cell carcinoma (BCC)

23 **70 kg; maximum 1% lidocaine with 1:200,000 adrenaline**

SAVED
49 ml

RECOMMENDED
49 ml

WHY, AND WHY THE DISTRACTORS ARE WRONG
7 mg/kg with adrenaline × 70 = 490 mg = 49 ml of 10 mg/ml. Plain lidocaine max is 3 mg/kg (21 ml). Use the BNF/ local policy

READ MORE
Local anaesthetics & allergy in skin surgery

24 **Warfarin after PE, INR 3.6, cyst excision**

SAVED
Option 3

RECOMMENDED
Rebook and ask him to liaise with his anticoagulation team/GP to get INR <3.5

WHY, AND WHY THE DISTRACTORS ARE WRONG
Don't stop warfarin unilaterally (PE indication). INR 3.6 is above the usual skin-surgery threshold (3.5, checked within 72 h). Missing doses or stopping for 3 days without the anticoagulation team risks VTE

READ MORE
Skin surgery: pre-operative checks, anticoagulants & cardiac devices

25 **54-year-old sunbather, enlarging, well-demarcated, red, scaly, crusted plaque on each lower leg**

SAVED
Actinic keratosis

RECOMMENDED
Bowen's disease (SCC in situ)

WHY, AND WHY THE DISTRACTORS ARE WRONG
A **well-demarcated** red scaly plaque on the lower legs of a woman is classic Bowen's. AK is a rough, ill-defined small macule or papule on the head and hands. UK management: 5-FU, imiquimod, PDT, cryotherapy, or C&C (BAD 2014 guideline)

READ MORE
Actinic keratoses, Seborrhoeic keratosis, Cutaneous squamous cell carcinoma (cSCC)

26 **6-year-old, yellow hairless scalp plaque since birth, thickening**

SAVED
Trichoblastoma

RECOMMENDED
Trichoblastoma

WHY, AND WHY THE DISTRACTORS ARE WRONG
Naevus sebaceous. The commonest secondary tumour is **trichoblastoma**, then syringocystadenoma papilliferum. BCC is rare (old data overestimated it). Excision is elective

READ MORE
Sebaceous naevus (naevus sebaceous)

27 **% of melanomas arising in a pre-existing naevus**

SAVED
25%

RECOMMENDED
25% (~20–30%)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Most melanomas arise de novo, so new or changing lesions matter most

READ MORE
Melanocytic naevi (moles) & the changing mole, Melanoma: recognition & referral

28 **Breslow 0.3 mm melanoma; 5-year survival**

SAVED
>90%

RECOMMENDED
>90%

WHY, AND WHY THE DISTRACTORS ARE WRONG
pT1a has a 5-year melanoma-specific survival of ~98–99%

READ MORE
Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment

29 **Recurrent pigmentation at a shave-excision scar within 3 months, confined to the scar**

SAVED
Pseudomelanoma

RECOMMENDED
Pseudomelanoma (recurrent naevus)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Pigment confined to the scar, early recurrence, benign original histology. **Always get the original histology** and refer if it spreads beyond the scar

READ MORE
Halo naevus, blue naevus, Spitz and Reed naevi

30 **Erb's point, the danger zone**

SAVED
Anterior-border option

RECOMMENDED
Halfway down the posterior border of sternocleidomastoid, ~2–3 cm above the clavicle; spinal accessory nerve, which causes trapezius weakness and shoulder drop (option 2)

WHY, AND WHY THE DISTRACTORS ARE WRONG
The accessory nerve runs superficially in the posterior triangle. The marginal mandibular nerve is at the jaw, not Erb's point. The anterior border options are wrong anatomically

READ MORE
Skin surgery: consent, complications & danger zones

Where the course answer differs from UK practice

- **M2 Q3 (BCC diagnosis):** shave biopsy is not ideal for nodular BCC. Use diagnostic excision (4 mm margin) or punch biopsy (BAD BCC guidance).
- **M2 Q8 (pacemaker):** UK dermatological surgery teaching says **bipolar**. Routine pre/post checks aren't needed for a stable pacemaker; ICDs differ.

Other exam reviews

- Exam review: Module 4 · inflammatory dermatoses
- Exam review: Module 5 · infections, infestations, photodermatoses & travel

Exam review

Exam review: Module 4 · inflammatory dermatoses

All 30 questions from the Module 4 end-of-module exam, with the recommended UK answer, why the distractors are wrong and links to the matching condition pages. Themes: Blistering mimics, IM (not IV) adrenaline, SJS vs TEN cut-off, pustular psoriasis vs DH, eczema herpeticum.

End-of-module exam · Apr 2024

ALSO COVERS exam SBA revision module 4 exam single best answer

About this review

Revision key for the DermUSW end-of-module exam for Module 4 (Apr 2024): 30 single-best-answer questions. **Saved** is the answer ticked before submission (not a graded course key). **Recommended** is the teaching answer on current UK guidance (NICE / CKS, BNF, BAD, BASHH, Resuscitation Council UK, UKHSA).

NOTE

Key

- ✓ saved and recommended agree
- ✗ the saved answer is probably wrong
- ⚠ debatable or image-dependent; the reasoning is given and any course-vs-UK difference is flagged

✓ AGREE	✗ LIKELY WRONG	⚠ DEBATABLE
20	7 (Q15, Q16, Q20, Q22, Q23, Q27, Q29)	3 (Q2, Q8, Q14)

Main themes of errors: Blistering mimics, **IM not IV adrenaline**, SJS vs TEN cut-off, pustular psoriasis vs DH, eczema herpeticum.

Each question links to the condition page that covers the topic, where one exists.

Questions & answers

1 **Trainee bricklayer, rough hands then vesicles, feet too (leather boots), better on holiday**

SAVED
Irritant + allergic CD

RECOMMENDED
ICD + ACD

WHY, AND WHY THE DISTRACTORS ARE WRONG
Wet cement is an irritant (alkaline) **plus chromate allergy** (cement and leather tanning, which explains the boots/feet). Gloves made of leather contain chromate too. No atopic history. Fungal infection doesn't improve on holiday

READ MORE
Contact dermatitis, Hand eczema (chronic, occupational & pompholyx)

2 **4-year-old, red swollen itchy cheeks, flexural scratching, red watery eyes, antibacterial soap made it worse**

SAVED
Establish a diagnosis first

RECOMMENDED
No ideal option. UK management would be **stop the soap emollient + soap substitute mild TCS (hydrocortisone 1%) for the face**, stepping to moderate for short periods if needed

WHY, AND WHY THE DISTRACTORS ARE WRONG
The picture is AE (flexures + face) ± irritant soap. "Establish a diagnosis first" is defensible only if the image is atypical (consider ACD, impetiginisation, or eczema herpeticum if punched-out erosions). **Potent TCS on a 4-year-old's face is not UK practice** (NICE CG57). Emollients alone are insufficient for an active flare. Oral antibiotics only if infected (NG190)

READ MORE
Eczema / dermatitis

3 **Linear vesicles/bullae on exposed limbs after clearing shrubs near woods**

SAVED
Poison ivy

RECOMMENDED
ACD to plants (poison ivy, urushiol)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Linear streaks where the plant brushed the skin. In the UK, think also of **phytophotodermatitis** (giant hogweed, rue, citrus) and primula. BP doesn't give linear streaks. Zoster is dermatomal

READ MORE
Phytophotodermatitis and cutaneous amyloidosis

4 **Pruritic wheals with heat, anxiety and exercise**

SAVED
Cholinergic urticaria

RECOMMENDED
Cholinergic urticaria

WHY, AND WHY THE DISTRACTORS ARE WRONG
Small 1–3 mm wheals with a flare, triggered by a rise in core temperature. Solar urticaria needs sun. "Chronic idiopathic" ignores the clear trigger

READ MORE
Urticaria & angioedema

5 **60-year-old, oral ulcers then flaccid bullae; biopsy = PV**

SAVED
Urgent referral + prednisolone

RECOMMENDED
Urgent dermatology referral and start oral prednisolone (course key)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Untreated PV progresses, and steroids are the bridge. **UK note:** specialists now give **rituximab + prednisolone** first line (BAD 2017). In primary care, start steroids only with on-call dermatology advice. Doxycycline is for BP. Routine referral is too slow

READ MORE
Pemphigus vulgaris & the pemphigus family

6 **ACD pathogenesis (hair dye)**

SAVED
Type IV

RECOMMENDED
Delayed type IV hypersensitivity

WHY, AND WHY THE DISTRACTORS ARE WRONG
It needs prior sensitisation and is T-cell-mediated, 48–96 h. A corrosive non-specific response is ICD. Type III is immune complex

READ MORE
Contact dermatitis

7 **Atopic woman, deep-seated itchy palmar vesicles for 6 weeks**

SAVED
Potent TCS 2 weeks

RECOMMENDED
Potent TCS once daily for ~2 weeks

WHY, AND WHY THE DISTRACTORS ARE WRONG
Pompholyx (dyshidrotic eczema): potent or very potent TCS short-term + emollients. Prednisolone only for severe flares (specialist). Dapsone has no role. Permethrin only if scabies

READ MORE
Hand eczema (chronic, occupational & pompholyx)

8 **2-year-old with AE flaring with pollen; which cell drives the clinical features**

SAVED
Mast cells

RECOMMENDED
Mast cells (course key for the immediate type I pollen reaction)

WHY, AND WHY THE DISTRACTORS ARE WRONG
IgE-armed mast cells degranulate on allergen exposure. **Note:** eczema itself is driven by **Th2 T cells** (IL-4/13/31), so "T cells" is defensible for an eczema flare. Read the stem: "manifestation after exposure to pollen" suggests type I

READ MORE
Eczema / dermatitis

9 **Wheals <24 h, recurring for 8 months**

SAVED
Chronic urticaria

RECOMMENDED
Chronic (spontaneous) urticaria

WHY, AND WHY THE DISTRACTORS ARE WRONG
More than 6 weeks = chronic. Urticarial vasculitis lasts >24 h with bruising. Autoimmune urticaria is a subtype that needs specialist tests

READ MORE
Urticaria & angioedema

10 **19-year-old, burning prodrome, painful vesicles on the back, 2 days, well**

SAVED
Oral aciclovir

RECOMMENDED
Oral aciclovir 800 mg five times daily for 7 days

WHY, AND WHY THE DISTRACTORS ARE WRONG
Shingles <72 h with **moderate–severe pain/rash** meets CKS criteria for antivirals even if young and truncal. Flucloxacillin is for bacterial infection. Don't wait for swabs

READ MORE
Herpes zoster (shingles)

11 **Raindrop plaques 2 weeks after a sore throat**

SAVED
Guttate psoriasis

RECOMMENDED
Guttate psoriasis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Post-streptococcal. PR has a herald patch and collarette. Drug reaction is possible but the morphology fits guttate

READ MORE
Psoriasis

12 **Eyelid ACD to new cosmetics; which metal**

SAVED
Nickel

RECOMMENDED
Nickel

WHY, AND WHY THE DISTRACTORS ARE WRONG
The commonest UK metal allergen (eyeshadow and mascara pigments, and from fingers). Cobalt and chromate are also possible

READ MORE
Contact dermatitis

13 **Itch and erythema beyond the sutures under an adhesive dressing after C-section**

SAVED
Adhesive dressing

RECOMMENDED
Adhesive dressing (acrylates/colophony)

WHY, AND WHY THE DISTRACTORS ARE WRONG
The pattern matches the plaster outline. Infection would centre on the wound

READ MORE
Contact dermatitis

14 **17-year-old, generalised AE flare including face and eyes, only using 1% HC**

SAVED
TCI

RECOMMENDED
Short course of oral steroid with emollient cover is the likelier course key for an acute generalised flare. UK first-line step-up would really be **moderate–potent TCS for the body + TCI or mild TCS for the face/eyelids** (not offered as a single option)

WHY, AND WHY THE DISTRACTORS ARE WRONG
TCIs (NICE TA82) are second line for moderate–severe AE when TCS has failed or isn't tolerated, and they are ideal for **eyelids/face**, so the saved answer is defensible for the face. Ciclosporin and oral retinoids are specialist. Antibiotics only if infected

READ MORE
Eczema / dermatitis

15 **21-year-old, itchy bumps small blisters on distal shins and forearms after a Turkey holiday; nitrofurantoin for 3 days; mucosa normal**

SAVED
PV

RECOMMENDED
Insect bite reaction (bullous papular urticaria)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Exposed distal sites, a holiday, itchy papules turning to blisters, well, normal mucosa. PV is mucosal, flaccid and occurs in an older age group. BP is elderly. Bullous drug eruption is less likely after a 3-day course with an exposed-site distribution

READ MORE
Papular urticaria & insect bites, Approach to blistering diseases

16 **Swollen lips, face and tongue with difficulty breathing**

SAVED

IV adrenaline

RECOMMENDED

IM adrenaline 500 micrograms (0.5 ml of 1 mg/ml) into the anterolateral thigh, repeat after 5 min (Resuscitation Council UK 2021)

WHY, AND WHY THE DISTRACTORS ARE WRONG

IV adrenaline is for specialists only (anaesthesia/ICU, titrated) because of arrhythmia risk. Steroids and antihistamines are not first-line in anaphylaxis

READ MORE

Urticaria & angioedema

17 **Herald patch then collarette ovals after URTI, asymptomatic**

SAVED

Reassure + emollients

RECOMMENDED

Reassure and advise regular emollients

WHY, AND WHY THE DISTRACTORS ARE WRONG

Pityriasis rosea is self-limiting over 6–12 weeks. Erythromycin evidence is weak. Use TCS only if itchy. Check syphilis serology if atypical or at risk

READ MORE

Pityriasis rosea

18 **Wheals with scratching, <1 h, chef, lansoprazole**

SAVED

Antihistamines

RECOMMENDED

Non-sedating H1 antihistamine (up-dose to 4× if needed)

WHY, AND WHY THE DISTRACTORS ARE WRONG

Symptomatic dermographism. Lansoprazole is not causal. Oral steroids are not indicated

READ MORE

Urticaria & angioedema

19 **14-month-old, nocturnal itch, web-space excoriations, feet vesicles, itchy sibling**

SAVED
Scabies

RECOMMENDED
Scabies

WHY, AND WHY THE DISTRACTORS ARE WRONG
Infant pattern (palms/soles, vesicles) + an affected household contact. Treat **all contacts on the same day** with permethrin 5% ×2, one week apart

READ MORE
Scabies

20 **55-year-old, gradual erythroderma with salmon patches, islands of sparing, follicular hyperkeratosis, thick yellow nails**

SAVED
GPP

RECOMMENDED
Pityriasis rubra pilaris

WHY, AND WHY THE DISTRACTORS ARE WRONG
Salmon-orange colour, islands of sparing, follicular papules and orange palmoplantar keratoderma are the hallmarks. GPP gives sheets of sterile **pustules** with fever (an emergency). Eczema herpeticum and PLEVA don't fit

READ MORE
Psoriasis

21 **Sharply defined red plaques with scale; Auspitz sign**

SAVED
Plaque psoriasis

RECOMMENDED
Plaque psoriasis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Auspitz pinpoint bleeding on removing scale. ACD is ill-defined

READ MORE
Psoriasis

22 **60-year-old, very itchy coin-shaped oozing scaly patches on extensor limbs; recurs in cold dry climate**

SAVED
Granuloma annulare

RECOMMENDED
Discoid (nummular) eczema

WHY, AND WHY THE DISTRACTORS ARE WRONG
Oozing, scaly, intensely itchy, extensors, worse with dryness. GA is a non-scaly, non-itchy ring of papules. Tinea is usually single or asymmetric with a scaly advancing edge (scrape it). NL is shin plaques with yellow atrophy (diabetes)

READ MORE
Eczema / dermatitis, Describing skin lesions: the language of dermatology

23 **45-year-old, sterile yellow-brown pustules on medial insteps of both feet, burning**

SAVED
DH

RECOMMENDED
Localised pustular psoriasis (palmoplantar pustulosis)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Sterile pustules that turn brown on palms or soles, strongly linked to **smoking** and to thyroid autoimmunity. DH is on the extensor limbs and buttocks with tiny vesicles, not plantar pustules. Impetigo herpetiformis is GPP in pregnancy

READ MORE
Palmoplantar pustulosis & palmoplantar psoriasis

24 **Inflamed DIP/PIP joints, RF-negative, erosions, raised CRP**

SAVED
Psoriatic arthropathy

RECOMMENDED
Psoriatic arthritis (stem omits the skin, but this is the best fit)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Seronegative, DIP-predominant, erosive. Screen with PEST and refer to rheumatology early

READ MORE
Psoriasis

25 **Atopic boy, papulopustules at the mouth after repeated 1% HC**

SAVED
Perioral dermatitis

RECOMMENDED
(Steroid-induced) perioral dermatitis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Papulopustules triggered by the TCS on top of lip-licking/angular fissuring. Stop the steroid. Topical metronidazole/azelaic acid, or oral tetracycline/erythromycin depending on age

READ MORE
Rosacea (incl. Demodex & steroid rosacea), Acne mimics: rosacea, perioral & folliculitis

26 **Hand/forearm vesicular ACD hours after mehndi (henna)**

SAVED
PPD

RECOMMENDED
Paraphenylenediamine (PPD) in "black henna"

WHY, AND WHY THE DISTRACTORS ARE WRONG
Natural henna rarely sensitises. PPD is added to darken it, and later hair dye reactions can be severe. Advise lifelong avoidance

READ MORE
Contact dermatitis

27 **Carbamazepine 1 month, flu-like prodrome, flaccid blisters, 40% BSA, Nikolsky positive, mucosae involved**

SAVED
SJS

RECOMMENDED
Toxic epidermal necrolysis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Detachment **>30% BSA = TEN** (SJS <10%, overlap 10–30%). Stop carbamazepine and send to ED/burns unit. EM is targetoid, acral and HSV-related. SSSS spares the mucosa

READ MORE
Drug-induced blistering (SJS/TEN, FDE, AGEP)

28 **73-year-old, urticated plaques tense blisters, mucosa normal**

SAVED
BP

RECOMMENDED
Bullous pemphigoid

WHY, AND WHY THE DISTRACTORS ARE WRONG
Elderly, prebullous urticarial phase, tense bullae, no mucosal involvement. EBA scars. LAD forms annular "string of pearls". PV is flaccid with mucosal involvement

READ MORE
Bullous pemphigoid (incl. MMP & pemphigoid gestationis)

29 **Atopic, unvaccinated child, febrile, grouped vesicles and crusted erosions around the eye/cheek**

SAVED
Secondary infection

RECOMMENDED
Eczema herpeticum

WHY, AND WHY THE DISTRACTORS ARE WRONG
Grouped monomorphic vesicles/punched-out erosions + fever on a background of AE = HSV. **Same-day** aciclovir and assessment, plus **same-day ophthalmology** (periocular). "Secondary infection" (bacterial) misses the emergency. Impetigo has honey crusts without fever

READ MORE
Eczema / dermatitis, Herpes simplex labialis (cold sores)

30 **Pits, onycholysis "oil drops", subungual debris, family history of psoriasis**

SAVED
Nail psoriasis

RECOMMENDED
Nail psoriasis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Pitting + salmon patches/oil drops + onycholysis. Fungal infection usually affects toenails first, and here the toenails are spared. Yellow nail syndrome is slow growth with no pits

READ MORE
Nail signs: psoriatic nails, Beau's lines & clubbing, Psoriasis

Where the course answer differs from UK practice

- **M4 Q5 (PV):** the course says "urgent referral and start prednisolone". UK specialists use **rituximab + prednisolone** first line (BAD 2017). In primary care, start steroids only after a specialist discussion.
- **M4 Q2 (child's facial eczema):** none of the options matches NICE CG57. Potent TCS on a child's face is not UK practice.
- **M4 Q14 (generalised AE flare):** the UK step-up is to a more potent TCS for the body plus TCI or mild TCS for the face. Oral steroid courses are reserved for severe flares, and TCIs are second line (NICE TA82).
- **M4 Q8 (pollen and AE):** "mast cells" is a type I framing. Eczematous inflammation is Th2 T-cell driven.

Other exam reviews

- Exam review: Module 2 · skin lesions, tumours & skin surgery
- Exam review: Module 5 · infections, infestations, photodermatoses & travel

Exam review

Exam review: Module 5 · infections, infestations, photodermatoses & travel

All 30 questions from the Module 5 end-of-module exam, with the recommended UK answer, why the distractors are wrong and links to the matching condition pages. Themes: Reiter's vs Behçet's, Lyme management, the Kaposi sarcoma virus, pyoderma gangrenosum.

End-of-module exam · Jun 2024

ALSO COVERS exam · SBA · revision · module 5 exam · single best answer

About this review

Revision key for the DermUSW end-of-module exam for Module 5 (Jun 2024): 30 single-best-answer questions. **Saved** is the answer ticked before submission (not a graded course key). **Recommended** is the teaching answer on current UK guidance (NICE / CKS, BNF, BAD, BASHH, Resuscitation Council UK, UKHSA).

NOTE

Key

- ✓ saved and recommended agree
- ✗ the saved answer is probably wrong
- ⚠ debatable or image-dependent; the reasoning is given and any course-vs-UK difference is flagged

✓ AGREE	✗ LIKELY WRONG	⚠ DEBATABLE
23	5 (Q14, Q16, Q20, Q22, Q25)	2 (Q13, Q18)

Main themes of errors: Reiter's vs Behçet's (swapped), Lyme management, KS virus, PG.

Each question links to the condition page that covers the topic, where one exists.

Questions & answers

1 **Itchy papular rash on legs/soles after swimming in the Nile, feverish**

SAVED
Schistosomiasis

RECOMMENDED
Schistosomiasis (cercarial dermatitis / acute Katayama)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Freshwater exposure in Egypt or Sudan. Refer to infectious diseases (serology from 6–12 weeks; praziquantel). Dengue has no freshwater link. Scabies is not this acute

READ MORE
The returning traveller with a skin problem

2 **Hand blisters and fragility, naproxen, normal porphyrins**

SAVED
Pseudoporphyria

RECOMMENDED
Pseudoporphyria

WHY, AND WHY THE DISTRACTORS ARE WRONG
Looks like PCT, but porphyrins are normal and there is an NSAID. Stop naproxen

READ MORE
Porphyria cutanea tarda (& pseudoporphyria)

3 **Crab lice in pubic hair and eyelashes**

SAVED
Pthirus

RECOMMENDED
Pthirus pubis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Pediculus humanus is head/body lice. Offer a **full STI screen** (BASHH)

READ MORE
Head lice (pediculosis capitis)

4 **Swimmer, malodorous white plantar skin with pits**

SAVED
Pitted keratolysis

RECOMMENDED
Pitted keratolysis (Kytococcus/corynebacteria)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Hyperhidrosis, occlusion. Topical clindamycin/erythromycin/fusidic acid or BPO, plus aluminium chloride. Erythrasma (coral-red under Wood's lamp) is in the flexures

READ MORE
—

5 **Recurrent grouped vesicles on the lip after a cold**

SAVED
HSV

RECOMMENDED
Herpes simplex labialis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Prodrome, recurrence at the same site. Angular cheilitis is fissuring at the angles of the mouth

READ MORE
Herpes simplex labialis (cold sores)

6 **Green-black hallux nail, onycholysis, KOH-negative, failed terbinafine**

SAVED
Pseudomonas

RECOMMENDED
Pseudomonas aeruginosa (green nail syndrome)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Pyocyanin pigment. Keep the nail dry, trim the onycholytic nail, topical antiseptic (e.g. dilute vinegar/chlorhexidine soaks) ± topical antibiotic. **Exclude subungual melanoma if the colour is black or brown**

READ MORE
Fungal nail infection & nail discolouration

7 **14-year-old, palpable purpura on legs, arthralgia, abdominal pain after a sore throat, normal platelets**

SAVED
HSP

RECOMMENDED
IgA vasculitis (HSP)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Palpable purpura with a normal platelet count. **Check urine and BP** for nephritis, with follow-up dipsticks. ITP has low platelets. Meningococcaemia is an unwell child with non-palpable purpura (emergency)

READ MORE
—

8 **Eyelash lice**

SAVED
Vaseline

RECOMMENDED
Occlusive white soft paraffin (petroleum jelly) to the lashes twice daily for ~8–10 days (BASHH)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Insecticides are not for use near the eye. Lindane is withdrawn. Tea tree oil irritates

READ MORE
Head lice (pediculosis capitis)

9 **Exaggerated sunburn minutes after sun exposure, recently started furosemide**

SAVED
Drug-induced photosensitivity

RECOMMENDED
Drug-induced (phototoxic) photosensitivity

WHY, AND WHY THE DISTRACTORS ARE WRONG
Furosemide (and amlodipine less so) are photosensitisers. Medication review

READ MORE
Drug-induced photosensitivity

10 **3-month-old, napkin rash with satellite lesions despite zinc oxide**

SAVED
Miconazole

RECOMMENDED
Topical imidazole (miconazole or clotrimazole) for 7–14 days, ± 1% HC if inflamed

WHY, AND WHY THE DISTRACTORS ARE WRONG
Satellite papules = Candida. Neomycin/betamethasone is too potent and sensitising for an infant. Oral nystatin only if there is oral thrush

READ MORE
Napkin (nappy) dermatitis

11 **Wavelength for vitamin D synthesis**

SAVED
UVB

RECOMMENDED
UVB (~290–315 nm)

WHY, AND WHY THE DISTRACTORS ARE WRONG
UVA and visible light don't convert 7-dehydrocholesterol. Supplement vitamin D (10 micrograms/day) in sun-avoiders and NMSC patients

READ MORE
Polymorphic light eruption

12 **Painless ulcer with raised edge, Colombian Amazon**

SAVED
Leishmaniasis

RECOMMENDED
Cutaneous leishmaniasis (New World, so a **mucosal** risk with L. braziliensis)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Refer to the HTD (London) or the Liverpool tropical service. Sporotrichosis spreads along lymphatics

READ MORE
Cutaneous leishmaniasis

13 **Serpiginous itchy track on the foot after a Caribbean beach**

SAVED
Topical thiabendazole

RECOMMENDED
Topical thiabendazole is the best of the listed options. UK practice is **oral ivermectin 200 micrograms/kg single dose (unlicensed) or albendazole 400 mg daily for 3 days** via the travel clinic or specialist

WHY, AND WHY THE DISTRACTORS ARE WRONG
Hookworm CLM. Itraconazole, neomycin and clobetasol don't kill larvae. Topical thiabendazole 10–15% is effective but is a UK special

READ MORE
Cutaneous larva migrans

14 **Painful rapidly enlarging undermined ulcer after minor trauma, Crohn's**

SAVED
PV

RECOMMENDED
Pyoderma gangrenosum

WHY, AND WHY THE DISTRACTORS ARE WRONG
Pathergy, violaceous undermined edge, IBD. Don't debride. Same-week dermatology; potent topical steroid/tacrolimus or systemic therapy. PV gives erosions, not deep ulcers. Ecthyma gangrenosum is Pseudomonas sepsis in the neutropenic

READ MORE
Pyoderma gangrenosum

15 **Shallow medial gaiter ulcer, varicosities, better with elevation**

SAVED
Venous ulcer

RECOMMENDED
Venous leg ulcer

WHY, AND WHY THE DISTRACTORS ARE WRONG
Check ABPI before compression (NICE CKS / NWCSP). Arterial ulcers are punched out, lateral or on the foot, and painful on elevation

READ MORE
Venous leg ulcer

16 **Post-sexual urethritis, then arthritis, conjunctivitis, keratoderma blennorrhagicum and circinate balanitis**

SAVED
Behçet's

RECOMMENDED
Reactive arthritis (Reiter's syndrome)

WHY, AND WHY THE DISTRACTORS ARE WRONG
"Can't see, can't pee, can't climb a tree" plus the classic skin signs. Refer to GUM (chlamydia/gonorrhoea) and rheumatology. Behçet's is **oral + genital ulcers** + uveitis (the answer to Q22)

READ MORE
—

17 **Cellulitis, fever 38.3 °C, systemically well, uncontrolled T2DM and obesity**

SAVED
Class 2

RECOMMENDED
Eron class 2

WHY, AND WHY THE DISTRACTORS ARE WRONG
Class 2 = systemically unwell **or** well with comorbidity that may complicate or delay resolution (diabetes, obesity, PVD). Class 3 = significant systemic upset or limb-threatening. Class 1 = no toxicity or comorbidity

READ MORE
Cellulitis & erysipelas

18 **Sore (not itchy) lesions on back, arms and legs after a country hotel in a forest (image)**

SAVED
Tick bites

RECOMMENDED
Bed bugs (Cimex) fits "hotel + back, arms and legs, multiple, often linear 'breakfast-lunch-dinner'". Tick bites are usually few, with an attached tick, in the flexures and waistband

WHY, AND WHY THE DISTRACTORS ARE WRONG
Depends on the image. Multiple grouped or linear papules on the trunk and limbs favour bed bugs. A single erythema migrans favours ticks. Flea bites are on the lower legs

READ MORE
Papular urticaria & insect bites

19 **65-year-old, painful unilateral facial/ear/neck vesicles for 2 days**

SAVED
Herpes zoster

RECOMMENDED
Dermatomal blistering rash: herpes zoster

WHY, AND WHY THE DISTRACTORS ARE WRONG
Aciclovir within 72 h (age >50). **Check eye (Hutchinson's sign)** and ear (Ramsay Hunt: facial palsy)

READ MORE
Herpes zoster (shingles)

20 **Erythema migrans after tick bites**

SAVED
Order serology

RECOMMENDED
Start antibiotics immediately; diagnose clinically

WHY, AND WHY THE DISTRACTORS ARE WRONG
NICE **NG95**: EM is diagnostic, so **don't test**. Serology is often negative early. **Doxycycline 100 mg twice daily (or 200 mg once daily) for 21 days**; amoxicillin if doxycycline is unsuitable. Biopsy and waiting are wrong

READ MORE
Lyme disease & erythema migrans

21 **Photo-distributed eczema sparing behind the ears and under the chin, abnormal UVB phototest**

SAVED
CAD

RECOMMENDED
Chronic actinic dermatitis

WHY, AND WHY THE DISTRACTORS ARE WRONG
An older man with sharp demarcation and **reduced MED to UVB (± UVA/visible)**. PMLE is self-limiting and seasonal

READ MORE
Chronic actinic dermatitis

22 **Oral + genital ulcers, iritis, arthritis**

SAVED
Recurrent aphthous ulcers

RECOMMENDED
Behçet's disease

WHY, AND WHY THE DISTRACTORS ARE WRONG
Recurrent oral ulcers + genital ulcers + eye disease (uveitis) are the ISG criteria. RAS has no genital or eye involvement. SJS is acute and drug-related

READ MORE
—

23 **Senegal migrant, foot swelling, nodules, draining sinuses, branched filaments**

SAVED
Mycetoma

RECOMMENDED
Mycetoma (actinomycetoma, given branched filaments)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Triad of swelling, sinuses and grains. Actinomycetoma is treated with antibiotics (co-trimoxazole + amikacin). Eumycetoma needs antifungals + surgery

READ MORE
Mycetoma (Madura foot)

24 **Nursing-home scabies outbreak**

SAVED
Permethrin + ivermectin

RECOMMENDED
Permethrin 5% plus oral ivermectin (UKHSA 2023 care-home outbreak guidance)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Mass treatment of residents and staff on the same day. Ivermectin for crusted scabies and outbreaks. Lindane and malathion aren't recommended. Crotamiton is weak

READ MORE
Scabies

25 **KS in HIV (CD4 150), causative organism**

SAVED
CMV

RECOMMENDED
Human herpesvirus 8 (KSHV)

WHY, AND WHY THE DISTRACTORS ARE WRONG
HHV-8 causes KS, primary effusion lymphoma and multicentric Castleman disease. CMV causes retinitis/colitis. Treatment is effective ART ± chemotherapy

READ MORE
—

26 **Painful dome-shaped swelling on the nape**

SAVED
Furuncle

RECOMMENDED
Furuncle (boil)

WHY, AND WHY THE DISTRACTORS ARE WRONG
A single follicle-centred abscess. Carbuncle is a multi-headed coalescence. Swab if recurrent (PVL). I&D if fluctuant

READ MORE
Boils & carbuncles

27 **18-year-old, umbilicated papules on one side of the chest**

SAVED
Reassure

RECOMMENDED
Reassure (molluscum is self-limiting)

WHY, AND WHY THE DISTRACTORS ARE WRONG
Resolves over 6–18 months. Cryotherapy and curettage scar and hurt. **Consider sexual transmission or immunosuppression in adults with genital/extensive disease**

READ MORE
—

28 **Herald patch, Christmas-tree pattern**

SAVED
PR

RECOMMENDED
Pityriasis rosea

WHY, AND WHY THE DISTRACTORS ARE WRONG
Guttate psoriasis has no herald patch. Secondary syphilis involves the palms and soles, so check serology if at risk

READ MORE
Pityriasis rosea

29 **Linear blisters after gardening among hogweed, sun-exposed forearm**

SAVED
Phytophotodermatitis

RECOMMENDED
Phytophotodermatitis

WHY, AND WHY THE DISTRACTORS ARE WRONG
Furocoumarins (giant hogweed) + UVA, leading to linear blisters and then pigmentation lasting months. Cool compresses, potent TCS, sun protection

READ MORE
Phytophotodermatitis and cutaneous amyloidosis

30 **Thailand, high fever, retro-orbital headache, diffuse rash, prostration**

SAVED
Dengue

RECOMMENDED
Dengue serology/NS1 + FBC (platelets, haematocrit) + coagulation

WHY, AND WHY THE DISTRACTORS ARE WRONG
Watch for warning signs of severe dengue. **Avoid NSAIDs/aspirin**. Use paracetamol. Exclude malaria the same day

READ MORE
Dengue: skin signs & safety

Where the course answer differs from UK practice

- **M5 Q13 (CLM):** thiabendazole is the best option offered. UK practice is oral ivermectin or albendazole.

Other exam reviews

- Exam review: Module 2 · skin lesions, tumours & skin surgery
- Exam review: Module 4 · inflammatory dermatoses

APPENDIX B

Shared red flags & study tips

Shared red flags: one-page summary

Keep these pages in mind during every skin consultation. The common conditions in this guide all have a dangerous version or a dangerous mimic; in babies (Chapters F–H) the first question is always "Is this baby well?".

Red flag	Recognise	Action
Eczema herpeticum	Punched-out erosions or vesicles on eczematous skin; painful; may be unwell	**Immediate hospital admission**
Erythroderma	Inflamed skin over most of the body surface; temperature, fluid and infection risk	**Same-day** specialist assessment
Generalised pustular psoriasis	Sheets of sterile pustules on red skin; systemically unwell	**Same-day** specialist assessment
Acne fulminans	Sudden ulcerating, crusted acne with systemic illness	**Same-day** on-call dermatology
Necrotising fasciitis / sepsis	Pain out of proportion, rapid spread, crepitus, dusky or purple bullae, toxicity	**Emergency** transfer
Periorbital / orbital cellulitis	Eyelid swelling and redness ± pain on eye movement, visual change, proptosis	**Same-day / emergency** pathway
Atypical persistent plaques	Treatment-resistant "eczema" or "psoriasis" in an older adult	Refer: exclude cutaneous lymphoma
Psoriatic arthritis	Joint pain or swelling, dactylitis, nail disease	Rheumatology referral
Neonatal herpes simplex (F)	Grouped vesicles/erosions in a neonate; fever, poor feeding, lethargy, eye involvement — baby may look well early	**Emergency** same-day paediatric admission; IV aciclovir pathway
Unwell infant with rash (F)	Fever, poor feeding, lethargy, prematurity; infant <1 month with significant rash; bullous impetigo in an infant	**Same-day** paediatric assessment
High-risk infantile haemangioma (G)	Periocular, beard/airway, lip, nasal tip, segmental facial, napkin-area, ulcerated; ≥ 5 lesions	**Urgent** specialist referral in the growth phase (same day if stridor or eye closure)
Facial V1 port-wine stain (G)	Forehead / upper-eyelid capillary malformation	Refer: Sturge–Weber assessment (eye, neurology)
Suspected epidermolysis bullosa (H)	Neonate or infant blistering or losing skin with minimal friction; hoarse cry or stridor	**Urgent** EB / paediatric dermatology referral; airway signs = emergency
Complicated aplasia cutis (H)	Large, deep, multiple or midline scalp defect; **hair collar sign**; limb defects	**Urgent** paediatric / neurosurgical pathway; imaging
Eczema herpeticum (A+)	Rapidly worsening painful eczema with monomorphic punched-out erosions, fever	**Same-day** assessment; systemic aciclovir; eye review if periocular
Erythrodermic / generalised pustular psoriasis (B+)	>90% skin red, or sheets of sterile pustules; fever, unwell	**Emergency** same-day dermatology / admission
Acne fulminans (I)	Sudden ulcerating, crusted acne with fever and joint pain (often on isotretinoin)	**Urgent** dermatology
Malignant acanthosis nigricans (I)	Sudden, extensive, rapidly spreading AN in a slim older adult; mucosal or palmar involvement; weight loss	Urgent investigation for underlying cancer (NG12)

Red flag	Recognise	Action
Illegal skin-lightening products (J)	Mercury, hydroquinone or potent steroid “creams”; ochronosis, striae, steroid damage	Stop; counsel; report to Trading Standards / MHRA
Older skin red flags (K)	Non-healing, bleeding or rapidly growing lesion; tense blisters (pemphigoid); itch with weight loss or night sweats	NG12 suspected-cancer pathway; dermatology; investigate systemic causes
Filler vascular occlusion / visual change (L)	Blanching, livedo, dusky patch, severe pain after filler; **any visual disturbance**	**Emergency**: injector immediately / A&E; vision loss = 999 / eye emergency
Iatrogenic botulism (L)	Ptosis, double vision, dysphagia, slurred speech, breathlessness after toxin injection	**Emergency** 999 / A&E; notifiable; Yellow Card
Crusted scabies (M)	Thick crumbly crusts on hands/feet in a frail or immunosuppressed person; itchy staff	**Urgent** same-day specialist advice; infection control / health protection
Disseminated Lyme disease (M)	Facial palsy, meningism, heart block/syncope, uveitis, multiple EM	Emergency or same-day referral; do not delay antibiotics (NG95)
Anaphylaxis / airway angioedema (O)	Wheals or swelling with breathing, airway or circulation problems	**Emergency**: 999, IM adrenaline (RCUK)
Hereditary / ACE-inhibitor angioedema (O)	Swelling without wheals, not responding to antihistamine; family history; ACE inhibitor	Airway = emergency; stop ACE inhibitor; refer immunology
Scarring alopecia (P)	Loss of follicular openings, shiny scarred scalp, pain, pustules or perifollicular scale	**Urgent** dermatology with photographs
Atypical anogenital lesion (Q)	Pigmented, ulcerated, indurated or bleeding “wart”; lichen sclerosus; immunosuppressed patient	Refer (sexual health / dermatology / NG12 as indicated); safeguarding in children
Subungual melanoma (R)	New, widening or irregular nail band; pigment onto the nail fold (Hutchinson sign); nail destruction	Refer on the suspected-cancer pathway (NG12)
Severe hidradenitis (S)	Tunnels, scarring, recurrent abscesses, systemic upset	Early dermatology referral (biologic assessment); drain acute abscess
SJS/TEN and EM major (T, U, V)	Target-like or dusky lesions with mucosal erosions, skin pain, blistering or skin loss, fever	**Emergency**: stop suspect drugs, 999 / same-day admission (burns/ICU)
Immunobullous disease (V)	Pemphigus with oral erosions; pemphigoid with eye, airway or pregnancy involvement	Same-day or urgent dermatology (ophthalmology / obstetrics as needed)
Critical limb ischaemia / diabetic foot (W)	Rest pain, pallor, absent pulses; foot ulcer with infection or ischaemia	**Emergency** vascular referral; same-day diabetic foot team
Febrile returning traveller (X)	Fever after the tropics, with or without rash; dengue warning signs	Same-day malaria test / acute medicine; no NSAIDs or aspirin
Ophthalmic or disseminated zoster (Y)	Forehead/nose-tip vesicles (Hutchinson sign), eye pain, visual change; zoster in an immunosuppressed patient	Same-day ophthalmology; urgent specialist advice
Invasive group A strep (Y)	Scarlet fever or skin infection with rapid deterioration, severe pain, sepsis signs	**Emergency**; notify UKHSA
Photosensitive rash with systemic features (Z)	Malar or photo-distributed rash with joint pain, ulcers, fatigue	Investigate for lupus; refer

Red flag	Recognise	Action
SJS/TEN (1C)	Skin pain, dusky or atypical targets, blisters or skin detachment, two or more mucosal sites, fever after a new drug	**Emergency**: stop all suspect drugs; same-day admission (burns/ICU); eye review
DRESS (1C)	Rash 2–8 weeks after a new drug (allopurinol, carbamazepine, lamotrigine) with fever, facial oedema, lymphadenopathy, eosinophilia, abnormal LFTs	**Same-day** hospital assessment; stop the drug
ACE-inhibitor / airway angioedema (1C)	Lip, tongue or throat swelling without wheals; voice change, stridor, difficulty swallowing	**Emergency** 999; stop the ACE inhibitor permanently
Vasculitis with organ involvement (1D)	Palpable purpura with blood or protein on urine dipstick, raised BP, abdominal pain, haemoptysis, neuropathy	Same-day medical / renal assessment
Non-blanching rash in an unwell patient (1D)	Fever, petechiae or purpura, drowsiness, cold peripheries	**Emergency**: suspected meningococcal disease, 999 and parenteral benzylpenicillin if delay (NICE NG240)
Fingertip lesions with fever (1D)	Osler nodes, Janeway lesions, splinters with fever or a new murmur	Same-day admission for blood cultures and echocardiography (endocarditis)
Adult dermatomyositis (1E)	Heliotrope rash, Gottron papules, proximal weakness, dysphagia, breathlessness	Urgent rheumatology/dermatology; CK; screen for underlying cancer
Adrenal crisis (1E)	Pigmentation with vomiting, hypotension, confusion, hyponatraemia or hyperkalaemia	**Emergency**: IM/IV hydrocortisone, 999 (NICE NG243)
Eruptive xanthomas (1E)	Crops of yellow papules with red halo; triglycerides very high	Same-day lipids; TG >20 mmol/L = urgent specialist advice (pancreatitis risk)
Pyoderma gangrenosum (1E)	Graze or wound that turns into a rapidly enlarging, very painful ulcer with a purple undermined edge, especially with IBD, arthritis or a blood disorder	Urgent dermatology; exclude infection; **do not debride**
Suspected melanoma (2D, 2E)	7-point checklist score 3 or more; new nodule; ugly duckling; change in a congenital naevus	Suspected-cancer pathway referral (NICE NG12); do not remove in primary care
Suspected SCC (2C)	Tender, indurated, rapidly growing or ulcerated lesion on sun-exposed skin, lip or ear; transplant patient	Suspected-cancer pathway referral (NICE NG12)
Possible sarcoma or DFSP (2A)	Lump growing, deep, >5 cm or painful; scar-like plaque that keeps enlarging	Urgent ultrasound / suspected-cancer referral (NICE NG12)
Local anaesthetic systemic toxicity (2F)	Perioral tingling, metallic taste, tinnitus, agitation, seizures, arrhythmia after local anaesthetic	**Emergency**: stop injecting, call for help, ABC, lipid emulsion per AAGBI guidance
Immunotherapy adverse events (2E)	Diarrhoea, breathlessness, jaundice, severe fatigue, headache or rash in a patient on checkpoint inhibitors	Same-day contact with the oncology acute line

Cross-cutting pearls

Theme	Remember
Formulate	Emollient base, gel vs ointment, scalp vs body. Adherence is the intervention.
Steroid ladder	Potency to match site and severity. Face, flexures and genitals: lower potency, shorter course.
Infection vs inflammation	Crusted eczema and red legs do not automatically need antibiotics.
Stewardship	No antibiotic monotherapy in acne; short SSTI courses; always a review date.
Skin of colour	Judge by scale, induration, pigment change and symptoms, not erythema alone.
Psychosocial	Screen for distress in acne, psoriasis and eczema; refer to mental health services when needed.
Infants	“Is this baby well?” first. Most neonatal rashes and birthmarks need explanation, documentation and safety-netting, not a prescription.
Always check	Current BNF / BNF for Children, SmPC, MHRA alerts, local formulary / microguide.

Study tips: building visual pattern recognition

Dermatology is a visual skill. Reading builds understanding, but only **looking at many real images** builds recognition. Real DermNet clinical photos are embedded throughout the chapters; AI illustrations remain for pathophysiology diagrams. Still use the live DermNet galleries to see more variation.

After each chapter: the 20-minute DermNet routine

1. **Open the DermNet links** at the end of the chapter and scroll through the image gallery for each topic.
2. **Describe before you diagnose:** for 10 images, say the morphology (macule, papule, plaque, vesicle, pustule, crust, scale), the distribution and the colour aloud.
3. **Compare look-alikes side by side** using the chapter's differentials table (e.g. tinea vs discoid eczema; psoriasis vs eczema; cellulitis vs venous eczema).
4. **Look deliberately at skin of colour** examples for every condition. Note how erythema and pigment change differ.
5. **Write one management line** for each image you looked at: self-care, first-line Rx, and when to refer.

Chapter	Gallery focus	Self-test question
A · Eczema	Acute vs chronic; flexural vs hand; infected eczema; eczema herpeticum	Which images would make you admit today?
B · Psoriasis	Plaque, guttate, flexural, scalp, nail	Where does the scale stop on the scalp?
C · Acne	Lesion counting; rosacea vs acne; scars; PIH	Mild–moderate or moderate–severe?
D · Fungal	Active edge; tinea incognito; Candida satellites; PV colour	Would a steroid help or harm here?
E · Bacterial	Honey crust; bullous impetigo; cellulitis mimics	Unilateral and acute, or bilateral and chronic?
F · Neonatal	ETN vs TNPM vs miliaria vs HSV; irritant vs candidal napkin rash	Is this baby well? Folds spared or involved?
G · Birthmarks	Haemangioma vs port-wine stain; blue-grey and pale patches; high-risk sites	Tumour or malformation? Which site needs referral?
H · EB & ACC	EB types; EB vs bullous impetigo/SSSS; aplasia cutis and hair collar sign	At which level does the skin split?
A+ / B+ · Deepen	Eczema age patterns, herpeticum; guttate, PsA, erythroderma	Which age pattern? Joints, nails, CV risk checked?
I · Acne & mimics	Acne vs rosacea vs Malassezia / Gram-negative folliculitis	Comedones present? Monomorphic itchy papules?
J · Pigmentation	Melasma vs ochronosis vs LPP vs PIH; piebaldism vs vitiligo	Epidermal or dermal? Born with it or acquired?
K · Older skin	Asteatotic vs venous eczema; AK vs SCC; seborrhoeic keratosis vs melanoma	Dry or systemic itch? Is any lesion changing?
L · Aesthetics	Bruise vs vascular occlusion after filler	Pain, colour, capillary refill, vision?
M · Infestations	Scabies burrows; nits vs dandruff; EM vs bite reaction; flea vs bed-bug bites	Who else itches? How long did the ring last?

Chapter	Gallery focus	Self-test question
N–Q	ACD vs ICD; urticaria vs vasculitis; AA vs TTM vs scarring; warts vs PPP/Fordyce	Exposure? <24 h? Follicles visible? Normal variant?
0 · Lesion language	Macule vs patch vs plaque; vesicle vs bulla; scale vs crust; Fitzpatrick types	Can you describe it so a colleague could picture it?
R–T	Paronychia vs onychomycosis vs melanonychia; HS vs boils; ICD vs ACD vs psoriasis on hands	Nail fold or plate? Tracts? Wet work?
U–V	EM vs urticaria; annular DDx; LP; pemphigus vs pemphigoid vs DH	Targets fixed >24 h? Flaccid or tense? Mouth?
W–Z	Ulcer types; leishmaniasis vs CLM; zoster vs HSV; PR vs guttate; PMLE vs drug phototoxicity	ABPI? Travel? Dermatome? Shaded skin spared?
1B–1C	TCS potency ladder and FTU; retinoids; acne pills; exanthem vs SJS/TEN vs DRESS; FDE	Right potency for the site? Drug started when? Mucosa?
1D–1E	Vasculitis by vessel size; UV vs urticaria; livedo; DM, Addison's, xanthomas, AN, sarcoid	Palpable? >24 h? Urine dipstick? First test?
2A–2F	DF vs DFSP; cyst vs lipoma; SK vs melanoma; BCC vs SCC; naevi; 7-point checklist; lidocaine maximum; BSDS	Stable or changing? NG12 route? Safe to operate?

Revision techniques that work

- **Spaced recall:** revisit the at-a-glance strip of each chapter after 1 day, 1 week and 1 month.
- **Explain the "why":** for each drug, name which part of the pathophysiology it targets.
- **Case reflection:** after each skin consultation, note the diagnosis, the differential you excluded, and what you would do differently.

In clinic

- Examine in good light and look at the whole skin, nails and scalp.
- With consent, photograph lesions to monitor response and to support advice-and-guidance referrals.
- Always document potency, site, quantity and duration, and a review or safety-net plan.

APPENDIX C

Course module overviews

Carried over from the original guide. Chapter letters refer to the original edition; the content now lives in the chapters above.

Modules 1 & 2 at a glance: therapeutics, drug reactions, vasculitis, systemic disease, skin lesions, cancer & surgery

Chapters 1B–1E continue Module 1 after Chapter 0 (the language of dermatology). Chapters 2A–2F cover Module 2: lumps and benign lesions, skin cancer and cutaneous lymphoma, naevi and melanoma, and skin surgery at knowledge level. Surgical technique is not taught here; the focus is on what a pharmacist prescriber needs to recognise, counsel, prescribe and refer.

Chapter	Sorting question	First-line UK approach	Don't miss
1B · Therapeutics	Which potency, vehicle and quantity for this age, site and severity?	TCS ladder with FTUs; emollients always; NG198 acne ladder; MHRA retinoid and isotretinoin rules	Steroid atrophy/tinea incognito; retinoid in pregnancy; infection on biologics
1C · Drug eruptions	Which drug, started when? Any skin pain, mucosal or organ involvement?	Stop the culprit; antihistamine/ emollient/TCS for simple exanthem; Yellow Card; precise allergy record	emergency SJS/TEN, DRESS, airway angioedema
1D · Vasculitis	Palpable? Lasting >24 h? Net-like? Is the patient systemically unwell?	Urine dipstick and BP today; bloods incl. ANCA/complement; stop culprit drug; biopsy early	emergency Meningococcal sepsis; renal involvement; endocarditis
1E · Systemic disease	What internal disease could this skin sign point to?	The first test (CK, cortisol, lipids, tTG-IgA, HbA1c, calcium, CXR) and the right speed of referral	urgent Dermatomyositis (cancer screen); adrenal crisis; TG >20
2A–2B · Lumps & benign lesions	Does it fit a benign pattern, and is it stable?	Recognise and reassure; dermoscopy if trained; no cosmetic removal on the NHS; histology for anything removed	DFSP; sarcoma (NG12); amelanotic melanoma; Leser–Trélat
2C · NMSC & lymphoma	Pearly and slow (BCC), or tender, indurated and fast (SCC)? Fixed 'eczema'?	NG12: SCC 2-week-wait, BCC usually routine; surgery first for nodular BCC; biopsy persistent atypical rash	urgent SCC in transplant patients; lip/ear SCC; MF
2D–2E · Naevi & melanoma	7-point checklist score? Ugly duckling? Nodule?	2-week-wait referral; photograph; specialist excision with 2 mm margin; NG14 staging and follow-up	urgent Nodular/amelanotic melanoma; irAEs on immunotherapy
2F · Skin surgery	Drugs, allergies, devices, anatomy: is it safe here and now?	Continue most antithrombotics (BSDS); lidocaine maxima (BNF); no endocarditis prophylaxis (CG64)	emergency LA toxicity; anaphylaxis

Course-vs-UK-guidance differences flagged in these chapters

- Nodular BCC: **surgical excision is first line** (BAD 2021); destructive or topical options are for selected low-risk, usually superficial, lesions.
- Congenital melanocytic naevi: melanoma risk **depends on size** (highest with projected adult size over 20 cm and multiple satellites); small and medium CMN carry a very low risk.
- Antithrombotics: the BSDS advises **continuing DOACs, warfarin (INR in range, checked) and antiplatelets for most minor skin surgery**; do not stop them on your own.
- Infective endocarditis: **no routine antibiotic prophylaxis** for skin or dental procedures (NICE CG64).
- Suspected melanoma: use the **NICE weighted 7-point checklist** (score 3 or more = suspected-cancer referral) alongside ABCDE.

- Melanoma survival: quote **AJCC 8th edition** stage-specific melanoma-specific survival figures.
- Lidocaine maximum doses (BNF): **200 mg plain; 500 mg with adrenaline** (adult), with weight-based limits in smaller adults.

Modules 3–5 at a glance: infestations, contact, urticaria, hair & perineal skin

Chapters M–Q take the same template into five more primary-care problems. Each one has a **single question that sorts most cases**, and each has a dangerous version you must not miss.

Chapter	Sorting question	First-line UK approach	Don't miss
M · Infestations & bites	Are close contacts itchy? Is there a live louse? Is the ring expanding over days?	Permethrin 5% ×2, 7 days apart, all contacts same day; wet combing / dimeticone; doxycycline 21 days for EM (NG95)	urgent Crusted scabies; Lyme with CNS/cardiac/eye signs
N · Contact dermatitis	Does the rash match an exposure, job or product?	Avoid, emollients and soap substitutes, short TCS; patch testing referral	Occupational disease; severe facial/eyelid swelling
O · Urticaria	Do individual wheals clear within 24 h? Any deep swelling without wheals?	Non-sedating antihistamine, up-dose to 4× under guidance; review ACE inhibitors	emergency Anaphylaxis / airway angioedema; HAE; urticarial vasculitis
P · Scalp & hair	Scarring or non-scarring? Patchy, diffuse or patterned?	Name the pattern, bloods where indicated, support and signpost	Scarring alopecia (refer early); tinea capitis
Q · Perineal	Normal variant, infection, inflammation — or neoplasia?	Reassure variants; STI-aware care; first-line HS measures	Atypical/ulcerated lesions; lichen sclerosus; safeguarding

Course-vs-UK-guidance differences flagged in these chapters

- Scabies: repeat permethrin after **7 days** (not 2 weeks); sexual contacts within the past month (CKS); hot wash ideally 60 °C or bag items for 3–4 days.
- Head lice: permethrin is **not** UK-recommended; use wet combing, dimeticone 4% or aqueous malathion.
- Psoriasis: the UK guideline is **NICE CG153** (with CKS and NICE technology appraisals).

Module 6 at a glance: infants, birthmarks & fragile skin

Chapters F–H cover the skin of newborns and infants. The thinking changes slightly: the first question is always **"Is this baby well?"**, most rashes need **explanation rather than a prescription**, and the rare dangerous conditions (neonatal HSV, high-risk haemangiomas, epidermolysis bullosa, deep aplasia cutis) need **recognition and fast referral** rather than primary-care treatment.

F · ETN
Real clinical photo © DermNet (F01) — see chapter & Image credits

F · Candidal napkin
Real clinical photo © DermNet (F12) — see chapter & Image credits

G · Haemangioma
Real clinical photo © DermNet (G01) — see chapter & Image credits

G · Mongolian spot
Real clinical photo © DermNet (G12) — see chapter & Image credits

H · EB
Real clinical photo © DermNet (H01) — see chapter & Image credits

	F · Neonatal & childhood rashes	G · Birthmarks	H · EB & aplasia cutis
Core idea	Benign pustules and rashes in **well** babies vs infection in unwell ones	Tumour (grows, then involutes) vs malformation (persists) vs pigment	Inherited fragility at a set skin level; congenital absence of skin
Key clues	Timing (birth vs days vs weeks); red vs non-red base; folds spared or involved	Present at birth? Growth rate? Site? Colour (red, blue-grey, pale, brown)?	Blisters from minimal friction; scarring, milia; scalp vertex defect; hair collar sign
Primary-care role	Reassure, cool, barrier care; short, mild topicals (BNFC)	Document and photograph; reassure; **refer high-risk sites early**	**Recognise and refer**; gentle handling; no adhesives; medicines support
Don't miss	emergency Neonatal HSV; septic or unwell infant	urgent Periocular, airway, ulcerated, segmental haemangioma; V1 port-wine stain	urgent Suspected new EB; airway symptoms; deep or midline ACC

Three questions before every infant skin decision

1. **Is this baby well?** Fever, poor feeding, lethargy, or vesicles in a neonate means same-day assessment.
2. **Does this need a prescription at all?** ETN, TNPM, miliaria, neonatal acne and most birthmarks need explanation and documentation.
3. **Is the site or the type high risk?** Periocular or airway haemangiomas, suspected EB and deep scalp defects go straight to specialists.

Chapters R–Z at a glance: nails, folds, hands & feet, blisters, ulcers, travel, viral rashes & light

Chapters R–Z finish Modules 3–5 with the same template. Chapter 0 (the language of dermatology) and the deepening chapters E+, I+ and L+ sit next to the chapters they build on.

Chapter	Sorting question	First-line UK approach	Don't miss
R · Nails	Nail fold, nail plate or pigment? Acute or chronic?	Dry, protect, treat the cause; confirm fungus before oral terbinafine; flucloxacillin for spreading acute paronychia	urgent Widening or irregular pigmented band, Hutchinson sign (NG12)
S · Flexural	Scale or no scale? Satellite pustules? Boils and tracts?	Dry the fold; mild TCS or vitamin D analogue for flexural psoriasis (CG153); tetracycline 12 weeks for HS	HS with tunnels/scarring; Paget disease; erythrasma or tinea
T · Hands & feet	Irritant, allergic, psoriatic or fungal? Vesicles or pustules?	Emollient + glove plan, potent TCS, patch testing if >3 months; scrape first	Occupational disease; secondary syphilis; SJS/TEN
U · Inflammatory DDx	Targets, rings, purple papules, scarring or drops?	Name the morphology; scrape rings; review drugs; biopsy what you cannot name	urgent EM major vs SJS/TEN; pemphigus; scarring DLE
V · Blistering	Flaccid erosions with mouth involvement, or tense blisters in an older person?	Same-day/urgent dermatology; stop the culprit drug (gliptin); doxycycline/clobetasol under specialist plan	emergency SJS/TEN; ocular MMP; pemphigoid gestationis
W · Ulcers & cellulitis	Venous, arterial, neuropathic — or PG? Is it really cellulitis?	ABPI then compression; NG141 antibiotics with review; diabetic foot pathway	emergency Critical ischaemia; necrotising fasciitis
X · Tropical & travel	Where, when, what exposure? Fever?	Travel history; same-day malaria exclusion if febrile; ivermectin/albendazole for CLM; refer to tropical medicine	emergency Febrile traveller; severe dengue (no NSAIDs)
Y · Viral & infective	Dermatomal? Herald patch? Sore throat with sandpaper rash?	Antivirals within 72 h for zoster; reassure PR (test syphilis if atypical); penicillin V 10 days for scarlet fever	urgent Ophthalmic zoster; invasive group A strep
Z · Photodermatoses	Does it spare shaded skin? New drug? Plant contact?	Medication review; broad-spectrum high-UVA sunscreen; short potent TCS; refer for phototesting	Lupus; PCT with liver disease/ haemochromatosis

Course-vs-UK-guidance differences flagged in these chapters

- Pemphigus vulgaris: UK specialists use **rituximab + prednisolone first line** (BAD 2017); azathioprine/MMF are not the only route.
- Bullous pemphigoid: **DPP-4 inhibitors (gliptins)** are the key drug trigger of drug-induced **pemphigoid** (not pemphigus).
- Cutaneous larva migrans: UK practice is oral **ivermectin or albendazole**; thiabendazole is not used.
- Impetigo: **hydrogen peroxide 1% cream** first for localised non-bullous impetigo (NICE NG153), not fusidic acid by default.
- Flexural and palmoplantar psoriasis: the UK guideline is **NICE CG153**.

Quick visual comparison: the five conditions at a glance

Start with **morphology** (what the lesion is), then **distribution** (where it is), then **symptoms** and **clues**. Most primary-care errors come from treating a red, scaly or crusted rash before deciding what kind of rash it is.

A · Eczema
Real clinical photo © DermNet — see chapter & Image credits

B · Psoriasis
Real clinical photo © DermNet — see chapter & Image credits

C · Acne vulgaris
Real clinical photo © DermNet — see chapter & Image credits

D · Tinea corporis
Real clinical photo © DermNet — see chapter & Image credits

E · Impetigo
Real clinical photo © DermNet — see chapter & Image credits

	Eczema	Psoriasis	Acne	Tinea / Candida	Impetigo / cellulitis
Core lesion	Ill-defined dry or weeping patches; lichenification when chronic	Sharply demarcated plaques with silvery-white scale	Comedones plus papules, pustules ± nodules	Tinea: annular, active scaly edge. Candida: beefy-red with satellites	Honey crusts (impetigo); spreading hot erythema (cellulitis)
Classic sites	Flexures (child), hands and face (adult), cheeks (infant)	Extensor elbows/ knees, scalp, lower back; nails	Face, neck, upper chest, back	Groin, feet, trunk; moist folds (Candida)	Face/limbs (impetigo); usually one leg (cellulitis)
Symmetry	Often symmetrical	Often symmetrical	Symmetrical, seborrhoeic areas	**Asymmetrical**	Cellulitis **unilateral**
Itch / pain	Intense itch	Mild–moderate itch; fissures painful	Tender inflamed lesions	Itchy	Sore; cellulitis painful and hot
Key cause	Barrier defect + Th2 inflammation	IL-23/Th17 immune activation rapid turnover	Sebum, follicular plugging, C. acnes, inflammation	Dermatophytes / yeasts in warm, moist skin	S. aureus / S. pyogenes through a breached barrier
Single best clue	Itch + flexures + atopy	Nail pitting; scalp beyond hairline	**Comedones** (absent in rosacea)	Active scaly edge; scrapings positive	Honey crust; acute, unilateral, systemic signs
First-line	Emollient + TCS matched to severity and site	Potent TCS + vitamin D analogue (trunk/limbs)	12-week fixed topical combination ± oral tetracycline	Topical terbinafine or imidazole	H_2O_2 1% cream (localised impetigo); flucloxacillin (cellulitis)
Don't miss	admit now Eczema herpeticum	same day Erythroderma, pustular psoriasis	same day Acne fulminans	Tinea incognito from steroid misuse	emergency Necrotising fasciitis, orbital cellulitis

Three questions before every skin prescription

1. **Is it inflammation or infection, or both?** Crusted eczema and red legs do not automatically need antibiotics.
2. **Could a steroid make this worse?** Tinea, impetigo, rosacea and perioral dermatitis can all be worsened or masked by steroids.

3. **Have I matched potency, vehicle and quantity to the site and the patient?** Adherence is the treatment.

APPENDIX D

Sources & further reading

Sources & further reading

Sources & further reading

Content was paraphrased and synthesised from the public sources below; no large verbatim extracts are reproduced. NICE CKS pages may be geo-restricted outside the UK. Real clinical photographs from DermNet are embedded with attribution for personal educational study; also view the live galleries at dermnetnz.org under DermNet's image licence.

DermNet: clinical photo pages

Atopic dermatitis	dermnetnz.org/topics/atopic-dermatitis
Dermatitis	dermnetnz.org/topics/dermatitis
Contact dermatitis	dermnetnz.org/topics/contact-dermatitis
Treatment of AD	dermnetnz.org/topics/treatment-of-atopic-dermatitis
Psoriasis	dermnetnz.org/topics/psoriasis
Acne	dermnetnz.org/topics/acne
Acne & follicular	dermnetnz.org/topics/acne-and-other-follicular-disorders
Tinea corporis	dermnetnz.org/topics/tinea-corporis
Tinea cruris	dermnetnz.org/topics/tinea-cruris
Tinea pedis	dermnetnz.org/topics/tinea-pedis
Pityriasis versicolor	dermnetnz.org/topics/pityriasis-versicolor
Candidal intertrigo	dermnetnz.org/topics/candidiasis-of-skin-folds
Intertrigo	dermnetnz.org/topics/intertrigo
Fungal infections hub	dermnetnz.org/topics/fungal-skin-infections
Impetigo	dermnetnz.org/topics/impetigo
Cellulitis	dermnetnz.org/topics/cellulitis
Cellulitis mimics	dermnetnz.org/topics/cellulitis-mimics
Erysipelas	dermnetnz.org/topics/erysipelas
Bacterial folliculitis	dermnetnz.org/topics/bacterial-folliculitis
Boil	dermnetnz.org/topics/boil

NICE guidelines

- NG198 Acne vulgaris: management. nice.org.uk/guidance/ng198
- NG141 Cellulitis and erysipelas: antimicrobial prescribing. nice.org.uk/guidance/ng141
- NG153 Impetigo: antimicrobial prescribing. nice.org.uk/guidance/ng153
- CG153 Psoriasis: assessment and management. nice.org.uk/guidance/cg153
- CG57 Atopic eczema in under 12s (historical; still referenced).

Sources & further reading

NICE CKS

- Eczema – atopic · Dermatitis – contact · Psoriasis · Acne vulgaris
- Fungal skin infection – body and groin · Fungal skin infection – foot · Pityriasis versicolor
- Cellulitis – acute · Impetigo · Boils, carbuncles & staphylococcal carriage
- All at cks.nice.org.uk/topics/… (full URLs in each chapter)

PCDS

- Clinical A–Z pcds.org.uk/clinical-a-z-list
- Atopic eczema pcds.org.uk/clinical-guidance/atopic-eczema
- Desktop treatment guide pcds.org.uk/desktop-treatment-guide

Other UK resources

- BNF (topical corticosteroids, psoriasis treatment summaries) bnf.nice.org.uk
- NHS: Acne, Pityriasis versicolor, Cellulitis nhs.uk/conditions
- British Association of Dermatologists patient information bad.org.uk
- MHRA: isotretinoin risk-minimisation guidance

Module 6 sources (Chapters F–H)

Toxic erythema of the newborn	dermnetnz.org/topics/toxic-erythema-of-the-newborn
TNPM	dermnetnz.org/topics/transient-neonatal-pustular-melanosis
Neonatal blisters & pustules	dermnetnz.org/topics/blisters-and-pustules-in-neonates
Miliaria	dermnetnz.org/topics/miliaria
Acne in children	dermnetnz.org/topics/acne-in-children
Napkin dermatitis	dermnetnz.org/topics/napkin-dermatitis
Infantile haemangioma	dermnetnz.org/topics/infantile-haemangioma
Capillary malformation	dermnetnz.org/topics/capillary-vascular-malformation
Naevus of Ota / Ito / Hori	dermnetnz.org/topics/naevus-of-ota-ito-hori
Dermal melanocytosis	dermnetnz.org/topics/lumbosacral-dermal-melanocytosis
Achromic naevus	dermnetnz.org/topics/achromic-naevus
Naevus anaemicus	dermnetnz.org/topics/naevus-anaemicus
Becker naevus	dermnetnz.org/topics/becker-naevus
Epidermal naevi	dermnetnz.org/topics/epidermal-naevi
Epidermolysis bullosa	dermnetnz.org/topics/epidermolysis-bullosa
Aplasia cutis	dermnetnz.org/topics/aplasia-cutis

- NICE CKS: Nappy rash cks.nice.org.uk/topics/nappy-rash; Eczema – atopic

- NICE ESUOM47: Infantile haemangioma — topical timolol nice.org.uk/advice/esuom47
- PCDS: Napkin eczema; Strawberry naevus; Birthmarks overview; Naevus of Ota and Ito pcds.org.uk
- BAD patient leaflet: Haemangioma of infancy bad.org.uk/pils/haemangioma-of-infancy
- GOSH Epidermolysis bullosa service gosh.nhs.uk
- DEBRA UK debra.org.uk
- BNF for Children bnfc.nice.org.uk
- Course material: DermUSW Module 6 Week 1 discussion, daily notes and quiz (topic scope only; paraphrased).

No dedicated NICE guideline covers infantile haemangioma; recommendations follow PCDS, BAD and specialist consensus. NICE CKS full text may be geo-restricted outside the UK — re-verify durations and doses in live CKS and the BNFC.

Expansion sources (Chapters A+, B+, I–Q)

Each condition in Chapters A+, B+ and I–Q ends with its own "Real photos & references" list (DermNet topic pages, NICE / CKS, BNF, PCDS, BAD, NHS, UKHSA, MHRA). Key UK guidance used:

- NICE CG57 Atopic eczema in under-12s; CKS Eczema – atopic; CKS Dermatitis – contact; BAD / PCDS contact dermatitis and patch-testing guidance
- NICE **CG153** Psoriasis: assessment and management (not NG199, which is C. difficile antimicrobial prescribing); CKS Psoriasis; NICE NG238 CVD risk assessment
- NICE NG198 Acne vulgaris; CKS Acne vulgaris, Rosacea; MHRA isotretinoin safety measures and Pregnancy Prevention Programme
- CKS Melasma (and PCDS pigmentation guidance); UK Cosmetic Products Regulations / Trading Standards on skin-lightening products
- CKS Pruritus – generalized, Venous eczema, Actinic keratosis, Seborrhoeic keratosis; NICE NG12 suspected cancer; ISTAP skin-tear classification
- GPhC guidance for pharmacist prescribers (face-to-face for non-surgical cosmetic medicines); UKHSA iatrogenic botulism alerts (2025); Botulinum Toxin and Cosmetic Fillers (Children) Act 2021; ACE Group World complication guidance (awareness only)
- CKS Scabies, Head lice, Insect bites and stings; BASHH 2025 scabies guideline; NICE NG95 Lyme disease; NICE NG182 insect bites (antimicrobial prescribing)
- BSACI / BAD urticaria guidance; CKS Urticaria, Angio-oedema and anaphylaxis; Resuscitation Council UK anaphylaxis 2021; NICE CG134
- BAD alopecia areata guideline; NICE TA926 (ritlecitinib) / baricitinib appraisals; CKS Alopecia areata
- BASHH anogenital warts guidance; UK HPV vaccination programme (UKHSA Green Book); BAD hidradenitis suppurativa guideline; CKS Hidradenitis suppurativa
- Course material: DermUSW Modules 3–6 discussions, cases and quizzes (topic scope only; paraphrased).

Some CKS pages were geo-restricted during compilation and were cross-checked against BNF, SmPCs and UK formularies. Re-verify doses, durations and licensing in the live BNF/BNFC and CKS.

Sources & further reading

Fold-in sources (Chapters 0, E+, I+, L+, R–Z and the exam appendix)

- CKS Rosacea; CKS Paronychia – acute and chronic; CKS Fungal nail infection; NICE NG12 (nail melanoma); BAD onychomycosis guideline
- BAD hidradenitis suppurativa guideline 2018; NICE TA392 / TA935 (adalimumab, secukinumab); NICE **CG153** psoriasis (face, flexures, palms and soles)
- CKS Dermatitis – contact; BAD/ESCD hand eczema guidance; HSE skin at work; NICE TA81 alitretinoin
- BAD guidelines: pemphigus vulgaris (2017), bullous pemphigoid (2012), SJS/TEN (2016); BLISTER trial; NICE NG20 coeliac disease
- NICE NG141 cellulitis; NICE NG153 impetigo; NICE NG19 diabetic foot; SIGN 120 / CKS leg ulcer – venous; PCDS pyoderma gangrenosum
- UKHSA / NaTHNaC travel health; WHO leishmaniasis, leprosy, mycetoma and dengue fact sheets; UKHSA scarlet fever guidance; CKS Shingles, Herpes simplex – oral, Pityriasis rosea
- BAD photodermatoses patient guidance; British Photodermatology Group; CKS Sun protection; MHRA drug photosensitivity advice; MHRA finasteride warnings (2024)
- Course material: DermUSW Modules 1 and 3–6 discussions, cases, quizzes and end-of-module exams (topic scope only; paraphrased).

Modules 1 & 2 sources (Chapters 1B–1E and 2A–2F)

- BNF treatment summaries: topical corticosteroids, emollients, acne, psoriasis; MHRA Drug Safety Updates (topical steroid withdrawal, emollient fire risk, oral retinoids and isotretinoin, valproate/pregnancy); NICE NG198 acne; NICE CG153 psoriasis; NICE technology appraisals for biologics and JAK inhibitors; BAD/PCDS guidance
- NICE CG183 drug allergy; MHRA Yellow Card; BAD SJS/TEN guideline (2016); RegiSCAR DRESS criteria; BSACI guidance; Resuscitation Council UK anaphylaxis (2021); CKS Erythema nodosum
- EULAR/ACR vasculitis classification; BSR guidance on ANCA-associated vasculitis; NICE NG240 meningococcal disease; ESC endocarditis guidance (modified Duke criteria); CKS Henoch–Schönlein purpura
- NICE NG12 suspected cancer; NICE NG20 coeliac disease; NICE NG243 adrenal insufficiency; NICE NG238 lipids; Society for Endocrinology adrenal crisis guidance; BTS sarcoidosis clinical statement (2021); BAD hyperhidrosis guidance
- BAD guidelines: basal cell carcinoma (2021), cutaneous SCC (2020), actinic keratosis (2017), Bowen's disease, congenital melanocytic naevi; NICE NG14 melanoma (2015, updated 2022); NICE 7-point checklist; AJCC 8th edition staging (Gershenwald 2017); RCPath melanoma dataset; BAD/PCDS teledermatology quality standards
- British Society for Dermatological Surgery (BSDS) antithrombotic guidance (2023); BNF lidocaine monograph; AAGBI management of severe local anaesthetic toxicity; NICE CG64 prophylaxis against infective endocarditis; NICE NG125 surgical site infection; NICE NG141 cellulitis
- DermNet topic pages as cited under each condition. Course material: DermUSW Modules 1 and 2 discussions, cases, quizzes and summaries (topic scope only; paraphrased).

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Sources & further reading

Compiled October 2026 for Mostafa Damghani, pharmacist independent prescriber. Educational use only. Verify all prescribing against the current BNF / BNF for Children, NICE / CKS and local formulary.

APPENDIX E

Image credits

Image credits

Image credits & attribution

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Adult core (eczema, psoriasis, acne, fungal, bacterial)

72 images · `assets/img/dermnet`

CODE	CAPTION	©	SOURCE PAGE	USED ON
A01	Acute flexural atopic dermatitis	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A02	Flexural atopic dermatitis in childhood	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A03	Lichenified flexural atopic dermatitis	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A04	Lichenification	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A05	Infantile atopic dermatitis on the cheeks	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A06	Infantile atopic dermatitis, face	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A07	Staphylococcal infection of atopic dermatitis	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A08	Eczema herpeticum	© DermNet	dermnetnz.org/topics/eczema-herpeticum	Eczema / dermatitis, Impetigo
A09	Hand dermatitis on the dorsal fingers in an atopic boy	© DermNet	dermnetnz.org/topics/hand-dermatitis	Eczema / dermatitis
A10	Atopic hand dermatitis	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
A11	Infected hand dermatitis	© DermNet	dermnetnz.org/topics/hand-dermatitis	Eczema / dermatitis
A12	Thickening, dryness, and scale on the dorsal fingers	© DermNet	dermnetnz.org/topics/hand-dermatitis	Eczema / dermatitis

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
A13	Hyperpigmentation due to atopic dermatitis	© Dr Richard Ashton	dermnetnz.org/topics/atopic-dermatitis	Eczema / dermatitis
B01	Thick silvery scale over the elbow	© DermNet	dermnetnz.org/topics/psoriasis	Psoriasis
B02	Well-defined scaly plaque psoriasis on the knee	© DermNet	dermnetnz.org/topics/chronic-plaque-psoriasis	Psoriasis
B03	Close up of well demarcated plaques with silvery scale in chronic plaque psoriasis	© DermNet	dermnetnz.org/topics/psoriasis	Psoriasis
B04	Well demarcated plaques with silvery scale in chronic plaque psoriasis	© DermNet	dermnetnz.org/topics/psoriasis	Psoriasis
B05	Chronic plaque psoriasis in skin of colour	© DermNet	dermnetnz.org/topics/psoriasis	Psoriasis
B06	Scalp psoriasis extending beyond the anterior hairline	© DermNet	dermnetnz.org/topics/scalp-psoriasis	Psoriasis
B07	Adherent scale in scalp psoriasis	© Te Whatu Ora	dermnetnz.org/topics/psoriasis	Psoriasis
B08	Irregular pits and onycholysis of a finger nail - note the yellow band proximal to the onycholysis	© DermNet	dermnetnz.org/topics/psoriasis	Psoriasis
B09	Onycholysis, proximal yellow band and nail plate splitting due to nail unit psoriasis	© DermNet	dermnetnz.org/topics/psoriasis	Psoriasis
B10	Pitting and onychorrhexis	© DermNet	dermnetnz.org/topics/nail-psoriasis	Psoriasis
B11	Guttate psoriasis	© DermNet	dermnetnz.org/topics/guttate-psoriasis	Psoriasis
B12	Small plaques of psoriasis appear almost like rain drops splashed over the skin	© DermNet	dermnetnz.org/topics/psoriasis	Psoriasis
B13	Axillary psoriasis	© Raimo Suhonen	dermnetnz.org/topics/psoriasis	Psoriasis
B14	Well demarcated salmon pink erythema which extends to the apex of the skin fold and into the natal cleft due to flexural psoriasis	© DermNet	dermnetnz.org/topics/psoriasis	Candidal intertrigo, Psoriasis, Tinea: corporis, cruris & pedis
C01	Open comedones	© Raimo Suhonen	dermnetnz.org/topics/comedonal-acne	Acne vulgaris
C02	Closed comedones	© Raimo Suhonen	dermnetnz.org/topics/comedonal-acne	Acne vulgaris
C03	Comedonal acne	© DermNet	dermnetnz.org/topics/comedonal-acne	Acne vulgaris
C04	Mild acne	© Te Whatu Ora	dermnetnz.org/topics/acne-vulgaris	Acne vulgaris

CODE	CAPTION	©	SOURCE PAGE	USED ON
C05	Acne vulgaris	© DermNet	dermnetnz.org/topics/acne-vulgaris	Acne vulgaris
C06	Moderate acne	© DermNet	dermnetnz.org/topics/acne-vulgaris	Acne vulgaris
C07	Severe acne	© Raimo Suhonen	dermnetnz.org/topics/acne-vulgaris	Acne vulgaris
C08	Nodulocystic acne: face	© DermNet	dermnetnz.org/topics/nodulocystic-acne	Acne vulgaris
C09	Nodulocystic acne	© DermNet	dermnetnz.org/topics/nodulocystic-acne	Acne vulgaris
C10	Acne vulgaris (back)	© DermNet	dermnetnz.org/topics/acne-vulgaris	Acne vulgaris
C11	Nodulocystic acne (back)	© DermNet	dermnetnz.org/topics/nodulocystic-acne	Acne vulgaris
C12	Acne with post-inflammatory pigmentation	© DermNet	dermnetnz.org/topics/acne-vulgaris	Acne vulgaris
D01	Tinea corporis	© DermNet	dermnetnz.org/topics/tinea-corporis-images	Tinea: corporis, cruris & pedis
D02	Tinea corporis	© DermNet	dermnetnz.org/topics/tinea-corporis-images	Tinea: corporis, cruris & pedis
D03	Tinea corporis	© DermNet	dermnetnz.org/topics/tinea-corporis-images	Tinea: corporis, cruris & pedis
D04	Sharp red scaly margin of tinea corporis	© DermNet	dermnetnz.org/topics/tinea-corporis	Tinea: corporis, cruris & pedis
D05	Athlete's foot	© DermNet	dermnetnz.org/topics/intertrigo	Cellulitis & erysipelas, Tinea: corporis, cruris & pedis
D06	Moccasin tinea pedis	© DermNet	dermnetnz.org/topics/tinea-pedis	Tinea: corporis, cruris & pedis
D07	Tinea pedis	© DermNet	dermnetnz.org/topics/tinea-pedis	Tinea: corporis, cruris & pedis
D08	Tinea pedis	© DermNet	dermnetnz.org/topics/tinea-pedis	Tinea: corporis, cruris & pedis
D09	Well-defined plaque	© DermNet	dermnetnz.org/topics/tinea-cruris	Tinea: corporis, cruris & pedis
D10	Raised border and central clearing	© DermNet	dermnetnz.org/topics/tinea-cruris	Tinea: corporis, cruris & pedis
D11	Raised border, central clearing	© DermNet	dermnetnz.org/topics/tinea-cruris	Candidal intertrigo, Psoriasis, Tinea: corporis, cruris & pedis

CODE	CAPTION	©	SOURCE PAGE	USED ON
D12	Pityriasis versicolor	© DermNet	dermnetnz.org/topics/pityriasis-versicolor	Pityriasis versicolor, Progressive macular hypomelanosis (PMH)
D13	Pityriasis versicolor	© DermNet	dermnetnz.org/topics/pityriasis-versicolor	Pityriasis versicolor
D14	Pityriasis versicolor	© DermNet	dermnetnz.org/topics/pityriasis-versicolor	Pityriasis versicolor, Progressive macular hypomelanosis (PMH)
D15	Intertrigo due to candida infection	© Te Whatu Ora	dermnetnz.org/topics/intertrigo	Candidal intertrigo, Psoriasis, Tinea: corporis, cruris & pedis
D16	Candida intertrigo	© DermNet	dermnetnz.org/images/candida-intertrigo-images	Candidal intertrigo
D17	Candida intertrigo	© DermNet	dermnetnz.org/images/candida-intertrigo-images	Candidal intertrigo
E01	Perioral honey-coloured crusts in impetigo	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
E02	Honey-coloured crusts on the chin in impetigo	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
E03	Honey-coloured crusted lesions in facial impetigo	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
E04	An intact blister and erosions in bullous impetigo	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
E05	Widespread bullous impetigo over the back	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
E06	Bullous impetigo	© DermNet	dermnetnz.org/topics/intertrigo	Impetigo
E07	Cellulitis	© DermNet	dermnetnz.org/topics/cellulitis	Cellulitis & erysipelas, Red leg: differentials & referral
E08	Erysipelas	© DermNet	dermnetnz.org/topics/erysipelas	Cellulitis & erysipelas
E09	Erysipelas	© DermNet	dermnetnz.org/topics/erysipelas	Cellulitis & erysipelas
E10	Superficial bacterial folliculitis	© Raimo Suhonen	dermnetnz.org/topics/folliculitis	Folliculitis
E11	Folliculitis	© DermNet	dermnetnz.org/topics/folliculitis	Folliculitis

CODE	CAPTION	©	SOURCE PAGE	USED ON
E12	Folliculitis in axilla	© DermNet	dermnetnz.org/topics/intertrigo	Folliculitis
E13	Boil	© DermNet	dermnetnz.org/topics/boil	Boils & carbuncles
E14	Surrounding cellulitis	© Raimo Suhonen	dermnetnz.org/topics/boil	Boils & carbuncles
E15	Boil in axilla	© DermNet	dermnetnz.org/topics/boil	Boils & carbuncles
E16	Carbuncle	© DermNet	dermnetnz.org/topics/boil	Boils & carbuncles

Module 5 · Week 1 — skin infections

14 images · assets/img/m5w1

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w1:L01	Common wart	© DermNet	dermnetnz.org/topics/viral-wart	Viral warts (HPV)
m5w1:L02	Plantar wart	© DermNet	dermnetnz.org/topics/viral-wart	Viral warts (HPV)
m5w1:L03	Plane warts	© DermNet	dermnetnz.org/topics/viral-wart	Viral warts (HPV)
m5w1:L04	Filiform wart	© DermNet	dermnetnz.org/topics/viral-wart	Viral warts (HPV)
m5w1:L05	Periungual / subungual wart	© DermNet	dermnetnz.org/topics/viral-wart	Viral warts (HPV)
m5w1:M01	Progressive macular hypomelanosis	© DermNet	dermnetnz.org/topics/progressive-macular-hypomelanosis	Progressive macular hypomelanosis (PMH)
m5w1:M02	Progressive macular hypomelanosis — back	© DermNet	dermnetnz.org/topics/progressive-macular-hypomelanosis	Progressive macular hypomelanosis (PMH)
m5w1:N01	Facial impetigo	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
m5w1:N02	Impetigo honey crust	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
m5w1:N03	Bullous impetigo	© DermNet	dermnetnz.org/topics/impetigo	Impetigo
m5w1:O01	Tinea cruris groin	© DermNet	dermnetnz.org/topics/tinea-cruris	Tinea: corporis, cruris & pedis
m5w1:O02	Tinea cruris raised border	© DermNet	dermnetnz.org/topics/tinea-cruris	Tinea: corporis, cruris & pedis
m5w1:O03	Candidal intertrigo with satellites	© DermNet	dermnetnz.org/topics/intertrigo	Candidal intertrigo
m5w1:O04	Intertrigo in skin fold	© DermNet	dermnetnz.org/topics/intertrigo	Candidal intertrigo

Image credits

Module 5 · Week 2 — cellulitis & ulcers

19 images · `assets/img/m5w2`

CODE	CAPTION	©	SOURCE PAGE	USED ON
`m5w2:P01`	Venous leg ulcer — medial gaiter	© DermNet	dermnetnz.org/topics/stasis-ulcer	Venous leg ulcer
`m5w2:P02`	Venous leg ulcer — gaiter	© DermNet	dermnetnz.org/topics/stasis-ulcer	Venous leg ulcer
`m5w2:P03`	Venous stasis / lipodermatosclerosis changes	© DermNet	dermnetnz.org/topics/stasis-ulcer	Red leg: differentials & referral, Venous leg ulcer
`m5w2:P04`	Venous leg ulcer shallow sloping	© DermNet	dermnetnz.org/topics/stasis-ulcer	Venous leg ulcer
`m5w2:Q01`	Cellulitis of the leg	© DermNet	dermnetnz.org/topics/cellulitis	Cellulitis & erysipelas, Red leg: differentials & referral
`m5w2:Q02`	Cellulitis peau d'orange	© DermNet	dermnetnz.org/topics/cellulitis	Cellulitis & erysipelas
`m5w2:Q03`	Bullous cellulitis	© DermNet	dermnetnz.org/topics/cellulitis	Cellulitis & erysipelas
`m5w2:Q04`	Facial erysipelas	© DermNet	dermnetnz.org/topics/erysipelas	Cellulitis & erysipelas
`m5w2:Q05`	Erysipelas involving ear (Milian context)	© DermNet	dermnetnz.org/topics/erysipelas	Cellulitis & erysipelas
`m5w2:Q06`	Erysipelas of the leg	© DermNet	dermnetnz.org/topics/erysipelas	Cellulitis & erysipelas
`m5w2:R01`	Pyoderma gangrenosum violaceous border	© DermNet	dermnetnz.org/topics/pyoderma-gangrenosum	Pyoderma gangrenosum
`m5w2:R02`	Pyoderma gangrenosum leg ulcer	© DermNet	dermnetnz.org/topics/pyoderma-gangrenosum	Pyoderma gangrenosum
`m5w2:R03`	Early / pathergic pyoderma gangrenosum	© DermNet	dermnetnz.org/topics/pyoderma-gangrenosum	Pyoderma gangrenosum
`m5w2:R04`	Multiple pyoderma gangrenosum ulcers	© DermNet	dermnetnz.org/topics/pyoderma-gangrenosum	Pyoderma gangrenosum
`m5w2:S01`	Leg ulcer (diabetic/ischaemic example from DermNet leg-ulcer page)	© DermNet	dermnetnz.org/topics/leg-ulcer	Arterial (ischaemic) leg ulcer
`m5w2:S02`	Diabetic ulcer	© DermNet	dermnetnz.org/topics/leg-ulcer	Neuropathic and diabetic foot ulcers
`m5w2:S03`	Diabetic foot ulcer plantar	© DermNet	dermnetnz.org/topics/diabetic-foot-ulcer	Neuropathic and diabetic foot ulcers

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w2:S04	Diabetic foot ulcer pressure site	© DermNet	dermnetnz.org/topics/diabetic-foot-ulcer	Neuropathic and diabetic foot ulcers
m5w2:S05	Neuropathic ulcer	© DermNet	dermnetnz.org/topics/leg-ulcer	Neuropathic and diabetic foot ulcers

Module 5 · Week 3 — infestations & bites

41 images · assets/img/m5w3

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w3:T01	A burrow on the palm - the mite can just be seen adjacent to the vesicle	© DermNet	dermnetnz.org/topics/scabies	Scabies
m5w3:T02	Interdigital scaling in the first web space of the hand	© DermNet	dermnetnz.org/topics/scabies	Scabies
m5w3:T03	Papular lesions in the 4th interdigital space on the hand	© DermNet	dermnetnz.org/topics/scabies-images	Scabies
m5w3:T04	Scabies burrows on the palm	© DermNet	dermnetnz.org/topics/scabies	Scabies
m5w3:T05	A dark scabies mite at the end of a burrow on dermoscopy	© DermNet	dermnetnz.org/topics/scabies-images	Scabies
m5w3:T06	Groin and abdominal papules and pustules in scabies	© DermNet	dermnetnz.org/topics/scabies-images	Scabies
m5w3:T07	Erythematous papules on the penis in scabies	© DermNet	dermnetnz.org/topics/scabies-images	Scabies
m5w3:T08	Scabies nodules in the axilla	© DermNet	dermnetnz.org/topics/scabies	Scabies
m5w3:T09	Scabies burrows on a babies foot	© DermNet	dermnetnz.org/topics/scabies	Scabies
m5w3:T10	Scabies with superadded streptococcal infection	© DermNet	dermnetnz.org/topics/scabies	Scabies
m5w3:T11	Thick crusted plaques on the thumb and palm due to crusted scabies in an elderly woman	© DermNet	dermnetnz.org/topics/scabies	Scabies
m5w3:T12	Hyperkeratosis and fissuring on the feet in advanced crusted scabies	© DermNet	dermnetnz.org/topics/crusted-scabies	Scabies
m5w3:U01	Louse	© DermNet	dermnetnz.org/topics/head-lice	Head lice (pediculosis capitis)
m5w3:U02	Louse	© DermNet	dermnetnz.org/topics/head-lice	Head lice (pediculosis capitis)
m5w3:U03	Head lice eggs	© DermNet	dermnetnz.org/topics/head-lice	Head lice (pediculosis capitis)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w3:U04	Nits and dermatitis	© DermNet	dermnetnz.org/topics/head-lice	Head lice (pediculosis capitis)
m5w3:U05	Nits and bites	© DermNet	dermnetnz.org/topics/head-lice	Head lice (pediculosis capitis)
m5w3:V01	Ixodes tick	© DermNet	dermnetnz.org/topics/lyme-disease	Lyme disease & erythema migrans
m5w3:V02	Erythema chronicum migrans	© DermNet	dermnetnz.org/topics/lyme-disease	Lyme disease & erythema migrans
m5w3:V03	Erythema chronicum migrans	© DermNet	dermnetnz.org/topics/lyme-disease	Lyme disease & erythema migrans
m5w3:V04	Erythema chronicum migrans	© DermNet	dermnetnz.org/topics/erythema-migrans-images	Lyme disease & erythema migrans
m5w3:V05	Erythema chronicum migrans	© DermNet	dermnetnz.org/topics/erythema-migrans-images	Lyme disease & erythema migrans
m5w3:V06	Inflammatory stage	© DermNet	dermnetnz.org/topics/acrodermatitis-chronica-atrophicans	Lyme disease & erythema migrans
m5w3:V07	Inflammatory stage	© DermNet	dermnetnz.org/topics/acrodermatitis-chronica-atrophicans	Lyme disease & erythema migrans
m5w3:V08	Atrophic stage	© DermNet	dermnetnz.org/topics/acrodermatitis-chronica-atrophicans	Lyme disease & erythema migrans
m5w3:W01	Cantharidin	© DermNet	dermnetnz.org/topics/cantharidin	Blister beetle & Paederus dermatitis
m5w3:W02	Cantharidin	© DermNet	dermnetnz.org/topics/cantharidin	Blister beetle & Paederus dermatitis
m5w3:W03	Kissing paederus dermatitis	© DermNet	dermnetnz.org/topics/paederus-dermatitis	Blister beetle & Paederus dermatitis
m5w3:W04	Paedarus littoralis	© DermNet	dermnetnz.org/topics/paederus-dermatitis	Blister beetle & Paederus dermatitis
m5w3:W05	Lytta vesicatoria	Stefanie Hamm	commons.wikimedia.org/wiki/File:Lytta-vesicatoria03.jpg	Blister beetle & Paederus dermatitis
m5w3:W06	Mylabris oleae (Meloidae)	Hectonichus	commons.wikimedia.org/wiki/File:Meloidae__-__Mylabris_oleae.JPG	Blister beetle & Paederus dermatitis
m5w3:X01	Papular urticaria	© DermNet	dermnetnz.org/topics/papular-urticaria	Papular urticaria & insect bites
m5w3:X02	Papular urticaria	© DermNet	dermnetnz.org/topics/papular-urticaria	Papular urticaria & insect bites

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w3:X03	Papular urticaria	© DermNet	dermnetnz.org/topics/papular-urticaria	Papular urticaria & insect bites
m5w3:X04	Papular urticaria	© DermNet	dermnetnz.org/topics/papular-urticaria	Papular urticaria & insect bites
m5w3:X05	Flea bites	© DermNet	dermnetnz.org/topics/flea-bite	Papular urticaria & insect bites
m5w3:X06	Flea bites	© DermNet	dermnetnz.org/topics/flea-bite	Papular urticaria & insect bites
m5w3:X07	Bed bug bite on the calf - note the central punctum	© DermNet	dermnetnz.org/topics/bed-bugs	Papular urticaria & insect bites
m5w3:X08	Bed bug bites on the calf	© DermNet	dermnetnz.org/topics/bed-bugs	Papular urticaria & insect bites
m5w3:X09	Fed adult	© DermNet	dermnetnz.org/topics/bed-bugs	Papular urticaria & insect bites
m5w3:X10	Bullous insect bite	© DermNet	dermnetnz.org/topics/arthropod-bites-and-stings	Papular urticaria & insect bites

Module 6 · Week 1 — infant & neonatal

39 images · assets/img/m6w1

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w1:F01	Toxic erythema of the newborn	© DermNet	dermnetnz.org/topics/toxic-erythema-of-the-newborn	Neonatal pustular rashes
m6w1:F02	Toxic erythema of the newborn	© DermNet	dermnetnz.org/topics/toxic-erythema-of-the-newborn	Neonatal pustular rashes
m6w1:F03	Toxic erythema of the newborn	© DermNet	dermnetnz.org/topics/skin-conditions-in-newborn-babies	Neonatal pustular rashes
m6w1:F04	Neonatal pustular melanosis	© DermNet	dermnetnz.org/topics/transient-neonatal-pustular-melanosis	Neonatal pustular rashes
m6w1:F05	Neonatal pustular melanosis	© DermNet	dermnetnz.org/topics/transient-neonatal-pustular-melanosis	Neonatal pustular rashes
m6w1:F06	Transient neonatal pustular melanosis	© DermNet	dermnetnz.org/topics/transient-neonatal-pustular-melanosis	Neonatal pustular rashes
m6w1:F07	Miliaria	© DermNet	dermnetnz.org/topics/miliaria	Neonatal pustular rashes
m6w1:F08	Miliaria	© DermNet	dermnetnz.org/topics/miliaria	Neonatal pustular rashes
m6w1:F09	Neonatal cephalic pustulosis	© DermNet	dermnetnz.org/topics/neonatal-cephalic-pustulosis	Neonatal & infantile acne

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w1:F10	Infantile acne	© DermNet	dermnetnz.org/topics/infantile-acne	Neonatal & infantile acne
m6w1:F11	Napkin dermatitis	© DermNet	dermnetnz.org/topics/napkin-dermatitis	Napkin (nappy) dermatitis
m6w1:F12	Napkin dermatitis thrush	© DermNet	dermnetnz.org/topics/napkin-dermatitis	Napkin (nappy) dermatitis
m6w1:F13	Napkin dermatitis	© DermNet	dermnetnz.org/topics/napkin-dermatitis	Napkin (nappy) dermatitis
m6w1:F14	Infantile atopic dermatitis on the cheeks	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Napkin (nappy) dermatitis
m6w1:F15	Infantile atopic dermatitis, face	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Napkin (nappy) dermatitis
m6w1:F16	Herpes simplex	© DermNet	dermnetnz.org/topics/blisters-and-pustules-in-neonates	Neonatal pustular rashes
m6w1:G01	Infantile haemangioma, face	© DermNet	dermnetnz.org/topics/infantile-haemangioma	Infantile haemangioma (strawberry mark)
m6w1:G02	Regressing infantile haemangioma, scalp	© DermNet	dermnetnz.org/topics/infantile-haemangioma	Infantile haemangioma (strawberry mark)
m6w1:G03	Infantile haemangioma, buttock	© DermNet	dermnetnz.org/topics/infantile-haemangioma	Infantile haemangioma (strawberry mark)
m6w1:G04	Preauricular infantile haemangioma	© DermNet	dermnetnz.org/topics/infantile-haemangioma-complications-and-treatment	Infantile haemangioma (strawberry mark)
m6w1:G05	Naevus of Ota	© DermNet	dermnetnz.org/topics/naevus-of-ota-ito-hori	Pigmented & epidermal birthmarks
m6w1:G06	Naevus of Ota	© DermNet	dermnetnz.org/topics/naevus-of-ota-ito-hori	Pigmented & epidermal birthmarks
m6w1:G07	Achromic naevus	© DermNet	dermnetnz.org/topics/achromic-naevus	Pigmented & epidermal birthmarks
m6w1:G08	A unilateral depigmented lesion on the arm - the child was otherwise well	© DermNet	dermnetnz.org/topics/achromic-naevus	Pigmented & epidermal birthmarks
m6w1:G09	Naevus anaemicus	© DermNet	dermnetnz.org/topics/naevus-anaemicus	Pigmented & epidermal birthmarks

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w1:G10	Hyperpigmentation and hypertrichosis over the chest in Becker naevus	© DermNet	dermnetnz.org/topics/becker-naevus	Pigmented & epidermal birthmarks
m6w1:G11	Pigmentation and hypertrichosis in Becker naevus	© Dr Richard Ashton	dermnetnz.org/topics/becker-naevus	Pigmented & epidermal birthmarks
m6w1:G12	Lumbosacral dermal melanocytosis. Mongolian spot	© DermNet	dermnetnz.org/topics/lumbosacral-dermal-melanocytosis	Pigmented & epidermal birthmarks
m6w1:G13	Mongolian spot — lumbosacral dermal melanocytosis	© DermNet	dermnetnz.org/topics/lumbosacral-dermal-melanocytosis	Pigmented & epidermal birthmarks
m6w1:G14	Epidermal naevus	© DermNet	dermnetnz.org/topics/epidermal-naevi	Pigmented & epidermal birthmarks
m6w1:G15	Epidermal naevus	© DermNet	dermnetnz.org/topics/epidermal-naevi	Pigmented & epidermal birthmarks
m6w1:G16	Port-wine stain	© DermNet	dermnetnz.org/topics/capillary-vascular-malformation	Port-wine stain (capillary malformation)
m6w1:G17	Port-wine stain	© Te Whatu Ora	dermnetnz.org/topics/capillary-vascular-malformation	Port-wine stain (capillary malformation)
m6w1:H01	Epidermolysis bullosa	© DermNet	dermnetnz.org/topics/blisters-and-pustules-in-neonates	Epidermolysis bullosa (EB)
m6w1:H02	Epidermolysis bullosa	© DermNet	dermnetnz.org/topics/epidermolysis-bullosa	Epidermolysis bullosa (EB)
m6w1:H03	Epidermolysis bullosa	© DermNet	dermnetnz.org/topics/epidermolysis-bullosa	Epidermolysis bullosa (EB)
m6w1:H04	Recessive dystrophic epidermolysis bullosa	© DermNet	dermnetnz.org/topics/dystrophic-epidermolysis-bullosa	Epidermolysis bullosa (EB)
m6w1:H05	Aplasia cutis	© DermNet	dermnetnz.org/topics/aplasia-cutis	Aplasia cutis congenita
m6w1:H06	Aplasia cutis	© DermNet	dermnetnz.org/topics/aplasia-cutis	Aplasia cutis congenita

Module 6 · Week 2 — acne & mimics

40 images · assets/img/m6w2

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w2:I01	Acne vulgaris	© DermNet	dermnetnz.org/topics/acne-vulgaris	—
m6w2:I02	Comedonal acne	© DermNet	dermnetnz.org/topics/comedones	—
m6w2:I03	Open comedones	© DermNet	dermnetnz.org/topics/comedones	—

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w2:I04	Closed comedones	© Te Whatu Ora	dermnetnz.org/topics/comedones	—
m6w2:I05	Mild acne	© Te Whatu Ora	dermnetnz.org/topics/acne-vulgaris	—
m6w2:I06	Moderate acne	© DermNet	dermnetnz.org/topics/acne-vulgaris	—
m6w2:I07	Severe acne	© Raimo Suhonen	dermnetnz.org/topics/acne-vulgaris	—
m6w2:I08	Nodulocystic acne: face	© DermNet	dermnetnz.org/topics/nodulocystic-acne	—
m6w2:I09	Nodulocystic acne	© DermNet	dermnetnz.org/topics/nodulocystic-acne	—
m6w2:I10	Nodulocystic acne: chest	© DermNet	dermnetnz.org/topics/nodulocystic-acne	—
m6w2:I11	Acne fulminans	© DermNet	dermnetnz.org/topics/acne-fulminans	Acne vulgaris
m6w2:I12	Acne fulminans	© Te Whatu Ora	dermnetnz.org/topics/acne-fulminans	Acne vulgaris
m6w2:I13	Acne fulminans	© DermNet	dermnetnz.org/topics/sapho-syndrome	Acne vulgaris
m6w2:I14	Cheilitis	© DermNet	dermnetnz.org/topics/isotretinoin	Acne vulgaris
m6w2:I15	Acne flare	© DermNet	dermnetnz.org/topics/isotretinoin	Acne vulgaris
m6w2:I16	Ice pick scarring on the cheeks	© Dr Richard Ashton	dermnetnz.org/topics/acne-scarring	Acne vulgaris
m6w2:I17	Post inflammatory pigmentation after facial acne in skin of colour	© DermNet	dermnetnz.org/topics/acne-scarring	Acne vulgaris
m6w2:I18	True keloids on the shoulders post acne in skin of colour	© DermNet	dermnetnz.org/topics/acne-scarring	Acne vulgaris
m6w2:I19	Adult acne	© DermNet	dermnetnz.org/topics/adult-acne	Acne vulgaris
m6w2:I20	Polycystic ovarian syndrome	© DermNet	dermnetnz.org/topics/polycystic-ovary-syndrome	Acne vulgaris
m6w2:I21	Hirsutism	© DermNet	dermnetnz.org/topics/polycystic-ovary-syndrome	Acne vulgaris
m6w2:I22	Acne in pregnancy	© DermNet	dermnetnz.org/topics/acne-in-pregnancy	Acne vulgaris
m6w2:J01	Acanthosis nigricans on the neck	© Te Whatu Ora	dermnetnz.org/topics/acanthosis-nigricans	Acanthosis nigricans
m6w2:J02	Axillary acanthosis nigricans	© DermNet	dermnetnz.org/topics/acanthosis-nigricans	Acanthosis nigricans
m6w2:J03	Acanthosis nigricans	© DermNet	dermnetnz.org/topics/acanthosis-nigricans	Acanthosis nigricans

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w2:K01	Papulopustular rosacea on the cheeks	© DermNet	dermnetnz.org/topics/rosacea	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K02	Erythematotelangiectatic and papulopustular rosacea on the cheeks	© DermNet	dermnetnz.org/topics/rosacea	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K03	Rhinophyma	© DermNet	dermnetnz.org/topics/rosacea	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K04	Papulopustular and ocular rosacea	© DermNet	dermnetnz.org/topics/rosacea	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K05	Pyoderma faciale, rosacea fulminans	© DermNet	dermnetnz.org/topics/pyoderma-faciale	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K06	Pyoderma faciale, rosacea fulminans	© DermNet	dermnetnz.org/topics/pyoderma-faciale	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K07	Steroid-induced acne vulgaris	© DermNet	dermnetnz.org/topics/steroid-acne	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K08	Steroid induced papulopustular rosacea	© DermNet	dermnetnz.org/topics/rosacea	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K09	Florid perioral dermatitis due to potent topical steroid use	© Dr Richard Ashton	dermnetnz.org/topics/periorificial-dermatitis	Acne mimics: rosacea, steroid eruptions and folliculitis, Perioral (periorificial) dermatitis
m6w2:K10	Gram-negative folliculitis	© DermNet	dermnetnz.org/topics/gram-negative-folliculitis	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K11	Malassezia folliculitis	© DermNet	dermnetnz.org/topics/malassezia-folliculitis	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K12	Malassezia folliculitis	© DermNet	dermnetnz.org/topics/malassezia-folliculitis	Acne mimics: rosacea, steroid eruptions and folliculitis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w2:K13	Pseudomonas folliculitis	© DermNet	dermnetnz.org/topics/spa-pool-folliculitis	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K14	Spa pool folliculitis	© DermNet	dermnetnz.org/topics/spa-pool-folliculitis	Acne mimics: rosacea, steroid eruptions and folliculitis
m6w2:K15	Superficial bacterial folliculitis	© Raimo Suhonen	dermnetnz.org/topics/folliculitis	Acne mimics: rosacea, steroid eruptions and folliculitis

Module 6 · Week 3 — pigmentation

56 images · assets/img/m6w3

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w3:L01	Melasma	© DermNet	dermnetnz.org/topics/melasma	Melasma & facial melanosis
m6w3:L02	Melasma	© DermNet	dermnetnz.org/topics/melasma	Melasma & facial melanosis
m6w3:L03	Melasma	© DermNet	dermnetnz.org/topics/melasma	Melasma & facial melanosis
m6w3:L04	Melasma	© DermNet	dermnetnz.org/topics/melasma	Melasma & facial melanosis
m6w3:L05	Melasma	© DermNet	dermnetnz.org/topics/melasma	Melasma & facial melanosis
m6w3:L06	Melasma	© DermNet	dermnetnz.org/topics/melasma	Melasma & facial melanosis
m6w3:L07	Ochronosis following long-term use of hydroquinone	© Te Whatu Ora	dermnetnz.org/topics/alkaptonuria-and-ochronosis	Melasma & facial melanosis
m6w3:L08	Ochronosis following long-term use of hydroquinone	© Te Whatu Ora	dermnetnz.org/topics/alkaptonuria-and-ochronosis	Melasma & facial melanosis
m6w3:L09	Dermoscopy of ochronosis	© Te Whatu Ora	dermnetnz.org/topics/alkaptonuria-and-ochronosis	Melasma & facial melanosis
m6w3:L11	Nail pigmentation due to hydroquinone	© DermNet	dermnetnz.org/topics/hydroquinone	Melasma & facial melanosis
m6w3:L12	A phototoxic drug eruption due to an oral non-steroidal anti-inflammatory drug	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Melasma & facial melanosis
m6w3:L13	A photosensitive drug eruption showing sparing under the watch and strap	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Melasma & facial melanosis
m6w3:M01	Photoallergic contact dermatitis due to a fragrance in an aftershave	© DermNet	dermnetnz.org/topics/photocontact-dermatitis	Contact & lichenoid pigmentation

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w3:M02	Contact allergy on neck	© Te Whatu Ora	dermnetnz.org/topics/fragrance-allergy	Contact & lichenoid pigmentation
m6w3:M03	Patch test strips in place after application	© DermNet	dermnetnz.org/topics/patch-tests	Contact & lichenoid pigmentation
m6w3:M04	Fragrance mix II +++	© Te Whatu Ora	dermnetnz.org/topics/fragrance-allergy	Contact & lichenoid pigmentation
m6w3:M05	Black skin staining due to the hair dye allergen PPD	© DermNet	dermnetnz.org/topics/patch-tests	Contact & lichenoid pigmentation
m6w3:M06	Lichen planus pigmentosus of face	© DermNet	dermnetnz.org/topics/lichen-planus	Contact & lichenoid pigmentation
m6w3:M07	Lichen planus pigmentosus of lower back	© DermNet	dermnetnz.org/topics/acquired-dermal-macular-hyperpigmentation	Contact & lichenoid pigmentation
m6w3:M08	Lichen planus pigmentosus of lower lip	© DermNet	dermnetnz.org/topics/lichen-planus	Contact & lichenoid pigmentation
m6w3:M09	Acquired dermal macular hyperpigmentation	© Raimo Suhonen	dermnetnz.org/topics/acquired-dermal-macular-hyperpigmentation	Contact & lichenoid pigmentation
m6w3:M10	Erythema dyschromicum perstans	© DermNet	dermnetnz.org/topics/erythema-dyschromicum-perstans	Contact & lichenoid pigmentation
m6w3:M11	Erythema dyschromicum perstans	© DermNet	dermnetnz.org/topics/erythema-dyschromicum-perstans	Contact & lichenoid pigmentation
m6w3:M12	Erythema dyschromicum perstans	© DermNet	dermnetnz.org/topics/erythema-dyschromicum-perstans	Contact & lichenoid pigmentation
m6w3:N01	Streaky erythema on the arm	© DermNet	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis
m6w3:N02	Linear redness and blisters from strimming	© DermNet	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis
m6w3:N03	Acute bullous phytophotodermatitis on the ankle	© Dr Richard Ashton	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis
m6w3:N04	Phototoxic contact dermatitis due to accidental lime juice contact on a sunny day	© DermNet	dermnetnz.org/topics/photocontact-dermatitis	Phytophotodermatitis
m6w3:N05	Postinflammatory pigmentation after resolution of phytophotodermatitis	© Dr Richard Ashton	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w3:N06	Postinflammatory pigmentation from lime juice phytophotoderatitis on the chest	© Te Whatu Ora	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis
m6w3:N07	Rippled pigmentation on the upper chest due to macular amyloidosis	© DermNet	dermnetnz.org/topics/cutaneous-amyloidosis	Cutaneous amyloidosis (macular and lichen)
m6w3:N08	Macular amyloidosis showing rippled pigmentation over the upper back	© DermNet	dermnetnz.org/topics/cutaneous-amyloidosis	Cutaneous amyloidosis (macular and lichen)
m6w3:N09	Repetitive scratching of an area of notalgia paraesthetica resulting in macular amyloidosis on the back	© DermNet	dermnetnz.org/topics/cutaneous-amyloidosis	Cutaneous amyloidosis (macular and lichen)
m6w3:N10	Scaly tan brown papules, some arranged lineraly in lichen amyloidosis on the leg	© DermNet	dermnetnz.org/topics/cutaneous-amyloidosis	Cutaneous amyloidosis (macular and lichen)
m6w3:N11	Extensive and confluent lesions of lichen amyloidosis on the shin	© DermNet	dermnetnz.org/topics/cutaneous-amyloidosis	Cutaneous amyloidosis (macular and lichen)
m6w3:O01	Postinflammatory hyperpigmentation (inflamed cyst)	© DermNet	dermnetnz.org/topics/postinflammatory-hyperpigmentation	Hypopigmentation & skin of colour
m6w3:O02	Healed eczema	© DermNet	dermnetnz.org/topics/postinflammatory-hyperpigmentation	Hypopigmentation & skin of colour
m6w3:O03	Violin playing (friction)	© Raimo Suhonen	dermnetnz.org/topics/postinflammatory-hyperpigmentation	Hypopigmentation & skin of colour
m6w3:O04	Lichen planus actinicus	© DermNet	dermnetnz.org/topics/postinflammatory-hyperpigmentation	Hypopigmentation & skin of colour
m6w3:O05	Irritant dermatitis	© DermNet	dermnetnz.org/topics/postinflammatory-hyperpigmentation	Hypopigmentation & skin of colour
m6w3:O06	Hyperpigmentation due to acne	© DermNet	dermnetnz.org/topics/hydroquinone	Hypopigmentation & skin of colour
m6w3:O07	Suntan in dark skin	© DermNet	dermnetnz.org/topics/pigmentation-disorders	Hypopigmentation & skin of colour
m6w3:O08	Hormonal pigmentation	© DermNet	dermnetnz.org/topics/pigmentation-disorders	Hypopigmentation & skin of colour
m6w3:O09	Piebaldism	© DermNet	dermnetnz.org/topics/piebaldism	Hypopigmentation & skin of colour
m6w3:O10	Piebaldism	© DermNet	dermnetnz.org/topics/piebaldism	Hypopigmentation & skin of colour

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w3:011	Piebaldism	© DermNet	dermnetnz.org/topics/piebaldism	Hypopigmentation & skin of colour
m6w3:012	Piebaldism	© DermNet	dermnetnz.org/topics/piebaldism	Hypopigmentation & skin of colour
m6w3:013	Symmetrical wrist vitiligo - a common location	© DermNet	dermnetnz.org/topics/vitiligo	Hypopigmentation & skin of colour
m6w3:014	Vitiligo over the knuckles - keobnerisation due to trauma often localises vitiligo	© Dr Lawrence Meyerson	dermnetnz.org/topics/vitiligo	Hypopigmentation & skin of colour
m6w3:015	Vitiligo on the lid with poliosis of the lashes	© Raimo Suhonen	dermnetnz.org/topics/vitiligo	Hypopigmentation & skin of colour
m6w3:016	Extensive symmetrical facial vitiligo	© Dr Lawrence Meyerson	dermnetnz.org/topics/vitiligo	Hypopigmentation & skin of colour
m6w3:017	Follicular repigmentation in a patch of vitiligo	© Te Whatu Ora	dermnetnz.org/topics/vitiligo	Hypopigmentation & skin of colour
m6w3:018	Pityriasis alba on the cheeks	© DermNet	dermnetnz.org/topics/pigmentation-disorders	Hypopigmentation & skin of colour
m6w3:019	Longstanding prurigo	© DermNet	dermnetnz.org/topics/pigmentation-disorders	Hypopigmentation & skin of colour
m6w3:020	Dermovate cream	Open access image	dermnetnz.org/topics/bleaching-agents	Hypopigmentation & skin of colour
m6w3:021	Stretch marks	© DermNet	dermnetnz.org/topics/bleaching-agents	Hypopigmentation & skin of colour

Module 6 · Week 4 — older skin

47 images · assets/img/m6w4

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w4:W4A-01	Eczema craquelé	© DermNet	dermnetnz.org/topics/asteatotic-eczema	Itch & xerosis in older adults
m6w4:W4A-02	Eczema craquelé	© DermNet	dermnetnz.org/topics/asteatotic-eczema	Itch & xerosis in older adults
m6w4:W4A-03	Eczema craquelé	© DermNet	dermnetnz.org/topics/asteatotic-eczema	Itch & xerosis in older adults
m6w4:W4A-04	Dermatitis from dry skin	© DermNet	dermnetnz.org/topics/dry-skin	Itch & xerosis in older adults
m6w4:W4A-05	Winter itch — excoriations	© DermNet	dermnetnz.org/topics/winter-itch	Itch & xerosis in older adults
m6w4:W4A-06	Excoriations and bruises	© DermNet	dermnetnz.org/topics/pruritus	Itch & xerosis in older adults
m6w4:W4A-07	Venous eczema	© DermNet	dermnetnz.org/topics/venous-eczema	Itch & xerosis in older adults, Red leg: differentials & referral

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w4:W4A-08	Lipodermatosclerosis	© Te Whatu Ora	dermnetnz.org/topics/venous-eczema	Itch & xerosis in older adults, Red leg: differentials & referral
m6w4:W4A-09	Discoid eczema	© DermNet	dermnetnz.org/topics/discoid-eczema	Itch & xerosis in older adults
m6w4:W4A-10	Localised lichen simplex on the lateral shin - a common area for this disorder	© DermNet	dermnetnz.org/topics/lichen-simplex	Itch & xerosis in older adults
m6w4:W4A-11	Lichen simplex on the ankles - accentuated skin markings, scale and pigmentation in skin of colour	© DermNet	dermnetnz.org/topics/lichen-simplex	Itch & xerosis in older adults
m6w4:W4A-12	Lichen simplex	© DermNet	dermnetnz.org/topics/lichen-simplex	Itch & xerosis in older adults
m6w4:W4A-13	Excoriated papules on the arms in nodular prurigo	© DermNet	dermnetnz.org/topics/nodular-prurigo	Itch & xerosis in older adults
m6w4:W4A-14	Brachioradial pruritus — lichenification from scratching	© DermNet	dermnetnz.org/topics/brachioradial-pruritus	Itch & xerosis in older adults
m6w4:W4A-15	Numerous palmar scabies burrows in an elderly lady	© DermNet	dermnetnz.org/topics/scabies	Itch & xerosis in older adults
m6w4:W4A-16	Thick crusted plaques on the thumb and palm due to crusted scabies in an elderly woman	© DermNet	dermnetnz.org/topics/scabies	Itch & xerosis in older adults
m6w4:W4A-17	Bullous pemphigoid	© DermNet	dermnetnz.org/topics/bullous-pemphigoid	Itch & xerosis in older adults
m6w4:W4B-01	Senile purpura	© DermNet	dermnetnz.org/topics/senile-purpura	Skin fragility & dermatoporosis
m6w4:W4B-02	Senile purpura	© DermNet	dermnetnz.org/topics/senile-purpura	Skin fragility & dermatoporosis
m6w4:W4B-03	Senile purpura	© DermNet	dermnetnz.org/topics/senile-purpura	Skin fragility & dermatoporosis
m6w4:W4B-04	Thin skin	© DermNet	dermnetnz.org/topics/ageing-skin	Skin fragility & dermatoporosis
m6w4:W4B-05	Skin fragility	© DermNet	dermnetnz.org/topics/ageing-skin	Skin fragility & dermatoporosis
m6w4:W4B-06	Stellate pseudoscars	© DermNet	dermnetnz.org/topics/ageing-skin	Skin fragility & dermatoporosis
m6w4:W4B-07	Steroid purpura	© DermNet	dermnetnz.org/topics/purpura	Skin fragility & dermatoporosis
m6w4:W4C-01	A pink base with a hyperkeratotic top on the nasal bridge - a common site for actinic keratoses	© Raimo Suhonen	dermnetnz.org/topics/actinic-keratosis	Actinic keratoses (solar keratoses)
m6w4:W4C-02	An actinic keratosis on the nose	© Te Whatu Ora	dermnetnz.org/topics/actinic-keratosis	Actinic keratoses (solar keratoses)
m6w4:W4C-03	Field change over the forehead with multiple superficial actinic keratoses	© DermNet	dermnetnz.org/topics/actinic-keratosis	Actinic keratoses (solar keratoses)

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w4:W4C-04	Hyperkeratotic actinic keratoses on the forehead	© DermNet	dermnetnz.org/topics/actinic-keratosis	Actinic keratoses (solar keratoses)
m6w4:W4C-05	Thicker actinic keratoses on the dorsal hand	© DermNet	dermnetnz.org/topics/actinic-keratosis	Actinic keratoses (solar keratoses)
m6w4:W4C-06	Actinic cheilitis	© DermNet	dermnetnz.org/topics/actinic-cheilitis	Actinic keratoses (solar keratoses)
m6w4:W4C-07	Seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis	Seborrhoeic keratosis
m6w4:W4C-08	Pigmented seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis	Seborrhoeic keratosis
m6w4:W4D-01	Atrophic photoageing	© DermNet	dermnetnz.org/topics/ageing-skin	Photoageing
m6w4:W4D-02	Hypertrophic photoageing	© DermNet	dermnetnz.org/topics/ageing-skin	Photoageing
m6w4:W4D-03	Solar elastosis	© DermNet	dermnetnz.org/topics/solar-elastosis	Photoageing
m6w4:W4D-04	Yellow lesions of solar elastosis on the nose	© DermNet	dermnetnz.org/topics/solar-elastosis	Photoageing
m6w4:W4D-05	Favre-Racouchot syndrome showing elastosis and solar comedones and cysts	© DermNet	dermnetnz.org/topics/solar-elastosis	Photoageing
m6w4:W4D-06	Solar comedones	© DermNet	dermnetnz.org/topics/solar-comedo	Photoageing
m6w4:W4D-07	Cutis rhomboidalis nuchae often seen in association with Favre Racouchot syndrome	© DermNet	dermnetnz.org/topics/solar-comedo	Photoageing
m6w4:W4D-08	Idiopathic guttate hypomelanosis	© DermNet	dermnetnz.org/topics/idiopathic-guttate-hypomelanosis	Photoageing
m6w4:W4D-09	Porcelain-white patch extending peripherally	© DermNet	dermnetnz.org/topics/idiopathic-guttate-hypomelanosis	Photoageing
m6w4:W4D-10	Poikiloderma of Civatte	© DermNet	dermnetnz.org/topics/poikiloderma-of-civatte	Photoageing
m6w4:W4D-11	Poikiloderma of Civatte	© DermNet	dermnetnz.org/topics/poikiloderma-of-civatte	Photoageing
m6w4:W4D-12	Telangiectases	© DermNet	dermnetnz.org/topics/ageing-skin	Photoageing
m6w4:W4D-13	Smoking and its effects on the skin – wrinkling	© DermNet	dermnetnz.org/topics/smoking-and-its-effects-on-the-skin	Photoageing
m6w4:W4D-14	Cherry angiomas	© DermNet	dermnetnz.org/topics/cherry-angioma	Photoageing
m6w4:W4D-15	Sebaceous hyperplasia	© DermNet	dermnetnz.org/topics/sebaceous-hyperplasia	Photoageing

Image credits

Module 6 · Week 5 — cosmetic (ageing)

34 images · `assets/img/m6w5`

CODE	CAPTION	©	SOURCE PAGE	USED ON
`m6w5:W5A-01`	Dynamic lines in smiling teenager	© DermNet	dermnetnz.org/topics/facial-lines-and-wrinkles	Facial ageing & neuromodulators
`m6w5:W5A-02`	Static lines (crow's feet) in 40's	© DermNet	dermnetnz.org/topics/facial-lines-and-wrinkles	Facial ageing & neuromodulators
`m6w5:W5A-03`	Worry lines	© DermNet	dermnetnz.org/topics/facial-lines-and-wrinkles	Facial ageing & neuromodulators
`m6w5:W5A-04`	Extensive perioral lines	© DermNet	dermnetnz.org/topics/facial-lines-and-wrinkles	Facial ageing & neuromodulators
`m6w5:W5A-05`	Deep furrows	© DermNet	dermnetnz.org/topics/facial-lines-and-wrinkles	Facial ageing & neuromodulators
`m6w5:W5A-06`	Hollow-looking eyes	© DermNet	dermnetnz.org/topics/facial-lines-and-wrinkles	Facial ageing & neuromodulators
`m6w5:W5A-07`	Ageing skin	© DermNet	dermnetnz.org/topics/ageing-skin	Facial ageing & neuromodulators
`m6w5:W5A-08`	Smoking and its effects on the skin	© DermNet	dermnetnz.org/topics/smoking-and-its-effects-on-the-skin	Facial ageing & neuromodulators
`m6w5:W5B-01`	Subject frowning prior to botulinum toxin injectons	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5B-02`	Subject relaxed prior to botulinum toxin injections	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5B-03`	Subject attempting to frown after botulinum toxin injections	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5B-04`	Botulinum toxin — crow's feet before injection	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5B-05`	Botulinum toxin — crow's feet after injection	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5B-06`	BOTOX® injections — bruising	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5B-07`	BOTOX® injections — eyelid ptosis	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5B-08`	BOTOX® injections — eyelid ptosis	© DermNet	dermnetnz.org/topics/botulinum-toxin	Facial ageing & neuromodulators
`m6w5:W5C-01`	A solar lentigo on the nose — it is regularly pigmented and macular, with a regular border	© DermNet	dermnetnz.org/topics/solar-lentigo	Solar lentigines
`m6w5:W5C-02`	A solar lentigo on the cheek	© DermNet	dermnetnz.org/topics/solar-lentigo	Solar lentigines

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w5:W5C-03	Multiple solar lentigines on a sun-damaged back	© DermNet	dermnetnz.org/topics/solar-lentigo	Solar lentigines
m6w5:W5C-04	Multiple solar lentigines on the back of the hand	© DermNet	dermnetnz.org/topics/solar-lentigo	Solar lentigines
m6w5:W5C-05	Dermoscopy of a solar lentigo	© DermNet	dermnetnz.org/topics/solar-lentigo	Solar lentigines
m6w5:W5C-06	Ink spot lentigo	© DermNet	dermnetnz.org/topics/lentigo	Solar lentigines
m6w5:W5C-07	Atypical solar lentigo	© Te Whatu Ora	dermnetnz.org/topics/atypical-solar-lentigo	Solar lentigines
m6w5:W5C-08	Lentigo maligna	© DermNet	dermnetnz.org/topics/lentigo-maligna-and-lentigo-maligna-melanoma	Solar lentigines
m6w5:W5C-09	Lentigo maligna	© Te Whatu Ora	dermnetnz.org/topics/lentigo-maligna-and-lentigo-maligna-melanoma	Solar lentigines
m6w5:W5C-10	Freckles	© DermNet	dermnetnz.org/topics/ephelis	Solar lentigines
m6w5:W5C-11	Melasma	© DermNet	dermnetnz.org/topics/melasma	Solar lentigines
m6w5:W5C-12	Seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis	Solar lentigines
m6w5:W5D-01	Effect of IPL on a freckle — close-up	© DermNet	dermnetnz.org/topics/intense-pulsed-light-therapy	Intense pulsed light (IPL)
m6w5:W5D-02	Effect of IPL on a freckle — dermoscopy	© DermNet	dermnetnz.org/topics/intense-pulsed-light-therapy	Intense pulsed light (IPL)
m6w5:W5D-03	Hypopigmentation following IPL	© DermNet	dermnetnz.org/topics/intense-pulsed-light-therapy	Intense pulsed light (IPL)
m6w5:W5D-04	Hypopigmentation following IPL	© DermNet	dermnetnz.org/topics/intense-pulsed-light-therapy	Intense pulsed light (IPL)
m6w5:W5E-01	Effect of bimatoprost solution on eyelashes	© DermNet	dermnetnz.org/topics/bimatoprost	Eyelash enhancement (bimatoprost)
m6w5:W5E-02	Effect of bimatoprost solution on eyelashes	© DermNet	dermnetnz.org/topics/bimatoprost	Eyelash enhancement (bimatoprost)

Module 6 · Week 6 — scars & hair

16 images · assets/img/m6w6

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w6:I01	Ice pick scarring on the cheeks	© DermNet	dermnetnz.org/topics/acne-scarring	Acne scars

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w6:I02	Rolling scars over the cheek	© DermNet	dermnetnz.org/topics/acne-scarring	Acne scars, Cosmetic procedures: peels, needling & lasers
m6w6:I03	Atrophic acne scarring	© DermNet	dermnetnz.org/topics/acne-scarring	Acne scars, Cosmetic procedures: peels, needling & lasers
m6w6:I04	Severe atrophic scarring from back nodulocystic acne	© DermNet	dermnetnz.org/topics/acne-scarring	Acne scars
m6w6:I05	Post inflammatory pigmentation after facial acne in skin of colour	© DermNet	dermnetnz.org/topics/acne-scarring	Acne scars, Cosmetic procedures: peels, needling & lasers
m6w6:I06	Mild atrophic acne scarring on the back	© DermNet	dermnetnz.org/topics/acne-scarring	Acne scars
m6w6:J01	Keloid scar on deltoid/shoulder	© DermNet	dermnetnz.org/topics/keloid-and-hypertrophic-scar	Keloids and hypertrophic scars
m6w6:J02	Post-acne keloid scars	© DermNet	dermnetnz.org/topics/keloid-and-hypertrophic-scar	Cosmetic procedures: peels, needling & lasers, Keloids and hypertrophic scars
m6w6:J03	Earlobe or facial keloid	© DermNet	dermnetnz.org/topics/keloid-and-hypertrophic-scar	Keloids and hypertrophic scars
m6w6:J04	Hypertrophic scar	© DermNet	dermnetnz.org/topics/keloid-and-hypertrophic-scar	Keloids and hypertrophic scars
m6w6:K01	Male pattern hair loss	© DermNet	dermnetnz.org/topics/male-pattern-hair-loss	Androgenetic alopecia (male & female)
m6w6:K02	Male pattern hair loss — vertex	© DermNet	dermnetnz.org/topics/male-pattern-hair-loss	Androgenetic alopecia (male & female)
m6w6:K03	Male pattern hair loss — recession	© DermNet	dermnetnz.org/topics/male-pattern-hair-loss	Androgenetic alopecia (male & female)
m6w6:K04	Female pattern hair loss	© DermNet	dermnetnz.org/topics/female-pattern-hair-loss	Androgenetic alopecia (male & female)
m6w6:K05	Female pattern hair loss — parting	© DermNet	dermnetnz.org/topics/female-pattern-hair-loss	Androgenetic alopecia (male & female)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m6w6:K06	Female pattern hair loss — diffuse thinning	© DermNet	dermnetnz.org/topics/female-pattern-hair-loss	Androgenetic alopecia (male & female)

Module 3 · Week 4 — perineal dermatoses

20 images · assets/img/m3w4

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w4:W4A01	Anogenital wart	© DermNet	dermnetnz.org/topics/sexually-acquired-human-papillomavirus	Perineal dermatoses
m3w4:W4A02	Anogenital wart	© DermNet	dermnetnz.org/topics/sexually-acquired-human-papillomavirus	Perineal dermatoses
m3w4:W4A03	Anogenital wart	© DermNet	dermnetnz.org/topics/sexually-acquired-human-papillomavirus	Perineal dermatoses
m3w4:W4A04	A cryotherapy ice ball on a vulval wart	© Te Whatu Ora	dermnetnz.org/images/genital-wart-images	Perineal dermatoses
m3w4:W4A05	Genital warts (pigmented)	© DermNet	dermnetnz.org/images/genital-wart-images	Perineal dermatoses
m3w4:W4B01	Hidradenitis suppurativa of groin	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa-images	Perineal dermatoses
m3w4:W4B02	Hidradenitis suppurativa of groin	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa-images	Perineal dermatoses
m3w4:W4B03	Hidradenitis suppurativa of groin	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa-images	Perineal dermatoses
m3w4:W4B04	Inflammatory lesion in hidradenitis suppurativa	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa-images	Perineal dermatoses
m3w4:W4B05	An acute painful inflammatory nodule in axillary HS	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa	Perineal dermatoses
m3w4:W4C01	Typical umbilicated papules	© DermNet	dermnetnz.org/topics/molluscum-contagiosum	Perineal dermatoses
m3w4:W4C02	Molluscum contagiosum	© DermNet	dermnetnz.org/topics/molluscum-contagiosum	Perineal dermatoses
m3w4:W4D01	Pearly penile papules	© DermNet	dermnetnz.org/topics/pearly-penile-papules-images	Perineal dermatoses

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w4:W4D02	Pearly penile papules	© DermNet	dermnetnz.org/topics/pearly-penile-papules-images	Perineal dermatoses
m3w4:W4E01	Fordyce spots	© Raimo Suhonen	dermnetnz.org/topics/fordyce-spots	Perineal dermatoses
m3w4:W4E02	Fordyce spots	© DermNet	dermnetnz.org/topics/fordyce-spots	Perineal dermatoses
m3w4:W4F01	Candida infection	© DermNet	dermnetnz.org/topics/candida	Perineal dermatoses
m3w4:W4F02	Candida infection (fold)	© DermNet	dermnetnz.org/topics/candida	Perineal dermatoses
m3w4:W4G01	Lichen sclerosus	© DermNet	dermnetnz.org/topics/lichen-sclerosus	Perineal dermatoses
m3w4:W4G02	Lichen sclerosus	© DermNet	dermnetnz.org/topics/lichen-sclerosus	Perineal dermatoses

Module 4 · Week 3 — contact dermatitis

21 images · assets/img/m4w3

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w3:A01	Allergic contact dermatitis (rubber accelerator example)	© DermNet	dermnetnz.org/topics/allergic-contact-dermatitis	Contact dermatitis
m4w3:A02	Acute allergic contact dermatitis	© DermNet	dermnetnz.org/topics/allergic-contact-dermatitis	Contact dermatitis
m4w3:A03	Irritant contact dermatitis of the hands	© DermNet	dermnetnz.org/topics/irritant-contact-dermatitis	Contact dermatitis
m4w3:A04	Chronic irritant hand dermatitis	© DermNet	dermnetnz.org/topics/irritant-contact-dermatitis	Contact dermatitis
m4w3:B01	Facial allergic contact dermatitis	© DermNet	dermnetnz.org/topics/allergic-contact-dermatitis	Contact dermatitis
m4w3:B02	Eyelid contact dermatitis	© DermNet	dermnetnz.org/topics/eyelid-contact-dermatitis	Contact dermatitis
m4w3:B03	Fragrance-related cosmetic contact reaction	© DermNet	dermnetnz.org/topics/contact-reactions-to-cosmetics	Contact dermatitis
m4w3:C01	Airborne contact dermatitis affecting eyelids	© DermNet	dermnetnz.org/topics/airborne-contact-dermatitis	Contact dermatitis
m4w3:C02	Fragrance allergy neck	© DermNet	dermnetnz.org/topics/fragrance-allergy	Contact dermatitis
m4w3:C03	Airborne dermatitis due to epoxy allergy	© DermNet	dermnetnz.org/topics/airborne-contact-dermatitis	Contact dermatitis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w3:D01	Red tattoo pigment (cinnabar) hypersensitivity reaction	© DermNet	dermnetnz.org/topics/tattoo-associated-skin-reactions	Contact dermatitis
m4w3:D02	Tattoo-associated eczematous reaction	© DermNet	dermnetnz.org/topics/tattoo-associated-skin-reactions	Contact dermatitis
m4w3:D03	Black henna tattoo reaction (PPD)	© DermNet	dermnetnz.org/topics/black-henna-tattoo-reaction	Contact dermatitis
m4w3:E01	Hand dermatitis (occupational pattern)	© DermNet	dermnetnz.org/topics/hand-dermatitis	Contact dermatitis
m4w3:E02	Palmar hand dermatitis	© DermNet	dermnetnz.org/topics/hand-dermatitis	Contact dermatitis
m4w3:E03	Allergic contact dermatitis related to rubber/latex gloves	© DermNet	dermnetnz.org/topics/latex-allergy	Contact dermatitis
m4w3:F01	Patch test panels applied to back	© DermNet	dermnetnz.org/topics/patch-tests	Contact dermatitis
m4w3:F02	Positive patch test to rubber accelerator	© DermNet	dermnetnz.org/topics/patch-tests	Contact dermatitis
m4w3:F03	++ reaction to fragrance mix I	© DermNet	dermnetnz.org/topics/fragrance-allergy	Contact dermatitis
m4w3:F04	Patch test equipment / chambers	© DermNet	dermnetnz.org/topics/patch-tests	Contact dermatitis
m4w3:G01	Allergic contact dermatitis to adhesive plaster	© DermNet	dermnetnz.org/topics/allergic-contact-dermatitis	Contact dermatitis

Module 4 · Week 5 — urticaria & angioedema

29 images · `assets/img/m4w5`

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w5:A01	Urticaria	© DermNet	dermnetnz.org/topics/acute-urticaria	Urticaria & angioedema
m4w5:A02	Acute urticaria	© Te Whatu Ora	dermnetnz.org/topics/acute-urticaria	Urticaria & angioedema
m4w5:A03	Urticaria	© Dr Richard Ashton	dermnetnz.org/topics/acute-urticaria	Urticaria & angioedema
m4w5:A04	Spontaneous urticaria	© Raimo Suhonen	dermnetnz.org/topics/chronic-spontaneous-urticaria	Urticaria & angioedema
m4w5:A05	Spontaneous urticaria	© Te Whatu Ora	dermnetnz.org/topics/chronic-spontaneous-urticaria	Urticaria & angioedema
m4w5:A06	Urticaria in skin of colour	© Te Whatu Ora	dermnetnz.org/topics/chronic-urticaria	Urticaria & angioedema
m4w5:A07	Wheal in dark skin	© DermNet	dermnetnz.org/topics/urticaria-an-overview	Urticaria & angioedema

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w5:B01	Dermographism: arm	© Dr Mashihul Hossain	dermnetnz.org/topics/dermographism	Urticaria & angioedema
m4w5:B02	Dermographism	© Raimo Suhonen	dermnetnz.org/topics/dermographism	Urticaria & angioedema
m4w5:B03	Cold urticaria	© DermNet	dermnetnz.org/topics/cold-urticaria	Urticaria & angioedema
m4w5:B04	Enlarging weal 5 minutes later	© DermNet	dermnetnz.org/topics/cold-urticaria	Urticaria & angioedema
m4w5:B05	The transient 5 mm wheals of cholinergic urticaria triggered by exercise	© DermNet	dermnetnz.org/topics/cholinergic-urticaria	Urticaria & angioedema
m4w5:B06	Pressure urticaria	© DermNet	dermnetnz.org/topics/pressure-urticaria	Urticaria & angioedema
m4w5:B07	Solar urticaria	© DermNet	dermnetnz.org/topics/solar-urticaria	Urticaria & angioedema
m4w5:B08	Latex reaction	© DermNet	dermnetnz.org/topics/contact-urticaria	Urticaria & angioedema
m4w5:C01	Urticaria pigmentosa in a child	© DermNet	dermnetnz.org/topics/maculopapular-cutaneous-mastocytosis	Urticaria & angioedema
m4w5:C02	Cutaneous mastocytosis in an infant - yellow-brown macules over the chest and back. The chest lesion has urticated 5 minutes after rubbing (Darier sign)	© DermNet	dermnetnz.org/topics/mastocytosis	Urticaria & angioedema
m4w5:C03	Urticaria pigmentosa: brown spots that urticate	© DermNet	dermnetnz.org/topics/maculopapular-cutaneous-mastocytosis	Urticaria & angioedema
m4w5:C04	Maculopapular cutaneous mastocytosis (urticaria pigmentosa); red-brown maculopapular lesions on the leg	© DermNet	dermnetnz.org/topics/mastocytosis	Urticaria & angioedema
m4w5:C05	Darier sign in mastocytoma	© DermNet	dermnetnz.org/topics/mastocytoma	Urticaria & angioedema
m4w5:D01	Severe lip angioedema	© DermNet	dermnetnz.org/topics/angioedema	Urticaria & angioedema
m4w5:D02	Lid angioedema - there was typical urticaria elsewhere on the body	© Raimo Suhonen	dermnetnz.org/topics/angioedema	Urticaria & angioedema
m4w5:D03	Angioedema of the right hand	© DermNet	dermnetnz.org/topics/angioedema	Urticaria & angioedema
m4w5:D04	Severe lingual angioedema in acquired C1 esterase inhibitor deficiency	© DermNet	dermnetnz.org/topics/anaphylaxis	Urticaria & angioedema
m4w5:D05	Clinical suspicion for hereditary angioedema	© DermNet	dermnetnz.org/topics/hereditary-angioedema	Urticaria & angioedema

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w5:E01	Urticarial vasculitis	© DermNet	dermnetnz.org/topics/urticarial-vasculitis	Urticaria & angioedema
m4w5:E02	Urticarial vasculitis	© DermNet	dermnetnz.org/topics/urticarial-vasculitis	Urticaria & angioedema
m4w5:E03	Urticarial vasculitis	© DermNet	dermnetnz.org/topics/urticarial-vasculitis	Urticaria & angioedema
m4w5:E04	Urticarial vasculitis in skin of colour	© DermNet	dermnetnz.org/topics/urticarial-vasculitis	Urticaria & angioedema

Module 3 · Week 3 — scalp & hair

47 images · assets/img/m3w3

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w3:A01	A single patch of alopecia areata	© DermNet	dermnetnz.org/topics/alopecia-areata	Alopecia areata
m3w3:A02	A single patch of alopecia areata on the vertex	© DermNet	dermnetnz.org/images/alopecia-areata-images	Alopecia areata
m3w3:A03	Several patches of alopecia areata on the scalp	© DermNet	dermnetnz.org/images/alopecia-areata-images	Alopecia areata
m3w3:A04	Exclamation mark hairs at edge of patch	© DermNet	dermnetnz.org/topics/alopecia-areata-in-children	Alopecia areata
m3w3:A05	Dermoscopic image of alopecia areata presenting with exclamation mark hairs	© DermNet	dermnetnz.org/topics/alopecia-areata	Alopecia areata
m3w3:A06	Yellow dots, black dots, and short hairs	© DermNet	dermnetnz.org/topics/alopecia-areata-in-children	Alopecia areata
m3w3:A07	Alopecia areata with fine hypopigmented regrowing hairs	© DermNet	dermnetnz.org/images/alopecia-areata-images	Alopecia areata
m3w3:A08	Patch alopecia areata with poliosis in a 14 year old	© DermNet	dermnetnz.org/images/alopecia-areata-images	Alopecia areata
m3w3:A09	Patchy alopecia areata in a child	© DermNet	dermnetnz.org/topics/alopecia-areata-in-children	Alopecia areata
m3w3:A10	Eyebrow loss in alopecia areata	© DermNet	dermnetnz.org/topics/alopecia-areata-in-children	Alopecia areata
m3w3:A11	Alopecia areata	© DermNet	dermnetnz.org/topics/madarosis	Alopecia areata
m3w3:A12	Alopecia areata: beard	© DermNet	dermnetnz.org/images/alopecia-areata-images	Alopecia areata
m3w3:A13	Ophiasic pattern alopecia areata	© DermNet	dermnetnz.org/topics/alopecia-areata	Alopecia areata

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m3w3:A14	Extensive patchy alopecia areata	© DermNet	dermnetnz.org/topics/alopecia-areata	Alopecia areata
m3w3:A15	Alopecia totalis	© DermNet	dermnetnz.org/topics/alopecia-areata	Alopecia areata
m3w3:A16	Alopecia areata nail pitting	© DermNet	dermnetnz.org/topics/alopecia-areata-in-children	Alopecia areata
m3w3:B01	Trichotillomania	© DermNet	dermnetnz.org/topics/trichotillomania	Trichotillomania
m3w3:B02	Trichotillomania	© DermNet	dermnetnz.org/topics/trichotillomania	Trichotillomania
m3w3:B03	Trichotillomania	© DermNet	dermnetnz.org/topics/trichotillomania	Trichotillomania
m3w3:B04	Trichotillomania	© DermNet	dermnetnz.org/topics/trichotillomania	Trichotillomania
m3w3:B05	Trichotillomania	© DermNet	dermnetnz.org/topics/trichotillomania	Trichotillomania
m3w3:B06	Trichotillomania	© DermNet	dermnetnz.org/topics/trichotillomania	Trichotillomania
m3w3:B07	Trichotillomania showing black dots; follicular openings with remnants of pigmented hairs broken at scalp level	© DermNet	dermnetnz.org/topics/trichoscopy	Trichotillomania
m3w3:B08	Trichotillomania showing V-sign that represents two hair shafts emerging from the same follicle and cut at the same level	© DermNet	dermnetnz.org/topics/trichoscopy	Trichotillomania
m3w3:C01	Erythema, scaling and alopecia in scalp discoid lupus erythematosus	© DermNet	dermnetnz.org/images/discoid-lupus-erythematosus-images	Scarring alopecia overview
m3w3:C02	Scarring alopecia due to discoid lupus erythematosus	© DermNet	dermnetnz.org/topics/discoid-lupus-erythematosus	Scarring alopecia overview
m3w3:C03	Scarring alopecia with follicular plugging in discoid lupus erythematosus	© DermNet	dermnetnz.org/images/discoid-lupus-erythematosus-images	Scarring alopecia overview
m3w3:C04	Scarring alopecia with loss of follicular openings, sclerosis and telangiectatic vessels due to discoid lupus erythematosus	© DermNet	dermnetnz.org/topics/trichoscopy	Scarring alopecia overview
m3w3:C05	Lichen planopilaris	© DermNet	dermnetnz.org/topics/lichen-planopilaris	Scarring alopecia overview
m3w3:C06	Lichen planopilaris	© DermNet	dermnetnz.org/topics/hair-loss	Scarring alopecia overview
m3w3:C07	Dermoscopic image showing perifollicular scales (red circles, peritubular cast (black arrows) and blue grey dots (yellow circles)	© DermNet	dermnetnz.org/topics/lichen-planopilaris	Scarring alopecia overview

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w3:C08	Frontal fibrosing alopecia	© DermNet	dermnetnz.org/topics/frontal-fibrosing-alopecia	Scarring alopecia overview
m3w3:C09	Note the perifollicular inflammation	© DermNet	dermnetnz.org/topics/frontal-fibrosing-alopecia	Scarring alopecia overview
m3w3:C10	Frontal fibrosing alopecia	© DermNet	dermnetnz.org/topics/madarosis	Scarring alopecia overview
m3w3:C11	Tufted hairs and perifollicular crusting	© DermNet	dermnetnz.org/topics/folliculitis-decalvans	Scarring alopecia overview
m3w3:C12	Central centrifugal cicatricial alopecia	© DermNet	dermnetnz.org/topics/central-centrifugal-cicatricial-alopecia	Scarring alopecia overview
m3w3:D01	Grey patch scalp ringworm	© DermNet	dermnetnz.org/topics/tinea-capitis	Telogen effluvium
m3w3:D02	Black dot scalp ringworm	© DermNet	dermnetnz.org/topics/tinea-capitis	Telogen effluvium
m3w3:D03	Tinea capitis	© DermNet	dermnetnz.org/topics/tinea-capitis	Telogen effluvium
m3w3:D04	Pattern balding	© DermNet	dermnetnz.org/topics/male-pattern-hair-loss	Telogen effluvium
m3w3:D05	Pattern balding	© DermNet	dermnetnz.org/topics/male-pattern-hair-loss	Telogen effluvium
m3w3:D06	Female pattern balding	© DermNet	dermnetnz.org/topics/hair-loss	Telogen effluvium
m3w3:D07	Severe female pattern balding	© DermNet	dermnetnz.org/topics/hair-loss	Telogen effluvium
m3w3:D08	Female pattern hair loss	© DermNet	dermnetnz.org/topics/female-pattern-hair-loss	Telogen effluvium
m3w3:D09	Telogen effluvium	© DermNet	dermnetnz.org/topics/telogen-effluvium	Telogen effluvium
m3w3:D10	Hair shedding	© DermNet	dermnetnz.org/topics/telogen-effluvium	Telogen effluvium
m3w3:D11	Traction alopecia	© DermNet	dermnetnz.org/topics/traction-alopecia	Telogen effluvium

Module 1 · Week 1 — language of dermatology

48 images · assets/img/m1w1

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w1:A01	Freckles	© DermNet	dermnetnz.org/topics/ephelis	Describing skin lesions: the language of dermatology

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w1:A02	Symmetrical wrist vitiligo - a common location	© DermNet	dermnetnz.org/topics/vitiligo	Describing skin lesions: the language of dermatology
m1w1:A03	Café-au-lait macule	© DermNet	dermnetnz.org/topics/cafe-au-lait-macule	Describing skin lesions: the language of dermatology
m1w1:A04	Flat topped, shiny, polygonal papules of lichen planus on the volar wrist	© DermNet	dermnetnz.org/topics/lichen-planus	Describing skin lesions: the language of dermatology, Lichen planus
m1w1:A05	Typical umbilicated papules	© DermNet	dermnetnz.org/topics/molluscum-contagiosum	Describing skin lesions: the language of dermatology
m1w1:A06	Well demarcated plaques with silvery scale in chronic plaque psoriasis	© DermNet	dermnetnz.org/topics/psoriasis	Describing skin lesions: the language of dermatology
m1w1:A07	Nodular basal cell carcinoma	© DermNet	dermnetnz.org/topics/basal-cell-carcinoma	Describing skin lesions: the language of dermatology
m1w1:A08	Nodular prurigo on the arms	© DermNet	dermnetnz.org/topics/nodular-prurigo	Describing skin lesions: the language of dermatology
m1w1:A09	Eczema herpeticum	© DermNet	dermnetnz.org/topics/eczema-herpeticum	Describing skin lesions: the language of dermatology
m1w1:A10	Chickenpox	© Te Whatu Ora	dermnetnz.org/topics/chickenpox	Describing skin lesions: the language of dermatology
m1w1:A11	Bullous pemphigoid (gallery image, no caption on page)	© DermNet	dermnetnz.org/topics/bullous-pemphigoid	Bullous pemphigoid (incl. MMP & pemphigoid gestationis), Describing skin lesions: the language of dermatology

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m1w1:A12	Blisters of bullous pemphigoid on foot, skin of colour	© DermNet	dermnetnz.org/topics/bullous-pemphigoid	Bullous pemphigoid (incl. MMP & pemphigoid gestationis), Describing skin lesions: the language of dermatology
m1w1:A13	Superficial bacterial folliculitis	© Raimo Suhonen	dermnetnz.org/topics/folliculitis	Describing skin lesions: the language of dermatology
m1w1:A14	Pustules on the palm due to PPP	© DermNet	dermnetnz.org/topics/palmoplantar-pustulosis	Describing skin lesions: the language of dermatology
m1w1:A15	Urticaria	© DermNet	dermnetnz.org/topics/urticaria-an-overview	Describing skin lesions: the language of dermatology
m1w1:A16	Epidermoid cyst with an obvious punctum	© Raimo Suhonen	dermnetnz.org/topics/epidermoid-cyst	Describing skin lesions: the language of dermatology
m1w1:B01	Thick silvery scale over the elbow	© DermNet	dermnetnz.org/topics/psoriasis	Describing skin lesions: the language of dermatology
m1w1:B02	Oval scaly patches in pityriasis rosea on the trunk	© DermNet	dermnetnz.org/topics/pityriasis-rosea	Describing skin lesions: the language of dermatology
m1w1:B03	Perioral honey-coloured crusts in impetigo	© DermNet	dermnetnz.org/topics/impetigo	Describing skin lesions: the language of dermatology
m1w1:B04	An intact blister and erosions in bullous impetigo	© DermNet	dermnetnz.org/topics/impetigo	Describing skin lesions: the language of dermatology
m1w1:B05	Pemphigus vulgaris	© DermNet	dermnetnz.org/topics/pemphigus-vulgaris	Describing skin lesions: the language of dermatology
m1w1:B06	Leg ulcer	© DermNet	dermnetnz.org/topics/venous-insufficiency	Describing skin lesions: the language of dermatology

Image credits

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m1w1:B07	Juvenile plantar dermatosis	© DermNet	dermnetnz.org/topics/juvenile-plantar-dermatosis	Describing skin lesions: the language of dermatology
m1w1:B08	Bilateral angular cheilitis with grey maceration at the angles of the mouth	© DermNet	dermnetnz.org/topics/angular-cheilitis	Describing skin lesions: the language of dermatology
m1w1:B09	Localised lichen simplex on the lateral shin - a common area for this disorder	© DermNet	dermnetnz.org/topics/lichen-simplex	Describing skin lesions: the language of dermatology
m1w1:B10	Skin thinning	© DermNet	dermnetnz.org/topics/topical-steroid	Describing skin lesions: the language of dermatology
m1w1:B11	Stretch marks	© DermNet	dermnetnz.org/topics/topical-steroid	Describing skin lesions: the language of dermatology
m1w1:B12	Excoriated papules on the arms in nodular prurigo	© DermNet	dermnetnz.org/topics/nodular-prurigo	Describing skin lesions: the language of dermatology
m1w1:B13	A hypertrophic scar on the upper back - the thickening does not extend beyond the confines of the injury	© DermNet	dermnetnz.org/topics/keloid-and-hypertrophic-scar	Describing skin lesions: the language of dermatology
m1w1:B14	A large 'dumbbell' keloid on the earlobe	© Dr Richard Ashton	dermnetnz.org/topics/keloid-and-hypertrophic-scar	Describing skin lesions: the language of dermatology
m1w1:C01	Granuloma annulare	© DermNet	dermnetnz.org/topics/granuloma-annulare	Annular (ring-shaped) rashes: differential, Describing skin lesions: the language of dermatology
m1w1:C02	Cutaneous adverse reaction to anticonvulsant, erythema multiforme target lesions	© DermNet	dermnetnz.org/topics/erythema-multiforme	Describing skin lesions: the language of dermatology
m1w1:C03	Papular lichen planus with a linear Koebner lesion due to a scratch	© DermNet	dermnetnz.org/topics/lichen-planus	Describing skin lesions: the language of dermatology, Lichen planus

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w1:C04	Herpes zoster	© DermNet	dermnetnz.org/topics/herpes-zoster	Describing skin lesions: the language of dermatology, Herpes zoster (shingles)
m1w1:C05	Well demarcated salmon pink erythema which extends to the apex of the skin fold due to flexural psoriasis	© Dr Ph Abimelec - dermatologue	dermnetnz.org/topics/psoriasis	Describing skin lesions: the language of dermatology
m1w1:C06	Dry atopic dermatitis in an extensor pattern	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Describing skin lesions: the language of dermatology
m1w1:D01	Skin structure	Open access image	dermnetnz.org/topics/the-structure-of-normal-skin	Skin layers, Fitzpatrick skin types & dermoscopy
m1w1:E01	Phototype I	© DermNet	dermnetnz.org/topics/skin-phototype	Skin layers, Fitzpatrick skin types & dermoscopy
m1w1:E02	Phototype IV	© DermNet	dermnetnz.org/topics/skin-phototype	Skin layers, Fitzpatrick skin types & dermoscopy
m1w1:E03	Phototype VI	© DermNet	dermnetnz.org/topics/skin-phototype	Skin layers, Fitzpatrick skin types & dermoscopy
m1w1:E04	Post-inflammatory hypopigmentation as guttate psoriasis is resolving	© DermNet	dermnetnz.org/topics/plaque-psoriasis	Skin layers, Fitzpatrick skin types & dermoscopy
m1w1:F01	Dermoscopy of a solar lentigo	© DermNet	dermnetnz.org/topics/solar-lentigo	Skin layers, Fitzpatrick skin types & dermoscopy
m1w1:F02	Molluscum contagiosum	© DermNet	dermnetnz.org/topics/molluscum-contagiosum	Skin layers, Fitzpatrick skin types & dermoscopy
m1w1:G01	Flexural atopic dermatitis in childhood	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Describing skin lesions: the language of dermatology
m1w1:G02	Chronic flexural atopic dermatitis	© Te Whatu Ora	dermnetnz.org/topics/atopic-dermatitis	Describing skin lesions: the language of dermatology

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w1:G03	Excoriated lichenified atopic dermatitis on the lower back in an adult	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Describing skin lesions: the language of dermatology
m1w1:G04	Lichen simplex on the ankles - accentuated skin markings, scale and pigmentation in skin of colour	© DermNet	dermnetnz.org/topics/lichen-simplex	Describing skin lesions: the language of dermatology
m1w1:G05	Eczema herpeticum	© DermNet	dermnetnz.org/topics/eczema-herpeticum	Describing skin lesions: the language of dermatology

Module 3 · Week 1 — rosacea

16 images · assets/img/m3w1

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w1:R01	Mild erythematotelangiectatic rosacea	© Te Whatu Ora	dermnetnz.org/images/rosacea-images	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R02	Erythematotelangiectatic rosacea on the nose cheeks and chin	© DermNet	dermnetnz.org/images/rosacea-images	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R03	Rosacea causing flushing	© DermNet	dermnetnz.org/topics/flushing	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R04	Papular rosacea on the cheeks	© DermNet	dermnetnz.org/images/rosacea-images	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R05	Papulopustular rosacea on the cheek	© Te Whatu Ora	dermnetnz.org/images/rosacea-images	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R06	Severe papulopustular rosacea with a rhinophyma	© Dr Lawrence Meyerson	dermnetnz.org/images/rosacea-images	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R07	Grade 2	© DermNet	dermnetnz.org/topics/rhinophyma	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R08	Papulopustular and ocular rosacea	© DermNet	dermnetnz.org/images/rosacea-images	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R09	Ocular rosacea	© DermNet	dermnetnz.org/topics/ocular-rosacea	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R10	Granulomatous rosacea and rosacea eye disease	© Te Whatu Ora	dermnetnz.org/images/rosacea-images	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R11	Demodicosis	© DermNet	dermnetnz.org/topics/demodex	Rosacea (incl. Demodex & steroid rosacea)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w1:R12	Rosacea due to demodicosis	© Te Whatu Ora	dermnetnz.org/topics/demodex	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R13	Dermoscopy of demodicosis	© DermNet	dermnetnz.org/topics/demodex	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R14	Steroid induced rosacea on the forehead	© DermNet	dermnetnz.org/topics/rosacea	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R15	Swelling and papulopustules in steroid induced rosacea	© Te Whatu Ora	dermnetnz.org/topics/rosacea	Rosacea (incl. Demodex & steroid rosacea)
m3w1:R16	Perioral erythematous papules sparing the immediate perivermillion skin in perioral dermatitis	© DermNet	dermnetnz.org/topics/periorificial-dermatitis	Rosacea (incl. Demodex & steroid rosacea)

Module 3 · Week 2 — nails

18 images · assets/img/m3w2

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w2:A01	Candida paronychia	© DermNet	dermnetnz.org/cme/fungal-infections/candida-infection	Paronychia (acute and chronic)
m3w2:A02	Chronic paronychia	© DermNet	dermnetnz.org/topics/paronychia	Paronychia (acute and chronic)
m3w2:A03	Paronychia nail-fold dystrophy	© DermNet	dermnetnz.org/topics/paronychia	Paronychia (acute and chronic)
m3w2:A04	Candidal onycholysis	© DermNet	dermnetnz.org/cme/fungal-infections/candida-infection	Fungal nail infection & nail discolouration, Paronychia (acute and chronic)
m3w2:A05	Candida nail infection	© DermNet	dermnetnz.org/topics/candida	Fungal nail infection & nail discolouration, Paronychia (acute and chronic)
m3w2:B01	Distal lateral onychomycosis	© DermNet	dermnetnz.org/topics/fungal-nail-infections	Fungal nail infection & nail discolouration
m3w2:B02	Onychomycosis toenails	© DermNet	dermnetnz.org/topics/fungal-nail-infections	Fungal nail infection & nail discolouration
m3w2:B03	White superficial onychomycosis	© DermNet	dermnetnz.org/topics/fungal-nail-infections	Fungal nail infection & nail discolouration
m3w2:B04	Subungual haematoma	© DermNet	dermnetnz.org/topics/subungual-haemorrhage	Fungal nail infection & nail discolouration

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w2:C01	Benign longitudinal melanonychia	© DermNet	dermnetnz.org/topics/melanonychia	Melanonychia & nail melanoma
m3w2:C02	Longitudinal melanonychia	© DermNet	dermnetnz.org/topics/melanonychia	Melanonychia & nail melanoma
m3w2:C03	Melanoma of the nail unit	© DermNet	dermnetnz.org/topics/melanoma-of-the-nail-unit	Melanonychia & nail melanoma
m3w2:C04	Nail melanoma periungual pigment	© DermNet	dermnetnz.org/topics/melanoma-of-the-nail-unit	Melanonychia & nail melanoma
m3w2:C05	Nail melanoma	© DermNet	dermnetnz.org/topics/nail-terminology	Melanonychia & nail melanoma
m3w2:D01	Psoriatic nail pitting	© DermNet	dermnetnz.org/topics/nail-psoriasis	Nail signs: psoriatic nails, Beau's lines & clubbing, Psoriasis
m3w2:D02	Nail psoriasis	© DermNet	dermnetnz.org/topics/nail-psoriasis	Nail signs: psoriatic nails, Beau's lines & clubbing, Psoriasis
m3w2:D03	Beau's lines	© DermNet	dermnetnz.org/topics/nail-terminology	Nail signs: psoriatic nails, Beau's lines & clubbing
m3w2:D04	Finger clubbing	© DermNet	dermnetnz.org/topics/skin-signs-of-respiratory-disease	Nail signs: psoriatic nails, Beau's lines & clubbing

Module 3 · Week 5 — flexural dermatoses

18 images · assets/img/m3w5

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w5:A01	Nodules and sinus formation in axillary HS	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa	Hidradenitis suppurativa
m3w5:A02	An acute painful inflammatory nodule in axillary HS	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa	Hidradenitis suppurativa
m3w5:A03	Severe axillary HS with scars and sinus formation	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa	Hidradenitis suppurativa
m3w5:A04	Marked comedones in axillary HS	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa	Hidradenitis suppurativa
m3w5:A05	Mild axillary HS showing scars and post-inflammatory erythema	© DermNet	dermnetnz.org/topics/hidradenitis-suppurativa	Hidradenitis suppurativa

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w5:B01	Intertrigo (inframammary fold)	© DermNet	dermnetnz.org/topics/candida	Intertrigo & the flexural differential
m3w5:B02	Athlete's foot / toe-web maceration (moist fold intertrigo)	© DermNet	dermnetnz.org/topics/intertrigo	Intertrigo & the flexural differential
m3w5:B03	Contact irritant dermatitis due to urine. Napkin dermatitis in adult	© DermNet	dermnetnz.org/topics/intertrigo	Intertrigo & the flexural differential
m3w5:C01	Flexural psoriasis	© DermNet	dermnetnz.org/topics/flexural-psoriasis	Psoriasis
m3w5:C02	Flexural psoriasis	© DermNet	dermnetnz.org/topics/flexural-psoriasis	Psoriasis
m3w5:C03	Flexural psoriasis	© DermNet	dermnetnz.org/topics/flexural-psoriasis-images	Psoriasis
m3w5:C04	Flexural psoriasis	© DermNet	dermnetnz.org/topics/flexural-psoriasis-images	Psoriasis
m3w5:D01	Intertrigo due to candida infection	© DermNet	dermnetnz.org/topics/intertrigo	Intertrigo & the flexural differential
m3w5:D02	Raised border and central clearing	© DermNet	dermnetnz.org/topics/tinea-cruris	Intertrigo & the flexural differential
m3w5:D03	Axillary erythrasma	© DermNet	dermnetnz.org/topics/intertrigo	Intertrigo & the flexural differential
m3w5:D04	Wood's light fluorescence	© DermNet	dermnetnz.org/topics/erythrasma	Intertrigo & the flexural differential
m3w5:D05	Flexural seborrhoeic dermatitis in the axilla	© DermNet	dermnetnz.org/topics/seborrhoeic-dermatitis	Intertrigo & the flexural differential
m3w5:D06	Hailey-Hailey disease in axilla	© DermNet	dermnetnz.org/topics/hailey-hailey-disease	Intertrigo & the flexural differential

Module 3 · Week 6 — palmar & plantar dermatoses

31 images · `assets/img/m3w6`

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w6:A01	Exfoliative keratolysis	© DermNet	dermnetnz.org/topics/keratolysis-exfoliativa	Keratolysis exfoliativa
m3w6:A02	Exfoliative keratolysis	© DermNet	dermnetnz.org/topics/keratolysis-exfoliativa	Keratolysis exfoliativa
m3w6:A03	Exfoliative keratolysis	© DermNet	dermnetnz.org/topics/keratolysis-exfoliativa	Keratolysis exfoliativa

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w6:B01	Palmar dermatitis	© DermNet	dermnetnz.org/topics/hand-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B02	Hyperkeratotic palmar eczema	© DermNet / Te Whatu Ora Health New Zealand (watermarked)	dermnetnz.org/topics/hand-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B03	Diffuse scale and redness on the palms due to eczema	© DermNet / Te Whatu Ora Health New Zealand (watermarked)	dermnetnz.org/topics/hand-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B04	Hand dermatitis due to wet work	© DermNet	dermnetnz.org/topics/irritant-contact-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B06	Postinflammatory hyperpigmentation following irritant hand dermatitis	© DermNet	dermnetnz.org/topics/irritant-contact-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B08	Thickening, dryness, and scale on the dorsal fingers	© DermNet / Te Whatu Ora Health New Zealand (watermarked)	dermnetnz.org/topics/hand-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B09	Reading the patch tests - there are several positives	© DermNet	dermnetnz.org/topics/hand-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B10	A positive patch test to a piece of a suspect rubber glove 48 hrs after application on the skin	© DermNet	dermnetnz.org/topics/allergic-contact-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B11	Infected hand dermatitis	© DermNet	dermnetnz.org/topics/hand-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:B12	Hyperlinear palm	© DermNet	dermnetnz.org/topics/atopic-dermatitis	Hand eczema (chronic, occupational & pompholyx)
m3w6:C01	Discrete vesicles and blisters on the palmar skin	© DermNet	dermnetnz.org/topics/dyshidrotic-eczema	Hand eczema (chronic, occupational & pompholyx)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w6:C02	Vesicles and blisters on the fingers in vesicular eczema	© DermNet	dermnetnz.org/topics/dyshidrotic-eczema	Hand eczema (chronic, occupational & pompholyx)
m3w6:C03	Extensive vesiculation and blistering on the soles - vesicular eczema is usually symmetrical	© DermNet	dermnetnz.org/topics/dyshidrotic-eczema	Hand eczema (chronic, occupational & pompholyx)
m3w6:D01	Pustules and rusty dots in a scaly plaque typical of PPP	© DermNet	dermnetnz.org/topics/palmoplantar-pustulosis	Palmoplantar pustulosis & palmoplantar psoriasis
m3w6:D02	Pustules on the palm due to PPP	© DermNet	dermnetnz.org/topics/palmoplantar-pustulosis	Palmoplantar pustulosis & palmoplantar psoriasis
m3w6:D03	Typical lesions of PPP on the soles - redness, scale, pustules, and rusty red dots	© DermNet	dermnetnz.org/topics/palmoplantar-pustulosis	Palmoplantar pustulosis & palmoplantar psoriasis
m3w6:E01	Palmar psoriasis	© DermNet	dermnetnz.org/topics/psoriasis-of-the-palms-and-soles	Palmoplantar pustulosis & palmoplantar psoriasis, Psoriasis
m3w6:E02	Plantar psoriasis	© DermNet	dermnetnz.org/topics/psoriasis-of-the-palms-and-soles	Palmoplantar pustulosis & palmoplantar psoriasis, Psoriasis
m3w6:E03	Palmoplantar psoriasis	© DermNet	dermnetnz.org/topics/psoriasis-of-the-palms-and-soles	Palmoplantar pustulosis & palmoplantar psoriasis, Psoriasis
m3w6:F01	Tinea manuum	© DermNet	dermnetnz.org/topics/tinea-manuum	Hand eczema (chronic, occupational & pompholyx), Tinea: corporis, cruris & pedis
m3w6:F02	Tinea manuum	© DermNet	dermnetnz.org/topics/tinea-manuum	Hand eczema (chronic, occupational & pompholyx), Tinea: corporis, cruris & pedis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m3w6:F03	Tinea manuum	© DermNet	dermnetnz.org/topics/tinea-manuum	Hand eczema (chronic, occupational & pompholyx), Tinea: corporis, cruris & pedis
m3w6:F04	Moccasin tinea pedis	© DermNet	dermnetnz.org/topics/tinea-pedis	Hand eczema (chronic, occupational & pompholyx), Tinea: corporis, cruris & pedis
m3w6:F05	Tinea pedis	© DermNet	dermnetnz.org/topics/tinea-pedis	Hand eczema (chronic, occupational & pompholyx), Tinea: corporis, cruris & pedis
m3w6:G01	Erythema multiforme	© DermNet	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme
m3w6:G02	Erythema multiforme minor	© DermNet	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme
m3w6:G03	Erythema multiforme	© DermNet / Te Whatu Ora Health New Zealand (watermarked)	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme
m3w6:G04	Cutaneous adverse reaction to anticonvulsant, erythema multiforme target lesions	© DermNet	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme

Module 4 · Week 1 — inflammatory rashes

25 images · assets/img/m4w1

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w1:A01	Erythema multiforme target lesions	© DermNet	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme
m4w1:A02	Erythema multiforme target lesion close-up	© DermNet	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme
m4w1:A03	Erythema multiforme mucosal involvement lips	© DermNet	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme
m4w1:A04	Erythema multiforme ocular mucosal involvement	© DermNet	dermnetnz.org/topics/erythema-multiforme	Erythema multiforme
m4w1:B01	Chronic plaque psoriasis with silver scale	© DermNet	dermnetnz.org/topics/psoriasis	—
m4w1:B02	Chronic plaque psoriasis extensor surface	© DermNet	dermnetnz.org/topics/psoriasis	—
m4w1:B03	Drug-induced psoriasis (example)	© DermNet	dermnetnz.org/topics/drug-induced-psoriasis	Psoriasis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w1:C01	Tinea corporis annular plaque	© DermNet	dermnetnz.org/topics/tinea-corporis	Annular (ring-shaped) rashes: differential, Tinea: corporis, cruris & pedis
m4w1:C02	Tinea corporis active scaly edge	© DermNet	dermnetnz.org/topics/tinea-corporis	Annular (ring-shaped) rashes: differential, Tinea: corporis, cruris & pedis
m4w1:C03	Granuloma annulare annular plaques	© DermNet	dermnetnz.org/topics/granuloma-annulare	Annular (ring-shaped) rashes: differential
m4w1:C04	Pityriasis rosea herald patch	© DermNet	dermnetnz.org/topics/pityriasis-rosea	Annular (ring-shaped) rashes: differential
m4w1:C05	Pityriasis rosea secondary rash trunk	© DermNet	dermnetnz.org/topics/pityriasis-rosea	Annular (ring-shaped) rashes: differential
m4w1:C06	Erythema annulare centrifugum	© DermNet	dermnetnz.org/topics/erythema-annulare-centrifugum	Annular (ring-shaped) rashes: differential
m4w1:D01	Papular lichen planus	© DermNet	dermnetnz.org/topics/lichen-planus	Lichen planus
m4w1:D02	Lichen planus polygonal papules	© DermNet	dermnetnz.org/topics/lichen-planus	Lichen planus
m4w1:D03	Oral lichen planus	© DermNet	dermnetnz.org/topics/lichen-planus	Lichen planus
m4w1:E01	Pemphigus vulgaris erosions and crusts	© DermNet	dermnetnz.org/topics/pemphigus-vulgaris	Pemphigus vulgaris & the pemphigus family
m4w1:E02	Pemphigus vulgaris flaccid blisters	© DermNet	dermnetnz.org/topics/pemphigus-vulgaris	Pemphigus vulgaris & the pemphigus family
m4w1:E03	Pemphigus vulgaris oral erosions	© DermNet	dermnetnz.org/topics/pemphigus-vulgaris	Pemphigus vulgaris & the pemphigus family
m4w1:F01	Discoid lupus erythematosus face	© DermNet	dermnetnz.org/topics/discoid-lupus-erythematosus	Discoid lupus erythematosus
m4w1:F02	Discoid lupus erythematosus scalp alopecia	© DermNet	dermnetnz.org/topics/discoid-lupus-erythematosus	Discoid lupus erythematosus
m4w1:F03	Discoid lupus erythematosus plaque	© DermNet	dermnetnz.org/topics/discoid-lupus-erythematosus	Discoid lupus erythematosus
m4w1:G01	Guttate psoriasis trunk	© DermNet	dermnetnz.org/topics/guttate-psoriasis	Pityriasis rosea, Psoriasis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w1:G02	Guttate psoriasis drop-like lesions	© DermNet	dermnetnz.org/topics/guttate-psoriasis	Pityriasis rosea, Psoriasis
m4w1:G03	Guttate psoriasis widespread	© DermNet	dermnetnz.org/topics/guttate-psoriasis	Pityriasis rosea, Psoriasis

Module 4 · Week 6 — blistering diseases

30 images · assets/img/m4w6

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w6:A01	Erosions and flaccid blisters in pemphigus vulgaris	© DermNet	dermnetnz.org/images/pemphigus-vulgaris-images	Pemphigus vulgaris & the pemphigus family
m4w6:A02	Eroded plaques in pemphigus vulgaris over the trunk	© DermNet	dermnetnz.org/images/pemphigus-vulgaris-images	Pemphigus vulgaris & the pemphigus family
m4w6:A03	Oral pemphigus vulgaris	© DermNet	dermnetnz.org/images/pemphigus-vulgaris-images	Pemphigus vulgaris & the pemphigus family
m4w6:A04	Chronic lip erosion in pemphigus vulgaris	© DermNet	dermnetnz.org/images/pemphigus-vulgaris-images	Pemphigus vulgaris & the pemphigus family
m4w6:A05	Suprabasal intraepidermal blister formation with acantholytic keratinocytes and	© Dr Richard Ashton	dermnetnz.org/images/pemphigus-vulgaris-images	Pemphigus vulgaris & the pemphigus family
m4w6:A06	Direct immunofluorescence of perilesional skin in pemphigus vulgaris - immunostaining with IgG producing intercellular	© Dr Richard Ashton	dermnetnz.org/images/pemphigus-vulgaris-images	Pemphigus vulgaris & the pemphigus family
m4w6:A07	Pemphigus foliaceus	© DermNet	dermnetnz.org/topics/pemphigus-foliaceus	Pemphigus vulgaris & the pemphigus family
m4w6:A08	Drug-induced pemphigus foliaceus	© DermNet	dermnetnz.org/topics/drug-induced-pemphigus	Pemphigus vulgaris & the pemphigus family
m4w6:B01	Bullous pemphigoid	© Raimo Suhonen	dermnetnz.org/images/bullous-pemphigoid-images	Bullous pemphigoid (incl. MMP & pemphigoid gestationis)
m4w6:B02	Bullous pemphigoid	© DermNet	dermnetnz.org/images/bullous-pemphigoid-images	Bullous pemphigoid (incl. MMP & pemphigoid gestationis)
m4w6:B03	Bullous pemphigoid	© Raimo Suhonen	dermnetnz.org/images/bullous-pemphigoid-images	Bullous pemphigoid (incl. MMP & pemphigoid gestationis)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w6:B04	Bullous pemphigoid	© Raimo Suhonen	dermnetnz.org/images/bullous-pemphigoid-images	Bullous pemphigoid (incl. MMP & pemphigoid gestationis)
m4w6:B05	Gingival erosions due to MMP	© DermNet	dermnetnz.org/topics/mucous-membrane-pemphigoid	Bullous pemphigoid (incl. MMP & pemphigoid gestationis)
m4w6:B06	Conjunctival symblepharon due to MMP	© Dr Richard Ashton	dermnetnz.org/topics/mucous-membrane-pemphigoid	Bullous pemphigoid (incl. MMP & pemphigoid gestationis)
m4w6:B07	Pemphigoid gestationis	© DermNet	dermnetnz.org/topics/pemphigoid-gestationis	Bullous pemphigoid (incl. MMP & pemphigoid gestationis)
m4w6:C01	Elbow, sacral and buttock lesions - the classical distribution of dermatitis herpetiformis	© Dr Richard Ashton	dermnetnz.org/topics/dermatitis-herpetiformis	Dermatitis herpetiformis & coeliac disease
m4w6:C02	Clustered vesicles have been excoriated on the elbows in dermatitis herpetiformis	© DermNet	dermnetnz.org/topics/dermatitis-herpetiformis	Dermatitis herpetiformis & coeliac disease
m4w6:C03	Excoriated vesicles on the buttocks and sacrum in dermatitis herpetiformis	© DermNet	dermnetnz.org/topics/dermatitis-herpetiformis	Dermatitis herpetiformis & coeliac disease
m4w6:C04	Excoriated plaques in dermatitis herpetiformis	© Dr Richard Ashton	dermnetnz.org/images/dermatitis-herpetiformis-images	Dermatitis herpetiformis & coeliac disease
m4w6:C05	Granular deposits of IgA shown on direct immunofluorescence of perilesional skin (courtesy of Dr Richard Groves, London)	© St John's Institute for Dermatology	dermnetnz.org/topics/dermatitis-herpetiformis	Dermatitis herpetiformis & coeliac disease
m4w6:D01	Porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/blistering-skin-conditions	Porphyria cutanea tarda (& pseudoporphyria)
m4w6:D02	Pseudoporphyria	© DermNet	dermnetnz.org/topics/pseudoporphyria	Porphyria cutanea tarda (& pseudoporphyria)
m4w6:D03	Blisters milia and scars due to EBA - they may mimic porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/epidermolysis-bullosa-acquisita	Porphyria cutanea tarda (& pseudoporphyria)
m4w6:E01	Toxic epidermal necrolysis	© DermNet	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	SJS/TEN and drug-induced blistering

CODE	CAPTION	©	SOURCE PAGE	USED ON
m4w6:E02	Stevens Johnson Syndrome	© DermNet	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	SJS/TEN and drug-induced blistering
m4w6:E03	Blistering fixed drug eruption	© Raimo Suhonen	dermnetnz.org/topics/fixed-drug-eruption	SJS/TEN and drug-induced blistering
m4w6:E04	Acute generalised exanthematous pustulosis (AGEP)	© DermNet	dermnetnz.org/topics/acute-generalised-exanthematous-pustulosis	SJS/TEN and drug-induced blistering
m4w6:E05	String of beads sign	© DermNet	dermnetnz.org/topics/linear-iga-bullous-disease	SJS/TEN and drug-induced blistering
m4w6:F01	Hailey-Hailey disease blisters	© DermNet	dermnetnz.org/topics/hailey-hailey-disease	Approach to blistering diseases
m4w6:F02	Superficial blistering over the axillae, face and chest due to staphylococcal scalded skin sydrome	© Dr Lawrence Meyerson	dermnetnz.org/topics/staphylococcal-scalded-skin-syndrome	Approach to blistering diseases

Module 5 · Week 4 — tropical & travel

32 images · assets/img/m5w4

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w4:W4A01	Leishmaniasis (untitled gallery image, cutaneous leishmaniasis series)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis, The returning traveller with a skin problem
m5w4:W4A02	Ulcer (New World leishmaniasis)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis
m5w4:W4A03	Ulcer (New World leishmaniasis)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis
m5w4:W4A04	Leishmaniasis (untitled gallery image, cutaneous leishmaniasis series)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis
m5w4:W4A05	Leishmaniasis (untitled gallery image, cutaneous leishmaniasis series)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis
m5w4:W4A06	Leishmaniasis (untitled gallery image, Old World leishmaniasis)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis
m5w4:W4A07	Leishmaniasis (untitled image, mucocutaneous leishmaniasis section)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis
m5w4:W4A08	Leishmaniasis (untitled image, New World leishmaniasis gallery)	© DermNet	dermnetnz.org/topics/leishmaniasis	Cutaneous leishmaniasis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w4:W4B01	The serpiginous track of larva migrans on the sole	© DermNet (Dr Ian Coulson)	dermnetnz.org/images/cutaneous-larva-migrans-images	Cutaneous larva migrans, The returning traveller with a skin problem
m5w4:W4B02	Typical snake-like tracks of larva migrans	© Dr Richard Ashton	dermnetnz.org/images/cutaneous-larva-migrans-images	Cutaneous larva migrans
m5w4:W4B03	Larva migrans on the hands	© Dr Lawrence Meyerson	dermnetnz.org/images/cutaneous-larva-migrans-images	Cutaneous larva migrans
m5w4:W4B04	Larva migrans on the hand	© Dr Lawrence Meyerson	dermnetnz.org/images/cutaneous-larva-migrans-images	Cutaneous larva migrans
m5w4:W4B05	Bullous reactions to larva migrans	© Dr Richard Ashton	dermnetnz.org/images/cutaneous-larva-migrans-images	Cutaneous larva migrans
m5w4:W4B06	Cutaneous larva migrans	© DermNet	dermnetnz.org/topics/cutaneous-larva-migrans	Cutaneous larva migrans
m5w4:W4C01	The rash of Dengue — confluent erythema with white islands ('white in a sea of red')	© DermNet	dermnetnz.org/images/dengue-fever-images	Dengue: skin signs & safety, The returning traveller with a skin problem
m5w4:W4C02	Exanthem with islands of sparing	© DermNet	dermnetnz.org/topics/dengue	Dengue: skin signs & safety
m5w4:W4C03	Dengue rash	© DermNet	dermnetnz.org/topics/dengue	Dengue: skin signs & safety
m5w4:W4D01	Leonine facies with loss of eyebrows and infiltrated facial skin	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D02	Lepromatous leprosy (untitled gallery image)	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D03	Leprosy (untitled gallery image, lepromatous series)	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D04	Asymmetrical distribution of erythematous infiltrated plaques in skin of colour	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D05	Dome-shaped papules and nodules in skin of colour	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D06	Well-defined erythematous plaque on the cheek in skin of colour	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions, The returning traveller with a skin problem

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w4:W4D07	Red lesions of variable size. Characteristic 'Swiss cheese' appearance (black arrows)	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D08	BT leprosy with type 1 reaction	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D09	Trophic ulcer on the heel	© DermNet	dermnetnz.org/topics/leprosy	Leprosy (Hansen disease) & lepra reactions
m5w4:W4D10	Erythema nodosum	© DermNet	dermnetnz.org/topics/erythema-nodosum	Leprosy (Hansen disease) & lepra reactions
m5w4:W4E01	Mycetoma foot	© DermNet	dermnetnz.org/topics/mycetoma	Mycetoma (Madura foot)
m5w4:W4E02	Mycetoma foot (courtesy Dr Sriyani Samaraweera)	© DermNet (courtesy Dr Sriyani Samaraweera)	dermnetnz.org/topics/mycetoma	Mycetoma (Madura foot)
m5w4:W4E03	Actinomycetoma due to nocardia	© DermNet	dermnetnz.org/topics/mycetoma	Mycetoma (Madura foot)
m5w4:W4E04	Actinomycetoma foot and legs	© DermNet	dermnetnz.org/topics/mycetoma	Mycetoma (Madura foot)
m5w4:W4E05	H&E stain of actinomycosis grains	© DermNet	dermnetnz.org/topics/mycetoma	Mycetoma (Madura foot)

Module 5 · Week 5 — viral & infective rashes

40 images · assets/img/m5w5

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w5:W5A01	Scrofuloderma	© DermNet	dermnetnz.org/topics/cutaneous-tuberculosis	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5A02	Scrofuloderma	© DermNet	dermnetnz.org/topics/cutaneous-tuberculosis	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5A03	Scrofuloderma	© DermNet	dermnetnz.org/topics/cutaneous-tuberculosis	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5A04	Lupus vulgaris	© DermNet	dermnetnz.org/topics/cutaneous-tuberculosis	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5A05	Cutaneous tuberculosis/warty tuberculosis	© DermNet	dermnetnz.org/topics/cutaneous-tuberculosis	Cutaneous tuberculosis (scrofuloderma & others)

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w5:W5A06	Cutaneous miliary tuberculosis	© DermNet	dermnetnz.org/topics/cutaneous-tuberculosis	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5A07	Cutaneous tuberculosis: Bazin disease	© DermNet	dermnetnz.org/topics/tuberculids	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5A08	Papulonecrotic tuberculid	© DermNet	dermnetnz.org/topics/tuberculids	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5A09	Mantoux test reaction	© DermNet	dermnetnz.org/topics/cutaneous-tuberculosis	Cutaneous tuberculosis (scrofuloderma & others)
m5w5:W5B01	Herpes zoster	© DermNet	dermnetnz.org/topics/herpes-zoster	Herpes zoster (shingles)
m5w5:W5B02	Herpes zoster	© DermNet	dermnetnz.org/topics/herpes-zoster	Herpes zoster (shingles)
m5w5:W5B03	Herpes Zoster	© DermNet	dermnetnz.org/topics/herpes-zoster	Herpes zoster (shingles)
m5w5:W5B04	Shingles	© DermNet	dermnetnz.org/topics/herpes-zoster	Herpes zoster (shingles)
m5w5:W5B05	Herpes zoster	© DermNet	dermnetnz.org/topics/herpes-zoster	Herpes zoster (shingles)
m5w5:W5B06	Herpes zoster	© DermNet	dermnetnz.org/topics/herpes-zoster	Herpes zoster (shingles)
m5w5:W5C01	The herald patch of pityriasis rosea	© DermNet	dermnetnz.org/images/pityriasis-rosea-images	Pityriasis rosea
m5w5:W5C02	Pityriasis rosea and a herald patch on the torso	© DermNet	dermnetnz.org/images/pityriasis-rosea-images	Pityriasis rosea
m5w5:W5C03	Oval annular lesions of pityriasis rosea with the long axis of the oval orientated around the ribs	© DermNet	dermnetnz.org/images/pityriasis-rosea-images	Pityriasis rosea
m5w5:W5C04	Annular collaretted lesions in truncal pityriasis rosea	© DermNet	dermnetnz.org/images/pityriasis-rosea-images	Pityriasis rosea
m5w5:W5C05	Pityriasis rosea and a herald patch in skin of colour	© DermNet	dermnetnz.org/images/pityriasis-rosea-images	Pityriasis rosea
m5w5:W5C06	Pityriasis rosea on the back	© DermNet	dermnetnz.org/images/pityriasis-rosea-images	Pityriasis rosea
m5w5:W5C07	Close up of guttate psoriasis	© DermNet	dermnetnz.org/topics/guttate-psoriasis	Pityriasis rosea

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w5:W5C08	Guttate psoriasis	© DermNet	dermnetnz.org/topics/guttate-psoriasis	Pityriasis rosea
m5w5:W5C09	Secondary syphilis	© DermNet	dermnetnz.org/topics/syphilis	Pityriasis rosea
m5w5:W5D01	Herpes simplex labialis	© Dr Richard Ashton	dermnetnz.org/images/herpes-simplex-images	Herpes simplex labialis (cold sores)
m5w5:W5D02	Herpes simplex labialis	© DermNet	dermnetnz.org/images/herpes-simplex-images	Herpes simplex labialis (cold sores)
m5w5:W5D03	Herpes labialis	© Dr Mashihul Hossain	dermnetnz.org/topics/herpes-simplex	Herpes simplex labialis (cold sores)
m5w5:W5D04	Herpes simplex on the cheek	© DermNet	dermnetnz.org/topics/herpes-simplex	Herpes simplex labialis (cold sores)
m5w5:W5D05	Extenive lip erosions due to chronic herpes simplex infection in HIV	© Dr Richard Ashton	dermnetnz.org/images/herpes-simplex-images	Herpes simplex labialis (cold sores)
m5w5:W5D06	Eczema herpeticum	© DermNet	dermnetnz.org/topics/eczema-herpeticum	Herpes simplex labialis (cold sores)
m5w5:W5D07	Herpetic whitlow	© DermNet	dermnetnz.org/topics/herpes-simplex	Herpes simplex labialis (cold sores)
m5w5:W5D08	Herpes simplex labialis associated with erythema multiforme	© DermNet	dermnetnz.org/images/herpes-simplex-images	Herpes simplex labialis (cold sores)
m5w5:W5E01	Scarlet fever	© DermNet	dermnetnz.org/topics/scarlet-fever	Scarlet fever & strawberry tongue
m5w5:W5E02	Scarlet fever	© DermNet	dermnetnz.org/topics/scarlet-fever	Scarlet fever & strawberry tongue
m5w5:W5E03	Scarlet fever	© DermNet	dermnetnz.org/topics/scarlet-fever	Scarlet fever & strawberry tongue
m5w5:W5E04	Scarlet fever	© DermNet	dermnetnz.org/topics/scarlet-fever	Scarlet fever & strawberry tongue
m5w5:W5E05	Scarlet fever	© DermNet	dermnetnz.org/topics/scarlet-fever	Scarlet fever & strawberry tongue
m5w5:W5E06	Peeling toes after scarlet fever	© DermNet	dermnetnz.org/topics/scarlet-fever	Scarlet fever & strawberry tongue

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w5:W5E07	(Wikimedia Commons, not DermNet) Strep throat with white strawberry tongue	© Whispyhistory	commons.wikimedia.org/wiki/File:Strep_throat_with_white_strawberry_tongue.jpg	Scarlet fever & strawberry tongue
m5w5:W5E08	(Wikimedia Commons, not DermNet) Scharlach (raspberry tongue, later stage of scarlet fever)	© Martin Kronawitter	commons.wikimedia.org/wiki/File:Scharlach.JPG	Scarlet fever & strawberry tongue

Module 5 · Week 6 — photodermatoses

29 images · assets/img/m5w6

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w6:A01	Papulovesicular polymorphic light eruption on the arm	© DermNet	dermnetnz.org/topics/polymorphic-light-eruption	Polymorphic light eruption
m5w6:A02	Erythematous papules on the recently sun-exposed 'V' area of the chest due to polymorphic light eruption	© DermNet	dermnetnz.org/topics/polymorphic-light-eruption	Polymorphic light eruption
m5w6:A03	Papular polymorphic light eruption early on in a summer vacation	© DermNet	dermnetnz.org/topics/polymorphic-light-eruption	Polymorphic light eruption
m5w6:A04	Extensive papular and plaque polymorphic light eruption	© DermNet	dermnetnz.org/topics/polymorphic-light-eruption	Polymorphic light eruption
m5w6:A05	Targetoid polymorphic light eruption on recently exposed arm skin	© DermNet	dermnetnz.org/topics/polymorphic-light-eruption	Polymorphic light eruption
m5w6:B01	Porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/porphyria-cutanea-tarda	Porphyria cutanea tarda (& pseudoporphyria)
m5w6:B02	Porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/porphyria-cutanea-tarda	Porphyria cutanea tarda (& pseudoporphyria)
m5w6:B03	Porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/porphyria-cutanea-tarda	Porphyria cutanea tarda (& pseudoporphyria)
m5w6:B04	Porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/porphyria-cutanea-tarda	Porphyria cutanea tarda (& pseudoporphyria)
m5w6:B05	Porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/porphyria-cutanea-tarda	Porphyria cutanea tarda (& pseudoporphyria)
m5w6:B06	Porphyria cutanea tarda	© DermNet	dermnetnz.org/topics/porphyria-cutanea-tarda	Milia, Porphyria cutanea tarda (& pseudoporphyria)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w6:B07	Pseudoporphyria	© DermNet	dermnetnz.org/topics/pseudoporphyria	Porphyria cutanea tarda (& pseudoporphyria)
m5w6:C01	Severe lichenified eczema with sparing under the collar area due to the photoprotection of double fabric	© DermNet	dermnetnz.org/topics/chronic-actinic-dermatitis	Chronic actinic dermatitis
m5w6:C02	Chronic actinic dermatitis - thickened eczema with sparing in the relatively photoprotected area behind and under the ear	© DermNet	dermnetnz.org/topics/chronic-actinic-dermatitis	Chronic actinic dermatitis
m5w6:C03	Chronic actinic dermatitis on the dorsal hand - note the cut off at the wrist and relative sparing between the fingers	© DermNet	dermnetnz.org/topics/chronic-actinic-dermatitis	Chronic actinic dermatitis
m5w6:C04	Severe chronic actinic dermatitis with sparing below the neck line and in the skin creases	© DermNet	dermnetnz.org/topics/chronic-actinic-dermatitis	Chronic actinic dermatitis
m5w6:C05	Pigmented chronic actinic dermatitis - note the cut-off at the neck line, and sparing below and behind the ears	© DermNet	dermnetnz.org/topics/chronic-actinic-dermatitis	Chronic actinic dermatitis
m5w6:D01	Doxycycline-induced phototoxic eruption on the face	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Drug-induced photosensitivity
m5w6:D02	A phototoxic reaction due to doxycycline - note the sparing between the light-shielded sides of the fingers	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Drug-induced photosensitivity
m5w6:D03	Photosensitivity due to a thiazide diuretic - note the sparing under his T-shirt area	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Drug-induced photosensitivity
m5w6:D04	Thiazide photosensitivity - note the sparing under the watch strap	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Drug-induced photosensitivity
m5w6:D05	Photosensitivity due to amiodarone affecting the nose and lids	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Drug-induced photosensitivity
m5w6:D06	An eczematous tamsulosin-induced photosensitive reaction	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Drug-induced photosensitivity
m5w6:D07	A phototoxic drug eruption due to an oral non-steroidal anti-inflammatory drug	© DermNet	dermnetnz.org/topics/drug-induced-photosensitivity	Drug-induced photosensitivity
m5w6:E01	Phytophotodermatitis from lime juice	© DermNet	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis
m5w6:E02	Streaked lesions in phytophotodermatitis	© DermNet	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis
m5w6:E03	Bullous phytophotodermatitis	© DermNet	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m5w6:E04	Phytophotodermatitis from hogweed	© DermNet	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis
m5w6:E05	Post inflammatory pigmentation after phytophotodermatitis	© DermNet	dermnetnz.org/topics/phytophotodermatitis	Phytophotodermatitis

Module 1 · Week 2 — therapeutics

28 images · assets/img/m1w2

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w2:A01	Steroid telangiectasia	© DermNet	dermnetnz.org/topics/steroid-rosacea	Topical corticosteroids: potency, quantities & safety
m1w2:A02	Bruising	© DermNet	dermnetnz.org/topics/topical-steroid	Topical corticosteroids: potency, quantities & safety
m1w2:A03	Steroid rosacea	© DermNet	dermnetnz.org/topics/steroid-rosacea	Topical corticosteroids: potency, quantities & safety
m1w2:A04	Perioral dermatitis due to the use of a steroid nebuliser mask	© DermNet	dermnetnz.org/topics/periorificial-dermatitis	Topical corticosteroids: potency, quantities & safety
m1w2:A05	Tinea incognita - limb	© DermNet	dermnetnz.org/topics/tinea-incognito	Topical corticosteroids: potency, quantities & safety
m1w2:A06	Localised pustular psoriasis	© DermNet	dermnetnz.org/topics/topical-steroid	Topical corticosteroids: potency, quantities & safety
m1w2:A07	Positive patch test to budesonide	© DermNet	dermnetnz.org/topics/allergy-to-topical-corticosteroid	Topical corticosteroids: potency, quantities & safety
m1w2:B01	Redness and swelling of the lids - steroid withdrawal after prolonged potent steroid use on the face.	© DermNet	dermnetnz.org/topics/topical-corticosteroid-withdrawal	Topical corticosteroids: potency, quantities & safety

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w2:B02	Redness and swelling of the forearm after abrupt discontinuation of prolonged potent topical steroid use. Palms are not affected (red sleeve sign).	© DermNet	dermnetnz.org/topics/topical-corticosteroid-withdrawal	Topical corticosteroids: potency, quantities & safety
m1w2:C01	Fingertip unit	© DermNet	dermnetnz.org/topics/topical-steroid	Topical corticosteroids: potency, quantities & safety
m1w2:C02	Topical steroids in differing vehicles	Open access image (DermNet)	dermnetnz.org/topics/topical-steroid	Topical corticosteroids: potency, quantities & safety
m1w2:C03	A selection of emollients from the light to the very greasy - 'the right one is the one the patient likes'	Open access image (DermNet)	dermnetnz.org/topics/emollients-and-moisturisers	Emollients & soap substitutes
m1w2:D01	Psoriasis pre-calcipotriol	© DermNet	dermnetnz.org/topics/calcipotriol	Vitamin D analogues, coal tar, dithranol & phototherapy
m1w2:D02	Psoriasis post-calcipotriol (after 6 weeks of treatment with calcipotriol ointment)	© DermNet	dermnetnz.org/topics/calcipotriol	Vitamin D analogues, coal tar, dithranol & phototherapy
m1w2:D03	Coaltar	© DermNet	dermnetnz.org/topics/coal-tar	Vitamin D analogues, coal tar, dithranol & phototherapy
m1w2:D04	PUVA hand unit	Open access image (DermNet)	dermnetnz.org/topics/puva-photochemotherapy	Vitamin D analogues, coal tar, dithranol & phototherapy
m1w2:D05	MED testing to determine initial dose of UVB	© DermNet	dermnetnz.org/topics/narrowband-uvb-phototherapy	Vitamin D analogues, coal tar, dithranol & phototherapy
m1w2:E01	Sunburn	© DermNet	dermnetnz.org/topics/isotretinoin	Retinoids (topical & oral)
m1w2:E02	Impetigo	© DermNet	dermnetnz.org/topics/isotretinoin	Retinoids (topical & oral)
m1w2:E03	Dryness due to a topical retinoid	© DermNet	dermnetnz.org/topics/topical-retinoids	Retinoids (topical & oral)
m1w2:E04	Acitretin nail thinning	© DermNet	dermnetnz.org/topics/acitretin	Retinoids (topical & oral)
m1w2:E05	Acitretin diffuse hair loss	© DermNet	dermnetnz.org/topics/acitretin	Retinoids (topical & oral)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w2:F01	Ferriman-Gallwey visual scale for assessing hirsutism	Open access image (DermNet)	dermnetnz.org/topics/hirsutism	Hirsutism & the Ferriman–Gallwey score
m1w2:F02	Facial hirsutism	© DermNet	dermnetnz.org/topics/hirsutism	Hirsutism & the Ferriman–Gallwey score
m1w2:F03	Hirsutism	© DermNet	dermnetnz.org/topics/hirsutism	Hirsutism & the Ferriman–Gallwey score
m1w2:F04	Folliculitis due to hair removal	© DermNet	dermnetnz.org/topics/hirsutism	Hirsutism & the Ferriman–Gallwey score
m1w2:G01	Dupilumab-induced ocular disease 8 weeks after treatment initiation	© DermNet	dermnetnz.org/topics/dupilumab-associated-ocular-surface-disease	Biologics, JAK inhibitors & conventional systemics
m1w2:G02	Chronic plaque psoriasis	© Raimo Suhonen	dermnetnz.org/topics/biologics	Biologics, JAK inhibitors & conventional systemics

Module 1 · Week 3 — drug eruptions

24 images · assets/img/m1w3

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w3:A01	Morbilliform drug eruption	© DermNet	dermnetnz.org/topics/morbilliform-drug-reaction	Morbilliform drug eruption & drug urticaria
m1w3:A02	Morbilliform drug eruption	© DermNet	dermnetnz.org/topics/morbilliform-drug-reaction	Morbilliform drug eruption & drug urticaria
m1w3:A03	Urticaria	© Dr Richard Ashton	dermnetnz.org/topics/drug-induced-urticaria	Morbilliform drug eruption & drug urticaria
m1w3:B01	Toxic epidermal necrolysis	© DermNet	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	Erythema multiforme, SJS/TEN and drug-induced blistering
m1w3:B02	Toxic epidermal necrolysis	© DermNet	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	Erythema multiforme, SJS/TEN and drug-induced blistering
m1w3:B03	Toxic epidermal necrolysis	© Raimo Suhonen	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	Erythema multiforme, SJS/TEN and drug-induced blistering
m1w3:B04	Stevens Johnson Syndrome	© DermNet	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	SJS/TEN and drug-induced blistering

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w3:B05	Symblepharon	© DermNet	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	SJS/TEN and drug-induced blistering
m1w3:B06	Nail shedding	© DermNet	dermnetnz.org/topics/stevens-johnson-syndrome-toxic-epidermal-necrolysis	SJS/TEN and drug-induced blistering
m1w3:C01	Drug eruption	© DermNet	dermnetnz.org/topics/drug-hypersensitivity-syndrome	DRESS (drug reaction with eosinophilia and systemic symptoms)
m1w3:C02	Acute generalised exanthematous pustulosis	© DermNet	dermnetnz.org/topics/acute-generalised-exanthematous-pustulosis	AGEP (acute generalised exanthematous pustulosis)
m1w3:C03	Acute generalised exanthematous pustulosis (AGEP)	© DermNet	dermnetnz.org/topics/acute-generalised-exanthematous-pustulosis	AGEP (acute generalised exanthematous pustulosis)
m1w3:D01	Lip angioedema	© DermNet	dermnetnz.org/topics/ace-inhibitor-induced-angioedema	ACE-inhibitor (bradykinin) angioedema
m1w3:E01	Erythema nodosum	© DermNet	dermnetnz.org/topics/erythema-nodosum	Erythema nodosum (incl. drug-induced)
m1w3:E02	Erythema nodosum	© Raimo Suhonen	dermnetnz.org/topics/erythema-nodosum	Erythema nodosum (incl. drug-induced)
m1w3:F01	Blue-grey pigmentation due to long term amiodarone use	© DermNet	dermnetnz.org/topics/drug-induced-pigmentation	Drug-induced pigmentation (amiodarone, minocycline)
m1w3:F02	Blue pigmentation due to long-term minocycline use for pyoderma gangrenosum	© DermNet	dermnetnz.org/topics/drug-induced-pigmentation	Drug-induced pigmentation (amiodarone, minocycline)
m1w3:G01	Photosensitivity	© DermNet	dermnetnz.org/topics/photosensitivity	Drug-induced photosensitivity
m1w3:G02	Photosensitivity	© DermNet	dermnetnz.org/topics/photosensitivity	Drug-induced photosensitivity
m1w3:H01	Pigmented fixed drug eruption on the lip	© DermNet	dermnetnz.org/topics/fixed-drug-eruption	Fixed drug eruption
m1w3:H02	Fixed drug eruption	© DermNet	dermnetnz.org/topics/fixed-drug-eruption	Fixed drug eruption
m1w3:H03	Targetoid lesion in fixed drug eruption	© DermNet	dermnetnz.org/topics/fixed-drug-eruption	Fixed drug eruption
m1w3:H04	Fixed drug eruption	© DermNet	dermnetnz.org/topics/fixed-drug-eruption	Fixed drug eruption

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w3:I01	Pseudoporphyria	© DermNet	dermnetnz.org/topics/non-steroidal-anti-inflammatory-drugs-and-their-skin-side-effects	Porphyria cutanea tarda (& pseudoporphyria)

Module 1 · Week 4 — vasculitis

27 images · assets/img/m1w4

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w4:A01	Cutaneous vasculitis	© DermNet	dermnetnz.org/topics/cutaneous-small-vessel-vasculitis	Cutaneous small-vessel (leukocytoclastic) vasculitis
m1w4:A02	Cutaneous vasculitis	© DermNet	dermnetnz.org/topics/cutaneous-small-vessel-vasculitis	Cutaneous small-vessel (leukocytoclastic) vasculitis
m1w4:A03	Hypersensitivity vasculitis	© Te Whatu Ora	dermnetnz.org/topics/cutaneous-small-vessel-vasculitis	Cutaneous small-vessel (leukocytoclastic) vasculitis
m1w4:A04	Cutaneous vasculitis	© DermNet	dermnetnz.org/topics/cutaneous-vasculitis	Cutaneous small-vessel (leukocytoclastic) vasculitis
m1w4:A05	Cutaneous vasculitis	© DermNet	dermnetnz.org/topics/cutaneous-vasculitis	Cutaneous small-vessel (leukocytoclastic) vasculitis
m1w4:A06	Vasculitis	© Raimo Suhonen	dermnetnz.org/topics/purpura	Cutaneous small-vessel (leukocytoclastic) vasculitis
m1w4:B01	Osler node on ring finger as well as more subtle nodes seen on the little and middle finger	© DermNet	dermnetnz.org/topics/osler-nodes-and-janeway-lesions	Osler nodes, Janeway lesions & splinter haemorrhages
m1w4:B02	Osler nodes on index finger. Note also the tiny splinter haemorrhages on fingernail also seen in end	© DermNet	dermnetnz.org/topics/osler-nodes-and-janeway-lesions	Osler nodes, Janeway lesions & splinter haemorrhages
m1w4:B03	Splinter haemorrhages	© DermNet	dermnetnz.org/topics/splinter-haemorrhage	Osler nodes, Janeway lesions & splinter haemorrhages
m1w4:B04	Splinter haemorrhages	© DermNet	dermnetnz.org/topics/splinter-haemorrhage	Osler nodes, Janeway lesions & splinter haemorrhages

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w4:B05	Psoriasis	© DermNet	dermnetnz.org/topics/splinter-haemorrhage	Osler nodes, Janeway lesions & splinter haemorrhages
m1w4:B06	Septic emboli	© DermNet	dermnetnz.org/topics/septic-emboli	Osler nodes, Janeway lesions & splinter haemorrhages
m1w4:C01	Urticarial vasculitis	© DermNet	dermnetnz.org/topics/urticarial-vasculitis	Urticarial vasculitis & HUVS
m1w4:C02	Urticarial vasculitis	© DermNet	dermnetnz.org/topics/urticarial-vasculitis	Urticarial vasculitis & HUVS
m1w4:D01	Cutaneous polyarteritis nodosa	© DermNet	dermnetnz.org/topics/cutaneous-polyarteritis-nodosa	Polyarteritis nodosa (systemic & cutaneous)
m1w4:D02	Cutaneous polyarteritis nodosa	© DermNet	dermnetnz.org/topics/cutaneous-polyarteritis-nodosa	Polyarteritis nodosa (systemic & cutaneous)
m1w4:D03	Livedo reticularis	© DermNet	dermnetnz.org/topics/livedo-reticularis	Livedo reticularis vs livedo racemosa
m1w4:D04	Livedo racemosa	© DermNet	dermnetnz.org/topics/livedo-reticularis	Livedo reticularis vs livedo racemosa
m1w4:D05	Cutis marmorata	© DermNet	dermnetnz.org/topics/livedo-reticularis	Livedo reticularis vs livedo racemosa
m1w4:D06	Atrophie blanche	© DermNet	dermnetnz.org/topics/livedoid-vasculopathy	Livedo reticularis vs livedo racemosa
m1w4:E01	Classical purpura	© DermNet	dermnetnz.org/topics/henoch-schoenlein-purpura	IgA vasculitis (Henoch–Schönlein purpura)
m1w4:E02	Ankle swelling (arthritis)	© DermNet	dermnetnz.org/topics/henoch-schoenlein-purpura	IgA vasculitis (Henoch–Schönlein purpura)
m1w4:E03	Henoch-Schönlein purpura	© DermNet	dermnetnz.org/topics/henoch-schoenlein-purpura	IgA vasculitis (Henoch–Schönlein purpura)
m1w4:F01	Acute haemorrhagic oedema	© DermNet	dermnetnz.org/topics/acute-haemorrhagic-oedema-of-infancy	Purpura: the differential diagnosis
m1w4:F02	Cryoglobulinaemia	© DermNet	dermnetnz.org/topics/cryoglobulinaemia	Purpura: the differential diagnosis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w4:F03	Lip ulceration due to granulomatosis with polyangiitis	© DermNet	dermnetnz.org/topics/granulomatosis-with-polyangiitis	Purpura: the differential diagnosis
m1w4:F04	Disseminated intravascular coagulation	© DermNet	dermnetnz.org/topics/purpura	Purpura: the differential diagnosis

Module 1 · Week 5 — skin signs of systemic disease

26 images · assets/img/m1w5

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w5:A01	Heliotrope rash	© DermNet	dermnetnz.org/topics/adult-onset-dermatomyositis	Dermatomyositis
m1w5:A02	Gottron papules over the joints of the hands	© DermNet	dermnetnz.org/topics/adult-onset-dermatomyositis	Dermatomyositis
m1w5:A03	Shawl sign over the shoulder	© DermNet	dermnetnz.org/topics/adult-onset-dermatomyositis	Dermatomyositis
m1w5:A04	Photosensitive rash on the V of chest	© DermNet	dermnetnz.org/topics/adult-onset-dermatomyositis	Dermatomyositis
m1w5:A05	Ragged cuticles and nailfold redness	© DermNet	dermnetnz.org/topics/adult-onset-dermatomyositis	Dermatomyositis
m1w5:A06	Gottron sign over the knees	© DermNet	dermnetnz.org/topics/adult-onset-dermatomyositis	Dermatomyositis
m1w5:B01	Metastatic adenocarcinoma of unknown origin (carcinoma erysipeloides)	© DermNet	dermnetnz.org/topics/skin-metastasis	Skin signs of lung cancer & internal malignancy
m1w5:B02	Tripe palms	© DermNet	dermnetnz.org/topics/tripe-palms	Skin signs of lung cancer & internal malignancy
m1w5:C01	Streoid striae	© DermNet	dermnetnz.org/topics/cushing-syndrome	Cushing's syndrome: skin signs
m1w5:C02	Atrophy and purpura	© Raimo Suhonen	dermnetnz.org/topics/cushing-syndrome	Cushing's syndrome: skin signs
m1w5:C03	Moon face	© DermNet	dermnetnz.org/topics/systemic-steroids	Cushing's syndrome: skin signs
m1w5:D01	Excessive sweating in armpits	© DermNet	dermnetnz.org/topics/hyperhidrosis	Hyperhidrosis
m1w5:D02	Palmar hyperhidrosis	© DermNet	dermnetnz.org/topics/drug-induced-hyperhidrosis	Hyperhidrosis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m1w5:E01	Eruptive xanthomas	© DermNet	dermnetnz.org/topics/xanthoma	Eruptive xanthomas & skin signs of lipids
m1w5:E02	Eruptive xanthomas	© Raimo Suhonen	dermnetnz.org/topics/xanthoma	Eruptive xanthomas & skin signs of lipids
m1w5:E03	Eruptive xanthomas	© DermNet	dermnetnz.org/topics/xanthoma	Eruptive xanthomas & skin signs of lipids
m1w5:F01	Acanthosis nigricans	© DermNet	dermnetnz.org/topics/tripe-palms	Acanthosis nigricans
m1w5:G01	Fever and a rash	© DermNet	dermnetnz.org/topics/sarcoidosis	Sarcoidosis (incl. Löfgren syndrome)
m1w5:G02	Bihilar lymphadenopathy in sarcoidosis	Open access image (DermNet)	dermnetnz.org/topics/sarcoidosis	Sarcoidosis (incl. Löfgren syndrome)
m1w5:G03	Lupus pernio in cutaneous sarcoidosis	© DermNet	dermnetnz.org/topics/sarcoidosis	Sarcoidosis (incl. Löfgren syndrome)
m1w5:G04	Scar sarcoid mimicing a keloid on a pierced ear lobe	© DermNet	dermnetnz.org/topics/sarcoidosis	Sarcoidosis (incl. Löfgren syndrome)
m1w5:G05	Plaque sarcoid on the cheek and nose	© DermNet	dermnetnz.org/topics/sarcoidosis	Sarcoidosis (incl. Löfgren syndrome)
m1w5:H01	Hairline lesions in dermatitis herpetiformis	© Dr Richard Ashton	dermnetnz.org/topics/dermatitis-herpetiformis	Dermatitis herpetiformis & coeliac disease
m1w5:I01	Multiple ulcers on the shin - the superior edge of the proximal ulcer showing the characterisitc purple edge	© DermNet	dermnetnz.org/topics/pyoderma-gangrenosum	Pyoderma gangrenosum
m1w5:I02	Severe pathergic pyoderma gangrenosum after breast cancer reconstructive surgery (TRAM procedure)	© DermNet	dermnetnz.org/topics/pyoderma-gangrenosum	Pyoderma gangrenosum
m1w5:I03	Healed post operative pyoderma gangrenosum after pulse intravenous methylprednisolone and ciclosporin	© DermNet	dermnetnz.org/topics/pyoderma-gangrenosum	Pyoderma gangrenosum

Module 2 · Week 1 — lumps & bumps

28 images · assets/img/m2w1

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w1:A01	Pinch sign of dermatofibroma	© DermNet	dermnetnz.org/topics/dermatofibroma	Dermatofibroma

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w1:A02	Dermatofibroma	© Raimo Suhonen	dermnetnz.org/topics/dermatofibroma	Dermatofibroma
m2w1:A03	Dermoscopy. Peripheral network, central white area: dermatofibroma	© DermNet	dermnetnz.org/topics/dermatofibroma-images	Dermatofibroma
m2w1:A04	Dermoscopy. Prominent central white area: dermatofibroma	© DermNet	dermnetnz.org/topics/dermatofibroma-images	Dermatofibroma
m2w1:B01	Lesion in 2014	© DermNet	dermnetnz.org/topics/dermatofibrosarcoma-protuberans	Dermatofibrosarcoma protuberans (DFSP)
m2w1:B02	Same lesion in 2015	© DermNet	dermnetnz.org/topics/dermatofibrosarcoma-protuberans	Dermatofibrosarcoma protuberans (DFSP)
m2w1:C01	A hairless occipital patch in a child - the nevus sebaceous may become thicker and verrucous at puberty	© DermNet	dermnetnz.org/topics/sebaceous-naevus	Sebaceous naevus (naevus sebaceous)
m2w1:C02	Nevus sebaceous, present since birth, slowly become rough and has developed a benign cyst within it	© DermNet	dermnetnz.org/topics/sebaceous-naevus	Sebaceous naevus (naevus sebaceous)
m2w1:C03	A nevus sebaceous in skin of colour - the typical yellow colour cannot be appreciated in darker skin types	© Dr Richard Ashton	dermnetnz.org/topics/sebaceous-naevus	Sebaceous naevus (naevus sebaceous)
m2w1:C04	A crusted area has developed at the edge of this nevus sebaceous - biopsy is needed to exclude either a benign or malignant lesion arising within	© DermNet	dermnetnz.org/topics/sebaceous-naevus	Sebaceous naevus (naevus sebaceous)
m2w1:D01	Palpating an epidermoid cyst - note this accentuates the punctum	© DermNet	dermnetnz.org/topics/epidermoid-cyst	Epidermoid, pilar and dermoid cysts (and Gardner syndrome)
m2w1:D02	A recurrent epidermoid cyst - note the scar from a previous incomplete excision	© DermNet	dermnetnz.org/topics/epidermoid-cyst	Epidermoid, pilar and dermoid cysts (and Gardner syndrome)
m2w1:D03	Multiple facial epidermal cysts in Gardner syndrome	© DermNet	dermnetnz.org/topics/gardner-syndrome	Epidermoid, pilar and dermoid cysts (and Gardner syndrome)
m2w1:D04	A scalp osteoma in Gardner syndrome	© Dr Lawrence Meyerson	dermnetnz.org/images/gardner-syndrome-images	Epidermoid, pilar and dermoid cysts (and Gardner syndrome)

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w1:D05	Pilar cyst	© DermNet	dermnetnz.org/topics/cutaneous-cysts-and-pseudocysts	Epidermoid, pilar and dermoid cysts (and Gardner syndrome)
m2w1:E01	A typical lipoma on the back	© DermNet	dermnetnz.org/topics/lipoma	Lipoma & the sarcoma lump pathway
m2w1:F01	Pyogenic granuloma	© DermNet	dermnetnz.org/topics/pyogenic-granuloma	Pyogenic granuloma
m2w1:F02	Pyogenic granuloma	© DermNet	dermnetnz.org/topics/pyogenic-granuloma	Pyogenic granuloma
m2w1:F03	Angiokeratoma circumscriptum	© DermNet	dermnetnz.org/topics/angiokeratoma	Angiokeratomas & Fabry disease
m2w1:F04	Angiokeratoma of Fordyce on scrotum	© DermNet	dermnetnz.org/topics/angiokeratoma	Angiokeratomas & Fabry disease
m2w1:F05	Angiokeratoma	© DermNet	dermnetnz.org/topics/angiokeratoma-images	Angiokeratomas & Fabry disease
m2w1:F06	Angiokeratomas in suspected Fabry disease	© DermNet	dermnetnz.org/topics/angiokeratoma-images	Angiokeratomas & Fabry disease
m2w1:G01	Neurofibroma	© DermNet	dermnetnz.org/topics/neurofibromatosis	Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more
m2w1:G02	Plexiform neurofibroma	© DermNet	dermnetnz.org/topics/neurofibromatosis	Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more
m2w1:G03	Pilomatricoma	© DermNet	dermnetnz.org/topics/pilomatricoma	Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more
m2w1:G04	Myxoid cyst	© DermNet	dermnetnz.org/topics/digital-myxoid-pseudocyst	Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more
m2w1:G05	Sebaceous hyperplasia	© DermNet	dermnetnz.org/topics/sebaceous-hyperplasia	Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more
m2w1:G06	Morphoea	© DermNet	dermnetnz.org/topics/morphoea	Other skin lumps: neurofibroma, pilomatricoma, myxoid cyst & more

Image credits

Module 2 · Week 2 — benign lesions

26 images · `assets/img/m2w2`

CODE	CAPTION	©	SOURCE PAGE	USED ON
`m2w2:A01`	Dermatofibroma pigmented	© DermNet	dermnetnz.org/topics/dermatofibroma-images	Dermatofibroma
`m2w2:A02`	Dermatofibroma	© DermNet	dermnetnz.org/topics/dermatofibroma	Dermatofibroma
`m2w2:A03`	Dermatofibroma	© DermNet	dermnetnz.org/topics/dermatofibroma	Dermatofibroma
`m2w2:A04`	Dermatoscopy of dermatofibroma	© DermNet	dermnetnz.org/topics/dermatofibroma-images	Dermatofibroma
`m2w2:A05`	Dermoscopy. Bland, structureless pigmentation: dermatofibroma	© DermNet	dermnetnz.org/topics/dermatofibroma-images	Dermatofibroma
`m2w2:B01`	Image provided by Dr Trevor Evans	© DermNet	dermnetnz.org/topics/chondrodermatitis-nodularis	Chondrodermatitis nodularis helicis & ear nodules
`m2w2:B02`	Chondrodermatitis	© DermNet	dermnetnz.org/topics/chondrodermatitis-nodularis	Chondrodermatitis nodularis helicis & ear nodules
`m2w2:B03`	Chondrodermatitis	© DermNet	dermnetnz.org/topics/chondrodermatitis-nodularis	Chondrodermatitis nodularis helicis & ear nodules
`m2w2:C01`	Gouty tophi	© DermNet	dermnetnz.org/topics/gout	Chondrodermatitis nodularis helicis & ear nodules
`m2w2:C02`	Keratoacanthoma	© DermNet	dermnetnz.org/topics/keratoacanthoma	Chondrodermatitis nodularis helicis & ear nodules
`m2w2:D01`	Seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-images	Seborrhoeic keratosis
`m2w2:D02`	Seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-images	Seborrhoeic keratosis
`m2w2:D03`	Seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis	Seborrhoeic keratosis
`m2w2:D04`	Seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis	Seborrhoeic keratosis
`m2w2:D05`	Seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis	Seborrhoeic keratosis
`m2w2:D06`	Irritated seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis	Seborrhoeic keratosis
`m2w2:D07`	Seborrhoeic keratoses	© DermNet	dermnetnz.org/topics/sign-of-leser-trelat	Seborrhoeic keratosis

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w2:E01	Seborrhoeic keratosis dermoscopy	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-dermoscopy-images	Seborrhoeic keratosis
m2w2:E02	Seborrhoeic keratosis dermoscopy	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-dermoscopy-images	Seborrhoeic keratosis
m2w2:E03	Seborrhoeic keratosis dermoscopy	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-dermoscopy-images	Seborrhoeic keratosis
m2w2:E04	Seborrhoeic keratosis dermoscopy	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-dermoscopy-images	Seborrhoeic keratosis
m2w2:E05	Dermoscopy of seborrhoeic keratosis	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-dermoscopy-images	Seborrhoeic keratosis
m2w2:E06	Multiple colours	© DermNet	dermnetnz.org/topics/seborrhoeic-keratosis-dermoscopy-images	Seborrhoeic keratosis
m2w2:F01	Clear cell acanthoma	© DermNet	dermnetnz.org/topics/clear-cell-acanthoma	Clear cell acanthoma
m2w2:F02	Clear cell acanthoma	© DermNet	dermnetnz.org/topics/clear-cell-acanthoma	Clear cell acanthoma
m2w2:F03	Clear cell acanthoma	© DermNet	dermnetnz.org/topics/clear-cell-acanthoma	Clear cell acanthoma

Module 2 · Week 3 — skin cancer & lymphoma

29 images · assets/img/m2w3

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w3:A01	Basal cell carcinoma affecting the nose	© Raimo Suhonen	dermnetnz.org/topics/basal-cell-carcinoma-affecting-the-nose-images	Basal cell carcinoma (BCC)
m2w3:A02	A typical ulcerated nodular basal cell carcinoma - note the pearly appearence and broken telangiectatic vessels	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:A03	A nodular basal cell carcinoma on the forehead - note the raised pearly edge, telangiectasia, and central depression	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:A04	An advanced and neglected basal cell carcinoma on the eyelid	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:A05	A nodular pigmented basal cell carcinoma in a Māori patient	© Te Whatu Ora	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w3:B01	A slowly-spreading pink and well-marginated superficial basal cell carcinoma	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:B02	A morphoeic basal cell carcinoma on the lip - note the branched telangiectasia within the scar-like lesion	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:B03	A pigmented basal cell carcinoma on the nose	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:C01	Dermoscopy shows the broken arborising telangiectasia and central featureless sclerosis	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:C02	Blue-grey ovoid nests in pigmented basal cell carcinoma dermoscopy	© DermNet	dermnetnz.org/topics/basal-cell-carcinoma-dermoscopy	Basal cell carcinoma (BCC)
m2w3:C03	Spoke wheels in pigmented basal cell carcinoma dermoscopy	© DermNet	dermnetnz.org/topics/basal-cell-carcinoma-dermoscopy	Basal cell carcinoma (BCC)
m2w3:C04	A nodular basal cell carcinoma - the peripheral palisading of the basaloid cells can be appreciated	© DermNet	dermnetnz.org/images/basal-cell-carcinoma-images	Basal cell carcinoma (BCC)
m2w3:D01	Loss of the ala nasi from picking in trigeminal trophic syndrome	© DermNet	dermnetnz.org/topics/trigeminal-trophic-syndrome	Basal cell carcinoma (BCC)
m2w3:D02	Excoriation of the ala nasi in trigeminal trophic syndrome	© Te Whatu Ora	dermnetnz.org/topics/trigeminal-trophic-syndrome	Basal cell carcinoma (BCC)
m2w3:D03	Multiple BCCs and scars	© DermNet	dermnetnz.org/topics/basal-cell-naevus-syndrome	Basal cell carcinoma (BCC)
m2w3:D04	Palmar pits	© DermNet	dermnetnz.org/topics/basal-cell-naevus-syndrome	Basal cell carcinoma (BCC)
m2w3:E01	An ulcerated and firm lesion on the lip - the patient was a smoker and the lesion was a squamous cell carcinoma	© DermNet	dermnetnz.org/images/cutaneous-squamous-cell-carcinoma-images	Cutaneous squamous cell carcinoma (cSCC)
m2w3:E02	A bleeding and ulcerated poorly-differentiated squamous cell carcinoma on the ear	© DermNet	dermnetnz.org/images/cutaneous-squamous-cell-carcinoma-images	Cutaneous squamous cell carcinoma (cSCC)
m2w3:E03	An ulcerated squamous cell carcinoma on the ear; there is enlargement of the pre-auricular lymph node due to spread of the tumour	© DermNet	dermnetnz.org/images/cutaneous-squamous-cell-carcinoma-images	Cutaneous squamous cell carcinoma (cSCC)
m2w3:E04	Cutaneous horn	© Te Whatu Ora	dermnetnz.org/topics/cutaneous-squamous-cell-carcinoma	Cutaneous squamous cell carcinoma (cSCC)

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w3:E05	A squamous cell carcinoma has arisen within a longstanding venous ulcer	© DermNet	dermnetnz.org/images/cutaneous-squamous-cell-carcinoma-images	Cutaneous squamous cell carcinoma (cSCC)
m2w3:E06	Carcinoma cuniculatum	© DermNet	dermnetnz.org/topics/cutaneous-squamous-cell-carcinoma	Cutaneous squamous cell carcinoma (cSCC)
m2w3:F01	Actinic cheilitis	© DermNet	dermnetnz.org/topics/actinic-cheilitis	Cutaneous squamous cell carcinoma (cSCC)
m2w3:F02	Squamous cell carcinoma in situ	© DermNet	dermnetnz.org/topics/intraepidermal-squamous-cell-carcinoma	Cutaneous squamous cell carcinoma (cSCC)
m2w3:G01	Patches and plaques of mycosis fungoides	© Raimo Suhonen	dermnetnz.org/topics/mycosis-fungoides	Mycosis fungoides & cutaneous lymphoma
m2w3:G02	Plaque stage mycosis fungoides	© Te Whatu Ora	dermnetnz.org/topics/mycosis-fungoides	Mycosis fungoides & cutaneous lymphoma
m2w3:G03	Tumour stage of CTCL	© DermNet	dermnetnz.org/topics/cutaneous-t-cell-lymphoma	Mycosis fungoides & cutaneous lymphoma
m2w3:G04	Sézary syndrome	© DermNet	dermnetnz.org/topics/cutaneous-t-cell-lymphoma	Mycosis fungoides & cutaneous lymphoma
m2w3:G05	Mycosis fungoides, histology	© DermNet	dermnetnz.org/topics/mycosis-fungoides	Mycosis fungoides & cutaneous lymphoma

Module 2 · Week 4 — naevi

26 images · assets/img/m2w4

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w4:A01	Junctional naevus	© Te Whatu Ora	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:A02	Mole: compound naevus	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:A03	Mole: dermal naevus	© Te Whatu Ora	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:A04	miescher	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:A05	Unna naevus	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w4:A06	common mole	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:B01	Dermoscopy: reticular naevus	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:B02	Globular naevus dermoscopy	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:B03	Dermoscopy: starburst naevus	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole
m2w4:B04	Dermoscopy of an atypical naevus with peripheral dots	© Te Whatu Ora	dermnetnz.org/topics/atypical-naevus-images	Melanocytic naevi (moles) & the changing mole
m2w4:B05	Dermoscopy of a dysplastic naevus	© Te Whatu Ora	dermnetnz.org/topics/atypical-naevus-images	Melanocytic naevi (moles) & the changing mole
m2w4:C01	Congenital naevus	© DermNet	dermnetnz.org/topics/congenital-melanocytic-naevi	Congenital melanocytic naevi (CMN)
m2w4:C02	Congenital naevus	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Congenital melanocytic naevi (CMN)
m2w4:C03	Hairy congenital naevi	© Te Whatu Ora	dermnetnz.org/topics/melanocytic-naevus	Congenital melanocytic naevi (CMN)
m2w4:C04	Congenital naevus	© Raimo Suhonen	dermnetnz.org/topics/congenital-melanocytic-naevi	Congenital melanocytic naevi (CMN)
m2w4:D01	Atypical naevi	© DermNet	dermnetnz.org/topics/atypical-naevus-images	Atypical naevus syndrome & familial melanoma
m2w4:D02	Atypical naevi	© DermNet	dermnetnz.org/topics/atypical-naevus-images	Atypical naevus syndrome & familial melanoma
m2w4:D03	A variably-pigmented, irregularly-marginated naevus - histology showed a dysplastic naevus	© Te Whatu Ora	dermnetnz.org/topics/atypical-naevus-images	Atypical naevus syndrome & familial melanoma
m2w4:D04	Asymmetric, variably-pigmented, and irregularly-marginated atypical naevus	© Te Whatu Ora	dermnetnz.org/topics/atypical-naevus-images	Atypical naevus syndrome & familial melanoma
m2w4:D05	dysplastic	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Atypical naevus syndrome & familial melanoma

Image credits

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w4:E01	Halo naevus	© DermNet	dermnetnz.org/topics/halo-naevus	Halo naevus (Sutton naevus)
m2w4:E02	Halo naevus dermoscopy	© DermNet	dermnetnz.org/topics/halo-naevus	Halo naevus (Sutton naevus)
m2w4:E03	A blue naevus - the monochrome slate blue colour is typical	© DermNet	dermnetnz.org/topics/blue-naevus	Blue naevus
m2w4:E04	Classic Spitz naevus	© DermNet	dermnetnz.org/topics/spitz-naevus	Spitz and Reed naevi
m2w4:E05	Pigmented Spitz naevus	© DermNet	dermnetnz.org/topics/spitz-naevus	Spitz and Reed naevi
m2w4:E06	Recurrent naevus	© DermNet	dermnetnz.org/topics/melanocytic-naevus	Melanocytic naevi (moles) & the changing mole

Module 2 · Week 5 — melanoma

26 images · assets/img/m2w5

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w5:A01	Nodular melanoma	© Te Whatu Ora	dermnetnz.org/topics/nodular-melanoma	Melanoma: recognition & referral
m2w5:A02	An ulcerated and crusted purple-red rapidly growing lesion on the back - histology showed a nodular melanoma	© Te Whatu Ora	dermnetnz.org/images/nodular-melanoma-images	Melanoma: recognition & referral
m2w5:A03	A nodular lesion with a medial satelllite lesion - the larger part is pink with dark brown and black speckles suggesting an amelanotic nodular malignant melanoma	© DermNet	dermnetnz.org/images/nodular-melanoma-images	Melanoma: recognition & referral
m2w5:A04	Ulcerated nodular melanoma	© DermNet	dermnetnz.org/images/nodular-melanoma-images	Melanoma: recognition & referral
m2w5:A05	Amelanotic nodular melanoma	© Raimo Suhonen	dermnetnz.org/images/nodular-melanoma-images	Melanoma: recognition & referral
m2w5:A06	Dermoscopy of nodular melanoma	© DermNet	dermnetnz.org/images/nodular-melanoma-images	Melanoma: recognition & referral
m2w5:A07	Dermoscopy shows a pink lesion with atypical vessels and speckles of pigment peripherally - it was an 8.3 mm Breslow thickness malignant melanoma	© DermNet	dermnetnz.org/images/nodular-melanoma-images	Melanoma: recognition & referral
m2w5:B01	Asymmetry of shape, structure and colour	© Te Whatu Ora	dermnetnz.org/topics/abcdes-of-melanoma	Melanoma: recognition & referral

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w5:B02	Border irregularity	© Te Whatu Ora	dermnetnz.org/topics/abcdes-of-melanoma	Melanoma: recognition & referral
m2w5:B03	Colour variation	© Te Whatu Ora	dermnetnz.org/topics/abcdes-of-melanoma	Melanoma: recognition & referral
m2w5:B04	2012	© DermNet	dermnetnz.org/topics/abcdes-of-melanoma	Melanoma: recognition & referral
m2w5:B05	2017	© DermNet	dermnetnz.org/topics/abcdes-of-melanoma	Melanoma: recognition & referral
m2w5:C01	Multiple areas of notching of the border of this ugly duckling lesion - a malignant melanoma	© Te Whatu Ora	dermnetnz.org/images/superficial-spreading-melanoma-images	Melanoma: recognition & referral
m2w5:C02	A changing unusual mole on the left shoulder - the 'ugly duckling' sign of melanoma	© DermNet	dermnetnz.org/images/nodular-melanoma-images	Melanoma: recognition & referral
m2w5:C03	Dermoscopy of a superficial spreading malignant melanoma - irregular edge and pigmentation and a blue-grey veil	© Te Whatu Ora	dermnetnz.org/images/superficial-spreading-melanoma-images	Melanoma: recognition & referral
m2w5:C04	A changing irregularly pigmented lesion with a new nodule at the inferior pole - a malignant melanoma	© Te Whatu Ora	dermnetnz.org/images/superficial-spreading-melanoma-images	Melanoma: recognition & referral
m2w5:D01	Lentigo maligna	© Te Whatu Ora	dermnetnz.org/topics/lentigo-maligna-and-lentigo-maligna-melanoma	Melanoma: recognition & referral
m2w5:D02	Lentigo maligna	© DermNet	dermnetnz.org/topics/lentigo-maligna-and-lentigo-maligna-melanoma	Melanoma: recognition & referral
m2w5:D03	Slowly enlarging irregularly pigmented lesion on the heel present for 3 years	© DermNet	dermnetnz.org/topics/acral-lentiginous-melanoma	Melanoma: recognition & referral
m2w5:D04	An advanced acral lentiginous melanoma in skin of colour on the heel	© DermNet	dermnetnz.org/images/acral-lentiginous-melanoma-images	Melanoma: recognition & referral
m2w5:D05	Parallel ridge pattern, asymmetrical structure and diffuse pigmentation	© DermNet	dermnetnz.org/images/acral-lentiginous-melanoma-images	Melanoma: recognition & referral
m2w5:D06	An irregularly marginated red lesion on the arm - an amelanotic melanoma	© Te Whatu Ora	dermnetnz.org/images/amelanotic-melanoma-images	Melanoma: recognition & referral

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w5:E01	In transit metastases from primary melanoma on foot	© DermNet	dermnetnz.org/topics/metastatic-melanoma	Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment
m2w5:E02	Nodal melanoma	© DermNet	dermnetnz.org/topics/metastatic-melanoma	Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment
m2w5:E03	Multiple blue nodules of cutaneous metastatic malignant melanoma	© DermNet	dermnetnz.org/topics/melanoma	Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment
m2w5:E04	Sentinel node biopsy persistent tattoo	© DermNet	dermnetnz.org/topics/sentinel-lymph-node-biopsy	Melanoma: Breslow, AJCC 8 staging & NICE NG14 treatment

Module 2 · Week 6 — skin surgery

24 images · assets/img/m2w6

CODE	CAPTION	©	SOURCE PAGE	USED ON
m2w6:A01	New lesion	© DermNet	dermnetnz.org/topics/pyogenic-granuloma	Pyogenic granuloma
m2w6:A02	2 weeks later	© DermNet	dermnetnz.org/topics/pyogenic-granuloma	Pyogenic granuloma
m2w6:A03	4 weeks after first photograph	© DermNet	dermnetnz.org/topics/pyogenic-granuloma	Pyogenic granuloma
m2w6:A04	Red clods separated by white lines in pyogenic granuloma	© DermNet	dermnetnz.org/topics/pyogenic-granuloma	Pyogenic granuloma
m2w6:A05	Pyogenic granuloma	© DermNet	dermnetnz.org/topics/pyogenic-granuloma	Pyogenic granuloma
m2w6:B01	Excision wound	© DermNet	dermnetnz.org/topics/excision-of-skin-lesions	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:B02	Sutured wound	© DermNet	dermnetnz.org/topics/excision-of-skin-lesions	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:B03	Healed wound	© DermNet	dermnetnz.org/topics/excision-of-skin-lesions	Skin surgery procedures: excision, curettage, cryotherapy & Mohs

Image credits

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m2w6:B04	Excision selecting relaxed tension lines	© DermNet	dermnetnz.org/topics/skin-tension-lines	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:C01	Surgical wound dehiscence	© DermNet	dermnetnz.org/topics/risks-and-complications-of-skin-surgery	Skin surgery: consent, complications & danger zones
m2w6:C02	Surgical wound dehiscence	© DermNet	dermnetnz.org/topics/risks-and-complications-of-skin-surgery	Skin surgery: consent, complications & danger zones
m2w6:C03	Hypertrophic scar	© DermNet	dermnetnz.org/topics/risks-and-complications-of-skin-surgery	Wound aftercare: sutures, activity, infection & scars
m2w6:C04	Atrophic scar	© DermNet	dermnetnz.org/topics/risks-and-complications-of-skin-surgery	Wound aftercare: sutures, activity, infection & scars
m2w6:C05	A large post-surgical presternal keloid	© DermNet	dermnetnz.org/images/keloid-and-hypertrophic-scar-images	Wound aftercare: sutures, activity, infection & scars
m2w6:C06	A keloid on the earlobe after piercing	© Dr Richard Ashton	dermnetnz.org/images/keloid-and-hypertrophic-scar-images	Wound aftercare: sutures, activity, infection & scars
m2w6:D01	A cryotherapy blister 48 hrs after treatment	© DermNet	dermnetnz.org/topics/cryotherapy	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:D02	Haemorrhagic blisters 3 days after treatment	© DermNet	dermnetnz.org/topics/cryotherapy	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:D03	Eyelid oedema 24 hrs after cryotherapy to the lesion adjacent to the nose	© DermNet	dermnetnz.org/topics/cryotherapy	Skin surgery procedures: excision, curettage, cryotherapy & Mohs

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m2w6:D04	Recurrent BCC post-cryotherapy	© DermNet	dermnetnz.org/topics/cryotherapy	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:E01	A positive latex prick test	© DermNet	dermnetnz.org/topics/latex-allergy	Local anaesthetics & allergy in skin surgery
m2w6:E02	An immediate type 1 reaction after exposure to a rubber balloon; the cheeks and lids have swollen in the last 10 minutes	© DermNet	dermnetnz.org/topics/latex-allergy	Local anaesthetics & allergy in skin surgery
m2w6:F01	Recurrent basal cell carcinoma	© DermNet	dermnetnz.org/topics/mohs-micrographic-surgery	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:F02	Skin flap reconstruction 12 months after a large BCC excision	© Te Whatu Ora	dermnetnz.org/topics/flaps	Skin surgery procedures: excision, curettage, cryotherapy & Mohs
m2w6:F03	The established full thickness skin graft after skin cancer removal on the nose	© DermNet	dermnetnz.org/topics/skin-grafting	Skin surgery procedures: excision, curettage, cryotherapy & Mohs

Card & differential photos (DermNet / Wikimedia Commons)

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v6:V001	Severe lip angioedema	© DermNet	dermnetnz.org/topics/angioedema	ACE-inhibitor (bradykinin) angioedema
v6:V002	Ecthyma	© DermNet	dermnetnz.org/topics/ecthyma	—
v6:V003	Janeway lesion	Warfieldian · CC BY-SA 4.0	commons.wikimedia.org/wiki/File:Janeway_lesion.JPG	—
v6:V004	Necrotising fasciitis	© DermNet	dermnetnz.org/topics/necrotising-fasciitis	—

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v6:V005	Superficial blistering over the axillae, face and chest due to staphylococcal scalded skin sydrome	© Dr Lawrence Meyerson	dermnetnz.org/topics/staphylococcal-scalded-skin-syndrome	—
v6:V006	Sunburn from a day at the pool - the skin below the straps is spared	© DermNet	dermnetnz.org/topics/sunburn	—
v6:V007	Drug-induced subacute lupus erythematosus	© DermNet	dermnetnz.org/topics/subacute-cutaneous-lupus-erythematosus	—
v6:V008	Cutaneous B-cell lymphoma	© DermNet	dermnetnz.org/topics/cutaneous-b-cell-lymphoma	—
v6:V009	Acute gout	© DermNet	dermnetnz.org/topics/gout	—
v6:V010	Deep vein thrombosis of the right leg	James Heilman, MD · CC BY-SA 3.0	commons.wikimedia.org/wiki/File:Deep_vein_thrombosis_of_the_right_leg.jpg	—
v6:V011	Elephantiasis nostras verruciformis	© Raimo Suhonen	dermnetnz.org/topics/lymphoedema	—
v6:V012	Acute lipodermatosclerosis	© DermNet	dermnetnz.org/topics/lipodermatosclerosis	—
v6:V013	Pressure sore	© DermNet	dermnetnz.org/topics/pressure-ulcer	—
v6:V014	Martorell ulcer	© DermNet	dermnetnz.org/topics/martorell-ulcer	—
v6:V015	Actinic prurigo	© DermNet	dermnetnz.org/topics/actinic-prurigo	—
v6:V016	Aphthous ulcer	© DermNet	dermnetnz.org/topics/aphthous-ulcer	—
v6:V017	Mycobacterium marinum	© DermNet	dermnetnz.org/topics/atypical-mycobacterial-infection	—
v6:V018	Vesicular stage of incontinentia pigmenti	© DermNet	dermnetnz.org/topics/incontinentia-pigmenti	—
v6:V019	Friction Blisters On Human Foot	AndryFrench · CC BY 3.0	commons.wikimedia.org/wiki/File:Friction_Blisters_On_Human_Foot.jpg	—
v6:V020	Body lice	Janice Harney Carr, Center for Disease Control · Public domain	commons.wikimedia.org/wiki/File:Body_lice.jpg	—
v6:V021	Dermatitis artefacta	© DermNet	dermnetnz.org/topics/dermatitis-artefacta	—

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v6:V022	Chilblains	© DermNet	dermnetnz.org/topics/chilblains	—
v6:V023	Confluent and reticulated papillomatosis	© DermNet	dermnetnz.org/topics/confluent-and-reticulated-papillomatosis	—
v6:V024	Excoriations	© DermNet	dermnetnz.org/topics/delusions-of-parasitosis	—
v6:V025	Erythropoietic protoporphyria	© DermNet	dermnetnz.org/topics/erythropoietic-protoporphyria	—
v6:V026	Extramammary Paget disease of the skin of the vulva	© DermNet	dermnetnz.org/topics/extramammary-paget-disease	—
v6:V027	Rash due to infectious mononucleosis plus amoxicillin	© DermNet	dermnetnz.org/topics/infectious-mononucleosis	—
v6:V028	Hailey-Hailey disease in axilla	© DermNet	dermnetnz.org/topics/hailey-hailey-disease	—
v6:V029	Juvenile xanthogranuloma	© DermNet	dermnetnz.org/topics/juvenile-xanthogranuloma	—
v6:V030	Plaque stage KS	© DermNet	dermnetnz.org/topics/kaposi-sarcoma	—
v6:V031	Kawasaki symptoms A	Kawasaki_symptoms.jpg: Dong Soo Kim derivative work: Natr (talk) · CC BY 2.0	commons.wikimedia.org/wiki/File:Kawasaki_symptoms_A.jpg	—
v6:V032	Keratosis pilaris	© DermNet	dermnetnz.org/topics/keratosis-pilaris	—
v6:V033	Rash Day 3	© DermNet	dermnetnz.org/topics/measles	—
v6:V034	Merkel cell carcinoma	© DermNet	dermnetnz.org/topics/merkel-cell-carcinoma	—
v6:V035	Paedarus littoralis	Open access image	dermnetnz.org/topics/paederus-dermatitis	—
v6:V036	Perianal streptococcal dermatitis and vulvovaginitis	© DermNet	dermnetnz.org/topics/perianal-streptococcal-dermatitis	—
v6:V037	Podoconiosis	Open access image	dermnetnz.org/topics/podoconiosis	—
v6:V038	Pretibial myxoedema	© DermNet	dermnetnz.org/topics/pretibial-myxoedema	—

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v6:V039	Chloronychia1	Ekem · CC0	commons.wikimedia.org/wiki/File:Chloronychia1.jpg	—
v6:V040	Rocky Mountain spotted fever PHIL 1962 lores	Unknown author · Public domain	commons.wikimedia.org/wiki/File:Rocky_Mountain_spotted_fever_PHIL_1962_lores.jpg	—
v6:V041	Schnitzler syndrome	© DermNet	dermnetnz.org/topics/schnitzler-syndrome	—
v6:V042	Perifollicular haemorrhage and ecchymoses	© DermNet	dermnetnz.org/topics/scurvy	—
v6:V043	Mastocytoma	© DermNet	dermnetnz.org/topics/mastocytoma	—
v6:V045	Variegate porphyria	© DermNet	dermnetnz.org/topics/variegate-porphyria	—
v6:V046	Venous lake	© DermNet	dermnetnz.org/topics/venous-lake	—
v6:V047	Zika virus infection	© DermNet	dermnetnz.org/topics/zika-virus	—
v6:V048	Ash leaf marks in tuberous sclerosis	© DermNet	dermnetnz.org/topics/tuberous-sclerosis	—
v6:V049	WomenWithAddison	James Heilman, MD · CC BY-SA 4.0	commons.wikimedia.org/wiki/File:WomenWithAddison.jpg	Addison's disease: skin signs
v6:V050	Site of tick removal	Denys Williams · CC BY-SA 3.0	commons.wikimedia.org/wiki/File:Site_of_tick_removal.jpg	—
v6:V051	Ingrown toenail	© DermNet	dermnetnz.org/topics/ingrown-toenail	Ingrowing toenail (onychocryptosis)
v6:V052	Ingrown toenail	© DermNet	dermnetnz.org/topics/ingrown-toenail	Ingrowing toenail (onychocryptosis)
v6:V053	Ingrown toenail	© DermNet	dermnetnz.org/topics/ingrown-toenail	Ingrowing toenail (onychocryptosis)
v6:V054	Periocular milia	© DermNet	dermnetnz.org/topics/milium	Milia
v6:V055	Neonatal centrofacial milia	© DermNet	dermnetnz.org/topics/milium	Milia
v6:V056	Milia after pemphigoid blisters	© DermNet	dermnetnz.org/topics/milium	Milia
v6:V057	Dermatitis induced by pembrolizumab	© DermNet	dermnetnz.org/topics/cutaneous-adverse-effects-of-checkpoint-inhibitors	Melanoma immunotherapy & targeted therapy: side-effects

Image credits

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v6:V058	Nivolumab-induced lichenoid dermatitis	© DermNet	dermnetnz.org/topics/cutaneous-adverse-effects-of-checkpoint-inhibitors	Melanoma immunotherapy & targeted therapy: side-effects
v6:V059	Pembrolizumab-induced psoriasis	© DermNet	dermnetnz.org/topics/cutaneous-adverse-effects-of-checkpoint-inhibitors	Melanoma immunotherapy & targeted therapy: side-effects
v6:V060	Excision wound	© DermNet	dermnetnz.org/topics/excision-of-skin-lesions	Skin surgery: pre-operative checks, anticoagulants & cardiac devices
v6:V061	Sutured wound	© DermNet	dermnetnz.org/topics/excision-of-skin-lesions	Skin surgery: pre-operative checks, anticoagulants & cardiac devices
v6:V062	Healed wound	© DermNet	dermnetnz.org/topics/excision-of-skin-lesions	Skin surgery: pre-operative checks, anticoagulants & cardiac devices
v6:V063	Skin examination	© DermNet	dermnetnz.org/topics/dermoscopy	Teledermatology & lesion photography
v6:V064	Photography	© DermNet	dermnetnz.org/topics/dermoscopy	Teledermatology & lesion photography
v6:V065	Perioral erythematous papules sparing the immediate perivermillion skin in perioral dermatitis	© DermNet	dermnetnz.org/topics/periorificial-dermatitis	Perioral (periorificial) dermatitis
v6:V066	Florid perioral dermatitis due to potent topical steroid use	© Dr Richard Ashton	dermnetnz.org/topics/periorificial-dermatitis	Perioral (periorificial) dermatitis
v6:V067	A blue naevus - the monochrome slate blue colour is typical	© DermNet	dermnetnz.org/topics/blue-naevus	Blue naevus
v6:V068	A blue naevus on the hand	© Te Whatu Ora	dermnetnz.org/topics/blue-naevus	Blue naevus
v6:V069	Drug eruption	© DermNet	dermnetnz.org/topics/drug-hypersensitivity-syndrome	DRESS (drug reaction with eosinophilia and systemic symptoms)
v6:V070	A lipoma on the shoulder	© DermNet	dermnetnz.org/topics/lipoma	Lipoma & the sarcoma lump pathway
v6:V071	A selection of emollients from the light to the very greasy - 'the right one is the one the patient likes'	Open access image	dermnetnz.org/topics/emollients-and-moisturisers	Emollients & soap substitutes

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